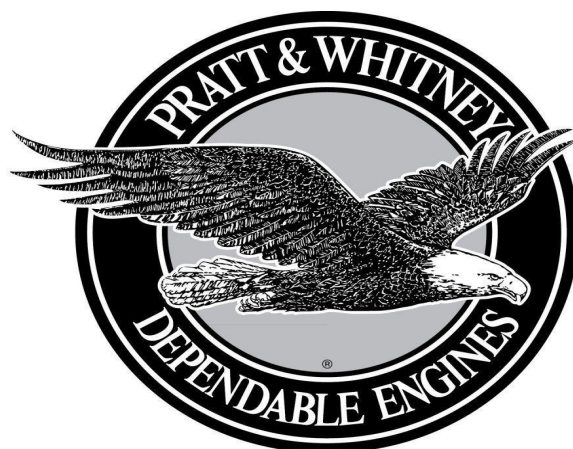


**ENGINEERING REPORT NO. 11709-00 Rev. C
PW127XT-S ENGINE APPLICATION SOFTWARE DESIGN DOCUMENT
SOFTWARE ID 127S0101**



**June 2026
EXPORT CONTROL CLASSIFICATION**

DATE OF MARKING(YYYY-MM-DD):	2026-06-15	BADGE(XX#####):	p939776
-------------------------------------	------------	------------------------	---------

☐

Check this box if the document
contains no technical data

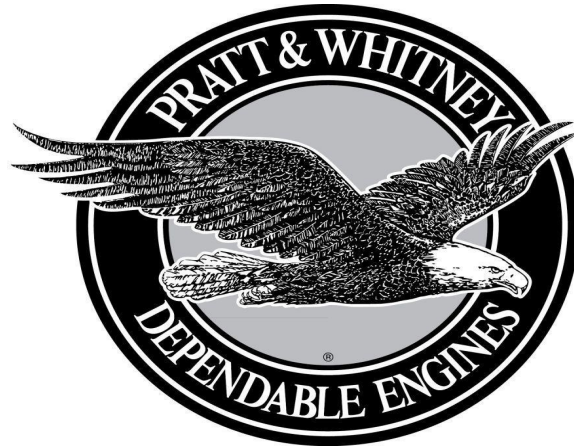
OR

Summarize the export classification of
all content as instructed below

Classifications must always be provided for U.S. and Outside U.S. boxes . When the person completing the grid is physically located outside of the U.S., the local classifications must also be provided.	CLASSIFICATION		
	LOCATION	U.S. REGULATION	
		EAR	ITAR
	U.S.	9E991	NA
	OUTSIDE U.S.	9E991	NA
		LOCAL REGULATION	
	CANADA	EIPA (ECL)	DPA (CG)
	NSR	NO	

**PRATT & WHITNEY CANADA CORP.
1000 Marie-Victorin, Longueuil, Quebec, Canada J4G 1A1**

**ENGINEERING REPORT NO. 11709-00 Rev. C
PW127XT-S ENGINE APPLICATION SOFTWARE DESIGN DOCUMENT
SOFTWARE ID 127S0101**



June 2026

APPROVAL		
Prepared By:	Michel Kandalajt	Submitted by Michel Kandalajt at June 15, 2026 14:42:52
Software Design Lead:	Sivaram Wijenddra	Approved by Sivaram Wijenddra at June 15, 2026 14:45:56
Controls Manager:	Jeffrey Weinstein	Approved by Helene Laflamme at June 16, 2026 13:39:37

**PRATT & WHITNEY CANADA CORP.
1000 Marie-Victorin, Longueuil, Quebec, Canada J4G 1A1**

PROPRIETARY NOTICE

Warning unpublished work © 2023 Collins Aerospace. Within this document, there is, so designated, property of Collins Aerospace and Pratt & Whitney Canada Corp. As documented in the Collins Aerospace/Pratt & Whitney Canada Corp. side letter agreement JKH-21-004 dated 3-19-2021, as modified by amendment 1 JKH-22-002 dated 2-23-2022. You may not possess, use, copy or disclose this document or any information in it, for any purpose, including without limitation, to design, manufacture or repair parts, or obtain any government approval to do so, without express written permission of Collins Aerospace and Pratt & Whitney Canada Corp. Neither receipt nor possession of this document alone, from any source, constitutes such permission. Possession, use, copying or disclosure by anyone without express written permission is not authorized and may result in criminal and/or civil liability.

NOTICE - DISCLOSURE OF INFORMATION

This document contains trade secrets, financial, commercial, scientific, technical or other confidential information, the further disclosure of which will result in material financial harm and/or prejudice to the competitive position of Pratt & Whitney Canada Corp. ("P&WC"). P&WC considers any and all information contained herein or disclosed pursuant hereto, taken separately or as a whole, as a "record" the disclosure of which should be refused pursuant to section 20(1) of the Access to Information Act. If the head of a Government institution or department intends to disclose any of this information, prior written consent must be obtained from Vice President - Counsel and Corporate Secretary, Pratt & Whitney Canada Corp., 1000 Marie-Victorin (01BE5), Longueuil, Quebec J4G 1A1.

REVISION INDEX				
ISSUE	REVI- SION	DATE	DESCRIPTION OF REVISION	PREPARED BY
00	-	28 February 2024	Software Design Document for the PW127XT-S Engine Software Version 127S0000	Helene Laflamme
00	A	19 February 2025	Software Design Document for the PW127XT-S Engine Software Version 127S0100	Javier Manotas
00	B	25 February 2026	Software Design Document for the PW127XT-S Engine Software Version 127S0100 (Note: This version is documentation update only as per DR_7434)	Michel Kandalaft
00	C	12 June 2026	Software Design Document for the PW127XT-S Engine Software Version 127S0101	Michel Kandalaft

Table of Contents

01.00 SCOPE	1
01.01 IDENTIFICATION	1
01.02 SYSTEM OVERVIEW	1
01.03 DOCUMENT OVERVIEW	1
02.00 APPLICABLE DOCUMENTS	1
02.01 GOVERNMENT DOCUMENTS	1
02.02 NON-GOVERNMENT DOCUMENTS	1
02.03 HAMILTON SUNDSTRAND DOCUMENTS	2
02.04 PRATT & WHITNEY CANADA DOCUMENTS	2
03.00 SOFTWARE STRUCTURE	2
03.01 PROGRAM DESCRIPTION	2
04.00 SOFTWARE IMPLEMENTATION	7
04.01 INPUT SIGNAL PROCESSING	7
04.01.01 Low-Level Analog Input Processing	7
04.01.01.01 Low Level Drift Compensation [I01-Sens_LL_Drift_Comp]	7
04.01.01.02 Strain Gauge Compensation [I02-Sens_LL_Strain_Gauge_Comp]	7
04.01.01.03 Cold Junction Temperature Input [I03-Sens_LL_Cold_Junct_Temp]	8
04.01.01.04 PSAMB Strain Gauge [I04-Sens_LL_PSAMB_Strain_Gauge]	8
04.01.01.05 Delta P Strain Gauge [I05-Sens_LL_DELTA_P_Strain_Gauge]	8
04.01.01.06 TQ RTD [I07-Sens_LL_TQ_RTD]	8
04.01.01.07 T18 RTD [I08-Sens_LL_T18_RTD]	8
04.01.02 High-Level Analog Input Processing	8
04.01.02.01 High Level Drift [I09-Sens_HL_Drift]	9
04.01.02.02 High Level Gain Compensation [I10-Sens_HL_Gain_Comp]	9
04.01.02.03 Torque Bias Trim [I11-Sens_HL_Q_Bias_Trim]	9
04.01.02.04 Torque Gain Trim [I12-Sens_HL_Q_Gain_Trim]	9
04.01.02.05 Engine Variant Selection Logic [I13-ENG VARIANT]	9
04.01.02.06 Power Lever Angle (PLA) Process Group	10
04.01.02.06.01 Power Lever Angle [I14-Sens_HL_PLA]	10
04.01.02.06.02 PLA Bias [I15-Sens_HL_PLA_Bias]	11
04.01.02.06.03 PLA Gain [I16-Sens_HL_PLA_Gain]	11
04.01.02.07 Stepper Motor Coil Sum Voltage Input [I38-Sens_HL_SM_Coil_Sum_Volt]	11
04.01.02.08 Bleed Wraparound Input [I45-Sens_HL_Bleed_W/A_Input]	11
04.01.03 Frequency Input Processing	11
04.01.03.01 NH - High Speed Turbine	12
04.01.03.01.01 Start Primary NH Conversion [I17-Sens_FR_Start_Pri_NH_Convert]	12
04.01.03.01.02 Read NH Probe Logic [I18-Sens_FR_Read_NH_Probe]	12
04.01.03.01.03 Dual Probe Logic [I19-Sens_FR_Dual_Probe]	12
04.01.03.01.04 NH Speed Filtering [I20-Sens_FR_NH_Speed_Filter]	12
04.01.03.01.05 Start Secondary NH Conversion [I22-Sens_FR_Start_Sec_NH_Convert]	12
04.01.03.01.06 Read NH Probe Two [I23-Sens_FR_Read_NH_Probe_Two]	13
04.01.03.01.07 NH Probe 2 Filtering [I24-Sens_FR_NH_Probe_2_Filter]	13
04.01.03.02 Torque and Torque Shaft speed Input Processing	13
04.01.03.02.01 Calculate Torque Shaft Deflection and Speed [I25-Sens_FR_Q_NPT_In]	13
04.01.03.02.02 Calculator or Torque from Shaft Deflection [I26-Sens_FR_Q_Shaft_Deflect]	13
04.01.03.02.03 Torque Shaft Speed Filtering [I27-Sens_FR_Q_Shaft_Speed_Filter]	13
04.01.03.02.04 Torque/speed Dual Input Fault Logic [I43-Sens_FR_Q/Speed_Dual_In_Fault]	13
04.01.04 Discrete Input Processing	14
04.01.04.01 Discrete Input Word #1 [I28-DISINW1]	14
04.01.04.02 Discrete Input Word #2 [I29-DISINW2]	15

04.01.04.03 EEC Fault Logic Input Discrete Word [I30-BIT_EEC_Disc_In]	15
04.01.05 UART Input Processing	15
04.01.05.01 N-Parameter Protocol UART Input Processing [I31-NPARAMIN]	15
04.01.05.02 UART Input Processing [I31-UARTIN]	16
04.01.06 ARINC Input Processing [I32-ARINC_HW]	16
04.01.07 Air Data Computer (ADC) Inputs	16
04.01.07.01 Total Air Temperature	16
04.01.07.01.01 ARINC Total Air Temperature Processing [I33-ARINC_In_TAT]	16
04.01.07.01.02 Total Air Temperature Source Selection Logic [I34-ARINC_In_TAT_Source_Sel]	17
04.01.07.02 Ambient Pressure	17
04.01.07.02.01 ARINC Pressure Altitude Processing [I35-ARINC_In_Press_Altitude]	17
04.01.07.02.02 Ambient Pressure Source Selection Logic [I36-ARINC_In_Press_Source_Sel]	18
04.01.07.03 Indicated Air Speed	19
04.01.07.03.01 ARINC Indicated Air Speed Processing [I37-ARINC_IN_IAS]	19
04.01.07.03.02 Indicated Airspeed Fault Logic [I46-ARINC_In_IAS_Fault]	20
04.02 EXECUTIVE PROGRAM	20
04.02.01.01 Reset, PSR/POR and Initialization Executive [E01-BIT_Proc_Reset_PSR_POR]	20
04.02.01.02 RAM Test [E02-BIT_Proc_Ram_Test]	22
04.02.01.03 Complete ROM and EEPROM Checksum [E03-BIT_Proc_ROM_EE_Chksum]	22
04.02.01.04 Instruction Test [E04-BIT_Proc_Instruct_Test]	22
04.02.01.05 RAM Initialization [E05-RAMINIT]	22
04.02.01.06 Torque Test Routine [E06-BIT_GA_Q_Test_Routine]	32
04.02.01.07 Initialization Stepper Motor Sum Voltage Check [E09-BIT_EEC_Init_SM_Sum_Volt]	32
04.02.01.08 Gate Array Selection Logic [E22-BIT_GA_GateArraySelect]	33
04.02.02 Background Executive [E07-Exec_Background]	33
04.02.03 Foreground Executive [E08-EXEC_FOREGND]	33
04.02.04 Foreground Tests	55
04.02.04.01 Partial RAM and BREG Test [E10-BIT_EEC_Partial_RAM_BREG]	55
04.02.04.02 Loss of Connector [E11-BIT_EEC_Loss_Connector]	55
04.02.04.03 MFC Identification Verification [E27-Maint_Misc_MFC_Id]	55
04.02.04.04 Wraparound Tests [E13-WATEST]	55
04.02.04.04.01 Bleed Wraparound Tests [E28-BIT_EEC_Bleed_W/A_Tests]	55
04.02.04.05 EEPROM Fault Storage [E14-FAULT_STORAGE]	56
04.02.04.05.01 Internal EEC Fault Storage [E26-INT_FAULTSTORE]	56
04.02.04.06 Background Functional Test [E15-BIT_Proc_Background]	56
04.02.04.07 Stepper Motor Phase Fault Test [E16-BIT_EEC_SM_Phase]	56
04.02.04.08 Internal EEC Fault Detection [E25-BIT_EEC_Int_Fault_Detect]	56
04.02.05 Fault Logic	57
04.02.05.01 Non-Critical Faults [E17-NONCRITICAL_FAULT]	57
04.02.05.02 Fault Summary [E18-FAULTSUM]	61
04.02.06 EEPROM Trim Data Storage [E19-BIT_Store_EEPROM_Trim_Data]	65
04.02.06.01 PLA Trim EEPROM Data Storage [E23-BIT_Store_PLA_Trim_Data]	67
04.02.07 Test Mode [E20-Maint_Dev_Test_Mode]	67
04.02.08 Reconfiguration Logic [E21-RECONFIG]	68
04.02.09 Program Identification Module [E30-IDENTIFI]	68
04.02.10 External Discrete Word [E31-Maint_Misc_Ext_Disc_Word]	73
04.03 SOFTWARE REQUIREMENTS	73
04.03.01 Fuel Control System - Outer Loop	73
04.03.01.01 NH Governing Loop	73
04.03.01.01.01 NH Minimum/Maximum Calculation [C27-NHMINMAX]	73
04.03.01.01.02 NH Governing Logic [C18-Gov_NH_Closed_Loop]	74
04.03.01.01.03 Closed Loop on Power [C08-Gov_SHP_Closed_Loop]	75
04.03.01.01.04 PLA Trim Calculation [C02-PLATRIM_CALC]	75

04.03.01.02 NPT Governing Logic [C12-Gov_NPT_Closed_Loop]	76
04.03.01.03 NH Acceleration Limiting Logic [C13-Trans_Accel_NH]	78
04.03.01.04 NH Limiting Logic [C14-Limit_NH]	78
04.03.01.05 NH Deceleration Limiting Logic [C15-Trans_Decel_NH]	78
04.03.01.06 Loop Selection Logic [C16-Gov_Inner_Loop_Sel]	78
04.03.02 Fuel Control system - Inner Loop	78
04.03.02.01 Overtorque Protection Logic [C07-LIMIT_OVERQ_PROTECT]	78
04.03.02.02 Propeller Underspeed Logic [C19-Limit_Prop_Underspeed]	79
04.03.02.03 Propeller Overspeed Logic [C20-Limit_Prop_Overspeed]	79
04.03.02.04 Horsepower Request Calculation [C11-Gov_SHP_Req_Calc]	79
04.03.02.05 NH Request Calculation [C17-Gov_Inner_NH_Req_Calc]	79
04.03.02.06 Stepper Motor Rate Request [C30-Gov_Inner_SM_Rate_Req]	79
04.03.03 Rating Logic	79
04.03.03.01 Thermally Limited Rated Power Algorithm [C03-TLIMRT_D]	79
04.03.03.02 Rated Power Calculation [C04-PWRCALC_D]	84
04.03.03.03 Horsepower Reference Calculation [C06-POWER_REQUEST]	85
04.03.03.04 Torque Bug Calculation [C05-Rating_Q_Bug_Calc]	87
04.03.04 Handling Bleed Valve Logic	88
04.03.04.01 EEC Mode Handling Bleed Valve Logic [C21-OUT_HBOV_EEC_MODE]	88
04.03.04.02 Manual Backup Handling Bleed Valve Logic [C22-Out_HBOV_Man_Mode]	88
04.03.05 Start Logic [C01-Start]	88
04.03.06 Miscellaneous Parameter Calculation [C26-MPARAM]	88
04.03.07 Torque Limiting Logic [C31-Limit_Q]	91
04.03.08 Engine Overtorque Test Process [C24-BIT_PreFly_Eng_OverQ_Test]	91
04.03.09 Auto Ignition Logic [C10-Out_Auto_Ignit]	92
04.04 OUTPUT SIGNAL PROCESSING	92
04.04.01 Stepper Motor Output Process [O01-Out_SM]	92
04.04.02 Handling Bleed Process [O02-HBOV]	92
04.04.03 Discrete Output Process [O03-OUT_DISC]	92
04.04.04 UART Output Selection Logic [UART_OUT_SEL]	93
04.04.04.01 UART Output Process [O04-UART_OUT]	94
04.04.04.02 UART Output Process [O04-UART_OUT1]	94
04.04.04.03 N-Parameter Protocol UART Output Processing [O04 - NPARAMOUT]	95
04.04.05 ARINC Output Process Group	95
04.04.05.01 ARINC Label 062 Output Process [O40-ARINCO_062]	95
04.04.05.02 ARINC Label 063 Output Process [O42-ARINCO_063]	99
04.04.05.03 ARINC Label 064 Output Process [O41-ARINCO_064]	102
04.04.05.04 ARINC Label 065 Output Process [O43-ARINCO_065]	105
04.04.05.05 ARINC Label 066 Output Process [O44-ARINCO_066]	108
04.04.05.06 ARINC Label 067 Output Process [O45-ARINCO_067]	111
04.04.05.07 ARINC Label 133 Output Process [O05-ARINCO_133]	115
04.04.05.08 ARINC Label 134 Output Process [O06-ARINCO_134]	115
04.04.05.09 Reserved	116
04.04.05.10 ARINC Label 203 Output Process [O07-ARINCO_203_ALTPS]	116
04.04.05.11 ARINC Label 203 Output Process [O07-ARINCO_203_ALT]	116
04.04.05.12 Reserved	117
04.04.05.13 Reserved	117
04.04.05.14 Reserved	117
04.04.05.15 Reserved	117
04.04.05.16 ARINC Label 211 Output Process [O09-ARINCO_211_TAT11]	117
04.04.05.17 ARINC Label 240 Output Process [O10-ARINCO_240]	118
04.04.05.18 ARINC Label 245 Output Process [O11-ARINCO_245]	118
04.04.05.19 ARINC Label 341 Output Process [O15-ARINCO_341]	119

04.04.05.20 ARINC Label 346 Output Process [O16-ARINCO_346]	120
04.04.05.21 ARINC Label 350 Output Process [O17-ARINCO_350]	120
04.04.05.22 ARINC Label 270 Output Process [O18-ARINCO_270]	121
04.04.05.23 ARINC Label 272 Output Process [O19-ARINCO_272]	125
04.04.05.24 ARINC Output Subroutine [O22-ARINC_Out_SubRoutine]	126
04.04.05.25 Reserved	126
04.04.05.26 ARINC Label 340 Output Process [O26 - ARINCO_340]	126
04.04.05.27 ARINC Label 342 Output Process [O27 - ARINCO_342]	126
04.05 DCU AND LOW CYCLE FATIGUE CAPABILITIES	128
04.05.01 DCU Capabilities	128
04.05.01.01 DCU Communications Protocol [DCUPROTO]	128
04.05.01.02 DCU Block Data Input Logic [DCUREAD]	129
04.05.01.03 DCU Block Data Output Logic [DCUWRITE]	143
04.05.01.04 DCU EEC Data Block Validation Logic [BLK09CHK]	158
04.05.01.05 DCU Engine Data Block Validation Logic [BLK00CHK]	159
04.05.01.06 DCU Component Data Block Validation Logic [BLK01CHK]	160
04.05.01.07 DCU Counter Data Block Validation Logic [BLK305CHK]	161
04.05.01.08 DCU ID Data Block Validation Logic [BLK304CHK]	170
04.05.01.09 DCU LCF Data Block Validation Logic [LCF_FACTCHK]	175
04.05.01.10 DCU Engine Data Block Configuration Logic [DCU_ENG_CFG]	181
04.05.01.11 DCU Component Data Block Configuration Logic [DCU_CP_CFG]	184
04.05.01.12 DCU EEC Data Block Configuration Logic [DCU_EEC_CFG]	185
04.05.01.13 DCU Engine Usage Data Block Configuration Logic [DCU_ENGUSE_CFG]	187
04.05.01.14 DCU Block Data Status Summary Logic [DCUDSTAT]	190
04.05.01.15 EEPROM Counters Configuration Logic [EEPCOUNTERS]	199
04.05.02 LCF Capabilities	204
04.05.02.01 LCF Cycle Counting Logic [CYCLE_COUNTING]	204
04.05.02.02 Engine Usage Historic Data Logic [EVENT_COUNTING]	204
04.05.02.03 Start / Flight Determination Logic [STRT_FLIGHT_DETERM]	207
04.05.02.04 Mission Determination Logic [MISSION_DETERM]	208
04.05.03 EEC Fault EEPROM Mapping [EEPROM_MAP]	209
05.00 MEMORY AND PROCESSING TIME ALLOCATION	214
05.01 MEMORY MARGINS	214
05.02 TIMING MARGINS	215
06.00 CODE TO SDD TO SRS TRACEABILITY	215
06.01 MODULE TO DESIGN TO SRS CAPABILITY TRACEABILITY (SORTED BY MODULE)	215
06.02 RESERVED	253
06.03 IMMATURE LLRS (DOCUMENTATION ONLY)	253
07.00 SRS TO SDD TO CODE TRACEABILITY	255
07.01 SRS CAPABILITY TO DESIGN TO MODULE TRACEABILITY (SORTED BY CAPABILITY)	255
07.02 RESERVED	290
08.00 MODULE SEQUENCING	290
08.01 POR/PSR MODULE SEQUENCING	290
08.02 BACKGROUND MODULE SEQUENCING	291
08.03 REAL-TIME INTERRUPT MODULE SEQUENCING	292
08.04 MODULE FLOW/STATE TABLE FOREGROUND MODULE SEQUENCING	292
08.04.01 Primary State Table	292
08.04.02 Secondary State Table	300
09.00 SET/USE TABLE	312
09.01 SET/USE TABLE (SORTED BY PARAMETER)	312
09.02 SET/USE TABLE (SORTED BY MODULE)	381
10.00 DATA DICTIONARY LISTING	393
10.01 GLOBAL PARAMETERS	393

10.02 SCOPED PARAMETERS	422
10.03 PROGRAM UNITS	444
11.00 Modules Affected	453
12.00 RESERVED	453
13.00 DESIGN MODELS (GPU DIAGRAMS)	453
14.00 DEACTIVATED CODE	1055
15.00 GLOSSARY	1055
16.00 CROSS REFERENCE TO RTCA/DO-178C AND DO-331	1058
17.00 Traceability Matrices	1059
17.01 LLR to HLR Trace Matrix	1059
17.02 HLR to LLR Trace Matrix	1072
18.00 DERIVED REQUIREMENTS JUSTIFICATION	1090
19.00 PARTIALLY DERIVED REQUIREMENTS JUSTIFICATION	1092
20.00 Requirements Allocation Matrix	1093

List of Tables

02.02 Table 1: Non-Government-Documents	1
02.03 Table 1: Hamilton-Sundstrand-Documents	2
02.04 Table 1: Pratt-and-Whitney-Canada-Documents	2
03.01 Table 1: Mux-code-word	6
Table:(LLR 12490064.001-1) Foreground State Table	39
Table:(LLR 12490064.001-2) Code Constants Definition	43
Table:(LLR 12490065.001-3) First Minor Cycle Subroutines	44
Table:(LLR 12490066.002-4) Table - 4: Second Minor Cycle Subroutine Order	48
Table:(LLR 12490067.001-5) Third Minor Cycle Subroutine Order	49
Table:(LLR 12490068.001-6) Fourth Minor Cycle Subroutine Order	52
Table:(LLR 12490596.000-1) Software ID Memory Address	71
Table:(LLR 12490124.001-1) Checksum Memory Address	71
Table:(LLR 12490350.000-1) Table 1 - DCU Read Subroutines	140
Table:(LLR 12490351.000-1) Table 2 - DCU Process Subroutines	141
Table:(LLR 12490352.000-1) Table 3 - DCU Block 0 Read Data Block	141
Table:(LLR 12490354.000-1) Table 4 - DCU Block 1 Read Data Block	142
Table:(LLR 12490355.000-1) Table 5 - DCU Block 304 Read Data Block	142
Table:(LLR 12490356.000-1) Table 6 - DCU Block 305 Read Data Block	143
Table:(LLR 12490357.000-1) Table 7 - DCU Block 9 Read Data Block	143
Table:(LLR 12490397.000-1) Table 1 - DCU Write Subroutine	156
Table:(LLR 12490398.000-1) Table 2 - DCU Block 0 Write Data Info	156
Table:(LLR 12490399.000-1) Table 3 - DCU Block 1 Write Data Info	157
Table:(LLR 12490400.000-1) Table 4 - DCU Block 305 Write Data Info	157
Table:(LLR 12490401.000-1) Table 5 - DCU Block 9 Write Data Info	158
Table:(Ov 12490494.000-1) DCU_BLOCK_00 Parameter Definition	183
Table:(LLR 12490501.000-1) Table 1:DCU_BLOCK_09 Parameter Definition	186
Table:(LLR 12490584.000-1) TABLE - EEC FAULT MEMORY OF DCU BLOCK 0	209
Table:(LLR 12490586.001-1) TABLE 1 : EEC Fault EEPROM Memory Map of DCU Block 1	210
Table:(LLR 12490574.001-1) TABLE 2 : EEC Fault EEPROM Memory Map of EUHD Block 2	212
Table:(LLR 12490576.001-1) TABLE 3 : EEC Fault EEPROM Memory Map of Block 3	213
Table:(LLR 12490578.000-1) TABLE - EEC FAULT EEPROM MEMORY MAP OF REMAINING EPPROM DATA	214
05.00 Table 1: Memory-and-Processing-Time-Allocation	214
05.01 Table 1: Memory Margin	215
05.02 Table 1: Timing Margin	215
06.01 Table 1: MODULE_DESIGN_SRS	216

06.03 Table 1: IMMATURE LLRS	254
07.01 Table 1: SRS_DESIGN_MODULE	255
08.01 Table 1: POR-PSR-Module-Sequencing	291
08.02 Table 1: Background-Module-Sequencing	292
08.03 Table 1: Real-Time-Interrupt-Module-Sequencing	292
09.01 Table 1: Set-Use-by-Parameter	312
09.02 Table 1: Set-Use-by-Module	381
10.01 Table 1: Global-Parameters	393
10.02 Table 1: Scoped-Parameters	422
10.03 Table 1: Program-Units	444
14.00 Table 1: Deactivated-code	1055
15.00 Table 1: Description-of-Terms-Used-Within-SDD	1055
16.00 Table 1: CROSS REFERENCE TO RTCA/DO-178C AND DO-331	1058

List of Figures

13.00 Figure 1: ARCOUT_A	453
13.00 Figure 1: ARCOUT_B	454
13.00 Figure 2: C01_START	455
13.00 Figure 3: C02_PLATRIM_1	456
13.00 Figure 3: C02_PLATRIM_2	457
13.00 Figure 4: C03_TLMRT_1	458
13.00 Figure 5: C03_TLMRT_2_1	459
13.00 Figure 5: C03_TLMRT_2_2	460
13.00 Figure 6: C03_TLMRT_3_1	461
13.00 Figure 6: C03_TLMRT_3_2	462
13.00 Figure 7: C03_TLMRT_4_1	463
13.00 Figure 7: C03_TLMRT_4_2	464
13.00 Figure 8: C03_TLMRT_5_1	465
13.00 Figure 8: C03_TLMRT_5_2	466
13.00 Figure 9: C03_TLMRT_6_1	467
13.00 Figure 9: C03_TLMRT_6_2	468
13.00 Figure 10: C03_TLMRT_7	469
13.00 Figure 11: C04_RPCAL_2_D	470
13.00 Figure 12: C05_TQBUG	471
13.00 Figure 13: C06_HPREF_1	472
13.00 Figure 14: C06_HPREF	473
13.00 Figure 15: C07_OVRTQPRO	474
13.00 Figure 16: C08_SHPCL	475
13.00 Figure 17: C10_ATOIGN	476
13.00 Figure 18: C11_SHPREQ_1	477
13.00 Figure 19: C11_SHPREQ_2	478
13.00 Figure 20: C12_NPTGOV_1	479
13.00 Figure 21: C12_NPTGOV_2	480
13.00 Figure 22: C13_NHACCELLIM	481
13.00 Figure 23: C14_NHLIM	482
13.00 Figure 24: C15_NHDECELLIM	483
13.00 Figure 25: C16_LOOPSEL	484
13.00 Figure 26: C17_NHREQUEST_1	485
13.00 Figure 27: C17_NHREQUEST_2	486
13.00 Figure 28: C18_NHGOV_1	487
13.00 Figure 29: C18_NHGOV_2	488

13.00 Figure 30: C19 _UNDERSPEED	489
13.00 Figure 31: C20 _OVRSPED	490
13.00 Figure 32: C21 _HBOVEM _1	491
13.00 Figure 33: C21 _HBOVEM _2	492
13.00 Figure 34: C22 _HBOVMM _1	493
13.00 Figure 35: C22 _HBOVMM _2	494
13.00 Figure 36: C22 _HBOVMM _3	495
13.00 Figure 37: C24 _OVRTQTST	496
13.00 Figure 38: C26 _MPARAM _1	497
13.00 Figure 39: C26 _MPARAM _2	498
13.00 Figure 40: C26 _MPARAM _3	499
13.00 Figure 41: C26 _MPARAM _4	500
13.00 Figure 42: C26 _MPARAM _5	501
13.00 Figure 43: C26 _MPARAM _6	502
13.00 Figure 44: C26 _MPARAM _7	503
13.00 Figure 45: C27 _NHMNMX _1	504
13.00 Figure 47: C30 _SMRTRQ	505
13.00 Figure 48: C31 _TQLIM	506
13.00 Figure 49: CMPACT _A	507
13.00 Figure 50: CMPACT _B	508
13.00 Figure 51: CMPACT _C	509
13.00 Figure 52: DBL _LL _SUBR _A	510
13.00 Figure 53: DBL _LL _SUBR _B	511
13.00 Figure 54: DBL _LL _SUBR _C	512
13.00 Figure 55: DBL _LL _SUBR _D	513
13.00 Figure 56: DBL _LL _SUBR _E	514
13.00 Figure 58: DERLAG _A	515
13.00 Figure 59: DERLAG _B	516
13.00 Figure 60: DIVPRT _A	517
13.00 Figure 61: DIVPRT _B	518
13.00 Figure 62: DPA _A	519
13.00 Figure 63: DPA _B	520
13.00 Figure 64: DPM _A	521
13.00 Figure 65: DPM _B	522
13.00 Figure 66: DPM _C	523
13.00 Figure 67: DPM _D	524
13.00 Figure 68: DPM _E	525
13.00 Figure 69: E01 _PORPSR _A	526
13.00 Figure 70: E01 _PORPSR _B	527
13.00 Figure 71: E01 _PORPSR _C	528
13.00 Figure 72: E01 _PORPSR _D	529
13.00 Figure 73: E02 _RAMTEST _A	530
13.00 Figure 74: E02 _RAMTEST _B	531
13.00 Figure 75: E02 _RAMTEST _C	532
13.00 Figure 76: E03 _CHECKSUM _A	533
13.00 Figure 77: E03 _CHECKSUM _B	534
13.00 Figure 78: E04 _INSTST _A	535
13.00 Figure 79: E05 _RAMINI _A	536
13.00 Figure 80: E05 _RAMINI _B	537
13.00 Figure 81: E05 _RAMINI _C	538
13.00 Figure 82: E05 _RAMINI _D	539
13.00 Figure 83: E05 _RAMINI _E	540
13.00 Figure 84: E05 _RAMINI _F	541

13.00 Figure 85: E05_RAMINI_G	542
13.00 Figure 86: E05_RAMINI_H	543
13.00 Figure 87: E05_RAMINI_I	544
13.00 Figure 88: E05_RAMINI_J	545
13.00 Figure 89: E05_RAMINI_K	546
13.00 Figure 90: E05_RAMINI_L	547
13.00 Figure 91: E05_RAMINI_M	548
13.00 Figure 92: E06_TQTEST_A	549
13.00 Figure 93: E06_TQTEST_B	550
13.00 Figure 94: E06_TQTEST_C	551
13.00 Figure 95: E07_BG_EXECUTIVE_A	552
13.00 Figure 96: E07_BG_EXECUTIVE_B	553
13.00 Figure 97: E08_FG_EXECUTIVE_A	554
13.00 Figure 98: E08_FG_EXECUTIVE_B	555
13.00 Figure 99: E08_FG_EXECUTIVE_C	556
13.00 Figure 100: E08_FG_EXECUTIVE_D	557
13.00 Figure 101: E08_FG_EXECUTIVE_E	558
13.00 Figure 102: E08_FG_EXECUTIVE_F	559
13.00 Figure 103: E08_FG_EXECUTIVE_G	560
13.00 Figure 104: E08_FG_EXECUTIVE_H	561
13.00 Figure 105: E08_FG_EXECUTIVE_I	562
13.00 Figure 106: E08_FG_SECSEL_A	563
13.00 Figure 107: E09_SMSUMV_A	564
13.00 Figure 108: E09_SMSUMV_B	565
13.00 Figure 109: E10_RAMTEST_A	566
13.00 Figure 110: E10_RAMTEST_B	567
13.00 Figure 111: E11_CONTEST_A	568
13.00 Figure 112: E13_WATEST_1_A	569
13.00 Figure 113: E13_WATEST_1_B	570
13.00 Figure 114: E13_WATEST_2_A	571
13.00 Figure 115: E13_WATEST_3_A	572
13.00 Figure 116: E13_WATEST_4_A	573
13.00 Figure 117: E13_WATEST_5_A	574
13.00 Figure 118: E13_WATEST_6_A	575
13.00 Figure 119: E13_WATEST_7_A	576
13.00 Figure 120: E14_FAULTSTORE_1_A	577
13.00 Figure 121: E14_FAULTSTORE_1_B	578
13.00 Figure 122: E14_FAULTSTORE_1_C	579
13.00 Figure 123: E14_FAULTSTORE_3_A	580
13.00 Figure 124: E14_FAULTSTORE_3_B	581
13.00 Figure 125: E14_FAULTSTORE_3_C	582
13.00 Figure 126: E14_FAULTSTORE_4_A	583
13.00 Figure 127: E14_FAULTSTORE_4_B	584
13.00 Figure 128: E14_FAULTSTORE_5_A	585
13.00 Figure 129: E14_FAULTSTORE_6_A	586
13.00 Figure 130: E14_FAULTSTORE_6_B	587
13.00 Figure 131: E15_BCKGNDTEST_A	588
13.00 Figure 132: E16_SMTEST_A	589
13.00 Figure 133: E16_SMTEST_B	590
13.00 Figure 134: E17_NCFLT_1_A	591
13.00 Figure 135: E17_NCFLT_1_B	592
13.00 Figure 136: E17_NCFLT_2_A	593
13.00 Figure 137: E17_NCFLT_2_B	594

13.00 Figure 138: E17_NCFLT_3_A	595
13.00 Figure 139: E17_NCFLT_3_B	596
13.00 Figure 140: E17_NCFLT_4_A	597
13.00 Figure 141: E17_NCFLT_4_B	598
13.00 Figure 142: E17_NCFLT_5_A	599
13.00 Figure 143: E17_NCFLT_5_B	600
13.00 Figure 144: E17_NCFLT_6_A	601
13.00 Figure 145: E17_NCFLT_6_B	602
13.00 Figure 146: E18_FAULTSUM_1_A	603
13.00 Figure 147: E18_FAULTSUM_1_B	604
13.00 Figure 148: E18_FAULTSUM_2_A	605
13.00 Figure 149: E18_FAULTSUM_2_B	606
13.00 Figure 150: E18_FAULTSUM_3_A	607
13.00 Figure 151: E18_FAULTSUM_3_B	608
13.00 Figure 152: E18_FAULTSUM_4_A	609
13.00 Figure 153: E18_FAULTSUM_4_B	610
13.00 Figure 154: E19_TRMSTORE_1_A	611
13.00 Figure 155: E19_TRMSTORE_2_A	612
13.00 Figure 156: E19_TRMSTORE_4_A	613
13.00 Figure 157: E19_TRMSTORE_5_A	614
13.00 Figure 158: E20_TESTMODE_A	615
13.00 Figure 159: E20_TESTMODE_B	616
13.00 Figure 160: E20_TESTMODE_C	617
13.00 Figure 161: E21_RECONFIG	618
13.00 Figure 162: E22_GASEL_A	619
13.00 Figure 163: E22_GASEL_B	620
13.00 Figure 164: E22_GASEL_C	621
13.00 Figure 165: E22_GASEL_D	622
13.00 Figure 166: E23_PLATRIM_A	623
13.00 Figure 167: E23_PLATRIM_B	624
13.00 Figure 168: E23_PLATRIM_C	625
13.00 Figure 169: E25_FAULTDET_A	626
13.00 Figure 170: E26_FAULTSTORE_1_A	627
13.00 Figure 171: E26_FAULTSTORE_1_B	628
13.00 Figure 172: E26_FAULTSTORE_1_C	629
13.00 Figure 173: E26_FAULTSTORE_2_A	630
13.00 Figure 174: E26_FAULTSTORE_2_B	631
13.00 Figure 175: E26_FAULTSTORE_2_C	632
13.00 Figure 176: E26_FAULTSTORE_3_A	633
13.00 Figure 177: E26_FAULTSTORE_3_B	634
13.00 Figure 178: E26_FAULTSTORE_3_C	635
13.00 Figure 179: E26_FAULTSTORE_3_D	636
13.00 Figure 180: E26_FAULTSTORE_4_A	637
13.00 Figure 181: E26_FAULTSTORE_4_B	638
13.00 Figure 182: E27_MFCID_A	639
13.00 Figure 183: E28_HBOVWA_1_A	640
13.00 Figure 184: E28_HBOVWA_1_B	641
13.00 Figure 185: E28_HBOVWA_2_A	642
13.00 Figure 186: E28_HBOVWA_2_B	643
13.00 Figure 187: E31_EXTDIS_A	644
13.00 Figure 188: E31_EXTDIS_B	645
13.00 Figure 189: E31_EXTDIS_C	646
13.00 Figure 190: E31_EXTDIS_D	647

13.00 Figure 191: E31_EXTDIS_E	648
13.00 Figure 192: E31_EXTDIS_F	649
13.00 Figure 193: E31_EXTDIS_G	650
13.00 Figure 194: EESUM_16_A	651
13.00 Figure 195: EESUM_8_A	652
13.00 Figure 196: FAILED_A	653
13.00 Figure 197: FAILLT_A	654
13.00 Figure 198: FAILLT_B	655
13.00 Figure 199: FILCON_A	656
13.00 Figure 200: FILCON_B	657
13.00 Figure 201: FILTER_A	658
13.00 Figure 202: FILTER_B	659
13.00 Figure 203: FILTER_C	660
13.00 Figure 204: GETWRD_A	661
13.00 Figure 205: GETWRD_B	662
13.00 Figure 206: GOTO_A	663
13.00 Figure 207: I01_LLDRIFT_1_A	664
13.00 Figure 208: I01_LLDRIFT_2_A	665
13.00 Figure 209: I01_LLDRIFT_2_B	666
13.00 Figure 210: I02_SGCOM_1_A	667
13.00 Figure 211: I02_SGCOM_2_A	668
13.00 Figure 212: I02_SGCOM_2_B	669
13.00 Figure 213: I03_RDCJT_1_A	670
13.00 Figure 214: I03_RDCJT_2_A	671
13.00 Figure 215: I03_RDCJT_2_B	672
13.00 Figure 216: I03_RDCJT_2_C	673
13.00 Figure 217: I03_RDCJT_2_D	674
13.00 Figure 218: I03_RDCJT_3_A	675
13.00 Figure 219: I03_RDCJT_3_B	676
13.00 Figure 220: I03_RDCJT_3_C	677
13.00 Figure 221: I04_PSAMBSG_1_A	678
13.00 Figure 222: I04_PSAMBSG_2_A	679
13.00 Figure 223: I04_PSAMBSG_2_B	680
13.00 Figure 224: I04_PSAMBSG_2_C	681
13.00 Figure 225: I04_PSAMBSG_3_A	682
13.00 Figure 226: I04_PSAMBSG_3_B	683
13.00 Figure 227: I04_PSAMBSG_4_A	684
13.00 Figure 228: I04_PSAMBSG_5_A	685
13.00 Figure 229: I05_DELTAPSG_1_A	686
13.00 Figure 230: I05_DELTAPSG_2_A	687
13.00 Figure 231: I05_DELTAPSG_2_B	688
13.00 Figure 232: I05_DELTAPSG_3_A	689
13.00 Figure 233: I05_DELTAPSG_3_B	690
13.00 Figure 234: I05_DELTAPSG_3_C	691
13.00 Figure 235: I05_DELTAPSG_4_A	692
13.00 Figure 236: I05_DELTAPSG_4_B	693
13.00 Figure 237: I05_DELTAPSG_4_C	694
13.00 Figure 238: I05_DELTAPSG_4_D	695
13.00 Figure 239: I07_READTQ_1_A	696
13.00 Figure 240: I07_READTQ_2_A	697
13.00 Figure 241: I07_READTQ_2_B	698
13.00 Figure 242: I07_READTQ_3_A	699
13.00 Figure 243: I07_READTQ_3_B	700

13.00 Figure 244: I07_READTQ_3_C	701
13.00 Figure 245: I08_RDT18_1_A	702
13.00 Figure 246: I08_RDT18_2_A	703
13.00 Figure 247: I08_RDT18_3_A	704
13.00 Figure 248: I08_RDT18_3_B	705
13.00 Figure 249: I08_RDT18_3_C	706
13.00 Figure 250: I08_RDT18_3_D	707
13.00 Figure 251: I08_RDT18_3_E	708
13.00 Figure 252: I08_RDT18_4_A	709
13.00 Figure 253: I08_RDT18_4_B	710
13.00 Figure 254: I09_HLDRIIFT_1_A	711
13.00 Figure 255: I09_HLDRIIFT_2_A	712
13.00 Figure 256: I09_HLDRIIFT_2_B	713
13.00 Figure 257: I10_GAINCOM_1_A	714
13.00 Figure 258: I10_GAINCOM_2_A	715
13.00 Figure 259: I10_GAINCOM_2_B	716
13.00 Figure 260: I11_TQBTRIM_1_A	717
13.00 Figure 261: I11_TQBTRIM_2_A	718
13.00 Figure 262: I11_TQBTRIM_2_B	719
13.00 Figure 263: I11_TQBTRIM_2_C	720
13.00 Figure 264: I12_TQGTRIM_1_A	721
13.00 Figure 265: I12_TQGTRIM_1_B	722
13.00 Figure 266: I12_TQGTRIM_1_C	723
13.00 Figure 267: I12_TQGTRIM_2_A	724
13.00 Figure 268: I12_TQGTRIM_2_B	725
13.00 Figure 269: I12_TQGTRIM_2_C	726
13.00 Figure 270: I13_ENGVARIANT_1_A	727
13.00 Figure 271: I13_ENGVARIANT_2	728
13.00 Figure 272: I14_RDPLA_1_A	729
13.00 Figure 273: I14_RDPLA_1_B	730
13.00 Figure 274: I14_RDPLA_1_C	731
13.00 Figure 275: I14_RDPLA_2_A	732
13.00 Figure 276: I14_RDPLA_2_B	733
13.00 Figure 277: I14_RDPLA_2_C	734
13.00 Figure 278: I14_RDPLA_2_D	735
13.00 Figure 279: I14_RDPLA_3_A	736
13.00 Figure 280: I14_RDPLA_3_B	737
13.00 Figure 281: I14_RDPLA_3_C	738
13.00 Figure 282: I14_RDPLA_3_D	739
13.00 Figure 283: I14_RDPLA_3_E	740
13.00 Figure 284: I14_RDPLA_4_A	741
13.00 Figure 285: I15_PLABIAS_1_A	742
13.00 Figure 286: I15_PLABIAS_2_A	743
13.00 Figure 287: I15_PLABIAS_2_B	744
13.00 Figure 288: I16_PLAGAIN_1_A	745
13.00 Figure 289: I16_PLAGAIN_1_B	746
13.00 Figure 290: I16_PLAGAIN_2_A	747
13.00 Figure 291: I16_PLAGAIN_2_B	748
13.00 Figure 292: I17_PNHCVN_1_A	749
13.00 Figure 293: I17_PNHCVN_1_B	750
13.00 Figure 294: I17_PNHCVN_1_C	751
13.00 Figure 295: I17_PNHCVN_2_A	752
13.00 Figure 296: I18_READNH_A	753

13.00 Figure 297: I18_READNH_B	754
13.00 Figure 298: I18_READNH_C	755
13.00 Figure 299: I18_READNH_D	756
13.00 Figure 300: I18_READNH_E	757
13.00 Figure 301: I18_READNH_F	758
13.00 Figure 302: I18_READNH_G	759
13.00 Figure 303: I18_READNH_H	760
13.00 Figure 304: I18_READNH_I	761
13.00 Figure 305: I18_READNH_J	762
13.00 Figure 306: I18_READNH_K	763
13.00 Figure 307: I19_NHDUAL_1_A	764
13.00 Figure 308: I19_NHDUAL_1_B	765
13.00 Figure 309: I19_NHDUAL_2_A	766
13.00 Figure 310: I20_NHSPEED_A	767
13.00 Figure 311: I22_SNHCNV_1_A	768
13.00 Figure 312: I22_SNHCNV_1_B	769
13.00 Figure 313: I22_SNHCNV_2_A	770
13.00 Figure 314: I23_RDSNHP_A	771
13.00 Figure 315: I23_RDSNHP_B	772
13.00 Figure 316: I23_RDSNHP_C	773
13.00 Figure 317: I23_RDSNHP_D	774
13.00 Figure 318: I23_RDSNHP_E	775
13.00 Figure 319: I23_RDSNHP_F	776
13.00 Figure 320: I23_RDSNHP_G	777
13.00 Figure 321: I23_RDSNHP_H	778
13.00 Figure 322: I23_RDSNHP_I	779
13.00 Figure 323: I23_RDSNHP_J	780
13.00 Figure 324: I23_RDSNHP_K	781
13.00 Figure 325: I24_NHPROBE2_A	782
13.00 Figure 326: I25_TQNPT_1_A	783
13.00 Figure 327: I25_TQNPT_2_A	784
13.00 Figure 328: I25_TQNPT_2_B	785
13.00 Figure 329: I25_TQNPT_2_C	786
13.00 Figure 330: I25_TQNPT_2_D	787
13.00 Figure 331: I25_TQNPT_2_E	788
13.00 Figure 332: I25_TQNPT_2_F	789
13.00 Figure 333: I25_TQNPT_3_A	790
13.00 Figure 334: I25_TQNPT_3_B	791
13.00 Figure 335: I25_TQNPT_3_C	792
13.00 Figure 336: I25_TQNPT_3_D	793
13.00 Figure 337: I25_TQNPT_3_E	794
13.00 Figure 338: I25_TQNPT_3_F	795
13.00 Figure 339: I25_TQNPT_4_A	796
13.00 Figure 340: I25_TQNPT_4_B	797
13.00 Figure 341: I25_TQNPT_4_C	798
13.00 Figure 342: I25_TQNPT_4_D	799
13.00 Figure 343: I25_TQNPT_4_E	800
13.00 Figure 344: I25_TQNPT_4_F	801
13.00 Figure 345: I25_TQNPT_4_G	802
13.00 Figure 346: I25_TQNPT_4_H	803
13.00 Figure 347: I25_TQNPT_4_I	804
13.00 Figure 348: I25_TQNPT_4_J	805
13.00 Figure 349: I25_TQNPT_4_K	806

13.00 Figure 350: I25 _TQNPT_4_L	807
13.00 Figure 351: I25 _TQNPT_4_M	808
13.00 Figure 352: I25 _TQNPT_4_N	809
13.00 Figure 353: I25 _TQNPT_4_O	810
13.00 Figure 354: I25 _TQNPT_5_A	811
13.00 Figure 355: I25 _TQNPT_5_B	812
13.00 Figure 356: I25 _TQNPT_5_C	813
13.00 Figure 357: I25 _TQNPT_5_D	814
13.00 Figure 358: I25 _TQNPT_5_E	815
13.00 Figure 359: I25 _TQNPT_5_F	816
13.00 Figure 360: I25 _TQNPT_5_G	817
13.00 Figure 361: I25 _TQNPT_5_H	818
13.00 Figure 362: I25 _TQNPT_5_I	819
13.00 Figure 363: I25 _TQNPT_5_J	820
13.00 Figure 364: I25 _TQNPT_5_K	821
13.00 Figure 365: I25 _TQNPT_5_L	822
13.00 Figure 366: I25 _TQNPT_5_M	823
13.00 Figure 367: I25 _TQNPT_5_N	824
13.00 Figure 368: I25 _TQNPT_5_O	825
13.00 Figure 369: I25 _TQNPT_6_A	826
13.00 Figure 370: I25 _TQNPT_6_B	827
13.00 Figure 371: I25 _TQNPT_7_A	828
13.00 Figure 372: I26 _TQCAL_A	829
13.00 Figure 373: I26 _TQCAL_B	830
13.00 Figure 374: I27 _TQSPEED_A	831
13.00 Figure 375: I28 _DISINW1_A	832
13.00 Figure 376: I28 _DISINW1_B	833
13.00 Figure 377: I28 _DISINW1_C	834
13.00 Figure 378: I28 _DISINW1_D	835
13.00 Figure 379: I28 _DISINW1_E	836
13.00 Figure 380: I29 _DISINW2_A	837
13.00 Figure 381: I30 _DISINFAULT_A	838
13.00 Figure 382: I31 _UARTIN_A	839
13.00 Figure 383: I31 _UARTIN_NPM_A	840
13.00 Figure 384: I31 _UARTIN_NPM_B	841
13.00 Figure 385: I31 _UARTIN_NPM_C	842
13.00 Figure 386: I31 _UARTIN_NPM_D	843
13.00 Figure 387: I32 _ARINCIN_1_A	844
13.00 Figure 388: I32 _ARINCIN_1_B	845
13.00 Figure 389: I32 _ARINCIN_1_C	846
13.00 Figure 390: I32 _ARINCIN_1_D	847
13.00 Figure 391: I32 _ARINCIN_1_E	848
13.00 Figure 392: I32 _ARINCIN_1_F	849
13.00 Figure 393: I32 _ARINCIN_2_A	850
13.00 Figure 394: I33 _ARITAT_A	851
13.00 Figure 395: I33 _ARITAT_B	852
13.00 Figure 396: I33 _ARITAT_C	853
13.00 Figure 397: I33 _ARITAT_D	854
13.00 Figure 398: I33 _ARITAT_E	855
13.00 Figure 399: I33 _ARITAT_F	856
13.00 Figure 410: I33 _ARITAT_G	857
13.00 Figure 411: I33 _ARITAT_H	858
13.00 Figure 412: I34 _TATSEL_A	859

13.00 Figure 413: I34 _TATSEL _B	860
13.00 Figure 414: I34 _TATSEL _C	861
13.00 Figure 415: I34 _TATSEL _D	862
13.00 Figure 416: I34 _TATSEL _E	863
13.00 Figure 417: I34 _TATSEL _F	864
13.00 Figure 418: I34 _TATSEL _G	865
13.00 Figure 419: I35 _ARIPA _A	866
13.00 Figure 420: I35 _ARIPA _B	867
13.00 Figure 421: I35 _ARIPA _C	868
13.00 Figure 422: I35 _ARIPA _D	869
13.00 Figure 423: I35 _ARIPA _E	870
13.00 Figure 424: I35 _ARIPA _F	871
13.00 Figure 425: I35 _ARIPA _G	872
13.00 Figure 426: I36 _PAMSEL _A	873
13.00 Figure 427: I36 _PAMSEL _B	874
13.00 Figure 428: I36 _PAMSEL _C	875
13.00 Figure 429: I36 _PAMSEL _D	876
13.00 Figure 430: I36 _PAMSEL _E	877
13.00 Figure 431: I36 _PAMSEL _F	878
13.00 Figure 432: I36 _PAMSEL _G	879
13.00 Figure 433: I36 _PAMSEL _H	880
13.00 Figure 434: I37 _ARIISA _1 _A	881
13.00 Figure 435: I37 _ARIISA _1 _B	882
13.00 Figure 436: I37 _ARIISA _1 _C	883
13.00 Figure 437: I37 _ARIISA _1 _D	884
13.00 Figure 438: I37 _ARIISA _1 _E	885
13.00 Figure 439: I37 _ARIISA _1 _F	886
13.00 Figure 440: I37 _ARIISA _1 _G	887
13.00 Figure 441: I37 _ARIISA _1 _H	888
13.00 Figure 442: I37 _ARIISA _2 _A	889
13.00 Figure 443: I37 _ARIISA _2 _B	890
13.00 Figure 444: I38 _SMCSV _1 _A	891
13.00 Figure 445: I38 _SMCSV _1 _B	892
13.00 Figure 446: I38 _SMCSV _2 _A	893
13.00 Figure 447: I38 _SMCSV _2 _B	894
13.00 Figure 448: I38 _SMCSV _2 _C	895
13.00 Figure 449: I43 _TQSPEED _1 _A	896
13.00 Figure 450: I43 _TQSPEED _1 _B	897
13.00 Figure 451: I43 _TQSPEED _2 _A	898
13.00 Figure 452: I43 _TQSPEED _3 _A	899
13.00 Figure 453: I43 _TQSPEED _3 _B	900
13.00 Figure 454: I45 _HBOVWA _1 _A	901
13.00 Figure 455: I45 _HBOVWA _2 _A	902
13.00 Figure 456: I45 _HBOVWA _2 _B	903
13.00 Figure 457: INA _A	904
13.00 Figure 458: INC _CTRS _2	905
13.00 Figure 459: INC _CTRS _3	906
13.00 Figure 460: INC _CTRS _4	907
13.00 Figure 461: INIFLT _A	908
13.00 Figure 462: INIFLT _B	909
13.00 Figure 463: INIFLT _C	910
13.00 Figure 464: INSTST _A	911
13.00 Figure 465: INSTST _B	912

13.00 Figure 466: INSTST _C	913
13.00 Figure 467: INSTST _D	914
13.00 Figure 468: INSTST _E	915
13.00 Figure 469: INSTST _F	916
13.00 Figure 470: INSTST _G	917
13.00 Figure 471: INSTST _H	918
13.00 Figure 472: INSTST _I	919
13.00 Figure 473: INSTST _J	920
13.00 Figure 474: INSTST _K	921
13.00 Figure 475: INSTST _L	922
13.00 Figure 476: INSTST _M	923
13.00 Figure 477: INSTST _N	924
13.00 Figure 478: INSTST _O	925
13.00 Figure 479: INSTST _P	926
13.00 Figure 480: INSTST _Q	927
13.00 Figure 481: INSTST _R	928
13.00 Figure 482: INSTST _S	929
13.00 Figure 483: INSTST _T	930
13.00 Figure 484: INSTST _U	931
13.00 Figure 485: INSTST _V	932
13.00 Figure 486: INSTST _W	933
13.00 Figure 487: INSTST _X	934
13.00 Figure 488: INSTST _Y	935
13.00 Figure 489: INSTST _Z	936
13.00 Figure 490: LDLG _SUBR _A	937
13.00 Figure 491: LDLG _SUBR _B	938
13.00 Figure 492: LDLG _SUBR _C	939
13.00 Figure 493: LOAD _A	940
13.00 Figure 494: LOAD _B	941
13.00 Figure 495: LOGHYS	942
13.00 Figure 496: LOOKUP13	943
13.00 Figure 497: MISSIONS	944
13.00 Figure 498: O01 _SMOUT _A	945
13.00 Figure 499: O01 _SMOUT _B	946
13.00 Figure 500: O01 _SMOUT _C	947
13.00 Figure 501: O02 _HBOVOUT _A	948
13.00 Figure 502: O03 _DISOUT	949
13.00 Figure 503: O03 _DISOUT _A	950
13.00 Figure 504: O04 _UARTOUTSEL	951
13.00 Figure 505: O04 _UARTOUT _DTS _A	952
13.00 Figure 506: O04 _UARTOUT _DTS _B	953
13.00 Figure 507: O04 _UARTOUT _DTS _C	954
13.00 Figure 508: O04 _UARTOUT _DTS _D	955
13.00 Figure 509: O04 _UARTOUT _GDTS _A	956
13.00 Figure 510: O04 _UARTOUT _GDTS _B	957
13.00 Figure 511: O04 _UARTOUT _GDTS _C	958
13.00 Figure 512: O04 _UARTOUT _GDTS _D	959
13.00 Figure 513: O04 _UARTOUT _NPM _A	960
13.00 Figure 514: O04 _UARTOUT _NPM _B	961
13.00 Figure 515: O05 _ARO133 _A	962
13.00 Figure 516: O06 _ARO134 _A	963
13.00 Figure 517: O07 _ARO203 _ALT11 _A	964
13.00 Figure 518: O07 _ARO203 _ALTPS _A	965

13.00 Figure 519: O09 _ARO211 _TAT11	966
13.00 Figure 520: O10 _ARO240 _A	967
13.00 Figure 521: O10 _ARO240 _B	968
13.00 Figure 522: O10 _ARO240 _C	969
13.00 Figure 523: O11 _ARO245 _A	970
13.00 Figure 524: O11 _ARO245 _B	971
13.00 Figure 525: O15 _ARO341 _A	972
13.00 Figure 526: O15 _ARO341 _B	973
13.00 Figure 527: O16 _ARO346 _A	974
13.00 Figure 528: O17 _ARO350 _A	975
13.00 Figure 529: O17 _ARO350 _B	976
13.00 Figure 530: O18 _ARO270 _A	977
13.00 Figure 531: O18 _ARO270 _B	978
13.00 Figure 532: O18 _ARO270 _C	979
13.00 Figure 533: O18 _ARO270 _D	980
13.00 Figure 534: O18 _ARO270 _E	981
13.00 Figure 535: O18 _ARO270 _F	982
13.00 Figure 536: O18 _ARO270 _G	983
13.00 Figure 537: O18 _ARO270 _H	984
13.00 Figure 538: O18 _ARO270 _HA	985
13.00 Figure 539: O18 _ARO270 _I	986
13.00 Figure 540: O19 _ARO272	987
13.00 Figure 541: O19 _ARO272 _A	988
13.00 Figure 543: O26 _ARO340 _A	989
13.00 Figure 544: O27 _ARO342 _A	990
13.00 Figure 545: O27 _ARO342 _B	991
13.00 Figure 546: O27 _ARO342 _A _Subfunc	992
13.00 Figure 547: OPMODE _1	993
13.00 Figure 548: OPMODE _2	993
13.00 Figure 549: OPMODE _3	994
13.00 Figure 550: OTA _A	995
13.00 Figure 551: OTA _B	996
13.00 Figure 552: PASSED _A	997
13.00 Figure 553: PASSLT _A	998
13.00 Figure 554: QUAD _A	999
13.00 Figure 555: QUAD _B	1000
13.00 Figure 556: QUAD _C	1001
13.00 Figure 557: RAMREAD0 _A	1002
13.00 Figure 558: RAMREAD1 _A	1003
13.00 Figure 559: RAMSTOR0 _A	1004
13.00 Figure 560: RAMSTOR1 _A	1005
13.00 Figure 561: RATE _LIMIT	1005
13.00 Figure 562: READ _A	1006
13.00 Figure 563: ROMSUM _A	1007
13.00 Figure 564: ROMSUM _B	1008
13.00 Figure 565: RTDWRITE	1009
13.00 Figure 566: SBIVAR _A	1010
13.00 Figure 567: SBIVAR _B	1011
13.00 Figure 568: SBIVAR _C	1012
13.00 Figure 569: SBIVAR _D	1013
13.00 Figure 570: SBIVAR _E	1014
13.00 Figure 571: SBIVAR _F	1015
13.00 Figure 572: SNDWRD _A	1016

13.00 Figure 573: SUNVAR_A	1017
13.00 Figure 574: SUNVAR_B	1018
13.00 Figure 575: SUNVAR_C	1019
13.00 Figure 576: SYSTEM_RESET	1019
13.00 Figure 577: TUSTIN_SUBR_A	1020
13.00 Figure 578: TUSTIN_SUBR_B	1021
13.00 Figure 579: TUSTIN_SUBR_C	1022
13.00 Figure 580: UART_IO_SELECT	1023
13.00 Figure 581: UNIVAR_A	1024
13.00 Figure 582: UNIVAR_B	1025
13.00 Figure 583: UNIVAR_C	1026
13.00 Figure 584: URTOUT_A	1027
13.00 Figure 585: URTOUT_B	1028
13.00 Figure 586: URTRED_A	1029
13.00 Figure 587: URTRED_B	1030
13.00 Figure 588: VARIABLE_LAG_TUSTIN	1031
13.00 Figure 589: VAR_TUSTIN_SUBR	1032
13.00 Figure 590: VAR_TUSTIN_SUBR_A	1033
13.00 Figure 591: VAR_TUSTIN_SUBR_B	1034
13.00 Figure 592: VAR_TUSTIN_SUBR_C	1035
13.00 Figure 593: WDGRST_A	1036
13.00 Figure 594: WRITE_A	1037
13.00 Figure 595: WRITE_B	1038
13.00 Figure 596: WRITE_C	1039
13.00 Figure 597: WRTEEP_A	1040
13.00 Figure 598: WRTEEP_B	1041
13.00 Figure 599: WRTEEP_C	1042
13.00 Figure 600: DCU_RANGE_CHECKS	1043
13.00 Figure 601: DCU_IS_OBSOLETE_A	1044
13.00 Figure 602: DCU_IS_OBSOLETE_B	1045
13.00 Figure 603: DCU_IS_OBSOLETE_C	1046
13.00 Figure 604: DCU_IS_OBSOLETE_D	1047
13.00 Figure 605: DCU_IS_OBSOLETE_E	1048
13.00 Figure 606: DCU_IS_OBSOLETE_F	1049
13.00 Figure 607: INC_CTRS_1_A	1050
13.00 Figure 608: INC_CTRS_1_B	1051
13.00 Figure 609: INC_CTRS_1_C	1052
13.00 Figure 610: INC_CTRS_1_D	1053
13.00 Figure 611: LOOKUP10	1054

01.00 SCOPE

01.01 IDENTIFICATION

This document is titled PW127XT-S Engine Application Software Design Document Software, ER 11709-00 Rev.C. The document containing the higher-level specifications from which this Computer Software Configuration Item (CSCI) was derived is PW127XT-S Engine Application Software Requirements Document Software, ER 11666-00 Rev.C.

01.02 SYSTEM OVERVIEW

The Electronic Engine Control of the PW127XT-S engine is a single channel, microprocessor-based controller. The Electronic Engine Control software is responsible for control of the PW127XT-S engine system, including managing input/output, failure detection and accommodation, control laws, ground monitoring, and data storage.

01.03 DOCUMENT OVERVIEW

The purpose of this document is to define the design and implementation of the software for an electronic control for the PW127XT-S. The Design shall include software structure, software implementation, memory processing time allocation, module to Capability (module/CCAP) cross reference, and the module flow/state table.

02.00 APPLICABLE DOCUMENTS

Revision numbers or letters for documents that are part of the Software Life Cycle Data can be found in the Life Cycle Data Table in the Software Accomplishment Summary, for the program. For all other documents, at the time of publication of this document, the latest issue of the document, unless otherwise specified, is applicable to the extent defined herein.

02.01 GOVERNMENT DOCUMENTS

None.

02.02 NON-GOVERNMENT DOCUMENTS

02.02 Table 1: Non-Government-Document

RTCA/DO-178B	Software Considerations in Airborne Systems and Equipment Certification
RTCA/DO-178C	Software Considerations in Airborne Systems and Equipment Certification
RTCA/DO-331	Model-Based Development and Verification Supplement to DO-178C and DO-278A

02.03 HAMILTON SUNDSTRAND DOCUMENTS

02.03 Table 1: Hamilton-Sundstrand-Documnet

HSER 7998 Rev. -	Assembler Instruction Manual for HS 16/1600 CMOS Processor
HSER 9644 Rev. B	MAC16/MACLIB Macro Library User's Manual Version 3.3
HSER 9645 Rev. -	MAC16 – HS1600 Macro Preprocessor User's Manual
HSER 10120-002	Control16 Macro Library User's Manual
HSER 13396 Rev. -	User's Manual for RASM
HSER 13397 Rev. -	User's Manual for RLNK
HSER 13855 Rev. D	GPU V3.2 Picture Language Reference Manual
HSER 15893 Rev. -	EEC206-100 Applications Manual
HSER 39419-001	Software Configuration Index for the EEC132-210 Electronic Engine Control for the Pratt & Whitney Canada PW127XT-L Engine
HSER 39423-001	Software Accomplishment Summary for the EEC132-210 Electronic Engine Control for the Pratt & Whitney Canada PW127XT-L Engine

02.04 PRATT & WHITNEY CANADA DOCUMENTS

02.04 Table 1: Pratt-and-Whitney-Canada-Documnet

ER 4852 Rev. 1	Automated Damage Tracking Systems Requirements Document for PWC Engine Applications
ER 5792 Rev. 3	Generic EEC Requirement Specification for Support of an Altair Data Transmission Unit
ER 11666-00 Rev.C	Software Requirements Document for PW127XT-S
ER 11609-01 Rev.-	Purchase Performance Specification No. 3060 Electronic Engine Control Software Requirement Document for PW127XT-S
ER 11427-00 Rev.-	Control System Requirements Document for PW127XT-S
WIM 1063	SOFTWARE DESIGN PROCESS FOR PW127XT-S ENGINE

03.00 SOFTWARE STRUCTURE

03.01 PROGRAM DESCRIPTION

The EEC132-210 control system provides a full range of control for the gas generator fuel flow requirements. Engine data is obtained from dedicated sensors and used to provide fuel metering, engine limit protection, and transient control operation. Provision is also included for cockpit display of rated power in units of torque as a target to which the pilot will match engine output torque. The control system also includes a compressor bleed valve control. There are three exception vectors provided in the CPU; one interrupt vector and two processor reset vectors. The interrupt vector location that is used for real time software scheduling occurs at a rate of 1.04 msec. Control outputs are disabled until this interrupt is enabled. Once enabled, the independent hardware watchdog timer and the loss of activity monitor are automatically enabled. The processor reset vectors are a function of processor reset and boot discrete position. These vectors are used for entry to boot mode or application mode following processor reset.

The EEC132-210 software consists of a collection of modules which are tied to processes through their names. Processes and modules are partitioned into four main categories indicated by the first letter in the process and module name. Four categories: I, O, C, and E comprise the process-specific modules. A fifth category is comprised of the general-purpose support routines used by the four main categories as well as other routines pertaining to DCU (Data Collection Unit) and LCF (Low Cycle Fatigue) functionality. A brief description of each of these sections follows.

INPUTS (I)

Each EEC132-210 processes a variety of inputs in order to perform its function. There is one Analog-to-Digital (A/D) converter and associated multiplexers under software control. This A/D is used to read in temperature and pressure inputs, gain and drift compensation signals, and the power lever signal via the RVDT interface. There are two Frequency-to-Digital (F/D) converters used to read in gas generator rotor speed and power turbine speed and torque inputs. A discrete interface is available for external discrete inputs and internal wrap-around inputs. There is also a Universal Asynchronous Receiver Transmitter (UART) to accommodate communication with an external device for test purposes. An ARINC 429 interface is used to receive data from an air data computer. All modules in this section are hand coded in HS1600 assembly language.

OUTPUTS (O)

Each EEC132-210 provides various outputs required for controlling fuel flow, bleed air, and various cockpit displays. There is a stepper motor interface used to control a fuel flow valve. A pulse width modulator (PWM) is used to modulate the handling bleed valve. Solenoid actuations are handled by the discrete output interface. ARINC 429 provides the means for transmitting the continuous engine/control parameters and maintenance discrete words for cockpit display. A data stream is also provided over the UART output which can be received by an external device. All modules in this section are hand-coded in HS1600 assembly language.

CONTROL LAWS (C)

This section contains the modules used to implement the gas generator control laws. Included are the outer loop control logic, the inner loop control logic, logic to compute the rated torque for display on the "Torque Bug" in the cockpit, and the start logic. All of the Control Law modules are generated from control or data flow diagrams using the GPU (pictures-to-code) compiler.

OTHER (E)

This section contains the modules which do not fall within the three above categories. Included are the initialization, background, test mode, and built-in test modules. All modules in this section are hand coded in HS1600 assembly language.

SUPPORT ROUTINES

The support routines include univariate and bivariate table processing subroutines, lead-lag filter, lag filter, derivative lag, and other utility subroutines.

DCU and LCF ROUTINES

These routines provide functionality to read, write, and process DCU data. The LCF logic performs calculations based on data read from the DCU. The functional requirements of the EEC132-210 require that the software perform certain time-based tasks, execute built-in tests in whatever remaining time is available after all time-based tasks have been completed, and communicate with a ground-test unit in a "dedicated" communications mode. This "dedicated" communications mode can be exited only by a power-supply reset or a processor reset. Based on these requirements, the software is divided into four major subsystems: initialization, background, foreground, and test mode.

INITIALIZATION

After a PSR (Power Supply Reset) or a POR (Power-On Reset) occurs, the EEC software performs the required hardware and software initialization. At the completion of the initialization tasks, the logic enables real-time interrupts and passes control to the Background Logic. A processor reset starts execution in the initialization routine. The following tasks are performed by the initialization routine in the sequence listed:

1. All EEC outputs are disabled.
2. The overcurrent detector is reset.
3. A test is performed to check the watchdog timer.
4. A complete RAM read/write test is performed.
5. A complete EEPROM checksum test is performed.
6. A checksum test of pressure sensor data stored in EEPROM is performed.
7. A complete instruction test is performed.
8. A test of the torque sensor interface is performed.
9. All of the EEC interfaces are initialized so that when the first timing interrupt occurs, the processor will initiate execution in State 1 of the foreground logic.
10. Various RAM locations are initialized.
11. The addresses of the 17 variables to be transmitted on the UART are stored in the UART output table in RAM. These addresses are found in EEPROM and ROM.
12. Trim data and fault history data are read from EEPROM and stored into RAM.
13. All speed converters are stopped at the start of initialization. The NH converter is started as the control sequences through the state table.
14. The torque converter is started.
15. The A/D converter is reset.
16. The initialization logic waits for the watchdog time discrete to go high and then updates the watchdog timer.
17. The initialization logic is terminated, interrupts are enabled, and control is transferred to the background logic.

One of the RAM locations initialized is an initialization flag. This flag is used in the foreground during the first three passes of the state table. The primary purpose of this flag is to allow every filter, integrator, and derivative to go to its steady state. It also allows time for the system to determine the engine configuration and rating via the engine trim plug and several discrete inputs.

INITIALIZATION NOTES:

1. A "keep alive" signal is issued to the watchdog timer every time the CPU detects that the time discrete has been set throughout the initialization routine.
2. Timing interrupts are not enabled during the initialization routine. This is because a processor reset disables them initially and the software does not enable them until initialization is completed.
3. No stepper motor command is output while the initialization flag is set.

BACKGROUND

The Background Logic contains the software functions which do not require execution at some pre-determined rate. Specifically, the background logic executes only after all time-critical tasks in the system have completed. Timing interrupts are enabled by the software at the start of background execution. The background execution consists of two built-in tests: an EEPROM sum test and an instruction test, which are repeatedly executed. The EEPROM sum test is divided into page segments of 1,024 memory locations each. The contents of all of the program memory pages and sensor data areas are summed. If there are no EEPROM memory errors, the 16 least significant bits of the sum will equal zero. The instruction test verifies proper execution of the CPU instructions. Background flags are set at the completion of each segment of the EEPROM sum and the instruction test. These flags are tested in the foreground to verify proper execution of the background. The background increments a foreground counter which is

cleared in the foreground. If this counter is not cleared in the foreground, the background will detect that the foreground is not functioning.

FOREGROUND

The Foreground Logic is responsible for controlling the calling of all tasks that must execute at a fixed rate. The primary foreground is responsible for tasks which require one millisecond accuracy, while the secondary performs the majority of the logic based on a twenty-millisecond minor cycle. The following tasks are performed by the primary foreground:

1. The processor status (AREG, BREG, and overflow flag) from the interrupted background or secondary foreground routine is saved. Upon completion of the foreground, the processor status will be restored so that execution of the background or secondary can continue.
2. If 16 timing interrupts have occurred since the last time the watchdog timer was reset, the watchdog timer is reset.
3. The processor/memory status word is checked for any critical faults. If any have been recorded, the status word is stored in EEPROM and control is transferred to the test mode.
4. ARINC and UART input data is processed.
5. The state counter is incremented.
6. The analog to digital conversion is read and reset to start its next conversion if the current state indicates that A/D servicing is necessary.
7. If not performing DCU communications, a single byte is transmitted on the UART in even-numbered time states when the output buffer is empty. Otherwise DCU I/O is executed.
8. One or more primary foreground routines which require one millisecond control are executed. In general, the routines executed are dependent upon the A/D input that was just read or ARINC output which must be sent every four milliseconds. Once complete, control is returned to the background or secondary foreground module which was previously interrupted.

The state table controls the execution rate of each of the modules in the engine control system as well as the conversion of the analog inputs. The state table contains 80 entries corresponding to the 80 states of the major cycle. Each entry contains a mux code word followed by a varying number of module labels for the modules to be executed during this time state.

The format for the mux code word is shown in "Mux-Code-Word" table below.

Controlling fuel flow to the engine requires performing a variety of logic functions at rates determined by the applicable controlling parameters. These software functions make up the "Foreground" logic. Note that all time-dependent logic is scheduled using the internally generated interrupt rate of 1.04 milliseconds. Some of these time-based tasks must execute every real-time interrupt; these logic modules make up the primary tasks and they are called directly from the primary real-time interrupt handler. There are other logic modules in the system that require execution at some multiple of the real-time interrupt; these are the secondary tasks. The EEC software, in accordance with this requirement, divides the logic tasks of the control unit into four groups. These categories determine the rates at which secondary modules can execute:

- 1.04 millisecond repetition rate
- 20.80 millisecond repetition rate (minor cycle)
- 41.60 millisecond repetition rate
- 83.20 millisecond repetition rate (major cycle)

The term "minor cycle" refers to the secondary logic tasks executing at the fastest possible rate (20.80 milliseconds); the term "major cycle" refers to the tasks executing at the slowest possible rate (83.20 milliseconds). All minor/major cycle time-dependent logic in the EEC is activated via a table-driven "Control Scheduler". This scheduler is actually a part of the real-time interrupt logic. This table-driven implementation gives the software more flexibility in scheduling the rates at which particular pieces of logic should execute; a change to the rate at which a module runs involves

merely a change to the applicable control table entry. Each time an interrupt occurs, a new time state is begun. Once the primary (one millisecond) tasks for the state have completed, the secondary (20 millisecond) tasks are performed. The background tasks are performed during any time remaining in each 20 millisecond time slice once the foreground tasks are completed. The control scheduler table is comprised of 80 "time" states, based upon the major cycle time of 83.20 milliseconds. These 80 time states are divided into parts as determined by a number of "rendezvous" conditions. A rendezvous condition can be defined by an I/O event that must occur before the remaining modules in the scheduler table can execute. Each I/O event in the PW127XT-S software can be related to a particular time state (interrupt count). Thus, an interrupt number may be used to specify a rendezvous condition. The modules within the control scheduler table between rendezvous conditions must complete before the latter rendezvous condition (interrupt count) occurs. The remaining time left after all modules within a particular rendezvous state have finished executing is spent in the background logic. There are two types of entries in the Control Scheduler Table: the rendezvous conditions themselves, and the addresses of the procedures to execute after a particular rendezvous has occurred. The rendezvous states within the Control Scheduler Table are differentiated from procedure addresses by setting bit 16 in the word; bits 1-15 contain the interrupt counter value at which the rendezvous condition is considered to have occurred. The background logic, as stated above, consists of functions that are not time-dependent; they execute in whatever time is available after all time-critical tasks have finished executing. The background logic runs continuously: as soon as one pass through the background logic completes, a second pass begins. Specifically, the background logic consists of a number of built-in tests to ensure that the hardware is functioning properly.

TEST MODE

The test mode logic allows the user to communicate with the control in a dedicated mode. Once established, the control software will remain in test mode until a PSR or a POR occurs. This software makes it possible to download diagnostics into the EEC and to examine the memory contents of the EEC. The EEC enters Test Mode in one of two ways:

1. The ground test unit sends a command to the EEC telling it to enter test mode.
2. The EEC software detects a critical failure and transfers control to the test mode logic.

While executing in the test mode state, all EEC outputs are disabled and the channel is under control of an external device. Communication to the external device is accomplished through the test UART using a command-response protocol. In order to get into this mode, a discrete is set, (BOOT discrete) and a POR initiates. Once in this mode, the software makes it possible to download diagnostics into the PW127XT-S and to examine the memory contents of the PW127XT-S.

03.01 Table 1: Mux-code-word

BIT	
16	-1
15	
14	- A/D MUX CODE
13	
12	
11,10	- A/D MUX CODE
9,8,7	- RVDVT MUX CODE
6,5,4	- HIGH LEVEL MUX CODE
3,2,1	- LOW LEVEL MUX CODE
Bit 16 is always set to one (1) to indicate that this word is a mux code word and the start of a new time state.	

04.00 SOFTWARE IMPLEMENTATION

In accordance with section 5.0 of DO-178C, Software Development Processes, single layer requirements such as hand coded modules may be used to provide coverage for both HLR (functional description) and LLR (implementation description).

There are instances where software requirement functionality may be highly correlated with the software implementation. The creation and maintenance of an artificial distinction between high-level software requirements and low-level software requirements under these circumstances becomes an impractical burden on the development and verification process. This problem is addressed on an exception basis, by the selective use of a single layer of software requirements to provide definition for the direct development of Source Code. These software requirements are treated as both high-level and low-level requirements for the purpose of software development and verification, and the applicable DO-178C guidance for both types of requirements is applied.

All designs are either implemented by design models or textual LLRs based on HLRs or are implemented directly from HLRs (Single Level of Requirements). In the case of the latter, the justification is provided.

All design models can be found in section Design Models (GPU Diagrams).

Iteration rates are implemented in the E08-EXEC_FOREGND.

04.01 INPUT SIGNAL PROCESSING

04.01.01 Low-Level Analog Input Processing

04.01.01.01 Low Level Drift Compensation [I01-Sens_LL_Drift_Comp]

This logic is needed to provide a correction value for hardware amplifier drift. Drift is measured by a 0 volt input to the low level amplifier. RAWDRF, the amplifier output, is then range checked, EOC checked and, if not acceptable, the logic sets appropriate fault bits. A delay on setting the drift fault improves tolerance to indirect lightning.

04.01.01.02 Strain Gauge Compensation [I02-Sens_LL_Strain_Gauge_Comp]

This logic is needed to provide a gain error correction resulting from strain gauge excitation variations. The gain correction is derived from a 50.21 mV source. RAWSG, the amplifier output, with a 12.556 mV bias, is range checked and drift corrected and scaled by NMGAIN. In the event of an EOC a range fault, specific fault flags are set and a default value NMSGC is assigned to SGC.

04.01.01.03 Cold Junction Temperature Input [I03-Sens_LL_Cold_Junct_Temp]

In this process the raw cold junction input, obtained from the A/D, is corrected for drift. A compensated cold junction value is processed through a polynomial. This value is processed through the polynomial to obtain a value for the cold junction in degrees F. Cold junction in degrees F is used in the inverse bivariate maps from which the static ambient pressure and ram pressure are obtained. Upon detection of a fault, a transient counter is started and the last good value of CJ is retained. If the fault persists, the counter reaches its limit and the fault is latched.

04.01.01.04 PSAMB Strain Gauge [I04-Sens_LL_PSAMB_Strain_Gauge]

In this process, the raw value for the nacelle static pressure is gain and drift corrected. The compensated pressure signal and cold junction temperature are processed through an inverse bivariate map to obtain nacelle static pressure, PSAMB. PSAMB is then filtered through a single order lag to obtain PSAMBF. This variable is then correlated to obtain ambient pressure, P0AMBF. PSAMBF is converted to pressure altitude in feet for use in the control.

04.01.01.05 Delta P Strain Gauge [I05-Sens_LL_DELTA_P_Strain_Gauge]

The raw input signal for the ram pressure is drift and gain compensated in this process. The compensated signal and cold junction temperature are processed through an inverse bivariate map to obtain the ram pressure. This ram pressure is then filtered using a third order (quadratic plus single order) filter. The final output from these filters is passed through a Select High (with respect to zero) in order to eliminate negative values. The filtered ram pressure is converted to a rate limited airspeed value. If a fault is detected, a delay is required before the fault is latched.

04.01.01.06 TQ RTD [I07-Sens_LL_TQ_RTD]

The TQ Resistance Temperature Device (RTD) measures the oil mist temperature of the torque shaft/gear box oil. This measurement is used in the calculation of engine output torque by temperature correcting the measured torque twist calculation.

04.01.01.07 T18 RTD [I08-Sens_LL_T18_RTD]

This process takes the measured resistive temperature device (RTD) value at station 1.8 and correlates it to the temperature at station 1.5.

04.01.02 High-Level Analog Input Processing

These inputs are selected by an 8-input A/D multiplexer. One of these is a clamp input required during A/D sequencing, and is not a software input. Two of the A/D mux inputs are sub-multiplexed inputs. One of these input groups is the Low-Level Analog Inputs defined in section 4.1.1. The other group is comprised of the High Level Buffered DC Analog Input Group, which includes the Trim Inputs and the AC Analog Input Group, which is a further sub-multiplexed input group. The Trim Inputs include Trim Gain and Offset inputs to compensate for variations in the Trim Excitation Reference. All of the High Level Buffered DC inputs are buffered through a common High Level DC unity gain amplifier to provide isolation from the A/D input. The AC Analog Inputs are converted to compatible A/D DC input levels through a common AC to DC converting amplifier with a fixed gain and offset.

04.01.02.01 High Level Drift [I09-Sens_HL_Drift]

High level drift is required to correct the high level analog signals whose errors are a function of the high level amplifier. The drift reference voltage, a ratio metric value of the high level analog excitation, is the reference source for detecting amplifier drift. This section range checks the raw sensed voltage, subtracts an offset, and outputs the drift parameter DRIFTC.

04.01.02.02 High Level Gain Compensation [I10-Sens_HL_Gain_Comp]

High level gain compensation corrects for the high level analog signal errors resulting from variations in the interface excitation voltage. This process logic performs a range check, drift correction, and scaling to create the gain compensation parameter GAINC.

04.01.02.03 Torque Bias Trim [I11-Sens_HL_Q_Bias_Trim]

Torque bias trim is required to offset an engine torque shaft physical bias. The required trim is selected for each control/engine combination with a trim resistor in the J4 trim plug. The EEC provides excitation and amplification for the trim bias. The raw input for the torque bias trim is drift and gain corrected in this process.

During initialization, the corrected torque bias trim is compared to the current bias trim stored in EEPROM. If these two values differ by more than one resistor step the new bias trim is written to EEPROM. During subsequent passes through this logic this new value is compared to the calculated current input. If the difference between these two values exceeds its tolerance and the transient fault timer is at its limit, then the torque bias system is considered failed and the fault latch bit for this process (TBTIF.16) is set. This will cause a LRU fault code to be recorded in EEPROM and the bypassing of sections of this process during subsequent executions of this process.

04.01.02.04 Torque Gain Trim [I12-Sens_HL_Q_Gain_Trim]

Torque gain trim is required to correct an engine torque shaft deviation from an expected spring rate. One of 64 possible resistors is selected and is installed in the J4 trim plug by the user. Excitation and amplification for the gain trim is provided by the EEC.

During initialization, the corrected torque gain trim is compared to the current gain trim stored in EEPROM. If these two values differ by more than one resistor step the new gain trim is written to EEPROM. During subsequent passes through this logic this new value is compared to the calculated current input. If the difference between these two values exceeds EEGDIF and a transient fault timer has expired, the torque gain system is considered to have failed and a fault latch bit for this process (TGTIF.16) is then set. This will cause a LRU fault code to be recorded in EEPROM and the bypassing of sections of process on doing subsequent passes of this process.

04.01.02.05 Engine Variant Selection Logic [I13-ENG VARIANT]

Ov 12490001.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The function of this process is to read in and validate the Engine Validity raw input signal from the Trim Plug and accommodate the Mode to either DEVMOD, NORMOD or INVALID, based on the selection logic. The Derate Level, based on engine rating discretes, is also defined and stored accordingly. This is allowed to occur once per POR, during initialization.

LLR 12490004.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module I13_ENGVARIANT_1 shall be implemented per design model I13_ENGVARIANT_1_A.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490004.000	0			

LLR 12490006.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module I13_ENGVARIANT_2 shall be implemented per design model I13_ENGVARIANT_2.CTRLFLOW

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490006.000	0			CL12470224.001 CL12470225.001 CL12470226.000 CL12470228.002 CL12470229.001 CL12470231.001 CL12470232.000 CL12470233.000 CL12470234.000 CL12470235.001 CL12473029.000

04.01.02.06 Power Lever Angle (PLA) Process Group

04.01.02.06.01 Power Lever Angle [I14-Sens_HL_PLA]

In this logic process the PLA inputs from each RVDT winding are processed. The two signals are range checked and drift corrected before being used to calculate degrees position. The signal is then range checked, rate limited, passed through hysteresis, and filtered to attenuate noise.

04.01.02.06.02 PLA Bias [I15-Sens_HL_PLA_Bias]

PLA bias is required to correct for gain and drift errors in the PLA signal processing. This is done with the same excitation gain correction. As a result, the drift calculation must include the gain element to adjust for variations in the excitation supply. In this process, the raw PLA bias input is range checked. The PLA gain is subtracted from the range checked PLA bias, and this new value is multiplied by the bias intercept. The sum of the PLA gain corrected value (multiplied by the AC bias intercept) and the range checked PLA bias is the PLA drift signal.

04.01.02.06.03 PLA Gain [I16-Sens_HL_PLA_Gain]

In this process the raw PLA gain input is range checked. This value is used in conjunction with the PLA bias input to form the PLA drift signal.

04.01.02.07 Stepper Motor Coil Sum Voltage Input [I38-Sens_HL_SM_Coil_Sum_Volt]

This process will input the Stepper Motor Sum Voltage through the A/D Converter. This voltage is proportional to the sum of the voltage across all four Stepper Motor Coils. During normal operation (Stepper Motor enabled), this process will convert the raw input from A/D counts to engineering units and pass it to the Stepper Motor Phase Fault Test (Reference Process E16-BIT_EEC_SM_Phase).

When the EEC is in Fail-Fixed mode (Stepper Motor disabled), this process will determine if the Sum Voltage hardware (BIT circuit) and the input interface hardware are operating properly by monitoring the Sum Voltage for EOC and input range faults.

04.01.02.08 Bleed Wraparound Input [I45-Sens_HL_Bleed_W/A_Input]

This process checks the validity of the Bleed Wraparound Input.

04.01.03 Frequency Input Processing

The EEC converts three frequency signals: two for NH signals and one for Torque and NPT. The frequency signals are multiplexed into their corresponding F/D converter.

NH conversion involves starting conversion and stopping the F/D within one minor cycle. The number of teeth and time counts are then processed to derive NH.

Torque and NPT conversion involves manually starting the converter, but hardware stops the conversion by counting a preset number of teeth. Conversion of torque is done by phase displacement method and, therefore, torque is determined by a duty cycle converter.

04.01.03.01 NH - High Speed Turbine

04.01.03.01.01 Start Primary NH Conversion [I17-Sens_FR_Start_Pri_NH_Convert]

The logic chooses the NH probe to be used as the primary data source. The primary probe supplies the NH data to be used throughout the system. If NH probe 1 data passes its validity checks, then it is chosen as the primary source. If NH probe 1 does not pass its validity checks, then NH probe 2 is chosen as the primary source.

If $NPT < NPTGEN$ or the engine is in feather, then the primary source logic switches between probes 1 and 2 to ensure an NH signal during engine start.

Once the probe to be used as the primary source is selected, the appropriate probe NH conversion is started.

04.01.03.01.02 Read NH Probe Logic [I18-Sens_FR_Read_NH_Probe]

The read NH probe logic converts teeth and time counters generated by hardware into an NH speed. The NH value is range checked with the appropriate failure flag set.

04.01.03.01.03 Dual Probe Logic [I19-Sens_FR_Dual_Probe]

This logic performs two tasks. First, probe one and two fault words are examined to determine if a dual probe failure exists. If a dual probe failure exists, the dual probe fault latch is set. Second, if the data from both probes is valid, the difference between the two NH values is calculated. If that difference is greater than $NHDIF$, an NH Cross Check fault is flagged. The difference calculation is bypassed if $NPT < NPTGEN$ or if the engine is in feather ($DISC1.13 = 1$). The crosscheck fault is not set until the $NHTCT$ counter has reached its limit of $KTRMX$.

04.01.03.01.04 NH Speed Filtering [I20-Sens_FR_NH_Speed_Filter]

This process filters the raw NH signal (NHI) using a first order filter and outputs the final value for use in the remainder of the control logic.

04.01.03.01.05 Start Secondary NH Conversion [I22-Sens_FR_Start_Sec_NH_Convert]

The logic decides if NH probe 2 is required as the secondary source. The secondary source is used only for cross checking with the primary source. If probe 2 is being used as the primary source, then no secondary source is required.

If probe 1 is being used as the primary source, probe 2 is selected as the secondary source.
If probe 2 is required as the secondary source, then the probe 2 conversion is started.

04.01.03.01.06 Read NH Probe Two [I23-Sens_FR_Read_NH_Probe_Two]

If NH probe 2 is selected as the secondary source, this logic functions. If NH probe 2 is selected as the primary source, then this logic is bypassed.

The logic performs range and validity checks on the probe 2 data. If all validity checks pass, then the unfiltered gas generator speed is calculated from the tooth and time counts. If a validity check fails, a fault counter is started. Once the fault counter reaches its maximum value, a speed fault latch is set. In the case of a low speed fault or an End of Conversion fault, a latch is only set when $NPT \geq NPTGEN$ and the engine is not in feather ($DISC1.13 = 0$).

04.01.03.01.07 NH Probe 2 Filtering [I24-Sens_FR_NH_Probe_2_Filter]

This logic takes the unfiltered gas generator speed calculated from NH probe 2 data and filters this speed signal.

04.01.03.02 Torque and Torque Shaft speed Input Processing

04.01.03.02.01 Calculate Torque Shaft Deflection and Speed [I25-Sens_FR_Q_NPT_In]

This process controls the operation of the torque input conversion hardware, and calculates torque shaft angular deflection and speed from the converter data. Converter status and data range checks are used to determine interface fault status. The input to be converted for each process execution is determined from the conversion history and interface fault status.

04.01.03.02.02 Calculator or Torque from Shaft Deflection [I26-Sens_FR_Q_Shaft_Deflect]

The measured torque shaft deflection is converted to shaft torque, with corrections for shaft temperature and gain/bias calibration trims for the specific engine. The shaft torque is then multiplied by the gear ratio from the torque shaft to the propeller shaft to provide a torque value in engineering units.

04.01.03.02.03 Torque Shaft Speed Filtering [I27-Sens_FR_Q_Shaft_Speed_Filter]

This process filters the torque shaft speed acquired from the torque/speed input interface.

04.01.03.02.04 Torque/speed Dual Input Fault Logic [I43-Sens_FR_Q/Speed_Dual_In_Fault]

Two torque/speed inputs are processed by the EEC torque input interface. This process determines if a dual input failure exists by a crosscheck of the two torque and two speed values for agreement, and by examining the individual input fault latches for combinations which indicate a total interface failure.

04.01.04 Discrete Input Processing

The EEC has the capability to receive thirty-two (32) discrete inputs. These are a mix of internal and external discrete signals. The discretes that are read through the 4-(8x1) discrete muxes are read as two 16-bit status words by input port commands.

04.01.04.01 Discrete Input Word #1 [I28-DISINW1]

Ov 12490008.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

Discrete Input Word #1 is read from Input Register 312 octal. The bits used in this application are extracted and logical inversions performed to provide a consistent positive logic sense for all discrete inputs. The processed bits are then stored in data word DISC1.

This process also contains the logic used to determine the Engine Rating Selection based on discrete input settings.

The Bleed #1 and #2 Discretes are used to determine the engine bleed level state. The propeller configuration is also determined by a discrete input.

LLR 12490011.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module I28_DISINW1 shall be implemented per design model I28_DISINW1_A.CPGC, I28_DISINW1_B.CPGC, I28_DISINW1_C.CPGC, I28_DISINW1_D.CPGC and I28_DISINW1_E.CPGC

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490011.001	0			CL12470527.002 CL12470528.001 CL12470529.001 CL12470530.001 CL12470531.001 CL12470532.001 CL12470535.001 CL12470538.001 CL12470541.003 CL12470542.000 CL12470543.000

04.01.04.02 Discrete Input Word #2 [I29-DISINW2]

Ov 12490017.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

Discrete Input Word #2 is read from Input Register 313 octal. The bits used in this application are extracted and logical inversions performed to provide a consistent positive logic sense for all input discretes. The processed discrete inputs are stored in DISC2.

LLR 12490020.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module I29-DISINW2 shall be implemented per design model(s): I29_DISINW2_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490020.000	0			CL12470553.000 CL12470555.001 CL12470556.001 CL12470557.001 CL12472905.000

04.01.04.03 EEC Fault Logic Input Discrete Word [I30-BIT_EEC_Disc_In]

The Fault Logic Input Discrete Word is read from Input Register '314. The discrete bits are extracted and logical inversions performed to provide a consistent positive logic sense for all input discretes. The processed bits are stored in FLTWRD.

04.01.05 UART Input Processing

04.01.05.01 N-Parameter Protocol UART Input Processing [I31-NPARAMIN]

This capability monitors an incoming UART data stream for the N parameter protocol "address list update" command. This command is sent by an external piece of test equipment to change the EEC data that is being monitored. Data reception by this capability is only performed if a software switch in EEPROM selects the N-Parameter protocol mode rather than the real time data stream mode. The "address list update" command data frame consists of an identifier, followed by the number of requested parameters, followed by alternating pairs of parameter address and size data in the order the external test equipment expects the returned data. The UART reception protocol is 1 start bit, 8 data bits, 1 stop bit, and odd parity, at 9600 baud.

04.01.05.02 UART Input Processing [I31-UARTIN]

This function provides a UART communication link to interrogate the EEC software, for data or to patch the software, as desired. When test is commanded, an ASCII "T" is transmitted to the EEC. If the "T" is received, also, check if DISC2.9, i.e., the Ground Test discrete is on. If both permissives occurred, then place the processor status in test mode. If one or the other permissive does not exist, then place the processor in normal operation mode.

04.01.06 ARINC Input Processing [I32-ARINC_HW]

THE ARINC 429 input process stores the decoded labels and their respective data into an appropriate RAM location for later use. Only the high speed processing is performed here to minimize the process execution time.

04.01.07 Air Data Computer (ADC) Inputs

ADC Input processing requirements are subdivided into process groups, each group defining the processing required for an ADC input parameter. The previous process, "ARINC Input Processing" defines the parameters acquired for this application.

04.01.07.01 Total Air Temperature

04.01.07.01.01 ARINC Total Air Temperature Processing [I33-ARINC_In_TAT]

This process takes the Total Air Temperature (TAT) value received over ARINC on Label 211 from the Air Data Computer and correlates it to the temperature at Station 1.5.

LLR 12490629.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module I33-ARINC_In_TAT shall be implemented per design models I33_ARITAT_A.CPGC, I33_ARITAT_B.CPGC, I33_ARITAT_C.CPGC, I33_ARITAT_D.CPGC, I33_ARITAT_E.CPGC, I33_ARITAT_F.CPGC, I33_ARITAT_G.CPGC and I33_ARITAT_H.CPGC

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490629.000	0			CL12470631.000 CL12470632.000 CL12470633.000 CL12470634.000 CL12470635.000 CL12470636.000 CL12470637.000 CL12470638.000 CL12470639.000 CL12470640.000 CL12470641.000 CL12470642.000 CL12470643.000 CL12470644.000 CL12473058.000

04.01.07.01.02 Total Air Temperature Source Selection Logic [I34-ARINC_In_TAT_Source_Sel]

This process selects between the air data computer (ADC) and the EEC total air temperature values depending on the status of each of these sources.

04.01.07.02 Ambient Pressure

04.01.07.02.01 ARINC Pressure Altitude Processing [I35-ARINC_In_Press_Altitude]

This process takes the pressure altitude value in feet received over ARINC on label 203 from the Air Data Computer (ADC) and correlates it to PSIA.

LLR 12490627.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module I35-ARINC_In_Press_Altitude shall be implemented per design models I35_ARIPA_A.CPGC, I35_ARIPA_B.CPGC, I35_ARIPA_C.CPGC, I35_ARIPA_D.CPGC, I35_ARIPA_E.CPGC, I35_ARIPA_F.CPGC and I35_ARIPA_G.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490627.000	0			CL12470678.000 CL12470679.000 CL12470680.000 CL12470681.000 CL12470682.000 CL12470683.000 CL12470684.000 CL12470685.000 CL12470686.000 CL12470687.000 CL12470688.000 CL12470689.000 CL12470690.000 CL12470691.000 CL12470692.000 CL12473056.000

04.01.07.02.02 Ambient Pressure Source Selection Logic [I36-ARINC_In_Press_Source_Sel]

This process selects between the Air Data Computer ambient pressure value and the EEC ambient pressure value, depending on the status of each source. When development mode is set, Air Data Computer ambient pressure is selected.

LLR 12490622.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The capability I36-ARINC_In_Press_Source_Sel shall be implemented per design models I36_PAMSEL_A.CPGC, I36_PAMSEL_B.CPGC, I36_PAMSEL_C.CPGC, I36_PAMSEL_D.CPGC, I36_PAMSEL_E.CPGC, I36_PAMSEL_F.CPGC, I36_PAMSEL_G.CPGC and I36_PAMSEL_H.CPGC

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490622.000	0			CL12470705.000 CL12470706.000 CL12470707.000 CL12470708.000 CL12470709.000 CL12470710.000 CL12470711.000 CL12470712.000 CL12470713.000 CL12472791.000 CL12472792.000

04.01.07.03 Indicated Air Speed

04.01.07.03.01 ARINC Indicated Air Speed Processing [I37-ARINC_IN_IAS]

This process takes the Air Data Computer (ADC) air speed value in knots, as received over ARINC 429 label 206, and uses it to compute ram pressure. Four data cases are addressed in this process:

1. Valid data available
2. Invalid data available
3. No data available
4. Intermittent data

When data from the Air Data Computer has been determined invalid, a default value is selected for Indicated Air Speed (IAS). Intermittent data logic allows five cycles from invalid to valid states before default data is latched. Once default data is latched, return to processing ADC Indicated Air Speed (IAS) data is permitted only when a POR, an INHIBIT RESET, or NH<NHSTDN occurs. Latching of faults is bypassed if NPT is less than or equal to ground idle speed.

LLR 12490628.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module I37-ARINC_IN_IAS shall be implemented per design models I37_ARIISA_1_A.CPGC, I37_ARIISA_1_B.CPGC, I37_ARIISA_1_C.CPGC, I37_ARIISA_1_D.CPGC, I37_ARIISA_1_E.CPGC, I37_ARIISA_1_F.CPGC, I37_ARIISA_1_G.CPGC, I37_ARIISA_1_H.CPGC, I37_ARIISA_2_A.CPGC and I37_ARIISA_2_B.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490628.000	0			CL12470725.000 CL12470726.000 CL12470728.000 CL12470730.000 CL12470732.000 CL12470734.000 CL12470736.000 CL12470738.000 CL12473057.000

04.01.07.03.02 Indicated Airspeed Fault Logic [I46-ARINC_In_IAS_Fault]

This process monitors both the Delta P sensor data and the ARINC Indicated Airspeed data. If both sources are failed then this process sets a dual airspeed fault which is used to disable the ARINC Label 206 (SDI 11) output.

04.02 EXECUTIVE PROGRAM

The Executive Program performs initialization, continuously monitors system health, incorporates a means of accessing memory for testing and provides fault status for diagnostic and maintenance purposes.

04.02.01.01 Reset, PSR/POR and Initialization Executive [E01-BIT_Proc_Reset_PSR_POR]

Ov 12490023.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process initializes the Stepper Motor, Disables the Multi-Level Indirects, resets the Torque Bug Meter, initializes the Bleed Valve, resets the Torque Indicator Meter and initializes the Discreate Outputs. It also initializes the A/D and F/D Converters, performs an Overcurrent Reset, initializes the UART and the RTD Default Parameter List. A Watchdog Timer Self Test is performed. Finally, the software performs a Ram_Test, EEPROM Checksum Test, Instruction Test, an Initial Stepper Motor Sum Voltage Check, a Gate Array Selection and a Gate Array Test. This process also initializes RAM.

Ov 12490024.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCDs: SCR: PW127L-EEC-SW-PWC-16080 SCO-D: PW127L-EEC-SW-PWC-16081

Ov 12490025.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design is incorporated in modules E01_PORPSR and RTDWRITE. E01_PORPSR calls E02_RAMTST, E03_CHECKSUM, E04_INSTST, E05_RAMINI, E06_TQTEST, E09_SMSUMV, E22_GASEL, and RTDWRITE. Designs changed from the prior baseline are identified in section 11.

Ov 12490026.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design of E01_PORPSR is defined by the following LLR:

LLR 12490027.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Design models E01_PORPSR_A, E01_PORPSR_B, E01_PORPSR_C, and E01_PORPSR_D shall be implemented.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490027.000	0			CL12470760.000 CL12470761.000 CL12470762.000 CL12470763.000 CL12470764.000 CL12470765.000 CL12470766.000 CL12470767.000 CL12470768.000 CL12470769.000 CL12470770.000 CL12470771.000 CL12470772.000 CL12470773.000 CL12470774.000 CL12470819.000

Ov 12490028.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design of RTDWRITE is defined by the following LLR:

LLR 12490029.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Design model RTDWRITE shall be implemented.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490029.000	0			CL12470774.000

04.02.01.02 RAM Test [E02-BIT_Proc_Ram_Test]

This Process Performs Hardware Checks on Address Lines and RAM Locations.

04.02.01.03 Complete ROM and EEROM Checksum [E03-BIT_Proc_ROM_EE_Chksum]

This process computes ROM checksum, ambient pressure compensation data checksums and delta pressure compensation data checksums from EEROM.

04.02.01.04 Instruction Test [E04-BIT_Proc_Instruct_Test]

This Process Performs CPU Instruction Tests and B Register Shift Tests.

04.02.01.05 RAM Initialization [E05-RAMINIT]

Ov 12490030.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process initializes all non-zero RAM locations, sets up the UART output table and pressure sensor compensation data. The Configuration is also determined so that the Foreground Logic can set up the correct State Tables.

- EE_READ_DATA

This subroutine performs the Test Mode DUMP operation. It reads two 16-bit addresses from the UART, and then transmits the contents of all the memory locations between those two addresses (inclusive) out the UART. After the entire block has been transmitted, it sends an ACK and returns to the Test Mode Executive Routine. If any input errors occur, the command operation is immediately aborted.

- INI_DCU_DATA

This routine initializes DCU related RAM memory.

- DCU_RANGE_CHECKS

This routine traverses through a table of items to perform a range check on. If an item is out of range, the appropriate flag is set.

LLR 12490047.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_A.CPGC.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490047.000	0			CL12470821.000 CL12470822.000 CL12470823.000 CL12470824.000 CL12470825.000 CL12470829.000 CL12470830.000 CL12470855.000

LLR 12490048.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490048.000	0			CL12470826.000

LLR 12490049.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_C.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490049.000	0			CL12470827.000

LLR 12490050.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_D.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490050.000	0			CL12470828.000

LLR 12490051.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_E.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490051.000	0			CL12470831.000 CL12470832.000 CL12470833.000 CL12470836.000

LLR 12490031.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_F.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490031.000	0			CL12470833.000

LLR 12490032.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_G.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490032.000	0			CL12470837.000

LLR 12490033.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_H.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490033.000	0			CL12470838.000

LLR 12490034.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_I.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490034.000	0			CL12470839.000 CL12470840.000

LLR 12490035.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_J.CPGC.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490035.000	0			CL12470835.000 CL12470842.000 CL12470843.000 CL12470844.000 CL12470846.000 CL12470847.000

LLR 12490036.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_K.CPGC.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490036.000	0			CL12470834.000 CL12470847.000 CL12470849.000 CL12470853.001

LLR 12490037.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_L.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490037.000	0			CL12470853.001

LLR 12490038.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module E05_RAMINI shall be implemented in part per design model E05_RAMINI_M.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490038.000	0			CL12470854.000 CL12470855.000 CL12471977.002

LLR 12490607.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module DCU_RANGE_CHECKS shall be implemented as per the DCU_RANGE_CHECKS.CPGC design model.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490607.000	Func		N/A	CL12470851.001 CL12472929.000 CL12472930.000 CL12472931.000 CL12472943.000 CL12472944.000 CL12472945.000 CL12472946.000

LLR 12490040.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall define Global Ram Signals and assign labels to the second 1K RAM locations in the PW127 Electronic Engine Control in PAGE1.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490040.001	0			CL12472786.000

LLR 12490042.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall define Global Ram Signals and assign labels to the first 1K RAM locations in the PW127 Electronic Engine Control in PAGE0.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490042.001	0			CL12472786.000

LLR 12490043.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall assign labels to the first EEPROM storage page for the PW127 Electronic Engine Control in PAGE10.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490043.001	0			CL12472786.000

LLR 12490044.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall assign labels to the EEPROM storage page for the PW127 Electronic Engine Control in PAGE11.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490044.001	0			CL12472786.000

LLR 12490608.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EE_CTRDAT_IF bits three, four, five and sixteen to TRUE, if any of the following conditions are met:

- LPIMP_CYCCRS is less K_CYCCRS_ZERO or,

- LPIMP_CYCCRS is greater than K_CYCCRS_32K or,
- HPIMP_CYCCRS is less K_CYCCRS_ZERO or,
- HPIMP_CYCCRS is greater than K_CYCCRS_32K or,
- HPTFCP_CYCCRS is less K_CYCCRS_ZERO or,
- HPTFCP_CYCCRS is greater than K_CYCCRS_32K or,
- HPTD_CYCCRS is less K_CYCCRS_ZERO or,
- HPTD_CYCCRS is greater than K_CYCCRS_32K or,
- HPTRCP_CYCCRS is less K_CYCCRS_ZERO or,
- HPTRCP_CYCCRS is greater than K_CYCCRS_32K or,
- ISS_CYCCRS is less than K_CYCCRS_ZERO or,
- ISS_CYCCRS is greater than K_CYCCRS_32K or,
- LPTD_CYCCRS is less K_CYCCRS_ZERO or,
- LPTD_CYCCRS is greater than K_CYCCRS_32K or,
- PT1D_CYCCRS is less K_CYCCRS_ZERO or,
- PT1D_CYCCRS is greater than K_CYCCRS_32K or,
- PT2D_CYCCRS is less K_CYCCRS_ZERO or,
- PT2D_CYCCRS is greater than K_CYCCRS_32K or,

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490608.000	Func		N/A	CL12470851.001

LLR 12490609.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EE_CTRDAT_IF bits three, four, six and sixteen to TRUE, if ACCSTRTS_ALL is greater than K_CYCCRS_65K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490609.000	Func	DERIVED	No	CL12472931.000

Derived Requirement Justification:

ACCSTRTS_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

LLR 12490610.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EE_CTRDAT_IF bits three, four, seven and sixteen to TRUE, if ACCINFLSTRTS_ALL is less than K_CYCCRS_ZERO or greater than K_CYCCRS_32K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490610.000	Func		N/A	CL12472930.000

LLR 12490611.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The software shall set EE_CTRDAT_IF bits three, four, eight and sixteen to TRUE, if ACCNUMFL_ALL is greater than K_CYCCRS_65K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490611.000	Func	DERIVED	No	CL12472929.000

Derived Requirement Justification:

ACCNUMFL_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

LLR 12490612.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The software shall set EE_CTRDAT_IF bits three, four, nine and sixteen to TRUE, if ENG_RUN_CRS_ALL is less than K_CYCCRS_ZERO or greater than K_CYCCRS_32K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490612.000	Func		N/A	CL12472946.000

LLR 12490613.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The software shall set EE_CTRDAT_IF bits three, four, ten and sixteen to TRUE, if ENGLTM_CRS_ALL is less than K_CYCCRS_ZERO or greater than K_CYCCRS_32K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490613.000	Func		N/A	CL12472945.000

LLR 12490615.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The software shall set EE_CTRDAT_IF bits three, four and eleven and sixteen to TRUE, if any of the following conditions are met:

- ACCSHUTD_ALL is greater than K_CYCCRS_65K or,
- MTO_DL3 is less K_CYCCRS_ZERO or,
- MTO_DL3 is greater than K_CYCCRS_32K or,
- MTO_DL2 is less K_CYCCRS_ZERO or,
- MTO_DL2 is greater than K_CYCCRS_32K or,
- MTO_DL1 is less K_CYCCRS_ZERO or,
- MTO_DL1 is greater than K_CYCCRS_32K or,
- NTO_DL3 is less K_CYCCRS_ZERO or,
- NTO_DL3 is greater than K_CYCCRS_32K or,
- NTO_DL2 is less K_CYCCRS_ZERO or,
- NTO_DL2 is greater than K_CYCCRS_32K or,
- NTO_DL1 is less K_CYCCRS_ZERO or,
- NTO_DL1 is greater than K_CYCCRS_32K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490615.000	Func	DERIVED	No	CL12472944.000

Derived Requirement Justification:

ACCSHUTD_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

LLR 12490614.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The software shall set EE_CTRDAT_IF bits three, four, twelve and sixteen to TRUE, if ACCSTRTSFLT_ALL is greater than K_CYCCRS_65K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490614.000	Func	DERIVED	No	CL12472943.000

Derived Requirement Justification:

ACCSTRTSFLT_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

04.02.01.06 Torque Test Routine [E06-BIT_GA_Q_Test_Routine]

This process performs general hardware checks on torque conversions by wrapping around a torque test signal. It also sets up the torque interface for control execution. This is done because it is now the last initialization routine to be executed.

04.02.01.07 Initialization Stepper Motor Sum Voltage Check [E09-BIT_EEC_Init_SM_Sum_Volt]

This process will input the sum of the voltage across the four Stepper Motor Coils and determine if the summing hardware is operating correctly. This test is only run once during initialization and utilizes a tight range check that can be performed only when the +28V excitation to the Stepper Motor is disabled.

If an input hardware or interface fault is detected, a flag will be set to disable the Stepper Motor Fault detection Logic in the Built in Test section. The EEC will still run but the pilot will have to detect Stepper Motor failures through poor engine response.

04.02.01.08 Gate Array Selection Logic [E22-BIT_GA_GateArraySelect]

The HBOV is controlled by either Pulse Width Modulator #1 or #3, depending on which IIO4 Gate Array is in the control. This process will determine the Gate Array type, and set a flag informing the Bleed Output Process which PWM to use. PWM #1 and #3's output drivers are set to zero mA after the logic has run.

04.02.02 Background Executive [E07-Exec_Background]

The background logic serves to check the functional integrity of the processor and its associated memory. The instruction and the ROM (Read Only Memory) checksum tests are performed as time permits within the basic 1.40 msec interrupt.

This Process Continuously Performs Checks On CPU, ROM Memory and Interrupt Line.

Timing interrupts are enabled by the software at the start of background execution. The background execution consists of two built in tests, a ROM SUM test and an INSTRUCTION test, which are repeatedly executed.

The ROM SUM test is divided into page (1024 memory location) segments. The contents of the ROM memory locations in the page are summed, including a forcing number. If there are no ROM memory errors, the 16 least significant bits of the sum will equal zero. The execution time of the test is less than 10 msec per page.

The INSTRUCTION test verifies proper execution of the CPU instructions.

Background flags are set at the completion of each segment of the ROM SUM and the INSTRUCTION test. Those flags are tested in the foreground to verify proper execution of the background. The background increments a foreground counter which is cleared in the foreground. If the flag is not cleared in the foreground, the background will detect that the foreground is not functioning.

04.02.03 Foreground Executive [E08-EXEC_FOREGND]

Ov 12490052.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The foreground logic performs all of the functions associated with the EEC input and output operation. It contains all the I/O (Input/Output) Control logic, control dynamics, and control decision logic. The foreground logic is performed in 80 timing interrupts or states. This major control cycle of 80 states is divided into 4 minor control cycles of 20 states each.

This process controls foreground execution involving test-mode, A/D Converter Servicing, and State Table Control.

Ov 12490069.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

- E08_FG_STATETABLE

This is the table of 1 msec tasks which are scheduled by the primary executive program (E08_FG_EXECUTIVE). Each state is entered every 1.04msec due to an interrupt. All analog inputs are available through an input mux. The input and A/D mux addresses are set by setting the appropriate data bits of two OTA commands: OTA 306 and OTA 407. The analog signal conditioning stages have a settling time of 1.0 msec, and the A/D mux has a settling time 20 msec. The analog to digital conversion consumes 0.8 msec. To ensure proper operation, there must be 1 msec between settling of the input mux code and setting up of the A/D mux code. The appropriate codes for the input muxes and the A/D mux are contained in the state table. To maximize efficiency, the input mux code selections should alternate between the AC and DC (high/low level) input mux when possible. This allows a conversion every 1 msec interrupt.

Ov 12490070.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

- E08_FG_EXECUTIVE

LLR 12490071.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490071.000	0			CL12470930.000

LLR 12490072.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490072.000	0			CL12470930.000

LLR 12490073.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_C.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490073.000	0			CL12470930.000

LLR 12490074.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_D.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490074.000	0			CL12470930.000

LLR 12490075.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_E.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490075.000	0			CL12470930.000

LLR 12490053.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_F.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490053.000	0			CL12470930.000

LLR 12490054.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_G.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490054.000	0			CL12470930.000

LLR 12490055.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_H.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490055.000	0			CL12470930.000

LLR 12490056.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_EXECUTIVE shall be implemented per design model(s): E08_FG_EXECUTIVE_I.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490056.000	0			CL12470930.000

Ov 12490057.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

- E08_FG_SCHEDULE

This routine executes the foreground routines scheduled by the secondary executive program (E08_FG_SCHEDULE). It is executed every minor cycle (20.8 ms). Upon entry, this routine uses MINCYC to determine which of the four minor cycle states it is in and then sequentially executes the routines associated with that current minor cycle state. In Minor Cycle 1 (MINCYC=1), RC is used to determine which Thermally Limited Rated Power Algorithm to use

RC Value Module Called

- 1 C03_TLIMRT_1
- 2 C03_TLIMRT_2_1
- 2 C03_TLIMRT_2_2

3 C03_TLIMRT_3_1
3 C03_TLIMRT_3_2
4 C03_TLIMRT_4_1
4 C03_TLIMRT_4_2
5 C03_TLIMRT_5_1
5 C03_TLIMRT_5_2
6 C03_TLIMRT_6_1
6 C03_TLIMRT_6_2

When all the routines in a minor cycle have been executed, the routine returns back to the background processing.

LLR 12490058.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The EEC system shall maintain a margin of at least 15% spare memory in each category: stack usage, RAM, Program Memory, and non-volatile memory (NVM).

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490058.000	0			CL12472786.000

LLR 12490059.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The software shall have a timing margin target of at least 15% with respect to a major cycle (83.2 msec).

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490059.000	0			CL12472787.000

Ov 12490060.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed Designs:

LLR 12490061.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E08_FG_SECSEL shall be implemented per design model(s): E08_FG_SECSEL_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490061.000	0			CL12470930.000

Ov 12490062.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed Low Level Requirements:

LLR 12490064.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall call the following routines with the associated codes per Table 1 and Table 2.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490064.001	0			Ov12470016.000 Ov12470053.000 Ov12470077.000 Ov12470101.000 Ov12470119.000 Ov12470144.000 Ov12470158.000 Ov12470172.000 Ov12470194.000 Ov12470214.000 CL12470224.001 Ov12470262.000 Ov12470282.000 Ov12470298.000 Ov12470311.000 Ov12470322.000 Ov12470332.000 Ov12470344.000 Ov12470374.000 Ov12470386.000 CL12470525.000 CL12470553.000 Ov12470569.000 CL12470577.000 CL12470930.000 CL12471043.000 CL12471788.000 Ov12471948.000 CL12471957.000 CL12471972.002 CL12472028.000 CL12472105.000 CL12472129.000 CL12472146.000 CL12472166.000 CL12472644.000 CL12472660.000

Table:(LLR 12490064.001-1) Foreground State Table

Table 1: Foreground State Table

State Number	Module(s)/Routine(s) Called	Code For Muxes
S1	I17_PNHCVN_1 I28_DISINW1	-1
	I29_DISINW1 I30_DISINFAULT	
S2	N/A - None	-1
S3	N/A - None	-1
State Number	Module(s)/Routine(s) Called	Code For Muxes
S4	I31_UARTIN	-1
S5	O06	-1
S6	N/A - None	-1
S7	N/A - None	-1
S8	N/A - None	-1
S9	O07_ARO203_ALT10	-1
S10	I17_PNHCVN_2	-1
S11	I18_READNH I38_SMCSV_1	-1
S12	I38_SMCSV_2	S+SLLG+RVDT+SPLA1+HLVL
	I14_RDPLA_1	
S13	O07_ARO203_ALT11	S+SLLG+RVDT+SPLA2+LLVL
S14	I02_SGCOM_1	S+ST18+RVDT+SPLA2+LLVL
S15	I14_RDPLA_2	S+ST18+SEVT+LLVL
S16	I08_RDT18_1	S+HBV
	I45_HBOVWA_1 I22_SNHCVN_1	
S17	O08_ARO206_IAS	S+SPS+RVDT+SPLAG+LLVL
S18	N/A - None	-1
S19	I22_SNHCVN_2	-1
S20	I23_RDSNHP	-1
	I17_PNHCVN_1	
S21	O08_ARO206_VK	-1
S22	N/A - None	-1
S23	N/A - None	-1
S24	N/A - None	-1
S25	O09_ARO211_TAT	-1
S26	N/A - None	-1
S27	N/A - None	-1
State Number	Module(s)/Routine(s) Called	Code For Muxes
S28	N/A - None	-1
S29	O09_ARO211_T18	-1
S30	I17_PNHCVN_2	-1
	I18_READNH I38_SMCSV_1	-1
S31	I38_SMCSV_2	S+SPS+RVDT+SPLAG+HLVL
	I16_PLAGAIN_1	
S33	O10	S+SPS+SHLG+LLVL

S34	I04_PSAMBSG_1	S+SCJC+SHLG+HLVL
S35	I10_GAINCOM_1	S+SCJC+SQ2BT+LLVL
S36	I03_RDCJT_1	S+SDELP+SQ2BT+HLVL
S37	I11_TQBTRIM_1 I22_SNHCVN_1 O11	S+SDELP+RVDT+SPLA1+LLVL
S38	N/A - None	-1
S39	I22_SNHCVN_2	-1
S40	I23_RDSNHP	-1
S41	I17_RNHCVN_1 O07_ARO203_ALTPS	-1
S42	N/A - None	-1
S43	N/A - None	-1
S44	N/A - None	-1
S45	O25_ARO254	-1
S46	N/A - None	-1
S47	N/A - None	-1
S48	N/A - None	-1
S49	O15_ARO341	-1
S50	I17_PNHCVN_2	-1
S51	I18_READNH I38_SMCSV_1	-1
State Number	Module(s)/Routine(s) Called	Code For Muxes
S52	I38_SMCSV_2	S+SDELP+RVDT+SPLA1+HLVL
S53	I14_RDPLA_1 O16	S+SDELP+RDVT+SPLA2+LLVL
S54	I05_DELTAPSG_1	S+SLLDFT+RVDT+SPLA2+HLVL
S55	I14_RDPLA_2	S+SLLDFT+RVDT+SPLAB+LLVL
S56	I01_LLDRIPT_1	S+SEVT+HLVL
S57	I13_ENGVARIANT_1 I22_SNHCVN_1 O17	S+SPS+RVDT+SPLAB+LLVL
S58	N/A - None	-1
S59	I22_SNHCVN_2	-1
S60	I23_RDSNHP	-1
S61	I17_PNHCVN_1 O18_ARO270	-1
S62	N/A - None	-1
S63	N/A - None	-1
S64	N/A - None	-1
S65	O40_ARO06X	-1
S66	N/A - None	-1
S67	N/A - None	-1
S68	N/A - None	-1
S69	O27_ARO342	-1
S70	I17_PNHCVN_2	-1

S71	I18_READNH I38_SMCSV_1	-1
S72	I38_SMCSV_2	S+SPS+RVDT+SPLAB+HLVL
S73	I15_PLABIAS_1 O26_ARO345	S+SPS+SHLDFT+SPLAB+LLVL
S74	I04_PSAMBSG_1	S+STQ+SHLDFT+HLVL
S75	I09_HLDRIFT_1	S+STQ+SQ2GT+LLVL
State Number	Module(s)/Routine(s) Called	Code For Muxes
S76	I07_READTQ_1	S+SLLG+SQ2GT+HLVL
S77	I12+TQGTRIM_1 I22_SNHCVN_1 O08_ARO206_IASDP	S+SLLG+RVDT+SPLA1+LLVL
S78	N/A - None	-1
S79	I22_SNHCVN_2	-1
S80	I23_RDSNHP E15_BCKGNDTEST I32_ARINCIN_2	-1

Table:(LLR 12490064.001-2) Code Constants Definition

Table 2: Code Constants Definition	
Constant Name	Value
SDELP	0
SPS	1
SLLG	2
ST18	3
SLLDFT	4
STQ	5
SCJC	6
ST6	7
SHLG	0
SHLDFT	'10
RVDT	'20
SQ2BT	'30
SQ2GT	'40
ST6TR	'50
SEVT	'60
SQ1GT	'70
SPLA1	'500
Constant Name	Value
SPLA2	'400
SPLAB	'600
SPLAG	'700

LLR 12490065.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute the following modules defined in Table 3 when Minor Cycle, MINCYC, is equal to ONE.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490065.001	0			CL12470930.000 CL12471081.000 CL12471093.000 CL12471138.000 CL12471179.000 CL12471180.000 CL12471247.000 CL12471248.000 CL12471501.000 CL12471537.000 CL12471551.000 CL12472283.000 CL12472404.000 CL12472425.000 CL12472447.000 CL12472552.000 CL12472591.001 CL12472674.000 CL12472675.002 CL12472706.000

Table:(LLR 12490065.001-3) First Minor Cycle Subroutines

TABLE 3: First Minor Cycle Subroutines

Order	Subroutine	Additional Info for when to occur
1	EE_WRITE_BLOCK	
2	I05_DELTAPSG_2	
3	I05_DELTAPSG_3	
4	I08_RDT18_3	
5	I08_RDT18_2	
6	I08_RDT18_4	
7	I09_HLDRIFT_2	
9	I12_TQGTRIM_2	
10	I19_NHDUAL_1	
11	I19_NHDUAL_2	
12	I20_NHSPEED	
13	I24_NHPROBE2	
14	I25_TQNPT_7	
15	I25_TQNPT_1	
16	I26_TQCAL	
17	I36_PAMSEL	
18	I37_ARIISA_1	
19	I37_ARIISA_2	
20	E10_RAMTEST	
Order	Subroutine	Additional Info for when to occur
21	E16_SMTEST	
22	E17_NCFLT_1	
23	E17_NCFLT_2	
24	E18_FAULTSUM_1	
25	E19_TRMSTORE_1	
26	E19_TRMSTORE_2	
27	E19_TRMSTORE_4	
28	E19_TRMSTORE_5	
29	E23_PLATRIM	
30	E26_FAULTSTORE_4	
31	O01_SMOUT	
32	O02_HBOVOUT	
33	C01_START	
34	C10_ATOIGN	
35	C26_MPARAM_1	
36	C26_MPARAM_4	
37	C26_MPARAM_5	
38	C26_MPARAM_6	
39	C26_MPARAM_7	
40	E21_RECONFIG	
41	C03_TLIMRT_1	This only occurs when RC is equal to ONE
42.A	C03_TLIMRT_2_1	This only occurs when RC is equal to TWO
42.B	C03_TLIMRT_2_2	This only occurs when RC is equal to TWO
42.C	C03_TLIMRT_3_1	This only occurs when RC is equal to THREE
42.D	C03_TLIMRT_3_2	This only occurs when RC is equal to THREE

42.E	C03_TLIMRT_4_1	This only occurs when RC is equal to FOUR
42.F	C03_TLIMRT_4_2	This only occurs when RC is equal to FOUR
42.G	C03_TLIMRT_5_1	This only occurs when RC is equal to FIVE
42.H	C03_TLIMRT_5_2	This only occurs when RC is equal to FIVE
42.I	C03_TLIMRT_6_1	This only occurs when RC is equal to SIX
42.J	C03_TLIMRT_6_2	This only occurs when RC is equal to SIX
43	C03_TLIMRT_7	
Order	Subroutine	Additional Info for when to occur
44	C13_NHACCELLIM	
45	C14_NHLIM	
46	C15_NHDECELLIM	
47	C31_TQLIM	
48	C19_UNDERSPEED	
49	C20_OVRSPEED	
50	C11_SHPREQ_2	
51	C08_SHPCL	
52	C12_NPTGOV_1	
53	C12_NPTGOV_2	
54	C17_NHREQUEST_1	
55	C17_NHREQUEST_2	
56	C16_LOOPSEL	
57	C18_NHGOV_1	
58	C18_NHGOV_2	
59	C30_SMRTRQ	
60	C22_HBOVMM_1	
61	C22_HBOVMM_2	
62	C22_HBOVMM_3	
63	C21_HBOVEM_1	
64	C21_HBOVEM_2	
65	OPMODE_1	
66	OPMODE_2	
67	OPMODE_3	
68	RUNTIMER	
Order	Subroutine	Additional Info for when to occur
69	FLTTIMER	
70	INC_CTRS_1	
80	INC_CTRS_4	

LLR 12490066.002

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall execute the following modules defined in Table 4 when Minor Cycle, MINCYC, is equal to TWO.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490066.002	0			CL12470930.000 CL12470980.000 CL12471081.000 CL12471179.000 CL12471180.000 CL12471501.000 CL12471537.000 CL12471551.000 CL12472283.000 CL12472404.000 CL12472425.000 CL12472447.000 CL12472552.000 CL12472605.000 CL12472706.000

Table:(LLR 12490066.002-4) Table - 4: Second Minor Cycle Subroutine Order

Table 4: Second Minor Cycle Subroutine Order	
Order	Subroutine
1	EE_WRITE_BLOCK
2	I02_SGCOM_2
3	I04_PSAMBSG_2
4	I04_PSAMBSG_3
5	I04_PSAMBSG_4
6	I04_PSAMBSG_5
7	I10_GAINCOM_2
8	I14_RDPLA_3
9	I14_RDPLA_4
10	I15_PLABIAS_2
11	I16_PLAGAIN_2
12	I19_NHDUAL_1
13	I19_NHDUAL_2
14	I20_NHSPEED
15	I24_NHPROBE2
16	I25_TQNPT_1
17	I27_TQSPEED
18	I35_ARIPA
19	I43_TQSPEED_1
20	I43_TQSPEED_2
21	I43_TQSPEED_3
22	E11_CONTEST
23	E14_FAULTSTORE_1
24	E16_SMTEST
25	E17_NCFLT_3
26	E17_NCFLT_4
27	E18_FAULTSUM_2
28	E26_FAULTSTORE_1
29	I33_ARITAT
30	I34_TATSEL
31	E13_WATEST_4
32	E13_WATEST_5
33	E13_WATEST_6
34	E13_WATEST_7
35	O01_SMOUT
36	O02_HBOVOUT
37	C01_START
38	C10_ATOIGN
39	C26_MPARAM_2
40	C26_MPARAM_6
41	C26_MPARAM_7
42	C02_PLATRIM_1
43	C02_PLATRIM_2
44	C13_NHACCELLIM
45	C14_NHLIM
46	C15_NHDECELLIM
47	C31_TQLIM
48	C11_SHPREQ_2
49	C08_SHPCL
50	C27_NHMNMX_1
51	C17_NHREQUEST_1
52	C17_NHREQUEST_2
53	C16_LOOPSEL
54	C18_NHGOV_1
55	C18_NHGOV_2
56	C30_SMRTRQ
57	C22_HBOVMM_1
58	C22_HBOVMM_2
59	C22_HBOVMM_3
60	C21_HBOVEM_1
61	C21_HBOVEM_2
62	MISSIONS
63	RUNTIMER
64	FLTTIMER
65	INC_CTRS_2

LLR 12490067.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall execute the following modules defined in Table 5 when Minor Cycle, MINCYC, is equal to THREE.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490067.001	0			CL12470908.000 CL12470930.000 CL12470970.000 CL12471081.000 CL12471093.000 CL12471179.000 CL12471180.000 CL12471247.000 CL12471248.000 CL12471416.000 CL12471501.000 CL12471537.000 CL12471551.000 CL12472283.000 CL12472404.000 CL12472425.000 CL12472447.000 CL12472528.000 CL12472552.000 CL12472706.000

Table:(LLR 12490067.001-5) Third Minor Cycle Subroutine Order

Table 5: Third Minor Cycle Subroutine Order

Order	Subroutine
1	EE_WRITE_BLOCK
2	I01_LLDRIFT_2
3	I05_DELTAPSG_4
4	I07_READTQ_2
5	I13_ENGVARIANT_2
6	I19_NHDUAL_1
7	I19_NHDUAL_2
8	I20_NHSPEED
Order	Subroutine
9	I24_NHPROBE2
10	I25_TQNPT_1
11	I26_TQCAL
12	I36_PAMSEL
13	I37_ARIISA_1
14	I37_ARIISA_2
15	I45_HBOVWA_2
16	I46_IASFAULT
17	E16_SMTEST
18	E17_NCFLT_5
19	E17_NCFLT_6
20	E18_FAULTSUM_3
21	E19_TRMSTORE_1
22	E19_TRMSTORE_2
23	E19_TRMSTORE_4
24	E19_TRMSTORE_5
25	E23_PLATRIM
26	E25_FAULTDET
27	E26_FAULTSTORE_2
28	E27_MFCID
29	E31_EXTDIS
30	O01_SMOUT
31	O02_HBOVOUT
32	C01_START
33	C10_ATOIGN
34	C26_MPARAM_1
Order	Subroutine
35	C26_MPARAM_3
36	C26_MPARAM_4
37	C26_MPARAM_5
38	C26_MPARAM_6
39	C26_MPARAM_7
40	C04_RPCAL_2_D
41	C05_TQBUG
42	C13_NHACCELLIM
43	C14_NHLIM

44	C15_NHDECELLIM
45	C31_TQLIM
46	C19_UNDEERSPEED
47	C20_OVRSPEED
48	C11_SHPREQ_2
49	C08_SHPCL
50	C12_NPTGOV_1
51	C12_NPTGOV_2
52	C17_NHREQUEST_1
53	C17_NHREQUEST_2
54	C16_LOOPSEL
55	C18_NHGOV_1
56	C18_NHGOV_2
57	C30_SMRTRQ
58	C22_HBOVMM_1
59	C22_HBOVMM_2
Order	Subroutine
60	C22_HBOVMM_3
61	C21_HBOVEM_1
62	C21_HBOVEM_2
63	C24_OVRTQTST
64	RUNTIMER
65	FLTTIMER
66	INC_CTRS_3
67	LCF_CHECK

LLR 12490068.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall execute the following modules defined in Table 6 when Minor Cycle, MINCYC, is equal to FOUR.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490068.001	0			CL12470930.000 CL12470980.000 CL12470992.000 CL12471081.000 CL12471179.000 CL12471180.000 CL12471324.000 CL12471463.000 CL12471464.000 CL12471501.000 CL12471537.000 CL12471551.000 CL12471660.000 CL12471763.000 CL12472239.000 CL12472253.000 CL12472270.000 CL12472283.000 CL12472296.000 CL12472315.000 CL12472361.000 CL12472385.000 CL12472404.000 CL12472425.000 CL12472447.000 CL12472484.000 CL12472505.000 CL12472552.000

Table:(LLR 12490068.001-6) Fourth Minor Cycle Subroutine Order

Table 6: Fourth Minor Cycle Subroutine Order

Order	Subroutine
1	EE_WRITE_BLOCK
2	I03_RDCJT_3
3	I03_RDCJT_2
4	I04_PSAMBSG_2
5	I04_PSAMBSG_3
6	I04_PSAMBSG_4
7	I04_PSAMBSG_5
8	I07_READTQ_3
9	I11_TQBTRIM_2
10	I14_RDPLA_3
11	I14_RDPLA_4
12	I19_NHDUAL_1
13	I19_NHDUAL_2
14	I20_NHSPEED
15	I24_NHPROBE2
16	I25_TQNPT_1
17	I27_TQSPEED
18	I35_ARIPA
19	I43_TQSPEED_1
20	I43_TQSPEED_2
21	I43_TQSPEED_3
22	E13_WATEST_1
23	E13_WATEST_2
24	E13_WATEST_3
25	E14_FAULTSTORE_1
26	E14_FAULTSTORE_2
27	E14_FAULTSTORE_3
28	E14_FAULTSTORE_4
29	E14_FAULTSTORE_5
30	E14_FAULTSTORE_6
31	E16_SMTEST
32	E18_FAULTSUM_4
33	E26_FAULTSTORE_1
34	E28_HBOVWA_1
35	E28_HBOVWA_2
36	O01_SMOUT
37	O02_HBOVOUT
38	O03_DISOUT
39	C01_START
40	C10_ATOIGN
41	C26_MPARAM_6
42	C26_MPARAM_7
43	C02_PLATRIM_1
44	C02_PLATRIM_2
45	C06_HPREF_1

46	C06_HPREF
47	C07_OVRTQPRO
48	C13_NHACCELLIM
49	C14_NHLIM
50	C15_NHDECELLIM
51	C31_TQLIM
52	C11_SHPREQ_1
53	C11_SHPREQ_2
54	C08_SHPCL
55	C27_NHNMNMX_1
56	C17_NHREQUEST_1
57	C17_NHREQUEST_2
58	C16_LOOPSEL
59	C18_NHGOV_1
60	C18_NHGOV_2
61	C30_SMRTRQ
62	C22_HBOVMM_1
63	C22_HBOVMM_2
64	C22_HBOVMM_3
65	C21_HBOVEM_1
66	C21_HBOVEM_2
67	RUNTIMER
68	FLTTIMER
69	UPDATE_EE_DATA
70	DCU_SCHEDULE
71	UART_IO_SELECT

04.02.04 Foreground Tests

This logic completes a system check, that is: RAM tests, connector integrity, loss of power, and UART interface tests. Tests are done at predetermined time intervals.

04.02.04.01 Partial RAM and BREG Test [E10-BIT_EEC_Partial_RAM_BREG]

This process performs a Partial RAM Test along with a B Register (BREG) Test during Foreground Execution.

04.02.04.02 Loss of Connector [E11-BIT_EEC_Loss_Connector]

This process examines specific bits of the non-critical fault words, NCFAL1 and NCFAL2, to logically determine the integrity of the J1 and J4 connections. Detection of either connectors separation from the EEC is determined by examining the NCFAL words for specific bit patterns. If detected, specific bits of CONNIF, the loss of Connector Interface Status Word, are set.

04.02.04.03 MFC Identification Verification [E27-Maint_Misc_MFC_Id]

This process verifies that the MFC installed on the aircraft is a PW127 and not a PW124B. If the MFC on the aircraft is determined to be a PW124B, then a fault is set.

04.02.04.04 Wraparound Tests [E13-WATEST]

There are seven wraparound (W/A) tests performed on the output discretes. These tests assume that the state of the hardware outputs must equal the state defined by the control discretes and control laws. The seven tests check the two High Side and the five Low Side Switches.

The test for High Side Switch #1 incorporates three separate tests with "Go/No Go" logic performed before each test. The Low Side W/A tests simply verify that the Output equals the Input. A W/A test is considered to have failed when the requested status does not match the W/A status. If the test fails with sufficient frequency as to have exceeded the maximum allowable fault count, the latch bit (bit 16) for the appropriate interface status word will be set. Lightning tolerance is improved and false W/A faults are avoided by recording the latched state only after a timeout of an associated transient fault timer.

For Low Side Switches and HSS#2, if the Output Discrete equals "1", the W/A discrete should equal "1". However, HSS#1 uses negative logic. The Output Discrete should be the opposite of the W/A.

04.02.04.04.01 Bleed Wraparound Tests [E28-BIT_EEC_Bleed_W/A_Tests]

This process verifies that the HBOV system is functioning. The HBOV system includes the output drivers, wraparound circuits and Bleed torque motor. The check on this system is considered a gross check, as it does not attempt to isolate the cause of an HBOV system failure.

The analog Bleed wraparound is checked at a specific Bleed set-point, unless the HBOV hardware (High Side or Low Side) is commanded off. The discrete wraparound is checked continuously, unless the EEC is Fail Fixed.

04.02.04.05 EEPROM Fault Storage [E14-FAULT_STORAGE]

This routine stores up to 8 system faults in EEPROM in the order in which they occur. Three bytes of information are stored for each fault: LRU Code Number, the related I/F status word, and the cold junction temperature. The pilot and maintenance personnel use this information to identify failed units.

This process will also clear faults from EEPROM when the appropriate conditions are met.

04.02.04.05.01 Internal EEC Fault Storage [E26-INT_FAULTSTORE]

This routine stores up to 8 internal EEC faults in EEPROM in the order which they occur. Three bytes of information are stored for each fault: LRU Code Number, the related I/F status word, and the cold junction temperature. Once these faults are stored, they may only be cleared manually using test equipment and UART.

04.02.04.06 Background Functional Test [E15-BIT_Proc_Background]

This process checks the background functional flag to be sure the background program is operating properly.

The logic checks the Background functionality flags. If both flags are not set to one, then increment the fault counter. If the flags are not both set for three consecutive passes, a processor/memory failure word is set to indicate a failure. If the Background functionality flags are both set, clear them so that the background logic must reset them. Also, check IC, the initial condition flag. If it does not equal zero, increment IC. IC is checked if the background functionality flags are not set.

04.02.04.07 Stepper Motor Phase Fault Test [E16-BIT_EEC_SM_Phase]

This process will detect and confirm Stepper Motor Faults due to coil opens or shorts. A critical Fault Flag will be set upon the confirmation of such a failure. This process will be bypassed if the sum voltage interface (BIT circuit) has failed.

A range check is performed on the Stepper Motor Coil sum voltage. The tolerance is set around the nominal (expected) voltage and is wide enough to eliminate false flags. If a fault is suspected, a confirmation test is begun. The logic will distinguish between intermittent and hard failures.

04.02.04.08 Internal EEC Fault Detection [E25-BIT_EEC_Int_Fault_Detect]

This process is responsible for monitoring those faults which are considered to be indicative of an internal EEC failure. When one or more internal EEC faults are detected, the internal EEC failure word, ECINTF, is set to indicate the condition.

04.02.05 Fault Logic

Summarizes systems faults into status words and performs fault action on system control as delegated by a fault summary table.

04.02.05.01 Non-Critical Faults [E17-NONCRITICAL_FAULT]

Ov 12490076.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process examines the latched system faults and prepares six non-critical fault words, NCFAL1 through NCFAL6. The non-critical fault words are then monitored in another process for fault accommodation.

Ov 12490085.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCDs: SCR: PW127L-EEC-SW-PWC-11362 SCO-D: PW127L-EEC-SW-PWC-12134
SCR: PW127L-EEC-SW-PWC-11551
SCR: PW127L-EEC-SW-PWC-11560

Ov 12490086.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

LLR 12490087.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_1 shall be implemented by E17_NCFLT_1_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490087.000	0			CL12472707.000 CL12472713.000

LLR 12490088.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_1 shall be implemented by E17_NCFLT_1_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490088.000	0			CL12472707.000 CL12472713.000

LLR 12490089.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_2 shall be implemented by E17_NCFLT_2_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490089.000	0			CL12472707.000 CL12472713.000

LLR 12490090.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_2 shall be implemented by E17_NCFLT_2_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490090.000	0			CL12472707.000 CL12472713.000

LLR 12490091.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_3 shall be implemented by E17_NCFLT_3_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490091.000	0			CL12472707.000 CL12472713.000

LLR 12490092.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_3 shall be implemented by E17_NCFLT_3_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490092.000	0			CL12472707.000 CL12472713.000

LLR 12490077.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_4 shall be implemented by E17_NCFLT_4_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490077.000	0			CL12472707.000 CL12472713.000

LLR 12490078.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_4 shall be implemented by E17_NCFLT_4_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490078.000	0			CL12472707.000 CL12472713.000

LLR 12490079.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_5 shall be implemented by E17_NCFLT_5_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490079.000	0			CL12472707.000 CL12472713.000

LLR 12490080.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_5 shall be implemented by E17_NCFLT_5_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490080.000	0			CL12472707.000 CL12472713.000

LLR 12490081.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_6 shall be implemented by E17_NCFLT_6_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490081.000	0			CL12472707.000 CL12472713.000

LLR 12490082.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_6 shall be implemented by E17_NCFLT_6_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490082.000	0			CL12472707.000 CL12472713.000

LLR 12490083.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E17_NCFLT_7 shall be implemented directly by HLRs PWC-6071, PWC-6072, PWC-6073, PWC-6074, PWC-6075, PWC-6077, and PWC-6081

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490083.000	0			CL12472734.000 CL12472737.000 CL12472740.000 CL12472743.000 CL12472746.000 CL12472749.000

Ov 12490084.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed Designs:

None

Changed Low Level Requirements

None

04.02.05.02 Fault Summary [E18-FAULTSUM]

Ov 12490093.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process monitors the non-critical fault words and sets the corresponding bits in the Fault Action Words. There are four different fault actions words. One for the hardware outputs, two that will disable the ARINC outputs, and one to act upon the control laws.

Ov 12490100.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCDs: SCR: PW127L-EEC-SW-PWC-11205 SCO-D: PW127L-EEC-SW-PWC-12136

SCR: PW127L-EEC-SW-PWC-11550

SCR: PW127L-EEC-SW-PWC-11560

Ov 12490101.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

LLR 12490102.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_1 shall be implemented per design model E18_FAULTSUM_1_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490102.000	0			CL12471082.000

LLR 12490103.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_1 shall be implemented per design model E18_FAULTSUM_1_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490103.000	0			CL12471082.000

LLR 12490104.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_2 shall be implemented per design model E18_FAULTSUM_2_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490104.000	0			CL12471084.000

LLR 12490105.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_2 shall be implemented per design model E18_FAULTSUM_2_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490105.000	0			CL12471084.000

LLR 12490106.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_3 shall be implemented per design model E18_FAULTSUM_3_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490106.000	0			CL12471085.000

LLR 12490107.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_3 shall be implemented per design model E18_FAULTSUM_3_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490107.000	0			CL12471085.000

LLR 12490094.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_4 shall be implemented per design model E18_FAULTSUM_4_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490094.000	0			CL12471083.000

LLR 12490095.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_4 shall be implemented per design model E18_FAULTSUM_4_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490095.000	0			CL12471083.000

LLR 12490096.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_5 shall be implemented directly per HLRs PWC-11401, PWC-11442, PWC-11444, and PWC-11445.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490096.000	0			CL12471082.000 CL12471083.000 CL12471084.000 CL12471085.000

LLR 12490097.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module E18_FAULTSUM_6 shall be implemented directly per HLRs PWC-11401, PWC-11442, PWC-11444, and PWC-11445.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490097.000	0			CL12471082.000 CL12471083.000 CL12471084.000 CL12471085.000

Ov 12490098.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed Designs:
None

Ov 12490099.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed Low Level Requirements:
None

04.02.06 EEPROM Trim Data Storage [E19-BIT_Store_EEPROM_Trim_Data]

Ov 12490108.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Identification of EEC detectable faults is achieved by polling the interface status words and creating a non-critical fault words (NCFAL1-NCFAL6). Each word identifies a section of the control system where the fault originated. The fault code, the interface status word and the cold junction temperature are stored in EEPROM when the fault has been identified. Eight faults are saved in the order they occurred. Since these faults are stored in EEPROM, a non-volatile memory device, subsequent +28V airframe power cycling does change the memory stored in memory.

This process writes the torque gain and bias trims, the Engine Rating, and the selected Pulse Width Modulator for Bleed to EEPROM.

Ov 12490109.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCDs: SCR: PW127L-SW-EEC-PWC-13745 SCO-D: PW127L-EEC-SW-PWC-14143

Ov 12490110.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

LLR 12490111.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The Design E19_TRMSTORE_1 shall be implemented per design model E19_TRMSTORE_1_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490111.000	0			CL12471094.000 CL12471095.000 CL12471096.000

LLR 12490112.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The Design E19_TRMSTORE_2 shall be implemented per design model E19_TRMSTORE_2_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490112.000	0			CL12471094.000 CL12471095.000 CL12471096.000

LLR 12490113.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The Design E19_TRMSTORE_4 shall be implemented per design model E19_TRMSTORE_4_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490113.000	0			CL12471094.000 CL12471095.000 CL12471096.000

LLR 12490114.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The Design E19_TRMSTORE_5 shall be implemented per design model E19_TRMSTORE_5_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490114.000	0			CL12471094.000 CL12471095.000 CL12471096.000

Ov 12490115.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed designs:
None

Ov 12490116.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed Low Level Requirements:
None

04.02.06.01 PLA Trim EEPROM Data Storage [E23-BIT_Store_PLA_Trim_Data]

This process will write the PLA Trim to EEPROM. Once a new Trim Value is calculated, the code will start a counter that will control the write process. This latch will be maintained even when the PLA Trim is completed, preventing a bypass of the write if EEPROM is busy with some other process.

Since PLA Trim is a two Byte word, each byte is written and then verified. If a write failure occurs, the interface flag is set, the counter is cleared, and the process is bypassed. After the PLA write is complete (or fails) a flag is set to prevent another trim calculation from occurring until power is cycled.

04.02.07 Test Mode [E20-Maint_Dev_Test_Mode]

Test Mode logic is a general purpose, interactive, diagnostic debugging routine which allows communication with the EEC CPU from an external computer (GDTS or DTS). When in test mode, the processor communicates with the external computer via the UART interface.

Test mode allows external communication via the UART.

- DUMP

This subroutine performs the Test Mode DUMP operation. It reads two 16-bit addresses from the UART, and then transmits the contents of all the memory locations between those two addresses (inclusive) out the UART. After the entire block has been transmitted, it sends an ACK and returns to the Test Mode executive routine. If any input errors occur, the command operation is immediately aborted.

04.02.08 Reconfiguration Logic [E21-RECONFIG]

Ov 12490117.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process will monitor the Rating of the Engine. If the logic detects a legal change engine rating, the EEC must reconfigure to accommodate the change.

Ov 12490118.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCD(s): SCR: PW127-EEC-SW-PWC-15837 SCO-D: PW127-EEC-SW-15862

Ov 12490119.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design is incorporated in modules E21_RECONFIG. Designs changed from the prior baseline are identified in section 11.

Ov 12490120.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design of E21_RECONFIG is defined by the following LLR:

LLR 12490121.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Design model E21_RECONFIG shall be implemented.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490121.000	0			CL12471139.000 CL12471140.000 CL12471141.000 CL12471142.000 CL12471143.000 CL12471144.000 CL12471145.000

04.02.09 Program Identification Module [E30-IDENTIFI]

Ov 12490122.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

This section defines the Program Identification defining the S/W which consists of Engine Program, Software Version and Software Checksum.

LLR 12490133.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute this process once on power up.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490133.000	0			CL12472711.001

LLR 12490134.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall use the number 19 for ENGPID to represent the letter "S" of the engine model.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490134.001	0			CL12472719.001

LLR 12490135.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall set KSBLID equal to 1.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490135.001	0			CL12472720.001

LLR 12490136.002

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall set KSSVID equal to 1.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490136.002	0			CL12472721.001

LLR 12490123.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set PWMSI2 equal to 0.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490123.001	0			CL12472722.001

LLR 12490595.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set PWMSI1 equal to 0.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490595.000	0			CL12473012.000

LLR 12490596.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set specific address for ENGPID, KSBLID, KSSVID, PWMSI1 and PWMSI2 in the program memory as per the table: Software ID Memory Address.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490596.000	0			CL12472973.000

Table:(LLR 12490596.000-1) Software ID Memory Address

Constants	Address in Program Memory Page/Location (octal)
ENGPID	37/1772
KSBLID	37/1773
KSSVID	37/1774
PWMSI1	37/1775
PWMSI2	37/1776

LLR 12490124.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall reserve two locations for a 32-bit checksum value computed for the software version using CKSUMH and CKSUML as per the table: Checksum Memory Address.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490124.001	0			CL12472723.001

Table:(LLR 12490124.001-1) Checksum Memory Address

Checksum	Address in Program Memory Page/Location (octal)
CKSUMH (Software 32-bit checksum MSH)	37/1770
CKSUML (Software 32-bit checksum LSH)	37/1771

LLR 12490125.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set SV_SW_EPI equal to ENGPID.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490125.000	0			CL12472726.000

LLR 12490126.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set SV_SW_SBI equal to KSBLID.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490126.000	0			CL12472727.000

LLR 12490127.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set SV_SW_SVI equal to KSSVID.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490127.000	0			CL12472728.000

LLR 12490128.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set SV_PWC_PATCHID equal to PWMSI2.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490128.000	0			CL12472725.000

04.02.10 External Discrete Word [E31-Maint_Misc_Ext_Disc_Word]

This process takes discrete information from DISC1, DISC2, DISCOT, and PLATDI and combines this into a single discrete word EXDISC which may be transmitted over UART.

LLR 12490623.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The Design E31_EXTDIS_G shall be implemented per design model E31_EXTDIS_G.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490623.000	Func	HLR	No	CL12470909.002

04.03 SOFTWARE REQUIREMENTS

04.03.01 Fuel Control System - Outer Loop

04.03.01.01 NH Governing Loop

04.03.01.01.01 NH Minimum/Maximum Calculation [C27-NHMINMAX]

This logic calculates a minimum and maximum NH range limit to satisfy the requirements of idle speed governing, low PLA power loop authority control and NH maximum speed limit. Hotel Mode and low power Uptrim (positive indication for autofeather test) are also accommodated. The PLA resetting logic limits the engine torque by setting PLA to FI when a feather condition is detected.

LLR 12490624.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C27_NHNMNMX_1 shall be implemented per design model C27_NHNMNMX_1.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490624.000	Func	HLR	No	CL12472903.000 CL12472909.000 CL12472910.000 CL12472914.000 CL12472928.000 CL12473031.000 CL12473032.000 CL12473033.000 CL12473034.000 CL12473035.000 CL12473036.000 CL12473037.000 CL12473038.000 CL12473039.000 CL12473040.000 CL12473041.000 CL12473042.000 CL12473043.000 CL12473044.000 CL12473046.000

04.03.01.01.02 NH Governing Logic [C18-Gov_NH_Closed_Loop]

Ov 12490137.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process calculates a stepper motor rate request required to govern high rotor speed (NH) at the calculated request (NHREQ) using closed loop compensation, for which the parameters are scheduled as a function of NHC, as well as station 1.5 temperature and pressure.

Ov 12490138.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCD(s): SCR: PW127-EEC-SW-PWC-16235 SCO-D: PW127-EEC-SW-16248

Ov 12490139.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design is incorporated in modules C18_NHGOV_1 and C18_NHGOV_2. Designs changed from the prior baseline are identified in section 11.

Ov 12490140.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design of C18_NHGOV_1 is defined by the following LLR:

LLR 12490141.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Design model C18_NHGOV_1.DATAFLOW shall be implemented by the flow.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490141.000	0			CL12471182.000 CL12471183.000 CL12471184.000 CL12471185.000 CL12471186.000 CL12471187.000 CL12471188.000 CL12471189.000 CL12471190.000

Ov 12490142.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design of C18_NHGOV_2 is defined by the following:

Ov 12490143.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Design model C18_NHGOV_2.DATAFLOW shall be implemented by the flow.

04.03.01.01.03 Closed Loop on Power [C08-Gov_SHP_Closed_Loop]

This control logic calculates high rotor speed reference (NHREF1) required to achieve the closed loop power request (HPREQ) using proportional plus integral control action.

NOTE: Some calculations performed in this process are in double precision. The parameters and calculations which are double precision in this process are: NHREF1, NHHPI, NHHPP, NHHPR and the unnamed variable into the addition with NHHPR to calculate NHHPI.

04.03.01.01.04 PLA Trim Calculation [C02-PLATRIM_CALC]

Ov 12490679.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The PLA trim calculation permits the pilot to remove any PLA rigging inaccuracies at the nominal quadrant PLA position. Upon cockpit discrete activation of the logic, it detects whether the PLA is within a suitable distance of the nominal PLA before latching an appropriate trim value. The PLA trim value is retained in non-volatile memory and shall offset the actual PLA calculated from the MFC sensor values while actual PLA is high.

LLR 12490678.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C02_PLATRIM_1 shall be implemented per design model C02_PLATRIM_1.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490678.000	Func	HLR	No	CL12471228.001 CL12471229.001 CL12471230.001 CL12472876.000 CL12472895.000 CL12472896.000 CL12473007.000 CL12473008.000

LLR 12490680.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C02_PLATRIM_2 shall be implemented per design model C02_PLATRIM_2.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490680.000	Func	HLR	No	CL12471227.001 CL12471232.001 CL12471233.001 CL12473009.000

04.03.01.02 NPT Governing Logic [C12-Gov_NPT_Closed_Loop]

Ov 12490144.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Propeller speed is normally governed by the EEC when PLA is less than 35 degrees. The intent of propeller governing is to provide both quiet ground handling and a higher thrust schedule for approach and landing. NPTErr is formed by subtracting NPT from the NPT reference and putting the error through a PID controller.

The NPT governing can be disabled under the following conditions:

1. PLA greater than 70 degrees and NPTErr less than NHERR
2. NPT less than 6000 RPM
3. Propeller is in the feathered state

4. NPOSDF is set
5. VK greater than 125 knots and PLA is greater than 30 degree
6. QTOT greater than 100%
7. DMPMFC is set and PLA is less than 30 degrees
8. DMPMFC alone is set

Ov 12490145.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCD(s): SCR: PW127-EEC-SW-PWC-16235 SCO-D: PW127-EEC-SW-16248

Ov 12490146.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design is incorporated in modules C12_NPTGOV_1 and C12_NPTGOV_2. Designs changed from the prior baseline are identified in section 11.

Ov 12490147.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design of C12_NPTGOV_1 is defined by the following LLR:

LLR 12490148.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Design model C12_NPTGOV_1.DATFLOW shall be implemented by the flow.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490148.000	0			CL12471250.000 CL12471251.000 CL12471252.000 CL12471253.000 CL12471254.000 CL12471255.000 CL12471256.000 CL12471257.000 CL12471258.000 CL12471259.000 CL12471260.000 CL12471261.000

Ov 12490149.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design of C12_NPTGOV_2 is defined by the following:

Ov 12490150.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Design model C12_NPTGOV_2.DATAFLOW shall be implemented by the flow.

04.03.01.03 NH Acceleration Limiting Logic [C13-Trans_Accel_NH]

This process calculates a high rotor speed request rate (DNHACC) that is used to constrain NH acceleration.

04.03.01.04 NH Limiting Logic [C14-Limit_NH]

This process calculates a high rotor speed request rate (DNHLIM) required to affect NH limiting.

04.03.01.05 NH Deceleration Limiting Logic [C15-Trans_Decel_NH]

This process calculates a high rotor speed request rate (DNHDEC) that is used to constrain NH deceleration.

04.03.01.06 Loop Selection Logic [C16-Gov_Inner_Loop_Sel]

This process calculates a discrete (ILOOP) that identifies which one of nine loops is being invoked. The HP governing loop integrator tracking logic is also calculated in this process.

04.03.02 Fuel Control system - Inner Loop

04.03.02.01 Overtorque Protection Logic [C07-LIMIT_OVERQ_PROTECT]

Ov 12490618.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process is designed to torque limit the Range Limited Power Reference (HPREF) based on QLM0, QHPL and the applied Dead-Band on NP, in order and calculate the Requested Torque Calculated From HPREF (QHPLCP) based on the blending factor obtained from PLA.

LLR 12490597.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C07_OVRTQPRO shall be implemented per design model C07_OVRTQPRO.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490597.000			N/A	CL12473020.000 CL12473021.000 CL12473022.000 CL12473023.000 CL12473024.000

04.03.02.02 Propeller Underspeed Logic [C19-Limit_Prop_Underspeed]

The propeller underspeed logic is used to boost engine speed in anticipation of a significant load increase to the propeller system. This increase in engine speed is required to maintain NPT above the point at which the generator will drop off-line. This condition exists when the propeller passes through its "disking" point (i.e., the propeller zero-load blade angle point).

This logic is active at all PLA positions below flight idle; however, the value of the boost is determined as a function of PLA and air speed. The profile of the boost is that of a derivative lag. The time constant for this lag is determined as a function of the difference between PLADB and the calculated disking point. The calculated boost, in RPM, is the output of a Tustin Filter.

04.03.02.03 Propeller Overspeed Logic [C20-Limit_Prop_Overspeed]

This process calculates NPT slam flags for upward and downward PLA slams, and sets five associated discrete flags. The calculation is based on NPT underspeed PLA signal (PLADB), NPT, and NPTDOT.

04.03.02.04 Horsepower Request Calculation [C11-Gov_SHP_Req_Calc]

This process calculates horsepower set-point request (HPREQ), using the HPREF calculated and the uptrim commanded flag (UPTFLG).

04.03.02.05 NH Request Calculation [C17-Gov_Inner_NH_Req_Calc]

This logic calculates a NH request based on the NH reference calculated by the primary HP loop and subject to the constraints imposed by high rotor acceleration, deceleration, NH minimum, NH maximum, as well as NPT governing.

04.03.02.06 Stepper Motor Rate Request [C30-Gov_Inner_SM_Rate_Req]

This process calculates stepper motor rate request (STEPS) based on primary rate command calculated by the NH governing logic as well as secondary requests calculated by the NPT BUMP/DUMP process.

04.03.03 Rating Logic

04.03.03.01 Thermally Limited Rated Power Algorithm [C03-TLIMRT_D]

Ov 12490151.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process calculates the thermally limited rated shaft horsepower. The calculation is based on ambient conditions, airspeed, bleed level, and engine rating.

LLR 12490156.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C03_TLIMRT_1 shall be implemented per design model C03_TLIMRT_1.CTRLFLOW.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490156.001	0			CL12472893.000 CL12473052.000 CL12473053.000 CL12473054.000 CL12473055.000

LLR 12490157.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C03_TLIMRT_2_1 shall be implemented per design model C03_TLIMRT_2_1.CTRLFLOW.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490157.001	0			CL12472959.000 CL12473052.000 CL12473053.000 CL12473054.000 CL12473055.000

LLR 12490681.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C03_TLIMRT_2_2 shall be implemented per design model C03_TLIMRT_2_2.CTRLFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490681.000	0			CL12472882.000 CL12472892.000

LLR 12490158.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C03_TLIMRT_3_1 shall be implemented per design model C03_TLIMRT_3_1.CTRLFLOW.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490158.001	0			CL12472958.000 CL12473052.000 CL12473053.000 CL12473054.000 CL12473055.000

LLR 12490682.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C03_TLIMRT_3_2 shall be implemented per design model C03_TLIMRT_3_2.CTRLFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490682.000	0			CL12472881.000 CL12472891.000

LLR 12490159.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C03_TLIMRT_4_1 shall be implemented per design model C03_TLIMRT_4_1.CTRLFLOW.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490159.001	0			CL12472957.000 CL12473052.000 CL12473053.000 CL12473054.000 CL12473055.000

LLR 12490683.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C03_TLIMRT_4_2 shall be implemented per design model C03_TLIMRT_4_2.CTRLFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490683.000	0			CL12472880.000 CL12472890.000

LLR 12490160.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C03_TLIMRT_5_1 shall be implemented per design model C03_TLIMRT_5_1.CTRLFLOW.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490160.001	0			CL12472956.000 CL12473052.000 CL12473053.000 CL12473054.000 CL12473055.000

LLR 12490684.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C03_TLIMRT_5_2 shall be implemented per design model C03_TLIMRT_5_2.CTRLFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490684.000	0			CL12472878.000 CL12472888.000

LLR 12490152.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C03_TLIMRT_6_1 shall be implemented per design model C03_TLIMRT_6_1.CTRLFLOW.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490152.001	0			CL12472955.000 CL12473052.000 CL12473053.000 CL12473054.000 CL12473055.000

LLR 12490685.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C03_TLIMRT_6_2 shall be implemented per design model C03_TLIMRT_6_2.CTRLFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490685.000	0			CL12472879.000 CL12472889.000

LLR 12490686.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C03_TLIMRT_7 shall be implemented per design model C03_TLIMRT_7.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490686.000	0			CL12472961.000 CL12472962.000 CL12472963.000 CL12472964.000 CL12472965.000 CL12473070.000 CL12473071.000

04.03.03.02 Rated Power Calculation [C04-PWRCALC_D]

Ov 12490162.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The rating logic is used to calculate the rated power levels for each flight mode. These power levels are used to generate the reference power for closed loop on power and cockpit rating display. To permit the overtorque test logic to be generic, the overtorque thresholds and time limits required are assigned to output variables in this process.

LLR 12490167.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C04_RPCAL_2_D shall be implemented per design model C04_RPCAL_2_D.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490167.000	0			CL12471421.001 CL12471426.000 CL12471427.000 CL12471428.000 CL12471429.000 CL12471430.000 CL12471431.000 CL12471432.000 CL12471433.000 CL12471434.000 CL12471435.000 CL12471436.000 CL12471438.000 CL12471439.000 CL12471440.000 CL12471441.000 CL12471442.000 CL12471443.000 CL12471444.000 CL12471445.000 CL12471446.000 CL12471447.000 CL12471448.000 CL12472932.000 CL12472933.000 CL12472934.000 CL12472937.000 CL12472940.000 CL12472941.000 CL12472942.000 CL12473047.000

04.03.03.03 Horsepower Reference Calculation [C06-POWER_REQUEST]

Ov 12490169.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

This process calculates pre-torque limited power request (QHPL) based on the Uptrim (DISC1.6), the MCT enable (DISC1.3), the Derating Level (ENGRAT) and the rate limited PLA.

The DMPMFC resets the rate limit over the PLA to avoid limiting the power requested during rapid high power to low power maneuver.

LLR 12490619.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C06_HPREF_1 shall be implemented per design model C06_HPREF_1.DATAFLOW

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490619.000	Func	HLR	No	CL12471451.001 CL12472999.000 CL12473000.000 CL12473001.000 CL12473002.000 CL12473065.000

Requirement Design Notes- (LLR 12490619.000)

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The default value of the Reset is covered by KFALSE.

LLR 12490173.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C06_HPREF shall be implemented per design model C06_HPREF.DATAFLOW

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490173.000	Func	HLR	No	CL12471418.001 CL12471470.002 CL12471471.001 CL12472797.000 CL12472798.000 CL12472799.000 CL12472800.000 CL12473003.000 CL12473004.000 CL12473005.000 CL12473006.000 CL12473049.000 CL12473072.000 CL12473073.000 CL12473074.000

Requirement Design Notes- (LLR 12490173.000)

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

When SELMRS transitions from TRUE to FALSE, the Tustin-Filter outputs the value of the input for 1 cycle before processing the Tustin-Filter logic. The last past value of the input passed to the Tustin-Filter is used to allow for a smooth transition when SELMRS transitions to FALSE. Refer to the logic in section 4.23 of HSER-13855 rev.D.

04.03.03.04 Torque Bug Calculation [C05-Rating_Q_Bug_Calc]

This process calculates the torque bugs corresponding to MCR, MCL, NTO and MTO detents based on the Rated Power and a rating specific NP. Constant propeller speed is selected for all Rated Powers with the exception of MCL and MCR. For the latter, actual range and rate limited NP speed is chosen for the value of NPQBUG. Actual NP is used in the calculation of torque bug in MCL and MCR because in these ratings different NPs are selectable. Using actual NP ensures that an overtorque condition will not be commanded by the bug. A TQBUG and its TQBGI are selected between the calculated torque bugs by considering the current PLA Position.

LLR 12490620.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C05_TQBUG shall be implemented per design model C05_TQBUG.DATAFLOW

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490620.000	Func	HLR	No	CL12472894.000 CL12472897.000 CL12472898.000 CL12472899.000 CL12472900.000 CL12472901.000

04.03.04 Handling Bleed Valve Logic

04.03.04.01 EEC Mode Handling Bleed Valve Logic [C21-OUT_HBOV_EEC_MODE]

The EEC handling bleed valve logic determines when to open and close the bleed servo valve as a function of NH, DNHSEL, PAMB, DELTA and PLANHM in order to protect against engine surges. This logic also selects manual control in the appropriate cases.

LLR 12490625.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module C21_HBOVEM_2 shall be implemented per design model C21_HBOVEM_2.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490625.000	0			CL12471503.000

04.03.04.02 Manual Backup Handling Bleed Valve Logic [C22-Out_HBOV_Man_Mode]

This process calculates HBOV torque motor current when the EEC is failed fixed or in manual. The calculation is done on the basis of PLA, NH and NHDOT. If PLA fails, the calculation is done based solely on NH and NHDOT.

04.03.05 Start Logic [C01-Start]

The start logic module moves the stepper motor, which closes the stepper motor orifice when the engine NH is below a predefined datum, and PLA is in range with FEATHER on.

04.03.06 Miscellaneous Parameter Calculation [C26-MPARAM]

Ov 12490175.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process converts processed inputs into parameters that are later used in various other processes. The outputs are:

- Filtered torques.
- Temperatures.
- NP/NPT and NPTDOT values.
- Actual and rate of change power values.
- Corrected and rate of change of NH.
- PLA rate of change.

This process also unpacks the following words:

DISC1 - EEC Discrete Input Word Number 1

DISC2 - EEC Discrete Input Word Number 2
FLACCW - Fault Action Control Word
FLACHW - Fault action Hardware Word
IASDAT - ADC IAS data valid flag
QTOTSW - Total Torque Interface Status Word

LLR 12490181.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C26_MPARAM_2 shall be implemented per design model C26_MPARAM_2.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490181.000	0			CL12471561.000

LLR 12490182.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C26_MPARAM_3 shall be implemented per design model C26_MPARAM_3.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490182.000	0			CL12471563.000

LLR 12490183.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C26_MPARAM_4 shall be implemented per design model C26_MPARAM_4.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490183.000	0			CL12471565.000

LLR 12490184.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C26_MPARAM_5 shall be implemented per design model C26_MPARAM_5.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490184.000	0			CL12471567.000

LLR 12490185.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C26_MPARAM_6 shall be implemented per design model C26_MPARAM_6.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490185.000	0			CL12471569.000

LLR 12490186.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module C26_MPARAM_7 shall be implemented per design model C26_MPARAM_7.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490186.000	0			CL12471571.000

LLR 12490177.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module C26_MPARAM_1 shall be implemented per design model C26_MPARAM_1.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490177.000	0			CL12471554.000 CL12471555.001 CL12471556.000 CL12471557.000 CL12471558.001 CL12471559.000

04.03.07 Torque Limiting Logic [C31-Limit_Q]

This process calculates a high rotor speed request rate (DNHTQ) required to affect torque limiting.

04.03.08 Engine Overtorque Test Process [C24-BIT_PreFly_Eng_OverQ_Test]

The engine overtorque test process flags maintenance personnel whenever the engine delivered torque (QTOT) exceeds the maintenance service torque limits for a pre-determined time period. It informs maintenance personnel that engine service is required. This process monitors QTOT and sets unique overtorque condition status words when any of the following overtorque conditions are proven:

1.Q22TQE.16 set if:

QTOT>= QLM1 for time>= OTTM8 or
QTOT>= QLM0 for time>= OTTM0 (Installation Manual Region B)

2.Q37TQE.16 set if:

QTOT>= QLM2 for time>= OTTM9 or
If QTOT>= QLM5 for time>= OTTM9 (Installation Manual Region C)

3.Q47TQE.16 (Installation Manual Region D) set if:

QTOT>= QLM1 for time>= OTTM1 or
QTOT>= QLM2 for time>= OTTM2 or
QTOT>= QLM3 for time>= OTTM3 or
QTOT>= QLM4 for time>= OTTM4 or
QTOT>= QLM5 for time>= OTTM5 or
QTOT>= QLM6 for time>= OTTM6 or
QTOT>= QLM7 for time>= OTTM7 or
QTOT>= QLM8 for time>= OTTM8 or
QTOT>= QLM9 for time>= OTTM9.

A proven engine overtorque condition will result in the status word being written to the EEPROM through the non critical fault words and the EEPROM fault maintenance block process.

04.03.09 Auto Ignition Logic [C10-Out_Auto_Ignit]

This process sets the auto-ignition discrete (IGNITE) in the event of a flameout in the absence of either feather or manual modes. A hysteresis bandwidth of 0.3% is provided to prevent cycling.

04.04 OUTPUT SIGNAL PROCESSING

04.04.01 Stepper Motor Output Process [O01-Out_SM]

This process drives the stepper motor by initiating the stepper motor request through an OTA 317. It selects the request from the control laws and start logic and converts the stepper motor request in steps/second to steps/cycle (the cycle time is the process update rate, 20.8 msec). All partial steps are accumulated until a full step is requested during the next program cycle. If the start logic has not completed, the steps/second to steps/cycle conversion is not performed.

04.04.02 Handling Bleed Process [O02-HBOV]

This process develops the commanded current for the handling bleed valve. A pulse width modulator is used to output this request. The handling bleed request is rescaled to 10 bits for output to either Pulse Width Modulator #1 or #3. Controls that utilize the newer IIO4 Gate Array, where PWM #3 is not inhibited or reset during reversion, will utilize PWM #3. The rest will use PWM #1 to output Bleed. The Fault Action Word (FLACHW.8) may set the Torque Motor Current to zero if commanded.

04.04.03 Discrete Output Process [O03-OUT_DISC]

Ov 12490187.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

This capability produces discrete output word "DISCOT". It is output via the discrete output ports specified by output register 300 and 307 octal.

LLR 12490190.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module O03_DISOUT shall be implemented per design model O03_DISOUT.CTRLFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490190.000	0			CL12471661.000

LLR 12490192.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module O03_DISOUT_A shall be implemented per design model O03_DISOUT_A.DATAFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490192.000	0			CL12471663.002 CL12471664.002

04.04.04 UART Output Selection Logic [UART_OUT_SEL]

Ov 12490194.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

If the DCU data processing is active and the communication protocol is set to DAS or N-parameter, then set the communication protocol to DCU/DAS or DCU/ N-parameter, respectively.

If the DCU is currently not requested and the protocol is DCU/DAS or DCU/ N-parameter, then the new protocol becomes DAS or N-parameter. If you are currently DCU Protocol and Ground Test is enabled, then the protocol becomes DAS. If you are currently DAS and Ground Test is not enabled, then the protocol becomes DCU.

Ov 12490195.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCD(s): SCR: PW127L-EEC-SW-PWC-16236 SCO-D: PW127L-EEC-SW-PWC-16242

Ov 12490196.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design is incorporated in module UART_IO_SELECT.

Designs changed from the prior baseline are identified in section 11.

LLR 12490197.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O04_UARTOUTSEL shall be implemented per design model(s): O04_UARTOUTSEL.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490197.000	0			CL12471773.000 CL12471774.000 CL12471775.000

Ov 12490198.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design of UART_IO_SELECT is defined by the following LLR:

LLR 12490199.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Design model UART_IO_SELECT shall be implemented.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490199.000	0			CL12471765.000 CL12471766.000 CL12471767.000 CL12471768.000 CL12471769.000 CL12471770.000

04.04.04.01 UART Output Process [O04-UART_OUT]

The EEC utilizes the UART interface to output data to a remote terminal. A total of thirty-eight (38) bytes of data are output over the UART every 83.2 msec, or one complete major control cycle. Data transfers occur every 2.08 msec starting with state #6 and continuing to state #80. UART data is transmitted only during even numbered control states.

The data to be transferred to the UART output includes the following:

1. Start Synchronization Byte
2. Status Byte
3. 17 Memory Locations (34 Bytes)
4. Stop Synchronization Byte
5. Check Sum Byte

04.04.04.02 UART Output Process [O04-UART_OUT1]

The EEC utilizes the UART interface to output data to a remote terminal. A total of thirty-eight (38) bytes of data are output over the UART every 83.2 milliseconds, or one complete major control cycle. Data transfers occur every 2.08 milliseconds starting with state #6 and continuing to state #80. UART data is transmitted only during even numbered control states.

The data to be transferred to the UART output includes the following:

1. Start Synchronization Byte
2. Status Byte
3. 17 Memory Locations (34 Bytes)
4. Stop Synchronization Byte
5. Check Sum Byte

04.04.04.03 N-Parameter Protocol UART Output Processing [O04 - NPARAMOUT]

This capability transmits data between the EEC and an external piece of equipment via a RS-422 UART serial data interface. The UART transmission protocol is 1 start bit, 8 data bits, 1 stop bit, and odd parity, at 9600 baud. Data transmission by this capability is performed only when a software switch stored in EEPROM selects the N-Parameter protocol mode rather than the real time data stream mode. The EEC repetitively transmits a data frame that provides the data in the order requested by the external equipment. The data frame consists of an alternating sync/counter-sync byte, followed by a 32-bit address sum that uniquely identifies the data frame, followed by the parameter data in the requested order. Bytes in a data frame are transmitted sequentially with a minimal gap time between each byte.

04.04.05 ARINC Output Process Group

This design contains logic to output ARINC labels 062 through 067. It achieves this by traversing through a set of tables that specify when a label is to be output, what type of data the label is associated with, and data sequencing information. Each label is output at an assigned interval of 0.5, 1.0, or 2.0 seconds. Upon entry, the routine determines which label is to be output according to its assigned interval. It then determines which data item in the assigned list of data items for the label is to be output. The next data item in the list is sent each time the label is output, except for one label which only moves on to the next data item when the pilot flips a switch. When the end of the list is reached, the sequencing wraps back around to the first data item in the list. If the label has a sequence number associated with it, it is incremented for each item sent. When the data wraps around, the sequence number is set back to zero. If there is a fault at the time a label is to be sent out, zeros are sent out for the data. Before being sent out, each data item is packed into a high and low ARINC word along with the ARINC label, the SDI, the SSM, and the sequence number if applicable. It then sends the label via ARCOUT.

04.04.05.01 ARINC Label 062 Output Process [O40-ARINCO_062]

Ov 12490200.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process transmits the Engine LCF Component Accumulated Cycle Data over ARINC in label 062. This transmission shall scroll through a list of the component words with the use of a sequence code. FLACA2.6 is used to determine the validity of the data. If the FLACA2.6 bit is clear, then the data shall be transmitted with a source destination identifier of '00' and a valid sign/status matrix value of '11'. If the data is invalid, this process shall transmit default values over ARINC with a source destination identifier of '00' and a Failure/Warning sign/status matrix value of '00'.

Ov 12490208.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed Designs:

None

Changed Low Level Requirements:

LLR 12490209.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute this process once per every 998.4 ms.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490209.000	0			CL12471788.000

LLR 12490210.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store the reverse of the label identifier for label 062 in AROLSH.1-8, 114 octal.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490210.000	0			CL12471789.000

LLR 12490211.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store '00' in the SDI bits of AROLSH.9-10.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490211.000	0			CL12471790.000

LLR 12490212.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall fill AROLSH.11-14 with the Scrolling Component Identification Bit Sequence bits as per TABLE 2.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490212.000	0			CL12471794.000

Ov 12490213.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to "Table 2 - ARINC Label 062 Scrolling Component ID Word Definition - Total Accumulated Cycles" in SRS capability O40-ARINCO_062 for information on Scrolling Component Identification Bit Sequence bits.

LLR 12490214.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store LPIMP_CYCCRS, HPIMP_CYCCRS, HPTFCP_CYCCRS, HPTD_CYCCRS, HPTRCP_CYCCRS, ISS_CYCCRS, LPTD_CYCCRS, PT1D_CYCCRS, or PT2D_CYCCRS in AROLSH.15-16 and AROMSH.1-13 when FLACA2.6 is not set, otherwise set to default.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490214.001	0			CL12471795.001

LLR 12490201.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall scroll between LPIMP_CYCCRS, HPIMP_CYCCRS, HPTFCP_CYCCRS, HPTD_CYCCRS, HPTRCP_CYCCRS, ISS_CYCCRS, LPTD_CYCCRS, PT1D_CYCCRS, and PT2D_CYCCRS in a cyclic pattern and then repeat.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490201.001			N/A	CL12471794.000

Ov 12490202.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

For example, when Data field #1 for ARINC label 062 is bit 11 = 0, bit 12 = 0, bit 13 = 0 and bit 14 = 0, Data Field #2 information contains the "LP Impeller Accumulated Total Cycles". When Data Field #1 for ARINC label 062 is bit 11 = 0, bit 12 = 0, bit 13 = 0 and bit 14 = 1, Data Field #2 will display the "2nd Stage Power Turbine Disk Accumulated Total Cycles".

Ov 12490203.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to "Table 1 " in SRS capability O40-ARINCO_062 for information on "Data Field 1" and "Data Field 2".

LLR 12490204.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store data equal to 0 in AROLSH.15-16 and AROMSH.1-13 when FLACA2.6 is set, and AROLSH.11-14 will continue to scroll through the list of "Scrolling Component Identification Bit Sequence".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490204.000	0			

LLR 12490205.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set AROMSH.14-15 equal to '00' when FLACA2.6 is set, otherwise set AROMSH.14-15 equal to '11'.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490205.000	0			CL12471797.000

LLR 12490206.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall use the ARCOUT subroutine to send the two 16 bit words AROMSH and AROLSH out over ARINC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490206.000	0			CL12471798.000

04.04.05.02 ARINC Label 063 Output Process [O42-ARINCO_063]

Ov 12490215.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process is responsible for transmitting QGTM, QBTM and SWID over ARINC with an identification label of 063. This transmission shall scroll through a list of the component words with the use of a sequence code. FLACA2.7 is used to determine the validity of the data. If the FLACA2.7 bit is clear, then the data shall be transmitted with a source destination identifier of '00' and a valid sign/status matrix value of '11'. If the data is invalid, this process shall transmit default values over ARINC with a source destination identifier of '00' and a Failure/Warning sign/status matrix value of '00'.

Ov 12490222.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed Designs:

None

Changed Low Level Requirements:

LLR 12490223.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall execute this process once per every 1,996.8 ms.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490223.001	0			CL12471972.002

LLR 12490224.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store the reverse of the label identifier for label 063 in AROLSH.1-8, 314 octal.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490224.000	0			CL12471975.000

LLR 12490225.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store '00' in the SDI bits of AROLSH.9-10.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490225.000	0			CL12471977.002

LLR 12490226.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall fill AROLSH.11-13 with the Torque Bias Trim Data, the Torque Gain Trim Data, and the Software ID Data.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490226.001	0			CL12471980.001

LLR 12490227.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store QBTM, QGTM, or SWID in AROLSH.14-16 and AROMSH.1-13 when FLACA2.7 is not set.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490227.001	0			CL12471981.002 CL12472067.001 CL12472068.001 CL12472069.001 CL12472070.001

LLR 12490228.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall scroll between QBTM, QGTM, and SWID in a cyclic pattern and then repeat.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490228.001	0			CL12471982.001

Ov 12490216.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

For example, when Data field #1 for ARINC label 063 is bit 11 = 0, bit 12 = 0 and bit 13 = 0, Data Field #2 information contains the "Torque Bias Trim Data". When Data Field #1 for ARINC label 063 is bit 11 = 0, bit 12 = 1 and bit 13 = 0, Data Field #2 will display the "Software ID Data".

Ov 12490217.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to "Table 1 - Engine Torque Trim and Engine Variant Trim Data - Label 063" in SRS capability O42-ARINCO_063 for information on "Data Field 1" and "Data Field 2".

LLR 12490218.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store data equal to 0 in AROLSH.14-16 and AROMSH.1-13 when FLACA2.7 is set, and AROLSH.11-13 will continue to scroll through the list of "Scrolling Engine Trim Data ID".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490218.000	0			CL12471985.001

LLR 12490219.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set AROMSH.14-15 equal to '00' when FLACA2.7 is set, otherwise set AROMSH.14-15 equal to '11'.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490219.000	0			CL12471987.001

LLR 12490220.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall use the ARCOU subroutine to send the two 16 bit words AROMSH and AROLSH out over ARINC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490220.000	0			CL12471989.000

04.04.05.03 ARINC Label 064 Output Process [O41-ARINCO_064]

Ov 12490229.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This design is common with the ARINC Label 062 Output Process and is described in Section 4.4.5.1.

Ov 12490237.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed Designs:

None

Changed Low Level Requirements:

LLR 12490238.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute this process once per every 1996.8 mS.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490238.000	0			CL12472105.000

LLR 12490239.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store the reverse of the label identifier for label 064 in AROLSH.1-8, 054 octal.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490239.000	0			CL12472107.000

LLR 12490240.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store '00' in the SDI bits of AROLSH.9-10.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490240.000	0			

LLR 12490241.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall fill AROLSH.11-13 with the Scrolling Engine Model ID bits as per TABLE 2.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490241.000	0			

Ov 12490242.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to Table: "Label 064 Scrolling Engine Usage Data IDs" in SRS capability O41-ARINCO_064 for information on Engine Usage Data ID bits.

LLR 12490243.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store ENG_RUN_CRS_ALL, ENGFLTM_CRS_ALL, ACCSTRTS_ALL, or ACCNUMFL_ALL in AROLSH.14-16 and AROMSH.1-13 when FLACA2.8 is not set.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490243.001	0			CL12472113.001

LLR 12490230.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall scroll between ENG_RUN_CRS_ALL, ENGFLTM_CRS_ALL, ACCSTRTS_ALL, and ACCNUMFL_ALL in a cyclic pattern and then repeat.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490230.001	0			CL12472114.001

Ov 12490231.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

For example, when Data field #1 for ARINC label 064 is bit 11 = 0, bit 12 = 0 and bit 13 = 0, Data Field #2 information contains the "Total Engine Run Time". When Data Field #1 for ARINC label 064 is bit 11 = 1, bit 12 = 1 and bit 13 = 0, Data Field #2 will display the "Total Accumulated Number of Flights".

Ov 12490232.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to Table: "EEC Supplied Flight Deck Data (Accumulated Engine Usage - Label 064)" in SRS capability O41-ARINCO_064 for information on "Data Field 1" and "Data Field 2".

LLR 12490233.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall store data equal to 0 in AROLSH.14-16 and AROMSH.1-13 when FLACA2.8 is set, and continue to scroll through the list of "Scrolling Engine Usage Data ID" in AROLSH.11-13.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490233.001	0			CL12472117.002

LLR 12490234.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set AROMSH.14-15 equal to '00' when FLACA2.8 is set, otherwise set AROMSH.14-15 equal to '11'.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490234.000	0			CL12472118.000

LLR 12490235.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall use the ARCOU subroutine to send the two 16 bit words AROMSH and AROLSH out over ARINC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490235.000	0			CL12472119.000

04.04.05.04 ARINC Label 065 Output Process [O43-ARINCO_065]

Ov 12490244.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This design is common with the ARINC Label 062 Output Process and is described in Section 4.4.5.1.

Ov 12490248.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCDs: SCR: PW127L-EEC-SW-PWC-11231 SCO-D: PW127L-EEC-SW-PWC-13431

Ov 12490249.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed Designs:

None

Changed Low Level Requirements:

LLR 12490250.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute this process once per every 499.2 ms.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490250.000	0			CL12472129.000

LLR 12490251.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store the reverse of the label identifier for label 065 in AROLSH.1-8, 254 octal.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490251.000	0			CL12472130.000

LLR 12490252.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store '00' in the SDI bits of AROLSH.9-10.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490252.000	0			CL12472131.000

LLR 12490253.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store Engine S/N Delimiter ("#EN"),
ENGSA12, ENGSA34, ENGSA56, Turbo Machine S/N Delimiter ("#TM"),
TRBMSA12, TRBMSA34, TRBMSA56, RGB S/N Delimiter ("#RGB"),
RBSA12, RBSA34 or RBSA56 two ASCII characters at a time into AROLSH.11-16 and AROMSH.1-13
when FLACA2.9 is FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490253.000	0			CL12472134.000

Ov 12490254.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to TABLE 1 and 2 in SRS capability O43-ARINCO_065 for more information on ARINC word Definition
and a sample Data Stream. Additionally ARMOSH11-13 are Spare and not used

LLR 12490255.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall scroll between Engine S/N Delimiter ("#EN"),
ENGSA12, ENGSA34, ENGSA56, Turbo Machine S/N Delimiter ("#TM"),
TRBMSA12, TRBMSA34, TRBMSA56, RGB S/N Delimiter ("#RGB"),
RBSA12, RBSA34 and RBSA56 in a cyclic pattern and then repeat.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490255.000	0			CL12472134.000

LLR 12490245.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store data equal to 0 in AROLSH.11-16 and AROMSH.1-13 when FLACA2.9 is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490245.000	0			CL12472134.000

LLR 12490246.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set AROMSH.14-15 equal to '00' when FLACA2.9 is set, otherwise set AROMSH.14-15 equal to '11'.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490246.000	0			CL12472136.000

LLR 12490247.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall use the ARCOUNT subroutine to send the two 16 bit words AROMSH and AROLSH out over ARINC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490247.000	0			CL12472137.000

04.04.05.05 ARINC Label 066 Output Process [O44-ARINCO_066]

Ov 12490256.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This design is common with the ARINC Label 062 Output Process and is described in Section 4.4.5.1.

Ov 12490260.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCDs: SCR: PW127L-EEC-SW-PWC-16236 SCO-D: PW127L-EEC-SW-PWC-16242

Ov 12490261.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed Designs:

None

Changed Low Level Requirements:

LLR 12490262.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute this process once per every 499.2 ms.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490262.000	0			CL12472146.000

LLR 12490263.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store the reverse of the label identifier for label 066 in AROLSH.1-8, 154 octal.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490263.000	0			CL12472147.000

LLR 12490264.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store '00' in the SDI bits of AROLSH.9-10.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490264.000	0			CL12472148.000

LLR 12490265.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store LP Impeller Delimiter ("C1#P"), LPIMP_PNA12, LPIMP_PNA34, LPIMP_PNA56, LPIMP_PNA78, HP Impeller Delimiter ("C2#P"), HPIMP_PNA12, HPIMP_PNA34, HPIMP_PNA56, HPIMP_PNA78, HP Turbine Front Cover Plate Delimiter ("C3#P"), HPTFCP_PNA12, HPTFCP_PNA34, HPTFCP_PNA56, HPTFCP_PNA78, HP Turbine Disk Delimiter ("C4#P"), HPTD_PNA12, HPTD_PNA34, HPTD_PNA56, HPTD_PNA78, HP Turbine Rear Cover Plate Delimiter ("C5#P"), HPTRCP_PNA12, HPTRCP_PNA34, HPTRCP_PNA56, HPTRCP_PNA78, Inter-stage Seal Delimiter ("C6#P"), ISS_PNA12, ISS_PNA34, ISS_PNA56,

ISS_PNA78, LP Turbine Disk Delimiter ("C7#P"), LPTD_PNA12, LPTD_PNA34, LPTD_PNA56, LPTD_PNA78, Power Turbine Disk 1 Delimiter ("C8#P"), PT1D_PNA12, PT1D_PNA34, PT1D_PNA56, PT1D_PNA78, Power Turbine Disk 2 Delimiter ("C9#P"), PT2D_PNA12, PT2D_PNA34, PT2D_PNA56 or PT1D_PNA78 two ASCII characters at a time into AROLSH.11-16 and AROMSH.1-13 when FLACA2.9 is FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490265.000	0			CL12472151.000

Ov 12490266.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to TABLE 1 and 2 in SRS capability O44-ARINCO_066 for more information on ARINC word Definition and the Component Part Numbers and P/N Delimiter definition. Only two ASCII characters are transmitted at a time. Additionally ARMOSH11-13 are Spare and not used

LLR 12490267.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall scroll between LPIMP_PNA12, LPIMP_PNA34, LPIMP_PNA56, LPIMP_PNA78, HP Impeller Delimiter ("C2#P"), HPIMP_PNA12, HPIMP_PNA34, HPIMP_PNA56, HPIMP_PNA78, HP Turbine Front Cover Plate Delimiter ("C3#P"), HPTFCP_PNA12, HPTFCP_PNA34, HPTFCP_PNA56, HPTFCP_PNA78, HP Turbine Disk Delimiter ("C4#P"), HPTD_PNA12, HPTD_PNA34, HPTD_PNA56, HPTD_PNA78, HP Turbine Rear Cover Plate Delimiter ("C5#P"), HPTRCP_PNA12, HPTRCP_PNA34, HPTRCP_PNA56, HPTRCP_PNA78, Inter-stage Seal Delimiter ("C6#P"), ISS_PNA12, ISS_PNA34, ISS_PNA56, ISS_PNA78, LP Turbine Disk Delimiter ("C7#P"), LPTD_PNA12, LPTD_PNA34, LPTD_PNA56, LPTD_PNA78, Power Turbine Disk 1 Delimiter ("C8#P"), PT1D_PNA12, PT1D_PNA34, PT1D_PNA56, PT1D_PNA78, Power Turbine Disk 2 Delimiter ("C9#P"), PT2D_PNA12, PT2D_PNA34, PT2D_PNA56 and PT1D_PNA78 in a cyclic pattern and then repeat.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490267.000	0			CL12472151.000

LLR 12490257.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store data equal to 0 in AROLSH.11-16 and AROMSH.1-13 when FLACA2.9 is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490257.000	0			CL12472151.000

LLR 12490258.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set AROMSH.14-15 equal to '00' when FLACA2.9 is set, otherwise set AROMSH.14-15 equal to '11'.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490258.000	0			CL12472155.000

LLR 12490259.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall use the ARCOUNT subroutine to send the two 16 bit words AROMSH and AROLSH out over ARINC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490259.000	0			CL12472156.000

04.04.05.06 ARINC Label 067 Output Process [O45-ARINCO_067]

Ov 12490268.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This design is common with the ARINC Label 062 Output Process and is described in Section 4.4.5.1.

Ov 12490272.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCDs: SCR: PW127L-EEC-SW-PWC-11231 SCO-D: PW127L-EEC-SW-PWC-13431

Ov 12490273.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed Designs:

None

Changed Low Level Requirements:

LLR 12490274.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute this process once per every 499.2 ms.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490274.000	0			CL12472166.000

LLR 12490275.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store the reverse of the label identifier for label 067 in AROLSH.1-8, 354 octal.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490275.000	0			CL12472167.000

LLR 12490276.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store '00' in the SDI bits of AROLSH.9-10.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490276.000	0			CL12472168.000

LLR 12490277.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store LP Impeller S/N Delimiter ("C1#SN"), LPIMP_SNA12, LPIMP_SNA34, LPIMP_SNA56, HP Impeller S/N Delimiter ("C2#SN"), HPIMP_SNA12, HPIMP_SNA34, HPIMP_SNA56, HP Turbine Front Cover Plate S/N Delimiter ("C3#SN"), HPTFCP_SNA12, HPTFCP_SNA34, HPTFCP_SNA56, HP Turbine Disk S/N Delimiter ("C4#SN"), HPTD_SNA12, HPTD_SNA34, HPTD_SNA56, HP Turbine Rear Cover Plate S/N Delimiter ("C5#SN"), HPTRCP_SNA12, HPTRCP_SNA34, HPTRCP_SNA56, Inter-stage Seal S/N Delimiter ("C6#SN"), ISS_SNA12, ISS_SNA34, ISS_SNA56, LP Turbine Disk S/N Delimiter ("C7#SN"), LPTD_SNA12, LPTD_SNA34, LPTD_SNA56, Power Turbine Disk 1 S/N Delimiter ("C8#SN"), PT1D_SNA12, PT1D_SNA34, PT1D_SNA56, Power Turbine Disk 2 S/N Delimiter ("C9#SN"), PT2D_SNA12, PT2D_SNA34, or PT2D_SNA56 two ASCII characters at a time into AROLSH.11-16 and AROMSH.1-13 when FLACA2.9 is FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490277.000	0			CL12472171.000

Ov 12490278.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to TABLE 1 and 2 in SRS capability O45-ARINCO_067 for more information on ARINC word Definition and the Component Serial Numbers and S/N Delimiter definition.
Additionally ARMOSH11-13 are Spare and not used

LLR 12490279.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall scroll between LP Impeller S/N Delimiter ("C1#SN"), LPIMP_SNA12, LPIMP_SNA34, LPIMP_SNA56, HP Impeller S/N Delimiter ("C2#SN"), HPIMP_SNA12, HPIMP_SNA34, HPIMP_SNA56, HP Turbine Front Cover Plate S/N Delimiter ("C3#SN"), HPTFCP_SNA12, HPTFCP_SNA34, HPTFCP_SNA56, HP Turbine Disk S/N Delimiter ("C4#SN"), HPTD_SNA12, HPTD_SNA34, HPTD_SNA56, HP Turbine Rear Cover Plate S/N Delimiter ("C5#SN"), HPTRCP_SNA12, HPTRCP_SNA34, HPTRCP_SNA56, Inter-stage Seal S/N Delimiter ("C6#SN"), ISS_SNA12, ISS_SNA34, ISS_SNA56, LP Turbine Disk S/N Delimiter ("C7#SN"), LPTD_SNA12, LPTD_SNA34, LPTD_SNA56, Power Turbine Disk 1 S/N Delimiter ("C8#SN"), PT1D_SNA12, PT1D_SNA34, PT1D_SNA56, Power Turbine Disk 2 S/N Delimiter ("C9#SN"), PT2D_SNA12, PT2D_SNA34, and PT2D_SNA56 in a cyclic pattern and then repeat.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490279.000	0			CL12472171.000

LLR 12490269.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall store data equal to 0 in AROLSH.11-16 and AROMSH.1-13 when FLACA2.9 is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490269.000	0			CL12472171.000

LLR 12490270.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set AROMSH.14-15 equal to '00' when FLACA2.9 is set, otherwise set AROMSH.14-15 equal to '11'.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490270.000	0			CL12472174.000

LLR 12490271.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall use the ARROUT subroutine to send the two 16 bit words AROMSH and AROLSH out over ARINC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490271.000	0			CL12472175.000

04.04.05.07 ARINC Label 133 Output Process [O05-ARINCO_133]

This process is responsible for transmitting PLA over ARINC with an identification label of 133. FLACA1.1 is used to determine the validity of the PLA data. If the bit is clear, then PLA is transmitted with a source destination identifier of "00" and a valid sign/status matrix value of "11". If the PLA data is invalid, this process will transmit a default value over ARINC with a source destination identifier of "00" and an invalid sign/status matrix value of "00".

LLR 12490635.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O05-ARINCO_133 shall be implemented per design model O05_ARO133_A.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490635.000	0			CL12472185.000 CL12472186.001 CL12472187.000 CL12472188.000 CL12472189.000

04.04.05.08 ARINC Label 134 Output Process [O06-ARINCO_134]

This process is responsible for transmitting PLAIN over ARINC with an identification label of 134. FLACA1.2 is used to determine the validity of the data. If the bit is clear, then the data is transmitted with a source destination identifier of "00" and a valid sign/status matrix value of "11". If the data is invalid, this process will transmit a default value over ARINC with a source destination identifier of "00" and an invalid sign/status matrix value of "00" for PLAIN.

LLR 12490640.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O06-ARINCO_134 shall be implemented per design model O06_ARO134_A.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490640.000	0			CL12472199.000 CL12472200.001 CL12472201.000 CL12472202.000 CL12472203.000

04.04.05.09 Reserved**04.04.05.10 ARINC Label 203 Output Process [O07-ARINCO_203_ALTPS]**

This process is responsible for transmitting ALTPS over ARINC with an identification label of 203. FLACA1.4 is used to determine the validity of the ALTPS data. If the bit is clear, then ALTPS is transmitted with a source destination identifier of '01' and a valid sign/status matrix value of "11". If the ALTPS data is invalid, this process will transmit a default value over ARINC with a source destination identifier of "01" and an invalid sign/status matrix value of "00".

LLR 12490650.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O07-ARINCO_203_ALTPS shall be implemented per design model O07_ARO203_ALTPS_A.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490650.000	0			CL12471823.000 CL12471824.001 CL12471825.000 CL12471826.000 CL12471827.000 CL12471828.000 CL12471829.000

04.04.05.11 ARINC Label 203 Output Process [O07-ARINCO_203_ALT]

This process is responsible for transmitting Selected Pressure Altitude over ARINC with an identification label of 203. FLACA2.4 is used to determine the validity of the Selected Pressure Altitude data. If the bit is clear, then Selected Pressure Altitude is transmitted with a source destination identifier of '11' or '00' (depending on the engine position) and a valid sign/status matrix value of "11". If the Selected Pressure Altitude data is invalid, this process will transmit a default value over ARINC with a source destination identifier of '11' or '00' (depending on the engine position) and an invalid sign/status matrix value of "00".

LLR 12490645.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O07-ARINCO_203_ALT shall be implemented per design model O07_ARO203_ALT11_A.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490645.000	0			CL12472213.000 CL12472214.001 CL12472215.000 CL12472216.000 CL12472217.000

04.04.05.12 Reserved

04.04.05.13 Reserved

04.04.05.14 Reserved

04.04.05.15 Reserved

04.04.05.16 ARINC Label 211 Output Process [O09-ARINCO_211_TAT11]

This process is responsible for transmitting Selected Total Air Temperature (THETAC) over ARINC with an identification label of 211. The source destination identifier for this label is determined based on the Engine Position. The data is transmitted with a valid sign/status matrix value of '11'.

LLR 12490630.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O09-ARINCO_211_TAT11 shall be implemented per design model O09_ARO211_TAT11_A.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490630.000	0			CL12473059.000 CL12473060.000 CL12473062.000 CL12473063.000

04.04.05.17 ARINC Label 240 Output Process [O10-ARINCO_240]

This process is responsible for transmitting LRU fault codes over ARINC as the LRU advance discrete is toggled. The data has an identification label of 240 and is transmitted with a source destination identifier of "00" and a valid sign/status matrix value of "11".

LLR 12490655.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O10-ARINCO_240 shall be implemented per design models O10_ARO240_A.CPGC, O10_ARO240_B.CPGC and O10_ARO240_C.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490655.000	0			CL12471927.000 CL12471928.001 CL12471929.000 CL12471930.000 CL12471931.000

04.04.05.18 ARINC Label 245 Output Process [O11-ARINCO_245]

This process is responsible for transmitting NH over ARINC with an identification label of 245. FLACA1.10 is used to determine the validity of the NH data. If the bit is clear, then NH is transmitted with a source destination identifier of "00" and a valid sign/status matrix value of "11". If the NH data is invalid, this process will transmit a default value over ARINC with a source destination identifier of "00" and an invalid sign/status matrix value of "00".

LLR 12490660.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O11-ARINCO_245 shall be implemented per design models O11_ARO245_A.CPGC and O11_ARO245_B.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490660.000	0			CL12471942.000 CL12471943.001 CL12471944.000 CL12471945.000 CL12471946.000

04.04.05.19 ARINC Label 341 Output Process [O15-ARINCO_341]

Ov 12490280.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process is responsible for transmitting QHPLCP over ARINC with an identification label of 341. FLACA1.15 is used to determine the validity of the QHPLCP data. If the bit is clear, then QHPLCP is transmitted with a source destination identifier of '00' and a valid sign/status matrix value of '11'. If the QHPLCP data is invalid, this process will transmit a default value over ARINC with a source destination identifier of '00' and an invalid sign/status matrix value of '00'.

LLR 12490283.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O15_ARO341 shall be implemented per design model O15_ARO341_A.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490283.000	0			CL12471957.000 CL12471961.000 CL12471962.000

LLR 12490284.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O15_ARO341 shall be implemented per design model O15_ARO341_B.CPGC.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490284.000	0			CL12471959.000 CL12471960.001 CL12471961.000 CL12471962.000 CL12471963.000

04.04.05.20 ARINC Label 346 Output Process [O16-ARINCO_346]

This process is responsible for transmitting NPTOUT over ARINC with an identification label of 346. FLACA1.16 and QTOTSW.10 are used to determine the validity of the NPTOUT data. If these bits are clear, then NPTOUT is transmitted with a source destination identifier of "00" and a valid sign/status matrix value of "11". If the NPTOUT data is invalid, this process will transmit a default value over ARINC with a source destination identifier of "00" and an invalid sign/status matrix value of "00".

LLR 12490665.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O16-ARINCO_346 shall be implemented per design model O16_ARO346_A.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490665.000	0			CL12471998.000 CL12471999.001 CL12472000.000 CL12472001.000 CL12472002.000

04.04.05.21 ARINC Label 350 Output Process [O17-ARINCO_350]

This process is responsible for transmitting LRU fault codes over ARINC at a rate of one fault every 166.4 ms. The data has an identification label of 350 and is transmitted with a source destination identifier of "00" and a valid sign/status matrix value of "11".

LLR 12490670.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O17-ARINCO_350 shall be implemented per design model O17_ARO350_A.CPGC and O17_ARO350_B.CPGC.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490670.000	0			CL12472013.000 CL12472014.001 CL12472015.000 CL12472016.000 CL12472017.000 CL12472018.000

04.04.05.22 ARINC Label 270 Output Process [O18-ARINCO_270]

Ov 12490286.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process is responsible for transmitting Discrete Data over ARINC with an identification label of 270. FLACA1.11 is used to determine the validity of the data. If the bit is clear, then the data is transmitted with a source destination identifier of '00' and a valid sign/status matrix value of '11'. If the data is invalid, this process will transmit default values over ARINC with a source destination identifier of '00' and an invalid sign/status matrix value of '00'.

LLR 12490293.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_A.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490293.000	Func	HLR	No	CL12472028.000 CL12472042.000 CL12472043.000 CL12472046.000 CL12472047.000 CL12472058.000 CL12472796.000

Requirement Design Notes- (LLR 12490293.000)

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

O18_ARO270 is executing O19_ARO272 instead when it is not time to execute O18_ARO270.

LLR 12490294.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_B.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490294.000	0			CL12470535.001 CL12472043.000 CL12472796.000

LLR 12490295.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_C.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490295.000	0			CL12472035.001 CL12472043.000

LLR 12490296.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_D.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490296.000	0			CL12472035.001 CL12472043.000

LLR 12490297.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_E.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490297.000	0			CL12472035.001 CL12472043.000

LLR 12490298.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_F.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490298.000	0			CL12472035.001 CL12472043.000

LLR 12490287.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_G.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490287.000	0			CL12472035.001 CL12472043.000

LLR 12490288.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_H.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490288.000	0			CL12472035.001 CL12472043.000

LLR 12490626.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_HA.CPGC

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490626.000	0			CL12472035.001 CL12472043.000

LLR 12490289.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O18_ARO270 shall be implemented per design model O18_ARO270_I.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490289.000	0			CL12472032.000 CL12472035.001 CL12472042.000 CL12472043.000 CL12472046.000 CL12472047.000 CL12472049.000

04.04.05.23 ARINC Label 272 Output Process [O19-ARINCO_272]

Ov 12490691.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process transmits Derating Levels over ARINC in label 272. The Derating Levels on label 272 are always transmitted with a source destination identifier. No Computed Data is transmitted via the sign/status matrix when Inhibit/Reset is true or EEC issues a Fail Fixed command. A valid sign/status matrix has a value of '11'.

LLR 12490303.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O19_ARO272 shall be implemented per design model O19_ARO272.CPGC.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490303.001	0			CL12472059.000 CL12472060.001 CL12472063.000 CL12472064.000 CL12472065.000

LLR 12490616.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O19_ARO272_A shall be implemented per design model(s): O19_ARO272_A.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490616.000	0			CL12473014.000 CL12473015.000

04.04.05.24 ARINC Output Subroutine [O22-ARINC_Out_SubRoutine]

This process is a sub-process of the ARINC label output processes. It is responsible for outputting two 16 bit words (AROMSH and AROLSH) over the ARINC interface and for checking correct operation of the output interface.

04.04.05.25 Reserved

04.04.05.26 ARINC Label 340 Output Process [O26 - ARINCO_340]

Ov 12490309.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This process is responsible for transmitting QOUT over ARINC with an identification label of 340. FLACA1.13, QTOTSW.10, CTIF.11, and FLACA2.3 are used to determine the validity of the QOUT data. If these bits are clear, then QOUT is transmitted with a source destination identifier of '00' and a valid sign/status matrix value of '11'. If the QOUT data is invalid, this process will transmit a default value over ARINC with a source destination identifier of '00' and either a valid ('11') or an invalid ('00') sign/status matrix value depending on the current state of FLACA2.3.

LLR 12490312.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O26_ARO340 shall be implemented per design model O26_ARO340_A.CPGC

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490312.001	0			CL12472660.000 CL12472662.001 CL12472663.001 CL12472664.000 CL12472665.000 CL12472666.000

04.04.05.27 ARINC Label 342 Output Process [O27 - ARINCO_342]

Ov 12490314.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

This process is responsible for transmitting a sequence of Torque Bugs with their identifiers over ARINC with an identification label of 342, in a scrolling manner. FLACA1.14 is used to determine the validity of the data. If the bit is clear, then STQBI and STQBUG are transmitted with a SDI of '00' and a valid SSM value of '11'. If the data is invalid, this process will transmit a default value over ARINC with a SDI of '00' and an invalid SSM value of '00'. If CTIF.11 is set, the default value is transmitted over ARINC and the SSM follows the determination from FLACA1.14.

LLR 12490317.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O27_ARO342 shall be implemented per design model O27_ARO342_A.CPGC

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490317.001	0			CL12472644.000 CL12472648.001 CL12472649.001 CL12472975.000

LLR 12490621.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module O27_ARO342_A shall be implemented per design model O27_ARO342_A.DATAFLOW

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490621.000	Func	HLR	No	CL12472902.000 CL12473050.000

LLR 12490318.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module O27_ARO342 shall be implemented per design model O27_ARO342_B.CPGC

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490318.001	0			CL12472644.000 CL12472646.001 CL12472647.001 CL12472648.001 CL12472649.001 CL12472650.001 CL12472975.000

04.05 DCU AND LOW CYCLE FATIGUE CAPABILITIES

DCU and Low Cycle Fatigue Input Signal Processing defines the processing required to access, read, and process DCU input data provided from the DCU to the EEC.

04.05.01 DCU Capabilities

04.05.01.01 DCU Communications Protocol [DCUPROTO]

This capability defines the communications protocol for reading and writing data to the Data Collection Unit (DCU).

The following routines are responsible for implementing this capability:

- DCU_Receive

This routine is responsible for processing data that is read from the DCU. A state machine is utilized to receive the DCU status, data bytes, and checksum. The status word received from the DCU is evaluated, and used for fault detection of the DCU and the data received from the DCU. When the required number of bytes have been received from the DCU, a completion flag will be set.

- DCU_Transfer

This routine is responsible for scheduling DCU device communications, including initialization and failure conditions. After a PSR, a two hundred millisecond timer is used to allow the DCU to complete its initialization before the EEC will declare the DCU failed if communications cannot be established. Once initialization is complete and communications are established, upon request, the EEC will transmit the command and data to the DCU. When the transmission is complete, the EEC will wait for the data to be received from the DCU. If the expected data is received within the time interval and with no errors, the routine will signify completion of the communications sequence, so that the requesting routine can process the data. If a busy status is received from the DCU, a counter to limit the time between retries of the communications is maintained by this routine in order to allow the DCU to complete its memory management. This timer is set to wait ten seconds between tries to communicate with the DCU up to a maximum number of seconds specified by the routine that made the request. After that time, a failure status is set so that the requesting routine can know communication has failed. This counter is reset to zero whenever a status is received from the DCU that does not indicate busy. If any other status bit is received, the DCU will cause the logic to retransmit the request to the DCU, for up to three retries before indicating a failure in communications. The exception to this logic is when an Illegal Block status is received, which means that a read request was made to a block that has never been written. In this case, the routine will not attempt any retries and signify completion immediately.

- DCU_Transmit

This routine is responsible for processing data to be written to the DCU. A state machine is utilized to transmit the DCU address, the command, the data bytes, and the checksum. When all required data has been transmitted, a flag is set indicating completion of the transmission sequence.

04.05.01.02 DCU Block Data Input Logic [DCUREAD]

Ov 12490320.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Following a PSR, a series of data blocks are read from the DCU. Requests are made to the DCU driver routines to read each block of data. The status word received from the DCU is used to determine reception of the data had completed. This capability is performed by the following processes:

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Read

This routine is the interface between the requesting routines and the lower level DCU device routines. It is used to initiate a read request. The DCU command buffer is initialized to perform a read request, the data buffer to receive the data bytes is specified, and various control information to be used by the lower-level DCU device routines is defined such as: bytes to be received, busy timeout length, and initial status. Once the buffers and control data are set up, the routine signals a read request.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

Ov 12490353.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The following are the LLRs related to DCU_SCHEDULE as it relates to DCUREAD

LLR 12490358.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall be in Read DCU Mode when DCU Case State, DCU_SCHED_STATE, is equal to READ_BLOCKS.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490358.000	0			CL12472333.000 CL12472334.000

LLR 12490359.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall be in Process DCU Mode when DCU Case State, DCU_SCHED_STATE, is equal to PROCESS_BLOCKS.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490359.000	0			CL12472333.000 CL12472334.000

Ov 12490360.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: This partially covers the linked HLR the state table that calls this process covers the rest of it.
READ_BLOCKS is equal to ZERO. PROCESS_BLOCKS is equal to ONE.

LLR 12490361.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Read pointer, DCU_RTABLE_PTR, equal to the DCU Read Table Start Address, DCU_RTABLE_ADR, and set DCU Process pointer, DCU_PTABLE_PTR, equal to the DCU Process Table Start Address, DCU_PTABLE_ADR, when all of the following conditions are met:

- DCU Available, DCU_IS_AVAILABLE, is equal to TRUE.
- DCU Read in Process, DCU_IN_PROCESS, is equal to FALSE.
- DCU Read pointer, DCU_RTABLE_PTR, is less than DCU Read Table Start Address, DCU_RTABLE_ADR, or DCU Read pointer, DCU_RTABLE_PTR, is greater than or equal to the Address of DCU Read table End DCU_RTABLE_END.
- DCU Case State, DCU_SCHED_STATE, is equal to ZERO.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490361.000	0			CL12472336.000 CL12472337.002

LLR 12490362.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute all of the following when DCU Case State, DCU_SCHED_STATE, is equal to 0, the value at DCU Read pointer, DCU_RTABLE_PTR, is NONZERO, DCU Available, DCU_IS_AVAILABLE, is equal to TRUE and DCU Read in Process, DCU_IN_PROCESS, is equal to FALSE:

- Call subroutine DCU_READ with DCU Read Pointer DCU_RTABLE_PTR.
- DCU Read in Process, READ_IN_PROCESS, is set TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490362.000	0			CL12472334.000 CL12472336.000 CL12472337.002 CL12472348.000

LLR 12490321.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute the following when DCU Process Pointer, DCU_PTABLE_PTR, is Greater than or equal to DCU Process Table Start Address, DCU_PTABLE_ADR, or DCU process pointer, DCU_PTABLE_PTR, is less than the Address of DCU process table End, DCU_PTABLE_END, and DCU Case State, DCU_SCHED_STATE, is equal to ONE:

- variable 27, SP27, set equal to the memory at DCU Process pointer DCU_PTABLE_PTR.
- Call the subroutine equal to the Data at variable 27 SP27.
- increment the DCU process pointer, DCU_PTABLE_PTR, by ONE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490321.000	0			CL12472339.000 CL12472345.000 CL12472348.000

Ov 12490322.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: The subroutine call above is what sets the DCU_DATA_READY signal to the correct values according to the SRS the calls are determined by Table 2. The Set bit 3 equal to 1 is actually done as part of DCUWRITE and a LLR is in that SDD section. Additionally the subroutines called are what does the Fault reporting.

LLR 12490323.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Hold table, DCU_HOLD_TABLE, equal to DCU process pointer, DCU_PTABLE_PTR, when all of the following conditions are met:

- DCU Process Pointer, DCU_PTABLE_PTR, is Greater than or equal to DCU Process Table Start Address, DCU_PTABLE_ADR, or DCU process pointer, DCU_PTABLE_PTR, is less than the Address of DCU process table End DCU_PTABLE_END.
- DCU Hold table, DCU_HOLD_TABLE, is equal to ONE.
- DCU Case State, DCU_SCHED_STATE, is equal to ONE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490323.000	0			CL12472336.000 CL12472337.002

LLR 12490324.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Read pointer, DCU_RTABLE_PTR, equal to the DCU Read Table Start Address, DCU_RTABLE_ADR, DCU Process pointer, DCU_PTABLE_PTR, set equal to the DCU Process Table Start Address, DCU_PTABLE_ADR, and DCU Case State, DCU_SCHED_STATE, is equal to ZERO when all of the following conditions are met:

- DCU Process Pointer, DCU_PTABLE_PTR, is less than DCU Process Table Start Address, DCU_PTABLE_ADR, or DCU process pointer, DCU_PTABLE_PTR, is greater than or equal to the Address of DCU Process Table End DCU_PTABLE_END.
- DCU Case State, DCU_SCHED_STATE, is equal to ONE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490324.000	0			CL12472336.000

LLR 12490325.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Case State, DCU_SCHED_STATE, is equal to ZERO and set DCU Process Pointer, DCU_PTABLE_PTR, equal to DCU Hold table, DCU_HOLD_TABLE, when all of the following conditions are met:

- DCU Process Pointer, DCU_PTABLE_PTR, is Greater than or equal to DCU Process Table Start Address, DCU_PTABLE_ADR, or DCU process pointer, DCU_PTABLE_PTR, is less than the Address of DCU process table End DCU_PTABLE_END.
- Data at DCU Process Pointer, DCU_PTABLE_PTR, is ZERO.
- DCU Hold table, DCU_HOLD_TABLE, is NONZERO.
- DCU Case State, DCU_SCHED_STATE, is equal to ONE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490325.000	0			CL12472336.000

LLR 12490326.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Case State, DCU_SCHED_STATE, is equal to ZERO and increment DCU Process Pointer, DCU_PTABLE_PTR, and DCU Read Pointer, DCU_RTABLE_PTR, by ONE when all of the following conditions are met:

- DCU Process Pointer, DCU_PTABLE_PTR, is Greater than or equal to DCU Process Table Start Address, DCU_PTABLE_ADR, or DCU process pointer, DCU_PTABLE_PTR, is less than the Address of DCU process table End DCU_PTABLE_END.
- Data at DCU Process Pointer, DCU_PTABLE_PTR, is ZERO.
- DCU Hold table, DCU_HOLD_TABLE, is ZERO.
- DCU Case State, DCU_SCHED_STATE, is equal to ONE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490326.000	0			CL12472336.000 CL12472337.002

LLR 12490327.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Case State, DCU_SCHED_STATE, to TWO and DCU Process pointer, DCU_PTABLE_PTR, is set equal to DCU Update Address, DCU_UPDATE_ADR, when DCU Case State, DCU_SCHED_STATE, is equal to 0, the value at DCU Read pointer, DCU_RTABLE_PTR, is ZERO, DCU Available, DCU_IS_AVAILABLE, is equal to TRUE and DCU Read in Process, DCU_IN_PROCESS, is equal to FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490327.000	0			CL12472336.000 CL12472337.002

Ov 12490328.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: This moves the DCU from the Read State to the Write State. It checks if DCU_RTABLE_PTR is equal to DCU_RTABLE_END.

LLR 12490329.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Read in Process, READ_IN_PROCESS, to FALSE and set DCU Case State, DCU_SCHED_STATE, to ONE when all of the following conditions are met:

- DCU Case State, DCU_SCHED_STATE, is equal to 0.
- DCU Available, DCU_IS_AVAILABLE, is equal to TRUE.
- DCU Read in Process, DCU_IN_PROCESS, is equal to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490329.000	0			CL12472336.000 CL12472337.002

Ov 12490330.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: The requirement above moves the DCU from the Read to Process state.

Ov 12490332.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: The following LLRs relate to DCU_READ as it related to DCUREAD capability.

LLR 12490333.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Low Byte, DCU_LO_BYTE, equal to ZERO.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490333.000	0			CL12472336.000 CL12472337.002

LLR 12490334.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Data Pointer, DCU_DATA_PTR, equal to DCU Command Address, DCU_COMMAND_ADR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490334.000	0			CL12472336.000 CL12472337.002

LLR 12490335.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The Design shall shift left EIGHT Bits of DCU Address, DCU_ADDRESS, into the Value of DCU data, DCU_DATA_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490335.000	0			CL12472336.000 CL12472337.002

LLR 12490336.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the data at the DCU Data Pointer, DCU_DATA_PTR bit 3, to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490336.000	0			CL12472336.000 CL12472337.002

LLR 12490337.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall increment DCU Data Pointer, DCU_DATA_PTR, by one.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490337.000	0			CL12472336.000 CL12472337.002

LLR 12490338.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the Data Residing at DCU Data Pointer, DCU_DATA_PTR, equal to the Data Residing at DCU Block Pointer DCU_BLOCK_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490338.000	0			CL12472336.000 CL12472337.002

LLR 12490339.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall increment DCU Block Pointer, DCU_BLOCK_PTR, by one.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490339.000	0			CL12472336.000 CL12472337.002

LLR 12490340.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the DCU Buffer Pointer, DCU_BUFFER_PTR, equal to the Data Residing at DCU Block Pointer DCU_BLOCK_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490340.000	0			CL12472336.000 CL12472337.002

LLR 12490341.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall increment DCU Block Pointer, DCU_BLOCK_PTR, by one.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490341.000	0			CL12472336.000 CL12472337.002

LLR 12490343.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the DCU Max Number of 10 Second Retries, DCU_MAX_BUSY, equal to the Data Residing at DCU Block Pointer DCU_BLOCK_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490343.000	0			CL12472336.000 CL12472337.002 CL12472341.000

LLR 12490344.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Busy Timer, DCU_BUSY_TIMER, equal to 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490344.000	0			CL12472336.000 CL12472337.002 CL12472341.000

LLR 12490345.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Retry Counter, DCU_RETRY_COUNT, equal to 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490345.000	0			CL12472336.000 CL12472337.002 CL12472341.000 CL12472343.000 CL12472344.000

LLR 12490346.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU I/O State Variable, DCU_IO_STATE, equal to DCU Transfer State, START_TRANSFER.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490346.000	0			CL12472334.000 CL12472336.000 CL12472337.002 CL12472342.000

Ov 12490347.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: This part allows the DCU protocol to be in the right state so this will allow us to start reading the DCU with the info above and do fault detection on the data block messages and allow the retrying of the DCU Reads.

LLR 12490348.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU is Available, DCU_IS_AVAILABLE, equal to FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490348.000	0			CL12472334.000 CL12472340.000

Ov 12490349.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The following LLRs relate to DCU_CONSTANTS as it relates to DCUREAD.

LLR 12490350.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Read Subroutine Table, DCU_RBLOCK_TABLE, as per "Table 1".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490350.000	0			CL12472336.000 CL12472337.002

Table:(LLR 12490350.000-1) Table 1 - DCU Read Subroutines

Position	Table Variable Name	Subroutine/Value	Related DCU Block
1	<i>DCU_BLOCK_A</i>	DCU_EECDAT_BLOCK	9
Position	Table Variable Name	Subroutine/Value	Related DCU Block
2	<i>DCU_BLOCK_B</i>	DCU_ENGDAT_BLOCK	0
3	<i>DCU_BLOCK_C</i>	DCU_CPDAT_BLOCK	1
4	<i>DCU_BLOCK_D</i>	DCU_CTRDAT_BLOCK	305
5	<i>DCU_DCUID_RBLOCK</i>	DCU_DCUID_BLOCK	304
6	<i>DCU_BLOCK_E</i>	<i>LCF_BLOCK</i>	310-336
7	<i>DCU_RTABLE_END</i>	0	N/A

Ov 12490592.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: LCF block is an area defined in PAGE1.SRC it defines the LCF Block RAM signals

LLR 12490351.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Process Subroutine Table, DCU_PBLOCK_TABLE, as per "Table 2".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490351.000	0			CL12472336.000 CL12472337.002

Table:(LLR 12490351.000-1) Table 2 - DCU Process Subroutines

Position	Table Variable Name	Subroutine/Value	Related DCU Block
1	DCU_TASK_RAA	DCU_EEC_DATA	9
2	DCU_TASKS_RAEND	0	NA
3	DCU_TASK_RBA	DCU_ENGINE_DATA	0
4	DCU_TASK_RBAEND	0	NA
5	DCU_TASK_RCA	DCU_COMPONT_DATA	1
6	DCU_TASKS_RCEND	0	NA
7	DCU_TASK_RDD	DCU_IS_OBSOLETE	305
8	DCU_TASK_RDA	DCU_COUNTER_DATA	305
9	DCU_TASK_RDB	DCU_ENGINE_DATA	0
Position	Table Variable Name	Subroutine/Value	Related DCU Block
10	DCU_TASK_RDC	DCU_COMPONT_DATA	1
11	DCU_TASK_RDE	DCU_COUNTER_DATA	305
12	DCU_TASKS_RDEND	0	NA
13	DCU_TASK_REA	DCU_DCUID_DATA	304
14	DCU_LCF_PBLOCK	DCU_LCF_DATA	310-336
15	DCU_TASKS_REEND	0	NA
16	DCU_TASK_RFA	DCU_LCF_DATA	310-336
17	DCU_TASK_RFB	DCU_LCF_DATA	310-336
18	DCU_TASKS_RFEND	0	NA
19	DCU_PTABLE_END	0	NA

LLR 12490352.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 0 Read Data Block, DCU_ENGDAT_BLOCK, as per "Table 3".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490352.000	0			CL12472336.000

Table:(LLR 12490352.000-1) Table 3 - DCU Block 0 Read Data Block

Position	Table Variable Name	Signal/Value	Description
1	DCU_ENGDAT_BN	0	DCU Block Number
2	DCU_ENGDAT_BF	DCU_ENGTMP_BUF	Temp Buffer in RAM
3	DCU_ENGDAT_BT	3	# of 10-sec Retries

LLR 12490354.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 1 Read Data Block, DCU_CPDAT_BLOCK, as per "Table 4".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490354.000	0			CL12472336.000

Table:(LLR 12490354.000-1) Table 4 - DCU Block 1 Read Data Block

Position	Table Variable Name	Signal/Value	Description
1	DCU_CPDAT_NB	1	DCU Block Number
2	DCU_CPDAT_BF	DCU_CPTMP_BUF	Temp Buffer in RAM
3	DCU_CPDAT_BT	3	# of 10-sec Retries

LLR 12490355.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 304 Read Data Block, DCU_DCUID_BLOCK, as per "Table 5".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490355.000	0			CL12472337.002

Table:(LLR 12490355.000-1) Table 5 - DCU Block 304 Read Data Block

Position	Table Variable Name	Signal/Value	Description
1	DCU_DCUID_NB	304	DCU Block Number
2	DCU_DCUID_BF	DCU_DATA_BUFFER	Temp Buffer in RAM
3	DCU_DCUID_BT	3	# of 10-sec Retries

LLR 12490356.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 305 Read Data Block, DCU_CTRDAT_BLOCK, as per "Table 6".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490356.000	0			CL12472336.000

Table:(LLR 12490356.000-1) Table 6 - DCU Block 305 Read Data Block

Position	Table Variable Name	Signal/Value	Description
1	DCU_CTRDAT_NB	305	DCU Block Number
2	DCU_CTRDAT_BF	DCU_DATA_BUFFER	Temp Buffer in RAM
3	DCU_CTRDAT_BT	3	# of 10-sec Retries

LLR 12490357.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 9 Read Data Block, DCU_EECDAT_BLOCK, as per "Table 7".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490357.000	0			CL12472336.000

Table:(LLR 12490357.000-1) Table 7 - DCU Block 9 Read Data Block

Position	Table Variable Name	Signal/Value	Description
1	DCU_EECDAT_NB	9	DCU Block Number
2	DCU_EECDAT_BF	DCU_DATA_BUFFER	Temp Buffer in RAM
Position	Table Variable Name	Signal/Value	Description
3	DCU_EECDAT_BT	3	# of 10-sec Retries

04.05.01.03 DCU Block Data Output Logic [DCUWRITE]

OV 12490363.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

During normal system operation, various data blocks are written to the DCU. Requests are made to the DCU driver routines to write these blocks of data. The status word received from the DCU is used to determine whether transmission of the data had completed. This capability is performed by the following processes:

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Get_Queue

This routine gets a data entry from the DCU write queue.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

- DCU_Write

This routine is the interface between the requesting routines and the lower-level DCU device routines. It is used to initiate a write request. The DCU command buffer initialized to perform a read request, the data buffer to receive the data bytes is specified, and various control information to be used by the lower-level DCU device routines is defined such as: bytes to be transmitted, busy timeout length, and initial status. Once the buffers and control data is set up, the routine signals a write request to the lower-level DCU device routines.

Ov 12490396.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: These are the LLRs relating to the DCUWRITE portion of DCU_Schedule.

LLR 12490402.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall be in Write DCU Mode when DCU Case State, DCU_SCHED_STATE, is equal to WRITE_BLOCKS.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490402.000	0			CL12472361.000

Ov 12490589.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: This partially covers the linked HLR the other portion is the state table that executes DCU_SCHEDULE.

LLR 12490403.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the Process Table Pointer, DCU_PTABLE_PTR, equal to the DCU Write Table Address, DCU_UPDATE_ADR, when the Process Table Pointer, DCU_PTABLE_PTR, is less than the DCU Write Table Address, DCU_UPDATE_ADR, or the Process Table Pointer, DCU_PTABLE_PTR, is greater than or equal to the memory Address of DCU Write End location DCU_UPDATE_END.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490403.000	0			CL12472368.000

LLR 12490404.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set temporary Variable 27, SP27, equal to Data located at the Process Table Pointer, DCU_PTABLE_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490404.000	0			CL12472368.000

LLR 12490405.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall call the subroutine identified by "Table 1", related to the current position of the Process Table Pointer, DCU_PTABLE_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490405.000	0			CL12472368.000 CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490406.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall increment the Process Table Pointer, DCU_PTABLE_PTR by ONE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490406.000	0			CL12472368.000

LLR 12490364.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Write in Progress, WRITE_IN_PROGRESS, to FALSE when DCU is Available Flag, DCU_IS_AVAILABLE, is TRUE and DCU Write in Progress, WRITE_IN_PROGRESS, is equal to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490364.000	0			CL12472368.000

LLR 12490365.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Flag, DCUSIF, when DCU is Available Flag, DCU_IS_AVAILABLE, is TRUE, DCU Write in Progress, WRITE_IN_PROGRESS, is equal to TRUE and DCU Status, DCU_STATUS is not equal to ZERO:

- DCUSIF bits 1-4 set to 12.
- DCUSIF bits 7-15 is OR'ed with DCU_STATUS bits 7-15.
- DCUSIF bit 16 set to 1.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490365.000	0			CL12472362.000 CL12472367.000

LLR 12490366.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall call subroutine DCU_GET_QUEUE with DCU Block Pointer, DCU_BLOCK_PTR, and DCU Queue Empty Flag DCU_QUEUE_EMPTY when DCU is Available Flag, DCU_IS_AVAILABLE, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490366.000	0			CL12472368.000

Ov 12490590.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: This will return one of tables from Table 2-5 into DCU_BLOCK_PTR.

LLR 12490367.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute the following when DCU Queue Empty Flag, DCU_QUEUE_EMPTY, is equal to ZERO and DCU Functionality Status word, DCU_FUNC_SW, is equal to ZERO and DCU is Available Flag, DCU_IS_AVAILABLE, is TRUE:

- Call the DCU_WRITE subroutine with DCU Block Pointer DCU_BLOCK_PTR.
- DCU Data ready, DCU_DATA_READY, bit 3 set to 0.
- Write in Progress, WRITE_IN_PROGRESS, is set to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490367.000	0			CL12472362.000 CL12472368.000 CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000 CL12472374.000

LLR 12490368.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Data ready, DCU_DATA_READY, bit 3 set to TRUE when DCU Queue Empty Flag, DCU_QUEUE_EMPTY, is equal to ONE and the Process Table Pointer, DCU_PTABLE_PTR, is greater than or equal to the Address of DCU Write End, DCU_UPDATE_END, and DCU is Available Flag, DCU_IS_AVAILABLE, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490368.000	0			CL12472374.000

Ov 12490369.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: These are the LLRs relating to DCU_Get_Queue.

LLR 12490370.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set A equal to the DCU Queue get Entry Pointer DCU_QUEUE_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490370.000	0			CL12472368.000

LLR 12490371.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set A equal to DCU Queue Address, DCU_QUEUE_ADR, when A is less than DCU Queue Address DCU_QUEUE_ADDRESS. Otherwise A is incremented by ONE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490371.000	0			CL12472368.000

LLR 12490372.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set A equal to DCU Queue Address, DCU_QUEUE_ADR, when A is greater than or equal to the Memory address of DCU Queue End DCU_QUEUE_END.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490372.000	0			CL12472368.000

LLR 12490373.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set Queue is Empty, QUEUE_IF_EMPTY, equal to A.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490373.000	0			CL12472368.000

LLR 12490375.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Write Queue Put Pointer, DCU_PQUEUE_PTR, equal to DCU Queue Address, DCU_QUEUE_ADR, when DCU Write Queue Put Pointer, DCU_PQUEUE_PTR, is less than DCU Queue Address DCU_QUEUE_ADDRESS.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490375.000	0			CL12472368.000

LLR 12490376.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set Queue is Empty, QUEUE_IF_EMPTY, to TRUE if Queue is Empty, QUEUE_IF_EMPTY, is equal to DCU Write Queue Put Pointer DCU_PQUEUE_PTR. Otherwise Queue is Empty, QUEUE_IF_EMPTY, is set FALSE and DCU Write Queue Get Pointer, DCU_GQUEUE_PTR, is set equal to Queue is Empty, QUEUE_IF_EMPTY, and ENTRY is set equal to the Data located at DCU Write Queue Get Pointer DCU_GQUEUE_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490376.000	0			CL12472368.000

Ov 12490377.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note These are the LLRs relating to DCU_Write

LLR 12490378.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Low Byte, DCU_LO_BYTE, equal to ZERO.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490378.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490379.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Data Pointer, DCU_DATA_PTR, equal to DCU Command Address, DCU_COMMAND_ADR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490379.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490380.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The Design shall shift left EIGHT Bits of DCU Address, DCU_ADDRESS, into the Value of DCU data, DCU_DATA_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490380.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490381.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the data at the DCU Data Pointer, DCU_DATA_PTR bit 1 and 3, to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490381.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490382.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall increment DCU Data Pointer, DCU_DATA_PTR, by one.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490382.000	0			CL12472368.000 CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490383.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the Data Residing at DCU Data Pointer, DCU_DATA_PTR, equal to the Data Residing at DCU Block Pointer DCU_BLOCK_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490383.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490384.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall increment DCU Block Pointer, DCU_BLOCK_PTR, by one.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490384.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490386.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the DCU Buffer Pointer, DCU_BUFFER_PTR, equal to the Data Residing at DCU Block Pointer DCU_BLOCK_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490386.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490387.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall increment DCU Block Pointer, DCU_BLOCK_PTR, by one.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490387.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490388.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the DCU Bytes Sent, DCU_BYTES_SENT, equal to the Data Residing at DCU Block Pointer DCU_BLOCK_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490388.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490389.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall increment DCU Block Pointer, DCU_BLOCK_PTR, by one.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490389.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490390.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the DCU Max Number of 10 Second Retries, DCU_MAX_BUSY, equal to the Data Residing at DCU Block Pointer DCU_BLOCK_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490390.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490391.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Busy Timer, DCU_BUSY_TIMER, equal to 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490391.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490392.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU Retry Counter, DCU_RETRY_COUNT, equal to 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490392.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

LLR 12490393.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU I/O State Variable, DCU_IO_STATE, equal to DCU Transfer State, START_TRANSFER.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490393.000	0			CL12472364.000 CL12472366.000 CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

Ov 12490591.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: This part allows the DCU protocol to be in the right state so this will allow us to start writing to DCU with the info above and do fault detection on the data block messages and allow the retrying of the DCU writes.

LLR 12490394.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU is Available, DCU_IS_AVAILABLE, equal to FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490394.000	0			CL12472369.000 CL12472370.000 CL12472371.000 CL12472372.000

Ov 12490395.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: These are the LLRs relating to the DCUWRITE portion of DCU_Constants.

LLR 12490397.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Write Subroutine table, DCU_UPDATE_TABLE, as per "Table 1".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490397.000	0			CL12472368.000

Table:(LLR 12490397.000-1) Table 1 - DCU Write Subroutine

Position	Table Variable Name	Subroutine/Value	Related DCU Block
1	DCU_TASK_WA	DCU_WRITE_ENGDAT	0
2	DCU_TASK_WB	DCU_WRITE_CPDAT	1
3	DCU_TASK_WC	DCU_WRITE_CTRDAT	305
4	DCU_TASK_WD	DCU_WRITE_EECDAT	9
5	DCU_UPDATE_END	0	N/A

LLR 12490398.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 0 Write Data Info, DCU_ENGOUT_BLOCK, as per "Table 2".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490398.000	0			CL12472369.000

Table:(LLR 12490398.000-1) Table 2 - DCU Block 0 Write Data Info

Position	Table Variable Name	Signal/Value	Description
1	DCU_ENGOUT_BLOCK	0	DCU Block Number
2	DCU_ENGOUT_BU	DCU_ENGDAT_BUF	Source of Data to Write
3	DCU_ENGOUT_BL	'22	Number of bytes to Write
4	DCU_ENGOUT_BT	3	Number of 10 second Retries
Position	Table Variable Name	Signal/Value	Description
5	DCU_ENGOUT_BF	'1000	DCU_FUNC_SW bit Write Fault

LLR 12490399.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 1 Write Data Info, DCU_CPOUT_BLOCK, as per "Table 3".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490399.000	0			CL12472370.000

Table:(LLR 12490399.000-1) Table 3 - DCU Block 1 Write Data Info

Position	Table Variable Name	Signal/Value	Description
1	DCU_CPOUT_BLOCK	1	DCU Block Number
2	DCU_CPOUT_BU	DCU_CPDATA_BUF	Source of Data to Write
3	DCU_CPOUT_BL	'176	Number of bytes to Write
4	DCU_CPOUT_BT	3	Number of 10 second Retries
5	DCU_CPOUT_BF	'2000	DCU_FUNC_SW bit Write Fault

LLR 12490400.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 305 Write Data Info, DCU_CTROUT_BLOCK, as per "Table 4".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490400.000	0			CL12472371.000

Table:(LLR 12490400.000-1) Table 4 - DCU Block 305 Write Data Info

Position	Table Variable Name	Signal/Value	Description
1	DCU_CTROUT_BLOCK	305	DCU Block Number
2	DCU_CTROUT_BU	DCU_CTRDATA_BUF	Source of Data to Write
3	DCU_CTROUT_BL	'174	Number of bytes to Write
Position	Table Variable Name	Signal/Value	Description
4	DCU_CTROUT_BT	3	Number of 10 second Retries
5	DCU_CTROUT_BF	'4000	DCU_FUNC_SW bit Write Fault

LLR 12490401.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall define the DCU Block 9 Write Data Info, DCU_EECOUT_BLOCK, as per "Table 5".

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490401.000	0			CL12472372.000

Table:(LLR 12490401.000-1) Table 5 - DCU Block 9 Write Data Info

Position	Table Variable Name	Signal/Value	Description
1	DCU_EECOUT_BLOCK	9	DCU Block Number
2	DCU_EECOUT_BU	DCU_EECDATA_BUF	Source of Data to Write
3	DCU_EECOUT_BL	'40	Number of bytes to Write
4	DCU_EECOUT_BT	3	Number of 10 second Retries
5	DCU_EECOUT_BF	'10000	DCU_FUNC_SW bit Write Fault

04.05.01.04 DCU EEC Data Block Validation Logic [BLK09CHK]

This logic is responsible for validating the EEC data block. The logic determines when the DCU is new, has been reprogrammed, or there is a mismatch between the EEC data block and the analogous data in EEPROM. This capability is implemented by the following processes:

- DCU_EEC_Data

This routine processes the EEC data block after it has been read in from the DCU. First it checks for any DCU faults. If there is a fault and the fault is due to a new DCU, it clears the fault and signals a new DCU was detected; otherwise, it sets the appropriate fault flags. If no faults were detected, it checks if the DCU is reprogrammed. If so, it signals a reprogrammed DCU has been detected. If the DCU is neither new nor reprogrammed, it performs a comparison of the EEC data to its equivalent in EEPROM. If it determines there is a mismatch, it sets the mismatch flag. If the DCU is new or reprogrammed or there is a mismatch found, this routine signals the data in the DCU for this block be updated.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Compare_Item

This routine compares two sets of data words. If there is a mismatch, a flag is set.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

04.05.01.05 DCU Engine Data Block Validation Logic [BLK00CHK]

This logic compares the engine data received from the DCU with the corresponding data stored in EEPROM memory for a mismatch. It also makes a request to update the information in the DCU or in EEPROM depending on the DCU status. This capability is performed by the following processes:

- DCU_Engine_Data

After the engine data block has been read in, this routine processes the data. First it checks if any faults exist. If there is a fault and the fault is due to a new DCU, it clears the fault and signals the data in the DCU for this block be updated.

Otherwise, it sets the appropriate fault flags. If no DCU faults exist, a comparison is done between the EEPROM data values and the DCU data values. If a mismatch is detected for any value, it signals a mismatch was found. If there are no faults and the DCU is new or reprogrammed or there is a mismatch found in the EEC data or engine data, this routine requests an update of the EEPROM data for this block.

- DCU_Compare_Data

This routine traverses through a table of items to compare. If a mismatch is found in a given item, the appropriate flag is set.

- DCU_Compare_Item

This routine compares two sets of data words. If there is a mismatch a flag is set.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Get_Data

This routine copies the DCU data from one area to another.

- EE_Get_Queue

This routine gets data from the EEPROM queue.

- EE_Put_Queue

This routine puts a data entry into the EEPROM queue.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state

it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

- EE_Write_Block

This routine writes blocks of data to EEPROM. Each EEPROM block contains up to 64 bytes. The data is written in groups of 2 bytes at a time. If the data spans over an EEPROM block boundary, data in the current EEPROM block is written and the remaining data is processed in the next block write.

04.05.01.06 DCU Component Data Block Validation Logic [BLK01CHK]

This logic compares the component data received from the DCU with the corresponding data stored in EEPROM memory for a mismatch. It also makes a request to update the information in the DCU or in EEPROM depending on the DCU status. This capability is performed by the following processes:

- DCU_Component_Data

After the component data block has been read in, this routine processes the data. First it checks if any faults exist. If there is a fault and the fault is due to a new DCU, it clears the fault and signals the data in the DCU for this block be updated. Otherwise, it sets the appropriate fault flags. If no DCU faults exist, a comparison is done between the EEPROM data values and the DCU data values. If a mismatch is detected for any value, it signals a mismatch was found. Also if there are no faults and the DCU is new or reprogrammed or there is a mismatch found in the EEC data or engine data, this routine requests an update of the EEPROM data for this block. A check is also performed to ensure the engine data and component data blocks are simultaneously present or not. If not, an inconsistency fault is set.

- DCU_Compare_Data

This routine traverses through a table of items to compare. If a mismatch is found in a given item, the appropriate flag is set.

- DCU_Compare_Item

This routine compares two sets of data words. If there is a mismatch, a flag is set.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Get_Data

This routine copies the DCU data from one area to another.

- EE_Get_Queue

This routine gets data from the EEPROM queue.

- EE_Put_Queue

This routine puts a data entry into the EEPROM queue.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

- EE_Write_Block

This routine writes blocks of data to EEPROM. Each EEPROM block contains up to 64 bytes. The data is written in groups of 2 bytes at a time. If the data spans over an EEPROM block boundary, data in the current EEPROM block is written and the remaining data is processed in the next block write.

04.05.01.07 DCU Counter Data Block Validation Logic [BLK305CHK]

Ov 12490407.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic validates the counter data received from the DCU. It also makes a request to update the information in the DCU or in EEPROM depending on the DCU status. This capability is performed by the following processes:

- DCU_Is_Obsolete

This routine determines if there is a mismatch between the DCU data and the engine data with respect to the Start counter, Shutdown counter and Resync counter. When the engine is not running, the software will resync the Start and Shutdown counters by selecting the larger of the two. When the engine is running, the software will resync the Start and Shutdown counters by either adding or subtracting one. If there is a mismatch, RAM or the DCU is overwritten with the adjusted Start/Shutdown/Resync values.

Ov 12490418.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design is incorporated in modules DCU_IS_OBSOLETE. Designs changed from the prior baseline are identified in section 13.

LLR 12490429.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module DCU_IS_OBSOLETE shall be implemented as per the following design models:

- DCU_IS_OBSOLETE_A.CPGC
- DCU_IS_OBSOLETE_B.CPGC
- DCU_IS_OBSOLETE_C.CPGC
- DCU_IS_OBSOLETE_D.CPGC
- DCU_IS_OBSOLETE_E.CPGC
- DCU_IS_OBSOLETE_F.CPGC

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490429.001	Func		N/A	CL12472458.000 CL12472459.000 CL12472460.000 CL12472461.000 CL12472462.000 CL12472463.000

Ov 12490440.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

- DCU_Counter_Data

After the counter data block has been read in, this routine processes the data. First it checks if any faults exist. If there is a fault and the fault is due to a new DCU, it clears the fault and, if the EEPROM data is valid, it signals the data in the DCU for this block be updated. Otherwise, it sets the appropriate fault flags. If no DCU faults exist, it performs a range check on the data and if any value is out of range a fault flag is set. If no DCU faults exist and the data in the block has been determined to be obsolete, it adjusts the counter data and requests the updates be stored in EEPROM as well as signals that the data in the DCU for this block be updated. If there are no data errors and the DCU is new or reprogrammed or there is a mismatch found in the EEC data, this routine requests an update of the EEPROM data for this block.

- DCU_Range_Checks

This routine traverses through a table of items to perform a range check on. If an item is out of range, the appropriate flag is set.

- DCU_Get_Data

This routine copies the DCU data from one area to another.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

- EE_Get_Queue

This routine gets data from the EEPROM queue.

- EE_Put_Queue

This routine puts a data entry into the EEPROM queue.

- EE_Write_Block

This routine writes blocks of data to EEPROM. Each EEPROM block contains up to 64 bytes. The data is written in groups of 2 bytes at a time. If the data spans over an EEPROM block boundary, data in the current EEPROM block is written and the remaining data is processed in the next block write.

LLR 12490604.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module DCU_RANGE_CHECKS shall be implemented as per the DCU_RANGE_CHECKS.CPGC design model.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490604.000	Func	HLR	No	CL12472454.001 CL12472948.000 CL12472949.000 CL12472950.000 CL12472951.000 CL12472952.000 CL12472953.000 CL12472954.000

Ov 12490448.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Textual Low Level Requirements:

LLR 12490688.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The software shall set DCU_CTRDAT_IF.5 to TRUE, when all the following conditions are met:

1. DCU_FLAGS_1.2 is FALSE and,
2. Any of the following conditions are met:
 - LPIMP_CYCCRS is less K_CYCCRS_ZERO or,
 - LPIMP_CYCCRS is greater than K_CYCCRS_32K or
 - HPIMP_CYCCRS is less K_CYCCRS_ZERO or,
 - HPIMP_CYCCRS is greater than K_CYCCRS_32K or
 - HPTFCP_CYCCRS is less K_CYCCRS_ZERO or,
 - HPTFCP_CYCCRS is greater than K_CYCCRS_32K or,
 - HPTD_CYCCRS is less K_CYCCRS_ZERO or,
 - HPTD_CYCCRS is greater than K_CYCCRS_32K or,
 - HPTRCP_CYCCRS is less K_CYCCRS_ZERO or,
 - HPTRCP_CYCCRS is greater than K_CYCCRS_32K or,
 - ISS_CYCCRS is less than K_CYCCRS_ZERO or,
 - ISS_CYCCRS is greater than K_CYCCRS_32K or,
 - LPTD_CYCCRS is less K_CYCCRS_ZERO or,
 - LPTD_CYCCRS is greater than K_CYCCRS_32K or,
 - PT1D_CYCCRS is less K_CYCCRS_ZERO or,
 - PT1D_CYCCRS is greater than K_CYCCRS_32K or
 - PT2D_CYCCRS is less K_CYCCRS_ZERO or,
 - PT2D_CYCCRS is greater than K_CYCCRS_32K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490688.000	0			CL12472454.001

LLR 12490413.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall set DCU_CTRDAT_IF.6 to TRUE, when all the following conditions are met:

- DCU_FLAGS_1.2 is FALSE and,
- ACCSTRTS_ALL is greater than K_CYCCRS_65K.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490413.001	Func	DERIVED	N/A	CL12472952.000

Derived Requirement Justification:

ACCSTRTS_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

LLR 12490408.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall set DCU_CTRDAT_IF.7 to TRUE, when all the following conditions are met:

1. DCU_FLAGS_1.2 is FALSE and,
2. Any of the following condition are met:
 - ACCINFLSTRTS_ALL is less than K_CYCCRS_ZERO or,
 - ACCINFLSTRTS_ALL is greater than K_CYCCRS_32K.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490408.001	Func		N/A	CL12472951.000

LLR 12490410.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall set DCU_CTRDAT_IF.8 to TRUE, when all the following conditions are met:

- DCU_FLAGS_1.2 is FALSE and,
- ACCNUMFL_ALL is greater than K_CYCCRS_65K.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490410.001	Func	DERIVED	N/A	CL12472950.000

Derived Requirement Justification:

ACCNUMFL_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

LLR 12490421.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall set DCU_CTRDAT_IF.9 to TRUE, when all the following conditions are met:

1. DCU_FLAGS_1.2, is FALSE and,
2. Any of the following conditions are met:
 - ENG_RUN_CRS_ALL is less than K_CYCCRS_ZERO or,
 - ENG_RUN_CRS_ALL is greater than K_CYCCRS_32K.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490421.001	Func		N/A	CL12472949.000

LLR 12490423.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall set DCU_CTRDAT_IF.10 to TRUE, when all the following conditions are met:

1. DCU_FLAGS_1.2, is FALSE and,
2. Any of the following conditions are met:
 - ENGFLTM_CRS_ALL is less than K_CYCCRS_ZERO or,
 - ENGFLTM_CRS_ALL is greater than K_CYCCRS_32K.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490423.001	Func		N/A	CL12472948.000

LLR 12490689.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The software shall set DCU_CTRDAT_IF.11 to TRUE, if all the following conditions are met:

1. DCU_FLAGS_1.2 is FALSE and,
2. Any of the following conditions are met:
 - ACCSHUTD_ALL is greater than K_CYCCRS_65K or,
 - NTO_DL3 is less than K_CYCCRS_ZERO or,
 - NTO_DL3 is greater than K_CYCCRS_32K or,
 - NTO_DL2 is less than K_CYCCRS_ZERO or,
 - NTO_DL2 is greater than K_CYCCRS_32K or,
 - NTO_DL1 is less than K_CYCCRS_ZERO or,
 - NTO_DL1 is greater than K_CYCCRS_32K or,
 - MTO_DL3 is less than K_CYCCRS_ZERO or,
 - MTO_DL3 is greater than K_CYCCRS_32K or,
 - MTO_DL2 is less than K_CYCCRS_ZERO or,
 - MTO_DL2 is greater than K_CYCCRS_32K or,
 - MTO_DL1 is less than K_CYCCRS_ZERO or,
 - MTO_DL1 is greater than K_CYCCRS_32K.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490689.000	Func	DERIVED	N/A	CL12472953.000

Derived Requirement Justification:

ACCSHUTD_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

LLR 12490415.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall set DCU_CTRDAT_IF.12 to TRUE, when all the following conditions are met:

- DCU_FLAGS_1.2 is FALSE and,
- ACCSTRTSFLT_ALL is greater than K_CYCCRS_65K.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490415.001	Func	DERIVED	N/A	CL12472954.000

Derived Requirement Justification:

ACCSTRTSFLT_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

LLR 12490449.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_FLAGS_1 bit 7 to 1 if DCU_FLAGS_1 bit 2 is equal to 1 AND DCU_FLAGS_1 bit 1 is equal to 0 AND EE_CTRDAT_IF is equal to 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490449.000	0			CL12472452.000

LLR 12490450.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall unpack DCU_BLOCK_305 into the DCU Fields identified by "TABLE 1: DCU_BLOCK_305 Parameter Definition" when DCU_FLAGS_1 bit 2 is equal to 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490450.000	0			CL12472453.000

LLR 12490438.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCF_UPD to TRUE when DCU_OBSOLETE is TRUE, otherwise DCU_LCF_UPD remains equal to the last past value.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490438.000	0			CL12472464.000

LLR 12490439.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set ACCSTR_SD_RESYNC equal to DCU_ADJUST_RSYNC when DCU_OBSOLETE is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490439.000	0			CL12472465.000

LLR 12490441.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set ACCSTRTS_ALL equal to DCU_ADJUST_STRTS when DCU_OBSOLETE is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490441.000	0			CL12472465.000

LLR 12490442.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set ACCSHUTD_ALL equal to DCU_ADJUST_SHUTD when DCU_OBSOLETE is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490442.000	0			CL12472465.000

LLR 12490443.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the RAM signals identified by "Table 1: DCU_BLOCK_305 Parameter Definition" equal to their associated DCU Fields, except for the Software Identification Signals identified in "Table 3: Software Identification Signals", if DCU_FLAGS_1 bit 7, is equal to 0. Otherwise the Ram signals are equal to their associated EEPROM fields.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490443.000	0			CL12472466.000

LLR 12490444.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the EEPROM signals identified by "Table 1: DCU_BLOCK_305 Parameter Definition" equal to their associated RAM signals if DCU_FLAGS_1 bit 1 is equal to 1 OR DCU_FLAGS_1 bit 3 is equal to 1.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490444.000	0			CL12472469.000

LLR 12490445.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_CTRDAT_IF bits 1-4 to 12, bit 16 to 1, when DCU_CTRDAT_IF bits 5-12 are not equal to 0. Otherwise DCU_CTRDAT_IF bits 1-4 remain 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490445.000	0			

Ov 12490446.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: See SRS capability BLK305CHK for references to "Table 1: DCU_BLOCK_305 Parameter Definition" and "Table 3: Software Identification Signals".

04.05.01.08 DCU ID Data Block Validation Logic [BLK304CHK]

Ov 12490452.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic validates the DCU ID data received from the DCU. This capability is performed by the following processes:

- DCU_DCUID_Data

After the DCU ID data block has been read in, this routine validates the data. First it checks if any DCU faults exist. If there is a fault, it sets the appropriate fault flags. If no DCU faults exist, it performs a checksum on the data. If the checksum fails, a fault flag is set. Otherwise, it proceeds to perform range checks on the data. If the range checks fail, a fault flag is set. If there are no range faults, it performs comparison checks on some of the data with expected values. If the comparison checks fail, a fault is set. If all the above checks are good, it signals that the DCU data read in up to this point is ready for use by other logic.

LLR 12490459.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_DCUID_IF bit 9 to 1, bits 1-4 to 12, and bit 16 to 1 and the process will end when DCU_CHECKSUM is equal to 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490459.000	0			CL12472487.000

LLR 12490460.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_DCUID_IF bit 12 to 1, bits 1-4 to 12, and bit 16 to 1 when the NH Speed for Engine OFF detection is less than NH Speed for Engine Run detection value comparison check, KNH_ENG_OFF, fails.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490460.000	0			CL12472488.000

LLR 12490461.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_DCUID_IF bit 6 to 1, bits 1-4 to 12, and bit 16 to 1 when KNH_ENG_OFF is less than KNH_ENG_OFF_MIN OR greater than KNH_ENG_OFF_MAX.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490461.000	0			CL12472489.000

LLR 12490462.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_DCUID_IF bit 7 to 1, bits 1-4 to 12, and bit 16 to 1 when KNH_ENG_RUN is less than KNH_ENG_RUN_MIN OR greater than KNH_ENG_RUN_MAX.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490462.000	0			CL12472490.000

LLR 12490463.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_DCUID_IF bit 8 to 1, bits 1-4 to 12, and bit 16 to 1 when KVK_FLT_ADC is less than KVK_FLT_ADC_MINOR greater than KVK_FLT_ADC_MAX.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490463.000	0			CL12472491.000

LLR 12490464.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCUI_DCUID_IF bit 9 to 1, bits 1-4 to 12, and bit 16 to 1 when KVK_FLT_DPSNR is less than KVK_FLTDPSNR_MIN OR greater than KVK_FLTDPSNR_MAX.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490464.000	0			CL12472492.000

LLR 12490465.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_DCUID_IF bit 10 to 1, bits 1-4 to 12, and bit 16 to 1 when the LCF Cycle Counting Engine Model ASCII Chars 1 and 2 Comparison check, CYC_ENG_MODEL12, fails.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490465.000	0			CL12472493.000

LLR 12490466.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCUI_DCUID_IF bit 10 to 1, bits 1-4 to 12, and bit 16 to 1 when the LCF Cycle Counting Engine Model ASCII Chars 3 and 4 Comparison check, CYC_ENG_MODEL34, fails.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490466.000	0			CL12472494.000

Ov 12490453.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

- DCU_Compare_Data

This routine traverses through a table of items to compare. If a mismatch is found in a given item, the appropriate flag is set.

- DCU_Compare_Item

This routine compares two sets of data words. If there is a mismatch a flag is set.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Get_Checksum

This routine computes the checksum for a block of data.

LLR 12490454.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the address of DCU_BUFFER_PTR equal to the address of DCU_BUFFER_ADR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490454.000	0			CL12472487.000

LLR 12490455.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_CHECKSUM equal to 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490455.000	0			CL12472487.000

LLR 12490456.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall perform the CHECKSUM operation by setting DCU_CHECKSUM equal to the value at DCU_BUFFER_PTR and increment that address of DCU_BUFFER_PTR while the address of DCU_BUFFER_PTR is less than the address of DCU_BUFFER_END, else the CHECKSUM operation is not performed.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490456.000	0			CL12472487.000

Ov 12490457.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

- DCU_Get_Data

This routine copies the DCU data from one area to another.

- DCU_Range_Checks

This routine traverses through a table of items to perform a range check on. If an item is out of range, the appropriate flag is set.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

04.05.01.09 DCU LCF Data Block Validation Logic [LCF_FACTCHK]

Ov 12490467.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic prepares for LCF data to be read from the DCU and validates the data after it is received from the DCU. This capability is performed by the following processes:

- DCU_LCF_Data

Before a block of LCF data is read from the DCU, this routine sets up information needed to read the block. After the block has been read, this routine verifies the data. First it checks for DCU faults and if there are any, it set the appropriate fault flags. If there are no DCU faults, it performs a checksum on the LCF data. If the checksum fails, it sets a checksum fault. If the checksum is good, it then proceeds to find a matching part number for the component that the LCF data pertains to. If it does not find a match, it sets up information to read the next block for that component. If no match is found for a particular component among the blocks pertaining to the component, a fault is set for that component. If it does find a match, it transfers the LCF data to a designated buffer area for that component. It then performs a range check on the data. If the range check fails, it sets a fault. After the data has been verified, it sets up information to read data for the next component.

LLR 12490478.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall process the LCF component block triplets for each component regardless of failure status of the processing of any preceding LCF component block triplets as indicated by DCU_LCFFACT_SW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490478.000	0			CL12472506.001 CL12472507.000 CL12472508.000

LLR 12490483.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall exit the part number search process and set bit 13 of DCU_LCFFACT_SW (DCU_LCFPN_NF) to 1 when DCU_STATUS is equal to 0 AND DCU_LCF_SUM is equal to 0 AND LCF_PN_NOTFOUND increments to 3.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490483.000	0			CL12472510.000 CL12472511.000 CL12472513.000 CL12472516.000

LLR 12490484.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 1 to 1 when the LPIMP block Triplet Fault, DCU_LCFSW_CFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490484.000	0			CL12472516.000

LLR 12490485.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 2 to 1 when the HPIMP block Triplet Fault, DCU_LCFSW_CFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490485.000	0			CL12472516.000

LLR 12490486.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 3 to 1 when the HPTFCP block Triplet Fault, DCU_LCFSW_CFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490486.000	0			CL12472516.000

LLR 12490487.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 4 to 1 when the HPTD block Triplet Fault, DCU_LCFSW_CFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490487.000	0			CL12472516.000

LLR 12490488.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 5 to 1 when the HPTRC block Triplet Fault, DCU_LCFSW_CFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490488.000	0			CL12472516.000

LLR 12490489.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 6 to 1 when the ISS block Triplet Fault, DCU_LCFSW_CFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490489.000	0			CL12472516.000

LLR 12490468.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 7 to 1 when the LPTD block Triplet Fault, DCU_LCFSW_CFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490468.000	0			CL12472516.000

LLR 12490469.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 8 to 1 when the PT1D block Triplet Fault, DCU_LCFSW_CFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490469.000	0			CL12472516.000

LLR 12490470.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 9 to 1 when the PT2D block Triplet Fault, DCU_LCFSW_CFAULT, is set

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490470.000	0			CL12472516.000

LLR 12490471.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 10 to 1 when the Triplet Block 1 fault, DCU_LCF_TFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490471.000	0			CL12472516.000

LLR 12490472.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 11 to 1 when the Triplet Block 2 fault, DCU_LCF_TFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490472.000	0			CL12472516.000

LLR 12490473.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 12 to 1 when the Triplet Block 3 fault, DCU_LCF_TFAULT, is set.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490473.000	0			CL12472516.000

LLR 12490474.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 14 to 1 when the Abbreviated Cycle Factor (ACF) range fault, DCU_LCFDAT_RANGE, is set. Otherwise the component cycle information is acceptable for LCF calculations in capability CYCLE_COUNTING.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490474.000	0			CL12472514.000 CL12472515.000 CL12472516.000

LLR 12490475.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 15 to 1 and exit the process when the Block data checksum fault, DCU_LCFDAT_SUM, is set. Otherwise further processing of the block reads will continue.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490475.000	0			CL12472509.000 CL12472516.000

LLR 12490476.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_LCFFACT_SW bit 16 to 1 when any other fault bit in DCU_LCFFACT_SW is set to 1.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490476.000	0			CL12472517.000

Ov 12490477.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

- DCU_Compare_Item

This routine compares two sets of data words. If there is a mismatch, a flag is set.

- DCU_Get_Checksum

This routine computes the checksum for a block of data.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Get_Data

This routine copies the DCU data from one area to another.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read;

in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

04.05.01.10 DCU Engine Data Block Configuration Logic [DCU_ENG_CFG]

Ov 12490490.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic makes requests to write engine data to the DCU. This capability is performed by the following processes:

- DCU_Write_ENGdat

This routine checks if a request should be made to write engine data to the DCU and, if so, makes the request.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Put_Queue

This routine puts a data entry into the DCU write queue.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

Ov 12490491.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCD(s): SCR: PW127L-EEC-SW-PWC-16236 SCO-D: PW127L-EEC-SW-PWC-16242

Ov 12490492.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed Designs:

None

Changed Low Level Requirements:

LLR 12490493.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The Design shall scale and locate the fields within DCU_BLOCK_00 at specified byte offsets, with RAM and EEPROM Data associated with each field.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490493.000	0			CL12472240.000

Ov 12490494.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to "Table 1: DCU_BLOCK_00 Parameter Definition" for the following information: DCU Byte, DCU Parameter Description, RAM Storage, EEPROM Storage, Scale Factor, and Units.

Table:(Ov 12490494.000-1) DCU _BLOCK _00 Parameter Definition

DCU Byte	DCU Parameter Description	RAM Storage	EEPROM Storage	Scale Factor	Units
0, 1	Engine S/N Chars 1&2	<i>ENG SNA12</i>	<i>ENG SNA12EE</i>	N/A	ASCII
2, 3	Engine S/N Chars 3&4	<i>ENG SNA34</i>	<i>ENG SNA34EE</i>	N/A	ASCII
4, 5	Engine S/N Chars 5&6	<i>ENG SNA56</i>	<i>ENG SNA56EE</i>	N/A	ASCII
6, 7	Turbomachine S/N Chars 1&2	<i>TRBMSNA12</i>	<i>TRBMSNA12EE</i>	N/A	ASCII
8, 9	Turbomachine S/N Chars 3&4	<i>TRBMSNA34</i>	<i>TRBMSNA34EE</i>	N/A	ASCII
10, 11	Turbomachine S/N Chars 5&6	<i>TRBMSNA56</i>	<i>TRBMSNA56EE</i>	N/A	ASCII
12, 13	RGB S/N Chars 1&2	<i>RGB SNA12</i>	<i>RGB SNA12EE</i>	N/A	ASCII
14, 15	RGB S/N Chars 3&4	<i>RGB SNA34</i>	<i>RGB SNA34EE</i>	N/A	ASCII
16, 17	RGB S/N Chars 5&6	<i>RGB SNA56</i>	<i>RGB SNA56EE</i>	N/A	ASCII
18, 19	Spare (reserved: Torque Bias Trim)	None	None	N/A	N/A
20, 21	Spare (reserved: Torque Gain Trim)	None	None	N/A	N/A
22, 23	Spare (reserved: MGT Bias Trim)	None	None	N/A	N/A
24 - 127	Spare	None	None	N/A	N/A

LLR 12490495.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute the following logic when DCU_FUNC_SW is equal to 0 and DCUIF is equal to 0 and DCU_FLAGS_1 bit 5 is TRUE:

- Copy RAM to DCU Fields: Copy RAM signals into the DCU Fields identified by "Block 0 Organization".
- Pack DCU Data: Pack the DCU fields identified by "Block 0 Organization" into DCU_BLOCK_00.
- Write DCU: Give a write request to capability DCUWRITE to write DCU_BLOCK_00 to the DCU.
- Clear DCU Initialization Request: Set DCU_FLAGS_1 bit 5 to FALSE.

Otherwise, the logic is not executed.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490495.000	0			CL12472241.000

04.05.01.11 DCU Component Data Block Configuration Logic [DCU_CP_CFG]

This logic makes requests to write component data to the DCU. This capability is performed by the following processes:

- DCU_Write_CPdat

This routine checks if a request should be made to write component data to the DCU and, if so, makes the request.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Put_Queue

This routine puts a data entry into the DCU write queue.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

04.05.01.12 DCU EEC Data Block Configuration Logic [DCU_EEC_CFG]

Ov 12490496.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic makes requests to write EEC data to the DCU. This capability is performed by the following processes:

- DCU_Write_EECdat

This routine checks if a request should be made to write EEC data to the DCU and, if so, makes the request.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Put_Queue

This routine puts a data entry into the DCU write queue.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

Ov 12490498.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Ov 12490499.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed Designs:

None

Ov 12490500.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Changed Low Level Requirements:

LLR 12490501.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall scale and locate the fields within DCU_BLOCK_09 at specified byte offsets, with RAM and EEPROM data associated with each field.

Note: Refer to "Table 1: DCU_BLOCK_09 Parameter Definition" for the following information: DCU Byte, DCU Parameter Description, RAM Storage, EEPROM Storage, Scale Factor, and Units.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490501.000	0			CL12472271.000

Table:(LLR 12490501.000-1) Table 1:DCU _BLOCK _09 Parameter Definition

DCU	DCU Parameter	RAM	EEPROM Storage	Scale	Units
Byte	Description	Storage		Factor	
0, 1	EEC P/N & S/N Chars 1 & 2	EECPN_SNA12	EECPN_SNA12EE	N/A	ASCII
2, 3	EEC P/N & S/N Chars 3 & 4	EECPN_SNA34	EECPN_SNA34EE	N/A	ASCII
4, 5	EEC P/N & S/N Chars 5 & 6	EECPN_SNA56	EECPN_SNA56EE	N/A	ASCII
6, 7	EEC P/N & S/N Chars 7 & 8	EECPN_SNA78	EECPN_SNA78EE	N/A	ASCII
8, 9	EEC P/N & S/N Chars 9 & 10	EECPN_SNA910	EECPN_SNA910EE	N/A	ASCII
10, 11	EEC P/N & S/N Chars 11 & 12	EECPN_SNA1112	EECPN_SNA1112EE	N/A	ASCII
12, 13	EEC P/N & S/N Chars 13 & 14	EECPN_SNA1314	EECPN_SNA1314EE	N/A	ASCII
14, 15	EEC P/N & S/N Chars 15 & 16	EECPN_SNA1516	EECPN_SNA1516EE	N/A	ASCII
16, 17	EEC P/N & S/N Chars 17 & 18	EECPN_SNA1718	EECPN_SNA1718EE	N/A	ASCII
18, 19	EEC P/N & S/N Chars 19 & 20	EECPN_SNA1920	EECPN_SNA1920EE	N/A	ASCII
20, 21	EEC P/N & S/N Chars 21 & 22	EECPN_SNA2122	EECPN_SNA2122EE	N/A	ASCII
22, 23	EEC P/N & S/N Chars 23 & 24	EECPN_SNA2324	EECPN_SNA2324EE	N/A	ASCII
24, 25	EEC P/N & S/N Chars 25 & 26	EECPN_SNA2526	EECPN_SNA2526EE	N/A	ASCII
26, 27	EEC P/N & S/N Chars 27 & 28	EECPN_SNA2728	EECPN_SNA2728EE	N/A	ASCII
28, 29	EEC P/N & S/N Chars 29 & 30	EECPN_SNA2930	EECPN_SNA2930EE	N/A	ASCII
30, 31	EEC P/N & S/N Chars 31 & 32	EECPN_SNA3132	EECPN_SNA3132EE	N/A	ASCII
32,33	Reprogrammed DCU flag (=1 when reprogrammed)	MODDCUDATA	N/A	Discrete	None
34 - 127	Spare	None	None	N/A	N/A

LLR 12490503.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute the following logic when DCU_FUNC_SW is equal to 0 and DCUSIF is equal to 0 and DCU_FLAGS_1 bit 4 is TRUE:

- Copy RAM to DCU Fields: Copy the RAM signals into the DCU Fields identified by "Block 9 Organization" as follows:
 - Set MODDCUDATA to 0 in the DCU EEC information data block.
 - Initialize the 32-byte EEC part number and serial number stored in the DCU EEC information data block to the 32-byte EEC part number and serial number stored in EEPROM (note that capability E05_RAMINIT copied the EEPROM values to RAM).
 - Pack DCU Data: Pack the DCU Fields identified by "Block 9 Organization" into DCU_BLOCK_09.
 - Write DCU: Give a "Write Request" to capability DCUWRITE to write DCU_BLOCK_09 to the DCU.
 - Clear DCU Initialization Request: Set DCU_FLAGS_1 bit 4 to FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490503.000	0			CL12472272.000

04.05.01.13 DCU Engine Usage Data Block Configuration Logic [DCU_ENGUSE_CFG]

Ov 12490504.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic makes requests to write engine usage data to the DCU. This capability is performed by the following processes:

- DCU_Write_CTRdat

This routine checks if a request should be made to write counter data to the DCU and, if so, makes the request.

- DCU_Constants

This module contains the tables and constants that are used by the various DCU routines.

- DCU_Put_Queue

This routine puts a data entry into the DCU write queue.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

- EE_Write_Block

This routine writes blocks of data to EEPROM. Each EEPROM block contains up to 64 bytes. The data is written in groups of 2 bytes at a time. If the data spans over an EEPROM block boundary, data in the current EEPROM block is written and the remaining data is processed in the next block write.

- EE_Get_Queue

This routine gets data from the EEPROM queue.

- Update_EE_Data

This routine makes a request to update the counter data in EEPROM and signals the counter data in the DCU should be updated. If the counter data is ready (i.e., has been read from the DCU) and the engine has just started, or if the engine has just shutdown, or if an LCF data adjustment is required due to resynch, then it requests the counter data be updated in EEPROM. After the EEPROM value has been updated, it signals that the DCU counter data be updated.

LLR 12490510.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall scale and locate the fields within the DCU_BLOCK_305 at specified byte offsets, with RAM and EEPROM data associated with each field.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490510.000	0			CL12472449.002

Ov 12490511.001

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: Refer to SRS to BLK305CHK for information found in Table 1: DCU_BLOCK_305 Parameter Definition.

LLR 12490512.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set DCU_FLAGS_1 bit 7 to TRUE when DCU_UPD_CTRDAT is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490512.000	0			CL12472285.000

LLR 12490513.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall read the engine counter data written to RAM and EEPROM when the following conditions are met, otherwise the logic is not executed and the process is exited:

- DCU_FUNC_SW bit 4 and bit 12 are equal to 0.
- DCUSIF is equal to 0.
- DCU_FLAGS_1 bit 7 transitions to TRUE OR ENGSTART is TRUE OR ENGSHUTDOWN is TRUE

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490513.000	0			CL12472285.000

LLR 12490514.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall pack the DCU Fields identified by "Block 305 Organization" into DCU_BLOCK_305 using DCU_CTRDATA_BUF when the counter data written to RAM and EEPROM are read.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490514.000	0			CL12472285.000

LLR 12490505.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall give a write request to capability DCUWRITE to write DCU_BLOCK_305 to the DCU using DCU_CTRDATA_BUF when the DCU Fields are packed.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490505.000	0			CL12472285.000

LLR 12490506.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall check if QUEUE_IS_FULL is FALSE then DCU_FLAGS_1 bit 7 will transition to FALSE, else DCU_FLAGS_1 will remain TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490506.000	0			CL12472285.000

04.05.01.14 DCU Block Data Status Summary Logic [DCUDSTAT]

Ov 12490515.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic is responsible for updating the DCU data status word.

Ov 12490593.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

- DCU_Component_Data

After the component data block has been read in, this routine processes the data. First it checks if any faults exist. If there is a fault and the fault is due to a new DCU, it clears the fault and signals the data in the DCU for this block be updated. Otherwise, it sets the appropriate fault flags. If no DCU faults exist, a comparison is done between the EEPROM data values and the DCU data values. If a mismatch is detected for any value, it signals a mismatch was found. Also if there are no faults and the DCU is new or reprogrammed or there is a mismatch found in the EEC data or engine data, this routine requests an update of the EEPROM data for this block. A check is also performed to ensure the engine data and component data blocks are simultaneously present or not. If not, an inconsistency fault is set.

- DCU_Counter_Data

After the counter data block has been read in, this routine processes the data. First it checks if any faults exist. If there is a fault and the fault is due to a new DCU, it clears the fault and, if the EEPROM data is valid, it signals the data in the DCU for this block be updated. Otherwise, it sets the appropriate fault flags. If no DCU faults exist, it performs a range check on the data and if any value is out of range a fault flag is set. If no DCU faults exist and the data in the block has been determined to be obsolete, it adjusts the counter data and requests the updates be stored in EEPROM as well as signals that the data in the DCU for this block be updated. If there are no data errors and the DCU is new or reprogrammed or there is a mismatch found in the EEC data, this routine requests an update of the EEPROM data for this block.

- DCU_DCUID_Data

After the DCU ID data block has been read in, this routine validates the data. First it checks if any DCU faults exist. If there is a fault, it sets the appropriate fault flags. If no DCU faults exist, it performs a checksum on the data. If the checksum fails, a fault flag is set. Otherwise, it proceeds to perform range checks on the data. If the range checks fail, a fault flag is set. If there are no range faults, it performs comparison checks on some of the data with expected values. If the comparison checks fail, a fault is set. If all the above checks are good, it signals that the DCU data read in up to this point is ready for use by other logic.

- DCU_EEC_Data

This routine processes the EEC data block after it has been read in from the DCU. First it checks for any DCU faults. If there is a fault and the fault is due to a new DCU, it clears the fault and signals a new DCU was detected, otherwise, it sets the appropriate fault flags. If no faults were detected, it checks if the DCU is reprogrammed. If so it signals a reprogrammed DCU has been detected. If the DCU is neither new nor reprogrammed, it performs a comparison of the EEC data to its equivalent in EEPROM. If it determines there is a mismatch, it set the mismatch flag. If the DCU is new or reprogrammed or there is a mismatch found, this routine signals the data in the DCU for this block be updated.

Ov 12490594.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

- DCU_Engine_Data

After the engine data block has been read in, this routine processes the data. First it checks if any faults exist. If there is a fault and the fault is due to a new DCU, it clears the fault and signals the data in the DCU for this block be updated. Otherwise, it sets the appropriate fault flags. If no DCU faults exist, a comparison is done between the EEPROM data values and the DCU data values. If a mismatch is detected for any value, it signals a mismatch was found. If there are no faults and the DCU is new or reprogrammed or there is a mismatch found in the EEC data or engine data, this routine requests an update of the EEPROM data for this block.

- DCU_LCF_Data

Before a block of LCF data is read from the DCU, this routine sets up information needed to read the block. After the block has been read, this routine verifies the data. First it checks for DCU faults and if there are any, it set the appropriate fault flags. If there are no DCU faults, it performs a checksum on the LCF data. If the checksum fails, it sets a checksum fault. If the checksum is good, it then proceeds to find a matching part number for the component that the LCF data pertains to. If it does not find a match, it sets up information to read the next block for that component. If no match is found for a particular component among the blocks pertaining to the component, a fault is set for that component. If it does find a match, it transfers the LCF data to a designated buffer area for that component. It then performs a range check on the data. If the range check fails it sets a fault. After the data has been verified, it sets up information to read data for the next component.

- DCU_Schedule

This routine schedules all the DCU routines including the routine to request block reads from the DCU. It does this via a state machine. In the read state it traverses through a table that specifies the order of DCU blocks to be read; in the process state it traverses through a table specifying the order of the process routines to run; and in the write state it traverses through a table that specifies the routines that are run to request blocks to be written and processes write faults. This routine also provides synchronization for switching between DCU protocol and the other communication protocols.

Ov 12490526.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

RCD(s): SCR(s):PW127L-EEC-SW-PWC-16236 SCO-D(s): PW127L-EEC-SW-PWC-16242

Ov 12490536.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed designs:

None

Low Level Requirements:

Ov 12490537.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The Low Level Requirements below are related to module DCU_SCHEDULE.

LLR 12490538.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU Status, DCU_STATUS, is NONZERO, DCU Available, DCU_IS_AVAILABLE, is TRUE and DCU Scheduler State, DCU_SCHED_STATE, is equal to WRITE_BLOCKS:

- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bits 6-15 equal to the data at DCU Block Pointer DCU_BLOCK_PTR.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490538.000	0			CL12472304.000 CL12472305.000

Ov 12490539.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: DCU_BLOCK_PTR is either bit 10 (block 0 write fault), bit 11 (block 1 write fault), bit 12 (block 305 write fault) or bit 13 (block 9 write fault).

Ov 12490540.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The Low Level Requirements below are related to module DCU_LCF_DATA.

LLR 12490541.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set SP3 equal to the data at the address of the LCF Read Faults LCF_READ_FAULTS plus CONFIG when DCU STATUS, DCU_STATUS, is NONZERO and DCU LCF Summary word, DCU_LCFFACT_SW, is greater than ZERO and DCU Reading LCF Flag, DCU_LCF_READING, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490541.000	0			CL12472303.000 CL12472305.000

Ov 12490542.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: This is '200 (bit 8 TRUE).

LLR 12490516.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU STATUS , DCU_STATUS, is NONZERO and DCU LCF Summary word, DCU_LCFFACT_SW, is greater than ZERO and DCU Reading LCF Flag, DCU_LCF_READING, is TRUE:

- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 8 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490516.000	0			CL12472303.000 CL12472305.000

LLR 12490517.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU STATUS , DCU_STATUS, is NONZERO and DCU LCF Summary word, DCU_LCFFACT_SW, is greater than ZERO, DCU Reading LCF Flag, DCU_LCF_READING, is FALSE and DCU Hold Table DCU_HOLD_TABLE is FALSE:

- set DCU Status Word, DCU_FUNC_SW, bit 16 to FALSE.
- set DCU Status Word, DCU_FUNC_SW, bit 8 to FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490517.000	0			CL12472303.000 CL12472305.000

Ov 12490518.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The Low Level Requirements below are related to module DCU_ENGINE_DATA.

LLR 12490519.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU STATUS, DCU_STATUS, is NONZERO and DCU Update Flag 2, DCU_FLAGS_2 bit 1 is FALSE:

- set DCU Status Word, DCU_FUNC_SW, bit 1 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490519.000	0			CL12472298.000 CL12472305.000

Ov 12490520.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The Low Level Requirements below are related to module DCU_EEC_DATA.

LLR 12490521.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU Status Flag, DCUSIF, is NONZERO:

- set DCU Status Word, DCU_FUNC_SW, bit 5 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490521.000	0			CL12472298.000 CL12472305.000

Ov 12490522.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The Low Level Requirements below are related to module DCU_DCUID_DATA.

LLR 12490523.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU Status Flag, DCU_STATUS, is ZERO and DCU_DCUID_IF is NONZERO:

- set DCU Status Word, DCU_FUNC_SW, bit 3 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490523.000	0			CL12472301.000 CL12472305.000

LLR 12490524.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU Status Flag, DCU_STATUS, is NONZERO:

- set DCU Status Word, DCU_FUNC_SW, bit 3 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490524.000	0			CL12472301.000 CL12472305.000

Ov 12490525.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The Low Level Requirements below are related to module DCU_COUNTER_DATA.

LLR 12490527.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set A to TRUE when any bits excluding 13-15 of the Counter Data Status word, DCU_CTRDAT_IF, are TRUE. Otherwise A is FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490527.000	0			CL12472302.000 CL12472305.000

LLR 12490528.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, bits 4 and 16 to TRUE when DCU FLAGS 2, DCU_FLAGS_2, bit 3 is FALSE, DCU Flags 1, DCU_FLAGS_1, bit 2 is FALSE, DCU Status Flag, DCU_STATUS, is ZERO and A is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490528.000	0			CL12472302.000 CL12472305.000

LLR 12490529.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, bits 4 and 16 to TRUE when DCU Status Flag, DCU_STATUS, bit 9 is FALSE, DCU FLAGS 2, DCU_FLAGS_2, bit 3 is FALSE DCU Status Flag, DCU_STATUS, is NONZERO.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490529.000	0			CL12472302.000 CL12472305.000

LLR 12490530.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, bits 15 and 16 to TRUE when DCU Status Flag, DCU_STATUS, bit 9 is FALSE, DCU FLAGS 2, DCU_FLAGS_2, bit 3 is FALSE and either of the following conditions are met:

- DCU Flags 1, DCU_FLAGS_1, bit 2 is TRUE.
- DCU Status Flag, DCU_STATUS, is NONZERO.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490530.000	0			CL12472300.000 CL12472305.000

Ov 12490531.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Note: The Low Level Requirements below are related to module DCU_COMPONENT_DATA.

LLR 12490532.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU Status Flag, DCU_STATUS, is ZERO, DCU Flags 2, DCU_FLAGS_2, bit 2 is FALSE and DCU Flags 1, DCU_FLAGS_1, bit 5 is TRUE:

- set DCU Status Word, DCU_FUNC_SW, bit 9 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490532.000	0			CL12472299.000 CL12472305.000

LLR 12490533.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU Status Flag, DCU_STATUS, is NONZERO, DCU Flags 2, DCU_FLAGS_2, bit 2 is FALSE, DCU Flags 1, DCU_FLAGS_1, bit 5 is TRUE and DCU Flags 1, DCU_FLAGS_1, bit 9 is TRUE:

- set DCU Status Word, DCU_FUNC_SW, bit 9 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490533.000	0			CL12472299.000 CL12472305.000

LLR 12490534.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU Status Flag, DCU_STATUS, is NONZERO, DCU Flags 2, DCU_FLAGS_2, bit 2 is FALSE, DCU Flags 1, DCU_FLAGS_1, bit 5 is TRUE and DCU Flags 1, DCU_FLAGS_1, bit 9 is FALSE:

- set DCU Status Word, DCU_FUNC_SW, bit 9 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490534.000	0			CL12472299.000 CL12472305.000

LLR 12490535.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set the following bits in DCU Status Word, DCU_FUNC_SW, when DCU Status Flag, DCU_STATUS, is NONZERO, and DCU Flags 2, DCU_FLAGS_2, bit 2 is FALSE:

- set DCU Status Word, DCU_FUNC_SW, bit 2 to TRUE.
- set DCU Status Word, DCU_FUNC_SW, bit 16 to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490535.000	0			CL12472298.000 CL12472305.000

04.05.01.15 EEPROM Counters Configuration Logic [EECOUNTERS]

Ov 12490543.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic queues the DCU counter data to be updated in EEPROM.

- Update_EE_Data

This routine makes a request to update the counter data in EEPROM and signals the counter data in the DCU should be updated. If the counter data is ready (i.e. has been read from the DCU) and the engine has just started, or if the engine has just shutdown, or if an LCF data adjustment is required due to resynch, then it requests the counter data be updated in EEPROM. After the EEPROM value has been updated, it signals that the DCU counter data be updated.

Ov 12490553.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Designs unchanged from prior baseline:

None

Changed designs:

None

Low Level Requirements:

LLR 12490554.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall execute this process when DCU Data Ready Status, DCU_DATA_READY bit 1, is equal to TRUE. Otherwise the process is exited.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490554.000	0			CL12472317.000 CL12472318.004 CL12472322.000 CL12472323.000

LLR 12490555.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 1, to TRUE when Engine Startup, ENGSTART, is equal to TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490555.000	0			CL12472317.000

LLR 12490556.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall call the EE_PUT_QUEUE subroutine with Block 305 Counter EE and RAM data, EE_CTRDAT_BLOCK, and Queue full Flag, SP3, to queue a write of the Counter data to EEPROM when EEPROM Update Flags, EE_UPDATE_FLAGS bit 1, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490556.000	0			CL12472317.000 CL12472318.004

Ov 12490557.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

NOTE: EE_CTRDAT_BLOCK is defined in BLK305CHK capability inside the module DCU_CONSTANTS. It holds the Start and End of the RAM Counter Data locations, the start of the EEPROM Counter data and the write to EEPROM complete Flag, DCU_CTRDAT_EEREQ.

LLR 12490558.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 1, to FALSE and Bit 2 to TRUE when Queue Full Flag, SP3, is FALSE and EEPROM Update Flags, EE_UPDATE_FLAGS bit 1, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490558.000	0			CL12472317.000

LLR 12490559.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 2, to FALSE and set Startup Counter Written to EEPROM Flag, STARTUP_WRITTEN, to TRUE when Completed EEPROM write Flag, DCU_CTRDAT_EEREQ, is TRUE and EEPROM Update Flags, EE_UPDATE_FLAGS bit 2, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490559.000	0			CL12472317.000

LLR 12490544.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 3, to TRUE when Engine Shutdown, ENGSHTUTDOWN, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490544.000	0			CL12472322.000

LLR 12490545.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall call the EE_PUT_QUEUE subroutine with Block 305 Counter EE and RAM data, EE_CTRDAT_BLOCK, and Queue full Flag, SP3, to queue a write of the Counter data to EEPROM when EEPROM Update Flags, EE_UPDATE_FLAGS bit 3, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490545.000	0			CL12472318.004 CL12472322.000

LLR 12490546.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 3, to FALSE and Bit 4 to TRUE when Queue full flag, SP3, is FALSE and EEPROM Update Flags, EE_UPDATE_FLAGS bit 3, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490546.000	0			CL12472322.000

LLR 12490547.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 4, to FALSE and set Shutdown Counter Written to EEPROM Flag, SHUTDOWN_WRITTEN, to TRUE when Completed EEPROM write Flag, DCU_CTRDAT_EEREQ, is TRUE and EEPROM Update Flags, EE_UPDATE_FLAGS bit 4, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490547.000	0			CL12472322.000

LLR 12490548.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 5, to TRUE when Block 305 DCU and EEPROM requires update flag, DCU_EECTRDAT_UPD, is TRUE and LCF counts require update flag, DCU_LCF_UPD, is FALSE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490548.000	0			CL12472323.000

LLR 12490549.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall call the EE_PUT_QUEUE subroutine with Block 305 Counter EE and RAM data, EE_CTRDAT_BLOCK, and Queue full Flag, SP3, to queue a write of the Counter data to EEPROM when EEPROM Update Flags, EE_UPDATE_FLAGS bit 5, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490549.000	0			CL12472318.004 CL12472323.000

LLR 12490550.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 5, to FALSE and Bit 6 to TRUE when Queue full flag, SP3, is FALSE and EEPROM Update Flags, EE_UPDATE_FLAGS bit 5, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490550.000	0			CL12472323.000

LLR 12490551.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall set EEPROM Update Flags, EE_UPDATE_FLAGS bit 6, to FALSE and set Resync Counter Written to EEPROM Flag, RESYNC_WRITTEN, to TRUE when Completed EEPROM write Flag, DCU_CTRDAT_EEREQ, is TRUE and EEPROM Update Flags, EE_UPDATE_FLAGS bit 6, is TRUE.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490551.000	0			CL12472323.000

04.05.02 LCF Capabilities

Cumulative engine cycle counts shall be calculated for each of 9 applicable LCF engine components. Engine usage historic data shall be maintained within DCU Block 305. Requirements are defined to perform engine start and flight cycle determination in support of the accumulated cycle counting function. Mission profiles are defined and it is determined which mission shall be selected with respect to cycle usage for each configuration. The EEC Fault EEPROM mapping for the DCU Block 0, DCU Block 1, DCU Block 305, and the location of the RTD addresses and the UART Protocol Selection Flag are defined.

04.05.02.01 LCF Cycle Counting Logic [CYCLE_COUNTING]

Cumulative engine cycle counts will be calculated for each of 9 applicable LCF engine components. Accumulated cycles are updated by the EEC at the end of each flight cycle. Final cycle factors for each engine component are determined based on the mission level achieved during the flight cycle. LCF counting is disabled if there is a DCU failure.

This capability is performed by the following processes:

- LCF_Check

This routine determines if it is time to update the LCF cycle counts and, if so, it does for each engine component. A special one-time LCF adjustment can occur due to a resynch of the DCU startup/shutdown counters at power-up.

- LCF_Cycle_Count

This routine calculates the LCF cycle counts for a given component. The calculation is based on the mission number and the LCF factors read in from the DCU for the component. After doing the calculation, the coarse and fine cycle count values are updated with the new cycle count value.

- LCF_Table

This module contains tables for the LCF calculations.

04.05.02.02 Engine Usage Historic Data Logic [EVENT_COUNTING]

Ov 12490560.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

Engine usage historic data shall be maintained within DCU Block 305. This data shall consist of event frequency and time counters that relate to aircraft and engine operation, including operation of the engine in a specific configuration.

This capability is performed by the following processes:

- Runtime

This routine keeps track of the engine run time in hours and fractions of hours.

- Flttimer

This routine keeps track of the flight time in hours and fractions of hours.

- Inc_Ctrs_1

This routine increments the following counters: Flight cycle, engine start, in-flight start, start followed by flight, and shutdown. It updates the individual counters based on configuration as well as the total counters for all configurations.

- Inc_Ctrs_2

This routine increments the engine run time counters. First it checks if the LCF data from the DCU is ready. If so, it checks if the engine is running. If so, it proceeds to increment the individual engine run time counters based on configuration and the total engine run counter.

- Inc_Ctrs_3

This routine increments the flight time counters. First it checks if the LCF data from the DCU is ready. If so, it checks if the engine is running and the aircraft is in flight. If so, it proceeds to increment the individual flight time counters based on configuration and the total flight time counter.

- Inc_Ctrs_4

This routine increments the take-off counters. First it checks if the LCF data from the DCU is ready. If so, it proceeds to increment the appropriate take-off counter depending on current engine rating.

- Cnt_Evnt

This routine calculates the total accumulated time in coarse and fine counts.

LLR 12490564.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module INC_CTRS_1 shall be implemented as per the following design models:

- INC_CTRS_1_A.CPGC
- INC_CTRS_1_B.CPGC
- INC_CTRS_1_C.CPGC
- INC_CTRS_1_D.CPGC

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490564.001	0			CL12472555.000 CL12472558.000 CL12472561.000 CL12472564.000 CL12472573.000

LLR 12490565.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module INC_CTRS_4 shall be implemented per design model INC_CTRS_4.CTRLFLOW.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490565.000	0			CL12472575.001 CL12472576.001 CL12472577.001 CL12472578.001 CL12472579.001 CL12472580.001

LLR 12490566.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module INC_CTRS_2 shall be implemented per design model INC_CTRS_2.CPGC.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490566.001			N/A	CL12472553.000 CL12472567.000

LLR 12490567.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The module INC_CTRS_3 shall be implemented per design model INC_CTRS_3.CPGC.

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490567.001	0			CL12472553.000 CL12472570.000

04.05.02.03 Start / Flight Determination Logic [STRT_FLIGHT_DETERM]

This logic defines requirements to perform engine start, flight cycle, and takeoff determination in support of the accumulated cycle counting function. The deltaP sensor can read a higher signal at low airspeeds such that, statically, the airspeed can read above the Flight/Ground Trigger point. To accommodate this, two separate airspeed triggers for Ground/Flight determination are defined as a function of the currently selected airspeed source. If no valid airspeed source is available, then gas generator speed will be used to determine Ground/Flight status. DCU Block 304 shall provide the constants for NH to determine Engine OFF and Engine RUN as well as providing airspeed constants for Ground/Flight determination.

This capability is performed by the following processes:

- Opmode_1

This routine determines whether the engine is running and if the aircraft is in flight.

- Opmode_2

This routine determines the aircraft flight cycle.

- Opmode_3

This routine determines whether a take-off has occurred.

LLR 12490676.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module OPMODE_1 shall be implemented per design model OPMODE_1.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490676.000	0			CL12472592.000

LLR 12490677.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module OPMODE_2 shall be implemented per design model OPMODE_2.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490677.000	0			CL12472592.000

LLR 12490675.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The module OPMODE_3 shall be implemented per design model OPMODE_3.DATAFLOW.

ALLOCATION: PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490675.000	0			CL12472843.000 CL12472853.000 CL12472865.000

04.05.02.04 Mission Determination Logic [MISSION_DETERM]

Ov 12490568.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

This logic defines the mission profiles and determines which mission will be selected with respect to cycle usage. These are basic (mission = 1, 2, and 3) and advanced missions (mission = 4, 5, and 6).

The PW127XT-S configuration has six defined missions.

- [Mission] = 0: No flight mission selected
- [Mission] = 1: Basic flight mission with [Derating Level] = 3
- [Mission] = 2: Basic flight mission with [Derating Level] = 2
- [Mission] = 3: Basic flight mission with [Derating Level] = 1
- [Mission] = 4: Advanced flight mission with [Derating Level] = 3
- [Mission] = 5: Advanced flight mission with [Derating Level] = 2
- [Mission] = 6: Advanced flight mission with [Derating Level] = 1

The initial mission level is 0. Once a higher level mission is achieved, it will be retained for the remainder of that flight cycle.

LLR 12490571.000

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: 5400

The module MISSIONS shall be implemented per design model MISSIONS.DATAFLOW

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490571.000	Func	HLR	No	CL12472607.001 CL12472608.000 CL12472609.001 CL12472614.001

04.05.03 EEC Fault EEPROM Mapping [EEPROM_MAP]

Ov 12490573.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

This logic defines the EEC Fault EEPROM mapping for the DCU Block 0, DCU Block 1, DCU Block 305, and the location of the RTD addresses and the UART Protocol Selection Flag.

LLR 12490584.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall map the EEC Fault EEPROM MEMORY of DCU Block 0 Information as per TABLE - EEC FAULT MEMORY OF DCU BLOCK 0.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490584.000	0			CL12472623.000

Table:(LLR 12490584.000-1) TABLE - EEC FAULT MEMORY OF DCU BLOCK 0

Fault EEPROM Address (Octal)	Function (parameter name)	Entry Description
23000	ENGSA12EE	Engine S/N (1st pair of ASCII characters - most significant)
23002	ENGSA34EE	Engine S/N (2nd pair of ASCII characters)
23004	ENGSA56EE	Engine S/N (3rd pair of ASCII characters - least significant)
23006	TRBMSA12EE	TM S/N (1st pair of ASCII characters - most significant)
23010	TRBMSA34EE	TM S/N (2nd pair of ASCII characters)
23012	TRBMSA56EE	TM S/N (3rd pair of ASCII characters - least significant)
23014	RGBSA12EE	RGB S/N (1st pair of ASCII characters - most significant)
23016	RGBSA34EE	RGB S/N (2nd pair of ASCII characters)
23020	RGBSA56EE	RGB S/N (3rd pair of ASCII characters - least significant)
23022	Spare (reserved for Torque Bias Trim)	Reserved for Torque Shaft Bias Trim Input
23024	Spare (reserved Torque Gain Trim)	Reserved for Torque Shaft Gain Trim Input
23026	Spare (reserved for ITT Bias Trim)	Reserved for ITT Trim Input
23030 - 23177	Spare (reserved)	

LLR 12490586.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall map the EEPROM as per "TABLE 1 : EEC Fault EEPROM Memory Map of DCU Block 1".

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490586.001	Func		N/A	CL12472625.001

Table:(LLR 12490586.001-1) TABLE 1 : EEC Fault EEPROM Memory Map of DCU Block 1

Fault EEPROM Address (Octal)	Function (parameter name)	Entry Description
23200	LPIMP_PNA12EE	LP Impeller Part Number (1st pair of ASCII characters - most significant)
23202	LPIMP_PNA34EE	LP Impeller Part Number (2nd pair of ASCII characters)
23204	LPIMP_PNA56EE	LP Impeller Part Number (3rd pair of ASCII characters)
23206	LPIMP_PNA78EE	LP Impeller Part Number (4th pair of ASCII characters - least significant)
23210	LPIMP_SNA12EE	LP Impeller S/N (1st pair of ASCII characters - most significant)
23212	LPIMP_SNA34EE	LP Impeller S/N (2nd pair of ASCII characters)
23214	LPIMP_SNA56EE	LP Impeller S/N (3rd pair of ASCII characters - least significant)
23216	HPIMP_PNA12EE	HP Impeller Part Number (1st pair of ASCII characters - most significant)
23220	HPIMP_PNA34EE	HP Impeller Part Number (2nd pair of ASCII characters)
23222	HPIMP_PNA56EE	HP Impeller Part Number (3rd pair of ASCII characters)
23224	HPIMP_PNA78EE	HP Impeller Part Number (4th pair of ASCII characters - least significant)
23226	HPIMP_SNA12EE	HP Impeller S/N (1st pair of ASCII characters - most significant)
23230	HPIMP_SNA34EE	HP Impeller S/N (2nd pair of ASCII characters)
23232	HPIMP_SNA56EE	HP Impeller S/N (3rd pair of ASCII characters - least significant)
23234	HPTFCP_PNA12EE	HPT Front Cover Plate Part Number (1st pair of ASCII characters - most significant)
23236	HPTFCP_PNA34EE	HPT Front Cover Plate Part Number (2nd pair of ASCII characters)
23240	HPTFCP_PNA56EE	HPT Front Cover Plate Part Number (3rd pair of ASCII characters)
23242	HPTFCP_PNA78EE	HPT Front Cover Plate Part Number (4th pair of ASCII characters - least significant)
23244	HPTFCP_SNA12EE	HPT Front Cover Plate S/N (1st pair of ASCII characters - most significant)
23246	HPTFCP_SNA34EE	HPT Front Cover Plate S/N (2nd pair of ASCII characters)
23250	HPTFCP_SNA56EE	HPT Front Cover Plate S/N (3rd pair of ASCII characters - least significant)
23252	HPTD_PNA12EE	HPT Disk Part Number (1st pair of ASCII characters - most significant)
23254	HPTD_PNA34EE	HPT Disk Part Number (2nd pair of ASCII characters)
23256	HPTD_PNA56EE	HPT Disk Part Number (3rd pair of ASCII characters)
23260	HPTD_PNA78EE	HPT Disk Part Number (4th pair of ASCII characters - least significant)
23262	HPTD_SNA12EE	HPT Disk S/N (1st pair of ASCII characters - most significant)
23264	HPTD_SNA34EE	HPT Disk S/N (2nd pair of ASCII characters)
23266	HPTD_SNA56EE	HPT Disk S/N (3rd pair of ASCII characters - least significant)
23270	HPTRCP_PNA12EE	HPT Rear Cover Plate Part Number (1st pair of ASCII characters - most significant)
23272	HPTRCP_PNA34EE	HPT Rear Cover Plate Part Number (2nd pair of ASCII characters)
23274	HPTRCP_PNA56EE	HPT Rear Cover Plate Part Number (3rd pair of ASCII characters)
23276	HPTRCP_PNA78EE	HPT Rear Cover Plate Part Number (4th pair of ASCII characters - least significant)
23300	HPTRCP_SNA12EE	HPT Rear Cover Plate S/N (1st pair of ASCII characters - most significant)
23302	HPTRCP_SNA34EE	HPT Rear Cover Plate S/N (2nd pair of ASCII characters)
23304	HPTRCP_SNA56EE	HPT Rear Cover Plate S/N (3rd pair of ASCII characters - least significant)
23306	ISS_PNA12EE	Interstage Seal Part Number (1st pair of ASCII characters - most significant)
23310	ISS_PNA34EE	Interstage Seal Part Number (2nd pair of ASCII characters)
23312	ISS_PNA56EE	Interstage Seal Part Number (3rd pair of ASCII characters)
23314	ISS_PNA78EE	Interstage Seal Part Number (4th pair of ASCII characters - least significant)
23316	ISS_SNA12EE	Interstage Seal S/N (1st pair of ASCII characters - most significant)
23320	ISS_SNA34EE	Interstage Seal S/N (2nd pair of ASCII characters)
23322	ISS_SNA56EE	Interstage Seal S/N (3rd pair of ASCII characters - least significant)
23324	LPTD_PNA12EE	LPT Disk Part Number (1st pair of ASCII characters - most significant)
23326	LPTD_PNA34EE	LPT Disk Part Number (2nd pair of ASCII characters)
23330	LPTD_PNA56EE	LPT Disk Part Number (3rd pair of ASCII characters)
23332	LPTD_PNA78EE	LPT Disk Part Number (4th pair of ASCII characters - least significant)
23334	LPTD_SNA12EE	LPT Disk S/N (1st pair of ASCII characters - most significant)
23336	LPTD_SNA34EE	LPT Disk S/N (2nd pair of ASCII characters)
23340	LPTD_SNA56EE	LPT Disk S/N (3rd pair of ASCII characters - least significant)
23342	PT1D_PNA12EE	PT1 Disk Part Number (1st pair of ASCII characters - most significant)
23344	PT1D_PNA34EE	PT1 Disk Part Number (2nd pair of ASCII characters)
23346	PT1D_PNA56EE	PT1 Disk Part Number (3rd pair of ASCII characters)
23350	PT1D_PNA78EE	PT1 Disk Part Number (4th pair of ASCII characters - least significant)
23352	PT1D_SNA12EE	PT1 Disk S/N (1st pair of ASCII characters - most significant)
23354	PT1D_SNA34EE	PT1 Disk S/N (2nd pair of ASCII characters)
23356	PT1D_SNA56EE	PT1 Disk S/N (3rd pair of ASCII characters - least significant)
23360	PT2D_PNA12EE	PT2 Disk Part Number (1st pair of ASCII characters - most significant)
23362	PT2D_PNA34EE	PT2 Disk Part Number (2nd pair of ASCII characters)
23364	PT2D_PNA56EE	PT2 Disk Part Number (3rd pair of ASCII characters)
23366	PT2D_PNA78EE	PT2 Disk Part Number (4th pair of ASCII characters - least significant)
23370	PT2D_SNA12EE	PT2 Disk S/N (1st pair of ASCII characters - most significant)
23372	PT2D_SNA34EE	PT2 Disk S/N (2nd pair of ASCII characters)
23374	PT2D_SNA56EE	PT2 Disk S/N (3rd pair of ASCII characters - least significant)
23376	Spare (reserved)	

LLR 12490574.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall map EEPROM as per "TABLE 2 : EEC Fault EEPROM Memory Map of EUHD Block 2"

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490574.001	Func		N/A	CL12472627.001

Table:(LLR 12490574.001-1) TABLE 2 : EEC Fault EEPROM Memory Map of EUHD Block 2

Fault EEPROM Address (Octal)	Function (parameter name)	Entry Description
23400	E_LPIMP_CYCCRS	LP Impeller Accumulated Cycles Coarse Count
23402	E_LPIMP_CYCFIN	LP Impeller Accumulated Cycles Fine Count
23404	E_HPIMP_CYCCRS	HP Impeller Accumulated Cycles Coarse Count
23406	E_HPIMP_CYCFIN	HP Impeller Accumulated Cycles Fine Count
23410	E_HPTFCP_CYCCRS	HPT Front Cover Plate Accumulated Cycles Coarse Count
23412	E_HPTFCP_CYCFIN	HPT Front Cover Plate Accumulated Cycles Fine Count
23414	E_HPTD_CYCCRS	HPT Disk Accumulated Cycles Coarse Count
23416	E_HPTD_CYCFIN	HPT Disk Accumulated Cycles Fine Count
23420	E_HPTRCP_CYCCRS	HPT Rear Cover Plate Accumulated Cycles Coarse Count
23422	E_HPTRCP_CYCFIN	HPT Rear Cover Plate Accumulated Cycles Fine Count
23424	E_ISS_CYCCRS	Inter-stage Seal Accumulated Cycles Coarse Count
23426	E_ISS_CYCFIN	Inter-stage Seal Accumulated Cycles Fine Count
23430	E_LPTD_CYCCRS	LPT Disk Accumulated Cycles Coarse Count
23432	E_LPTD_CYCFIN	LPT Disk Accumulated Cycles Fine Count
23434	E_PT1D_CYCCRS	PT1 Disk Accumulated Cycles Coarse Count
23436	E_PT1D_CYCFIN	PT1 Disk Accumulated Cycles Fine Count
23440	E_PT2D_CYCCRS	PT2 Disk Accumulated Cycles Coarse Count
23442	E_PT2D_CYCFIN	PT2 Disk Accumulated Cycles Fine Count
23444	E_ACCSTRTS_ALL	Accumulated Starts
23446	E_ACCINFLSTRT_AL	Accumulated Inflight Starts
23450	E_ACCNUMFL_ALL	Accumulated Number of Flights
23452	Spare (reserved)	Spare (reserved)
23454	Spare (reserved)	Spare (reserved)
23456	Spare (reserved)	Spare (reserved)
23460	Spare (reserved)	Spare (reserved)
23462	Spare (reserved)	Spare (reserved)
23464	Spare (reserved)	Spare (reserved)
23466	Spare (reserved)	Spare (reserved)
23470	Spare (reserved)	Spare (reserved)
23472	Spare (reserved)	Spare (reserved)
23474	E_ENG_RUN_CR5_ALL	Total Accumulated Engine Run Time Coarse Count
23476	E_ENG_RUN_FIN_ALL	Total Accumulated Engine Run Time Fine Count
23500	Spare (reserved)	Spare (reserved)
23502	Spare (reserved)	Spare (reserved)
23504	Spare (reserved)	Spare (reserved)
23506	Spare (reserved)	Spare (reserved)
23510	Spare (reserved)	Spare (reserved)
23512	Spare (reserved)	Spare (reserved)
23514	E_ENGFLTM_CR5_AL	Total Accumulated Flight Time Coarse Count
23516	E_ENGFLTM_FIN_AL	Total Accumulated Flight Time Fine Count
23520	Spare (reserved)	Spare (reserved)
23522	Spare (reserved)	Spare (reserved)
23524	Spare (reserved)	Spare (reserved)
23526	Spare (reserved)	Spare (reserved)
23530	Spare (reserved)	Spare (reserved)
23532	Spare (reserved)	Spare (reserved)
23534	E_ACCSHUTD_ALL	Accumulated Shut Downs Counter
23536	E_ACCSTRTSFLT_AL	Accumulated Starts followed by a flight
23540	Spare (reserved)	Spare (reserved)
23542	Spare (reserved)	Spare (reserved)
23544	Spare (reserved)	Spare (reserved)
23546	E_ACCSTR_SD_RESYNC	No of Acc. Starts/Shutdown Resynch Counter
23550	E_SV_SW_EPI	EEC Application Software Engine Program Identification
23552	E_SV_SW_SBI	EEC Application Software Baseline Identification
23554	E_SV_SW_SVI	EEC Application Software Sub-version Identification
23556	E_SV_PWC_PATCHID	EEC Application Software PWC Patch Identification
23560	E_NTO_DL3	Accumulated Normal Takeoffs at DERATE Power Rating
23562	E_NTO_DL2	Accumulated Normal Takeoffs at NORM Power Rating
23564	E_NTO_DL1	Accumulated Normal Takeoffs at UPRATE Power Rating
23566	E_MTO_DL3	Accumulated Maximum Takeoffs at DERATE Power Rating
23570	E_MTO_DL2	Accumulated Maximum Takeoffs at NORM Power Rating
23572	E_MTO_DL1	Accumulated Maximum Takeoffs at UPRATE Power Rating
23574	Spare (reserved)	Spare (reserved)

LLR 12490576.001

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR

The design shall map EEPROM as per "TABLE 3 : EEC Fault EEPROM Memory Map of Block 3".

ALLOCATION: Collins IP, PWC IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490576.001	Func		N/A	CL12472629.001

Table:(LLR 12490576.001-1) TABLE 3 : EEC Fault EEPROM Memory Map of Block 3

Fault EEPROM Address (Octal)	Function (parameter name)	Entry Description
23600	RTD Address 1	
23602	RTD Address 2	
23604	RTD Address 3	
23606	RTD Address 4	
23610	RTD Address 5	
23612	RTD Address 6	
23614	RTD Address 7	
23616	RTD Address 8	
23620	RTD Address 9	
23622	RTD Address 10	
23624	RTD Address 11	
23626	RTD Address 12	
23630	RTD Address 13	
23632	RTD Address 14	
23634	RTD Address 15	
23636	RTD Address 16	
23640	RTD Address 17	
23642 - 23733	Spare (reserved)	
23734 - 23735	RTD_CHECK	
23736	ERF	Engine Rating
23737	NPARAM	UART Communication Protocol Mode Byte in EEPROM 0 - DCU Communications Protocol Mode (production) 1 - DAS Communication Protocol Mode 2 - N-Parameter Communication Protocol Mode 3 - DCU / DAS Communication Modes 4 - DCU / N-Parameter Communication Modes >=5 - DCU Communication Protocol Mode
23740	EECPN_SNA12EE	EEC Part and Serial Number (32 bytes)
23742	EECPN_SNA34EE	
23744	EECPN_SNA56EE	
23746	EECPN_SNA78EE	
23750	EECPN_SNA910EE	
23752	EECPN_SNA1112EE	
23754	EECPN_SNA1314EE	
23756	EECPN_SNA1516EE	
23760	EECPN_SNA1718EE	
23762	EECPN_SNA1920EE	
23764	EECPN_SNA2122EE	
23766	EECPN_SNA2324EE	
23770	EECPN_SNA2526EE	
23772	EECPN_SNA2728EE	
23774	EECPN_SNA2930EE	
23776	EECPN_SNA3132EE	

LLR 12490578.000

Export Classification: US: EAR 9E991, Outside US: EAR NSR, Canadian EIPA: NSR

The design shall map the EEC FAULT EEPROM memory of the remaining EEPROM data as per TABLE - EEC FAULT EEPROM MEMORY MAP OF REMAINING EEPROM DATA.

ALLOCATION: Collins IP

HLR Tag	Functional	HLR Type	HLR Safety	SR Tag(s)
LLR 12490578.000	0			CL12472631.000

Table:(LLR 12490578.000-1) TABLE - EEC FAULT EEPROM MEMORY MAP OF REMAINING EPPROM DATA

Fault EEPROM Address (Octal)	Function (parameter name)	Entry Description
20000-20577	Sensor Data	
20600, 20601	Reserved	
20602-20703	Fault Recording Area	
20704-22777	Reserved	

05.00 MEMORY AND PROCESSING TIME ALLOCATION

The memory availability for the Hamilton Sundstrand EEC132-210 Engine Electronic Control is as follows:

- 46K x 16 bits of Write-Protected EEPROM for main program memory
- 2K x 16 bits of RAM
- 2K x 8 bits of EEPROM for data storage

The processing time for the software is based on four minor cycles of 20.80 msec for a total of 83.20 msec.

05.00 Table 1: Memory-and-Processing-Time-Allocation

Sectored Lower Memory			
Start(Octal)	End(Octal)	Size(Pages)	Description
144000	147777	2	Interrupt Vectors/Boot
20000	23777	2	Fault EEPROM
0	3777	2	RAM
4000	17777	6	Application Code
24000	143777	40	Application Code
Note: The page size is equal to 1024 words (1K x 16 bits).			

05.01 MEMORY MARGINS

The EEC system should maintain a margin of at least 15% spare memory in each category: RAM, Program Memory, and non-volatile memory (NVM). The HS1600 processor does not have a stack. A memory map file is created during the software build and is used to gauge the percent usage and percent availability.

The following table contains the memory margins of the software:

05.01 Table 1: Memory Margin

Memory Element	Percent Consumed/Reserved	Total Consumed (words)	Total Available (words)
Program Memory	70.60%	33254	47101
RAM	97.12%	1989	2048
Write-Enabled, Non-Volatile Memory	46.48%	952	2048

05.02 TIMING MARGINS

The software shall have a timing margin target of at least 15% with respect to a major cycle (83.2 msec).

The processing time for the software is based on eighty 1.04 msec interrupts, four minor cycles of 20.8 msecs, and one major cycle of 83.2 msecs. Critical timing analysis is done to ensure the worst-case processing time of each of minor and major cycle is less than the time available.

There are two possible scenarios if the real time interrupt of 1.04 msecs is not maintained. In the first scenario, if one real-time interrupt overruns by a small amount, it will have an insignificant effect on the processing; the timing may be slightly off but there is no system impact. In the second scenario, if several real-time interrupts overrun or one realtime interrupt overruns by a large amount, the watchdog timer would not be updated during the appropriate window and the hardware watchdog timer will trigger and cause a POR.

The following table contains the timing margins of the software for each minor frame:

05.02 Table 1: Timing Margin

	Foreground(usec)	Secondary FG(usec)	Total Max Theoretical(usec)	Spare(usec)	% Spare
MINOR1.SRC	6090	11255.21	17345.2	3454.8	16.60%
MINOR2.SRC	5952	9718.69	15670.7	5129.3	24.70%
MINOR3.SRC	6049.4	11666.17	17715.5	3084.5	14.80%
MINOR4.SRC	6243	10950.9	17193.9	3606.1	17.30%
	24334.4	45591	67925.4	15274.6	18.40%

06.00 CODE TO SDD TO SRS TRACEABILITY

06.01 MODULE TO DESIGN TO SRS CAPABILITY TRACEABILITY (SORTED BY MODULE)

The following table represents the Module to Design Model to SRS Capability Traceability.

Column 1 identifies the Source Code Module.

Column 2 identifies the Design to which the Source Code Module traces. If no Design Model exists, then the Design is the Design section/capability.

Column 3 identifies the SRS Capability to which the Design traces.

Note: A blank entry in this table indicates an item as described above does not exist.

This table is sorted first by column 1, then by column 2, then by column 3, each in ascending order.

06.01 Table 1: MODULE _DESIGN _SRS

Module	Design	SRS Capability
ARCOU.TSRC	ARCOU.TA.CPGC	O22-ARINC_Out_SubRoutine
ARCOU.TSRC	ARCOU.TB.CPGC	O22-ARINC_Out_SubRoutine
C01_START.S00	C01_START.DATAFLOW	C01-Start
C02_PLATIM.T1.S00	C02_PLATIM.T1.DATAFLOW	C02-PLATIM_CALC
C02_PLATIM.T2.SRC	C02_PLATIM.T2.DATAFLOW	C02-PLATIM_CALC
C03_TLMRT.T1.SRC	C03_TLMRT.T1.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T2.T1.SRC	C03_TLMRT.T2.T1.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T2.T2.SRC	C03_TLMRT.T2.T2.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T3.T1.SRC	C03_TLMRT.T3.T1.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T3.T2.SRC	C03_TLMRT.T3.T2.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T4.T1.SRC	C03_TLMRT.T4.T1.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T4.T2.SRC	C03_TLMRT.T4.T2.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T5.T1.SRC	C03_TLMRT.T5.T1.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T5.T2.SRC	C03_TLMRT.T5.T2.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T6.T1.SRC	C03_TLMRT.T6.T1.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T6.T2.SRC	C03_TLMRT.T6.T2.CTRLFLOW	C03 - TLMRT_D
C03_TLMRT.T7.SRC	C03_TLMRT.T7.DATAFLOW	C03 - TLMRT_D
C04_RPCAL.T2.D.SRC	C04_RPCAL.T2.D.DATAFLOW	C04 - PWRCALC_D
C05_TQBUG.SRC	C05_TQBUG.DATAFLOW	C05-Rating_Q_Bug_Calc
C06_HPREF.SRC	C06_HPREF.DATAFLOW	C06-POWER_REQUEST
C06_HPREF.T1.SRC	C06_HPREF.T1.DATAFLOW	C06-POWER_REQUEST
C07_OVRTQPRO.SRC	C07_OVRTQPRO.DATAFLOW	C07-LIMIT_OVERQ_PROTECT
C08_SHPCL.SRC	C08_SHPCL.DATAFLOW	C08-Gov_SHP_Closed_Loop
C10_ATOIGN.SRC	C10_ATOIGN.DATAFLOW	C10-Out_Auto_Ignit
C11_SHPREQ.T1.SRC	C11_SHPREQ.T1.DATAFLOW	C11-Gov_SHP_Req_Calc
C11_SHPREQ.T2.SRC	C11_SHPREQ.T2.DATAFLOW	C11-Gov_SHP_Req_Calc
C12_NPTGOV.T1.SRC	C12_NPTGOV.T1.DATAFLOW	C12-Gov_NPT_Closed_Loop
C12_NPTGOV.T2.SRC	C12_NPTGOV.T2.DATAFLOW	C12-Gov_NPT_Closed_Loop
C13_NHACCELLIM.SRC	C13_NHACCELLIM.DATAFLOW	C13-Trans_Accel_NH
C14_NHLIM.SRC	C14_NHLIM.DATAFLOW	C14-Limit_NH
C16_LOOPSEL.SRC	C16_LOOPSEL.DATAFLOW	C16-Gov_Inner_Loop_Sel
C17_NHREQUEST.T1.SRC	C17_NHREQUEST.T1.DATAFLOW	C17-Gov_Inner_NH_Req_Calc
C17_NHREQUEST.T2.SRC	C17_NHREQUEST.T2.DATAFLOW	C17-Gov_Inner_NH_Req_Calc
C18_NHGOV.T1.SRC	C18_NHGOV.T1.DATAFLOW	C18-Gov_NH_Closed_Loop
C18_NHGOV.T2.SRC	C18_NHGOV.T2.DATAFLOW	C18-Gov_NH_Closed_Loop
C19_UNDESPED.SRC	C19_UNDESPED.DATAFLOW	C19-Limit_Prop_Underspeed
C20_OVRSPEED.SRC	C20_OVRSPEED.DATAFLOW	C20-Limit_Prop_Overspeed
C21_HBOVEM.T1.SRC	C21_HBOVEM.T1.DATAFLOW	C21-OUT_HBOV_EEC_MODE
C21_HBOVEM.T2.SRC	C21_HBOVEM.T2.DATAFLOW	C21-OUT_HBOV_EEC_MODE
C22_HBOVMM.T1.SRC	C22_HBOVMM.T1.DATAFLOW	C22-Out_HBOV_Man_Mode
C22_HBOVMM.T2.SRC	C22_HBOVMM.T2.DATAFLOW	C22-Out_HBOV_Man_Mode
C22_HBOVMM.T3.SRC	C22_HBOVMM.T3.DATAFLOW	C22-Out_HBOV_Man_Mode
C24_OVRTQTST.SRC	C24_OVRTQTST.CTRLFLOW	C24-BIT_PreFly_Eng_OverQ_Test

Module	Design	SRS Capability
C26_MPARAM_1.SRC	C26_MPARAM_1.DATAFLOW	C26-MPARAM
C26_MPARAM_2.SRC	C26_MPARAM_2.DATAFLOW	C26-MPARAM
C26_MPARAM_3.SRC	C26_MPARAM_3.DATAFLOW	C26-MPARAM
C26_MPARAM_4.S00	C26_MPARAM_4.DATAFLOW	C26-MPARAM
C26_MPARAM_5.S00	C26_MPARAM_5.DATAFLOW	C26-MPARAM
C26_MPARAM_6.S00	C26_MPARAM_6.DATAFLOW	C26-MPARAM
C26_MPARAM_7.S00	C26_MPARAM_7.DATAFLOW	C26-MPARAM
C27_NHNMNMX_1.SRC	C27_NHNMNMX_1.DATAFLOW	C27-NHMINMAX
C30_SMRTREQ.SRC	C30_SMRTREQ.DATAFLOW	C30-Gov_Inner_SM_Rate_Req
C31_TQLIM.SRC	C31_TQLIM.DATAFLOW	C31-Limit_Q
CHKSUM.SRC	E07-Exec_Background	E07-Exec_Background
CMPACT.SRC	CMPACT_A.CPGC	E05-RAMINIT
CMPACT.SRC	CMPACT_B.CPGC	E05-RAMINIT
CMPACT.SRC	CMPACT_C.CPGC	E05-RAMINIT
CNT_EVNT.S00	EVENT_COUNTING	EVENT_COUNTING
DBL_LL_SUBR.SRC	DBL_LL_SUBR_A.CPGC	C17-Gov_Inner_NH_Req_Calc
DBL_LL_SUBR.SRC	DBL_LL_SUBR_A.CPGC	C18-Gov_NH_Closed_Loop
DBL_LL_SUBR.SRC	DBL_LL_SUBR_B.CPGC	C17-Gov_Inner_NH_Req_Calc
DBL_LL_SUBR.SRC	DBL_LL_SUBR_B.CPGC	C18-Gov_NH_Closed_Loop
DBL_LL_SUBR.SRC	DBL_LL_SUBR_C.CPGC	C17-Gov_Inner_NH_Req_Calc
DBL_LL_SUBR.SRC	DBL_LL_SUBR_C.CPGC	C18-Gov_NH_Closed_Loop
DBL_LL_SUBR.SRC	DBL_LL_SUBR_D.CPGC	C17-Gov_Inner_NH_Req_Calc
DBL_LL_SUBR.SRC	DBL_LL_SUBR_D.CPGC	C18-Gov_NH_Closed_Loop
DBL_LL_SUBR.SRC	DBL_LL_SUBR_E.CPGC	C17-Gov_Inner_NH_Req_Calc
DBL_LL_SUBR.SRC	DBL_LL_SUBR_E.CPGC	C18-Gov_NH_Closed_Loop
DCU_COMPARE_DATA.S00	BLK00CHK	BLK00CHK
DCU_COMPARE_DATA.S00	BLK01CHK	BLK01CHK
DCU_COMPARE_DATA.S00		BLK304CHK
DCU_COMPARE_ITEM.S00	BLK00CHK	BLK00CHK
DCU_COMPARE_ITEM.S00	BLK01CHK	BLK01CHK
DCU_COMPARE_ITEM.S00	BLK09CHK	BLK09CHK
DCU_COMPARE_ITEM.S00	LCF_FACTCHK	LCF_FACTCHK
DCU_COMPARE_ITEM.S00		BLK304CHK
DCU_COMPONT_DATA.S00	BLK01CHK	BLK01CHK
DCU_COMPONT_DATA.S00		DCUDSTAT
DCU_CONSTANTS.S00	BLK00CHK	BLK00CHK
DCU_CONSTANTS.S00	BLK01CHK	BLK01CHK
DCU_CONSTANTS.S00	BLK09CHK	BLK09CHK
DCU_CONSTANTS.S00	DCU_CP_CFG	DCU_CP_CFG
DCU_CONSTANTS.S00		DCUREAD
DCU_CONSTANTS.S00		DCUWRITE
DCU_CONSTANTS.S00		BLK305CHK
DCU_CONSTANTS.S00		BLK304CHK

Module	Design	SRS Capability
DCU_CONSTANTS.S00		LCF_FACTCHK
DCU_CONSTANTS.S00		DCU_ENG_CFG
DCU_CONSTANTS.S00		DCU_EEC_CFG
DCU_CONSTANTS.S00		DCU_ENGUSE_CFG
DCU_COUNTER_DATA.S00		E30 - IDENTIFI
DCU_COUNTER_DATA.S00		BLK305CHK
DCU_COUNTER_DATA.S00		DCUDSTAT
DCU_DCUID_DATA.S00		BLK304CHK
DCU_DCUID_DATA.S00		DCUDSTAT
DCU_EEC_DATA.S00	BLK09CHK	BLK09CHK
DCU_EEC_DATA.S00		DCUDSTAT
DCU_ENGINE_DATA.S00	BLK00CHK	BLK00CHK
DCU_ENGINE_DATA.S00		DCUDSTAT
DCU_GET_CHECKSUM.S00	LCF_FACTCHK	LCF_FACTCHK
DCU_GET_CHECKSUM.S00		BLK304CHK
DCU_GET_DATA.S00	BLK00CHK	BLK00CHK
DCU_GET_DATA.S00	BLK01CHK	BLK01CHK
DCU_GET_DATA.S00	BLK305CHK	BLK305CHK
DCU_GET_DATA.S00	LCF_FACTCHK	LCF_FACTCHK
DCU_GET_QUEUE.S00		DCUWRITE
DCU_IS_OBSOLETE.SRC	DCU_IS_OBSOLETE_A.CPGC	BLK305CHK
DCU_IS_OBSOLETE.SRC	DCU_IS_OBSOLETE_B.CPGC	BLK305CHK
DCU_IS_OBSOLETE.SRC	DCU_IS_OBSOLETE_C.CPGC	BLK305CHK
DCU_IS_OBSOLETE.SRC	DCU_IS_OBSOLETE_D.CPGC	BLK305CHK
DCU_IS_OBSOLETE.SRC	DCU_IS_OBSOLETE_E.CPGC	BLK305CHK
DCU_IS_OBSOLETE.SRC	DCU_IS_OBSOLETE_F.CPGC	BLK305CHK
DCU_LCF_DATA.S00		LCF_FACTCHK
DCU_LCF_DATA.S00		DCUDSTAT
DCU_PUT_QUEUE.S00	DCU_CP_CFG	DCU_CP_CFG
DCU_PUT_QUEUE.S00		DCU_ENG_CFG
DCU_PUT_QUEUE.S00		DCU_EEC_CFG
DCU_RANGE_CHECKS.SRC	E05-RAMINIT	E05-RAMINIT
DCU_RANGE_CHECKS.SRC	DCU_RANGE_CHECKS.CPGC	BLK305CHK
DCU_RANGE_CHECKS.SRC	DCU_RANGE_CHECKS.CPGC	E05-RAMINIT
DCU_RECEIVE.S00		DCU_RECEIVE
DCU_SCHEDULE.S00	BLK00CHK	BLK00CHK
DCU_SCHEDULE.S00	BLK01CHK	BLK01CHK
DCU_SCHEDULE.S00	BLK09CHK	BLK09CHK
DCU_SCHEDULE.S00	BLK305CHK	BLK305CHK
DCU_SCHEDULE.S00	DCU_CP_CFG	DCU_CP_CFG
DCU_SCHEDULE.S00	LCF_FACTCHK	LCF_FACTCHK
DCU_SCHEDULE.S00		DCUREAD
DCU_SCHEDULE.S00		DCUDSTAT

Module	Design	SRS Capability
DCU_SCHEDULE.S00		DCUWRITE
DCU_SCHEDULE.S00		DCU_EEC_CFG
DCU_SCHEDULE.S00		DCU_ENGUSE_CFG
DCU_SCHEDULE.S00		DCU_ENG_CFG
DCU_TRANSFER.S00	DCUPROTO	DCUPROTO
DCU_TRANSMIT.S00	DCUPROTO	DCUPROTO
DCU_WRITE.S00		DCUWRITE
DCU_WRITE_CPDAT.S00	DCU_CP_CFG	DCU_CP_CFG
DCU_WRITE_CTRDAT.S00	CYCLE_COUNTING	CYCLE_COUNTING
DCU_WRITE_CTRDAT.S00		DCU_ENGUSE_CFG
DCU_WRITE_EECDAT.S00		DCU_EEC_CFG
DCU_WRITE_ENGDAT.S00		DCU_ENG_CFG
DERLAG.SRC	DERLAG_A.CPGC	C11-Gov_SHP_Req_Calc
DERLAG.SRC	DERLAG_A.CPGC	C12-Gov_NPT_Closed_Loop
DERLAG.SRC	DERLAG_B.CPGC	C11-Gov_SHP_Req_Calc
DERLAG.SRC	DERLAG_B.CPGC	C12-Gov_NPT_Closed_Loop
DIVPRT.SRC	DIVPRT_A.CPGC	C03 - TLIMRT_D
DIVPRT.SRC	DIVPRT_A.CPGC	C07-LIMIT_OVERQ_PROTECT
DIVPRT.SRC	DIVPRT_A.CPGC	C10-Out_Auto_Ignit
DIVPRT.SRC	DIVPRT_A.CPGC	C12-Gov_NPT_Closed_Loop
DIVPRT.SRC	DIVPRT_A.CPGC	C16-Gov_Inner_Loop_Sel
DIVPRT.SRC	DIVPRT_A.CPGC	C17-Gov_Inner_NH_Req_Calc
DIVPRT.SRC	DIVPRT_A.CPGC	C18-Gov_NH_Closed_Loop
DIVPRT.SRC	DIVPRT_A.CPGC	C19-Limit_Prop_Underspeed
DIVPRT.SRC	DIVPRT_A.CPGC	C21-OUT_HBOV_EEC_MODE
DIVPRT.SRC	DIVPRT_A.CPGC	C22-Out_HBOV_Man_Mode
DIVPRT.SRC	DIVPRT_A.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DIVPRT.SRC	DIVPRT_A.CPGC	C26-MPARAM
DIVPRT.SRC	DIVPRT_A.CPGC	C27-NHMINMAX
DIVPRT.SRC	DIVPRT_A.CPGC	C31-Limit_Q
DIVPRT.SRC	DIVPRT_B.CPGC	C07-LIMIT_OVERQ_PROTECT
DIVPRT.SRC	DIVPRT_B.CPGC	C10-Out_Auto_Ignit
DIVPRT.SRC	DIVPRT_B.CPGC	C12-Gov_NPT_Closed_Loop
DIVPRT.SRC	DIVPRT_B.CPGC	C16-Gov_Inner_Loop_Sel
DIVPRT.SRC	DIVPRT_B.CPGC	C17-Gov_Inner_NH_Req_Calc
DIVPRT.SRC	DIVPRT_B.CPGC	C18-Gov_NH_Closed_Loop
DIVPRT.SRC	DIVPRT_B.CPGC	C21-OUT_HBOV_EEC_MODE
DIVPRT.SRC	DIVPRT_B.CPGC	C22-Out_HBOV_Man_Mode
DIVPRT.SRC	DIVPRT_B.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DIVPRT.SRC	DIVPRT_B.CPGC	C26-MPARAM
DIVPRT.SRC	DIVPRT_B.CPGC	C27-NHMINMAX
DPA.SRC	DPA_A.CPGC	C03 - TLIMRT_D
DPA.SRC	DPA_A.CPGC	C07-LIMIT_OVERQ_PROTECT

Module	Design	SRS Capability
DPA.SRC	DPA_A.CPGC	C08-Gov_SHP_Closed_Loop
DPA.SRC	DPA_A.CPGC	C10-Out_Auto_Ignit
DPA.SRC	DPA_A.CPGC	C11-Gov_SHP_Req_Calc
DPA.SRC	DPA_A.CPGC	C12-Gov_NPT_Closed_Loop
DPA.SRC	DPA_A.CPGC	C13-Trans_Accel_NH
DPA.SRC	DPA_A.CPGC	C16-Gov_Inner_Loop_Sel
DPA.SRC	DPA_A.CPGC	C17-Gov_Inner_NH_Req_Calc
DPA.SRC	DPA_A.CPGC	C18-Gov_NH_Closed_Loop
DPA.SRC	DPA_A.CPGC	C19-Limit_Prop_Underspeed
DPA.SRC	DPA_A.CPGC	C20-Limit_Prop_Overspeed
DPA.SRC	DPA_A.CPGC	C21-OUT_HBOV_EEC_MODE
DPA.SRC	DPA_A.CPGC	C22-Out_HBOV_Man_Mode
DPA.SRC	DPA_A.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DPA.SRC	DPA_A.CPGC	C26-MPARAM
DPA.SRC	DPA_A.CPGC	C27-NHMINMAX
DPA.SRC	DPA_A.CPGC	C30-Gov_Inner_SM_Rate_Req
DPA.SRC	DPA_A.CPGC	C31-Limit_Q
DPA.SRC	DPA_A.CPGC	I31-NPARAMIN
DPA.SRC	DPA_B.CPGC	C03 - TLIMRT_D
DPA.SRC	DPA_B.CPGC	C07-LIMIT_OVERQ_PROTECT
DPA.SRC	DPA_B.CPGC	C08-Gov_SHP_Closed_Loop
DPA.SRC	DPA_B.CPGC	C11-Gov_SHP_Req_Calc
DPA.SRC	DPA_B.CPGC	C12-Gov_NPT_Closed_Loop
DPA.SRC	DPA_B.CPGC	C13-Trans_Accel_NH
DPA.SRC	DPA_B.CPGC	C16-Gov_Inner_Loop_Sel
DPA.SRC	DPA_B.CPGC	C17-Gov_Inner_NH_Req_Calc
DPA.SRC	DPA_B.CPGC	C18-Gov_NH_Closed_Loop
DPA.SRC	DPA_B.CPGC	C19-Limit_Prop_Underspeed
DPA.SRC	DPA_B.CPGC	C20-Limit_Prop_Overspeed
DPA.SRC	DPA_B.CPGC	C21-OUT_HBOV_EEC_MODE
DPA.SRC	DPA_B.CPGC	C22-Out_HBOV_Man_Mode
DPA.SRC	DPA_B.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DPA.SRC	DPA_B.CPGC	C26-MPARAM
DPA.SRC	DPA_B.CPGC	C27-NHMINMAX
DPA.SRC	DPA_B.CPGC	C30-Gov_Inner_SM_Rate_Req
DPA.SRC	DPA_B.CPGC	C31-Limit_Q
DPA.SRC	DPA_B.CPGC	I31-NPARAMIN
DPM.SRC	DPM_A.CPGC	C03 - TLIMRT_D
DPM.SRC	DPM_A.CPGC	C07-LIMIT_OVERQ_PROTECT
DPM.SRC	DPM_A.CPGC	C10-Out_Auto_Ignit
DPM.SRC	DPM_A.CPGC	C12-Gov_NPT_Closed_Loop
DPM.SRC	DPM_A.CPGC	C16-Gov_Inner_Loop_Sel
DPM.SRC	DPM_A.CPGC	C17-Gov_Inner_NH_Req_Calc

Module	Design	SRS Capability
DPM.SRC	DPM_A.CPGC	C18-Gov_NH_Closed_Loop
DPM.SRC	DPM_A.CPGC	C19-Limit_Prop_Underspeed
DPM.SRC	DPM_A.CPGC	C20-Limit_Prop_Overspeed
DPM.SRC	DPM_A.CPGC	C21-OUT_HBOV_EEC_MODE
DPM.SRC	DPM_A.CPGC	C22-Out_HBOV_Man_Mode
DPM.SRC	DPM_A.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DPM.SRC	DPM_A.CPGC	C26-MPARAM
DPM.SRC	DPM_A.CPGC	C27-NHMINMAX
DPM.SRC	DPM_A.CPGC	C30-Gov_Inner_SM_Rate_Req
DPM.SRC	DPM_A.CPGC	C31-Limit_Q
DPM.SRC	DPM_B.CPGC	C03 - TLIMRT_D
DPM.SRC	DPM_B.CPGC	C07-LIMIT_OVERQ_PROTECT
DPM.SRC	DPM_B.CPGC	C10-Out_Auto_Ignit
DPM.SRC	DPM_B.CPGC	C12-Gov_NPT_Closed_Loop
DPM.SRC	DPM_B.CPGC	C16-Gov_Inner_Loop_Sel
DPM.SRC	DPM_B.CPGC	C17-Gov_Inner_NH_Req_Calc
DPM.SRC	DPM_B.CPGC	C18-Gov_NH_Closed_Loop
DPM.SRC	DPM_B.CPGC	C19-Limit_Prop_Underspeed
DPM.SRC	DPM_B.CPGC	C20-Limit_Prop_Overspeed
DPM.SRC	DPM_B.CPGC	C21-OUT_HBOV_EEC_MODE
DPM.SRC	DPM_B.CPGC	C22-Out_HBOV_Man_Mode
DPM.SRC	DPM_B.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DPM.SRC	DPM_B.CPGC	C26-MPARAM
DPM.SRC	DPM_B.CPGC	C27-NHMINMAX
DPM.SRC	DPM_B.CPGC	C30-Gov_Inner_SM_Rate_Req
DPM.SRC	DPM_B.CPGC	C31-Limit_Q
DPM.SRC	DPM_C.CPGC	C03 - TLIMRT_D
DPM.SRC	DPM_C.CPGC	C07-LIMIT_OVERQ_PROTECT
DPM.SRC	DPM_C.CPGC	C12-Gov_NPT_Closed_Loop
DPM.SRC	DPM_C.CPGC	C16-Gov_Inner_Loop_Sel
DPM.SRC	DPM_C.CPGC	C17-Gov_Inner_NH_Req_Calc
DPM.SRC	DPM_C.CPGC	C18-Gov_NH_Closed_Loop
DPM.SRC	DPM_C.CPGC	C19-Limit_Prop_Underspeed
DPM.SRC	DPM_C.CPGC	C20-Limit_Prop_Overspeed
DPM.SRC	DPM_C.CPGC	C21-OUT_HBOV_EEC_MODE
DPM.SRC	DPM_C.CPGC	C22-Out_HBOV_Man_Mode
DPM.SRC	DPM_C.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DPM.SRC	DPM_C.CPGC	C26-MPARAM
DPM.SRC	DPM_C.CPGC	C27-NHMINMAX
DPM.SRC	DPM_C.CPGC	C30-Gov_Inner_SM_Rate_Req
DPM.SRC	DPM_C.CPGC	C31-Limit_Q
DPM.SRC	DPM_D.CPGC	C03 - TLIMRT_D
DPM.SRC	DPM_D.CPGC	C07-LIMIT_OVERQ_PROTECT

Module	Design	SRS Capability
DPM.SRC	DPM_D.CPGC	C10-Out_Auto_Ignit
DPM.SRC	DPM_D.CPGC	C12-Gov_NPT_Closed_Loop
DPM.SRC	DPM_D.CPGC	C16-Gov_Inner_Loop_Sel
DPM.SRC	DPM_D.CPGC	C17-Gov_Inner_NH_Req_Calc
DPM.SRC	DPM_D.CPGC	C18-Gov_NH_Closed_Loop
DPM.SRC	DPM_D.CPGC	C19-Limit_Prop_Underspeed
DPM.SRC	DPM_D.CPGC	C20-Limit_Prop_Overspeed
DPM.SRC	DPM_D.CPGC	C21-OUT_HBOV_EEC_MODE
DPM.SRC	DPM_D.CPGC	C22-Out_HBOV_Man_Mode
DPM.SRC	DPM_D.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DPM.SRC	DPM_D.CPGC	C26-MPARAM
DPM.SRC	DPM_D.CPGC	C27-NHMINMAX
DPM.SRC	DPM_D.CPGC	C30-Gov_Inner_SM_Rate_Req
DPM.SRC	DPM_D.CPGC	C31-Limit_Q
DPM.SRC	DPM_E.CPGC	C03 - TLIMRT_D
DPM.SRC	DPM_E.CPGC	C07-LIMIT_OVERQ_PROTECT
DPM.SRC	DPM_E.CPGC	C10-Out_Auto_Ignit
DPM.SRC	DPM_E.CPGC	C12-Gov_NPT_Closed_Loop
DPM.SRC	DPM_E.CPGC	C16-Gov_Inner_Loop_Sel
DPM.SRC	DPM_E.CPGC	C17-Gov_Inner_NH_Req_Calc
DPM.SRC	DPM_E.CPGC	C18-Gov_NH_Closed_Loop
DPM.SRC	DPM_E.CPGC	C19-Limit_Prop_Underspeed
DPM.SRC	DPM_E.CPGC	C20-Limit_Prop_Overspeed
DPM.SRC	DPM_E.CPGC	C21-OUT_HBOV_EEC_MODE
DPM.SRC	DPM_E.CPGC	C22-Out_HBOV_Man_Mode
DPM.SRC	DPM_E.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
DPM.SRC	DPM_E.CPGC	C26-MPARAM
DPM.SRC	DPM_E.CPGC	C27-NHMINMAX
DPM.SRC	DPM_E.CPGC	C30-Gov_Inner_SM_Rate_Req
DPM.SRC	DPM_E.CPGC	C31-Limit_Q
DPM.SRC	DPM_V.CPGC	C10-Out_Auto_Ignit
DUMP.SRC	E20-Maint_Dev_Test_Mode	E20-Maint_Dev_Test_Mode
E01_PORPSR.SRC	E01_PORPSR_A.CPGC	E01-BIT_Proc_Reset_PSR_POR
E01_PORPSR.SRC	E01_PORPSR_A.CPGC	E05-RAMINIT
E01_PORPSR.SRC	E01_PORPSR_B.CPGC	E01-BIT_Proc_Reset_PSR_POR
E01_PORPSR.SRC	E01_PORPSR_C.CPGC	E01-BIT_Proc_Reset_PSR_POR
E01_PORPSR.SRC	E01_PORPSR_D.CPGC	E01-BIT_Proc_Reset_PSR_POR
E02_RAMTEST.SRC	E02_RAMTEST_A.CPGC	E02-BIT_Proc_Ram_Test
E02_RAMTEST.SRC	E02_RAMTEST_B.CPGC	E02-BIT_Proc_Ram_Test
E02_RAMTEST.SRC	E02_RAMTEST_C.CPGC	E02-BIT_Proc_Ram_Test
E03_CHECKSUM.SRC	E03_CHECKSUM_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum
E03_CHECKSUM.SRC	E03_CHECKSUM_B.CPGC	E03-BIT_Proc_ROM_EE_Chksum
E04_INSTST.SRC	E04_INSTST_A.CPGC	E04-BIT_Proc_Instruct_Test

Module	Design	SRS Capability
E05_RAMINI.SRC	E05_RAMINI_F.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_G.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_H.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_I.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_J.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_K.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_L.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_M.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_A.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_B.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_C.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_D.CPGC	E05-RAMINIT
E05_RAMINI.SRC	E05_RAMINI_E.CPGC	E05-RAMINIT
E06_TQTEST.SRC	E06_TQTEST_A.CPGC	E06-BIT_GA_Q_Test_Routine
E06_TQTEST.SRC	E06_TQTEST_B.CPGC	E06-BIT_GA_Q_Test_Routine
E06_TQTEST.SRC	E06_TQTEST_C.CPGC	E06-BIT_GA_Q_Test_Routine
E07_BG_EXECUTIVE.SRC	E07_BG_EXECUTIVE_A.CPGC	C10-Out_Auto_Ignit
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_F.CPGC	E08 - EXEC_FOREGND
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_G.CPGC	E08 - EXEC_FOREGND
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_H.CPGC	E08 - EXEC_FOREGND
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_I.CPGC	E08 - EXEC_FOREGND
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_A.CPGC	E08 - EXEC_FOREGND
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_B.CPGC	E08 - EXEC_FOREGND
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_C.CPGC	E08 - EXEC_FOREGND
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_D.CPGC	E08 - EXEC_FOREGND
E08_FG_EXECUTIVE.SRC	E08_FG_EXECUTIVE_E.CPGC	E08 - EXEC_FOREGND
E08_FG_SCHEDULE.SRC		BLK00CHK
E08_FG_SCHEDULE.SRC		BLK01CHK
E08_FG_SCHEDULE.SRC		C01-Start
E08_FG_SCHEDULE.SRC		C04 - PWRCALC_D
E08_FG_SCHEDULE.SRC		C12-Gov_NPT_Closed_Loop
E08_FG_SCHEDULE.SRC		C18-Gov_NH_Closed_Loop
E08_FG_SCHEDULE.SRC		C21-OUT_HBOV_EEC_MODE
E08_FG_SCHEDULE.SRC		C26-MPARAM
E08_FG_SCHEDULE.SRC		CYCLE_COUNTING
E08_FG_SCHEDULE.SRC		DCU_ENGUSE_CFG
E08_FG_SCHEDULE.SRC		E08 - EXEC_FOREGND
E08_FG_SCHEDULE.SRC		E17 - NONCRITICAL_FAULT
E08_FG_SCHEDULE.SRC		E18-FAULTSUM
E08_FG_SCHEDULE.SRC		E19-BIT_Store_EEPROM_Trim_Data
E08_FG_SCHEDULE.SRC		E27-Maint_Misc_MFC_Id
E08_FG_SCHEDULE.SRC		E31-Maint_Misc_Ext_Disc_Word
E08_FG_SCHEDULE.SRC		EVENT_COUNTING

Module	Design	SRS Capability
E08_FG_SCHEDULE.SRC		BLK09CHK
E08_FG_SCHEDULE.SRC		BLK304CHK
E08_FG_SCHEDULE.SRC		BLK305CHK
E08_FG_SCHEDULE.SRC		C06-POWER_REQUEST
E08_FG_SCHEDULE.SRC		C07-LIMIT_OVERQ_PROTECT
E08_FG_SCHEDULE.SRC		DCU_CP_CFG
E08_FG_SCHEDULE.SRC		DCU_EEC_CFG
E08_FG_SCHEDULE.SRC		DCU_ENG_CFG
E08_FG_SCHEDULE.SRC		DCUDSTAT
E08_FG_SCHEDULE.SRC		DCUWRITE
E08_FG_SCHEDULE.SRC		E13-WATEST
E08_FG_SCHEDULE.SRC		E28-BIT_EEC_Bleed_W/A_Tests
E08_FG_SCHEDULE.SRC		EECOUNTERS
E08_FG_SCHEDULE.SRC		LCF_FACTCHK
E08_FG_SCHEDULE.SRC		O03 - OUT_DISC
E08_FG_SCHEDULE.SRC		UART_OUT_SEL
E08_FG_SCHEDULE.SRC	C03_TLMRT_6_D.DATAFLOW	C03 - TLMRT_D
E08_FG_SCHEDULE.SRC		C03 - TLMRT_D
E08_FG_SCHEDULE.SRC		E21-RECONFIG
E08_FG_SCHEDULE.SRC		STRT_FLIGHT_DETERM
E08_FG_SCHEDULE.SRC		MISSION_DETERM
E08_FG_SCHEDULE.SRC	BLK09CHK	BLK09CHK
E08_FG_SCHEDULE.SRC	C01-Start	C01-Start
E08_FG_SCHEDULE.SRC	CYCLE_COUNTING	CYCLE_COUNTING
E08_FG_SCHEDULE.SRC	E13-WATEST	E13-WATEST
E08_FG_SCHEDULE.SRC	STRT_FLIGHT_DETERM	STRT_FLIGHT_DETERM
E08_FG_SECSEL.SRC	E08_FG_SECSEL_A.CPGC	E08 - EXEC_FOREGND
E08_FG_STATETABLE.SRC	I31-UARTIN	I31-UARTIN
E08_FG_STATETABLE.SRC		I13-ENG VARIANT
E08_FG_STATETABLE.SRC		I28-DISINW1
E08_FG_STATETABLE.SRC		I29-DISINW2
E08_FG_STATETABLE.SRC		I31-UARTIN
E08_FG_STATETABLE.SRC		O15-ARINCO_341
E08_FG_STATETABLE.SRC		O18-ARINCO_270
E08_FG_STATETABLE.SRC		O40-ARINCO_062
E08_FG_STATETABLE.SRC		O41-ARINCO_064
E08_FG_STATETABLE.SRC		O42-ARINCO_063
E08_FG_STATETABLE.SRC		O43-ARINCO_065
E08_FG_STATETABLE.SRC		O44-ARINCO_066
E08_FG_STATETABLE.SRC		O45-ARINCO_067
E08_FG_STATETABLE.SRC	E08 - EXEC_FOREGND	E08 - EXEC_FOREGND
E09_SMSUMV.SRC	E09_SMSUMV_A.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
E09_SMSUMV.SRC	E09_SMSUMV_B.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt

Module	Design	SRS Capability
E10_RAMTEST.SRC	E10_RAMTEST_A.CPGC	E10-BIT_EEC_Partial_RAM_BREG
E10_RAMTEST.SRC	E10_RAMTEST_B.CPGC	E10-BIT_EEC_Partial_RAM_BREG
E11_CONTEST.SRC	E11_CONTEST_A.CPGC	E11-BIT_EEC_Loss_Connector
E13_WATEST_1.SRC	E13_WATEST_1_A.CPGC	E13-WATEST
E13_WATEST_1.SRC	E13_WATEST_1_B.CPGC	E13-WATEST
E13_WATEST_2.SRC	E13_WATEST_2_A.CPGC	E13-WATEST
E13_WATEST_3.SRC	E13_WATEST_3_A.CPGC	E13-WATEST
E13_WATEST_4.SRC	E13_WATEST_4_A.CPGC	E13-WATEST
E13_WATEST_5.SRC	E13_WATEST_5_A.CPGC	E13-WATEST
E13_WATEST_6.SRC	E13_WATEST_6_A.CPGC	E13-WATEST
E13_WATEST_7.SRC	E13_WATEST_7_A.CPGC	E13-WATEST
E14_FAULTSTORE_1.SRC	E14_FAULTSTORE_1_A.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_1.SRC	E14_FAULTSTORE_1_B.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_1.SRC	E14_FAULTSTORE_1_C.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_2.SRC	E14-FAULT_STORAGE	E14-FAULT_STORAGE
E14_FAULTSTORE_3.SRC	E14_FAULTSTORE_3_A.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_3.SRC	E14_FAULTSTORE_3_B.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_3.SRC	E14_FAULTSTORE_3_C.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_4.SRC	E14_FAULTSTORE_4_A.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_4.SRC	E14_FAULTSTORE_4_B.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_5.SRC	E14_FAULTSTORE_5_A.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_6.SRC	E14_FAULTSTORE_6_A.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_6.SRC	E14_FAULTSTORE_6_B.CPGC	E14-FAULT_STORAGE
E14_FAULTSTORE_8.SRC	E14-FAULT_STORAGE	E14-FAULT_STORAGE
E15_BCKGNDTEST.SRC	E15_BCKGNDTEST_A.CPGC	E07-Exec_Background
E15_BCKGNDTEST.SRC	E15_BCKGNDTEST_A.CPGC	E15-BIT_Proc_Background
E15_BCKGNDTEST.SRC	E15_BCKGNDTEST_B.CPGC	E07-Exec_Background
E16_SMTEST.SRC	E16_SMTEST_A.CPGC	E16-BIT_EEC_SM_Phase
E16_SMTEST.SRC	E16_SMTEST_B.CPGC	E16-BIT_EEC_SM_Phase
E17_NCFLT_1.SRC	E17_NCFLT_1_A.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_1.SRC	E17_NCFLT_1_B.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_2.SRC	E17_NCFLT_2_A.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_2.SRC	E17_NCFLT_2_B.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_3.SRC	E17_NCFLT_3_A.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_3.SRC	E17_NCFLT_3_B.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_4.SRC	E17_NCFLT_4_A.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_4.SRC	E17_NCFLT_4_B.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_5.SRC	E17_NCFLT_5_A.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_5.SRC	E17_NCFLT_5_B.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_6.SRC	E17_NCFLT_6_A.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_6.SRC	E17_NCFLT_6_B.CPGC	E17 - NONCRITICAL_FAULT
E17_NCFLT_7.SRC	E17 - NONCRITICAL_FAULT	E17 - NONCRITICAL_FAULT
E18_FAULTSUM_1.SRC	E18_FAULTSUM_1_A.CPGC	E18-FAULTSUM

Module	Design	SRS Capability
E18_FAULTSUM_1.SRC	E18_FAULTSUM_1_B.CPGC	E18-FAULTSUM
E18_FAULTSUM_2.SRC	E18_FAULTSUM_2_A.CPGC	E18-FAULTSUM
E18_FAULTSUM_2.SRC	E18_FAULTSUM_2_B.CPGC	E18-FAULTSUM
E18_FAULTSUM_3.SRC	E18_FAULTSUM_3_A.CPGC	E18-FAULTSUM
E18_FAULTSUM_3.SRC	E18_FAULTSUM_3_B.CPGC	E18-FAULTSUM
E18_FAULTSUM_4.SRC	E18_FAULTSUM_4_A.CPGC	E18-FAULTSUM
E18_FAULTSUM_4.SRC	E18_FAULTSUM_4_B.CPGC	E18-FAULTSUM
E18_FAULTSUM_5.SRC		E18-FAULTSUM
E18_FAULTSUM_6.SRC		E18-FAULTSUM
E19_TRMSTORE_1.SRC	E19_TRMSTORE_1_A.CPGC	E19-BIT_Store_EEPROM_Trim_Data
E19_TRMSTORE_2.SRC	E19_TRMSTORE_2_A.CPGC	E19-BIT_Store_EEPROM_Trim_Data
E19_TRMSTORE_4.SRC	E19_TRMSTORE_4_A.CPGC	E19-BIT_Store_EEPROM_Trim_Data
E19_TRMSTORE_5.SRC	E19_TRMSTORE_5_A.CPGC	E19-BIT_Store_EEPROM_Trim_Data
E20_TESTMODE.SRC	E20_TESTMODE_A.CPGC	E20-Maint_Dev_Test_Mode
E20_TESTMODE.SRC	E20_TESTMODE_B.CPGC	E20-Maint_Dev_Test_Mode
E20_TESTMODE.SRC	E20_TESTMODE_C.CPGC	E20-Maint_Dev_Test_Mode
E21_RECONFIG.S00	E21_RECONFIG.CTRLFLOW	E21-RECONFIG
E22_GASEL.SRC	E22_GASEL_A.CPGC	E22-BIT_GA_GateArraySelect
E22_GASEL.SRC	E22_GASEL_B.CPGC	E22-BIT_GA_GateArraySelect
E22_GASEL.SRC	E22_GASEL_C.CPGC	E22-BIT_GA_GateArraySelect
E22_GASEL.SRC	E22_GASEL_D.CPGC	E22-BIT_GA_GateArraySelect
E23_PLATRIM.SRC	E23_PLATRIM_A.CPGC	E23-BIT_Store_PLA_Trim_Data
E23_PLATRIM.SRC	E23_PLATRIM_B.CPGC	E23-BIT_Store_PLA_Trim_Data
E23_PLATRIM.SRC	E23_PLATRIM_C.CPGC	E23-BIT_Store_PLA_Trim_Data
E25_FAULTDET.SRC	E25_FAULTDET_A.CPGC	E25-BIT_EEC_Int_Fault_Detect
E26_FAULTSTORE_1.SRC	E26_FAULTSTORE_1_A.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_1.SRC	E26_FAULTSTORE_1_B.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_1.SRC	E26_FAULTSTORE_1_C.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_2.SRC	E26_FAULTSTORE_2_A.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_2.SRC	E26_FAULTSTORE_2_B.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_2.SRC	E26_FAULTSTORE_2_C.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_3.SRC	E26_FAULTSTORE_3_A.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_3.SRC	E26_FAULTSTORE_3_B.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_3.SRC	E26_FAULTSTORE_3_C.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_3.SRC	E26_FAULTSTORE_3_D.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_4.SRC	E26_FAULTSTORE_4_A.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_4.SRC	E26_FAULTSTORE_4_B.CPGC	E26-INT_FAULTSTORE
E26_FAULTSTORE_5.SRC	E26-INT_FAULTSTORE	E26-INT_FAULTSTORE
E27_MFCID.SRC	E27_MFCID_A.CPGC	E27-Maint_Misc_MFC_Id
E28_HBOVWA_1.SRC	E28_HBOVWA_1_A.CPGC	E28-BIT_EEC_Bleed_W/A_Tests
E28_HBOVWA_1.SRC	E28_HBOVWA_1_B.CPGC	E28-BIT_EEC_Bleed_W/A_Tests
E28_HBOVWA_2.SRC	E28_HBOVWA_2_A.CPGC	E28-BIT_EEC_Bleed_W/A_Tests
E28_HBOVWA_2.SRC	E28_HBOVWA_2_B.CPGC	E28-BIT_EEC_Bleed_W/A_Tests

Module	Design	SRS Capability
E31_EXTDIS.SRC	E31_EXTDIS_A.CPGC	E31-Maint_Misc_Ext_Disc_Word
E31_EXTDIS.SRC	E31_EXTDIS_B.CPGC	E31-Maint_Misc_Ext_Disc_Word
E31_EXTDIS.SRC	E31_EXTDIS_C.CPGC	E31-Maint_Misc_Ext_Disc_Word
E31_EXTDIS.SRC	E31_EXTDIS_D.CPGC	E31-Maint_Misc_Ext_Disc_Word
E31_EXTDIS.SRC	E31_EXTDIS_E.CPGC	E31-Maint_Misc_Ext_Disc_Word
E31_EXTDIS.SRC	E31_EXTDIS_F.CPGC	E31-Maint_Misc_Ext_Disc_Word
E31_EXTDIS.SRC	E31_EXTDIS_G.CPGC	E31-Maint_Misc_Ext_Disc_Word
EE_GET_QUEUE.S00	BLK00CHK	BLK00CHK
EE_GET_QUEUE.S00	BLK01CHK	BLK01CHK
EE_GET_QUEUE.S00	BLK305CHK	BLK305CHK
EE_PUT_QUEUE.S00	BLK00CHK	BLK00CHK
EE_PUT_QUEUE.S00	BLK01CHK	BLK01CHK
EE_PUT_QUEUE.S00	BLK305CHK	BLK305CHK
EE_PUT_QUEUE.S00		EECOUNTERS
EE_READ_DATA.S00	E05-RAMINIT	E05-RAMINIT
EE_WRITE_BLOCK.S00	BLK00CHK	BLK00CHK
EE_WRITE_BLOCK.S00	BLK01CHK	BLK01CHK
EE_WRITE_BLOCK.S00	BLK305CHK	BLK305CHK
EESUM_16.SRC	EESUM_16_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum
EESUM_8.SRC	EESUM_8_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum
FAILED.SRC	FAILED_A.CPGC	E27-Maint_Misc_MFC_Id
FAILED.SRC	FAILED_A.CPGC	I08-Sens_LL_T18_RTD
FAILED.SRC	FAILED_A.CPGC	I14-Sens_HL_PLA
FAILED.SRC	FAILED_A.CPGC	I25-Sens_FR_Q_NPT_In
FAILLT.SRC	FAILLT_A.CPGC	E13-WATEST
FAILLT.SRC	FAILLT_A.CPGC	I01-Sens_LL_Drift_Comp
FAILLT.SRC	FAILLT_A.CPGC	I02-Sens_LL_Strain_Gauge_Comp
FAILLT.SRC	FAILLT_A.CPGC	I03-Sens_LL_Cold_Junct_Temp
FAILLT.SRC	FAILLT_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
FAILLT.SRC	FAILLT_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
FAILLT.SRC	FAILLT_A.CPGC	I07-Sens_LL_TQ_RTD
FAILLT.SRC	FAILLT_A.CPGC	I08-Sens_LL_T18_RTD
FAILLT.SRC	FAILLT_A.CPGC	I09-Sens_HL_Drift
FAILLT.SRC	FAILLT_A.CPGC	I10-Sens_HL_Gain_Comp
FAILLT.SRC	FAILLT_A.CPGC	I11-Sens_HL_Q_Bias_Trim
FAILLT.SRC	FAILLT_A.CPGC	I12-Sens_HL_Q_Gain_Trim
FAILLT.SRC	FAILLT_A.CPGC	I14-Sens_HL_PLA
FAILLT.SRC	FAILLT_A.CPGC	I15-Sens_HL_PLA_Bias
FAILLT.SRC	FAILLT_A.CPGC	I16-Sens_HL_PLA_Gain
FAILLT.SRC	FAILLT_A.CPGC	I19-Sens_FR_Dual_Probe
FAILLT.SRC	FAILLT_A.CPGC	I45-Sens_HL_Bleed_W/A_Input
FAILLT.SRC	FAILLT_B.CPGC	E13-WATEST
FAILLT.SRC	FAILLT_B.CPGC	I01-Sens_LL_Drift_Comp

Module	Design	SRS Capability
FAILLT.SRC	FAILLT_B.CPGC	I02-Sens_LL_Strain_Gauge_Comp
FAILLT.SRC	FAILLT_B.CPGC	I03-Sens_LL_Cold_Junct_Temp
FAILLT.SRC	FAILLT_B.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
FAILLT.SRC	FAILLT_B.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
FAILLT.SRC	FAILLT_B.CPGC	I07-Sens_LL_TQ_RTD
FAILLT.SRC	FAILLT_B.CPGC	I08-Sens_LL_T18_RTD
FAILLT.SRC	FAILLT_B.CPGC	I09-Sens_HL_Drift
FAILLT.SRC	FAILLT_B.CPGC	I10-Sens_HL_Gain_Comp
FAILLT.SRC	FAILLT_B.CPGC	I11-Sens_HL_Q_Bias_Trim
FAILLT.SRC	FAILLT_B.CPGC	I12-Sens_HL_Q_Gain_Trim
FAILLT.SRC	FAILLT_B.CPGC	I14-Sens_HL_PLA
FAILLT.SRC	FAILLT_B.CPGC	I15-Sens_HL_PLA_Bias
FAILLT.SRC	FAILLT_B.CPGC	I16-Sens_HL_PLA_Gain
FAILLT.SRC	FAILLT_B.CPGC	I19-Sens_FR_Dual_Probe
FILCON.SRC	FILCON_A.CPGC	C18-Gov_NH_Closed_Loop
FILCON.SRC	FILCON_B.CPGC	C18-Gov_NH_Closed_Loop
FILTER.SRC	FILTER_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
FILTER.SRC	FILTER_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
FILTER.SRC	FILTER_A.CPGC	I07-Sens_LL_TQ_RTD
FILTER.SRC	FILTER_A.CPGC	I08-Sens_LL_T18_RTD
FILTER.SRC	FILTER_A.CPGC	I14-Sens_HL_PLA
FILTER.SRC	FILTER_A.CPGC	I20-Sens_FR_NH_Speed_Filter
FILTER.SRC	FILTER_A.CPGC	I24-Sens_FR_NH_Probe_2_Filter
FILTER.SRC	FILTER_A.CPGC	I27-Sens_FR_Q_Shaft_Speed_Filter
FILTER.SRC	FILTER_B.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
FILTER.SRC	FILTER_B.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
FILTER.SRC	FILTER_B.CPGC	I07-Sens_LL_TQ_RTD
FILTER.SRC	FILTER_B.CPGC	I08-Sens_LL_T18_RTD
FILTER.SRC	FILTER_B.CPGC	I14-Sens_HL_PLA
FILTER.SRC	FILTER_B.CPGC	I20-Sens_FR_NH_Speed_Filter
FILTER.SRC	FILTER_B.CPGC	I24-Sens_FR_NH_Probe_2_Filter
FILTER.SRC	FILTER_B.CPGC	I27-Sens_FR_Q_Shaft_Speed_Filter
FILTER.SRC	FILTER_C.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
FILTER.SRC	FILTER_C.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
FILTER.SRC	FILTER_C.CPGC	I07-Sens_LL_TQ_RTD
FILTER.SRC	FILTER_C.CPGC	I08-Sens_LL_T18_RTD
FILTER.SRC	FILTER_C.CPGC	I20-Sens_FR_NH_Speed_Filter
FILTER.SRC	FILTER_C.CPGC	I24-Sens_FR_NH_Probe_2_Filter
FILTER.SRC	FILTER_C.CPGC	I27-Sens_FR_Q_Shaft_Speed_Filter
FLTIMER.S00	EVENT_COUNTING	EVENT_COUNTING
GETWRD.SRC	GETWRD_A.CPGC	E01-BIT_Proc_Reset_PSR_POR
GETWRD.SRC	GETWRD_A.CPGC	E02-BIT_Proc_Ram_Test
GETWRD.SRC	GETWRD_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum

Module	Design	SRS Capability
GETWRD.SRC	GETWRD_A.CPGC	E05-RAMINIT
GETWRD.SRC	GETWRD_A.CPGC	E06-BIT_GA_Q_Test_Routine
GETWRD.SRC	GETWRD_A.CPGC	E08 - EXEC_FOREGND
GETWRD.SRC	GETWRD_A.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
GETWRD.SRC	GETWRD_A.CPGC	E20-Maint_Dev_Test_Mode
GETWRD.SRC	GETWRD_A.CPGC	E22-BIT_GA_GateArraySelect
GETWRD.SRC	GETWRD_A.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
GETWRD.SRC	GETWRD_A.CPGC	I18-Sens_FR_Read_NH_Probe
GETWRD.SRC	GETWRD_A.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
GETWRD.SRC	GETWRD_A.CPGC	I23-Sens_FR_Read_NH_Probe_Two
GETWRD.SRC	GETWRD_A.CPGC	I25-Sens_FR_Q_NPT_In
GETWRD.SRC	GETWRD_A.CPGC	I28-DISINW1
GETWRD.SRC	GETWRD_A.CPGC	I29-DISINW2
GETWRD.SRC	GETWRD_A.CPGC	I30-BIT_EEC_Disc_In
GETWRD.SRC	GETWRD_A.CPGC	I31-NPARAMIN
GETWRD.SRC	GETWRD_A.CPGC	I31-UARTIN
GETWRD.SRC	GETWRD_A.CPGC	I32-ARINC_HW
GETWRD.SRC	GETWRD_A.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
GETWRD.SRC	GETWRD_A.CPGC	O01-Out_SM
GETWRD.SRC	GETWRD_A.CPGC	O02-HBOV
GETWRD.SRC	GETWRD_A.CPGC	O04 - NPARAMOUT
GETWRD.SRC	GETWRD_A.CPGC	O04-UART_OUT
GETWRD.SRC	GETWRD_A.CPGC	O04-UART_OUT1
GETWRD.SRC	GETWRD_A.CPGC	O22-ARINC_Out_SubRoutine
GETWRD.SRC	GETWRD_B.CPGC	E01-BIT_Proc_Reset_PSR_POR
GETWRD.SRC	GETWRD_B.CPGC	E02-BIT_Proc_Ram_Test
GETWRD.SRC	GETWRD_B.CPGC	E03-BIT_Proc_ROM_EE_Chksum
GETWRD.SRC	GETWRD_B.CPGC	E05-RAMINIT
GETWRD.SRC	GETWRD_B.CPGC	E06-BIT_GA_Q_Test_Routine
GETWRD.SRC	GETWRD_B.CPGC	E08 - EXEC_FOREGND
GETWRD.SRC	GETWRD_B.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
GETWRD.SRC	GETWRD_B.CPGC	E20-Maint_Dev_Test_Mode
GETWRD.SRC	GETWRD_B.CPGC	E22-BIT_GA_GateArraySelect
GETWRD.SRC	GETWRD_B.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
GETWRD.SRC	GETWRD_B.CPGC	I18-Sens_FR_Read_NH_Probe
GETWRD.SRC	GETWRD_B.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
GETWRD.SRC	GETWRD_B.CPGC	I23-Sens_FR_Read_NH_Probe_Two
GETWRD.SRC	GETWRD_B.CPGC	I25-Sens_FR_Q_NPT_In
GETWRD.SRC	GETWRD_B.CPGC	I29-DISINW2
GETWRD.SRC	GETWRD_B.CPGC	I30-BIT_EEC_Disc_In
GETWRD.SRC	GETWRD_B.CPGC	I31-NPARAMIN
GETWRD.SRC	GETWRD_B.CPGC	I31-UARTIN
GETWRD.SRC	GETWRD_B.CPGC	I32-ARINC_HW

Module	Design	SRS Capability
GETWRD.SRC	GETWRD_B.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
GETWRD.SRC	GETWRD_B.CPGC	O01-Out_SM
GETWRD.SRC	GETWRD_B.CPGC	O02-HBOV
GETWRD.SRC	GETWRD_B.CPGC	O04 - NPARAMOUT
GETWRD.SRC	GETWRD_B.CPGC	O04-UART_OUT
GETWRD.SRC	GETWRD_B.CPGC	O04-UART_OUT1
GETWRD.SRC	GETWRD_B.CPGC	O22-ARINC_Out_SubRoutine
GOTO.SRC	GOTO_A.CPGC	E20-Maint_Dev_Test_Mode
I01_LLDRIFT_1.SRC	I01_LLDRIFT_1_A.CPGC	I01-Sens_LL_Drift_Comp
I01_LLDRIFT_2.SRC	I01_LLDRIFT_2_A.CPGC	I01-Sens_LL_Drift_Comp
I01_LLDRIFT_2.SRC	I01_LLDRIFT_2_B.CPGC	I01-Sens_LL_Drift_Comp
I02_SGCOM_1.SRC	I02_SGCOM_1_A.CPGC	I02-Sens_LL_Strain_Gauge_Comp
I02_SGCOM_2.SRC	I02_SGCOM_2_A.CPGC	I02-Sens_LL_Strain_Gauge_Comp
I02_SGCOM_2.SRC	I02_SGCOM_2_B.CPGC	I02-Sens_LL_Strain_Gauge_Comp
I03_RDCJT_1.SRC	I03_RDCJT_1_A.CPGC	I03-Sens_LL_Cold_Junct_Temp
I03_RDCJT_2.SRC	I03_RDCJT_2_A.CPGC	I03-Sens_LL_Cold_Junct_Temp
I03_RDCJT_2.SRC	I03_RDCJT_2_B.CPGC	I03-Sens_LL_Cold_Junct_Temp
I03_RDCJT_2.SRC	I03_RDCJT_2_C.CPGC	I03-Sens_LL_Cold_Junct_Temp
I03_RDCJT_2.SRC	I03_RDCJT_2_D.CPGC	I03-Sens_LL_Cold_Junct_Temp
I03_RDCJT_3.SRC	I03_RDCJT_3_A.CPGC	I03-Sens_LL_Cold_Junct_Temp
I03_RDCJT_3.SRC	I03_RDCJT_3_B.CPGC	I03-Sens_LL_Cold_Junct_Temp
I03_RDCJT_3.SRC	I03_RDCJT_3_C.CPGC	I03-Sens_LL_Cold_Junct_Temp
I04_PSAMBSG_1.SRC	I04_PSAMBSG_1_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
I04_PSAMBSG_2.SRC	I04_PSAMBSG_2_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
I04_PSAMBSG_2.SRC	I04_PSAMBSG_2_B.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
I04_PSAMBSG_2.SRC	I04_PSAMBSG_2_C.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
I04_PSAMBSG_3.SRC	I04_PSAMBSG_3_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
I04_PSAMBSG_3.SRC	I04_PSAMBSG_3_B.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
I04_PSAMBSG_4.SRC	I04_PSAMBSG_4_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
I04_PSAMBSG_5.SRC	I04_PSAMBSG_5_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
I05_DELTAPSG_1.SRC	I05_DELTAPSG_1_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_2.SRC	I05_DELTAPSG_2_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_2.SRC	I05_DELTAPSG_2_B.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_3.SRC	I05_DELTAPSG_3_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_3.SRC	I05_DELTAPSG_3_B.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_3.SRC	I05_DELTAPSG_3_C.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_4.SRC	I05_DELTAPSG_4_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_4.SRC	I05_DELTAPSG_4_B.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_4.SRC	I05_DELTAPSG_4_C.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I05_DELTAPSG_4.SRC	I05_DELTAPSG_4_D.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
I07_READTQ_1.SRC	I07_READTQ_1_A.CPGC	I07-Sens_LL_TQ_RTD
I07_READTQ_2.SRC	I07_READTQ_2_A.CPGC	I07-Sens_LL_TQ_RTD
I07_READTQ_2.SRC	I07_READTQ_2_B.CPGC	I07-Sens_LL_TQ_RTD

Module	Design	SRS Capability
I07_READTQ_3.SRC	I07_READTQ_3_A.CPGC	I07-Sens_LL_TQ_RTD
I07_READTQ_3.SRC	I07_READTQ_3_B.CPGC	I07-Sens_LL_TQ_RTD
I07_READTQ_3.SRC	I07_READTQ_3_C.CPGC	I07-Sens_LL_TQ_RTD
I08_RDT18_1.SRC	I08_RDT18_1_A.CPGC	I08-Sens_LL_T18_RTD
I08_RDT18_2.SRC	I08_RDT18_2_A.CPGC	I08-Sens_LL_T18_RTD
I08_RDT18_3.SRC	I08_RDT18_3_A.CPGC	I08-Sens_LL_T18_RTD
I08_RDT18_3.SRC	I08_RDT18_3_B.CPGC	I08-Sens_LL_T18_RTD
I08_RDT18_3.SRC	I08_RDT18_3_C.CPGC	I08-Sens_LL_T18_RTD
I08_RDT18_3.SRC	I08_RDT18_3_D.CPGC	I08-Sens_LL_T18_RTD
I08_RDT18_3.SRC	I08_RDT18_3_E.CPGC	I08-Sens_LL_T18_RTD
I08_RDT18_4.SRC	I08_RDT18_4_A.CPGC	I08-Sens_LL_T18_RTD
I08_RDT18_4.SRC	I08_RDT18_4_B.CPGC	I08-Sens_LL_T18_RTD
I09_HLDRIFT_1.SRC	I09_HLDRIFT_1_A.CPGC	I09-Sens_HL_Drift
I09_HLDRIFT_2.SRC	I09_HLDRIFT_2_A.CPGC	I09-Sens_HL_Drift
I09_HLDRIFT_2.SRC	I09_HLDRIFT_2_B.CPGC	I09-Sens_HL_Drift
I10_GAINCOM_1.SRC	I10_GAINCOM_1_A.CPGC	I10-Sens_HL_Gain_Comp
I10_GAINCOM_2.SRC	I10_GAINCOM_2_A.CPGC	I10-Sens_HL_Gain_Comp
I10_GAINCOM_2.SRC	I10_GAINCOM_2_B.CPGC	I10-Sens_HL_Gain_Comp
I11_TQBTRIM_1.SRC	I11_TQBTRIM_1_A.CPGC	I11-Sens_HL_Q_Bias_Trim
I11_TQBTRIM_2.SRC	I11_TQBTRIM_2_A.CPGC	I11-Sens_HL_Q_Bias_Trim
I11_TQBTRIM_2.SRC	I11_TQBTRIM_2_B.CPGC	I11-Sens_HL_Q_Bias_Trim
I11_TQBTRIM_2.SRC	I11_TQBTRIM_2_C.CPGC	I11-Sens_HL_Q_Bias_Trim
I12_TQGTRIM_1.SRC	I12_TQGTRIM_1_A.CPGC	C05-Rating_Q_Bug_Calc
I12_TQGTRIM_1.SRC	I12_TQGTRIM_1_B.CPGC	C05-Rating_Q_Bug_Calc
I12_TQGTRIM_1.SRC	I12_TQGTRIM_1_C.CPGC	C05-Rating_Q_Bug_Calc
I12_TQGTRIM_2.SRC	I12_TQGTRIM_2_A.CPGC	I12-Sens_HL_Q_Gain_Trim
I12_TQGTRIM_2.SRC	I12_TQGTRIM_2_B.CPGC	I12-Sens_HL_Q_Gain_Trim
I12_TQGTRIM_2.SRC	I12_TQGTRIM_2_C.CPGC	I12-Sens_HL_Q_Gain_Trim
I13_ENGVARIANT_1.SRC	I13_ENGVARIANT_1_A.CPGC	
I13_ENGVARIANT_2.SRC	I13_ENGVARIANT_2.CTRLFLOW	I13-ENGVARIENT
I14_RDPLA_1.SRC	I14_RDPLA_1_A.CPGC	I14-Sens_HL_PLA
I14_RDPLA_1.SRC	I14_RDPLA_1_B.CPGC	I14-Sens_HL_PLA
I14_RDPLA_1.SRC	I14_RDPLA_1_C.CPGC	I14-Sens_HL_PLA
I14_RDPLA_2.SRC	I14_RDPLA_2_A.CPGC	I14-Sens_HL_PLA
I14_RDPLA_2.SRC	I14_RDPLA_2_B.CPGC	I14-Sens_HL_PLA
I14_RDPLA_2.SRC	I14_RDPLA_2_C.CPGC	I14-Sens_HL_PLA
I14_RDPLA_2.SRC	I14_RDPLA_2_D.CPGC	I14-Sens_HL_PLA
I14_RDPLA_3.SRC	I14_RDPLA_3_A.CPGC	I14-Sens_HL_PLA
I14_RDPLA_3.SRC	I14_RDPLA_3_B.CPGC	I14-Sens_HL_PLA
I14_RDPLA_3.SRC	I14_RDPLA_3_C.CPGC	I14-Sens_HL_PLA
I14_RDPLA_3.SRC	I14_RDPLA_3_D.CPGC	I14-Sens_HL_PLA
I14_RDPLA_3.SRC	I14_RDPLA_3_E.CPGC	I14-Sens_HL_PLA
I14_RDPLA_4.SRC	I14_RDPLA_4_A.CPGC	I14-Sens_HL_PLA

Module	Design	SRS Capability
I15_PLABIAS_1.SRC	I15_PLABIAS_1_A.CPGC	I15-Sens_HL_PLA_Bias
I15_PLABIAS_2.SRC	I15_PLABIAS_2_A.CPGC	I15-Sens_HL_PLA_Bias
I15_PLABIAS_2.SRC	I15_PLABIAS_2_B.CPGC	I15-Sens_HL_PLA_Bias
I16_PLAGAIN_1.SRC	I16_PLAGAIN_1_A.CPGC	I16-Sens_HL_PLA_Gain
I16_PLAGAIN_1.SRC	I16_PLAGAIN_1_B.CPGC	I16-Sens_HL_PLA_Gain
I16_PLAGAIN_2.SRC	I16_PLAGAIN_2_A.CPGC	I16-Sens_HL_PLA_Gain
I16_PLAGAIN_2.SRC	I16_PLAGAIN_2_B.CPGC	I16-Sens_HL_PLA_Gain
I17_PNHCV_1.SRC	I17_PNHCV_1_A.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
I17_PNHCV_1.SRC	I17_PNHCV_1_B.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
I17_PNHCV_1.SRC	I17_PNHCV_1_C.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
I17_PNHCV_2.SRC	I17_PNHCV_2_A.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
I18_READNH.SRC	I18_READNH_A.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_B.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_C.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_D.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_E.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_F.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_G.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_H.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_I.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_J.CPGC	I18-Sens_FR_Read_NH_Probe
I18_READNH.SRC	I18_READNH_K.CPGC	I18-Sens_FR_Read_NH_Probe
I19_NHDUAL_1.SRC	I19_NHDUAL_1_A.CPGC	I19-Sens_FR_Dual_Probe
I19_NHDUAL_1.SRC	I19_NHDUAL_1_B.CPGC	I19-Sens_FR_Dual_Probe
I19_NHDUAL_2.SRC	I19_NHDUAL_2_A.CPGC	I19-Sens_FR_Dual_Probe
I20_NHSPEED.SRC	I20_NHSPEED_A.CPGC	I20-Sens_FR_NH_Speed_Filter
I22_SNHCV_1.SRC	I22_SNHCV_1_A.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
I22_SNHCV_1.SRC	I22_SNHCV_1_B.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
I22_SNHCV_2.SRC	I22_SNHCV_2_A.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
I23_RDSNHP.SRC	I23_RDSNHP_A.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_B.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_C.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_D.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_E.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_F.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_G.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_H.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_I.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_J.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I23_RDSNHP.SRC	I23_RDSNHP_K.CPGC	I23-Sens_FR_Read_NH_Probe_Two
I24_NHPROBE2.SRC	I24_NHPROBE2_A.CPGC	I24-Sens_FR_NH_Probe_2_Filter
I25_TQNPT_1.SRC	I25_TQNPT_1_A.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_2.SRC	I25_TQNPT_2_A.CPGC	I25-Sens_FR_Q_NPT_In

Module	Design	SRS Capability
I25_TQNPT_2.SRC	I25_TQNPT_2_B.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_2.SRC	I25_TQNPT_2_C.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_2.SRC	I25_TQNPT_2_D.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_2.SRC	I25_TQNPT_2_E.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_2.SRC	I25_TQNPT_2_F.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_3.SRC	I25_TQNPT_3_A.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_3.SRC	I25_TQNPT_3_B.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_3.SRC	I25_TQNPT_3_C.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_3.SRC	I25_TQNPT_3_D.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_3.SRC	I25_TQNPT_3_E.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_3.SRC	I25_TQNPT_3_F.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_A.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_B.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_C.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_D.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_E.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_F.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_G.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_H.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_I.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_J.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_K.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_L.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_M.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_N.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_4.SRC	I25_TQNPT_4_O.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_A.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_B.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_C.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_D.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_E.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_F.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_G.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_H.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_I.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_J.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_K.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_L.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_M.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_N.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_5.SRC	I25_TQNPT_5_O.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_6.SRC	I25_TQNPT_6_A.CPGC	I25-Sens_FR_Q_NPT_In
I25_TQNPT_6.SRC	I25_TQNPT_6_B.CPGC	I25-Sens_FR_Q_NPT_In

Module	Design	SRS Capability
I25_TQNPT_7.SRC	I25_TQNPT_7.A.CPGC	I25-Sens_FR_Q_NPT_In
I26_TQCAL.SRC	I26_TQCAL_A.CPGC	I26-Sens_FR_Q_Shaft_Deflect
I26_TQCAL.SRC	I26_TQCAL_B.CPGC	I26-Sens_FR_Q_Shaft_Deflect
I27_TQSPEED.SRC	I27_TQSPEED_A.CPGC	I27-Sens_FR_Q_Shaft_Speed_Filter
I28_DISINW1.SRC	I28_DISINW1_A.CPGC	I28-DISINW1
I28_DISINW1.SRC	I28_DISINW1_B.CPGC	I28-DISINW1
I28_DISINW1.SRC	I28_DISINW1_C.CPGC	I28-DISINW1
I28_DISINW1.SRC	I28_DISINW1_D.CPGC	I28-DISINW1
I28_DISINW1.SRC	I28_DISINW1_E.CPGC	I28-DISINW1
I29_DISINW2.SRC	I29_DISINW2_A.CPGC	I29-DISINW2
I30_DISINFAULT.SRC	I30_DISINFAULT_A.CPGC	I30-BIT_EEC_Disc_In
I31_UARTIN.SRC	I31_UARTIN_A.CPGC	I31-UARTIN
I31_UARTIN_NPM.SRC	I31_UARTIN_NPM_A.CPGC	I31-NPARAMIN
I31_UARTIN_NPM.SRC	I31_UARTIN_NPM_B.CPGC	I31-NPARAMIN
I31_UARTIN_NPM.SRC	I31_UARTIN_NPM_C.CPGC	I31-NPARAMIN
I31_UARTIN_NPM.SRC	I31_UARTIN_NPM_D.CPGC	I31-NPARAMIN
I32_ARINCIN_1.SRC	I32_ARINCIN_1_A.CPGC	I32-ARINC_HW
I32_ARINCIN_1.SRC	I32_ARINCIN_1_B.CPGC	I32-ARINC_HW
I32_ARINCIN_1.SRC	I32_ARINCIN_1_C.CPGC	I32-ARINC_HW
I32_ARINCIN_1.SRC	I32_ARINCIN_1_D.CPGC	I32-ARINC_HW
I32_ARINCIN_1.SRC	I32_ARINCIN_1_E.CPGC	I32-ARINC_HW
I32_ARINCIN_1.SRC	I32_ARINCIN_1_F.CPGC	I32-ARINC_HW
I32_ARINCIN_2.SRC	I32_ARINCIN_2_A.CPGC	I32-ARINC_HW
I33_ARITAT.SRC	I33_ARITAT_A.CPGC	I33-ARINC_In_TAT
I33_ARITAT.SRC	I33_ARITAT_B.CPGC	I33-ARINC_In_TAT
I33_ARITAT.SRC	I33_ARITAT_C.CPGC	I33-ARINC_In_TAT
I33_ARITAT.SRC	I33_ARITAT_D.CPGC	I33-ARINC_In_TAT
I33_ARITAT.SRC	I33_ARITAT_E.CPGC	I33-ARINC_In_TAT
I33_ARITAT.SRC	I33_ARITAT_F.CPGC	I33-ARINC_In_TAT
I33_ARITAT.SRC	I33_ARITAT_G.CPGC	I33-ARINC_In_TAT
I33_ARITAT.SRC	I33_ARITAT_H.CPGC	I33-ARINC_In_TAT
I34_TATSEL.SRC	I34_TATSEL_A.CPGC	I34-ARINC_In_TAT_Source_Sel
I34_TATSEL.SRC	I34_TATSEL_B.CPGC	I34-ARINC_In_TAT_Source_Sel
I34_TATSEL.SRC	I34_TATSEL_C.CPGC	I34-ARINC_In_TAT_Source_Sel
I34_TATSEL.SRC	I34_TATSEL_D.CPGC	I34-ARINC_In_TAT_Source_Sel
I34_TATSEL.SRC	I34_TATSEL_E.CPGC	I34-ARINC_In_TAT_Source_Sel
I34_TATSEL.SRC	I34_TATSEL_F.CPGC	I34-ARINC_In_TAT_Source_Sel
I34_TATSEL.SRC	I34_TATSEL_G.CPGC	I34-ARINC_In_TAT_Source_Sel
I35_ARIPA.SRC	I35_ARIPA_A.CPGC	I35-ARINC_In_Press_Altitude
I35_ARIPA.SRC	I35_ARIPA_B.CPGC	I35-ARINC_In_Press_Altitude
I35_ARIPA.SRC	I35_ARIPA_C.CPGC	I35-ARINC_In_Press_Altitude
I35_ARIPA.SRC	I35_ARIPA_D.CPGC	I35-ARINC_In_Press_Altitude
I35_ARIPA.SRC	I35_ARIPA_E.CPGC	I35-ARINC_In_Press_Altitude

Module	Design	SRS Capability
I35_ARIPA.SRC	I35_ARIPA_F.CPGC	I35-ARINC_In_Press_Altitude
I35_ARIPA.SRC	I35_ARIPA_G.CPGC	I35-ARINC_In_Press_Altitude
I36_PAMSEL.SRC	I36_PAMSEL_A.CPGC	I36-ARINC_In_Press_Source_Sel
I36_PAMSEL.SRC	I36_PAMSEL_B.CPGC	I36-ARINC_In_Press_Source_Sel
I36_PAMSEL.SRC	I36_PAMSEL_C.CPGC	I36-ARINC_In_Press_Source_Sel
I36_PAMSEL.SRC	I36_PAMSEL_D.CPGC	I36-ARINC_In_Press_Source_Sel
I36_PAMSEL.SRC	I36_PAMSEL_E.CPGC	I36-ARINC_In_Press_Source_Sel
I36_PAMSEL.SRC	I36_PAMSEL_F.CPGC	I36-ARINC_In_Press_Source_Sel
I36_PAMSEL.SRC	I36_PAMSEL_G.CPGC	I36-ARINC_In_Press_Source_Sel
I36_PAMSEL.SRC	I36_PAMSEL_H.CPGC	I36-ARINC_In_Press_Source_Sel
I37_ARIISA_1.SRC	I37_ARIISA_1_A.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_1.SRC	I37_ARIISA_1_B.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_1.SRC	I37_ARIISA_1_C.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_1.SRC	I37_ARIISA_1_D.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_1.SRC	I37_ARIISA_1_E.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_1.SRC	I37_ARIISA_1_F.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_1.SRC	I37_ARIISA_1_G.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_1.SRC	I37_ARIISA_1_H.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_2.SRC	I37_ARIISA_2_A.CPGC	I37-ARINC_IN_IAS
I37_ARIISA_2.SRC	I37_ARIISA_2_B.CPGC	I37-ARINC_IN_IAS
I38_SMCSV_1.SRC	I38_SMCSV_1_A.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
I38_SMCSV_1.SRC	I38_SMCSV_1_B.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
I38_SMCSV_2.SRC	I38_SMCSV_2_A.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
I38_SMCSV_2.SRC	I38_SMCSV_2_B.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
I38_SMCSV_2.SRC	I38_SMCSV_2_C.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
I43_TQSPEED_1.SRC	I43_TQSPEED_1_A.CPGC	I43-Sens_FR_Q/Speed_Dual_In_Fault
I43_TQSPEED_1.SRC	I43_TQSPEED_1_B.CPGC	I43-Sens_FR_Q/Speed_Dual_In_Fault
I43_TQSPEED_2.SRC	I43_TQSPEED_2_A.CPGC	I43-Sens_FR_Q/Speed_Dual_In_Fault
I43_TQSPEED_3.SRC	I43_TQSPEED_3_A.CPGC	I43-Sens_FR_Q/Speed_Dual_In_Fault
I43_TQSPEED_3.SRC	I43_TQSPEED_3_B.CPGC	I43-Sens_FR_Q/Speed_Dual_In_Fault
I45_HBOVWA_1.SRC	I45_HBOVWA_1_A.CPGC	I45-Sens_HL_Bleed_W/A_Input
I45_HBOVWA_2.SRC	I45_HBOVWA_2_A.CPGC	I45-Sens_HL_Bleed_W/A_Input
I45_HBOVWA_2.SRC	I45_HBOVWA_2_B.CPGC	I45-Sens_HL_Bleed_W/A_Input
I46_IASFAULT.SRC	I46-ARINC_In_IAS_Fault	I46-ARINC_In_IAS_Fault
INA.SRC	INA_A.CPGC	E01-BIT_Proc_Reset_PSR_POR
INA.SRC	INA_A.CPGC	E02-BIT_Proc_Ram_Test
INA.SRC	INA_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum
INA.SRC	INA_A.CPGC	E05-RAMINIT
INA.SRC	INA_A.CPGC	E06-BIT_GA_Q_Test_Routine
INA.SRC	INA_A.CPGC	E08 - EXEC_FOREGND
INA.SRC	INA_A.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
INA.SRC	INA_A.CPGC	E20-Maint_Dev_Test_Mode
INA.SRC	INA_A.CPGC	E22-BIT_GA_GateArraySelect

Module	Design	SRS Capability
INA.SRC	INA_A.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
INA.SRC	INA_A.CPGC	I18-Sens_FR_Read_NH_Probe
INA.SRC	INA_A.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
INA.SRC	INA_A.CPGC	I23-Sens_FR_Read_NH_Probe_Two
INA.SRC	INA_A.CPGC	I25-Sens_FR_Q_NPT_In
INA.SRC	INA_A.CPGC	I28-DISINW1
INA.SRC	INA_A.CPGC	I29-DISINW2
INA.SRC	INA_A.CPGC	I30-BIT_EEC_Disc_In
INA.SRC	INA_A.CPGC	I31-NPARAMIN
INA.SRC	INA_A.CPGC	I31-UARTIN
INA.SRC	INA_A.CPGC	I32-ARINC_HW
INA.SRC	INA_A.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
INA.SRC	INA_A.CPGC	O01-Out_SM
INA.SRC	INA_A.CPGC	O02-HBOV
INA.SRC	INA_A.CPGC	O04 - NPARAMOUT
INA.SRC	INA_A.CPGC	O04-UART_OUT
INA.SRC	INA_A.CPGC	O04-UART_OUT1
INA.SRC	INA_A.CPGC	O22-ARINC_Out_SubRoutine
INC_CTRS_1.SRC	INC_CTRS_1_A.CPGC	EVENT_COUNTING
INC_CTRS_1.SRC	INC_CTRS_1_B.CPGC	EVENT_COUNTING
INC_CTRS_1.SRC	INC_CTRS_1_C.CPGC	EVENT_COUNTING
INC_CTRS_1.SRC	INC_CTRS_1_D.CPGC	EVENT_COUNTING
INC_CTRS_2.SRC	INC_CTRS_2.CPGC	EVENT_COUNTING
INC_CTRS_3.SRC	INC_CTRS_3.CPGC	EVENT_COUNTING
INC_CTRS_4.SRC	INC_CTRS_4.CTRLFLOW	EVENT_COUNTING
INI_DCJ_DATA.S00	E05-RAMINIT	E05-RAMINIT
INIFLT.SRC	INIFLT_A.CPGC	E05-RAMINIT
INIFLT.SRC	INIFLT_B.CPGC	E05-RAMINIT
INIFLT.SRC	INIFLT_C.CPGC	E05-RAMINIT
INSTST.SRC	INSTST_A.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_A.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_B.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_B.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_C.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_C.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_D.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_D.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_E.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_E.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_F.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_F.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_G.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_G.CPGC	E07-Exec_Background

Module	Design	SRS Capability
INSTST.SRC	INSTST_H.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_H.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_I.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_I.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_J.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_J.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_K.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_K.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_L.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_L.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_M.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_M.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_N.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_N.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_O.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_O.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_P.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_P.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_Q.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_Q.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_R.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_R.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_S.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_S.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_T.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_T.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_U.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_U.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_V.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_V.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_W.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_W.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_X.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_X.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_Y.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_Y.CPGC	E07-Exec_Background
INSTST.SRC	INSTST_Z.CPGC	E04-BIT_Proc_Instruct_Test
INSTST.SRC	INSTST_Z.CPGC	E07-Exec_Background
LCF_CHECK.S00	CYCLE_COUNTING	CYCLE_COUNTING
LCF_CYCLE_COUNT.S00	CYCLE_COUNTING	CYCLE_COUNTING
LCF_TABLE.S00	CYCLE_COUNTING	CYCLE_COUNTING
LDLG_SUBR.SRC	LDLG_SUBR_A.CPGC	C17-Gov_Inner_NH_Req_Calc
LDLG_SUBR.SRC	LDLG_SUBR_B.CPGC	C17-Gov_Inner_NH_Req_Calc

Module	Design	SRS Capability
LDLG_SUBR.SRC	LDLG_SUBR_C.CPGC	C17-Gov_Inner_NH_Req_Calc
LOAD.SRC	LOAD_A.CPGC	E20-Maint_Dev_Test_Mode
LOAD.SRC	LOAD_B.CPGC	E20-Maint_Dev_Test_Mode
MISSIONS.SRC	MISSIONS.DATAFLOW	MISSION_DETERM
O01_SMOUT.SRC	O01_SMOUT_A.CPGC	O01-Out_SM
O01_SMOUT.SRC	O01_SMOUT_B.CPGC	O01-Out_SM
O01_SMOUT.SRC	O01_SMOUT_C.CPGC	O01-Out_SM
O02_HBOVOUT.SRC	O02_HBOVOUT_A.CPGC	O02-HBOV
O03_DISOUT.S00	O03_DISOUT.CTRLFLOW	O03 - OUT_DISC
O03_DISOUT_A.SRC	O03_DISOUT_A.DATAFLOW	O03 - OUT_DISC
O04_UARTOUT_DTS.SRC	O04_UARTOUT_DTS_A.CPGC	O04-UART_OUT
O04_UARTOUT_DTS.SRC	O04_UARTOUT_DTS_B.CPGC	O04-UART_OUT
O04_UARTOUT_DTS.SRC	O04_UARTOUT_DTS_C.CPGC	O04-UART_OUT
O04_UARTOUT_DTS.SRC	O04_UARTOUT_DTS_D.CPGC	O04-UART_OUT
O04_UARTOUT_GDTS.SRC	O04_UARTOUT_GDTS_A.CPGC	O04-UART_OUT1
O04_UARTOUT_GDTS.SRC	O04_UARTOUT_GDTS_B.CPGC	O04-UART_OUT1
O04_UARTOUT_GDTS.SRC	O04_UARTOUT_GDTS_C.CPGC	O04-UART_OUT1
O04_UARTOUT_GDTS.SRC	O04_UARTOUT_GDTS_D.CPGC	O04-UART_OUT1
O04_UARTOUT_NPM.SRC	O04_UARTOUT_NPM_A.CPGC	O04 - NPARAMOUT
O04_UARTOUT_NPM.SRC	O04_UARTOUT_NPM_B.CPGC	O04 - NPARAMOUT
O04_UARTOUTSEL.SRC	O04_UARTOUTSEL.CPGC	UART_OUT_SEL
O05_ARO133.SRC	O05_ARO133_A.CPGC	O05-ARINCO_133
O06_ARO134.SRC	O06_ARO134_A.CPGC	O06-ARINCO_134
O07_ARO203_ALT11.SRC	O07_ARO203_ALT11_A.CPGC	O07-ARINCO_203_ALT
O07_ARO203_ALTPS.SRC	O07_ARO203_ALTPS_A.CPGC	O07-ARINCO_203_ALTPS
O09_ARO211_TAT11.SRC	O09_ARO211_TAT11_A.CPGC	O09_ARO211_TAT11
O10_ARO240.SRC	O10_ARO240_A.CPGC	O10-ARINCO_240
O10_ARO240.SRC	O10_ARO240_B.CPGC	O10-ARINCO_240
O11_ARO245.SRC	O11_ARO245_A.CPGC	O11-ARINCO_245
O11_ARO245.SRC	O11_ARO245_B.CPGC	O11-ARINCO_245
O15_ARO341.SRC	O15_ARO341_A.CPGC	O15-ARINCO_341
O15_ARO341.SRC	O15_ARO341_B.CPGC	O15-ARINCO_341
O16_ARO346.SRC	O16_ARO346_A.CPGC	O16-ARINCO_346
O17_ARO350.SRC	O17_ARO350_A.CPGC	O17-ARINCO_350
O17_ARO350.SRC	O17_ARO350_B.CPGC	O17-ARINCO_350
O18_ARO270.SRC	O18_ARO270_G.CPGC	O18-ARINCO_270
O18_ARO270.SRC	O18_ARO270_H.CPGC	O18-ARINCO_270
O18_ARO270.SRC	O18_ARO270_I.CPGC	O18-ARINCO_270
O18_ARO270.SRC	O18_ARO270_A.CPGC	O18-ARINCO_270
O18_ARO270.SRC	O18_ARO270_B.CPGC	O18-ARINCO_270
O18_ARO270.SRC	O18_ARO270_C.CPGC	O18-ARINCO_270
O18_ARO270.SRC	O18_ARO270_D.CPGC	O18-ARINCO_270
O18_ARO270.SRC	O18_ARO270_E.CPGC	O18-ARINCO_270

Module	Design	SRS Capability
O18_ARO270.SRC	O18_ARO270_F.CPGC	O18-ARINCO_270
O18_ARO270.SRC	O18_ARO270_HA.CPGC	O18-ARINCO_270
O19_ARO272.SRC	O19_ARO272.CPGC	O19-ARINCO_272
O19_ARO272_A.SRC	O19_ARO272_A.DATAFLOW	O19-ARINCO_272
O26_ARO340.SRC	O26_ARO340_A.CPGC	O26-ARINCO_340
O27_ARO342.SRC	O27_ARO342_A.CPGC	O27_ARINCO_342
O27_ARO342.SRC	O27_ARO342_B.CPGC	O27_ARINCO_342
O27_ARO342_A.SRC	O27_ARO342_A.DATAFLOW	O27_ARINCO_342
O40_ARO06X.S00		O40-ARINCO_062
O40_ARO06X.S00		O42-ARINCO_063
O40_ARO06X.S00		O41-ARINCO_064
O40_ARO06X.S00		O43-ARINCO_065
O40_ARO06X.S00		O44-ARINCO_066
O40_ARO06X.S00		O45-ARINCO_067
O40_ARO06X_TABLE.S00		O40-ARINCO_062
O40_ARO06X_TABLE.S00		O42-ARINCO_063
O40_ARO06X_TABLE.S00		O41-ARINCO_064
O40_ARO06X_TABLE.S00		O43-ARINCO_065
O40_ARO06X_TABLE.S00		O44-ARINCO_066
O40_ARO06X_TABLE.S00		O45-ARINCO_067
OPMODE_1.S00	OPMODE_1.DATAFLOW	STRT_FLIGHT_DETERM
OPMODE_2.S00	OPMODE_2.DATAFLOW	STRT_FLIGHT_DETERM
OPMODE_3.SRC	OPMODE_3.DATAFLOW	STRT_FLIGHT_DETERM
OTA.SRC	O01-Out_SM	O01-Out_SM
OTA.SRC	OTA_A.CPGC	E01-BIT_Proc_Reset_PSR_POR
OTA.SRC	OTA_A.CPGC	E02-BIT_Proc_Ram_Test
OTA.SRC	OTA_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum
OTA.SRC	OTA_A.CPGC	E05-RAMINIT
OTA.SRC	OTA_A.CPGC	E06-BIT_GA_Q_Test_Routine
OTA.SRC	OTA_A.CPGC	E08 - EXEC_FOREGND
OTA.SRC	OTA_A.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
OTA.SRC	OTA_A.CPGC	E20-Maint_Dev_Test_Mode
OTA.SRC	OTA_A.CPGC	E25-BIT_EEC_Int_Fault_Detect
OTA.SRC	OTA_A.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
OTA.SRC	OTA_A.CPGC	I18-Sens_FR_Read_NH_Probe
OTA.SRC	OTA_A.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
OTA.SRC	OTA_A.CPGC	I23-Sens_FR_Read_NH_Probe_Two
OTA.SRC	OTA_A.CPGC	I25-Sens_FR_Q_NPT_In
OTA.SRC	OTA_A.CPGC	I28-DISINW1
OTA.SRC	OTA_A.CPGC	I29-DISINW2
OTA.SRC	OTA_A.CPGC	I30-BIT_EEC_Disc_In
OTA.SRC	OTA_A.CPGC	I31-NPARAMIN
OTA.SRC	OTA_A.CPGC	I31-UARTIN

Module	Design	SRS Capability
OTA.SRC	OTA_A.CPGC	I32-ARINC_HW
OTA.SRC	OTA_A.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
OTA.SRC	OTA_A.CPGC	O01-Out_SM
OTA.SRC	OTA_A.CPGC	O02-HBOV
OTA.SRC	OTA_A.CPGC	O04 - NPARAMOUT
OTA.SRC	OTA_A.CPGC	O04-UART_OUT
OTA.SRC	OTA_A.CPGC	O04-UART_OUT1
OTA.SRC	OTA_A.CPGC	O22-ARINC_Out_SubRoutine
OTA.SRC	OTA_B.CPGC	E01-BIT_Proc_Reset_PSR_POR
OTA.SRC	OTA_B.CPGC	E02-BIT_Proc_Ram_Test
OTA.SRC	OTA_B.CPGC	E03-BIT_Proc_ROM_EE_Chksum
OTA.SRC	OTA_B.CPGC	E05-RAMINIT
OTA.SRC	OTA_B.CPGC	E06-BIT_GA_Q_Test_Routine
OTA.SRC	OTA_B.CPGC	E08 - EXEC_FOREGND
OTA.SRC	OTA_B.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
OTA.SRC	OTA_B.CPGC	E20-Maint_Dev_Test_Mode
OTA.SRC	OTA_B.CPGC	E25-BIT_EEC_Int_Fault_Detect
OTA.SRC	OTA_B.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
OTA.SRC	OTA_B.CPGC	I18-Sens_FR_Read_NH_Probe
OTA.SRC	OTA_B.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
OTA.SRC	OTA_B.CPGC	I23-Sens_FR_Read_NH_Probe_Two
OTA.SRC	OTA_B.CPGC	I25-Sens_FR_Q_NPT_In
OTA.SRC	OTA_B.CPGC	I28-DISINW1
OTA.SRC	OTA_B.CPGC	I29-DISINW2
OTA.SRC	OTA_B.CPGC	I30-BIT_EEC_Disc_In
OTA.SRC	OTA_B.CPGC	I31-NPARAMIN
OTA.SRC	OTA_B.CPGC	I31-UARTIN
OTA.SRC	OTA_B.CPGC	I32-ARINC_HW
OTA.SRC	OTA_B.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
OTA.SRC	OTA_B.CPGC	O01-Out_SM
OTA.SRC	OTA_B.CPGC	O02-HBOV
OTA.SRC	OTA_B.CPGC	O04 - NPARAMOUT
OTA.SRC	OTA_B.CPGC	O04-UART_OUT
OTA.SRC	OTA_B.CPGC	O04-UART_OUT1
OTA.SRC	OTA_B.CPGC	O22-ARINC_Out_SubRoutine
PAGE0.SRC	E05-RAMINIT	E05-RAMINIT
PAGE0.SRC		Internal Interfaces
PAGE1.SRC	E05-RAMINIT	E05-RAMINIT
Page1.SRC		Internal Interfaces
PAGE10.SRC	E05-RAMINIT	E05-RAMINIT
PAGE10.SRC		Internal Interfaces
PAGE11.SRC	E05-RAMINIT	E05-RAMINIT
Page11.SRC		Internal Interfaces

Module	Design	SRS Capability
Page11.SRC		EEPROM_MAP
PASSED.SRC	PASSED_A.CPGC	E27-Maint_Misc_MFC_Id
PASSED.SRC	PASSED_A.CPGC	I08-Sens_LL_T18_RTD
PASSED.SRC	PASSED_A.CPGC	I14-Sens_HL_PLA
PASSED.SRC	PASSED_A.CPGC	I25-Sens_FR_Q_NPT_In
PASSTL.SRC	PASSTL_A.CPGC	E13-WATEST
PASSTL.SRC	PASSTL_A.CPGC	I01-Sens_LL_Drift_Comp
PASSTL.SRC	PASSTL_A.CPGC	I02-Sens_LL_Strain_Gauge_Comp
PASSTL.SRC	PASSTL_A.CPGC	I03-Sens_LL_Cold_Junct_Temp
PASSTL.SRC	PASSTL_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
PASSTL.SRC	PASSTL_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
PASSTL.SRC	PASSTL_A.CPGC	I07-Sens_LL_TQ_RTD
PASSTL.SRC	PASSTL_A.CPGC	I08-Sens_LL_T18_RTD
PASSTL.SRC	PASSTL_A.CPGC	I09-Sens_HL_Drift
PASSTL.SRC	PASSTL_A.CPGC	I10-Sens_HL_Gain_Comp
PASSTL.SRC	PASSTL_A.CPGC	I11-Sens_HL_Q_Bias_Trim
PASSTL.SRC	PASSTL_A.CPGC	I12-Sens_HL_Q_Gain_Trim
PASSTL.SRC	PASSTL_A.CPGC	I15-Sens_HL_PLA_Bias
PASSTL.SRC	PASSTL_A.CPGC	I16-Sens_HL_PLA_Gain
PASSTL.SRC	PASSTL_A.CPGC	I19-Sens_FR_Dual_Probe
PASSTL.SRC	PASSTL_A.CPGC	I45-Sens_HL_Bleed_W/A_Input
PID.SRC		E30 - IDENTIFI
QUAD.SRC	QUAD_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
QUAD.SRC	QUAD_B.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
QUAD.SRC	QUAD_C.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
RAMREAD0.SRC	RAMREAD0_A.CPGC	E02-BIT_Proc_Ram_Test
RAMREAD1.SRC	RAMSTOR1_A.CPGC	E02-BIT_Proc_Ram_Test
RAMSTOR0.SRC	RAMSTOR0_A.CPGC	E02-BIT_Proc_Ram_Test
RAMSTOR1.SRC	RAMSTOR1_A.CPGC	E02-BIT_Proc_Ram_Test
READ.SRC	READ_A.CPGC	E20-Maint_Dev_Test_Mode
ROMSUM.SRC	ROMSUM_A.CPGC	E07-Exec_Background
RTDWRITE.S00	RTDWRITE.CTRLFLOW	E01-BIT_Proc_Reset_PSR_POR
RUNTIMER.S00	EVENT_COUNTING	EVENT_COUNTING
SBIVAR.SRC	SBIVAR_A.CPGC	C02-PLATRIM_CALC
SBIVAR.SRC	SBIVAR_A.CPGC	C03 - TLIMRT_D
SBIVAR.SRC	SBIVAR_A.CPGC	C07-LIMIT_OVERQ_PROTECT
SBIVAR.SRC	SBIVAR_A.CPGC	C08-Gov_SHP_Closed_Loop
SBIVAR.SRC	SBIVAR_A.CPGC	C10-Out_Auto_Ignit
SBIVAR.SRC	SBIVAR_A.CPGC	C11-Gov_SHP_Req_Calc
SBIVAR.SRC	SBIVAR_A.CPGC	C12-Gov_NPT_Closed_Loop
SBIVAR.SRC	SBIVAR_A.CPGC	C13-Trans_Accel_NH
SBIVAR.SRC	SBIVAR_A.CPGC	C14-Limit_NH
SBIVAR.SRC	SBIVAR_A.CPGC	C16-Gov_Inner_Loop_Sel

Module	Design	SRS Capability
SBIVAR.SRC	SBIVAR_A.CPGC	C17-Gov_Inner_NH_Req_Calc
SBIVAR.SRC	SBIVAR_A.CPGC	C18-Gov_NH_Closed_Loop
SBIVAR.SRC	SBIVAR_A.CPGC	C19-Limit_Prop_Underspeed
SBIVAR.SRC	SBIVAR_A.CPGC	C20-Limit_Prop_Overspeed
SBIVAR.SRC	SBIVAR_A.CPGC	C21-OUT_HBOV_EEC_MODE
SBIVAR.SRC	SBIVAR_A.CPGC	C22-Out_HBOV_Man_Mode
SBIVAR.SRC	SBIVAR_A.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
SBIVAR.SRC	SBIVAR_A.CPGC	C26-MPARAM
SBIVAR.SRC	SBIVAR_A.CPGC	C27-NHMINMAX
SBIVAR.SRC	SBIVAR_A.CPGC	C30-Gov_Inner_SM_Rate_Req
SBIVAR.SRC	SBIVAR_A.CPGC	C31-Limit_Q
SBIVAR.SRC	SBIVAR_A.CPGC	I12-Sens_HL_Q_Gain_Trim
SBIVAR.SRC	SBIVAR_B.CPGC	C02-PLATRIM_CALC
SBIVAR.SRC	SBIVAR_B.CPGC	C03 - TLIMRT_D
SBIVAR.SRC	SBIVAR_B.CPGC	C07-LIMIT_OVERQ_PROTECT
SBIVAR.SRC	SBIVAR_B.CPGC	C08-Gov_SHP_Closed_Loop
SBIVAR.SRC	SBIVAR_B.CPGC	C10-Out_Auto_Ignit
SBIVAR.SRC	SBIVAR_B.CPGC	C11-Gov_SHP_Req_Calc
SBIVAR.SRC	SBIVAR_B.CPGC	C12-Gov_NPT_Closed_Loop
SBIVAR.SRC	SBIVAR_B.CPGC	C13-Trans_Accel_NH
SBIVAR.SRC	SBIVAR_B.CPGC	C14-Limit_NH
SBIVAR.SRC	SBIVAR_B.CPGC	C16-Gov_Inner_Loop_Sel
SBIVAR.SRC	SBIVAR_B.CPGC	C17-Gov_Inner_NH_Req_Calc
SBIVAR.SRC	SBIVAR_B.CPGC	C18-Gov_NH_Closed_Loop
SBIVAR.SRC	SBIVAR_B.CPGC	C19-Limit_Prop_Underspeed
SBIVAR.SRC	SBIVAR_B.CPGC	C20-Limit_Prop_Overspeed
SBIVAR.SRC	SBIVAR_B.CPGC	C21-OUT_HBOV_EEC_MODE
SBIVAR.SRC	SBIVAR_B.CPGC	C22-Out_HBOV_Man_Mode
SBIVAR.SRC	SBIVAR_B.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
SBIVAR.SRC	SBIVAR_B.CPGC	C26-MPARAM
SBIVAR.SRC	SBIVAR_B.CPGC	C27-NHMINMAX
SBIVAR.SRC	SBIVAR_B.CPGC	C30-Gov_Inner_SM_Rate_Req
SBIVAR.SRC	SBIVAR_B.CPGC	C31-Limit_Q
SBIVAR.SRC	SBIVAR_B.CPGC	I12-Sens_HL_Q_Gain_Trim
SBIVAR.SRC	SBIVAR_C.CPGC	C02-PLATRIM_CALC
SBIVAR.SRC	SBIVAR_C.CPGC	C03 - TLIMRT_D
SBIVAR.SRC	SBIVAR_C.CPGC	C05-Rating_Q_Bug_Calc
SBIVAR.SRC	SBIVAR_C.CPGC	C07-LIMIT_OVERQ_PROTECT
SBIVAR.SRC	SBIVAR_C.CPGC	C08-Gov_SHP_Closed_Loop
SBIVAR.SRC	SBIVAR_C.CPGC	C10-Out_Auto_Ignit
SBIVAR.SRC	SBIVAR_C.CPGC	C11-Gov_SHP_Req_Calc
SBIVAR.SRC	SBIVAR_C.CPGC	C12-Gov_NPT_Closed_Loop
SBIVAR.SRC	SBIVAR_C.CPGC	C13-Trans_Accel_NH

Module	Design	SRS Capability
SBIVAR.SRC	SBIVAR_C.CPGC	C14-Limit_NH
SBIVAR.SRC	SBIVAR_C.CPGC	C16-Gov_Inner_Loop_Sel
SBIVAR.SRC	SBIVAR_C.CPGC	C17-Gov_Inner_NH_Req_Calc
SBIVAR.SRC	SBIVAR_C.CPGC	C18-Gov_NH_Closed_Loop
SBIVAR.SRC	SBIVAR_C.CPGC	C19-Limit_Prop_Underspeed
SBIVAR.SRC	SBIVAR_C.CPGC	C20-Limit_Prop_Overspeed
SBIVAR.SRC	SBIVAR_C.CPGC	C21-OUT_HBOV_EEC_MODE
SBIVAR.SRC	SBIVAR_C.CPGC	C22-Out_HBOV_Man_Mode
SBIVAR.SRC	SBIVAR_C.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
SBIVAR.SRC	SBIVAR_C.CPGC	C26-MPARAM
SBIVAR.SRC	SBIVAR_C.CPGC	C27-NHMINMAX
SBIVAR.SRC	SBIVAR_C.CPGC	C30-Gov_Inner_SM_Rate_Req
SBIVAR.SRC	SBIVAR_C.CPGC	C31-Limit_Q
SBIVAR.SRC	SBIVAR_C.CPGC	I12-Sens_HL_Q_Gain_Trim
SBIVAR.SRC	SBIVAR_D.CPGC	C02-PLATRIM_CALC
SBIVAR.SRC	SBIVAR_D.CPGC	C03 - TLIMRT_D
SBIVAR.SRC	SBIVAR_D.CPGC	C05-Rating_Q_Bug_Calc
SBIVAR.SRC	SBIVAR_D.CPGC	C07-LIMIT_OVERQ_PROTECT
SBIVAR.SRC	SBIVAR_D.CPGC	C08-Gov_SHP_Closed_Loop
SBIVAR.SRC	SBIVAR_D.CPGC	C10-Out_Auto_Ignit
SBIVAR.SRC	SBIVAR_D.CPGC	C11-Gov_SHP_Req_Calc
SBIVAR.SRC	SBIVAR_D.CPGC	C12-Gov_NPT_Closed_Loop
SBIVAR.SRC	SBIVAR_D.CPGC	C13-Trans_Accel_NH
SBIVAR.SRC	SBIVAR_D.CPGC	C14-Limit_NH
SBIVAR.SRC	SBIVAR_D.CPGC	C16-Gov_Inner_Loop_Sel
SBIVAR.SRC	SBIVAR_D.CPGC	C17-Gov_Inner_NH_Req_Calc
SBIVAR.SRC	SBIVAR_D.CPGC	C18-Gov_NH_Closed_Loop
SBIVAR.SRC	SBIVAR_D.CPGC	C19-Limit_Prop_Underspeed
SBIVAR.SRC	SBIVAR_D.CPGC	C20-Limit_Prop_Overspeed
SBIVAR.SRC	SBIVAR_D.CPGC	C21-OUT_HBOV_EEC_MODE
SBIVAR.SRC	SBIVAR_D.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
SBIVAR.SRC	SBIVAR_D.CPGC	C26-MPARAM
SBIVAR.SRC	SBIVAR_D.CPGC	C27-NHMINMAX
SBIVAR.SRC	SBIVAR_D.CPGC	C30-Gov_Inner_SM_Rate_Req
SBIVAR.SRC	SBIVAR_D.CPGC	C31-Limit_Q
SBIVAR.SRC	SBIVAR_D.CPGC	I12-Sens_HL_Q_Gain_Trim
SBIVAR.SRC	SBIVAR_E.CPGC	C02-PLATRIM_CALC
SBIVAR.SRC	SBIVAR_E.CPGC	C03 - TLIMRT_D
SBIVAR.SRC	SBIVAR_E.CPGC	C05-Rating_Q_Bug_Calc
SBIVAR.SRC	SBIVAR_E.CPGC	C07-LIMIT_OVERQ_PROTECT
SBIVAR.SRC	SBIVAR_E.CPGC	C08-Gov_SHP_Closed_Loop
SBIVAR.SRC	SBIVAR_E.CPGC	C10-Out_Auto_Ignit
SBIVAR.SRC	SBIVAR_E.CPGC	C11-Gov_SHP_Req_Calc

Module	Design	SRS Capability
SBIVAR.SRC	SBIVAR_E.CPGC	C12-Gov_NPT_Closed_Loop
SBIVAR.SRC	SBIVAR_E.CPGC	C13-Trans_Accel_NH
SBIVAR.SRC	SBIVAR_E.CPGC	C14-Limit_NH
SBIVAR.SRC	SBIVAR_E.CPGC	C16-Gov_Inner_Loop_Sel
SBIVAR.SRC	SBIVAR_E.CPGC	C17-Gov_Inner_NH_Req_Calc
SBIVAR.SRC	SBIVAR_E.CPGC	C18-Gov_NH_Closed_Loop
SBIVAR.SRC	SBIVAR_E.CPGC	C19-Limit_Prop_Underspeed
SBIVAR.SRC	SBIVAR_E.CPGC	C20-Limit_Prop_Overspeed
SBIVAR.SRC	SBIVAR_E.CPGC	C21-OUT_HBOV_EEC_MODE
SBIVAR.SRC	SBIVAR_E.CPGC	C22-Out_HBOV_Man_Mode
SBIVAR.SRC	SBIVAR_E.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
SBIVAR.SRC	SBIVAR_E.CPGC	C26-MPARAM
SBIVAR.SRC	SBIVAR_E.CPGC	C27-NHMINMAX
SBIVAR.SRC	SBIVAR_E.CPGC	C30-Gov_Inner_SM_Rate_Req
SBIVAR.SRC	SBIVAR_E.CPGC	C31-Limit_Q
SBIVAR.SRC	SBIVAR_E.CPGC	I12-Sens_HL_Q_Gain_Trim
SBIVAR.SRC	SBIVAR_F.CPGC	C02-PLATRIM_CALC
SBIVAR.SRC	SBIVAR_F.CPGC	C03 - TLIMRT_D
SBIVAR.SRC	SBIVAR_F.CPGC	C05-Rating_Q_Bug_Calc
SBIVAR.SRC	SBIVAR_F.CPGC	C07-LIMIT_OVERQ_PROTECT
SBIVAR.SRC	SBIVAR_F.CPGC	C08-Gov_SHP_Closed_Loop
SBIVAR.SRC	SBIVAR_F.CPGC	C10-Out_Auto_Ignit
SBIVAR.SRC	SBIVAR_F.CPGC	C11-Gov_SHP_Req_Calc
SBIVAR.SRC	SBIVAR_F.CPGC	C12-Gov_NPT_Closed_Loop
SBIVAR.SRC	SBIVAR_F.CPGC	C13-Trans_Accel_NH
SBIVAR.SRC	SBIVAR_F.CPGC	C14-Limit_NH
SBIVAR.SRC	SBIVAR_F.CPGC	C16-Gov_Inner_Loop_Sel
SBIVAR.SRC	SBIVAR_F.CPGC	C17-Gov_Inner_NH_Req_Calc
SBIVAR.SRC	SBIVAR_F.CPGC	C18-Gov_NH_Closed_Loop
SBIVAR.SRC	SBIVAR_F.CPGC	C19-Limit_Prop_Underspeed
SBIVAR.SRC	SBIVAR_F.CPGC	C20-Limit_Prop_Overspeed
SBIVAR.SRC	SBIVAR_F.CPGC	C21-OUT_HBOV_EEC_MODE
SBIVAR.SRC	SBIVAR_F.CPGC	C22-Out_HBOV_Man_Mode
SBIVAR.SRC	SBIVAR_F.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
SBIVAR.SRC	SBIVAR_F.CPGC	C26-MPARAM
SBIVAR.SRC	SBIVAR_F.CPGC	C27-NHMINMAX
SBIVAR.SRC	SBIVAR_F.CPGC	C30-Gov_Inner_SM_Rate_Req
SBIVAR.SRC	SBIVAR_F.CPGC	C31-Limit_Q
SBIVAR.SRC	SBIVAR_F.CPGC	I12-Sens_HL_Q_Gain_Trim
SNDWRD.SRC	O01-Out_SM	O01-Out_SM
SNDWRD.SRC	SNDWRD_A.CPGC	E01-BIT_Proc_Reset_PSR_POR
SNDWRD.SRC	SNDWRD_A.CPGC	E02-BIT_Proc_Ram_Test
SNDWRD.SRC	SNDWRD_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum

Module	Design	SRS Capability
SNDWRD.SRC	SNDWRD_A.CPGC	E05-RAMINIT
SNDWRD.SRC	SNDWRD_A.CPGC	E06-BIT_GA_Q_Test_Routine
SNDWRD.SRC	SNDWRD_A.CPGC	E08 - EXEC_FOREGND
SNDWRD.SRC	SNDWRD_A.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
SNDWRD.SRC	SNDWRD_A.CPGC	E20-Maint_Dev_Test_Mode
SNDWRD.SRC	SNDWRD_A.CPGC	E22-BIT_GA_GateArraySelect
SNDWRD.SRC	SNDWRD_A.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
SNDWRD.SRC	SNDWRD_A.CPGC	I19-Sens_FR_Dual_Probe
SNDWRD.SRC	SNDWRD_A.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
SNDWRD.SRC	SNDWRD_A.CPGC	I25-Sens_FR_Q_NPT_In
SNDWRD.SRC	SNDWRD_A.CPGC	I28-DISINW1
SNDWRD.SRC	SNDWRD_A.CPGC	I29-DISINW2
SNDWRD.SRC	SNDWRD_A.CPGC	I30-BIT_EEC_Disc_In
SNDWRD.SRC	SNDWRD_A.CPGC	I31-NPARAMIN
SNDWRD.SRC	SNDWRD_A.CPGC	I31-UARTIN
SNDWRD.SRC	SNDWRD_A.CPGC	I32-ARINC_HW
SNDWRD.SRC	SNDWRD_A.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
SNDWRD.SRC	SNDWRD_A.CPGC	O01-Out_SM
SNDWRD.SRC	SNDWRD_A.CPGC	O02-HBOV
SNDWRD.SRC	SNDWRD_A.CPGC	O04-UART_OUT1
SNDWRD.SRC	SNDWRD_A.CPGC	O22-ARINC_Out_SubRoutine
SUNVAR.SRC	SUNVAR_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
SUNVAR.SRC	SUNVAR_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
SUNVAR.SRC	SUNVAR_A.CPGC	I08-Sens_LL_T18_RTD
SUNVAR.SRC	SUNVAR_A.CPGC	I34-ARINC_In_TAT_Source_Sel
SUNVAR.SRC	SUNVAR_A.CPGC	I35-ARINC_In_Press_Altitude
SUNVAR.SRC	SUNVAR_A.CPGC	I36-ARINC_In_Press_Source_Sel
SUNVAR.SRC	SUNVAR_B.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
SUNVAR.SRC	SUNVAR_B.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
SUNVAR.SRC	SUNVAR_B.CPGC	I08-Sens_LL_T18_RTD
SUNVAR.SRC	SUNVAR_B.CPGC	I34-ARINC_In_TAT_Source_Sel
SUNVAR.SRC	SUNVAR_B.CPGC	I35-ARINC_In_Press_Altitude
SUNVAR.SRC	SUNVAR_B.CPGC	I36-ARINC_In_Press_Source_Sel
SUNVAR.SRC	SUNVAR_C.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
SUNVAR.SRC	SUNVAR_C.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
SUNVAR.SRC	SUNVAR_C.CPGC	I08-Sens_LL_T18_RTD
SUNVAR.SRC	SUNVAR_C.CPGC	I34-ARINC_In_TAT_Source_Sel
SUNVAR.SRC	SUNVAR_C.CPGC	I35-ARINC_In_Press_Altitude
SUNVAR.SRC	SUNVAR_C.CPGC	I36-ARINC_In_Press_Source_Sel
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C02-PLATRIM_CALC
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C04 - PWRCALC_D
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C05-Rating_Q_Bug_Calc
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C19-Limit_Prop_Underspeed

Module	Design	SRS Capability
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C20-Limit_Prop_Overspeed
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C21-OUT_HBOV_EEC_MODE
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C22-Out_HBOV_Man_Mode
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C26-MPARAM
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	C27-NHMINMAX
TUSTIN_SUBR.SRC	TUSTIN_SUBR_A.CPGC	I12-Sens_HL_Q_Gain_Trim
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	C02-PLATRIM_CALC
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	C04 - PWRCALC_D
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	C19-Limit_Prop_Underspeed
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	C20-Limit_Prop_Overspeed
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	C21-OUT_HBOV_EEC_MODE
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	C22-Out_HBOV_Man_Mode
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	C26-MPARAM
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	C27-NHMINMAX
TUSTIN_SUBR.SRC	TUSTIN_SUBR_B.CPGC	I12-Sens_HL_Q_Gain_Trim
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C02-PLATRIM_CALC
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C04 - PWRCALC_D
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C05-Rating_Q_Bug_Calc
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C19-Limit_Prop_Underspeed
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C20-Limit_Prop_Overspeed
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C21-OUT_HBOV_EEC_MODE
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C22-Out_HBOV_Man_Mode
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C26-MPARAM
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	C27-NHMINMAX
TUSTIN_SUBR.SRC	TUSTIN_SUBR_C.CPGC	I12-Sens_HL_Q_Gain_Trim
UART_IO_SELECT.S00	UART_IO_SELECT.CTRLFLOW	UART_OUT_SEL
UNIVAR.SRC	UNIVAR_A.CPGC	C02-PLATRIM_CALC
UNIVAR.SRC	UNIVAR_A.CPGC	C03 - TLIMRT_D
UNIVAR.SRC	UNIVAR_A.CPGC	C05-Rating_Q_Bug_Calc
UNIVAR.SRC	UNIVAR_A.CPGC	C07-LIMIT_OVERQ_PROTECT
UNIVAR.SRC	UNIVAR_A.CPGC	C08-Gov_SHP_Closed_Loop
UNIVAR.SRC	UNIVAR_A.CPGC	C10-Out_Auto_Ignit
UNIVAR.SRC	UNIVAR_A.CPGC	C11-Gov_SHP_Req_Calc
UNIVAR.SRC	UNIVAR_A.CPGC	C12-Gov_NPT_Closed_Loop
UNIVAR.SRC	UNIVAR_A.CPGC	C13-Trans_Accel_NH
UNIVAR.SRC	UNIVAR_A.CPGC	C14-Limit_NH
UNIVAR.SRC	UNIVAR_A.CPGC	C16-Gov_Inner_Loop_Sel
UNIVAR.SRC	UNIVAR_A.CPGC	C17-Gov_Inner_NH_Req_Calc
UNIVAR.SRC	UNIVAR_A.CPGC	C18-Gov_NH_Closed_Loop
UNIVAR.SRC	UNIVAR_A.CPGC	C19-Limit_Prop_Underspeed
UNIVAR.SRC	UNIVAR_A.CPGC	C20-Limit_Prop_Overspeed
UNIVAR.SRC	UNIVAR_A.CPGC	C21-OUT_HBOV_EEC_MODE
UNIVAR.SRC	UNIVAR_A.CPGC	C22-Out_HBOV_Man_Mode

Module	Design	SRS Capability
UNIVAR.SRC	UNIVAR_A.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
UNIVAR.SRC	UNIVAR_A.CPGC	C26-MPARAM
UNIVAR.SRC	UNIVAR_A.CPGC	C27-NHMINMAX
UNIVAR.SRC	UNIVAR_A.CPGC	C30-Gov_Inner_SM_Rate_Req
UNIVAR.SRC	UNIVAR_A.CPGC	C31-Limit_Q
UNIVAR.SRC	UNIVAR_A.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
UNIVAR.SRC	UNIVAR_A.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
UNIVAR.SRC	UNIVAR_A.CPGC	I08-Sens_LL_T18_RTD
UNIVAR.SRC	UNIVAR_A.CPGC	I33-ARINC_In_TAT
UNIVAR.SRC	UNIVAR_A.CPGC	I34-ARINC_In_TAT_Source_Sel
UNIVAR.SRC	UNIVAR_A.CPGC	I35-ARINC_In_Press_Altitude
UNIVAR.SRC	UNIVAR_A.CPGC	I36-ARINC_In_Press_Source_Sel
UNIVAR.SRC	UNIVAR_B.CPGC	C02-PLATRIM_CALC
UNIVAR.SRC	UNIVAR_B.CPGC	C03 - TLIMRT_D
UNIVAR.SRC	UNIVAR_B.CPGC	C05-Rating_Q_Bug_Calc
UNIVAR.SRC	UNIVAR_B.CPGC	C07-LIMIT_OVERQ_PROTECT
UNIVAR.SRC	UNIVAR_B.CPGC	C08-Gov_SHP_Closed_Loop
UNIVAR.SRC	UNIVAR_B.CPGC	C10-Out_Auto_Ignit
UNIVAR.SRC	UNIVAR_B.CPGC	C11-Gov_SHP_Req_Calc
UNIVAR.SRC	UNIVAR_B.CPGC	C12-Gov_NPT_Closed_Loop
UNIVAR.SRC	UNIVAR_B.CPGC	C13-Trans_Accel_NH
UNIVAR.SRC	UNIVAR_B.CPGC	C14-Limit_NH
UNIVAR.SRC	UNIVAR_B.CPGC	C16-Gov_Inner_Loop_Sel
UNIVAR.SRC	UNIVAR_B.CPGC	C17-Gov_Inner_NH_Req_Calc
UNIVAR.SRC	UNIVAR_B.CPGC	C18-Gov_NH_Closed_Loop
UNIVAR.SRC	UNIVAR_B.CPGC	C19-Limit_Prop_Underspeed
UNIVAR.SRC	UNIVAR_B.CPGC	C20-Limit_Prop_Overspeed
UNIVAR.SRC	UNIVAR_B.CPGC	C21-OUT_HBOV_EEC_MODE
UNIVAR.SRC	UNIVAR_B.CPGC	C22-Out_HBOV_Man_Mode
UNIVAR.SRC	UNIVAR_B.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
UNIVAR.SRC	UNIVAR_B.CPGC	C26-MPARAM
UNIVAR.SRC	UNIVAR_B.CPGC	C27-NHMINMAX
UNIVAR.SRC	UNIVAR_B.CPGC	C30-Gov_Inner_SM_Rate_Req
UNIVAR.SRC	UNIVAR_B.CPGC	C31-Limit_Q
UNIVAR.SRC	UNIVAR_B.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
UNIVAR.SRC	UNIVAR_B.CPGC	I05-Sens_LL_DELTA_P_Strain_Gauge
UNIVAR.SRC	UNIVAR_B.CPGC	I08-Sens_LL_T18_RTD
UNIVAR.SRC	UNIVAR_B.CPGC	I33-ARINC_In_TAT
UNIVAR.SRC	UNIVAR_B.CPGC	I34-ARINC_In_TAT_Source_Sel
UNIVAR.SRC	UNIVAR_B.CPGC	I35-ARINC_In_Press_Altitude
UNIVAR.SRC	UNIVAR_B.CPGC	I36-ARINC_In_Press_Source_Sel
UNIVAR.SRC	UNIVAR_C.CPGC	C02-PLATRIM_CALC
UNIVAR.SRC	UNIVAR_C.CPGC	C03 - TLIMRT_D

Module	Design	SRS Capability
UNIVAR.SRC	UNIVAR_C.CPGC	C05-Rating_Q_Bug_Calc
UNIVAR.SRC	UNIVAR_C.CPGC	C07-LIMIT_OVERQ_PROTECT
UNIVAR.SRC	UNIVAR_C.CPGC	C08-Gov_SHP_Closed_Loop
UNIVAR.SRC	UNIVAR_C.CPGC	C10-Out_Auto_Ignit
UNIVAR.SRC	UNIVAR_C.CPGC	C11-Gov_SHP_Req_Calc
UNIVAR.SRC	UNIVAR_C.CPGC	C12-Gov_NPT_Closed_Loop
UNIVAR.SRC	UNIVAR_C.CPGC	C13-Trans_Accel_NH
UNIVAR.SRC	UNIVAR_C.CPGC	C14-Limit_NH
UNIVAR.SRC	UNIVAR_C.CPGC	C16-Gov_Inner_Loop_Sel
UNIVAR.SRC	UNIVAR_C.CPGC	C17-Gov_Inner_NH_Req_Calc
UNIVAR.SRC	UNIVAR_C.CPGC	C18-Gov_NH_Closed_Loop
UNIVAR.SRC	UNIVAR_C.CPGC	C19-Limit_Prop_Underspeed
UNIVAR.SRC	UNIVAR_C.CPGC	C20-Limit_Prop_Overspeed
UNIVAR.SRC	UNIVAR_C.CPGC	C21-OUT_HBOV_EEC_MODE
UNIVAR.SRC	UNIVAR_C.CPGC	C22-Out_HBOV_Man_Mode
UNIVAR.SRC	UNIVAR_C.CPGC	C24-BIT_PreFly_Eng_OverQ_Test
UNIVAR.SRC	UNIVAR_C.CPGC	C26-MPARAM
UNIVAR.SRC	UNIVAR_C.CPGC	C27-NHMINMAX
UNIVAR.SRC	UNIVAR_C.CPGC	C30-Gov_Inner_SM_Rate_Req
UNIVAR.SRC	UNIVAR_C.CPGC	C31-Limit_Q
UNIVAR.SRC	UNIVAR_C.CPGC	I04-Sens_LL_PSAMB_Strain_Gauge
UNIVAR.SRC	UNIVAR_C.CPGC	I08-Sens_LL_T18_RTD
UNIVAR.SRC	UNIVAR_C.CPGC	I33-ARINC_In_TAT
UNIVAR.SRC	UNIVAR_C.CPGC	I34-ARINC_In_TAT_Source_Sel
UNIVAR.SRC	UNIVAR_C.CPGC	I35-ARINC_In_Press_Altitude
UNIVAR.SRC	UNIVAR_C.CPGC	I36-ARINC_In_Press_Source_Sel
UPDATE_EE_DATA.S00	BLK305CHK	BLK305CHK
UPDATE_EE_DATA.S00	CYCLE_COUNTING	CYCLE_COUNTING
UPDATE_EE_DATA.S00		EECOUNTERS
URTOUT.SRC	URTOUT_A.CPGC	E01-BIT_Proc_Reset_PSR_POR
URTOUT.SRC	URTOUT_A.CPGC	E02-BIT_Proc_Ram_Test
URTOUT.SRC	URTOUT_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum
URTOUT.SRC	URTOUT_A.CPGC	E05-RAMINIT
URTOUT.SRC	URTOUT_A.CPGC	E06-BIT_GA_Q_Test_Routine
URTOUT.SRC	URTOUT_A.CPGC	E08 - EXEC_FOREGND
URTOUT.SRC	URTOUT_A.CPGC	E20-Maint_Dev_Test_Mode
URTOUT.SRC	URTOUT_A.CPGC	E22-BIT_GA_GateArraySelect
URTOUT.SRC	URTOUT_A.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
URTOUT.SRC	URTOUT_A.CPGC	I18-Sens_FR_Read_NH_Probe
URTOUT.SRC	URTOUT_A.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
URTOUT.SRC	URTOUT_A.CPGC	I23-Sens_FR_Read_NH_Probe_Two
URTOUT.SRC	URTOUT_A.CPGC	I25-Sens_FR_Q_NPT_In
URTOUT.SRC	URTOUT_A.CPGC	I28-DISINW1

Module	Design	SRS Capability
URTOUT.SRC	URTOUT_A.CPGC	I29-DISINW2
URTOUT.SRC	URTOUT_A.CPGC	I30-BIT_EEC_Disc_In
URTOUT.SRC	URTOUT_A.CPGC	I31-NPARAMIN
URTOUT.SRC	URTOUT_A.CPGC	I31-UARTIN
URTOUT.SRC	URTOUT_A.CPGC	I32-ARINC_HW
URTOUT.SRC	URTOUT_A.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
URTOUT.SRC	URTOUT_A.CPGC	O01-Out_SM
URTOUT.SRC	URTOUT_A.CPGC	O02-HBOV
URTOUT.SRC	URTOUT_A.CPGC	O04 - NPARAMOUT
URTOUT.SRC	URTOUT_A.CPGC	O04-UART_OUT
URTOUT.SRC	URTOUT_A.CPGC	O04-UART_OUT1
URTOUT.SRC	URTOUT_A.CPGC	O22-ARINC_Out_SubRoutine
URTOUT.SRC	URTOUT_B.CPGC	E01-BIT_Proc_Reset_PSR_POR
URTOUT.SRC	URTOUT_B.CPGC	E02-BIT_Proc_Ram_Test
URTOUT.SRC	URTOUT_B.CPGC	E03-BIT_Proc_ROM_EE_Chksum
URTOUT.SRC	URTOUT_B.CPGC	E05-RAMINIT
URTOUT.SRC	URTOUT_B.CPGC	E06-BIT_GA_Q_Test_Routine
URTOUT.SRC	URTOUT_B.CPGC	E08 - EXEC_FOREGND
URTOUT.SRC	URTOUT_B.CPGC	E20-Maint_Dev_Test_Mode
URTOUT.SRC	URTOUT_B.CPGC	E22-BIT_GA_GateArraySelect
URTOUT.SRC	URTOUT_B.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
URTOUT.SRC	URTOUT_B.CPGC	I18-Sens_FR_Read_NH_Probe
URTOUT.SRC	URTOUT_B.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
URTOUT.SRC	URTOUT_B.CPGC	I23-Sens_FR_Read_NH_Probe_Two
URTOUT.SRC	URTOUT_B.CPGC	I25-Sens_FR_Q_NPT_In
URTOUT.SRC	URTOUT_B.CPGC	I28-DISINW1
URTOUT.SRC	URTOUT_B.CPGC	I29-DISINW2
URTOUT.SRC	URTOUT_B.CPGC	I30-BIT_EEC_Disc_In
URTOUT.SRC	URTOUT_B.CPGC	I31-NPARAMIN
URTOUT.SRC	URTOUT_B.CPGC	I31-UARTIN
URTOUT.SRC	URTOUT_B.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
URTOUT.SRC	URTOUT_B.CPGC	O01-Out_SM
URTOUT.SRC	URTOUT_B.CPGC	O02-HBOV
URTOUT.SRC	URTOUT_B.CPGC	O04 - NPARAMOUT
URTOUT.SRC	URTOUT_B.CPGC	O04-UART_OUT
URTOUT.SRC	URTOUT_B.CPGC	O04-UART_OUT1
URTOUT.SRC	URTOUT_B.CPGC	O22-ARINC_Out_SubRoutine
URTOUT.SRC	URTRED_B.CPGC	E03-BIT_Proc_ROM_EE_Chksum
URTRED.SRC	URTRED_A.CPGC	E01-BIT_Proc_Reset_PSR_POR
URTRED.SRC	URTRED_A.CPGC	E02-BIT_Proc_Ram_Test
URTRED.SRC	URTRED_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum
URTRED.SRC	URTRED_A.CPGC	E05-RAMINIT
URTRED.SRC	URTRED_A.CPGC	E06-BIT_GA_Q_Test_Routine

Module	Design	SRS Capability
URTRED.SRC	URTRED_A.CPGC	E08 - EXEC_FOREGND
URTRED.SRC	URTRED_A.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
URTRED.SRC	URTRED_A.CPGC	E20-Maint_Dev_Test_Mode
URTRED.SRC	URTRED_A.CPGC	E22-BIT_GA_GateArraySelect
URTRED.SRC	URTRED_A.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
URTRED.SRC	URTRED_A.CPGC	I18-Sens_FR_Read_NH_Probe
URTRED.SRC	URTRED_A.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
URTRED.SRC	URTRED_A.CPGC	I23-Sens_FR_Read_NH_Probe_Two
URTRED.SRC	URTRED_A.CPGC	I25-Sens_FR_Q_NPT_In
URTRED.SRC	URTRED_A.CPGC	I28-DISINW1
URTRED.SRC	URTRED_A.CPGC	I29-DISINW2
URTRED.SRC	URTRED_A.CPGC	I30-BIT_EEC_Disc_In
URTRED.SRC	URTRED_A.CPGC	I31-NPARAMIN
URTRED.SRC	URTRED_A.CPGC	I31-UARTIN
URTRED.SRC	URTRED_A.CPGC	I32-ARINC_HW
URTRED.SRC	URTRED_A.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
URTRED.SRC	URTRED_A.CPGC	O01-Out_SM
URTRED.SRC	URTRED_A.CPGC	O02-HBOV
URTRED.SRC	URTRED_A.CPGC	O04 - NPARAMOUT
URTRED.SRC	URTRED_A.CPGC	O04-UART_OUT
URTRED.SRC	URTRED_A.CPGC	O04-UART_OUT1
URTRED.SRC	URTRED_A.CPGC	O22-ARINC_Out_SubRoutine
URTRED.SRC	URTRED_B.CPGC	E01-BIT_Proc_Reset_PSR_POR
URTRED.SRC	URTRED_B.CPGC	E02-BIT_Proc_Ram_Test
URTRED.SRC	URTRED_B.CPGC	E05-RAMINIT
URTRED.SRC	URTRED_B.CPGC	E06-BIT_GA_Q_Test_Routine
URTRED.SRC	URTRED_B.CPGC	E08 - EXEC_FOREGND
URTRED.SRC	URTRED_B.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
URTRED.SRC	URTRED_B.CPGC	E20-Maint_Dev_Test_Mode
URTRED.SRC	URTRED_B.CPGC	E22-BIT_GA_GateArraySelect
URTRED.SRC	URTRED_B.CPGC	I17-Sens_FR_Start_Pri_NH_Convert
URTRED.SRC	URTRED_B.CPGC	I18-Sens_FR_Read_NH_Probe
URTRED.SRC	URTRED_B.CPGC	I22-Sens_FR_Start_Sec_NH_Convert
URTRED.SRC	URTRED_B.CPGC	I23-Sens_FR_Read_NH_Probe_Two
URTRED.SRC	URTRED_B.CPGC	I25-Sens_FR_Q_NPT_In
URTRED.SRC	URTRED_B.CPGC	I28-DISINW1
URTRED.SRC	URTRED_B.CPGC	I29-DISINW2
URTRED.SRC	URTRED_B.CPGC	I30-BIT_EEC_Disc_In
URTRED.SRC	URTRED_B.CPGC	I31-NPARAMIN
URTRED.SRC	URTRED_B.CPGC	I31-UARTIN
URTRED.SRC	URTRED_B.CPGC	I32-ARINC_HW
URTRED.SRC	URTRED_B.CPGC	I38-Sens_HL_SM_Coil_Sum_Volt
URTRED.SRC	URTRED_B.CPGC	O01-Out_SM

Module	Design	SRS Capability
URTRED.SRC	URTRED_B.CPGC	O02-HBOV
URTRED.SRC	URTRED_B.CPGC	O04 - NPARAMOUT
URTRED.SRC	URTRED_B.CPGC	O04-UART_OUT
URTRED.SRC	URTRED_B.CPGC	O04-UART_OUT1
URTRED.SRC	URTRED_B.CPGC	O22-ARINC_Out_SubRoutine
VAR_TUSTIN_SUBR.SRC	VAR_TUSTIN_SUBR_A.CPGC	C19-Limit_Prop_Underspeed
VAR_TUSTIN_SUBR.SRC	VAR_TUSTIN_SUBR_A.CPGC	C26-MPARAM
VAR_TUSTIN_SUBR.SRC	VAR_TUSTIN_SUBR_B.CPGC	C19-Limit_Prop_Underspeed
VAR_TUSTIN_SUBR.SRC	VAR_TUSTIN_SUBR_B.CPGC	C26-MPARAM
VAR_TUSTIN_SUBR.SRC	VAR_TUSTIN_SUBR_C.CPGC	C19-Limit_Prop_Underspeed
VAR_TUSTIN_SUBR.SRC	VAR_TUSTIN_SUBR_C.CPGC	C26-MPARAM
WDGRST.SRC	WDGRST_A.CPGC	E01-BIT_Proc_Reset_PSR_POR
WDGRST.SRC	WDGRST_A.CPGC	E02-BIT_Proc_Ram_Test
WDGRST.SRC	WDGRST_A.CPGC	E03-BIT_Proc_ROM_EE_Chksum
WDGRST.SRC	WDGRST_A.CPGC	E05-RAMINIT
WDGRST.SRC	WDGRST_A.CPGC	E06-BIT_GA_Q_Test_Routine
WDGRST.SRC	WDGRST_A.CPGC	E09-BIT_EEC_Init_SM_Sum_Volt
WDGRST.SRC	WDGRST_A.CPGC	E20-Maint_Dev_Test_Mode
WDGRST.SRC	WDGRST_A.CPGC	E22-BIT_GA_GateArraySelect
WRITE.SRC	WRITE_A.CPGC	E20-Maint_Dev_Test_Mode
WRITE.SRC	WRITE_B.CPGC	E20-Maint_Dev_Test_Mode
WRITE.SRC	WRITE_C.CPGC	E20-Maint_Dev_Test_Mode
WRTEEP.SRC	WRTEEP_A.CPGC	E20-Maint_Dev_Test_Mode
WRTEEP.SRC	WRTEEP_B.CPGC	E20-Maint_Dev_Test_Mode
WRTEEP.SRC	WRTEEP_C.CPGC	E20-Maint_Dev_Test_Mode

06.02 RESERVED

06.03 IMMATURE LLRS (DOCUMENTATION ONLY)

The following table represents LLRs that exist for documentation purposes and are considered immature. These LLRs have not been developed and verified to DO-178C standards.

Column 1 identifies the Immature LLR.

Column 2 identifies the SDD Section in which the Immature LLR resides.

This table is sorted first by column 1, and then by column 2, each in ascending order.

06.03 Table 1: IMMATURE LLRS

LLRs	SDD Section
LLR-12490077.000	E17 - NONCRITICAL_FAULT
LLR-12490078.000	E17 - NONCRITICAL_FAULT
LLR-12490079.000	E17 - NONCRITICAL_FAULT
LLR-12490080.000	E17 - NONCRITICAL_FAULT
LLR-12490081.000	E17 - NONCRITICAL_FAULT
LLR-12490082.000	E17 - NONCRITICAL_FAULT
LLR-12490083.000	E17 - NONCRITICAL_FAULT
LLR-12490087.000	E17 - NONCRITICAL_FAULT
LLR-12490088.000	E17 - NONCRITICAL_FAULT
LLR-12490089.000	E17 - NONCRITICAL_FAULT
LLR-12490090.000	E17 - NONCRITICAL_FAULT
LLR-12490091.000	E17 - NONCRITICAL_FAULT
LLR-12490092.000	E17 - NONCRITICAL_FAULT
LLR-12490111.000	E19-BIT_Store_EEPROM_Trim_Data
LLR-12490112.000	E19-BIT_Store_EEPROM_Trim_Data
LLR-12490113.000	E19-BIT_Store_EEPROM_Trim_Data
LLR-12490114.000	E19-BIT_Store_EEPROM_Trim_Data
LLR-12490181.000	C26-MPARAM
LLR-12490182.000	C26-MPARAM
LLR-12490183.000	C26-MPARAM
LLR-12490184.000	C26-MPARAM
LLR-12490185.000	C26-MPARAM
LLR-12490186.000	C26-MPARAM
LLR-12490190.000	O03 - OUT_DISC
LLR-12490197.000	UART_OUT_SEL
LLR-12490321.000	DCUREAD
LLR-12490368.000	DCUWRITE

07.00 SRS TO SDD TO CODE TRACEABILITY

07.01 SRS CAPABILITY TO DESIGN TO MODULE TRACEABILITY (SORTED BY CAPABILITY)

The following table represents the SRS Capability to Design Model to Module Traceability.

Column 1 identifies the SRS Capability.

Column 2 identifies the Design that traces to the HLR and SRS Capability. The Design will be one of the following:

- Design (if there is a Design model, either .CTRLFLOW, .DATAFLOW, or .CPGC will be identified)
- Blank
- Otherwise, Design section/capability is identified

Column 3 identifies the Source Code Module that traces to the Design.

Note: A blank entry in this table indicates an item as described above does not exist.

This table is sorted first by column 1, then by column 2, then by column 3, each in ascending order.

07.01 Table 1: SRS _DESIGN _MODULE

SRS Capability	Design	Module
BLK00CHK	BLK00CHK	DCU_COMPARE_DATA.S00
BLK00CHK	BLK00CHK	DCU_COMPARE_ITEM.S00
BLK00CHK	BLK00CHK	DCU_CONSTANTS.S00
BLK00CHK	BLK00CHK	DCU_ENGINE_DATA.S00
BLK00CHK	BLK00CHK	DCU_GET_DATA.S00
BLK00CHK	BLK00CHK	DCU_SCHEDULE.S00
BLK00CHK		E08_FG_SCHEDULE.SRC
BLK00CHK	BLK00CHK	EE_GET_QUEUE.S00
BLK00CHK	BLK00CHK	EE_PUT_QUEUE.S00
BLK00CHK	BLK00CHK	EE_WRITE_BLOCK.S00
BLK01CHK	BLK01CHK	DCU_COMPARE_DATA.S00
BLK01CHK	BLK01CHK	DCU_COMPARE_ITEM.S00
BLK01CHK	BLK01CHK	DCU_COMPONT_DATA.S00
BLK01CHK	BLK01CHK	DCU_CONSTANTS.S00
BLK01CHK	BLK01CHK	DCU_GET_DATA.S00
BLK01CHK	BLK01CHK	DCU_SCHEDULE.S00
BLK01CHK		E08_FG_SCHEDULE.SRC
BLK01CHK	BLK01CHK	EE_GET_QUEUE.S00
BLK01CHK	BLK01CHK	EE_PUT_QUEUE.S00
BLK01CHK	BLK01CHK	EE_WRITE_BLOCK.S00
BLK09CHK	BLK09CHK	DCU_COMPARE_ITEM.S00
BLK09CHK	BLK09CHK	DCU_CONSTANTS.S00
BLK09CHK	BLK09CHK	DCU_EEC_DATA.S00
BLK09CHK	BLK09CHK	DCU_SCHEDULE.S00
BLK09CHK		E08_FG_SCHEDULE.SRC
BLK09CHK	BLK09CHK	E08_FG_SCHEDULE.SRC
BLK304CHK		DCU_COMPARE_DATA.S00
BLK304CHK		DCU_COMPARE_ITEM.S00
BLK304CHK		DCU_CONSTANTS.S00
BLK304CHK		DCU_DCUID_DATA.S00
BLK304CHK		DCU_GET_CHECKSUM.S00
BLK304CHK		E08_FG_SCHEDULE.SRC
BLK305CHK		DCU_CONSTANTS.S00
BLK305CHK		DCU_COUNTER_DATA.S00
BLK305CHK	BLK305CHK	DCU_GET_DATA.S00
BLK305CHK	DCU_IS_OBSOLETE_A.CPGC	DCU_IS_OBSOLETE.SRC
BLK305CHK	DCU_IS_OBSOLETE_B.CPGC	DCU_IS_OBSOLETE.SRC
BLK305CHK	DCU_IS_OBSOLETE_C.CPGC	DCU_IS_OBSOLETE.SRC
BLK305CHK	DCU_IS_OBSOLETE_D.CPGC	DCU_IS_OBSOLETE.SRC
BLK305CHK	DCU_IS_OBSOLETE_E.CPGC	DCU_IS_OBSOLETE.SRC
BLK305CHK	DCU_IS_OBSOLETE_F.CPGC	DCU_IS_OBSOLETE.SRC
BLK305CHK	DCU_RANGE_CHECKS.CPGC	DCU_RANGE_CHECKS.SRC
BLK305CHK	BLK305CHK	DCU_SCHEDULE.S00
BLK305CHK		E08_FG_SCHEDULE.SRC
BLK305CHK	BLK305CHK	EE_GET_QUEUE.S00
BLK305CHK	BLK305CHK	EE_PUT_QUEUE.S00

SRS Capability	Design	Module
BLK305CHK	BLK305CHK	EE_WRITE_BLOCK.S00
BLK305CHK	BLK305CHK	UPDATE_EE_DATA.S00
C01-Start	C01_START.DATAFLOW	C01_START.S00
C01-Start		E08_FG_SCHEDULE.SRC
C01-Start	C01-Start	E08_FG_SCHEDULE.SRC
C02-PLATRIM_CALC	C02_PLATRIM_1.DATAFLOW	C02_PLATRIM_1.S00
C02-PLATRIM_CALC	C02_PLATRIM_2.DATAFLOW	C02_PLATRIM_2.SRC
C02-PLATRIM_CALC	SBIVAR_A.CPGC	SBIVAR.SRC
C02-PLATRIM_CALC	SBIVAR_B.CPGC	SBIVAR.SRC
C02-PLATRIM_CALC	SBIVAR_C.CPGC	SBIVAR.SRC
C02-PLATRIM_CALC	SBIVAR_D.CPGC	SBIVAR.SRC
C02-PLATRIM_CALC	SBIVAR_E.CPGC	SBIVAR.SRC
C02-PLATRIM_CALC	SBIVAR_F.CPGC	SBIVAR.SRC
C02-PLATRIM_CALC	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C02-PLATRIM_CALC	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
C02-PLATRIM_CALC	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
C02-PLATRIM_CALC	UNIVAR_A.CPGC	UNIVAR.SRC
C02-PLATRIM_CALC	UNIVAR_B.CPGC	UNIVAR.SRC
C02-PLATRIM_CALC	UNIVAR_C.CPGC	UNIVAR.SRC
C03 - TLIMRT_D	C03_TLIMRT_1.CTRLFLOW	C03_TLIMRT_1.SRC
C03 - TLIMRT_D	C03_TLIMRT_2_1.CTRLFLOW	C03_TLIMRT_2_1.SRC
C03 - TLIMRT_D	C03_TLIMRT_2_2.CTRLFLOW	C03_TLIMRT_2_2.SRC
C03 - TLIMRT_D	C03_TLIMRT_3_1.CTRLFLOW	C03_TLIMRT_3_1.SRC
C03 - TLIMRT_D	C03_TLIMRT_3_2.CTRLFLOW	C03_TLIMRT_3_2.SRC
C03 - TLIMRT_D	C03_TLIMRT_4_1.CTRLFLOW	C03_TLIMRT_4_1.SRC
C03 - TLIMRT_D	C03_TLIMRT_4_2.CTRLFLOW	C03_TLIMRT_4_2.SRC
C03 - TLIMRT_D	C03_TLIMRT_5_1.CTRLFLOW	C03_TLIMRT_5_1.SRC
C03 - TLIMRT_D	C03_TLIMRT_5_2.CTRLFLOW	C03_TLIMRT_5_2.SRC
C03 - TLIMRT_D	C03_TLIMRT_6_1.CTRLFLOW	C03_TLIMRT_6_1.SRC
C03 - TLIMRT_D	C03_TLIMRT_6_2.CTRLFLOW	C03_TLIMRT_6_2.SRC
C03 - TLIMRT_D	C03_TLIMRT_7.DATAFLOW	C03_TLIMRT_7.SRC
C03 - TLIMRT_D	DIVPRT_A.CPGC	DIVPRT.SRC
C03 - TLIMRT_D	DPA_A.CPGC	DPA.SRC
C03 - TLIMRT_D	DPA_B.CPGC	DPA.SRC
C03 - TLIMRT_D	DPM_A.CPGC	DPM.SRC
C03 - TLIMRT_D	DPM_B.CPGC	DPM.SRC
C03 - TLIMRT_D	DPM_C.CPGC	DPM.SRC
C03 - TLIMRT_D	DPM_D.CPGC	DPM.SRC
C03 - TLIMRT_D	DPM_E.CPGC	DPM.SRC
C03 - TLIMRT_D	C03_TLIMRT_6_D.DATAFLOW	E08_FG_SCHEDULE.SRC
C03 - TLIMRT_D		E08_FG_SCHEDULE.SRC
C03 - TLIMRT_D	SBIVAR_A.CPGC	SBIVAR.SRC
C03 - TLIMRT_D	SBIVAR_B.CPGC	SBIVAR.SRC
C03 - TLIMRT_D	SBIVAR_C.CPGC	SBIVAR.SRC
C03 - TLIMRT_D	SBIVAR_D.CPGC	SBIVAR.SRC
C03 - TLIMRT_D	SBIVAR_E.CPGC	SBIVAR.SRC

SRS Capability	Design	Module
C03 - TLIMRT_D	SBIVAR_F.CPGC	SBIVAR.SRC
C03 - TLIMRT_D	UNIVAR_A.CPGC	UNIVAR.SRC
C03 - TLIMRT_D	UNIVAR_B.CPGC	UNIVAR.SRC
C03 - TLIMRT_D	UNIVAR_C.CPGC	UNIVAR.SRC
C04 - PWRCALC_D	C04_RPCAL_2_D.DATAFLOW	C04_RPCAL_2_D.SRC
C04 - PWRCALC_D		E08_FG_SCHEDULE.SRC
C04 - PWRCALC_D	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C04 - PWRCALC_D	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
C04 - PWRCALC_D	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
C05-Rating_Q_Bug_Calc	C05_TQBUG.DATAFLOW	C05_TQBUG.SRC
C05-Rating_Q_Bug_Calc	I12_TQGTRIM_1_A.CPGC	I12_TQGTRIM_1.SRC
C05-Rating_Q_Bug_Calc	I12_TQGTRIM_1_B.CPGC	I12_TQGTRIM_1.SRC
C05-Rating_Q_Bug_Calc	I12_TQGTRIM_1_C.CPGC	I12_TQGTRIM_1.SRC
C05-Rating_Q_Bug_Calc	SBIVAR_C.CPGC	SBIVAR.SRC
C05-Rating_Q_Bug_Calc	SBIVAR_D.CPGC	SBIVAR.SRC
C05-Rating_Q_Bug_Calc	SBIVAR_E.CPGC	SBIVAR.SRC
C05-Rating_Q_Bug_Calc	SBIVAR_F.CPGC	SBIVAR.SRC
C05-Rating_Q_Bug_Calc	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C05-Rating_Q_Bug_Calc	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
C05-Rating_Q_Bug_Calc	UNIVAR_A.CPGC	UNIVAR.SRC
C05-Rating_Q_Bug_Calc	UNIVAR_B.CPGC	UNIVAR.SRC
C05-Rating_Q_Bug_Calc	UNIVAR_C.CPGC	UNIVAR.SRC
C06-POWER_REQUEST	C06_HPREF.DATAFLOW	C06_HPREF.SRC
C06-POWER_REQUEST	C06_HPREF_1.DATAFLOW	C06_HPREF_1.SRC
C06-POWER_REQUEST		E08_FG_SCHEDULE.SRC
C07-LIMIT_OVERQ_PROTECT	C07_OVRTQPRO.DATAFLOW	C07_OVRTQPRO.SRC
C07-LIMIT_OVERQ_PROTECT	DIVPRT_A.CPGC	DIVPRT.SRC
C07-LIMIT_OVERQ_PROTECT	DIVPRT_B.CPGC	DIVPRT.SRC
C07-LIMIT_OVERQ_PROTECT	DPA_A.CPGC	DPA.SRC
C07-LIMIT_OVERQ_PROTECT	DPA_B.CPGC	DPA.SRC
C07-LIMIT_OVERQ_PROTECT	DPM_A.CPGC	DPM.SRC
C07-LIMIT_OVERQ_PROTECT	DPM_B.CPGC	DPM.SRC
C07-LIMIT_OVERQ_PROTECT	DPM_C.CPGC	DPM.SRC
C07-LIMIT_OVERQ_PROTECT	DPM_D.CPGC	DPM.SRC
C07-LIMIT_OVERQ_PROTECT	DPM_E.CPGC	DPM.SRC
C07-LIMIT_OVERQ_PROTECT		E08_FG_SCHEDULE.SRC
C07-LIMIT_OVERQ_PROTECT	SBIVAR_A.CPGC	SBIVAR.SRC
C07-LIMIT_OVERQ_PROTECT	SBIVAR_B.CPGC	SBIVAR.SRC
C07-LIMIT_OVERQ_PROTECT	SBIVAR_C.CPGC	SBIVAR.SRC
C07-LIMIT_OVERQ_PROTECT	SBIVAR_D.CPGC	SBIVAR.SRC
C07-LIMIT_OVERQ_PROTECT	SBIVAR_E.CPGC	SBIVAR.SRC
C07-LIMIT_OVERQ_PROTECT	SBIVAR_F.CPGC	SBIVAR.SRC
C07-LIMIT_OVERQ_PROTECT	UNIVAR_A.CPGC	UNIVAR.SRC
C07-LIMIT_OVERQ_PROTECT	UNIVAR_B.CPGC	UNIVAR.SRC
C07-LIMIT_OVERQ_PROTECT	UNIVAR_C.CPGC	UNIVAR.SRC
C08-Gov_SHP_Closed_Loop	C08_SHPCL.DATAFLOW	C08_SHPCL.SRC

SRS Capability	Design	Module
C08-Gov_SHP_Closed_Loop	DPA_A.CPGC	DPA.SRC
C08-Gov_SHP_Closed_Loop	DPA_B.CPGC	DPA.SRC
C08-Gov_SHP_Closed_Loop	SBIVAR_A.CPGC	SBIVAR.SRC
C08-Gov_SHP_Closed_Loop	SBIVAR_B.CPGC	SBIVAR.SRC
C08-Gov_SHP_Closed_Loop	SBIVAR_C.CPGC	SBIVAR.SRC
C08-Gov_SHP_Closed_Loop	SBIVAR_D.CPGC	SBIVAR.SRC
C08-Gov_SHP_Closed_Loop	SBIVAR_E.CPGC	SBIVAR.SRC
C08-Gov_SHP_Closed_Loop	SBIVAR_F.CPGC	SBIVAR.SRC
C08-Gov_SHP_Closed_Loop	UNIVAR_A.CPGC	UNIVAR.SRC
C08-Gov_SHP_Closed_Loop	UNIVAR_B.CPGC	UNIVAR.SRC
C08-Gov_SHP_Closed_Loop	UNIVAR_C.CPGC	UNIVAR.SRC
C10-Out_Auto_Ignit	C10_ATOIGN.DATAFLOW	C10_ATOIGN.SRC
C10-Out_Auto_Ignit	DIVPRT_A.CPGC	DIVPRT.SRC
C10-Out_Auto_Ignit	DIVPRT_B.CPGC	DIVPRT.SRC
C10-Out_Auto_Ignit	DPA_A.CPGC	DPA.SRC
C10-Out_Auto_Ignit	DPM_A.CPGC	DPM.SRC
C10-Out_Auto_Ignit	DPM_B.CPGC	DPM.SRC
C10-Out_Auto_Ignit	DPM_D.CPGC	DPM.SRC
C10-Out_Auto_Ignit	DPM_E.CPGC	DPM.SRC
C10-Out_Auto_Ignit	DPM_V.CPGC	DPM.SRC
C10-Out_Auto_Ignit	E07_BG_EXECUTIVE_A.CPGC	E07_BG_EXECUTIVE.SRC
C10-Out_Auto_Ignit	SBIVAR_A.CPGC	SBIVAR.SRC
C10-Out_Auto_Ignit	SBIVAR_B.CPGC	SBIVAR.SRC
C10-Out_Auto_Ignit	SBIVAR_C.CPGC	SBIVAR.SRC
C10-Out_Auto_Ignit	SBIVAR_D.CPGC	SBIVAR.SRC
C10-Out_Auto_Ignit	SBIVAR_E.CPGC	SBIVAR.SRC
C10-Out_Auto_Ignit	SBIVAR_F.CPGC	SBIVAR.SRC
C10-Out_Auto_Ignit	UNIVAR_A.CPGC	UNIVAR.SRC
C10-Out_Auto_Ignit	UNIVAR_B.CPGC	UNIVAR.SRC
C10-Out_Auto_Ignit	UNIVAR_C.CPGC	UNIVAR.SRC
C11-Gov_SHP_Req_Calc	C11_SHPREQ_1.DATAFLOW	C11_SHPREQ_1.SRC
C11-Gov_SHP_Req_Calc	C11_SHPREQ_2.DATAFLOW	C11_SHPREQ_2.SRC
C11-Gov_SHP_Req_Calc	DERLAG_A.CPGC	DERLAG.SRC
C11-Gov_SHP_Req_Calc	DERLAG_B.CPGC	DERLAG.SRC
C11-Gov_SHP_Req_Calc	DPA_A.CPGC	DPA.SRC
C11-Gov_SHP_Req_Calc	DPA_B.CPGC	DPA.SRC
C11-Gov_SHP_Req_Calc	SBIVAR_A.CPGC	SBIVAR.SRC
C11-Gov_SHP_Req_Calc	SBIVAR_B.CPGC	SBIVAR.SRC
C11-Gov_SHP_Req_Calc	SBIVAR_C.CPGC	SBIVAR.SRC
C11-Gov_SHP_Req_Calc	SBIVAR_D.CPGC	SBIVAR.SRC
C11-Gov_SHP_Req_Calc	SBIVAR_E.CPGC	SBIVAR.SRC
C11-Gov_SHP_Req_Calc	SBIVAR_F.CPGC	SBIVAR.SRC
C11-Gov_SHP_Req_Calc	UNIVAR_A.CPGC	UNIVAR.SRC
C11-Gov_SHP_Req_Calc	UNIVAR_B.CPGC	UNIVAR.SRC
C11-Gov_SHP_Req_Calc	UNIVAR_C.CPGC	UNIVAR.SRC
C12-Gov_NPT_Closed_Loop	C12_NPTGOV_1.DATAFLOW	C12_NPTGOV_1.SRC

SRS Capability	Design	Module
C12-Gov_NPT_Closed_Loop	C12_NPTGOV_1.DATAFLOW	C12_NPTGOV_2.SRC
C12-Gov_NPT_Closed_Loop	DERLAG_A.CPGC	DERLAG.SRC
C12-Gov_NPT_Closed_Loop	DERLAG_B.CPGC	DERLAG.SRC
C12-Gov_NPT_Closed_Loop	DIVPRT_A.CPGC	DIVPRT.SRC
C12-Gov_NPT_Closed_Loop	DIVPRT_B.CPGC	DIVPRT.SRC
C12-Gov_NPT_Closed_Loop	DPA_A.CPGC	DPA.SRC
C12-Gov_NPT_Closed_Loop	DPA_B.CPGC	DPA.SRC
C12-Gov_NPT_Closed_Loop	DPM_A.CPGC	DPM.SRC
C12-Gov_NPT_Closed_Loop	DPM_B.CPGC	DPM.SRC
C12-Gov_NPT_Closed_Loop	DPM_C.CPGC	DPM.SRC
C12-Gov_NPT_Closed_Loop	DPM_D.CPGC	DPM.SRC
C12-Gov_NPT_Closed_Loop	DPM_E.CPGC	DPM.SRC
C12-Gov_NPT_Closed_Loop		E08_FG_SCHEDULE.SRC
C12-Gov_NPT_Closed_Loop	SBIVAR_A.CPGC	SBIVAR.SRC
C12-Gov_NPT_Closed_Loop	SBIVAR_B.CPGC	SBIVAR.SRC
C12-Gov_NPT_Closed_Loop	SBIVAR_C.CPGC	SBIVAR.SRC
C12-Gov_NPT_Closed_Loop	SBIVAR_D.CPGC	SBIVAR.SRC
C12-Gov_NPT_Closed_Loop	SBIVAR_E.CPGC	SBIVAR.SRC
C12-Gov_NPT_Closed_Loop	SBIVAR_F.CPGC	SBIVAR.SRC
C12-Gov_NPT_Closed_Loop	UNIVAR_A.CPGC	UNIVAR.SRC
C12-Gov_NPT_Closed_Loop	UNIVAR_B.CPGC	UNIVAR.SRC
C12-Gov_NPT_Closed_Loop	UNIVAR_C.CPGC	UNIVAR.SRC
C13-Trans_Accel_NH	C13_NHACCELLIM.DATAFLOW	C13_NHACCELLIM.SRC
C13-Trans_Accel_NH	DPA_A.CPGC	DPA.SRC
C13-Trans_Accel_NH	DPA_B.CPGC	DPA.SRC
C13-Trans_Accel_NH	SBIVAR_A.CPGC	SBIVAR.SRC
C13-Trans_Accel_NH	SBIVAR_B.CPGC	SBIVAR.SRC
C13-Trans_Accel_NH	SBIVAR_C.CPGC	SBIVAR.SRC
C13-Trans_Accel_NH	SBIVAR_D.CPGC	SBIVAR.SRC
C13-Trans_Accel_NH	SBIVAR_E.CPGC	SBIVAR.SRC
C13-Trans_Accel_NH	SBIVAR_F.CPGC	SBIVAR.SRC
C13-Trans_Accel_NH	UNIVAR_A.CPGC	UNIVAR.SRC
C13-Trans_Accel_NH	UNIVAR_B.CPGC	UNIVAR.SRC
C13-Trans_Accel_NH	UNIVAR_C.CPGC	UNIVAR.SRC
C14-Limit_NH	C14_NHLIM.DATAFLOW	C14_NHLIM.SRC
C14-Limit_NH	SBIVAR_A.CPGC	SBIVAR.SRC
C14-Limit_NH	SBIVAR_B.CPGC	SBIVAR.SRC
C14-Limit_NH	SBIVAR_C.CPGC	SBIVAR.SRC
C14-Limit_NH	SBIVAR_D.CPGC	SBIVAR.SRC
C14-Limit_NH	SBIVAR_E.CPGC	SBIVAR.SRC
C14-Limit_NH	SBIVAR_F.CPGC	SBIVAR.SRC
C14-Limit_NH	UNIVAR_A.CPGC	UNIVAR.SRC
C14-Limit_NH	UNIVAR_B.CPGC	UNIVAR.SRC
C14-Limit_NH	UNIVAR_C.CPGC	UNIVAR.SRC
C16-Gov_Inner_Loop_Sel	C16_LOOPSEL.DATAFLOW	C16_LOOPSEL.SRC
C16-Gov_Inner_Loop_Sel	DIVPRT_A.CPGC	DIVPRT.SRC

SRS Capability	Design	Module
C16-Gov_Inner_Loop_Sel	DIVPRT_B.CPGC	DIVPRT.SRC
C16-Gov_Inner_Loop_Sel	DPA_A.CPGC	DPA.SRC
C16-Gov_Inner_Loop_Sel	DPA_B.CPGC	DPA.SRC
C16-Gov_Inner_Loop_Sel	DPM_A.CPGC	DPM.SRC
C16-Gov_Inner_Loop_Sel	DPM_B.CPGC	DPM.SRC
C16-Gov_Inner_Loop_Sel	DPM_C.CPGC	DPM.SRC
C16-Gov_Inner_Loop_Sel	DPM_D.CPGC	DPM.SRC
C16-Gov_Inner_Loop_Sel	DPM_E.CPGC	DPM.SRC
C16-Gov_Inner_Loop_Sel	SBIVAR_A.CPGC	SBIVAR.SRC
C16-Gov_Inner_Loop_Sel	SBIVAR_B.CPGC	SBIVAR.SRC
C16-Gov_Inner_Loop_Sel	SBIVAR_C.CPGC	SBIVAR.SRC
C16-Gov_Inner_Loop_Sel	SBIVAR_D.CPGC	SBIVAR.SRC
C16-Gov_Inner_Loop_Sel	SBIVAR_E.CPGC	SBIVAR.SRC
C16-Gov_Inner_Loop_Sel	SBIVAR_F.CPGC	SBIVAR.SRC
C16-Gov_Inner_Loop_Sel	UNIVAR_A.CPGC	UNIVAR.SRC
C16-Gov_Inner_Loop_Sel	UNIVAR_B.CPGC	UNIVAR.SRC
C16-Gov_Inner_Loop_Sel	UNIVAR_C.CPGC	UNIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	C17_NHREQUEST_1.DATAFLOW	C17_NHREQUEST_1.SRC
C17-Gov_Inner_NH_Req_Calc	C17_NHREQUEST_2.DATAFLOW	C17_NHREQUEST_2.SRC
C17-Gov_Inner_NH_Req_Calc	DBL_LL_SUBR_A.CPGC	DBL_LL_SUBR.SRC
C17-Gov_Inner_NH_Req_Calc	DBL_LL_SUBR_B.CPGC	DBL_LL_SUBR.SRC
C17-Gov_Inner_NH_Req_Calc	DBL_LL_SUBR_C.CPGC	DBL_LL_SUBR.SRC
C17-Gov_Inner_NH_Req_Calc	DBL_LL_SUBR_D.CPGC	DBL_LL_SUBR.SRC
C17-Gov_Inner_NH_Req_Calc	DBL_LL_SUBR_E.CPGC	DBL_LL_SUBR.SRC
C17-Gov_Inner_NH_Req_Calc	DIVPRT_A.CPGC	DIVPRT.SRC
C17-Gov_Inner_NH_Req_Calc	DIVPRT_B.CPGC	DIVPRT.SRC
C17-Gov_Inner_NH_Req_Calc	DPA_A.CPGC	DPA.SRC
C17-Gov_Inner_NH_Req_Calc	DPA_B.CPGC	DPA.SRC
C17-Gov_Inner_NH_Req_Calc	DPM_A.CPGC	DPM.SRC
C17-Gov_Inner_NH_Req_Calc	DPM_B.CPGC	DPM.SRC
C17-Gov_Inner_NH_Req_Calc	DPM_C.CPGC	DPM.SRC
C17-Gov_Inner_NH_Req_Calc	DPM_D.CPGC	DPM.SRC
C17-Gov_Inner_NH_Req_Calc	DPM_E.CPGC	DPM.SRC
C17-Gov_Inner_NH_Req_Calc	LDLG_SUBR_A.CPGC	LDLG_SUBR.SRC
C17-Gov_Inner_NH_Req_Calc	LDLG_SUBR_B.CPGC	LDLG_SUBR.SRC
C17-Gov_Inner_NH_Req_Calc	LDLG_SUBR_C.CPGC	LDLG_SUBR.SRC
C17-Gov_Inner_NH_Req_Calc	SBIVAR_A.CPGC	SBIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	SBIVAR_B.CPGC	SBIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	SBIVAR_C.CPGC	SBIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	SBIVAR_D.CPGC	SBIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	SBIVAR_E.CPGC	SBIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	SBIVAR_F.CPGC	SBIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	UNIVAR_A.CPGC	UNIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	UNIVAR_B.CPGC	UNIVAR.SRC
C17-Gov_Inner_NH_Req_Calc	UNIVAR_C.CPGC	UNIVAR.SRC
C18-Gov_NH_Closed_Loop	C18_NHGOV_1.DATAFLOW	C18_NHGOV_1.SRC

SRS Capability	Design	Module
C18-Gov_NH_Closed_Loop	C18_NHGOV_2.DATAFLOW	C18_NHGOV_2.SRC
C18-Gov_NH_Closed_Loop	DBL_LL_SUBR_A.CPGC	DBL_LL_SUBR.SRC
C18-Gov_NH_Closed_Loop	DBL_LL_SUBR_B.CPGC	DBL_LL_SUBR.SRC
C18-Gov_NH_Closed_Loop	DBL_LL_SUBR_C.CPGC	DBL_LL_SUBR.SRC
C18-Gov_NH_Closed_Loop	DBL_LL_SUBR_D.CPGC	DBL_LL_SUBR.SRC
C18-Gov_NH_Closed_Loop	DBL_LL_SUBR_E.CPGC	DBL_LL_SUBR.SRC
C18-Gov_NH_Closed_Loop	DIVPRT_A.CPGC	DIVPRT.SRC
C18-Gov_NH_Closed_Loop	DIVPRT_B.CPGC	DIVPRT.SRC
C18-Gov_NH_Closed_Loop	DPA_A.CPGC	DPA.SRC
C18-Gov_NH_Closed_Loop	DPA_B.CPGC	DPA.SRC
C18-Gov_NH_Closed_Loop	DPM_A.CPGC	DPM.SRC
C18-Gov_NH_Closed_Loop	DPM_B.CPGC	DPM.SRC
C18-Gov_NH_Closed_Loop	DPM_C.CPGC	DPM.SRC
C18-Gov_NH_Closed_Loop	DPM_D.CPGC	DPM.SRC
C18-Gov_NH_Closed_Loop	DPM_E.CPGC	DPM.SRC
C18-Gov_NH_Closed_Loop		E08_FG_SCHEDULE.SRC
C18-Gov_NH_Closed_Loop	FILCON_A.CPGC	FILCON.SRC
C18-Gov_NH_Closed_Loop	FILCON_B.CPGC	FILCON.SRC
C18-Gov_NH_Closed_Loop	SBIVAR_A.CPGC	SBIVAR.SRC
C18-Gov_NH_Closed_Loop	SBIVAR_B.CPGC	SBIVAR.SRC
C18-Gov_NH_Closed_Loop	SBIVAR_C.CPGC	SBIVAR.SRC
C18-Gov_NH_Closed_Loop	SBIVAR_D.CPGC	SBIVAR.SRC
C18-Gov_NH_Closed_Loop	SBIVAR_E.CPGC	SBIVAR.SRC
C18-Gov_NH_Closed_Loop	SBIVAR_F.CPGC	SBIVAR.SRC
C18-Gov_NH_Closed_Loop	UNIVAR_A.CPGC	UNIVAR.SRC
C18-Gov_NH_Closed_Loop	UNIVAR_B.CPGC	UNIVAR.SRC
C18-Gov_NH_Closed_Loop	UNIVAR_C.CPGC	UNIVAR.SRC
C19-Limit_Prop_Underspeed	C19_UNDERSPEED.DATAFLOW	C19_UNDERSPEED.SRC
C19-Limit_Prop_Underspeed	DIVPRT_A.CPGC	DIVPRT.SRC
C19-Limit_Prop_Underspeed	DPA_A.CPGC	DPA.SRC
C19-Limit_Prop_Underspeed	DPA_B.CPGC	DPA.SRC
C19-Limit_Prop_Underspeed	DPM_A.CPGC	DPM.SRC
C19-Limit_Prop_Underspeed	DPM_B.CPGC	DPM.SRC
C19-Limit_Prop_Underspeed	DPM_C.CPGC	DPM.SRC
C19-Limit_Prop_Underspeed	DPM_D.CPGC	DPM.SRC
C19-Limit_Prop_Underspeed	DPM_E.CPGC	DPM.SRC
C19-Limit_Prop_Underspeed	SBIVAR_A.CPGC	SBIVAR.SRC
C19-Limit_Prop_Underspeed	SBIVAR_B.CPGC	SBIVAR.SRC
C19-Limit_Prop_Underspeed	SBIVAR_C.CPGC	SBIVAR.SRC
C19-Limit_Prop_Underspeed	SBIVAR_D.CPGC	SBIVAR.SRC
C19-Limit_Prop_Underspeed	SBIVAR_E.CPGC	SBIVAR.SRC
C19-Limit_Prop_Underspeed	SBIVAR_F.CPGC	SBIVAR.SRC
C19-Limit_Prop_Underspeed	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C19-Limit_Prop_Underspeed	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
C19-Limit_Prop_Underspeed	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
C19-Limit_Prop_Underspeed	UNIVAR_A.CPGC	UNIVAR.SRC

SRS Capability	Design	Module
C19-Limit_Prop_Underspeed	UNIVAR_B.CPGC	UNIVAR.SRC
C19-Limit_Prop_Underspeed	UNIVAR_C.CPGC	UNIVAR.SRC
C19-Limit_Prop_Underspeed	VAR_TUSTIN_SUBR_A.CPGC	VAR_TUSTIN_SUBR.SRC
C19-Limit_Prop_Underspeed	VAR_TUSTIN_SUBR_B.CPGC	VAR_TUSTIN_SUBR.SRC
C19-Limit_Prop_Underspeed	VAR_TUSTIN_SUBR_C.CPGC	VAR_TUSTIN_SUBR.SRC
C20-Limit_Prop_Overspeed	C20_OVRSPEED.DATAFLOW	C20_OVRSPEED.SRC
C20-Limit_Prop_Overspeed	DPA_A.CPGC	DPA.SRC
C20-Limit_Prop_Overspeed	DPA_B.CPGC	DPA.SRC
C20-Limit_Prop_Overspeed	DPM_A.CPGC	DPM.SRC
C20-Limit_Prop_Overspeed	DPM_B.CPGC	DPM.SRC
C20-Limit_Prop_Overspeed	DPM_C.CPGC	DPM.SRC
C20-Limit_Prop_Overspeed	DPM_D.CPGC	DPM.SRC
C20-Limit_Prop_Overspeed	DPM_E.CPGC	DPM.SRC
C20-Limit_Prop_Overspeed	SBIVAR_A.CPGC	SBIVAR.SRC
C20-Limit_Prop_Overspeed	SBIVAR_B.CPGC	SBIVAR.SRC
C20-Limit_Prop_Overspeed	SBIVAR_C.CPGC	SBIVAR.SRC
C20-Limit_Prop_Overspeed	SBIVAR_D.CPGC	SBIVAR.SRC
C20-Limit_Prop_Overspeed	SBIVAR_E.CPGC	SBIVAR.SRC
C20-Limit_Prop_Overspeed	SBIVAR_F.CPGC	SBIVAR.SRC
C20-Limit_Prop_Overspeed	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C20-Limit_Prop_Overspeed	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
C20-Limit_Prop_Overspeed	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
C20-Limit_Prop_Overspeed	UNIVAR_A.CPGC	UNIVAR.SRC
C20-Limit_Prop_Overspeed	UNIVAR_B.CPGC	UNIVAR.SRC
C20-Limit_Prop_Overspeed	UNIVAR_C.CPGC	UNIVAR.SRC
C21-OUT_HBOV_EEC_MODE	C21_HBOVEM_1.DATAFLOW	C21_HBOVEM_1.SRC
C21-OUT_HBOV_EEC_MODE	C21_HBOVEM_2.DATAFLOW	C21_HBOVEM_2.SRC
C21-OUT_HBOV_EEC_MODE	DIVPRT_A.CPGC	DIVPRT.SRC
C21-OUT_HBOV_EEC_MODE	DIVPRT_B.CPGC	DIVPRT.SRC
C21-OUT_HBOV_EEC_MODE	DPA_A.CPGC	DPA.SRC
C21-OUT_HBOV_EEC_MODE	DPA_B.CPGC	DPA.SRC
C21-OUT_HBOV_EEC_MODE	DPM_A.CPGC	DPM.SRC
C21-OUT_HBOV_EEC_MODE	DPM_B.CPGC	DPM.SRC
C21-OUT_HBOV_EEC_MODE	DPM_C.CPGC	DPM.SRC
C21-OUT_HBOV_EEC_MODE	DPM_D.CPGC	DPM.SRC
C21-OUT_HBOV_EEC_MODE	DPM_E.CPGC	DPM.SRC
C21-OUT_HBOV_EEC_MODE		E08_FG_SCHEDULE.SRC
C21-OUT_HBOV_EEC_MODE	SBIVAR_A.CPGC	SBIVAR.SRC
C21-OUT_HBOV_EEC_MODE	SBIVAR_B.CPGC	SBIVAR.SRC
C21-OUT_HBOV_EEC_MODE	SBIVAR_C.CPGC	SBIVAR.SRC
C21-OUT_HBOV_EEC_MODE	SBIVAR_D.CPGC	SBIVAR.SRC
C21-OUT_HBOV_EEC_MODE	SBIVAR_E.CPGC	SBIVAR.SRC
C21-OUT_HBOV_EEC_MODE	SBIVAR_F.CPGC	SBIVAR.SRC
C21-OUT_HBOV_EEC_MODE	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C21-OUT_HBOV_EEC_MODE	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
C21-OUT_HBOV_EEC_MODE	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC

SRS Capability	Design	Module
C21-OUT_HBOV_EEC_MODE	UNIVAR_A.CPGC	UNIVAR.SRC
C21-OUT_HBOV_EEC_MODE	UNIVAR_B.CPGC	UNIVAR.SRC
C21-OUT_HBOV_EEC_MODE	UNIVAR_C.CPGC	UNIVAR.SRC
C22-Out_HBOV_Man_Mode	C22_HBOVMM_1.DATAFLOW	C22_HBOVMM_1.SRC
C22-Out_HBOV_Man_Mode	C22_HBOVMM_2.DATAFLOW	C22_HBOVMM_2.SRC
C22-Out_HBOV_Man_Mode	C22_HBOVMM_3.DATAFLOW	C22_HBOVMM_3.SRC
C22-Out_HBOV_Man_Mode	DIVPRT_A.CPGC	DIVPRT.SRC
C22-Out_HBOV_Man_Mode	DIVPRT_B.CPGC	DIVPRT.SRC
C22-Out_HBOV_Man_Mode	DPA_A.CPGC	DPA.SRC
C22-Out_HBOV_Man_Mode	DPA_B.CPGC	DPA.SRC
C22-Out_HBOV_Man_Mode	DPM_A.CPGC	DPM.SRC
C22-Out_HBOV_Man_Mode	DPM_B.CPGC	DPM.SRC
C22-Out_HBOV_Man_Mode	DPM_C.CPGC	DPM.SRC
C22-Out_HBOV_Man_Mode	DPM_D.CPGC	DPM.SRC
C22-Out_HBOV_Man_Mode	DPM_E.CPGC	DPM.SRC
C22-Out_HBOV_Man_Mode	SBIVAR_A.CPGC	SBIVAR.SRC
C22-Out_HBOV_Man_Mode	SBIVAR_B.CPGC	SBIVAR.SRC
C22-Out_HBOV_Man_Mode	SBIVAR_C.CPGC	SBIVAR.SRC
C22-Out_HBOV_Man_Mode	SBIVAR_E.CPGC	SBIVAR.SRC
C22-Out_HBOV_Man_Mode	SBIVAR_F.CPGC	SBIVAR.SRC
C22-Out_HBOV_Man_Mode	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C22-Out_HBOV_Man_Mode	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
C22-Out_HBOV_Man_Mode	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
C22-Out_HBOV_Man_Mode	UNIVAR_A.CPGC	UNIVAR.SRC
C22-Out_HBOV_Man_Mode	UNIVAR_B.CPGC	UNIVAR.SRC
C22-Out_HBOV_Man_Mode	UNIVAR_C.CPGC	UNIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	C24_OVRTQTST.CTRLFLOW	C24_OVRTQTST.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DIVPRT_A.CPGC	DIVPRT.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DIVPRT_B.CPGC	DIVPRT.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DPA_A.CPGC	DPA.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DPA_B.CPGC	DPA.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DPM_A.CPGC	DPM.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DPM_B.CPGC	DPM.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DPM_C.CPGC	DPM.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DPM_D.CPGC	DPM.SRC
C24-BIT_PreFly_Eng_OverQ_Test	DPM_E.CPGC	DPM.SRC
C24-BIT_PreFly_Eng_OverQ_Test	SBIVAR_A.CPGC	SBIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	SBIVAR_B.CPGC	SBIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	SBIVAR_C.CPGC	SBIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	SBIVAR_D.CPGC	SBIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	SBIVAR_E.CPGC	SBIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	SBIVAR_F.CPGC	SBIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	UNIVAR_A.CPGC	UNIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	UNIVAR_B.CPGC	UNIVAR.SRC
C24-BIT_PreFly_Eng_OverQ_Test	UNIVAR_C.CPGC	UNIVAR.SRC
C26-MPARAM	C26_MPARAM_1.DATAFLOW	C26_MPARAM_1.SRC

SRS Capability	Design	Module
C26-MPARAM	C26_MPARAM_2.DATAFLOW	C26_MPARAM_2.SRC
C26-MPARAM	C26_MPARAM_3.DATAFLOW	C26_MPARAM_3.SRC
C26-MPARAM	C26_MPARAM_4.DATAFLOW	C26_MPARAM_4.S00
C26-MPARAM	C26_MPARAM_5.DATAFLOW	C26_MPARAM_5.S00
C26-MPARAM	C26_MPARAM_6.DATAFLOW	C26_MPARAM_6.S00
C26-MPARAM	C26_MPARAM_7.DATAFLOW	C26_MPARAM_7.S00
C26-MPARAM	DIVPRT_A.CPGC	DIVPRT.SRC
C26-MPARAM	DIVPRT_B.CPGC	DIVPRT.SRC
C26-MPARAM	DPA_A.CPGC	DPA.SRC
C26-MPARAM	DPA_B.CPGC	DPA.SRC
C26-MPARAM	DPM_A.CPGC	DPM.SRC
C26-MPARAM	DPM_B.CPGC	DPM.SRC
C26-MPARAM	DPM_C.CPGC	DPM.SRC
C26-MPARAM	DPM_D.CPGC	DPM.SRC
C26-MPARAM	DPM_E.CPGC	DPM.SRC
C26-MPARAM		E08_FG_SCHEDULE.SRC
C26-MPARAM	SBIVAR_A.CPGC	SBIVAR.SRC
C26-MPARAM	SBIVAR_B.CPGC	SBIVAR.SRC
C26-MPARAM	SBIVAR_C.CPGC	SBIVAR.SRC
C26-MPARAM	SBIVAR_D.CPGC	SBIVAR.SRC
C26-MPARAM	SBIVAR_E.CPGC	SBIVAR.SRC
C26-MPARAM	SBIVAR_F.CPGC	SBIVAR.SRC
C26-MPARAM	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C26-MPARAM	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
C26-MPARAM	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
C26-MPARAM	UNIVAR_A.CPGC	UNIVAR.SRC
C26-MPARAM	UNIVAR_B.CPGC	UNIVAR.SRC
C26-MPARAM	UNIVAR_C.CPGC	UNIVAR.SRC
C26-MPARAM	VAR_TUSTIN_SUBR_A.CPGC	VAR_TUSTIN_SUBR.SRC
C26-MPARAM	VAR_TUSTIN_SUBR_B.CPGC	VAR_TUSTIN_SUBR.SRC
C26-MPARAM	VAR_TUSTIN_SUBR_C.CPGC	VAR_TUSTIN_SUBR.SRC
C27-NHMINMAX	C27_NHMINMX_1.DATAFLOW	C27_NHMINMX_1.SRC
C27-NHMINMAX	DIVPRT_A.CPGC	DIVPRT.SRC
C27-NHMINMAX	DIVPRT_B.CPGC	DIVPRT.SRC
C27-NHMINMAX	DPA_A.CPGC	DPA.SRC
C27-NHMINMAX	DPA_B.CPGC	DPA.SRC
C27-NHMINMAX	DPM_A.CPGC	DPM.SRC
C27-NHMINMAX	DPM_B.CPGC	DPM.SRC
C27-NHMINMAX	DPM_C.CPGC	DPM.SRC
C27-NHMINMAX	DPM_D.CPGC	DPM.SRC
C27-NHMINMAX	DPM_E.CPGC	DPM.SRC
C27-NHMINMAX	SBIVAR_A.CPGC	SBIVAR.SRC
C27-NHMINMAX	SBIVAR_B.CPGC	SBIVAR.SRC
C27-NHMINMAX	SBIVAR_C.CPGC	SBIVAR.SRC
C27-NHMINMAX	SBIVAR_D.CPGC	SBIVAR.SRC
C27-NHMINMAX	SBIVAR_E.CPGC	SBIVAR.SRC

SRS Capability	Design	Module
C27-NHMINMAX	SBIVAR_F.CPGC	SBIVAR.SRC
C27-NHMINMAX	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
C27-NHMINMAX	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
C27-NHMINMAX	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
C27-NHMINMAX	UNIVAR_A.CPGC	UNIVAR.SRC
C27-NHMINMAX	UNIVAR_B.CPGC	UNIVAR.SRC
C27-NHMINMAX	UNIVAR_C.CPGC	UNIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	C30_SMRTRQ.DATAFLOW	C30_SMRTRQ.SRC
C30-Gov_Inner_SM_Rate_Req	DPA_A.CPGC	DPA.SRC
C30-Gov_Inner_SM_Rate_Req	DPA_B.CPGC	DPA.SRC
C30-Gov_Inner_SM_Rate_Req	DPM_A.CPGC	DPM.SRC
C30-Gov_Inner_SM_Rate_Req	DPM_B.CPGC	DPM.SRC
C30-Gov_Inner_SM_Rate_Req	DPM_C.CPGC	DPM.SRC
C30-Gov_Inner_SM_Rate_Req	DPM_D.CPGC	DPM.SRC
C30-Gov_Inner_SM_Rate_Req	DPM_E.CPGC	DPM.SRC
C30-Gov_Inner_SM_Rate_Req	SBIVAR_A.CPGC	SBIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	SBIVAR_B.CPGC	SBIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	SBIVAR_C.CPGC	SBIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	SBIVAR_D.CPGC	SBIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	SBIVAR_E.CPGC	SBIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	SBIVAR_F.CPGC	SBIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	UNIVAR_A.CPGC	UNIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	UNIVAR_B.CPGC	UNIVAR.SRC
C30-Gov_Inner_SM_Rate_Req	UNIVAR_C.CPGC	UNIVAR.SRC
C31-Limit_Q	C31_TQLIM.DATAFLOW	C31_TQLIM.SRC
C31-Limit_Q	DIVPRT_A.CPGC	DIVPRT.SRC
C31-Limit_Q	DPA_A.CPGC	DPA.SRC
C31-Limit_Q	DPA_B.CPGC	DPA.SRC
C31-Limit_Q	DPM_A.CPGC	DPM.SRC
C31-Limit_Q	DPM_B.CPGC	DPM.SRC
C31-Limit_Q	DPM_C.CPGC	DPM.SRC
C31-Limit_Q	DPM_D.CPGC	DPM.SRC
C31-Limit_Q	DPM_E.CPGC	DPM.SRC
C31-Limit_Q	SBIVAR_A.CPGC	SBIVAR.SRC
C31-Limit_Q	SBIVAR_B.CPGC	SBIVAR.SRC
C31-Limit_Q	SBIVAR_C.CPGC	SBIVAR.SRC
C31-Limit_Q	SBIVAR_D.CPGC	SBIVAR.SRC
C31-Limit_Q	SBIVAR_E.CPGC	SBIVAR.SRC
C31-Limit_Q	SBIVAR_F.CPGC	SBIVAR.SRC
C31-Limit_Q	UNIVAR_A.CPGC	UNIVAR.SRC
C31-Limit_Q	UNIVAR_B.CPGC	UNIVAR.SRC
C31-Limit_Q	UNIVAR_C.CPGC	UNIVAR.SRC
CYCLE_COUNTING	CYCLE_COUNTING	DCU_WRITE_CTRDAT.S00
CYCLE_COUNTING		E08_FG_SCHEDULE.SRC
CYCLE_COUNTING	CYCLE_COUNTING	E08_FG_SCHEDULE.SRC
CYCLE_COUNTING	CYCLE_COUNTING	LCF_CHECK.S00

SRS Capability	Design	Module
CYCLE_COUNTING	CYCLE_COUNTING	LCF_CYCLE_COUNT.S00
CYCLE_COUNTING	CYCLE_COUNTING	LCF_TABLE.S00
CYCLE_COUNTING	CYCLE_COUNTING	UPDATE_EE_DATA.S00
DCU_CP_CFG	DCU_CP_CFG	DCU_CONSTANTS.S00
DCU_CP_CFG	DCU_CP_CFG	DCU_PUT_QUEUE.S00
DCU_CP_CFG	DCU_CP_CFG	DCU_SCHEDULE.S00
DCU_CP_CFG	DCU_CP_CFG	DCU_WRITE_CPDAT.S00
DCU_CP_CFG		E08_FG_SCHEDULE.SRC
DCU_EEC_CFG		DCU_CONSTANTS.S00
DCU_EEC_CFG		DCU_PUT_QUEUE.S00
DCU_EEC_CFG		DCU_SCHEDULE.S00
DCU_EEC_CFG		DCU_WRITE_EECDAT.S00
DCU_EEC_CFG		E08_FG_SCHEDULE.SRC
DCU_ENG_CFG		DCU_CONSTANTS.S00
DCU_ENG_CFG		DCU_PUT_QUEUE.S00
DCU_ENG_CFG		DCU_SCHEDULE.S00
DCU_ENG_CFG		DCU_WRITE_ENGDAT.S00
DCU_ENG_CFG		E08_FG_SCHEDULE.SRC
DCU_ENGUSE_CFG		DCU_CONSTANTS.S00
DCU_ENGUSE_CFG		DCU_SCHEDULE.S00
DCU_ENGUSE_CFG		DCU_WRITE_CTRDAT.S00
DCU_ENGUSE_CFG		E08_FG_SCHEDULE.SRC
DCU_RECEIVE		DCU_RECEIVE.S00
DCUDSTAT		DCU_COMPONT_DATA.S00
DCUDSTAT		DCU_COUNTER_DATA.S00
DCUDSTAT		DCU_DCUID_DATA.S00
DCUDSTAT		DCU_EEC_DATA.S00
DCUDSTAT		DCU_ENGINE_DATA.S00
DCUDSTAT		DCU_LCF_DATA.S00
DCUDSTAT		DCU_SCHEDULE.S00
DCUDSTAT		E08_FG_SCHEDULE.SRC
DCUPROTO	DCUPROTO	DCU_TRANSFER.S00
DCUPROTO	DCUPROTO	DCU_TRANSMIT.S00
DCUREAD		DCU_CONSTANTS.S00
DCUREAD		DCU_SCHEDULE.S00
DCUWRITE		DCU_CONSTANTS.S00
DCUWRITE		DCU_GET_QUEUE.S00
DCUWRITE		DCU_SCHEDULE.S00
DCUWRITE		DCU_WRITE.S00
DCUWRITE		E08_FG_SCHEDULE.SRC
E01-BIT_Proc_Reset_PSR_POR	E01_PORPSR_A.CPGC	E01_PORPSR.SRC
E01-BIT_Proc_Reset_PSR_POR	E01_PORPSR_B.CPGC	E01_PORPSR.SRC
E01-BIT_Proc_Reset_PSR_POR	E01_PORPSR_C.CPGC	E01_PORPSR.SRC
E01-BIT_Proc_Reset_PSR_POR	E01_PORPSR_D.CPGC	E01_PORPSR.SRC
E01-BIT_Proc_Reset_PSR_POR	GETWRD_A.CPGC	GETWRD.SRC
E01-BIT_Proc_Reset_PSR_POR	GETWRD_B.CPGC	GETWRD.SRC

SRS Capability	Design	Module
E01-BIT_Proc_Reset_PSR_POR	INA_A.CPGC	INA.SRC
E01-BIT_Proc_Reset_PSR_POR	OTA_A.CPGC	OTA.SRC
E01-BIT_Proc_Reset_PSR_POR	OTA_B.CPGC	OTA.SRC
E01-BIT_Proc_Reset_PSR_POR	RTDWRITE.CTRLFLOW	RTDWRITE.S00
E01-BIT_Proc_Reset_PSR_POR	SNDWRD_A.CPGC	SNDWRD.SRC
E01-BIT_Proc_Reset_PSR_POR	URTOUT_A.CPGC	URTOUT.SRC
E01-BIT_Proc_Reset_PSR_POR	URTOUT_B.CPGC	URTOUT.SRC
E01-BIT_Proc_Reset_PSR_POR	URTRED_A.CPGC	URTRED.SRC
E01-BIT_Proc_Reset_PSR_POR	URTRED_B.CPGC	URTRED.SRC
E01-BIT_Proc_Reset_PSR_POR	WDGRST_A.CPGC	WDGRST.SRC
E02-BIT_Proc_Ram_Test	E02_RAMTEST_A.CPGC	E02_RAMTEST.SRC
E02-BIT_Proc_Ram_Test	E02_RAMTEST_B.CPGC	E02_RAMTEST.SRC
E02-BIT_Proc_Ram_Test	E02_RAMTEST_C.CPGC	E02_RAMTEST.SRC
E02-BIT_Proc_Ram_Test	GETWRD_A.CPGC	GETWRD.SRC
E02-BIT_Proc_Ram_Test	GETWRD_B.CPGC	GETWRD.SRC
E02-BIT_Proc_Ram_Test	INA_A.CPGC	INA.SRC
E02-BIT_Proc_Ram_Test	OTA_A.CPGC	OTA.SRC
E02-BIT_Proc_Ram_Test	OTA_B.CPGC	OTA.SRC
E02-BIT_Proc_Ram_Test	RAMREAD0_A.CPGC	RAMREAD0.SRC
E02-BIT_Proc_Ram_Test	RAMSTOR1_A.CPGC	RAMREAD1.SRC
E02-BIT_Proc_Ram_Test	RAMSTOR0_A.CPGC	RAMSTOR0.SRC
E02-BIT_Proc_Ram_Test	RAMSTOR1_A.CPGC	RAMSTOR1.SRC
E02-BIT_Proc_Ram_Test	SNDWRD_A.CPGC	SNDWRD.SRC
E02-BIT_Proc_Ram_Test	URTOUT_A.CPGC	URTOUT.SRC
E02-BIT_Proc_Ram_Test	URTOUT_B.CPGC	URTOUT.SRC
E02-BIT_Proc_Ram_Test	URTRED_A.CPGC	URTRED.SRC
E02-BIT_Proc_Ram_Test	URTRED_B.CPGC	URTRED.SRC
E02-BIT_Proc_Ram_Test	WDGRST_A.CPGC	WDGRST.SRC
E03-BIT_Proc_ROM_EE_Chksum	E03_CHECKSUM_A.CPGC	E03_CHECKSUM.SRC
E03-BIT_Proc_ROM_EE_Chksum	E03_CHECKSUM_B.CPGC	E03_CHECKSUM.SRC
E03-BIT_Proc_ROM_EE_Chksum	EESUM_16_A.CPGC	EESUM_16.SRC
E03-BIT_Proc_ROM_EE_Chksum	EESUM_8_A.CPGC	EESUM_8.SRC
E03-BIT_Proc_ROM_EE_Chksum	GETWRD_A.CPGC	GETWRD.SRC
E03-BIT_Proc_ROM_EE_Chksum	GETWRD_B.CPGC	GETWRD.SRC
E03-BIT_Proc_ROM_EE_Chksum	INA_A.CPGC	INA.SRC
E03-BIT_Proc_ROM_EE_Chksum	OTA_A.CPGC	OTA.SRC
E03-BIT_Proc_ROM_EE_Chksum	OTA_B.CPGC	OTA.SRC
E03-BIT_Proc_ROM_EE_Chksum	SNDWRD_A.CPGC	SNDWRD.SRC
E03-BIT_Proc_ROM_EE_Chksum	URTOUT_A.CPGC	URTOUT.SRC
E03-BIT_Proc_ROM_EE_Chksum	URTOUT_B.CPGC	URTOUT.SRC
E03-BIT_Proc_ROM_EE_Chksum	URTRED_B.CPGC	URTOUT.SRC
E03-BIT_Proc_ROM_EE_Chksum	URTRED_A.CPGC	URTRED.SRC
E03-BIT_Proc_ROM_EE_Chksum	WDGRST_A.CPGC	WDGRST.SRC
E04-BIT_Proc_Instruct_Test	E04_INSTST_A.CPGC	E04_INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_A.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_B.CPGC	INSTST.SRC

SRS Capability	Design	Module
E04-BIT_Proc_Instruct_Test	INSTST_C.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_D.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_E.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_F.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_G.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_H.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_I.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_J.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_K.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_L.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_M.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_N.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_O.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_P.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_Q.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_R.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_S.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_T.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_U.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_V.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_W.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_X.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_Y.CPGC	INSTST.SRC
E04-BIT_Proc_Instruct_Test	INSTST_Z.CPGC	INSTST.SRC
E05-RAMINIT	CMPACT_A.CPGC	CMPACT.SRC
E05-RAMINIT	CMPACT_B.CPGC	CMPACT.SRC
E05-RAMINIT	CMPACT_C.CPGC	CMPACT.SRC
E05-RAMINIT	E05-RAMINIT	DCU_RANGE_CHECKS.SRC
E05-RAMINIT	DCU_RANGE_CHECKS.CPGC	DCU_RANGE_CHECKS.SRC
E05-RAMINIT	E01_PORPSR_A.CPGC	E01_PORPSR.SRC
E05-RAMINIT	E05_RAMINI_F.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_G.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_H.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_I.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_J.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_K.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_L.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_M.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_A.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_B.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_C.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_D.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05_RAMINI_E.CPGC	E05_RAMINI.SRC
E05-RAMINIT	E05-RAMINIT	EE_READ_DATA.S00
E05-RAMINIT	GETWRD_A.CPGC	GETWRD.SRC
E05-RAMINIT	GETWRD_B.CPGC	GETWRD.SRC

SRS Capability	Design	Module
E05-RAMINIT	INA_A.CPGC	INA.SRC
E05-RAMINIT	E05-RAMINIT	INI_DCU_DATA.S00
E05-RAMINIT	INIFLT_A.CPGC	INIFLT.SRC
E05-RAMINIT	INIFLT_B.CPGC	INIFLT.SRC
E05-RAMINIT	INIFLT_C.CPGC	INIFLT.SRC
E05-RAMINIT	OTA_A.CPGC	OTA.SRC
E05-RAMINIT	OTA_B.CPGC	OTA.SRC
E05-RAMINIT	E05-RAMINIT	PAGE0.SRC
E05-RAMINIT	E05-RAMINIT	PAGE1.SRC
E05-RAMINIT	E05-RAMINIT	PAGE10.SRC
E05-RAMINIT	E05-RAMINIT	PAGE11.SRC
E05-RAMINIT	SNDWRD_A.CPGC	SNDWRD.SRC
E05-RAMINIT	URTOUT_A.CPGC	URTOUT.SRC
E05-RAMINIT	URTOUT_B.CPGC	URTOUT.SRC
E05-RAMINIT	URTRED_A.CPGC	URTRED.SRC
E05-RAMINIT	URTRED_B.CPGC	URTRED.SRC
E05-RAMINIT	WDGRST_A.CPGC	WDGRST.SRC
E06-BIT_GA_Q_Test_Routine	E06_TQTEST_A.CPGC	E06_TQTEST.SRC
E06-BIT_GA_Q_Test_Routine	E06_TQTEST_B.CPGC	E06_TQTEST.SRC
E06-BIT_GA_Q_Test_Routine	E06_TQTEST_C.CPGC	E06_TQTEST.SRC
E06-BIT_GA_Q_Test_Routine	GETWRD_A.CPGC	GETWRD.SRC
E06-BIT_GA_Q_Test_Routine	GETWRD_B.CPGC	GETWRD.SRC
E06-BIT_GA_Q_Test_Routine	INA_A.CPGC	INA.SRC
E06-BIT_GA_Q_Test_Routine	OTA_A.CPGC	OTA.SRC
E06-BIT_GA_Q_Test_Routine	OTA_B.CPGC	OTA.SRC
E06-BIT_GA_Q_Test_Routine	SNDWRD_A.CPGC	SNDWRD.SRC
E06-BIT_GA_Q_Test_Routine	URTOUT_A.CPGC	URTOUT.SRC
E06-BIT_GA_Q_Test_Routine	URTOUT_B.CPGC	URTOUT.SRC
E06-BIT_GA_Q_Test_Routine	URTRED_A.CPGC	URTRED.SRC
E06-BIT_GA_Q_Test_Routine	URTRED_B.CPGC	URTRED.SRC
E06-BIT_GA_Q_Test_Routine	WDGRST_A.CPGC	WDGRST.SRC
E07-Exec_Background	E07-Exec_Background	CHKSUM.SRC
E07-Exec_Background	E15_BCKGNDTEST_A.CPGC	E15_BCKGNDTEST.SRC
E07-Exec_Background	E15_BCKGNDTEST_B.CPGC	E15_BCKGNDTEST.SRC
E07-Exec_Background	INSTST_A.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_B.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_C.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_D.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_E.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_F.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_G.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_H.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_I.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_J.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_K.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_L.CPGC	INSTST.SRC

SRS Capability	Design	Module
E07-Exec_Background	INSTST_M.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_N.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_O.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_P.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_Q.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_R.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_S.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_T.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_U.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_V.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_W.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_X.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_Y.CPGC	INSTST.SRC
E07-Exec_Background	INSTST_Z.CPGC	INSTST.SRC
E07-Exec_Background	ROMSUM_A.CPGC	ROMSUM.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_F.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_G.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_H.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_I.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_A.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_B.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_C.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_D.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND	E08_FG_EXECUTIVE_E.CPGC	E08_FG_EXECUTIVE.SRC
E08 - EXEC_FOREGND		E08_FG_SCHEDULE.SRC
E08 - EXEC_FOREGND	E08_FG_SECSEL_A.CPGC	E08_FG_SECSEL.SRC
E08 - EXEC_FOREGND	E08 - EXEC_FOREGND	E08_FG_STATETABLE.SRC
E08 - EXEC_FOREGND	GETWRD_A.CPGC	GETWRD.SRC
E08 - EXEC_FOREGND	GETWRD_B.CPGC	GETWRD.SRC
E08 - EXEC_FOREGND	INA_A.CPGC	INA.SRC
E08 - EXEC_FOREGND	OTA_A.CPGC	OTA.SRC
E08 - EXEC_FOREGND	OTA_B.CPGC	OTA.SRC
E08 - EXEC_FOREGND	SNDWRD_A.CPGC	SNDWRD.SRC
E08 - EXEC_FOREGND	URTOUT_A.CPGC	URTOUT.SRC
E08 - EXEC_FOREGND	URTOUT_B.CPGC	URTOUT.SRC
E08 - EXEC_FOREGND	URTRED_A.CPGC	URTRED.SRC
E08 - EXEC_FOREGND	URTRED_B.CPGC	URTRED.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	E09_SMSUMV_A.CPGC	E09_SMSUMV.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	E09_SMSUMV_B.CPGC	E09_SMSUMV.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	GETWRD_A.CPGC	GETWRD.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	GETWRD_B.CPGC	GETWRD.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	INA_A.CPGC	INA.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	OTA_A.CPGC	OTA.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	OTA_B.CPGC	OTA.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	SNDWRD_A.CPGC	SNDWRD.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	URTRED_A.CPGC	URTRED.SRC

SRS Capability	Design	Module
E09-BIT_EEC_Init_SM_Sum_Volt	URTRED_B.CPGC	URTRED.SRC
E09-BIT_EEC_Init_SM_Sum_Volt	WDGRST_A.CPGC	WDGRST.SRC
E10-BIT_EEC_Partial_RAM_BREG	E10_RAMTEST_A.CPGC	E10_RAMTEST.SRC
E10-BIT_EEC_Partial_RAM_BREG	E10_RAMTEST_B.CPGC	E10_RAMTEST.SRC
E11-BIT_EEC_Loss_Connector	E11_CONTEST_A.CPGC	E11_CONTEST.SRC
E13-WATEST		E08_FG_SCHEDULE.SRC
E13-WATEST	E13-WATEST	E08_FG_SCHEDULE.SRC
E13-WATEST	E13_WATEST_1_A.CPGC	E13_WATEST_1.SRC
E13-WATEST	E13_WATEST_1_B.CPGC	E13_WATEST_1.SRC
E13-WATEST	E13_WATEST_2_A.CPGC	E13_WATEST_2.SRC
E13-WATEST	E13_WATEST_3_A.CPGC	E13_WATEST_3.SRC
E13-WATEST	E13_WATEST_4_A.CPGC	E13_WATEST_4.SRC
E13-WATEST	E13_WATEST_5_A.CPGC	E13_WATEST_5.SRC
E13-WATEST	E13_WATEST_6_A.CPGC	E13_WATEST_6.SRC
E13-WATEST	E13_WATEST_7_A.CPGC	E13_WATEST_7.SRC
E13-WATEST	FAILLT_A.CPGC	FAILLT.SRC
E13-WATEST	FAILLT_B.CPGC	FAILLT.SRC
E13-WATEST	PASSLT_A.CPGC	PASSLT.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_1_A.CPGC	E14_FAULTSTORE_1.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_1_B.CPGC	E14_FAULTSTORE_1.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_1_C.CPGC	E14_FAULTSTORE_1.SRC
E14-FAULT_STORAGE	E14-FAULT_STORAGE	E14_FAULTSTORE_2.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_3_A.CPGC	E14_FAULTSTORE_3.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_3_B.CPGC	E14_FAULTSTORE_3.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_3_C.CPGC	E14_FAULTSTORE_3.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_4_A.CPGC	E14_FAULTSTORE_4.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_4_B.CPGC	E14_FAULTSTORE_4.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_5_A.CPGC	E14_FAULTSTORE_5.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_6_A.CPGC	E14_FAULTSTORE_6.SRC
E14-FAULT_STORAGE	E14_FAULTSTORE_6_B.CPGC	E14_FAULTSTORE_6.SRC
E14-FAULT_STORAGE	E14-FAULT_STORAGE	E14_FAULTSTORE_8.SRC
E15-BIT_Proc_Background	E15_BCKGNDTEST_A.CPGC	E15_BCKGNDTEST.SRC
E16-BIT_EEC_SM_Phase	E16_SMTEST_A.CPGC	E16_SMTEST.SRC
E16-BIT_EEC_SM_Phase	E16_SMTEST_B.CPGC	E16_SMTEST.SRC
E17 - NONCRITICAL_FAULT		E08_FG_SCHEDULE.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_1_A.CPGC	E17_NCFLT_1.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_1_B.CPGC	E17_NCFLT_1.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_2_A.CPGC	E17_NCFLT_2.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_2_B.CPGC	E17_NCFLT_2.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_3_A.CPGC	E17_NCFLT_3.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_3_B.CPGC	E17_NCFLT_3.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_4_A.CPGC	E17_NCFLT_4.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_4_B.CPGC	E17_NCFLT_4.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_5_A.CPGC	E17_NCFLT_5.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_5_B.CPGC	E17_NCFLT_5.SRC
E17 - NONCRITICAL_FAULT	E17_NCFLT_6_A.CPGC	E17_NCFLT_6.SRC

SRS Capability	Design	Module
E17 - NONCRITICAL_FAULT	E17_NCFLT_6_B.CPGC	E17_NCFLT_6.SRC
E17 - NONCRITICAL_FAULT	E17 - NONCRITICAL_FAULT	E17_NCFLT_7.SRC
E18-FAULTSUM		E08_FG_SCHEDULE.SRC
E18-FAULTSUM	E18_FAULTSUM_1_A.CPGC	E18_FAULTSUM_1.SRC
E18-FAULTSUM	E18_FAULTSUM_1_B.CPGC	E18_FAULTSUM_1.SRC
E18-FAULTSUM	E18_FAULTSUM_2_A.CPGC	E18_FAULTSUM_2.SRC
E18-FAULTSUM	E18_FAULTSUM_2_B.CPGC	E18_FAULTSUM_2.SRC
E18-FAULTSUM	E18_FAULTSUM_3_A.CPGC	E18_FAULTSUM_3.SRC
E18-FAULTSUM	E18_FAULTSUM_3_B.CPGC	E18_FAULTSUM_3.SRC
E18-FAULTSUM	E18_FAULTSUM_4_A.CPGC	E18_FAULTSUM_4.SRC
E18-FAULTSUM	E18_FAULTSUM_4_B.CPGC	E18_FAULTSUM_4.SRC
E18-FAULTSUM		E18_FAULTSUM_5.SRC
E18-FAULTSUM		E18_FAULTSUM_6.SRC
E19-BIT_Store_EEPROM_Trim_Data		E08_FG_SCHEDULE.SRC
E19-BIT_Store_EEPROM_Trim_Data	E19_TRMSTORE_1_A.CPGC	E19_TRMSTORE_1.SRC
E19-BIT_Store_EEPROM_Trim_Data	E19_TRMSTORE_2_A.CPGC	E19_TRMSTORE_2.SRC
E19-BIT_Store_EEPROM_Trim_Data	E19_TRMSTORE_4_A.CPGC	E19_TRMSTORE_4.SRC
E19-BIT_Store_EEPROM_Trim_Data	E19_TRMSTORE_5_A.CPGC	E19_TRMSTORE_5.SRC
E20-Maint_Dev_Test_Mode	E20-Maint_Dev_Test_Mode	DUMP.SRC
E20-Maint_Dev_Test_Mode	E20_TESTMODE_A.CPGC	E20_TESTMODE.SRC
E20-Maint_Dev_Test_Mode	E20_TESTMODE_B.CPGC	E20_TESTMODE.SRC
E20-Maint_Dev_Test_Mode	E20_TESTMODE_C.CPGC	E20_TESTMODE.SRC
E20-Maint_Dev_Test_Mode	GETWRD_A.CPGC	GETWRD.SRC
E20-Maint_Dev_Test_Mode	GETWRD_B.CPGC	GETWRD.SRC
E20-Maint_Dev_Test_Mode	GOTO_A.CPGC	GOTO.SRC
E20-Maint_Dev_Test_Mode	INA_A.CPGC	INA.SRC
E20-Maint_Dev_Test_Mode	LOAD_A.CPGC	LOAD.SRC
E20-Maint_Dev_Test_Mode	LOAD_B.CPGC	LOAD.SRC
E20-Maint_Dev_Test_Mode	OTA_A.CPGC	OTA.SRC
E20-Maint_Dev_Test_Mode	OTA_B.CPGC	OTA.SRC
E20-Maint_Dev_Test_Mode	READ_A.CPGC	READ.SRC
E20-Maint_Dev_Test_Mode	SNDWRD_A.CPGC	SNDWRD.SRC
E20-Maint_Dev_Test_Mode	URTOUT_A.CPGC	URTOUT.SRC
E20-Maint_Dev_Test_Mode	URTOUT_B.CPGC	URTOUT.SRC
E20-Maint_Dev_Test_Mode	URTRED_A.CPGC	URTRED.SRC
E20-Maint_Dev_Test_Mode	URTRED_B.CPGC	URTRED.SRC
E20-Maint_Dev_Test_Mode	WDGRST_A.CPGC	WDGRST.SRC
E20-Maint_Dev_Test_Mode	WRITE_A.CPGC	WRITE.SRC
E20-Maint_Dev_Test_Mode	WRITE_B.CPGC	WRITE.SRC
E20-Maint_Dev_Test_Mode	WRITE_C.CPGC	WRITE.SRC
E20-Maint_Dev_Test_Mode	WRTEEP_A.CPGC	WRTEEP.SRC
E20-Maint_Dev_Test_Mode	WRTEEP_B.CPGC	WRTEEP.SRC
E20-Maint_Dev_Test_Mode	WRTEEP_C.CPGC	WRTEEP.SRC
E21-RECONFIG		E08_FG_SCHEDULE.SRC
E21-RECONFIG	E21_RECONFIG.CTRLFLOW	E21_RECONFIG.S00
E22-BIT_GA_GateArraySelect	E22_GASEL_A.CPGC	E22_GASEL.SRC

SRS Capability	Design	Module
E22-BIT_GA_GateArraySelect	E22_GASEL_B.CPGC	E22_GASEL.SRC
E22-BIT_GA_GateArraySelect	E22_GASEL_C.CPGC	E22_GASEL.SRC
E22-BIT_GA_GateArraySelect	E22_GASEL_D.CPGC	E22_GASEL.SRC
E22-BIT_GA_GateArraySelect	GETWRD_A.CPGC	GETWRD.SRC
E22-BIT_GA_GateArraySelect	GETWRD_B.CPGC	GETWRD.SRC
E22-BIT_GA_GateArraySelect	INA_A.CPGC	INA.SRC
E22-BIT_GA_GateArraySelect	SNDWRD_A.CPGC	SNDWRD.SRC
E22-BIT_GA_GateArraySelect	URTOUT_A.CPGC	URTOUT.SRC
E22-BIT_GA_GateArraySelect	URTOUT_B.CPGC	URTOUT.SRC
E22-BIT_GA_GateArraySelect	URTRED_A.CPGC	URTRED.SRC
E22-BIT_GA_GateArraySelect	URTRED_B.CPGC	URTRED.SRC
E22-BIT_GA_GateArraySelect	WDGRST_A.CPGC	WDGRST.SRC
E23-BIT_Store_PLA_Trim_Data	E23_PLATRIM_A.CPGC	E23_PLATRIM.SRC
E23-BIT_Store_PLA_Trim_Data	E23_PLATRIM_B.CPGC	E23_PLATRIM.SRC
E23-BIT_Store_PLA_Trim_Data	E23_PLATRIM_C.CPGC	E23_PLATRIM.SRC
E25-BIT_EEC_Int_Fault_Detect	E25_FAULTDET_A.CPGC	E25_FAULTDET.SRC
E25-BIT_EEC_Int_Fault_Detect	OTA_A.CPGC	OTA.SRC
E25-BIT_EEC_Int_Fault_Detect	OTA_B.CPGC	OTA.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_1_A.CPGC	E26_FAULTSTORE_1.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_1_B.CPGC	E26_FAULTSTORE_1.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_1_C.CPGC	E26_FAULTSTORE_1.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_2_A.CPGC	E26_FAULTSTORE_2.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_2_B.CPGC	E26_FAULTSTORE_2.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_2_C.CPGC	E26_FAULTSTORE_2.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_3_A.CPGC	E26_FAULTSTORE_3.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_3_B.CPGC	E26_FAULTSTORE_3.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_3_C.CPGC	E26_FAULTSTORE_3.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_3_D.CPGC	E26_FAULTSTORE_3.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_4_A.CPGC	E26_FAULTSTORE_4.SRC
E26-INT_FAULTSTORE	E26_FAULTSTORE_4_B.CPGC	E26_FAULTSTORE_4.SRC
E26-INT_FAULTSTORE	E26-INT_FAULTSTORE	E26_FAULTSTORE_5.SRC
E27-Maint_Misc_MFC_Id		E08_FG_SCHEDULE.SRC
E27-Maint_Misc_MFC_Id	E27_MFCID_A.CPGC	E27_MFCID.SRC
E27-Maint_Misc_MFC_Id	FAILED_A.CPGC	FAILED.SRC
E27-Maint_Misc_MFC_Id	PASSED_A.CPGC	PASSED.SRC
E28-BIT_EEC_Bleed_W/A_Tests		E08_FG_SCHEDULE.SRC
E28-BIT_EEC_Bleed_W/A_Tests	E28_HBOVWA_1_A.CPGC	E28_HBOVWA_1.SRC
E28-BIT_EEC_Bleed_W/A_Tests	E28_HBOVWA_1_B.CPGC	E28_HBOVWA_1.SRC
E28-BIT_EEC_Bleed_W/A_Tests	E28_HBOVWA_2_A.CPGC	E28_HBOVWA_2.SRC
E28-BIT_EEC_Bleed_W/A_Tests	E28_HBOVWA_2_B.CPGC	E28_HBOVWA_2.SRC
E30 - IDENTIFI		DCU_COUNTER_DATA.S00
E30 - IDENTIFI		PID.SRC
E31-Maint_Misc_Ext_Disc_Word		E08_FG_SCHEDULE.SRC
E31-Maint_Misc_Ext_Disc_Word	E31_EXTDIS_A.CPGC	E31_EXTDIS.SRC
E31-Maint_Misc_Ext_Disc_Word	E31_EXTDIS_B.CPGC	E31_EXTDIS.SRC
E31-Maint_Misc_Ext_Disc_Word	E31_EXTDIS_C.CPGC	E31_EXTDIS.SRC

SRS Capability	Design	Module
E31-Maint_Misc_Ext_Disc_Word	E31_EXTDIS_D.CPGC	E31_EXTDIS.SRC
E31-Maint_Misc_Ext_Disc_Word	E31_EXTDIS_E.CPGC	E31_EXTDIS.SRC
E31-Maint_Misc_Ext_Disc_Word	E31_EXTDIS_F.CPGC	E31_EXTDIS.SRC
E31-Maint_Misc_Ext_Disc_Word	E31_EXTDIS_G.CPGC	E31_EXTDIS.SRC
EECOUNTERS		E08_FG_SCHEDULE.SRC
EECOUNTERS		EE_PUT_QUEUE.S00
EECOUNTERS		UPDATE_EE_DATA.S00
EEPROM_MAP		Page11.SRC
EVENT_COUNTING	EVENT_COUNTING	CNT_EVNT.S00
EVENT_COUNTING		E08_FG_SCHEDULE.SRC
EVENT_COUNTING	EVENT_COUNTING	FLTTIMER.S00
EVENT_COUNTING	INC_CTRS_1_A.CPGC	INC_CTRS_1.SRC
EVENT_COUNTING	INC_CTRS_1_B.CPGC	INC_CTRS_1.SRC
EVENT_COUNTING	INC_CTRS_1_C.CPGC	INC_CTRS_1.SRC
EVENT_COUNTING	INC_CTRS_1_D.CPGC	INC_CTRS_1.SRC
EVENT_COUNTING	INC_CTRS_2.CPGC	INC_CTRS_2.SRC
EVENT_COUNTING	INC_CTRS_3.CPGC	INC_CTRS_3.SRC
EVENT_COUNTING	INC_CTRS_4.CTRLFLOW	INC_CTRS_4.SRC
EVENT_COUNTING	EVENT_COUNTING	RUNTIMER.S00
I01-Sens_LL_Drift_Comp	FAILLT_A.CPGC	FAILLT.SRC
I01-Sens_LL_Drift_Comp	FAILLT_B.CPGC	FAILLT.SRC
I01-Sens_LL_Drift_Comp	I01_LLDRIFT_1_A.CPGC	I01_LLDRIFT_1.SRC
I01-Sens_LL_Drift_Comp	I01_LLDRIFT_2_A.CPGC	I01_LLDRIFT_2.SRC
I01-Sens_LL_Drift_Comp	I01_LLDRIFT_2_B.CPGC	I01_LLDRIFT_2.SRC
I01-Sens_LL_Drift_Comp	PASSLT_A.CPGC	PASSLT.SRC
I02-Sens_LL_Strain_Gauge_Comp	FAILLT_A.CPGC	FAILLT.SRC
I02-Sens_LL_Strain_Gauge_Comp	FAILLT_B.CPGC	FAILLT.SRC
I02-Sens_LL_Strain_Gauge_Comp	I02_SGCOM_1_A.CPGC	I02_SGCOM_1.SRC
I02-Sens_LL_Strain_Gauge_Comp	I02_SGCOM_2_A.CPGC	I02_SGCOM_2.SRC
I02-Sens_LL_Strain_Gauge_Comp	I02_SGCOM_2_B.CPGC	I02_SGCOM_2.SRC
I02-Sens_LL_Strain_Gauge_Comp	PASSLT_A.CPGC	PASSLT.SRC
I03-Sens_LL_Cold_Junct_Temp	FAILLT_A.CPGC	FAILLT.SRC
I03-Sens_LL_Cold_Junct_Temp	FAILLT_B.CPGC	FAILLT.SRC
I03-Sens_LL_Cold_Junct_Temp	I03_RDCJT_1_A.CPGC	I03_RDCJT_1.SRC
I03-Sens_LL_Cold_Junct_Temp	I03_RDCJT_2_A.CPGC	I03_RDCJT_2.SRC
I03-Sens_LL_Cold_Junct_Temp	I03_RDCJT_2_B.CPGC	I03_RDCJT_2.SRC
I03-Sens_LL_Cold_Junct_Temp	I03_RDCJT_2_C.CPGC	I03_RDCJT_2.SRC
I03-Sens_LL_Cold_Junct_Temp	I03_RDCJT_2_D.CPGC	I03_RDCJT_2.SRC
I03-Sens_LL_Cold_Junct_Temp	I03_RDCJT_3_A.CPGC	I03_RDCJT_3.SRC
I03-Sens_LL_Cold_Junct_Temp	I03_RDCJT_3_B.CPGC	I03_RDCJT_3.SRC
I03-Sens_LL_Cold_Junct_Temp	I03_RDCJT_3_C.CPGC	I03_RDCJT_3.SRC
I03-Sens_LL_Cold_Junct_Temp	PASSLT_A.CPGC	PASSLT.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	FAILLT_A.CPGC	FAILLT.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	FAILLT_B.CPGC	FAILLT.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	FILTER_A.CPGC	FILTER.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	FILTER_B.CPGC	FILTER.SRC

SRS Capability	Design	Module
I04-Sens_LL_PSAMB_Strain_Gauge	FILTER_C.CPGC	FILTER.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	I04_PSAMBSG_1_A.CPGC	I04_PSAMBSG_1.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	I04_PSAMBSG_2_A.CPGC	I04_PSAMBSG_2.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	I04_PSAMBSG_2_B.CPGC	I04_PSAMBSG_2.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	I04_PSAMBSG_2_C.CPGC	I04_PSAMBSG_2.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	I04_PSAMBSG_3_A.CPGC	I04_PSAMBSG_3.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	I04_PSAMBSG_3_B.CPGC	I04_PSAMBSG_3.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	I04_PSAMBSG_4_A.CPGC	I04_PSAMBSG_4.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	I04_PSAMBSG_5_A.CPGC	I04_PSAMBSG_5.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	PASSLT_A.CPGC	PASSLT.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	SUNVAR_A.CPGC	SUNVAR.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	SUNVAR_B.CPGC	SUNVAR.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	SUNVAR_C.CPGC	SUNVAR.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	UNIVAR_A.CPGC	UNIVAR.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	UNIVAR_B.CPGC	UNIVAR.SRC
I04-Sens_LL_PSAMB_Strain_Gauge	UNIVAR_C.CPGC	UNIVAR.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	FAILLT_A.CPGC	FAILLT.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	FAILLT_B.CPGC	FAILLT.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	FILTER_A.CPGC	FILTER.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	FILTER_B.CPGC	FILTER.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	FILTER_C.CPGC	FILTER.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_1_A.CPGC	I05_DELTAPSG_1.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_2_A.CPGC	I05_DELTAPSG_2.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_2_B.CPGC	I05_DELTAPSG_2.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_3_A.CPGC	I05_DELTAPSG_3.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_3_B.CPGC	I05_DELTAPSG_3.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_3_C.CPGC	I05_DELTAPSG_3.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_4_A.CPGC	I05_DELTAPSG_4.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_4_B.CPGC	I05_DELTAPSG_4.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_4_C.CPGC	I05_DELTAPSG_4.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	I05_DELTAPSG_4_D.CPGC	I05_DELTAPSG_4.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	PASSLT_A.CPGC	PASSLT.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	QUAD_A.CPGC	QUAD.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	QUAD_B.CPGC	QUAD.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	QUAD_C.CPGC	QUAD.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	SUNVAR_A.CPGC	SUNVAR.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	SUNVAR_B.CPGC	SUNVAR.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	SUNVAR_C.CPGC	SUNVAR.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	UNIVAR_A.CPGC	UNIVAR.SRC
I05-Sens_LL_DELTA_P_Strain_Gauge	UNIVAR_B.CPGC	UNIVAR.SRC
I07-Sens_LL_TQ_RTD	FAILLT_A.CPGC	FAILLT.SRC
I07-Sens_LL_TQ_RTD	FAILLT_B.CPGC	FAILLT.SRC
I07-Sens_LL_TQ_RTD	FILTER_A.CPGC	FILTER.SRC
I07-Sens_LL_TQ_RTD	FILTER_B.CPGC	FILTER.SRC
I07-Sens_LL_TQ_RTD	FILTER_C.CPGC	FILTER.SRC
I07-Sens_LL_TQ_RTD	I07_READTQ_1_A.CPGC	I07_READTQ_1.SRC

SRS Capability	Design	Module
I07-Sens_LL_TQ_RTD	I07_READTQ_2_A.CPGC	I07_READTQ_2.SRC
I07-Sens_LL_TQ_RTD	I07_READTQ_2_B.CPGC	I07_READTQ_2.SRC
I07-Sens_LL_TQ_RTD	I07_READTQ_3_A.CPGC	I07_READTQ_3.SRC
I07-Sens_LL_TQ_RTD	I07_READTQ_3_B.CPGC	I07_READTQ_3.SRC
I07-Sens_LL_TQ_RTD	I07_READTQ_3_C.CPGC	I07_READTQ_3.SRC
I07-Sens_LL_TQ_RTD	PASSLT_A.CPGC	PASSLT.SRC
I08-Sens_LL_T18_RTD	FAILED_A.CPGC	FAILED.SRC
I08-Sens_LL_T18_RTD	FAILLT_A.CPGC	FAILLT.SRC
I08-Sens_LL_T18_RTD	FAILLT_B.CPGC	FAILLT.SRC
I08-Sens_LL_T18_RTD	FILTER_A.CPGC	FILTER.SRC
I08-Sens_LL_T18_RTD	FILTER_B.CPGC	FILTER.SRC
I08-Sens_LL_T18_RTD	FILTER_C.CPGC	FILTER.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_1_A.CPGC	I08_RDT18_1.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_2_A.CPGC	I08_RDT18_2.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_3_A.CPGC	I08_RDT18_3.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_3_B.CPGC	I08_RDT18_3.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_3_C.CPGC	I08_RDT18_3.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_3_D.CPGC	I08_RDT18_3.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_3_E.CPGC	I08_RDT18_3.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_4_A.CPGC	I08_RDT18_4.SRC
I08-Sens_LL_T18_RTD	I08_RDT18_4_B.CPGC	I08_RDT18_4.SRC
I08-Sens_LL_T18_RTD	PASSED_A.CPGC	PASSED.SRC
I08-Sens_LL_T18_RTD	PASSLT_A.CPGC	PASSLT.SRC
I08-Sens_LL_T18_RTD	SUNVAR_A.CPGC	SUNVAR.SRC
I08-Sens_LL_T18_RTD	SUNVAR_B.CPGC	SUNVAR.SRC
I08-Sens_LL_T18_RTD	SUNVAR_C.CPGC	SUNVAR.SRC
I08-Sens_LL_T18_RTD	UNIVAR_A.CPGC	UNIVAR.SRC
I08-Sens_LL_T18_RTD	UNIVAR_B.CPGC	UNIVAR.SRC
I08-Sens_LL_T18_RTD	UNIVAR_C.CPGC	UNIVAR.SRC
I09-Sens_HL_Drift	FAILLT_A.CPGC	FAILLT.SRC
I09-Sens_HL_Drift	FAILLT_B.CPGC	FAILLT.SRC
I09-Sens_HL_Drift	I09_HLDRIFT_1_A.CPGC	I09_HLDRIFT_1.SRC
I09-Sens_HL_Drift	I09_HLDRIFT_2_A.CPGC	I09_HLDRIFT_2.SRC
I09-Sens_HL_Drift	I09_HLDRIFT_2_B.CPGC	I09_HLDRIFT_2.SRC
I09-Sens_HL_Drift	PASSLT_A.CPGC	PASSLT.SRC
I10-Sens_HL_Gain_Comp	FAILLT_A.CPGC	FAILLT.SRC
I10-Sens_HL_Gain_Comp	FAILLT_B.CPGC	FAILLT.SRC
I10-Sens_HL_Gain_Comp	I10_GAINCOM_1_A.CPGC	I10_GAINCOM_1.SRC
I10-Sens_HL_Gain_Comp	I10_GAINCOM_2_A.CPGC	I10_GAINCOM_2.SRC
I10-Sens_HL_Gain_Comp	I10_GAINCOM_2_B.CPGC	I10_GAINCOM_2.SRC
I10-Sens_HL_Gain_Comp	PASSLT_A.CPGC	PASSLT.SRC
I11-Sens_HL_Q_Bias_Trim	FAILLT_A.CPGC	FAILLT.SRC
I11-Sens_HL_Q_Bias_Trim	FAILLT_B.CPGC	FAILLT.SRC
I11-Sens_HL_Q_Bias_Trim	I11_TQBTRIM_1_A.CPGC	I11_TQBTRIM_1.SRC
I11-Sens_HL_Q_Bias_Trim	I11_TQBTRIM_2_A.CPGC	I11_TQBTRIM_2.SRC
I11-Sens_HL_Q_Bias_Trim	I11_TQBTRIM_2_B.CPGC	I11_TQBTRIM_2.SRC

SRS Capability	Design	Module
I11-Sens_HL_Q_Bias_Trim	I11_TQBTRIM_2_C.CPGC	I11_TQBTRIM_2.SRC
I11-Sens_HL_Q_Bias_Trim	PASSLT_A.CPGC	PASSLT.SRC
I12-Sens_HL_Q_Gain_Trim	FAILLT_A.CPGC	FAILLT.SRC
I12-Sens_HL_Q_Gain_Trim	FAILLT_B.CPGC	FAILLT.SRC
I12-Sens_HL_Q_Gain_Trim	I12_TQGTRIM_2_A.CPGC	I12_TQGTRIM_2.SRC
I12-Sens_HL_Q_Gain_Trim	I12_TQGTRIM_2_B.CPGC	I12_TQGTRIM_2.SRC
I12-Sens_HL_Q_Gain_Trim	I12_TQGTRIM_2_C.CPGC	I12_TQGTRIM_2.SRC
I12-Sens_HL_Q_Gain_Trim	PASSLT_A.CPGC	PASSLT.SRC
I12-Sens_HL_Q_Gain_Trim	SBIVAR_A.CPGC	SBIVAR.SRC
I12-Sens_HL_Q_Gain_Trim	SBIVAR_B.CPGC	SBIVAR.SRC
I12-Sens_HL_Q_Gain_Trim	SBIVAR_C.CPGC	SBIVAR.SRC
I12-Sens_HL_Q_Gain_Trim	SBIVAR_D.CPGC	SBIVAR.SRC
I12-Sens_HL_Q_Gain_Trim	SBIVAR_E.CPGC	SBIVAR.SRC
I12-Sens_HL_Q_Gain_Trim	SBIVAR_F.CPGC	SBIVAR.SRC
I12-Sens_HL_Q_Gain_Trim	TUSTIN_SUBR_A.CPGC	TUSTIN_SUBR.SRC
I12-Sens_HL_Q_Gain_Trim	TUSTIN_SUBR_B.CPGC	TUSTIN_SUBR.SRC
I12-Sens_HL_Q_Gain_Trim	TUSTIN_SUBR_C.CPGC	TUSTIN_SUBR.SRC
I13-ENG VARIANT		E08_FG_STATETABLE.SRC
I13-ENG VARIANT	I13_ENGVARIANT_2.CTRLFLOW	I13_ENGVARIANT_2.SRC
I14-Sens_HL_PLA	FAILED_A.CPGC	FAILED.SRC
I14-Sens_HL_PLA	FAILLT_A.CPGC	FAILLT.SRC
I14-Sens_HL_PLA	FAILLT_B.CPGC	FAILLT.SRC
I14-Sens_HL_PLA	FILTER_A.CPGC	FILTER.SRC
I14-Sens_HL_PLA	FILTER_B.CPGC	FILTER.SRC
I14-Sens_HL_PLA	I14_RDPLA_1_A.CPGC	I14_RDPLA_1.SRC
I14-Sens_HL_PLA	I14_RDPLA_1_B.CPGC	I14_RDPLA_1.SRC
I14-Sens_HL_PLA	I14_RDPLA_1_C.CPGC	I14_RDPLA_1.SRC
I14-Sens_HL_PLA	I14_RDPLA_2_A.CPGC	I14_RDPLA_2.SRC
I14-Sens_HL_PLA	I14_RDPLA_2_B.CPGC	I14_RDPLA_2.SRC
I14-Sens_HL_PLA	I14_RDPLA_2_C.CPGC	I14_RDPLA_2.SRC
I14-Sens_HL_PLA	I14_RDPLA_2_D.CPGC	I14_RDPLA_2.SRC
I14-Sens_HL_PLA	I14_RDPLA_3_A.CPGC	I14_RDPLA_3.SRC
I14-Sens_HL_PLA	I14_RDPLA_3_B.CPGC	I14_RDPLA_3.SRC
I14-Sens_HL_PLA	I14_RDPLA_3_C.CPGC	I14_RDPLA_3.SRC
I14-Sens_HL_PLA	I14_RDPLA_3_D.CPGC	I14_RDPLA_3.SRC
I14-Sens_HL_PLA	I14_RDPLA_3_E.CPGC	I14_RDPLA_3.SRC
I14-Sens_HL_PLA	I14_RDPLA_4_A.CPGC	I14_RDPLA_4.SRC
I14-Sens_HL_PLA	PASSED_A.CPGC	PASSED.SRC
I15-Sens_HL_PLA_Bias	FAILLT_A.CPGC	FAILLT.SRC
I15-Sens_HL_PLA_Bias	FAILLT_B.CPGC	FAILLT.SRC
I15-Sens_HL_PLA_Bias	I15_PLABIAS_1_A.CPGC	I15_PLABIAS_1.SRC
I15-Sens_HL_PLA_Bias	I15_PLABIAS_2_A.CPGC	I15_PLABIAS_2.SRC
I15-Sens_HL_PLA_Bias	I15_PLABIAS_2_B.CPGC	I15_PLABIAS_2.SRC
I15-Sens_HL_PLA_Bias	PASSLT_A.CPGC	PASSLT.SRC
I16-Sens_HL_PLA_Gain	FAILLT_A.CPGC	FAILLT.SRC
I16-Sens_HL_PLA_Gain	FAILLT_B.CPGC	FAILLT.SRC

SRS Capability	Design	Module
I16-Sens_HL_PLA_Gain	I16_PLAGAIN_1_A.CPGC	I16_PLAGAIN_1.SRC
I16-Sens_HL_PLA_Gain	I16_PLAGAIN_1_B.CPGC	I16_PLAGAIN_1.SRC
I16-Sens_HL_PLA_Gain	I16_PLAGAIN_2_A.CPGC	I16_PLAGAIN_2.SRC
I16-Sens_HL_PLA_Gain	I16_PLAGAIN_2_B.CPGC	I16_PLAGAIN_2.SRC
I16-Sens_HL_PLA_Gain	PASSLT_A.CPGC	PASSLT.SRC
I17-Sens_FR_Start_Pri_NH_Convert	GETWRD_A.CPGC	GETWRD.SRC
I17-Sens_FR_Start_Pri_NH_Convert	GETWRD_B.CPGC	GETWRD.SRC
I17-Sens_FR_Start_Pri_NH_Convert	I17_PNHCVN_1_A.CPGC	I17_PNHCVN_1.SRC
I17-Sens_FR_Start_Pri_NH_Convert	I17_PNHCVN_1_B.CPGC	I17_PNHCVN_1.SRC
I17-Sens_FR_Start_Pri_NH_Convert	I17_PNHCVN_1_C.CPGC	I17_PNHCVN_1.SRC
I17-Sens_FR_Start_Pri_NH_Convert	I17_PNHCVN_2_A.CPGC	I17_PNHCVN_2.SRC
I17-Sens_FR_Start_Pri_NH_Convert	INA_A.CPGC	INA.SRC
I17-Sens_FR_Start_Pri_NH_Convert	OTA_A.CPGC	OTA.SRC
I17-Sens_FR_Start_Pri_NH_Convert	OTA_B.CPGC	OTA.SRC
I17-Sens_FR_Start_Pri_NH_Convert	SNDWRD_A.CPGC	SNDWRD.SRC
I17-Sens_FR_Start_Pri_NH_Convert	URTOUT_A.CPGC	URTOUT.SRC
I17-Sens_FR_Start_Pri_NH_Convert	URTOUT_B.CPGC	URTOUT.SRC
I17-Sens_FR_Start_Pri_NH_Convert	URTRED_A.CPGC	URTRED.SRC
I17-Sens_FR_Start_Pri_NH_Convert	URTRED_B.CPGC	URTRED.SRC
I18-Sens_FR_Read_NH_Probe	GETWRD_A.CPGC	GETWRD.SRC
I18-Sens_FR_Read_NH_Probe	GETWRD_B.CPGC	GETWRD.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_A.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_B.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_C.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_D.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_E.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_F.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_G.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_H.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_I.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_J.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	I18_READNH_K.CPGC	I18_READNH.SRC
I18-Sens_FR_Read_NH_Probe	INA_A.CPGC	INA.SRC
I18-Sens_FR_Read_NH_Probe	OTA_A.CPGC	OTA.SRC
I18-Sens_FR_Read_NH_Probe	OTA_B.CPGC	OTA.SRC
I18-Sens_FR_Read_NH_Probe	URTOUT_A.CPGC	URTOUT.SRC
I18-Sens_FR_Read_NH_Probe	URTOUT_B.CPGC	URTOUT.SRC
I18-Sens_FR_Read_NH_Probe	URTRED_A.CPGC	URTRED.SRC
I18-Sens_FR_Read_NH_Probe	URTRED_B.CPGC	URTRED.SRC
I19-Sens_FR_Dual_Probe	FAILLT_A.CPGC	FAILLT.SRC
I19-Sens_FR_Dual_Probe	FAILLT_B.CPGC	FAILLT.SRC
I19-Sens_FR_Dual_Probe	I19_NHDUAL_1_A.CPGC	I19_NHDUAL_1.SRC
I19-Sens_FR_Dual_Probe	I19_NHDUAL_1_B.CPGC	I19_NHDUAL_1.SRC
I19-Sens_FR_Dual_Probe	I19_NHDUAL_2_A.CPGC	I19_NHDUAL_2.SRC
I19-Sens_FR_Dual_Probe	PASSLT_A.CPGC	PASSLT.SRC
I19-Sens_FR_Dual_Probe	SNDWRD_A.CPGC	SNDWRD.SRC

SRS Capability	Design	Module
I20-Sens_FR_NH_Speed_Filter	FILTER_A.CPGC	FILTER.SRC
I20-Sens_FR_NH_Speed_Filter	FILTER_B.CPGC	FILTER.SRC
I20-Sens_FR_NH_Speed_Filter	FILTER_C.CPGC	FILTER.SRC
I20-Sens_FR_NH_Speed_Filter	I20_NHSPEED_A.CPGC	I20_NHSPEED.SRC
I22-Sens_FR_Start_Sec_NH_Convert	GETWRD_A.CPGC	GETWRD.SRC
I22-Sens_FR_Start_Sec_NH_Convert	GETWRD_B.CPGC	GETWRD.SRC
I22-Sens_FR_Start_Sec_NH_Convert	I22_SNHCVN_1_A.CPGC	I22_SNHCVN_1.SRC
I22-Sens_FR_Start_Sec_NH_Convert	I22_SNHCVN_1_B.CPGC	I22_SNHCVN_1.SRC
I22-Sens_FR_Start_Sec_NH_Convert	I22_SNHCVN_2_A.CPGC	I22_SNHCVN_2.SRC
I22-Sens_FR_Start_Sec_NH_Convert	INA_A.CPGC	INA.SRC
I22-Sens_FR_Start_Sec_NH_Convert	OTA_A.CPGC	OTA.SRC
I22-Sens_FR_Start_Sec_NH_Convert	OTA_B.CPGC	OTA.SRC
I22-Sens_FR_Start_Sec_NH_Convert	SNDWRD_A.CPGC	SNDWRD.SRC
I22-Sens_FR_Start_Sec_NH_Convert	URTOUT_A.CPGC	URTOUT.SRC
I22-Sens_FR_Start_Sec_NH_Convert	URTOUT_B.CPGC	URTOUT.SRC
I22-Sens_FR_Start_Sec_NH_Convert	URTRED_A.CPGC	URTRED.SRC
I22-Sens_FR_Start_Sec_NH_Convert	URTRED_B.CPGC	URTRED.SRC
I23-Sens_FR_Read_NH_Probe_Two	GETWRD_A.CPGC	GETWRD.SRC
I23-Sens_FR_Read_NH_Probe_Two	GETWRD_B.CPGC	GETWRD.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_A.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_B.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_C.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_D.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_E.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_F.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_G.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_H.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_I.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_J.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	I23_RDSNHP_K.CPGC	I23_RDSNHP.SRC
I23-Sens_FR_Read_NH_Probe_Two	INA_A.CPGC	INA.SRC
I23-Sens_FR_Read_NH_Probe_Two	OTA_A.CPGC	OTA.SRC
I23-Sens_FR_Read_NH_Probe_Two	OTA_B.CPGC	OTA.SRC
I23-Sens_FR_Read_NH_Probe_Two	URTOUT_A.CPGC	URTOUT.SRC
I23-Sens_FR_Read_NH_Probe_Two	URTOUT_B.CPGC	URTOUT.SRC
I23-Sens_FR_Read_NH_Probe_Two	URTRED_A.CPGC	URTRED.SRC
I23-Sens_FR_Read_NH_Probe_Two	URTRED_B.CPGC	URTRED.SRC
I24-Sens_FR_NH_Probe_2_Filter	FILTER_A.CPGC	FILTER.SRC
I24-Sens_FR_NH_Probe_2_Filter	FILTER_B.CPGC	FILTER.SRC
I24-Sens_FR_NH_Probe_2_Filter	FILTER_C.CPGC	FILTER.SRC
I24-Sens_FR_NH_Probe_2_Filter	I24_NHPROBE2_A.CPGC	I24_NHPROBE2.SRC
I25-Sens_FR_Q_NPT_In	FAILED_A.CPGC	FAILED.SRC
I25-Sens_FR_Q_NPT_In	GETWRD_A.CPGC	GETWRD.SRC
I25-Sens_FR_Q_NPT_In	GETWRD_B.CPGC	GETWRD.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_1_A.CPGC	I25_TQNPT_1.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_2_A.CPGC	I25_TQNPT_2.SRC

SRS Capability	Design	Module
I25-Sens_FR_Q_NPT_In	I25_TQNPT_2_B.CPGC	I25_TQNPT_2.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_2_C.CPGC	I25_TQNPT_2.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_2_D.CPGC	I25_TQNPT_2.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_2_E.CPGC	I25_TQNPT_2.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_2_F.CPGC	I25_TQNPT_2.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_3_A.CPGC	I25_TQNPT_3.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_3_B.CPGC	I25_TQNPT_3.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_3_C.CPGC	I25_TQNPT_3.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_3_D.CPGC	I25_TQNPT_3.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_3_E.CPGC	I25_TQNPT_3.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_3_F.CPGC	I25_TQNPT_3.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_A.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_B.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_C.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_D.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_E.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_F.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_G.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_H.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_I.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_J.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_K.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_L.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_M.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_N.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_4_O.CPGC	I25_TQNPT_4.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_A.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_B.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_C.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_D.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_E.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_F.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_G.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_H.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_I.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_J.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_K.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_L.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_M.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_N.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_5_O.CPGC	I25_TQNPT_5.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_6_A.CPGC	I25_TQNPT_6.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_6_B.CPGC	I25_TQNPT_6.SRC
I25-Sens_FR_Q_NPT_In	I25_TQNPT_7_A.CPGC	I25_TQNPT_7.SRC
I25-Sens_FR_Q_NPT_In	INA_A.CPGC	INA.SRC
I25-Sens_FR_Q_NPT_In	OTA_A.CPGC	OTA.SRC

SRS Capability	Design	Module
I25-Sens_FR_Q_NPT_In	OTA_B.CPGC	OTA.SRC
I25-Sens_FR_Q_NPT_In	PASSED_A.CPGC	PASSED.SRC
I25-Sens_FR_Q_NPT_In	SNDWRD_A.CPGC	SNDWRD.SRC
I25-Sens_FR_Q_NPT_In	URTOUT_A.CPGC	URTOUT.SRC
I25-Sens_FR_Q_NPT_In	URTOUT_B.CPGC	URTOUT.SRC
I25-Sens_FR_Q_NPT_In	URTRED_A.CPGC	URTRED.SRC
I25-Sens_FR_Q_NPT_In	URTRED_B.CPGC	URTRED.SRC
I26-Sens_FR_Q_Shift_Deflect	I26_TQCAL_A.CPGC	I26_TQCAL.SRC
I26-Sens_FR_Q_Shift_Deflect	I26_TQCAL_B.CPGC	I26_TQCAL.SRC
I27-Sens_FR_Q_Shift_Speed_Filter	FILTER_A.CPGC	FILTER.SRC
I27-Sens_FR_Q_Shift_Speed_Filter	FILTER_B.CPGC	FILTER.SRC
I27-Sens_FR_Q_Shift_Speed_Filter	FILTER_C.CPGC	FILTER.SRC
I27-Sens_FR_Q_Shift_Speed_Filter	I27_TQSPEED_A.CPGC	I27_TQSPEED.SRC
I28-DISINW1		E08_FG_STATETABLE.SRC
I28-DISINW1	GETWRD_A.CPGC	GETWRD.SRC
I28-DISINW1	I28_DISINW1_A.CPGC	I28_DISINW1.SRC
I28-DISINW1	I28_DISINW1_B.CPGC	I28_DISINW1.SRC
I28-DISINW1	I28_DISINW1_C.CPGC	I28_DISINW1.SRC
I28-DISINW1	I28_DISINW1_D.CPGC	I28_DISINW1.SRC
I28-DISINW1	I28_DISINW1_E.CPGC	I28_DISINW1.SRC
I28-DISINW1	INA_A.CPGC	INA.SRC
I28-DISINW1	OTA_A.CPGC	OTA.SRC
I28-DISINW1	OTA_B.CPGC	OTA.SRC
I28-DISINW1	SNDWRD_A.CPGC	SNDWRD.SRC
I28-DISINW1	URTOUT_A.CPGC	URTOUT.SRC
I28-DISINW1	URTOUT_B.CPGC	URTOUT.SRC
I28-DISINW1	URTRED_A.CPGC	URTRED.SRC
I28-DISINW1	URTRED_B.CPGC	URTRED.SRC
I29-DISINW2		E08_FG_STATETABLE.SRC
I29-DISINW2	GETWRD_A.CPGC	GETWRD.SRC
I29-DISINW2	GETWRD_B.CPGC	GETWRD.SRC
I29-DISINW2	I29_DISINW2_A.CPGC	I29_DISINW2.SRC
I29-DISINW2	INA_A.CPGC	INA.SRC
I29-DISINW2	OTA_A.CPGC	OTA.SRC
I29-DISINW2	OTA_B.CPGC	OTA.SRC
I29-DISINW2	SNDWRD_A.CPGC	SNDWRD.SRC
I29-DISINW2	URTOUT_A.CPGC	URTOUT.SRC
I29-DISINW2	URTOUT_B.CPGC	URTOUT.SRC
I29-DISINW2	URTRED_A.CPGC	URTRED.SRC
I29-DISINW2	URTRED_B.CPGC	URTRED.SRC
I30-BIT_EEC_Disc_In	GETWRD_A.CPGC	GETWRD.SRC
I30-BIT_EEC_Disc_In	GETWRD_B.CPGC	GETWRD.SRC
I30-BIT_EEC_Disc_In	I30_DISINFALT_A.CPGC	I30_DISINFALT.SRC
I30-BIT_EEC_Disc_In	INA_A.CPGC	INA.SRC
I30-BIT_EEC_Disc_In	OTA_A.CPGC	OTA.SRC
I30-BIT_EEC_Disc_In	OTA_B.CPGC	OTA.SRC

SRS Capability	Design	Module
I30-BIT_EEC_Disc_In	SNDWRD_A.CPGC	SNDWRD.SRC
I30-BIT_EEC_Disc_In	URTOUT_A.CPGC	URTOUT.SRC
I30-BIT_EEC_Disc_In	URTOUT_B.CPGC	URTOUT.SRC
I30-BIT_EEC_Disc_In	URTRED_A.CPGC	URTRED.SRC
I30-BIT_EEC_Disc_In	URTRED_B.CPGC	URTRED.SRC
I31-NPARAMIN	DPA_A.CPGC	DPA.SRC
I31-NPARAMIN	DPA_B.CPGC	DPA.SRC
I31-NPARAMIN	GETWRD_A.CPGC	GETWRD.SRC
I31-NPARAMIN	GETWRD_B.CPGC	GETWRD.SRC
I31-NPARAMIN	I31_UARTIN_NPM_A.CPGC	I31_UARTIN_NPM.SRC
I31-NPARAMIN	I31_UARTIN_NPM_B.CPGC	I31_UARTIN_NPM.SRC
I31-NPARAMIN	I31_UARTIN_NPM_C.CPGC	I31_UARTIN_NPM.SRC
I31-NPARAMIN	I31_UARTIN_NPM_D.CPGC	I31_UARTIN_NPM.SRC
I31-NPARAMIN	INA_A.CPGC	INA.SRC
I31-NPARAMIN	OTA_A.CPGC	OTA.SRC
I31-NPARAMIN	OTA_B.CPGC	OTA.SRC
I31-NPARAMIN	SNDWRD_A.CPGC	SNDWRD.SRC
I31-NPARAMIN	URTOUT_A.CPGC	URTOUT.SRC
I31-NPARAMIN	URTOUT_B.CPGC	URTOUT.SRC
I31-NPARAMIN	URTRED_A.CPGC	URTRED.SRC
I31-NPARAMIN	URTRED_B.CPGC	URTRED.SRC
I31-UARTIN	I31-UARTIN	E08_FG_STATETABLE.SRC
I31-UARTIN		E08_FG_STATETABLE.SRC
I31-UARTIN	GETWRD_A.CPGC	GETWRD.SRC
I31-UARTIN	GETWRD_B.CPGC	GETWRD.SRC
I31-UARTIN	I31_UARTIN_A.CPGC	I31_UARTIN.SRC
I31-UARTIN	INA_A.CPGC	INA.SRC
I31-UARTIN	OTA_A.CPGC	OTA.SRC
I31-UARTIN	OTA_B.CPGC	OTA.SRC
I31-UARTIN	SNDWRD_A.CPGC	SNDWRD.SRC
I31-UARTIN	URTOUT_A.CPGC	URTOUT.SRC
I31-UARTIN	URTOUT_B.CPGC	URTOUT.SRC
I31-UARTIN	URTRED_A.CPGC	URTRED.SRC
I31-UARTIN	URTRED_B.CPGC	URTRED.SRC
I32-ARINC_HW	GETWRD_A.CPGC	GETWRD.SRC
I32-ARINC_HW	GETWRD_B.CPGC	GETWRD.SRC
I32-ARINC_HW	I32_ARINCIN_1_A.CPGC	I32_ARINCIN_1.SRC
I32-ARINC_HW	I32_ARINCIN_1_B.CPGC	I32_ARINCIN_1.SRC
I32-ARINC_HW	I32_ARINCIN_1_C.CPGC	I32_ARINCIN_1.SRC
I32-ARINC_HW	I32_ARINCIN_1_D.CPGC	I32_ARINCIN_1.SRC
I32-ARINC_HW	I32_ARINCIN_1_E.CPGC	I32_ARINCIN_1.SRC
I32-ARINC_HW	I32_ARINCIN_1_F.CPGC	I32_ARINCIN_1.SRC
I32-ARINC_HW	I32_ARINCIN_2_A.CPGC	I32_ARINCIN_2.SRC
I32-ARINC_HW	INA_A.CPGC	INA.SRC
I32-ARINC_HW	OTA_A.CPGC	OTA.SRC
I32-ARINC_HW	OTA_B.CPGC	OTA.SRC

SRS Capability	Design	Module
I32-ARINC_HW	SNDWRD_A.CPGC	SNDWRD.SRC
I32-ARINC_HW	URTOUT_A.CPGC	URTOUT.SRC
I32-ARINC_HW	URTRED_A.CPGC	URTRED.SRC
I32-ARINC_HW	URTRED_B.CPGC	URTRED.SRC
I33-ARINC_In_TAT	I33_ARITAT_A.CPGC	I33_ARITAT.SRC
I33-ARINC_In_TAT	I33_ARITAT_B.CPGC	I33_ARITAT.SRC
I33-ARINC_In_TAT	I33_ARITAT_C.CPGC	I33_ARITAT.SRC
I33-ARINC_In_TAT	I33_ARITAT_D.CPGC	I33_ARITAT.SRC
I33-ARINC_In_TAT	I33_ARITAT_E.CPGC	I33_ARITAT.SRC
I33-ARINC_In_TAT	I33_ARITAT_F.CPGC	I33_ARITAT.SRC
I33-ARINC_In_TAT	I33_ARITAT_G.CPGC	I33_ARITAT.SRC
I33-ARINC_In_TAT	I33_ARITAT_H.CPGC	I33_ARITAT.SRC
I33-ARINC_In_TAT	UNIVAR_A.CPGC	UNIVAR.SRC
I33-ARINC_In_TAT	UNIVAR_B.CPGC	UNIVAR.SRC
I33-ARINC_In_TAT	UNIVAR_C.CPGC	UNIVAR.SRC
I34-ARINC_In_TAT_Source_Sel	I34_TATSEL_A.CPGC	I34_TATSEL.SRC
I34-ARINC_In_TAT_Source_Sel	I34_TATSEL_B.CPGC	I34_TATSEL.SRC
I34-ARINC_In_TAT_Source_Sel	I34_TATSEL_C.CPGC	I34_TATSEL.SRC
I34-ARINC_In_TAT_Source_Sel	I34_TATSEL_D.CPGC	I34_TATSEL.SRC
I34-ARINC_In_TAT_Source_Sel	I34_TATSEL_E.CPGC	I34_TATSEL.SRC
I34-ARINC_In_TAT_Source_Sel	I34_TATSEL_F.CPGC	I34_TATSEL.SRC
I34-ARINC_In_TAT_Source_Sel	I34_TATSEL_G.CPGC	I34_TATSEL.SRC
I34-ARINC_In_TAT_Source_Sel	SUNVAR_A.CPGC	SUNVAR.SRC
I34-ARINC_In_TAT_Source_Sel	SUNVAR_B.CPGC	SUNVAR.SRC
I34-ARINC_In_TAT_Source_Sel	SUNVAR_C.CPGC	SUNVAR.SRC
I34-ARINC_In_TAT_Source_Sel	UNIVAR_A.CPGC	UNIVAR.SRC
I34-ARINC_In_TAT_Source_Sel	UNIVAR_B.CPGC	UNIVAR.SRC
I34-ARINC_In_TAT_Source_Sel	UNIVAR_C.CPGC	UNIVAR.SRC
I35-ARINC_In_Press_Altitude	I35_ARIPA_A.CPGC	I35_ARIPA.SRC
I35-ARINC_In_Press_Altitude	I35_ARIPA_B.CPGC	I35_ARIPA.SRC
I35-ARINC_In_Press_Altitude	I35_ARIPA_C.CPGC	I35_ARIPA.SRC
I35-ARINC_In_Press_Altitude	I35_ARIPA_D.CPGC	I35_ARIPA.SRC
I35-ARINC_In_Press_Altitude	I35_ARIPA_E.CPGC	I35_ARIPA.SRC
I35-ARINC_In_Press_Altitude	I35_ARIPA_F.CPGC	I35_ARIPA.SRC
I35-ARINC_In_Press_Altitude	I35_ARIPA_G.CPGC	I35_ARIPA.SRC
I35-ARINC_In_Press_Altitude	SUNVAR_A.CPGC	SUNVAR.SRC
I35-ARINC_In_Press_Altitude	SUNVAR_B.CPGC	SUNVAR.SRC
I35-ARINC_In_Press_Altitude	SUNVAR_C.CPGC	SUNVAR.SRC
I35-ARINC_In_Press_Altitude	UNIVAR_A.CPGC	UNIVAR.SRC
I35-ARINC_In_Press_Altitude	UNIVAR_B.CPGC	UNIVAR.SRC
I35-ARINC_In_Press_Altitude	UNIVAR_C.CPGC	UNIVAR.SRC
I36-ARINC_In_Press_Source_Sel	I36_PAMSEL_A.CPGC	I36_PAMSEL.SRC
I36-ARINC_In_Press_Source_Sel	I36_PAMSEL_B.CPGC	I36_PAMSEL.SRC
I36-ARINC_In_Press_Source_Sel	I36_PAMSEL_C.CPGC	I36_PAMSEL.SRC
I36-ARINC_In_Press_Source_Sel	I36_PAMSEL_D.CPGC	I36_PAMSEL.SRC
I36-ARINC_In_Press_Source_Sel	I36_PAMSEL_E.CPGC	I36_PAMSEL.SRC

SRS Capability	Design	Module
I36-ARINC_In_Press_Source_Sel	I36_PAMSEL_F.CPGC	I36_PAMSEL.SRC
I36-ARINC_In_Press_Source_Sel	I36_PAMSEL_G.CPGC	I36_PAMSEL.SRC
I36-ARINC_In_Press_Source_Sel	I36_PAMSEL_H.CPGC	I36_PAMSEL.SRC
I36-ARINC_In_Press_Source_Sel	SUNVAR_A.CPGC	SUNVAR.SRC
I36-ARINC_In_Press_Source_Sel	SUNVAR_B.CPGC	SUNVAR.SRC
I36-ARINC_In_Press_Source_Sel	SUNVAR_C.CPGC	SUNVAR.SRC
I36-ARINC_In_Press_Source_Sel	UNIVAR_A.CPGC	UNIVAR.SRC
I36-ARINC_In_Press_Source_Sel	UNIVAR_B.CPGC	UNIVAR.SRC
I36-ARINC_In_Press_Source_Sel	UNIVAR_C.CPGC	UNIVAR.SRC
I37-ARINC_IN_IAS	I37_ARIISA_1_A.CPGC	I37_ARIISA_1.SRC
I37-ARINC_IN_IAS	I37_ARIISA_1_B.CPGC	I37_ARIISA_1.SRC
I37-ARINC_IN_IAS	I37_ARIISA_1_C.CPGC	I37_ARIISA_1.SRC
I37-ARINC_IN_IAS	I37_ARIISA_1_D.CPGC	I37_ARIISA_1.SRC
I37-ARINC_IN_IAS	I37_ARIISA_1_E.CPGC	I37_ARIISA_1.SRC
I37-ARINC_IN_IAS	I37_ARIISA_1_F.CPGC	I37_ARIISA_1.SRC
I37-ARINC_IN_IAS	I37_ARIISA_1_G.CPGC	I37_ARIISA_1.SRC
I37-ARINC_IN_IAS	I37_ARIISA_1_H.CPGC	I37_ARIISA_1.SRC
I37-ARINC_IN_IAS	I37_ARIISA_2_A.CPGC	I37_ARIISA_2.SRC
I37-ARINC_IN_IAS	I37_ARIISA_2_B.CPGC	I37_ARIISA_2.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	GETWRD_A.CPGC	GETWRD.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	GETWRD_B.CPGC	GETWRD.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	I38_SMCSV_1_A.CPGC	I38_SMCSV_1.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	I38_SMCSV_1_B.CPGC	I38_SMCSV_1.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	I38_SMCSV_2_A.CPGC	I38_SMCSV_2.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	I38_SMCSV_2_B.CPGC	I38_SMCSV_2.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	I38_SMCSV_2_C.CPGC	I38_SMCSV_2.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	INA_A.CPGC	INA.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	OTA_A.CPGC	OTA.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	OTA_B.CPGC	OTA.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	SNDWRD_A.CPGC	SNDWRD.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	URTOUT_A.CPGC	URTOUT.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	URTOUT_B.CPGC	URTOUT.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	URTRED_A.CPGC	URTRED.SRC
I38-Sens_HL_SM_Coil_Sum_Volt	URTRED_B.CPGC	URTRED.SRC
I43-Sens_FR_Q/Speed_Dual_In_Fault	I43_TQSPEED_1_A.CPGC	I43_TQSPEED_1.SRC
I43-Sens_FR_Q/Speed_Dual_In_Fault	I43_TQSPEED_1_B.CPGC	I43_TQSPEED_1.SRC
I43-Sens_FR_Q/Speed_Dual_In_Fault	I43_TQSPEED_2_A.CPGC	I43_TQSPEED_2.SRC
I43-Sens_FR_Q/Speed_Dual_In_Fault	I43_TQSPEED_3_A.CPGC	I43_TQSPEED_3.SRC
I43-Sens_FR_Q/Speed_Dual_In_Fault	I43_TQSPEED_3_B.CPGC	I43_TQSPEED_3.SRC
I45-Sens_HL_Bleed_W/A_Input	FAILLT_A.CPGC	FAILLT.SRC
I45-Sens_HL_Bleed_W/A_Input	I45_HBOVWA_1_A.CPGC	I45_HBOVWA_1.SRC
I45-Sens_HL_Bleed_W/A_Input	I45_HBOVWA_2_A.CPGC	I45_HBOVWA_2.SRC
I45-Sens_HL_Bleed_W/A_Input	I45_HBOVWA_2_B.CPGC	I45_HBOVWA_2.SRC
I45-Sens_HL_Bleed_W/A_Input	PASSLT_A.CPGC	PASSLT.SRC
I46-ARINC_In_IAS_Fault	I46-ARINC_In_IAS_Fault	I46_IASFAULT.SRC
Internal Interfaces		PAGE0.SRC

SRS Capability	Design	Module
Internal Interfaces		Page1.SRC
Internal Interfaces		PAGE10.SRC
Internal Interfaces		Page11.SRC
LCF_FACTCHK	LCF_FACTCHK	DCU_COMPARE_ITEM.S00
LCF_FACTCHK		DCU_CONSTANTS.S00
LCF_FACTCHK	LCF_FACTCHK	DCU_GET_CHECKSUM.S00
LCF_FACTCHK	LCF_FACTCHK	DCU_GET_DATA.S00
LCF_FACTCHK		DCU_LCF_DATA.S00
LCF_FACTCHK	LCF_FACTCHK	DCU_SCHEDULE.S00
LCF_FACTCHK		E08_FG_SCHEDULE.SRC
MISSION_DETERM		E08_FG_SCHEDULE.SRC
MISSION_DETERM	MISSIONS.DATAFLOW	MISSIONS.SRC
O01-Out_SM	GETWRD_A.CPGC	GETWRD.SRC
O01-Out_SM	GETWRD_B.CPGC	GETWRD.SRC
O01-Out_SM	INA_A.CPGC	INA.SRC
O01-Out_SM	O01_SMOUT_A.CPGC	O01_SMOUT.SRC
O01-Out_SM	O01_SMOUT_B.CPGC	O01_SMOUT.SRC
O01-Out_SM	O01_SMOUT_C.CPGC	O01_SMOUT.SRC
O01-Out_SM	O01-Out_SM	OTA.SRC
O01-Out_SM	OTA_A.CPGC	OTA.SRC
O01-Out_SM	OTA_B.CPGC	OTA.SRC
O01-Out_SM	O01-Out_SM	SNDWRD.SRC
O01-Out_SM	SNDWRD_A.CPGC	SNDWRD.SRC
O01-Out_SM	URTOUT_A.CPGC	URTOUT.SRC
O01-Out_SM	URTOUT_B.CPGC	URTOUT.SRC
O01-Out_SM	URTRED_A.CPGC	URTRED.SRC
O01-Out_SM	URTRED_B.CPGC	URTRED.SRC
O02-HBOV	GETWRD_A.CPGC	GETWRD.SRC
O02-HBOV	GETWRD_B.CPGC	GETWRD.SRC
O02-HBOV	INA_A.CPGC	INA.SRC
O02-HBOV	O02_HBOVOUT_A.CPGC	O02_HBOVOUT.SRC
O02-HBOV	OTA_A.CPGC	OTA.SRC
O02-HBOV	OTA_B.CPGC	OTA.SRC
O02-HBOV	SNDWRD_A.CPGC	SNDWRD.SRC
O02-HBOV	URTOUT_A.CPGC	URTOUT.SRC
O02-HBOV	URTOUT_B.CPGC	URTOUT.SRC
O02-HBOV	URTRED_A.CPGC	URTRED.SRC
O02-HBOV	URTRED_B.CPGC	URTRED.SRC
O03 - OUT_DISC		E08_FG_SCHEDULE.SRC
O03 - OUT_DISC	O03_DISOUT.CTRLFLOW	O03_DISOUT.S00
O03 - OUT_DISC	O03_DISOUT_A.DATAFLOW	O03_DISOUT_A.SRC
O04 - NPARAMOUT	GETWRD_A.CPGC	GETWRD.SRC
O04 - NPARAMOUT	GETWRD_B.CPGC	GETWRD.SRC
O04 - NPARAMOUT	INA_A.CPGC	INA.SRC
O04 - NPARAMOUT	O04_UARTOUT_NPM_A.CPGC	O04_UARTOUT_NPM.SRC
O04 - NPARAMOUT	O04_UARTOUT_NPM_B.CPGC	O04_UARTOUT_NPM.SRC

SRS Capability	Design	Module
O04 - NPARAMOUT	OTA_A.CPGC	OTA.SRC
O04 - NPARAMOUT	OTA_B.CPGC	OTA.SRC
O04 - NPARAMOUT	URTOUT_A.CPGC	URTOUT.SRC
O04 - NPARAMOUT	URTOUT_B.CPGC	URTOUT.SRC
O04 - NPARAMOUT	URTRED_A.CPGC	URTRED.SRC
O04 - NPARAMOUT	URTRED_B.CPGC	URTRED.SRC
O04-UART_OUT	GETWRD_A.CPGC	GETWRD.SRC
O04-UART_OUT	GETWRD_B.CPGC	GETWRD.SRC
O04-UART_OUT	INA_A.CPGC	INA.SRC
O04-UART_OUT	O04_UARTOUT_DTS_A.CPGC	O04_UARTOUT_DTS.SRC
O04-UART_OUT	O04_UARTOUT_DTS_B.CPGC	O04_UARTOUT_DTS.SRC
O04-UART_OUT	O04_UARTOUT_DTS_C.CPGC	O04_UARTOUT_DTS.SRC
O04-UART_OUT	O04_UARTOUT_DTS_D.CPGC	O04_UARTOUT_DTS.SRC
O04-UART_OUT	OTA_A.CPGC	OTA.SRC
O04-UART_OUT	OTA_B.CPGC	OTA.SRC
O04-UART_OUT	URTOUT_A.CPGC	URTOUT.SRC
O04-UART_OUT	URTOUT_B.CPGC	URTOUT.SRC
O04-UART_OUT	URTRED_A.CPGC	URTRED.SRC
O04-UART_OUT	URTRED_B.CPGC	URTRED.SRC
O04-UART_OUT1	GETWRD_A.CPGC	GETWRD.SRC
O04-UART_OUT1	GETWRD_B.CPGC	GETWRD.SRC
O04-UART_OUT1	INA_A.CPGC	INA.SRC
O04-UART_OUT1	O04_UARTOUT_GDTS_A.CPGC	O04_UARTOUT_GDTS.SRC
O04-UART_OUT1	O04_UARTOUT_GDTS_B.CPGC	O04_UARTOUT_GDTS.SRC
O04-UART_OUT1	O04_UARTOUT_GDTS_C.CPGC	O04_UARTOUT_GDTS.SRC
O04-UART_OUT1	O04_UARTOUT_GDTS_D.CPGC	O04_UARTOUT_GDTS.SRC
O04-UART_OUT1	OTA_A.CPGC	OTA.SRC
O04-UART_OUT1	OTA_B.CPGC	OTA.SRC
O04-UART_OUT1	SNDWRD_A.CPGC	SNDWRD.SRC
O04-UART_OUT1	URTOUT_A.CPGC	URTOUT.SRC
O04-UART_OUT1	URTOUT_B.CPGC	URTOUT.SRC
O04-UART_OUT1	URTRED_A.CPGC	URTRED.SRC
O04-UART_OUT1	URTRED_B.CPGC	URTRED.SRC
O05-ARINCO_133	O05_ARO133_A.CPGC	O05_ARO133.SRC
O06-ARINCO_134	O06_ARO134_A.CPGC	O06_ARO134.SRC
O07-ARINCO_203_ALT	O07_ARO203_ALT11_A.CPGC	O07_ARO203_ALT11.SRC
O07-ARINCO_203_ALTPS	O07_ARO203_ALTPS_A.CPGC	O07_ARO203_ALTPS.SRC
O09_ARO211_TAT11	O09_ARO211_TAT11_A.CPGC	O09_ARO211_TAT11.SRC
O10-ARINCO_240	O10_ARO240_A.CPGC	O10_ARO240.SRC
O10-ARINCO_240	O10_ARO240_B.CPGC	O10_ARO240.SRC
O11-ARINCO_245	O11_ARO245_A.CPGC	O11_ARO245.SRC
O11-ARINCO_245	O11_ARO245_B.CPGC	O11_ARO245.SRC
O15-ARINCO_341		E08_FG_STATETABLE.SRC
O15-ARINCO_341	O15_ARO341_A.CPGC	O15_ARO341.SRC
O15-ARINCO_341	O15_ARO341_B.CPGC	O15_ARO341.SRC
O16-ARINCO_346	O16_ARO346_A.CPGC	O16_ARO346.SRC

SRS Capability	Design	Module
O17-ARINCO_350	O17_ARO350_A.CPGC	O17_ARO350.SRC
O17-ARINCO_350	O17_ARO350_B.CPGC	O17_ARO350.SRC
O18-ARINCO_270		E08_FG_STATETABLE.SRC
O18-ARINCO_270	O18_ARO270_G.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_H.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_I.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_A.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_B.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_C.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_D.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_E.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_F.CPGC	O18_ARO270.SRC
O18-ARINCO_270	O18_ARO270_HA.CPGC	O18_ARO270.SRC
O19-ARINCO_272	O19_ARO272.CPGC	O19_ARO272.SRC
O19-ARINCO_272	O19_ARO272_A.DATAFLOW	O19_ARO272_A.SRC
O22-ARINC_Out_SubRoutine	ARCOUT_A.CPGC	ARCOUT.SRC
O22-ARINC_Out_SubRoutine	ARCOUT_B.CPGC	ARCOUT.SRC
O22-ARINC_Out_SubRoutine	GETWRD_A.CPGC	GETWRD.SRC
O22-ARINC_Out_SubRoutine	GETWRD_B.CPGC	GETWRD.SRC
O22-ARINC_Out_SubRoutine	INA_A.CPGC	INA.SRC
O22-ARINC_Out_SubRoutine	OTA_A.CPGC	OTA.SRC
O22-ARINC_Out_SubRoutine	OTA_B.CPGC	OTA.SRC
O22-ARINC_Out_SubRoutine	SNDWRD_A.CPGC	SNDWRD.SRC
O22-ARINC_Out_SubRoutine	URTOUT_A.CPGC	URTOUT.SRC
O22-ARINC_Out_SubRoutine	URTOUT_B.CPGC	URTOUT.SRC
O22-ARINC_Out_SubRoutine	URTRED_A.CPGC	URTRED.SRC
O22-ARINC_Out_SubRoutine	URTRED_B.CPGC	URTRED.SRC
O26-ARINCO_340	O26_ARO340_A.CPGC	O26_ARO340.SRC
O27_ARINCO_342	O27_ARO342_A.CPGC	O27_ARO342.SRC
O27_ARINCO_342	O27_ARO342_B.CPGC	O27_ARO342.SRC
O27_ARINCO_342	O27_ARO342_A.DATAFLOW	O27_ARO342_A.SRC
O40-ARINCO_062		E08_FG_STATETABLE.SRC
O40-ARINCO_062		O40_ARO06X.S00
O40-ARINCO_062		O40_ARO06X_TABLE.S00
O41-ARINCO_064		E08_FG_STATETABLE.SRC
O41-ARINCO_064		O40_ARO06X.S00
O41-ARINCO_064		O40_ARO06X_TABLE.S00
O42-ARINCO_063		E08_FG_STATETABLE.SRC
O42-ARINCO_063		O40_ARO06X.S00
O42-ARINCO_063		O40_ARO06X_TABLE.S00
O43-ARINCO_065		E08_FG_STATETABLE.SRC
O43-ARINCO_065		O40_ARO06X.S00
O43-ARINCO_065		O40_ARO06X_TABLE.S00
O44-ARINCO_066		E08_FG_STATETABLE.SRC
O44-ARINCO_066		O40_ARO06X.S00
O44-ARINCO_066		O40_ARO06X_TABLE.S00

SRS Capability	Design	Module
O45-ARINCO_067		E08_FG_STATETABLE.SRC
O45-ARINCO_067		O40_ARO06X.S00
O45-ARINCO_067		O40_ARO06X_TABLE.S00
STRT_FLIGHT_DETERM		E08_FG_SCHEDULE.SRC
STRT_FLIGHT_DETERM	STRT_FLIGHT_DETERM	E08_FG_SCHEDULE.SRC
STRT_FLIGHT_DETERM	OPMODE_1.DATAFLOW	OPMODE_1.S00
STRT_FLIGHT_DETERM	OPMODE_2.DATAFLOW	OPMODE_2.S00
STRT_FLIGHT_DETERM	OPMODE_3.DATAFLOW	OPMODE_3.SRC
UART_OUT_SEL		E08_FG_SCHEDULE.SRC
UART_OUT_SEL	O04_UARTOUTSEL.CPGC	O04_UARTOUTSEL.SRC
UART_OUT_SEL	UART_IO_SELECT.CTRLFLOW	UART_IO_SELECT.S00
	I13_ENGVARIANT_1_A.CPGC	I13_ENGVARIANT_1.SRC

07.02 RESERVED

08.00 MODULE SEQUENCING

08.01 POR/PSR MODULE SEQUENCING

The table below lists the calling architecture of modules called during POR/PSR, any System Registers that are used, the calling level of the module, and a comment to explain any module calls to a lower level than itself.

08.01 Table 1: POR-PSR-Module-Sequencing

Modules	Subroutine	Sub-subroutines	Sub-Sub-subroutine	System Register	Module Level	Comments
E01_PORPSR						
	E02_RAMTEST			SP4		RAM test
		WDGRST		SP1		Service Watchdog Timer
		RAMSTOR0		SP1		1st RAM test: Write pattern 1 (all-ones) on page 0
		WDGRST		SP1		
		RAMREAD0		SP1		1st RAM test: Read pattern 1 (all-ones) on page 0
		WDGRST		SP1		
		RAMSTOR1		SP1		1st RAM test: Write pattern 1 (all-ones) on page 1
		WDGRST		SP1		
		RAMREAD1		SP1		1st RAM test: Read pattern 1 (all-ones) on page 1
		WDGRST		SP1		
		RAMSTOR0		SP1		2nd RAM test: Write pattern 0 (all-zeros) on page 0
		WDGRST		SP1		
		RAMREAD0		SP1		2nd RAM test: Read pattern 0 (all-zeros) on page 0
		WDGRST		SP1		
		RAMSTOR1		SP1		2nd RAM test: Write pattern 0 (all-zeros) on page 1
		WDGRST		SP1		
		RAMREAD1		SP1		2nd RAM test: Read pattern 0 (all-zeros) on page 1
		WDGRST		SP1		
	E03_CHECKSUM			SP4		Checksum tests: Program Memory, P0 Sensor Data, Delta-P Sensor Data
		WDGRST		SP1		Service Watchdog Timer
		EESUM_16		ROMRET		Checksum routine for 16-bit memory (Program Memory)
		WDGRST		SP1		
		EESUM_8		ROMRET		Checksum routine for 8-bit memory (P0 Sensor Data)
		EESUM_8		ROMRET		Checksum routine for 8-bit memory (Delta-P Sensor Data)
	E04_INSTST			SP4		Instruction test
		INSTST		INSRET		Perform instruction test and return result
	E09_SMSUMV			SP4		Stepper Motor Sum Voltage test
		WDGRST		SP1		
		I38_SMCSV_1		PRETRN		Start A/D conversion
	RTDWRITE			RETRN2		RTD EEPROM Table
		WDGRST		SP1		
		WDGRST		SP1		
		WDGRST		SP1		
		WDGRST		SP1		
	E05_RAMINI			SP4		RAM Initialization
		WDGRST		SP1		
		INI_DCUI_DATA		RETRN1	LEV2	Initialize DCU data
		EE_READ_DATA		RETRN2	LEV3	Read data from EEPROM
		EE_READ_DATA		RETRN2	LEV3	Read data from EEPROM
		EE_READ_DATA		RETRN2	LEV3	Read data from EEPROM
		EE_READ_DATA		RETRN2	LEV3	Read data from EEPROM
		DCU_RANGE_CHECKS		RETRN2	LEV3	Range-check DCU data
		INIFLT		RETURN		Initialize fault variables and the temp fault words
		CMFACT		SBRTN1		Compact RAM Fault area
		WDGRST		SP1		
	E22_GASEL			SP4		Gate Array selection
		WDGRST		SP1		
		WDGRST		SP1		
		WDGRST		SP1		
		WDGRST		SP1		
	QTEST			SP4		Torque Interface Test
		WDGRST		SP1		
	WDGRST			SP1		Service Watchdog Timer
	@BACADR					Jump to Background; never returns

08.02 BACKGROUND MODULE SEQUENCING

The table below lists the calling architecture of modules called during Background, any System Registers that are used, the calling level of the module, and a comment to explain any module calls to a level other than LEV6 or LEV7, or any system registers used besides the dedicated return.

08.02 Table 1: Background-Module-Sequencing

Modules	Subroutines	System Register	Module Level	Comments
E07_BG_EXECUTIVE				Never returns
	INSTST	INSRET		Perform instruction test and return result
	ROMSUM	ROMRET		ROM Page Checksum Test

08.03 REAL-TIME INTERRUPT MODULE SEQUENCING

The table below lists the calling architecture of modules called during the Real-time Interrupt, any System Registers that are used, the calling level of the module, and a comment to explain any module calls to a level lower than itself or any system registers used besides the dedicated return.

08.03 Table 1: Real-Time-Interrupt-Module-Sequencing

Modules	Subroutines	Sub-Subroutines	System Register	Module Level	Comments
E08_FG_EXECUTIVE			ASAVE, BSAVE, OSAVE	LEV0	Saves and restores A,B, O registers
	ARINC_INPUT		PRETRN		Process ARINC input
	DCU_TRANSFER		RTRET2	RTLEV2	DCU transmit & receive
		DCU_TRANSMIT	RTRET3	RTLEV3	DCU transmit bytes
		DCU_RECEIVE	RTRET3	RTLEV3	DCU receive bytes
	I31_UARTIN_NPM		PRETRN		Process N-Param commands
	O04_UARTOUTSEL		PSBRTN		Choose type of UART output
		O04_UARTOUT_DTS	PRETRN		UART output in DTS mode
		O04_UARTOUT_GDTS	PRETRN		UART output in GDTS mode
		O04_UARTOUT_NPM	PRETRN		UART output for N-Param
	@PRETRN				Primary state table; increments and branches to CNTPNT pointer
		(primary state table)			
	SECSEL				Secondary state table
		(secondary state table)			
	@RETBKG				Returns to background

08.04 MODULE FLOW/STATE TABLE FOREGROUND MODULE SEQUENCING

08.04.01 Primary State Table

* NOTICE TO ALL PERSONS RECEIVING THIS DATA:

* WARNING UNPUBLISHED WORK ≪ 2023 COLLINS AEROSPACE. WITHIN THIS DOCUMENT,
* THERE IS, SO DESIGNATED, PROPERTY OF COLLINS AEROSPACE AND PRATT & WHITNEY
* CANADA CORP. AS DOCUMENTED IN THE COLLINS AEROSPACE/PRATT & WHITNEY CANADA
* CORP. SIDE LETTER AGREEMENT JKH-21-004 DATED 3-19-2021, AS MODIFIED BY
* AMENDMENT 1 JKH-22-002 DATED 2-23-2022. YOU MAY NOT POSSESS, USE, COPY OR
* DISCLOSE THIS DOCUMENT OR ANY INFORMATION IN IT, FOR ANY PURPOSE, INCLUDING
* WITHOUT LIMITATION, TO DESIGN, MANUFACTURE OR REPAIR PARTS, OR OBTAIN ANY
* GOVERNMENT APPROVAL TO DO SO, WITHOUT EXPRESS WRITTEN PERMISSION OF COLLINS
* AEROSPACE AND PRATT & WHITNEY CANADA CORP. NEITHER RECEIPT NOR POSSESSION OF
* THIS DOCUMENT ALONE, FROM ANY SOURCE, CONSTITUTES SUCH PERMISSION. POSSESSION,
* USE, COPYING OR DISCLOSURE BY ANYONE WITHOUT EXPRESS WRITTEN PERMISSION IS
* NOT AUTHORIZED AND MAY RESULT IN CRIMINAL AND/OR CIVIL LIABILITY.

* SUBJECT TO THE EAR, ECCN [9E991].

* THIS INFORMATION IS SUBJECT TO THE EXPORT CONTROL LAWS OF THE UNITED
* STATES, SPECIFICALLY INCLUDING THE EXPORT ADMINISTRATION REGULATIONS
* (EAR), 15 C.F.R. PART 730 ET. SEQ. TRANSFER, RETRANSFER, OR
* DISCLOSURE OF THIS DATA BY ANY MEANS TO A NON-U.S. PERSON (INDIVIDUAL
* OR COMPANY), WHETHER IN THE UNITED STATES OR ABROAD, WITHOUT ANY
* REQUIRED EXPORT LICENSE OR OTHER APPROVAL FROM THE U.S GOVERNMENT IS
* PROHIBITED, INCLUDING WITHOUT LIMITATION ANY DIVERSION TO A MILITARY
* END USER OR USE IN A MILITARY END USE APPLICATION.

*:
*: UNIT NAME : E08_FG_STATETABLE
*: FOREGROUND STATE TABLE OF 1 MS. TASKS
*:

*: REFERENCES : REFER TO SDD
*:

*: DESCRIPTION : THIS IS THE TABLE OF 1 MS. TASKS WHICH ARE SCHEDULED BY
*: THE PRIMARY EXECUTIVE PROGRAM (E08_FG_EXECUTIVE). EACH
*: STATE IS ENTERED EVERY 1.04 MS DUE TO AN INTERRUPT.
*: ALL ANALOG INPUTS ARE AVAILABLE THROUGH AN INPUT MUX.
*: THE INPUT AND A/D MUX ADDRESSES ARE SET BY SETTING THE
*: APPROPRIATE DATA BITS OF TWO OTA COMMANDS, OTA 306 AND
*: OTA 407. THE ANALOG SIGNAL CONDITIONING STAGES HAVE A
*: SETTLING TIME OF 1.0 MILLISECONDS. THE A/D MUX HAS A
*: SETTLING TIME OF 20 MICROSECONDS. THE ANALOG TO DIGITAL
*: CONVERSION CONSUMES 0.8 MILLISECONDS. TO ASSURE PROPER
*: OPERATION THERE MUST BE 1 MILLISECOND BETWEEN SETTLING
*: OF THE INPUT MUX CODE AND SETTING UP OF THE A/D MUX CODE.
*: THE APPROPRIATE CODES FOR THE INPUT MUX'S AND THE A/D MUX
*: ARE CONTAINED IN THE STATE TABLE.
*: TO MAXIMIZE EFFICIENCY, THE INPUT MUX CODE SELECTIONS
*: SHOULD ALTERNATE BETWEEN THE AC AND DC (HIGH/LOW LEVEL)
*: INPUT MUX WHEN POSSIBLE. THIS ALLOWS A CONVERSION EVERY
*: 1 MILLISECOND INTERRUPT.
*:

*: ITERATION RATE : MS
*:

*: CONSTRAINTS : 1) E15 PROCESS MUST BE LOCATED IN STATE 80.
*: 2) ARINC OUTPUT MODULES MUST BE SEPARATED BY AT LEAST
*: THREE MS. THEREFORE THEY SHOULD BE CALLED EVERY 4TH STATE.
*: 3) ALL MODULES IN THIS TABLE ARE PRIMARY MODULES AND
*: MUST USE TEMPORARY/RETURN VARIABLES DEDICATED TO
*: PRIMARY MODULES.
*: 4) I17_PNHCNV_1 MUST BE CALLED AT THE START OF TIME
*: STATES 1, 21, 41, AND 61.
*: 5) I17_PNHCNV_2 MUST BE CALLED AT THE START OF TIME
*: STATES 10, 30, 50, AND 70.
*: 6) I18_READNH MUST BE EXECUTED AT LEAST 1 TIME STATE
*: AFTER STOPPING THE F/D FOR NH1 CONVERSION.
*: 7) I23_RDSNHP MUST BE EXECUTED AT LEAST 1 TIME STATE
*: AFTER STOPPING THE F/D FOR NH2 CONVERSION.
*: 8) I31_UARTIN MUST BE CALLED IN STATE 4 ONLY.
*: 9) I38_SMCSV_2 MUST BE CALLED AT LEAST 0.8 MILLISECONDS

```
*: AFTER I38_SMCSV_1 (START A/D CONVERSION).
*: 10)THE LABEL STATE_TABLE IS PROVIDED TO AVOID NEEDING
*: TO MODIFY THE CALLING MODULE WHEN THIS MODULE CHANGES.
*: 11)SOME GENERIC LABELS ARE USED IN THE TABLE IN ORDER
*: TO REDUCE FUTURE EDITS.
*:
```

*
*

*** MODULE AND SUBROUTINES ***

*

INT E08_FG_STATETABLE,STATE_TABLE,TBLEND

*

```
EXTI I28_DISINW1,I29_DISINW2,I30_DISINFAULT
EXTI I31_UARTIN,I17_PNHCVN_1
EXTI I32_ARINCIN_2
EXTI I18_READNH,I22_SNHCVN_1,I22_SNHCVN_2
EXTI I14_RDPLA_1,I14_RDPLA_2,I17_PNHCVN_2
EXTI I02_SGCOM_1,I23_RDSNHP
EXTI I08_RDT18_1,I10_GAINCOM_1,I04_PSAMBSG_1
EXTI I03_RDCJT_1,I15_PLABIAS_1
EXTI I01_LLDRIFT_1,I11_TQBTRIM_1,I05_DELTAPSG_1
EXTI I16_PLAGAIN_1,I07_READTQ_1
EXTI I38_SMCSV_1,I38_SMCSV_2,I13_ENGVARIANT_1
EXTI I09_HLDRIFT_1,I12_TQGTRIM_1,I45_HBOVWA_1
```

```
EXTI O07_ARO203_ALT11,O07_ARO203_ALTPS
EXTI O09_ARO211_TAT11
EXTI O05,O06,O10,O11,O16,O17
EXTI O15_ARO341,O26_ARO340,O27_ARO342
```

```
EXTI O40_ARO06X,O18_ARO270
EXTI E15_BCKGNDTEST
```

*

*** SOURCE CODE ***

*

*

E08_FG_STATETABLE

STATE_TABLE ; GENERIC ENTRY POINT FOR CALLING ROUTINES

*

```
S1 DEC -1 ; TIME STATE 1 - ARINC OUTPUT STATE
DAC I17_PNHCVN_1 ; START NH
DAC I28_DISINW1 ; READ DISCRETE INPUT 1
DAC I29_DISINW2 ; READ DISCRETE INPUT 2
DAC I30_DISINFAULT ; READ FAULT DISCRETE INPUT
DAC O05 ; ARINC LABEL 133 OUTPUT
```

*

S2 DEC -1 ; TIME STATE 2

*

```

S3  DEC  -1      ; TIME STATE 3
*
S4  DEC  -1      ; TIME STATE 4
    DAC  I31_UARTIN      ; READ UART
*
S5  DEC  -1      ; TIME STATE 5 - ARINC OUTPUT STATE
    DAC  O06      ; ARINC LABEL 134 OUTPUT
*
S6  DEC  -1      ; TIME STATE 6
*
S7  DEC  -1      ; TIME STATE 7
*
S8  DEC  -1      ; TIME STATE 8
*
S9  DEC  -1      ; TIME STATE 9 - ARINC OUTPUT STATE
*
S10 DEC  -1      ; TIME STATE 10
    DAC  I17_PNHCVN_2    ; STOP NH
*
S11 DEC  -1      ; TIME STATE 11
    DAC  I18_READNH      ; READ NH
    DAC  I38_SMCSV_1      ; START STEPPER SUM VOLTAGE
*
S12 OCT  S+SLLG+RVDT+SPLA1+HLVL
      ;; HL MUX = RVDT ;; RVDT MUX = PLA1
    DAC  I38_SMCSV_2      ; READ STEPPER SUM VOLTAGE
      ; .. MUST BE .8 MS FROM START
*
S13 OCT  S+SLLG+RVDT+SPLA2+LLVL ; ARINC OUTPUT STATE
      ;;; LL MUX = LOW LEVEL GAIN
    DAC  I14_RDPLA_1      ; PLA1 = A/D INPUT
    DAC  O07_ARO203_ALT11 ; ARINC LABEL 203 OUTPUT (SDI=00,11)
*
S14 OCT  S+ST18+RVDT+SPLA2+HLVL
      ; HL MUX = RVDT ;; RVDT MUX = PLA2
    DAC  I02_SGCOM_1      ; RAWSG = A/D INPUT
*
S15 OCT  S+ST18+SEVT+LLVL
      ;;;; LL MUX = T1.8
    DAC  I14_RDPLA_2      ; PLA2 = A/D INPUT
*
S16 OCT  S+HBV
      ;;; HBV W/A
    DAC  I08_RDT18_1      ; RAWT18 = A/D INPUT
*
S17 OCT  S+SPS+RVDT+SPLAG+LLVL ; ARINC OUTPUT STATE
      ;;;; LL MUX = SPARE
    DAC  I45_HBOVWA_1      ; RAWHBV = A/D INPUT
    DAC  I22_SNHCVN_1      ; START NH 2
*
S18 DEC  -1      ; TIME STATE 18
*
S19 DEC  -1      ; TIME STATE 19

```



```

DAC I22_SNHCNV_2 ; STOP NH 2
*
S20 DEC -1 ; TIME STATE 20
DAC I23_RDSNHP ; READ NH 2
*
S21 DEC -1 ; TIME STATE 21 ; ARINC OUTPUT STATE
DAC I17_PNHCNV_1 ; START NH
*
S22 DEC -1 ; TIME STATE 22
*
S23 DEC -1 ; TIME STATE 23
*
S24 DEC -1 ; TIME STATE 24
*
S25 DEC -1 ; TIME STATE 25
*
S26 DEC -1 ; TIME STATE 26
*
S27 DEC -1 ; TIME STATE 27
*
S28 DEC -1 ; TIME STATE 28
*
S29 DEC -1 ; TIME STATE 29 ; ARINC OUTPUT STATE
DAC O09_ARO211_TAT11 ; ARINC LABEL 211 OUTPUT (SDI=00,11)
*
S30 DEC -1 ; TIME STATE 30
DAC I17_PNHCNV_2 ; STOP NH
*
S31 DEC -1 ; TIME STATE 31
DAC I18_READNH ; READ NH
DAC I38_SMCSV_1 ; START STEPPER SUM VOLTAGE
*
S32 OCT S+SPS+RVDT+SPLAG+HLVL ; TIME STATE 32
;;; HL MUX = RVDT ;; RVDT MUX = PLA GAIN
DAC I38_SMCSV_2 ; READ STEPPER SUM VOLTAGE
; .. MUST BE .8 MS FROM START
*
S33 OCT S+SPS+SHLG+LLVL ; TIME STATE 33 ; ARINC OUTPUT STATE
;;; LL MUX = PS
DAC I16_PLAGAIN_1 ; RAWPLG = A/D IPUT
DAC O10 ; ARINC LABEL 240 OUTPUT
*
S34 OCT S+SCJC+SHLG+HLVL ; TIME STATE 34
;;; HL MUX = HIGH LEVEL GAIN
DAC I04_PSAMBSG_1 ; RAWPS = A/D INPUT
*
S35 OCT S+SCJC+SQ2BT+LLVL ; TIME STATE 35
;;; LL MUX = COLD JUNCTION
DAC I10_GAINCOM_1 ; RAWGAN = A/D INPUT
*
S36 OCT S+SDELP+SQ2BT+HLVL ; TIME STATE 36
;;; HL MUX = Q2 BIAS TRIM
DAC I03_RDCJT_1 ; RAWCJ = A/D INPUT

```

```

*
S37  OCT   S+SDELP+RVDT+SPLA1+LLVL ; TIME STATE 37 ; ARINC OUTPUT STATE
      ;;; SELECT LL MUX (UNUSED SELECTION)
      DAC   I11_TQBTRIM_1           ; RAWTB2 = A/D INPUT
      DAC   I22_SNHCNV_1           ; START NH 2
      DAC   O11                     ; ARINC LABEL 245 OUTPUT

*
S38  DEC   -1                       ; TIME STATE 38

*
S39  DEC   -1                       ; TIME STATE 39
      DAC   I22_SNHCNV_2           ; STOP NH 2

*
S40  DEC   -1                       ; TIME STATE 40
      DAC   I23_RDSNHP             ; READ NH 2

*
S41  DEC   -1                       ; TIME STATE 41 ; ARINC OUTPUT STATE
      DAC   I17_PNHCNV_1           ; START NH
      DAC   O07_ARO203_ALTPS

*
S42  DEC   -1                       ; TIME STATE 42

*
S43  DEC   -1                       ; TIME STATE 43

*
S44  DEC   -1                       ; TIME STATE 44

*
S45  DEC   -1                       ; TIME STATE 45 ; ARINC OUTPUT STATE

*
S46  DEC   -1                       ; TIME STATE 46

*
S47  DEC   -1                       ; TIME STATE 47

*
S48  DEC   -1                       ; TIME STATE 48

*
S49  DEC   -1                       ; TIME STATE 49 ; ARINC OUTPUT STATE
      DAC   O15_ARO341             ; ARINC LABEL 341 OUTPUT

*
S50  DEC   -1                       ; TIME STATE 50
      DAC   I17_PNHCNV_2           ; STOP NH

*
S51  DEC   -1                       ; TIME STATE 51
      DAC   I18_READNH             ; READ NH
      DAC   I38_SMCSV_1            ; START STEPPER SUM VOLTAGE

*
S52  OCT   S+SDELP+RVDT+SPLA1+HLVL ; TIME STATE 52
      ;;; HL MUX = RVDT;; RVDT MUX = PLA1
      DAC   I38_SMCSV_2           ; READ STEPPER SUM VOLTAGE
      ; .. MUST BE .8 MS FROM START

*
S53  OCT   S+SDELP+RVDT+SPLA2+LLVL ; TIME STATE 53 ; ARINC OUTPUT STATE
      ;;; LL MUX = DELTAP
      DAC   I14_RDPLA_1           ; PLA1 = A/D INPUT
      DAC   O16                   ; ARINC LABEL 346 OUTPUT

*

```

```

S54  OCT   S+SLLDFT+RVDT+SPLA2+HLVL   ; TIME STATE 54
      ;;; HL MUX = RVDT;; RVDT MUX = PLA2
      DAC   I05_DELTAPSG_1           ; DELTAP = A/D INPUT
*
S55  OCT   S+SLLDFT+RVDT+SPLAB+LLVL   ; TIME STATE 55
      ;;; LL MUX = LOW LEVEL DRIFT
      DAC   I14_RDPLA_2             ; PLA2 = A/D INPUT
*
S56  OCT   S+SEVT+HLVL               ; TIME STATE 56
      ;;; HL MUX = SEVT
      DAC   I01_LLDRIFT_1           ; RAWDFT = A/D INPUT
*
S57  OCT   S+SPS+RVDT+SPLAB+LLVL   ; TIME STATE 57 ; ARINC OUTPUT STATE
      ;;; SELECT LL MUX (UNUSED SELECTION)
      DAC   I13_ENGVARIANT_1         ; SEVT = A/D INPUT
      DAC   I22_SNHCNV_1             ; START NH 2
      DAC   O17                     ; ARINC LABEL 350 OUTPUT
*
S58  DEC   -1                       ; TIME STATE 58
*
S59  DEC   -1                       ; TIME STATE 59
      DAC   I22_SNHCNV_2             ; STOP NH 2
*
S60  DEC   -1                       ; TIME STATE 60
      DAC   I23_RDSNHP               ; READ NH 2
*
S61  DEC   -1                       ; TIME STATE 61 ; ARINC OUTPUT STATE
      DAC   I17_PNHCNV_1             ; START NH
      DAC   O18_ARO270               ; DAC O18 ARINC LABEL 270 OUTPUT
*
S62  DEC   -1                       ; TIME STATE 62
*
S63  DEC   -1                       ; TIME STATE 63
*
S64  DEC   -1                       ; TIME STATE 64
*
S65  DEC   -1                       ; TIME STATE 65 ; ARINC OUTPUT STATE
      DAC   O40_ARO06X               ; ARINC LABELS 62-67 OUTPUT
*
S66  DEC   -1                       ; TIME STATE 66
*
S67  DEC   -1                       ; TIME STATE 67
*
S68  DEC   -1                       ; TIME STATE 68
*
S69  DEC   -1                       ; TIME STATE 69 ; ARINC OUTPUT STATE
      DAC   O27_ARO342               ; ARINC LABEL 342 OUTPUT
*
S70  DEC   -1                       ; TIME STATE 70
      DAC   I17_PNHCNV_2             ; STOP NH
*
S71  DEC   -1                       ; TIME STATE 71
      DAC   I18_READNH               ; READ NH

```

```

DAC I38_SMCSV_1 ; START STEPPER SUM VOLTAGE
*
S72 OCT S+SPS+RVDT+SPLAB+HLVL ; TIME STATE 72
    ;;; HL MUX = RVDT ;; RVDT MUX = PLA BIAS
DAC I38_SMCSV_2 ; READ STEPPER SUM VOLTAGE
    ; .. MUST BE .8 MS FROM START
*
S73 OCT S+SPS+SHLDFT+SPLAB+LLVL ; TIME STATE 73 ; ARINC OUTPUT STATE
    ;;; LL MUX = PS
DAC I15_PLABIAS_1 ; RAWPLB = A/D INPUT
DAC O26_ARO340 ; ARINC LABEL 340 OUTPUT
*
S74 OCT S+STQ+SHLDFT+HLVL ; TIME STATE 74
    ;;; HL MUX = HIGH LEVEL DRIFT
DAC I04_PSAMBSG_1 ; RAWPS = A/D INPUT
*
S75 OCT S+STQ+SQ2GT+LLVL ; TIME STATE 75
    ;;; LL MUX = TQ
DAC I09_HLDRIFT_1 ; RAWDFC = A/D INPUT
*
S76 OCT S+SLLG+SQ2GT+HLVL ; TIME STATE 76
    ;;; HL MUX = Q2 GAIN TRIM
DAC I07_READTQ_1 ; RAWTQ = A/D INPUT
*
S77 OCT S+SLLG+RVDT+SPLA1+LLVL ; TIME STATE 77 ; ARINC OUTPUT STATE
    ;;; SELECT LL MUX (UNUSED SELECTION)
DAC I12_TQGTRIM_1 ; RAWTG2 = A/D INPUT
DAC I22_SNHCNV_1 ; START NH 2
*
S78 DEC -1 ; TIME STATE 78
*
S79 DEC -1 ; TIME STATE 79
DAC I22_SNHCNV_2 ; STOP NH 2
*
S80 DEC -1 ; TIME STATE 80
DAC I23_RDSNHP ; READ NH 2
DAC E15_BCKGNDTEST ; .. UPDATE INITIALIZATION COUNTER
DAC I32_ARINCIN_2
*
DEC -1 ; END OF STATE 80
*
TBLEND DAC * ; END OF STATE TABLE
*
*
*****
*** CONSTANTS ***
*****
*
* ANALOG MUX CODE DEFINITIONS
*
S EQU '144000 ; START ADDRESS OF BOOT CODE
*
* A/D MUX CODES

```

*
LLVL EQU 0 ; SELECT LOW LEVEL INTERFACE
HBV EQU '20000 ; SELECT HBV INTERFACE
HLVL EQU '3000 ; SELECT HIGH LEVEL INTERFACE

*
* LOW LEVEL INTERFACE MUX CODES

*
SDELP EQU 0 ; RAM PRESSURE
SPS EQU 1 ; PS
SLLG EQU 2 ; LOW LEVEL GAIN
ST18 EQU 3 ; T1.8
SLLDFT EQU 4 ; LOW LEVEL DRIFT
STQ EQU 5 ; TORQUE SHAFT TEMPERATURE
SCJC EQU 6 ; COLD JUNCTION TEMPERATURE
ST6 EQU 7 ; T6 OPTION (UNUSED)

*
* HIGH LEVEL INTERFACE MUX CODES

*
SHLG EQU 0 ; HIGH LEVEL INTERFACE GAIN
SHLDFT EQU '10 ; HIGH LEVEL INTERFACE DRIFT
RVDT EQU '20 ; RVDT MUX
SQ2BT EQU '30 ; TORQUE #2 BIAS TRIM
SQ2GT EQU '40 ; TORQUE #2 GAIN TRIM
ST6TR EQU '50 ; T6 TRIM OPTION
SEVT EQU '60 ; ENGINE VARIANT TRIM
SQ1GT EQU '70 ; TORQUE #1 GAIN TRIM (UNUSED)

*
* RVDT MUX CODES

*
SPLA1 EQU '500 ; PLA SIGNAL #1
SPLA2 EQU '400 ; PLA SIGNAL #2
SPLAB EQU '600 ; RVDT INTERFACE BIAS
SPLAG EQU '700 ; RVDT INTERFACE GAIN

08.04.02 Secondary State Table

* NOTICE TO ALL PERSONS RECEIVING THIS DATA:
* WARNING UNPUBLISHED WORK < 2023 COLLINS AEROSPACE. WITHIN THIS DOCUMENT,
* THERE IS, SO DESIGNATED, PROPERTY OF COLLINS AEROSPACE AND PRATT & WHITNEY
* CANADA CORP. AS DOCUMENTED IN THE COLLINS AEROSPACE/PRATT & WHITNEY CANADA
* CORP. SIDE LETTER AGREEMENT JKH-21-004 DATED 3-19-2021, AS MODIFIED BY
* AMENDMENT 1 JKH-22-002 DATED 2-23-2022. YOU MAY NOT POSSESS, USE, COPY OR
* DISCLOSE THIS DOCUMENT OR ANY INFORMATION IN IT, FOR ANY PURPOSE,
* INCLUDING WITHOUT LIMITATION, TO DESIGN, MANUFACTURE OR REPAIR PARTS,
* OR OBTAIN ANY GOVERNMENT APPROVAL TO DO SO, WITHOUT EXPRESS WRITTEN
* PERMISSION *OF COLLINS AEROSPACE AND PRATT & WHITNEY CANADA CORP.
* NEITHER RECEIPT NOR POSSESSION OF THIS DOCUMENT ALONE, FROM ANY *SOURCE,
* CONSTITUTES SUCH PERMISSION. POSSESSION, USE, COPYING OR DISCLOSURE
* BY ANYONE WITHOUT EXPRESS WRITTEN *PERMISSION IS NOT AUTHORIZED AND MAY
* RESULT IN CRIMINAL AND/OR CIVIL LIABILITY.

* SUBJECT TO THE EAR, ECCN [9E991].
* THIS INFORMATION IS SUBJECT TO THE EXPORT CONTROL LAWS OF THE UNITED
* STATES, SPECIFICALLY INCLUDING THE EXPORT ADMINISTRATION REGULATIONS
* (EAR), 15 C.F.R. PART 730 ET. SEQ. TRANSFER, RETRANSFER, OR
* DISCLOSURE OF THIS DATA BY ANY MEANS TO A NON-U.S. PERSON (INDIVIDUAL
* OR COMPANY), WHETHER IN THE UNITED STATES OR ABROAD, WITHOUT ANY
* REQUIRED EXPORT LICENSE OR OTHER APPROVAL FROM THE U.S GOVERNMENT IS
* PROHIBITED, INCLUDING WITHOUT LIMITATION ANY DIVERSION TO A MILITARY
* END USER OR USE IN A MILITARY END USE APPLICATION.

*:
*: UNIT NAME : E08_FG_SCHEDULE
*: SECONDARY FOREGROUND EXECUTIVE
*:
*: REFERENCES : REFER TO SDD
*:
*: DESCRIPTION : THIS MODULE SCHEDULES ALL THE FOREGROUND LOGIC EXCEPT
*: THOSE WHICH ARE EXTREMELY TIME CRITICAL SUCH AS A/D AND
*: F/D INPUTS AND UART OUTPUTS. IT WILL BE INTERRUPTED EVERY
*: 1.04 MS FOR THE PRIMARY EXECUTIVE MODULE TO DO ITS TASKS.
*: EACH MINOR CYCLE OF 20.8 MS IS PERFORMED AND THEN
*: BACKGROUND TASKS USE ANY ADDITIONAL TIME.
*:
*: ITERATION RATE : 20.8 MS
*:
*: CONSTRAINTS : 1. THE TOTAL EXECUTION TIME OF EACH MINOR CYCLE PATH
*: MUST NOT EXCEED 20.8 MS.
*: 2. THE MODULE CALLING ORDER IS SPECIFIED BY PWC.
*: 3. I03_RDCJT_3 must be called before I03_RDCJT_2
*:
*:

INT FOREX1
INT E08_FG_SCHEDULE

*

EXTD MINCYC,RC
EXTD FORRTN

*

EXTI I01_LLDRIFT_2
EXTI I02_SGCOM_2
EXTI I03_RDCJT_2
EXTI I03_RDCJT_3
EXTI I04_PSAMBSG_2
EXTI I04_PSAMBSG_3
EXTI I04_PSAMBSG_4
EXTI I04_PSAMBSG_5
EXTI I05_DELTAPSG_2
EXTI I05_DELTAPSG_3
EXTI I05_DELTAPSG_4
EXTI I07_READTQ_2
EXTI I07_READTQ_3

EXTI I08_RDT18_2
EXTI I08_RDT18_3
EXTI I08_RDT18_4
EXTI I09_HLDRIFT_2
EXTI I10_GAINCOM_2
EXTI I11_TQBTRIM_2
EXTI I12_TQGTRIM_2
EXTI I13_ENGVARIANT_2
EXTI I14_RDPLA_3
EXTI I14_RDPLA_4
EXTI I15_PLABIAS_2
EXTI I16_PLAGAIN_2
EXTI I19_NHDUAL_1
EXTI I19_NHDUAL_2
EXTI I20_NHSPEED
EXTI I24_NHPROBE2
EXTI I25_TQNPT_1
EXTI I25_TQNPT_7
EXTI I26_TQCAL
EXTI I27_TQSPEED
EXTI I33_ARITAT
EXTI I34_TATSEL
EXTI I35_ARIPA
EXTI I36_PAMSEL
EXTI I37_ARIISA_1
EXTI I37_ARIISA_2
EXTI I43_TQSPEED_1
EXTI I43_TQSPEED_2
EXTI I43_TQSPEED_3
EXTI I45_HBOVWA_2
EXTI I46_IASFAULT

*

EXTI E10_RAMTEST
EXTI E11_CONTEST
EXTI E13_WATEST_1
EXTI E13_WATEST_2
EXTI E13_WATEST_3
EXTI E13_WATEST_4
EXTI E13_WATEST_5
EXTI E13_WATEST_6
EXTI E13_WATEST_7
EXTI E14_FAULTSTORE_1
EXTI E14_FAULTSTORE_2
EXTI E14_FAULTSTORE_3
EXTI E14_FAULTSTORE_4
EXTI E14_FAULTSTORE_5
EXTI E14_FAULTSTORE_6
EXTI E16_SMTEST
EXTI E17_NCFLT_1
EXTI E17_NCFLT_2
EXTI E17_NCFLT_3
EXTI E17_NCFLT_4
EXTI E17_NCFLT_5

EXTI E17_NCFLT_6
EXTI E18_FAULTSUM_1
EXTI E18_FAULTSUM_2
EXTI E18_FAULTSUM_3
EXTI E18_FAULTSUM_4
EXTI E19_TRMSTORE_1
EXTI E19_TRMSTORE_2
EXTI E19_TRMSTORE_4
EXTI E19_TRMSTORE_5
EXTI E21_RECONFIG
EXTI E23_PLATRIM
EXTI E26_FAULTSTORE_1
EXTI E26_FAULTSTORE_2
EXTI E26_FAULTSTORE_4
EXTI E25_FAULTDET
EXTI E27_MFCID
EXTI E28_HBOVWA_1
EXTI E28_HBOVWA_2
EXTI E31_EXTDIS

*

EXTI C01_START
EXTI C02_PLATRIM_1
EXTI C02_PLATRIM_2
EXTI C03_TLMRT_1
EXTI C03_TLMRT_2_1
EXTI C03_TLMRT_2_2
EXTI C03_TLMRT_3_1
EXTI C03_TLMRT_3_2
EXTI C03_TLMRT_4_1
EXTI C03_TLMRT_4_2
EXTI C03_TLMRT_5_1
EXTI C03_TLMRT_5_2
EXTI C03_TLMRT_6_1
EXTI C03_TLMRT_6_2
EXTI C03_TLMRT_7
EXTI C04_RPCAL_2_D
EXTI C05_TQBUG
EXTI C06_HPREF_1
EXTI C06_HPREF
EXTI C07_OVRTQPRO
EXTI C08_SHPCL
EXTI C10_ATOIGN
EXTI C11_SHPREQ_1
EXTI C11_SHPREQ_2
EXTI C12_NPTGOV_1
EXTI C12_NPTGOV_2
EXTI C13_NHACCELLIM
EXTI C14_NHLIM
EXTI C15_NHDECELLIM
EXTI C16_LOOPSEL
EXTI C17_NHREQUEST_1
EXTI C17_NHREQUEST_2
EXTI C18_NHGOV_1


```
EXTI  C18_NHGOV_2
EXTI  C19_UNDEERSPEED
EXTI  C20_OVRSPEED
EXTI  C21_HBOVEM_1
EXTI  C21_HBOVEM_2
EXTI  C22_HBOVMM_1
EXTI  C22_HBOVMM_2
EXTI  C22_HBOVMM_3
EXTI  C24_OVRTQTST
EXTI  C26_MPARAM_1
EXTI  C26_MPARAM_2
EXTI  C26_MPARAM_3
EXTI  C26_MPARAM_4
EXTI  C26_MPARAM_5
EXTI  C26_MPARAM_6
EXTI  C26_MPARAM_7
EXTI  C27_NHNMNMX_1
EXTI  C30_SMRTRQ
EXTI  C31_TQLIM
```

*

```
EXTI  O01_SMOUT
EXTI  O02_HBOVOUT
EXTI  O03_DISOUT
```

*

```
EXTI  DCU_SCHEDULE
EXTI  UPDATE_EE_DATA
EXTI  EE_WRITE_BLOCK
EXTI  UART_IO_SELECT
```

*

```
EXTI  INC_CTRS_1
EXTI  INC_CTRS_2
EXTI  INC_CTRS_3
EXTI  INC_CTRS_4
EXTI  LCF_CHECK
EXTI  MISSIONS
EXTI  OPMODE_1
EXTI  OPMODE_2
EXTI  OPMODE_3
EXTI  RUNTIMER
EXTI  FLTTIMER
```

*

*

E08_FG_SCHEDULE

FOREX1

```
STA  FORRTN    ; SAVE RETURN ADDRESS
LAA  MINCYC    ; CHECK MINOR CYCLE INDICATOR
SNE.D
BRU  FORMN1    ; FIRST
SNE.D
BRU  FORMN2    ; SECOND
SNE.D
BRU  FORMN3    ; THIRD
```

*

FORMN4 ; FOURTH MINOR CYCLE STATES 61 - 80

BSA EE_WRITE_BLOCK
BSA I03_RDCJT_3
BSA I03_RDCJT_2
BSA I04_PSAMBSG_2
BSA I04_PSAMBSG_3
BSA I04_PSAMBSG_4
BSA I04_PSAMBSG_5
BSA I07_READTQ_3
BSA I11_TQBTRIM_2
BSA I14_RDPLA_3
BSA I14_RDPLA_4
BSA I19_NHDUAL_1
BSA I19_NHDUAL_2
BSA I20_NHSPEED
BSA I24_NHPROBE2
BSA I25_TQNPT_1
BSA I27_TQSPEED
BSA I35_ARIPA
BSA I43_TQSPEED_1
BSA I43_TQSPEED_2
BSA I43_TQSPEED_3
BSA E13_WATEST_1
BSA E13_WATEST_2
BSA E13_WATEST_3
BSA E14_FAULTSTORE_1
BSA E14_FAULTSTORE_2
BSA E14_FAULTSTORE_3
BSA E14_FAULTSTORE_4
BSA E14_FAULTSTORE_5
BSA E14_FAULTSTORE_6
BSA E16_SMTEST
BSA E18_FAULTSUM_4
BSA E26_FAULTSTORE_1
BSA E28_HBOVWA_1
BSA E28_HBOVWA_2
BSA O01_SMOUT
BSA O02_HBOVOUT
BSA O03_DISOUT

*

BSA C01_START
BSA C10_ATOIGN
BSA C26_MPARAM_6
BSA C26_MPARAM_7

BSA C02_PLATRIM_1
BSA C02_PLATRIM_2

BSA C06_HPREF_1
BSA C06_HPREF
BSA C07_OVRTQPRO

BSA C13_NHACCELLIM
BSA C14_NHLIM
BSA C15_NHDECELLIM
BSA C31_TQLIM

BSA C11_SHPREQ_1

BSA C11_SHPREQ_2
BSA C08_SHPCL

BSA C27_NHMMMX_1

BSA C17_NHREQUEST_1
BSA C17_NHREQUEST_2
BSA C16_LOOPSEL
BSA C18_NHGOV_1
BSA C18_NHGOV_2
BSA C30_SMRTRQ
BSA C22_HBOVMM_1
BSA C22_HBOVMM_2
BSA C22_HBOVMM_3
BSA C21_HBOVEM_1
BSA C21_HBOVEM_2

*

BSA RUNTIMER
BSA FLTTIMER
BSA UPDATE_EE_DATA
BSA DCU_SCHEDULE
BSA UART_IO_SELECT

*

BRU FOREXT ; RETURN TO BACKGROUND THROUGH PRIMARY EXECUTIVE

FORMN3 ; THIRD MINOR CYCLE STATES 41 - 60

BSA EE_WRITE_BLOCK
BSA I01_LLDRIPT_2
BSA I05_DELTAPSG_4
BSA I07_READTQ_2
BSA I13_ENGVARIANT_2
BSA I19_NHDUAL_1
BSA I19_NHDUAL_2
BSA I20_NHSPEED
BSA I24_NHPROBE2
BSA I25_TQNPT_1
BSA I26_TQCAL
BSA I36_PAMSEL
BSA I37_ARIISA_1
BSA I37_ARIISA_2
BSA I45_HBOVWA_2
BSA I46_IASFAULT
BSA E16_SMTEST
BSA E17_NCFLT_5
BSA E17_NCFLT_6
BSA E18_FAULTSUM_3

BSA E19_TRMSTORE_1
BSA E19_TRMSTORE_2
BSA E19_TRMSTORE_4
BSA E19_TRMSTORE_5
BSA E23_PLATRIM
BSA E25_FAULTDET
BSA E26_FAULTSTORE_2
BSA E27_MFCID
BSA E31_EXTDIS
BSA O01_SMOUT
BSA O02_HBOVOUT

*

BSA C01_START
BSA C10_ATOIGN

BSA C26_MPARAM_1
BSA C26_MPARAM_3
BSA C26_MPARAM_4
BSA C26_MPARAM_5
BSA C26_MPARAM_6
BSA C26_MPARAM_7

BSA C04_RPCAL_2_D

BSA C05_TQBUG

BSA C13_NHACCELLIM
BSA C14_NHLIM
BSA C15_NHDECELLIM
BSA C31_TQLIM

BSA C19_UNDERSPEED
BSA C20_OVRSPEED

BSA C11_SHPREQ_2
BSA C08_SHPCL

BSA C12_NPTGOV_1
BSA C12_NPTGOV_2

BSA C17_NHREQUEST_1
BSA C17_NHREQUEST_2
BSA C16_LOOPSEL
BSA C18_NHGOV_1
BSA C18_NHGOV_2
BSA C30_SMRTRQ
BSA C22_HBOVMM_1
BSA C22_HBOVMM_2
BSA C22_HBOVMM_3
BSA C21_HBOVEM_1
BSA C21_HBOVEM_2
BSA C24_OVRTQTST

*

BSA RUNTIMER
BSA FLTTIMER
BSA INC_CTRS_3
BSA LCF_CHECK

*

BRU FOREXT ; RETURN TO BACKGROUND THROUGH PRIMARY EXECUTIVE

FORMN2 ; SECOND MINOR CYCLE STATES 21 - 40

BSA EE_WRITE_BLOCK
BSA I02_SGCOM_2
BSA I04_PSAMBSG_2
BSA I04_PSAMBSG_3
BSA I04_PSAMBSG_4
BSA I04_PSAMBSG_5
BSA I10_GAINCOM_2
BSA I14_RDPLA_3
BSA I14_RDPLA_4
BSA I15_PLABIAS_2
BSA I16_PLAGAIN_2
BSA I19_NHDUAL_1
BSA I19_NHDUAL_2
BSA I20_NHSPEED
BSA I24_NHPROBE2
BSA I25_TQNPT_1
BSA I27_TQSPEED
BSA I35_ARIPA
BSA I43_TQSPEED_1
BSA I43_TQSPEED_2
BSA I43_TQSPEED_3
BSA E11_CONTEST
BSA E14_FAULTSTORE_1
BSA E16_SMTEST
BSA E17_NCFLT_3
BSA E17_NCFLT_4
BSA E18_FAULTSUM_2
BSA E26_FAULTSTORE_1
BSA I33_ARITAT
BSA I34_TATSEL
BSA E13_WATEST_4
BSA E13_WATEST_5
BSA E13_WATEST_6
BSA E13_WATEST_7
BSA O01_SMOUT
BSA O02_HBOVOUT

*

BSA C01_START
BSA C10_ATOIGN

BSA C26_MPARAM_2

BSA C26_MPARAM_6
BSA C26_MPARAM_7

BSA C02_PLATRIM_1
BSA C02_PLATRIM_2

BSA C13_NHACCELLIM
BSA C14_NHLIM
BSA C15_NHDECELLIM
BSA C31_TQLIM
BSA C11_SHPREQ_2
BSA C08_SHPCL

BSA C27_NHMMMX_1

BSA C17_NHREQUEST_1
BSA C17_NHREQUEST_2
BSA C16_LOOPSEL
BSA C18_NHGOV_1
BSA C18_NHGOV_2
BSA C30_SMRTRQ
BSA C22_HBOVMM_1
BSA C22_HBOVMM_2
BSA C22_HBOVMM_3
BSA C21_HBOVEM_1
BSA C21_HBOVEM_2

*

BSA MISSIONS
BSA RUNTIMER
BSA FLTTIMER
BSA INC_CTRS_2

*

BRU FOREXT ; RETURN TO BACKGROUND THROUGH PRIMARY EXECUTIVE

FORMN1 ; FIRST MINOR CYCLE STATES 1 - 20

BSA EE_WRITE_BLOCK
BSA I05_DELTAPSG_2
BSA I05_DELTAPSG_3
BSA I08_RDT18_3
BSA I08_RDT18_2
BSA I08_RDT18_4
BSA I09_HLDRIFT_2
BSA I12_TQGTRIM_2
BSA I19_NHDUAL_1
BSA I19_NHDUAL_2
BSA I20_NHSPEED
BSA I24_NHPROBE2
BSA I25_TQNPT_7
BSA I25_TQNPT_1
BSA I26_TQCAL
BSA I36_PAMSEL
BSA I37_ARIISA_1
BSA I37_ARIISA_2
BSA E10_RAMTEST
BSA E16_SMTEST
BSA E17_NCFLT_1

BSA E17_NCFLT_2
BSA E18_FAULTSUM_1
BSA E19_TRMSTORE_1
BSA E19_TRMSTORE_2
BSA E19_TRMSTORE_4
BSA E19_TRMSTORE_5
BSA E23_PLATRIM
BSA E26_FAULTSTORE_4
BSA O01_SMOUT
BSA O02_HBOVOUT

*

BSA C01_START
BSA C10_ATOIGN

BSA C26_MPARAM_1
BSA C26_MPARAM_4
BSA C26_MPARAM_5

BSA C26_MPARAM_6
BSA C26_MPARAM_7

*

BSA E21_RECONFIG

*

LAA RC
SOA ; RC-1
SNE ; RC-1!=0
BRU FM1
SG.D ; Checks for RC-1 > 0
BRU FM2
SG.D
BRU FM3
SG.D
BRU FM4
SG.D
BRU FM5
BSA C03_TLIMRT_6_1
BSA C03_TLIMRT_6_2
BRU FM7

FM1

BSA C03_TLIMRT_1
BRU FM7

FM2

BSA C03_TLIMRT_2_1
BSA C03_TLIMRT_2_2
BRU FM7

FM3

BSA C03_TLIMRT_3_1
BSA C03_TLIMRT_3_2
BRU FM7

FM4

BSA C03_TLIMRT_4_1
BSA C03_TLIMRT_4_2

```

BRU      FM7
FM5
  BSA    C03_TLIMRT_5_1
  BSA    C03_TLIMRT_5_2
FM7
  BSA    C03_TLIMRT_7
CONFGT
  BSA    C13_NHACCELLIM
  BSA    C14_NHLIM
  BSA    C15_NHDECELLIM
  BSA    C31_TQLIM

  BSA    C19_UNDEVSPEED
  BSA    C20_OVRSPEED

  BSA    C11_SHPREQ_2
  BSA    C08_SHPCL

  BSA    C12_NPTGOV_1
  BSA    C12_NPTGOV_2

  BSA    C17_NHREQUEST_1
  BSA    C17_NHREQUEST_2
  BSA    C16_LOOPSEL
  BSA    C18_NHGOV_1
  BSA    C18_NHGOV_2
  BSA    C30_SMRTRQ
  BSA    C22_HBOVMM_1
  BSA    C22_HBOVMM_2
  BSA    C22_HBOVMM_3
  BSA    C21_HBOVEM_1
  BSA    C21_HBOVEM_2
*
  BSA    OPMODE_1
  BSA    OPMODE_2
  BSA    OPMODE_3
  BSA    RUNTIMER
  BSA    FLTTIMER
  BSA    INC_CTRS_1
  BSA    INC_CTRS_4
*
FOREXT
  BRU    @FORRTN      ; SECONDARY EXIT POINT

```


09.00 SET/USE TABLE

09.01 SET/USE TABLE (SORTED BY PARAMETER)

Set / Use Report for software version 127S0101 of the PW127XT-S

09.01 Table 1: Set-Use-by-Parameter

Parameter	SET BY	USED BY
A134SW		PAGE0
A203SW	I35_ARIPA	PAGE0, I36_PAMSEL
A206SW	I37_ARIISA_1	PAGE0, I46_IASFAULT
A211SW	I33_ARITAT	PAGE0, I34_TATSEL
A271SW		PAGE0
ACCEL	C13_NHACCELLIM	PAGE1
ACCEL1	C13_NHACCELLIM	PAGE1
ACCELS	C13_NHACCELLIM	PAGE1
ACCELG	C13_NHACCELLIM	PAGE1
ACCELP	C13_NHACCELLIM	PAGE1
ACCINFLSTRTS_ALL	INC_CTRS_1	PAGE1, DCU_CONSTANTS
ACCLSW	C13_NHACCELLIM	PAGE1, C08_SHPCL
ACCNUMFL_ALL	INC_CTRS_1	PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
ACCSHUTD_ALL	DCU_COUNTER_DATA, INC_CTRS_1	PAGE1, DCU_CONSTANTS, DCU_IS_OBSOLETE
ACCSTRTSFLT_ALL	INC_CTRS_1	PAGE1, DCU_CONSTANTS
ACSTRTS_ALL	DCU_COUNTER_DATA, INC_CTRS_1	PAGE1, DCU_CONSTANTS, DCU_IS_OBSOLETE, O40_ARO06X_TABLE
ACCSTR_SD_RESYNC	DCU_COUNTER_DATA	PAGE1, DCU_IS_OBSOLETE
ADCIAS	I37_ARIISA_1, I37_ARIISA_2	PAGE0
ADINPT	E08_FG_EXECUTIVE	PAGE0, I01_LLDRIPT_1, I02_SGCOM_1, I03_RDCJT_1, I04_PSAMBSG_1, I05_DELTAPSG_1, I07_READTQ_1, I08_RDT18_1, I09_HLDRIPT_1, I10_GAINCOM_1, I11_TQBTRIM_1, I12_TQGTRIM_1, I13_ENGVARIANT_1, I14_RDPLA_1, I14_RDPLA_2, I15_PLABIAS_1, I16_PLAGAIN_1, I38_SMCSV_2, I45_HBOVWA_1
ADSTAT	E08_FG_EXECUTIVE	PAGE0
AIGRST	C10_ATOIGN	PAGE0, C17_NHREQUEST_1, C18_NHGOV_2, C30_SMRTRQ
ALT	C26_MPARAM_3	PAGE1, C26_MPARAM_2, O07_ARO203_ALT11
ALT203	I35_ARIPA	PAGE0
ALTDAT	I35_ARIPA	PAGE0, I36_PAMSEL
ALTIF	I36_PAMSEL	PAGE0, OPMODE_1
ALTPS	I04_PSAMBSG_5	PAGE0, O07_ARO203_ALTPS
ALTRGC	I35_ARIPA	PAGE0
ALTTYM	I36_PAMSEL	PAGE0
ALTXF	I36_PAMSEL	PAGE0, OPMODE_1
AMBTIM	I35_ARIPA	PAGE0
AMODE1	O17_ARO350	PAGE0
AMODE2	O10_ARO240	PAGE0
AR203H	I32_ARINCIN_1, I35_ARIPA	PAGE0
AR203L	I32_ARINCIN_1, I35_ARIPA	PAGE0
AR206H	I32_ARINCIN_1, I37_ARIISA_1	PAGE0
AR206L	I32_ARINCIN_1, I37_ARIISA_1	PAGE0
AR211H	I32_ARINCIN_1, I33_ARITAT	PAGE0
AR211L	I32_ARINCIN_1, I33_ARITAT	PAGE0
ARCCNT	I32_ARINCIN_1, I32_ARINCIN_2	PAGE0
ARCOUT		ARCOUT, O05_ARO133, O06_ARO134, O07_ARO203_ALT11, O07_ARO203_ALTPS, O09_ARO211_TAT11, O10_ARO240, O11_ARO245, O15_ARO341, O16_ARO346, O17_ARO350, O18_ARO270, O19_ARO272, O26_ARO340, O27_ARO342, O40_ARO06X
ARILSH	I32_ARINCIN_1	PAGE0
ARIMSH	I32_ARINCIN_1	PAGE0
ARINC_INPUT		I32_ARINCIN_1, E08_FG_EXECUTIVE
ARINSW	I32_ARINCIN_1, I32_ARINCIN_2	PAGE0, I33_ARITAT, I34_TATSEL, I35_ARIPA, I36_PAMSEL, I37_ARIISA_1
AROIF	ARCOUT	PAGE0, E25_FAULTDET
AROLSH	O05_ARO133, O06_ARO134, O07_ARO203_ALT11, O07_ARO203_ALTPS, O09_ARO211_TAT11, O10_ARO240, O11_ARO245, O15_ARO341, O16_ARO346, O17_ARO350, O18_ARO270, O19_ARO272, O26_ARO340, O27_ARO342, O40_ARO06X	PAGE0, ARCOUT
AROMSH	O05_ARO133, O06_ARO134, O07_ARO203_ALT11, O07_ARO203_ALTPS, O09_ARO211_TAT11, O10_ARO240, O11_ARO245, O15_ARO341, O16_ARO346, O17_ARO350, O18_ARO270, O19_ARO272, O26_ARO340, O27_ARO342, O40_ARO06X	PAGE0, ARCOUT
ARO_DATA_PTR	O40_ARO06X	PAGE0
ARO_FAULTED	O40_ARO06X	PAGE1
ARO_ITEM_MAX	O40_ARO06X	PAGE1
ARO_ITEM_PTR	O40_ARO06X	PAGE1
ARO_LABEL_TYPE	O40_ARO06X	PAGE1
ARO_LISTA_SEQ		PAGE1, O40_ARO06X_TABLE
ARO_LISTB_SEQ		PAGE1, O40_ARO06X_TABLE
ARO_LISTC_SEQ		PAGE1, O40_ARO06X_TABLE
ARO_LISTD_LOBYTE		PAGE1, O40_ARO06X_TABLE
ARO_LISTD_SEQ		PAGE1, O40_ARO06X_TABLE

Parameter	SET BY	USED BY
ARO_LISTE_LOBYTE		PAGE1, O40_ARO06X_TABLE
ARO_LISTE_SEQ		PAGE1, O40_ARO06X_TABLE
ARO_LISTF_LOBYTE		PAGE1, O40_ARO06X_TABLE
ARO_LISTF_SEQ		PAGE1, O40_ARO06X_TABLE
ARO_OUTPUT_TIMER	O40_ARO06X	PAGE1
ARO_OUT_COUNTER	O40_ARO06X	PAGE1
ARO_PLACE HOLDER	O40_ARO06X	PAGE1
ARO_SEQUENCE_PTR	O40_ARO06X	PAGE0
ARO_SEQ_BUTTON	O40_ARO06X	PAGE1
ARO_TABLE_ADR		O40_ARO06X_TABLE, O40_ARO06X
ARO_TABLE_END		O40_ARO06X_TABLE, O40_ARO06X
ARO_TABLE_PTR	O40_ARO06X	PAGE0
ASAVE	E08_FG_EXECUTIVE	PAGE0
ASAVE2	E08_FG_EXECUTIVE	PAGE0
ASLMFL	C21_HBOVEM_1	PAGE1, C21_HBOVEM_2
ASLMSP	C22_HBOVMM_2	PAGE1
ASORCM		PAGE0
ATEMO	C07_OVRTQPRO, C08_SHPCL, C22_HBOVMM_1, UNIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UPDATE_EE_DATA

Parameter	SET BY	USED BY
ATEM1		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
ATEM2		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
AUTOFL	C10_ATOIGN	PAGE1
B134CT		PAGE0
B203CT	I36_PAMSEL	PAGE0
B206CT	I37_ARIISA_1, I37_ARIISA_2	PAGE0
B211CT	I34_TATSEL	PAGE0
BACGND		E07_BG_EXECUTIVE, E01_PORPSR, E08_FG_EXECUTIVE
BAD134		PAGE0
BAD203	I36_PAMSEL	PAGE0, I35_ARIPA
BAD206	I37_ARIISA_1, I37_ARIISA_2	PAGE0
BAD211	I34_TATSEL	PAGE0, I33_ARITAT

Parameter	SET BY	USED BY
BIPOLY		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_2, C18_NHGOV_1, C19_UNDESPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C31_TQIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
BKGEND		INSTST, E08_FG_EXECUTIVE
BKGTM	E07_BG_EXECUTIVE, E08_FG_EXECUTIVE	PAGE0
BLDELO	C06_HPREF_1	PAGE1
BLDR26	E26_FAULTSTORE_2, E26_FAULTSTORE_4	PAGE0
BLDREQ	C22_HBOVMM_3	PAGE1
BLEED	I28_DISINW1	PAGE1, C03_TLMRT_2_2, C03_TLMRT_3_2, C03_TLMRT_4_2, C03_TLMRT_5_2, C03_TLMRT_6_2, C06_HPREF_1
BOOST		PAGE0, C19_UNDESPPEED
BOOT1	DUMP, GOTO, INA, LOAD, OTA, READ, WRITE, WRTEEP	PAGE0
BOOT2	DUMP, GETWRD, INA, READ, WRITE, WRTEEP	PAGE0, GOTO, SNDWRD
BOOT3	DUMP, LOAD, READ, WRITE, WRTEEP	PAGE0
BOOT4	DUMP, LOAD, WRTEEP	PAGE0
BOOT5	GETWRD, SNDWRD, WRTEEP	PAGE0
BOOT6	URTOUT, URTRED, WRTEEP	PAGE0
BSAVE	E08_FG_EXECUTIVE	PAGE0
BSAVE2	E08_FG_EXECUTIVE	PAGE0
BSTGN	C19_UNDESPPEED	PAGE1
BSTNPT	C19_UNDESPPEED	PAGE1
BSTOFF	C19_UNDESPPEED	PAGE1
BSTPDB	C19_UNDESPPEED	PAGE1
BSTPLA	C19_UNDESPPEED	PAGE1
BUILDR	E14_FAULTSTORE_3, E14_FAULTSTORE_4	PAGE0
BYPASS	E14_FAULTSTORE_2	PAGE0
BYTEF	O04_UARTOUT_NPM	PAGE1
C01_START		C01_START, E08_FG_SCHEDULE
C02_PLATRIM_1		C02_PLATRIM_1, E08_FG_SCHEDULE
C02_PLATRIM_2		C02_PLATRIM_2, E08_FG_SCHEDULE
C03_TLMRT_1		C03_TLMRT_1, E08_FG_SCHEDULE
C03_TLMRT_2_1		C03_TLMRT_2_1, E08_FG_SCHEDULE
C03_TLMRT_2_2		C03_TLMRT_2_2, E08_FG_SCHEDULE
C03_TLMRT_3_1		C03_TLMRT_3_1, E08_FG_SCHEDULE
C03_TLMRT_3_2		C03_TLMRT_3_2, E08_FG_SCHEDULE
C03_TLMRT_4_1		C03_TLMRT_4_1, E08_FG_SCHEDULE
C03_TLMRT_4_2		C03_TLMRT_4_2, E08_FG_SCHEDULE
C03_TLMRT_5_1		C03_TLMRT_5_1, E08_FG_SCHEDULE
C03_TLMRT_5_2		C03_TLMRT_5_2, E08_FG_SCHEDULE
C03_TLMRT_6_1		C03_TLMRT_6_1, E08_FG_SCHEDULE
C03_TLMRT_6_2		C03_TLMRT_6_2, E08_FG_SCHEDULE
C03_TLMRT_7		C03_TLMRT_7, E08_FG_SCHEDULE
C04_RPCAL_2_D		C04_RPCAL_2_D, E08_FG_SCHEDULE
C05_TQBUG		C05_TQBUG, E08_FG_SCHEDULE
C06_HPREF		C06_HPREF, E08_FG_SCHEDULE
C06_HPREF_1		C06_HPREF_1, E08_FG_SCHEDULE
C07_OVRTQPRO		C07_OVRTQPRO, E08_FG_SCHEDULE
C08_SHPCL		C08_SHPCL, E08_FG_SCHEDULE
C10_ATOIGN		C10_ATOIGN, E08_FG_SCHEDULE
C11_SHPREQ_1		C11_SHPREQ_1, E08_FG_SCHEDULE
C11_SHPREQ_2		C11_SHPREQ_2, E08_FG_SCHEDULE
C12_NPTGOV_1		C12_NPTGOV_1, E08_FG_SCHEDULE
C12_NPTGOV_2		C12_NPTGOV_2, E08_FG_SCHEDULE
C13_NHACCELLIM		C13_NHACCELLIM, E08_FG_SCHEDULE
C14_NHLIM		C14_NHLIM, E08_FG_SCHEDULE
C15_NHDECELLIM		C15_NHDECELLIM, E08_FG_SCHEDULE
C16_LOOPSEL		C16_LOOPSEL, E08_FG_SCHEDULE
C17_NHREQUEST_1		C17_NHREQUEST_1, E08_FG_SCHEDULE
C17_NHREQUEST_2		C17_NHREQUEST_2, E08_FG_SCHEDULE
C18_NHGOV_1		C18_NHGOV_1, E08_FG_SCHEDULE
C18_NHGOV_2		C18_NHGOV_2, E08_FG_SCHEDULE

Parameter	SET BY	USED BY
C19 UNDERSPEED		C19 UNDERSPEED, E08_FG_SCHEDULE
C20 OVRSPD		C20 OVRSPD, E08_FG_SCHEDULE
C21_HBOVEM_1		C21_HBOVEM_1, E08_FG_SCHEDULE
C21_HBOVEM_2		C21_HBOVEM_2, E08_FG_SCHEDULE
C22_HBOVMM_1		C22_HBOVMM_1, E08_FG_SCHEDULE
C22_HBOVMM_2		C22_HBOVMM_2, E08_FG_SCHEDULE
C22_HBOVMM_3		C22_HBOVMM_3, E08_FG_SCHEDULE
C24_OVRTQTST		C24_OVRTQTST, E08_FG_SCHEDULE
C26_MPARAM_1		C26_MPARAM_1, E08_FG_SCHEDULE
C26_MPARAM_2		C26_MPARAM_2, E08_FG_SCHEDULE
C26_MPARAM_3		C26_MPARAM_3, E08_FG_SCHEDULE
C26_MPARAM_4		C26_MPARAM_4, E08_FG_SCHEDULE
C26_MPARAM_5		C26_MPARAM_5, E08_FG_SCHEDULE
C26_MPARAM_6		C26_MPARAM_6, E08_FG_SCHEDULE
C26_MPARAM_7		C26_MPARAM_7, E08_FG_SCHEDULE
C27_NHNMNX_1		C27_NHNMNX_1, E08_FG_SCHEDULE
C30_SMRTRQ		C30_SMRTRQ, E08_FG_SCHEDULE
C31_TQLIM		C31_TQLIM, E08_FG_SCHEDULE
CCORFM1		PAGE1
CHKSUM		CHKSUM
CJ	I03_RDCIT_2	PAGE0, E14_FAULTSTORE_4, E26_FAULTSTORE_4
CJC	I03_RDCIT_3	PAGE0, I03_RDCIT_2
CJIF	E05_RAMINI, I01_LLDRIFT_2, I03_RDCIT_3	PAGE0, E25_FAULTDET
CJMV	I03_RDCIT_2	PAGE0
CJCT		PAGE0, I03_RDCIT_3
CKSUM	O04_UARTOUT_DTS, O04_UARTOUT_GDTS	PAGE0
CKSUMH		PID
CKSUML		PID, RTDWRITE
CMPACT		CMPACT, INFLT
CNT203	I35_ARIPA	PAGE0
CNT206	I37_ARIISA_1	PAGE0
CNT211	I33_ARITAT	PAGE0
CNTA	E06_TQTEST, I25_TQNPT_2, I25_TQNPT_3	PAGE0, I25_TQNPT_4, I25_TQNPT_5
CNTB	E06_TQTEST, I25_TQNPT_2, I25_TQNPT_3	PAGE0, I25_TQNPT_4, I25_TQNPT_5
CNTC	E06_TQTEST, I25_TQNPT_2, I25_TQNPT_3	PAGE0, I25_TQNPT_4, I25_TQNPT_5
CNTPNT	E05_RAMINI, E08_FG_EXECUTIVE	PAGE0
CNT_EVTN		CNT_EVTN, INC_CTRS_2, INC_CTRS_3
CONFIG	E05_RAMINI	PAGE0, DCU_LCF_DATA, E08_FG_SECSL, E17_NCFLT_1, E17_NCFLT_2, E17_NCFLT_3, E17_NCFLT_4, E17_NCFLT_5, E17_NCFLT_6, E18_FAULTSUM_1, E18_FAULTSUM_2, E18_FAULTSUM_3, E18_FAULTSUM_4, FLTTIMER, INFLT, RUNTIMER
CONNIF	E11_CONTEST	PAGE0
COUNT	E14_FAULTSTORE_3	PAGE0, E14_FAULTSTORE_4
CPDAT_MISMATCH		PAGE1, DCU_CONSTANTS
CRPAMD	I36_PAMSEL	PAGE0
CRPFTR	I03_RDCIT_2	PAGE0, I04_PSAMBSG_3, I05_DELTAPSG_2
CRT15D	I34_TATSEL	PAGE0
CTIF	E05_RAMINI, I09_HLDRIFT_2, I10_GAINCOM_2, I13_ENGVARIANT_2	PAGE0, O26_ARO340, O27_ARO342
CXIF		PAGE0
CYC_ENG_MODEL12		PAGE1
CYC_ENG_MODEL34		PAGE1
D1	I28_DISINW1	PAGE0
D1TMR	I28_DISINW1	PAGE1
D1Z	I28_DISINW1	PAGE0
D1ZZ	I28_DISINW1	PAGE0
DBLCYC	E08_FG_EXECUTIVE	PAGE0, O05_ARO133, O06_ARO134, O07_ARO203, ALT11, O07_ARO203_ALTPS, O09_ARO211_TAT11, O10_ARO240, O11_ARO245, O15_ARO341, O16_ARO346, O17_ARO350, O18_ARO270, O26_ARO340, O27_ARO342
DBL_LL_SUBR		DBL_LL_SUBR, C17_NHREQUEST_1, C18_NHGOV_2
DCU3M_PNA12		PAGE1
DCU3M_PNA34		PAGE1
DCU3M_PNA56		PAGE1
DCU3M_PNA78		PAGE1
DCU3M_PNA910		PAGE1
DCUSIF	DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_LCF_DATA, DCU_SCHEDULE	PAGE1
DCU_ADDRESS		DCU_CONSTANTS, DCU_READ, DCU_RECEIVE, DCU_WRITE
DCU_ADJUST_RSYNC	DCU_IS_OBSOLETE	PAGE1, DCU_COUNTER_DATA
DCU_ADJUST_SHUTD	DCU_IS_OBSOLETE	PAGE1, DCU_COUNTER_DATA

Parameter	SET BY	USED BY
DCU_ADJUST_STRTS	DCU_IS_OBSOLETE	PAGE1, DCU_COUNTER_DATA
DCU_BLOCK_PTR	DCU_SCHEDULE	PAGE0
DCU_BUFFER_ADR		DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_GET_CHECKSUM, DCU_IS_OBSOLETE
DCU_BUFFER_END		PAGE1, DCU_GET_CHECKSUM
DCU_BUFFER_PTR	DCU_COMPARE_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_GET_CHECKSUM, DCU_LCF_DATA, DCU_RANGE_CHECKS, DCU_READ, DCU_WRITE	PAGE0, DCU_RECEIVE, DCU_TRANSMIT
DCU_BUSY_TIMER	DCU_READ, DCU_TRANSFER, DCU_WRITE	PAGE1
DCU_BYTES_INPUT	DCU_RECEIVE, DCU_TRANSFER	PAGE1
DCU_BYTES_SENT	DCU_READ, DCU_TRANSMIT, DCU_WRITE	PAGE1, DCU_RECEIVE
DCU_CHECKSM	DCU_RECEIVE	PAGE1
DCU_CHECKSUM	DCU_GET_CHECKSUM, DCU_RECEIVE, DCU_TRANSFER, DCU_TRANSMIT	PAGE1, DCU_CONSTANTS, DCU_DCUID_DATA, DCU_LCF_DATA
DCU_CHECKSUM_ADR		DCU_CONSTANTS, DCU_TRANSMIT
DCU_CMD_BUFFER		PAGE1, DCU_CONSTANTS
DCU_CMD_INVALID		DCU_CONSTANTS, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA
DCU_COMMAND_ADR		DCU_CONSTANTS, DCU_READ, DCU_TRANSFER, DCU_WRITE
DCU_COMMAND_END		PAGE1, DCU_TRANSMIT
DCU_COMPARE_DATA		DCU_COMPARE_DATA, DCU_COMPONT_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA
DCU_COMPARE_ITEM		DCU_COMPARE_ITEM, DCU_COMPARE_DATA, DCU_EEC_DATA, DCU_LCF_DATA
DCU_COMPONT_DATA		DCU_COMPONT_DATA, DCU_CONSTANTS
DCU_COUNTER_DATA		DCU_COUNTER_DATA, DCU_CONSTANTS
DCU_CPCMP_END		DCU_CONSTANTS, DCU_COMPONT_DATA
DCU_CPCMP_TABLE		DCU_CONSTANTS, DCU_COMPONT_DATA
DCU_CPDATA_ADR		DCU_CONSTANTS, DCU_COMPONT_DATA, DCU_LCF_DATA, INI_DCUI_DATA
DCU_CPDATA_BUF		PAGE1, DCU_CONSTANTS
DCU_CPDATA_COPY		DCU_CONSTANTS, DCU_COMPONT_DATA
DCU_CPDATA_END		PAGE1, DCU_CONSTANTS, INI_DCUI_DATA
DCU_CPDAT_CMP		DCU_CONSTANTS, DCU_COMPONT_DATA
DCU_CPOUT_BLOCK		DCU_CONSTANTS, DCU_WRITE_CPDAT
DCU_CPTMP_ADR		DCU_CONSTANTS, DCU_COMPONT_DATA
DCU_CPTMP_BUF		PAGE1, DCU_CONSTANTS
DCU_CPTMP_END		PAGE1, DCU_COMPONT_DATA
DCU_CTRDATA_ADR		DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_IS_OBSOLETE, INI_DCUI_DATA
DCU_CTRDATA_BUF		PAGE1, DCU_CONSTANTS
DCU_CTRDATA_COPY		DCU_CONSTANTS, DCU_COUNTER_DATA
DCU_CTRDATA_END		PAGE1, DCU_CONSTANTS, DCU_COUNTER_DATA, INI_DCUI_DATA
DCU_CTRDAT_EEREQ		PAGE1, DCU_CONSTANTS, UPDATE_EE_DATA
DCU_CTRDAT_IF	DCU_COUNTER_DATA, DCU_IS_OBSOLETE	PAGE1, DCU_CONSTANTS
DCU_CTRDAT_RANGE		DCU_CONSTANTS, DCU_COUNTER_DATA, INI_DCUI_DATA
DCU_CTREND_RANGE		DCU_CONSTANTS, DCU_COUNTER_DATA, INI_DCUI_DATA
DCU_CTRROUT_BLOCK		DCU_CONSTANTS, DCU_WRITE_CTRDAT
DCU_DATA	DCU_RECEIVE	PAGE1
DCU_DATA_BUFFER		PAGE1, DCU_CONSTANTS
DCU_DATA_BYTE	DCU_TRANSMIT	PAGE1
DCU_DATA_END	DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_LCF_DATA, DCU_TRANSMIT	PAGE1
DCU_DATA_FILLING	DCU_TRANSFER, DCU_TRANSMIT	PAGE1
DCU_DATA_PTR	DCU_COMPARE_DATA, DCU_DCUID_DATA, DCU_LCF_DATA, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE	PAGE0
DCU_DATA_READY	DCU_DCUID_DATA, DCU_LCF_DATA, DCU_SCHEDULE, INC_CTRS_2, INC_CTRS_3	PAGE1, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, OPMODE_1, UART_IO_SELECT, UPDATE_EE_DATA
DCU_DATA_TIMER	DCU_RECEIVE, DCU_TRANSFER	PAGE1
DCU_DCUCMP_END		DCU_CONSTANTS, DCU_DCUID_DATA
DCU_DCUCMP_TABLE		DCU_CONSTANTS, DCU_DCUID_DATA
DCU_DCUIDAT_READY		DCU_CONSTANTS, DCU_DCUID_DATA, UPDATE_EE_DATA
DCU_DCUIEND_RANGE		DCU_CONSTANTS, DCU_DCUID_DATA
DCU_DCUID_ADR		DCU_CONSTANTS, DCU_DCUID_DATA
DCU_DCUID_BUF		PAGE1, DCU_CONSTANTS
DCU_DCUID_DATA		DCU_DCUID_DATA, DCU_CONSTANTS
DCU_DCUID_END		PAGE1, DCU_DCUID_DATA
DCU_DCUID_IF	DCU_DCUID_DATA	PAGE1, DCU_CONSTANTS
DCU_DCUID_LT		DCU_CONSTANTS, DCU_DCUID_DATA
DCU_DCUID LTCMP		DCU_CONSTANTS, DCU_DCUID_DATA
DCU_DCUID_RANGE		DCU_CONSTANTS, DCU_DCUID_DATA

Parameter	SET BY	USED BY
DCU_DCUID_RBLOCK		DCU_CONSTANTS
DCU_DCUID_SUM		DCU_CONSTANTS, DCU_DCUID_DATA
DCU_DELAY_TIMER	DCU_TRANSFER	PAGE1
DCU_EECDATA_ADR		DCU_CONSTANTS, DCU_EEC_DATA
DCU_EECDATA_BUF		PAGE1, DCU_CONSTANTS
DCU_EECDATA_END		PAGE1
DCU_EECOUT_BLOCK		DCU_CONSTANTS, DCU_WRITE_EECDAT
DCU_EECPN_END		PAGE1, DCU_EEC_DATA, INI_DCUI_DATA
DCU_EECTRDAT_UPD	DCU_COUNTER_DATA, UPDATE_EE_DATA	PAGE1
DCU_EEC_DATA		DCU_EEC_DATA, DCU_CONSTANTS
DCU_EEC_PN		PAGE1, DCU_EEC_DATA, INI_DCUI_DATA
DCU_ENGCMP_END		DCU_CONSTANTS, DCU_ENGINE_DATA
DCU_ENGCMP_TABLE		DCU_CONSTANTS, DCU_ENGINE_DATA
DCU_ENGDAT_ADR		DCU_CONSTANTS, DCU_ENGINE_DATA, INI_DCUI_DATA
DCU_ENGDAT_BUF		PAGE1, DCU_CONSTANTS
DCU_ENGDAT_COPY		DCU_CONSTANTS, DCU_ENGINE_DATA
DCU_ENGDAT_END		PAGE1, DCU_CONSTANTS, INI_DCUI_DATA
DCU_ENGDAT_CMP		DCU_CONSTANTS, DCU_ENGINE_DATA
DCU_ENGINE_DATA		DCU_ENGINE_DATA, DCU_CONSTANTS
DCU_ENGOUT_BLOCK		DCU_CONSTANTS, DCU_WRITE_ENGDAT
DCU_ENGTMP_ADR		DCU_CONSTANTS, DCU_ENGINE_DATA
DCU_ENGTMP_BUF		PAGE1, DCU_CONSTANTS
DCU_ENGTMP_END		PAGE1, DCU_ENGINE_DATA
DCU_FLAGS_1	DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT	PAGE1, DCU_IS_OBSOLETE
DCU_FLAGS_2	DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_ENGINE_DATA	PAGE1
DCU_FUNC_SW	DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_LCF_DATA, DCU_SCHEDULE, E05_RAMINI	PAGE0, LCF_CHECK, OPMODE_1, UART_IO_SELECT
DCU_GET_CHECKSUM		DCU_GET_CHECKSUM, DCU_DCUID_DATA, DCU_LCF_DATA
DCU_GET_DATA		DCU_GET_DATA, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_LCF_DATA
DCU_GET_QUEUE		DCU_GET_QUEUE, DCU_SCHEDULE
DCU_GOT_CPDAT		DCU_CONSTANTS, DCU_COMPONT_DATA
DCU_GOT_CTRDATA		DCU_CONSTANTS, DCU_COUNTER_DATA
DCU_GOT_ENGDAT		DCU_CONSTANTS, DCU_ENGINE_DATA
DCU_GQUEUE_PTR	DCU_GET_QUEUE, DCU_PUT_QUEUE	PAGE0
DCU_HOLD_TABLE	DCU_LCF_DATA, DCU_SCHEDULE	PAGE1
DCU_INPUT_STATE	DCU_RECEIVE, DCU_TRANSFER	PAGE1
DCU_IO_COMPLETE	DCU_RECEIVE, DCU_TRANSFER, DCU_TRANSMIT	PAGE1
DCU_IO_QUIET		DCU_CONSTANTS, DCU_SCHEDULE, UART_IO_SELECT
DCU_IO_STATE	DCU_READ, DCU_TRANSFER, DCU_WRITE	PAGE1
DCU_IS_AVAILABLE	DCU_READ, DCU_TRANSFER, DCU_WRITE	PAGE1, DCU_SCHEDULE
DCU_IS_OBSOLETE		DCU_IS_OBSOLETE, DCU_CONSTANTS
DCU_LCFBLOCK_END		PAGE1, DCU_LCF_DATA
DCU_LCFBLOCK_MAX		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFBLOCK_MIN		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFBLOCK_NO	DCU_LCF_DATA	PAGE1
DCU_LCFBLOCK_NXT	DCU_LCF_DATA	PAGE1
DCU_LCFBUF_PTR		PAGE1, DCU_LCF_DATA
DCU_LCFCP_BLOCKS		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFCP_ENTRY		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFDATA_ADR		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFDATA_BUF		PAGE1, DCU_CONSTANTS
DCU_LCFDATA_PTR	DCU_LCF_DATA	PAGE1
DCU_LCFDAT_RANGE		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFDAT_READY		DCU_CONSTANTS, DCU_LCF_DATA, LCF_CHECK
DCU_LCFDAT_SUM		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFENTRY_LEN		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFENTRY_MAX		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFFACT_SW	DCU_LCF_DATA	PAGE1
DCU_LCFPN_NF		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCFPN_PTR	DCU_LCF_DATA	PAGE1
DCU_LCFSW_CFAULT		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCF_BLOCK		PAGE1, DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCF_BLOCKS		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCF_COMPONT	DCU_LCF_DATA	PAGE1
DCU_LCF_DATA		DCU_LCF_DATA, DCU_CONSTANTS
DCU_LCF_NUM_ACF		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCF_NUM_FCF		DCU_CONSTANTS, DCU_LCF_DATA, LCF_CYCLE_COUNT

Parameter	SET BY	USED BY
DCU_LCF_PBLOCK		DCU_CONSTANTS
DCU_LCF_READING	DCU_LCF_DATA	PAGE1
DCU_LCF_TABLE		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCF_TFAULT		DCU_CONSTANTS, DCU_LCF_DATA
DCU_LCF_TIMEOUT		PAGE1
DCU_LCF_UPD	DCU_COUNTER_DATA, LCF_CHECK	PAGE1, LCF_CYCLE_COUNT, UPDATE_EE_DATA
DCU_LO_BYTE	DCU_READ, DCU_RECEIVE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE	PAGE1
DCU_MAX_BUSY	DCU_READ, DCU_WRITE	PAGE1, DCU_TRANSFER
DCU_MISMATCH		DCU_CONSTANTS, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_IS_OBSOLETE
DCU_MODIFIED		DCU_CONSTANTS, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_IS_OBSOLETE
DCU_NEW		DCU_CONSTANTS, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_IS_OBSOLETE
DCU_NO_BLOCK		DCU_CONSTANTS, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA
DCU_NUM_MISSIONS		DCU_CONSTANTS, LCF_CYCLE_COUNT
DCU_OBSOLETE	DCU_IS_OBSOLETE	PAGE1, DCU_COUNTER_DATA
DCU_OUTPUT_STATE	DCU_TRANSFER, DCU_TRANSMIT	PAGE1
DCU_PORT_STATUS	DCU_TRANSMIT	PAGE1
DCU_PQUEUE_PTR	DCU_GET_QUEUE, DCU_PUT_QUEUE	PAGE0
DCU_PTABLE_ADR		DCU_CONSTANTS, DCU_SCHEDULE
DCU_PTABLE_END		DCU_CONSTANTS, DCU_SCHEDULE
DCU_PTABLE_PTR	DCU_SCHEDULE	PAGE0
DCU_PUT_QUEUE		DCU_PUT_QUEUE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT
DCU_QUEUE_ADR		DCU_CONSTANTS, DCU_GET_QUEUE, DCU_PUT_QUEUE
DCU_QUEUE_EMPTY		PAGE1, DCU_SCHEDULE
DCU_QUEUE_END		PAGE1, DCU_GET_QUEUE, DCU_PUT_QUEUE
DCU_RANGE_CHECKS		DCU_RANGE_CHECKS, DCU_COUNTER_DATA, DCU_DCUID_DATA, INI_DCU_DATA
DCU_READ		DCU_READ, DCU_SCHEDULE
DCU_RECEIVE		DCU_RECEIVE, DCU_TRANSFER
DCU_RETRY_COUNT	DCU_READ, DCU_TRANSFER, DCU_WRITE	PAGE1
DCU_RETRY_TIMER	DCU_TRANSFER	PAGE1
DCU_RTABLE_ADR		DCU_CONSTANTS, DCU_SCHEDULE
DCU_RTABLE_END		DCU_CONSTANTS, DCU_SCHEDULE
DCU_RTABLE_PTR	DCU_SCHEDULE	PAGE0
DCU_SCHEDULE		DCU_SCHEDULE, E08_FG_SCHEDULE
DCU_SCHED_STATE	DCU_SCHEDULE	PAGE1
DCU_STATUS	DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_RECEIVE, DCU_TRANSFER	PAGE1, DCU_DCUID_DATA, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_SCHEDULE
DCU_STSHT_MM_MSK		DCU_CONSTANTS, DCU_COUNTER_DATA
DCU_STSHT_OFF_MM		DCU_CONSTANTS, DCU_IS_OBSOLETE
DCU_STSHT_ON_MM		DCU_CONSTANTS, DCU_IS_OBSOLETE
DCU_TRANSFER		DCU_TRANSFER, E08_FG_EXECUTIVE
DCU_TRANSMIT		DCU_TRANSMIT, DCU_TRANSFER
DCU_UPDATE_ADR		DCU_CONSTANTS, DCU_SCHEDULE
DCU_UPDATE_END		DCU_CONSTANTS, DCU_SCHEDULE
DCU_UPD_CPDAT		DCU_CONSTANTS, DCU_COMPONT_DATA, DCU_WRITE_CPDAT
DCU_UPD_CTRDAT		DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_WRITE_CTRDAT
DCU_UPD_EECDAT		DCU_CONSTANTS, DCU_EEC_DATA, DCU_WRITE_EECDAT
DCU_UPD_ENGDAT		DCU_CONSTANTS, DCU_COMPONT_DATA, DCU_ENGINE_DATA, DCU_WRITE_ENGDAT
DCU_WRITE		DCU_WRITE, DCU_SCHEDULE
DCU_WRITE_CPDAT		DCU_WRITE_CPDAT, DCU_CONSTANTS
DCU_WRITE_CTRDAT		DCU_WRITE_CTRDAT, DCU_CONSTANTS
DCU_WRITE_EECDAT		DCU_WRITE_EECDAT, DCU_CONSTANTS
DCU_WRITE_ENGDAT		DCU_WRITE_ENGDAT, DCU_CONSTANTS
DCU_WRITE_QUEUE		PAGE1, DCU_CONSTANTS
DCU_XXX_EEREQ		PAGE1, DCU_CONSTANTS
DECEL1	C15_NHDECELLIM	PAGE1
DECEL2	C15_NHDECELLIM	PAGE1
DECEL4	C15_NHDECELLIM	PAGE1
DECELC	C15_NHDECELLIM	PAGE1
DECFAC	C15_NHDECELLIM	PAGE1
DELHP	C11_SHPREQ_1	PAGE1
DELHP1	C11_SHPREQ_1	PAGE1
DELISA	C26_MPAPARAM_2	PAGE1
DELPAM	I36_PAMSEL	PAGE0
DELPC	I05_DELTAPSG_4	PAGE0, I05_DELTAPSG_2

Parameter	SET BY	USED BY
DELPIF	E05_RAMINI, I01_LLDRIFT_2, I02_SGCOM_2, I05_DELTAPSG_4	PAGE0, E25_FAULTDET, I05_DELTAPSG_2, I46_IASFAULT, OPMODE_1
DELRAM	E14_FAULTSTORE_5, E14_FAULTSTORE_6	PAGE0
DELT15	I34_TATSEL	PAGE0
DELTA	C26_MPARAM_3	PAGE0, C08_SHPCL, C13_NHACCELLIM, C15_NHDECELLIM, C18_NHGOV_1, C21_HBOVEM_1, C22_HBOVMM_1, C27_NHMMMX_1
DELTA1	C08_SHPCL	PAGE1
DELTA2	C08_SHPCL	PAGE1
DELTAP	I05_DELTAPSG_2, I05_DELTAPSG_4	PAGE0, I05_DELTAPSG_3
DELTCT		PAGE0, I05_DELTAPSG_4
DERAT1	C26_MPARAM_1	PAGE1, E21_RECONFIG, I13_ENGVARIANT_2
DERAT2	C26_MPARAM_1	PAGE1, E21_RECONFIG, I13_ENGVARIANT_2
DERLAG		DERLAG, C11_SHPREQ_2, C12_NPTGOV_1
DEVMODEN	C26_MPARAM_1	PAGE1, I13_ENGVARIANT_2
DHPREQ	C11_SHPREQ_2	PAGE1
DISC1	I28_DISINW1	PAGE0, C05_TQBUG, C06_HPREF, C06_HPREF_1, C26_MPARAM_1, E13_WATEST_1, E31_EXTDIS, I17_PNHCONV_1, I18_READNH, I19_NHDUAL_2, I23_RDSNHP, I25_TQNPT_2, I25_TQNPT_3, I25_TQNPT_4, I25_TQNPT_5, I43_TQSPEED_1, I43_TQSPEED_2, O03_DISOUT, O03_DISOUT_A, O10_ARO240, O18_ARO270, O19_ARO272, O19_ARO272_A, O40_ARO06X, OPMODE_3, RTDWRITE
DISC2	I29_DISINW2	PAGE0, C26_MPARAM_1, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, E27_MFCID, E28_HBOVWA_1, E31_EXTDIS, I31_UARTIN, O18_ARO270, UART_IO_SELECT
DISCOT	E05_RAMINI, O03_DISOUT	PAGE0, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, E16_SMTEST, E31_EXTDIS, I38_SMCSV_2, O18_ARO270, O19_ARO272
DIVPRT		DIVPRT, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_2, C18_NHGOV_1, C19_UNDEERSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DLHPRF	C11_SHPREQ_1	PAGE1, C11_SHPREQ_2
DLNDF	C13_NHACCELLIM	PAGE1, C10_ATOIGN
DLTANH	C18_NHGOV_2	PAGE0
DLTNHL	C18_NHGOV_2	PAGE0
DMPPMFC	C20_OVRSPEED	PAGE1, C06_HPREF, C11_SHPREQ_2, C12_NPTGOV_2
DMPPC1	C20_OVRSPEED	PAGE1
DMPPC2	C20_OVRSPEED	PAGE1
DMPPCU	C20_OVRSPEED	PAGE1, C11_SHPREQ_2, C30_SMRTRQ
DNHACC	C13_NHACCELLIM	PAGE1, C16_LOOPSEL, C17_NHREQUEST_1
DNHDEC	C15_NHDECELLIM	PAGE1, C16_LOOPSEL, C17_NHREQUEST_1
DNHDFG	C20_OVRSPEED	PAGE1
DNHERR	C17_NHREQUEST_1	PAGE1, C12_NPTGOV_2, C16_LOOPSEL
DNHIL2	C12_NPTGOV_1	PAGE1
DNHILL	C17_NHREQUEST_1	PAGE0
DNHINL	C17_NHREQUEST_1	PAGE0, C16_LOOPSEL
DNHLIM	C14_NHLIM	PAGE1, C16_LOOPSEL, C17_NHREQUEST_1
DNHLM	C14_NHLIM	PAGE1
DNHNPD	C12_NPTGOV_1	PAGE1
DNHNPP	C12_NPTGOV_1	PAGE1
DNHNPT	C12_NPTGOV_2	PAGE1, C16_LOOPSEL, C17_NHREQUEST_1
DNHOSD	C20_OVRSPEED	PAGE1
DNHOSU	C20_OVRSPEED	PAGE1
DNHREQ	C17_NHREQUEST_1	PAGE0
DNHRQL	C17_NHREQUEST_1	PAGE0
DNHSEF		PAGE0, C21_HBOVEM_1
DNHSEL	C17_NHREQUEST_1	PAGE0, C17_NHREQUEST_2, C21_HBOVEM_1, RTDWRITE
DNHTQ	C31_TQLIM	PAGE1, C16_LOOPSEL, C17_NHREQUEST_1
DNHTQ1	C31_TQLIM	PAGE1
DNHUFG	C20_OVRSPEED	PAGE1
DNHWL1	C17_NHREQUEST_1	PAGE0
DNHWLL	C17_NHREQUEST_1	PAGE0

Parameter	SET BY	USED BY
DNSEL1	C17_NHREQUEST_1	PAGE1, C16_LOOPSEL
DNSEL2	C17_NHREQUEST_1	PAGE1, C16_LOOPSEL
DNSEL3	C17_NHREQUEST_1	PAGE1, C16_LOOPSEL
DNSEL4	C17_NHREQUEST_1	PAGE1, C16_LOOPSEL
DNSEL5	C17_NHREQUEST_1	PAGE1, C16_LOOPSEL
DP18F		PAGE0, I05_DELTAPSG_3
DPA		DPA, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRQ, C31_TQLIM, DBL_LL_SUBR, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPA1	C08_SHPCL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_2, C30_SMRTRQ, DBL_LL_SUBR, LCF_CYCLE_COUNT	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C18_NHGOV_1, C19_UNDESPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C31_TQLIM, DPA, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPA1B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPA2	C08_SHPCL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_2, C30_SMRTRQ, DBL_LL_SUBR, LCF_CYCLE_COUNT	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C18_NHGOV_1, C19_UNDESPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C31_TQLIM, DPA, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPA2B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3

Parameter	SET BY	USED BY
DPAB		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPATMP	DPA	PAGE0
DPBLK		PAGE10, E03_CHECKSUM
DPDATA		PAGE10, E05_RAMINI
DPL134		PAGE0
DPM		DPM, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNX_1, C30_SMRTRQ, C31_TQLIM, DBL_LL_SUBR, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPM1	C03_TLMRT_2_1, C03_TLMRT_3_1, C03_TLMRT_4_1, C03_TLMRT_5_1, C03_TLMRT_6_1, C05_TQBUG, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_2, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_5, C30_SMRTRQ, DBL_LL_SUBR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_2, C03_TLMRT_3_2, C03_TLMRT_4_2, C03_TLMRT_5_2, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C18_NHGOV_1, C19_UNDEERSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_4, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNX_1, C31_TQLIM, DPM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPM1B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPM2	C03_TLMRT_2_1, C03_TLMRT_3_1, C03_TLMRT_4_1, C03_TLMRT_5_1, C03_TLMRT_6_1, C05_TQBUG, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_2, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_5, C30_SMRTRQ, DBL_LL_SUBR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_2, C03_TLMRT_3_2, C03_TLMRT_4_2, C03_TLMRT_5_2, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C18_NHGOV_1, C19_UNDEERSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_4, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNX_1, C31_TQLIM, DPM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3

Parameter	SET BY	USED BY
DPM2B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPM3	DPM	PAGE0
DPM4	DPM	PAGE0
DPM5	DPM	PAGE0
DPM6	DPM	PAGE0
DPMB		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
DPMK	DPM	PAGE0
DPMRET	DPM	PAGE0
DPO	C26_MPARAM_3	PAGE1
DPOS	C26_MPARAM_3	PAGE0
DPOVRP	I05_DELTAPSG_3	PAGE0
DPSIZE		I05_DELTAPSG_2
DPTAB1		PAGE0, E05_RAMINI, I03_RDCJT_2
DPTAB9		PAGE0, I03_RDCJT_2
DPTABH	I03_RDCJT_2	PAGE0, I05_DELTAPSG_2
DPTABL	I03_RDCJT_2	PAGE0, I05_DELTAPSG_2
DQOUTI	C26_MPARAM_5	PAGE0
DRFCIF	I09_HLDRIFT_2	PAGE0, E17_NCFLT_2
DRFT1C	I01_LLDRIFT_2	PAGE0, I02_SGCOM_2, I03_RDCJT_3, I04_PSAMBSG_2, I05_DELTAPSG_4, I07_READTQ_3, I08_RDT18_3
DRFTCT		PAGE0, I01_LLDRIFT_2
DRFTIF	I01_LLDRIFT_2	PAGE0, E14_FAULTSTORE_4, E17_NCFLT_1, E25_FAULTDET, E26_FAULTSTORE_4
DRHTCT		PAGE0, I09_HLDRIFT_2
DRIFTC	I09_HLDRIFT_2	PAGE0, I10_GAINCOM_2, I11_TQ8TRIM_2, I12_TQGTTRIM_2, I13_ENGVARIANT_2
DROOP	C19_UNDERSPEED	PAGE1, C12_NPTGOV_1
DSLMLF	C21_HBOVEM_1	PAGE1, C21_HBOVEM_2
DSLMLS	C21_HBOVEM_1	PAGE1
DSLMSF	C22_HBOVMM_2	PAGE1
DSMNH4	C30_SMRTRQ	PAGE0
DSMNHK	C30_SMRTRQ	PAGE0
DSMNLH	C30_SMRTRQ	PAGE0
DSMP1	C18_NHGOV_2	PAGE0
DSMPFL	C30_SMRTRQ	PAGE0, C18_NHGOV_2
DSMPFR	C30_SMRTRQ	PAGE0, C18_NHGOV_2
DSMPRL	C18_NHGOV_2	PAGE0
DSMPRQ	C18_NHGOV_2	PAGE0
DSNH4L	C30_SMRTRQ	PAGE0
DTSCFG	E05_RAMINI	PAGE1, O04_UARTOUTSEL
DUMP		DUMP, E20_TESTMODE
DUMPGF	C20_OVRSPEED	PAGE1, C15_NHDECELLIM
DXBUMP	C19_UNDERSPEED	PAGE1, C30_SMRTRQ
DXIN	C18_NHGOV_2	PAGE0
DXINL	C18_NHGOV_2	PAGE0
DXOUTA	C18_NHGOV_2	PAGE0, C30_SMRTRQ
DXOUTL	C18_NHGOV_2	PAGE0
E01_PORPSR		E01_PORPSR
E02_RAMTEST		E02_RAMTEST
E03_CHECKSUM		E03_CHECKSUM, E01_PORPSR
E04_INSTST		E04_INSTST, E01_PORPSR
E05_RAMINI		E05_RAMINI

Parameter	SET BY	USED BY
E06_TQTEST		E06_TQTEST
E07_BG_EXECUTIVE		E07_BG_EXECUTIVE
E08_FG_EXECUTIVE		E08_FG_EXECUTIVE
E08_FG_SCHEDULE		E08_FG_SCHEDULE
E08_FG_SECSEL		E08_FG_SECSEL
E08_FG_STATETABLE		E08_FG_STATETABLE
E09_SMSUMV		E09_SMSUMV, E01_PORPSR
E10_RAMTEST		E10_RAMTEST, E08_FG_SCHEDULE
E11_CONTEST		E11_CONTEST, E08_FG_SCHEDULE
E13_WATEST_1		E13_WATEST_1, E08_FG_SCHEDULE
E13_WATEST_2		E13_WATEST_2, E08_FG_SCHEDULE
E13_WATEST_3		E13_WATEST_3, E08_FG_SCHEDULE
E13_WATEST_4		E13_WATEST_4, E08_FG_SCHEDULE
E13_WATEST_5		E13_WATEST_5, E08_FG_SCHEDULE
E13_WATEST_6		E13_WATEST_6, E08_FG_SCHEDULE
E13_WATEST_7		E13_WATEST_7, E08_FG_SCHEDULE
E14_FAULTSTORE_1		E14_FAULTSTORE_1, E08_FG_SCHEDULE
E14_FAULTSTORE_2		E14_FAULTSTORE_2, E08_FG_SCHEDULE
E14_FAULTSTORE_3		E14_FAULTSTORE_3, E08_FG_SCHEDULE
E14_FAULTSTORE_4		E14_FAULTSTORE_4, E08_FG_SCHEDULE
E14_FAULTSTORE_5		E14_FAULTSTORE_5, E08_FG_SCHEDULE
E14_FAULTSTORE_6		E14_FAULTSTORE_6, E08_FG_SCHEDULE
E14_FAULTSTORE_8		E14_FAULTSTORE_8
E15_BCKGNDTEST		E15_BCKGNDTEST, E08_FG_STATETABLE
E16_SMTEST		E16_SMTEST, E08_FG_SCHEDULE
E17_NCFLT_1		E17_NCFLT_1, E08_FG_SCHEDULE
E17_NCFLT_2		E17_NCFLT_2, E08_FG_SCHEDULE
E17_NCFLT_3		E17_NCFLT_3, E08_FG_SCHEDULE
E17_NCFLT_4		E17_NCFLT_4, E08_FG_SCHEDULE
E17_NCFLT_5		E17_NCFLT_5, E08_FG_SCHEDULE
E17_NCFLT_6		E17_NCFLT_6, E08_FG_SCHEDULE
E17_NCFLT_7		E17_NCFLT_7
E18_FAULTSUM_1		E18_FAULTSUM_1, E08_FG_SCHEDULE
E18_FAULTSUM_2		E18_FAULTSUM_2, E08_FG_SCHEDULE
E18_FAULTSUM_3		E18_FAULTSUM_3, E08_FG_SCHEDULE
E18_FAULTSUM_4		E18_FAULTSUM_4, E08_FG_SCHEDULE
E18_FAULTSUM_5		E18_FAULTSUM_5
E18_FAULTSUM_6		E18_FAULTSUM_6
E19_TRMSTORE_1		E19_TRMSTORE_1, E08_FG_SCHEDULE
E19_TRMSTORE_2		E19_TRMSTORE_2, E08_FG_SCHEDULE
E19_TRMSTORE_4		E19_TRMSTORE_4, E08_FG_SCHEDULE
E19_TRMSTORE_5		E19_TRMSTORE_5, E08_FG_SCHEDULE
E20_TESTMODE		E20_TESTMODE, E08_FG_EXECUTIVE
E21_RECONFIG		E21_RECONFIG, E08_FG_SCHEDULE
E22_GASEL		E22_GASEL, E01_PORPSR
E23_PLATRIM		E23_PLATRIM, E08_FG_SCHEDULE
E25_FAULTDET		E25_FAULTDET, E08_FG_SCHEDULE
E26_FAULTSTORE_1		E26_FAULTSTORE_1, E08_FG_SCHEDULE
E26_FAULTSTORE_2		E26_FAULTSTORE_2, E08_FG_SCHEDULE
E26_FAULTSTORE_3		E26_FAULTSTORE_3
E26_FAULTSTORE_4		E26_FAULTSTORE_4, E08_FG_SCHEDULE
E26_FAULTSTORE_5		E26_FAULTSTORE_5
E27_MFCID		E27_MFCID, E08_FG_SCHEDULE
E28_HBOVWA_1		E28_HBOVWA_1, E08_FG_SCHEDULE
E28_HBOVWA_2		E28_HBOVWA_2, E08_FG_SCHEDULE
E2CNT	E14_FAULTSTORE_1	PAGE0
E31		E31_EXTDIS
E31_EXTDIS		E31_EXTDIS, E08_FG_SCHEDULE
EBSY26	E26_FAULTSTORE_1	PAGE0
ECFAIL	E08_FG_EXECUTIVE	PAGE10
ECINTF	E25_FAULTDET	PAGE0
ECNT26	E26_FAULTSTORE_2, E26_FAULTSTORE_3	PAGE0
EDATIF	E03_CHECKSUM	PAGE0, E25_FAULTDET, I04_PSAMBSG_2, I05_DELTAPSG_2, I05_DELTAPSG_4
EEBPWM	E19_TRMSTORE_5	PAGE0
EEBQB	E19_TRMSTORE_1	PAGE0
EEBQG	E19_TRMSTORE_2	PAGE0
EEBUSY	E14_FAULTSTORE_1, E19_TRMSTORE_1, E19_TRMSTORE_2, E19_TRMSTORE_4, E19_TRMSTORE_5, E23_PLATRIM, E26_FAULTSTORE_1, EE_WRITE_BLOCK	PAGE0, E14_FAULTSTORE_4, E14_FAULTSTORE_6, E26_FAULTSTORE_4, EE_READ_DATA
EECPN_SNA1112		PAGE1
EECPN_SNA1112EE		PAGE11

Parameter	SET BY	USED BY
EECPN_SNA12		PAGE1
EECPN_SNA12EE		PAGE11
EECPN_SNA1314		PAGE1
EECPN_SNA1314EE		PAGE11
EECPN_SNA1516		PAGE1
EECPN_SNA1516EE		PAGE11
EECPN_SNA1718		PAGE1
EECPN_SNA1718EE		PAGE11
EECPN_SNA1920		PAGE1
EECPN_SNA1920EE		PAGE11
EECPN_SNA2122		PAGE1
EECPN_SNA2122EE		PAGE11
EECPN_SNA2324		PAGE1
EECPN_SNA2324EE		PAGE11
EECPN_SNA2526		PAGE1
EECPN_SNA2526EE		PAGE11
EECPN_SNA2728		PAGE1
EECPN_SNA2728EE		PAGE11
EECPN_SNA2930		PAGE1
EECPN_SNA2930EE		PAGE11
EECPN_SNA3132		PAGE1
EECPN_SNA3132EE		PAGE11
EECPN_SNA34		PAGE1
EECPN_SNA34EE		PAGE11
EECPN_SNA56		PAGE1
EECPN_SNA56EE		PAGE11
EECPN_SNA78		PAGE1
EECPN_SNA78EE		PAGE11
EECPN_SNA910		PAGE1
EECPN_SNA910EE		PAGE11
EECPOR	OPMODE_1	PAGE1, OPMODE_2
EEERF	E19_TRMSTORE_4	PAGE0
EEG		PAGE10, E19_TRMSTORE_5, E22_GASEL
EEGARC	E19_TRMSTORE_5, E22_GASEL	PAGE0
EEGARM	E22_GASEL	PAGE0
EESUM_16		EESUM_16, E03_CHECKSUM
EESUM_8		EESUM_8, E03_CHECKSUM
EE_CPDAT_BUF		PAGE11, DCU_CONSTANTS, INI_DCU_DATA
EE_CPDAT_BLOCK		DCU_CONSTANTS, DCU_COMPONT_DATA
EE_CTRDAT_BUF		PAGE11, DCU_CONSTANTS, INI_DCU_DATA
EE_CTRDAT_BLOCK		DCU_CONSTANTS, DCU_COUNTER_DATA, UPDATE_EE_DATA
EE_CTRDAT_IF	INI_DCU_DATA	PAGE1, DCU_COUNTER_DATA, DCU_IS_OBSOLETE
EE_DATA_ADR	EE_WRITE_BLOCK	PAGE1
EE_DATA_PTR	EE_WRITE_BLOCK	PAGE0
EE_EECPN		PAGE11, INI_DCU_DATA
EE_ENGDAT_BUF		PAGE11, DCU_CONSTANTS, INI_DCU_DATA
EE_ENGDAT_BLOCK		DCU_CONSTANTS, DCU_ENGINE_DATA
EE_GET_QUEUE		EE_GET_QUEUE, EE_WRITE_BLOCK
EE_QUEUE_PTR	EE_GET_QUEUE, EE_PUT_QUEUE	PAGE0
EE_PQUEUE_PTR	EE_GET_QUEUE, EE_PUT_QUEUE	PAGE0
EE_PUT_QUEUE		EE_PUT_QUEUE, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_ENGINE_DATA, UPDATE_EE_DATA
EE_QUEUE_ADR		DCU_CONSTANTS, EE_GET_QUEUE, EE_PUT_QUEUE
EE_QUEUE_EMPTY		PAGE1, EE_WRITE_BLOCK
EE_QUEUE_END		PAGE1, EE_GET_QUEUE, EE_PUT_QUEUE
EE_READ_DATA		EE_READ_DATA, INI_DCU_DATA
EE_REQUEST_DONE	EE_WRITE_BLOCK	PAGE1
EE_UPDATE_FLAGS	UPDATE_EE_DATA	PAGE1
EE_WRITE_BLOCK		EE_WRITE_BLOCK, E08_FG_SCHEDULE
EE_WRITE_QUEUE		PAGE1, DCU_CONSTANTS
EE_WRITING	EE_WRITE_BLOCK	PAGE1
EFLT26		PAGE10, E26_FAULTSTORE_1, E26_FAULTSTORE_3
EFLT26		PAGE10, E05_RAMINI, E14_FAULTSTORE_1, WRTEEP
EIDX26	E26_FAULTSTORE_1	PAGE0
ENDCYC	OPMODE_2	PAGE1
ENGDAT_MISMATCH		PAGE1, DCU_CONSTANTS
ENGFLTCYC	OPMODE_2	PAGE1, INC_CTRS_1, LCF_CHECK, MISSIONS
ENGFLTM_CRS_ALL		PAGE1, DCU_CONSTANTS, INC_CTRS_3, O40_ARO06X_TABLE
ENGFLTM_CRS_A_SS	INC_CTRS_3	PAGE1
ENGFLTM_FIN_ALL		PAGE1, INC_CTRS_3
ENGFLTM_FIN_A_SS	INC_CTRS_3	PAGE1
ENGPID		PID, DCU_COUNTER_DATA

Parameter	SET BY	USED BY
ENGPOS	C26 MPARAM 1	PAGE1, O07 ARO203 ALT11, O09 ARO211 TAT11
ENGRAT	E05_RAMINI, E21_RECONFIG, I13_ENGVARIANT_2	PAGE1, C04_RPCAL_2_D, C06_HPREF_1, INC_CTRS_4, MISSIONS, O19_ARO272_A
ENGRATC	E05_RAMINI, E19_TRMSTORE_4, E21_RECONFIG, I13_ENGVARIANT_2	PAGE1
ENGRATM	E21_RECONFIG, I13_ENGVARIANT_2	PAGE1
ENGRATS	C04_RPCAL_2_D	PAGE1
ENGRAT_BIT1	O19_ARO272_A	PAGE1, O19_ARO272
ENGRAT_BIT2	O19_ARO272_A	PAGE1, O19_ARO272
ENGRUN	OPMODE_1	PAGE1, DCU_IS_OBSOLETE, FLTTIMER, INC_CTRS_2, INC_CTRS_3, MISSIONS, OPMODE_2, OPMODE_3, RUNTIMER
ENGSHUTDOWN	OPMODE_1	PAGE1, INC_CTRS_1, LCF_CHECK, OPMODE_2, UPDATE_EE_DATA
ENGUNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
ENGUNA12EE		PAGE11
ENGUNA34		PAGE1, O40_ARO06X_TABLE
ENGUNA34EE		PAGE11
ENGUNA56		PAGE1, O40_ARO06X_TABLE
ENGUNA56EE		PAGE11
ENGSTART	OPMODE_1	PAGE1, INC_CTRS_1, OPMODE_2, UPDATE_EE_DATA
ENG_RUN_CRS_ALL		PAGE1, DCU_CONSTANTS, INC_CTRS_2, O40_ARO06X_TABLE
ENG_RUN_CRS_A_SS	INC_CTRS_2	PAGE1
ENG_RUN_FIN_ALL		PAGE1, INC_CTRS_2
ENG_RUN_FIN_A_SS	INC_CTRS_2	PAGE1
ERDELO	C06_HPREF_1	PAGE1
ERF		PAGE11, E19_TRMSTORE_4
ERNSEC	RUNTIMER	PAGE1
ERNSEC_SCL_PV	RUNTIMER	PAGE1
EROMB1	E14_FAULTSTORE_1	PAGE0
EROMIF	E05_RAMINI, E14_FAULTSTORE_1, E19_TRMSTORE_1, E19_TRMSTORE_2, E19_TRMSTORE_4, E19_TRMSTORE_5, E22_GASEL, E23_PLATRIM, E26_FAULTSTORE_1	PAGE0, C02_PLATRIM_1, E21_RECONFIG, E25_FAULTDET, I11_TQBTTRIM_2, I12_TQGTTRIM_2, I13_ENGVARIANT_2
ERRCNT	E14_FAULTSTORE_3, E14_FAULTSTORE_6, INIFLT	PAGE0
ERUNTC	RUNTIMER	PAGE1, INC_CTRS_2
ERUNTF	RUNTIMER	PAGE1, INC_CTRS_2
ESBFAF	OPMODE_2	PAGE1, INC_CTRS_1
ETLOCK	E23_PLATRIM	PAGE0, C02_PLATRIM_1, E31_EXTDIS, O18_ARO270
ETRIM		PAGE10
ETSTMTR	E14_FAULTSTORE_2	PAGE0
EVDFIF	I13_ENGVARIANT_2	PAGE0
EVFLTS	I13_ENGVARIANT_2	PAGE0
EVRAT	E21_RECONFIG	PAGE1
EVRATTMR	E21_RECONFIG	PAGE1
EWFAIL	C02_PLATRIM_1	PAGE1
EXDISC	E31_EXTDIS	PAGE1
E_26_INIT		E26_FAULTSTORE_3, E26_FAULTSTORE_2
E_ACCINFLSTRT_AL		PAGE11
E_ACCNUMFL_ALL		PAGE11
E_ACCSHUTD_ALL		PAGE11
E_ACCSTRTSFLT_AL		PAGE11
E_ACCSTRTS_ALL		PAGE11
E_ACCSTR_SD_RESYNC		PAGE11
E_ENGFLTM_CRS_AL		PAGE11
E_ENGFLTM_FIN_AL		PAGE11
E_ENG_RN_CRS_ALL		PAGE11
E_ENG_RN_FIN_ALL		PAGE11
E_HPIMP_CYCCRS		PAGE11
E_HPIMP_CYCFIN		PAGE11
E_HPTD_CYCCRS		PAGE11
E_HPTD_CYCFIN		PAGE11
E_HPTFCP_CYCCRS		PAGE11
E_HPTFCP_CYCFIN		PAGE11
E_HPTRCP_CYCCRS		PAGE11
E_HPTRCP_CYCFIN		PAGE11
E_ISS_CYCCRS		PAGE11
E_ISS_CYCFIN		PAGE11
E_LPIMP_CYCCRS		PAGE11
E_LPIMP_CYCFIN		PAGE11
E_LPTD_CYCCRS		PAGE11
E_LPTD_CYCFIN		PAGE11
E_MTO_DL1		PAGE11
E_MTO_DL2		PAGE11
E_MTO_DL3		PAGE11

Parameter	SET BY	USED BY
E_NTO_DL1		PAGE11
E_NTO_DL2		PAGE11
E_NTO_DL3		PAGE11
E_PT1D_CYCCRS		PAGE11
E_PT1D_CYCFIN		PAGE11
E_PT2D_CYCCRS		PAGE11
E_PT2D_CYCFIN		PAGE11
E_SV_PWC_PATCHID		PAGE11
E_SV_SW_EPI		PAGE11
E_SV_SW_SBI		PAGE11
E_SV_SW_SVI		PAGE11
F1_A1		E18_FAULTSUM_5
F1_A2		E18_FAULTSUM_5
F1_CW		E18_FAULTSUM_5
F1_HW		E18_FAULTSUM_5
F2_A1		E18_FAULTSUM_6, E18_FAULTSUM_5
F2_A2		E18_FAULTSUM_6, E18_FAULTSUM_5
F2_CW		E18_FAULTSUM_6, E18_FAULTSUM_5
F2_HW		E18_FAULTSUM_6, E18_FAULTSUM_5
FAILED		FAILED, E27_MFCID, I08_RDT18_3, I14_RDPLA_3, I25_TQNPT_2, I25_TQNPT_3, I25_TQNPT_4, I25_TQNPT_5
FAILLT		FAILLT, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, I01_LLDRIFT_2, I02_SGCOM_2, I03_RDCJT_3, I04_PSAMBSG_2, I05_DELTAPSG_4, I07_READTQ_3, I08_RDT18_3, I09_HLDRIFT_2, I10_GAINCOM_2, I11_TQ8TRIM_2, I12_TQ8TRIM_2, I15_PLA9IAS_2, I16_PLAGAIN_2, I19_NHDUAL_2, I45_HBOVWA_2
FETHDI	C26_MPARAM_1	PAGE1, C01_START, C10_ATOIGN, C12_NPTGOV_2, C13_NHACCELLIM, C15_NHDECELLIM, C27_NHMMNX_1, E13_WATEST_7
FGTM	E07_BG_EXECUTIVE, E15_BCKGNDTEST, INSTST	PAGE0
FGTMCT	E15_BCKGNDTEST	PAGE0
FILCON		FILCON, C18_NHGOV_2
FILTER		FILTER, I04_PSAMBSG_3, I05_DELTAPSG_3, I07_READTQ_2, I08_RDT18_3, I14_RDPLA_4, I20_NHSPEED, I24_NHPRBE2, I27_TQSPEED
FLACA1	E18_FAULTSUM_2	PAGE0, O05_ARO133, O06_ARO134, O07_ARO203_ALTPS, O11_ARO245, O15_ARO341, O16_ARO346, O18_ARO270, O26_ARO340, O27_ARO342
FLACA2	E18_FAULTSUM_3	PAGE0, O07_ARO203_ALT11, O26_ARO340, O40_ARO06X
FLACCW	E18_FAULTSUM_4	PAGE0, C26_MPARAM_1
FLACHW	E18_FAULTSUM_1	PAGE0, E13_WATEST_1, E28_HBOVWA_1, E28_HBOVWA_2, O01_SMOU, O02_HBOVOUT, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A
FLIGHT	OPMODE_1	PAGE1, FLTTIMER, INC_CTRS_1, INC_CTRS_3, MISSIONS, OPMODE_2, OPMODE_3
FLIGHTFLG1	OPMODE_1	PAGE1
FLIGHTFLG2	OPMODE_1	PAGE1
FLTCC	OPMODE_2	PAGE1
FLTCYC1	OPMODE_2	PAGE1
FLTMK1		E17_NCFLT_7, E17_NCFLT_1
FLTMK2		E17_NCFLT_7, E17_NCFLT_2
FLTMK3		E17_NCFLT_7, E17_NCFLT_3
FLTMK4		E17_NCFLT_7, E17_NCFLT_4
FLTMK5		E17_NCFLT_7, E17_NCFLT_5
FLTMK6		E17_NCFLT_7, E17_NCFLT_6
FLTSEC	FLTTIMER	PAGE1
FLTSEC_SCL_PV	FLTTIMER	PAGE1
FLTSTOP	OPMODE_1	PAGE1, OPMODE_2
FLTSTRT	OPMODE_1	PAGE1, OPMODE_2, OPMODE_3
FLTTAB		PAGE0, CMPACT, E05_RAMINI, E14_FAULTSTORE_1, E14_FAULTSTORE_5, E14_FAULTSTORE_6, INIFLT, O10_ARO240, O17_ARO350
FLTTC	FLTTIMER	PAGE1, INC_CTRS_3
FLTTF	FLTTIMER	PAGE1, INC_CTRS_3
FLTTIMER		FLTTIMER, E08_FG_SCHEDULE
FLTWRO	I30_DISINFALT	PAGE0, E13_WATEST_1, E16_SMTST, I38_SMCVS_2
FLT_A1_TBL		E18_FAULTSUM_5, E18_FAULTSUM_2
FLT_A2_TBL		E18_FAULTSUM_5, E18_FAULTSUM_3
FLT_CONFIG_P	FLTTIMER	PAGE1
FLT_CW_TBL		E18_FAULTSUM_5, E18_FAULTSUM_4
FLT_HW_TBL		E18_FAULTSUM_5, E18_FAULTSUM_1
FLT_MS_RAW	FLTTIMER	PAGE1
FLT_TIMER_RESET	FLTTIMER, INC_CTRS_3	PAGE1
FOREX1		E08_FG_SCHEDULE, E08_FG_SECSL
FORRTN	E08_FG_SCHEDULE	PAGE0
FSW_CPDATA_READ		DCU_CONSTANTS, DCU_COMPONT_DATA

Parameter	SET BY	USED BY
FSW_CTRDATA_READ		DCU_CONSTANTS,DCU_COUNTER_DATA
FSW_CTR_EEC_FLT		DCU_CONSTANTS,DCU_COUNTER_DATA
FSW_DCUID_READ		DCU_CONSTANTS,DCU_DCUID_DATA
FSW_DCU_INACTIVE		DCU_CONSTANTS,E05_RAMINI,UART_IO_SELECT
FSW_EECDATA_READ		DCU_CONSTANTS,DCU_EEC_DATA
FSW_ENGDATA_READ		DCU_CONSTANTS,DCU_ENGINE_DATA
FSW_INCONSISTANT		DCU_CONSTANTS,DCU_COMPONT_DATA
FSW_LCFDATA_READ		DCU_CONSTANTS,DCU_LCF_DATA
FWINDX	E26_FAULTSTORE_2	PAGE0
FWORD	E14_FAULTSTORE_3,INIFLT	PAGE0
FWRD26	E26_FAULTSTORE_2,E26_FAULTSTORE_3	PAGE0
FW_LATCH		DCU_CONSTANTS,DCU_COMPONT_DATA,DCU_COUNTER_DATA, DCU_DCUID_DATA,DCU_EEC_DATA,DCU_ENGINE_DATA,DCU_LCF_DATA, DCU_SCHEDULE,E05_RAMINI,INI_DCU_DATA,LCF_CHECK
FW_MAX_COUNTER		DCU_CONSTANTS,DCU_COMPONT_DATA,DCU_COUNTER_DATA, DCU_DCUID_DATA,DCU_EEC_DATA,DCU_ENGINE_DATA,DCU_LCF_DATA, DCU_SCHEDULE,INI_DCU_DATA
GAINC	E05_RAMINI,I10_GAINCOM_2	PAGE0,I11_TQBTRIM_2,I12_TQGTTRIM_2,I13_ENGVARIANT_2
GAINIF	E05_RAMINI,I09_HLDRIFT_2,I10_GAINCOM_2	PAGE0
GARRAY	E22_GASEL	PAGE0,O02_HBOVOUT
GETWRD		GETWRD,DUMP,GOTO,INA,LOAD,OTA,READ,WRITE,WRTEEP
GNTCT		PAGE0,I10_GAINCOM_2
GOTO		GOTO,E20_TESTMODE
HBDTCT	E28_HBOVWA_1	PAGE0
HBOVIF	I19_NHDUAL_1	PAGE0
HBRTCT	E28_HBOVWA_2	PAGE0
HBRTCT	I45_HBOVWA_2	PAGE0
HBVBDF	C26_MPARAM_1	PAGE1,C22_HBOVMM_3
HBVDE	C21_HBOVEM_2	PAGE1
HBVNDP	C26_MPARAM_1	PAGE1,C21_HBOVEM_2
HBVPDF	C26_MPARAM_1	PAGE1,C22_HBOVMM_1
HBVSS	C22_HBOVMM_3	PAGE1
HBWA1C	E28_HBOVWA_1	PAGE0
HBWA2C	E28_HBOVWA_2	PAGE0
HBWAIF	E22_GASEL,E28_HBOVWA_1,E28_HBOVWA_2,I45_HBOVWA_2	PAGE0,E17_NCFILT_6
HIPTX	SUNVAR	PAGE0
HIPTY		PAGE0
HITEMP	I08_RDT18_3	PAGE0
HOTMDI	C26_MPARAM_1	PAGE1,C27_NHMMNMX_1
HOTMOD	C27_NHMMNMX_1	PAGE1,C21_HBOVEM_2
HPDMP	C11_SHPREQ_2	PAGE1
HPDOT	C26_MPARAM_5	PAGE1
HPERR	C08_SHPCL	PAGE1
HPFB	C26_MPARAM_5	PAGE1,C08_SHPCL,MISSIONS,RTDWRITE
HPFBC	C26_MPARAM_5	PAGE1,C08_SHPCL
HPIMP_ACF		PAGE1,LCF_TABLE
HPIMP_ACF_ADR		LCF_TABLE
HPIMP_CYCCRS		PAGE1,DCU_CONSTANTS,LCF_CHECK,O40_ARO06X_TABLE
HPIMP_CYCFIN		PAGE1,LCF_CHECK
HPIMP_FCFM1		PAGE1,LCF_TABLE
HPIMP_FCFM1_ADR		LCF_TABLE
HPIMP_FCFM2		PAGE1,LCF_TABLE
HPIMP_FCFM2_ADR		LCF_TABLE
HPIMP_FCFM3		PAGE1,LCF_TABLE
HPIMP_FCFM3_ADR		LCF_TABLE
HPIMP_FCFM4		PAGE1,LCF_TABLE
HPIMP_FCFM4_ADR		LCF_TABLE
HPIMP_FCFM5		PAGE1,LCF_TABLE
HPIMP_FCFM5_ADR		LCF_TABLE
HPIMP_FCFM6		PAGE1,LCF_TABLE
HPIMP_FCFM6_ADR		LCF_TABLE
HPIMP_FCFMAX	LCF_CHECK	PAGE1
HPIMP_PNA12		PAGE1,DCU_CONSTANTS,O40_ARO06X_TABLE
HPIMP_PNA12EE		PAGE11
HPIMP_PNA34		PAGE1,O40_ARO06X_TABLE
HPIMP_PNA34EE		PAGE11
HPIMP_PNA56		PAGE1,O40_ARO06X_TABLE
HPIMP_PNA56EE		PAGE11
HPIMP_PNA78		PAGE1,O40_ARO06X_TABLE
HPIMP_PNA78EE		PAGE11
HPIMP_SNA12		PAGE1,O40_ARO06X_TABLE

Parameter	SET BY	USED BY
HPIMP_SNA12EE		PAGE11
HPIMP_SNA34		PAGE1, O40_ARO06X_TABLE
HPIMP_SNA34EE		PAGE11
HPIMP_SNA56		PAGE1, O40_ARO06X_TABLE
HPIMP_SNA56EE		PAGE11
HPKI	C08_SHPCL	PAGE1, C17_NHREQUEST_2
HPKIC	C08_SHPCL	PAGE0
HPKP	C08_SHPCL	PAGE1, C17_NHREQUEST_2
HPKPC1	C08_SHPCL	PAGE0
HPKPC2	C08_SHPCL	PAGE0
HPKPC3	C08_SHPCL	PAGE0
HPREF	C07_OVRTQPRO	PAGE1, C11_SHPREQ_1, C11_SHPREQ_2, C31_TQUM
HPREFZ	C11_SHPREQ_1	PAGE1
HPREQ	C11_SHPREQ_2	PAGE1, C08_SHPCL, RTDWRITE
HPTD_ACF		PAGE1, LCF_TABLE
HPTD_ACF_ADR		LCF_TABLE
HPTD_CYCCRS		PAGE1, DCU_CONSTANTS, LCF_CHECK, O40_ARO06X_TABLE
HPTD_CYCFIN		PAGE1, LCF_CHECK
HPTD_FCFM1		PAGE1, LCF_TABLE
HPTD_FCFM1_ADR		LCF_TABLE
HPTD_FCFM2		PAGE1, LCF_TABLE
HPTD_FCFM2_ADR		LCF_TABLE
HPTD_FCFM3		PAGE1, LCF_TABLE
HPTD_FCFM3_ADR		LCF_TABLE
HPTD_FCFM4		PAGE1, LCF_TABLE
HPTD_FCFM4_ADR		LCF_TABLE
HPTD_FCFM5		PAGE1, LCF_TABLE
HPTD_FCFM5_ADR		LCF_TABLE
HPTD_FCFM6		PAGE1, LCF_TABLE
HPTD_FCFM6_ADR		LCF_TABLE
HPTD_FCFMAX	LCF_CHECK	PAGE1
HPTD_PNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
HPTD_PNA12EE		PAGE11
HPTD_PNA34		PAGE1, O40_ARO06X_TABLE
HPTD_PNA34EE		PAGE11
HPTD_PNA56		PAGE1, O40_ARO06X_TABLE
HPTD_PNA56EE		PAGE11
HPTD_PNA78		PAGE1, O40_ARO06X_TABLE
HPTD_PNA78EE		PAGE11
HPTD_SNA12		PAGE1, O40_ARO06X_TABLE
HPTD_SNA12EE		PAGE11
HPTD_SNA34		PAGE1, O40_ARO06X_TABLE
HPTD_SNA34EE		PAGE11
HPTD_SNA56		PAGE1, O40_ARO06X_TABLE
HPTD_SNA56EE		PAGE11
HPTFCP_ACF		PAGE1, LCF_TABLE
HPTFCP_ACF_ADR		LCF_TABLE
HPTFCP_CYCCRS		PAGE1, DCU_CONSTANTS, LCF_CHECK, O40_ARO06X_TABLE
HPTFCP_CYCFIN		PAGE1, LCF_CHECK
HPTFCP_FCFM1		PAGE1, LCF_TABLE
HPTFCP_FCFM1_ADR		LCF_TABLE
HPTFCP_FCFM2		PAGE1, LCF_TABLE
HPTFCP_FCFM2_ADR		LCF_TABLE
HPTFCP_FCFM3		PAGE1, LCF_TABLE
HPTFCP_FCFM3_ADR		LCF_TABLE
HPTFCP_FCFM4		PAGE1, LCF_TABLE
HPTFCP_FCFM4_ADR		LCF_TABLE
HPTFCP_FCFM5		PAGE1, LCF_TABLE
HPTFCP_FCFM5_ADR		LCF_TABLE
HPTFCP_FCFM6		PAGE1, LCF_TABLE
HPTFCP_FCFM6_ADR		LCF_TABLE
HPTFCP_FCFMAX	LCF_CHECK	PAGE1
HPTFCP_PNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
HPTFCP_PNA12EE		PAGE11
HPTFCP_PNA34		PAGE1, O40_ARO06X_TABLE
HPTFCP_PNA34EE		PAGE11
HPTFCP_PNA56		PAGE1, O40_ARO06X_TABLE
HPTFCP_PNA56EE		PAGE11
HPTFCP_PNA78		PAGE1, O40_ARO06X_TABLE
HPTFCP_PNA78EE		PAGE11
HPTFCP_SNA12		PAGE1, O40_ARO06X_TABLE
HPTFCP_SNA12EE		PAGE11

Parameter	SET BY	USED BY
HPTFCP_SNA34		PAGE1, O40_ARO06X_TABLE
HPTFCP_SNA34EE		PAGE11
HPTFCP_SNA56		PAGE1, O40_ARO06X_TABLE
HPTFCP_SNA56EE		PAGE11
HPTRCP_ACF		PAGE1, LCF_TABLE
HPTRCP_ACF_ADR		LCF_TABLE
HPTRCP_CYCCRS		PAGE1, DCU_CONSTANTS, LCF_CHECK, O40_ARO06X_TABLE
HPTRCP_CYCFIN		PAGE1, LCF_CHECK
HPTRCP_FCFM1		PAGE1, LCF_TABLE
HPTRCP_FCFM1_ADR		LCF_TABLE
HPTRCP_FCFM2		PAGE1, LCF_TABLE
HPTRCP_FCFM2_ADR		LCF_TABLE
HPTRCP_FCFM3		PAGE1, LCF_TABLE
HPTRCP_FCFM3_ADR		LCF_TABLE
HPTRCP_FCFM4		PAGE1, LCF_TABLE
HPTRCP_FCFM4_ADR		LCF_TABLE
HPTRCP_FCFM5		PAGE1, LCF_TABLE
HPTRCP_FCFM5_ADR		LCF_TABLE
HPTRCP_FCFM6		PAGE1, LCF_TABLE
HPTRCP_FCFM6_ADR		LCF_TABLE
HPTRCP_FCFMAX	LCF_CHECK	PAGE1
HPTRCP_PNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
HPTRCP_PNA12EE		PAGE11
HPTRCP_PNA34		PAGE1, O40_ARO06X_TABLE
HPTRCP_PNA34EE		PAGE11
HPTRCP_PNA56		PAGE1, O40_ARO06X_TABLE
HPTRCP_PNA56EE		PAGE11
HPTRCP_PNA78		PAGE1, O40_ARO06X_TABLE
HPTRCP_PNA78EE		PAGE11
HPTRCP_SNA12		PAGE1, O40_ARO06X_TABLE
HPTRCP_SNA12EE		PAGE11
HPTRCP_SNA34		PAGE1, O40_ARO06X_TABLE
HPTRCP_SNA34EE		PAGE11
HPTRCP_SNA56		PAGE1, O40_ARO06X_TABLE
HPTRCP_SNA56EE		PAGE11
HPZM1	C26_MPARAM_5	PAGE1
HPZM2	C26_MPARAM_5	PAGE1
HPZM3	C26_MPARAM_5	PAGE1
HPZM4	C26_MPARAM_5	PAGE1
HS1TCT		PAGE0, E13_WATEST_1
HS2TCT		PAGE0, E13_WATEST_2
HSS1IF		PAGE0, E13_WATEST_1
HSS2IF		PAGE0, E13_WATEST_2
I01_LLDRIFT_1		I01_LLDRIFT_1, E08_FG_STATETABLE
I01_LLDRIFT_2		I01_LLDRIFT_2, E08_FG_SCHEDULE
I02_SGCOM_1		I02_SGCOM_1, E08_FG_STATETABLE
I02_SGCOM_2		I02_SGCOM_2, E08_FG_SCHEDULE
I03_RDCJT_1		I03_RDCJT_1, E08_FG_STATETABLE
I03_RDCJT_2		I03_RDCJT_2, E08_FG_SCHEDULE
I03_RDCJT_3		I03_RDCJT_3, E08_FG_SCHEDULE
I04_PSAMBSG_1		I04_PSAMBSG_1, E08_FG_STATETABLE
I04_PSAMBSG_2		I04_PSAMBSG_2, E08_FG_SCHEDULE
I04_PSAMBSG_3		I04_PSAMBSG_3, E08_FG_SCHEDULE
I04_PSAMBSG_4		I04_PSAMBSG_4, E08_FG_SCHEDULE
I04_PSAMBSG_5		I04_PSAMBSG_5, E08_FG_SCHEDULE
I05_DELTAPSG_1		I05_DELTAPSG_1, E08_FG_STATETABLE
I05_DELTAPSG_2		I05_DELTAPSG_2, E08_FG_SCHEDULE
I05_DELTAPSG_3		I05_DELTAPSG_3, E08_FG_SCHEDULE
I05_DELTAPSG_4		I05_DELTAPSG_4, E08_FG_SCHEDULE
I07_READTQ_1		I07_READTQ_1, E08_FG_STATETABLE
I07_READTQ_2		I07_READTQ_2, E08_FG_SCHEDULE
I07_READTQ_3		I07_READTQ_3, E08_FG_SCHEDULE
I08_RDT18_1		I08_RDT18_1, E08_FG_STATETABLE
I08_RDT18_2		I08_RDT18_2, E08_FG_SCHEDULE
I08_RDT18_3		I08_RDT18_3, E08_FG_SCHEDULE
I08_RDT18_4		I08_RDT18_4, E08_FG_SCHEDULE
I09_HLDRIFT_1		I09_HLDRIFT_1, E08_FG_STATETABLE
I09_HLDRIFT_2		I09_HLDRIFT_2, E08_FG_SCHEDULE
I10_GAINCOM_1		I10_GAINCOM_1, E08_FG_STATETABLE
I10_GAINCOM_2		I10_GAINCOM_2, E08_FG_SCHEDULE
I11_TQ8TRIM_1		I11_TQ8TRIM_1, E08_FG_STATETABLE
I11_TQ8TRIM_2		I11_TQ8TRIM_2, E08_FG_SCHEDULE

Parameter	SET BY	USED BY
I12 TQGRIM 1		I12 TQGRIM 1, E08 FG STATETABLE
I12 TQGRIM 2		I12 TQGRIM 2, E08 FG SCHEDULE
I13 ENGVARIENT 1		I13 ENGVARIENT 1, E08 FG STATETABLE
I13 ENGVARIENT 2		I13 ENGVARIENT 2, E08 FG SCHEDULE
I14 RDPLA 1		I14 RDPLA 1, E08 FG STATETABLE
I14 RDPLA 2		I14 RDPLA 2, E08 FG STATETABLE
I14 RDPLA 3		I14 RDPLA 3, E08 FG SCHEDULE
I14 RDPLA 4		I14 RDPLA 4, E08 FG SCHEDULE
I15 PLABIAS 1		I15 PLABIAS 1, E08 FG STATETABLE
I15 PLABIAS 2		I15 PLABIAS 2, E08 FG SCHEDULE
I16 PLAGAIN 1		I16 PLAGAIN 1, E08 FG STATETABLE
I16 PLAGAIN 2		I16 PLAGAIN 2, E08 FG SCHEDULE
I17 PNHCNV 1		I17 PNHCNV 1, E08 FG STATETABLE
I17 PNHCNV 2		I17 PNHCNV 2, E08 FG STATETABLE
I18 READNH		I18 READNH, E08 FG STATETABLE
I19 NHDUAL 1		I19 NHDUAL 1, E08 FG SCHEDULE
I19 NHDUAL 2		I19 NHDUAL 2, E08 FG SCHEDULE
I20 NHSPEED		I20 NHSPEED, E08 FG SCHEDULE
I22 SNHCNV 1		I22 SNHCNV 1, E08 FG STATETABLE
I22 SNHCNV 2		I22 SNHCNV 2, E08 FG STATETABLE
I23 RDSNHP		I23 RDSNHP, E08 FG STATETABLE
I24 NHPROBE2		I24 NHPROBE2, E08 FG SCHEDULE
I25 TQNPT 1		I25 TQNPT 1, E08 FG SCHEDULE
I25 TQNPT 2		I25 TQNPT 2
I25 TQNPT 3		I25 TQNPT 3
I25 TQNPT 4		I25 TQNPT 4
I25 TQNPT 5		I25 TQNPT 5
I25 TQNPT 6		I25 TQNPT 6
I25 TQNPT 7		I25 TQNPT 7, E08 FG SCHEDULE
I26 TQCAL		I26 TQCAL, E08 FG SCHEDULE
I27 TQSPEED		I27 TQSPEED, E08 FG SCHEDULE
I28 DISINW1		I28 DISINW1, E08 FG STATETABLE
I29 DISINW2		I29 DISINW2, E08 FG STATETABLE
I30 DISINFALT		I30 DISINFALT, E08 FG STATETABLE
I31 UARTIN		I31 UARTIN, E08 FG STATETABLE
I31 UARTIN_NPM		I31 UARTIN_NPM, E08 FG EXECUTIVE
I32 ARINCIN 1		I32 ARINCIN 1
I32 ARINCIN 2		I32 ARINCIN 2, E08 FG STATETABLE
I33 ARITAT		I33 ARITAT, E08 FG SCHEDULE
I34 TATSEL		I34 TATSEL, E08 FG SCHEDULE
I35 ARIPA		I35 ARIPA, E08 FG SCHEDULE
I36 PAMSEL		I36 PAMSEL, E08 FG SCHEDULE
I37 ARIISA 1		I37 ARIISA 1, E08 FG SCHEDULE
I37 ARIISA 2		I37 ARIISA 2, E08 FG SCHEDULE
I38 SMCSV 1		I38 SMCSV 1, E08 FG STATETABLE, E09 SMSUMV
I38 SMCSV 2		I38 SMCSV 2, E08 FG STATETABLE
I43 TQSPEED 1		I43 TQSPEED 1, E08 FG SCHEDULE
I43 TQSPEED 2		I43 TQSPEED 2, E08 FG SCHEDULE
I43 TQSPEED 3		I43 TQSPEED 3, E08 FG SCHEDULE
I45 HBOVWA 1		I45 HBOVWA 1, E08 FG STATETABLE
I45 HBOVWA 2		I45 HBOVWA 2, E08 FG SCHEDULE
I46 IASFAULT		I46 IASFAULT, E08 FG SCHEDULE
IADPF	OPMODE 1	PAGE1
IAS	I37 ARIISA 2	PAGE0, C26 MPARAM 3
IAS206	I37 ARIISA 1	PAGE0
IASDAT	I37 ARIISA 1	PAGE0, C26 MPARAM 1
IASDP	I05 DELTAPSG 3	PAGE0
IASFLG	I37 ARIISA 1	PAGE0, I37 ARIISA 2
IASIF	I46 IASFAULT	PAGE0
IASLM	C26 MPARAM 3	PAGE0
IAS5	C26 MPARAM 3	PAGE0
IAS5F	C26 MPARAM 1	PAGE0, C26 MPARAM 3, OPMODE 1
IASTIM	I37 ARIISA 1	PAGE0
IBLED2	C21 HBOVEM 2	PAGE1
IBLED3	C21 HBOVEM 2	PAGE0
IBLEDE	C21 HBOVEM 2	PAGE1
IBLEDM	C22 HBOVMM 3	PAGE0, C21 HBOVEM 2
IBLEDS	C21 HBOVEM 2	PAGE1
IBLEED	C21 HBOVEM 2	PAGE0, E28_HBOVWA_2, O02_HBOVOUT, RTDWRITE

Parameter	SET BY	USED BY
IC	E05_RAMINI, E15_BCKGNDTEST	PAGE0, C01_START, C05_TQBUG, C06_HPREF, C06_HPREF_1, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C17_NHREQUEST_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVM_1, C21_HBOVM_2, C22_HBOVM_1, C22_HBOVM_2, C22_HBOVM_3, C27_NHMMX_1, C31_TQIM, DERLAG, E14_FAULTSTORE_1, E21_RECONFIG, E26_FAULTSTORE_1, E26_FAULTSTORE_2, E27_MFCID, FILTER, I05_DELTAPSG_3, I07_READTQ_2, I11_TQ8TRIM_2, I12_TQ8TRIM_2, I13_ENGVARIANT_2, I14_RDPLA_1, I14_RDPLA_2, I14_RDPLA_3, I18_READNH, I23_RDSNHP, I25_TQNPT_4, I25_TQNPT_5, I25_TQNPT_6, I25_TQNPT_7, I28_DISINW1, I34_TATSEL, I36_PAMSEL, I37_ARISA_1, I37_ARISA_2, MISSIONS, O03_DISOUT, OPMODE_1, OPMODE_2, OPMODE_3, QUAD, TUSTIN_SUBR, VAR_TUSTIN_SUBR
IGNITE	C10_ATOIGN	PAGE1, O03_DISOUT
ILOOP	C16_LOOPSEL	PAGE0, C10_ATOIGN, C17_NHREQUEST_2
INA		INA, E20_TESTMODE
INC_CTRS_1		INC_CTRS_1, E08_FG_SCHEDULE
INC_CTRS_2		INC_CTRS_2, E08_FG_SCHEDULE
INC_CTRS_3		INC_CTRS_3, E08_FG_SCHEDULE
INC_CTRS_4		INC_CTRS_4, E08_FG_SCHEDULE
INC_MTO	OPMODE_3	PAGE1, INC_CTRS_4
INC_NTO	OPMODE_3	PAGE1, INC_CTRS_4
INIFLT		INIFLT, E05_RAMINI
INIRAM		E05_RAMINI, E01_PORPSR
INITF	CMPACT, INIFLT	PAGE0
INI_DCUI_DATA		INI_DCUI_DATA, E05_RAMINI
INRAM1	INSTST	PAGE0
INRAM2	INSTST	PAGE0
INS1	INA, OTA	PAGE0
INS2	INA, OTA	PAGE0
INS3	INA, OTA	PAGE0
INSRET	INSTST	PAGE0
INSTST		INSTST, E04_INSTST, E07_BG_EXECUTIVE
INTIAS	I05_DELTAPSG_3	PAGE0
INTTQ		PAGE0, I07_READTQ_2
ISS_ACF		PAGE1, LCF_TABLE
ISS_ACF_ADR		LCF_TABLE
ISS_CYCCRS		PAGE1, DCU_CONSTANTS, LCF_CHECK, O40_ARO06X_TABLE
ISS_CYCFIN		PAGE1, LCF_CHECK
ISS_FCFM1		PAGE1, LCF_TABLE
ISS_FCFM1_ADR		LCF_TABLE
ISS_FCFM2		PAGE1, LCF_TABLE
ISS_FCFM2_ADR		LCF_TABLE
ISS_FCFM3		PAGE1, LCF_TABLE
ISS_FCFM3_ADR		LCF_TABLE
ISS_FCFM4		PAGE1, LCF_TABLE
ISS_FCFM4_ADR		LCF_TABLE
ISS_FCFM5		PAGE1, LCF_TABLE
ISS_FCFM5_ADR		LCF_TABLE
ISS_FCFM6		PAGE1, LCF_TABLE
ISS_FCFM6_ADR		LCF_TABLE
ISS_FCFMAX	LCF_CHECK	PAGE1
ISS_PNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
ISS_PNA12EE		PAGE11
ISS_PNA34		PAGE1, O40_ARO06X_TABLE
ISS_PNA34EE		PAGE11
ISS_PNA56		PAGE1, O40_ARO06X_TABLE
ISS_PNA56EE		PAGE11
ISS_PNA78		PAGE1, O40_ARO06X_TABLE
ISS_PNA78EE		PAGE11
ISS_SNA12		PAGE1, O40_ARO06X_TABLE
ISS_SNA12EE		PAGE11
ISS_SNA34		PAGE1, O40_ARO06X_TABLE
ISS_SNA34EE		PAGE11
ISS_SNA56		PAGE1, O40_ARO06X_TABLE
ISS_SNA56EE		PAGE11
KNH_ENG_OFF		PAGE1, DCU_CONSTANTS, OPMODE_1
KNH_ENG_OFF_MAX		DCU_CONSTANTS
KNH_ENG_OFF_MIN		DCU_CONSTANTS
KNH_ENG_RUN		PAGE1, DCU_CONSTANTS, OPMODE_1
KNH_ENG_RUN_MAX		DCU_CONSTANTS
KNH_ENG_RUN_MIN		DCU_CONSTANTS

Parameter	SET BY	USED BY
KSLID		PID, DCU_COUNTER_DATA
KSSVID		PID, DCU_COUNTER_DATA
KVK_FLTDPNSNR_MAX		DCU_CONSTANTS
KVK_FLTDPNSNR_MIN		DCU_CONSTANTS
KVK_FLT_ADC		PAGE1, DCU_CONSTANTS, OPMODE_1
KVK_FLT_ADC_MAX		DCU_CONSTANTS
KVK_FLT_ADC_MIN		DCU_CONSTANTS
KVK_FLT_DPSNR		PAGE1, DCU_CONSTANTS, OPMODE_1
LCF_CHECK		LCF_CHECK, E08_FG_SCHEDULE
LCF_CYCLE_COUNT		LCF_CYCLE_COUNT, LCF_CHECK
LCF_FCF_MAX_PTR	LCF_CYCLE_COUNT	PAGE0
LCF_FCF_MAX_TBL		PAGE1, LCF_CYCLE_COUNT
LCF_NUMER	LCF_CYCLE_COUNT	PAGE1
LCF_PN_LENGTH		DCU_CONSTANTS, DCU_LCF_DATA
LCF_PN_NOTFOUND	DCU_LCF_DATA	PAGE1
LCF_READ_FAULTS		DCU_CONSTANTS, DCU_LCF_DATA
LCF_VALUE1	LCF_CYCLE_COUNT	PAGE0
LCF_VALUE2	LCF_CYCLE_COUNT	PAGE0
LDLG_SUBR		LDLG_SUBR, C31_TQLIM
LIM0TM	C24_OVRTQTST	PAGE1
LIM1TM	C24_OVRTQTST	PAGE1
LIM2TM	C24_OVRTQTST	PAGE1
LIM3TM	C24_OVRTQTST	PAGE1
LIM4TM	C24_OVRTQTST	PAGE1
LIM5TM	C24_OVRTQTST	PAGE1
LIM6TM	C24_OVRTQTST	PAGE1
LIM7TM	C24_OVRTQTST	PAGE1
LIM8TM	C24_OVRTQTST	PAGE1
LIM9TM	C24_OVRTQTST	PAGE1
LIMFLO	C24_OVRTQTST	PAGE1
LIMFL1	C24_OVRTQTST	PAGE1
LIMFL2	C24_OVRTQTST	PAGE1
LIMFL3	C24_OVRTQTST	PAGE1
LIMFL4	C24_OVRTQTST	PAGE1
LIMFL5	C24_OVRTQTST	PAGE1
LIMFL6	C24_OVRTQTST	PAGE1
LIMFL7	C24_OVRTQTST	PAGE1
LIMFL8	C24_OVRTQTST	PAGE1
LIMFL9	C24_OVRTQTST	PAGE1
LOAD		LOAD, E20_TESTMODE
LOOKUP10		PAGE1
LOPTX	SUNVAR	PAGE0
LOPTY		PAGE0
LOWB	O04_UARTOUT_DTS, O04_UARTOUT_GDTS	PAGE0
LPIMP_ACF		PAGE1, LCF_TABLE
LPIMP_ACF_ADR		LCF_TABLE, LCF_CYCLE_COUNT
LPIMP_CYCCRS		PAGE1, DCU_CONSTANTS, LCF_CHECK, O40_ARO06X_TABLE
LPIMP_CYCFIN		PAGE1, LCF_CHECK
LPIMP_FCFM1		PAGE1, LCF_TABLE
LPIMP_FCFM1_ADR		LCF_TABLE, LCF_CYCLE_COUNT
LPIMP_FCFM2		PAGE1, LCF_TABLE
LPIMP_FCFM2_ADR		LCF_TABLE
LPIMP_FCFM3		PAGE1, LCF_TABLE
LPIMP_FCFM3_ADR		LCF_TABLE
LPIMP_FCFM4		PAGE1, LCF_TABLE
LPIMP_FCFM4_ADR		LCF_TABLE
LPIMP_FCFM5		PAGE1, LCF_TABLE
LPIMP_FCFM5_ADR		LCF_TABLE
LPIMP_FCFM6		PAGE1, LCF_TABLE
LPIMP_FCFM6_ADR		LCF_TABLE
LPIMP_FCFMAX	LCF_CHECK	PAGE1
LPIMP_PNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
LPIMP_PNA12EE		PAGE11
LPIMP_PNA34		PAGE1, O40_ARO06X_TABLE
LPIMP_PNA34EE		PAGE11
LPIMP_PNA56		PAGE1, O40_ARO06X_TABLE
LPIMP_PNA56EE		PAGE11
LPIMP_PNA78		PAGE1, O40_ARO06X_TABLE
LPIMP_PNA78EE		PAGE11
LPIMP_SNA12		PAGE1, O40_ARO06X_TABLE
LPIMP_SNA12EE		PAGE11
LPIMP_SNA34		PAGE1, O40_ARO06X_TABLE

Parameter	SET BY	USED BY
LPIMP_SNA34EE		PAGE11
LPIMP_SNA56		PAGE1, O40_ARO06X_TABLE
LPIMP_SNA56EE		PAGE11
LPTD_ACF		PAGE1, LCF_TABLE
LPTD_ACF_ADR		LCF_TABLE
LPTD_CYCCRS		PAGE1, DCU_CONSTANTS, LCF_CHECK, O40_ARO06X_TABLE
LPTD_CYCFIN		PAGE1, LCF_CHECK
LPTD_FCFM1		PAGE1, LCF_TABLE
LPTD_FCFM1_ADR		LCF_TABLE
LPTD_FCFM2		PAGE1, LCF_TABLE
LPTD_FCFM2_ADR		LCF_TABLE
LPTD_FCFM3		PAGE1, LCF_TABLE
LPTD_FCFM3_ADR		LCF_TABLE
LPTD_FCFM4		PAGE1, LCF_TABLE
LPTD_FCFM4_ADR		LCF_TABLE
LPTD_FCFM5		PAGE1, LCF_TABLE
LPTD_FCFM5_ADR		LCF_TABLE
LPTD_FCFM6		PAGE1, LCF_TABLE
LPTD_FCFM6_ADR		LCF_TABLE
LPTD_FCFMAX	LCF_CHECK	PAGE1
LPTD_PNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
LPTD_PNA12EE		PAGE11
LPTD_PNA34		PAGE1, O40_ARO06X_TABLE
LPTD_PNA34EE		PAGE11
LPTD_PNA56		PAGE1, O40_ARO06X_TABLE
LPTD_PNA56EE		PAGE11
LPTD_PNA78		PAGE1, O40_ARO06X_TABLE
LPTD_PNA78EE		PAGE11
LPTD_SNA12		PAGE1, O40_ARO06X_TABLE
LPTD_SNA12EE		PAGE11
LPTD_SNA34		PAGE1, O40_ARO06X_TABLE
LPTD_SNA34EE		PAGE11
LPTD_SNA56		PAGE1, O40_ARO06X_TABLE
LPTD_SNA56EE		PAGE11
LRINDX	E26_FAULTSTORE_2	PAGE0, E26_FAULTSTORE_4
LRUCDE	E14_FAULTSTORE_3, INIFLT	PAGE0, E14_FAULTSTORE_4
LRUCDI	E26_FAULTSTORE_2, E26_FAULTSTORE_3	PAGE0, E26_FAULTSTORE_4
LRUTB26		E26_FAULTSTORE_5, E26_FAULTSTORE_3
LRUTBL		E14_FAULTSTORE_8, INIFLT
LS1TCT		PAGE0, E13_WATEST_3
LS2TCT		PAGE0, E13_WATEST_4
LS3TCT		PAGE0, E13_WATEST_5
LS4TCT		PAGE0, E13_WATEST_6
LS5TCT		PAGE0, E13_WATEST_7
LSS1IF		PAGE0, E13_WATEST_3
LSS2IF		PAGE0, E13_WATEST_4
LSS2OUT	O03_DISOUT_A	PAGE1, O03_DISOUT
LSS3IF		PAGE0, E13_WATEST_5
LSS3OUT	O03_DISOUT_A	PAGE1, O03_DISOUT
LSS4IF		PAGE0, E13_WATEST_6
LSS5IF		PAGE0, E13_WATEST_7
LTP	I04_PSAMBSG_3, I05_DELTAPSG_2	PAGE0
MACH	C26_MPARAM_3	PAGE0, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_3_1, C03_TLMRT_4_1, C03_TLMRT_5_1, C03_TLMRT_6_1, C26_MPARAM_2
MASK1	E14_FAULTSTORE_3, INIFLT	PAGE0
MASKI1	E26_FAULTSTORE_2, E26_FAULTSTORE_3	PAGE0
MASLMF	C22_HBOVMM_2	PAGE0, C22_HBOVMM_1
MAXRAT	C01_START	PAGE0, O01_SMOUT
MCLBM	C03_TLMRT_3_2	PAGE1
MCLRGL	C06_HPREF	PAGE1, C05_TQBUG
MCLRTL	C06_HPREF	PAGE1
MCLRTP	C04_RPCAL_2_D	PAGE1, C06_HPREF
MCLTB	C03_TLMRT_3_1	PAGE1, C03_TLMRT_3_2
MCLTC	C03_TLMRT_3_2	PAGE1, C03_TLMRT_7
MCLTL	C03_TLMRT_7	PAGE1, C04_RPCAL_2_D
MCLTQB	C05_TQBUG	PAGE1, O27_ARO342_A
MCRBM	C03_TLMRT_2_2	PAGE1
MCCRGL	C06_HPREF	PAGE1, C05_TQBUG
MCRRTL	C06_HPREF	PAGE1
MCRRTTP	C04_RPCAL_2_D	PAGE1, C06_HPREF
MCRTB	C03_TLMRT_2_1	PAGE1, C03_TLMRT_2_2
MCRTC	C03_TLMRT_2_2	PAGE1, C03_TLMRT_7

Parameter	SET BY	USED BY
MCRTL	C03_TLMRT_7	PAGE1, C04_RPCAL_2_D
MCRTQB	C05_TQBUG	PAGE1, O27_ARO342_A
MCTBM	C03_TLMRT_4_2	PAGE1
MCTRGL	C06_HPREF	PAGE1, C05_TQBUG
MCTRTP	C04_RPCAL_2_D	PAGE1, C06_HPREF
MCTSHPD	C03_TLMRT_7	PAGE1
MCTTB	C03_TLMRT_4_1	PAGE1, C03_TLMRT_4_2
MCTTC	C03_TLMRT_4_2	PAGE1, C03_TLMRT_7
MCTTL	C03_TLMRT_7	PAGE1, C04_RPCAL_2_D
MDSLMF	C22_HBOVMM_2	PAGE0, C22_HBOVMM_1, C22_HBOVMM_3
MFCIF		PAGE0, E27_MFCID
MFCINT	C10_ATOIGN	PAGE1, C21_HBOVEM_2
MIDPTX	SUNVAR	PAGE0
MIDPTY		PAGE0
MINCYC	E08_FG_EXECUTIVE	PAGE0, E08_FG_SCHEDULE
MISSION	MISSIONS	PAGE1, LCF_CYCLE_COUNT
MISSIONS		MISSIONS, E08_FG_SCHEDULE
MNTZ1	C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_3_1, C03_TLMRT_4_1, C03_TLMRT_5_1, C03_TLMRT_6_1	PAGE1
MNTZ2	C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_3_1, C03_TLMRT_4_1, C03_TLMRT_5_1, C03_TLMRT_6_1	PAGE1
MODDCUDATA		PAGE1, DCU_EEC_DATA
MODE	I13_ENGVARIANT_2	PAGE1, I36_PAMSEL
MOST	I32_ARINCIN_1	PAGE0
MRSMLF	C22_HBOVMM_2	PAGE0, C22_HBOVMM_1, C22_HBOVMM_3
MSNFLG	MISSIONS	PAGE1
MSNFLG1	MISSIONS	PAGE1
MSNFLG2	MISSIONS	PAGE1
MSTATE	I31_UARTIN_NPM	PAGE0
MTDELO	C06_HPREF_1	PAGE1
MTLHI	C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_3_1, C03_TLMRT_4_1, C03_TLMRT_5_1, C03_TLMRT_6_1	PAGE1
MTLLO	C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_3_1, C03_TLMRT_4_1, C03_TLMRT_5_1, C03_TLMRT_6_1	PAGE1
MTOBM	C03_TLMRT_5_2	PAGE1
MTODRL	C06_HPREF	PAGE1, C05_TQBUG
MTODRTP	C04_RPCAL_2_D	PAGE1, C06_HPREF
MTORGL	C06_HPREF	PAGE1, C05_TQBUG
MTORTL	C06_HPREF	PAGE1
MTORTP	C04_RPCAL_2_D	PAGE1, C06_HPREF, MISSIONS
MTOT9	C04_RPCAL_2_D	PAGE1
MTOTB	C03_TLMRT_5_1	PAGE1, C03_TLMRT_5_2
MTOTC	C03_TLMRT_5_2	PAGE1, C03_TLMRT_7
MTOTL	C03_TLMRT_7	PAGE1, C04_RPCAL_2_D
MTOTQB	C05_TQBUG	PAGE1, O27_ARO342_A
MTO_DL1	INC_CTRS_4	PAGE1, DCU_CONSTANTS
MTO_DL2	INC_CTRS_4	PAGE1, DCU_CONSTANTS
MTO_DL3	INC_CTRS_4	PAGE1, DCU_CONSTANTS
MTUBM	C03_TLMRT_6_2	PAGE1
MTUT9	C04_RPCAL_2_D	PAGE1
MTUTB	C03_TLMRT_6_1	PAGE1, C03_TLMRT_6_2
MTUTC	C03_TLMRT_6_2	PAGE1, C03_TLMRT_7
MTUTL	C03_TLMRT_7	PAGE1, C04_RPCAL_2_D
N	E14_FAULTSTORE_3	PAGE0
NCFAL1	E17_NCFLT_1	PAGE0, E11_CONTEST, E14_FAULTSTORE_3, E18_FAULTSUM_1, E18_FAULTSUM_2, E18_FAULTSUM_3, E18_FAULTSUM_4, E26_FAULTSTORE_2, E26_FAULTSTORE_3, INIFLT
NCFAL2	E17_NCFLT_2	PAGE0, E11_CONTEST
NCFAL3	E17_NCFLT_3	PAGE0
NCFAL4	E17_NCFLT_4	PAGE0
NCFAL5	E17_NCFLT_5	PAGE0
NCFAL6	E17_NCFLT_6	PAGE0
NCFL1	E14_FAULTSTORE_5	PAGE0, E14_FAULTSTORE_3, INIFLT
NCFL2	E14_FAULTSTORE_5	PAGE0
NCFL3	E14_FAULTSTORE_5	PAGE0
NCFL4	E14_FAULTSTORE_5	PAGE0
NCFL5	E14_FAULTSTORE_5	PAGE0
NCFL6	E14_FAULTSTORE_5	PAGE0
NCFLI1		PAGE0, E26_FAULTSTORE_2, E26_FAULTSTORE_3
NCFLI2		PAGE0
NCFLI3		PAGE0
NCFLI4		PAGE0

Parameter	SET BY	USED BY
NCFLJ5		PAGE0
NCFLJ6		PAGE0
NCFLT1	E17_NCFLT_1	PAGE0
NCFLT2	E17_NCFLT_2	PAGE0
NCFLT3	E17_NCFLT_3	PAGE0
NCFLT4	E17_NCFLT_4	PAGE0
NCFLT5	E17_NCFLT_5	PAGE0
NCFLT6	E17_NCFLT_6	PAGE0
NCFTB26		E26_FAULTSTORE_5, E26_FAULTSTORE_2
NCFTBL		E14_FAULTSTORE_8, E14_FAULTSTORE_3
NCMSK1	E17_NCFLT_1	PAGE0
NCMSK2	E17_NCFLT_2	PAGE0
NCMSK3	E17_NCFLT_3	PAGE0
NCMSK4	E17_NCFLT_4	PAGE0
NCMSK5	E17_NCFLT_5	PAGE0
NCMSK6	E17_NCFLT_6	PAGE0
NGOOD1	I25_TQNPT_2, I25_TQNPT_4, I25_TQNPT_5	PAGE0
NGOOD2	I25_TQNPT_3, I25_TQNPT_4, I25_TQNPT_5	PAGE0
NH	I20_NHSPEED	PAGE0, C02_PLATIM_1, C10_ATOIGN, C13_NHACCELLUM, C14_NHLM, C15_NHDECELLUM, C17_NHREQUEST_1, C17_NHREQUEST_2, C19_UNDESPED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_2, C22_HBOVMM_3, C26_MPARAM_6, E14_FAULTSTORE_2, E27_MFCID, I08_RDT18_3, I19_NHDUAL_2, I25_TQNPT_2, I25_TQNPT_3, I34_TATSEL, I36_PAMSEL, I37_ARIISA_1, O11_ARO245, OPMODE_1, RTDWRITE
NH1GD	I18_READNH	PAGE0
NH1IF	I17_PNHCVN_1, I18_READNH	PAGE0, I19_NHDUAL_1, I19_NHDUAL_2
NH1STA	I18_READNH	PAGE0
NH2	I24_NHPROBE2	PAGE0, I19_NHDUAL_2
NH2GD	I23_RDSNHP	PAGE0
NH2IF	I17_PNHCVN_1, I18_READNH, I22_SNHCNV_1, I23_RDSNHP	PAGE0, I19_NHDUAL_1, I19_NHDUAL_2
NH2STA	I23_RDSNHP	PAGE0
NHBLF	C27_NHMMNMX_1	PAGE1
NHC	C26_MPARAM_6	PAGE0, C12_NPTGOV_1, C18_NHGOV_1
NHDOT	C26_MPARAM_6	PAGE0, C10_ATOIGN, C22_HBOVMM_1, C22_HBOVMM_2
NHDOTL	C21_HBOVEM_1	PAGE0
NHDOTM	C22_HBOVMM_2	PAGE0
NHECS1	I18_READNH	PAGE0, I17_PNHCVN_1, I19_NHDUAL_1, I19_NHDUAL_2
NHECS2	I18_READNH, I23_RDSNHP	PAGE0, I17_PNHCVN_1, I19_NHDUAL_1, I19_NHDUAL_2, I22_SNHCNV_1
NHERR	C17_NHREQUEST_1	PAGE0
NHERRL	C17_NHREQUEST_1	PAGE0
NHF	C19_UNDESPED	PAGE1
NHFB	C17_NHREQUEST_1	PAGE0, C12_NPTGOV_1, C18_NHGOV_2, C26_MPARAM_7
NHFBOT	C26_MPARAM_7	PAGE0, C12_NPTGOV_1
NHFBL	C17_NHREQUEST_1	PAGE0, C18_NHGOV_2
NHFZM1	C26_MPARAM_7	PAGE0
NHFZM2	C26_MPARAM_7	PAGE0
NHFZM3	C26_MPARAM_7	PAGE0
NHFZM4	C26_MPARAM_7	PAGE0
NHHMAX	C27_NHMMNMX_1	PAGE1
NHHMIN	C27_NHMMNMX_1	PAGE1
NHHOT	C27_NHMMNMX_1	PAGE1
NHHOT1	C27_NHMMNMX_1	PAGE1
NHHPI	C08_SHPCL	PAGE0, C17_NHREQUEST_2
NHHPL	C08_SHPCL	PAGE0
NHHPP	C08_SHPCL	PAGE0
NHHPR	C17_NHREQUEST_2	PAGE0, C08_SHPCL
NHHPRL		PAGE0
NHHPRZ	C17_NHREQUEST_2	PAGE0
NHHPZL		PAGE0
NHHSCH	C27_NHMMNMX_1	PAGE1
NHI	I18_READNH	PAGE0, I20_NHSPEED
NHI2	I23_RDSNHP	PAGE0, I24_NHPROBE2
NHIDL	C27_NHMMNMX_1	PAGE0
NHINDI	C26_MPARAM_1	PAGE0, C10_ATOIGN, C21_HBOVEM_2, E13_WATEST_7, E28_HBOVWA_1
NHIP	I20_NHSPEED	PAGE0
NHIP2	I24_NHPROBE2	PAGE0
NHKALT	C27_NHMMNMX_1	PAGE1
NHKC	C18_NHGOV_1	PAGE1, C18_NHGOV_2, C30_SMRTRQ
NHKCP	C30_SMRTRQ	PAGE0
NHKP	C18_NHGOV_2	PAGE0, C30_SMRTRQ
NHKPL	C18_NHGOV_2	PAGE0
NHL	C19_UNDESPED	PAGE0

Parameter	SET BY	USED BY
NHL1IF	I18_READNH	PAGE0, C01_START, I19_NHDUAL_2, O01_SMOUT
NHL2IF	I18_READNH, I23_RDSNHP	PAGE0, C01_START, I19_NHDUAL_2, O01_SMOUT
NHLD1	C18_NHGOV_1	PAGE1, C18_NHGOV_2
NHLD1C	C18_NHGOV_1	PAGE1
NHLD2	C18_NHGOV_1	PAGE1, C18_NHGOV_2
NHLD2C	C18_NHGOV_1	PAGE1
NHLERR	C14_NHLM	PAGE0
NHLG2	C18_NHGOV_1	PAGE1
NHLG2C	C18_NHGOV_1	PAGE1
NHLG3	C18_NHGOV_1	PAGE0, C18_NHGOV_2
NHLG4	C18_NHGOV_1	PAGE0, C18_NHGOV_2, C30_SMRTRQ
NHLL	C19_UNDRSPEED	PAGE1
NHLNHF	C19_UNDRSPEED	PAGE1
NHLOSD		PAGE0, C20_OVRSPEED
NHLOSU		PAGE0, C20_OVRSPEED
NHLOW	C01_START	PAGE0, C10_ATOIGN, E13_WATEST_7
NHLRST	C19_UNDRSPEED	PAGE0
NHLTAU	C19_UNDRSPEED	PAGE1
NHLTRM	C19_UNDRSPEED	PAGE1
NHM	C27_NHMMNMX_1	PAGE1
NHMAX	C27_NHMMNMX_1	PAGE0, C16_LOOPSEL, C17_NHREQUEST_1
NHMAX1	C27_NHMMNMX_1	PAGE1
NHMAXC	C27_NHMMNMX_1	PAGE0
NHMBDB	C22_HBOVMM_3	PAGE0
NHMIN	C27_NHMMNMX_1	PAGE0, C16_LOOPSEL, C17_NHREQUEST_1
NHMIN1	C27_NHMMNMX_1	PAGE1
NHMIN2	C27_NHMMNMX_1	PAGE1
NHMX1	C27_NHMMNMX_1	PAGE1
NHMX2	C27_NHMMNMX_1	PAGE1
NHOSD	C20_OVRSPEED	PAGE0
NHOSU	C20_OVRSPEED	PAGE0
NHP	I18_READNH	PAGE0
NHP2	I23_RDSNHP	PAGE0
NHREF	C17_NHREQUEST_1	PAGE0, RTDWRITE
NHREF1	C08_SHPCL	PAGE0, C16_LOOPSEL, C17_NHREQUEST_1
NHREFL	C17_NHREQUEST_1	PAGE0
NHREQ	C17_NHREQUEST_1	PAGE0, C18_NHGOV_2, C31_TQIM, RTDWRITE
NHREQQL	C17_NHREQUEST_1	PAGE0
NHREQU	C17_NHREQUEST_1	PAGE0
NHREQZ	C17_NHREQUEST_1	PAGE0, C17_NHREQUEST_2
NHREZL	C17_NHREQUEST_1	PAGE0
NHRF1L	C08_SHPCL	PAGE0, C17_NHREQUEST_1
NHRQ2L	C17_NHREQUEST_1	PAGE0
NHRQ2Z	C17_NHREQUEST_1	PAGE0, C12_NPTGOV_1
NHRQUL	C17_NHREQUEST_1	PAGE0
NHSEL	I17_PNHCVN_1	PAGE0, I18_READNH, I22_SNHCVN_1, I23_RDSNHP
NHT	I18_READNH	PAGE0
NHT2	I23_RDSNHP	PAGE0
NHTCT		PAGE0, I19_NHDUAL_2
NHTIF	I19_NHDUAL_1	PAGE0, E17_NCFLT_3
NHTMP1	I18_READNH	PAGE0
NHTMP2	I23_RDSNHP	PAGE0
NHXIF		PAGE0, I19_NHDUAL_2
NHZM1	C26_MPARAM_6	PAGE1
NHZM2	C26_MPARAM_6	PAGE1
NHZM3	C26_MPARAM_6	PAGE1
NHZM4	C26_MPARAM_6	PAGE1
NLCPL	C03_TLIMRT_1	PAGE1, C04_RPCAL_2_D
NNHI	I18_READNH	PAGE0
NNHI2	I23_RDSNHP	PAGE0
NNSI1	I25_TQNPT_4	PAGE0
NNSI2	I25_TQNPT_5	PAGE0
NP	C26_MPARAM_4	PAGE0, C07_OVRTQPRO, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_5, I33_ARITAT
NP1IF	I25_TQNPT_4	PAGE0, I25_TQNPT_2, I25_TQNPT_6, I43_TQSPEED_2, I43_TQSPEED_3
NP1LIF	I25_TQNPT_2, I25_TQNPT_4	PAGE0, E17_NCFLT_5, I25_TQNPT_6, I43_TQSPEED_2, I43_TQSPEED_3
NP2IF	I25_TQNPT_5	PAGE0, I25_TQNPT_3, I25_TQNPT_6, I43_TQSPEED_2, I43_TQSPEED_3
NP2LIF	I25_TQNPT_3, I25_TQNPT_5	PAGE0, E17_NCFLT_4, I25_TQNPT_6, I43_TQSPEED_2, I43_TQSPEED_3
NPARAM	E05_RAMINI, UART_IO_SELECT	PAGE0, E08_FG_EXECUTIVE, I31_UARTIN, O04_UARTOUTSEL
NPARAM		PAGE11, E05_RAMINI
NPLOW	C26_MPARAM_1	PAGE0, C26_MPARAM_5

Parameter	SET BY	USED BY
NPMBF1		PAGE1, E05 RAMINI, I31 UARTIN_NPM
NPMBF2		PAGE1, I31 UARTIN_NPM
NPMBSL	I31 UARTIN_NPM	PAGE1
NPMCKH	I31 UARTIN_NPM	PAGE1
NPMCKL	I31 UARTIN_NPM	PAGE1
NPMENI	I31 UARTIN_NPM	PAGE1
NPMENO	O04 UARTOUT_NPM	PAGE1
NPMENOX	E05 RAMINI, I31 UARTIN_NPM	PAGE1, O04 UARTOUT_NPM
NPMPAD	I31 UARTIN_NPM	PAGE1
NPMPIT	I31 UARTIN_NPM	PAGE0
NPMPTO	O04 UARTOUT_NPM	PAGE0
NPMPTOX	E05 RAMINI, I31 UARTIN_NPM	PAGE1, O04 UARTOUT_NPM
NPMPITT	I31 UARTIN_NPM	PAGE0
NPOSDF	C26 MPARAM_1	PAGE0, C12 NPTGOV_2
NPPLAR	C12 NPTGOV_2	PAGE0
NPQBUG	C05 TQBUG	PAGE1, C07_OVRTQPRO
NPRGL	C26 MPARAM_4	PAGE0, C05 TQBUG
NPSE	C07_OVRTQPRO	PAGE0, C31 TQLIM
NPT	C26 MPARAM_4	PAGE0, C04_RPCAL_2_D, C12_NPTGOV_1, C12_NPTGOV_2, C19_UNDESPED, C20_OVRSPEED, C27_NHMMNMX_1, I17_PNHCMV_1, I18_READNH, I19_NHDUAL_2, I23_RDSNHP, I25_TQNPT_2, I25_TQNPT_3, I25_TQNPT_4, I25_TQNPT_5, I25_TQNPT_6, I32_ARINCIN_2, I33_ARITAT, I34_TATSEL, I35_ARIPA, I36_PAMSEL, I37_ARIISA_1, I37_ARIISA_2, I43_TQSPEED_1, I43_TQSPEED_2, RTDWRITE
NPTDIS	C12 NPTGOV_2	PAGE0, C19 UNDESPED
NPTDOT	C26 MPARAM_4	PAGE0, C19 UNDESPED, C20_OVRSPEED
NPTER1	C12 NPTGOV_1	PAGE0, C16_LOOPSEL
NPTER2	C12 NPTGOV_1	PAGE0, C16_LOOPSEL
NPTER5	C12 NPTGOV_1	PAGE0
NPTERP	C12 NPTGOV_1	PAGE0, C12_NPTGOV_2
NPTERR	C12 NPTGOV_1	PAGE0
NPTISW	C16_LOOPSEL	PAGE0, C18_NHGOV_1
NPTKD	C12 NPTGOV_1	PAGE0
NPTKP	C12 NPTGOV_1	PAGE0
NPTOUT	C26 MPARAM_4	PAGE0, O16_ARO346
NPTREF	C12 NPTGOV_1	PAGE0, C19 UNDESPED
NPTRF1	C12 NPTGOV_1	PAGE0
NPTSW	I25_TQNPT_2, I25_TQNPT_3, I43_TQSPEED_3	PAGE0, I43_TQSPEED_2, OPMODE_1
NPXCN	I43_TQSPEED_2	PAGE0
NPXIF	I43_TQSPEED_2	PAGE0
NPZM1	C26 MPARAM_4	PAGE1
NPZM2	C26 MPARAM_4	PAGE1
NPZM3	C26 MPARAM_4	PAGE1
NPZM4	C26 MPARAM_4	PAGE1
NS		PAGE0, C26_MPARAM_4, I27_TQSPEED
NSI	I25_TQNPT_6	PAGE0, I27_TQSPEED
NSI1	I25_TQNPT_4	PAGE0, I25_TQNPT_6, I43_TQSPEED_2
NSI2	I25_TQNPT_5	PAGE0, I25_TQNPT_6, I43_TQSPEED_2
NSIHPS		PAGE0, I27_TQSPEED
NSTAT1	I25_TQNPT_4	PAGE0
NSTAT2	I25_TQNPT_5	PAGE0
NSYNC	E05 RAMINI, O04 UARTOUT_NPM	PAGE1
NTEMP1	I25_TQNPT_4	PAGE0
NTEMP2	I25_TQNPT_5	PAGE0
NTORGL	C06_HPREF	PAGE1
NTORTP	C04_RPCAL_2_D	PAGE1, C06_HPREF
NTOTQB	C05_TQBUG	PAGE1, O27_ARO342_A
NTODL1	INC_CTRS_4	PAGE1, DCU_CONSTANTS
NTODL2	INC_CTRS_4	PAGE1, DCU_CONSTANTS
NTODL3	INC_CTRS_4	PAGE1, DCU_CONSTANTS
O01_SMOUT		O01_SMOUT, E08_FG_SCHEDULE
O02_HBOVOUT		O02_HBOVOUT, E08_FG_SCHEDULE
O03_DISOUT		O03_DISOUT, E08_FG_SCHEDULE
O03_DISOUT_A		O03_DISOUT_A, O03_DISOUT
O04_UARTOUTSEL		O04_UARTOUTSEL, E08_FG_EXECUTIVE
O04_UARTOUT_DTS		O04_UARTOUT_DTS, O04_UARTOUTSEL
O04_UARTOUT_GDTS		O04_UARTOUT_GDTS, O04_UARTOUTSEL
O04_UARTOUT_NPM		O04_UARTOUT_NPM, O04_UARTOUTSEL
O05		O05_ARO133, E08_FG_STATETABLE
O05_ARO133		O05_ARO133
O06		O06_ARO134, E08_FG_STATETABLE
O06_ARO134		O06_ARO134

Parameter	SET BY	USED BY
O0701B		O07_ARO203_ALTPS
O07_ARO203_ALT11		O07_ARO203_ALT11, E08_FG_STATETABLE
O07_ARO203_ALTPS		O07_ARO203_ALTPS, E08_FG_STATETABLE
O0901B		O09_ARO211_TAT11
O09_ARO211_TAT11		O09_ARO211_TAT11, E08_FG_STATETABLE
O10		O10_ARO240, E08_FG_STATETABLE
O10_ARO240		O10_ARO240
O11		O11_ARO245, E08_FG_STATETABLE
O11_ARO245		O11_ARO245
O15_ARO341		O15_ARO341, E08_FG_STATETABLE
O16		O16_ARO346, E08_FG_STATETABLE
O16_ARO346		O16_ARO346
O17		O17_ARO350, E08_FG_STATETABLE
O17_ARO350		O17_ARO350
O18		O18_ARO270
O18_ARO270		O18_ARO270, E08_FG_STATETABLE
O19		O19_ARO272, O18_ARO270
O19_ARO272		O19_ARO272
O19_ARO272_A		O19_ARO272_A, O19_ARO272
O26_ARO340		O26_ARO340, E08_FG_STATETABLE
O27_ARO342		O27_ARO342, E08_FG_STATETABLE
O27_ARO342_A		O27_ARO342_A, O27_ARO342
O40_ARO06X		O40_ARO06X, E08_FG_STATETABLE
OKCLR	E14_FAULTSTORE_2, E14_FAULTSTORE_5	PAGE0, E14_FAULTSTORE_3
OPMODE_1		OPMODE_1, E08_FG_SCHEDULE
OPMODE_2		OPMODE_2, E08_FG_SCHEDULE
OPMODE_3		OPMODE_3, E08_FG_SCHEDULE
ORETRN	O18_ARO270	PAGE1
OSAVE	E08_FG_EXECUTIVE	PAGE0
OSAVE2	E08_FG_EXECUTIVE	PAGE0
OTA		OTA, E20_TESTMODE
OTDDT	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM0	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM1	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM10	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM2	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM3	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM4	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM5	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM6	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM7	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM8	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDM9	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OTDMN	C04_RPCAL_2_D	PAGE1, C24_OVRTQTST
OUTSTP	O01_SMOUT	PAGE0
POAMBF	I04_PSAMBSG_4	PAGE0, I36_PAMSEL
POAMBP		PAGE0, I04_PSAMBSG_3
POBLK		PAGE10, E03_CHECKSUM
PODATA		PAGE10, E05_RAMINI
POTAB1		PAGE0, E05_RAMINI, I03_RDCIT_2
POTAB9		PAGE0, I03_RDCIT_2
POTABH	I03_RDCIT_2	PAGE0, I04_PSAMBSG_3
POTABL	I03_RDCIT_2	PAGE0, I04_PSAMBSG_3
POVEL	I05_DELTAPSG_3	PAGE0, C26_MPAMAM_3
POVELP		PAGE0, I05_DELTAPSG_3
P12_PAST_RESET		PAGE0
P18QF1		PAGE0, I05_DELTAPSG_3
P18QF2		PAGE0, I05_DELTAPSG_3
P203	I35_ARIPA	PAGE0, I36_PAMSEL
P20_TIMER		PAGE0
P23_PAST_IN_VAL		PAGE0
P29_C1		PAGE0, C18_NHGOV_2
P29_C2		PAGE0, C18_NHGOV_2
P29_INTER_PAST		PAGE0, C18_NHGOV_2
P39_C1	C17_NHREQUEST_1	PAGE0
P39_C2	C17_NHREQUEST_1	PAGE0
PABLD	C21_HBOVEM_2	PAGE1

Parameter	SET BY	USED BY
PAMB	I36_PAMSEL	PAGE0, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C21_HBOVEM_2, C26_MPARAM_3, C27_NHMMNMX_1, I05_DELTAPSG_3, I08_RDT18_2, RTDWRITE
PAMBI	I36_PAMSEL	PAGE0
PASSED		PASSED, E27_MFCID, I08_RDT18_3, I14_RDPLA_3, I25_TQNPT_4, I25_TQNPT_5
PASSLT		PASSLT, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, I01_LLDRIPT_2, I02_SGCOM_2, I03_RDCIT_3, I04_PSAMBSG_2, I05_DELTAPSG_4, I07_READTQ_3, I08_RDT18_3, I09_HLDRIPT_2, I10_GAINCOM_2, I11_TQBTRIM_2, I12_TQGTTRIM_2, I15_PLABIAS_2, I16_PLAGAIN_2, I19_NHDUAL_2, I45_HBOVWA_2
PBTCT		PAGE0, I15_PLABIAS_2
PGTCT		PAGE0, I16_PLAGAIN_2
PID		PID
PLA	C02_PLATRIM_2	PAGE0, C01_START, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLUM, C15_NHDECELLUM, C19_UNDEERSPEED, C22_HBOVMM_1, C27_NHMMNMX_1, E14_FAULTSTORE_2, I28_DISINW1, I34_TATSEL, I36_PAMSEL, MISSIONS, O03_DISOUT_A, O05_ARO133, O18_ARO270, OPMODE_3, RTDWRITE
PLA1	I14_RDPLA_1, I14_RDPLA_2	PAGE0, I14_RDPLA_3
PLA1IF	I14_RDPLA_1	PAGE0, I14_RDPLA_3
PLA2	I14_RDPLA_2	PAGE0, I14_RDPLA_3
PLA2IF	I14_RDPLA_2	PAGE0, I14_RDPLA_3
PLAABS	C06_HPREF_1	PAGE0
PLAADJ	C02_PLATRIM_2	PAGE0, C02_PLATRIM_1
PLAB	I15_PLABIAS_2	PAGE0
PLABIF	I15_PLABIAS_2	PAGE0
PLAC	I14_RDPLA_3	PAGE0, C19_UNDEERSPEED
PLACT	I14_RDPLA_3	PAGE0
PLADB	C19_UNDEERSPEED	PAGE1, C20_OVRSPEED
PLADB2	C21_HBOVEM_2	PAGE0
PLADIS	C02_PLATRIM_2	PAGE0
PLADOT	C26_MPARAM_7	PAGE0
PLADRF	I15_PLABIAS_2	PAGE0, I14_RDPLA_2, I14_RDPLA_3
PLAERF		PAGE0, C02_PLATRIM_2
PLAERR	C02_PLATRIM_2	PAGE0
PLAG	E05_RAMINI, I16_PLAGAIN_2	PAGE0, I14_RDPLA_2, I15_PLABIAS_2
PLAGCH	C02_PLATRIM_1	PAGE1
PLAGIF	I16_PLAGAIN_2	PAGE0
PLAGRB	C02_PLATRIM_1	PAGE0, C02_PLATRIM_2, E23_PLATRIM
PLAHYS	I14_RDPLA_3	PAGE0, C26_MPARAM_7, I14_RDPLA_4
PLAIF	E05_RAMINI, I14_RDPLA_3, I15_PLABIAS_2, I16_PLAGAIN_2	PAGE0
PLAIN		PAGE0, C02_PLATRIM_1, C02_PLATRIM_2, I14_RDPLA_4, O06_ARO134
PLAIP		PAGE0, I14_RDPLA_4
PLAMFL	C02_PLATRIM_1	PAGE1, C02_PLATRIM_2
PLANHM	C27_NHMMNMX_1	PAGE1, C21_HBOVEM_2
PLARAF	C22_HBOVMM_1	PAGE1
PLARDF	C22_HBOVMM_1	PAGE1
PLARFG	C22_HBOVMM_1	PAGE1, C21_HBOVEM_1, C22_HBOVMM_2
PLARLM	C06_HPREF	PAGE1
PLAROM	C02_PLATRIM_1, E05_RAMINI	PAGE0, C02_PLATRIM_2
PLATDI	C26_MPARAM_1	PAGE0, C02_PLATRIM_1, E14_FAULTSTORE_2, E31_EXTDIS, O18_ARO270
PLATOT	I14_RDPLA_3	PAGE0
PLATRC	C02_PLATRIM_1	PAGE0, E23_PLATRIM
PLATRM		PAGE10, E05_RAMINI, E23_PLATRIM
PLAVAL	C02_PLATRIM_1	PAGE1, C02_PLATRIM_2
PLAVAZ	C02_PLATRIM_1	PAGE1
PLAZM1	C26_MPARAM_7	PAGE1
PLAZM2	C26_MPARAM_7	PAGE1
PLAZM3	C26_MPARAM_7	PAGE1
PLAZM4	C26_MPARAM_7	PAGE1
PLINRG	C02_PLATRIM_1	PAGE1, E31_EXTDIS, O18_ARO270
PMEMIF	E01_PORPSR, E02_RAMTEST, E03_CHECKSUM, E04_INSTST, E07_BG_EXECUTIVE, E08_FG_EXECUTIVE, E10_RAMTEST, E15_BCKGNDTEST	PAGE0
POSIT	O01_SMOUT	PAGE0
PRATIO	C26_MPARAM_3	PAGE0

Parameter	SET BY	USED BY
PREMCLRTL	C06 HPREF	PAGE1
PREMTORTL	C06 HPREF	PAGE1
PRESELRTL	C06 HPREF	PAGE1
PRETRN	E08_FG_EXECUTIVE, E15_BCKGNDTEST, I01_LLDRIFT_1, I02_SGCOM_1, I03_RDCJT_1, I04_PSAMBSG_1, I05_DELTAPSG_1, I07_READTQ_1, I08_RDT18_1, I09_HLDRIFT_1, I10_GAINCOM_1, I11_TQBTRIM_1, I12_TQGTRIM_1, I13_ENGVARIANT_1, I14_RDPLA_1, I14_RDPLA_2, I15_PLABIAS_1, I16_PLAGAIN_1, I17_PNHCVN_1, I17_PNHCVN_2, I18_READNH, I22_SNHCVN_1, I22_SNHCVN_2, I23_RDSNHP, I28_DISINW1, I29_DISINW2, I30_DISINFALT, I31_UARTIN, I31_UARTIN_NPM, I32_ARINCIN_1, I32_ARINCIN_2, I38_SMCSV_1, I38_SMCSV_2, I45_HBOVWA_1, O04_UARTOUT_DTS, O04_UARTOUT_GDTS, O04_UARTOUT_NPM, O05_ARO133, O06_ARO134, O07_ARO203_ALT11, O07_ARO203_ALTPS, O09_ARO211_TAT11, O10_ARO240, O11_ARO245, O15_ARO341, O16_ARO346, O17_ARO350, O18_ARO270, O19_ARO272, O26_ARO340, O27_ARO342	PAGE0
PRT410	E05_RAMINI, I31_UARTIN, I31_UARTIN_NPM	PAGE0, O04_UARTOUT_DTS, O04_UARTOUT_GDTS
PSAMB	I04_PSAMBSG_3	PAGE0
PSAMBF		PAGE0, I04_PSAMBSG_3, I04_PSAMBSG_4, I04_PSAMBSG_5
PSBRTN	ARCOUT, O04_UARTOUTSEL	PAGE0
PSC	I04_PSAMBSG_2	PAGE0, I04_PSAMBSG_3
PSCORR	I04_PSAMBSG_4	PAGE0
PSDAT	I04_PSAMBSG_2	PAGE0, I04_PSAMBSG_3, I36_PAMSEL
PSIF	E05_RAMINI, I01_LLDRIFT_2, I02_SGCOM_2, I04_PSAMBSG_2	PAGE0, E25_FAULTDET
PSP1	I18_READNH, O04_UARTOUT_NPM	PAGE0
PSP2		PAGE0
PSTCT	I04_PSAMBSG_2	PAGE0
PT15	C26_MPARAM_3	PAGE0, RTDWRITE
PT1D_ACF		PAGE1, LCF_TABLE
PT1D_ACF_ADR		LCF_TABLE
PT1D_CYCCRS		PAGE1, DCU_CONSTANTS, LCF_CHECK, O40_ARO06X_TABLE
PT1D_CYCFIN		PAGE1, LCF_CHECK
PT1D_FCFM1		PAGE1, LCF_TABLE
PT1D_FCFM1_ADR		LCF_TABLE
PT1D_FCFM2		PAGE1, LCF_TABLE
PT1D_FCFM2_ADR		LCF_TABLE
PT1D_FCFM3		PAGE1, LCF_TABLE
PT1D_FCFM3_ADR		LCF_TABLE
PT1D_FCFM4		PAGE1, LCF_TABLE
PT1D_FCFM4_ADR		LCF_TABLE
PT1D_FCFM5		PAGE1, LCF_TABLE
PT1D_FCFM5_ADR		LCF_TABLE
PT1D_FCFM6		PAGE1, LCF_TABLE
PT1D_FCFM6_ADR		LCF_TABLE
PT1D_FCFMAX	LCF_CHECK	PAGE1
PT1D_PNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
PT1D_PNA12EE		PAGE11
PT1D_PNA34		PAGE1, O40_ARO06X_TABLE
PT1D_PNA34EE		PAGE11
PT1D_PNA56		PAGE1, O40_ARO06X_TABLE
PT1D_PNA56EE		PAGE11
PT1D_PNA78		PAGE1, O40_ARO06X_TABLE
PT1D_PNA78EE		PAGE11
PT1D_SNA12		PAGE1, O40_ARO06X_TABLE
PT1D_SNA12EE		PAGE11
PT1D_SNA34		PAGE1, O40_ARO06X_TABLE
PT1D_SNA34EE		PAGE11
PT1D_SNA56		PAGE1, O40_ARO06X_TABLE
PT1D_SNA56EE		PAGE11
PT1PT0	C26_MPARAM_3	PAGE1
PT2D_ACF		PAGE1, LCF_TABLE
PT2D_ACF_ADR		LCF_TABLE
PT2D_CYCCRS		PAGE1, DCU_CONSTANTS, LCF_CHECK, O40_ARO06X_TABLE
PT2D_CYCFIN		PAGE1, LCF_CHECK
PT2D_FCFM1		PAGE1, LCF_TABLE
PT2D_FCFM1_ADR		LCF_TABLE
PT2D_FCFM2		PAGE1, LCF_TABLE
PT2D_FCFM2_ADR		LCF_TABLE
PT2D_FCFM3		PAGE1, LCF_TABLE
PT2D_FCFM3_ADR		LCF_TABLE

Parameter	SET BY	USED BY
PT2D_FCFM4		PAGE1, LCF_TABLE
PT2D_FCFM4_ADR		LCF_TABLE
PT2D_FCFM5		PAGE1, LCF_TABLE
PT2D_FCFM5_ADR		LCF_TABLE
PT2D_FCFM6		PAGE1, LCF_TABLE
PT2D_FCFM6_ADR		LCF_TABLE
PT2D_FCFMAX	LCF_CHECK	PAGE1
PT2D_PNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
PT2D_PNA12EE		PAGE11
PT2D_PNA34		PAGE1, O40_ARO06X_TABLE
PT2D_PNA34EE		PAGE11
PT2D_PNA56		PAGE1, O40_ARO06X_TABLE
PT2D_PNA56EE		PAGE11
PT2D_PNA78		PAGE1, O40_ARO06X_TABLE
PT2D_PNA78EE		PAGE11
PT2D_SNA12		PAGE1, O40_ARO06X_TABLE
PT2D_SNA12EE		PAGE11
PT2D_SNA34		PAGE1, O40_ARO06X_TABLE
PT2D_SNA34EE		PAGE11
PT2D_SNA56		PAGE1, O40_ARO06X_TABLE
PT2D_SNA56EE		PAGE11
PTEMP	DCU_TRANSFER, I14_RDPLA_2, I17_PNHCNV_1, I18_READNH, I22_SNHCV_1, I23_RDSNHP, I28_DISINW1, I32_ARINCIN_1, I38_SMCSV_2, O04_UARTOUT_DTS, O04_UARTOUT_GDTS, O10_ARO240, O17_ARO350, O18_ARO270	PAGE0
PTEMP1	I18_READNH, I23_RDSNHP, I28_DISINW1, I32_ARINCIN_1	PAGE0
PTEMP2	I18_READNH	PAGE0
PTEMP3	I18_READNH, I23_RDSNHP	PAGE0
PTEMP4	I18_READNH	PAGE0
PTR		PAGE0, I33_ARITAT
PWMSI1		PID
PWMSI2		PID, DCU_COUNTER_DATA
Q1IF	I25_TQNPT_4	PAGE0, I25_TQNPT_2, I25_TQNPT_6, I43_TQSPEED_1, I43_TQSPEED_3
Q1LIF	I25_TQNPT_4	PAGE0, I25_TQNPT_2, I25_TQNPT_6, I43_TQSPEED_1, I43_TQSPEED_3
Q1OTQE		PAGE0
Q1SW	I25_TQNPT_2, I25_TQNPT_4	PAGE1, I25_TQNPT_3, I25_TQNPT_6
Q22TQE	C24_OVRTQTST	PAGE0
Q2IF	I25_TQNPT_5	PAGE0, I25_TQNPT_3, I25_TQNPT_6, I43_TQSPEED_1, I43_TQSPEED_3
Q2LIF	I25_TQNPT_5	PAGE0, I25_TQNPT_3, I25_TQNPT_6, I43_TQSPEED_1, I43_TQSPEED_3
Q2OTQE		PAGE0
Q2SEL		PAGE0
Q2SW	I25_TQNPT_3, I25_TQNPT_5	PAGE1, I25_TQNPT_2, I25_TQNPT_6
Q37TQE	C24_OVRTQTST	PAGE0
Q3OTQE		PAGE0
Q47TQE	C24_OVRTQTST	PAGE0
Q4OTQE		PAGE0
Q5OTQE		PAGE0
Q6OTQE		PAGE0
QB		PAGE10, E05_RAMINI, E19_TRMSTORE_1
QBTM	I11_QBTRIM_2	PAGE0, I26_TQCAL, O40_ARO06X_TABLE, RTDWRITE
QBTRMC	E19_TRMSTORE_1, I11_QBTRIM_2	PAGE0
QBTRME	E05_RAMINI, I11_QBTRIM_2	PAGE0
QBTRMI	I11_QBTRIM_2	PAGE0
QCNT	I25_TQNPT_2, I25_TQNPT_3	PAGE0, I25_TQNPT_4, I25_TQNPT_5
QCNTRL	E06_TQTEST, I25_TQNPT_6	PAGE0, I25_TQNPT_1, I25_TQNPT_2, I25_TQNPT_3
QCONV1		I25_TQNPT_4, I25_TQNPT_2
QCONV2		I25_TQNPT_5, I25_TQNPT_3
QDEF	I25_TQNPT_6	PAGE1, I26_TQCAL
QDEF1	I25_TQNPT_4	PAGE0, I25_TQNPT_6, I43_TQSPEED_1
QDEF2	I25_TQNPT_5	PAGE0, I25_TQNPT_6, I26_TQCAL, I43_TQSPEED_1
QG		PAGE10, E05_RAMINI, E19_TRMSTORE_2
QGOOD1	I25_TQNPT_2, I25_TQNPT_4, I25_TQNPT_5	PAGE1
QGOOD2	I25_TQNPT_3, I25_TQNPT_4, I25_TQNPT_5	PAGE1
QGTM	I12_TQGTRIM_2	PAGE0, I26_TQCAL, O40_ARO06X_TABLE, RTDWRITE
QGTMC	E19_TRMSTORE_2, I12_TQGTRIM_2	PAGE0
QGTME	E05_RAMINI, I12_TQGTRIM_2	PAGE0
QGTMI	I12_TQGTRIM_2	PAGE0
QHPFAC	C07_OVRTQPRO	PAGE1
QHPL	C06_HPREF	PAGE1, C07_OVRTQPRO
QHPLCP	C07_OVRTQPRO	PAGE1, O15_ARO341
QINSEL		I25_TQNPT_6, I25_TQNPT_2, I25_TQNPT_3
QLMO	C04_RPCAL_2_D	PAGE1, C07_OVRTQPRO, C12_NPTGOV_2, C24_OVRTQTST

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i.

Parameter	SET BY	USED BY
READ		READ, E20_TESTMODE
READQ1		I25_TQNP1_2, I25_TQNP1_1
READQ2		I25_TQNP1_3, I25_TQNP1_1
READ_IN_PROCESS	DCU_SCHEDULE	PAGE1
REQSW	C17_NHREQUEST_1	PAGE1, C17_NHREQUEST_2
RESLAM	C21_HBOVEM_1	PAGE1, C21_HBOVEM_2
RESYNC_WRITTEN	DCU_WRITE_CTRDAT, UPDATE_EE_DATA	PAGE1
RETBKG	E08_FG_EXECUTIVE	PAGE0
RETRN1	CNT_EVNT, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_READ, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, DIVPRT, EE_GET_QUEUE, FLTTIMER, INI_DCU_DATA, O03_DISOUT_A, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRO, C31_TQLIM, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_CONSTANTS, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, E21_RECONFIG, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE
RETRN2	DCU_COMPARE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, EE_PUT_QUEUE, EE_READ_DATA, LCF_CYCLE_COUNT, RTDWRITE	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRO, C31_TQLIM, CNT_EVNT, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
RETRN3	DCU_COMPARE_ITEM	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
RETRN4		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
RETRN5	O19_ARO272_A, O27_ARO342_A	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEVSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
RETRN6	C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEVSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, MISSIONS, OPMODE_1, OPMODE_2, OPMODE_3	PAGE0, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
RETSAV		PAGE0

Parameter	SET BY	USED BY
RETURN	DCU_SCHEDULE, E10_RAMTEST, E11_CONTEST, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, E14_FAULTSTORE_1, E14_FAULTSTORE_2, E14_FAULTSTORE_3, E14_FAULTSTORE_4, E14_FAULTSTORE_5, E14_FAULTSTORE_6, E16_SMTEST, E17_NCFLT_1, E17_NCFLT_2, E17_NCFLT_3, E17_NCFLT_4, E17_NCFLT_5, E17_NCFLT_6, E18_FAULTSUM_1, E18_FAULTSUM_2, E18_FAULTSUM_3, E18_FAULTSUM_4, E19_TRMSTORE_1, E19_TRMSTORE_2, E19_TRMSTORE_4, E19_TRMSTORE_5, E21_RECONFIG, E23_PLATIRM, E25_FAULTDET, E26_FAULTSTORE_1, E26_FAULTSTORE_2, E26_FAULTSTORE_4, E27_MFCID, E28_HBOVWA_1, E28_HBOVWA_2, E31_EXTDIS, EE_WRITE_BLOCK, I01_LLDRIFT_2, I02_SGCOM_2, I03_RDCIT_2, I03_RDCIT_3, I04_PSAMBSG_2, I04_PSAMBSG_3, I04_PSAMBSG_4, I04_PSAMBSG_5, I05_DELTAPSG_2, I05_DELTAPSG_3, I05_DELTAPSG_4, I07_READTQ_2, I07_READTQ_3, I08_RDT18_2, I08_RDT18_3, I08_RDT18_4, I09_HLDRIFT_2, I10_GAINCOM_2, I11_TQBTIRM_2, I12_TQGTIRM_2, I13_ENGVARIANT_2, I14_RDPLA_3, I14_RDPLA_4, I15_PLABIAS_2, I16_PLAGAIN_2, I19_NHDUAL_1, I19_NHDUAL_2, I20_NHSPEED, I24_NHPROBE2, I25_TQNPT_1, I25_TQNPT_7, I26_TQCAL, I27_TQSPEED, I33_ARITAT, I34_TATSEL, I35_ARIPA, I36_PAMSEL, I37_ARISA_1, I37_ARISA_2, I43_TQSPEED_1, I43_TQSPEED_2, I43_TQSPEED_3, I45_HBOVWA_2, I46_IASFAULT, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INIFLT, LCF_CHECK, O01_SMOUT, O02_HBOVOUT, O03_DISOUT	PAGE0, C01_START, C02_PLATIRM_1, C02_PLATIRM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVTNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPOINT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, FLTTIMER, INI_DCU_DATA, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
RFLT26		PAGE0, E26_FAULTSTORE_1, E26_FAULTSTORE_3
RGBSNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
RGBSNA12EE		PAGE11
RGBSNA34		PAGE1, O40_ARO06X_TABLE
RGBSNA34EE		PAGE11
RGBSNA56		PAGE1, O40_ARO06X_TABLE
RGBSNA56EE		PAGE11
RI5FAC		PAGE1
RM1252	E02_RAMTEST	PAGE0, E10_RAMTEST
RM525	E02_RAMTEST	PAGE0, E10_RAMTEST
ROMPT1	E03_CHECKSUM, EESUM_16	PAGE0
ROMPT2	EESUM_16, ROMSUM	PAGE0
ROMPTR	ROMSUM	PAGE0
ROMRET	EESUM_16, EESUM_8, ROMSUM	PAGE0
ROMSUM		ROMSUM, E07_BG_EXECUTIVE
RPTR26	E26_FAULTSTORE_3, E26_FAULTSTORE_4	PAGE0
RSPDSL	C30_SMRTRQ	PAGE0
RSPDSM	C30_SMRTRQ	PAGE0
RSPNH4	C30_SMRTRQ	PAGE0
RSTEPU	C30_SMRTRQ	PAGE0
RTDDAT	I08_RDT18_3	PAGE0, I34_TATSEL
RTDTSW		PAGE0
RTDWRITE		RTDWRITE, E01_PORPSR
RTD_CHECK		PAGE11, RTDWRITE
RTHDDL	C26_MPARAM_3	PAGE1, C18_NHGOV_1
RTHETA	C26_MPARAM_2	PAGE1, C08_SHPCL, C26_MPARAM_3, C26_MPARAM_6
RTHMDL	C26_MPARAM_3	PAGE1, C26_MPARAM_5

Parameter	SET BY	USED BY
RTRET1		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
RTRET2	DCU_TRANSFER, O40_ARO06X	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
RTRET3	DCU_RECEIVE, DCU_TRANSMIT	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_SCHEDULE, DCU_TRANSFER, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
RTRET4		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
RUNTIMER		RUNTIMER, E08_FG_SCHEDULE
RUN_CONFIG_P	RUNTIMER	PAGE1
RUN_MS_RAW	RUNTIMER	PAGE1
RUN_TIMER_RESET	INC_CTRS_2, RUNTIMER	PAGE1

Parameter	SET BY	USED BY
SBIVAR		SBIVAR, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRO, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
SBRTN1	CMPACT, E26_FAULTSTORE_3, I25_TQNPT_2, I25_TQNPT_3	PAGE0
SBRTN2	DBL_LL_SUBR, DERLAG, FILCON, FILTER, I25_TQNPT_4, I25_TQNPT_5, I25_TQNPT_6, LDLG_SUBR, QUAD, SUNVAR, TUSTIN_SUBR, VAR_TUSTIN_SUBR	PAGE0
SBRTN3	DPA, FAILED, FAILLT, PASSED, PASSLT	PAGE0
SBRTN4		PAGE0
SBRTN5		PAGE0
SECRET	E08_FG_SECSEL	PAGE0
SECRTN		PAGE0
SECSEL		E08_FG_SECSEL, E08_FG_EXECUTIVE
SELMRS	C06_HPREF_1	PAGE1, C06_HPREF
SELRTL	C06_HPREF	PAGE1
SGC	E05_RAMINI, I02_SGCOM_2	PAGE0, I04_PSAMBSG_2, I05_DELTAPSG_4
SGCIF	E05_RAMINI, I01_LLDRIPT_2, I02_SGCOM_2	PAGE0, E25_FAULTDET, I04_PSAMBSG_2, I05_DELTAPSG_4
SGCTCT		PAGE0, I02_SGCOM_2
SHUTDOWN_WRITTEN	DCU_WRITE_CTRDAT, UPDATE_EE_DATA	PAGE1
SLMASP	C21_HBOVEM_1	PAGE1
SLMCOR	C21_HBOVEM_1	PAGE1
SLMEND	C22_HBOVMM_1	PAGE1
SLMRST	C22_HBOVMM_1	PAGE1, C22_HBOVMM_2
SLMTIM	C22_HBOVMM_1	PAGE1
SMCFIF	E16_SMTEST	PAGE0
SMCNT	E16_SMTEST	PAGE0
SMINTF	E16_SMTEST	PAGE0
SMSUMV	I38_SMCSV_2	PAGE0, E16_SMTEST
SMSV	E09_SMSUMV, I38_SMCSV_2	PAGE0
SMSVIF	E09_SMSUMV, I38_SMCSV_2	PAGE0, E16_SMTEST
SMTST	E16_SMTEST	PAGE0
SNDWRD		SNDWRD, DUMP, INA, READ, WRITE, WRTEEP
SNPRP1	C26_MPARAM_4	PAGE1
SPO	E02_RAMTEST, E03_CHECKSUM, EESUM_8, FILCON, VAR_TUSTIN_SUBR	PAGE0, RAMREAD0, RAMREAD1, RAMSTOR0, RAMSTOR1

Parameter	SET BY	USED BY
SP1	C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C05_TQBUG, C06_HPREF, C07_OVRTQPRO, C08_SHPCL, C11_SHPREQ_2, C12_NPTGOV_1, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_2, C22_HBOVMM_3, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_5, C26_MPARAM_6, C27_NHNMNMX_1, C31_TQLIM, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_READ, DCU_SCHEDULE, DCU_WRITE, DERLAG, E03_CHECKSUM, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, E27_MFCID, EESUM_8, EE_GET_QUEUE, EE_WRITE_BLOCK, FAILED, FAILLT, I01_LLDRIFT_2, I02_SGCOM_2, I03_RDCIT_3, I04_PSAMBSG_2, I04_PSAMBSG_4, I04_PSAMBSG_5, I05_DELTAPSG_3, I05_DELTAPSG_4, I07_READTQ_3, I08_RDT18_2, I08_RDT18_3, I09_HLDRIFT_2, I10_GAINCOM_2, I11_TQBTRIM_2, I12_TQGTRIM_2, I14_RDPLA_3, I15_PLABIAS_2, I16_PLAGAIN_2, I19_NHDUAL_2, I25_TQNPT_2, I25_TQNPT_3, I25_TQNPT_4, I25_TQNPT_5, I33_ARITAT, I34_TATSEL, I35_ARIPA, I36_PAMSEL, I45_HBOVWA_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, LCF_CYCLE_COUNT, PASSED, PASSLT, RAMREAD0, RAMREAD1, RAMSTOR0, RAMSTOR1, SBIVAR, UNIVAR, WDGRST	PAGE0, C01_START, C02_PLATRIM_1, C04_RPCAL_2_D, C06_HPREF_1, C10_ATOIGN, C11_SHPREQ_1, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_1, C22_HBOVMM_1, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_4, C26_MPARAM_7, C30_SMRTRQ, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_RECEIVE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, DIVPRT, E21_RECONFIG, EE_PUT_QUEUE, EE_READ_DATA, FILCON, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LDLG_SUBR, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, TUSTIN_SUBR, UART_IO_SELECT, UPDATE_EE_DATA, VAR_TUSTIN_SUBR
SP10	DBL_LL_SUBR, DCU_COMPARE_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, I25_TQNPT_4, I25_TQNPT_5, INI_DCU_DATA, LCF_CYCLE_COUNT, QUAD	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, LCF_CHECK, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP11	DCU_COMPARE_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_RANGE_CHECKS, INI_DCU_DATA, QUAD	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP12	DCU_COMPARE_DATA	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP13	DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_EEC_DATA, DCU_LCF_DATA	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTQ, C31_TQLIM, CNT_EVNT, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUID_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP14	DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_EEC_DATA, DCU_LCF_DATA	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTQ, C31_TQLIM, CNT_EVNT, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUID_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP15	DCU_COMPARE_DATA, DCU_EEC_DATA, DCU_LCF_DATA	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP16	DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_EEC_DATA, DCU_LCF_DATA, DCU_RANGE_CHECKS	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_PUT_QUEUE, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP17	DCU_RANGE_CHECKS, UNIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRO, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UPDATE_EE_DATA
SP18	DCU_RANGE_CHECKS, SBIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRO, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP19	DCU_IS_OBSOLETE, DCU_RANGE_CHECKS, SBIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP1B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
SP1DBL	C17_NHREQUEST_1, C18_NHGOV_2	PAGE0, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_2, C18_NHGOV_1, C19_UNDESPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C26_MPARAM_2, C31_TQLIM, DBL_LL_SUBR
SP2	C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C05_TQBUG, C06_HPREF, C07_OVRTQPRO, C08_SHPCL, C11_SHPREQ_2, C12_NPTGOV_1, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_3, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_5, C26_MPARAM_6, C27_NHMMMX_1, C31_TQLIM, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_SCHEDULE, E03_CHECKSUM, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, E27_MFCD, EESUM_8, EE_GET_QUEUE, EE_WRITE_BLOCK, FLTTIMER, I01_LLDRIFT_2, I02_SGCOM_2, I03_RDCIT_3, I04_PSAMBSG_2, I05_DELTAPSG_3, I05_DELTAPSG_4, I07_READTQ_3, I08_RDT18_3, I09_HLDRIFT_2, I10_GAINCOM_2, I11_TQBTTRIM_2, I12_TQGTTRIM_2, I14_RDPLA_3, I15_PLABIAS_2, I16_PLAGAIN_2, I19_NHDUAL_2, I25_TQNPT_2, I25_TQNPT_3, I25_TQNPT_4, I25_TQNPT_5, I33_ARITAT, I45_HBOVWA_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, LCF_CYCLE_COUNT, QUAD, RAMREAD0, RAMREAD1, RAMSTOR0, RAMSTOR1, RUNTIMER, SBIVAR, VAR_TUSTIN_SUBR	PAGE0, C01_START, C02_PLATRIM_1, C04_RPCAL_2_D, C06_HPREF_1, C10_ATOIGN, C11_SHPREQ_1, C12_NPTGOV_2, C16_LOOPSEL, C22_HBOVMM_1, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_4, C26_MPARAM_7, C30_SMRTRQ, CNT_EVNT, DBL_LL_SUBR, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, DERLAG, DIVPRT, E02_RAMTEST, E21_RECONFIG, EE_PUT_QUEUE, EE_READ_DATA, FAILED, FAILLT, FILCON, I13_ENGVARIANT_2, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LDLG_SUBR, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, TUSTIN_SUBR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP20	DCU_COMPARE_DATA, DCU_IS_OBSOLETE, SBIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVTNT, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP21	SBIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVTNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP22	SBIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP23	SBIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP24	SBIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP25		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP26		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP27	DCU_SCHEDULE	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP28		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP29	DCU_IS_OBSOLETE	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP2B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
SP2DBL		PAGE0, C10_ATOIGN, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_2, C18_NHGOV_1, C19_UNDESPED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C26_MPARAM_2, C31_TQLIM

Parameter	SET BY	USED BY
SP3	C02_PLATIRM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C06_HPREF, C07_OVRTQPRO, C08_SHPCL, C11_SHPREQ_2, C12_NPTGOV_1, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C17_NHREQUEST_1, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_2, C22_HBOVMM_3, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_5, C27_NHMMMX_1, C31_TQLIM, DCU_COMPONT_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_IS_OBSOLETE, DCU_LCF_DATA, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, EESUM_8, FAILLT, I01_LLDRIFT_2, I02_SGCOM_2, I03_RDCJT_3, I04_PSAMBSG_2, I04_PSAMBSG_3, I04_PSAMBSG_4, I04_PSAMBSG_5, I05_DELTAPSG_3, I05_DELTAPSG_4, I07_READTQ_2, I07_READTQ_3, I08_RDT18_2, I08_RDT18_3, I09_HLDRIFT_2, I10_GAINCOM_2, I11_TQBTRIM_2, I12_TQGTTRIM_2, I14_RDPLA_4, I15_PLABIAS_2, I16_PLAGAIN_2, I19_NHDUAL_2, I20_NHSPEED, I24_NHPROBE2, I27_TQSPEED, I33_ARITAT, I34_TATSEL, I35_ARIPA, I36_PAMSEL, I45_HBOVWA_2, INC_CTRS_2, INC_CTRS_3, PASSLT, RAMREAD0, RAMREAD1, RAMSTOR0, RAMSTOR1	PAGE0, C01_START, C02_PLATIRM_1, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF_1, C10_ATOIGN, C11_SHPREQ_1, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_2, C22_HBOVMM_1, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_4, C26_MPARAM_6, C26_MPARAM_7, C30_SMRTRQ, CNT_EVTNT, DBL_LL_SUBR, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_EEC_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, DERLAG, E02_RAMTEST, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FILCON, FILTER, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, LDLG_SUBR, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, QUAD, RTDWRITE, RUNTIMER, SBIVAR, TUSTIN_SUBR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA, VAR_TUSTIN_SUBR
SP30	DCU_IS_OBSOLETE	PAGE0, C01_START, C02_PLATIRM_1, C02_PLATIRM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVTNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP31		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP32		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP33		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP34		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP35		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP36	DCU_IS_OBSOLETE	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP4	C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C06_HPREF, C07_OVRTQPRO, C08_SHPCL, C11_SHPREQ_2, C12_NPTGOV_1, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C17_NHREQUEST_1, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEVSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_2, C22_HBOVMM_3, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_5, C27_NHMMNMX_1, C31_TQLIM, E02_RAMTEST, E03_CHECKSUM, E04_INSTST, E05_RAMINI, E06_TQTEST, E09_SMSUMV, E13_WATEST_1, E13_WATEST_2, E13_WATEST_3, E13_WATEST_4, E13_WATEST_5, E13_WATEST_6, E13_WATEST_7, E22_GASEL, FILCON, FILTER, I01_LLDRIFT_2, I02_SGCOM_2, I03_RDCJT_3, I04_PSAMBSG_2, I04_PSAMBSG_3, I05_DELTAPSG_3, I05_DELTAPSG_4, I07_READTQ_2, I07_READTQ_3, I08_RDT18_3, I09_HLDRIFT_2, I10_GAINCOM_2, I11_TQBTRIM_2, I12_TQGTRIM_2, I14_RDPLA_4, I15_PLABIAS_2, I16_PLAGAIN_2, I19_NHDUAL_2, I20_NHSPEED, I24_NHPROBE2, I27_TQSPEED, I33_ARITAT, I45_HBOVWA_2, INC_CTRS_2, INC_CTRS_3, QUAD, TUSTIN_SUBR	PAGE0, C01_START, C02_PLATRIM_1, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF_1, C10_ATOIGN, C11_SHPREQ_1, C12_NPTGOV_2, C16_LOOPSEL, C17_NHREQUEST_2, C22_HBOVMM_1, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_4, C26_MPARAM_6, C26_MPARAM_7, C30_SMRTQ, CNT_EVNT, DBL_LL_SUBR, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, DERLAG, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FAILT, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, LDLG_SUBR, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA, VAR_TUSTIN_SUBR
SP5	C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C06_HPREF, C08_SHPCL, C11_SHPREQ_2, C12_NPTGOV_1, C17_NHREQUEST_1, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEVSPEED, C20_OVRSPEED, C21_HBOVEM_1, C22_HBOVMM_2, C26_MPARAM_5, C27_NHMMNMX_1, C31_TQLIM, CNT_EVNT, DERLAG, FILCON, I04_PSAMBSG_3, I04_PSAMBSG_4, I04_PSAMBSG_5, I05_DELTAPSG_3, I07_READTQ_2, I08_RDT18_2, I08_RDT18_3, I14_RDPLA_4, I20_NHSPEED, I24_NHPROBE2, I27_TQSPEED, I34_TATSEL, I35_ARIPA, I36_PAMSEL, INC_CTRS_2, INC_CTRS_3, TUSTIN_SUBR, VAR_TUSTIN_SUBR	PAGE0, C01_START, C02_PLATRIM_1, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF_1, C07_OVRTQPRO, C10_ATOIGN, C11_SHPREQ_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_2, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_6, C26_MPARAM_7, C30_SMRTQ, DBL_LL_SUBR, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FILTER, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, LDLG_SUBR, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, QUAD, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP6	C17_NHREQUEST_1, C18_NHGOV_2, C19_UNDEVSPEED, C26_MPARAM_5, C31_TQLIM, CNT_EVNT, DBL_LL_SUBR, FILCON, FILTER, I04_PSAMBSG_3, I05_DELTAPSG_3, I07_READTQ_2, I08_RDT18_3, I14_RDPLA_4, I20_NHSPEED, I24_NHPROBE2, I27_TQSPEED, INC_CTRS_2, INC_CTRS_3, LDLG_SUBR, VAR_TUSTIN_SUBR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_2, C18_NHGOV_1, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTQ, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SP7	DBL_LL_SUBR, DCU_COMPARE_DATA, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_GET_DATA, DCU_LCF_DATA, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, EE_READ_DATA, FILCON, FILTER, IOS_DELTAPSG_3, INI_DCU_DATA, LCF_CHECK, LDLG_SUBR, TUSTIN_SUBR, UPDATE_EE_DATA, VAR_TUSTIN_SUBR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNX_1, C30_SMRTQ, C31_TQLIM, CNT_EVTN, DCU_COMPARE_ITEM, DCU_CONSTANTS, DCU_EEC_DATA, DCU_GET_CHECKSUM, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, QUAD, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR
SP8	CNT_EVTN, DBL_LL_SUBR, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, EE_PUT_QUEUE, EE_READ_DATA, IOS_DELTAPSG_3, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, LDLG_SUBR, UPDATE_EE_DATA	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNX_1, C30_SMRTQ, C31_TQLIM, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_CONSTANTS, DCU_EEC_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, E21_RECONFIG, EE_GET_QUEUE, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, QUAD, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR

Parameter	SET BY	USED BY
SP9	DCU_COMPARE_DATA, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_ENGINE_DATA, DCU_GET_DATA, DCU_LCF_DATA, DCU_RANGE_CHECKS, I05_DELTAPSG_3, I25_TQNPT_4, I25_TQNPT_5, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, LDLG_SUBR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_ITEM, DCU_CONSTANTS, DCU_EEC_DATA, DCU_GET_CHECKSUM, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_PUT_QUEUE, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, QUAD, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SP9DBL	DBL_LL_SUBR	PAGE0
SSNHFG	C22_HBOVMM_1	PAGE1
SSTIME	C22_HBOVMM_1	PAGE1
STACK		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
STARTUP_WRITTEN	DCU_WRITE_CTRDAT, UPDATE_EE_DATA	PAGE1
STAT1	I14_RDPLA_1	PAGE0, I14_RDPLA_2, I14_RDPLA_3
STAT2	I14_RDPLA_2	PAGE0, I14_RDPLA_3
STATE	E05_RAMINI, E08_FG_EXECUTIVE	PAGE0, O04_UARTOUT_DTS, O04_UARTOUT_GDTS
STATE_TABLE		E08_FG_STATETABLE, E05_RAMINI, E08_FG_EXECUTIVE
STATUS	E05_RAMINI, I31_UARTIN	PAGE0, E08_FG_EXECUTIVE, O04_UARTOUT_DTS, O04_UARTOUT_GDTS
STCNTP	C01_START	PAGE1
STEPS	C30_SMRTQ	PAGE1, E16_SMTEST
STEPSR	O01_SMOUT	PAGE0, C30_SMRTQ
STOP	O01_SMOUT	PAGE0
STOPIL	O01_SMOUT	PAGE0, C17_NHREQUEST_1, I20_NHSPEED, I24_NHPROBE2
STPCMD	E16_SMTEST	PAGE0, O01_SMOUT
STPILZ	O01_SMOUT	PAGE0
STPRLU	C30_SMRTQ	PAGE0
STQBI	O27_ARO342_A	PAGE1, O27_ARO342
STQBUG	O27_ARO342_A	PAGE1, O27_ARO342

Parameter	SET BY	USED BY
STRPLA	E23_PLATRIM	PAGE0
STRTCYC	OPMODE_2	PAGE1
STRTSW	C01_START	PAGE0, O01_SMOUT
SUBVR1	SBIVAR, UNIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRO, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UPDATE_EE_DATA
SUBVR2	SBIVAR, UNIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRO, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCU_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SUBVR3	SBIVAR, UNIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UPDATE_EE_DATA
SUBVR4	SBIVAR, UNIVAR	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, UART_IO_SELECT, UPDATE_EE_DATA

Parameter	SET BY	USED BY
SUBVR5		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SUBVR6		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SUNVAR		SUNVAR, I04_PSAMBSG_4, I04_PSAMBSG_5, I05_DELTAPSG_3, I08_RDT18_2, I34_TATSEL, I35_ARIPA, I36_PAMSEL
SV_PWC_PATCHID	DCU_COUNTER_DATA	PAGE1
SV_SW_EPI	DCU_COUNTER_DATA	PAGE1
SV_SW_SBI	DCU_COUNTER_DATA	PAGE1
SV_SW_SVI	DCU_COUNTER_DATA	PAGE1
SWID	E05_RAMINI	PAGE1, O40_ARO06X_TABLE
SWIDHI	E05_RAMINI	PAGE1
SWIDLO	E05_RAMINI	PAGE1

Parameter	SET BY	USED BY
SYS1	C19_UNDEVSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_3, C31_TQLIM, DCU_LCF_DATA, DCU_TRANSFER, OPMODE_1	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C20_OVRSPEED, C22_HBOVMM_1, C22_HBOVMM_2, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRQ, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
SYS2		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEVSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRQ, C31_TQLIM, CNT_EVNT, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, DCU_COMPONT_DATA, DCU_CONSTANTS, DCU_COUNTER_DATA, DCU_DCUID_DATA, DCU_EEC_DATA, DCU_ENGINE_DATA, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_GET_QUEUE, DCU_IS_OBSOLETE, DCU_LCF_DATA, DCU_PUT_QUEUE, DCU_RANGE_CHECKS, DCU_READ, DCU_RECEIVE, DCU_SCHEDULE, DCU_TRANSFER, DCU_TRANSMIT, DCU_WRITE, DCU_WRITE_CPDAT, DCU_WRITE_CTRDAT, DCU_WRITE_EECDAT, DCU_WRITE_ENGDAT, E21_RECONFIG, EE_GET_QUEUE, EE_PUT_QUEUE, EE_READ_DATA, EE_WRITE_BLOCK, FLTTIMER, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_2, INC_CTRS_3, INC_CTRS_4, INI_DCUI_DATA, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, O40_ARO06X, O40_ARO06X_TABLE, OPMODE_1, OPMODE_2, OPMODE_3, RTDWRITE, RUNTIMER, SBIVAR, UART_IO_SELECT, UNIVAR, UPDATE_EE_DATA
T15	I34_TATSEL	PAGE0, C26_MPARAM_2, C27_NHMMNMX_1, RTDWRITE
T15211	I33_ARITAT	PAGE0, I34_TATSEL
T15F	I08_RDT18_4	PAGE0, I34_TATSEL
T15K	C26_MPARAM_2	PAGE1
T15R	C26_MPARAM_2	PAGE1
T1815F		PAGE0, I08_RDT18_3, I08_RDT18_4
T18C	I08_RDT18_3	PAGE0
T18CEL	I08_RDT18_4	PAGE0
T18F	I08_RDT18_3	PAGE0
T18IF	E05_RAMINI, I01_LLDRIFT_2, I08_RDT18_3	PAGE0
T18P		PAGE0, I08_RDT18_3
T18T15	I08_RDT18_2	PAGE0, I08_RDT18_4
T18TCT	I08_RDT18_3	PAGE0
TAISA	C26_MPARAM_2	PAGE1
TAKEOFF	OPMODE_3	PAGE1
TAKEOFFRST	OPMODE_3	PAGE1

Parameter	SET BY	USED BY
TAMB	C26_MPARAM_2	PAGE1, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2
TAT	C26_MPARAM_2	PAGE1
TATADC	I33_ARITAT	PAGE0
TATDAT	I33_ARITAT	PAGE0, I34_TATSEL
TATIF	I34_TATSEL	PAGE0
TATRGC	I33_ARITAT	PAGE0
TATTIM	I33_ARITAT	PAGE0
TATXIF	I34_TATSEL	PAGE0
TATXYM	I34_TATSEL	PAGE0
TBLEND		E08_FG_STATETABLE, E08_FG_EXECUTIVE
TBTCT		PAGE0, I11_TQBTRIM_2
TBTIF	E05_RAMINI, I09_HLDRIFT_2, I10_GAINCOM_2, I11_TQBTRIM_2	PAGE0
TEMP	C17_NHREQUEST_1, CMPACT, E05_RAMINI, E14_FAULTSTORE_1, E14_FAULTSTORE_3, E14_FAULTSTORE_4, E16_SMTST, E17_NCFLT_1, E17_NCFLT_2, E17_NCFLT_3, E17_NCFLT_4, E17_NCFLT_5, E17_NCFLT_6, E18_FAULTSUM_1, E18_FAULTSUM_2, E18_FAULTSUM_3, E18_FAULTSUM_4, E26_FAULTSTORE_1, E26_FAULTSTORE_2, E26_FAULTSTORE_3, E26_FAULTSTORE_4, E31_EXTDIS, I02_SGCOM_2, I05_DELTAPSG_4, I07_READTQ_3, I14_RDPLA_3, I25_TQNPT_2, I25_TQNPT_3, I34_TATSEL, I36_PAMSEL, INIFLT, O02_HBOVOUT, O07_ARO203_ALT11, O19_ARO272	PAGE0
TEMP1	C17_NHREQUEST_1, CMPACT, DBL_LL_SUBR, E05_RAMINI, E14_FAULTSTORE_1, E14_FAULTSTORE_6, E17_NCFLT_1, E17_NCFLT_2, E17_NCFLT_3, E17_NCFLT_4, E17_NCFLT_5, E17_NCFLT_6, E18_FAULTSUM_1, E18_FAULTSUM_2, E18_FAULTSUM_3, E18_FAULTSUM_4, E26_FAULTSTORE_1, E26_FAULTSTORE_2, E26_FAULTSTORE_3, I03_RDCJT_2, I03_RDCJT_3, I04_PSAMBSG_2, I04_PSAMBSG_3, I05_DELTAPSG_2, I08_RDT18_3, I10_GAINCOM_2, I25_TQNPT_2, I25_TQNPT_3, I34_TATSEL, I36_PAMSEL, I43_TQSPED_3, INIFLT, RTDWRITE	PAGE0
TEMP2	C17_NHREQUEST_1, CMPACT, E05_RAMINI, E18_FAULTSUM_1, E18_FAULTSUM_2, E18_FAULTSUM_3, E18_FAULTSUM_4, E26_FAULTSTORE_3, I03_RDCJT_2, I04_PSAMBSG_3, I05_DELTAPSG_2, I43_TQSPED_3, INIFLT, RTDWRITE	PAGE0
TEMP3	C17_NHREQUEST_1, E05_RAMINI, E18_FAULTSUM_1, E18_FAULTSUM_2, E18_FAULTSUM_3, E18_FAULTSUM_4, E26_FAULTSTORE_3, I03_RDCJT_2, I04_PSAMBSG_3, I05_DELTAPSG_2, I43_TQSPED_3, RTDWRITE	PAGE0
TEMP4	E18_FAULTSUM_1, E18_FAULTSUM_2, E18_FAULTSUM_3, E18_FAULTSUM_4, I43_TQSPED_3	PAGE0
TEMP5		PAGE0
TEMP6		PAGE0
TERM	C19_UNDERSPEED	PAGE1
TGTCT		PAGE0, I12_TQGRIM_2
TGTIF	E05_RAMINI, I09_HLDRIFT_2, I10_GAINCOM_2, I12_TQGRIM_2	PAGE0
THETA	C26_MPARAM_2	PAGE1
THETAC	C26_MPARAM_2	PAGE1, C27_NHNMNMX_1, O09_ARO211_TAT11
THETAI	I34_TATSEL	PAGE0
TI	I25_TQNPT_4, I25_TQNPT_5	PAGE1
TIMACC	C13_NHACCELLIM	PAGE1
TM	I25_TQNPT_4, I25_TQNPT_5	PAGE0
TOMOD	C03_TLMRT_7	PAGE1, C04_RPCAL_2_D
TOPAMD	I36_PAMSEL	PAGE0
TOT15D	I34_TATSEL	PAGE0
TOTTIM	I25_TQNPT_4, I25_TQNPT_5	PAGE0
TP	I04_PSAMBSG_3, I05_DELTAPSG_2	PAGE0
TQ	I26_TQCAL	PAGE0, RTDWRITE
TQBG1	C05_TQBUG	PAGE1, O27_ARO342_A
TQBUG	C05_TQBUG	PAGE0, I34_TATSEL
TQC	I07_READTQ_2	PAGE0, I26_TQCAL
TQCI	I07_READTQ_3	PAGE0, I07_READTQ_2
TQERR	C31_TQLIM	PAGE1
TQF1TR	C31_TQLIM	PAGE0
TQIF	E05_RAMINI, I01_LLDRIFT_2, I07_READTQ_3	PAGE0
TQIP		PAGE0, I07_READTQ_2
TQLIM2	C07_OVRTQPRO	PAGE1, C31_TQLIM
TQLIM3	C31_TQLIM	PAGE1

Parameter	SET BY	USED BY
TQTCT		PAGE0, I07_READTQ_3
TRBMSNA12		PAGE1, DCU_CONSTANTS, O40_ARO06X_TABLE
TRBMSNA12EE		PAGE11
TRBMSNA34		PAGE1, O40_ARO06X_TABLE
TRBMSNA34EE		PAGE11
TRBMSNA56		PAGE1, O40_ARO06X_TABLE
TRBMSNA56EE		PAGE11
TRIMEN	C02_PLATRIM_1	PAGE1
TRMMOD	C02_PLATRIM_2	PAGE1
TRMTYM	C02_PLATRIM_1	PAGE1
TRPLA	C06_HPREF_1	PAGE1
TS	I25_TQNPT_4, I25_TQNPT_5	PAGE0
TSTATE	O04_UARTOUT_NPM	PAGE0
TT1T02	C26_MPARAM_2	PAGE1
TT1T0T	I33_ARITAT	PAGE1
TTTEMP1	I18_READNH	PAGE0
TTTEMP2	I23_RDSNHP	PAGE0
TUSTIN_SUBR		TUSTIN_SUBR, C02_PLATRIM_2, C06_HPREF, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVMM_1, C22_HBOVMM_2, C26_MPARAM_5
UARTIF	O04_UARTOUT_DTS, O04_UARTOUT_GDTS	PAGE0, E25_FAULTDET
UART_IO_SELECT		UART_IO_SELECT, E08_FG_SCHEDULE
UHOLD	O04_UARTOUT_DTS, O04_UARTOUT_GDTS	PAGE0
UMODE	O04_UARTOUT_DTS, O04_UARTOUT_GDTS	PAGE0
UNIVAR		UNIVAR, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVMM_1, C21_HBOVMM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRQ, C31_TQLIM, I13_ENGVARIANT_2, I33_ARITAT, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
UOTBL		PAGE0, E05_RAMINI, O04_UARTOUT_DTS, O04_UARTOUT_GDTS
UPDATE_EE_DATA		UPDATE_EE_DATA, E08_FG_SCHEDULE
UPDELO	C06_HPREF_1	PAGE1
UPTFGL	C13_NHACCELLIM	PAGE1
UPTFLG	C11_SHPREQ_1	PAGE1, C11_SHPREQ_2, C13_NHACCELLIM, C31_TQLIM
UPTRSW	C11_SHPREQ_1	PAGE1
URTCFG		PAGE10, E05_RAMINI
URTOUT		URTOUT, DUMP, E20_TESTMODE, INA, LOAD, OTA, READ, SNDWRD, WRITE, WRTEEP
URTRD		URTRD, E20_TESTMODE, GETWRD
URTTBL		PAGE11, E05_RAMINI, RTDWRITE, WRTEEP
UWORD	O04_UARTOUT_DTS, O04_UARTOUT_GDTS	PAGE0
VAR_TUSTIN_SUBR		VAR_TUSTIN_SUBR, C19_UNDEERSPEED, C26_MPARAM_5
VK	C26_MPARAM_3	PAGE1, C08_SHPCL, C12_NPTGOV_1, C12_NPTGOV_2, C19_UNDEERSPEED, OPMODE_1
VK1	C08_SHPCL	PAGE1
VK2	C08_SHPCL	PAGE1
VKF	OPMODE_1	PAGE1
WCNT26	E26_FAULTSTORE_2	PAGE0, E26_FAULTSTORE_4
WCNT	E05_RAMINI, E08_FG_EXECUTIVE	PAGE0
WDGRST		WDGRST, E01_PORPSR, E02_RAMTEST, E03_CHECKSUM, E05_RAMINI, E06_TQTEST, E09_SMSUMV, E22_GASEL, RTDWRITE, WRITE, WRTEEP
WDTTIP	E01_PORPSR	PAGE0
WRDCNT	E14_FAULTSTORE_3	PAGE0, E14_FAULTSTORE_4
WRITE		WRITE, E20_TESTMODE
WRITE_IN_PROCESS	DCU_SCHEDULE	PAGE1
WRTEEP		WRTEEP, E20_TESTMODE
XH	SUNVAR	PAGE0
XIN	SUNVAR	PAGE0
XTAB		PAGE0, SUNVAR
YHI		PAGE0
YIN		PAGE0

Parameter	SET BY	USED BY
YTAB		PAGE0
ZALTD	I36 PAMSEL	PAGE0
ZLRUAD	O10 ARO240	PAGE1
ZNS1	I25 TQNPT 4	PAGE1
ZNS2	I25 TQNPT 5	PAGE1
ZPOINT	SUNVAR	PAGE0
ZQ1	I25 TQNPT 4	PAGE1
ZQ2	I25 TQNPT 5	PAGE1
ZTAB		PAGE0, SUNVAR
ZTATDT	I34 TATSEL	PAGE0
ZTEM1	C01_START, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_3_1, C03_TLMRT_4_1, C03_TLMRT_5_1, C03_TLMRT_6_1, C03_TLMRT_7, C04_RPCAL_2_D, C06_HPREF, C07_OVRTQPRO, C10_ATOIGN, C17_NHREQUEST_2, C18_NHGOV_2, C21_HBOVEM_2, C22_HBOVMM_1, C26_MPARAM_2, C26_MPARAM_3, C30_SMRTRQ, C31_TQLIM, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, OPMODE_1, OPMODE_3	PAGE0, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_2_2, C03_TLMRT_3_2, C03_TLMRT_4_2, C03_TLMRT_5_2, C03_TLMRT_6_2, C05_TQBUG, C06_HPREF_1, C08_SHPCL, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLUM, C14_NHLM, C15_NHDECELLUM, C16_LOOPSEL, C17_NHREQUEST_1, C18_NHGOV_1, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, O27_ARO342_A, OPMODE_2
ZTEM1B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM2	C01_START, C03_TLMRT_1, C18_NHGOV_2, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, OPMODE_1, OPMODE_3	PAGE0, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLUM, C14_NHLM, C15_NHDECELLUM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRQ, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O27_ARO342_A, OPMODE_2
ZTEM2B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3

Parameter	SET BY	USED BY
ZTEM3	C10_ATOIGN, C17_NHREQUEST_2, O03_DISOUT_A, OPMODE_1	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O19_ARO272_A, O27_ARO342_A, OPMODE_2, OPMODE_3
ZTEM3B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM4	OPMODE_1	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_2, OPMODE_3
ZTEM4B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM5		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDESPED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHNMNMX_1, C30_SMRTRQ, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3

Parameter	SET BY	USED BY
ZTEM5B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM6		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRO, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM6B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM7		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, C30_SMRTRO, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM7B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLMRT_1, C03_TLMRT_2_1, C03_TLMRT_2_2, C03_TLMRT_3_1, C03_TLMRT_3_2, C03_TLMRT_4_1, C03_TLMRT_4_2, C03_TLMRT_5_1, C03_TLMRT_5_2, C03_TLMRT_6_1, C03_TLMRT_6_2, C03_TLMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3

Parameter	SET BY	USED BY
ZTEM8		PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRO, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM8B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM9	C04_RPCAL_2_D, O27_ARO342_A	PAGE0, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C08_SHPCL, C10_ATOIGN, C11_SHPREQ_1, C11_SHPREQ_2, C12_NPTGOV_1, C12_NPTGOV_2, C13_NHACCELLIM, C14_NHLIM, C15_NHDECELLIM, C16_LOOPSEL, C17_NHREQUEST_1, C17_NHREQUEST_2, C18_NHGOV_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C22_HBOVMM_3, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_2, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, C30_SMRTRO, C31_TQLIM, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEM9B		PAGE1, C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C03_TLIMRT_1, C03_TLIMRT_2_1, C03_TLIMRT_2_2, C03_TLIMRT_3_1, C03_TLIMRT_3_2, C03_TLIMRT_4_1, C03_TLIMRT_4_2, C03_TLIMRT_5_1, C03_TLIMRT_5_2, C03_TLIMRT_6_1, C03_TLIMRT_6_2, C03_TLIMRT_7, C04_RPCAL_2_D, C05_TQBUG, C06_HPREF, C06_HPREF_1, C07_OVRTQPRO, C24_OVRTQTST, C26_MPARAM_1, C26_MPARAM_3, C26_MPARAM_4, C26_MPARAM_5, C26_MPARAM_6, C26_MPARAM_7, C27_NHMMNMX_1, I13_ENGVARIANT_2, INC_CTRS_1, INC_CTRS_4, LCF_CHECK, LCF_CYCLE_COUNT, MISSIONS, O03_DISOUT, O03_DISOUT_A, O19_ARO272_A, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3
ZTEMA	C01_START, C02_PLATRIM_1, C02_PLATRIM_2, C05_TQBUG, C06_HPREF, C06_HPREF_1, C10_ATOIGN, C12_NPTGOV_1, C12_NPTGOV_2, C17_NHREQUEST_1, C18_NHGOV_2, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C26_MPARAM_3, C30_SMRTRO, C31_TQLIM, MISSIONS, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3	PAGE0
ZTEMB	C01_START, C05_TQBUG, C06_HPREF, C06_HPREF_1, C08_SHPCL, C12_NPTGOV_2, C19_UNDEERSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C30_SMRTRO, C31_TQLIM, MISSIONS, O27_ARO342_A, OPMODE_1, OPMODE_2, OPMODE_3	PAGE0, C11_SHPREQ_2
ZTEMC	C01_START, C05_TQBUG, C06_HPREF_1, C11_SHPREQ_2, C12_NPTGOV_1, C19_UNDEERSPEED, C20_OVRSPEED, C21_HBOVEM_1, C21_HBOVEM_2, C22_HBOVMM_1, C22_HBOVMM_2, C31_TQLIM, MISSIONS, OPMODE_1, OPMODE_2, OPMODE_3	PAGE0

Parameter	SET BY	USED BY
ZTEMD	C01_START, C05_TQBUG, C06_HPREF_1, C08_SHPCL, C11_SHPREQ_2, C19_UNDESPPEED, C21_HBOVEM_1, C22_HBOVMM_1, C22_HBOVMM_2, C31_TQLIM, MISSIONS, OPMODE_1	PAGE0
ZTEME	C06_HPREF_1, C21_HBOVEM_1, C22_HBOVMM_1, C31_TQLIM, OPMODE_1	PAGE0, C12_NPTGOV_1
ZTEMF	C06_HPREF_1, C21_HBOVEM_1, C31_TQLIM, OPMODE_1	PAGE0
ZTEMG	C06_HPREF_1, OPMODE_1	PAGE0
ZTEMH	OPMODE_1	PAGE0
ZTEMI		PAGE0
ZTEMJ		PAGE0
ZTRMEN	C02_PLATRIM_1	PAGE1
ZXLO	SUNVAR	PAGE0
ZXLYI		PAGE0
ZXLYL		PAGE0

09.02 SET/USE TABLE (SORTED BY MODULE)

Set / Use Report for software version 127S0101 of the PW127XT-S

09.02 Table 1: Set-Use-by-Module

Module	Sets Parameter	Uses Parameter
ARCOUT	AROF, PSBRTN	ARCOUT, AROLSH, AROMSH
C01_START		ATEMO, ATEMI, ATEM2, BIPOLY, C01_START, DIVPRT, DPA, DPA1, DPA1B, DPA2, DPA2B, DPAB, DPM, DPM1, DPM1B, DPM2, DPM2B, DPMB, EROFHC, IC, NLIH1F, NLIH2F, PLA, RETRNI, RETRNI2, RETRNI3, RETRNI4, RETRNI5, RETRNI6, RETRNI7, RETRNI8, RETRNI9, RETRNI10, RETRNI11, RETRNI12, RETRNI13, RETRNI14, RETRNI15, RETRNI16, RETRNI17, RETRNI18, RETRNI19, RETRNI20, RETRNI21, RETRNI22, RETRNI23, RETRNI24, RETRNI25, RETRNI26, RETRNI27, RETRNI28, RETRNI29, RETRNI30, RETRNI31, RETRNI32, RETRNI33, RETRNI34, RETRNI35, RETRNI36, RETRNI37, RETRNI38, RETRNI39, RETRNI40, RETRNI41, RETRNI42, RETRNI43, RETRNI44, RETRNI45, RETRNI46, RETRNI47, RETRNI48, RETRNI49, RETRNI50, RETRNI51, RETRNI52, RETRNI53, RETRNI54, RETRNI55, RETRNI56, RETRNI57, RETRNI58, RETRNI59, RETRNI60, RETRNI61, RETRNI62, RETRNI63, RETRNI64, RETRNI65, RETRNI66, RETRNI67, RETRNI68, RETRNI69, RETRNI70, RETRNI71, RETRNI72, RETRNI73, RETRNI74, RETRNI75, RETRNI76, RETRNI77, RETRNI78, RETRNI79, RETRNI80, RETRNI81, RETRNI82, RETRNI83, RETRNI84, RETRNI85, RETRNI86, RETRNI87, RETRNI88, RETRNI89, RETRNI90, RETRNI91, RETRNI92, RETRNI93, RETRNI94, RETRNI95, RETRNI96, RETRNI97, RETRNI98, RETRNI99, RETRNI100, RETRNI101, RETRNI102, RETRNI103, RETRNI104, RETRNI105, RETRNI106, RETRNI107, RETRNI108, RETRNI109, RETRNI110, RETRNI111, RETRNI112, RETRNI113, RETRNI114, RETRNI115, RETRNI116, RETRNI117, RETRNI118, RETRNI119, RETRNI120, RETRNI121, RETRNI122, RETRNI123, RETRNI124, RETRNI125, RETRNI126, RETRNI127, RETRNI128, RETRNI129, RETRNI130, RETRNI131, RETRNI132, RETRNI133, RETRNI134, RETRNI135, RETRNI136, RETRNI137, RETRNI138, RETRNI139, RETRNI140, RETRNI141, RETRNI142, RETRNI143, RETRNI144, RETRNI145, RETRNI146, RETRNI147, RETRNI148, RETRNI149, RETRNI150, RETRNI151, RETRNI152, RETRNI153, RETRNI154, RETRNI155, RETRNI156, RETRNI157, RETRNI158, RETRNI159, RETRNI160, RETRNI161, RETRNI162, RETRNI163, RETRNI164, RETRNI165, RETRNI166, RETRNI167, RETRNI168, RETRNI169, RETRNI170, RETRNI171, RETRNI172, RETRNI173, RETRNI174, RETRNI175, RETRNI176, RETRNI177, RETRNI178, RETRNI179, RETRNI180, RETRNI181, RETRNI182, RETRNI183, RETRNI184, RETRNI185, RETRNI186, RETRNI187, RETRNI188, RETRNI189, RETRNI190, RETRNI191, RETRNI192, RETRNI193, RETRNI194, RETRNI195, RETRNI196, RETRNI197, RETRNI198, RETRNI199, RETRNI200, RETRNI201, RETRNI202, RETRNI203, RETRNI204, RETRNI205, RETRNI206, RETRNI207, RETRNI208, RETRNI209, RETRNI210, RETRNI211, RETRNI212, RETRNI213, RETRNI214, RETRNI215, RETRNI216, RETRNI217, RETRNI218, RETRNI219, RETRNI220, RETRNI221, RETRNI222, RETRNI223, RETRNI224, RETRNI225, RETRNI226, RETRNI227, RETRNI228, RETRNI229, RETRNI230, RETRNI231, RETRNI232, RETRNI233, RETRNI234, RETRNI235, RETRNI236, RETRNI237, RETRNI238, RETRNI239, RETRNI240, RETRNI241, RETRNI242, RETRNI243, RETRNI244, RETRNI245, RETRNI246, RETRNI247, RETRNI248, RETRNI249, RETRNI250, RETRNI251, RETRNI252, RETRNI253, RETRNI254, RETRNI255, RETRNI256, RETRNI257, RETRNI258, RETRNI259, RETRNI260, RETRNI261, RETRNI262, RETRNI263, RETRNI264, RETRNI265, RETRNI266, RETRNI267, RETRNI268, RETRNI269, RETRNI270, RETRNI271, RETRNI272, RETRNI273, RETRNI274, RETRNI275, RETRNI276, RETRNI277, RETRNI278, RETRNI279, RETRNI280, RETRNI281, RETRNI282, RETRNI283, RETRNI284, RETRNI285, RETRNI286, RETRNI287, RETRNI288, RETRNI289, RETRNI290, RETRNI291, RETRNI292, RETRNI293, RETRNI294, RETRNI295, RETRNI296, RETRNI297, RETRNI298, RETRNI299, RETRNI300, RETRNI301, RETRNI302, RETRNI303, RETRNI304, RETRNI305, RETRNI306, RETRNI307, RETRNI308, RETRNI309, RETRNI310, RETRNI311, RETRNI312, RETRNI313, RETRNI314, RETRNI315, RETRNI316, RETRNI317, RETRNI318, RETRNI319, RETRNI320, RETRNI321, RETRNI322, RETRNI323, RETRNI324, RETRNI325, RETRNI326, RETRNI327, RETRNI328, RETRNI329, RETRNI330, RETRNI331, RETRNI332, RETRNI333, RETRNI334, RETRNI335, RETRNI336, RETRNI337, RETRNI338, RETRNI339, RETRNI340, RETRNI341, RETRNI342, RETRNI343, RETRNI344, RETRNI345, RETRNI346, RETRNI347, RETRNI348, RETRNI349, RETRNI350, RETRNI351, RETRNI352, RETRNI353, RETRNI354, RETRNI355, RETRNI356, RETRNI357, RETRNI358, RETRNI359, RETRNI360, RETRNI361, RETRNI362, RETRNI363, RETRNI364, RETRNI365, RETRNI366, RETRNI367, RETRNI368, RETRNI369, RETRNI370, RETRNI371, RETRNI372, RETRNI373, RETRNI374, RETRNI375, RETRNI376, RETRNI377, RETRNI378, RETRNI379, RETRNI380, RETRNI381, RETRNI382, RETRNI383, RETRNI384, RETRNI385, RETRNI386, RETRNI387, RETRNI388, RETRNI389, RETRNI390, RETRNI391, RETRNI392, RETRNI393, RETRNI394, RETRNI395, RETRNI396, RETRNI397, RETRNI398, RETRNI399, RETRNI400, RETRNI401, RETRNI402, RETRNI403, RETRNI404, RETRNI405, RETRNI406, RETRNI407, RETRNI408, RETRNI409, RETRNI410, RETRNI411, RETRNI412, RETRNI413, RETRNI414, RETRNI415, RETRNI416, RETRNI417, RETRNI418, RETRNI419, RETRNI420, RETRNI421, RETRNI422, RETRNI423, RETRNI424, RETRNI425, RETRNI426, RETRNI427, RETRNI428, RETRNI429, RETRNI430, RETRNI431, RETRNI432, RETRNI433, RETRNI434, RETRNI435, RETRNI436, RETRNI437, RETRNI438, RETRNI439, RETRNI440, RETRNI441, RETRNI442, RETRNI443, RETRNI444, RETRNI445, RETRNI446, RETRNI447, RETRNI448, RETRNI449, RETRNI450, RETRNI451, RETRNI452, RETRNI453, RETRNI454, RETRNI455, RETRNI456, RETRNI457, RETRNI458, RETRNI459, RETRNI460, RETRNI461, RETRNI462, RETRNI463, RETRNI464, RETRNI465, RETRNI466, RETRNI467, RETRNI468, RETRNI469, RETRNI470, RETRNI471, RETRNI472, RETRNI473, RETRNI474, RETRNI475, RETRNI476, RETRNI477, RETRNI478, RETRNI479, RETRNI480, RETRNI481, RETRNI482, RETRNI483, RETRNI484, RETRNI485, RETRNI486, RETRNI487, RETRNI488, RETRNI489, RETRNI490, RETRNI491, RETRNI492, RETRNI493, RETRNI494, RETRNI495, RETRNI496, RETRNI497, RETRNI498, RETRNI499, RETRNI500, RETRNI501, RETRNI502, RETRNI503, RETRNI504, RETRNI505, RETRNI506, RETRNI507, RETRNI508, RETRNI509, RETRNI510, RETRNI511, RETRNI512, RETRNI513, RETRNI514, RETRNI515, RETRNI516, RETRNI517, RETRNI518, RETRNI519, RETRNI520, RETRNI521, RETRNI522, RETRNI523, RETRNI524, RETRNI525, RETRNI526, RETRNI527, RETRNI528, RETRNI529, RETRNI530, RETRNI531, RETRNI532, RETRNI533, RETRNI534, RETRNI535, RETRNI536, RETRNI537, RETRNI538, RETRNI539, RETRNI540, RETRNI541, RETRNI542, RETRNI543, RETRNI544, RETRNI545, RETRNI546, RETRNI547, RETRNI548, RETRNI549, RETRNI550, RETRNI551, RETRNI552, RETRNI553, RETRNI554, RETRNI555, RETRNI556, RETRNI557, RETRNI558, RETRNI559, RETRNI560, RETRNI561, RETRNI562, RETRNI563, RETRNI564, RETRNI565, RETRNI566, RETRNI567, RETRNI568, RETRNI569, RETRNI570, RETRNI571, RETRNI572, RETRNI573,

Module	Sets Parameter	Uses Parameter
C26_MPARAM_7		ATEM0, ATEM1, ATEM2, BIPOLY, C26_MPARAM_7, DIVPRT, DPA, DPA1, DPA1B, DPA2, DPA2B, DPA8, DPA, DPM1, DPM1B, DPM2, DPM2B, DPM8, NHFB, PLAIHS, RETRN1, RETRN2, RETRN3, RETRN4, RETRNS, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SBIVAR, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP1B, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2, UNIVAR, ZTEM1, ZTEM1B, ZTEM2, ZTEM2B, ZTEM3, ZTEM3B, ZTEMA, ZTEMA8, ZTEMS, ZTEM5B, ZTEMA6, ZTEMA8, ZTEM7, ZTEM7B, ZTEMA8, ZTEMA8, ZTEM9, ZTEM9B
C27_NHMMNMX_1	NHBTOT, NHZNM1, NHZNM2, NHZNM3, NHZNM4, PLADOT, PLAZNM1, PLAZNM2, PLAZNM3, PLAZNM4, RETRNE	ATEM0, ATEM1, ATEM2, BIPOLY, C27_NHMMNMX_1, DELTA, DIVPRT, DPA, DPA1, DPA1B, DPA2, DPA2B, DPA8, DPA, DPM1, DPM1B, DPM2, DPM2B, DPM8, PETHD, HOTMOD, IC, NPT, PAMB, PIA, RETRN1, RETRN2, RETRN3, RETRN4, RETRNS, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SBIVAR, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP1B, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2, T15, THETAC, UNIVAR, ZTEM1, ZTEM1B, ZTEM2, ZTEM2B, ZTEM3, ZTEM3B, ZTEMA, ZTEMA8, ZTEMS, ZTEM5B, ZTEMA6, ZTEMA8, ZTEM7, ZTEM7B, ZTEMA8, ZTEMA8, ZTEM9, ZTEM9B
C30_SMRTRQ	DPA1, DPA2, DPA1A, DPA2, DSMNH4, DSMNHK, DSMNH1, DSMNPL, DSMNPR, DSMNH4L, NHKCP, RETRNE, RSPOS1, RSPOS4, RSPNHA, RSTEP1, STEP5, STPRLL, ZTEM1, ZTEMA, ZTEMB	AGRST, ATEM0, ATEM1, ATEM2, C30_SMRTRQ, DMPPCP, DPA, DPA, DKBUMP, DKOUTA, NHK, NHK, NHLG4, RETRN1, RETRN2, RETRN3, RETRN4, RETRNS, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SBIVAR, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP1B, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2, UNIVAR, ZTEMA2, ZTEMA2, ZTEMA, ZTEMS, ZTEME6, ZTEME7, ZTEME8, ZTEME9
C31_TQLIM	DNHTO, DNHTO1, Q1TOT, RETRNE, SP1, SP2, SP3, SP4, SP5, SP6, SYS1, TOEIR, TOFITR, TQLIM2, ZTEMA1, ZTEMA, ZTEMB, ZTEMC, ZTEMO, ZTEME, ZTEMF	ATEM0, ATEM1, ATEM2, BIPOLY, C31_TQLIM, DIVPRT, DPA, DPA1, DPA2, DPA, DPM1, DPM2, HPREF, IC, LDIG, SUBR, NHREQ, NPSE, Q1TOT, RETRN1, RETRN2, RETRN3, RETRN4, RETRNS, RETURN, RTRET1, RTRET2, RTRET3, RETR4, SBIVAR, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP1B, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS2, TQLIM2, UNIVAR, UPTLIG, ZTEMA2, ZTEMA3, ZTEMA4, ZTEMS, ZTEME6, ZTEME7, ZTEME8, ZTEME9
CHKSUM		CHKSUM
CMFACT	INITF, RAMPTR, SBRTN1, TEMP, TEMP1, TEMP2	CMFACT, FLTTAB
CNT_EVTN		ATEM0, ATEM1, ATEM2, CNT_EVTN, RETRN1, RETRN2, RETRN3, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
DBL_LL_SUBR	DPA1, DPA2, DPA1A, DPA2, SBRTN2, SP10, SP6, SP7, SP8, SP9DBL, TEMP1	DBL_LL_SUBR, DPA, DPA, SP1DBL, SP2, SP3, SP4, SP5
DCU_COMPARE_DATA	DCU_BUFFER_PTR, DCU_DATA_PTR, RETRN2, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP20, SP7, SP9	ATEM0, ATEM1, ATEM2, DCU_COMPARE_DATA, DCU_COMPARE_ITEM, RETRN1, RETRN3, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RETRN2, RETRN3, RETRN4, SP1, SP17, SP18, SP19, SP2, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
DCU_COMPARE_ITEM		ATEM0, ATEM1, ATEM2, DCU_COMPARE_ITEM, RETRN1, RETRN2, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RTRET2, RETRN3, RETRN4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
DCU_COMPONT_DATA		ATEM0, ATEM1, ATEM2, DCU_CMD_INVALID, DCU_COMPARE_DATA, DCU_COMPONT_DATA, DCU_CPOMP_END, DCU_CPOMP_TABLE, DCU_CPDATA_ADR, DCU_CPDATA_COPY, DCU_CPDATA_CMP, DCU_CPTEMP_ADR, DCU_CPTEMP_END, DCU_GET_DATA, DCU_GOT_CPDATA, DCU_MISMATCH, DCU_MODIFIED, DCU_NEW, DCU_NO_BLOCK, DCU_UPD_CPDATA, DCU_UPD_ENGDAT, EE_CPDATA_BLOCK, EE_PUT_QUEUE, FSW_CPDATA_READ, FSW_INCONSISTANT, FW_LATCH, FW_MAX_COUNTER, RETRN2, RETRN3, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
DCU_CONSTANTS	DCUSIF, DCU_FLAGS_1, DCU_FLAGS_2, DCU_FUNC_SW, DCU_STATUS, RETRN1, SP3, SP7, SP8, SP9	ACCINFLSTRS, ALL, ACCNUMFL, ALL, ACCSHUTO, ALL, ACCSTRTS, ALL, ATEM0, ATEM1, ATEM2, CPDATA_MISMATCH, DCU_ADDRESS, DCU_BUFFER_ADR, DCU_CHECKSUM, DCU_CHECKSUM_ADR, DCU_CMD_BUFFER, DCU_CMD_INVALID, DCU_COMMAND_ADR, DCU_COMPONT_DATA, DCU_COUNTER_DATA, DCU_CPOMP_END, DCU_CPOMP_TABLE, DCU_CPDATA_ADR, DCU_CPDATA_COPY, DCU_CPDATA_CMP, DCU_CPTEMP_ADR, DCU_CPTEMP_END, DCU_GET_DATA, DCU_GOT_CPDATA, DCU_MISMATCH, DCU_MODIFIED, DCU_NEW, DCU_NO_BLOCK, DCU_UPD_CPDATA, DCU_UPD_ENGDAT, EE_CPDATA_BLOCK, EE_PUT_QUEUE, FSW_CPDATA_READ, FSW_INCONSISTANT, FW_LATCH, FW_MAX_COUNTER, RETRN2, RETRN3, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
DCU_COUNTER_DATA	ACCSHUTO, ALL, ACCSTRTS, ALL, ACCSTR_50, RESINC, DCUSIF, DCU_CTRDAT_4, DCU_DATA_END, DCU_EECRTDAT_UPD, DCU_FLAGS_1, DCU_FLAGS_2, DCU_FUNC_SW, DCU_LCF_UPD, DCU_STATUS, RETRN1, SP10, SP11, SP7, SP8, SP9, SV, PWC_PATCHID, SV_SW_EPL, SV_SW_SBL, SV_SW_SVI	ATEM0, ATEM1, ATEM2, DCU_ADJUST, RSYNC, DCU_ADJUST, SHUTO, DCU_ADJUST, STRTS, DCU_BUFFER_ADR, DCU_CMD_INVALID, DCU_COUNTER_DATA, DCU_CTRDAT_ADR, DCU_CTRDATA_COPY, DCU_CTRDATA_END, DCU_CTRDAT_RANGE, DCU_CTREND_RANGE, DCU_GET_DATA, DCU_GOT_CTRDATA, DCU_MISMATCH, DCU_MODIFIED, DCU_NEW, DCU_NO_BLOCK, DCU_OBSELETE, DCU_RANGE_CHECKS, DCU_TSHIT_MM, MSK, DCU_UPD_CTRDAT, EE_CTRDAT_BLOCK, EE_CTRDAT_IF, EE_PUT_QUEUE, ENGPIF, FSW_CTRDATA_READ, FSW_CTR_EEC, FIT, FW_LATCH, FW_MAX_COUNTER, KSLUD, KSSVID, PWSIS2, RETRN2, RETRN3, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
DCU_DCUID_DATA	DCUSIF, DCU_BUFFER_PTR, DCU_DATA_END, DCU_DATA_PTR, DCU_DATA_READY, DCU_DCUID_IF, DCU_FUNC_SW, RETRN1, SP10, SP11, SP7, SP8, SP9	ATEM0, ATEM1, ATEM2, DCU_BUFFER_ADR, DCU_CHECKSUM, DCU_COMPARE_DATA, DCU_DCUCMP_END, DCU_DCUCMP_TABLE, DCU_DCUDAT_READY, DCU_DCUCND_RANGE, DCU_DCUID_ADR, DCU_DCUID_DATA, DCU_DCUID_END, DCU_DCUID_L1, DCU_DCUID_L1TMP, DCU_DCUID_RANGE, DCU_DCUID_SUM, DCU_GET_CHECKSUM, DCU_GET_DATA, DCU_RANGE_CHECKS, DCU_STATUS, FSW_DCUID_READ, FW_LATCH, FW_MAX_COUNTER, RETRN2, RETRN3, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
DCU_EEC_DATA	DCUSIF, DCU_BUFFER_PTR, DCU_FLAGS_1, DCU_FUNC_SW, DCU_STATUS, RETRN1, SP13, SP14, SP15, SP16	ATEM0, ATEM1, ATEM2, DCU_BUFFER_ADR, DCU_CMD_INVALID, DCU_COMPARE_ITEM, DCU_EECDATA_ADR, DCU_EECPN_END, DCU_EEC_DATA, DCU_EEC_PN, DCU_MISMATCH, DCU_MODIFIED, DCU_NEW, DCU_NO_BLOCK, DCU_UPD_EECDAT, FSW_EECDATA_READ, FW_LATCH, FW_MAX_COUNTER, MODUCDATA, RETRN2, RETRN3, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
DCU_ENGINE_DATA	DCUSIF, DCU_FLAGS_1, DCU_FLAGS_2, DCU_FUNC_SW, DCU_STATUS, RETRN1, SP3, SP7, SP8, SP9	ATEM0, ATEM1, ATEM2, DCU_CMD_INVALID, DCU_COMPARE_DATA, DCU_ENGCMP_END, DCU_ENGCMP_TABLE, DCU_ENGDATA_ADR, DCU_ENGDATA_COPY, DCU_ENGDAT_CMP, DCU_ENGDAT_CMP_END, DCU_ENGCMP_ADR, DCU_ENGCMP_END, DCU_GET_DATA, DCU_GOT_ENGDATA, DCU_MISMATCH, DCU_MODIFIED, DCU_NEW, DCU_NO_BLOCK, DCU_UPD_ENGDAT, EE_ENGDAT_BLOCK, EE_PUT_QUEUE, FSW_ENGDATA_READ, FW_LATCH, FW_MAX_COUNTER, RETRN2, RETRN3, RETRN4, RETRNS, RETRNE, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2

Module	Sets Parameter	Uses Parameter
EE_WRITE_BLOCK	EEBUSV, EE_DATA_ADR, EE_DATA_PTR, EE_REQUEST_DONE, EE_WRITING_RAM_DATA_ADR, RAM_DATA_END, RAM_DATA_PTR, RETURN, SP1, SP2	ATEM0, ATEM1, ATEM2, EE_GET_QUEUE, EE_QUEUE_EMPTY, EE_WRITE_BLOCK, RETRN1, RETRN2, RETRN3, RETRN4, RETRNS, RETRNG, RETRET1, RETRET2, RETRET3, RETRET4, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
EEUIM_16	ROMPT1, ROMPT2, ROMRET	EEUIM_16
EEUIM_8	ROMRET, SP0, SP1, SP2, SP3	EEUIM_8
FAILED	SBRTN3, SP1	FAILED, SP2
FAILT	SBRTN3, SP1, SP3	FAILT, SP2, SP4
FILCON	SBRTN2, SP0, SP4, SP5, SP6, SP7	FILCON, SP1, SP2, SP3
FILTER	SBRTN2, SP4, SP6, SP7	FILTER, IC, SP3, SP5
FLTTIMER	FLTSEC, FLTSEC_SCL, PV, FLTTC, FLTIF, FLT_CONFIG_P, FLT_MS_RAW, FLT_TIMER_RESET, RETRN1, SP2	ATEM0, ATEM1, ATEM2, CONFIG, ENGRUN, FLIGHT, FLTTIMER, RETRN2, RETRN3, RETRN4, RETRNS, RETRNG, RETURN, RETRET1, RETRET2, RETRET3, RETRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2
GETWRD	BOOT2, BOOT5	GETWRD, URTRED
GOTO	BOOT5	BOOT2, GETWRD, GOTO
I01_LIDRIFT_1	PRETRN, RAWDOF	ADINPT, I01_LIDRIFT_1
I01_LIDRIFT_2	CIF, DELPFI, DRFTIC, DRFTIF, PSF, RETURN, SGCIF, SP1, SP2, SP3, SP4, T18FI, TQIF	DRFTCT, FAILT, I01_LIDRIFT_2, PASSLT, RAWDOF
I02_SGCOM_1	PRETRN, RAWSG	ADINPT, I02_SGCOM_1
I02_SGCOM_2	DELPFI, PSF, RETURN, SGCG, SP1, SP2, SP3, SP4	DRFTIC, FAILT, I02_SGCOM_2, PASSLT, RAWSG, SGCTCT
I03_RDCT_1	PRETRN, RAWCI	ADINPT, I03_RDCT_1
I03_RDCT_2	CJ, CIMV, CRPFT, DPTABH, DPTAB1, POTABH, POTAB1, RETURN, TEMP1, TEMP2, TEMP3	CJ, DPTAB1, DPTAB2, I03_RDCT_2, POTAB1, POTAB2
I03_RDCT_3	CJ, CIF, RETURN, SP1, SP2, SP3, SP4, TEMP1	CTCT, DRFTIC, FAILT, I03_RDCT_3, PASSLT, RAWCI
I04_PSAMBSG_1	PRETRN, RAWPS	ADINPT, I04_PSAMBSG_1
I04_PSAMBSG_2	PSC, PSDAT, PSF, PSTCT, RETURN, SP1, SP2, SP3, SP4	DRFTIC, EDATIF, FAILT, I04_PSAMBSG_2, PASSLT, RAWPS, SGCG, SGCIF
I04_PSAMBSG_3	LTP, PSAMB, RETURN, SP1, SP4, SP5, SP6, TEMP1, TEMP2, TEMP3, TP	CRPFT, FILTER, I04_PSAMBSG_3, PSAMB, POTABH, POTAB1, PSAMB, PSC, PSDAT
I04_PSAMBSG_4	PSAMB, PSOCOR, RETURN, SP1, SP3, SP5	I04_PSAMBSG_4, PSAMB, SUNVAR
I04_PSAMBSG_5	ALTP, RETURN, SP1, SP3, SP5	I04_PSAMBSG_5, PSAMB, SUNVAR
I05_DELTAPSG_1	PRETRN, RAWPLP	ADINPT, I05_DELTAPSG_1
I05_DELTAPSG_2	DELTAP, LTP, RETURN, TEMP1, TEMP2, TEMP3, TP	CRPFT, DELPC, DELPFI, DPSIZE, DPTABH, DPTAB1, EDATIF, I05_DELTAPSG_2
I05_DELTAPSG_3	DPOVRP, IASDP, INTIAS, POVEL, RETURN, SP1, SP2, SP3, SP4, SP5, SP6, SP7, SP8, SP9	DELTAP, DP18F, FILTER, I05_DELTAPSG_3, IC, POVEL, P18OF1, P18OF2, PAMB, QUAD, SUNVAR
I05_DELTAPSG_4	DELPFI, DELPFI, DELTAP, RETURN, SP1, SP2, SP3, SP4	DELCT, DRFTIC, EDATIF, FAILT, I05_DELTAPSG_4, PASSLT, RATESW, RAWPLP, SGCG, SGCIF
I07_READTO_1	PRETRN, RAWTO	ADINPT, I07_READTO_1
I07_READTO_2	RETURN, SP3, SP4, SP5, TQC	FILTER, I07_READTO_2, IC, INTTO, TQCI, TQIP
I07_READTO_3	RETURN, SP1, SP2, SP3, SP4, TEMP, TQCI, TQIF	DRFTIC, FAILT, I07_READTO_3, PASSLT, RAWTO, TQCTCT
I08_ROT18_1	PRETRN, RAWT18	ADINPT, I08_ROT18_1
I08_ROT18_2	RETURN, SP3, SP5, T18T15	I08_ROT18_2, PAMB, SUNVAR
I08_ROT18_3	HITEMP, RETURN, RTDATT, SP1, SP2, SP3, SP4, SP5, SP6, T18C, T18FI, T18FI, T18TCT, TEMP1	DRFTIC, FAILED, FAILT, FILTER, I08_ROT18_3, NH, PASSED, PASSLT, RAWT18, T18T5F, T18P
I08_ROT18_4	RETURN, T15F, T18CFL	I08_ROT18_4, T18T5F, T18T15
I09_HLDRIFT_1	PRETRN, RAWDOF	ADINPT, I09_HLDRIFT_1
I09_HLDRIFT_2	CTIF, DRFCF, DRFTIC, GAINFI, RETURN, SP1, SP2, SP3, SP4, T8TIF, TGTIF	DRHTCT, FAILT, I09_HLDRIFT_2, PASSLT, RAWDOF
I10_GAINCOM_1	PRETRN, RAWGAN	ADINPT, I10_GAINCOM_1
I10_GAINCOM_2	CTIF, GAINC, GAINFI, RETURN, SP1, SP2, SP3, SP4, T8TIF, TEMP1, TGTIF	DRIFTCT, FAILT, GINTCT, I10_GAINCOM_2, PASSLT, RAWGAN
I11_TQBTIM_1	PRETRN, RAWTB	ADINPT, I11_TQBTIM_1
I11_TQBTIM_2	QBTM, QBTC, QBTCM, QBTCM, RETURN, SP1, SP2, SP3, SP4, TBTIF	DRIFTCT, EROMIF, FAILT, GAINC, I11_TQBTIM_2, IC, PASSLT, RAWTB, TBTCT
I12_TQGTIRM_1	PRETRN, RAWTG	ADINPT, I12_TQGTIRM_1
I12_TQGTIRM_2	QGTM, QGTMC, QGTMC, QGTMC, RETURN, SP1, SP2, SP3, TGTIF	DRIFTCT, EROMIF, FAILT, GAINC, I12_TQGTIRM_2, IC, PASSLT, RAWTG, TGTCT
I13_ENGVARIANT_1	PRETRN, RAWEV	ADINPT, I13_ENGVARIANT_1
I13_ENGVARIANT_2		ATEM0, ATEM1, ATEM2, BIPOLY, DERAT1, DERAT2, DEVMODEN, DVPRPT, DPA, DPA1, DPA1B, DPA2, DPA2B, DPAB, DPM, DPM1, DPM1B, DPM2, DPM2B, DPMB, DRFTCT, EROMIF, GAINC, I13_ENGVARIANT_2, IC, RAWEV, RETRN1, RETRN2, RETRN3, RETRN4, RETRNS, RETRNG, RETRET1, RETRET2, RETRET3, RETRET4, SBVAR, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2, UNIVAR, ZTEM1, ZTEM1B, ZTEM2, ZTEM2B, ZTEM3, ZTEM3B, ZTEM4, ZTEM4B, ZTEM5, ZTEM5B, ZTEM6, ZTEM6B, ZTEM7, ZTEM7B, ZTEM8, ZTEM8B, ZTEM9, ZTEM9B
I14_RDPLA_1	PLA1, PLAFI, PRETRN, RAWPL1, STAT1	ADINPT, I14_RDPLA_1, IC
I14_RDPLA_2	PLA1, PLAFI, PLAFI, PRETRN, PTEMP, RAWPL2, STAT2	ADINPT, I14_RDPLA_2, IC, PLADRF, PLAG, STAT1
I14_RDPLA_3	PLAC, PLACT, PLAHYS, PLAFI, PLATOT, RETURN, SP2	FAILED, I14_RDPLA_3, IC, PASSED, PLA1, PLAFI, PLAFI, PLADRF, PLADRF, STAT1, STAT2
I14_RDPLA_4	RETURN, SP3, SP4, SP5, SP6	FILTER, I14_RDPLA_4, PLAHYS, PLAIN, PLAP
I15_PLABIAS_1	PRETRN, RAWPLB	ADINPT, I15_PLABIAS_1
I15_PLABIAS_2	PLAB, PLABFI, PLADRF, PLAFI, RETURN, SP1, SP2, SP3	FAILT, I15_PLABIAS_2, PASSLT, PRCTCT, PLAG, RAWPLB
I16_PLAGAIN_1	PRETRN, RAWPLG	ADINPT, I16_PLAGAIN_1
I16_PLAGAIN_2	PLAG, PLAGFI, PLAFI, RETURN, SP1, SP2, SP3, SP4	FAILT, I16_PLAGAIN_2, PASSLT, PGCTCT, RAWPLG
I17_PNHCHV_1	NH1FI, NH2FI, NHSEL, PRETRN, PTEMP	DISC1, I17_PNHCHV_1, NHCECS1, NHCECS2, NPT
I17_PNHCHV_2	PRETRN	I17_PNHCHV_2
I18_READNH	NH1GDO, NH1FI, NH1STA, NH2FI, NHCECS1, NHCECS2, NH1, NH1FI, NH2FI, NHP, NHT, NHTMP1, NNH1, PRETRN, PSP1, PTEMP, PTEMP1, PTEMP2, PTEMP3, PTEMP4, HBOVFI, NHTIF, RETURN	DISC1, I18_READNH, IC, NHSEL, NPT
I19_NHDUAL_1		I19_NHDUAL_1, NH1FI, NH2FI, NHCECS1, NHCECS2
I19_NHDUAL_2	RETURN, SP1, SP2, SP3, SP4	DISC1, FAILT, I19_NHDUAL_2, NH, NH1FI, NH2, NH2FI, NHCECS1, NHCECS2, NH1FI, NH2FI, NHTCT, NHOX, NPT, PASSLT
I20_NHSPED	NH, NHP, RETURN, SP3, SP4, SP5, SP6	FILTER, I20_NHSPED, NH1, STQPI
I22_SNHCHV_1	NH2FI, PRETRN, PTEMP	I22_SNHCHV_1, NHCECS2, NHSEL
I22_SNHCHV_2	PRETRN	I22_SNHCHV_2
I23_ROSNHP	NH2GDO, NH2FI, NH2STA, NHCECS2, NH2, NH2FI, NHP2, NHT2, NHTMP2, NNH2, PRETRN, PTEMP, PTEMP1, PTEMP2	DISC1, I23_ROSNHP, IC, NHSEL, NPT
I24_NHPROBE2	NH2, NHP2, RETURN, SP3, SP4, SP5, SP6	FILTER, I24_NHPROBE2, NH2, STQPI
I25_TQNPT_1	RETURN	I25_TQNPT_1, QCNTRL, READQ1, READQ2
I25_TQNPT_2	CNTA, CNTH, CNTC, NGOOD1, NP1FI, NP1SW, Q1SW, QCNCT, GGOOD1, QTIMER, Q1OT1SW, SBRTN1, SP1, SP2	DISC1, FAILED, I25_TQNPT_2, NH, NP2FI, NPT, Q1FI, Q1FI, Q1SW, QCNTRL, QCONV1, QINSEL, READQ1
I25_TQNPT_3	CNTA, CNTH, CNTC, NGOOD2, NP2FI, NP2SW, Q2SW, QCNCT, GGOOD2, QTIMER, Q2OT1SW, SBRTN1, SP1, SP2	DISC1, FAILED, I25_TQNPT_3, NH, NP2FI, NPT, Q1SW, Q2FI, QCNTRL, QCONV2, QINSEL, READQ2
I25_TQNPT_4	NGOOD1, NGOOD2, NNS1, NP1FI, NP1SW, NS1, NSTAT1, NTEMP1, Q1FI, Q1FI, Q1SW, QDEF1, GGOOD1, GGOOD2, QGDEF1, QSTAT1, QTEMP1, RANNS1, RAWK1, SBRTN2, SP1, SP10, SP2, SP3, T1, TM	CNTA, CNTH, CNTC, DISC1, FAILED, I25_TQNPT_4, IC, NPT, PASSED, QCNCT, QCONV1, QRAETH, QRAETH, QRAETH
I25_TQNPT_5	NGOOD1, NGOOD2, NNS2, NP2FI, NP2SW, NS2, NSTAT2, NTEMP2, Q2FI, Q2FI, Q2SW, QDEF2, GGOOD1, GGOOD2, QGDEF2, QSTAT2, QTEMP2, RANNS2, RAWK2, SBRTN2, SP1, SP10, SP2, SP3, T1, TM	CNTA, CNTH, CNTC, DISC1, FAILED, I25_TQNPT_5, IC, NPT, PASSED, QCNCT, QCONV2, QRAETH, QRAETH, QRAETH
I25_TQNPT_6	NSI, QCNTRL, QDEF, Q1OT1SW, SBRTN2	I25_TQNPT_6, IC, NP1FI, NP1SW, NP2FI, NP2SW, NPT, NS1, NS2, Q1FI, Q1FI, Q1SW, Q2FI, Q2FI, Q2SW, QDEF1, QDEF2, QINSEL
I26_TQCAL	QRAETH, QRAETH, QRAETH, RETURN	I26_TQCAL_2, IC
I27_TQSPED	QSTIFF, Q1OT1, RETURN, TO	I26_TQCAL, QBTM, QDEF, QDEF2, QGTM, TQIC
I28_DISINW1	RETURN, SP3, SP4, SP5, SP6	FILTER, I27_TQSPED, NS, NS1, NSHPS
I28_DISINW1	BLEED, D1, D1TMR, D1Z, D1Z2, DISC1, PRETRN, PTEMP, PTEMP1, RATESW, IC	I28_DISINW1, IC, PLA
I29_DISINW2	DISC1, PRETRN	I29_DISINW2
I30_DISINW2	FLTWRO, PRETRN	I30_DISINW2
I31_UARTIN	PRETRN, PRT410, STATUS	DISC2, I31_UARTIN, NPARAM

[illegible]

[illegible]

[illegible]

Module	Sets Parameter	Uses Parameter
UART_IO_SELECT		ATEM0, ATEM1, ATEM2, DCU_DATA_READY, DCU_FUNC_SW, DCU_IO_QUIET, DISC2, FSW_DCU_INACTIVE, RETRN2, RETRN3, RETRN4, RETRNS, RETRNG, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2, UART_IO_SELECT
	NPARAM, RETRN1	
UNIVAR	ATEM0, RETRN1, SP1, SP17, SUBVR1, SUBVR2, SUBVR3, SUBVR4	ATEM1, ATEM2, RETRN2, RETRN3, RETRN4, RETRNS, RETRNG, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP7, SP8, SP9, STACK, SUBVR5, SUBVR6, SYS1, SYS2, UNIVAR
UPDATE_EE_DATA	DCU_EECTRDAT_UPD, EE_UPDATE_FLAGS, RESYNC_WRITTEN, RETRN1, SHUTDOWN_WRITTEN, SP7, SP8, STARTUP_WRITTEN	ATEM0, ATEM1, ATEM2, DCU_CTRDAT_EEREQ, DCU_DATA_READY, DCU_DCUDAT_READY, DCU_LCF_UPD, EE_CTRDAT_BLOCK, EE_PUT_QUEUE, ENGSHUTDOWN, ENGSTART, RETRN2, RETRN3, RETRN4, RETRNS, RETRNG, RETURN, RTRET1, RTRET2, RTRET3, RTRET4, SP1, SP10, SP11, SP12, SP13, SP14, SP15, SP16, SP17, SP18, SP19, SP2, SP20, SP21, SP22, SP23, SP24, SP25, SP26, SP27, SP28, SP29, SP3, SP30, SP31, SP32, SP33, SP34, SP35, SP36, SP4, SP5, SP6, SP9, STACK, SUBVR1, SUBVR2, SUBVR3, SUBVR4, SUBVR5, SUBVR6, SYS1, SYS2, UPDATE_EE_DATA
URTOUT	BOOT6	URTOUT
URTRED	BOOT6	URTRED
VAR_TUSTIN_SUBR	SBRTN2, SP0, SP2, SP5, SP6, SP7	IC, SP1, SP3, SP4, VAR_TUSTIN_SUBR
WDGRST	SP1	WDGRST
WRITE	BOOT1, BOOT2, BOOT3	GETWRD, SNDWRD, URTOUT, WDGRST, WRITE
WRTEEP	BOOT1, BOOT2, BOOT3, BOOT4, BOOT5, BOOT6	EFLTTB, GETWRD, SNDWRD, URTOUT, URTTBL, WDGRST, WRTEEP

10.00 DATA DICTIONARY LISTING

10.01 GLOBAL PARAMETERS

10.01 Table 1: Global-Parameters

Name	Description	Type	Units	Value	Min Val	Max Val
@@_PW127L_DD	DATA DICTIONARY FOR THE PW127L PROGRAM					
A134SW	ARINC LABEL 134 STATUS WORD	Integer_Data				
A203SW	ARINC LABEL 203 STATUS WORD	Integer_Data				
A206SW	ARINC LABEL 206 STATUS WORD	Integer_Data				
A211SW	ARINC LABEL 211 INTERFACE STATUS WORD	Integer_Data				
A271SW	ARINC LABEL 271 STATUS WORD	Integer_Data				
ACCEL	ACCELERATION SCHEDULE SELECTED	Analog_Data	rpm/sec		0	6045
ACCEL1	CORRECTED ACCEL LIMIT CALCULATED	Analog_Data	rpm/sec		750	3000
ACCEL5	NPT GOV. SELECTED	Analog_Data	rpm/sec		0.0	4000
ACCELG	NPT GOV. ACCEL SCHEDULE	Analog_Data	rpm/sec		0.0	4000
ACCELP	NOMINAL ACCEL LIMIT CALCULATED	Analog_Data	RPM/Sec		487.5	3000
ACCINFLSTRTS_ALL	Accumulated Inflight Starts	Analog_Data	Counts		0	32767
ACCINFLSTRTS_E	Accumulated Starts Followed by Flight (PW127E Engine Model)	Analog_Data	Counts		0	32767
ACCINFLSTRT_ALL	Accumulated Inflight Starts (All Engine Models)	Analog_Data	Counts		0	32000
ACCLSW	ENABLE MAX UPTRIM ACCEL LIMIT	Boolean_Data				
ACCNUMFL_ALL	Accumulated Number of Flights	Analog_Data	Counts		0	65535
ACCNUMFL_E	Accumulated Number of Flights (PW127E Engine Model)	Analog_Data	Counts		0	32767
ACCSHUTD_ALL	Accumulated Shut Down	Analog_Data	Counts		0	65535
ACCSTRTSFLT_ALL	Accumulated Starts Followed by Flight	Analog_Data	Counts		0	65535
ACCSTRTSFLT_E	Accumulated Starts Followed by Flight (PW127E Engine Model)	Analog_Data	Counts		0	32767
ACCSTRTS_ALL	Accumulated Starts	Analog_Data	Counts		0	65535
ACCSTRTS_E	Accumulated Starts (PW127E Engine Model)	Analog_Data	Counts		0	32767
ACCSTR_SD_RESYNC	No of Acc. Starts/Shutdown Resynch Counter	Analog_Data	Counts		0	32767
ADCIAS	INDICATED AIR SPEED	Analog_Data	KNOTS		0	350
ADELNP	ABSOLUTE DIFFERENCE BETWEEN QNP AND ACTUAL NP	Analog_Data	RPM		0.0	1200.0
ADINPT	Latest reading from the A/D Converter	Analog_Data	Counts			
ADSTAT	A/D Status Word	Integer_Data				
AIGRST	1=AUTOIGNITION INTEGRATOR RESET ENABLE	Boolean_Data				
ALT	ALTITUDE CALCULATED	Analog_Data	ft		-2000.0	35000.0
ALT203	ADC PRESSURE ALTITUDE	Analog_Data	FEET		-65536	65535
ALTDAT	ADC AMBIENT PRESSURE DATA VALIDITY FLAG	Integer_Data			0	2
ALTIF	DUAL AMBIENT PRESSURE SOURCE FAILURE STATUS WORD	Integer_Data				
ALTPS	Pressure Altitude from the pressure sensor	Analog_Data	FEET		-2000	35000
ALTRGC	RANGE CHECKED ADC ALTITUDE VALUE	Analog_Data	FEET		-2000	32002
ALTYM	AMBIENT PRESSURE CROSSCHECK FAULT TIMER	Integer_Data	Counter		0	121
ALTIXF	AMBIENT PRESSURE CROSSCHECK STATUS WORD	Integer_Data				
AM8TIM	ADC PRESS. ALTITUDE INVALID DATA TIMER	Integer_Data	Counter		0	120
AMODE1	Index into ram fault maintenance block	Integer_Data				
AMODE2	Index into ram fault maintenance block	Integer_Data				
AR203H	MS WORD OF ARINC LABEL 203 DATA	Integer_Data				
AR203L	LS WORD OF ARINC LABEL 203 DATA	Integer_Data				
AR206H	MS WORD OF ARINC LABEL 206 DATA	Integer_Data				
AR206L	LS WORD OF ARINC LABEL 206 DATA	Integer_Data				
AR211H	PARITY BIT IN MS WORD OF ARINC LABEL 211 DATA	Integer_Data				
AR211L	PADS AND/OR DATA IN LS WORD OF ARINC LABEL 211 DATA	Integer_Data				
ARCCNT	ARINC DATA NOT AVAILABLE TIMER	Integer_Data	Counter		0	4808
AREG	Result of the Checksum	Integer_Data				
ARENGV	ARINC ENGINE VARIANT RATING MODE	Integer_Data			0	4
ARILSH	LS WORD OF ARINC INPUT DATA	Integer_Data				
ARIMSH	MS WORD OF ARINC INPUT DATA	Integer_Data				
ARINIT	ARINC RATED POWER DATA READ SUCCESSFULLY	Boolean_Data				
ARINSW	A206S	Integer_Data				
ARINSW_14	NO ARINC RECEIVED FLAG IN INTERFACE STATUS WORD	Integer_Data				
ARINSW_16	FAULT LATCH FLAG IN INTERFACE STATUS WORD	Integer_Data				
ARMASI	APPROACH SPEED CONTROL ARM VALUE	Boolean_Data				
AROIF	ARINC Output Interface Status Word	Integer_Data				
AROLSH	LS word of the ARINC Message	Integer_Data				
AROMSH	MS word of the ARINC Message	Integer_Data				
ARO_LISTA_SEQ	CURRENT SEQUENCE NUMBER FOR FIRST 06X LABEL (Index into the list of Label 062 data)	Integer_Data			0	15
ARO_LISTB_SEQ	CURRENT SEQUENCE NUMBER FOR SECOND 06X LABEL (Index into the list of Label 063 data)	Integer_Data			0	7
ARO_LISTC_SEQ	CURRENT SEQUENCE NUMBER FOR THIRD 06X LABEL (Index into the list of Label 064 data)	Integer_Data			0	7
ARO_LISTD_LOBYTE	LOW BYTE FLAG FOR FOURTH 06X ARINC LABEL	Boolean_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
ARO_LISTD_SEQ	CURRENT SEQUENCE NUMBER FOR FOURTH 06X LABEL (Index into the list of Label 065 data)	Integer_Data			0	14
ARO_LISTE_LOBYTE	LOW BYTE FLAG FOR FIFTH 06X ARINC LABEL	Boolean_Data				
ARO_LISTE_SEQ	CURRENT SEQUENCE NUMBER FOR FIFTH 06X LABEL (Index into the list of Label 066 data)	Integer_Data			0	62
ARO_LISTF_LOBYTE	LOW BYTE FLAG FOR SIXTH 06X ARINC LABEL	Boolean_Data				
ARO_LISTF_SEQ	CURRENT SEQUENCE NUMBER FOR SIXTH 06X LABEL (Index into the list of Label 067 data)	Integer_Data			0	53
ARO_OUT_COUNTER	Mask used along with the invocation rate mask. Its values are 00, 01, 10, 11. When 'AND'd with the invocation masks (00, 01, 11), a value of 0 will allow for signals to execute every 6, 12 or 24 invocations.	Integer_Data				
ARO_TABLE_ADR	START ADDRESS OF ARINC 06X LABELS TABLE	Integer_Data				
ARO_TABLE_END	END ADDRESS OF ARINC 06X LABELS TABLE	Integer_Data				
ARTEMP	RAW ARINC ENGINE VARIANT RATING MODE VALUE	Integer_Data			0	4
ASAVE	COPY OF THE A REGISTER	Integer_Data				
ASAVE2	COPY OF THE A REGISTER	Integer_Data				
ASLMFL	1=SLAM ACCEL DETECTED	Boolean_Data				
ASLMSP	ACCEL SLAM SETPOINT	Analog_Data	rpm/sec	0	0.0	3000.0
ATABLE	START ADDRESS OF CONFIGURATION TABLE	Integer_Data				
AUTOFL	1=PERMIT NORMAL FUNCTION OF AUTOIGNITION LOGIC	Boolean_Data				
B134CT	INTERMITTENT ARINC 134 DATA FAULT COUNTER	Integer_Data	Counter		0	6
B203CT	INTERMITTENT ARINC 203 DATA FAULT COUNTER	Integer_Data	Counter		0	6
B206CT	INTERMITTENT ARINC 206 DATA FAULT COUNTER	Integer_Data	COUNTER		0	6
B211CT	INTERMITTENT ARINC 211 DATA FAULT COUNTER	Integer_Data			0	6
BAD134	ARINC 134 INTERMITTENT DATA FAILURE	Boolean_Data				
BAD203	ARINC 203 INTERMITTENT DATA FAILURE	Boolean_Data				
BAD206	ARINC 206 INTERMITTENT DATA FAILURE	Integer_Data			0	1
BAD211	ARINC 211 INTERMITTENT DATA FAILURE	Boolean_Data				
BAD_CHECKSUM	DCU BAD CHECKSUM DATA BIT	Integer_Data		14		
BCBIT	Bit to set BAD_CHECKSUM Bit in DCU_STATUS	Integer_Data		14		
BKGTM	Foreground-functional Timer	Integer_Data			0	3
BLDELO	1=CHANGE IN BLCODE DELAYED OFF	Boolean_Data				
BLDR26	INDICATES A NEW 3 BYTE BLOCK HAS TO BE BUILT	Boolean_Data				
BLDREQ	BLEED VALUE REQUEST	Analog_Data	MA		0.0	75.0
BLEED	Selected BLEED based upon discrete inputs	Integer_Data			0	2
BOOST	NPT U/S BOOST VALUE BEFORE NHF GAIN	Analog_Data	rpm		0.0	9000.0
BOOT2	The word read from the UART	Integer_Data				
BOOT3	DATA BLOCK INDEX	Integer_Data				
BOOT4	LAST ADDRESS OF DATA BLOCK	Integer_Data				
BSAVE	COPY OF THE B REGISTER	Integer_Data				
BSAVE2	COPY OF THE B REGISTER	Integer_Data				
BSTGN	NPT U/S OUTPUT FROM MAP BSTTTB	Analog_Data	---		1.0	1.5
BSTNPT	NPT U/S BOOST TERMINATION VALUE	Analog_Data	rpm		-150	200.0
BSTOFF	NPT U/S NPT VALUE REQUIRED TO TERMINATE BOOST	Analog_Data	rpm		10000.0	21000.0
BSTPDB	NPT U/S DIFF. BETWEEN BSTPLA AND PLADB VALUE	Analog_Data	deg		-30.0	45.0
BSTPLA	NPT U/S OUTPUT FROM MAP BSTPTB	Analog_Data	deg		18.0	25.0
BUILDR	INDICATES A NEW 3 BYTE BLOCK HAS TO BE BUILT	Boolean_Data				
BYPASS	ONE TIME INITIALIZATION FLAG	Boolean_Data				
BYTEF	LSB AND MSB BYTE FLAG	Integer_Data				
CCORFM3	MTO/ATO Correction for PW127F/M/N from PW127 Baseline	Analog_Data				
CJ	COLD JUNCTION TEMPERATURE IN DEGREES F	Analog_Data	DG F		-73.2	256
CJC	Compensated Cold Junction Temp. Reading	Analog_Data	mV		-4.86	58.51
CJIF	Cold Junction Temp. interface status word	Integer_Data				
CJMV	Cold Junction Correction Factor	Analog_Data	mV		-2.19	5.19
CJCT	Cold Junction Temperature Transient Timer	Integer_Data	counts		0	25
CKSUM	UART Message Block Checksum	Integer_Data				
CKSUMH	HIGH WORD OF PROGRAM MEMORY CHECKSUM	Integer_Data				
CKSUML	LOW WORD OF PROGRAM MEMORY CHECKSUM	Integer_Data				
CNT134	ARINC 134 DATA_NOT_AVAILABLE COUNTER	Integer_Data	Counter		0	61
CNT203	ARINC 203 DATA NOT AVAILABLE TIMER	Integer_Data	Counter		0	120
CNT206	ARINC 206 DATA_NOT_AVAILABLE TIMER	Integer_Data			0	122
CNT211	ARINC 211 DATA_NOT_AVAILABLE TIMER	Integer_Data	COUNTER		0	60
CNT271	ARINC 271 DATA_NOT_AVAILABLE COUNTER	Integer_Data	Counter		0	61
CNTA	DATA READ FROM TORQUE COUNTER A	Analog_Data			0	0.04369
CNTB	DATA READ FROM TORQUE COUNTER B	Analog_Data			0	0.04369
CNTC	DATA READ FROM TORQUE CONVERTER CONTROL PORT	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
CNTERP	ERSP ARINC 211 DATA NOT AVAILABLE COUNTER	Integer_Data	Counter		0	61
CNTPNT	Table Pointer for the Control State Table	Integer_Data				
CONFIG	CONFIGURATION - BASED ON ENGINE RATING AND PROPELLER (Values are in octal) 1 Spare 2 Spare 3 Spare 4 Spare 5 PW127S Engine	Integer_Data			1	5
CONNIF	Loss of connector test interface status word	Integer_Data				
COUNT	1 COUNT PER BIT FOR ONE OF THE 6 NCFAL WORDS	Integer_Data				
CRPAMD	AMBIENT PRESSURE MIXING FACTOR, WHEN RATESW=4 OR 5	Analog_Data	PSIA/PSIA		0	1
CRPFTR	PRESSURE SENSOR Y-AXIS CREEP FACTOR	Analog_Data			0	1
CRT15D	FADER RATIO TO SELECT BETWEEN THE ADC TAT AND EEC TEMP.	Analog_Data	DEG F/DEGF		0	1
CTIF	Trim resistor engine variant config I/F status word (Initial value = '140) 1 4 Fault Counter 5 (spare) 6 Drift Correction Data Invalid Flag 7 Gain Correction Data Invalid Flag 8 (spare) 9 (spare) 10 (spare) 11 Configuration Conflict Flag 12 (spare) 13 (spare) 14 Range Check Flag 15 EOC Fault Flag 16 Fault Latch	Integer_Data				
CTRDAT_READY	Counter data from DCU ready	Boolean_Data				
CYCCRS	COARSE CYCLE COUNT	Analog_Data				
CYCFIN	FINE CYCLE COUNT	Analog_Data				
D1	Current DISC1 value read from input port	Integer_Data				
D1TMR	Discrete 1 Timer	Integer_Data			-27	0
D1TMR6	Bit 6 Delay timer	Integer_Data			-27	0
D1Z	Past value of D1	Integer_Data				
D1ZZ	Past value of D1Z	Integer_Data				
DATA_IO_STATE	STATE VALUE FOR SENDING, RECEIVING COMMANDS AND DATA TO/FROM THE DCU	Integer_Data				
DBLCYC	Double Major cycle indicator Only bit 1 is set	Integer_Data				
DCM_DCUID_LTCMP	START ADDRESS OF DCU ID DATA LESS THAN COMPARE TABLE	Integer_Data				
DCUSIF	DCU STATUS FLAG 1-4: Fault counter 5: (spare) 6: (spare) 7: DCU memory full reported by DCU (reserved) 8: Communications fault reported by DCU 9: Illegal block address fault reported by DCU 10: Internal status error reported by DCU 11: Last command invalid fault reported by DCU 12: Data integrity error reported by DCU 13: Busy status timeout fault 14: Message checksum error 15: No response	Integer_Data				
DCU_ADDRESS	DCU DEVICE ADDRESS	Integer_Data		1		
DCU_ADJUST_RSYNC	DCU obsolete counter adjustment value for Resynchs	Integer_Data				
DCU_ADJUST_SHUTD	DCU obsolete counter adjustment value for Shutdowns	Integer_Data				
DCU_ADJUST_STRTS	DCU obsolete counter adjustment value for Starts	Integer_Data				
DCU_BLOCK_PTR	DCU BLOCK POINTER	Integer_Data				
DCU_BLOCK_SIZE	DCU BLOCK SIZE IN BYTES	Integer_Data		128		
DCU_BUFFER_ADR	START ADDRESS OF DCU I/O DATA BUFFER	Integer_Data				
DCU_BUFFER_END	End address of DCU I/O data buffer	Integer_Data				
DCU_BUFFER_PTR	POINTER TO THE DCU BUFFER	Integer_Data				
DCU_BUSY_TIMER	DCU Busy Timer	Integer_Data				
DCU_BYTES_INPUT	NUMBER OF DATA BYTE RECEIVED FROM THE DCU	Integer_Data			0	128
DCU_BYTES_SENT	NUMBER OF BYTES SENT FROM DCU	Integer_Data				
DCU_CHECKSM	Computed checksum of DCU data block	Integer_Data				
DCU_CHECKSUM	DCU DATA CHECKSUM	Integer_Data			0	%O'177777'
DCU_CHECKSUM_ADR	ADDRESS OF DCU CHECKSUM VARIABLE	Integer_Data				
DCU_CMD_INVALID	DCU STATUS WORD COMMAND INVALID BIT	Integer_Data		%O'2000'		
DCU_COMMAND_ADR	START ADDRESS OF DCU COMMAND BUFFER	Integer_Data				
DCU_COMMAND_END	END ADDRESS OF DCU COMMAND BUFFER	Integer_Data				
DCU_CPCMP_END	END ADDRESS OF COMPONENT DATA COMPARE TABLE	Integer_Data				
DCU_CPCMP_TABLE	START ADDRESS OF COMPONENT DATA COMPARE TABLE	Integer_Data				
DCU_CPDATA_ADR	START ADDRESS OF COMPONENT DATA BUFFER	Integer_Data				
DCU_CPDATA_COPY	DCU FLAGS WORD 2 COPY COMPONENT DATA BIT	Integer_Data		2		
DCU_CPDATA_END	END ADDRESS OF COMPONENT DATA BUFFER	Integer_Data				
DCU_CPDAT_CMP	DCU FLAGS WORD 2 COMPONENT DATA MISMATCH BIT	Integer_Data		%O'200'		
DCU_CPOUT_BLOCK	START ADDRESS OF DCU COMPONENT DATA OUTPUT BLOCK	Integer_Data				
DCU_CPTMP_ADR	START ADDRESS OF TEMPORARY COMPONENT DATA BUFFER	Integer_Data				
DCU_CPTMP_END	END ADDRESS OF TEMPORARY COMPONENT DATA BUFFER	Integer_Data				
DCU_CTRDATA_ADR	START ADDRESS OF COUNTER DATA BLOCK	Integer_Data				
DCU_CTRDATA_COPY	DCU FLAGS WORD 2 COPY COUNTER DATA BIT	Integer_Data		4		
DCU_CTRDATA_END	END ADDRESS OF DCU COUNTER DATA BUFFER	Integer_Data				
DCU_CTRDAT_EEREQ	Request completed flag for writting DCU couter block data to EEPROM	Boolean_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
DCU_CTRDAT_IF	DCU COUNTER DATA(305) BLOCK STATUS WORD 1-4: Fault counter 5: Engine Component LCF counters range fault 6: Number of Engine Starts counters range fault 7: Number of In-flight Engine Starts counters range fault 8: Number of Flights counters range fault 9: Engine Run Time counters range fault 10: Engine Flight Time counters range fault 11: Accumulated Number of Engine Shutdowns counter or Takeoff counters range fault 12: Number of Engine Starts followed by a flight counters range fault 13: Accumulated starts vs shutdowns miscompare with Engine Off 14: Accumulated starts vs shutdowns miscompare with Engine Running 15: (spare) 16: Fault latch	Integer_Data				
DCU_CTRDAT_RANGE	START ADDRESS OF COUNTER DATA RANGE TABLE	Integer_Data				
DCU_CTREND_RANGE	END ADDRESS OF COUNTER DATA RANGE TABLE	Integer_Data				
DCU_CTROUT_BLOCK	START ADDRESS OF DCU COUNTER DATA OUTPUT BLOCK	Integer_Data				
DCU_DATA	DCU DATA VARIABLE	Integer_Data			0	0177777
DCU_DATA_END	END ADDRESS FOR DCU DATA TRANSFERS	Integer_Data			0	%O'77777'
DCU_DATA_FILLING	DCU ZERO FILLING OUTPUT FLAG	Boolean_Data				
DCU_DATA_PTR	POINTER TO THE DCU DATA	Integer_Data				
DCU_DATA_READY	DCU Data Ready Status 1 Blocks 0,1,204,205,9 Data Available 2 All LCF Data Available 3 DCU Inactive/Available for other Communication modes 4-16 (Reserved for software)	Integer_Data				
DCU_DATA_TIMER	DCU INPUT BYTE TIMEOUT TIMER	Integer_Data	MS		0	200
DCU_DCUCMP_END	END ADDRESS OF DCU ID DATA COMPARE TABLE	Integer_Data				
DCU_DCUCMP_TABLE	START ADDRESS OF DCU ID DATA COMPARE TABLE	Integer_Data				
DCU_DCUDAT_READY	DCU STATUS WORD DATA READY BIT	Integer_Data		1		
DCU_DCUEND_RANGE	END ADDRESS OF DCU ID DATA RANGE TABLE	Integer_Data				
DCU_DCUID_ADR	START ADDRESS OF DCU ID DATA BUFFER	Integer_Data				
DCU_DCUID_END	END ADDRESS OF DCU ID DATA	Integer_Data				
DCU_DCUID_IF	DCU DCU ID BLOCK STATUS WORD: 1-4: Fault Counter 5: Spare 6: NH Speed for Engine Off detection range fault 7: NH Speed for Engine Run detection range fault 8: ADC Supplied Ground/Flight Trigger Constant range fault 9: Engine deltaP Sensor Supplied Ground/Flight Trigger Constant range fault 10: LCF Cycle Counting Engine Model Number Fault 11: DCU Block 304 data checksum error 12: NH Speeds for Engine Run/Off detection reasonability fault 13-15: Spare 16: Fault latch	Integer_Data				
DCU_DCUID_LT	DCU ID DATA BLOCK STATUS LESS THAN COMPARE ERROR BIT	Integer_Data		%O'4000'		
DCU_DCUID_LTCMP	START ADDRESS OF DCU ID BLOCK LESS THAN COMPARE TABLE	Integer_Data				
DCU_DCUID_RANGE	START ADDRESS OF DCU ID DATA RANGE TABLE	Integer_Data				
DCU_DCUID_RBLOCK	ADDRESS OF DCU ID READ BLOCK IN DCU READ BLOCK TABLE	Integer_Data				
DCU_DCUID_SUM	DCU ID DATA BLOCK STATUS CHECKSUM ERROR BIT	Integer_Data		%O'2000'		
DCU_DELAY_TIMER	DCU DELAY TIMER	Integer_Data	MS		0	200
DCU_EECDATA_ADR	START ADDRESS OF THE EEC DATA BLOCK AREA	Integer_Data				
DCU_EECOUT_BLOCK	START ADDRESS OF DCU EEC DATA OUTPUT BLOCK	Integer_Data				
DCU_EECPN_END	END ADDRESS OF THE EEC PART NUMBER	Integer_Data				
DCU_EECTRDAT_UPD	Block 205 in DCU and EEPROM Requires Update Flag	Boolean_Data				
DCU_EEC_PN	START ADDRESS OF THE EEC PART NUMBER	Integer_Data				
DCU_ENGCMP_END	END ADDRESS OF ENGINE DATA COMPARE TABLE	Integer_Data				
DCU_ENGCMP_TABLE	START ADDRESS OF ENGINE DATA COMPARE TABLE	Integer_Data				
DCU_ENGDAT_ADR	START ADDRESS OF ENGINE DATA BLOCK	Integer_Data				
DCU_ENGDAT_COPY	DCU FLAGS WORD 2 COPY ENGINE DATA BIT	Integer_Data		1		
DCU_ENGDAT_END	END ADDRESS OF DCU ENGINE DATA BUFFER	Integer_Data				
DCU_ENGDAT_CMP	DCU ENGINE DATA BLOCK STATUS COMPARISON MISMATCH BIT	Integer_Data		%O'100'		
DCU_ENGOUT_BLOCK	START ADDRESS OF DCU ENGINE DATA OUTPUT BLOCK	Integer_Data				
DCU_ENGTMP_ADR	START ADDRESS OF TEMPORARY ENGINE DATA BUFFER	Integer_Data				
DCU_ENGTMP_END	END ADDRESS OF TEMPORARY ENGINE DATA BUFFER	Integer_Data				
DCU_FLAGS_1	DCU CONTROL FLAGS WORD 1 1 DCU modified/reprogrammed status flag 2 DCU Is New Status Flag 3 EEC/DCU mismatch of EEC P/N or S/N status flag 4 DCU EEC information update flag 5 DCU engine information update flag 6 DCU engine component information update flag 7 DCU engine usage update flag 8 Write CTRDAT to EEPROM, then DCU 9-16	Integer_Data				
DCU_FLAGS_2	DCU CONTROL FLAGS WORD 2 1 Copy engine data 2 Copy component data 3 Copy counter data 4 Got engine data 5 Got component data 6 Got counter data 7 Engine data mismatch 8 Component data mismatch 9-16 (spare)	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
DCU_FUNC_SW	DCU data functionality status summary word: 1 DCU engine data read fault (block 0) 2 DCU component data read fault (block 1) 3 DCU identification data read fault (block 304) 4 DCU engine usage data read fault (block 305) 5 DCU EEC data read fault (block 9) 6 (spare) 7 (spare) 8 data read fault (blocks 310-336) 9 New DCU blocks existence fault (blocks 0 & 1) 10 DCU engine data write fault (block 0) 11 DCU component data write fault (block 1) 12 DCU engine usage data write fault (block 305) 13 DCU EEC data write fault (block 9) 14 DCU communication deselected 15 New DCU blocks existence fault (blocks	Integer_Data				
DCU_GOT_CPDATA	DCU FLAGS WORD 2 GOT COMPONENT DATA BIT	Integer_Data		%0'20'		
DCU_GOT_CTRDATA	DCU FLAGS WORD 2 GOT COUNTER DATA BIT	Integer_Data		%0'40'		
DCU_GOT_ENGDATA	DCU FLAGS WORD 2 GOT ENGINE DATA BIT	Integer_Data		%0'10'		
DCU_GQUEUE_PTR	DCU WRITE QUEUE GET ENTRY POINTER	Integer_Data				
DCU_HOLD_TABLE	DCU HOLD TABLE POINTER FLAG	Integer_Data				
DCU_INPUT_PORT	Port address of the DCU UART input	Integer_Data		%0'410'		
DCU_INPUT_STATE	DCU INPUT DRIVER STATE VARIABLE	Integer_Data			0	4
DCU_IO_COMPLETE	DCU I/O IS COMPLETE FLAG	Boolean_Data				
DCU_IO_QUIET	DCU PROTOCOL NOT CURRENTLY REQUESTED BIT	Integer_Data		4		
DCU_IO_STATE	CURRENT DCU I/O STATE VARIABLE	Integer_Data			0	6
DCU_IS_AVAILABLE	FLAG TO INDICATE WHETHER THE DCU IS AVAILABLE OR NOT	Boolean_Data				
DCU_LCFBLOCK_END	END ADDRESS OF THE DCU LCF BLOCK	Integer_Data				
DCU_LCFBLOCK_MAX	MAXIMUM DCU LCF BLOCK NUMBER PLUS ONE	Integer_Data		237		
DCU_LCFBLOCK_MIN	MINIMUM DCU LCF BLOCK NUMBER	Integer_Data		210		
DCU_LCFBLOCK_NO	CURRENT BLOCK NUMBER OF LCF BLOCK TO READ	Integer_Data				
DCU_LCFBLOCK_NXT	NEXT DCU LCF BLOCK TRIPLET NUMBER	Integer_Data				
DCU_LCFBUF_PTR	POINTER INTO THE DCU LCF BUFFER	Integer_Data				
DCU_LCFCP_BLOCKS	NUMBER OF BLOCKS PER DCU LCF COMPONENT	Integer_Data		3		
DCU_LCFCP_ENTRY	DCU LCF DATA ENTRY LENGTH	Integer_Data		7		
DCU_LCFDATA_ADR	START ADDRESS OF DCU LCF DATA BLOCK	Integer_Data				
DCU_LCFDATA_LEN	LENGTH OF DCU LCF DATA	Integer_Data		8		
DCU_LCFDATA_PTR	POINTER TO DCU LCF DATA BUFFER	Integer_Data				
DCU_LCFDAT_RANGE	START ADDRESS OF DCU LCF RANGE TABLE	Integer_Data				
DCU_LCFDAT_READY	DCU LCF DATA READY BIT	Integer_Data		2		
DCU_LCFDAT_SUM	DCU LCF DATA BLOCK STATUS CHECKSUM ERROR BIT	Integer_Data		%0'40000'		
DCU_LCFENTRY_LEN	LENGTH OF A DCU LCF ENTRY	Integer_Data		11		
DCU_LCFENTRY_MAX	MAXIMUM DCU LCF ENTRY POSITION	Integer_Data		44		
DCU_LCFFACT_SW	DCU LCF DATA BLOCK STATUS WORD: 1 LPIMP Block Triplet Fault 2 HPIMP Block Triplet Fault 3 HPTFCP Block Triplet Fault 4 HPTD Block Triplet Fault 5 HPTRCP Block Triplet Fault 6 ISS Block Triplet Fault 7 LPTD Block Triplet Fault 8 PT1D Block Triplet Fault 9 PT2D Block Triplet Fault 10 Triplet Block 1 Fault 11 Triplet Block 2 Fault 12 Triplet Block 3 Fault 13 Component Part Number Not Found Fault 14 Abbreviated Cycle Factor (ACF) Range Fault 15 Block Data Checksum Fault 16 Fault Latch	Integer_Data				
DCU_LCFPN_NF	DCU LCF DATA BLOCK STATUS PART NUMBER NOT FOUND BIT	Integer_Data		%0'10000'		
DCU_LCFPN_PTR	POINTER TO CURRENT LCF COMPONENT PART NUMBER	Integer_Data				
DCU_LCFSW_CFAULT	START ADDRESS OF DCU LCF COMPONENT FAULT TABLE	Integer_Data				
DCU_LCF_BLOCK	START ADDRESS OF THE DCU LCF BLOCK	Integer_Data			10	96
DCU_LCF_BLOCKS	NUMBER OF DCU LCF BLOCKS PER CONFIGURATION	Integer_Data		27		
DCU_LCF_COMPONT	CURRENT DCU LCF COMPONENT NUMBER	Integer_Data			0	8
DCU_LCF_MISSIONS	START ADDRESS OF DCU LCF MISSIONS TABLE	Integer_Data				
DCU_LCF_NUM_ACF	Number of ACF entries per PN-group	Integer_Data		1		
DCU_LCF_NUM_FCF	Number of FCF entries per PN-group	Integer_Data		6		
DCU_LCF_PBLOCK	ADDRESS OF LCF PROCESS BLOCK IN DCU PROCESS BLOCK TABLE	Integer_Data				
DCU_LCF_READING	DCU READING LCF BLOCK FLAG	Boolean_Data				
DCU_LCF_TABLE	START ADDRESS OF THE DCU LCF TABLE	Integer_Data				
DCU_LCF_TFAULT	START ADDRESS OF DCU LCF TRIPLET FAULT TABLE	Integer_Data				
DCU_LCF_UPD	LCF Counts Require Update Flag	Boolean_Data				
DCU_LO_BYTE	LOW BYTE OF THE DCU DATA	Integer_Data				
DCU_MAX_BUSY	DCU BUSY TIMEOUT FLAG	Integer_Data				
DCU_MISMATCH	DCU FLAGS WORD 1 DATA MISMATCH BIT	Integer_Data		4		
DCU_MODIFIED	DCU FLAGS WORD 1 MODIFIED DCU BIT	Integer_Data		1		
DCU_NEW	DCU FLAGS WORD 1 NEW DCU BIT	Integer_Data		2		
DCU_NO_BLOCK	DCU STATUS WORD NO DATA BIT	Integer_Data		%0'400'		
DCU_NUM_MISSIONS	Number of Missions	Integer_Data		6		
DCU_OBSOLETE	DCU obsolete flag	Boolean_Data				
DCU_OUTPUT_PORT	Port address of the DCU UART output	Integer_Data		%0'410'		

Name	Description	Type	Units	Value	Min Val	Max Val
DCU_OUTPUT_STATE	CURRENT DCU OUTPUT STATE VARIABLE	Integer_Data			0	2
DCU_PORT_STATUS	UART PORT STATUS	Integer_Data				
DCU_PQUEUE_PTR	DCU WRITE QUEUE PUT ENTRY POINTER	Integer_Data				
DCU_PTABLE_ADR	START ADDRESS OF DCU PROCESS BLOCK TABLE	Integer_Data				
DCU_PTABLE_END	END ADDRESS OF DCU PROCESS BLOCK TABLE	Integer_Data				
DCU_PTABLE_PTR	POINTER INTO DCU PROCESS BLOCK TABLE	Integer_Data				
DCU_QUEUE_ADR	START ADDRESS OF THE DCU WRITE QUEUE	Integer_Data				
DCU_QUEUE_EMPTY	DCU WRITE QUEUE EMPTY FLAG	Boolean_Data				
DCU_QUEUE_END	END ADDRESS OF THE DCU WRITE QUEUE	Integer_Data				
DCU_RETRY_COUNT	DCU READ RETRY COUNTER	Integer_Data				
DCU_RETRY_TIMER	DCU RETRY TIMER	Integer_Data	SEC		0	10
DCU_RTABLE_ADR	START ADDRESS OF DCU READ BLOCK TABLE	Integer_Data				
DCU_RTABLE_END	END ADDRESS OF DCU READ BLOCK TABLE	Integer_Data				
DCU_RTABLE_PTR	POINTER INTO DCU READ BLOCK TABLE	Integer_Data				
DCU_SCHED_STATE	DCU SCHEDULE STATE	Integer_Data			0	2
DCU_SHUTDOWN_MM	DCU COUNTER DATA BLOCK STATUS SHUTDOWN MISMATCH	Integer_Data		%O'20000'		
DCU_STARTSD_MM	DCU COUNTER DATA BLOCK STATUS START AND SHUTDOWN MISMATCH BIT	Integer_Data		%O'40000'		
DCU_START_MM	DCU COUNTER DATA BLOCK STATUS STARTS MISMATCH BIT	Integer_Data		%O'10000'		
DCU_STATUS	DCU command status byte mapped with communication faults: 1-6: (not used) 7: DCU memory full (reserved) 8: Communications fault 9: Illegal block address fault 10: Internal status error 11: Last command invalid fault 12: Data integrity error 13: Busy status 14: Message checksum error 15: No response timeout or UART receiver fault 16: (not	Integer_Data				
DCU_STSHT_MM_MSK	Mask to ignore bits 13-15 in DCU_CTRDAT_IF	Integer_Data		%O'107777'		
DCU_STSHT_OFF_MM	Bit to set in DCU_CTRDAT_IF when Start/Shutdown miscompare with engine off	Integer_Data		%O'10000'		
DCU_STSHT_ON_MM	Bit to set in DCU_CTRDAT_IF when Start/Shutdown miscompare with engine on	Integer_Data		%O'20000'		
DCU_UPDATE_ADR	START ADDRESS OF DCU WRITE BLOCK TABLE	Integer_Data				
DCU_UPDATE_END	END ADDRESS OF DCU WRITE BLOCK TABLE	Integer_Data				
DCU_UPD_CPDAT	DCU FLAGS WORD 1 UPDATE COMPONENT DATA BIT	Integer_Data		%O'40'		
DCU_UPD_CTRDAT	DCU FLAGS WORD 1 UPDATE COUNTER DATA BIT	Integer_Data		%O'100'		
DCU_UPD_EECDAT	DCU FLAGS 1 WORD UPDATE EEC DATA BIT	Integer_Data		%O'10'		
DCU_UPD_ENGDAT	DCU FLAG WORD 1 UPDATE ENGINE DATA BIT	Integer_Data		%O'20'		
DECEL1	NOMINAL DECEL CALCULATED	Analog_Data	rpm/sec		-5500	-40
DECEL2	NPT GOV. DECEL SELECTED	Analog_Data	rpm/sec		-6400	0
DECEL4	NPT GOV. DECEL SCHEDULE	Analog_Data	rpm/sec		-6400	0
DECELC	CORRECTED DECEL SCHEDULE	Analog_Data	rpm/sec		-6400	-360
DECFACTOR	DECEL SCHEDULE CORRECTION FACTOR	Analog_Data			0.1	0.9
DELHP	DELTA HORSE POWER	Analog_Data	shp	0	-4000.0	4000.0
DELHP1	ASSIGNED DELTA HORSE POWER	Analog_Data	shp	0	-4000.0	4000.0
DELISA	DELTA BETWEEN ISA TEMPERATURE AND TAMB	Analog_Data	deg C		-70.0	40.0
DELPAM	DIFFERENCE BETWEEN ADC & STRAIN GAUGE PRESSURE	Analog_Data	PSIA		-13.0420	13.1960
DELPC	COMPENSATED RAM PRESSURE READING	Analog_Data	mA		-119.996	60.004
DELPIF	Ram pressure interface status word	Integer_Data				
DELRAM	FAULT BLOCK DELETION REQUESTED	Boolean_Data				
DELT15	DIFFERENCE BETWEEN ADC AND RTD TEMP. READINGS	Analog_Data	DEG F		-170	170
DELTA	STATION 1.5 TOTAL PRESSURE CORRECTED	Analog_Data			0.2	1.2
DELTA1	DELTA CORR'N FOR CORR'D PWR TO CORR'D NH GAIN	Analog_Data			0.8333	3.7037
DELTA2	DELTA CORR'N FOR CORR'D PWR TO CORR'D NH DOMINANT LAG	Analog_Data			0.9615	1.1494
DELTAP	UNFILTERED RAM PRESSURE FROM POSITION 1.8	Analog_Data	PSI		0	5
DELTCT	DELTAP Transient timer	Integer_Data	counts		0	25
DERAT1	Derating Discrete Input 1	Boolean_Data			False	True
DERAT2	Derating Discrete Input 2	Boolean_Data			False	True
DERATE	SELECTED DERATE VALUE	Integer_Data			1	4
DEVMODEN	DEVELOPMENT MODE ENABLE	Boolean_Data				
DHPREQ	DERIVATIVE LAG OUTPUT OF POWER UPTRIM	Analog_Data	SHp		-2000.0	4000.0
DISC1	Discrete Input Word #1 1 EEC Inhibit/Reset Request 2 Propeller Brake Discrete 3 MCT Enable 4 (spare) 5 (spare) 6 Uptrim (Opp Prop Feathered) 7 Bleed Valve #1 Open 8 Bleed Valve #2 Open 9 (spare) 10 LRU Display Advance Discrete 11 Engine Position 12 Development Mode Enable 13 Propeller Feathered 14 Derating Discrete 1 15 Derating	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
DISC2	Discrete Input Word #2 1 UPTRIM Lamp Wraparound 2 HBOV Torque Motor Overcurrent 3 EEPROM Write Enable Request 4 MFC Identifier 5 PLA Trim Discrete input 6 Auto-Ignition Relay Wraparound 7 (Spare) 8 Reserved for Overspeed Option 9 Ground Test Enable Request 10 Propeller Speed Reference Confirmation Output 2 Wraparound 11 Reserved for Spare HSS 12 Fail Fixed/MFCU Stepper Motor Output Wraparound 13 HBOV Torque Motor Output Wraparound 14 Propeller Speed Reference Confirmation Output 1 Wraparound 15 ECSF Lamp	Integer_Data				
DISCOT	Discrete output word 1 Propeller Speed Reference Confirmation Output 2 2 Reserved for Pseudo HSS 3 Uptrim Lamp 4 HBOV Torque Motor Output 5 Reserved for Overcurrent Reset 6 Auto Ignition Relay 7 (Spare) 8 (Spare) 9 Fail Fixed/MFCU Stepper Motor Output 10 ECSF Lamp 11 Reserved for Fail Fixed Initialization Discrete 12 Propeller Speed Reference Confirmation Output 1 13 (Spare) 14 (Spare) 15	Integer_Data				
DLHPRF	TRANSIENT CHANGE IN HPREF CALCULATED	Analog_Data	shp	0	0	2000.0
DLNDF	ACCEL SCHEDULE CORRECTION FACTOR	Analog_Data			0.65	2.0
DLPTR	START OF DELTA P SENSOR DATA IN EEPROM	Integer_Data				
DLTANH	NH GOV. LOOP ERROR	Analog_Data	rpm		-24000	24000
DMPMFC	NPT O/S MFC DUMP FLAG	Boolean_Data				
DMPPC1	1=DUMP LOGIC FOR PLA SLAMS FROM REVERSE THROUGH DISKING	Boolean_Data				
DMPPC2	1=DUMP LOGIC FOR PLA SLAMS FROM HIGH PLA INTO REVERSE	Boolean_Data				
DMPPCU	NPT O/S PCU DUMP FLAG	Boolean_Data				
DNHACC	NH ACCEL LIMIT CALCULATED	Analog_Data	rpm/sec		0.0	6400
DNHDEC	NH DECELERATION LIMIT CALCULATED	Analog_Data	rpm/sec		-15000	0
DNHDFG	NPT O/S DOWNWARD DUMP FLAG	Boolean_Data				
DNHERR	NH REQUEST RATE CALCULATED	Analog_Data	rpm/sec		-15000	15000
DNHIL2	UN-RANGE LIMITED NPT TRACKING ALLOWABLE ERROR	Analog_Data	RPM/SEC		-15000	15000
DNHINL	NH REQUEST RATE FOR INNER LOOP TRACKING	Analog_Data	RPM/SEC		0.0	15000.0
DNHLIM	NH LIMIT RATE REQUEST CALCULATED	Analog_Data	rpm/sec	0	-6400	6400
DNHLM	NH LIMIT RATE REQUEST	Analog_Data	rpm/sec	0	-3000	3000
DNHNPD	NPT GOV DERIVATIVE CONTRIBUTION	Analog_Data	rpm/sec		-6400.0	6400.0
DNHNPP	NPT GOV PROPORTIONAL CONTRIBUTION	Analog_Data	rpm/sec		-6400.0	6400.0
DNHNPT	NH RATE FOR NPT GOVERNING	Analog_Data	rpm/sec		-15000.0	15000.0
DNHOSD	NPT O/S PLA SLAM DOWNWARD DER LAG	Analog_Data			-700	700
DNHOSU	NPT O/S PLA SLAM UPWARD DER LAG	Analog_Data			-1500	1500
DNHREQ	INCREMENT IN NH REQUEST CALCULATED	Analog_Data	rpm		-350	350
DNHSEF	NH REQUEST RATE FILTERED	Analog_Data	RPM/SEC		-15000.0	15000.0
DNHSEL	NH REQUEST RATE SELECTED	Analog_Data	rpm/sec		-15000	15000
DNHTQ	TORQUE LIMIT NH RATE REQUEST CALCULATED	Analog_Data	RPM/SEC		-15000	15000
DNHTQ1	TORQUE LIMIT NH RATE REQUEST BEFORE UPTRIM SWITCH	Analog_Data	RPM/SEC		-15000	15000
DNHUFG	NPT O/S UPWARD DUMP FLAG	Boolean_Data				
DNHWL1	NON RANGE-LIMD NH REQ RATE FOR INNER LOOP TRACKING	Analog_Data	RPM/SEC		-15000.0	30000.0
DNSEL1	NH REQUEST RATE FIRST SELECTION	Analog_Data	rpm/sec		-15000	15000
DNSEL2	NH REQUEST RATE 2ND SELECTION	Analog_Data	rpm/sec		-15000	15000
DNSEL3	NH REQUEST RATE 3RD SELECTION	Analog_Data	rpm/sec		-15000	15000
DNSEL4	NH REQUEST RATE 4TH SELECTION	Analog_Data	rpm/sec		-15000.0	15000.0
DNSEL5	NH REQUEST RATE 5TH SELECTION	Analog_Data	rpm/sec		-15000.0	15000.0
DP18F	Filtered ram pressure at station 1.8	Analog_Data	PSI		0	5.0
DPL134	ARINC DELTA PLA REQUEST	Analog_Data	deg		-15	15
DPO	DELTA PRESSURE	Analog_Data	psid		0.0	4.0
DPOS	DELTA PRESSURE FROM SENSOR	Analog_Data	PSIA		0.0	3.8
DPOVRP	Ratio of ram to ambient pressure	Analog_Data			0	1.0
DPTAB1	DELTA-P STRAIN GUAGE TABLE	Integer_Data				
DQOUTI	ABS. VALUE OF QOUT FILTER TORQUE DIFFERENCE	Analog_Data	---		0.0	15.0
DR	UART DATA READY STATUS BIT	Integer_Data		11		
DRBIT	Data Ready Bit in DCU_DATA	Integer_Data		11		
DRFCIF	High Level drift Interface Status Word	Integer_Data				
DRFT1C	Low Level A/D Interface Drift Value	Analog_Data	mV		0.126	24.986
DRFTCT	Low level drift transient timer	Integer_Data	COUNTS		0	25
DRFTIF	Low level drift Interface Status Word	Integer_Data				
DRHTCT	High Level Drift Transient Timer	Integer_Data	counts		0	25
DRFTC	High Level A/D Interface Drift Value	Analog_Data	VDC		-0.0338	0.0262
DROOP	1=FLAG INDICATING BOOST IS ACTIVE AND PLA>20 DEG.	Boolean_Data				
DSLML	1=DECEL SLAM	Boolean_Data				
DSLMSR	1=DECEL SLAM RESET HBOV SS REACHED	Boolean_Data				
DSLMS	DECEL SLAM SETPOINT	Analog_Data	rpm/sec	0	-3000.0	0.0

Name	Description	Type	Units	Value	Min Val	Max Val
DSMNH4	BACK CALCD INCR IN SMP REQ FROM P+I IN INNER LOOP	Analog_Data	STEP/CYCLE		-1280.0	1280.0
DSMNHK	BACK CALCD INCR IN SMP REQ FROM I OF P+I IN INNER LOOP	Analog_Data	STEP/CYCLE		-1280.0	1280.0
DSMP1	NH GOV. OUTPUT OF LEAD-LAG # 1	Analog_Data	rpm		-24000	24000
DSMPFR	COMPENSATOR STEP REQUEST FEEDBACK/RESET	Analog_Data	STEP/CYCLE		-1280.0	1280.0
DSMPRQ	COMPENSATOR STEP REQUEST	Analog_Data	rpm		0.0	1280
DTSCFG	DTS/GDTS UART configuration flag.	Integer_Data			0	1
DUMPFQ	NPT O/S DUMP FLAG	Boolean_Data				
DXBUMP	NPT U/S BOOST CALCULATED	Analog_Data	step/sec		0.0	1200.0
DXIN	INPUT TO DYNAMIC COMPENSATION	Analog_Data	rpm		-24000	24000
DXOUTA	NH GOV. STEP REQUEST	Analog_Data	step/sec		-800	800
E2CNT	OFFSET INTO EEPROM, AND RAM FAULT TABLES	Integer_Data				
EBSY26	Local EEPROM busy flag	Boolean_Data				
ECFAIL	EEPROM fail storage word	Integer_Data				
ECINTF	INTERNAL EEC INTERFACE STATUS WORD	Integer_Data				
ECNT26	ERROR COUNT: INDICATES # OF FAULTS	Integer_Data			0	8
EDATIF	EEPROM Data Comp Status Word	Integer_Data				
EEBPWM	EEPROM SELECTED PWM WRITE FLAG	Boolean_Data				
EEBQB	TORQUE BIAS TRIM EEPROM WRITE FLAG	Boolean_Data				
EEBQG	TORQUE GAIN TRIM EEPROM WRITE FLAG	Boolean_Data				
EEBUSY	GLOBAL EEPROM BUSY FLAG	Boolean_Data				
EECPN_SNA1112	EEC P/N and S/N ASCII Chars 11 & 12	Integer_Data				
EECPN_SNA1112EE	EEC P/N and S/N ASCII Chars 11 & 12 in EEPROM	Integer_Data				
EECPN_SNA12	EEC P/N and S/N ASCII Chars 1 & 2	Integer_Data				
EECPN_SNA12EE	EEC P/N and S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
EECPN_SNA1314	EEC P/N and S/N ASCII Chars 13 & 14	Integer_Data				
EECPN_SNA1314EE	EEC P/N and S/N ASCII Chars 13 & 14 in EEPROM	Integer_Data				
EECPN_SNA1516	EEC P/N and S/N ASCII Chars 15 & 16	Integer_Data				
EECPN_SNA1516EE	EEC P/N and S/N ASCII Chars 15 & 16 in EEPROM	Integer_Data				
EECPN_SNA1718	EEC P/N and S/N ASCII Chars 17 & 18	Integer_Data				
EECPN_SNA1718EE	EEC P/N and S/N ASCII Chars 17 & 18 in EEPROM	Integer_Data				
EECPN_SNA1920	EEC P/N and S/N ASCII Chars 19 & 20	Integer_Data				
EECPN_SNA1920EE	EEC P/N and S/N ASCII Chars 19 & 20 in EEPROM	Integer_Data				
EECPN_SNA2122	EEC P/N and S/N ASCII Chars 21 & 22	Integer_Data				
EECPN_SNA2122EE	EEC P/N and S/N ASCII Chars 21 & 22 in EEPROM	Integer_Data				
EECPN_SNA2324	EEC P/N and S/N ASCII Chars 23 & 24	Integer_Data				
EECPN_SNA2324EE	EEC P/N and S/N ASCII Chars 23 & 24 in EEPROM	Integer_Data				
EECPN_SNA2526	EEC P/N and S/N ASCII Chars 25 & 26	Integer_Data				
EECPN_SNA2526EE	EEC P/N and S/N ASCII Chars 25 & 26 in EEPROM	Integer_Data				
EECPN_SNA2728	EEC P/N and S/N ASCII Chars 27 & 28	Integer_Data				
EECPN_SNA2728EE	EEC P/N and S/N ASCII Chars 27 & 28 in EEPROM	Integer_Data				
EECPN_SNA2930	EEC P/N and S/N ASCII Chars 29 & 30	Integer_Data				
EECPN_SNA2930EE	EEC P/N and S/N ASCII Chars 29 & 30 in EEPROM	Integer_Data				
EECPN_SNA3132	EEC P/N and S/N ASCII Chars 31 & 32	Integer_Data				
EECPN_SNA3132EE	EEC P/N and S/N ASCII Chars 31 & 32 in EEPROM	Integer_Data				
EECPN_SNA34	EEC P/N and S/N ASCII Chars 3 & 4	Integer_Data				
EECPN_SNA34EE	EEC P/N and S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
EECPN_SNA56	EEC P/N and S/N ASCII Chars 5 & 6	Integer_Data				
EECPN_SNA56EE	EEC P/N and S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
EECPN_SNA78	EEC P/N and S/N ASCII Chars 7 & 8	Integer_Data				
EECPN_SNA78EE	EEC P/N and S/N ASCII Chars 7 & 8 in EEPROM	Integer_Data				
EECPN_SNA910	EEC P/N and S/N ASCII Chars 9 & 10	Integer_Data				
EECPN_SNA910EE	EEC P/N and S/N ASCII Chars 9 & 10 in EEPROM	Integer_Data				
EECPOR	EEC POR Detected - Single-shot pulse occurs during the first pass after initialization is complete	Boolean_Data				
EEERF	Engine rating select write flag	Boolean_Data				
EEG	THE EEPROM LOCATION FOR SELECTED PWM	Integer_Data				
EEGARC	THE TYPE OF I104 GATE ARRAY, STORED IN EEPROM BUFFER	Integer_Data				
EEGARM	THE TYPE OF I104 GATE ARRAY STORED IN EEPROM	Integer_Data				
EE_CPDAT_BUF	START ADDRESS OF EEPROM COMPONENT DATA BUFFER	Integer_Data				
EE_CPDAT_BLOCK	START ADDRESS OF COMPONENT DATA EEPROM BUFFER	Integer_Data				
EE_CTRDAT_BUF	START ADDRESS OF EEPROM COUNTER DATA BUFFER	Integer_Data				
EE_CTRDAT_BLOCK	START ADDRESS OF EEPROM COUNTER DATA BLOCK	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
EE_CTRDAT_IF	EEPROM COUNTER DATA STATUS WORD 1-4: Fault counter 5: Engine Component LCF counters range fault 6: Number of Engine Starts counters range fault 7: Number of In-flight Engine Starts counters range fault 8: Number of Flights counters range fault 9: Engine Run Time counters range fault 10: Engine Flight Time counters range fault 11: Accumulated Number of Engine Shutdowns counter or Takeoff counters range fault 12: Number of Engine Starts followed by a flight counters range fault 13: (spare) 14: (spare) 15: (spare) 16: Fault latch	Integer_Data				
EE_DATA_ADR	START ADDRESS OF EEPROM DATA AREA	Integer_Data				
EE_DATA_PTR	POINTER TO EEPROM DATA AREA	Integer_Data				
EE_EECPN	START ADDRESS OF EEC PART NUMBER IN EEPROM	Integer_Data				
EE_ENGDAT_BUF	START ADDRESS OF EEPROM ENGINE DATA BUFFER	Integer_Data				
EE_ENGDAT_BLOCK	ADDRESS OF ENGINE DATA BLOCK FOR EEPROM STORAGE	Integer_Data				
EE_GQUEUE_PTR	EEPROM queue get entry pointer	Integer_Data				
EE_PQUEUE_PTR	EEPROM queue put entry pointer	Integer_Data				
EE_QUEUE_ADR	Start address of EEPROM queue	Integer_Data		EE_WRITE_QUEUE		
EE_QUEUE_EMPTY	EEPROM WRITE QUEUE IS EMPTY FLAG	Boolean_Data				
EE_QUEUE_END	End address of the EEPROM queue	Integer_Data				
EE_REQUEST_DONE	EEPROM WRITE REQUEST COMPLETE FLAG POINTER	Integer_Data				
EE_UPDATE_FLAGS	EE UPDATE FLAGS 1 Request EEPROM write CTRDAT due to Engine Startup 2 EEPROM has been queued to write (for Startup) 3 Request EEPROM write CTRDAT due to Engine Shutdown 4 EEPROM has been queued to write (for Shutdown) 5 Request EEPROM write CTRDAT due to Resync 6 EEPROM has been queued to write (for Resync) 7-16 (Spare)	Integer_Data				
EE_WRITING	EEPROM WRITE IN PROGRESS FLAG	Boolean_Data				
EIDX26	Offset into EEPROM, and RAM fault tables	Integer_Data			0	24
ENAA5I	APPROACH SPEED CONTROL ENABLE COMMAND FROM COCKPIT	Boolean_Data				
ENDCYC	Engine Flight Cycle Ended	Boolean_Data				
ENGFLTCYC	Engine Flight Cycle - Single-shot pulse occurs at the end of a flight cycle	Boolean_Data				
ENGFLTCYC_PV	Past value of Engine Flight Cycle - Single-shot pulse occurs at the end of a flight cycle	Boolean_Data				
ENGFLTM_CRS_ALL	Total Accumulated Flight Time Coarse Count	Analog_Data	Hours		0	32767
ENGFLTM_CRS_E	Total Accumulated Flight Time Coarse Count (PW127E Engine Model)	Analog_Data	Hours		0	32767
ENGFLTM_CRS_F_SS	Snapshot of Total Accumulated Flight Time Coarse Count (PW127F Engine Model)	Analog_Data	Hours		0	32767
ENGFLTM_CRS_M_SS	Snapshot of Total Accumulated Flight Time Coarse Count (PW127M Engine Model)	Analog_Data	Hours		0	32767
ENGFLTM_CRS_N_SS	Snapshot of Total Accumulated Flight Time Coarse Count (PW127N Engine Model)	Analog_Data	Hours		0	32767
ENGFLTM_FIN_ALL	Total Accumulated Flight Time Fine Count	Analog_Data	Fractional Hours		0	1
ENGFLTM_FIN_E	Total Accumulated Flight Time Fine Count (PW127E Engine Model)	Analog_Data	Fractional Hours		0	1
ENGPOS	ENGINE POSITION	Boolean_Data				
ENGRAT	Derating Level Configuration 0 = Uprate 1 = Norm 2 = (reserved) 3 =	Integer_Data			0	3
ENGRATC	Derating Level Configuration; to be stored in EEPROM (-1 = buffer empty) 0 = Uprate 1 = Norm 2 = (reserved) 3 = Derate	Integer_Data			-1	3
ENGRATM	Derating Level Configuration; last value stored in EEPROM 0 = Uprate 1 = Norm 2 = (reserved) 3 = Derate	Integer_Data			0	3
ENGRATS	Derating Level Configuration 1 = Uprate 2 = Norm 3 = Derate	Integer_Data			1	3
ENGRAT_BIT1	Derating Level confirmation bit 1	Boolean_Data				
ENGRAT_BIT2	Derating Level confirmation bit 2	Boolean_Data				
ENGRUN	Engine OFF/RUN Flag False = OFF True = RUN	Boolean_Data				
ENGSHUTDOWN	Engine Shutdown Detected - Single-shot pulse occurs each time the engine is shutdown	Boolean_Data				
ENGSHUTDOWN_PV	Past value of Engine SHUTDOWN Flag False = SHUTDOWN True = RUN	Boolean_Data				
ENG5NA12	Engine Serial Number ASCII Chars 1 & 2	Integer_Data				
ENG5NA12EE	Engine Serial Number ASCII Chars 1 & 2 in EEPROM	Integer_Data				
ENG5NA34	Engine Serial Number ASCII Chars 3 & 4	Integer_Data				
ENG5NA34EE	Engine Serial Number ASCII Chars 3 & 4 in EEPROM	Integer_Data				
ENG5NA56	Engine Serial Number ASCII Chars 5 & 6	Integer_Data				
ENG5NA56EE	Engine Serial Number ASCII Chars 5 & 6 in EEPROM	Integer_Data				
ENGSTART	Engine Start - Single-shot pulse occurs each time the engine is started	Boolean_Data				
ENGSTART_PV	Past value of Engine Start	Boolean_Data				
ENG_RUN_CRS_ALL	Total Accumulated Engine Run Time Coarse Count	Analog_Data	Hours		0	32767
ENG_RUN_CRS_E	Accumulated Engine Run Time in PW127E Configuration Coarse Count	Analog_Data	Hours		0	32767
ENG_RUN_CRS_F_SS	Snapshot of Total Accumulated Engine Run Time Coarse Count (PW127F Engine Model)	Analog_Data	Hours		0	32767

Name	Description	Type	Units	Value	Min Val	Max Val
ENG_RUN_CRS_M_SS	Snapshot of Total Accumulated Engine Run Time Coarse Count (PW127M Engine Model)	Analog_Data	Hours		0	32767
ENG_RUN_CRS_N_SS	Snapshot of Total Accumulated Engine Run Time Coarse Count (PW127N Engine Model)	Analog_Data	Hours		0	32767
ENG_RUN_FIN_ALL	Total Accumulated Engine Run Time Fine Count	Analog_Data	Fractional Hours		0	1
ENG_RUN_FIN_E	Accumulated Engine Run Time in PW127E Configuration Fine Count	Analog_Data	Fractional Hours		0	1
ERDAT	ERSP ARINC 211 DATA VALIDITY FLAG	Integer_Data			0	1
ERDELO	Change in ENGRAT delayed off	Boolean_Data				
ERF	Engine rating value in EEPROM	Integer_Data			0	3
ERNSEC	Engine Run Time (seconds)	Analog_Data	seconds		0	3600
ERNSEC_SCL_PV	Past value of ERNSEC	Analog_Data	seconds		0	3600
EROMB1	LOCAL EEPROM BUSY FLAG	Boolean_Data				
EROMIF	EEPROM INTERFACE FAULT WORD 1 - 7 Not Used 8 Engine Rating Write Verification Fault 9 Not Used 10 PLA TRIM READ/WRITE FAULT 11 GATE ARRAY SELECTION VALUE READ/WRITE FAULT 12 EEPROM LRU WRITE FAULT FLAG 13 EEPROM LRU WRITE FAULT FLAG FOR INTERNAL EEC FAILURES 14 Not Used 15 TORQUE TRIM WRITE VERIFICATION FAULT	Integer_Data				
ERRCNT	ERROR COUNT: INDICATES # OF FAULTS	Integer_Data	Counter		0	8
ERSPSW	ERSP ARINC LABEL 211 STATUS WORD	Integer_Data				
ERUNTC	ENGINE RUN TIME COARSE (HOURS)	Analog_Data			0	32767
ERUNTF	ENGINE RUN TIME FINE (FRACTION OF HOURS)	Analog_Data	HOURS		0	1.0
ESFBAF	Engine start followed by a flight each time a ground/flight transition occurs after an engine start	Boolean_Data				
ETLOCK	ENGINE TRIM ALREADY DONE	Boolean_Data				
ETSTMR	ENGINE-TRIM Switch Timer	Integer_Data				
EVDFIF	Loss of Configuration Data 16 - Fault Latch; Engine Variant Default In	Integer_Data				
EVFLGC	Value of Engine Variant from trim resistor stored in EEPROM write buffer (Values are in octal) 0 BUFFER EMPTY 1 TRIM A - CONFIGURATION 1 2 TRIM B - CONFIGURATION 2 (ILLEGAL) 3 TRIM C - CONFIGURATION 3 4 TRIM D - CONFIGURATION 4 41 TRIM A - CONFIGURATION 41 43 TRIM C - CONFIGURATION 43	Integer_Data	Variant		0	43
EVFLTS	Engine Validity Selection Compensated Analog Voltage	Analog_Data	Volts		-10.0	10.0
EVRAT	Engine Rating Configuration 0 = Power Level 1 Rating 1 = Power Level 2 Rating 2 = (reserved) 3 = Power Level 3 Rating	Integer_Data			0	3
EVRATTMR	Time to wait after rating inputs change before allowing a change to the rating.	Analog_Data	sec		0.0	5.0
EVTCT	Engine Variant Transient Fault Timer	Integer_Data	sec		0	2.1
EWFAIL	1=EEPROM WRITE FAILED	Boolean_Data		True		
EXDISC	External Discrete Word 1 EEC Inhibit/Reset Request 2 Propeller Brake Discrete 3 Propeller Feathered 4 Ground Test Enable Request 5 ECSF Lamp 6 Fail Fixed/MCFU Stepper Motor Output 7 MCT Enable 8 (Spare) 9 UPTRIM (Opp Prop Feathered) 10 Bleed Valve #1 Open 11 Bleed Valve #2 Open 12 PLAIN In Range for PLA Trim 13 Uptrim Lamp 14 (Spare) 15 PLA Trim Commanded from Cockpit 16 Engine Trim Already Done	Integer_Data				
E_ACCINFLSTRTS_E	Accumulated Inflight Starts (PW127E Engine Model) in EEPROM	Analog_Data	Counts		0	32767
E_ACCINFLSTRT_AL	Accumulated Inflight Starts in EEPROM	Analog_Data	Counts		0	32767
E_ACCNUMFL_ALL	Accumulated Number of Flights in EEPROM	Analog_Data	Counts		0	65535
E_ACCNUMFL_E	Accumulated Number of Flights (PW127E Engine Model) in EEPROM	Analog_Data	Counts		0	32767
E_ACCSHUTD_ALL	Accumulated Shut Downs Counter in EEPROM	Analog_Data	Counts		0	65535
E_ACCSTRTSFLT_AL	Accumulated Starts Followed by Flight in EEPROM	Analog_Data	Counts		0	65535
E_ACCSTRTSFLT_E	Accumulated Starts Followed by Flight (PW127E Engine Model) in EEPROM	Analog_Data	Counts		0	32767
E_ACCSTRTS_ALL	Accumulated Starts in EEPROM	Analog_Data	Counts		0	65535
E_ACCSTRTS_E	Accumulated Starts (PW127E Engine Model) in EEPROM	Analog_Data	Counts		0	32767
E_ACCSTR_SD_RESYNC	No of Acc. Starts/Shutdown Resynch Counter in EEPROM	Analog_Data	Counts		0	32767
E_ENGLFTM_CRS_AL	Total Accumulated Flight Time Coarse Count in EEPROM	Analog_Data	Hours		0	32767
E_ENGLFTM_CRS_E	Total Accumulated Flight Time Coarse Count (PW127E Engine Model) in EEPROM	Analog_Data	Hours		0	32767
E_ENGLFTM_FIN_AL	Total Accumulated Flight Time Fine Count in EEPROM	Analog_Data	Fractional Hours		0	1
E_ENGLFTM_FIN_E	Total Accumulated Flight Time Fine Count (PW127E Engine Model) in EEPROM	Analog_Data	Fractional Hours		0	1
E_ENG_RN_CRS_ALL	Total Accumulated Engine Run Time Coarse Count in EEPROM	Analog_Data	Hours		0	32767
E_ENG_RN_FIN_ALL	Total Accumulated Engine Run Time Fine Count in EEPROM	Analog_Data	Fractional Hours		0	1
E_ENG_RUN_CRS_E	Accumulated Engine Run Time in PW127E Configuration Coarse Count in EEPROM	Analog_Data	Hours		0	32767
E_ENG_RUN_FIN_E	Accumulated Engine Run Time in PW127E Configuration Fine Count in EEPROM	Analog_Data	Fractional Hours		0	1

Name	Description	Type	Units	Value	Min Val	Max Val
E_HPIMP_CYCCRS	HP Impeller Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_HPIMP_CYCFIN	HP Impeller Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_HPTD_CYCCRS	HPT Disk Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_HPTD_CYCFIN	HPT Disk Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_HPTFCP_CYCCRS	HPT Front Cover Plate Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_HPTFCP_CYCFIN	HPT Front Cover Plate Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_HPTRCP_CYCCRS	HPT Rear Cover Plate Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_HPTRCP_CYCFIN	HPT Rear Cover Plate Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_ISS_CYCCRS	Inter-stage Seal Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_ISS_CYCFIN	Inter-stage Seal Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_LPIMP_CYCCRS	LP Impeller Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_LPIMP_CYCFIN	LP Impeller Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_LPTD_CYCCRS	LPT Disk Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_LPTD_CYCFIN	LPT Disk Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_MTO_DL1	Accumulated Maximum Takeoff Counter at UPRATE Power Rating in EEPROM	Analog_Data	Count		0	32767
E_MTO_DL2	Accumulated Maximum Takeoff Counter at NORM Power Rating in EEPROM	Analog_Data	Counts		0	32767
E_MTO_DL3	Accumulated Maximum Takeoff Counter at DERATE Power Rating in EEPROM	Analog_Data	Counts		0	32767
E_NTO_DL1	Accumulated Normal Takeoff Counter at UPRATE Power Rating in EEPROM	Analog_Data	Counts		0	32767
E_NTO_DL2	Accumulated Normal Takeoff Counter at NORM Power Rating in EEPROM	Analog_Data	Counts		0	32767
E_NTO_DL3	Accumulated Normal Takeoff Counter at DERATE Power Rating in EEPROM	Analog_Data	Counts		0	32767
E_PT1D_CYCCRS	PT1 Disk Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_PT1D_CYCFIN	PT1 Disk Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_PT2D_CYCCRS	PT2 Disk Accumulated Cycles Coarse Count in EEPROM	Analog_Data	Counts		0	32767
E_PT2D_CYCFIN	PT2 Disk Accumulated Cycles Fine Count in EEPROM	Analog_Data	Fractional Counts		0	1
E_SV_PWC_PATCHID	EEC Application Software PWC Patch Identification in EEPROM	Integer_Data				
E_SV_SW_EPI	EEC Application Software Engine Program Identification in EEPROM	Integer_Data				
E_SV_SW_SBI	EEC Application Software Baseline Identification in EEPROM	Integer_Data				
E_SV_SW_SVI	EEC Application Software Sub-version Identification in EEPROM	Integer_Data				
FAILFX	1=FAILED FIXED	Boolean_Data				
FETHDI	1=FEATHER MODE	Boolean_Data				
FGTM	Background functionality flags	Integer_Data			0	1
FGTMCT	Background Fault Counter	Integer_Data			0	2
FLACA1	ARINC #1 Fault Action Word 1 ARINC 133 OUTPUT DISABLED 2 ARINC 134 OUTPUT DISABLED 3 NOT USED 4 ARINC 203 OUTPUT DISABLED - SOURCE IS SENSOR 5 NOT USED 6 NOT USED 7 NOT USED 8 NOT USED 9 ARINC 240 OUTPUT DISABLED - NOT USED 10 ARINC 245 OUTPUT DISABLED 11 ARINC 270 OUTPUT DISABLED 12 ARINC 271 OUTPUT DISABLED - NOT USED 13 ARINC 340 OUTPUT DATA = DEFAULT SSM = VALID 14 ARINC 342 AND 344 OUTPUT DISABLED 15 ARINC 341 OUTPUT DISABLED 16 ARINC 346 OUTPUT DISABLED	Integer_Data				
FLACA2	BUILT IN TEST FAULT ACTION SUMMARY WORD 2 - ARINC OUTPUTS 1 NOT USED 2 NOT USED 3 ARINC 340 OUTPUT DISABLED (SSM = INVALID) DATA 4 ARINC 203 OUTPUT DISABLED (SDI = 00 or 11) 5 NOT USED 6 ARINC 062 OUTPUT DISABLED 7 ARINC 063 OUTPUT DISABLED 8 ARINC 064 OUTPUT DISABLED 9 ARINC 065, 066 and 067 OUTPUT	Integer_Data				
FLACCW	BUILT IN TEST FAULT ACTION SUMMARY WORD - CONTROL PROCESSES 1 HBOV1 - DISABLE THE NORMAL HBOV CONTROL 2 HBOV2 - DISABLE THE MANUAL HBOV CONTROL 3 HBOV3 - DISABLE HBOV PLA ENABLE FUNCTION 4 - 13 NOT USED 14 NPGOV - DISABLE THE NPT GOVERNING LOOP 15 ASCDAT - DISABLE APPROACH SPEED CONTROL-NOT USED 16	Integer_Data				
FLACHW	Hardware Fault Action Word 1 Disable High Side Switch #1 2 Disable High Side Switch #2 3 Turn on Low Side Switch #1 4 Disable Low Side Switch #2 5 Disable Low Side Switch #3 6 Disable Low Side Switch #4 7 Disable Low Side Switch #5 8 Disable Pulse Width Modulator #3 9 Disable Pulse Width Modulator #2 [Not Used] 10 Disable Pulse Width Modulator #1 [Not Used] 11 [Not Used] 12 [Not Used] 13 [Not Used] 14 Disable Pseudo HSS [Not Used] 15 [Not Used] 16 [Not Used]	Integer_Data				
FLIGHT	Ground/Flight Flag False = Ground True = Flight	Boolean_Data				
FLIGHTFLG1	Temporary Indicated Air Speed Flight Flag	Boolean_Data				
FLIGHTFLG2	Temporary Indicated Air Speed Flight Flag	Boolean_Data				
FLTCC	Number of flight cycles detected during an engine run sequence.	Integer_Data	counts		0	100
FLTCYC1	Temporary Engine Flight Cycle Flag	Boolean_Data				
FLTPTR	START ADDRESS OF FAULT DATA in EEPROM	Integer_Data				
FLTSEC	Flight Time (seconds)	Analog_Data	seconds			

Name	Description	Type	Units	Value	Min Val	Max Val
FLTSTOP	Engine Flight Transitioned from RUN to OFF	Boolean_Data				
FLTSTRT	Engine Flight Transitioned from OFF to RUN	Boolean_Data				
FLTTC	FLIGHT TIME COARSE (HOURS)	Analog_Data			0	32767
FLTTF	FLIGHT TIME FINE (FRACTION OF HOURS)	Analog_Data	HOURS		0	1.0
FLTWRD	Fault Discrete Input Word	Integer_Data				
FLT_MS_RAW	Raw second counter for Flight Time logic	Analog_Data	seconds		0	681.5744
FLT_TIMER_RESET	Flight timer reset timers flag	Boolean_Data				
FLXTMP	RATING LOGIC - EXTERNAL THETA REFERENCE TEMP. - ATO	Analog_Data	deg F		-76	139
FORRTN	Module Return variable	Analog_Data				
FSW_CPDATA_READ	DCU BLOCK STATUS COMPONENT DATA READ ERROR BIT	Integer_Data		2		
FSW_CTRDATA_READ	DCU BLOCK STATUS COMPONENT DATA READ ERROR BIT	Integer_Data				
FSW_DCUID_READ	DCU ID BLOCK STATUS WORD READ ERROR BIT	Integer_Data		4		
FSW_DCU_INACTIVE	DCU DEVICE PROTOCOL NOT ACTIVE	Integer_Data		%O'20000'		
FSW_EECDATA_READ	DCU EEC DATA BLOCK STATUS READ ERROR BIT	Integer_Data		%O'20'		
FSW_ENGDATA_READ	DCU ENGINE DATA BLOCK STATUS READ ERROR BIT	Integer_Data		1		
FSW_INCONSISTANT	DCU BLOCK STATUS WORD INCONSISTANT FAULT BIT	Integer_Data		%O'400'		
FSW_LCFDATA_READ	DCU LCF DATA BLOCK STATUS READ ERROR BITS	Integer_Data		%O'340'		
FWINDX	FAULT WORD INDEX	Integer_Data			0	24
FWORD	TEMP HOLDER OF A FAULT WORD	Integer_Data				
FWRD26	HOLDER OF FAULT WORD BEING CHECKED	Integer_Data				
FW_LATCH	DCU STATUS WORD LATCH BIT	Integer_Data		%O'100000'		
FW_MAX_COUNTER	DCU STATUS WORD MAX FAULT COUNTER	Integer_Data		12		
GAINC	HIGH LEVEL A/D INTERFACE GAIN FACTOR	Analog_Data	VDC/VDC		0.9863	1.017
GAINIF	High Level Gain interface status word	Integer_Data				
GARRAY	VALUE WILL DEFINE THE TYPE OF IIO4 GATE ARRAY IN THE EEC	Integer_Data				
GNCTCT	High Level Gain Transient Fault Timer	Integer_Data			0	25
HBDTCT	BLEED DISCRETE WRAPAROUND TRANSIENT TIMER	Integer_Data			0	26
HBOVIF	Loss of HBOV Control Status Word	Integer_Data				
HBRCTCT	BLEED CURRENT WRAPAROUND OUT OF RANGE TRANSIENT TIMER	Integer_Data			0	25
HBTCT	HBOV W/A TRANSIENT FAULT TIMER	Integer_Data	counts		0	25
HVBDF	1=DISABLED HBOV BACKUP BLEED VALVE CONTROL	Boolean_Data				
HBVDE	DECR RATE LIM ON HBOV CURR SCHED FROM PLADBG	Analog_Data	MA/SEC		-500.0	0.0
HBVNDF	1=DISABLED HBOV NORMAL BLEED VALVE CONTROL	Boolean_Data				
HBVPDF	1=DISABLED HBOV PLA ENABLE FLAG	Boolean_Data				
HBVSS	MANUAL HBOV STEADY STATE REFERENCE	Analog_Data	mA		0	75
HBWA1C	BLEED DISCRETE WRAPAROUND FAULT COUNTER	Integer_Data			0	21
HBWA2C	BLEED ANALOG WRAPAROUND FAULT COUNTER	Integer_Data			0	21
HBWAIF	HBOV W/A INTERFACE STATUS WORD	Integer_Data				
HOTMDI	1=PROP BRAKE SELECTED)	Boolean_Data				
HOTMOD	1=HOTEL MODE ENABLE DISCRETE	Boolean_Data		True		
HPDMP	HORSE POWER DUMP	Analog_Data	shp		-1250.0	0.0
HPDOT	HPFB DERIVATIVE CALCULATED	Analog_Data	shp/sec		-2000.0	2000.0
HPERR	HP ERROR	Analog_Data	shp	0	-5000	5000
HPFB	HORSE POWER FEED BACK CALCULATED	Analog_Data	shp		0.0	5000.0
HPFBC	HORSE POWER FEED BACK CORRECTED	Analog_Data	shp		0.0	5000.0
HPIMP_ACF	Selected HP Impeller ACF	Analog_Data	None		0	16
HPIMP_CYCCRS	HP Impeller Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
HPIMP_CYCFIN	HP Impeller Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1
HPIMP_FCFM1	Selected HP Impeller FCF Mission #1	Analog_Data	None		0	16
HPIMP_FCFM2	Selected HP Impeller FCF Mission #2	Analog_Data	None		0	16
HPIMP_FCFM3	Selected HP Impeller FCF Mission #3	Analog_Data	None		0	16
HPIMP_FCFM4	Selected HP Impeller FCF Mission #4	Analog_Data	None		0	16
HPIMP_FCFM5	Selected HP Impeller FCF Mission #5	Analog_Data	None		0	16
HPIMP_FCFM6	Selected HP Impeller FCF Mission #6	Analog_Data	None		0	16
HPIMP_FCFMAX	Stored Value of the Maximum High-Pressure Impeller FCF During an Engine Run Cycle	Analog_Data	None		0	16
HPIMP_PNA12	HP Impeller P/N ASCII Chars 1 & 2	Integer_Data				
HPIMP_PNA12EE	HP Impeller P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
HPIMP_PNA34	HP Impeller P/N ASCII Chars 3 & 4	Integer_Data				
HPIMP_PNA34EE	HP Impeller P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
HPIMP_PNA56	HP Impeller P/N ASCII Chars 5 & 6	Integer_Data				
HPIMP_PNA56EE	HP Impeller P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
HPIMP_PNA78	HP Impeller P/N ASCII Chars 7 & 8 (ASC8 is a DASH)	Integer_Data				
HPIMP_PNA78EE	HP Impeller P/N ASCII Chars 7 & 8 (ASC8 is a DASH) in EEPROM	Integer_Data				
HPIMP_SNA12	HP Impeller S/N ASCII Chars 1 & 2	Integer_Data				
HPIMP_SNA12EE	HP Impeller S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
HPIMP_SNA34	HP Impeller S/N ASCII Chars 3 & 4	Integer_Data				
HPIMP_SNA34EE	HP Impeller S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
HPIMP_SNA56	HP Impeller S/N ASCII Chars 5 & 6	Integer_Data				
HPIMP_SNA56EE	HP Impeller S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
HPKI	HP LOOP INTEGRAL GAIN	Analog_Data	rpm/sec-sh	0	0.0	16.0
HPKIC	HP LOOP INTEGRAL GAIN CORRECTED	Analog_Data	RPM/SEC*H		0.0	10.0
HPKP	HP LOOP PROPORTIONAL GAIN	Analog_Data	rpm/HP	0	0.0	22.0
HPKPC1	HP LOOP PROPORTIONAL GAIN CORRECTED BY RTHETA & DELTA1	Analog_Data	RPM/HP		0.0	15.0
HPKPC2	HP LOOP PROPORTIONAL GAIN CORRECTED BY RTHETA	Analog_Data	RPM/HP		0.0	16.5
HPKPC3	HP LOOP PROPORTIONAL GAIN CORRECTED	Analog_Data	RPM/HP		0.0	15.0
HPREF	RANGE LIMITED POWER REFERENCE	Analog_Data	shp		0.0	3162.0
HPREFZ	PAST VALUE OF HPREF	Analog_Data	shp	0	-4000.0	4000.0
HPREQ	HORSEPOWER REQUEST CALCULATED	Analog_Data	shp		-2530.0	4500.0
HPTD_ACF	Selected HP Turbine Disk ACF	Analog_Data	None		0	16
HPTD_CYCCRS	HPT Disk Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
HPTD_CYCFIN	HPT Disk Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1
HPTD_FCFM1	Selected HP Turbine Disk FCF Mission #1	Analog_Data	None		0	16
HPTD_FCFM2	Selected HP Turbine Disk FCF Mission #2	Analog_Data	None		0	16
HPTD_FCFM3	Selected HP Turbine Disk FCF Mission #3	Analog_Data	None		0	16
HPTD_FCFM4	Selected HP Turbine Disk FCF Mission #4	Analog_Data	None		0	16
HPTD_FCFM5	Selected HP Turbine Disk FCF Mission #5	Analog_Data	None		0	16
HPTD_FCFM6	Selected HP Turbine Disk FCF Mission #6	Analog_Data	None		0	16
HPTD_FCFMAX	Stored Value of the Maximum High-Pressure Turbine Disk FCF During an Engine Run Cycle	Analog_Data	None		0	16
HPTD_PNA12	HPT Disk P/N ASCII Chars 1 & 2	Integer_Data				
HPTD_PNA12EE	HPT Disk P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
HPTD_PNA34	HPT Disk P/N ASCII Chars 3 & 4	Integer_Data				
HPTD_PNA34EE	HPT Disk P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
HPTD_PNA56	HPT Disk P/N ASCII Chars 5 & 6	Integer_Data				
HPTD_PNA56EE	HPT Disk P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
HPTD_PNA78	HPT Disk P/N ASCII Chars 7 & 8 (ASC8 is a DASH)	Integer_Data				
HPTD_PNA78EE	HPT Disk P/N ASCII Chars 7 & 8 (ASC8 is a DASH) in EEPROM	Integer_Data				
HPTD_SN12EE	HPT Disk S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
HPTD_SN34EE	HPT Disk S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
HPTD_SN56EE	HPT Disk S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
HPTD_SNA12	HPT Disk S/N ASCII Chars 1 & 2	Integer_Data				
HPTD_SNA34	HPT Disk S/N ASCII Chars 3 & 4	Integer_Data				
HPTD_SNA56	HPT Disk S/N ASCII Chars 5 & 6	Integer_Data				
HPTFCP_ACF	Selected HP Turbine Front Cover Plate ACF	Analog_Data	None		0	16
HPTFCP_CYCCRS	HPT Front Cover Plate Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
HPTFCP_CYCFIN	HPT Front Cover Plate Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1
HPTFCP_FCFM1	Selected HP Turbine Front Cover Plate FCF Mission #1	Analog_Data	None		0	16
HPTFCP_FCFM2	Selected HP Turbine Front Cover Plate FCF Mission #2	Analog_Data	None		0	16
HPTFCP_FCFM3	Selected HP Turbine Front Cover Plate FCF Mission #3	Analog_Data	None		0	16
HPTFCP_FCFM4	Selected HP Turbine Front Cover Plate FCF Mission #4	Analog_Data	None		0	16
HPTFCP_FCFM5	Selected HP Turbine Front Cover Plate FCF Mission #5	Analog_Data	None		0	16
HPTFCP_FCFM6	Selected HP Turbine Front Cover Plate FCF Mission #6	Analog_Data	None		0	16
HPTFCP_FCFMAX	Stored Value of the Maximum High-Pressure Turbine Front Cover Plate FCF During an Engine Run Cycle	Analog_Data	None		0	16
HPTFCP_PNA12	HPT Front Cover Plate P/N ASCII Chars 1 & 2	Integer_Data				
HPTFCP_PNA12EE	HPT Front Cover Plate P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
HPTFCP_PNA34	HPT Front Cover Plate P/N ASCII Chars 3 & 4	Integer_Data				
HPTFCP_PNA34EE	HPT Front Cover Plate P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
HPTFCP_PNA56	HPT Front Cover Plate P/N ASCII Chars 5 & 6	Integer_Data				
HPTFCP_PNA56EE	HPT Front Cover Plate P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
HPTFCP_PNA78	HPT Front Cover Plate P/N ASCII Chars 7 & 8	Integer_Data				
HPTFCP_PNA78EE	HPT Front Cover Plate P/N ASCII Chars 7 & 8 in EEPROM	Integer_Data				
HPTFCP_SNA12	HPT Front Cover Plate S/N ASCII Chars 1 & 2	Integer_Data				
HPTFCP_SNA12EE	HPT Front Cover Plate S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
HPTFCP_SNA34	HPT Front Cover Plate S/N ASCII Chars 3 & 4	Integer_Data				
HPTFCP_SNA34EE	HPT Front Cover Plate S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
HPTFCP_SNA56	HPT Front Cover Plate S/N ASCII Chars 5 & 6	Integer_Data				
HPTFCP_SNA56EE	HPT Front Cover Plate S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
HPTRCP_ACF	Selected HP Turbine Rear Cover Plate ACF	Analog_Data	None		0	16
HPTRCP_CYCCRS	HPT Rear Cover Plate Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
HPTRCP_CYCFIN	HPT Rear Cover Plate Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1

Name	Description	Type	Units	Value	Min Val	Max Val
HPTRCP_FCFM1	Selected HP Turbine Rear Cover Plate FCF Mission #1	Analog_Data	None		0	16
HPTRCP_FCFM2	Selected HP Turbine Rear Cover Plate FCF Mission #2	Analog_Data	None		0	16
HPTRCP_FCFM3	Selected HP Turbine Rear Cover Plate FCF Mission #3	Analog_Data	None		0	16
HPTRCP_FCFM4	Selected HP Turbine Rear Cover Plate FCF Mission #4	Analog_Data	None		0	16
HPTRCP_FCFM5	Selected HP Turbine Rear Cover Plate FCF Mission #5	Analog_Data	None		0	16
HPTRCP_FCFM6	Selected HP Turbine Rear Cover Plate FCF Mission #6	Analog_Data	None		0	16
HPTRCP_FCFMAX	Stored Value of the Maximum High-Pressure Turbine Rear Cover Plate FCF During an Engine Run Cycle	Analog_Data	None		0	16
HPTRCP_PNA12	HPT Rear Cover Plate P/N ASCII Chars 1 & 2	Integer_Data				
HPTRCP_PNA12EE	HPT Rear Cover Plate P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
HPTRCP_PNA34	HPT Rear Cover Plate P/N ASCII Chars 3 & 4	Integer_Data				
HPTRCP_PNA34EE	HPT Rear Cover Plate P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
HPTRCP_PNA56	HPT Rear Cover Plate P/N ASCII Chars 5 & 6	Integer_Data				
HPTRCP_PNA56EE	HPT Rear Cover Plate P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
HPTRCP_PNA78	HPT Rear Cover Plate P/N ASCII Chars 7 & 8	Integer_Data				
HPTRCP_PNA78EE	HPT Rear Cover Plate P/N ASCII Chars 7 & 8 in EEPROM	Integer_Data				
HPTRCP_SNA12	HPT Rear Cover Plate S/N ASCII Chars 1 & 2	Integer_Data				
HPTRCP_SNA12EE	HPT Rear Cover Plate S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
HPTRCP_SNA34	HPT Rear Cover Plate S/N ASCII Chars 3 & 4	Integer_Data				
HPTRCP_SNA34EE	HPT Rear Cover Plate S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
HPTRCP_SNA56	HPT Rear Cover Plate S/N ASCII Chars 5 & 6	Integer_Data				
HPTRCP_SNA56EE	HPT Rear Cover Plate S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
HPZM1	HPFB DELAYED ONCE	Analog_Data	shp		0.0	5000.0
HPZM2	HPFB DELAYED TWO TIMES	Analog_Data	shp		0.0	5000.0
HPZM3	HPFB DELAYED THREE TIMES	Analog_Data	shp		0.0	5000.0
HPZM4	HPFB DELAYED FOUR TIMES	Analog_Data	shp		0.0	5000.0
HS1TCT	HSS #1 transient fault timer	Integer_Data	Counter		0	26
HS2TCT	HSS #2 transient fault timer	Integer_Data	Counter		0	25
HSS1IF	High side switch #1 interface status word	Integer_Data				
HSS2IF	High side switch #2 interface status word	Integer_Data				
IADPF	IAS derived from deltaP is not valid	Boolean_Data				
IAS	INDICATED AIR SPEED	Analog_Data	KTS		0	350
IAS206	ADC RAW IAS INPUT	Analog_Data	KNOTS		-512	512
IASDAT	ADC IAS Data Valid Flag 1 Data Valid 2 - 16 Not Used	Integer_Data			0	1
IASDP	Indicated air speed	Analog_Data	KNOTS		0	330.0
IASFLG	FLAG INDICATES THAT DEFAULT IAS DATA SHOULD BE USED	Integer_Data			0	2
IASIF	DUAL AIRSPEED STATUS WORD	Integer_Data				
IASLM	INDICATED AIR SPEED LIMITED	Analog_Data	kts		0.0	310.0
IAS	IAS CALCULATED FROM DELTA PRESSURE	Analog_Data	KTS		0.0	330.0
IASSF	1=IAS INPUT FAILED	Boolean_Data				
IASTIM	ADC IAS INVALID DATA TIMER	Integer_Data			0	122
IBLED2	BLEED SCHEDULED FROM NHC	Analog_Data	MA		0.0	75.0
IBLED3	HBOV CURRENT SCHEDULED FROM PLADB2	Analog_Data	MA		0.0	140.0
IBLEDE	EEC MODE BLEED VALUE	Analog_Data	MA		0.0	75.0
IBLEDM	BACKUP MODE BLEED	Analog_Data	MA	0	0.0	75.0
IBLEDS	BLEED VALUE SELECTED	Analog_Data	MA		0.0	75.0
IBLEED	HANDLING BLEED VALVE POSITION	Analog_Data	MA		0.0	75.0
IC	Initial-Condition Flag	Integer_Data			-3	0
IGNITE	1=AUTO IGNITION ENABLED	Boolean_Data				
ILOOP	Loop Identification Discrete	Integer_Data				
INACTIVE	DCU inactive DCU transfer state	Integer_Data		1		
INC_MTO	Flag for Maximum Takeoff	Boolean_Data			False	True
INC_NTO	Flag for Normal Takeoff	Boolean_Data			False	True
INITF	INITIALIZATION FLAG	Boolean_Data				
INITIALIZE	DCU initialize DCU transfer state	Integer_Data		0		
INS1	TEMPORARY STORAGE OF INA INSTRUCTION	Integer_Data			13312	13823
INS2	TEMPORARY STORAGE OF RETURN INSTRUCTION	Integer_Data			-30720	-30209
INS3	TEMPORARY STORAGE OF RETURN ADDRESS	Integer_Data			0	32767
INTIAS	Interim value of IASDP	Analog_Data	KNOTS		0	350
INTTQ	NON-RATE LIMITED TQ VALUE	Analog_Data	mV		-4.8	57.8
ISS_ACF	Selected Inter-stage Seal ACF	Analog_Data	None		0	16
ISS_CYCCRS	Inter-stage Seal Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
ISS_CYCFIN	Inter-stage Seal Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1
ISS_FCFM1	Selected Inter-stage Seal FCF Mission #1	Analog_Data	None		0	16
ISS_FCFM2	Selected Inter-stage Seal FCF Mission #2	Analog_Data	None		0	16
ISS_FCFM3	Selected Inter-stage Seal FCF Mission #3	Analog_Data	None		0	16

Name	Description	Type	Units	Value	Min Val	Max Val
ISS_FCFM4	Selected Inter-stage Seal FCF Mission #4	Analog_Data	None		0	16
ISS_FCFM5	Selected Inter-stage Seal FCF Mission #5	Analog_Data	None		0	16
ISS_FCFM6	Selected Inter-stage Seal FCF Mission #6	Analog_Data	None		0	16
ISS_FCFMAX	Stored Value of the Maximum Inter-Stage Seal FCF During an Engine Run Cycle	Analog_Data	None		0	16
ISS_PNA12	Inter-Stage Seal P/N ASCII Chars 1 & 2	Integer_Data				
ISS_PNA12EE	Inter-Stage Seal P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
ISS_PNA34	Inter-Stage Seal P/N ASCII Chars 3 & 4	Integer_Data				
ISS_PNA34EE	Inter-Stage Seal P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
ISS_PNA56	Inter-Stage Seal P/N ASCII Chars 5 & 6	Integer_Data				
ISS_PNA56EE	Inter-Stage Seal P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
ISS_PNA78	Inter-Stage Seal P/N ASCII Chars 7 & 8 (ASC8 is a DASH)	Integer_Data				
ISS_PNA78EE	Inter-Stage Seal P/N ASCII Chars 7 & 8 (ASC8 is a DASH) in EEPROM	Integer_Data				
ISS_SNA12	Inter-Stage Seal S/N ASCII Chars 1 & 2	Integer_Data				
ISS_SNA12EE	Inter-Stage Seal S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
ISS_SNA34	Inter-Stage Seal S/N ASCII Chars 3 & 4	Integer_Data				
ISS_SNA34EE	Inter-Stage Seal S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
ISS_SNA56	Inter-Stage Seal S/N ASCII Chars 5 & 6	Integer_Data				
ISS_SNA56EE	Inter-Stage Seal S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
KNH_ENG_OFF	Value used to determine if the engine is in OFF Mode.	Analog_Data	RPM			
KNH_ENG_RUN	Value used to determine if the engine is in RUN Mode.	Analog_Data	RPM			
KVK_FLT_ADC	Trip Point For VK (ADC Supplied Input) Airspeed Selection	Analog_Data	Knots		0.0	350.0
KVK_FLT_DPSNR	Trip Point For VK (EEC's deltaP Sensor Input) Airspeed Selection	Analog_Data	Knots		0.0	350.0
LCF_FCF_MAX_PTR	Pointer within LCF_FCF_MAX_TBL	Integer_Data				
LCF_NUMER	NUMERATOR FOR LCF CYCLE COUNT CALCULATIONS	Analog_Data				
LCF_PN_LENGTH	LENGTH OF EEC PART NUMBER	Integer_Data		4		
LCF_PN_NOTFOUND	LCF PART NUMBER NOT FOUND FLAG	Integer_Data			0	3
LCF_READ_FAULTS	START ADDRESS OF DCU LCF BLOCK READ FAULTS	Integer_Data				
LCF_VALUE1	LOWER WORD FOR LCF CYCLE COUNT CALCULATIONS	Integer_Data				
LCF_VALUE2	UPPER WORD OF LCF CYCLE COUNT CALCULATIONS	Integer_Data				
LIM0TM	TIME AT WHICH TORQUE LIMIT 0 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIM1TM	TIME AT WHICH TORQUE LIMIT 1 WAS EXCEEDED	Analog_Data	sec	0	0.0	1000
LIM2TM	TIME AT WHICH TORQUE LIMIT 2 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIM3TM	TIME AT WHICH TORQUE LIMIT 3 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIM4TM	TIME AT WHICH TORQUE LIMIT 4 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIM5TM	TIME AT WHICH TORQUE LIMIT 5 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIM6TM	TIME AT WHICH TORQUE LIMIT 6 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIM7TM	TIME AT WHICH TORQUE LIMIT 7 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIM8TM	TIME AT WHICH TORQUE LIMIT 8 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIM9TM	TIME AT WHICH TORQUE LIMIT 9 WAS EXCEEDED	Analog_Data	sec		0.0	1000
LIMFLO	FLAG INDICATES TORQUE LIMIT 0 EXCEEDED	Boolean_Data				
LIMFL1	FLAG INDICATES TORQUE LIMIT 1 EXCEEDED	Boolean_Data				
LIMFL2	FLAG INDICATES TORQUE LIMIT 2 EXCEEDED	Boolean_Data				
LIMFL3	FLAG INDICATES TORQUE LIMIT 3 EXCEEDED	Boolean_Data				
LIMFL4	FLAG INDICATES TORQUE LIMIT 4 EXCEEDED	Boolean_Data				
LIMFL5	FLAG INDICATES TORQUE LIMIT 5 EXCEEDED	Boolean_Data				
LIMFL6	FLAG INDICATES TORQUE LIMIT 6 EXCEEDED	Boolean_Data				
LIMFL7	FLAG INDICATES TORQUE LIMIT 7 EXCEEDED	Boolean_Data				
LIMFL8	FLAG INDICATES TORQUE LIMIT 8 EXCEEDED	Boolean_Data				
LIMFL9	FLAG INDICATES TORQUE LIMIT 9 EXCEEDED	Boolean_Data				
LOWB	LS Byte / MS Byte Flag	Integer_Data			0	1
LO_BYTE_MASK	Mask for lower 8 bits of a 16 bit word	Integer_Data		%0'377'		
LPIMP_ACF	Selected LP Impeller ACF	Analog_Data	None		0	16
LPIMP_CYCCRS	LP Impeller Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
LPIMP_CYCFIN	LP Impeller Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1
LPIMP_FCFM1	Selected LP Impeller FCF Mission #1	Analog_Data	None		0	16
LPIMP_FCFM2	Selected LP Impeller FCF Mission #2	Analog_Data	None		0	16
LPIMP_FCFM3	Selected LP Impeller FCF Mission #3	Analog_Data	None		0	16
LPIMP_FCFM4	Selected LP Impeller FCF Mission #4	Analog_Data	None		0	16
LPIMP_FCFM5	Selected LP Impeller FCF Mission #5	Analog_Data	None		0	16
LPIMP_FCFM6	Selected LP Impeller FCF Mission #6	Analog_Data	None		0	16
LPIMP_FCFMAX	Stored Value of the Maximum Low-Pressure Impeller FCF During an Engine Run Cycle	Analog_Data	None		0	16
LPIMP_PNA12	LP Impeller P/N ASCII Chars 1 & 2	Integer_Data				
LPIMP_PNA12EE	LP Impeller P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
LPIMP_PNA34	LP Impeller P/N ASCII Chars 3 & 4	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
LPIMP_PNA34EE	LP Impeller P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
LPIMP_PNA56	LP Impeller P/N ASCII Chars 5 & 6	Integer_Data				
LPIMP_PNA56EE	LP Impeller P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
LPIMP_PNA78	LP Impeller P/N ASCII Chars 7 & 8 (ASC8 is a DASH)	Integer_Data				
LPIMP_PNA78EE	LP Impeller P/N ASCII Chars 7 & 8 (ASC8 is a DASH) in EEPROM	Integer_Data				
LPIMP_SNA12	LP Impeller S/N ASCII Chars 1 & 2	Integer_Data				
LPIMP_SNA12EE	LP Impeller S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
LPIMP_SNA34	LP Impeller S/N ASCII Chars 3 & 4	Integer_Data				
LPIMP_SNA34EE	LP Impeller S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
LPIMP_SNA56	LP Impeller S/N ASCII Chars 5 & 6	Integer_Data				
LPIMP_SNA56EE	LP Impeller S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
LPTD_ACF	Selected LP Turbine Disk ACF	Analog_Data	None		0	16
LPTD_CYCCRS	LPT Disk Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
LPTD_CYCFIN	LPT Disk Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1
LPTD_FCFM1	Selected LP Turbine Disk FCF Mission #1	Analog_Data	None		0	16
LPTD_FCFM2	Selected LP Turbine Disk FCF Mission #2	Analog_Data	None		0	16
LPTD_FCFM3	Selected LP Turbine Disk FCF Mission #3	Analog_Data	None		0	16
LPTD_FCFM4	Selected LP Turbine Disk FCF Mission #4	Analog_Data	None		0	16
LPTD_FCFM5	Selected LP Turbine Disk FCF Mission #5	Analog_Data	None		0	16
LPTD_FCFM6	Selected LP Turbine Disk FCF Mission #6	Analog_Data	None		0	16
LPTD_FCFMAX	Stored Value of the Maximum Low-Pressure Turbine Disk FCF During an Engine Run Cycle	Analog_Data	None		0	16
LPTD_PNA12	LPT Disk P/N ASCII Chars 1 & 2	Integer_Data				
LPTD_PNA12EE	LPT Disk P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
LPTD_PNA34	LPT Disk P/N ASCII Chars 3 & 4	Integer_Data				
LPTD_PNA34EE	LPT Disk P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
LPTD_PNA56	LPT Disk P/N ASCII Chars 5 & 6	Integer_Data				
LPTD_PNA56EE	LPT Disk P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
LPTD_PNA78	LPT Disk P/N ASCII Chars 7 & 8 (ASC8 is a DASH)	Integer_Data				
LPTD_PNA78EE	LPT Disk P/N ASCII Chars 7 & 8 (ASC8 is a DASH) in EEPROM	Integer_Data				
LPTD_SNA12	LPT Disk S/N ASCII Chars 1 & 2	Integer_Data				
LPTD_SNA12EE	LPT Disk S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
LPTD_SNA34	LPT Disk S/N ASCII Chars 3 & 4	Integer_Data				
LPTD_SNA34EE	LPT Disk S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
LPTD_SNA56	LPT Disk S/N ASCII Chars 5 & 6	Integer_Data				
LPTD_SNA56EE	LPT Disk S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
LRINDX	OFFSET INTO THE LRU-CODE TABLE	Integer_Data			0	24
LRUCDE	LRU FAULT CODE	Integer_Data	BCD		3	152
LRUCDI	LRU FAULT CODE	Integer_Data				
LRUTBL	NCFAL position from LRU table for all configurations	Integer_Data				
LS1TCT	LSS #1 transient fault timer	Integer_Data	Counter		0	25
LS2TCT	LSS #2 transient fault timer	Integer_Data	Counter		0	25
LS3TCT	LSS #3 transient fault timer	Integer_Data	Counter		0	25
LS4TCT	LSS #4 transient fault timer	Integer_Data	Counter		0	25
LS5TCT	LSS #5 transient fault timer	Integer_Data	Counter		0	25
LSS1IF	Low side switch #1 interface status word	Integer_Data				
LSS2IF	Low side switch #2 interface status word	Integer_Data				
LSS2OUT	LSS2 Combined Output Signal	Boolean_Data				
LSS3IF	Low side switch #3 interface status word	Integer_Data				
LSS3OUT	LSS3 Combined Output Signal	Boolean_Data		False		
LSS4IF	Low side switch #4 interface status word	Integer_Data				
LSS5IF	Low side switch #5 interface status word	Integer_Data				
MACH	MACH NUMBER	Analog_Data	---		0.0	0.7103
MASK1	MASK WITH 1 BIT SHIFTED LEFT EVERY COUNT	Integer_Data				
MASKI1	MASK WITH 1 BIT SHIFTED LEFT EVERY COUNT	Integer_Data				
MASLMF	BACKUP MODE SLAM FLAG	Boolean_Data				
MAXRAT	MAX RATE OF STEPPER MOTOR TRAVEL IN STEPS PER CYCLE	Integer_Data	steps/cycl		0	10
MCLBM	MCL Bleed Modifier Table Output	Analog_Data	---		-2.0	2.0
MCLRGL	MCL Range Limited	Analog_Data	shp		200.0	2192.0
MCLRTL	Range limited and filtered MCL Rated Power	Analog_Data	shp		200.0	2192.0
MCLRTP	MCL Rated Power	Analog_Data	shp		0.0	2192.0
MCLTB	MCL Thermal Limit Table Output	Analog_Data	SHP		0.0	8000.0
MCLTC	MCL Thermal Limit with Bleed Modifier	Analog_Data	SHP		0.0	8000.0
MCLTL	Max Climb Thermal Limit	Analog_Data	SHP		0.0	8000.0
MCLTQB	MCL TORQUE BUG	Analog_Data	ft-lbf		648.0	12792.0
MCRBM	MCR Bleed Modifier Table Output	Analog_Data	---		-2.0	2.0

Name	Description	Type	Units	Value	Min Val	Max Val
MCRRGL	MCR Range Limited	Analog_Data	shp		200.0	2132.0
MCRRTL	Range limited and filtered MCR Rated Power	Analog_Data	shp		200.0	2132.0
MCRRTP	MCR Rated Power	Analog_Data	shp		0.0	2132.0
MCRTB	MCR Thermal Limit Table Output	Analog_Data	SHP		0.0	8000.0
MCRTC	MCR Thermal Limit with Bleed Modifier	Analog_Data	SHP		0.0	8000.0
MCRTL	Max Cruise Thermal Limit	Analog_Data	SHP		0.0	8000.0
MCRTQB	MCR TORQUE BUG	Analog_Data	ft-lbf		648.0	12442.0
MCTBM	MCT Bleed Modifier Table Output	Analog_Data	---		-2.0	2.0
MCTRGL	MCT Range Limited	Analog_Data	shp		200.0	2500.0
MCTRTP	MCT Rated Power	Analog_Data	shp		0.0	2500.0
MCTSHPD	MCT Blending Fader Map Output	Analog_Data	SHP/SHP		0.0	1.0
MCTTB	MCT Thermal Limit Table Output	Analog_Data	SHP		0.0	8000.0
MCTTC	MCT Thermal Limit with Bleed Modifier	Analog_Data	SHP		0.0	8000.0
MCTTL	Max Continuous Thermal Limit	Analog_Data	SHP		0.0	8000.0
MDSLMF	BACKUP MODE DECEL SLAM FLAG	Boolean_Data				
MFCIF	MFC INTERFACE STATUS WORD	Integer_Data				
MFCINT	1= MFC IN INTERFERENCE	Boolean_Data				
MINCYC	Minor cycle indicator	Integer_Data			1	4
MISSION	Selected Mission for engine configuration with respect to cycle usage.	Integer_Data			0	6
MNTZ1	Thermal Limit Table read MN value 1	Analog_Data	---		0.0	1.0
MNTZ2	Thermal Limit Table read MN value 2	Analog_Data	---		0.0	1.0
MODCON	Constant used to alternate the value of MODCNG	Analog_Data		1		
MODDCUDATA	Reprogrammed DCU Flag	Boolean_Data				
MODE	Engine Mode Selection (Values are in octal) 0 INVALID 1 DEVMOD 2 NORMOD	Integer_Data			0	2
MOST	ARINC INPUT LS/MS WORD FLAG	Integer_Data	Counter		0	3
MRLSLMF	BACKUP MODE RESLAM FLAG	Boolean_Data				
MSNFLG	Intermediate Selected Mission for engine configuration with respect to cycle usage	Integer_Data			0	6
MSNFLG1	Reset to advanced mission timer for engine running	Boolean_Data				
MSNFLG2	Advanced Mission Confirmation Flag	Boolean_Data				
MSTATE	NPARAM INPUT STATE	Integer_Data				
MTDELO	1=Change in MTC Enable delayed off	Boolean_Data				
MTLHI	Thermal Limit Table read high value	Analog_Data	SHP		0.0	8000.0
MTLLO	Thermal Limit Table read low value	Analog_Data	SHP		0.0	8000.0
MTOBM	MTO Bleed Modifier Table Output	Analog_Data	---		-2.0	2.0
MTODRL	MTO Detent Range Limited	Analog_Data	shp		200.0	2750.0
MTODRTP	MTO Detent Rated Power	Analog_Data	shp		0.0	2750.0
MTORGL	MTO Range Limited	Analog_Data	shp		200.0	2750.0
MTORTL	Range limited and filtered MTO Rated Power	Analog_Data	shp		200.0	2750.0
MTORTP	MTO Rated Power	Analog_Data	shp		0.0	2750.0
MTOT9	0.9 of Max Takeoff Thermal Limit	Analog_Data	SHP		0.0	7200.0
MTOTB	MTO Thermal Limit Table Output	Analog_Data	SHP		0.0	8000.0
MTOTC	MTO Thermal Limit with Bleed Modifier	Analog_Data	SHP		0.0	8000.0
MTOTL	Max Takeoff Thermal Limit	Analog_Data	SHP		0.0	8000.0
MTOTQB	MTO TORQUE BUG	Analog_Data	ft-lbf		875.0	12037.0
MTO_DL1	Accumulated Maximum Takeoff Counter at UPRATE Power Rating	Analog_Data	Count		0	32767
MTO_DL2	Accumulated Maximum Takeoff Counter at NORM Power Rating	Analog_Data	Count		0	32767
MTO_DL3	Accumulated Maximum Takeoff Counter at DERATE Power Rating	Analog_Data	Count		0	32767
MTUBM	MTO Uprate Bleed Modifier Table Output	Analog_Data	---		-2.0	2.0
MTUT9	0.9 of Max Takeoff Uprate Thermal Limit	Analog_Data	SHP		0.0	7200.0
MTUTB	MTO Uprate Thermal Limit Table Output	Analog_Data	SHP		0.0	8000.0
MTUTC	MTO Uprate Thermal Limit with Bleed Modifier	Analog_Data	SHP		0.0	8000.0
MTUTL	Max Takeoff Uprate Thermal Limit	Analog_Data	SHP		0.0	8000.0
MUXPTR	POINTER TO START OF STATE TABLE	Integer_Data				
NCFAL1	Non-Critical Fault Summary Word #1	Integer_Data				
NCFAL2	Non-Critical Fault Summary Word #2	Integer_Data				
NCFAL3	Non-Critical Fault Summary Word #3	Integer_Data				
NCFAL4	Non-Critical Fault Summary Word #4	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
NCFAL5	Non-Critical Fault Summary Word #5 1 NPT PROBE #1 LOW SPEED FAILURE 2 NPT PROBE #1 HIGH SPEED FAILURE 3 TORQUE PROBE #1 HIGH SPEED FAILURE 4 TORQUE PROBE #1 LOW SPEED FAILURE 5 LOW SIDE SWITCH #4 W/A FAILURE 6 HIGH SIDE SWITCH #2 W/A FAILURE 7 LOW SIDE SWITCH #3 W/A FAILURE 8 LOW SIDE SWITCH #2 W/A FAILURE 9 NOT USED 10 DUAL NPT CROSSCHECK FAILURE 11 CONFIGURATION TRIM FAILURE 12 CONFIGURATION CROSSCHECK FAILURE - NOT USED 13 LOW SIDE SWITCH #5 W/A FAILURE 14 LOSS OF HBOV CONTROL 15 OVERTORQUE #5 FAILURE - NOT USED 16	Integer_Data				
NCFAL6	Non-Critical Fault Summary Word #6	Integer_Data				
NCFLI1	TEMPORARY COPY OF NCFAL1	Integer_Data				
NCFLI2	TEMPORARY COPY OF NCFAL2	Integer_Data				
NCFLI3	TEMPORARY COPY OF NCFAL3	Integer_Data				
NCFLI4	TEMPORARY COPY OF NCFAL4	Integer_Data				
NCFLI5	TEMPORARY COPY OF NCFAL5	Integer_Data				
NCFLI6	TEMPORARY COPY OF NCFAL6	Integer_Data				
NCFLT1	Temporary Fault Summary Word #1	Integer_Data				
NCFLT2	Temporary Fault Summary Word #2	Integer_Data				
NCFLT3	Temporary Fault Summary Word #3	Integer_Data				
NCFLT4	Temporary Fault Summary Word #4	Integer_Data				
NCFLT5	Temporary Fault Summary Word #5	Integer_Data				
NCFLT6	Temporary Fault Summary Word #6	Integer_Data				
NCFTBL	LRU from NCFAL position table	Integer_Data				
NCMSK1	Fault Summary Word #1 Mask	Integer_Data				
NCMSK2	Fault Summary Word #2 Mask	Integer_Data				
NCMSK3	Fault Summary Word #3 Mask	Integer_Data				
NCMSK4	Fault Summary Word #4 Mask	Integer_Data				
NCMSK5	Fault Summary Word #5 Mask	Integer_Data				
NCMSK6	Fault Summary Word #6 Mask	Integer_Data				
NDBIT	Bit to set NO_DATA Bit in DCU_STATUS	Integer_Data		15		
NGOOD1	Previous NS #1 data is valid	Boolean_Data				
NGOOD2	Previous NS #2 data is valid	Boolean_Data				
NH	Gas Generator Speed	Analog_Data	RPM		0	36820
NH1GD	Primary NH above shutdown flag	Boolean_Data				
NH1IF	NH PROBE #1 Interface Status Word	Integer_Data				
NH1STA	Primary NH transient state counter	Integer_Data			0	1
NH2	NH PROBE #2 Gas Generator Speed	Analog_Data	RPM		0	36820
NH2GB	Secondary NH above shutdown flag	Boolean_Data				
NH2IF	NH Probe #2 Interface Status Word: High Speed	Integer_Data				
NH2STA	Secondary NH transient state counter	Integer_Data			0	1
NHBLF	Blending Factor	Analog_Data	rpm		0.0	1.0
NHC	NH SPEED CORRECTED	Analog_Data	rpm		0.0	43318.0
NHDOT	NH DERIVATIVE CALCULATED	Analog_Data	rpm/sec		-15000.0	15000.0
NHDOTL	HBOV NHDOT LAGGED VALUE	Analog_Data	RPM/SEC		-15000.0	15000.0
NHDOTM	HBOV NHDOT LAGGED VALUE	Analog_Data	rpm/sec	0	-15000	15000
NHECS1	NH probe #1 ECS Fault Status Word	Integer_Data				
NHECS2_16	NH probe #2 ECS Fault Status Word	Integer_Data				
NHERR	NH REQUEST LOOP ERROR	Analog_Data	rpm		-2000	2000
NHF	NPT U/S GAIN BASED ON NH	Analog_Data	st/rpm/s		0.0	0.15
NHFB	NH FEEDBACK SIGNAL	Analog_Data	rpm		0.0	40000.0
NHFBDT	DERIVATIVE OF NHFB	Analog_Data	RPM/SEC		-15000.0	15000.0
NHFZM1	NHFB DELAYED ONCE	Analog_Data	RPM		0.0	40000.0
NHFZM2	NHFB DELAYED TWO TIMES	Analog_Data	RPM		0.0	40000.0
NHFZM3	NHFB DELAYED THREE TIMES	Analog_Data	RPM		0.0	40000.0
NHFZM4	NHFB DELAYED FOUR TIMES	Analog_Data	RPM		0.0	40000.0
NHHMAX	MAXIMUM ALLOWABLE HOTEL MODE NH	Analog_Data	rpm		23365.0	29200.0
NHHMIN	TEMPERATURE COMPENSATED MIN ALLOWABLE NH FOR HOTEL MODE	Analog_Data	rpm		21978.0	23500.0
NHHOT	HOTEL MODE NH REFERENCE	Analog_Data	rpm		19000.0	40000.0
NHHOT1	SELECTED MAX NH LIMIT FOR HOTEL MODE NH	Analog_Data	rpm		21978.0	34600.0
NHHPI	HP LOOP INTEGRAL PATH CONTRIBUTION	Analog_Data	rpm	0	-44000	44000
NHHPP	HP LOOP PROPORTIONAL PATH CONTRIBUTION	Analog_Data	rpm	0	-44000	44000
NHHPR	HP LOOP RESET	Analog_Data	rpm		0.0	44000
NHHPRZ	HP LOOP RESET PAST VALUE	Analog_Data	rpm		0.0	44000
NHHSCH	SCHEDULED HOTEL MODE NH	Analog_Data	rpm		21978.0	34600.0
NHI	Unfiltered NH Speed Reading	Analog_Data	RPM		0	36820
NHI2	Unfiltered NH Speed Reading FROM PROBE #2	Analog_Data	RPM		0	36820

Name	Description	Type	Units	Value	Min Val	Max Val
NHIDLC	CORRECTED MINIMUM NH	Analog_Data	rpm		21978.0	32400.0
NHINDI	1=INHIBIT EEC STEPPER MOTOR OUTPUT	Boolean_Data				
NHIP2	Past value of NH12 / 2	Analog_Data	RPM		0	18410
NHKALT	HOTEL MODE NH SCHEDULE ALTITUDE CORR FACTOR	Analog_Data	rpm/rpm		0.0	0.10
NHKC	COMPENSATOR GAIN	Analog_Data	step/rpm		0.0	0.112837
NHKCP	RATIO OF INTEG. TO PROPR GAIN FOR INNER LOOP RESET CALC	Analog_Data	---		0.0	0.125
NHKP	COMPENSATOR PROPORTIONAL GAIN	Analog_Data	rpm		0.0	0.465453
NHL	NPT U/S CORRECTION BASED ON PLA	Analog_Data	rpm		0.0	6000.0
NHL1IF	NH PROBE #1 Interface Status Word	Integer_Data				
NHL2IF	NH Probe #2 Infrface Status Word: Low Speed	Integer_Data				
NHLD1	COMPENSATOR LEAD TIME CST 1	Analog_Data	sec		0.1	4.2
NHLD1C	COMPENSATOR CORR. LEAD TIME CST 1	Analog_Data	sec		0.2	0.7
NHLD2	COMPENSATOR LEAD TIME CST 2	Analog_Data	sec		0.1	66
NHLD2C	COMPENSATOR CORR. LEAD TIME CST 2	Analog_Data	sec		0.2	11
NHLERR	NH LIMIT ERROR	Analog_Data	rpm	0	-1500	1500
NHLG2	COMPENSATOR LAG TIME CST 2	Analog_Data	1/sec		0.0	10.0
NHLG2C	COMPENSATOR CORR. LAG TIME CST 2	Analog_Data	1/sec		0.0	7.0
NHLG3	COMPENSATOR LAG TIME CONSTANT 1	Analog_Data	SEC		0.0	1.0
NHLG4	COMPENSATOR LAG TIME CONSTANT 2	Analog_Data	1/SEC		0.00	10.0
NHLL	NPT U/S NHL DERIVATIVE LAG VALUE	Analog_Data	rpm		0.0	9000.0
NHLNHF	NPT U/S FILTERED BOOST VALUE CONVERTED	Analog_Data	step/sec		0.0	1350.0
NHLOSD	NPT O/S NHODTB TABLE OUTPUT LAGGED	Analog_Data			0	700
NHLOSU	NPT O/S NHOUTB TABLE OUTPUT LAGGED	Analog_Data			0	1500
NHLOW	1=NH IN AUTO IGNITION RANGE	Boolean_Data				
NHLRST	1=PLA CROSSING HI/LO POWER PLA THRESHOLD	Boolean_Data				
NHLTAU	NPT U/S TIME CST FOR DERIVATIVE LAG	Analog_Data	Sec		0.06	0.6
NHLTRM	NPT U/S UNFILTERED BOOST VALUE	Analog_Data	rpm		0.0	9000.0
NHM	BASELINE NH REFERENCE	Analog_Data	rpm		23365.0	26558.0
NHMAX	NH MAXIMUM SPEED CALCULATED	Analog_Data	rpm		19000	43170
NHMAX1	INTERMEDIATE NH MAXIMUM SPEED CALCULATED	Analog_Data	rpm		19000.0	42670.0
NHMAXC	CORRECTED MAXIMUM NH	Analog_Data	rpm		21978.0	40000.0
NHMBDB	DEAD BANDED NH SPEED	Analog_Data	(RPM)		0.0	36820.0
NHMIN	NH MINIMUM SPEED CALCULATED	Analog_Data	rpm		19000.0	40500.0
NHMIN1	INTERMEDIATE NH MINIMUM SPEED CALCULATED	Analog_Data	rpm		19000.0	40000.0
NHMIN1	CORRECTED MINIMUM NH	Analog_Data	rpm		21978.0	32400.0
NHMIN2	MINIMUM NH From Table MP52TB	Analog_Data	rpm		21978.0	32400.0
NHMX1	CORRECTED MAXIMUM NH	Analog_Data	rpm		21978.0	40000.0
NHMX2	MAXIMUM NH From Table NHR2TB	Analog_Data	rpm		21978.0	32400.0
NHOSD	NPT O/S NHODTB TABLE OUTPUT	Analog_Data			0	700
NHOSU	NPT O/S NHOUTB TABLE OUTPUT	Analog_Data			0	1500
NHP	F/D Period Counter	Analog_Data	COUNTS		0	32768
NHP2	F/D Period Counter FOR PROBE #2	Analog_Data	Counts		0	32768
NHREF	NH REFERENCE CALCULATED	Analog_Data	rpm		0.0	44000
NHREF1	HP GOVERNING NH REFERENCE CALCULATED	Analog_Data	rpm	0	-44000	44000
NHREQ	NH REQUEST CALCULATED	Analog_Data	rpm		21815	44000
NHREQU	UNLIMITED NHREQ	Analog_Data	rpm		0.0	34500
NHREQZ	NH REQUEST DELAYED VALUE	Analog_Data	rpm		21815	44000
NHRQZ2	FEED BACKWARD PAST VALUE OF NHREQ	Analog_Data	rpm		21815	44000.0
NHSEL	NH sensor flag	Boolean_Data				
NHT	F/D Tooth COUNTER	Analog_Data	COUNTS		0	32768
NHT2	F/D Tooth COUNTER FOR PROBE #2	Analog_Data	Counts		0	32768
NHTCT	NH crosscheck transient timer	Integer_Data	Counter		0	101
NHTIF	Dual NH Probe Failure	Integer_Data				
NHTMP1	Rate limited raw primary NH value	Analog_Data	RPM		0	44000
NHTMP2	Rate limited raw secondary NH value	Analog_Data	RPM		0	44000
NHXIF	NH probe cross check fault word	Integer_Data				
NHZM1	NH DELAYED ONCE	Analog_Data	rpm		0	40000
NHZM2	NH DELAYED TWO TIMES	Analog_Data	rpm		0	40000
NHZM3	NH DELAYED THREE TIMES	Analog_Data	rpm		0	40000
NHZM4	NH DELAYED FOUR TIMES	Analog_Data	rpm		0	40000
NLCPL	NL Corrected Limit Table Output	Analog_Data	SHP		1000.0	3000.0
NNHI	Validated high speed turbine data	Analog_Data	RPM		0	36820
NNHI2	Validated high speed turbine data	Analog_Data	RPM		0	36820
NNSI1	Validated NS #1 data	Analog_Data	RPM		0	5755
NNSI2	Validated NS #2 data	Analog_Data	RPM		0	5755
NO_DATA	DCU NO DATA STATUS BIT	Integer_Data		15		

Name	Description	Type	Units	Value	Min Val	Max Val
NP	PROPELLOR SPEED	Analog_Data	rpm		0.0	1620.0
NP1IF	Torque Shaft Speed Input #1 High Range Fault word	Integer_Data				
NP1LIF	Torque Shaft Speed Input #1 Status Word: Low Speed	Integer_Data				
NP2IF	Torque Shaft Speed Input #2 High Range Fault word	Integer_Data				
NP2LIF	Torque Shaft Speed Input #2 Status Word: Low Speed	Integer_Data				
NPARAM	UART Communication Protocol Mode Word 0: DCU Communications Protocol Mode (production) 1: DAS Communication Protocol Mode (i.e., EEPROM RTD Output) 2: N-Parameter Communication Protocol Mode 3: DCU / DAS Communication Modes 4: DCU / N-Parameter Communication Modes 5+: DCU Communication Protocol Mode	Integer_Data				
NPARAM	NPARAM FLAG VALUE IN EEPROM	Integer_Data				
NPLow	1=NP LOWER THAN 24%	Boolean_Data				
NPMBF1	NPARAM INPUT BUFFER #1	Integer_Data				
NPMBF2	NPARAM INPUT BUFFER #2	Integer_Data				
NPMBSL	NPARAM INPUT BUFFER SELECTION FLAG	Integer_Data				
NPMCKH	HIGH WORD OF NPARAM CHECKSUM	Integer_Data				
NPMCKL	LOW WORD OF NPARAM CHECKSUM	Integer_Data				
NPMENI	END OF NPARAM INPUT BUFFER	Integer_Data				
NPMENO	NPARAM END OF BUFFER POINTER	Integer_Data				
NPMENOX	NPARAM BUFFER END	Integer_Data				
NPMPAD	NPARAM ADDRESS BYTE	Integer_Data				
NPMPTI	NPARAM INPUT BUFFER POINTER	Integer_Data				
NPMPTO	NPARAM TRANSMIT BUFFER POINTER	Integer_Data				
NPMPTOX	NPARAM BUFFER START	Integer_Data				
NPMPTT	START OF NPARAM INPUT BUFFER	Integer_Data				
NPOSDF	1=NPT GOVERNING DISABLED	Boolean_Data				
NPPLAR	NPT LOOP RE-ENABLE ACTIVE	Boolean_Data				
NPQBUG	RANGE AND RATE LIMITED NP SPEED USED TO CALC TQBUG.	Analog_Data	rpm		900.0	1620.0
NPRGL	LOW RANGE LIMITED PROP SPEED	Analog_Data	rpm		0.0	1620.0
NPSE	NP SELECTED BY THE Q/T PROT. LOGIC	Analog_Data	(RPM)		0.0	1620.0
NPT	POWER TURBINE SPEED	Analog_Data	rpm		0.0	27000.0
NPTDIS	1=NPT GOVERNING DISABLED	Boolean_Data				
NPTDOT	NPT DERIVATIVE CALCULATED	Analog_Data	rpm/sec		-6400.0	6400.0
NPTER1	NPT LOOP ERROR BEFORE NPT TRACKING	Analog_Data	rpm/sec		-15000	15000
NPTER2	NPT TRACKING ALLOWABLE ERROR	Analog_Data	rpm/sec		-15000	15000
NPTER5	NPT LOOP ERROR BEFORE NPT TRACKING	Analog_Data	RPM/SEC		-15000.0	15000.0
NPTERP	NPT ERROR COMPENSATED	Analog_Data	rpm/sec		-6400.0	6400.0
NPTERR	NPT LOOP ERROR	Analog_Data	rpm		-8000.0	8000.0
NPTISW	1 = NPT GOV INNER LOOP TIME CONSTANT SELECTED	Boolean_Data				
NPTKD	NPT GOV DERIVATIVE GAIN	Analog_Data			0.0	3.0
NPTKP	NPT GOV PROPORTIONAL GAIN	Analog_Data			0.0	3.0
NPTLTR	NP BLENDING FACTOR	Analog_Data			0.0	1.0
NPTOUT	POWER TURBINE SPEED FROM TQ SENSOR 2 FOR ARINC OUTPUT	Analog_Data	rpm		0.0	27000.0
NPTREF	NPT REFERENCE SELECTED	Analog_Data	rpm		0.0	32000.0
NPTRF1	NPT REFERENCE SELECTED	Analog_Data	rpm		0.0	32000
NPTSW	Torque Shaft Speed input interface Status Word	Integer_Data				
NPXCN	Power Turbine dual input speed crosscheck fault counter	Integer_Data	COUNTS		0	361
NPXIF	Power Turbine dual input speed crosscheck fault	Integer_Data				
NPZM1	NPT DELAYED ONCE	Analog_Data	rpm		0	27000
NPZM2	NPT DELAYED TWO TIMES	Analog_Data	rpm		0	27000
NPZM3	NPT DELAYED THREE TIMES	Analog_Data	rpm		0	27000
NPZM4	NPT DELAYED FOUR TIMES	Analog_Data	rpm		0	27000
NS	Filtered Torque Shaft Speed for Control	Analog_Data	RPM		0	5755
NS1OUT	Unfiltered layshaft speed from torque sensor #1	Analog_Data	RPM		0	5755
NSI	Unfiltered select layshaft speed	Analog_Data	RPM		0	5755
NSI1	Torque Shaft Speed Obtained From Input #1	Analog_Data	RPM		0	5755
NSI2	Torque Shaft Speed Obtained From Input #2	Analog_Data	RPM		0	5755
NSIHPS	Past value of Filtered Torque Shaft Speed for Control	Analog_Data	RPM		0	2877.5
NSTAT1	NS #1 transient state counter	Integer_Data			0	2
NSTAT2	NS #2 transient state counter	Integer_Data	counts		0	2
NSYNC	NPARAM SYNCHRONIZATION WORD	Integer_Data				
NTEMP1	Rate limited Raw NS #1 value	Analog_Data	RPM			
NTEMP2	Rate limited Raw NS #2 value	Analog_Data	RPM			
NTORGL	NTO Range Limited	Analog_Data	shp		200.0	2192.0
NTORTP	NTO Rated Power	Analog_Data	shp		0.0	2192.0
NTOTQB	NTO TORQUE BUG	Analog_Data	ft-lbf		875.0	12037.0

Name	Description	Type	Units	Value	Min Val	Max Val
NT0_DL1	Accumulated Normal Takeoff Counter at UPRATE Power Rating	Analog_Data	Count		0	32767
NT0_DL2	Accumulated Normal Takeoff Counter at NORM Power Rating	Analog_Data	Count		0	32767
NT0_DL3	Accumulated Normal Takeoff Counter at DERATE Power Rating	Analog_Data	Count		0	32767
OE	UART OVERRUN ERROR STATUS BIT	Integer_Data		9		
OEBIT	Overrun Error Bit in DCU_DATA	Integer_Data		9		
OKCLR	INDICATES WHETHER TO CLEAR FAULT BLOCKS	Boolean_Data				
OSAVE	COPY OF THE OVERFLOW FLAG	Integer_Data				
OSAVE2	COPY OF THE OVERFLOW FLAG	Integer_Data				
OTNPT	NPT VALUE FOR TORQUE LIMIT INCREASE	Analog_Data	RPM		0.0	21000.0
OTTDT	O/T TEST PROCESS CYCLE TIME	Analog_Data	SEC		0.0	1.0
OTTM0	O/T TEST TIMER 0	Analog_Data	SEC		0.0	1000.0
OTTM1	O/T TEST TIMER 1	Analog_Data	SEC		0.0	400.0
OTTM10	O/T TEST TIMER 10	Analog_Data	SEC		0.0	40.0
OTTM2	O/T TEST TIMER 2	Analog_Data	SEC		0.0	400.0
OTTM3	O/T TEST TIMER 3	Analog_Data	SEC		0.0	300.0
OTTM4	O/T TEST TIMER 4	Analog_Data	SEC		0.0	300.0
OTTM5	O/T TEST TIMER 5	Analog_Data	SEC		0.0	200.0
OTTM6	O/T TEST TIMER 6	Analog_Data	SEC		0.0	200.0
OTTM7	O/T TEST TIMER 7	Analog_Data	SEC		0.0	100.0
OTTM8	O/T TEST TIMER 8	Analog_Data	SEC		0.0	100.0
OTTM9	O/T TEST TIMER 9	Analog_Data	SEC		0.0	1.0
OTTMN	O/T TEST TIMER MIN VALUE	Analog_Data	SEC		0.0	1.0
OUTSTP	No. steps written to STEPPER MOTOR	Analog_Data	steps/cycle			
POAMBF	Filtered Ambient Pressure	Analog_Data	PSI		1.5	17
POAMBP	Past value of Filtered Nacelle Static Pressure	Analog_Data	PSIA		0.75	8.25
POPTR	START OF PRESSURE SENSOR DATA IN EEPROM	Integer_Data				
POTAB1	PS STRAIN GUAGE TABLE	Integer_Data				
POVEL	Ram pressure at station 1.5	Analog_Data	PSI		0	5.0
POVELP	Past value of ram pressure at station 1.5	Analog_Data	PSI		0	2.1
P18QF1	QUAD filter intermediate value #1	Analog_Data	PSI		0	5
P18QF2	QUAD filter intermediate value #2	Analog_Data	PSI		0	5
P203	AMBIENT PRESS. FROM THE AIR DATA COMPUTER	Analog_Data	PSIA		3.458	15.79
PABLD	BLEED SCHEDULED FROM PAMB	Analog_Data	MA		0.0	75.0
PAMB	SELECTED AMBIENT PRESSURE	Analog_Data	PSIA		1.5	16.5
PAMBI	SELECTED AMBIENT PRESSURE, WITHOUT RATE LIMIT	Analog_Data	PSIA		1.5	16.5
PBTCT	PLA bias Transient Fault Timer	Integer_Data	COUNTS		0	25
PE	UART PARITY ERROR STATUS BIT	Integer_Data		10		
PEBIT	Parity Error Bit in DCU_DATA	Integer_Data		10		
PGTCT	PLA Gain Transient Fault Timer	Integer_Data	Counts		0	25
PLA	FILTERED AND TRIMMED PLA FOR USE IN CONTROL LAWS	Analog_Data	deg		-26.0	125.0
PLA1	RANGE CHECKED PLA INPUT #1	Analog_Data	VRMS		0.525	4.1125
PLA1IF	PLA #1 INPUT STATUS WORD	Integer_Data				
PLA2	RANGE CHECKED PLA INPUT #2	Analog_Data	VRMS		0.6275	4.2150
PLA2IF	PRIMARY RANGE CHECK FLAG IN PLA #2 INPUT STATUS WORD	Integer_Data				
PLAABS	ABSOLUTE VALUE OF CHANGE IN PLA	Analog_Data	DEG		0.0	120.0
PLAADJ	PLA ERROR STORED IN EEPROM	Analog_Data	deg		-3.0	3.0
PLAB	RVDT Interface Bias Value	Analog_Data	VRMS		0.107	0.2026
PLABIF	RVDT Interface Bias Status Word	Integer_Data				
PLAC	Compensated Power Lever Angle Reading	Analog_Data	DEG		-21	120
PLACT	Scaled Power Lever Angle Reading	Analog_Data	DEG		-21	120
PLADAT	DELTA PLA & ASC ENABLE DATA VALIDITY FLAG	Integer_Data			0	1
PLADB	NPT U/S PLA SIGNAL	Analog_Data	deg		0.0	120.0
PLADB2	DEAD BANDED PLA SIGNAL FOR HBOV CURRENT	Analog_Data	DEG		0.0	120.0
PLADJS	PLA ERROR	Analog_Data	deg		-3.0	3.0
PLADOT	PLA DERIVATIVE CALCULATED	Analog_Data	deg/sec		-8200.0	8200.0
PLADRF	HBOV DECEL FLAG ENABLED = 1	Boolean_Data		False	False	True
PLAERF	PLA TRIM VALUE	Analog_Data	deg			
PLAERR	PLA ADJUSTMENT	Analog_Data	deg		-3.0	3.0
PLAG	PLA gain correction	Analog_Data	VRMS		2.9486	4.0268
PLAGCH	1=CHANGE DETECTED IN PLAGRB	Boolean_Data		True		
PLAGIF	RVDT Interface Status Word	Integer_Data				
PLAGRB	1=PLA ADJUSTMENT COMPLETE FLAG	Boolean_Data			False	True
PLAHYS	Unfiltered power lever angle	Analog_Data	DEG		-21	120
PLAIF	PLA interface status word	Integer_Data				
PLAIN	Filtered power lever angle	Analog_Data	DEG		-21	120
PLAIP	Past Filter input / 2	Analog_Data	DEG		-10.5	60

Name	Description	Type	Units	Value	Min Val	Max Val
PLAMFL	1=PLA ADJUST IN PROGRESS	Boolean_Data		True		
PLANHM	PLA INPUT TO NH MIN/MAX & EEC MODE HBOV LOGIC	Analog_Data	DEG		-26.0	125.0
PLAR1	RATE LIMITED PLA	Analog_Data	DEG		0	128
PLAR2	RATE LIMITED PLA	Analog_Data	DEG		0	128.0
PLARAF	HBOV ACCEL/RESLAM FLAG ENABLED = 1	Boolean_Data		False	False	True
PLARDF	HBOV DECEL FLAG ENABLED = 1	Boolean_Data		False	False	True
PLARFG	HBOV TRANSIENT FLAG ENABLED = 1	Boolean_Data		False	False	True
PLARLM	RATE LIMITED PLA FOR POWER/PLA SCHEDULE	Analog_Data	DEG		-26.0	125.0
PLAROM	EEPROM OUTPUT VALUE OF PLA TRIM	Analog_Data	deg		-3.0	3.0
PLATDI	1=PLA TRIM COMMANDED FROM COCKPIT	Boolean_Data				
PLATIM	DELTA PLA & ASC ENABLE INVALID DATA TIMER	Integer_Data	Counter		0	61
PLATMH	ADDRESS OF PLA TRIM IN EEPROM (MSB)	Integer_Data				
PLATML	ADDRESS OF PLA TRIM IN EEPROM (LSB)	Integer_Data				
PLATOT	Total PLA RVDT Input Signal	Analog_Data	VRMS		1.0025	8.4775
PLATRC	TEMPORARY TRIM VALUE IN PLA RAM	Analog_Data	deg		-3.0	3.0
PLATRM	PLA TRIM EEPROM WRITE BUFFER	Analog_Data	Degrees		-5	5
PLAVAL	TEMPORARY TRIM VALUE BEFORE EEPROM WRITE	Analog_Data	deg		-3.0	3.0
PLAVAZ	PAST VALUE OF PLAVAL	Analog_Data	deg		-5.0	5.0
PLAZM1	PLA DELAYED ONCE	Analog_Data	deg		-21	120
PLAZM2	PLA DELAYED TWO TIMES	Analog_Data	deg		-21	120
PLAZM3	PLA DELAYED THREE TIMES	Analog_Data	deg		-21	120
PLAZM4	PLA DELAYED FOUR TIMES	Analog_Data	deg		-21	120
PLINRG	1=PLAIN IN RANGE FOR PLA TRIM	Boolean_Data		True		
PMEMIF	Processor/Memory Failure	Integer_Data				
POSIT	STEPPER MOTOR Position Counter	Analog_Data	steps/cycle			
PRATIO	INLET TO AMBIENT PRESSURE RATIO	Analog_Data	psia/psia		1.0	1.5
PREMCLRTL	Preliminary MCLRTL	Analog_Data	shp		200.0	2192.0
PREMTORTL	Preliminary MTORTL	Analog_Data	shp		200.0	2750.0
PRESELRTL	Preliminary SELRTL	Analog_Data	shp		200.0	2750.0
PROCESS_STATUS	DCU process status transfer state	Integer_Data		5		
PRT410	Data read from input Port 410	Analog_Data				
PSAMB	Nacelle Static Pressure	Analog_Data	PSIA		1.5	16.5
PSAMBF	Filtered Nacelle Static Pressure	Analog_Data	PSIA		1.5	16.5
PSC	Corrected NACELLE Static Pressure	Analog_Data	mV		-1.8804	60
PSCORR	CONVERSION RATIO TO CORRELATE NACELLE STATIC PRESSURE TO AMBIENT PRESSURE	Analog_Data	PSIA/PSIA		0	2
PSDAT	Strain gauge sensor data valid flag	Boolean_Data				
PSIF	NACELLE STATIC Pressure Interface status word	Integer_Data				
PSP1	TRANSMIT DATA WORD	Integer_Data				
PSTCT	Ambient pressure transient timer	Integer_Data	counts		0	52
PT15	STATION 1.5 TOTAL PRESSURE	Analog_Data	psia		5.0	18.0
PT1D_ACF	Selected 1st Stage PT Disk ACF	Analog_Data	None		0	16
PT1D_CYCCRS	PT1 Disk Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
PT1D_CYCFIN	PT1 Disk Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1
PT1D_FCFM1	Selected 1st Stage PT Disk FCF Mission #1	Analog_Data	None		0	16
PT1D_FCFM2	Selected 1st Stage PT Disk FCF Mission #2	Analog_Data	None		0	16
PT1D_FCFM3	Selected 1st Stage PT Disk FCF Mission #3	Analog_Data	None		0	16
PT1D_FCFM4	Selected 1st Stage PT Disk FCF Mission #4	Analog_Data	None		0	16
PT1D_FCFM5	Selected 1st Stage PT Disk FCF Mission #5	Analog_Data	None		0	16
PT1D_FCFM6	Selected 1st Stage PT Disk FCF Mission #6	Analog_Data	None		0	16
PT1D_FCFMAX	Stored Value of the Maximum Power Turbine #1 Disk FCF During an Engine Run Cycle	Analog_Data	None		0	16
PT1D_PNA12	PT1 Disk P/N ASCII Chars 1 & 2	Integer_Data				
PT1D_PNA12EE	PT1 Disk P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
PT1D_PNA34	PT1 Disk P/N ASCII Chars 3 & 4	Integer_Data				
PT1D_PNA34EE	PT1 Disk P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
PT1D_PNA56	PT1 Disk P/N ASCII Chars 5 & 6	Integer_Data				
PT1D_PNA56EE	PT1 Disk P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
PT1D_PNA78	PT1 Disk P/N ASCII Chars 7 & 8 (ASC8 is a DASH)	Integer_Data				
PT1D_PNA78EE	PT1 Disk P/N ASCII Chars 7 & 8 (ASC8 is a DASH) in EEPROM	Integer_Data				
PT1D_SNA12	PT1 Disk S/N ASCII Chars 1 & 2	Integer_Data				
PT1D_SNA12EE	PT1 Disk S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
PT1D_SNA34	PT1 Disk S/N ASCII Chars 3 & 4	Integer_Data				
PT1D_SNA34EE	PT1 Disk S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
PT1D_SNA56	PT1 Disk S/N ASCII Chars 5 & 6	Integer_Data				
PT1D_SNA56EE	PT1 Disk S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
PT1PT0	PROP PRESSURE RISE RATIO	Analog_Data	---		1.000	1.0225
PT2D_ACF	Selected 2nd Stage PT Disk ACF	Analog_Data	None		0	16
PT2D_CYCCRS	PT2 Disk Accumulated Cycles Coarse Count	Analog_Data	Counts		0	32767
PT2D_CYCFIN	PT2 Disk Accumulated Cycles Fine Count	Analog_Data	Fractional Counts		0	1
PT2D_FCFM1	Selected 2nd Stage PT Disk FCF Mission #1	Analog_Data	None		0	16
PT2D_FCFM2	Selected 2nd Stage PT Disk FCF Mission #2	Analog_Data	None		0	16
PT2D_FCFM3	Selected 2nd Stage PT Disk FCF Mission #3	Analog_Data	None		0	16
PT2D_FCFM4	Selected 2nd Stage PT Disk FCF Mission #4	Analog_Data	None		0	16
PT2D_FCFM5	Selected 2nd Stage PT Disk FCF Mission #5	Analog_Data	None		0	16
PT2D_FCFM6	Selected 2nd Stage PT Disk FCF Mission #6	Analog_Data	None		0	16
PT2D_FCFMAX	Stored Value of the Maximum Power Turbine #2 Disk FCF During an Engine Run Cycle	Analog_Data	None		0	16
PT2D_PNA12	PT2 Disk P/N ASCII Chars 1 & 2	Integer_Data				
PT2D_PNA12EE	PT2 Disk P/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
PT2D_PNA34	PT2 Disk P/N ASCII Chars 3 & 4	Integer_Data				
PT2D_PNA34EE	PT2 Disk P/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
PT2D_PNA56	PT2 Disk P/N ASCII Chars 5 & 6	Integer_Data				
PT2D_PNA56EE	PT2 Disk P/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
PT2D_PNA78	PT2 Disk P/N ASCII Chars 7 & 8 (ASC8 is a DASH)	Integer_Data				
PT2D_PNA78EE	PT2 Disk P/N ASCII Chars 7 & 8 (ASC8 is a DASH) in EEPROM	Integer_Data				
PT2D_SNA12	PT2 Disk S/N ASCII Chars 1 & 2	Integer_Data				
PT2D_SNA12EE	PT2 Disk S/N ASCII Chars 1 & 2 in EEPROM	Integer_Data				
PT2D_SNA34	PT2 Disk S/N ASCII Chars 3 & 4	Integer_Data				
PT2D_SNA34EE	PT2 Disk S/N ASCII Chars 3 & 4 in EEPROM	Integer_Data				
PT2D_SNA56	PT2 Disk S/N ASCII Chars 5 & 6	Integer_Data				
PT2D_SNA56EE	PT2 Disk S/N ASCII Chars 5 & 6 in EEPROM	Integer_Data				
PTEMP	Temporary variable	Integer_Data				
PTEMP1	Temporary variable	Integer_Data				
PWMSI1	PWC IN-HOUSE MODIFIED SOFTWARE IDENTIFICATION WORD 1	Integer_Data		0		
Q1IF	Torque Input #1 High Range Fault word	Integer_Data				
Q1LIF	Torque Probe #1 Low Range Fault word	Integer_Data				
Q1SW	Torque Input #1 Interface Status Word	Integer_Data				
Q22TQE	OVERTORQUE CONDITION #1 SET	Integer_Data				
Q2IF	Torque Input #2 High Speed Fault word	Integer_Data				
Q2LIF	Torque Probe #2 Low Speed Fault word	Integer_Data				
Q2SW	Torque Input #2 Interface Status Word	Integer_Data				
Q37TQE	OVERTORQUE CONDITION #2 SET	Integer_Data				
Q47TQE	OVERTORQUE CONDITION #10 SET	Integer_Data				
QB	TORQUE BIAS LOCATION WRITTEN TO (IN EEPROM)	Analog_Data	VDC			
QBTM	Torque Bias Trim	Analog_Data	FT-LB		-529.8	546.371
QBPMC	EEPROM WRITE BUFFER	Analog_Data	VDC		0.5	8.23
QBME	Torque #2 Bias Trim (from EEPROM)	Analog_Data	V		0.5	8.23
QBIMI	CORRECTED Q BIAS TRIM INPUT	Analog_Data	V		0	8.23
QCNT	TORQUE CONVERSION TIME COUNTER	Integer_Data	counts		0	4
QCNTL	Torque conversion input mux select	Integer_Data				
QDEF	Selected shaft deflection value	Analog_Data	DEG		-1.0	11.5
QDEF1	TORQUE PROBE #1 Deflection Angle	Analog_Data	DEG		-1	11.5
QDEF2	TORQUE PROBE #2 Deflection Angle	Analog_Data	DEG		-1	11.5
QG	TORQUE GAIN LOCATION WRITTEN TO (IN EEPROM)	Analog_Data	VDC			
QGOOD1	Previous Q #1 data is valid	Boolean_Data				
QGOOD2	Previous Q #2 data is valid	Boolean_Data				
QGTM	Torque Gain Trim	Analog_Data			0.95	1.05
QGTMC	EEPROM WRITE BUFFER	Analog_Data	VDC		0.5586	8.039
QGTME	Torque #2 Gain Trim (from EEPROM)	Analog_Data	V		0.0	8.2330
QGTMI	Latest corrected Q Gain Trim A/D Reading	Analog_Data	VDC		0.5	8.23
QHPFAC	FACTOR TO BLEND FROM REQUEST TO ACTUAL TORQUE	Analog_Data			0.0	1.0
QHPL	COMMANDED POWER	Analog_Data	SHP		0	4000
QHPLCP	REQUESTED TORQUE CALCULATED FROM HPREF	Analog_Data	FT-LB		-3158.0	23576.0
QHPLX1	PLA BREAKPOINT #1 FOR QHPLT MAP	Analog_Data	DEG	0		
QHPLX2	PLA BREAKPOINT #2 FOR QHPLT MAP	Analog_Data	DEG	35		
QHPLX3	PLA BREAKPOINT #3 FOR QHPLT MAP	Analog_Data	DEG	53		
QHPLX4	PLA BREAKPOINT #4 FOR QHPLT MAP	Analog_Data	DEG	58		
QHPLX5	PLA BREAKPOINT #5 FOR QHPLT MAP	Analog_Data	DEG	69		
QHPLX6	PLA BREAKPOINT #6 FOR QHPLT MAP	Analog_Data	DEG	73		
QHPLX7	PLA BREAKPOINT #7 FOR QHPLT MAP	Analog_Data	DEG	75		
QHPLX8	PLA BREAKPOINT #8 FOR QHPLT MAP	Analog_Data	DEG	77		

Name	Description	Type	Units	Value	Min Val	Max Val
QHPLX9	PLA BREAKPOINT #9 FOR QHPLT MAP	Analog_Data	DEG	81.5		
QHPLXA	PLA BREAKPOINT #10 FOR QHPLT MAP	Analog_Data	DEG	86		
QHPLXB	PLA BREAKPOINT #11 FOR QHPLT MAP	Analog_Data	DEG	90		
QHPLXC	PLA BREAKPOINT #12 FOR QHPLT MAP	Analog_Data	DEG	93		
QHPLXD	PLA BREAKPOINT #13 FOR QHPLT MAP	Analog_Data	DEG	95		
QHPLY1	QHPL BREAKPOINT #1 FOR QHPLT MAP	Analog_Data	SHP		0	0
QHPLY2	QHPL BREAKPOINT #2 FOR QHPLT MAP	Analog_Data	SHP		0	0
QHPLY3	QHPL BREAKPOINT #3 FOR QHPLT MAP	Analog_Data	SHP		0	1200
QHPLY4	QHPL BREAKPOINT #4 FOR QHPLT MAP	Analog_Data	SHP		0	2000
QHPLY5	QHPL BREAKPOINT #5 FOR QHPLT MAP	Analog_Data	SHP		-220	4000
QHPLY6	QHPL BREAKPOINT #6 FOR QHPLT MAP	Analog_Data	SHP		0	4000
QHPLY7	QHPL BREAKPOINT #7 FOR QHPLT MAP	Analog_Data	SHP		0	4000
QHPLY8	QHPL BREAKPOINT #8 FOR QHPLT MAP	Analog_Data	SHP		0	4000
QHPLY9	QHPL BREAKPOINT #9 FOR QHPLT MAP	Analog_Data	SHP		0	4220
QHPLYA	QHPL BREAKPOINT #10 FOR QHPLT MAP	Analog_Data	SHP		0	4220
QHPLYB	QHPL BREAKPOINT #11 FOR QHPLT MAP	Analog_Data	SHP		0	4220
QHPLYC	QHPL BREAKPOINT #12 FOR QHPLT MAP	Analog_Data	SHP		0	4220
QHPLYD	QHPL BREAKPOINT #13 FOR QHPLT MAP	Analog_Data	SHP		0	4500
QLM0	O/T TORQUE LIMIT 0	Analog_Data	FT-LBS		-3610.0	27081.0
QLM0N	O/T TORQUE LIMIT 0 FOR NPT < OTNPT	Analog_Data	FT-LBS		-3610.0	27081.0
QLM1	O/T TORQUE LIMIT 1	Analog_Data	FT-LBS		-3610.0	27081.0
QLM2	O/T TORQUE LIMIT 2	Analog_Data	FT-LBS		-3610.0	27081.0
QLM3	O/T TORQUE LIMIT 3	Analog_Data	FT-LBS		-3610.0	27081.0
QLM4	O/T TORQUE LIMIT 4	Analog_Data	FT-LBS		-3610.0	27081.0
QLM5	O/T TORQUE LIMIT 5	Analog_Data	FT-LBS		-3610.0	27081.0
QLM6	O/T TORQUE LIMIT 6	Analog_Data	FT-LBS		-3610.0	27081.0
QLM7	O/T TORQUE LIMIT 7	Analog_Data	FT-LBS		-3610.0	27081.0
QLM8	O/T TORQUE LIMIT 8	Analog_Data	FT-LBS		-3610.0	27081.0
QLM9	O/T TORQUE LIMIT 9	Analog_Data	FT-LBS		-3610.0	27081.0
QNP	RATING DEPENDANT NP FOR TORQUE LIMIT CALC.	Analog_Data	rpm		900	1300
QOUT	PROPELLER SHAFT TORQUE FOR ARINC OUTPUT	Analog_Data	lbf-ft		0.0	27081.0
QOUTFL	OUTPUT TORQUE TIME CONSTANT	Analog_Data	Sec		0.1	1.0
QQDEF1	Validated Q #1 data	Analog_Data	DEG		-1	11.5
QQDEF2	Validated Q #2 data	Analog_Data	DEG		-1	11.5
QRATEH	Positive torque deflection transient rate limit	Analog_Data	DEG/CYCLE			
QRATEL	Negative torque deflection transient rate limit	Analog_Data	DEG/CYCLE			
QRATET	Torque lightning transient limit	Analog_Data	DEG/CYCLE			
QSTAT1	Q #1 transient state counter	Integer_Data	COUNTS		0	2
QSTAT2	Q #2 transient state counter	Integer_Data	COUNTS		0	2
QSTIFF	Young's Modulus	Analog_Data	FT_LB/DG		2814.4	2927.9
QTEMP1	Rate limited Raw Q #1 value	Analog_Data	DEG			
QTEMP2	Rate limited Raw Q #2 value	Analog_Data	DEG			
QTIMER	TORQUE SYSTEM DELAY TIMER	Integer_Data			0	961
QTMR	TOTAL TIME FOR OCCURRENCE OF ANY O/T FAULT	Analog_Data	sec		0.0	1859
QTOT	PROPELLER SHAFT TORQUE FOR CONTROL	Analog_Data	lbf-ft		-3610	27081
QTOTL	TORQUE LIMIT LOOP TORQUE FEEDBACK	Analog_Data	LBF-FT		0.0	30000
QTOTSW	Torque Input Interface Status Word 1 - 7 Not Used 8 Initialization Torque Test Failed 9 Not Used 10 Valid Conversion Sequence Not Started 11 Converter Invalid Read Fault 12 - 15 Not Used 16 Fault Latch	Integer_Data				
QTOTU	Unfiltered Torque #2	Analog_Data	FT_LB		-3610	30000
QTSTSW	Initialization Torque test status word	Integer_Data				
QUEUE_IS_EMPTY	EEPROM queue is empty flag	Boolean_Data				
QXCN	Torque dual input crosscheck fault counter	Integer_Data			0	120
QXGOOD	Torque crosscheck good updates counter	Integer_Data			0	5
QXIF	Torque dual input crosscheck fault word	Integer_Data				
RAETIM	ADC IAS DATA RANGE CHECK TIMER	Integer_Data			0	121
RAMPTR	HOLDS THE NEXT AVAILABLE SLOT IN RAM BLOCK	Integer_Data				
RAM_DATA_ADR	START ADDRESS OF RAM DATA TO BE WRITTEN TO EEPROM	Integer_Data				
RAM_DATA_END	END ADDRESS OF RAM DATA TO BE WRITTEN TO EEPROM	Integer_Data				
RAM_DATA_PTR	POINTER TO RAM DATA BUFFER	Integer_Data				
RANTIM	ADC TEMPERATURE DATA RANGE CHECK TIMER	Integer_Data	COUNTER		0	60
RATDAT	PMP DISCRETE DATA VALIDITY FLAG	Integer_Data			0	1
RATESW	Selected Rated Power mode (initial value = 2) 1 = Maximum Take-off Power 2 = Normal Take-off Power 3 = Maximum Continuous Power 4 = Maximum Climb Power 5 = Maximum Cruise Power	Integer_Data			0	5
RAWCJ	Raw Cold Junction Temperature A/D Reading	Analog_Data	mV		-12.556	67.291

Name	Description	Type	Units	Value	Min Val	Max Val
RAWDFC	Raw High Level Interface Drift reading	Analog_Data	VDC		0	10
RAWDFT	Raw Low Level Interface Drift reading	Analog_Data	mV		-12.556	67.291
RAWDLP	Raw RAM Pressure A/D Reading	Analog_Data	mV		-12.556	67.291
RAWEVT	Raw Engine Validity From Trim Resistor 1 - 15 Raw Engine Validity Selection Analog Data 16 End of Conversion Flag	Analog_Data	Volts		-9.0	9.0
RAWGAN	Raw High Level Interface Gain Reading	Analog_Data	VDC		-0.000305	10
RAWHBV	Raw Bleed PWM W/A Reading.	Analog_Data			0	10
RAWNS1	Raw NS #1 value	Analog_Data	RPM			
RAWNS2	Raw NS #2 value	Analog_Data	RPM			
RAWPL1	Raw PLA1 A/D reading	Analog_Data	VRMS		0	4.5709
RAWPL2	Raw PLA2 A/D reading	Analog_Data	VRMS		0	4.5709
RAWPLB	Raw PLAB Reading	Analog_Data	VRMS		0	4.5709
RAWPLG	Raw PLAG Reading	Analog_Data	VRMS		0	4.5709
RAWPS	Raw NACELLE Static Pressure	Analog_Data	mV		-12.556	67.291
RAWQ1	Raw Q #1 value	Analog_Data	DEG			
RAWQ2	Raw Q #2 value	Analog_Data	DEG			
RAWSG	Raw Low Level Interface Gain reading	Analog_Data	mV		-12.556	67.291
RAWSMS	RAW STEPPER MOTOR COIL SUM VOLTAGE IN COUNTS	Analog_Data	VOLTS		10.0	0.0
RAWT18	Raw Inlet temperature (T1.8)	Analog_Data	mV		-12.556	67.291
RAWTB	Raw Torque Bias Trim	Analog_Data	VDC		0	10
RAWTG	Raw Torque Gain Trim A/D Reading	Analog_Data	VDC		0	10
RAWTQ	Raw TQ Reading	Analog_Data	mV		-12.556	67.291
RC	Selected Limit Calculation, per each major cycle 1 = NL Corrected Limit 2 = Max Cruise Thermal Limit 3 = Max Climb Thermal Limit 4 = Max Continuous Thermal Limit 5 = Max Takeoff Thermal Limit 6 = Max Takeoff Uprate Thermal Lt	Integer_Data			1	6
RCCON	Constant used to alternate the value of RC between selected and	Analog_Data		1		
READING_ADDRESS	DCU reading address receive state	Integer_Data		0		
READING_CHECKSUM	DCU reading checksum receive state	Integer_Data		3		
READING_DATA	DCU reading data receive state	Integer_Data		2		
READING_STATUS	DCU reading status receive state	Integer_Data		1		
READ_BLOCK_CMD	DCU read block command value	Integer_Data		4		
READ_IN_PROCESS	DCU READ IN PROCESS FLAG	Boolean_Data				
RECEIVE	DCU receive data transfer data	Integer_Data		4		
RECEIVE_STATES	Number of DCU receive states	Integer_Data		5		
REQSW	1=SET NHREQ TO NH DUE TO FAIL FIX OR START COND.	Boolean_Data				
RESLAM	1=RESLAM DETECTED	Boolean_Data				
RESYNCH_WRITTEN	RESYNCH COUNTER WRITTEN TO EEPROM FLAG	Boolean_Data				
RGBSNA12	Gearbox Serial Number ASCII Chars 1 & 2	Integer_Data				
RGBSNA12EE	Gearbox Serial Number ASCII Chars 1 & 2 in EEPROM	Integer_Data				
RGBSNA34	Gearbox Serial Number ASCII Chars 3 & 4	Integer_Data				
RGBSNA34EE	Gearbox Serial Number ASCII Chars 3 & 4 in EEPROM	Integer_Data				
RGBSNA56	Gearbox Serial Number ASCII Chars 5 & 6	Integer_Data				
RGBSNA56EE	Gearbox Serial Number ASCII Chars 5 & 6 in EEPROM	Integer_Data				
RI5FAC	MCR - PT15 CORRECTION FACTOR FOR CONFIG 3.	Analog_Data			1.0	1.08
RM1252	RAM Check Word #2	Integer_Data		'052525		
RM525	RAM Check Word #1	Integer_Data		'125252		
ROMPT1	Address Pointer	Integer_Data				
RPTR26	HOLDS THE NEXT AVAILABLE SLOT IN RAM BLOCK	Integer_Data				
RSPDSM	CALCULATED INCREASE IN SMP MINUS DSMNHK	Analog_Data	STEP/CYCLE		-1280.0	1280.0
RSPNH4	BACK CALCULATED SMP LOOP ERROR	Analog_Data	STEP/CYCLE		-1280.0	1280.0
RSTPU	STEPPER MOTOR RATE REQUEST PER UPDATES	Analog_Data	STEP/CYCLE		-10.0	10.0
RTDDAT	RTD DATA VALID FLAG	Boolean_Data				
RTDTCT	RATED POWER DATA INVALID FAULT COUNTER	Integer_Data	Counter		0	60
RTDTSW	RATED POWER DATA STATUS WORD	Integer_Data				
RTHDDL	RTHETA DIVIDED BY DELTA	Analog_Data	---		0.6	6.0
RTHETA	SQRT STATION 1.5 TEMP. CORRECTED	Analog_Data			0.85	1.0930
RTHMDL	RTHETA MULTIPLIED BY DELTA	Analog_Data	---		0.0	1.4
RUN_MS_RAW	Raw second counter for engine run time logic	Analog_Data	seconds		0	681.5744
RUN_TIMER_RESET	Engine run timer reset timers flag	Boolean_Data				
SEL85	POWER REFERENCE FOR 85 DEG. PLA	Analog_Data	SHp		0.0	4220.0
SELMRS	1=RESET RATED POWER LAG	Boolean_Data				
SELRTL	Range limited and filtered MCT/NT0/MTO Rated power	Analog_Data	shp		200.0	2750.0
SGC	LOW LEVEL A/D INTERFACE GAIN	Analog_Data			0.8964	1.0933
SGCIF	Strain gauge gain interface status word	Integer_Data				
SGCTCT	Strain Gauge Compensation Transient Timer	Integer_Data	Counts		0	25

Name	Description	Type	Units	Value	Min Val	Max Val
SHUTDOWN_WRITTEN	SHUTDOWN COUNTER WRITTEN TO EEPROM FLAG	Boolean_Data				
SLMASP	ACCEL SLAM SET POINT	Analog_Data	RPM/SEC		0.0	3000.0
SLMCOR	HBOV LOGIC ALTITUDE CORRECTION FACTOR	Analog_Data			0	1.2
SLMEND	AT LEAST ONE SLAM FLAG SET FOR KSLFGT SEC	Boolean_Data				
SLMRST	SLAM RESET FLAG	Boolean_Data				
SLMTIM	TIME SINCE A SLAM FLAG WAS SET	Analog_Data	SEC		0.0	30.0
SMCFIF	Stepper motor coil fault interface fault word	Integer_Data				
SMCNT	INTERMITTENT FAULT COUNTER	Integer_Data			0	50
SMINTF	Stepper motor intermittent fault word	Integer_Data				
SMSUMV	STEPPER MOTOR SUM VOLTAGE	Analog_Data	VOLTS		-10.0	1.765
SMSV	RAW STEPPER MOTOR COIL SUM VOLTAGE	Analog_Data	VOLTS		-10.0	1.765
SMSVIF	STEPPER MOTOR SUM VOLTAGE STATUS WORD	Integer_Data				
SMTST	STEPPER MOTOR COIL FAULT CONFIRMATION TEST PHASE COUNTER	Integer_Data			0	3
SNPRP1	LIMITED PROPELLER SPEED	Analog_Data	rpm		900.0	1380.0
SP5	MS word of the Derivative	Integer_Data				
SS211H	ARINC LABEL 211 HIGH BYTE BIT 15 SET (SS211H=1)	Boolean_Data				
SS211L	ARINC LABEL 211 HIGH BYTE BIT 14 SET (SS211L=1)	Boolean_Data				
SS271H	ARINC LABEL 271 HIGH BYTE BIT 15 SET (SS271H=1)	Boolean_Data				
SS271L	ARINC LABEL 271 HIGH BYTE BIT 14 SET (SS271L=1)	Boolean_Data				
SSNHFG	NHDOT LESS THAN SS THRESHOLD FOR KNHDRT SEC	Boolean_Data				
SSTIME	TIME SINCE STEADY STATE DETECTED	Analog_Data	SEC		0.0	2.0
STARTUP_WRITTEN	STARTUP COUNTER WRITTEN TO EEPROM FLAG	Boolean_Data				
START_TRANSFER	DCU start transfer transfer state	Integer_Data		2		
STAT1	PLA #1 STATUS INDICATOR	Integer_Data			0	3
STAT2	PLA #2 STATUS INDICATOR	Integer_Data			0	3
STATE	Control Loop State Counter	Integer_Data			0	80
STATUS	Processor Status Word	Integer_Data				
STCNTP	POSITIVE DIRECTION COUNTER FOR STEPPER MOTOR	Integer_Data	COUNTS		0	350
STEPS	STEPPER MOTOR OUTPUT REQUEST CALCULATED	Analog_Data	steps/sec		-549	800
STEPSR	Residual Step Rate Factor	Analog_Data	steps/sec			
STOP	TEMP. VAR. SET WHEN THE INNER LOOP INTEG. IS TO BE RESET	Integer_Data			0	1
STOPII	STOP-INNER-LOOP Flag	Integer_Data			0	1
STPCMD	STEPPER MOTOR RATE REQUEST FROM PHASE FAULT TEST	Analog_Data	Steps/Sec		-549	800
STPII	PAST VALUE OF STOP	Integer_Data			0	1
STPRLU	RANGE LIMITED STEPS/CYCLE COMMAND FROM CONTROL LAWS	Analog_Data	STEP/CYCLE		-10.0	10.0
STQBI	Torque Bug Identifier: 0. MTO Detent Torque Bug 1. NTO Detent Torque Bug 2. MCL Detent Torque Bug 3. MCR Detent Torque Bug	Integer_Data			0	3
STQBUG	TORQUE BUG	Analog_Data			0.0	107.0
STRPLA	PLA TRIM DATA EEPROM STORE IN PROGRESS COUNTER	Integer_Data			0	3
STRTCYC	Temporary Engine Flight Cycle Determination Flag	Boolean_Data				
STRTSW	START LOGIC STATUS WORD	Integer_Data				
SV_PWC_PATCHID	PWC PATCH ID	Integer_Data				
SV_SW_EPI	ENGINE PROGRAM IDENTIFICATION	Integer_Data				
SV_SW_SBI	BASELINE IDENTIFICATION	Integer_Data				
SV_SW_SVI	SUB_VERSION IDENTIFICATION	Integer_Data				
SWID	Software Version ID Word	Integer_Data				
SWIDHI	Software Version ID HI Word	Integer_Data				
SWIDLO	Software Version ID Low Word	Integer_Data				
T15	RATE LIMITED TEMPERATURE AT STATION 1.5	Analog_Data	deg F		-86.4	170
T15211	TOTAL AIR TEMP. AT STATION 1.5 FROM THE ADC	Analog_Data	DEG F		-57.43	214.48
T15F	TEMPERATURE AT STATION 1.5	Analog_Data	DEG F		-86.4	170
T15K	STATION 1.5 TEMP. IN KELVIN	Analog_Data	deg K		173.0	373.0
T15R	STATION 1.5 TEMP. IN RANKINE	Analog_Data	deg R		310.0	670.0
T1815F	FILTERED STATION 1.8 TEMPERATURE	Analog_Data	DEG F		-74.39	170
T18C	Compensated T1.8 A/D reading	Analog_Data	mV		-3.42	64.12
T18CEL	TEMPERATURE AT STATION 1.8	Analog_Data	DEG C		-59.11	76.67
T18F	Temperature at station 1.8	Analog_Data	DEG F		-74.39	170
T18IF	Engine inlet temp. interface status word	Integer_Data				
T18P	Past INPUT/2 used for T1815F Filter	Analog_Data	DEG F		-37.195	85
T18T15	RATIO FOR CORRELATING STA 1.8 TO STA 1.5 TEMP	Analog_Data			0	2
T18TCT	T18 Transient Fault Timer	Integer_Data	counts		0	26
TAISA	ISA TEMPERATURE	Analog_Data	deg C		-60.0	30.0
TAKEOFF	Flag for Takeoff Conditions Met	Boolean_Data			False	True
TAKEOFFRST	Reset Flag for Takeoff Timer	Boolean_Data			False	True
TAMB	AMBIENT STATIC AIR TEMP FROM SELECTED T15	Analog_Data	deg C		-80.0	70.0
TAT	AMBIENT TOTAL AIR TEMP FROM SELECTED T15	Analog_Data	deg K		150.0	435.0

Name	Description	Type	Units	Value	Min Val	Max Val
TATADC	TOTAL AIR TEMPERATURE FROM THE ADC	Analog_Data	DEG C		-512	511.75
TATDAT	ADC TOTAL AIR TEMPERATURE DATA VALID FLAG	Integer_Data				
TATIF	ADC/EEC TEMPERATURE FAULT STATUS WORD	Integer_Data				
TATRGC	RANGE CHECKED ADC TEMPERATURE VALUE	Analog_Data	DEG C		-50	99
TATTIM	ADC TOTAL AIR TEMP. INVALID DATA FAULT TIMER	Integer_Data	COUNTER		0	60
TATXIF	ADC AND EEC TEMP. CROSS CHECK FAULT STATUS WORD	Integer_Data				
TAXTYM	ADC/RTD TAT CROSSCHECK FAULT TIMER	Integer_Data			0	62
TBTCT	Torque Bias Transient Fault Timer	Integer_Data	counts		0	25
TBTIF	Torque bias trim interface status word	Integer_Data				
TEMP	PLACE HOLDER	Integer_Data				
TERM	1=NPT U/S TERMINATE BOOST AND LAG BOOST EFFECT	Boolean_Data				
TGTCT	Torque Gain Transient Fault Timer	Integer_Data	counts		0	25
TGTIF	Torque gain trim interface status word	Integer_Data				
THEREF	RAW ARINC FLEX TEMPERATURE DATA IN DEG. F	Analog_Data	DEG F		-81.4	147.2
THETA	RELATIVE ABSOLUTE TEMPERATURE RATIO	Analog_Data			0.6207	1.209
THETAC	STATION 1.5 TEMP. IN CELSIUS	Analog_Data	deg C		-65.0	71.11
THETAI	UNFILTERED STATION 1.5 TEMPERATURE	Analog_Data	DEG F		-86.40	170
THRFTCT	FLEX TEMPERATURE INVALID FAULT COUNTER	Integer_Data	Counter		0	60
THRFWSW	FLEX TEMPERATURE DATA STATUS WORD	Integer_Data				
TI	Difference between TS and TM	Analog_Data	SEC			
TIMACC	ELAPSED TIME FOR UPTFLG TIMER	Analog_Data	sec		0.0	1.0
TIMERE	TIMER FOR DELAY OF PLAT127E SELECTION	Integer_Data	SEC		0.0	0.5
TIMERF	TIMER FOR DELAY OF PLAT127F_M SELECTION	Integer_Data	SEC		0.0	0.5
TM	MS 16 BITS OF TORQUE MARK time counter	Analog_Data	SEC			
TOMOD	BLENDED MAX TAKE OFF POWER LIMIT	Analog_Data	SHP		2500.0	2750.0
TOPAMD	AMBIENT PRESSURE MIXING FACTOR, WHEN RATESW=1,2,3,6,OR 7	Analog_Data	PSIA/PSIA		0	1
TOT15D	FADER RATIO TO SELECT BETWEEN THE ADC TAT AND EEC TEMP.	Analog_Data	DEG F/DEGF		0	1
TOTTIM	TM + TS	Analog_Data	SEC			
TQ	Torque Shaft Temperature	Analog_Data	DEG F		-75.6429	253.061
TQBGI	Selected Torque Bug Identifier: 0. MTO Detent Torque Bug 1. NTO Detent Torque Bug 2. MCL Detent Torque Bug 3. MCR Detent Torque	Integer_Data			0	3
TQBUG	TORQUE BUG OUTPUT	Analog_Data	ft-lbf		648.0	12792.0
TQC	Filtered Corrected Torque shaft temperature	Analog_Data	mV		-4.83	57.75
TQCI	Corrected TQ Reading	Analog_Data	mV		-4.8	57.8
TQERR	TORQUE LIMIT LOOP ERROR	Analog_Data	LBF-FT		-30000	30000
TQFTR	TQLIM2 FILTER OUTPUT	Analog_Data	FT-LB		8000	13000
TQIF	Torque shaft temp. interface status word	Integer_Data				
TQLIM1	RATING DEPENDENT TORQUE LIMIT	Analog_Data	ft-lb		8000	12036.0
TQLIM2	CORRECTED RATING DEPENDENT TORQUE LIMIT	Analog_Data	ft-lb		9628.8	13841.4
TQLIM3	TORQUE LIMIT LOOP REFERENCE	Analog_Data	LBF-FT		0.0	30000
TQTCT	TQ transient timer	Integer_Data	COUNTS		0	25
TRANSFER_STATES	Number of DCU transfer states	Integer_Data		7		
TRANSMIT	DCU transmit data transfer state	Integer_Data		3		
TRANSMIT_BLOCK	DCU transmit block transmit state	Integer_Data		1		
TRANSMIT_CHECKSM	DCU transmit checksum transmit state	Integer_Data		2		
TRANSMIT_COMMAND	DCU transmit command transmit state	Integer_Data		0		
TRANSMIT_STATES	Number of DCU transmit states	Integer_Data		3		
TRBMSNA12	Turbomachine Serial Number ASCII Chars 1 & 2	Integer_Data				
TRBMSNA12EE	Turbomachine Serial Number ASCII Chars 1 & 2 in EEPROM	Integer_Data				
TRBMSNA34	Turbomachine Serial Number ASCII Chars 3 & 4	Integer_Data				
TRBMSNA34EE	Turbomachine Serial Number ASCII Chars 3 & 4 in EEPROM	Integer_Data				
TRBMSNA56	Turbomachine Serial Number ASCII Chars 5 & 6	Integer_Data				
TRBMSNA56EE	Turbomachine Serial Number ASCII Chars 5 & 6 in EEPROM	Integer_Data				
TREFC	RAW ARINC FLEX TEMPERATURE DATA IN DEG. C	Analog_Data	DEG C		-63	64
TRIMEN	1=ENABLE TRIM	Boolean_Data		True		
TRMMOD	PLA TRIM GAIN FACTOR	Analog_Data			0.0	1.0
TRMTYM	PLA TRIM ENABLE TIMER	Analog_Data	sec		0	8
TRPLA	Transition PLA	Analog_Data	DEG		-26.0	125.0
TS	MS 16 BITS OF TORQUE SPACE time counter	Analog_Data	SEC			
TSTATE	NPARAM TRANSMIT STATE	Integer_Data				
TT1T02	PROP PRESSURE RISE RATIO	Analog_Data	deg K / deg C		1.0	1.00797
TT1TOT	PROPELLER PRESSURE RISE RATIO	Analog_Data	DEGK/DEGK		1.0	1.00797
TTEMP1	Raw primary NH value	Analog_Data	RPM		0	44000
TTEMP2	Raw secondary NH value	Analog_Data	RPM		0	44000
UARTIF	UART transmitter interface word	Integer_Data				
UHOLD	UART OUTPUT DATA BUFFER	Integer_Data				

Name	Description	Type	Units	Value	Min Val	Max Val
UMODE	UART Output Mode	Integer_Data			0	4
UOTBL	UART output parameter address	Integer_Data				
UPDELO	1=Change in UPTRIM Transitions True to False delayed off	Boolean_Data				
UPTFGL	UPTRIM TIMER LATCHED	Boolean_Data				
UPTFLG	1=UPTRIM COMMANDED	Boolean_Data				
UPTRSW	1=TRANSITION TO UPTRIM TAKING PLACE	Boolean_Data				
URTCFG	UART configuration flag: 0 = DTS, 1 = GDTS	Integer_Data			0	1
URTPTR	START OF UART ADDRESS TABLE	Integer_Data				
USPEVF	UNSPECIFIED CONFIGURATION VALUE	Integer_Data		2		
UWORD	UART Data Word Counter	Integer_Data			0	17
VERIFY_CHECKSUM	DCU verify checksum receive state	Integer_Data		4		
VK	AIR SPEED	Analog_Data	kts		0.0	500.0
VK1	RATE LIMITED AIRSPEED USED FOR PWR LOOP GAIN SCHEDULING	Analog_Data	KTS		0.0	500.0
VK2	AIRSPEED USED FOR POWER LOOP GAIN SCHEDULING	Analog_Data	KTS		0.0	500.0
VKF	IAS Failure Flag	Boolean_Data				
WAITING	DCU waiting transfer state	Integer_Data		6		
WCNT26	1 COUNT PER BIT FOR ONE OF THE NCFALx WORDS	Integer_Data				
WDCNT	Watchdog Timer Interrupt Service Counter	Integer_Data			-17	0
WDTTIP	Watchdog Timer Self-Test Flag	Integer_Data				
WRDCNT	OFFSET INTO THE LRU-CODE TABLE	Integer_Data				
WRITE_BLOCK_CMD	DCU write block command value	Integer_Data		5		
WRITE_IN_PROCESS	DCU WRITE IN PROCESS FLAG	Boolean_Data				
ZALTDI	PAST VALUE OF ADC PRESS. ALT. DATA VALID FLAG	Integer_Data			0	2
ZLRUAD	Past Value of LRU Advance Discrete	Integer_Data				
ZNS1	Past converted value of NS #1	Analog_Data	RPM			
ZNS2	Past converted value of NS #2	Analog_Data	RPM			
ZQ1	Past converted valud of Q #1	Analog_Data	DEG			
ZQ2	Past converted valud of Q #2	Analog_Data	DEG			
ZTATDI	PAST VALUE OF THE ADC TAT DATA VALIDITY FLAG TATDAT	Integer_Data				
ZTRMEN	LAST PASS VALUE OF TRIMEN	Boolean_Data			False	True
test						

10.02 SCOPED PARAMETERS

10.02 Table 1: Scoped-Parameters

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
ARCOUT	ARCB13	Bit 13	Integer_Data		%O'10000'		
ARCOUT	ARCB15	Bit 15	Integer_Data		%O'40000'		
ARCOUT	ARCCBT	Mask to clear bits 1-4	Integer_Data		%O'177760'		
ARCOUT	ARCCLR	Mask to clear bits 8-15	Integer_Data		%O'100177'		
ARCOUT	ARCF3	Fault Counter Increment Value	Integer_Data		%O'3'		
ARCOUT	ARCK12	MAX Limit for I/F Fault Counter	Integer_Data		12		
ARCOUT	ARCM16	Bit 16 Mask	Integer_Data		%O'100000'		
BD134	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
C01_START	K0	ZERO	Integer_Data		0		
C01_START	KMXSTP	MAX ALLOWABLE STEPS FOR STEPPER MOTOR	Integer_Data	COUNTS	350		
C01_START	KSPLAH	MAX PLA FOR START LOGIC	Analog_Data	deg	45.0		
C01_START	KSPLAL	MIN PLA FOR START LOGIC	Analog_Data	deg	20.0		
C01_START	KSTCRT	START LOGIC COUNTER RESET	Integer_Data	counts	0		
C01_START	KSTEPS	STEPPER MOTOR CONSTANT	Integer_Data	steps/cycl	0		
C01_START	KSTPR	STEPPER MOTOR CONSTANT	Integer_Data	steps/cycl	10		
C02_PLATRIM_1	KNHLOW	Value of NH Speed to disable PLA Trim, LRU Fault clear and engine model configuration.	Analog_Data	rpm	10000.0		
C02_PLATRIM_1	KROMDF	PLAROM DEFAULT VALUE	Analog_Data	deg	0.0		
C02_PLATRIM_1	KSTRTY	PLA TRIM TIMER START VALUE	Analog_Data	sec	0.0		
C02_PLATRIM_1	KT1	MIN PLA FOR TRIM SEQUENCE	Analog_Data	deg	88.0		
C02_PLATRIM_1	KT2	MAX PLA FOR TRIM SEQUENCE	Analog_Data	deg	94.0		
C02_PLATRIM_1	KTRMEN	PLA TRIM STORE TIME	Analog_Data	sec	7.0		
C02_PLATRIM_1	KTRSTR	PLA TRIM SEQUENCE START TIME	Analog_Data	sec	2.0		
C02_PLATRIM_1	KTYINC	PLA TRIM TIMER INCREMENT	Analog_Data	sec	0.0416		
C02_PLATRIM_2	KPLREF	ENGINE TRIM REFERENCE PLA VALUE	Analog_Data	deg	91.0		
C02_PLATRIM_2	KTRMMN	MINIMUM PLA ERROR LIMIT	Analog_Data	deg	-3.0		
C02_PLATRIM_2	KTRMMX	MAXIMUM PLA ERROR LIMIT	Analog_Data	deg	3.0		
C02_PLATRIM_2	KTRMTC	TIME CST VALUE FOR ENGINE TRIM FILTER	Analog_Data	SEC	1.0		
C02_PLATRIM_2	TRMMTB	PLAIN TO TRIM GAIN FACTOR CONVERSION	Maps				
C03_TLIMRT_1	K1	INCREMENT TO BLEED.	Integer_Data	---	1		
C03_TLIMRT_1	KMN0	Smallest Value of MACH	Analog_Data	---	0.0		
C03_TLIMRT_1	KMN25	MACH Value 0.25	Analog_Data	---	0.25		
C03_TLIMRT_1	KMN5	MACH Value 0.5	Analog_Data	---	0.5		
C03_TLIMRT_1	KNLCOFF	MACH Value 0.5	Analog_Data	---	0.5		
C03_TLIMRT_1	KTOMOD	MAX TAKE OFF POWER LIMIT	Analog_Data	shp	2750.0		
C03_TLIMRT_1	NLCPM1	NL CORRECTED LIMIT TABLE MACH=0.0	Maps				
C03_TLIMRT_1	NLCPM2	NL CORRECTED LIMIT TABLE MACH=0.25	Maps				
C03_TLIMRT_1	NLCPM3	NL CORRECTED LIMIT TABLE MACH=0.5	Maps				
C03_TLIMRT_2_1	K1FP	Increment to Bleed Modifier	Analog_Data		1.0	-1.0	1.0
C03_TLIMRT_2_1	KMN0	Smallest Value of MACH	Analog_Data	---	0.0		
C03_TLIMRT_2_1	KMN4	MACH Value 0.4	Analog_Data	---	0.4		
C03_TLIMRT_2_1	KMN44	MNTZ1 squared	Analog_Data	---	0.16		
C03_TLIMRT_2_1	KMN7	MACH Value 0.7	Analog_Data	---	0.7		
C03_TLIMRT_2_1	KMN77	MNTZ2 squared	Analog_Data	---	0.49		
C03_TLIMRT_2_1	MCRMT1	MCR THERMAL POWER TABLES MACH=0.0	Maps				
C03_TLIMRT_2_1	MCRMT2	MCR THERMAL POWER TABLES MACH=0.4	Maps				
C03_TLIMRT_2_1	MCRMT3	MCR THERMAL POWER TABLES MACH=0.7	Maps				
C03_TLIMRT_2_2	K0	ZERO	Integer_Data		0		
C03_TLIMRT_2_2	K1FP	Increment to Bleed Modifier	Analog_Data		1.0	-1.0	1.0
C03_TLIMRT_2_2	K2	VALUE of 2	Integer_Data		2		
C03_TLIMRT_2_2	MCRMM0	MCR BLEED MODIFIER TABLE BLEED=0	Maps				
C03_TLIMRT_2_2	MCRMM2	MCR BLEED MODIFIER TABLE BLEED=2	Maps				
C03_TLIMRT_3_1	KMN0	Smallest Value of MACH	Analog_Data	---	0.0		
C03_TLIMRT_3_1	KMN4	MACH Value 0.4	Analog_Data	---	0.4		
C03_TLIMRT_3_1	KMN44	MNTZ1 squared	Analog_Data	---	0.16		
C03_TLIMRT_3_1	KMN7	MACH Value 0.7	Analog_Data	---	0.7		
C03_TLIMRT_3_1	KMN77	MNTZ2 squared	Analog_Data	---	0.49		
C03_TLIMRT_3_1	MCLTM1	MCL THERMAL POWER TABLES MACH=0.0	Maps				
C03_TLIMRT_3_1	MCLTM2	MCL THERMAL POWER TABLES MACH=0.4	Maps				
C03_TLIMRT_3_1	MCLTM3	MCL THERMAL POWER TABLES MACH=0.7	Maps				
C03_TLIMRT_3_2	K0	ZERO	Integer_Data		0		
C03_TLIMRT_3_2	K1FP	Increment to Bleed Modifier	Analog_Data		1.0	-1.0	1.0
C03_TLIMRT_3_2	K2	VALUE of 2	Integer_Data		2		
C03_TLIMRT_3_2	MCLMM0	MCL BLEED MODIFIER TABLE BLEED=0	Maps				
C03_TLIMRT_3_2	MCLMM2	MCL BLEED MODIFIER TABLE BLEED=2	Maps				
C03_TLIMRT_4_1	KMN0	Smallest Value of MACH	Analog_Data	---	0.0		
C03_TLIMRT_4_1	KMN4	MACH Value 0.4	Analog_Data	---	0.4		
C03_TLIMRT_4_1	KMN44	MNTZ1 squared	Analog_Data	---	0.16		
C03_TLIMRT_4_1	KMN7	MACH Value 0.7	Analog_Data	---	0.7		
C03_TLIMRT_4_1	KMN77	MNTZ1 squared	Analog_Data	---	0.49		
C03_TLIMRT_4_1	MCTTM1	MCT THERMAL POWER TABLES MACH=0.0	Maps				
C03_TLIMRT_4_1	MCTTM2	MCT THERMAL POWER TABLES MACH=0.4	Maps				
C03_TLIMRT_4_1	MCTTM3	MCT THERMAL POWER TABLES MACH=0.7	Maps				
C03_TLIMRT_4_2	K0	ZERO	Integer_Data		0		
C03_TLIMRT_4_2	K1	Increment to Bleed Modifier	Integer_Data		1		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
C03_TLIMRT_4_2	K1FP	Increment to Bleed Modifier	Analog_Data		1.0	-1.0	1.0
C03_TLIMRT_4_2	MCTMM0	MCT BLEED MODIFIER TABLE BLEED=0	Maps				
C03_TLIMRT_4_2	MCTMM1	MCT BLEED MODIFIER TABLE BLEED=1	Maps				
C03_TLIMRT_5_1	KMN0	Smallest Value of MACH	Analog_Data	---	0.0		
C03_TLIMRT_5_1	KMN2	MACH Value 0.2	Analog_Data	---	0.2		
C03_TLIMRT_5_1	KMN22	MNTZ2 squared	Analog_Data	---	0.04		
C03_TLIMRT_5_1	KMN3	MACH Value 0.3	Analog_Data	---	0.3		
C03_TLIMRT_5_1	KMN33	MNTZ1 squared	Analog_Data	---	0.09		
C03_TLIMRT_5_1	MTOTM1	MTO THERMAL POWER TABLES MACH=0.0	Maps				
C03_TLIMRT_5_1	MTOTM2	MTO THERMAL POWER TABLES MACH=0.2	Maps				
C03_TLIMRT_5_1	MTOTM3	MTO THERMAL POWER TABLES MACH=0.3	Maps				
C03_TLIMRT_5_2	K1	Increment to Bleed Modifier	Integer_Data		1		
C03_TLIMRT_5_2	K1FP	Increment to Bleed Modifier	Analog_Data		1.0	-1.0	1.0
C03_TLIMRT_5_2	K2	VALUE of 2	Integer_Data		2		
C03_TLIMRT_5_2	MTOMM1	MTO BLEED MODIFIER TABLE BLEED=1	Maps				
C03_TLIMRT_5_2	MTOMM2	MTO BLEED MODIFIER TABLE BLEED=2	Maps				
C03_TLIMRT_6_1	KMN0	Smallest Value of MACH	Analog_Data	---	0.0		
C03_TLIMRT_6_1	KMN2	MACH Value 0.2	Analog_Data	---	0.2		
C03_TLIMRT_6_1	KMN22	MNTZ1 squared	Analog_Data	---	0.04		
C03_TLIMRT_6_1	KMN3	MACH Value 0.3	Analog_Data	---	0.3		
C03_TLIMRT_6_1	KMN33	MNTZ2 squared	Analog_Data	---	0.09		
C03_TLIMRT_6_1	MTUTM1	MTO UPRATE THERMAL POWER TABLES MACH=0.0	Maps				
C03_TLIMRT_6_1	MTUTM2	MTO UPRATE THERMAL POWER TABLES MACH=0.2	Maps				
C03_TLIMRT_6_1	MTUTM3	MTO UPRATE THERMAL POWER TABLES MACH=0.3	Maps				
C03_TLIMRT_6_2	K1	Increment to Bleed Modifier	Integer_Data		1		
C03_TLIMRT_6_2	K1FP	Increment to Bleed Modifier	Analog_Data		1.0	-1.0	1.0
C03_TLIMRT_6_2	K2	VALUE of 2	Integer_Data		2		
C03_TLIMRT_6_2	MTUMM1	MTO UPRATE BLEED MODIFIER TABLE BLEED=1	Maps				
C03_TLIMRT_6_2	MTUMM2	MTO UPRATE BLEED MODIFIER TABLE BLEED=2	Maps				
C03_TLIMRT_6_D	KPL1RAT	Engine Rating Power Level 1	Integer_Data		0		
C03_TLIMRT_6_D	KPL3RAT	Engine Rating Power Level for PW127F	Integer_Data		3		
C03_TLIMRT_7	KMCTOD	MAX CONTINUING POWER LIMIT	Analog_Data	shp	2500.0		
C03_TLIMRT_7	KTOMOD	MAX TAKE OFF POWER LIMIT	Analog_Data	shp	2750.0		
C03_TLIMRT_7	MCTDM	MCT POWER LIMITS FADER MAP	Maps				
C04_RPCAL_2_D	K90	VALUE OF 0.9	Analog_Data	---	0.9		
C04_RPCAL_2_D	KATO0D	NORMAL TAKE OFF POWER LIMIT	Analog_Data	shp	2475.0		
C04_RPCAL_2_D	KMCL0D	MAX CLIMB POWER LIMIT	Analog_Data	shp	2192.0		
C04_RPCAL_2_D	KMCR0D	MAX CRUISE POWER LIMIT	Analog_Data	shp	2132.0		
C04_RPCAL_2_D	KMCT0D	MAX CONTINUING POWER LIMIT	Analog_Data	shp	2500.0		
C04_RPCAL_2_D	KOTM10	O/T TEST TIMER 10 MAX VALUE	Analog_Data	SEC	0.208		
C04_RPCAL_2_D	KOTNPD	NPT VALUE FOR TORQUE LIMIT INCREASE	Analog_Data	RPM	18800.0		
C04_RPCAL_2_D	KOTTD	O/T TEST PROCESS CYCLE TIME	Analog_Data	SEC	0.0832		
C04_RPCAL_2_D	KOTTM0	O/T TEST TIMER 0 MAX VALUE	Analog_Data	SEC	600.0		
C04_RPCAL_2_D	KOTTM1	O/T TEST TIMER 1 MAX VALUE	Analog_Data	SEC	300.0		
C04_RPCAL_2_D	KOTTM2	O/T TEST TIMER 2 MAX VALUE	Analog_Data	SEC	256.6		
C04_RPCAL_2_D	KOTTM3	O/T TEST TIMER 3 MAX VALUE	Analog_Data	SEC	217.9		
C04_RPCAL_2_D	KOTTM4	O/T TEST TIMER 4 MAX VALUE	Analog_Data	SEC	179.1		
C04_RPCAL_2_D	KOTTM5	O/T TEST TIMER 5 MAX VALUE	Analog_Data	SEC	140.4		
C04_RPCAL_2_D	KOTTM6	O/T TEST TIMER 6 MAX VALUE	Analog_Data	SEC	101.7		
C04_RPCAL_2_D	KOTTM7	O/T TEST TIMER 7 MAX VALUE	Analog_Data	SEC	62.9		
C04_RPCAL_2_D	KOTTM8	O/T TEST TIMER 8 MAX VALUE	Analog_Data	SEC	20.0		
C04_RPCAL_2_D	KOTTM9	O/T TEST TIMER 9 MAX VALUE	Analog_Data	SEC	0.208		
C04_RPCAL_2_D	KOTTMN	O/T TEST TIMER MIN VALUE	Analog_Data	SEC	0.0		
C04_RPCAL_2_D	KPL2RAT	Engine Rating for Power Level 2	Integer_Data		1		
C04_RPCAL_2_D	KPL3RAT	Engine Rating for Power Level 3	Integer_Data		3		
C04_RPCAL_2_D	KQLM0D	O/T 115% TORQUE LIMIT 0	Analog_Data	FT-LBS	12036.0		
C04_RPCAL_2_D	KQLM1	O/T TORQUE LIMIT 1	Analog_Data	FT-LBS	12800.0		
C04_RPCAL_2_D	KQLM2	O/T TORQUE LIMIT 2	Analog_Data	FT-LBS	14440.0		
C04_RPCAL_2_D	KQLM3	O/T TORQUE LIMIT 3	Analog_Data	FT-LBS	15000.0		
C04_RPCAL_2_D	KQLM4	O/T TORQUE LIMIT 4	Analog_Data	FT-LBS	15500.0		
C04_RPCAL_2_D	KQLM5	O/T TORQUE LIMIT 5	Analog_Data	FT-LBS	16000.0		
C04_RPCAL_2_D	KQLM6	O/T TORQUE LIMIT 6	Analog_Data	FT-LBS	16500.0		
C04_RPCAL_2_D	KQLM7	O/T TORQUE LIMIT 7	Analog_Data	FT-LBS	17000.0		
C04_RPCAL_2_D	KQLM8	O/T TORQUE LIMIT 8	Analog_Data	FT-LBS	17500.0		
C04_RPCAL_2_D	KQLM9	O/T TORQUE LIMIT 9	Analog_Data	FT-LBS	18054.0		
C04_RPCAL_2_D	KRCTO	RATING CODE FOR TAKE OFF	Integer_Data		1		
C04_RPCAL_2_D	KTOPL3	Max Takeoff Power Limit at Engine Power Level Derate	Analog_Data	shp	2435		
C05_TQBUG	KFALSE	FALSE IN RESETS	Boolean_Data		False		
C05_TQBUG	KHPCON	HORSE POWER CONVERSION FACTOR	Analog_Data	ft-lbf/shp	5252.11		
C05_TQBUG	KMCLTBI	MCL Torque Bug Identifier	Integer_Data		2		
C05_TQBUG	KMCLUB	MCL Detent Upper Boundary	Analog_Data	deg	85.0		
C05_TQBUG	KMCRBTBI	MCR Torque Bug Identifier	Integer_Data		3		
C05_TQBUG	KMCRUB	MCR Detent Upper Boundary	Analog_Data	deg	77.0		
C05_TQBUG	KMNPMM	NP RANGE MINIMUM VALUE	Analog_Data	rpm	900.0		
C05_TQBUG	KMNPMM	NP RANGE MAXIMUM VALUE	Analog_Data	rpm	1620.0		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
C05_TQBUG	KMTOTBI	MTO Torque Bug Identifier	Integer_Data		0		
C05_TQBUG	KNPDEC	NP CHANGE MIN RATE LIMIT	Analog_Data	rpm/sec	-45.0		
C05_TQBUG	KNPINC	NP CHANGE MAX RATE LIMIT	Analog_Data	rpm/sec	45.0		
C05_TQBUG	KNPTO	FIXED NP FOR 100% NP RATING	Analog_Data	rpm	1200		
C05_TQBUG	KNTOTBI	NTO Torque Bug Identifier	Integer_Data		1		
C05_TQBUG	KNTOUB	NTO Detent Upper Boundary	Analog_Data	deg	93.0		
C0601003004	CONOHT	103%	Analog_Data		1.03		
C0601003004	CONPTF	50%	Analog_Data		0.5		
C0601003004	COPTFF	5.5%	Analog_Data		0.055		
C0601003004	COPTZF	5.0%	Analog_Data		0.05		
C0601003004	COTF	POWER LIMIT FOR PLA > 90	Analog_Data		2805.0		
C0601003004	FPLCN	LOCATION OF FIRST PLA BREAKPOINT	Maps				
C0601003004	KDECPL	PLA RATE DECREMENT DEGREE PER SEC.	Analog_Data		-2.496		
C0601003004	KINCPN	INCREASING PLA RATE LIMIT	Analog_Data		1.664		
C0601003004	KQHY3	MULTIPLIER FOR QHPLY3	Analog_Data		0.30		
C0601003004	KQHY4	MULTIPLIER FOR QHPLY4	Analog_Data		0.50		
C0601003004	KQHYD	MULTIPLIER FOR QHPLYD	Analog_Data		1.15		
C0601003004	LPLCN	LOCATION AFTER LAST PLA BREAKPOINT	Maps				
C0601003004	PLANIN	PLA SWITCH POINT FOR LIMITING POWER REF.	Analog_Data		92		
C0601003004	PLCN1	PLA BREAKPOINT 1	Analog_Data		0		
C0601003004	PLCN2	PLA BREAKPOINT 2	Analog_Data		35		
C0601003004	PLCN3	PLA BREAKPOINT 3	Analog_Data		53		
C0601003004	PLCN4	PLA BREAKPOINT 4	Analog_Data		58		
C0601003004	PLCN5	PLA BREAKPOINT 5	Analog_Data		69		
C0601003004	PLCN6	PLA BREAKPOINT 6	Analog_Data		73		
C0601003004	PLCN7	PLA BREAKPOINT 7	Analog_Data		75		
C0601003004	PLCN8	PLA BREAKPOINT 8	Analog_Data		77		
C0601003004	PLCN9	PLA BREAKPOINT 9	Analog_Data		81.5		
C0601003004	PLCNA	PLA BREAKPOINT 10	Analog_Data		86		
C0601003004	PLCNB	PLA BREAKPOINT 11	Analog_Data		90		
C0601003004	PLCNC	PLA BREAKPOINT 12	Analog_Data		93		
C0601003004	PLCND	PLA BREAKPOINT 13	Analog_Data		95		
C0601003004	QHPLT	ADDR OF PLARLM BREAKPOINT TABLE IN RAM	Maps				
C0601003004	QHPLTL	LENGTH OF BREAKPOINT TABLE	Integer_Data				
C0601003004	QHPLTT	ADDR OF QHPL1 BREAKPOINT TABLE IN RAM	Maps				
C06_HPREF_1	KBLTIM	TIME TO PERMIT RATED POWER LAG AFTER BLEED CHANGE	Analog_Data	sec	12.0		
C06_HPREF_1	KBSTIM	TIME TO PERMIT RATED POWER LAG AFTER DERATING LEVEL SELECTION	Analog_Data	sec	12.0		
C06_HPREF_1	KFALSE	FALSE	Boolean_Data		False		
C06_HPREF_1	KMTTIM	TIME TO PERMIT RATED POWER LAG AFTER MCT ENABLE CHANGE	Analog_Data	sec	12.0		
C06_HPREF_1	KPLAMV	PLA MOVEMENT REQUIRED TO DISABLE RATED POWER LAG	Analog_Data	deg	5.0		
C06_HPREF_1	KRCTO	RATING CODE FOR TAKE OFF	Integer_Data		1		
C06_HPREF_1	KUPTIM	TIME TO PERMIT RATED POWER LAG AFTER UPTRIM TRANSITIONS TRUE TO FALSE	Analog_Data	sec	12.0		
C06_HPREF	K2	Constant value of 2	Analog_Data		2.0		
C06_HPREF	KDECPL	PLA DECREASING RATE OF CHANGE LIMIT	Analog_Data	DEG/SEC	-80.0		
C06_HPREF	KFLGCN	Flight Idle Stop Center	Analog_Data	deg	35.0		
C06_HPREF	KFRSLB	Full Reverse Stop Lower Boundary	Analog_Data	deg	0.0		
C06_HPREF	KINCPN	PLA INCREASING RATE OF CHANGE LIMIT	Analog_Data	DEG/SEC	80.0		
C06_HPREF	KMCLLB	MCL Detent Lower Boundary	Analog_Data	deg	81.0		
C06_HPREF	KMCLUB	MCL Detent Upper Boundary	Analog_Data	deg	85.0		
C06_HPREF	KMCRLB	MCR Detent Lower Boundary	Analog_Data	deg	73.0		
C06_HPREF	KMCRUB	MCR Detent Upper Boundary	Analog_Data	deg	77.0		
C06_HPREF	KMTOLB	MTO Detent Lower Boundary	Analog_Data	deg	97.0		
C06_HPREF	KMTOUB	MTO Detent Upper Boundary	Analog_Data	deg	100.0		
C06_HPREF	KNTOLB	NTO Detent Lower Boundary	Analog_Data	deg	89.0		
C06_HPREF	KNTOUB	NTO Detent Upper Boundary	Analog_Data	deg	93.0		
C06_HPREF	KRATTC	TIME CST FOR RATED POWER LAG	Analog_Data	SEC	3.0		
C06_HPREF	KTOMN	MINIMUM VALUE FOR MAX TAKE OFF POWER	Analog_Data	shp	200.0		
C06_HPREF	TMP5	Temporary Variable to help GPU compile	Analog_Data				
C0701003004	KHPCON	HORSE POWER CONVERSION FACTOR	Analog_Data	rpm-ftlb/shp	5252.11		
C0701003004	KNPCL	FIXED 86% NP FOR MCL RATING	Analog_Data	rpm	1032.0		
C0701003004	KNPCR	FIXED 86% NP FOR CRUISE RATING	Analog_Data	rpm	1032.0		
C0701003004	KNPTO	FIXED NP FOR 100% NP RATING	Analog_Data	rpm	1200.0		
C0701003004	KONE	ONE CST	Analog_Data		1.0		
C0701003004	KTL10A	MAX TAKE OFF TORQUE LIMIT	Analog_Data	ft-lb	12036.0		
C0701003004	KTL20A	NORMAL TAKE OFF TORQUE LIMIT	Analog_Data	ft-lb	12036.0		
C0701003004	KTL30A	MAX CONTINUOUS TORQUE LIMIT	Analog_Data	ft-lb	12036.0		
C0701003004	KTL40A	MAX CLIMB TORQUE LIMIT	Analog_Data	ft-lb	12036.0		
C0701003004	KTL50A	MAX CRUISE TORQUE LIMIT	Analog_Data	ft-lb	12036.0		
C0701003004	QHPFTB	PW127 TABLE OF TORQUE REQUEST OUTPUT BLENDING	Maps				
C0701006011	KHPCON	HORSE POWER CONVERSION FACTOR	Analog_Data	rpm-ftlb/shp	5252.11		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
C0701006011	KONE	ONE CST	Analog_Data		1.0		
C0701006011	QHPFTB	PW127 TABLE OF TORQUE REQUEST OUTPUT BLENDING	Maps				
C07_OVRTQPRO	KHPCON	HORSE POWER CONVERSION FACTOR	Analog_Data	rpm-ftlb/shp	5252.11		
C07_OVRTQPRO	KONE	ONE CST	Analog_Data		1.0		
C07_OVRTQPRO	NPMU0A	NPT BLENDING FACTOR TABLE	Maps				
C07_OVRTQPRO	QHPFTB	PW127 TABLE OF TORQUE REQUEST OUTPUT BLENDING	Maps				
C08_SHPCL	DELT1T	DELTA CORRECTION TABLE FOR CORRECTED POWER TO CORRECTED NH GAIN - DATA INVERTED	Maps				
C08_SHPCL	DELT2T	DELTA CORRECTION TABLE FOR CORRECTED POWER TO CORRECTED NH DOMINANT LAG - DATA INVERTED	Maps				
C08_SHPCL	HPKITB	HP LOOP INTEGRAL GAIN TABLE	Maps				
C08_SHPCL	HPKPTB	HP LOOP PROPORTIONAL GAIN TABLE	Maps				
C08_SHPCL	KC08DT	HP GOVERNING PROCESS C08 CYCLE TIME	Analog_Data	sec	0.0208		
C08_SHPCL	KFALSE	FALSE IN RESETS	Boolean_Data		False		
C08_SHPCL	KUTM	RATE LIMIT RESET TIME DELAY	Analog_Data	SEC	2.0		
C08_SHPCL	KVKRHU	ASCEND RATE LI FOR AIRSPEED USED IN PWR GOV	Analog_Data	KTS/SEC	100.0		
C08_SHPCL	KVKRLU	DESC RATE LIM FOR AIRSPEED USED IN PWR GOV	Analog_Data	KTS/SEC	-500.0		
C08_SHPCL	KVKUPT	DEFAULT AIRSPEED FOR UPTRIMS	Analog_Data	KTS	60.0		
C10_ATOIGN	K0	ZERO	Integer_Data		0		
C10_ATOIGN	KAINHD	AUTOIGNITION NHDOT RESET SETPOINT	Analog_Data	RPM/SEC	-100.0		
C10_ATOIGN	KLOOP8	NH TRACKING LOOP IDENT. CST	Integer_Data		8		
C10_ATOIGN	KMFCND	NHDOT SET PT TO RESET MFC INTERF. LOGIC.	Analog_Data	RPM/SEC	600.0		
C10_ATOIGN	KMFCNH	NH SET PT TO DISABLE MFC INTERF. LOGIC	Analog_Data	RPM	27000.0		
C10_ATOIGN	KNHBND	AUTOIGNITION HYSTERESIS BAND WIDTH	Analog_Data	rpm	100.0		
C10_ATOIGN	KNHTRP	AUTOIGNITION HYSTERESIS TRIP POINT	Analog_Data	rpm	21450.0		
C11_SHPREQ_1	KDLHPR	TRANSIENT CHANGE RESET VALUE	Analog_Data	shp	0.0		
C11_SHPREQ_1	KRCTO	RATING CODE FOR TAKE OFF	Integer_Data		1		
C11_SHPREQ_2	KHP0	ZERO HP DUMP	Analog_Data	shp	0.0		
C11_SHPREQ_2	KHPDMP	HP DUMP CONSTANT VALUE	Analog_Data	shp	-1250.0		
C11_SHPREQ_2	KUPTGN	UPTRIM REQUEST DERIVATIVE GAIN	Analog_Data	sec	0.64		
C11_SHPREQ_2	KUPTTC	UPTRIM REQUEST DERIVATIVE TIME CST	Analog_Data	sec	0.4		
C1202005006	KFALSE	FALSE IN RESETS	Boolean_Data		False		
C1202005006	KIASTO	TAKE OFF MINIMUM AIR SPEED	Analog_Data	knts	125.0		
C1202005006	KLOPLA	GROUND HANDLING PLA THRESHOLD	Analog_Data	deg	30.0		
C1202005006	KNGVMN	PLA VALUE AT WHICH NPT GOVERNING IS RE-ENABLED	Analog_Data	deg	68.0		
C1202005006	KNGVMX	PLA VALUE AT WHICH NPT GOVERNING IS DISABLED	Analog_Data	deg	70.0		
C1202005006	KNPCLT	NPT LOOP RE-ENABLE DELAY	Analog_Data	sec	5.0		
C1202005006	KNPDMT	NPT LOOP ENABLE DELAY	Analog_Data	sec	1.0		
C1202005006	KNPTC1	NPT SPEED REFERENCE FOR NPT LOOP ENABLE	Analog_Data	rpm	6000.0		
C1202005006	KNPTC2	NPT LOOP DISABLE VALUE	Analog_Data	rpm/sec	-15000.0		
C1202005006	KNPTDY	NPT LOOP ENABLE DELAY	Analog_Data	sec	5.0		
C1202005006	KPLAD	PLA VALUE AT WHICH NPT GOV IS DISABLED	Analog_Data	deg	30.0		
C1202005006	KQTOA	MAX TQ VALUE ALLOWED BEFORE NPT GOV DISABLED	Analog_Data	ft-lbs	12036.0		
C12_NPTGOV_1	K12DT1	INVERSE OF PROCESS C-17 UPDATE RATE	Analog_Data	1/SEC	48.077		
C12_NPTGOV_1	KDNHBP	RATE ADDED TO NHFBT DURING BOOST IN FORWARD	Analog_Data	RPM/SEC	400.0		
C12_NPTGOV_1	KFALSE	FALSE	Boolean_Data		False		
C12_NPTGOV_1	KMNNPT	NPT TRACKING RANGE LIMIT MINIMUM	Analog_Data	RPM/SEC	-15000		
C12_NPTGOV_1	KMXDNH	NPT TRACKING RANGE LIMIT MAXIMUM	Analog_Data	RPM/SEC	15000		
C12_NPTGOV_1	KNHTE1	NPT TRACKING MAX DIFFERENCES	Analog_Data	RPM	5000		
C12_NPTGOV_1	KNPRFD	DECREASING RATE LIMIT ON MPTREF	Analog_Data	RPM/SEC	-2000		
C12_NPTGOV_1	KNPRFI	INCREASING RATE LIMIT ON NPTREF	Analog_Data	RPM/SEC	2000		
C12_NPTGOV_1	KNPTTC	NPT GOV DERIVATIVE LAG TIME CONSTANT	Analog_Data	SEC	0.08		
C12_NPTGOV_1	NPTDTB	NPT GOVERNING DERIVATIVE GAIN SCHEDULE	Maps				
C12_NPTGOV_1	NPTPTB	NPT GOVERNING PROPORTIONAL GAIN SCHEDULE	Maps				
C12_NPTGOV_1	NPTXTB	NPT GOVERNING SCHEDULE	Maps				
C12_NPTGOV_2	KFALSE	FALSE IN RESETS	Boolean_Data		False		
C12_NPTGOV_2	KIASTO	TAKE OFF MINIMUM AIR SPEED	Analog_Data	knts	125.0		
C12_NPTGOV_2	KLOPLA	GROUND HANDLING PLA THRESHOLD	Analog_Data	deg	30.0		
C12_NPTGOV_2	KNGVMN	PLA VALUE AT WHICH NPT GOVERNING IS RE-ENABLED	Analog_Data	deg	68.0		
C12_NPTGOV_2	KNGVMX	PLA VALUE AT WHICH NPT GOVERNING IS DISABLED	Analog_Data	deg	70.0		
C12_NPTGOV_2	KNPCLT	NPT LOOP RE-ENABLE DELAY	Analog_Data	sec	5.0		
C12_NPTGOV_2	KNPDMT	NPT LOOP ENABLE DELAY	Analog_Data	sec	1.0		
C12_NPTGOV_2	KNPTC1	NPT SPEED REFERENCE FOR NPT LOOP ENABLE	Analog_Data	rpm	6000.0		
C12_NPTGOV_2	KNPTC2	NPT LOOP DISABLE VALUE	Analog_Data	rpm/sec	-15000.0		
C12_NPTGOV_2	KNPTDY	NPT LOOP ENABLE DELAY	Analog_Data	sec	5.0		
C12_NPTGOV_2	KPLAD	PLA VALUE AT WHICH NPT GOV IS DISABLED	Analog_Data	deg	30.0		
C13_NHACCELLIM	ACL4TB	NPT GOV ACCEL SCHEDULE TABLE	Maps				
C13_NHACCELLIM	DLNDBT	ACCEL LIMIT CORRECTION FACTOR TABLE	Maps				
C13_NHACCELLIM	KACCL4	ACCELERATION CST	Analog_Data	rpm/sec	0.0		
C13_NHACCELLIM	KACCMX	UPTRIM ACCELERATION LIMIT	Analog_Data	rpm/sec	6400.0		
C13_NHACCELLIM	KUPTMR	UPTRIM TIMER LIMIT	Analog_Data	sec	1.0		
C13_NHACCELLIM	NHDTB	CORRECTED ACCEL SCHEDULE TABLE	Maps				
C14_NHLIM	KNHLIM	HIGH ROTOR SPEED LIMIT	Analog_Data	rpm	35298		
C14_NHLIM	KNHML	NH LIMIT GAIN	Analog_Data	rpm/rpm-se	2.0		
C14_NHLIM	NHLMTB	NH LIMIT RATE REQUEST CONV. TABLE	Maps				

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
C15_NHDECELLIM	DCL4TB	NPT GOV. DECEL SCHEDULE TABLE	Maps				
C15_NHDECELLIM	DECFTB	DECEL ALTITUDE CORRECTION FACTOR TABLE	Maps				
C15_NHDECELLIM	DECLTB	CORRECTED DECEL SCHEDULE TABLE	Maps				
C15_NHDECELLIM	KDECL4	DECELERATION CONSTANT	Analog_Data	rpm/sec	0.0		
C15_NHDECELLIM	KDECMX	DUMP LOGIC MAX DECEL RATE	Analog_Data	rpm/sec	-15000		
C16_LOOPSEL	KLOOP0	DECEL LOOP IDENT. CST	Integer_Data	---	0		
C16_LOOPSEL	KLOOP1	ACCEL LOOP IDENT. CST	Integer_Data	---	1		
C16_LOOPSEL	KLOOP2	HP GOV. LOOP IDENT. CST	Integer_Data	---	2		
C16_LOOPSEL	KLOOP3	NPT GOV. LOOP IDENT. CST	Integer_Data	---	3		
C16_LOOPSEL	KLOOP4	NH LIM. LOOP IDENT. CST	Integer_Data	---	4		
C16_LOOPSEL	KLOOP5	NH MIN LOOP IDENT. CST	Integer_Data	---	5		
C16_LOOPSEL	KLOOP6	NH MAX LOOP IDENT. CST	Integer_Data	---	6		
C16_LOOPSEL	KLOOP7	NH MAX LOOP IDENT. CST	Integer_Data	---	7		
C16_LOOPSEL	KLOOP8	NH TRACKING LOOP IDENT. CST	Integer_Data	---	8		
C16_LOOPSEL	KLOOP9	NPT TRACKING LOOP IDENT. CST	Integer_Data	---	9		
C17_NHREQUEST_1	K17DT1	INVERSE OF PROCESS UPDATE TIME	Analog_Data	---	48.077		
C17_NHREQUEST_1	KDTD2	NH REQUEST CYCLE TIME	Analog_Data	sec	0.0208		
C17_NHREQUEST_1	KERMUL	NH REQUEST LOOP GAIN	Analog_Data	rpm/rpm-sec	8.0		
C17_NHREQUEST_1	KMNDNH	MIN NH RATE OF CHANGE FROM TRACKING LOGIC	Analog_Data	RPM/SEC	0.0		
C17_NHREQUEST_1	KMXDNH	NPT TRACKING RANGE LIMIT MAXIMUM	Analog_Data	RPM/SEC	15000.0		
C17_NHREQUEST_1	KNHFB	NHFB LEAD TIME CONSTANT	Analog_Data	SEC	0.16		
C17_NHREQUEST_1	KNHFBG	NHFB LAG TIME CONSTANT	Analog_Data	SEC	0.04		
C17_NHREQUEST_1	KNHTE	INNER LOOP TRACKING ERROR FOR PWR GOV.	Analog_Data	RPM	1200		
C17_NHREQUEST_1	KREQMN	NHREQ RANGE LIMIT MINIMUM	Analog_Data	rpm	0.0		
C17_NHREQUEST_1	KREQMX	NHREQ RANGE LIMIT MAXIMUM	Analog_Data	rpm	34500		
C17_NHREQUEST_2	KC17DT	NH REQUEST PROCESS CYCLE TIME	Analog_Data	sec	0.0208		
C17_NHREQUEST_2	KERMUL	NH REQUEST LOOP GAIN	Analog_Data	rpm/rpm-se	0.125		
C17_NHREQUEST_2	KLOOP2	HP GOV. LOOP IDENT. CST	Integer_Data	---	2		
C18_NHGOV_1	KNHGL1	COMPENSATOR LAG TIME CST 1 FOR PWR GOV	Analog_Data	SEC	0.04		
C18_NHGOV_1	KNHGL2	COMPENSATOR FACTOR FOR NPT GOV	Analog_Data		1.333		
C18_NHGOV_1	KNHGL3	COMPENSATOR LAG TIME CNSTS FOR NPT GOV	Analog_Data	SEC	0.0333		
C18_NHGOV_1	KNHCTB	COMPENSATOR GAIN TABLE	Maps				
C18_NHGOV_1	NHL1TB	COMPENSATOR LEAD TIME CST 1 TABLE	Maps				
C18_NHGOV_1	NHL2TB	COMPENSATOR LEAD TIME CST 2 TABLE	Maps				
C18_NHGOV_1	NHLGTB	COMPENSATOR LAG TIME CST 2 TABLE	Maps				
C18_NHGOV_2	KC18DT	UPDATE TIME FOR PROCESS C-18	Analog_Data	SEC	0.0208		
C18_NHGOV_2	KNHDB	COMPENSATOR INPUT DEADBAND	Analog_Data	rpm	20.0		
C19_UNDERSPEED	BSTGOA	NPT U/S BOOST AS A FUNCTION OF AIR SPEED	Maps				
C19_UNDERSPEED	BSTPOA	NPT U/S MAP OF PLA DISKING POINTS	Maps				
C19_UNDERSPEED	BSTTOA	NPT U/S BOOST TERMINATION VALUE	Maps				
C19_UNDERSPEED	C1901DT	ITERATION RATE OF THE MODULE	Analog_Data	Sec	0.0416		
C19_UNDERSPEED	KBST35	BOOST LOGIC BOOST ENABLE CONSTANT	Analog_Data	Deg	22		
C19_UNDERSPEED	KBSTAU	BOOST TERMINATION TAU	Analog_Data	sec	0.3		
C19_UNDERSPEED	KBSTMN	NPT U/S CORRECTION RESET	Analog_Data	rpm	0.0		
C19_UNDERSPEED	KDXBC1	MIN DXBUMP TO SET DROOP FLAG=1	Analog_Data	STEP/SEC	50.0		
C19_UNDERSPEED	KDXBC2	MIN PLA TO SET DROOP FLAG=1	Analog_Data	DEG	20.0		
C19_UNDERSPEED	KFALSE	FALSE	Boolean_Data		False		
C19_UNDERSPEED	KNHLMX	MAXIMUM VALUE OF BOOST ALLOWED INTO STEPPER LOGIC	Analog_Data	step/sec	800.0		
C19_UNDERSPEED	KNPTCF	NPTDOT CUTOFF VALUE - UNDERSPEED LOGIC	Analog_Data	RPM/SEC	300.0		
C19_UNDERSPEED	KNPTTH	THRESHOLD OF NPT FOR UNDERSPEED ZEROING	Analog_Data	rpm	19000.0		
C19_UNDERSPEED	KPLAH	NPT U/S PLA HYSTERESIS	Analog_Data	deg	1.5		
C19_UNDERSPEED	KSTPMN	NPT U/S BOOST RESET	Analog_Data	step	0.0		
C19_UNDERSPEED	KTERMV	BOOST CANCEL VALUE	Analog_Data	rpm	0.0		
C19_UNDERSPEED	NHF0A	NPT U/S MAP TO CONVERT BOOST TO STEP/SEC	Maps				
C19_UNDERSPEED	NHLC0A	NPT U/S MAP OF DERIVATIVE LAG TIME CONSTANT	Maps				
C19_UNDERSPEED	NHLT0A	NPT U/S INITIAL SCHEDULE OF BOOST VALUES	Maps				
C20_OVRSPEED	KDNPOA	NPT O/S NPT RATE SETPOINT	Analog_Data	rpm/sec	800		
C20_OVRSPEED	KFALSE	FALSE IN RESETS	Boolean_Data		False		
C20_OVRSPEED	KNPDOS	MIN NPT RATE TO TRIGGER DMPPC2 LOGIC	Analog_Data	RPM/SEC	100.0		
C20_OVRSPEED	KNPOST	DELAY OFF TIME FOR HIGH PLA FLAG	Analog_Data	SEC	1.5		
C20_OVRSPEED	KNPTOA	NPT O/S NPT MINIMUM SPEED	Analog_Data	rpm	15000		
C20_OVRSPEED	KNPTOS	MIN NPT TO TRIGGER DMPPC2 LOGIC	Analog_Data	RPM	19800.0		
C20_OVRSPEED	KOS0A	NPT O/S DOWNWARD DER LAG SETPOINT	Analog_Data		-250.0		
C20_OVRSPEED	KOS10A	NPT O/S UPWARD DER LAG SETPOINT 1	Analog_Data		-250.0		
C20_OVRSPEED	KOS20A	NPT O/S UPWARD DER LAG SETPOINT 2	Analog_Data		-70		
C20_OVRSPEED	KOSD0A	NPT O/S PLA DOWNWARD SLAM TIME CST	Analog_Data	sec	0.5		
C20_OVRSPEED	KOSU0A	NPT O/S PLA UPWARD SLAM TIME CST	Analog_Data	sec	0.5		
C20_OVRSPEED	KPLA01	HIGH PLA CONSTANT FOR DMPPC2 LOGIC	Analog_Data	DEG	70.0		
C20_OVRSPEED	KPLA02	LOW PLA CONSTANT FOR DMPPC2 LOGIC	Analog_Data	DEG	25.0		
C20_OVRSPEED	NHOD0A	NPT O/S PLA SLAM DOWNWARD MAP	Maps				
C20_OVRSPEED	NHOU0A	NPT O/S PLA SLAM UPWARD MAP	Maps				
C2101005006	KDSLSP	DECEL SLAM SET POINT	Analog_Data	RPM/SEC	-2800.0		
C2101005006	KFALSE	FALSE	Boolean_Data		False		
C2101005006	KNHDT1	STEADY STATE TIME - SLAM ACCEL	Analog_Data	SEC	0.5		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
C2101005006	KNHDT2	STEADY STATE TIME - RESLAM	Analog_Data	SEC	0.5		
C2101005006	KNHDTCT	HBOV RESLAM FILTER TIME CST	Analog_Data	SEC	0.5		
C2101005006	KPOS4	POSITIVE CONSTANT	Analog_Data	rpm/sec	0.0		
C2101005006	KRSLM	NHDOT SET POINT FOR RESLAM	Analog_Data	rpm/sec	-400.0		
C2101005006	SLMATB	ACCEL SLAM SET POINT	Maps				
C2101005006	SLMCTB	ALT CORRECTION FACTOR FOR HBOV SET POINT	Maps				
C2102007008	KCLOSE	HBOV CLOSE VALUE	Analog_Data	MA	0.0		
C2102007008	KFALSE	FALSE	Boolean_Data		False		
C2102007008	KHBMFC	DECR RATE LIM ON HBOV WHEN MFC IN INTERF.	Analog_Data	MA/SEC	-10.0		
C2102007008	KHBVOP	OPEN BLEED VALVE VALUE	Analog_Data	MA	65.0		
C2102007008	KMFCDL	MFC INTERFERENCE DELAY OFF.	Analog_Data	SEC	1.0		
C2102007008	KPLAH2	HBOV CURRENT PLA BANDWIDTH	Analog_Data	DEG	1.5		
C2102007008	KPLAIN	INCR RATE LIM ON HBOV CURR SCHED FROM PLADB2	Analog_Data	MA/SEC	3000.0		
C2102007008	KSTART	THRESHOLD FOR HBOV CLOSING DURING START	Analog_Data	RPM	18000.0		
C2102007008	KZERO	ZERO IN RANGE CHECKS	Analog_Data	(---)	0		
C2102007008	PBDETB	DECREASING HBOV CURRENT RATE LIMIT	Maps				
C2102007008	PBLDTB	BLEED VERSUS PAMB TABLE	Maps				
C2102007008	PLADTB	BLEED VERSUS PLADB2 TABLE	Maps				
C21_HBOVEM_1	KDNHCT	DNHSEL FILTER TIME CONSTANT	Analog_Data	SEC	0.32		
C21_HBOVEM_1	KDLSLP	DECEL SLAM SET POINT	Analog_Data	RPM/SEC	-2800.0		
C21_HBOVEM_1	KFALSE	FALSE	Boolean_Data		False		
C21_HBOVEM_1	KNHDT1	STEADY STATE TIME - SLAM ACCEL	Analog_Data	SEC	0.5		
C21_HBOVEM_1	KNHDT2	STEADY STATE TIME - RESLAM	Analog_Data	SEC	0.5		
C21_HBOVEM_1	KNHDTCT	HBOV RESLAM FILTER TIME CST	Analog_Data	SEC	0.5		
C21_HBOVEM_1	KPOS4	POSITIVE CONSTANT	Analog_Data	rpm/sec	0.0		
C21_HBOVEM_1	KRSLM	NHDOT SET POINT FOR RESLAM	Analog_Data	rpm/sec	-400.0		
C21_HBOVEM_1	SLMATB	ACCEL SLAM SET POINT	Maps				
C21_HBOVEM_1	SLMCTB	ALT CORRECTION FACTOR FOR HBOV SET POINT	Maps				
C21_HBOVEM_2	KCLOSE	HBOV CLOSE VALUE	Analog_Data	MA	0.0		
C21_HBOVEM_2	KFALSE	FALSE	Boolean_Data		False		
C21_HBOVEM_2	KHBMFC	DECR RATE LIM ON HBOV WHEN MFC IN INTERF.	Analog_Data	MA/SEC	-10.0		
C21_HBOVEM_2	KHBVOP	OPEN BLEED VALVE VALUE	Analog_Data	MA	65.0		
C21_HBOVEM_2	KMFCDL	MFC INTERFERENCE DELAY OFF.	Analog_Data	SEC	1.0		
C21_HBOVEM_2	KPLAH2	HBOV CURRENT PLA BANDWIDTH	Analog_Data	DEG	1.5		
C21_HBOVEM_2	KPLAIN	INCR RATE LIM ON HBOV CURR SCHED FROM PLADB2	Analog_Data	MA/SEC	3000.0		
C21_HBOVEM_2	KSTART	THRESHOLD FOR HBOV CLOSING DURING START	Analog_Data	RPM	18000.0		
C21_HBOVEM_2	KZERO	ZERO IN RANGE CHECKS	Analog_Data	(---)	0		
C21_HBOVEM_2	PBDETB	DECREASING HBOV CURRENT RATE LIMIT	Maps				
C21_HBOVEM_2	PBLDTB	BLEED VERSUS PAMB TABLE	Maps				
C21_HBOVEM_2	PLADTB	BLEED VERSUS PLADB2 TABLE	Maps				
C2201000000	KNHDT	STEADY STATE TIMER	Analog_Data	sec	1.0		
C2201000000	KSLFGT	SLAM RESET TIMER	Analog_Data	sec	20.0		
C2201000000	KSLRST	STEADY STATE DETECTION VALUE	Analog_Data	rpm/sec	233.0		
C2202002002	ASLMTB	ACCEL SLAM SETPOINT TABLE	Maps				
C2202002002	DSLMTB	DECEL SLAM SETPOINT TABLE	Maps				
C2202002002	KNHDT	HBOV RESLAM FILTER TIME CST	Analog_Data	sec	0.5		
C2202002002	KPOS4	POSITIVE CONSTANT	Analog_Data	(rpm/sec)	0.0		
C2202002002	KRSLAM	NHDOT SET POINT FOR RESLAM	Analog_Data	rpm/sec	-400		
C22_HBOVMM_1	KFALSE	VALUE OF FALSE	Boolean_Data		False		
C22_HBOVMM_1	KFALSE	VALUE OF FALSE	Boolean_Data		False		
C22_HBOVMM_1	KNEG1	SIGN CHANGE CONSTANT	Analog_Data		-1.0		
C22_HBOVMM_1	KNHDT	STEADY STATE TIMER	Analog_Data	sec	1.0		
C22_HBOVMM_1	KPLARL	HBOV PLA ENABLE RATE LIMIT	Analog_Data	DEG/SEC	8.0		
C22_HBOVMM_1	KPLASP	HBOV PLA ENABLE THRESHOLD	Analog_Data	DEG	5.0		
C22_HBOVMM_1	KPLAUR	HBOV PLA ENABLE NO RATE LIMIT	Analog_Data	DEG/SEC	1500.0		
C22_HBOVMM_1	KSLFGT	SLAM RESET TIMER	Analog_Data	sec	20.0		
C22_HBOVMM_1	KSLRST	STEADY STATE DETECTION VALUE	Analog_Data	rpm/sec	233.0		
C22_HBOVMM_1	TEMP	TEMPORARY STORAGE	Analog_Data				
C22_HBOVMM_2	ASLMTB	ACCEL SLAM SETPOINT TABLE	Maps				
C22_HBOVMM_2	DSLMTB	DECEL SLAM SETPOINT TABLE	Maps				
C22_HBOVMM_2	KNHDT	HBOV RESLAM FILTER TIME CST	Analog_Data	sec	0.5		
C22_HBOVMM_2	KPOS4	POSITIVE CONSTANT	Analog_Data	(rpm/sec)	0.0		
C22_HBOVMM_2	KRSLAM	NHDOT SET POINT FOR RESLAM	Analog_Data	rpm/sec	-400		
C22_HBOVMM_3	KFALSE	FALSE IN RESETS	Boolean_Data		False		
C22_HBOVMM_3	KHBVCL	CLOSE BLEED VALVE VALUE	Analog_Data	MA	0.0		
C22_HBOVMM_3	KHBVOP	OPEN BLEED VALVE VALUE	Analog_Data	MA	65.0		
C22_HBOVMM_3	KNHHS	DEADBAND OF HYSTERISIS FOR NH	Analog_Data	(RPM)	333.0		
C22_HBOVMM_3	SSHBTB	MANUAL STEADY STATE HBOV MAP	Maps				
C2602001001	KT1P8	TEMP. GAIN FACTOR	Analog_Data	degF/deg	1.800		
C2602001001	KT273	TEMP. REFERENCE IN KELVIN	Analog_Data	deg K	273.15		
C2602001001	KT32	TEMP. CONVERSION FACTOR	Analog_Data	deg F	32.0		
C2602001001	KT460	TEMPERATURE DATUM	Analog_Data	deg F	459.67		
C2602001001	KT519	TEMP. REFERENCE IN RANKINE	Analog_Data	degR	518.67		
C2602001001	SQRT	RTHETA vs. THETA	Maps				
C2603003003	ALTTB	PAMB TO ALTITUDE CONVERSION	Maps				

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
C2603003003	DPOTB	IAS TO DPO TABLE	Maps				
C2603003003	IASTB	DPOS PRESSURE TO IASS CONVERSION	Maps				
C2603003003	KIASBS	MINIMUM AIR SPEED	Analog_Data	kts	50		
C2603003003	KIASTO	TAKE OFF MINIMUM AIR SPEED	Analog_Data	knts	125.0		
C2603003003	KIRF	INTAKE RECOVERY FACTOR	Analog_Data		0.9925		
C2603003003	KP15	REFERENCE PRESSURE	Analog_Data	psia	14.7		
C2603003003	MACHTB	PRATIO TO MACH TABLE	Maps				
C2603003003	PT15TB	NP TO PROP PRES. RATIO TABLE	Maps				
C26_MPARAM_2	KALTTA	ALTITUDE TO ISA TEMPERATURE GAIN	Analog_Data	deg C/ feet	-0.00198		
C26_MPARAM_2	KELVIN	CONVERSION BIAS FROM DEGREES CELSIUS TO KELVIN	Analog_Data	deg C	273.15		
C26_MPARAM_2	KFC	CELSIUS TO FAHRENHEIT CONVERSION GAIN	Analog_Data	deg F / deg C	1.8		
C26_MPARAM_2	KISASL	SEA LEVEL ISA TEMPERATURE	Analog_Data	deg C	15.0		
C26_MPARAM_2	KRANK	FAHRENHEIT TO RANKINE CONVERSION OFFSET	Analog_Data	deg F	459.67		
C26_MPARAM_2	KT1P8	TEMP. GAIN FACTOR	Analog_Data	degF/deg	1.800		
C26_MPARAM_2	KT273	TEMP. REFERENCE IN KELVIN	Analog_Data	deg K	273.15		
C26_MPARAM_2	KT32	TEMP. CONVERSION FACTOR	Analog_Data	deg F	32.0		
C26_MPARAM_2	KT460	TEMPERATURE DATUM	Analog_Data	deg F	459.67		
C26_MPARAM_2	KT519	TEMP. REFERENCE IN RANKINE	Analog_Data	degR	518.67		
C26_MPARAM_2	KTAT1	TOTAL TO STATIC CONVERSION CONSTANT #1	Analog_Data		1.0		
C26_MPARAM_2	KTAT2	TOTAL TO STATIC CONVERSION CONSTANT #2	Analog_Data		0.2		
C26_MPARAM_2	SQRT	RTHETA vs. THETA	Maps				
C26_MPARAM_2	TTOTTB	PROP TOTAL TEMPERATURE RATIO STN 0 TO 1	Maps				
C26_MPARAM_3	ALTTB	PAMB TO ALTITUDE CONVERSION	Maps				
C26_MPARAM_3	DPOTB	IAS TO DPO TABLE	Maps				
C26_MPARAM_3	IASTB	DPOS PRESSURE TO IASS CONVERSION	Maps				
C26_MPARAM_3	KIASBS	MINIMUM AIR SPEED	Analog_Data	kts	50		
C26_MPARAM_3	KIAS_TO	TAKE OFF MINIMUM AIR SPEED	Analog_Data	knts	50.0		
C26_MPARAM_3	KIRF	INTAKE RECOVERY FACTOR	Analog_Data		0.9925		
C26_MPARAM_3	KP15	REFERENCE PRESSURE	Analog_Data	psia	14.7		
C26_MPARAM_3	MACHTB	PRATIO TO MACH TABLE	Maps				
C26_MPARAM_3	PT15TB	NP TO PROP PRES. RATIO TABLE	Maps				
C26_MPARAM_4	K10DT2	TEN TIMES THE 0.0416 SEC UPDATE RATE	Analog_Data	sec	0.416		
C26_MPARAM_4	KMNPMMN	NP RANGE MINIMUM VALUE	Analog_Data	rpm	900.0		
C26_MPARAM_4	KMNPMMX	NP RANGE MAXIMUM VALUE	Analog_Data	rpm	1620.0		
C26_MPARAM_4	KNPGR1	PWT TO LAY SHAFT GEAR RATIO	Analog_Data	rpm/rpm	4.695652		
C26_MPARAM_4	KNPGR2	PROP TO LAY SHAFT GEAR RATIO	Analog_Data	rpm/rpm	0.2817		
C26_MPARAM_5	C2605DT	ITERATION RATE OF MODULE	Analog_Data	Sec	0.0416		
C26_MPARAM_5	K10DT2	TEN TIMES THE 0.0416 SEC UPDATE RATE	Analog_Data	sec	0.416		
C26_MPARAM_5	KFALSE	FALSE IN RESETS	Boolean_Data		False		
C26_MPARAM_5	KHPCON	HORSE POWER CONVERSION FACTOR	Analog_Data	RPM-FTLB	5252.11		
C26_MPARAM_5	KQOUMN	BOTTOM CLAMP FOR QOUT	Analog_Data	ft-lb	0.0		
C26_MPARAM_5	KQTAU3	TORQUE FILTER #3 TIME CST	Analog_Data	sec	0.111		
C26_MPARAM_5	KQZERO	ZERO TORQUE	Analog_Data	ft-lbs	0.0		
C26_MPARAM_5	KTQ100	1% OF MAX TORQUE	Analog_Data	FT-LB	120.36		
C26_MPARAM_5	TQFLTb	TORQUE FILTER TIME CONSTANT TABLE	Maps				
C26_MPARAM_6	K10CDT	PROCESS CYCLE TIME TIMES 10	Analog_Data	sec	0.208		
C26_MPARAM_7	K10CDT	PROCESS CYCLE TIME TIMES 10	Analog_Data	sec	0.208		
C2701001001	ALTKTB	HOTEL MODE NH SCHEDULE ALTITUDE CORR.	Maps				
C2701001001	KMAX	MAXIMUM NH ALLOWABLE	Analog_Data	RPM	34500		
C2701001001	KNHHTD	HOTEL MODE DECREASING RATE LIMIT	Analog_Data	rpm/sec	-150.0		
C2701001001	KNHHTI	HOTEL MODE INCREASING RATE LIMIT	Analog_Data	rpm/sec	150.0		
C2701001001	KNPTH	MAXIMUM NPT PERMITTED IN HOTEL MODE	Analog_Data	rpm	6000.0		
C2701001001	MP51TB	NH MINIMUM TABLE	Maps				
C2701001001	NHHOTB	HOTEL MODE TEMPERATURE CORRECTION FACTOR	Maps				
C2701001001	NHIDTB	IDLE SPEED SCHEDULE #1 TABLE	Maps				
C2701001001	NHMTTB	NH GOVERNING HOTEL MODE NH SCHEDULE	Maps				
C2701001001	NHRVTB	NH MAXIMUM TABLE	Maps				
C2701006001	ALTKTB	HOTEL MODE NH SCHEDULE ALTITUDE CORR.	Maps				
C2701006001	KMAX	MAXIMUM NH ALLOWABLE	Analog_Data	RPM	34500		
C2701006001	KNHHTD	HOTEL MODE DECREASING RATE LIMIT	Analog_Data	rpm/sec	-150.0		
C2701006001	KNHHTI	HOTEL MODE INCREASING RATE LIMIT	Analog_Data	rpm/sec	150.0		
C2701006001	KNPTH	MAXIMUM NPT PERMITTED IN HOTEL MODE	Analog_Data	rpm	6000.0		
C2701006001	MP51TB	NH MINIMUM TABLE	Maps				
C2701006001	NHHOTB	HOTEL MODE TEMPERATURE CORRECTION FACTOR	Maps				
C2701006001	NHIDTB	IDLE SPEED SCHEDULE #1 TABLE	Maps				
C2701006001	NHMTTB	NH GOVERNING HOTEL MODE NH SCHEDULE	Maps				
C2701006001	NHRVTB	NH MAXIMUM TABLE	Maps				
C2701009009	ALTKTB	HOTEL MODE NH SCHEDULE ALTITUDE CORR.	Maps				
C2701009009	KMAX	MAXIMUM NH ALLOWABLE	Analog_Data	RPM	34500		
C2701009009	KNHHTD	HOTEL MODE DECREASING RATE LIMIT	Analog_Data	rpm/sec	-150.0		
C2701009009	KNHHTI	HOTEL MODE INCREASING RATE LIMIT	Analog_Data	rpm/sec	150.0		
C2701009009	KNPTH	MAXIMUM NPT PERMITTED IN HOTEL MODE	Analog_Data	rpm	6000.0		
C2701009009	MP51TB	NH MINIMUM TABLE #1 (Valid for Ambient Temperatures above -15C)	Maps				

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
C2701009009	MP52TB	NH MINIMUM TABLE #2 (Valid for Ambient Temperatures below -45C)	Maps				
C2701009009	NHBLTB	Blending Table for -45C<=Ambient Temperature<=-15C	Maps				
C2701009009	NHHOTB	HOTEL MODE TEMPERATURE CORRECTION FACTOR	Maps				
C2701009009	NHIDTB	IDLE SPEED SCHEDULE	Maps				
C2701009009	NHMTTB	NH GOVERNING HOTEL MODE NH SCHEDULE	Maps				
C2701009009	NHR2TB	NH MAXIMUM TABLE #2 (Valid for Ambient Temperatures below -45C)	Maps				
C2701009009	NHRVTB	NH MAXIMUM TABLE (Valid for Ambient Temperatures above -15C)	Maps				
C27_NHMMNMX_1	ALTKTB	HOTEL MODE NH SCHEDULE ALTITUDE CORR.	Maps				
C27_NHMMNMX_1	KMAX	MAXIMUM NH ALLOWABLE	Analog_Data	RPM	34500		
C27_NHMMNMX_1	KNHHTD	HOTEL MODE DECREASING RATE LIMIT	Analog_Data	rpm/sec	-150.0		
C27_NHMMNMX_1	KNHHTI	HOTEL MODE INCREASING RATE LIMIT	Analog_Data	rpm/sec	150.0		
C27_NHMMNMX_1	KNPHTH	MAXIMUM NPT PERMITTED IN HOTEL MODE	Analog_Data	rpm	6000.0		
C27_NHMMNMX_1	KNPTLO	77% POWER TURBINE SPEED	Analog_Data	RPM	15400.0		
C27_NHMMNMX_1	KPLAID	PLA RESETTING VALUE WHEN IN FEATHER	Analog_Data	DEG	35.0		
C27_NHMMNMX_1	MP51TB	NH MINIMUM TABLE #1 (Valid for Ambient Temperatures above -15C)	Maps				
C27_NHMMNMX_1	MP52TB	NH MINIMUM TABLE #2 (Valid for Ambient Temperatures below -45C)	Maps				
C27_NHMMNMX_1	NHBLTB	Blending Table for -45C<=Ambient Temperature<=-15C	Maps				
C27_NHMMNMX_1	NHHOTB	HOTEL MODE TEMPERATURE CORRECTION FACTOR	Maps				
C27_NHMMNMX_1	NHIDTB	IDLE SPEED SCHEDULE	Maps				
C27_NHMMNMX_1	NHMTTB	NH GOVERNING HOTEL MODE NH SCHEDULE	Maps				
C27_NHMMNMX_1	NHR2TB	NH MAXIMUM TABLE #2 (Valid for Ambient Temperatures below -45C)	Maps				
C27_NHMMNMX_1	NHRVTB	NH MAXIMUM TABLE (Valid for Ambient Temperatures above -15C)	Maps				
C3001002002	KC18DT	UPDATE TIME FOR PROCESS C-18	Analog_Data	SEC	0.0208		
C3001002002	KC30DT	UPDATE TIME FOR PROCESS C-30	Analog_Data	SEC	0.0208		
C3001002002	KDXMAX	MAXIMUM STEPPER MOTOR RATE REQUEST	Analog_Data	steps/sec	500		
C3001002002	KDXMIN	MINIMUM STEPPER MOTOR RATE REQUEST	Analog_Data	steps/sec	-500		
C3001002002	KSTEPS	Overspeed Correction Stepper Motor Request	Analog_Data	steps/sec	-500		
C3001002002	KSTPRN	STEPPER MOTOR DECREASE RATE PER CYCLE	Analog_Data	STEP/CYCLE	-10.0		
C3001002002	KSTPRX	STEPPER MOTOR INCREASE RATE PER CYCLE	Analog_Data	STEP/CYCLE	10.0		
C30_SMRTQR	KAIGRS		Analog_Data		0.0		
C30_SMRTQR	KC18DT	UPDATE TIME FOR PROCESS C-18	Analog_Data	SEC	0.0208		
C30_SMRTQR	KC30DT	UPDATE TIME FOR PROCESS C-30	Analog_Data	SEC	0.0208		
C30_SMRTQR	KDXMAX	MAXIMUM STEPPER MOTOR RATE REQUEST	Analog_Data	steps/sec	500		
C30_SMRTQR	KDXMIN	MINIMUM STEPPER MOTOR RATE REQUEST	Analog_Data	steps/sec	-500		
C30_SMRTQR	KSTEPS	Overspeed Correction Stepper Motor Request	Analog_Data	steps/sec	-500		
C30_SMRTQR	KSTPRN	STEPPER MOTOR DECREASE RATE PER CYCLE	Analog_Data	STEP/CYCLE	-10.0		
C30_SMRTQR	KSTPRX	STEPPER MOTOR INCREASE RATE PER CYCLE	Analog_Data	STEP/CYCLE	10.0		
C31_TQLIM	KDNHLM	LIMITED NH RATE REQUEST	Analog_Data	RPM/SEC	0.0		
C31_TQLIM	KDNHLMN	MIN NH RATE REQUEST	Analog_Data	RPM/SEC	-15000.0		
C31_TQLIM	KDNHMX	MAX NH RATE REQUEST	Analog_Data	RPM/SEC	15000.0		
C31_TQLIM	KFALSE	FALSE IN RESETS	Boolean_Data		False		
C31_TQLIM	KHPCON	HORSEPOWER CONVERSION FACTOR	Analog_Data	RPM-FTLB	5252.11		
C31_TQLIM	KNHRO	LOWER RANGE OF REQUESTED NHREQ	Analog_Data	RPM	21000.0		
C31_TQLIM	KNPHI	UPPER RANGE LIMIT ON NPSE	Analog_Data	RPM	1800.0		
C31_TQLIM	KNPLOW	LOWER RANGE LIMIT ON NPSE	Analog_Data	RPM	240.0		
C31_TQLIM	KQTLT	DELAY TIME FOR LOW TORQUE FLAG	Analog_Data	SEC	1.0		
C31_TQLIM	KTQL2T	TQLIM2 FILTER TIME CONSTANT	Analog_Data	SEC	1.5		
C31_TQLIM	KTQLD	TORQUE LIMIT LOOP LEAD TIME CONSTANT	Analog_Data	SEC	0.3333		
C31_TQLIM	KTQLG	TORQUE LIMIT LOOP LAG TIME CONSTANT	Analog_Data	SEC	0.0400		
C31_TQLIM	KTQLM1	5% TORQUE CONSTANT	Analog_Data	LBF-FT	602.0		
C31_TQLIM	KTQLM2	116% TORQUE CONSTANT	Analog_Data	LBF-FT	1805.0		
C31_TQLIM	KTQLM3	20% TORQUE CONSTANT	Analog_Data	LBF-FT	2407.0		
C31_TQLIM	QLMTB	TORQUE LIMIT RATE REQUEST CONV. TABLE	Maps				
CMPACT	CMPMN3	NEGATIVE THREE	Integer_Data		'%O'177775'		
CMPACT	CTHREE	# RAM LOCATIONS PER FAULT BLOCK	Integer_Data		3		
CNT_EVNT	CRS_ALL	TOTAL ACCUMULATED TIME COARSE COUNTS	Analog_Data			0	32767
CNT_EVNT	CRS_ALL_SS	SNAPSHOT OF TOTAL ACCUMULATED TIME COARSE COUNTS	Analog_Data			0	32767
CNT_EVNT	FINE_ALL	TOTAL ACCUMULATED TIME FINE COUNTS	Analog_Data			0	1
CNT_EVNT	FINE_ALL_SS	SNAPSHOT OF TOTAL ACCUMULATED TIME FINE COUNTS	Analog_Data			0	1
CNT_EVNT	FINE_ALL_T	TEMPORARY TOTAL ACCUMULATED TIME FINE COUNTS	Analog_Data			0	1
CNT_EVNT	K1 CRS	COURSE COUNTER INCREMENT	Analog_Data		1.0		
CNT_EVNT	K1 FIN	FINE COUNTER INCREMENT	Analog_Data		1.0		
DBL_LL_SUBR	S4FNEG	Negative Full Scale Value	Integer_Data		'%O'100001'		
DBL_LL_SUBR	S4FPOS	Positive Full Scale Value	Integer_Data		'%O'77777'		
DBL_LL_SUBR	S4INIT	Initialization output value	Integer_Data		'%O'100000'		
DCU_RECEIVE	DATA_INTEGRITY	DCU DATA INTEGRITY STATUS BIT	Integer_Data		'%O'4000'		
DCU_TRANSFER	DCU_DATA_PURGE	DCU DATA PURGE STATUS BIT	Integer_Data		'%O'10000'		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
DCU_TRANSFER	DCU_DELAY_MAX	MAXIMUM DELAY TIME FOR DCU	Integer_Data	MS	208		
DCU_TRANSFER	MAX_RETRY_TIMER	MAXIMUM RETRY TIME FOR DCU	Integer_Data	SEC	9.9996		
DCU_TRANSMIT	DCU_DATA_BYTE	DCU INPUT DATA BYTE VALUE	Integer_Data				
DCU_TRANSMIT	FIFO_NOT_FULL	UART FIFO NOT FULL BIT	Integer_Data		%0'10000'		
DERLAG	DRLNFS	- FULL SCALE	Integer_Data		%0'100001'		
DERLAG	DRLPFS	+ FULL SCALE	Integer_Data		%0'77777'		
DUMP	SPUP	Upper address of the Reserved Memory area + 1	Integer_Data		%0'2000'		
E01_PORPSR	INIB13	Mask bit 13	Integer_Data		%0'10000'		
E01_PORPSR	INID18	18 Millisecond Delay loop count	Integer_Data		9009		
E01_PORPSR	INIM1	Pseudo-Random Data Pattern	Integer_Data		%0'55675'		
E02_RAMTEST	PAT1	Alternating 1-0 Pattern	Integer_Data		'125252		
E02_RAMTEST	PAT2	Alternating 0-1 Pattern	Integer_Data		'052525		
E02_RAMTEST	PAT3	All Ones Pattern	Integer_Data		'177777		
E02_RAMTEST	PAT4	All Zeros Pattern	Integer_Data		'000000		
E03_CHECKSUM	INRB14	Fault bits for DP EEPROM checksum fault	Integer_Data		%0'120000'		
E03_CHECKSUM	INRB15	Fault bits for P0 EEPROM checksum fault	Integer_Data		%0'140000'		
E03_CHECKSUM	TBSTRT	Table of Program EEPROM Address	Maps				
E04_INSTST	INCK1	Octal 125252	Integer_Data		%0'125252'		
E04_INSTST	INCK2	Octal 52525	Integer_Data		%0'52525'		
E05_RAMINI	INID0	DISCRETE OUTPUT INITIALIZATION BIT PATTERN	Integer_Data		%0'12'		
E05_RAMINI	IRC15	MASK BITS 1-4	Integer_Data		15		
E05_RAMINI	IRC3	THREE	Integer_Data		3		
E05_RAMINI	IRGAIN	INITIAL HIGH LEVEL A/D INTERFACE GAIN FACTOR	Analog_Data		1.0		
E05_RAMINI	IRPLAG	INITIAL RVDT INTERFACE GAIN VALUE	Analog_Data		3.488		
E05_RAMINI	IRSGC	INITIAL LOW LEVEL A/D INTERFACE GAIN	Analog_Data		1.0		
E05_RAMINI	KPL2RAT	Engine Rating Power Level 2	Integer_Data				
E05_RAMINI	MAXDGT	Maximum+1 legal value for the ones-digit	Integer_Data		%0'12'		
E05_RAMINI	MAXLRU	Maximum+1 legal value for an LRU-CODE	Integer_Data		%0'231'		
E05_RAMINI	PIDPTR	Software Version ID Address Pointer	Integer_Data		%0'77772'		
E05_RAMINI	PLAFAL	PLA trim read fault flags	Integer_Data		%0'101000'		
E05_RAMINI	PLAMAX	MAXIMUM ALLOWABLE VALUE OF PLA TRIM	Analog_Data		3.0		
E06_TQTEST	QDTMX	Delta-time check limit + 1 count	Analog_Data		2		
E06_TQTEST	QTEB11	DELTA-TIME FAULT BIT	Integer_Data		%0'002000'		
E06_TQTEST	QTEB12	RANGE CHECK FAULT BIT	Integer_Data		%0'004000'		
E06_TQTEST	QTEB16	FAULT LATCH	Integer_Data		%0'100000'		
E06_TQTEST	QTMN	Minimum time range check value	Analog_Data		3837		
E06_TQTEST	QTMX	Maximum time range check value + 1 count	Analog_Data		3844		
E06_TQTEST	TWMSCT	12 millisecond delay count	Integer_Data		6000		
E08_FG_EXECUTIVE	ADMSK	Mask A/D MUX code (High level and HBV)	Integer_Data		%0'23000'		
E08_FG_EXECUTIVE	BEND	Ending address of the BACKGROUND region + 1	Integer_Data				
E08_FG_EXECUTIVE	BSTRT	Starting address of the BACKGROUND region	Integer_Data				
E08_FG_EXECUTIVE	C81	Eighty-one	Integer_Data		81		
E08_FG_EXECUTIVE	CM777	OCTAL 077777	Integer_Data		%0'077777'		
E08_FG_EXECUTIVE	CTLB16	MASK BIT 16	Integer_Data		%0'100000'		
E08_FG_EXECUTIVE	HLVMSK	Mask to test bits 10 & 11	Integer_Data		%0'3000'		
E08_FG_EXECUTIVE	MUXTAB	Foreground Control State Table	Integer_Data		STATE_TABLE		
E08_FG_EXECUTIVE	SBOOT	START ADDRESS OF BOOT CODE	Integer_Data		%0'144000'		
E08_FG_EXECUTIVE	TABEND	End of state table	Integer_Data		TBLEND		
E08_FG_EXECUTIVE	TMODE	Address of test mode routine	Integer_Data				
E08_FG_SECSL	CONFIG_HI_PLUS1	Max configuration value + 1	Integer_Data		6		
E08_FG_SECSL	CONFIG_LO	Min configuration value	Integer_Data		5		
E09_SMSUMV	INB146	BITS 14 AND 16 SET	Integer_Data		%0'120000'		
E09_SMSUMV	INB156	BITS 15 AND 16 SET	Integer_Data		%0'140000'		
E09_SMSUMV	INCNT2	0.8 millisecond delay count	Integer_Data		400		
E09_SMSUMV	SVITMN	SUM VOLTAGE MIN VALUE FOR INITIALIZATION TEST	Analog_Data	Volts	-7.85		
E09_SMSUMV	SVITMX	SUM VOLTAGE MAX VALUE FOR INITIALIZATION TEST (+1 CNT)	Analog_Data	Volts	-2.34955		
E10_RAMTEST	CON252	Octal value 125252	Integer_Data		%0'125252'		
E10_RAMTEST	CON525	Octal value 52525	Integer_Data		%0'52525'		
E11_CONTEST	CMASK5	Mask to check Torque Trim Faults, in NCFAL2	Integer_Data		%0'000600'		
E11_CONTEST	CMASK6	Mask to check TQ & T1.8 Faults, in NCFAL1	Integer_Data		%0'30'		
E11_CONTEST	CMASK7	Mask to check PLA1 & PLA2 Faults, in NCFAL2	Integer_Data		%0'14'		
E11_CONTEST	CMASK8	Mask to isolate bits 2, 8, 9 of NCFAL2	Integer_Data		%0'000602'		
E11_CONTEST	LOC816	Mask to set J1 connector fault	Integer_Data		%0'100200'		
E14_FAULTSTORE_1	CPEENG	MASK BITS 16 & 12 (MAINT. FAULT WRITE/VERIFY FAULT)	Integer_Data		%0'104000'		
E14_FAULTSTORE_1	MXE2CT	MAXIMUM NUMBER OF FAULT TABLE ENTRIES	Integer_Data		24		
E14_FAULTSTORE_2	ETLIM	Max limit for ENGINE-TRIM Switch timer	Integer_Data		36		
E14_FAULTSTORE_2	IDLMAX	Max PLA value for IDLE setting	Analog_Data		35		
E14_FAULTSTORE_2	IDLMIN	Min PLA value for IDLE setting + 1 count	Analog_Data		10		
E14_FAULTSTORE_2	NHLM	High limit for NH	Analog_Data		10000.0		
E14_FAULTSTORE_3	BITS16	1 COUNT PER BIT	Integer_Data		16		
E14_FAULTSTORE_3	MAXEFZ	MAXIMUM NUMBER OF FAULTS ALLOWED	Integer_Data		8		
E14_FAULTSTORE_3	MXWDCT	MAXIMUM OFFSET INTO LRU-CODE TABLE	Integer_Data		96		
E14_FAULTSTORE_4	BLDMSB	MASK TO ISOLATE MOST SIGNIFICANT BYTE.	Integer_Data		%0'177400'		
E14_FAULTSTORE_4	BLDMSK	BITS 8 THRU 15 SET, REST CLEAR	Integer_Data		%0'077600'		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
E16_SMTEST	INFCNF	INTERMITTENT FAULT COUNTER THRESHOLD VALUE	Integer_Data		5		
E16_SMTEST	NOMS1	NOMINAL SM SUM VOLT.-TOLERANCE(NORMAL MODE)	Analog_Data		-7.1		
E16_SMTEST	NOMS2	NOMINAL SM SUM VOLT.+TOLERANCE(NORMAL MODE)+1	Analog_Data		-3.09967		
E16_SMTEST	NOMS3	NOMINAL SM SUM VOLT.+TOLER.(CONFIRMATION TEST)	Analog_Data		-2.35		
E16_SMTEST	NOMS4	NOMINAL SM SUM VOLT.-TOLER.(CONFIRMATION TEST)	Analog_Data		-7.85		
E16_SMTEST	SMB09	BIT 9 SET	Integer_Data		%O'400'		
E16_SMTEST	SMB16	BIT 16 SET	Integer_Data		%O'100000'		
E16_SMTEST	SMCFMX	MAX STEPPER MOTOR COIL FAULT COUNT	Integer_Data		24		
E16_SMTEST	STP98	COMMAND (+2) STEPS	Analog_Data		98.0		
E17_NCFLT_7	FLTMK1	Non Critical Failure mask table for NCFAL1	Integer_Data				
E17_NCFLT_7	FLTMK2	Non Critical Failure mask table for NCFAL2	Integer_Data				
E17_NCFLT_7	FLTMK3	Non Critical Failure mask table for NCFAL3	Integer_Data				
E17_NCFLT_7	FLTMK4	Non Critical Failure mask table for NCFAL4	Integer_Data				
E17_NCFLT_7	FLTMK5	Non Critical Failure mask table for NCFAL5	Integer_Data				
E17_NCFLT_7	FLTMK6	Non Critical Failure mask table for NCFAL6	Integer_Data				
E19_TRMSTORE_1	EWQBMK	MASK TO SET EEPROM FAILURE [BITS 15,16 OF EROMIF]	Integer_Data		%O'140000'		
E19_TRMSTORE_2	EWQGMK	MASK TO SET EEPROM FAILURE [BITS 15,16 OF EROMIF]	Integer_Data		%O'140000'		
E19_TRMSTORE_3	ARSLMK	MASK TO SET EEPROM FAILURE	Integer_Data		%O'100400'		
E19_TRMSTORE_4	EVFLMK	MASK TO SET EEPROM FAILURE	Integer_Data		%O'100200'		
E19_TRMSTORE_5	PWMLMK	MASK TO SET EEPROM FAILURE	Integer_Data		%O'102000'		
E20_TESTMODE	HS	Protocol HANDSHAKE Code	Integer_Data				
E20_TESTMODE	O21	Delta between Command Code values	Integer_Data				
E21_RECONFIG	DT_INCREASE	Time to increase for each invocation	Analog_Data	sec	0.0832		
E21_RECONFIG	KPL1RAT	Engine Rating Power Level 1	Integer_Data		0		
E21_RECONFIG	KPL2RAT	Engine Rating Power Level 2	Integer_Data		1		
E21_RECONFIG	KPL3RAT	Engine Rating Power Level 3	Integer_Data		3		
E21_RECONFIG	KVRATTM	Time to wait after rating inputs change before allowing a change to ENGRAT	Analog_Data	sec	0.49		
E21_RECONFIG	ZERO_OUTPUT	Timer value set to 0	Analog_Data	sec	0.0		
E22_GASEL	GAB016	MASK BIT 16	Integer_Data		%O'100000'		
E22_GASEL	GAB116	BITS 11 AND 16 SET	Integer_Data		%O'102000'		
E22_GASEL	GAB136	BITS 13 AND 16 SET	Integer_Data		%O'110000'		
E22_GASEL	GAB146	BITS 14 AND 16 SET	Integer_Data		%O'120000'		
E22_GASEL	GAB156	BITS 15 AND 16 SET	Integer_Data		%O'140000'		
E22_GASEL	GACNT1	0.02 MILLISECOND DELAY COUNTER	Integer_Data		10		
E22_GASEL	GACNT2	0.8 MILLISECOND DELAY COUNTER	Integer_Data		400		
E22_GASEL	GARMAX	MAXIMUM VALUE OF THE GATE ARRAY (+1 CNT)	Integer_Data		3		
E22_GASEL	GARMIN	MINIMUM VALUE OF THE GATE ARRAY	Integer_Data		0		
E22_GASEL	HBVMAX	MAXIMUM VALUE OF THE HBOV (+1 CNT)	Analog_Data		2.0000305		
E22_GASEL	HBVMIN	MINIMUM VALUE OF THE HBOV W/B	Analog_Data		1.4		
E22_GASEL	IHBVWA	CURRENT SENT THROUGH THE HBOV PWM FOR THE TYPE OF PWM	Integer_Data		%O'263'		
E23_PLATIRM	PLAWLO	MASK THE LSB OF PLA DATA	Integer_Data		%O'377'		
E25_FAULTDET	ECB16	BIT 16 SET	Integer_Data		%O'100000'		
E26_FAULTSTORE_1	CPEENG	Mask bits 16 & 13	Integer_Data		%O'110000'		
E26_FAULTSTORE_1	MXE2CT	Maximum number of fault table entries	Integer_Data		24		
E26_FAULTSTORE_2	BITS16	1 COUNT PER BIT	Integer_Data		16		
E26_FAULTSTORE_2	MAXE2F	MAXIMUM NUMBER OF FAULTS ALLOWED	Integer_Data		8		
E26_FAULTSTORE_2	MXWDCT	MAXIMUM OFFSET INTO LRU-CODE TABLE	Integer_Data		96		
E26_FAULTSTORE_3	IMSK10	Mask for BCD tens digit	Integer_Data		%Q'360'		
E26_FAULTSTORE_3	INIC3	3	Integer_Data		3		
E26_FAULTSTORE_3	NOLRU	Value used for unassigned LRU codes	Integer_Data		99		
E26_FAULTSTORE_4	BLDMSB	MASK TO ISOLATE MOST SIGNIFICANT BYTE.	Integer_Data		%Q'177400'		
E26_FAULTSTORE_4	BLDMSK	BITS 8 THRU 15 SET, REST CLEAR	Integer_Data		%Q'77600'		
E27_MFCID	KMFC16	BIT 16 MASK	Integer_Data		%O'100000'		
E27_MFCID	KMFCNH	NH THRESHOLD FOR VERIFICATION OF MFC ID	Analog_Data	RPM	18000		
E28_HBOVWA_1	E28C11	BIT 11 CLEAR MASK	Integer_Data		%O'175777'		
E28_HBOVWA_1	E28S11	BIT 11 SET MASK	Integer_Data		%O'2000'		
E28_HBOVWA_1	E28S16	BIT 16 SET MASK	Integer_Data		%O'100000'		
E28_HBOVWA_1	E28TRE	CONSTANT 3	Integer_Data		3		
E28_HBOVWA_1	HW1MXC	MAX TIME COUNTS TILL HBOV W/A FAILURE	Integer_Data	COUNTS	18		
E28_HBOVWA_1	KTRMX	TRANSIENT DETECTION TIMER THRESHOLD	Integer_Data		25		
E28_HBOVWA_2	E28003	CONSTANT 3	Integer_Data		3		
E28_HBOVWA_2	E28125	BIT 12 CLEAR MASK	Integer_Data		%O'173777'		
E28_HBOVWA_2	E28B12	BIT 12 SET MASK	Integer_Data		%O'4000'		
E28_HBOVWA_2	E28B16	BIT 16 SET MASK	Integer_Data		%O'100000'		
E28_HBOVWA_2	HW2MXC	MAX TIME COUNTS TILL HBOV ANALOG W/A FAILURE	Integer_Data	COUNTS	18		
E28_HBOVWA_2	IBLDSP	COMMANDED BLEED VALUE AT WHICH POINT ANALOG W/A TEST RUN	Analog_Data	mA	64.5		
E28_HBOVWA_2	KTRMX	TRANSIENT DETECTION TIMER THRESHOLD	Integer_Data	COUNTS	25		
E28_HBOVWA_2	RHBVMIN	MINIMUM BLEED W/A THRESHOLD	Analog_Data	VDC	4.5		
EESUM_16	ROMCNT	Loop count value for checksum calculation	Integer_Data		-1024		
EVFSEL	EVFB13	BIT 13 SET	Integer_Data		%O'10000'		
EVFSEL	EVFB14	BIT 14 SET	Integer_Data		%O'20000'		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
EVFSEL	EVFMN1	MIN. RANGE CHECK VALUE FOR EVFLG = 1	Analog_Data		4.0654		
EVFSEL	EVFMN2	MIN. RANGE CHECK VALUE FOR EVFLG = 2	Analog_Data		6.4566		
EVFSEL	EVFMN3	MIN. RANGE CHECK VALUE FOR EVFLG = 3	Analog_Data		5.2628		
EVFSEL	EVFMN4	MIN. RANGE CHECK VALUE FOR EVFLG = 4	Analog_Data		7.6608		
EVFSEL	EVFMX1	MAX. RANGE CHECK VALUE FOR EVFLG = 1	Analog_Data		4.5654		
EVFSEL	EVFMX2	MAX. RANGE CHECK VALUE FOR EVFLG = 2	Analog_Data		6.95662		
EVFSEL	EVFMX3	MAX. RANGE CHECK VALUE FOR EVFLG = 3	Analog_Data		5.76281		
EVFSEL	EVFMX4	MAX. RANGE CHECK VALUE FOR EVFLG = 4	Analog_Data		8.16082		
FAILED	CLRBTS	Mask to clear bits 1-4	Integer_Data		%O'177760'		
FAILED	FA3	Fault Counter Increment Value	Integer_Data		3		
FAILED	IFLB16	BIT 16 MASK	Integer_Data		%O'100000'		
FILCON	SSC1K	Positive Full Scale Value ('77777)	Integer_Data		%O'77777'		
FILCON	SSC2K	Octal 100000	Integer_Data		%O'100000'		
FILTER	FLTBI6	NEGATIVE 32768	Integer_Data		%O'100000'		
FILTER	FLTNF5	NEGATIVE 32767	Integer_Data		%O'100001'		
FILTER	FLTPF5	PLUS 32767	Integer_Data		%O'77777'		
FLTTIMER	FLTSEC_SCL_PV	PAST VALUE OF SCALED FLTSEC	Analog_Data				
FLTTIMER	FLT_CONFIG_P	Past value of config for flight timer logic	Integer_Data			0	3
FLTTIMER	KMS_ZERO	Zero flight time raw second counter value	Analog_Data		0.0		
FLTTIMER	SEC_RESCALE	Flight time second timer rescale value	Analog_Data		1.0		
FLTTIMER	SEC_ZERO	Zero flight time second timer value	Analog_Data		0.0		
FLTTIMER	TC_ZERO	Zero flight time course counter	Analog_Data		0.0		
I01_LLDRIFT_2	DFTIMN	Primary Range Check MIN Limit	Analog_Data	mV	0.126		
I01_LLDRIFT_2	DFTIMX	Primary Range Check MAX Limit	Analog_Data	mV	24.986		
I01_LLDRIFT_2	NMDRFI	Reference Drift Value	Analog_Data	mV	0		
I01_LLDRIFT_2	NMDRFT	Nominal Low Level Interface Drift Value	Analog_Data	mV	12.556		
I02_SGCOM_2	NMGAIN	Low Level Interface Gain Reference Value	Analog_Data		50.21		
I02_SGCOM_2	NMSGC	Nominal Low Level Interface Gain	Analog_Data		1		
I02_SGCOM_2	SGCB14	Primary Range Check Bit mask	Integer_Data		%O'20000'		
I02_SGCOM_2	SGCB15	End of conversion Bit mask	Integer_Data		%O'40000'		
I02_SGCOM_2	SGMN	Primary Range Check MIN Limit	Analog_Data		46.0499		
I02_SGCOM_2	SGMX	Primary Range Check MAX Limit	Analog_Data		79.601		
I03_RDCJT_2	CJNGLM	NEGATIVE LIMIT FOR 1 LEFT SHIFT	Integer_Data		%O'140001'		
I03_RDCJT_2	CJPSLM	POSITIVE LIMIT FOR 1 LEFT SHIFT	Integer_Data		%O'37777'		
I03_RDCJT_2	DPSIZ	NUMBER OF X POINTS FOR DELTA-P	Integer_Data		6		
I03_RDCJT_2	DPT1	Addr. of first table of DELTA-P compensation data	Analog_Data				
I03_RDCJT_2	POSIZ	NUMBER OF X POINTS FOR PSAMB	Integer_Data		11		
I03_RDCJT_2	POT1	Address of first table of PS compensation data	Analog_Data				
I03_RDCJT_2	SC1	X**3 Coefficient for Polynomial CJMVP	Analog_Data		-2.63932		
I03_RDCJT_2	SC2	X**2 Coefficient for Polynomial CJMVP	Analog_Data		0.407684		
I03_RDCJT_2	SC3	X**1 Coefficient for Polynomial CJMVP	Analog_Data		0.102961		
I03_RDCJT_2	SC4	X**0 Coefficient for Polynomial CJMVP	Analog_Data		1.699697		
I03_RDCJT_2	SC5	X**2 Coefficient for Polynomial CJPOLY	Analog_Data		0.00290577		
I03_RDCJT_2	SC6	X**1 Coefficient for Polynomial CJPOLY	Analog_Data		5.102899		
I03_RDCJT_2	SC7	X**0 Coefficient for Polynomial CJPOLY	Analog_Data		-48.50262		
I03_RDCJT_2	TYTAB	Temperature Breakpoint Table	Analog_Data				
I03_RDCJT_3	CJCMN	Compensated Range Check MIN limit	Analog_Data		-4.86		
I03_RDCJT_3	CJCMX	Compensated Range Check MAX limit + 1 Count	Analog_Data		58.511		
I03_RDCJT_3	CJCNMN	Nominal CJC Value	Analog_Data		25.971		
I03_RDCJT_3	CJIMN	Primary Range Check MIN limit	Analog_Data		0.136		
I03_RDCJT_3	CJIMX	Primary Range Check MAX limit + 1 Count	Analog_Data		79.5993		
I03_RDCJT_3	RDSB13	Mask to set bit 13	Integer_Data		%O'10000'		
I03_RDCJT_3	RDSB14	Mask to set bit 14	Integer_Data		%O'20000'		
I03_RDCJT_3	RDSB15	Mask to set bit 15	Integer_Data		%O'40000'		
I04_PSAMBSG_2	KTRMAX	KTRMX + 1 count	Integer_Data		52		
I04_PSAMBSG_2	KTRMX	Transient Detection timer max count	Integer_Data		51		
I04_PSAMBSG_2	KTWO	VALUE OF 2	Integer_Data		2		
I04_PSAMBSG_2	PSCLO	NACELLE STATIC PRESSURE CORRECTED LOW RANGE WITH SG I/F FAULT LATCHED.	Analog_Data		0.0		
I04_PSAMBSG_2	PSCMN	NACELLE static pressure corrected range check MIN.	Analog_Data	mV	-1.8804		
I04_PSAMBSG_2	PSCMX	NACELLE static pressure corrected range check maximum + 1 Count	Analog_Data	mV	60.004		
I04_PSAMBSG_2	PSIMN	NACELLE static pressure Primary Range Check MIN.	Analog_Data	mV	1.9709		
I04_PSAMBSG_2	PSIMX	NACELLE static pressure Primary Range Check MAX + 1 Count	Analog_Data	mV	79.601		
I04_PSAMBSG_2	RPSB13	BIT MASK BIT 13	Integer_Data		%O'10000'		
I04_PSAMBSG_2	RPSB14	BIT MASK BIT 14	Integer_Data		%O'20000'		
I04_PSAMBSG_2	RPSB15	BIT MASK BIT 15	Integer_Data		%O'40000'		
I04_PSAMBSG_2	RPSB16	BIT MASK BIT 16	Integer_Data		%O'100000'		
I04_PSAMBSG_3	POSIZE	Number of points in the PSAMB X-table	Integer_Data		11		
I04_PSAMBSG_3	PSDEF	Default PSAMB value, in case Comp. data is bad	Analog_Data		14.67		
I04_PSAMBSG_3	TP0A	POAMBF Filter Coefficient	Analog_Data		0.0340727		
I04_PSAMBSG_3	XMAX	Maximum pressure value	Analog_Data		16.5		
I04_PSAMBSG_3	XMIN	Minimum pressure value	Analog_Data				
I04_PSAMBSG_4	PST0A	Tables for PSCORR vs. PSAMBF map	Analog_Data				

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
I04_PSAMBSG_5	ALTT	POAMBF to ALTPS map	Analog_Data				
I05_DELTAPSG_2	DPSIZE	Number of points in the DELTA-P X-table	Integer_Data		6		
I05_DELTAPSG_2	XMAXD	LAST ADDRESS OF X-TABLE	Analog_Data				
I05_DELTAPSG_2	XMIND	FIRST ADDRESS OF X-TABLE	Analog_Data				
I05_DELTAPSG_3	DELRAH	DELTAP max increase rate	Analog_Data		0.832		
I05_DELTAPSG_3	DELRAL	DELTAP max decrease rate limit	Analog_Data		1.248		
I05_DELTAPSG_3	DPQA	QUAD. FILTER COEFF. A	Analog_Data		0.001091		
I05_DELTAPSG_3	DPQC	QUAD. FILTER COEFF. C	Analog_Data		1.4		
I05_DELTAPSG_3	DPQD	QUAD. FILTER COEFF. D	Analog_Data		0.064512		
I05_DELTAPSG_3	DPQF	QUAD. FILTER COEFF. F	Analog_Data		0.06764		
I05_DELTAPSG_3	TDPA	FILTER CONSTANT	Analog_Data		0.0669844		
I05_DELTAPSG_4	DLPB13	Bit 13	Integer_Data		%O'10000'		
I05_DELTAPSG_4	DLPB14	Bit 14	Integer_Data		%O'20000'		
I05_DELTAPSG_4	DLPB15	Bit 15	Integer_Data		%O'40000'		
I05_DELTAPSG_4	DLPCL0	Ram Pressure Compensated Low Range	Analog_Data		0.0		
I05_DELTAPSG_4	DLPCLMN	Compensated Range Check MIN limit - 1 Count	Analog_Data		-119.99634		
I05_DELTAPSG_4	DLPCLMX	Compensated Range Check MAX limit + 1 Count	Analog_Data		60.004		
I05_DELTAPSG_4	DLPIMN	Primary Range Check MIN limit - 1 Count	Analog_Data		-79.844		
I05_DELTAPSG_4	DLPIMX	Primary Range Check MAX limit + 1 Count	Analog_Data		79.601		
I05_DELTAPSG_4	NMDP1	Nominal delta P value #1	Analog_Data		0.936		
I05_DELTAPSG_4	NMDP2	Nominal delta P value #2	Analog_Data		0.612		
I05_DELTAPSG_4	NMDP3	Nominal delta P value #3	Analog_Data		0.372		
I07_READTQ_2	TQRATH	TQ positive rate limit	Analog_Data		0.47665		
I07_READTQ_2	TQRATL	Absolute TQ negative rate limit	Analog_Data		0.47665		
I07_READTQ_2	TTQLGA	TQC Filter Coefficient	Analog_Data		0.066984		
I07_READTQ_3	NMTQC	Nominal value for TQCI	Analog_Data		43.15		
I07_READTQ_3	RDTB13	BIT 13 SET	Integer_Data		%O'10000'		
I07_READTQ_3	RDTB14	BIT 14 SET	Integer_Data		%O'20000'		
I07_READTQ_3	RDTB15	BIT 15 SET	Integer_Data		%O'40000'		
I07_READTQ_3	TQCMN	Compensated Data Range Check MIN limit	Analog_Data		-4.83		
I07_READTQ_3	TQCMX	Compensated Data Range Check MAX limit + 1 Count	Analog_Data		57.7507		
I07_READTQ_3	TQCOR	TQ Interface Corrective Bias	Analog_Data		0.07407		
I07_READTQ_3	TQIMN	Primary Range Check MIN limit	Analog_Data		0.136		
I07_READTQ_3	TQIMX	Primary Range Check MAX limit + 1 Count	Analog_Data		79.599		
I08_RDT18_2	TCR1	TABLES FOR T1.8 TO T1.5 CONVERSION	Maps				
I08_RDT18_3	ADCSPP	GAS GENERATOR NOMINAL GROUND IDLE SPEED	Analog_Data		21978		
I08_RDT18_3	T18B13	BIT 13 MASK	Integer_Data		%O'10000'		
I08_RDT18_3	T18B14	BIT 14 MASK	Integer_Data		%O'20000'		
I08_RDT18_3	T18B15	BIT 15 MASK	Integer_Data		%O'40000'		
I08_RDT18_3	T18B16	BIT 16 MASK	Integer_Data		%O'100000'		
I08_RDT18_3	T18CMN	Compensated Range Check MIN limit	Analog_Data		-3.42		
I08_RDT18_3	T18CMX	Compensated Range Check MAX limit + 1 Count	Analog_Data		64.1203		
I08_RDT18_3	T18COR	Station 1.8 Temperature Interface Corrective Bias	Analog_Data		0.06408		
I08_RDT18_3	T18IMN	Primary Range Check MIN limit	Analog_Data		0.136		
I08_RDT18_3	T18IMX	Primary Range Check MAX limit + 1 Count	Analog_Data		79.6009		
I08_RDT18_3	T18P0	X**0 Coefficient for the T18PLY Polynomial	Analog_Data		-61.1059855		
I08_RDT18_3	T18P1	X**1 Coefficient for the T18PLY Polynomial	Analog_Data		3.89103341		
I08_RDT18_3	T18P2	X**2 Coefficient for the T18PLY Polynomial	Analog_Data		0.00184913		
I08_RDT18_3	T18PH5	HALF OF FULL SCALE	Analog_Data		169.989		
I08_RDT18_3	T18TAU	Filter Coefficient	Analog_Data		0.0539564		
I08_RDT18_3	TMPLCR	MASK TO CLEAR ALL BUT COUNTER	Integer_Data		12		
I08_RDT18_4	FCBIAS	BIAS USED IN DEG F TO DEG C CONVERSION	Analog_Data		32		
I08_RDT18_4	FCGAIN	GAIN USED IN DEG F TO DEG C CONVERSION	Analog_Data		0.55556		
I08_RDT18_4	T18NVF	NEGATIVE OVERFLOW LIMIT	Integer_Data		%O'160001'		
I08_RDT18_4	T18PVF	POSITIVE OVERFLOW LIMIT	Integer_Data		%O'17777'		
I08_RDT18_4	THETAK	DEG F TO DEG R conversion Bias	Analog_Data		459.67		
I09_HLDRIFT_2	DFCDVC	MASK TO CLEAR BIT 6	Integer_Data		%O'17773'		
I09_HLDRIFT_2	DFCEOC	BIT 15 MASK	Integer_Data		%O'40000'		
I09_HLDRIFT_2	DFCRNG	BIT 14 MASK	Integer_Data		%O'20000'		
I09_HLDRIFT_2	DRFCMN	Primary range check MIN limit	Analog_Data		0.49		
I09_HLDRIFT_2	DRFCMX	Primary range check MAX limit + 1 count	Analog_Data		0.55024		
I09_HLDRIFT_2	NMDRFA	Nominal High Level Interface Drift Value	Analog_Data		0		
I09_HLDRIFT_2	NMDRFC	High Level Interface Reference Drift Value	Analog_Data		0.5238		
I10_GAINCOM_2	GAINMN	Primary Range Check MIN Limit	Analog_Data		8.7464		
I10_GAINCOM_2	GAINMX	Primary Range Check MAX Limit + 1 count	Analog_Data		8.9503		
I10_GAINCOM_2	GANEOC	BIT 15 MASK	Integer_Data		%O'40000'		
I10_GAINCOM_2	GANGCV	MASK TO CLEAR BIT 7	Integer_Data		%O'17767'		
I10_GAINCOM_2	GANNRG	BIT 14 MASK	Integer_Data		%O'20000'		
I10_GAINCOM_2	NMGNC	Nominal High Level Interface Gain Factor	Analog_Data		1.0		
I10_GAINCOM_2	NMGPG	NOMINAL HIGH LEVEL GAIN EXCITATION	Analog_Data		8.8627		
I11_QBTRIM_2	EEBDIF	MAX ALLOWED DIFF. BETWEEN QBTIME AND QBTFM	Analog_Data		0.35039		
I11_QBTRIM_2	QBCL8R	MASK TO CLEAR LOW BYTE	Integer_Data		%O'177400'		
I11_QBTRIM_2	QBKM13	MASK BIT 13	Integer_Data		%O'10000'		
I11_QBTRIM_2	QBKM14	MASK BIT 14	Integer_Data		%O'20000'		
I11_QBTRIM_2	QBKM15	MASK BIT 15	Integer_Data		%O'40000'		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
I11_TQBTRIM_2	QBTBF	Trim scaling Bias Factor	Analog_Data		-599.41		
I11_TQBTRIM_2	QBTGF	Trim scaling gain Factor	Analog_Data		139.22		
I11_TQBTRIM_2	QBTMN	Primary Range Check MIN limit	Analog_Data		0.4216		
I11_TQBTRIM_2	QBTMX	Primary Range Check MAX limit + 1 Count	Analog_Data		8.37987		
I12_TQGTRIM_2	EEGDIF	MAX ALLOWABLE DIFFERENCE BETWEEN QGTME AND QGTMI	Analog_Data		0.35039		
I12_TQGTRIM_2	QGCLR8	MASK TO CLEAR LOW BYTE	Integer_Data		%O'177400'		
I12_TQGTRIM_2	QGMK13	MASK BIT 13	Integer_Data		%O'10000'		
I12_TQGTRIM_2	QGMK14	MASK BIT 14	Integer_Data		%O'20000'		
I12_TQGTRIM_2	QGMK15	MASK BIT 15	Integer_Data		%O'40000'		
I12_TQGTRIM_2	QGTBF	Trim scaling Bias	Analog_Data		0.942533		
I12_TQGTRIM_2	QGTGF	Trim scaling Gain	Analog_Data		0.013368		
I12_TQGTRIM_2	QGTMN	Primary Range Check MIN limit	Analog_Data		0.4216		
I12_TQGTRIM_2	QGTMX	Primary Range Check MAX limit + 1 Count	Analog_Data		8.37986		
I13_ENGVARIANT_2	EVFPMN	Minimum Range Check for Engine Validity Value From The Trim Plug	Analog_Data	Volts	2.415	-10.0	10.0
I13_ENGVARIANT_2	EVFPMX	Maximum Range Check For Engine Validity Selection From Trim Plug	Analog_Data	Volts	2.915	-10.0	10.0
I13_ENGVARIANT_2	KOCMNTOL	Minimum Open Circuit Range Check For Engine Validity Value From The Trim Plug	Analog_Data	Volts	-0.25	-10.0	0.0
I13_ENGVARIANT_2	KOCMXTOL	Maximum Open Circuit Range Check For Engine Validity Value From The Trim Plug	Analog_Data	Volts	0.25	0.0	10.0
I13_ENGVARIANT_2	KPL1RAT	Engine Rating Power Level 1	Integer_Data		0		
I13_ENGVARIANT_2	KPL2RAT	Engine Rating Power Level 2	Integer_Data		1		
I13_ENGVARIANT_2	KPL3RAT	Engine Rating Power Level 3	Integer_Data		3		
I14_RDPLA_1	NMPLA1	Nominal PLA1 Value	Analog_Data		1.550		
I14_RDPLA_1	PL1CBT	Mask to clear bits 1-4	Integer_Data		%O'177760'		
I14_RDPLA_1	PL1FA3	Fault Counter Increment Value	Integer_Data		%O'3'		
I14_RDPLA_1	PL1K12	MAX Limit for I/F Fault Counter	Integer_Data		12		
I14_RDPLA_1	PL1M16	Bit 16 Mask	Integer_Data		%O'100000'		
I14_RDPLA_1	PLA1CL	Mask to clear bits 8-15	Integer_Data		%O'100177'		
I14_RDPLA_1	PLA1MN	Primary Range Check MIN limit	Analog_Data		0.5250		
I14_RDPLA_1	PLA1MX	Primary Range Check MAX limit + 1 Count	Analog_Data		4.11253		
I14_RDPLA_1	PLTRAN	PLA Transient detection level	Analog_Data		1.7057		
I14_RDPLA_1	STATLM	Lightning Transient Limit	Integer_Data		3		
I14_RDPLA_2	KDRMS	Drift adjustment for PLA synthesis	Analog_Data		0.64124		
I14_RDPLA_2	KRMS	GAIN adjustment for PLA synthesis	Analog_Data		1.35875		
I14_RDPLA_2	NMPLA2	Nominal PLA2 Value	Analog_Data		3.190		
I14_RDPLA_2	PL2CBT	Mask to clear bits 1-4	Integer_Data		%O'177760'		
I14_RDPLA_2	PL2FA3	Fault Counter Increment Value	Integer_Data		%O'3'		
I14_RDPLA_2	PL2K12	MAX Limit for I/F Fault Counter	Integer_Data		12		
I14_RDPLA_2	PL2M16	Bit 16 Mask	Integer_Data		%O'100000'		
I14_RDPLA_2	PLA2CL	Mask to clear bits 8-15	Integer_Data		%O'100177'		
I14_RDPLA_2	PLA2MN	Primary Range Check MIN limit	Analog_Data		0.6275		
I14_RDPLA_2	PLA2MX	Primary Range Check MAX limit + 1 Count	Analog_Data		4.21505		
I14_RDPLA_2	PLTRAN	PLA Transient detection level	Analog_Data		1.7057		
I14_RDPLA_2	STATLM	Lightning Transient detection level	Integer_Data		3		
I14_RDPLA_3	BPLA	PLA Scaling Bias Angle	Analog_Data		50		
I14_RDPLA_3	PLAHVL	One-half PLA Hysteresis Bandwidth Value	Analog_Data		0.125		
I14_RDPLA_3	PLAMN	PLAC Range Check MIN limit	Analog_Data		-21.0		
I14_RDPLA_3	PLAMX	PLAC Range Check MAX limit + 1 Count	Analog_Data		120.004		
I14_RDPLA_3	PLARAH	Increasing PLA Rate limit	Analog_Data		4.16		
I14_RDPLA_3	PLARAL	Decreasing PLA Rate limit	Analog_Data		4.16		
I14_RDPLA_3	PLASC1	RVDT Angle Scaling Factor	Analog_Data		115.61		
I14_RDPLA_4	TPLLGA	First-Order PLA Lag Filter constant	Analog_Data		0.34032		
I15_PLABIAS_2	BINTCP	RVDT Interface Drift Scale Factor	Analog_Data		0.04646		
I15_PLABIAS_2	NMPLAB	Nominal Value for PLAB	Analog_Data		0.1550		
I15_PLABIAS_2	PLABMN	Primary Range Check MIN limit	Analog_Data		0.107		
I15_PLABIAS_2	PLABMX	Primary Range Check MAX limit + 1 Count	Analog_Data		0.20274		
I16_PLAGAIN_2	NMPLAG	Nominal RVDT Interface Gain Value	Analog_Data		3.488		
I16_PLAGAIN_2	PLAGMN	Primary Range Check MIN limit	Analog_Data		2.9486		
I16_PLAGAIN_2	PLAGMX	Primary Range Check MAX limit + 1 Count	Analog_Data		4.02694		
I17_PNHCVN_1	NPTEN2	NPT VALUE TO ENABLE NH LOW RANGE CHECK	Analog_Data		5999		
I17_PNHCVN_1	STNMK1	Mask to set bits 10 & 16	Integer_Data		%O'101000'		
I18_READNH	NHIMN	Scaled Speed Min Limit	Analog_Data		10000		
I18_READNH	NHIMX	Scaled Speed Max Limit + 1 Count	Analog_Data		36820.2		
I18_READNH	NHPMN	Period Counter Min limit	Analog_Data		25000		
I18_READNH	NHPMX	Period Counter Max Limit + 1 Count	Analog_Data		30002		
I18_READNH	NHSF	NH Speed Scale Factor	Integer_Data		15303		
I18_READNH	NHTMN	Tooth Counter Min Limit	Integer_Data		24		
I18_READNH	NHTMX	Tooth Counter Max Limit + 1 Count	Integer_Data		251		
I18_READNH	NHTRH	NH max increase rate limit(10000 RPM/SEC*0.0208 SEC)	Analog_Data		208		
I18_READNH	NHTRL	NH max decrease rate ABS(-10000 RPM/SEC * 0.0208 SEC)	Analog_Data		208		
I18_READNH	NPTGEN	NPT VALUE TO ENABLE NH LOW RANGE CHECK	Analog_Data		6000		
I18_READNH	NRATE	NH Transient rate threshold	Analog_Data		208		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
I18_READNH	RNH777	MASK TO ISOLATE TOOTH COUNTS	Integer_Data		%O'777'		
I18_READNH	RNHB12	BIT 12 SET, REST CLEAR	Integer_Data		%O'4000'		
I18_READNH	RNHB13	BIT 13 SET, REST CLEAR	Integer_Data		%O'10000'		
I18_READNH	RNHB14	BIT 14 SET, REST CLEAR	Integer_Data		%O'20000'		
I18_READNH	RNHB15	BIT 15 SET, REST CLEAR	Integer_Data		%O'40000'		
I18_READNH	RNHB16	BIT 16 SET, REST CLEAR	Integer_Data		%O'100000'		
I18_READNH	RNHCBT	Mask to clear bits 1-4	Integer_Data		%O'177760'		
I18_READNH	RNHFA3	Fault Counter Increment Value	Integer_Data		%O'3'		
I18_READNH	RNHK12	Max. limit for Fault Counter	Integer_Data		12		
I18_READNH	RNHM16	Bit 16 Mask	Integer_Data		%O'100000'		
I18_READNH	RNHPPF5	POSITIVE FULL SCALE	Integer_Data		%O'77777'		
I19_NHDUAL_1	NHTB10	BIT 10 SET	Integer_Data		%O'1000'		
I19_NHDUAL_1	NHTB16	BIT 16 SET	Integer_Data		%O'100000'		
I19_NHDUAL_2	NHDIF	NH CROSSCHECK DIFFERENCE LIMIT	Analog_Data		300		
I19_NHDUAL_2	NPTGEN	NPT VALUE TO ENABLE NH CROSSCHECK	Analog_Data		6000		
I19_NHDUAL_2	NXKB13	MASK BIT 13	Integer_Data		%O'10000'		
I20_NHSPEED	NHFCFA	Lag-Filter Coefficient	Analog_Data		0.564822		
I22_SNHCONV_1	SMSK1	MASK TO SET BITS 10 & 16	Integer_Data		%O'101000'		
I23_RDSNHP	NHIMN2	Scaled Speed Min Limit	Analog_Data		10000		
I23_RDSNHP	NHIMX2	Scaled Speed Max Limit + 1 Count	Analog_Data		36820.2		
I23_RDSNHP	NHPMN2	Period Counter Min limit	Analog_Data		3120		
I23_RDSNHP	NHPMX2	Period Counter Max Limit + 1 Count	Analog_Data		9361		
I23_RDSNHP	NHSF2	NH Speed Scale Factor	Integer_Data		15303		
I23_RDSNHP	NHTMN2	Tooth Counter Min Limit	Integer_Data		5		
I23_RDSNHP	NHTMX2	Tooth Counter Max Limit + 1 Count	Integer_Data		57		
I23_RDSNHP	NHTRH	NH max increase rate limit(10000 RPM/SEC*0.0208 SEC)	Analog_Data		208		
I23_RDSNHP	NHTRL	NH max decrease rate ABS(-10000 RPM/SEC * 0.0208 SEC)	Analog_Data		208		
I23_RDSNHP	NPTEN3	NPT VALUE TO ENABLE NH LOW RANGE CHECK	Analog_Data		6000		
I23_RDSNHP	NRATE	NH Transient rate threshold	Analog_Data		208		
I23_RDSNHP	R2H777	MASK TO ISOLATE TOOTH COUNTS	Integer_Data		%O'777'		
I23_RDSNHP	R2HB12	BIT 12 SET, REST CLEAR	Integer_Data		%O'4000'		
I23_RDSNHP	R2HB13	BIT 13 SET, REST CLEAR	Integer_Data		%O'10000'		
I23_RDSNHP	R2HB14	BIT 14 SET, REST CLEAR	Integer_Data		%O'20000'		
I23_RDSNHP	R2HB15	BIT 15 SET, REST CLEAR	Integer_Data		%O'40000'		
I23_RDSNHP	R2HB16	BIT 16 SET, REST CLEAR	Integer_Data		%O'100000'		
I23_RDSNHP	R2HCBT	Mask to clear bits 1-4	Integer_Data		%O'177760'		
I23_RDSNHP	R2HFA3	Fault Counter Increment Value	Integer_Data		%O'3'		
I23_RDSNHP	R2HK12	Max. limit for Fault Counter	Integer_Data		12		
I23_RDSNHP	R2HM16	Bit 16 Mask	Integer_Data		%O'100000'		
I23_RDSNHP	R2HPPF5	POSITIVE FULL SCALE	Integer_Data		%O'77777'		
I24_NHPROBE2	NHFCFA2	Lag-Filter Coefficient	Analog_Data		0.564822		
I25_TQNPT_2	NHQCVT	NH AT WHICH TORQUE INPUT CONVERSION IS VALID(+1CNT)	Analog_Data		24000.73242		
I25_TQNPT_2	NPTQSD	NPT threshold for Torque conversion shut down	Analog_Data		5000		
I25_TQNPT_2	QDELAY	TORQUE SYSTEM DELAY CONSTANT (-1CNT)	Integer_Data		961		
I25_TQNPT_2	RDQB15	MASK BIT 15	Integer_Data		%O'40000'		
I25_TQNPT_2	RDQBDA	MASK BIT 13	Integer_Data		%O'10000'		
I25_TQNPT_2	RDQBGE	MASK BIT 14	Integer_Data		%O'20000'		
I25_TQNPT_2	RDQC14	MASK TO CLEAR BITS 1 - 4	Integer_Data		%O'177760'		
I25_TQNPT_3	NHQCVT	NH AT WHICH TORQUE INPUT CONVERSION IS VALID(+1CNT)	Analog_Data		24000.73242		
I25_TQNPT_3	NPTQSD	NPT threshold for Torque conversion shut down	Analog_Data		5000		
I25_TQNPT_3	QDELAY	TORQUE SYSTEM DELAY CONSTANT (-1CNT)	Integer_Data		961		
I25_TQNPT_3	RDQB15	MASK BIT 15	Integer_Data		%O'40000'		
I25_TQNPT_3	RDQBDA	MASK BIT 13	Integer_Data		%O'10000'		
I25_TQNPT_3	RDQBGE	MASK BIT 14	Integer_Data		%O'20000'		
I25_TQNPT_3	RDQC14	MASK TO CLEAR BITS 1 - 4	Integer_Data		%O'177760'		
I25_TQNPT_4	DEFMAX	Deflection Angle MAX limit + 1 count	Analog_Data	DEG	11.5		
I25_TQNPT_4	DEFMIN	Deflection Angle MIN limit	Analog_Data	DEG	-1.0		
I25_TQNPT_4	K50PC	NPT above which Torque & NS trans.logic is enabled+1 cnt	Analog_Data	RPM	10000.97656		
I25_TQNPT_4	NPTQSD	NPT threshold for torque conversion shut down	Analog_Data	RPM	5000		
I25_TQNPT_4	NQSF	Shaft Speed Scale Factor	Analog_Data		60.0		
I25_TQNPT_4	NSHMAX	Shaft-Speed MAX limit (HIGH SPEED)	Analog_Data	RPM	5755.0		
I25_TQNPT_4	NSHMIN	Shaft-Speed MIN limit (HIGH SPEED)	Analog_Data	RPM	1400		
I25_TQNPT_4	NSLMAX	SHAFT SPEED MAX LIMIT (LOW SPEED)	Analog_Data	RPM	1600		
I25_TQNPT_4	NSLMIN	SHAFT SPEED MIN LIMIT (LOW SPEED)	Analog_Data	RPM	1023.12		
I25_TQNPT_4	NSRAH	Positive NS transient update rate limit	Analog_Data	RPM/SEC	132		
I25_TQNPT_4	NSRAL	Negative NS transient update rate limit	Analog_Data	RPM/SEC	132		
I25_TQNPT_4	NSRATE	NS transient rate threshold + 1 count	Analog_Data	RPM/SEC	132.08		
I25_TQNPT_4	QCVB10	BIT 10	Integer_Data		%O'1000'		
I25_TQNPT_4	QCVB15	BIT 15	Integer_Data		%O'40000'		
I25_TQNPT_4	QCVB16	MASK BIT 16	Integer_Data		%O'100000'		
I25_TQNPT_4	QCVFV	MAX. VALUE OF GOOD COUNTER	Integer_Data		5		
I25_TQNPT_4	QCVMSK	MASK BIT 13	Integer_Data		%O'010000'		
I25_TQNPT_4	QCVPF5	32767 COUNTS - POSITIVE FULL SCALE	Integer_Data		%O'077777'		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
I25_TQNPT_4	QCVPHS	POSITIVE HALF SCALE	Integer_Data		%O'037777'		
I25_TQNPT_4	QDEFS	Deflection Angle Scale Factor	Analog_Data	DEG/CT/CT	18		
I25_TQNPT_4	QOFFS	Torque shaft null deflection conversion bias	Analog_Data	DEG	-1.0		
I25_TQNPT_4	STATNS	NS lightning transient limit	Integer_Data		2		
I25_TQNPT_4	STATQ	Torque lightning transient limit	Integer_Data		2		
I25_TQNPT_5	DEFMAX	TORQUE SHAFT DEFLECTION HIGH RANGE LIMIT.	Analog_Data	DEG	11.5		
I25_TQNPT_5	DEFMIN	Deflection Angle MIN limit	Analog_Data	DEG	-1.0		
I25_TQNPT_5	NPTQSD	NPT threshold for torque conversion shut down	Analog_Data	RPM	5000		
I25_TQNPT_5	NQSF	Shaft Speed Scale Factor	Analog_Data		60.0		
I25_TQNPT_5	NSHMAX	Shaft-Speed MAX limit (HIGH SPEED)	Analog_Data	RPM	5755.0		
I25_TQNPT_5	NSHMIN	Shaft-Speed MIN limit (HIGH SPEED)	Analog_Data	RPM	1400.0		
I25_TQNPT_5	NSLMAX	SHAFT SPEED MAX LIMIT (LOW SPEED)	Analog_Data	RPM	1600		
I25_TQNPT_5	NSLMIN	SHAFT SPEED MIN LIMIT (LOW SPEED)	Analog_Data	RPM	1023.12		
I25_TQNPT_5	NSRAH	Positive NS transient update rate limit	Analog_Data		132		
I25_TQNPT_5	NSRAL	Negative NS transient update rate limit	Analog_Data		132		
I25_TQNPT_5	NSRATE	NS transient rate threshold + 1 count	Analog_Data		132.08		
I25_TQNPT_5	QCVB10	BIT 10	Integer_Data		%O'1000'		
I25_TQNPT_5	QCVB15	BIT 15	Integer_Data		%O'40000'		
I25_TQNPT_5	QCVB16	MASK BIT 16	Integer_Data		%O'100000'		
I25_TQNPT_5	QCVFIV	MAX. VALUE OF GOOD COUNTER	Integer_Data		%O'5'		
I25_TQNPT_5	QCVMSK	MASK BIT 13	Integer_Data		%O'010000'		
I25_TQNPT_5	QCVPF5	32767 COUNTS - POSITIVE FULL SCALE	Integer_Data		%O'077777'		
I25_TQNPT_5	QCVPHS	POSITIVE HALF SCALE	Integer_Data		%O'037777'		
I25_TQNPT_5	QDEFS	Deflection Angle Scale Factor	Analog_Data	DEG/CT/CT	18		
I25_TQNPT_5	QOFFS	Torque shaft null deflection conversion bias	Analog_Data	DEG	-1		
I25_TQNPT_5	STATNS	NS lightning transient limit	Integer_Data		2		
I25_TQNPT_5	STATQ	Torque lightning transient limit	Integer_Data		2		
I25_TQNPT_6	K75PC	NPT THRESHOLD (+1CNT)	Analog_Data		15040.039		
I25_TQNPT_6	QICB10	BIT 10 CLEAR MASK	Integer_Data		%O'176777'		
I25_TQNPT_6	QINB16	BIT 16 MASK	Integer_Data		%O'100000'		
I25_TQNPT_6	QINC89	BIT 9 CLEAR MASK	Integer_Data		%O'177377'		
I25_TQNPT_7	QRAH2	Pos. torque deflection transient rate limit: PW127	Analog_Data		0.230464		
I25_TQNPT_7	QRAL2	Neg. torque deflection transient rate limit: PW127	Analog_Data		0.230464		
I25_TQNPT_7	QRATEB	Torque lightning transient limit + 1 count: PW127	Analog_Data		0.6245		
I26_TQCAL	KNSNPX	Torque output gain	Analog_Data		1.0		
I26_TQCAL	KQSC10	Young's modulus bias	Analog_Data		2919.13		
I26_TQCAL	KQSC7	Shaft Temperature mv to deg f Gain	Analog_Data		5.236567		
I26_TQCAL	KQSC8	Shaft Temperature mv to deg f Bias	Analog_Data		-49.35028		
I26_TQCAL	KQSC9	Young's modulus gain	Analog_Data		-1.8136		
I27_TQSPEED	KNSTC1	Torque Shaft Speed Filter Constant	Analog_Data		0.61995		
I28_DISINW1	KTRMX	Transient detection timer threshold.	Integer_Data		-27		
I28_DISINW1	MASKA	INVERT BIT MASK	Integer_Data		%O'007047'		
I28_DISINW1	MASKC	CONFIRMED BIT MASK	Integer_Data		%O'070306'		
I28_DISINW1	MASKN	NON-CONFIRMED BIT MASK (ONES COMPLEMENT OF MASKC)	Integer_Data		%O'107471'		
I29_DISINW2	MSK2	Mask to invert all ACTIVE-LOW bits in Port #2	Integer_Data		%O'63475'		
I30_DISINFALT	MSK3	Mask to invert all ACTIVE-LOW bits in Port #3	Integer_Data		%O'300'		
I32_ARINCIN_1	ARCN13	MAX COUNTS FOR MOST COUNTER	Integer_Data		3		
I32_ARINCIN_1	ARIB11	MASK FOR BIT 11	Integer_Data		%O'2000'		
I32_ARINCIN_1	ARIB12	MASK FOR BIT 12	Integer_Data		%O'4000'		
I32_ARINCIN_1	ARIB14	MASK FOR BIT 14	Integer_Data		%O'20000'		
I32_ARINCIN_1	LBL134	ASC COMMAND/DELTA PLA LABEL	Integer_Data		%O'72'		
I32_ARINCIN_1	LBL203	PRESSURE ALTITUDE DATA LABEL	Integer_Data		%O'301'		
I32_ARINCIN_1	LBL206	INDICATED AIR SPEED DATA LABEL	Integer_Data		%O'141'		
I32_ARINCIN_1	LBL211	TOTAL AIR TEMPERATURE DATA LABEL	Integer_Data		%O'221'		
I32_ARINCIN_1	LBL271	PMP RATING DATA LABEL	Integer_Data		%O'235'		
I32_ARINCIN_2	ARIFLT	MASK FOR BITS 14 & 16	Integer_Data				
I32_ARINCIN_2	ARIMX	MAX LIMIT FOR ARINC INPUT MESSAGE TIMER	Integer_Data		60		
I32_ARINCIN_2	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000.0		
I33_ARITAT	A211MX	LACK OF LABEL 211 DATA TIME LIMIT	Integer_Data		60		
I33_ARITAT	ARITSM	ARINC STATUS MATRIX MASK	Integer_Data		%O'60000'		
I33_ARITAT	ARTM12	BIT 12 MASK	Integer_Data		%O'4000'		
I33_ARITAT	ARTM13	BIT 13 MASK	Integer_Data		%O'10000'		
I33_ARITAT	ARTM14	BIT 14 MASK	Integer_Data		%O'20000'		
I33_ARITAT	ARTM15	BIT 15 MASK	Integer_Data		%O'40000'		
I33_ARITAT	ARTM16	BIT 16 MASK	Integer_Data		%O'100000'		
I33_ARITAT	FCGAIN	CONVERSION FACTOR FROM DEG. K TO R	Analog_Data		1.8		
I33_ARITAT	FIVSEC	RANGE CHECK TIME LIMIT	Integer_Data		60		
I33_ARITAT	KELVIN	CONVERSION BIAS FROM DEG. C TO K	Analog_Data		273.15		
I33_ARITAT	MN211	ADC TOTAL AIR TEMP. LOW RANGE CHECK VALUE	Analog_Data		-50		
I33_ARITAT	MX211	ADC TOT. AIR TEMP. HIGH RANGE CHECK VALUE +1 COUNT	Analog_Data		99.0156		
I33_ARITAT	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
I33_ARITAT	POSLIM	POSTIVE FULL SCALE FOR LEFT SHIFT BY 1	Integer_Data		%O'37777'		
I33_ARITAT	POSLM	POSTIVE FULL SCALE FOR LEFT SHIFT BY 2	Integer_Data		%O'17777'		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
I33_ARITAT	TATIME	ADC TOTAL AIR TEMP FAULT TIME LIMIT	Integer_Data		60		
I33_ARITAT	THETAK	CONVERSION BIAS FROM DEG. R TO F	Analog_Data		459.67		
I33_ARITAT	TT1SEC	LACK OF LABEL 211 DATA TIME LIMIT	Integer_Data		12		
I34_TATSEL	KSOAK	MIN. TORQUE VALUE FOR VALID TAT CROSSCHECK FAULT	Analog_Data		0.9		
I34_TATSEL	MTTDGF	-30.2 DEGREES F	Analog_Data		-30.2		
I34_TATSEL	NHSTDN	GAS GEN/ENGINE SHUTDOWN THRESHOLD	Analog_Data		10000		
I34_TATSEL	NMTHTF	DEFAULT TEMP FOR THETA1	Analog_Data		59		
I34_TATSEL	T15PLA	PLA TRHESHOLD OF 40 DEGREES	Analog_Data		40		
I34_TATSEL	TATRTH	POSITIVE RATE LIMIT	Analog_Data		0.416		
I34_TATSEL	TSLFSN	FULL SCALE NEGATIVE	Integer_Data		%O'100001'		
I34_TATSEL	TSLFSP	FULL SCALE POSITIVE	Integer_Data		%O'77777'		
I34_TATSEL	TSLM16	BIT 16 MASK	Integer_Data		%O'100000'		
I34_TATSEL	TSNONE	FADER RATIO CALCULATION FACTOR	Analog_Data		1		
I34_TATSEL	TSLSIX	VALID/INVALID TEMPERATURE TRANSITION LIMIT	Integer_Data		6		
I34_TATSEL	TSLSPD	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
I34_TATSEL	TTTDGF	23.3 DEGREES F	Analog_Data		23.3		
I34_TATSEL	TXTIME	ADC, RTD CROSSCHECK TIMER LIMIT	Integer_Data		61		
I35_ARIPA	A203MX	LACK OF LABEL 203 DATA TIME LIMIT	Integer_Data		120		
I35_ARIPA	AL1SEC	LACK OF LABEL 203 DATA TIME LIMIT	Integer_Data		24		
I35_ARIPA	ALTC12	MASK TO CLEAR BIT 12	Integer_Data		%O'173777'		
I35_ARIPA	ALTMAX	ARINC PRESSURE ALT. HIGH RANGE CHECK VALUE	Analog_Data		32002		
I35_ARIPA	ALTMIN	ARINC PRESSURE ALTITUDE LOW RANGE CHECK VALUE	Analog_Data		-2000		
I35_ARIPA	AMTIME	ARINC PRESSURE ALTITUDE FAULT TIME LIMIT	Integer_Data		120		
I35_ARIPA	ARALSM	STATUS MATRIX MASK (BITS 14,15)	Integer_Data		%O'60000'		
I35_ARIPA	ARLM12	BIT 12 MASK	Integer_Data		%O'4000'		
I35_ARIPA	ARLM13	BIT 13 MASK	Integer_Data		%O'10000'		
I35_ARIPA	ARLM14	BIT 14 MASK	Integer_Data		%O'20000'		
I35_ARIPA	ARLM15	BIT 15 MASK	Integer_Data		%O'40000'		
I35_ARIPA	ARLM16	BIT 16 MASK	Integer_Data		%O'100000'		
I35_ARIPA	ARLNSH	NEGATIVE LIMIT FOR LEFT SHIFT 1	Integer_Data		140001		
I35_ARIPA	ARLP5H	POSITIVE LIMIT FOR LEFT SHIFT 1	Integer_Data		%O'37777'		
I35_ARIPA	FVESEC	RANGE CHECK TIME LIMIT	Integer_Data		120		
I35_ARIPA	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
I36_PAMSEL	ALTIME	PRESS. ALTITUDE CROSSCHECK TIMER LIMIT	Integer_Data		121		
I36_PAMSEL	ALTRTH	POSITIVE RATE LIMIT FOR PAMB	Analog_Data		0.00555		
I36_PAMSEL	ALRTRL	NEGATIVE RATE LIMIT FOR PAMB	Analog_Data		0.00277		
I36_PAMSEL	PFFFPP	DEFAULT AMB. PRESS. RATS=5	Analog_Data		5.454		
I36_PAMSEL	NHSTDN	GAS GEN/ENGINE SHUTDOWN THRESHOLD	Analog_Data		10000		
I36_PAMSEL	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
I36_PAMSEL	PAMM16	BIT 16 MASK	Integer_Data		%O'100000'		
I36_PAMSEL	PAMONE	FADER RATIO CALCULATION FACTOR	Analog_Data		1		
I36_PAMSEL	PAMSIX	VALID/INVALID ADC AMBIENT PRESSURE TRANSITION LIMIT	Integer_Data		6		
I36_PAMSEL	SEALEV	NOMINAL PRESSURE AT SEA LEVEL, WHEN RATESW=1,2,3,6,OR 7	Analog_Data		14.7		
I36_PAMSEL	TLIM	PLA PERMISSIVE FOR PRESS. SOURCE XCHK	Analog_Data		40		
I36_PAMSEL	TPEPSI	DEFAULT AMB. PRESS. RATESW=4	Analog_Data		10.11		
I37_ARIISA_1	A206MN	TIME LIMIT TO FLAG LACK OF LABEL 206 DATA	Integer_Data		12		
I37_ARIISA_1	A206MX	TIME LIMIT FOR LACK OF LABEL 206 DATA FAULT	Integer_Data		121		
I37_ARIISA_1	ARSB12	BIT 12	Integer_Data		%O'4000'		
I37_ARIISA_1	ARSB13	BIT 13	Integer_Data		%O'10000'		
I37_ARIISA_1	ARSB15	BIT 15	Integer_Data		%O'40000'		
I37_ARIISA_1	ARSB16	BIT 16	Integer_Data		%O'100000'		
I37_ARIISA_1	ARSMVD	MASK FOR VALID DATA STATUS MATRIX	Integer_Data		%O'60000'		
I37_ARIISA_1	FIVSCD	RANGE CHECK TIME LIMIT	Integer_Data		121		
I37_ARIISA_1	IASLIM	INVALID DATA TIME LIMIT	Integer_Data		121		
I37_ARIISA_1	ISN512	MIN LIMIT FOR LEFT SHIFT OF IAS206	Analog_Data		-511.97		
I37_ARIISA_1	ISP512	MAX LIMIT FOR LEFT SHIFT OF IAS206	Analog_Data		511.97		
I37_ARIISA_1	MN206	IAS LOW RANGE CHECK	Analog_Data		0		
I37_ARIISA_1	MX206	IAS HIGH RANGE CHECK	Analog_Data		350.015625		
I37_ARIISA_1	NHSTDN	ENGINE SHUTDOWN NH THRESHOLD	Analog_Data		10000		
I37_ARIISA_1	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
I37_ARIISA_2	B206MX	MAX. VALUE OF INTERMITTENT FAULT COUNTER	Integer_Data		6		
I37_ARIISA_2	IASATO	DEFAULT IAS, ATO MODES	Analog_Data		125.0		
I37_ARIISA_2	IASGA	DEFAULT IAS, GA MODES	Analog_Data		125.0		
I37_ARIISA_2	IASMCL	DEFAULT IAS, MCL MODES	Analog_Data		160.0		
I37_ARIISA_2	IASMCR	DEFAULT IAS, MCR MODE	Analog_Data		200.0		
I37_ARIISA_2	IASMCT	DEFAULT IAS, MCT MODES	Analog_Data		160.0		
I37_ARIISA_2	IASRTH	RATE LIMIT FOR POSITIVE IAS CHANGE	Analog_Data		0.416		
I37_ARIISA_2	IASRTL	RATE LIMIT FOR NEGATIVE IAS CHANGE	Analog_Data		0.624		
I37_ARIISA_2	IASTO	DEFAULT IAS, TO, MODES	Analog_Data		125.0		
I37_ARIISA_2	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
I38_SMCSV_1	INCNT1	0.02 millisecond delay count	Integer_Data		10		
I38_SMCSV_2	SMSVMN	SUM VOLTAGE MIN VALUE FOR NORMAL MODE	Analog_Data		-7.85		
I38_SMCSV_2	SMSVMX	SUM VOLTAGE MAX VALUE FOR NORMAL MODE	Analog_Data		-2.35		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
I38_SMCSV_2	SMVB09	BIT 9 SET	Integer_Data		%O'400'		
I38_SMCSV_2	SMVB14	BIT 14 SET	Integer_Data		%O'20000'		
I38_SMCSV_2	SMVB15	BIT 15 SET	Integer_Data		%O'40000'		
I38_SMCSV_2	SMVCBT	Mask to clear bits 1-4	Integer_Data		%O'177760'		
I38_SMCSV_2	SMVFA3	Fault Counter Increment Value	Integer_Data		%O'3'		
I38_SMCSV_2	SMVK12	MAX Limit for I/F Fault Counter	Integer_Data		12		
I38_SMCSV_2	SMVM16	Bit 16 Mask	Integer_Data		%O'100000'		
I43_TQSPEED_1	KQXMX	Torque dual input crosscheck tolerance	Analog_Data		0.2518		
I43_TQSPEED_1	QXCB16	Mask bit 16	Integer_Data		%O'100000'		
I43_TQSPEED_1	QXCEN	NPT value to enable SPEED/TORQUE crosscheck	Analog_Data		15040		
I43_TQSPEED_1	QXKB13	Mask bit 13	Integer_Data		%O'10000'		
I43_TQSPEED_1	QXMX	Torque dual input crosscheck Fault time threshold	Integer_Data		120		
I43_TQSPEED_2	KNPXMV	Torque dual input crosscheck Fault time threshold	Analog_Data		213		
I43_TQSPEED_2	NPXB13	Mask bit 13	Integer_Data		%O'10000'		
I43_TQSPEED_2	NPXB16	Mask bit 16	Integer_Data		%O'100000'		
I43_TQSPEED_2	NPXMV	Power Turbine dual input speed crosscheck tolerance	Integer_Data		361		
I43_TQSPEED_2	QXCEN	NPT value to enable SPEED/TORQUE crosscheck	Analog_Data		15040		
I43_TQSPEED_3	QNPB14	Mask bit 14	Integer_Data		%O'20000'		
I43_TQSPEED_3	QNP516	Mask to set bit 16	Integer_Data		%O'100000'		
I45_HBOVWA_2	I45B15	BIT 15 SET MASK	Integer_Data		%O'40000'		
I45_HBOVWA_2	PCLR15	BIT 15 CLEAR MASK	Integer_Data		%O'137777'		
INA	M777	Mask to isolate bits 1-9	Integer_Data		%O'777'		
INC_CTRS_2	ENG_RUN_CRS_A_SS	Snapshot of Total Accumulated Engine Run Time Coarse Count (All Engine Models)	Analog_Data	Hours		0	32767
INC_CTRS_2	ENG_RUN_FIN_A_SS	Snapshot of Total Accumulated Engine Run Time Fine Count (All Engine Models)	Analog_Data	Fractional Hours		0	1
INC_CTRS_3	ENGFLTM_CRS_A_SS	Snapshot Total Accumulated Flight Time Coarse Count (All Engine Models)	Analog_Data	Hours		0	32767
INC_CTRS_3	ENGFLTM_FIN_A_SS	Snapshot of Total Accumulated Flight Time Fine Count (All Engine Models)	Analog_Data	Fractional Hours		0	1
INC_CTRS_4	KPL1RAT	Engine Rating for Power Level 1	Integer_Data		0		
INC_CTRS_4	KPL2RAT	Engine Rating for Power Level 2	Integer_Data		1		
INC_CTRS_4	KPL3RAT	Engine Rating Power Level 3	Integer_Data		3		
INIFLT	IMSK10	Mask for BCD tens digit	Integer_Data		%O'360'		
INIFLT	INIC3	3	Integer_Data		3		
INIFLT	NOLRU	Value used for unassigned LRU codes	Integer_Data		99		
INSTST	INSC1	Octal 125252	Integer_Data		%O'125252'		
INSTST	INSC12	Octal 1777	Integer_Data		%O'1777'		
INSTST	INSC14	Octal 16162	Integer_Data		%O'16162'		
INSTST	INSC15	Octal 34345	Integer_Data		%O'34345'		
INSTST	INSC16	Octal 152524	Integer_Data		%O'152524'		
INSTST	INSC17	Octal 25253	Integer_Data		%O'25253'		
INSTST	INSC18	Octal 177776	Integer_Data		%O'177776'		
INSTST	INSC19	Octal 107777	Integer_Data		%O'107777'		
INSTST	INSC2	One	Integer_Data		1		
INSTST	INSC20	Octal 177525	Integer_Data		%O'177525'		
INSTST	INSC21	Octal 52437	Integer_Data		%O'52437'		
INSTST	INSC22	Octal 53777	Integer_Data		%O'53777'		
INSTST	INSC23	Octal 172525	Integer_Data		%O'172525'		
INSTST	INSC24	Octal 72525	Integer_Data		%O'72725'		
INSTST	INSC25	Octal 52531	Integer_Data		%O'52531'		
INSTST	INSC26	Octal 177652	Integer_Data		%O'177652'		
INSTST	INSC27	Octal 65252	Integer_Data		%O'65252'		
INSTST	INSC28	Octal 120000	Integer_Data		%O'120000'		
INSTST	INSC29	Octal 37776	Integer_Data		%O'37776'		
INSTST	INSC3	Two	Integer_Data		2		
INSTST	INSC30	Octal 100000	Integer_Data		%O'100000'		
INSTST	INSC4	Octal 125253	Integer_Data		%O'125253'		
INSTST	INSC5	Octal 52525	Integer_Data		%O'52525'		
INSTST	INSC6	Octal 52526	Integer_Data		%O'52526'		
INSTST	INSC7	Zero	Integer_Data		0		
INSTST	INSC8	Octal 177777	Integer_Data		%O'177777'		
INSTST	INSC9	Octal 152525	Integer_Data		%O'152525'		
INV134	IVB16	MASK WITH BIT 16 SET	Integer_Data		%O'100000'		
INV134	IVB1C	BIT 1 CLEAR	Integer_Data		%O'177776'		
INV134	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
INV134	PLTIME	DELTA PLA & ASC ENABLE DATA RANGE FAULT TIME LIMIT	Integer_Data		60		
INV271	IV3B16	MASK WITH BIT 16 SET	Integer_Data		%O'100000'		
INV271	IVB1CL	BIT 1 CLEAR	Integer_Data		%O'177776'		
INV271	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
INV271	RDTIME	RATED POWER DATA RANGE FAULT TIME LIMIT	Integer_Data		60		
INVFLX	IVFB16	MASK WITH BIT 16 SET	Integer_Data		%O'100000'		
INVFLX	NMFLEX	NOMINAL FLEX TEMPERATURE VALUE	Analog_Data		-76		
INVFLX	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
INVLX	THRFCN	FLEX TEMPERATURE INVALID FAULT TIME LIMIT	Integer_Data		60		
INVRAT	IVRB1	BIT 1 CLEAR	Integer_Data		%O'177776'		
INVRAT	IVRB16	MASK WITH BIT 16 SET	Integer_Data		%O'100000'		
INVRAT	NMFLX	NOMINAL FLEX TEMPERATURE VALUE	Analog_Data		-76		
INVRAT	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
INVRAT	RDTIME	RATED POWER DATA RANGE FAULT TIME LIMIT	Integer_Data		60		
KATHY	K1	ENGINE CONFIG 1	Integer_Data		%O'1'		
KATHY	K2	ENGINE CONFIG 2	Integer_Data		%O'2'		
KATHY	K3	ENGINE CONFIG 3	Integer_Data		%O'3'		
KATHY	K4	ENGINE CONFIG 4	Integer_Data		%O'4'		
KATHY	K41	ENGINE CONFIG 41	Integer_Data		%O'41'		
KATHY	K43	ENGINE CONFIG 43	Integer_Data		%O'43'		
LCF_CHECK	HPIMP	COMPONENT ORDER VALUE FOR HPIMP	Integer_Data		1		
LCF_CHECK	HPTD	COMPONENT ORDER VALUE FOR HPID	Integer_Data		2		
LCF_CHECK	HPTFCP	COMPONENT ORDER VALUE FOR HPTFCP	Integer_Data		3		
LCF_CHECK	HPTRCP	COMPONENT ORDER VALUE FOR HPTRCP	Integer_Data		4		
LCF_CHECK	ISS	COMPONENT ORDER VALUE FOR ISS	Integer_Data		5		
LCF_CHECK	LPIMP	COMPONENT ORDER VALUE FOR LPIMP	Integer_Data		0		
LCF_CHECK	LPTD	COMPONENT ORDER VALUE FOR LPTD	Integer_Data		6		
LCF_CHECK	PT1D	COMPONENT ORDER VALUE FOR PT1D	Integer_Data		7		
LCF_CHECK	PT2D	COMPONENT ORDER VALUE FOR PT2D	Integer_Data		8		
LCF_CYCLE_COUNT	IX	INDEX INTO CYCLE COUNT TABLE	Integer_Data			0	8
LDLG_SUBR	S4FNFG	Negative Full Scale Value	Integer_Data		%O'100001'		
LDLG_SUBR	S4FPOS	Positive Full Scale Value	Integer_Data		%O'77777'		
LDLG_SUBR	S4INIT	Initialization output value	Integer_Data		%O'100000'		
LOAD	SPUP1	Reserved Memory area Upper Address + 1	Integer_Data		%O'2000'		
LOOKUP10	X	Independent lookup value	Analog_Data				
LOOKUP10	X1	Independent lookup value 1	Analog_Data				
LOOKUP10	X10	Independent lookup value 10	Analog_Data				
LOOKUP10	X2	Independent lookup value 2	Analog_Data				
LOOKUP10	X3	Independent lookup value 3	Analog_Data				
LOOKUP10	X4	Independent lookup value 4	Analog_Data				
LOOKUP10	X5	Independent lookup value 5	Analog_Data				
LOOKUP10	X6	Independent lookup value 6	Analog_Data				
LOOKUP10	X7	Independent lookup value 7	Analog_Data				
LOOKUP10	X8	Independent lookup value 8	Analog_Data				
LOOKUP10	X9	Independent lookup value 9	Analog_Data				
LOOKUP10	Y	Resulting dependent value	Analog_Data				
LOOKUP10	Y1	Dependent lookup value 1	Analog_Data				
LOOKUP10	Y10	Dependent lookup value 10	Analog_Data				
LOOKUP10	Y2	Dependent lookup value 2	Analog_Data				
LOOKUP10	Y3	Dependent lookup value 3	Analog_Data				
LOOKUP10	Y4	Dependent lookup value 4	Analog_Data				
LOOKUP10	Y5	Dependent lookup value 5	Analog_Data				
LOOKUP10	Y6	Dependent lookup value 6	Analog_Data				
LOOKUP10	Y7	Dependent lookup value 7	Analog_Data				
LOOKUP10	Y8	Dependent lookup value 8	Analog_Data				
LOOKUP10	Y9	Dependent lookup value 9	Analog_Data				
LOOKUP13	X	Independent lookup value	Analog_Data				
LOOKUP13	X1	Independent lookup value 1	Analog_Data				
LOOKUP13	X10	Independent lookup value X10	Analog_Data				
LOOKUP13	X11	Independent lookup value X11	Analog_Data				
LOOKUP13	X12	Independent lookup value X12	Analog_Data				
LOOKUP13	X13	Independent lookup value X13	Analog_Data				
LOOKUP13	X2	Independent lookup value X2	Analog_Data				
LOOKUP13	X3	Independent lookup value X3	Analog_Data				
LOOKUP13	X4	Independent lookup value X4	Analog_Data				
LOOKUP13	X5	Independent lookup value X5	Analog_Data				
LOOKUP13	X6	Independent lookup value X6	Analog_Data				
LOOKUP13	X7	Independent lookup value X7	Analog_Data				
LOOKUP13	X8	Independent lookup value X8	Analog_Data				
LOOKUP13	X9	Independent lookup value X9	Analog_Data				
LOOKUP13	Y	Resulting dependent value	Analog_Data				
LOOKUP13	Y1	Dependent lookup value 1	Analog_Data				
LOOKUP13	Y10	Dependent lookup value 10	Analog_Data				
LOOKUP13	Y11	Dependent lookup value 11	Analog_Data				
LOOKUP13	Y12	Dependent lookup value 12	Analog_Data				
LOOKUP13	Y13	Dependent lookup value 13	Analog_Data				
LOOKUP13	Y2	Dependent lookup value 2	Analog_Data				
LOOKUP13	Y3	Dependent lookup value 3	Analog_Data				
LOOKUP13	Y4	Dependent lookup value 4	Analog_Data				
LOOKUP13	Y5	Dependent lookup value 5	Analog_Data				
LOOKUP13	Y6	Dependent lookup value 6	Analog_Data				
LOOKUP13	Y7	Dependent lookup value 7	Analog_Data				
LOOKUP13	Y8	Dependent lookup value 8	Analog_Data				

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
LOOKUP13	Y9	Dependent lookup value 9	Analog_Data				
MISSIONS	KMCLUB	MCL Detent Upper Boundary	Analog_Data	deg	85.0		
MISSIONS	KMXTOLVL	Derated Maximum Take-Off Rated Power	Analog_Data		0.95		
MISSIONS	KMXTOTIM	Time the Derated Maximum Take-Off Rated Power is held for Mission Determination	Analog_Data	seconds	5.0		
MISSIONS	KNTOUB	NTD Detent Upper Boundary	Analog_Data	deg	93.0		
MISSIONS	KPL1RAT	Engine Rating For Power Level 1	Integer_Data		0		
MISSIONS	KPL2RAT	Engine Rating For Power Level 2	Integer_Data		1		
NO134	A134MN		Integer_Data		6		
NO134	A134MX	LACK OF LABEL 134 DATA TIME LIMIT	Integer_Data		60		
NO134	N3B1	BIT 1 CLEAR	Integer_Data		%O'177776'		
NO134	N3B15	MASK WITH BIT 15 SET	Integer_Data		%O'40000'		
NO134	N3B16	MASK WITH BIT 16 SET	Integer_Data		%O'100000'		
NO134	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
NO211	E211MN	ONE SECOND GRACE PERIOD FOR LACK OF LABEL 211 DATA	Integer_Data		12		
NO211	E211MX	LACK OF LABEL 211 DATA TIME LIMIT	Integer_Data		60		
NO211	NFB1	BIT 1 CLEAR	Integer_Data		%O'177776'		
NO211	NFB14	MASK WITH BIT 14 SET	Integer_Data		%O'20000'		
NO211	NFB16	MASK WITH BIT 16 SET	Integer_Data		%O'100000'		
NO211	NMFLEX	NOMINAL FLEX TEMPERATURE VALUE	Analog_Data		-76		
NO211	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
NO271	A271MN	ONE SECOND GRACE PERIOD FOR LACK OF LABEL 271 DATA	Integer_Data		12		
NO271	A271MX	LACK OF LABEL 271 DATA TIME LIMIT	Integer_Data		60		
NO271	N3B14	MASK WITH BIT 14 SET	Integer_Data		%O'20000'		
NO271	N3B16	MASK WITH BIT 16 SET	Integer_Data		%O'100000'		
NO271	NB1CL	BIT 1 CLEAR	Integer_Data		%O'177776'		
NO271	NPTFLT	NOMINAL GROUND IDLE POWER TURBINE SHAFT SPEED	Analog_Data		12000		
O01_SMOUT	DT1	Module update interval	Analog_Data		0.0208		
O01_SMOUT	IDT1	1 / DT1	Analog_Data		48.07692		
O01_SMOUT	KPSTMN	LOW RANGE VALUE FOR STEPPER MOTOR Position Count	Integer_Data		-600		
O01_SMOUT	STPK1	Max no. steps allowed per program cycle	Integer_Data		10		
O02_HBOVOUT	KBLEED	Bleed PWM Interface Hardware Current Bias	Analog_Data		1.4382		
O03_DISOUT_A	KFALSE	FALSE	Boolean_Data		False		
O03_DISOUT_A	KMCLUB	MCL Detent Upper Boundary	Analog_Data	deg	85.0		
O03_DISOUT	KRCTO	Rating Code for Maximum Takeoff Power	Integer_Data		1		
O04_UARTOUT_DTS	U377	Mask to clear bits 9-16	Integer_Data		%O'377'		
O04_UARTOUT_DTS	URTB12	MASK UART RCVR BUFR EMPTY FLAG	Integer_Data		%O'4000'		
O04_UARTOUT_DTS	URTB15	XMTR NOT EMPTY FAULT FLAG	Integer_Data		%O'40000'		
O04_UARTOUT_DTS	URTCBT	MASK to clear bits 1-4	Integer_Data		%O'177760'		
O04_UARTOUT_DTS	URTCRL	MASK to clear bits 8-15	Integer_Data		%O'100177'		
O04_UARTOUT_DTS	URTF3	Fault Counter Increment Value	Integer_Data		%O'3'		
O04_UARTOUT_DTS	URTK12	MAX Limit for I/F Fault Counter	Integer_Data		12		
O04_UARTOUT_DTS	URTM16	Bit 16 MASK	Integer_Data		%O'100000'		
O04_UARTOUT_GDTS	U377	Mask to clear bits 9-16	Integer_Data		%O'377'		
O04_UARTOUT_GDTS	URTB12	MASK UART RCVR BUFR EMPTY FLAG	Integer_Data		%O'4000'		
O04_UARTOUT_GDTS	URTB15	XMTR NOT EMPTY FAULT FLAG	Integer_Data		%O'40000'		
O04_UARTOUT_GDTS	URTCBT	MASK to clear bits 1-4	Integer_Data		%O'177760'		
O04_UARTOUT_GDTS	URTCRL	MASK to clear bits 8-15	Integer_Data		%O'100177'		
O04_UARTOUT_GDTS	URTF3	Fault Counter Increment Value	Integer_Data		%O'3'		
O04_UARTOUT_GDTS	URTK12	MAX Limit for I/F Fault Counter	Integer_Data		12		
O04_UARTOUT_GDTS	URTM16	Bit 16 MASK	Integer_Data		%O'100000'		
O05_ARO133	ARPLA	Factor to rescale PLA from [128 DEG] to [180 DEG]	Analog_Data		1		
O05_ARO133	LBL133	Message LABEL and SDI for PLA	Integer_Data		%O'332'		
O05_ARO133	SSM133	Normal status matrix	Integer_Data		%O'60000'		
O06_ARO134	ARPLA	Factor to rescale PLAIN from [128 DEG] to [180 DEG]	Analog_Data		1		
O06_ARO134	LBL134	Message LABEL and SDI for PLAIN	Integer_Data		%O'72'		
O06_ARO134	SSM134	Normal status matrix	Integer_Data		%O'60000'		
O07_ARO203_ALT11	LBL203	Message LABEL and SDI for ALT, SDI (1,1) when ENGPOS is 1	Integer_Data		%O'1701'		
O07_ARO203_ALT11	LBL203NS	Message LABEL and SDI for ALT, SDI (0,0) when ENGPOS is 1	Integer_Data		%O'301'		
O07_ARO203_ALT11	SSM203	Normal status matrix	Integer_Data		%O'60000'		
O07_ARO203_ALTPS	LBL203	Message LABEL and SDI for ALTPS	Integer_Data		%O'701'		
O07_ARO203_ALTPS	SSM203	Normal status matrix	Integer_Data		%O'60000'		
O09_ARO211_TAT11	LBL211	Message LABEL and SDI for TAT211, SDI (1,1) when ENGPOS is 1	Integer_Data		%O'1621'		
O09_ARO211_TAT11	LBL211NS	Message LABEL and SDI for TAT211, SDI (0,0) when ENGPOS is 0	Integer_Data		%O'221'		
O09_ARO211_TAT11	RSC211	Rescale for TAT	Analog_Data		1		
O09_ARO211_TAT11	SSM211	Normal status matrix	Integer_Data		%O'60000'		
O10_ARO240	AFD10	Count at end of cycle	Integer_Data		10		
O10_ARO240	AFD9	Count at end of message frame	Integer_Data		9		
O10_ARO240	ELABL	Message Label ('240) and SDI field	Integer_Data		%O'5'		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
O10_ARO240	SSM240	Valid sign status matrix	Integer_Data		%O'60000'		
O10_ARO240	STOPM		Integer_Data		%O'100'		
O10_ARO240	STRTM		Integer_Data		%O'40'		
O11_ARO245	LBL245	Message LABEL and SDI for NH	Integer_Data		%O'245'		
O11_ARO245	RSCNH	Factor to rescale NH from [44000 RPM] to [256 %]	Analog_Data		0.003003		
O11_ARO245	SSM245	Normal status matrix	Integer_Data		%O'60000'		
O15_ARO341	LBL341	Message LABEL and SDI for QHPLCP	Integer_Data		%O'207'		
O15_ARO341	QHP256	Max QHPLCP Limit	Analog_Data		255.98		
O15_ARO341	QHPF15	FLACA1.15 - Arinc 341 Ouput Disabled	Integer_Data		%O'40000'		
O15_ARO341	RSCQH	Factor to rescale QHPLCP from [30000 FT_LBS] to [256 %]	Analog_Data		0.00952		
O15_ARO341	SSM341	Normal status matrix	Integer_Data		%O'60000'		
O16_ARO346	LBL346	Message LABEL and SDI for NPTOUT	Integer_Data		%O'147'		
O16_ARO346	RSCNP	Factor to rescale NPTOUT from [32000 RPM] to [256%]	Analog_Data		0.005		
O16_ARO346	SSM346	Normal status matrix	Integer_Data		%O'60000'		
O17_ARO350	AFD10	Count at end of cycle	Integer_Data		10		
O17_ARO350	AFD9	Count at end of message frame	Integer_Data		9		
O17_ARO350	ELABL	Message Label ('350) and SDI field	Integer_Data		%O'27'		
O17_ARO350	SSM350	Valid sign status matrix	Integer_Data		%O'60000'		
O17_ARO350	STOPM		Integer_Data		%O'100'		
O17_ARO350	STRTM		Integer_Data		%O'40'		
O18_ARO270	DSL270	Bit 11 of FLACA1 - Label 270 output disabled	Integer_Data		%O'2000'		
O18_ARO270	I270HK	Default value (M5 transmission word)	Integer_Data		0		
O18_ARO270	LBL270	Message Label and SDI	Integer_Data		%O'35'		
O18_ARO270	SSM270	Normal Status Matrix	Integer_Data		%O'60000'		
O19_ARO272_A	KPL2RAT	Engine Rating for Power Level 2	Integer_Data		1		
O19_ARO272_A	KPL3RAT	Engine Rating Power Level 3	Integer_Data		3		
O19_ARO272	LBL272	Message LABEL and SDI	Integer_Data		%O'135'		
O26_ARO340	CTB11	CTIF BIT 11	Integer_Data		%O'2000'		
O26_ARO340	LBL340	Message LABEL and SDI for QOUT	Integer_Data		%O'7'		
O26_ARO340	QTF13	FLACA1.13 - Arinc 340 Ouput Disabled	Integer_Data		%O'10000'		
O26_ARO340	RSCLOT	Factor to rescale QOUT from [30000 FT_LBS] to [256 %]	Analog_Data		0.0083084		
O26_ARO340	SSM340	Normal status matrix	Integer_Data		%O'60000'		
O27_ARO342_A	KMCRBTBI	MCR Torque Bug Identifier	Integer_Data		3		
O27_ARO342_A	KMTOTBI	MTO Torque Bug Identifier	Integer_Data		0		
O27_ARO342_A	KRSCLCQ	Rescales Torque Bug to 256%	Analog_Data		0.0083084		
O40_ARO06X	ARO_DATA_PTR	ARINC 06X LABELS DATA POINTER	Integer_Data				
O40_ARO06X	ARO_FAULTED	ARINC 06X LABELS FAULTED FLAG	Integer_Data				
O40_ARO06X	ARO_ITEM_MAX	ARINC 06X LABELS DATA ITEM MAXIMUM ADDRESS	Integer_Data				
O40_ARO06X	ARO_ITEM_PTR	ARINC 06X LABELS DATA ITEM POINTER	Integer_Data				
O40_ARO06X	ARO_LABEL_TYPE	ARINC LABEL TYPE FOR 06X LABELS	Integer_Data			0	3
O40_ARO06X	ARO_OUTPUT_TIMER	OUTPUT CYCLE TIMER FOR ARINC 06X LABELS	Integer_Data			0	6
O40_ARO06X	ARO_OUT_COUNTER	OUTPUT COUNTER FOR ARINC 06X LABELS	Integer_Data			0	3
O40_ARO06X	ARO_PLACE HOLDER	ARINC 06X LABELS TABLE POINTER PLACE HOLDER	Integer_Data				
O40_ARO06X	ARO_SEQUENCE_PTR	ARINC 06X LABELS SEQUENCE NUMBER POINTER	Integer_Data				
O40_ARO06X	ARO_SEQ_BUTTON	ARINC 06X LABELS SEQUENCE BUTTON FLAG	Integer_Data				
O40_ARO06X	ARO_TABLE_PTR	ARINC 06X LABELS TABLE POINTER	Integer_Data				
OPMODE_1	KADCFLTBW	Band Width For VK (ADC Supplied Input) Airspeed Selection	Analog_Data	Knots	10.0		
OPMODE_1	KDNH_ENG_OFF	Default NH Speed for Engine OFF detection	Analog_Data	RPM	6660.0	0.0	36820.0
OPMODE_1	KDNH_ENG_RUN	Default NH Speed for Engine OFF detection	Analog_Data	RPM	21000.0	0.0	36820.0
OPMODE_1	KDPFLTBW	Band Width For VK (EEC's deltaP Sensor Input) Airspeed Selection	Analog_Data	Knots	15.0		
OPMODE_1	KDVK_FLT_ADC	Default Airspeed from ADC for Ground/Flight Flag setting	Analog_Data	KNOTS	75.0	0.0	350.0
OPMODE_1	KDVK_FLT_DPSNR	Default Airspeed from ADC for Ground/Flight Flag setting	Analog_Data	KNOTS	127.5	0.0	350.0
OPMODE_1	KFLTITIM	Time the Indicated Air Speed Flight Flag is held for Flight Determination	Analog_Data	seconds	5.0		
OPMODE_1	KZERO	Zero seconds	Analog_Data	Seconds	16.0	0	16
OPMODE_3	KNTOTUB	NTO Detent Upper Boundary	Analog_Data	deg	93.0		
OPMODE_3	KRCMCL	Rating Code for Maximum Climb Power	Integer_Data		4		
OPMODE_3	KRCTO	Rating Code for Maximum Takeoff Power	Integer_Data		1		
OPMODE_3	KTOTIM	Time Value for Takeoff Determination	Analog_Data	seconds	5.0		
OTA	O777	Mask to isolate bits 1-9	Integer_Data				
PASSED	PCLALL	Mask to clear bits 8-15	Integer_Data		%O'100177'		
PID	ENGPID	ENGINE PROGRAM ID	Integer_Data		19		
PID	KSBLID	Software Major Version	Integer_Data		1		
PID	KSSVID	Software Minor Version	Integer_Data		1		
PID	PWMSI1	Software Interim build Version	Integer_Data		0		
PID	PWMSI2	Software Test Version	Integer_Data		0		
QUAD	QUADMF	(-32767) COUNTS	Integer_Data		%O'100001'		
QUAD	QUADPF	32767 COUNTS	Integer_Data		%O'77777'		
RAMREAD0	RAMDN2	Address of last RAM + 1	Integer_Data		%O'2000'		
RAMREAD1	RAMDN2	Address of last RAM + 1	Integer_Data		%O'4000'		
RAMREAD1	RAMST2	Address of first RAM	Integer_Data		%O'2000'		
RAMSTOR0	RAMDN1	Address of last RAM + 1	Integer_Data		%O'2000'		
RAMSTOR1	RAMDN1	Address of last RAM + 1	Integer_Data		%O'4000'		

Scope	Name	Description	Type	Units	Value	Min Val	Max Val
RAMSTOR1	RAMST1	Address of PAGE 1	Integer_Data		%O'2000'		
RAT271	MSKRT	USED TO MASK BITS 1,2,4-7 OF AR271H	Integer_Data		%O'173'		
ROMSUM	FLTROM	Address of fault EEPROM (Page 10)	Integer_Data		%O'20000'		
ROMSUM	MAXROM	Start address of boot memory	Integer_Data		%O'144000'		
ROMSUM	MINROM	Address of first loc. of available program memory	Integer_Data		%O'4000'		
ROMSUM	PG12AD	Address adjustment to skip fault EEPROM (Page 12)	Integer_Data		%O'4000'		
ROMSUM	ROMCNT	Loop count value for checksum calculation	Integer_Data		-1024		
ROMSUM	ROMMSK	Mask to clear bits 1-10	Integer_Data		%O'176000'		
RUNTIMER	ERNSEC_SCL_PV	Engine Run Timer, seconds past value	Analog_Data	seconds			
RUNTIMER	ERNSEC_STEMP	Engine Run Time seconds rescale temp value	Analog_Data				
RUNTIMER	K3600	3600 Seconds	Analog_Data	seconds	3600.0	0	3600
RUNTIMER	KHR1	One Hour	Analog_Data	hours	1.0	0	32768
RUNTIMER	KMS_INC	Raw seconds increment	Analog_Data	seconds	0.0208	0.0	681.5744
RUNTIMER	KMS_MAX	Raw seconds max value	Analog_Data	sec	12.9792	0.0	13.0
RUNTIMER	KMS_MIN	Raw seconds min value	Analog_Data	seconds	0.0	0.0	681.5744
RUNTIMER	KMS_ZERO	Zero engine run time raw second counter value	Analog_Data		0.0		
RUNTIMER	KSBINV	1/8 Second, increment	Analog_Data	seconds	0.125	0	1.0
RUNTIMER	KSEC1	One second	Analog_Data	none	1.0	0	32768
RUNTIMER	KSEC2HR	1/3600 seconds to hours conversion factor	Analog_Data		0.00027777778	0	0.00048828125
RUNTIMER	KSF1	Rescale factor from 681.5744 to 512	Analog_Data	none	1.0		
RUNTIMER	KSF2	Rescale factor from 32768 to (3600*16)	Analog_Data	none	1.0		
RUNTIMER	KZERO	Zero seconds	Analog_Data	seconds	0.0	0	4095
RUNTIMER	RUN_CONFIG_P	Past value of config for the engine run timer logic	Integer_Data			0	4
RUNTIMER	SEC_RESCALE	Engine run time second timer rescale value	Analog_Data		1.0		
RUNTIMER	SEC_ZERO	Zero engine runtime second timer value	Analog_Data		0.0		
RUNTIMER	TC_ZERO	Zero engine runtime course counter	Analog_Data		0.0		
TUSTIN_SUBR	FLTB16	NEGATIVE 32768	Integer_Data		%O'100000'		
TUSTIN_SUBR	FLTNFS	NEGATIVE 32767	Integer_Data		%O'100001'		
TUSTIN_SUBR	FLTPFS	PLUS 32767	Integer_Data		%O'77777'		
URTRED	M1777	Mask to isolate bits 1-10	Integer_Data		%O'1777'		
URTRED	M3777	Mask to isolate bits 1-11	Integer_Data		%O'3777'		
VAL134	VALB11	MASK WITH BIT 11 SET	Integer_Data		%O'2000'		
VAL134	VALB16	MASK WITH BIT 16 SET	Integer_Data		%O'100000'		
VAL211	ARTRF	THEREF MASK FOR AR211H	Integer_Data		%O'17700'		
VAL211	FCBIAS	CONVERSION BIAS FROM DEG C TO DEG F	Analog_Data		32		
VAL211	FCGAIN	CONVERSION GAIN FROM DEG C TO DEG F	Analog_Data		1.8		
VAL211	THRFMN	LOW RANGE OF THEREF	Analog_Data		-76		
VAL211	THRFMX	HIGH RANGE OF THEREF	Analog_Data		139		
VAL211	VALB1	BIT 1 MASK	Integer_Data		1		
VAL211	VALC14	BIT 14 CLEAR	Integer_Data		%O'157777'		
VAL211	VALS14	BIT 14 SET	Integer_Data		%O'20000'		
VAL271	VALB01	BIT 1 MASK	Integer_Data		1		
VAL271	VALC13	BIT 13 CLEAR	Integer_Data		%O'167777'		
VAL271	VALC14	BIT 14 CLEAR	Integer_Data		%O'157777'		
VAL271	VALS14	BIT 14 SET	Integer_Data		%O'20000'		
VARIABLE_LAG_TUSTIN	DT_VAL	MODULE ITERATION RATE	Analog_Data				
VARIABLE_LAG_TUSTIN	IN_VAL	VARIABLE_LAG_TUSTIN INPUT	Analog_Data				
VARIABLE_LAG_TUSTIN	LAG	FILTER TIME CONSTANT (SEC). IF THE VALUE OF THE TIME CONSTANT IS LESS THAN DT, FILTER PASSES THE INPUT VALUE UNCHANGED.	Analog_Data				
VARIABLE_LAG_TUSTIN	OUT_VAL	VARIABLE_LAG_TUSTIN OUTPUT	Analog_Data				
VARIABLE_LAG_TUSTIN	PAST_IN_VAL	VARIABLE_LAG_TUSTIN PAST VALUE	Analog_Data				
VARIABLE_LAG_TUSTIN	RESET	VARIABLE_LAG_TUSTIN RESET	Boolean_Data		False		
VARIABLE_LAG_TUSTIN	TAU	FILTER TIME CONSTANT (SEC). IF THE VALUE OF THE TIME CONSTANT IS LESS THAN DT, FILTER PASSES THE INPUT VALUE UNCHANGED.	Analog_Data				
VAR_TUSTIN_SUBR	DT_VAL	MODULE ITERATION RATE	Analog_Data				
VAR_TUSTIN_SUBR	IN_VAL	VARIABLE_LAG_TUSTIN INPUT	Analog_Data				
VAR_TUSTIN_SUBR	OUT_VAL	VARIABLE_LAG_TUSTIN OUTPUT	Analog_Data				
VAR_TUSTIN_SUBR	PAST_IN_VAL	VARIABLE_LAG_TUSTIN PAST VALUE	Analog_Data				
VAR_TUSTIN_SUBR	RESET	VARIABLE_LAG_TUSTIN RESET	Boolean_Data		False		
VAR_TUSTIN_SUBR	TAU	FILTER TIME CONSTANT (SEC). IF THE VALUE OF THE TIME CONSTANT IS LESS THAN DT, FILTER PASSES THE INPUT VALUE UNCHANGED.	Analog_Data				
WRITE	WRIT1	Address of the first EEPROM location	Integer_Data		%O'20000'		
WRITE	WRIT2	Address of the last EEPROM location + 1	Integer_Data		%O'40000'		
WRITE	WSPLOW	Address of first protected location in RAM	Analog_Data				
WRITE	WSPUP	Address of last location in RAM + 1	Integer_Data		%O'2000'		
WRTEEP	E2END	ADDRESS OF LAST EEPROM LOCATION - 1	Integer_Data		%O'22776'		
WRTEEP	E2STR	ADDRESS OF FIRST EEPROM LOCATION	Integer_Data		%O'20000'		

10.03 PROGRAM UNITS

10.03 Table 1: Program-Units

Unit	Description
ARCOU	ARINC Output Subroutine
BD134	DELTA PLA INTERMITTENT FAILURE
C01_START	START LOGIC
C02_PLATRIM_1	pla trim calculation
C02_PLATRIM_2	PLA Trim Calculation
C03_TLMRT_1	THERMALLY LIMITED RATED POWER ALGORITHM -1
C03_TLMRT_2	THERMALLY LIMITED RATED POWER ALGORITHM -2
C03_TLMRT_2_1	THERMALLY LIMITED RATED POWER ALGORITHM, PART II
C03_TLMRT_2_2	THERMALLY LIMITED RATED POWER ALGORITHM, PART II
C03_TLMRT_3	THERMALLY LIMITED RATED POWER ALGORITHM -3
C03_TLMRT_3_1	THERMALLY LIMITED RATED POWER ALGORITHM,PART III
C03_TLMRT_3_2	THERMALLY LIMITED RATED POWER ALGORITHM,PART III
C03_TLMRT_4	THERMALLY LIMITED RATED POWER ALGORITHM -4
C03_TLMRT_4_1	THERMALLY LIMITED RATED POWER ALGORITHM,PART IV
C03_TLMRT_4_2	THERMALLY LIMITED RATED POWER ALGORITHM,PART IV
C03_TLMRT_5	THERMALLY LIMITED RATED POWER ALGORITHM -5
C03_TLMRT_5_1	THERMALLY LIMITED RATED POWER ALGORITHM,PART V
C03_TLMRT_5_2	THERMALLY LIMITED RATED POWER ALGORITHM,PART V
C03_TLMRT_6	THERMALLY LIMITED RATED POWER ALGORITHM -6
C03_TLMRT_6_1	THERMALLY LIMITED RATED POWER ALGORITHM,PART VI
C03_TLMRT_6_2	THERMALLY LIMITED RATED POWER ALGORITHM,PART VI
C03_TLMRT_7	THERMALLY LIMITED RATED POWER ALGORITHM -7
C04_RPCAL_2_D	Rated power calculation
C05_TQBUG	Torque Bug Calculation
C0601003004	CALCULATE COMMANDED POWER
C06_HPREF	CALCULATE COMMANDED POWER
C06_HPREF_1	Rated Power Filtering
C0701003004	Overtorque Protection Logic
C0701006011	Overtorque Protection Logic
C07_OVRTQPRO	Overtorque Protection Logic
C08_SHPCL	CLOSED LOOP ON POWER.
C10_ATOIGN	Autoignition hysteresis trip point
C11_SHPREQ_1	Horsepower request calculation
C11_SHPREQ_2	Horsepower request calculation
C1202005006	NPT GOVERNING LOGIC
C12_NPTGOV_1	NPT GOVERNING LOGIC
C12_NPTGOV_2	NPT GOVERNING LOGIC
C13_NHACCELLIM	NH ACCELERATION LIMITING LOGIC
C14_NHLIM	NH LIMITING LOGIC
C15_NHDECELLIM	NH DECELERATION LIMITING LOGIC
C16_LOOPSEL	LOOP SELECTION LOGIC
C17_NHREQUEST_1	NH REQUEST CALCULATION
C17_NHREQUEST_2	NH REQUEST CALCULATION
C18_NHGOV_1	NH GOVERNING LOGIC
C18_NHGOV_2	NH GOVERNING LOGIC
C19_UNDEERSPEED	PROPELLER UNDEERSPEED LOGIC
C20_OVRSPEED	PROPELLER OVERSPEED LOGIC

Unit	Description
C2101005006	EEC MODE HANDLING BLEED VALVE LOGIC (SCRB)
C2102007008	EEC MODE HANDLING BLEED VALVE LOGIC
C21_HBOVEM_1	EEC MODE HANDLING BLEED VALVE LOGIC (SCRB)
C21_HBOVEM_2	EEC MODE HANDLING BLEED VALVE LOGIC
C2201000000	MANUAL BACKUP HANDLING BLEED VALVE LOGIC
C2202002002	MANUAL BACKUP HANDLING BLEED VALVE LOGIC
C22_HBOVMM_1	MANUAL BACKUP HANDLING BLEED VALVE LOGIC
C22_HBOVMM_2	MANUAL BACKUP HANDLING BLEED VALVE LOGIC
C22_HBOVMM_3	MANUAL BACKUP HANDLING BLEED VALVE LOGIC
C2401007003	ENGINE OVERTORQUE TEST PROCESS
C2401008003	ENGINE OVERTORQUE TEST PROCESS
C24_OVRTQTST	ENGINE OVERTORQUE TEST PROCESS
C2601003003	MISCELLANEOUS PARAMETER CALCULATION
C2602001001	MISCELLANEOUS PARAMETER CALCULATION
C2603003003	MISCELLANEOUS PARAMETER CALCULATION
C26_MPARAM_1	MISCELLANEOUS PARAMETER CALCULATION
C26_MPARAM_2	MISCELLANEOUS PARAMETER CALCULATION
C26_MPARAM_3	MISCELLANEOUS PARAMETER CALCULATION
C26_MPARAM_4	MISCELLANEOUS PARAMETER CALCULATION
C26_MPARAM_5	MISCELLANEOUS PARAMETER CALCULATION
C26_MPARAM_6	MISCELLANEOUS PARAMETER CALCULATION
C26_MPARAM_7	MISCELLANEOUS PARAMETER CALCULATION
C2701001001	NH MINIMUM / MAXIMUM CALCULATION
C2701006001	NH MINIMUM / MAXIMUM CALCULATION
C2701009009	NH MINIMUM / MAXIMUM CALCULATION
C27_NHNMNMX_1	NH MINIMUM / MAXIMUM CALCULATION
C3001002002	STEPPER MOTOR RATE REQUEST
C30_SMRTRQ	STEPPER MOTOR RATE REQUEST
C31_TQLIM	TORQUE LIMITING LOGIC
CHKSUM	Allocate Page Checksum Locations
CMPACT	COMPACT RAM FAULT MAINTENANCE BLOCK
CNT_EVNT	Calculate Accumulated Engine Run Time
DBL_LL_SUBR	Double Precision Lead/Lag Filter Subroutine
DCU_COMPARE_DATA	DCU compare data
DCU_COMPARE_ITEM	DCU compare routine
DCU_COMPONT_DATA	DCU process component data block
DCU_COUNTER_DATA	DCU process counter data block
DCU_DCUID_DATA	DCU process DCU ID data block
DCU_EEC_DATA	DCU EEC HW part number data block
DCU_ENGINE_DATA	DCU process engine data block
DCU_GET_CHECKSUM	DCU constant block checksum routine
DCU_GET_DATA	Get DCU data
DCU_IS_OBSOLETE	Obsolete DCU data determination routine
DCU_LCF_DATA	DCU process LCF data blocks
DCU_RANGE_CHECKS	DCU range check routine
DCU_READ	DCU read routine
DCU_RECEIVE	DCU receive routine

Unit	Description
DCU_SCHEDULE	DCU schedule routine
DCU_TRANSFER	DCU transfer routine
DCU_TRANSMIT	DCU transmit routine
DCU_WRITE	DCU write routine
DCU_WRITE_CPDAT	DCU write component data block
DCU_WRITE_CTRDAT	Write DCU counter data block
DCU_WRITE_EECDAT	Write DCU EEC HW part data block
DCU_WRITE_ENGDAT	DCU write engine data block
DERLAG	Derivative-Lag Filter Subroutine
DIVPRT	Protected Divide Subroutine
DPA	Double-Precision Addition Subroutine
DPM	Double Precision Multiply Subroutine
DUMP	DUMP-BLOCK Command Processor
E01_PORPSR	POR/PSR Initialization Routine
E02_RAMTEST	Complete RAM Test
E03_CHECKSUM	Complete Program EEPROM Checksum Test
E04_INSTST	Initialization CPU Tests
E05_RAMINI	Initialize all Non-Zero RAM Locations
E06_TQTEST	Torque Interface Test
E07_BG_EXECUTIVE	Background Processing Executive
E08_FG_EXECUTIVE	Foreground Processing Executive
E08_FG_SCHEDULE	Secondary Foreground Executive for Configurations 1, 2, & 3
E08_FG_SECSEL	Select Secondary schedule
E08_FG_STATETABLE	Foreground State Table of 1 ms. tasks
E09_SMSUMV	INITIAL STEPPER MOTOR SUM VOLTAGE CHECK
E10_RAMTEST	Continuous RAM and B-Register Shift Tests
E11_CONTEST	Loss-of-Connector Logic
E13_WATEST_1	HIGH SIDE SWITCH #1 W/A TEST LOGIC
E13_WATEST_2	HIGH SIDE SWITCH #2 W/A TEST LOGIC
E13_WATEST_3	LOW SIDE SWITCH #1 W/A TEST LOGIC
E13_WATEST_4	LOW SIDE SWITCH #2 W/A TEST LOGIC
E13_WATEST_5	LOW SIDE SWITCH #3 W/A TEST LOGIC
E13_WATEST_6	LOW SIDE SWITCH #4 W/A TEST LOGIC
E13_WATEST_7	LOW SIDE SWITCH #5 W/A TEST LOGIC
E14_FAULTSTORE_1	COPY RAM FAULT MAINTENANCE BLOCK TO EEPROM
E14_FAULTSTORE_2	CLEAR LOGIC FOR FAULT MAINTENANCE
E14_FAULTSTORE_3	ADD A NEW FAULT TO THE RAM FAULT BLOCK
E14_FAULTSTORE_4	BUILD RAM FAULT BLOCK
E14_FAULTSTORE_5	CLEAR TEMPORARY FAULT WORDS
E14_FAULTSTORE_6	DELETE LRU CODE FROM RAM FAULT BLOCK
E14_FAULTSTORE_8	Non-Critical fault LRU tables
E15_BCKGNDTEST	Background-Functional Check
E16_SMTEST	STEPPER MOTOR PHASE FAULT TEST
E17_NCFLT_1	Non-Critical Fault Summary Logic -- Part 1
E17_NCFLT_2	Non-Critical Fault Summary Logic -- Part 2
E17_NCFLT_3	Non-Critical Fault Summary Logic -- Part 3
E17_NCFLT_4	Non-Critical Fault Summary Logic -- Part 4

Unit	Description
E17_NCFLT_5	Non-Critical Fault Summary Logic -- Part 5
E17_NCFLT_6	Non-Critical Fault Summary Logic -- Part 6
E17_NCFLT_7	Built In Test Fault data Tables for E_17 process
E18_FAULTSUM_1	Fault Action Logic for FLACHW
E18_FAULTSUM_2	Fault Action Logic for FLACA1
E18_FAULTSUM_3	Fault Action Logic for FLACA2
E18_FAULTSUM_4	Fault Action Logic for FLACCW
E18_FAULTSUM_5	Fault Action Summary Tables #1
E18_FAULTSUM_6	Fault Action Summary Tables #2
E19_TRMSTORE_1	TORQUE BIAS TRIM DATA STORAGE LOGIC
E19_TRMSTORE_2	TORQUE GAIN TRIM DATA STORAGE LOGIC
E19_TRMSTORE_3	ENGINE VARIANT FROM ARINC STORAGE LOGIC
E19_TRMSTORE_4	ENGINE RATING FROM STORAGE LOGIC
E19_TRMSTORE_5	SELECTED PWM EEPROM DATA STORAGE LOGIC
E20_TESTMODE	Test Mode Executive Routine
E21_RECONFIG	Reconfiguration logic
E22_GASEL	GATE ARRAY SELECTION LOGIC
E23_PLATRIM	PLA TRIM EEPROM DATA STORAGE LOGIC
E25_FAULTDET	INTERNAL EEC FAULT DETECTION
E26_FAULTSTORE_1	COPY INTERNAL FAULT BLOCK FROM RAM TO EEPROM
E26_FAULTSTORE_2	INDICATE AN INTERNAL EEC FAULT OCCURED
E26_FAULTSTORE_3	INITIALIZE INTERNAL EEC FAULT VARIABLES
E26_FAULTSTORE_4	BUILD INTERNAL EEC FAULT BLOCK
E26_FAULTSTORE_5	Internal EEC fault tables
E27_MFCID	MFC IDENTIFICATION VERIFICATION
E28_HBOVWA_1	HBOV DISCRETE WRAPAROUND TEST
E28_HBOVWA_2	HBOV ANALOG WRAPAROUND TEST
E31_EXTDIS	Create External Discrete Word
EESUM_16	EEPROM Checksum Routine for program memory
EESUM_8	EEPROM Checksum Routine for 8 Bit memory
EE_GET_QUEUE	Get data from EEPROM queue
EE_PUT_QUEUE	Put data into EEPROM queue
EE_READ_DATA	EEPROM read data routine
EE_WRITE_BLOCK	EEPROM write block routine
EVFSEL	ENGINE VARIANT SELECTION LOGIC
FAILED	Increment an Interface fault Counter
FAILLT	Increment an Interface fault Counter with Lightning Delay
FILCON	Filter Constant Calculation Subroutine
FILTER	First-Order Lag Filter Subroutine
FLTTIMER	Flight time calculation
GETWRD	Read a 16-bit value from the UART
GOTO	GOTO Command Processor
I01_LLDRIFT_1	Read Low Level A/D Interface Drift
I01_LLDRIFT_2	Low Level Drift Compensation
I02_SGCOM_1	Read Low Level A/D Interface Gain
I02_SGCOM_2	Low Level A/D Interface Gain Input Processing
I03_RDCJT_1	Read Cold Junction Temperature

Unit	Description
I03_RDCJT_2	Scale the Cold Junction Temperature Reading
I03_RDCJT_3	Cold Junction Temperature Input Processing
I04_PSAMBSG_1	Engine Inlet Pressure Input
I04_PSAMBSG_2	Engine Inlet Pressure Input Processing
I04_PSAMBSG_3	Scale the Static Nacelle Ambient Pressure Reading
I04_PSAMBSG_4	Calculate Ambient Pressure
I04_PSAMBSG_5	Schedule altitude as a function of ambient pressure
I05_DELTAPSG_1	Read RAM Pressure
I05_DELTAPSG_2	Scale the Ram Pressure Reading
I05_DELTAPSG_3	Compensate & Filter the Ram Pressure Reading
I05_DELTAPSG_4	DELTA P Strain Gauge Logic
I07_READTQ_1	Read Torque Shaft Temperature
I07_READTQ_2	Filter/Rate limit the Torque Shaft Temperature Reading
I07_READTQ_3	Torque Shaft Temperature Input Processing
I08_RDT18_1	Read Station 1.8 Temperature
I08_RDT18_2	Ratio of Station 1.8 Temperature to Station 1.5 Temperature
I08_RDT18_3	Station 1.8 Temperature Input Processing
I08_RDT18_4	Calculate Station 1.5 Temperature from Station 1.8
I09_HLDRIFT_1	Read High Level Interface Drift
I09_HLDRIFT_2	High Level Interface Drift Input Processing
I10_GAINCOM_1	Read High Level Interface Gain
I10_GAINCOM_2	High Level Interface Gain Input Processing
I11_TQBTRIM_1	Read Torque Bias Trim
I11_TQBTRIM_2	Torque Bias Trim Input Processing
I12_TQGTRIM_1	Read Torque Gain Trim
I12_TQGTRIM_2	Torque Gain Trim Input Processing
I13_ENGVARIANT_1	READ A/D FOR ENGINE VARIANT
I13_ENGVARIANT_2	Engine variant fault logic
I14_RDPLA_1	PLA RVDT Signal #1 Input Processing
I14_RDPLA_2	PLA RVDT Signal #2 Input Processing
I14_RDPLA_3	Scale the PLA RVDT Reading
I14_RDPLA_4	Filter the Power Lever Angle Reading
I15_PLABIAS_1	Read RVDT Interface Bias
I15_PLABIAS_2	RVDT Interface Bias Input Processing
I16_PLAGAIN_1	Read RVDT Interface Gain
I16_PLAGAIN_2	RVDT Interface Gain Input Processing
I17_PNHCVN_1	Start Primary Probe NH Conversion
I17_PNHCVN_2	Stop the Gas Generator Speed (NH) Conversion
I18_READNH	Gas Generator Speed (NH) Input Logic
I19_NHDUAL_1	Dual NH Sensor Fault Summary Logic
I19_NHDUAL_2	NH PROBE CROSSCHECK LOGIC
I20_NHSPEED	Filter the Gas Generator Speed Reading
I22_SNHCVN_1	START GAS GENERATOR SPEED PROBE #2 CONVERSION
I22_SNHCVN_2	Stop the Gas Generator Speed (NH) Conversion
I23_RDSNHP	SECONDARY Gas Generator Speed (NH) Input Logic
I24_NHPROBE2	Filter the Probe Two Gas Generator Speed Reading
I25_TQNPT_1	CALCULATE TORQUE SHAFT DEFLECTION AND SPEED

Unit	Description
I25_TQNPT_2	Torque and Power Turbine Speed Input #1 Processing
I25_TQNPT_3	Torque and Power Turbine Speed Input #2 Processing
I25_TQNPT_4	Calculate Shaft Deflection and Speed input # 1 Subroutine
I25_TQNPT_5	Calculate Shaft Deflection and Speed input # 2 Subroutine
I25_TQNPT_6	TORQUE/SPEED ASSIGNMENT & PROBE SELECTION
I25_TQNPT_7	Torque Initialization
I26_TQCAL	Calculate Torque from shaft deflection
I27_TQSPEED	TORQUE SHAFT SPEED FILTERING
I28_DISINW1	Discrete Input Word #1 Logic
I29_DISINW2	Discrete Input Word #2 Logic
I30_DISINFALT	Fault Discrete Input Logic
I31_UARTIN	UART Input Control Logic
I31_UARTIN_NPM	UART N-PARAMETER TABLE INPUT LOGIC
I32_ARINCIN_1	ARINC 429 INPUT PROCESSING
I32_ARINCIN_2	ARINC 429 INPUT PROCESSING
I33_ARITAT	ARINC TOTAL AIR TEMPERATURE PROCESSING
I34_TATSEL	TOTAL AIR TEMPERATURE SOURCE SELECTION LOGIC
I35_ARIPA	ARINC PRESSURE ALTITUDE PROCESSING
I36_PAMSEL	AMBIENT PRESSURE SOURCE SELECTION LOGIC
I37_ARIISA_1	ARINC INDICATED AIR SPEED
I37_ARIISA_2	INDICATED AIR SPEED SELECTION LOGIC
I38_SMCSV_1	STEPPER MOTOR SUM VOLTAGE CONVERSION START
I38_SMCSV_2	STEPPER MOTOR SUM VOLTAGE INPUT
I43_TQSPEED_1	TORQUE DUAL INPUT CROSSCHECK LOGIC
I43_TQSPEED_2	POWER TURBINE DUAL INPUT SPEED CROSSCHECK LOGIC
I43_TQSPEED_3	DUAL TORQUE/SPEED INPUT FAULT LOGIC
I45_HBOVWA_1	Read Bleed PWM A/D Interface
I45_HBOVWA_2	BLEED PWM A/D INPUT PROCESSING
I46_IASFAULT	Indicated Airspeed Fault Logic
INA	READ-INPUT-PORT Command Processor
INC_COUNTERS	INCREMENT START/FLIGHT COUNTERS
INC_COUNTERS_2	INCREMENT ENGINE RUN AND FLIGHT COUNTERS
INC_CTRS_1	INCREMENT START/FLIGHT COUNTERS, PART 1
INC_CTRS_2	INCREMENT START/FLIGHT COUNTERS, PART 2
INC_CTRS_3	INCREMENT START/FLIGHT COUNTERS, PART 3
INC_CTRS_4	INCREMENT START/FLIGHT COUNTERS, PART 4
INIFLT	INITIALIZE FAULT VARIABLES AND TEMP FAULT WORDS
INI_DCU_DATA	Initialization of DCU data from EEPROM
INSTST	CPU Instruction Tests
INV134	INVALID LABEL 134 DATA RECEIVED
INV271	INVALID LABEL 271 DATA RECEIVED
INVFLX	INVALID FLEX TEMPERATURE DATA RECEIVED
INVRAT	INVALID RATING DATA RECEIVED
LCF_CHECK	Performs LCF logic
LCF_CYCLE_COUNT	Calculates LCF Cycle Count
LCF_TABLE	LCF constant address tables
LDLG_SUBR	Lead/Lag Filter Subroutine

Unit	Description
LOAD	LOAD-BLOCK Command Processor
LOOKUP10	
LOOKUP13	
MISSIONS	Mission Determination Logic
MISSION_N	PW127N Mission Determination Logic
NO134	NO LABEL 134 DATA RECEIVED
NO211	NO LABEL 211 DATA RECEIVED
NO271	NO LABEL 271 DATA RECEIVED
O01_SMOUT	STEPPER MOTOR LOGIC
O02_HBOVOUT	Calculate and output Bleed Valve current
O03_DISOUT	Discrete Output Logic
O03_DISOUT_A	Determine LSS#2 and LSS#3 discrete outputs
O04_UARTOUTSEL	UART OUTPUT SELECTION LOGIC
O04_UARTOUT_DTS	UART Output Control Logic
O04_UARTOUT_GDTS	UART Output Control Logic
O04_UARTOUT_NPM	UART N-PARAMETER TABLE OUTPUT LOGIC
O05_ARO133	Transmit Label 133 (PLA)
O06_ARO134	Transmit Label 134 (PLAIN)
O07_ARO203_ALT11	Transmit Label 203 (ALT)
O07_ARO203_ALTPS	Transmit Label 203 (ALTPS)
O09_ARO211_TAT11	Transmit Label 211 (TAT)
O10_ARO240	Transmit Label 240 (LRU fault codes)
O11_ARO245	Transmit Label 245 (NH)
O15_ARO341	Transmit Label 341 (QHPLCP)
O16_ARO346	Transmit Label 346 (NPTOUT)
O17_ARO350	Transmit Label 350 (LRU fault codes)
O18_ARO270	Transmit Label 270
O19_ARO272	Transmit Label 272
O19_ARO272_A	Determine ENGRAT_BIT1 and ENGRAT_BIT1 discretes
O26_ARO340	Transmit Label 340 (QOUT)
O27_ARO342_A	Sequence Torque Bug calculation
O40_ARO06X	ARINC 06X LABELS OUTPUT ROUTINE
OPMODE_1	Aircraft Operating Mode Determination
OPMODE_2	Aircraft Flight Cycle Determination
OPMODE_3	Takeoff Determination
OTA	WRITE-TO-OUTPUT-PORT Command Processor
PAGE0	Ram Page #1
PAGE1	Ram Page #2
PAGE10	EEPROM Storage Page #1
PAGE11	EEPROM Storage Page #2
PASSED	Decrement Interface Fault Counter
PASSLT	Decrement Interface Fault Counter with Lightning Delay
PID	Program ID
QUAD	QUADRATIC FILTER SUBROUTINE
RAMREAD0	Page 0 RAM Pattern Verification Subroutine
RAMREAD1	Page 1 RAM Pattern Verification Subroutine
RAMSTOR0	Page 0 RAM Pattern Storage Subroutine

Unit	Description
RAMSTOR1	Page 1 RAM Pattern Storage Subroutine
RAT211	RATING MODE SELECTION LOGIC
RAT271	RATING MODE SELECTION LOGIC
RDTWRITE	RTD EEPROM Table Write Routine
READ	READ-WORD Command Processor
ROMSUM	Page Checksum Test Subroutine
RTDWRITE	RTD EEPROM Table Write Routine
RUNTIMER	Run Time Calculation
SBIVAR	BIVARIATE MAP LOOKUP ROUTINE
SCHEDULER	DUMMY SCHEDULER for MODEL
SNDWRD	Write a 16-bit Value to the UART
SUNVAR	Univariate Table Look-up Subroutine
TUSTIN_SUBR	First-Order Tustin Lag Filter Subroutine
UNIVAR	UNIVARIATE MAP LOOKUP
UPDATE_EE_DATA	Update DCU counter data in EEPROM
URTOUT	Write a byte to the UART
URTRED	Read a byte from the UART
VAL134	VALID LABEL 134 DATA RECEIVED
VAL211	VALID LABEL 211 DATA RECEIVED
VAL271	VALID LABEL 271 DATA RECEIVED
VARIABLE_LAG_TUSTIN	Variable Lag Tustin
VAR_TUSTIN_SUBR	First-Order Tustin Lag Filter Subroutine with variable TAU
WDGRST	Watchdog Timer Reset Subroutine
WRITE	WRITE-WORD Command Processor
WRTEEP	WRITE-WORD or WRITE-BYTE to EEPROM Command Processor

Please refer the SBR-B for the modules that are modified, created, or obsoleted since the baseline software version 127S0000.

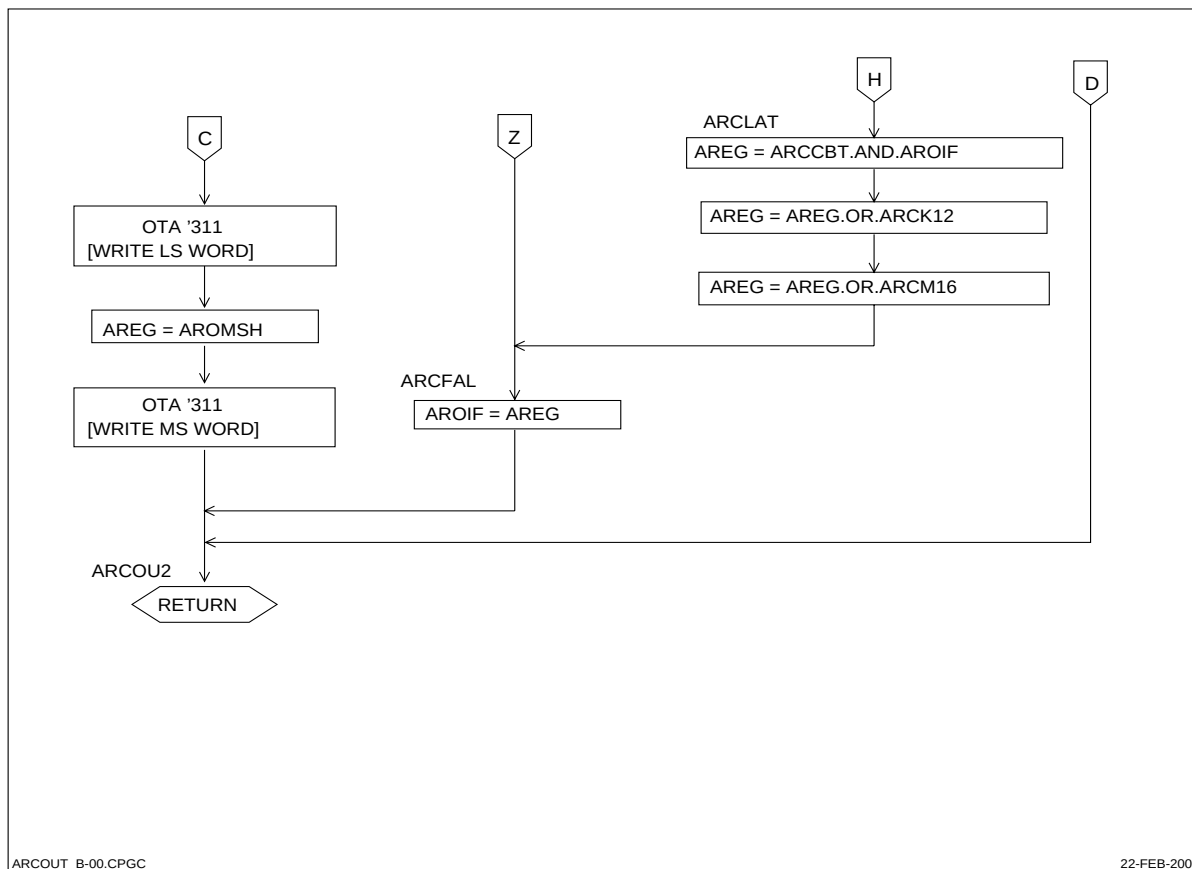
13.00 DESIGN MODELS (GPU DIAGRAMS)

```

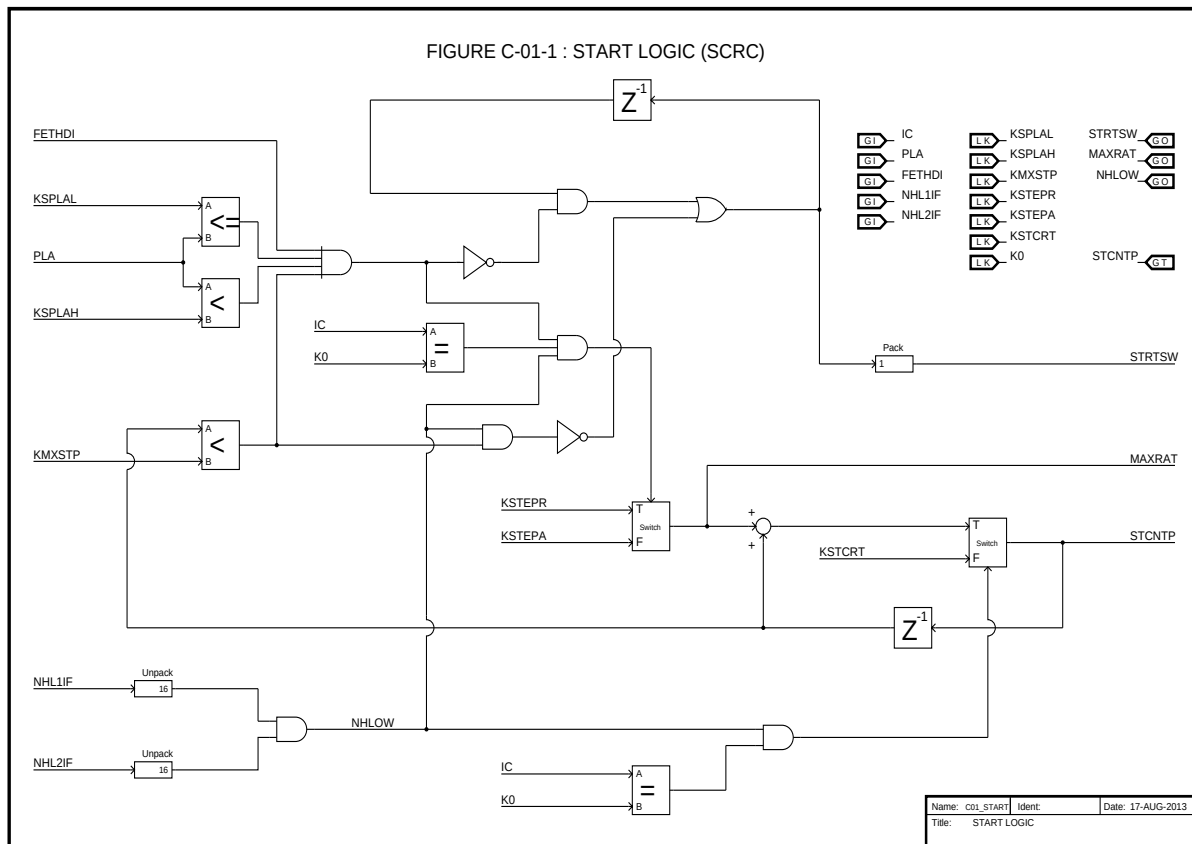
graph TD
    Start([ENTER]) --> PSBRTN[PSBRTN = SUBROUTINE RETURN ADDRESS]
    PSBRTN --> Bit16{BIT 16 OF AROIF CLEAR}
    Bit16 -- N --> ExitD([D])
    Bit16 -- Y --> INA[INA '314  
[READ ARINC STATUS]]
    INA --> Bit13{BIT 13 OF AREG SET}
    Bit13 -- N --> ARCOU1[ARCOU1  
SET BIT 15 OF AROIF]
    ARCOU1 --> AREG1[AREG = AROIF.AND.'17]
    AREG1 --> AREG2[AREG = AREG + ARCFA3]
    AREG2 --> AREG_LT{AREG.LT.  
ARCK12}
    AREG_LT -- N --> ExitH([H])
    AREG_LT -- Y --> AREG_ADD[AREG = AROIF + ARCFA3]
    AREG_ADD --> ExitZ([Z])
    Bit13 -- Y --> AREG_AND17[AREG = AROIF.AND.'17]
    AREG_AND17 --> AREG_EQ0{AREG  
EQ.0}
    AREG_EQ0 -- N --> ARCD[ARCD  
AREG=AROIF-1]
    ARCD --> AREG_AND17
    AREG_EQ0 -- Y --> AREG_AND_ARCCLR[AREG = AROIF.AND.ARCCLR]
    AREG_AND_ARCCLR --> ARCU1
    AREG_AND_ARCCLR --> ARCU2[ARCU2  
AROIF = AREG]
    ARCU2 --> ARCU3[ARCU3  
AREG = AROLSH]
    ARCU3 --> ExitC([C])

```

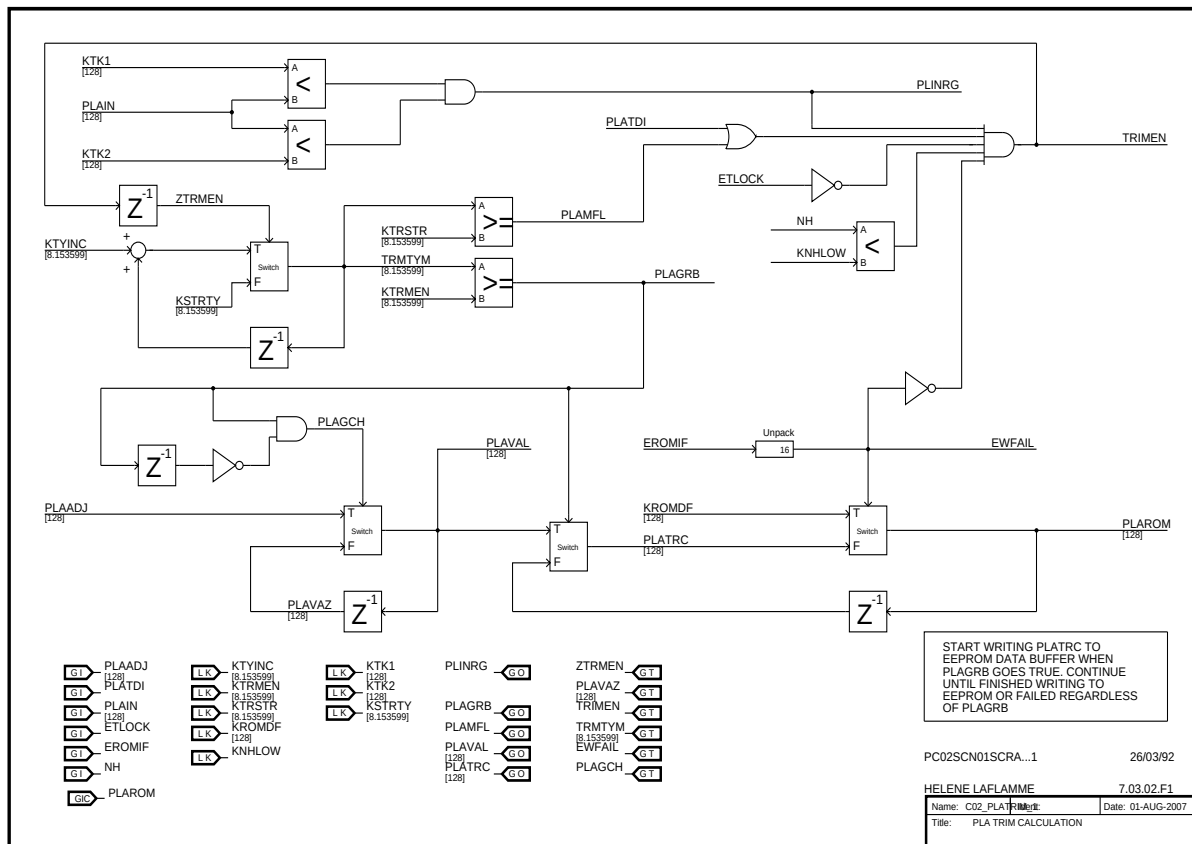
13.00 Figure 1: ARCOUT _B



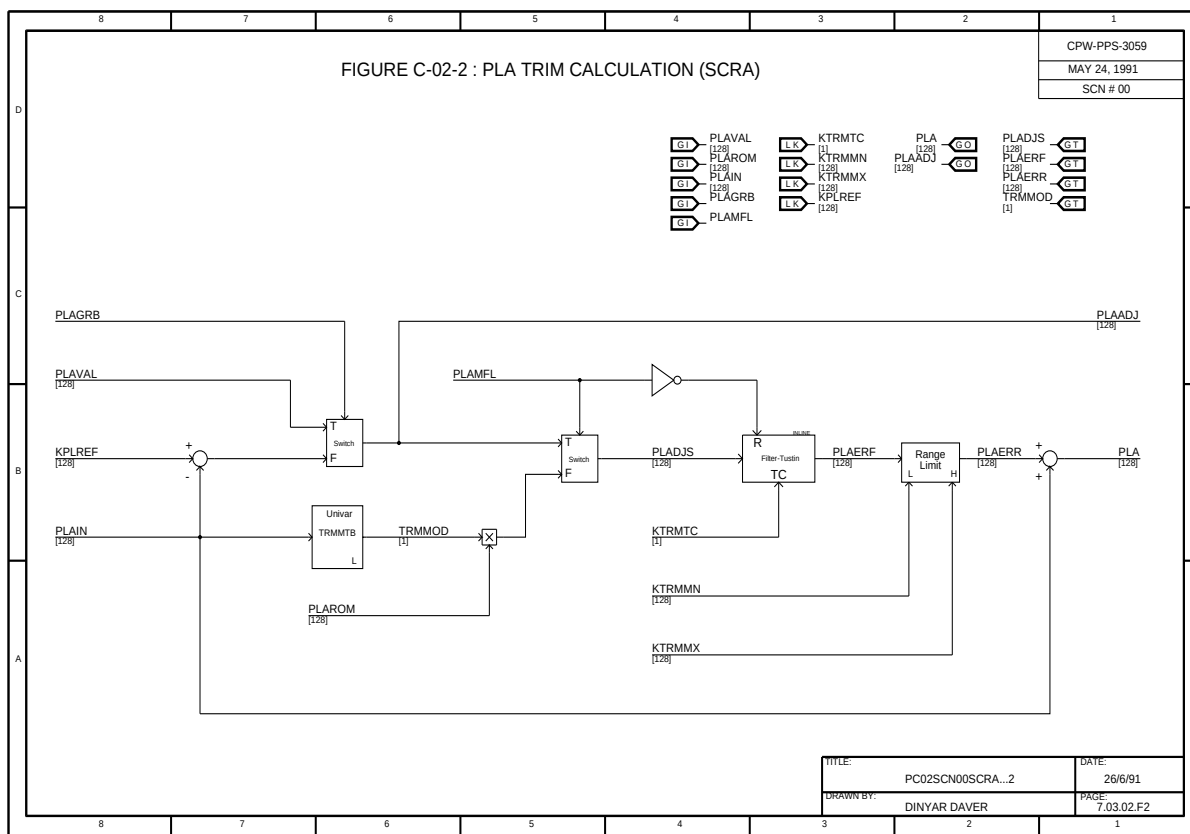
13.00 Figure 2: C01 _START



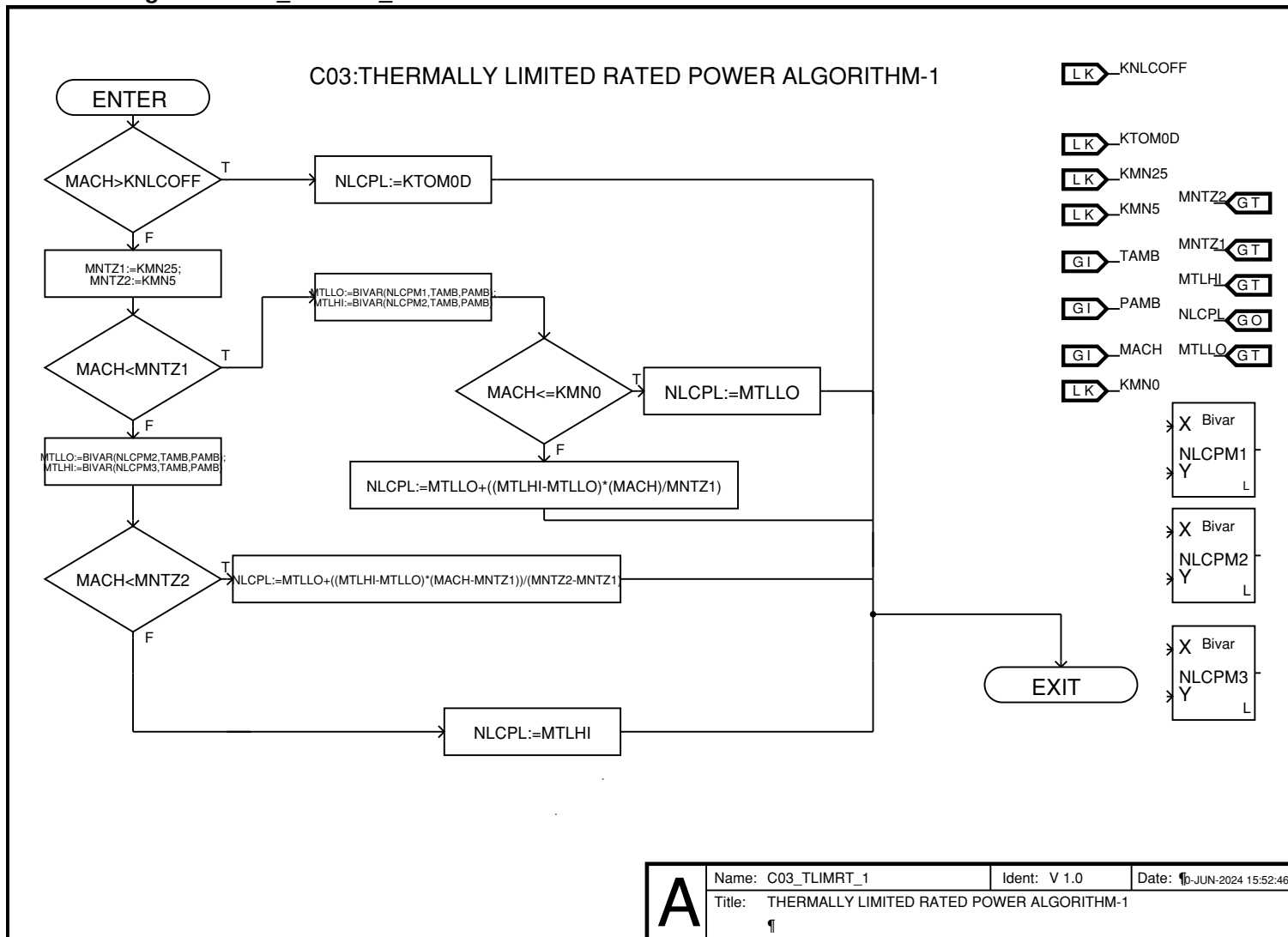
13.00 Figure 3: C02_PLATRIM_1



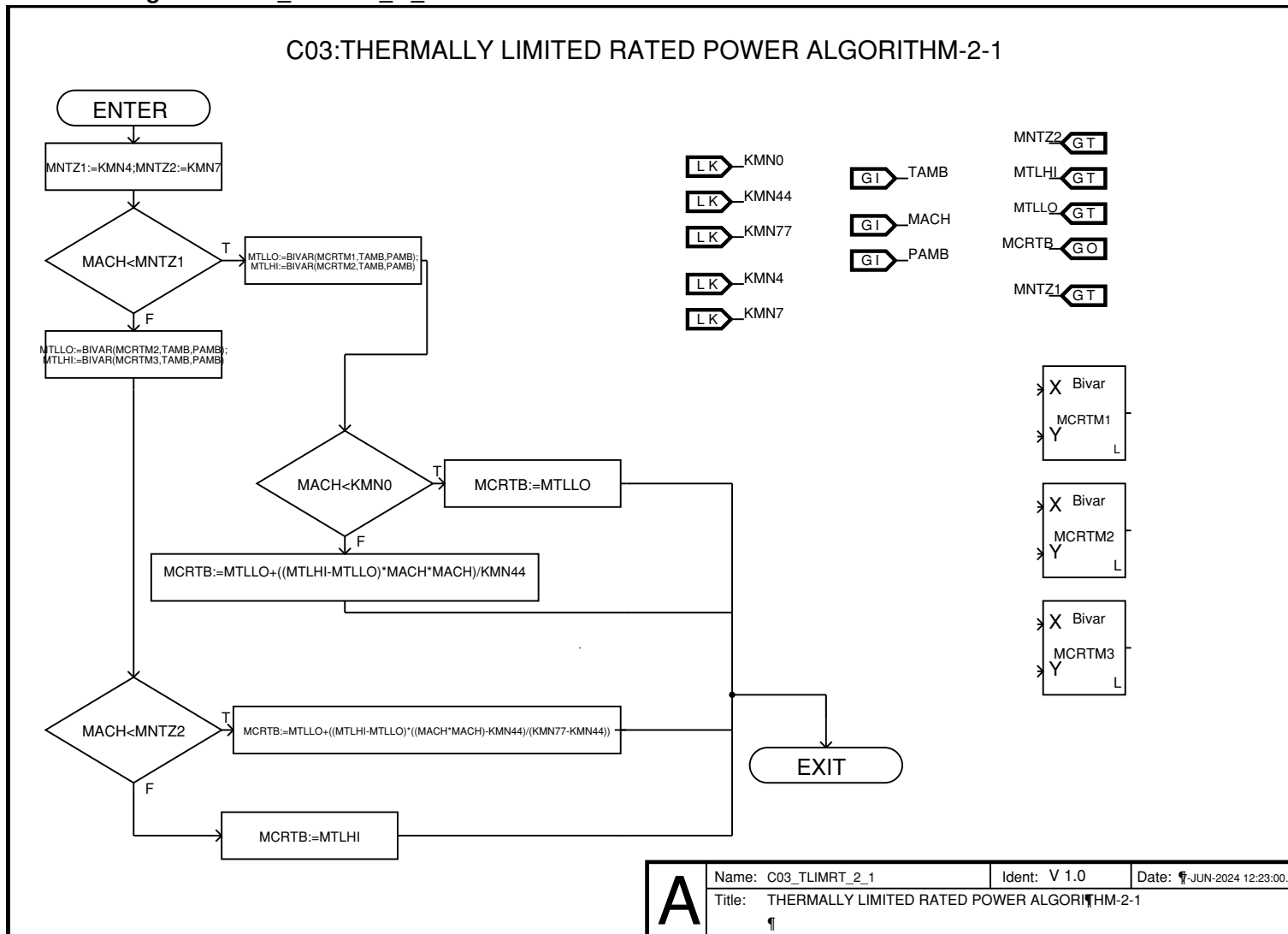
13.00 Figure 3: C02_PLATRIM_2



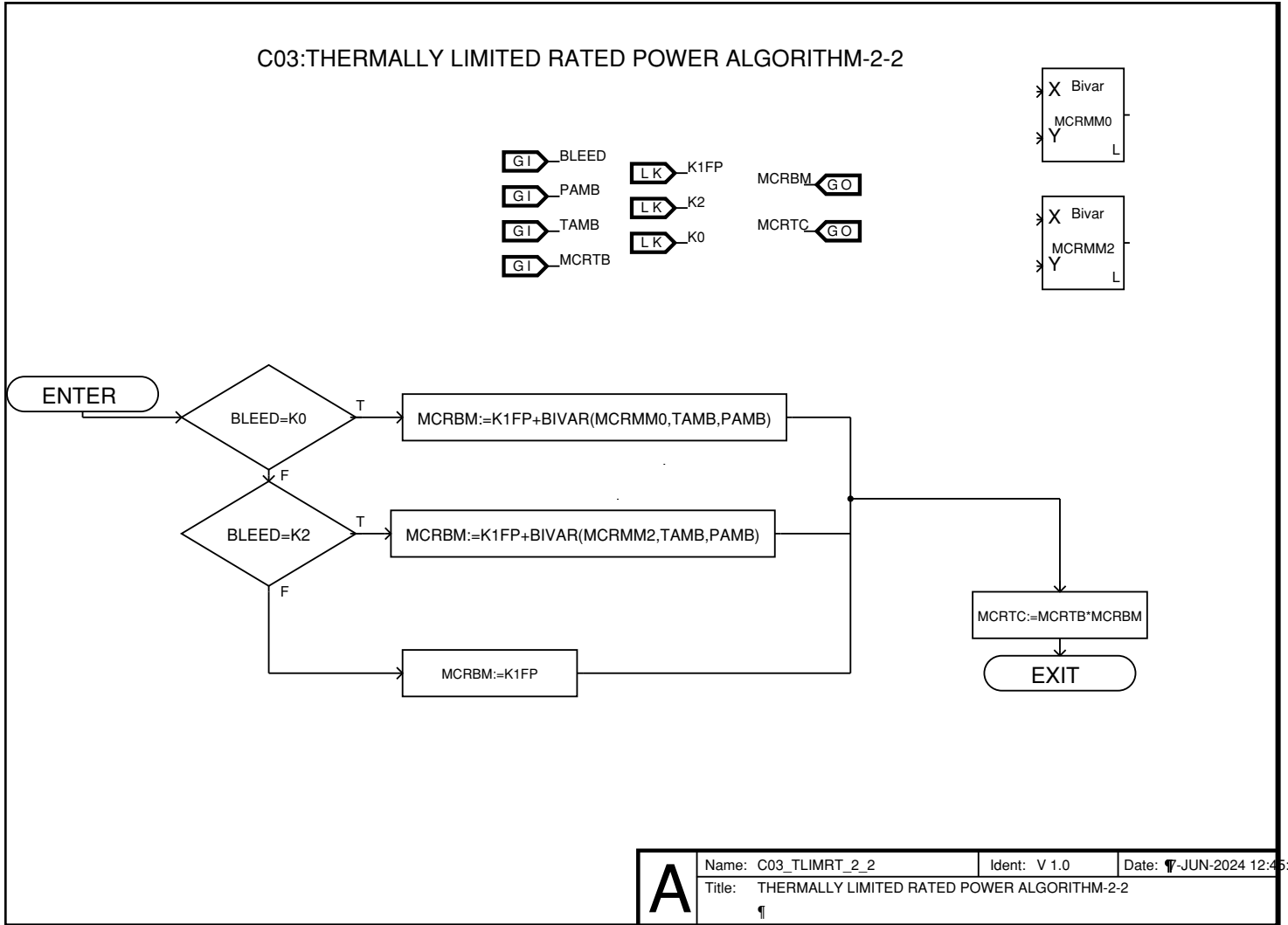
13.00 Figure 4: C03_TLIMRT_1



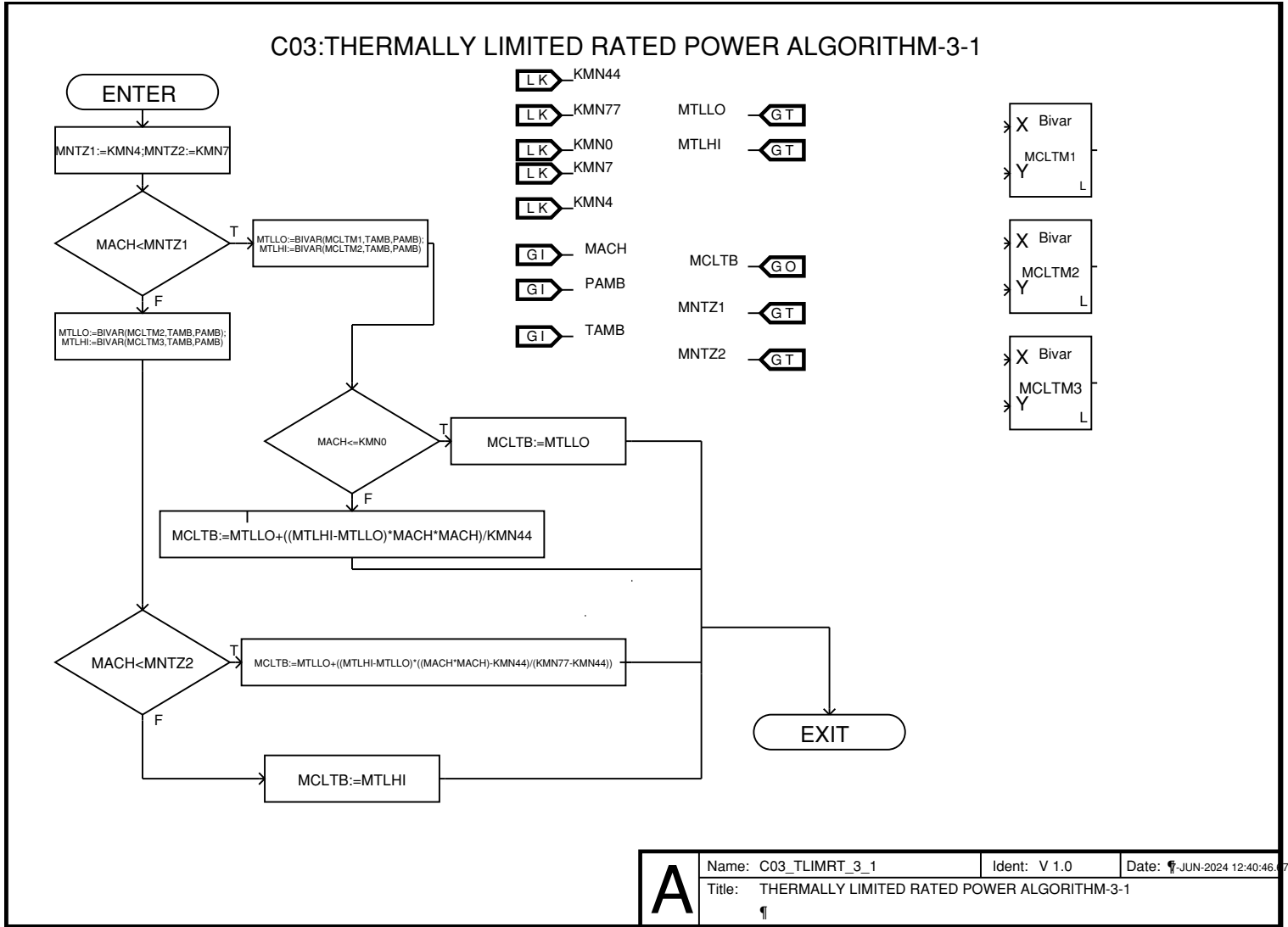
13.00 Figure 5: C03_TLIMRT_2_1



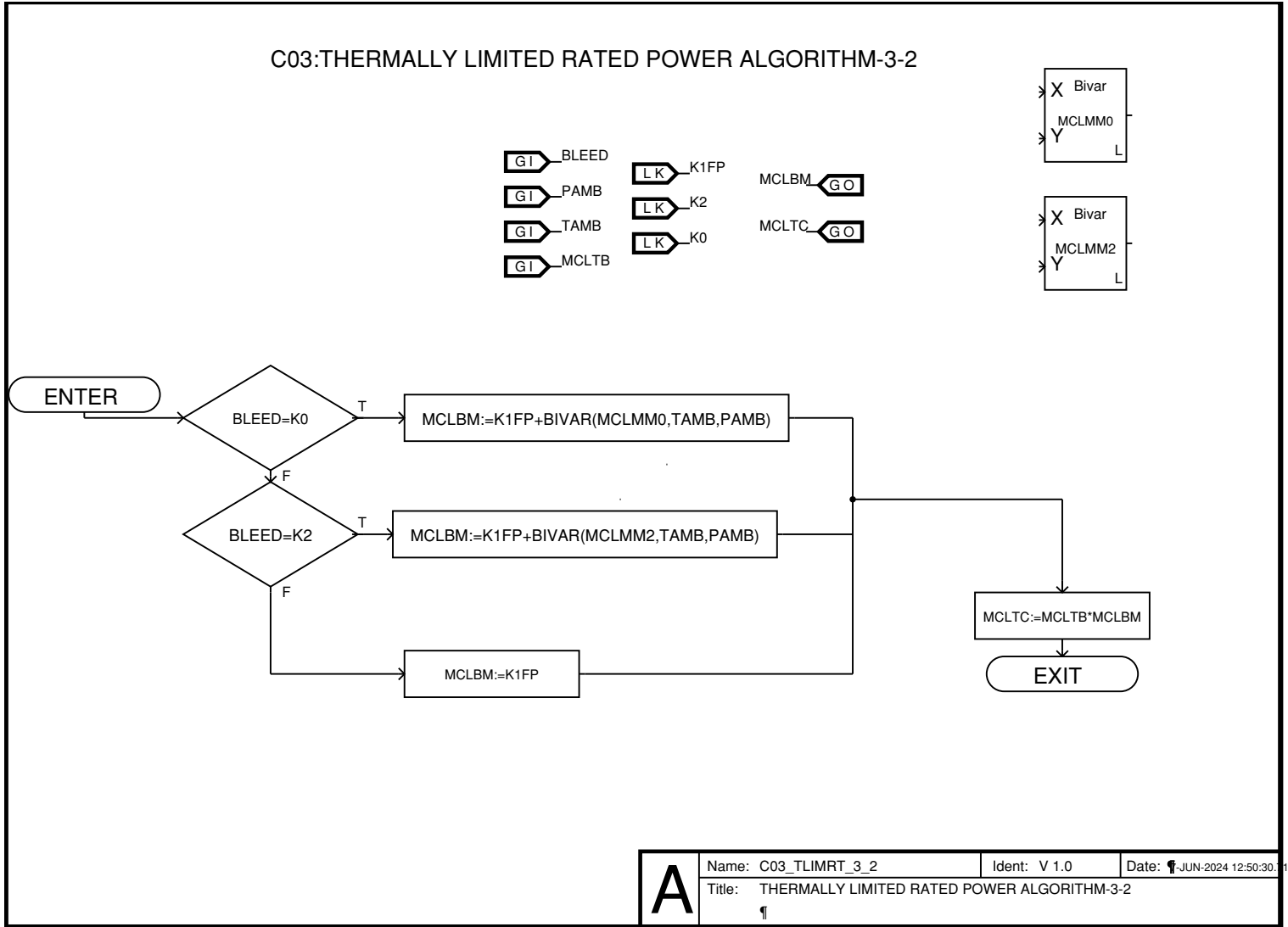
13.00 Figure 5: C03_TLIMRT_2_2



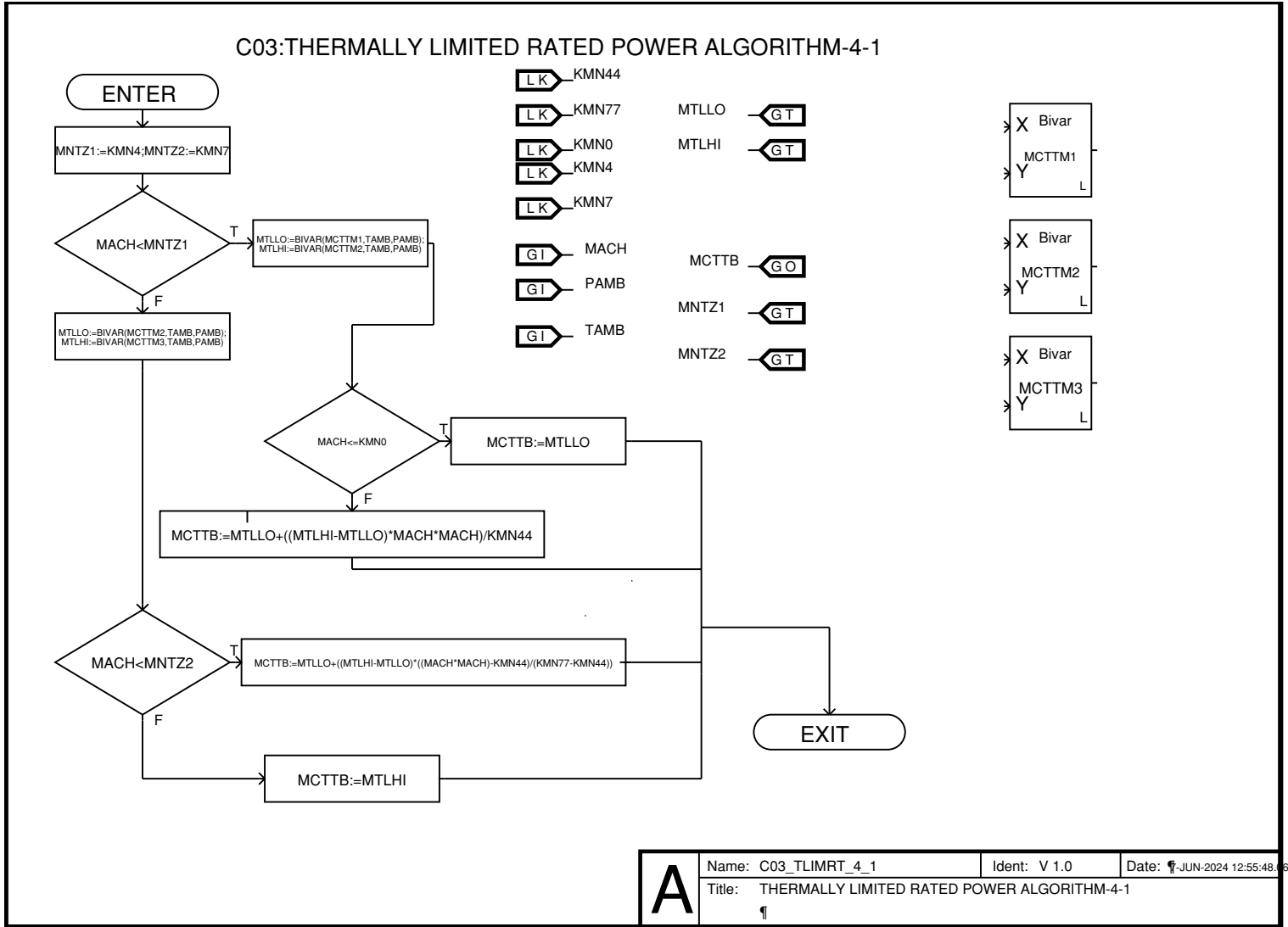
13.00 Figure 6: C03_TLIMRT_3_1



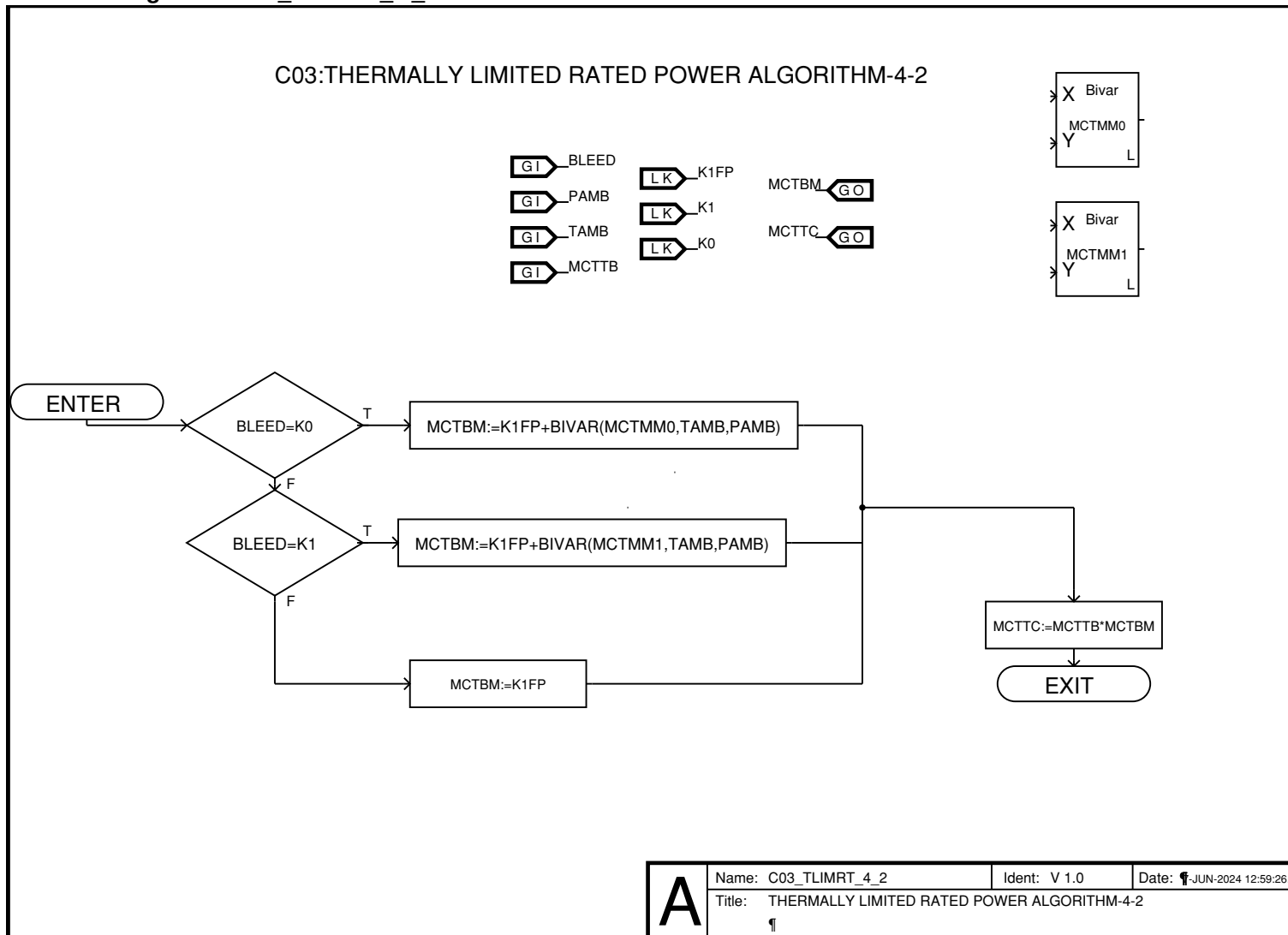
13.00 Figure 6: C03_TLIMRT_3_2



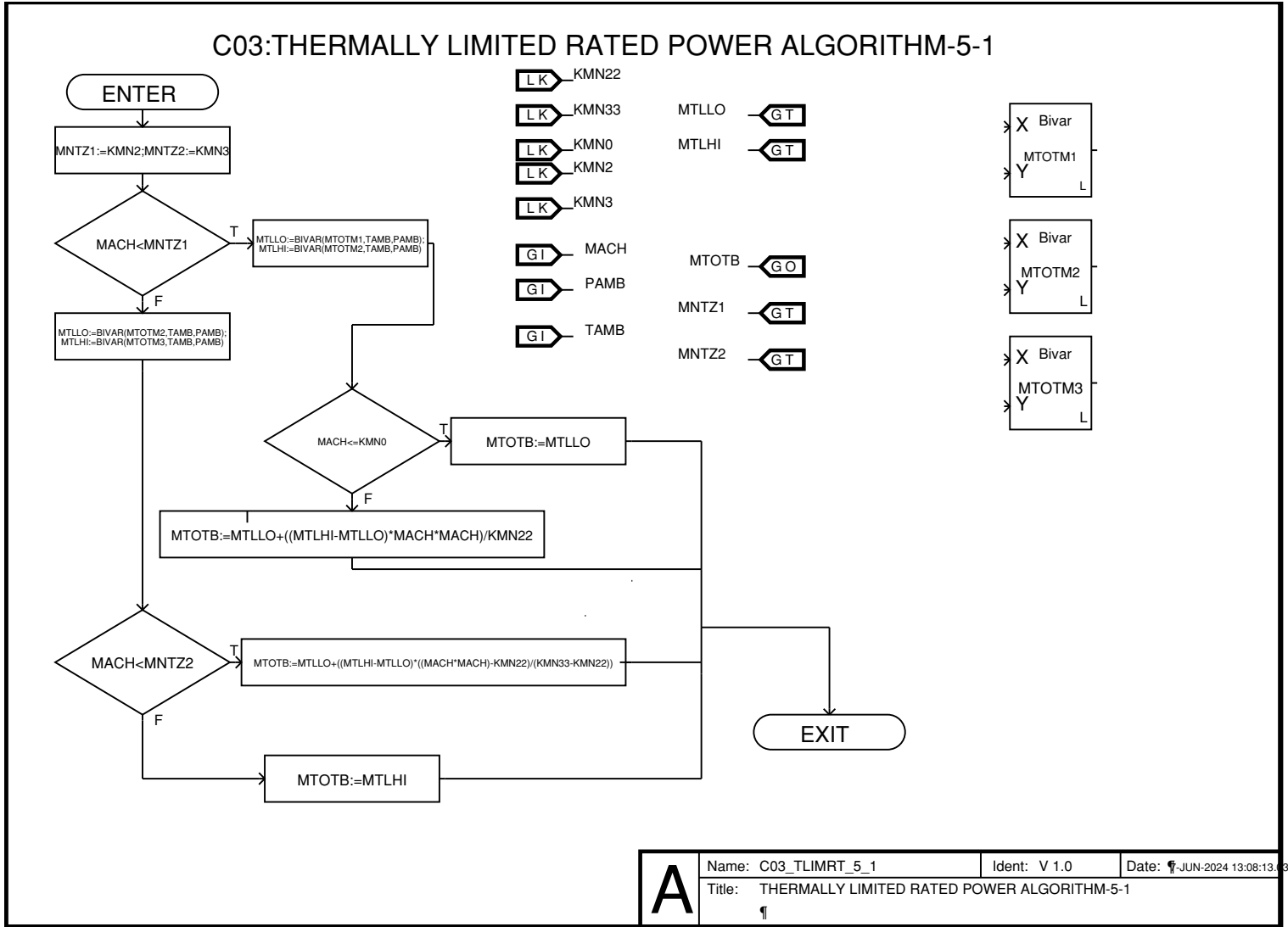
13.00 Figure 7: C03_TLIMRT_4_1



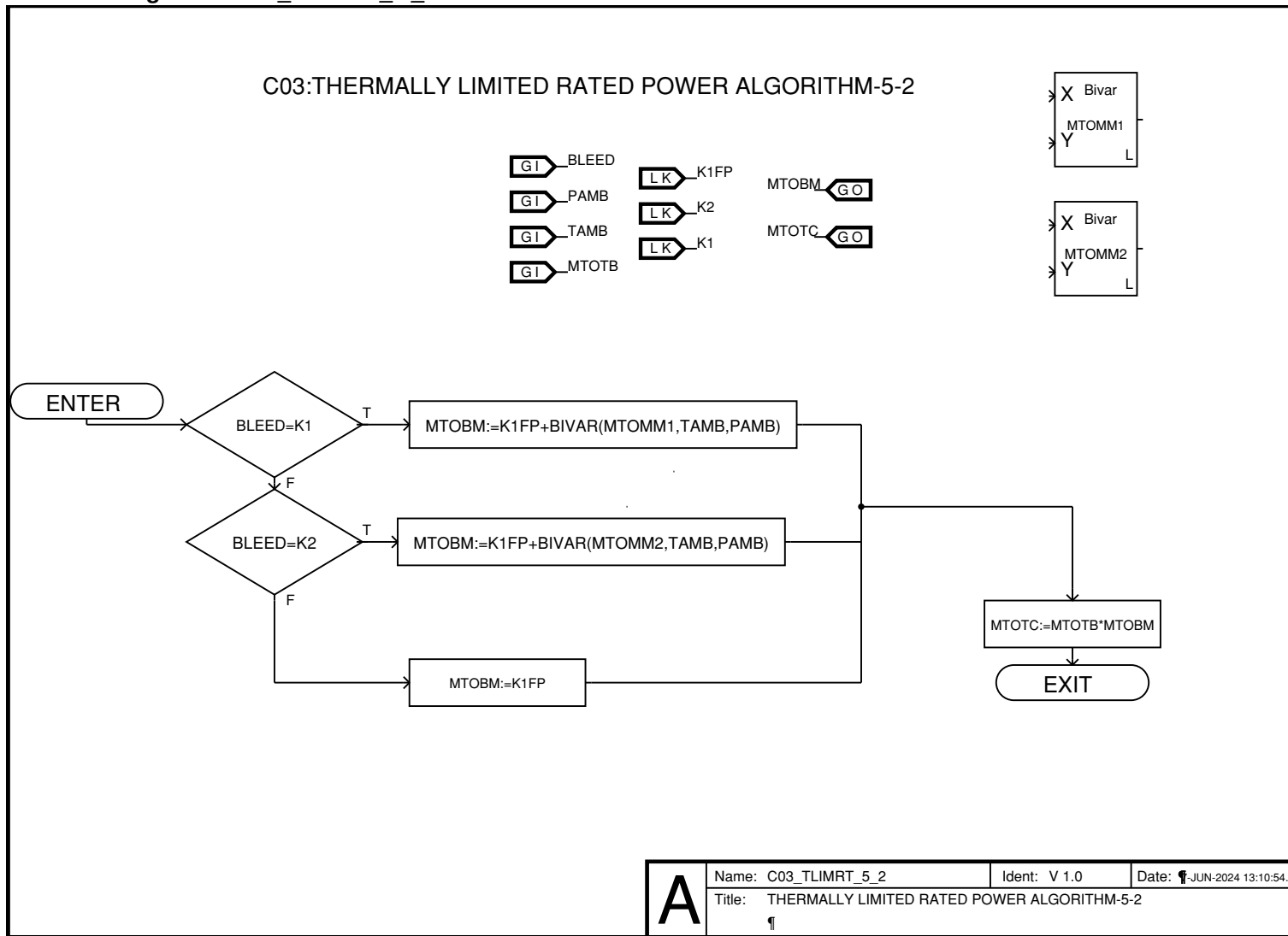
13.00 Figure 7: C03_TLIMRT_4_2



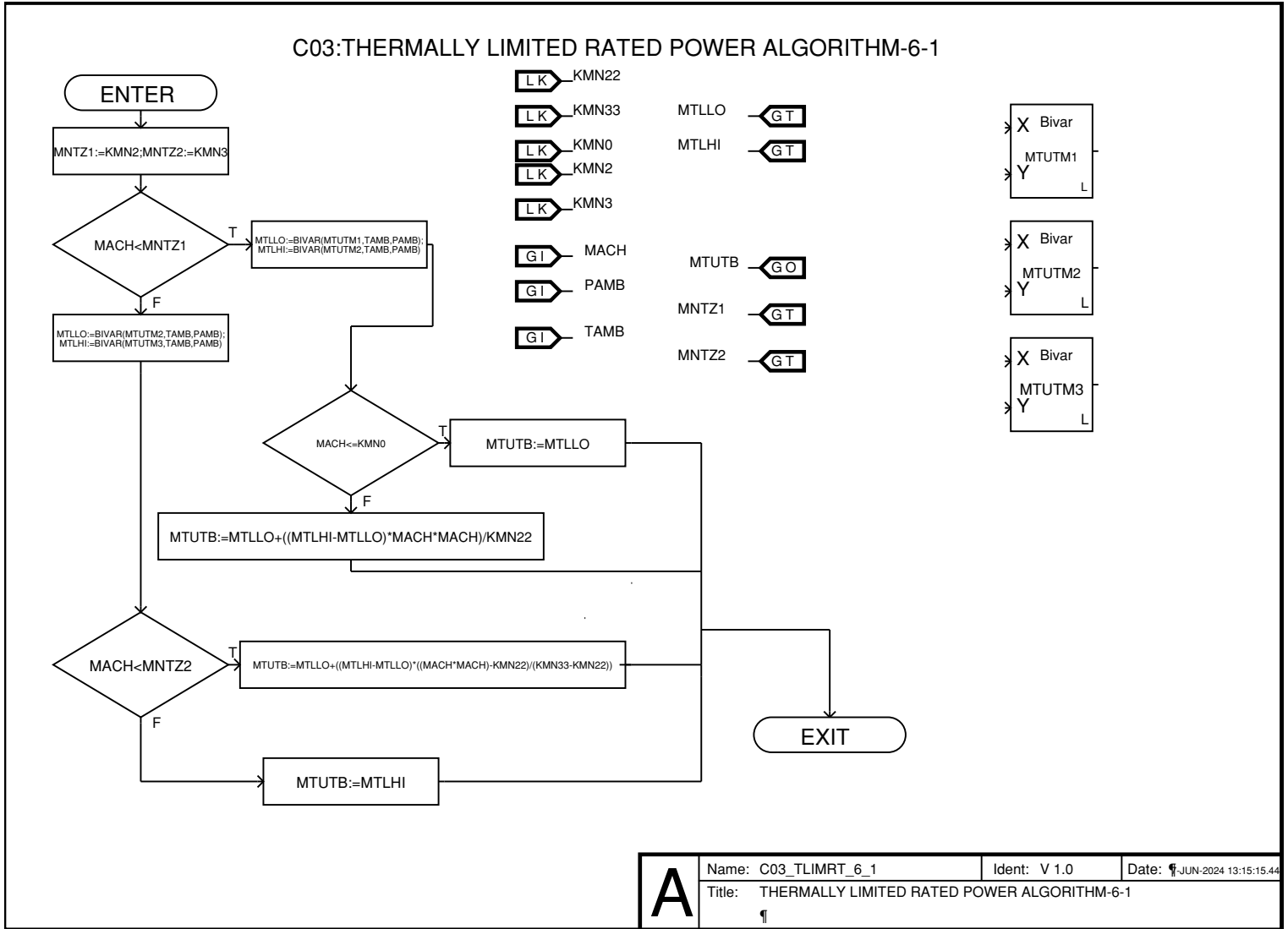
13.00 Figure 8: C03_TLIMRT_5_1



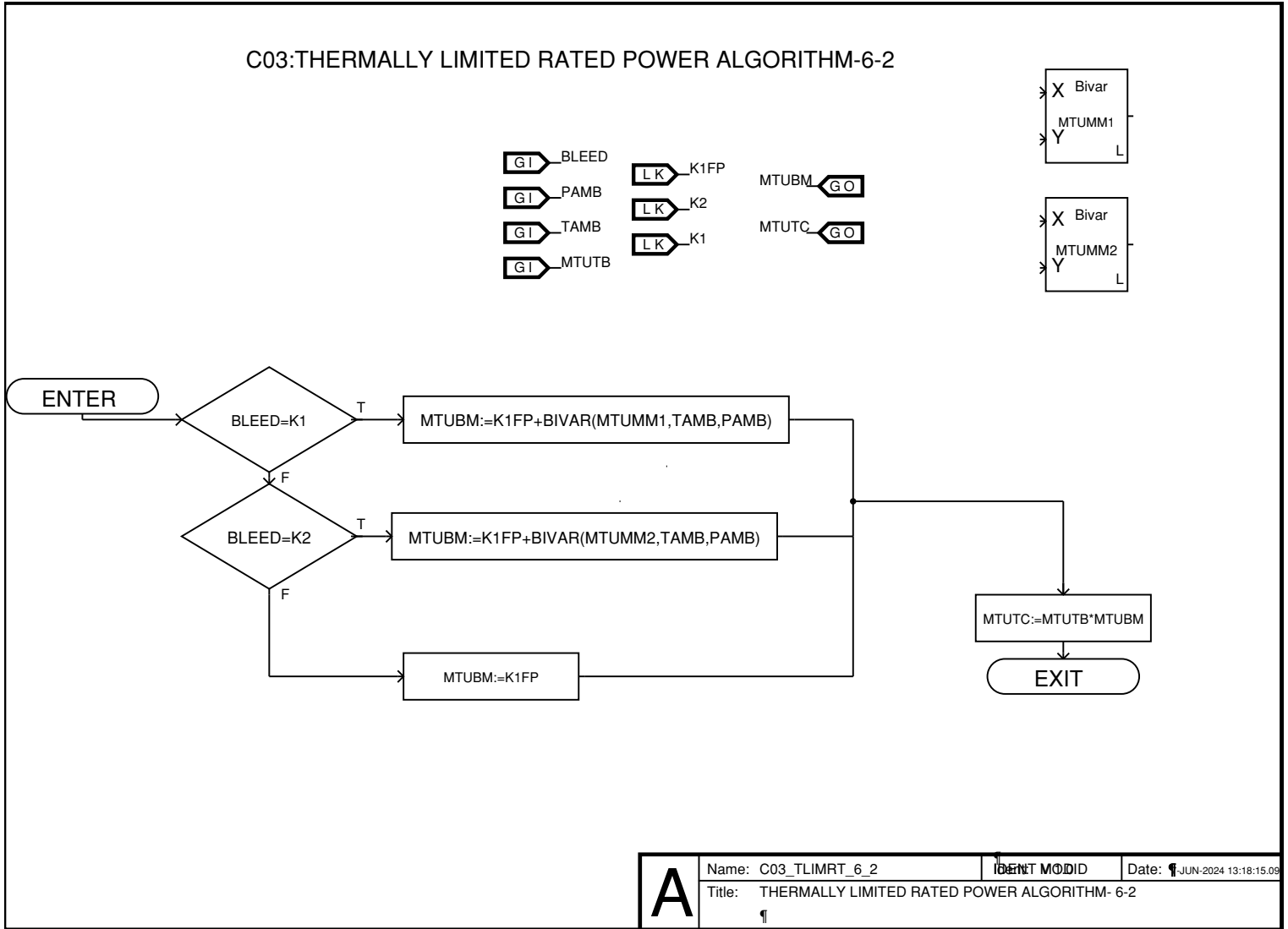
13.00 Figure 8: C03_TLIMRT_5_2



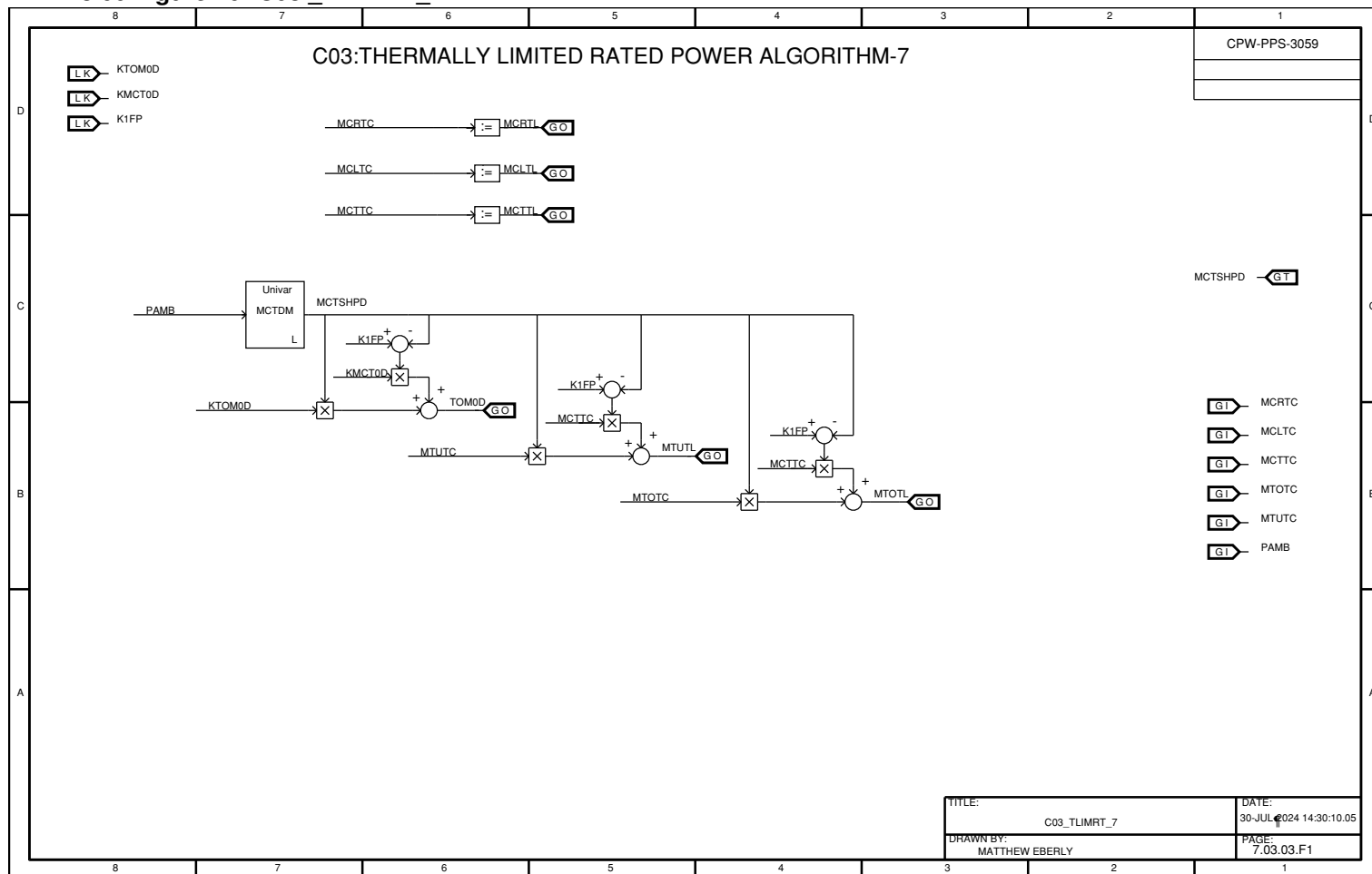
13.00 Figure 9: C03_TLIMRT_6_1



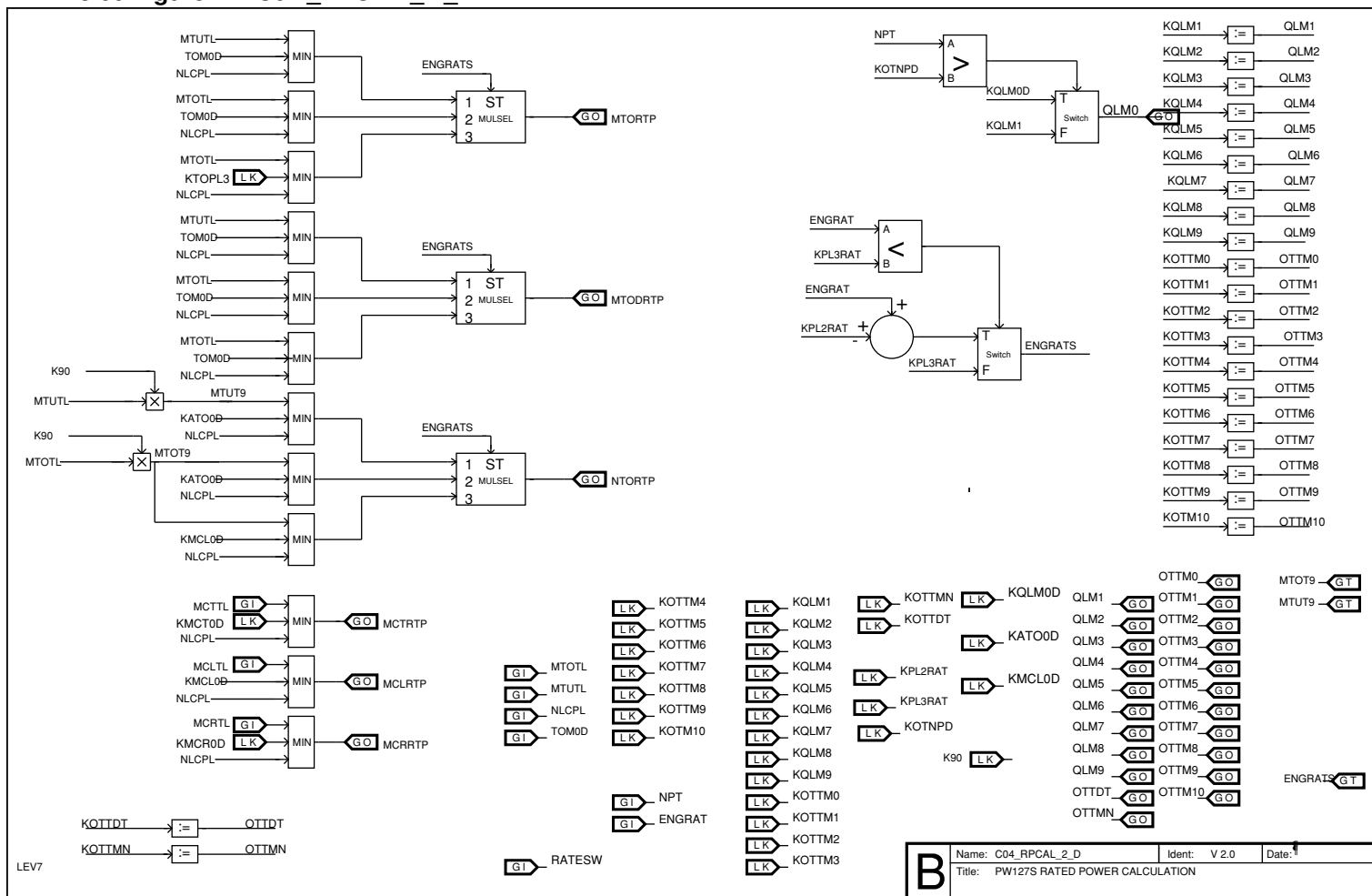
13.00 Figure 9: C03_TLIMRT_6_2



13.00 Figure 10: C03_TLIMRT_7

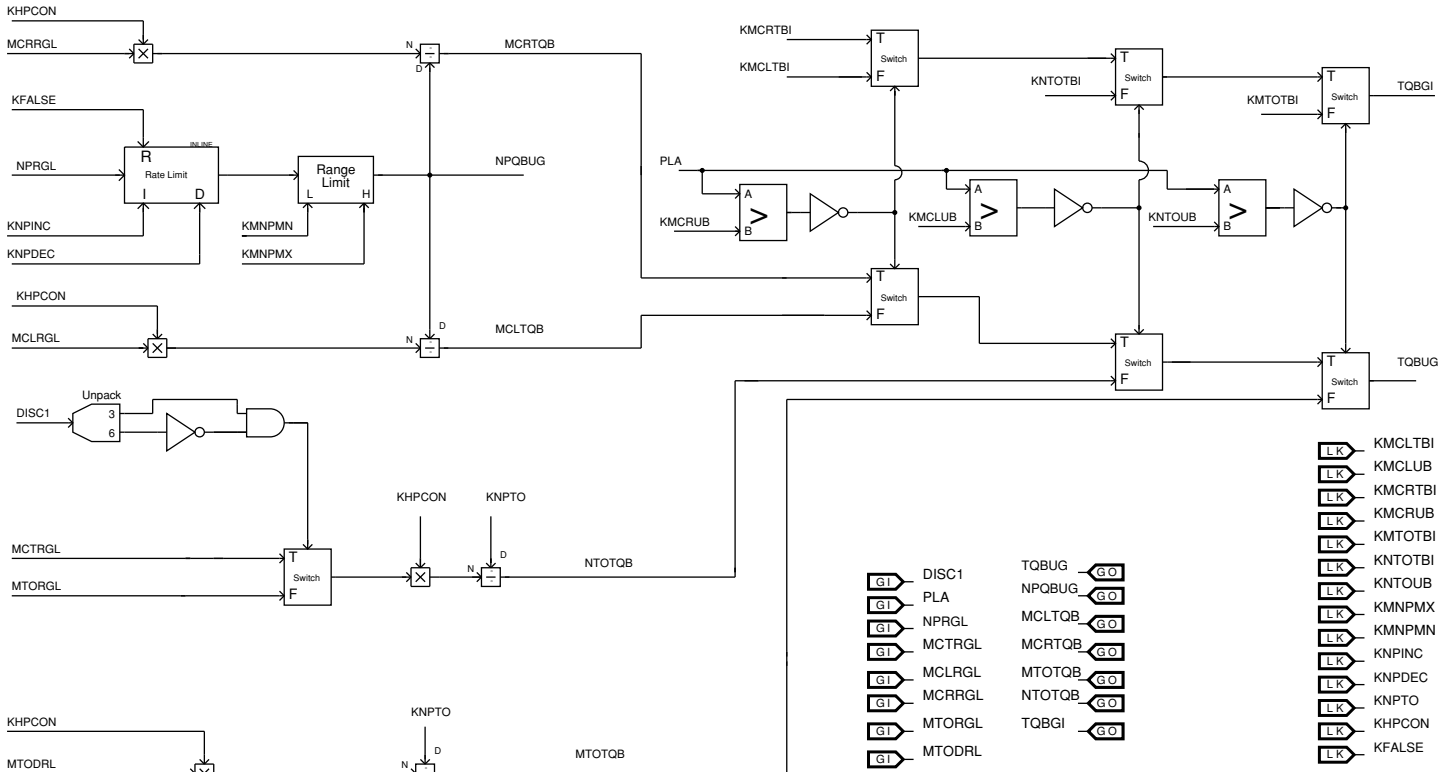


13.00 Figure 11: C04_RPCAL_2_D



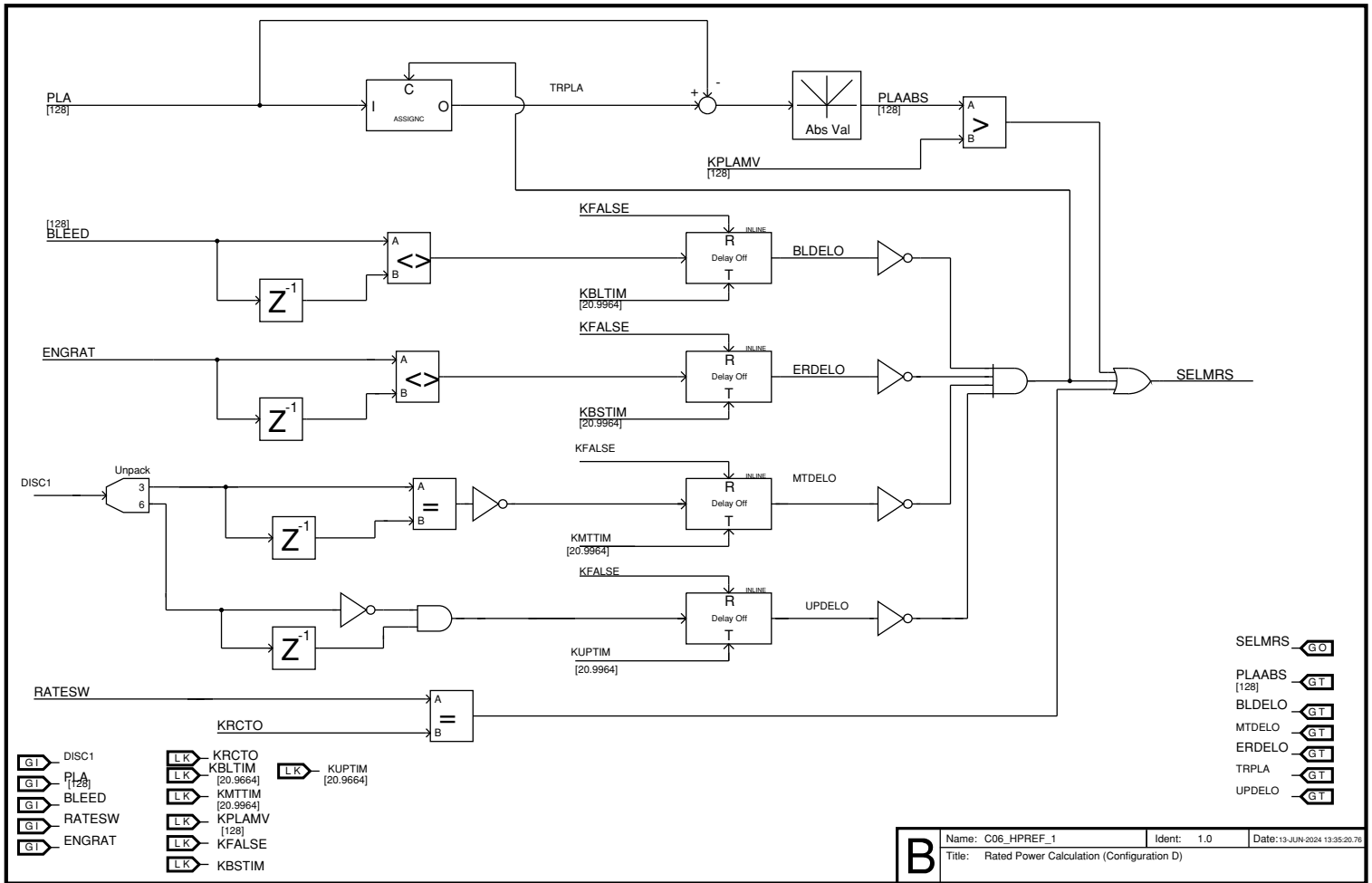
13.00 Figure 12: C05_TQBUG

FIGURE C-05-1: TORQUE BUG CALCULATION

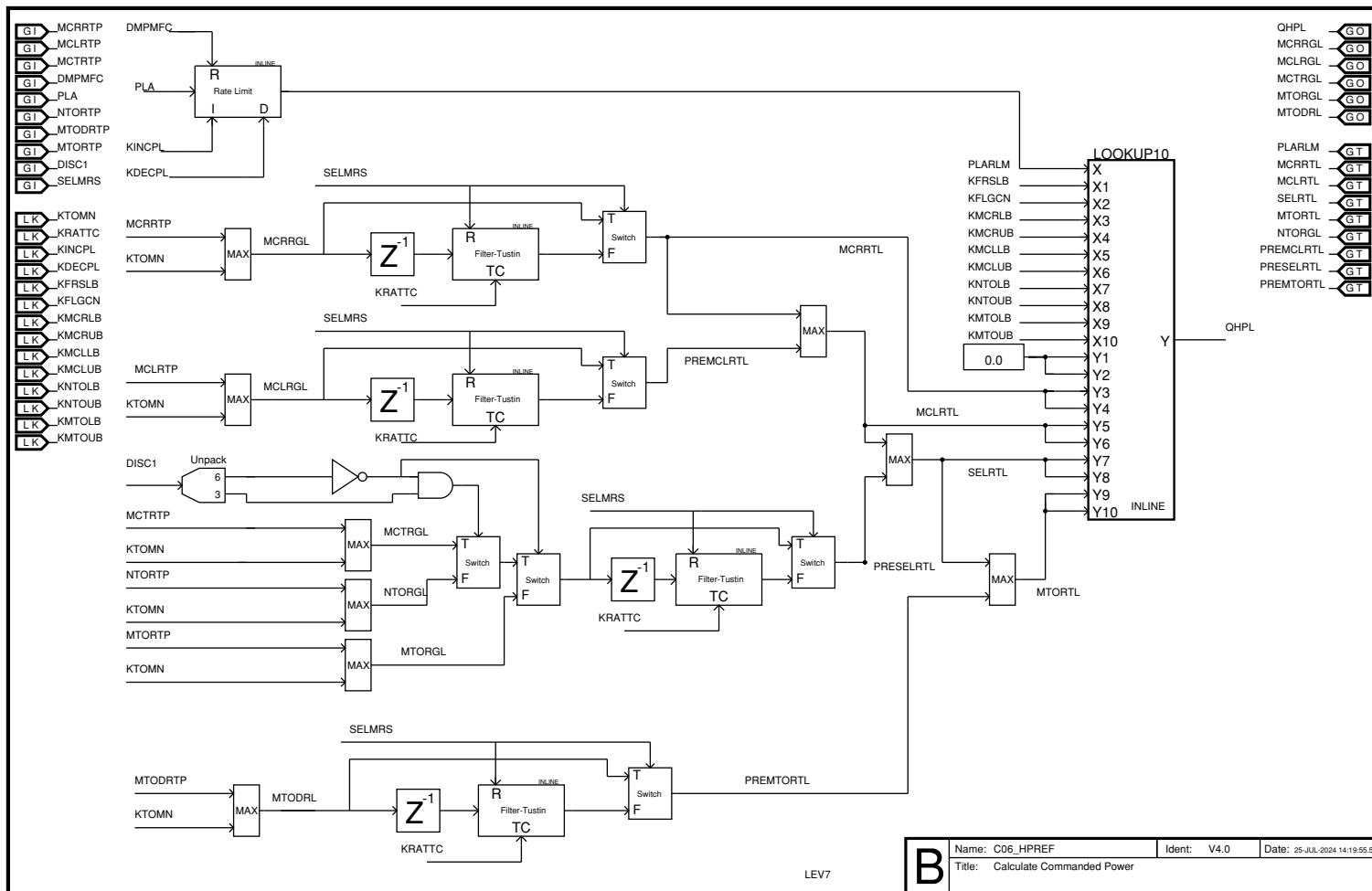


Name: C05_TQBUG Ident: 2.0 Date: 18-JUL-2024 17:19:41.17
Title: TORQUE BUG CALCULATION

13.00 Figure 13: C06_HPREF_1

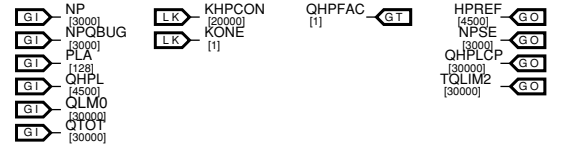
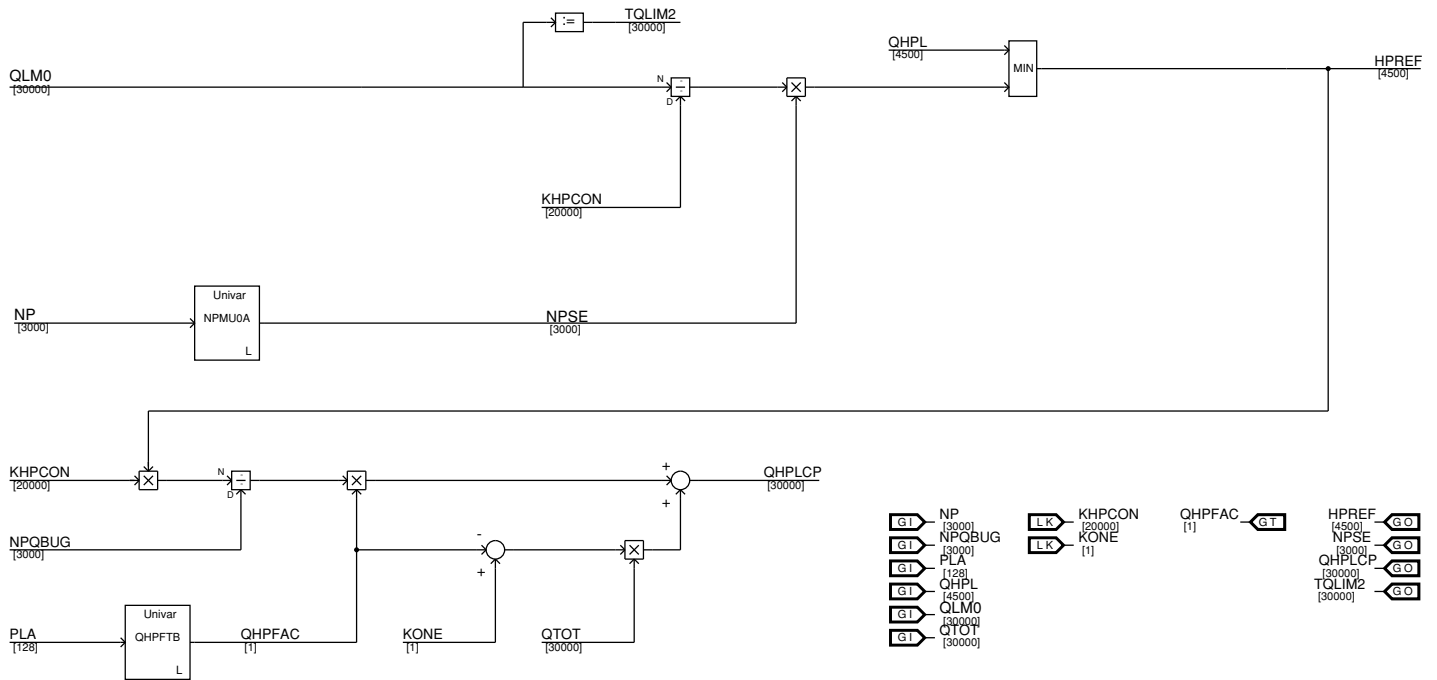


13.00 Figure 14: C06_HPREF



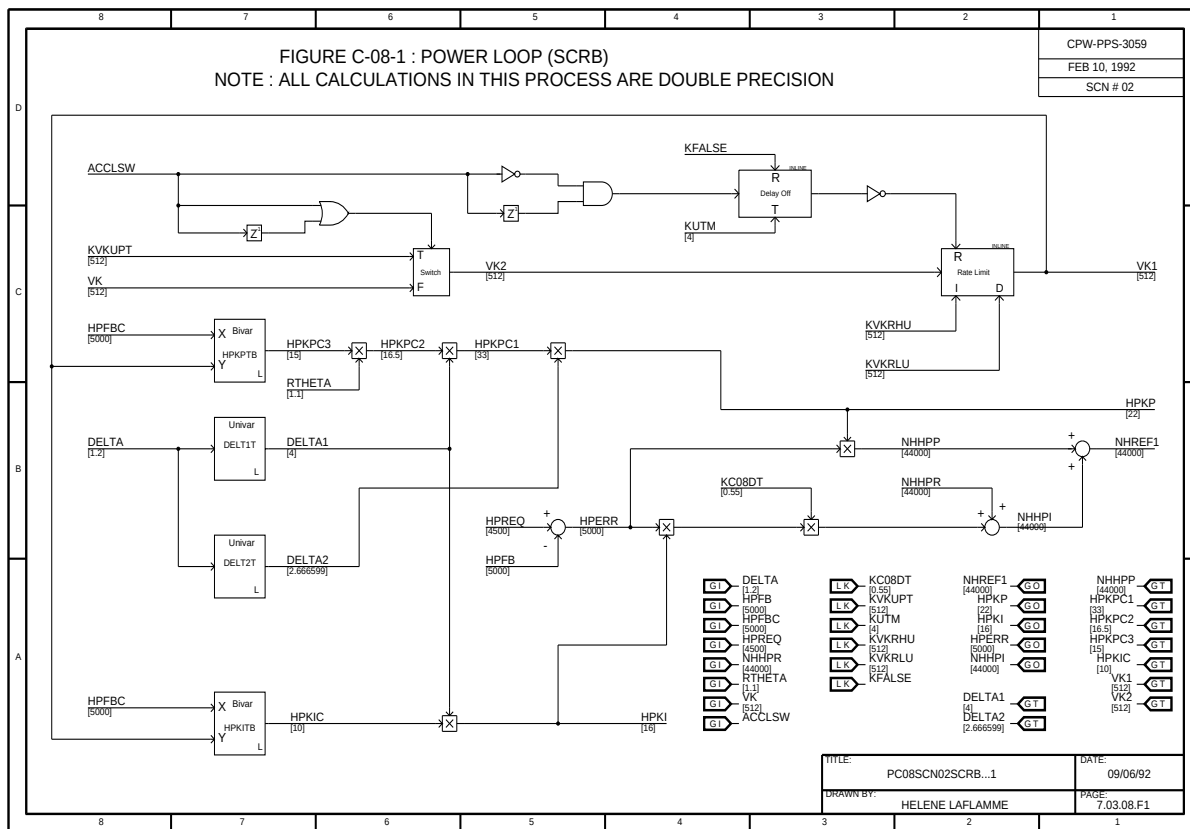
13.00 Figure 15: C07_OVRTQPRO

FIGURE C - 07 - 01 : OVERTORQUE PROTECTION LOGIC

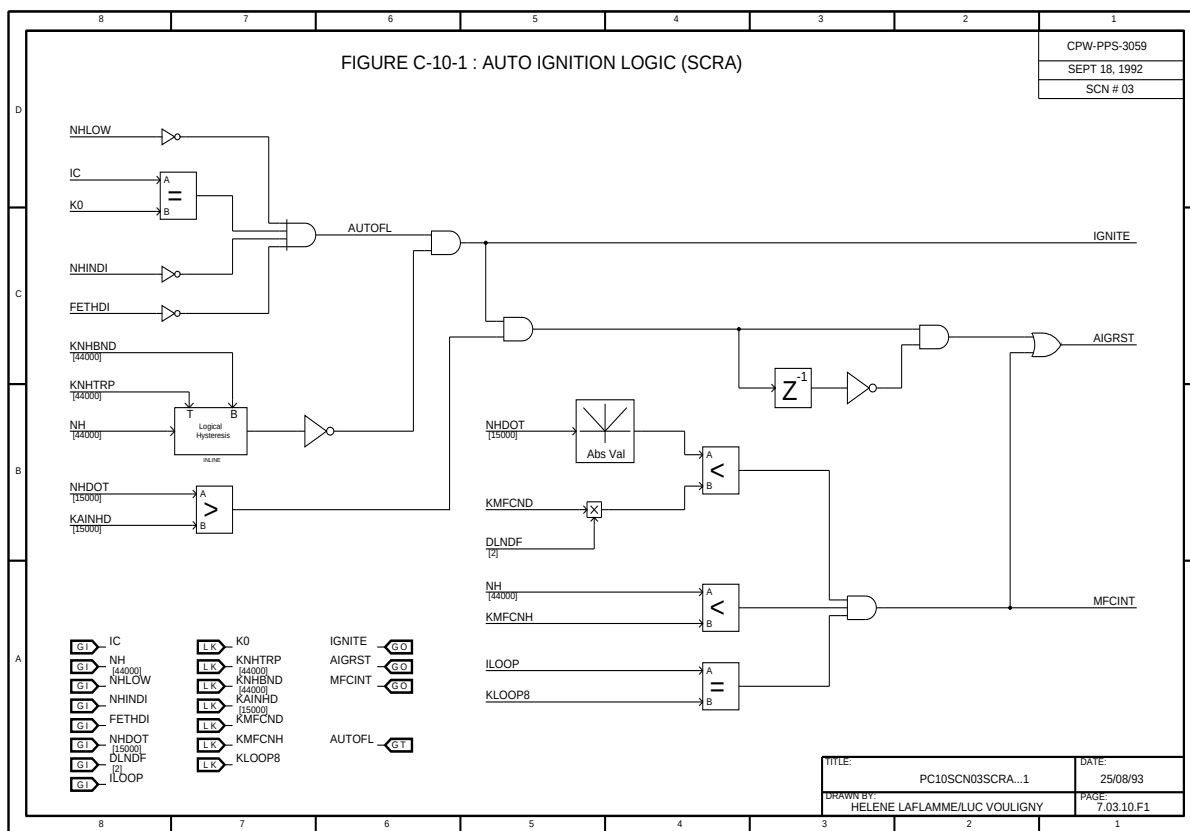


Name: C07_OVR	Date: 21-JUN-2024 16:25:53.37
Title: OVERTORQUE PROTECTION LOGIC	

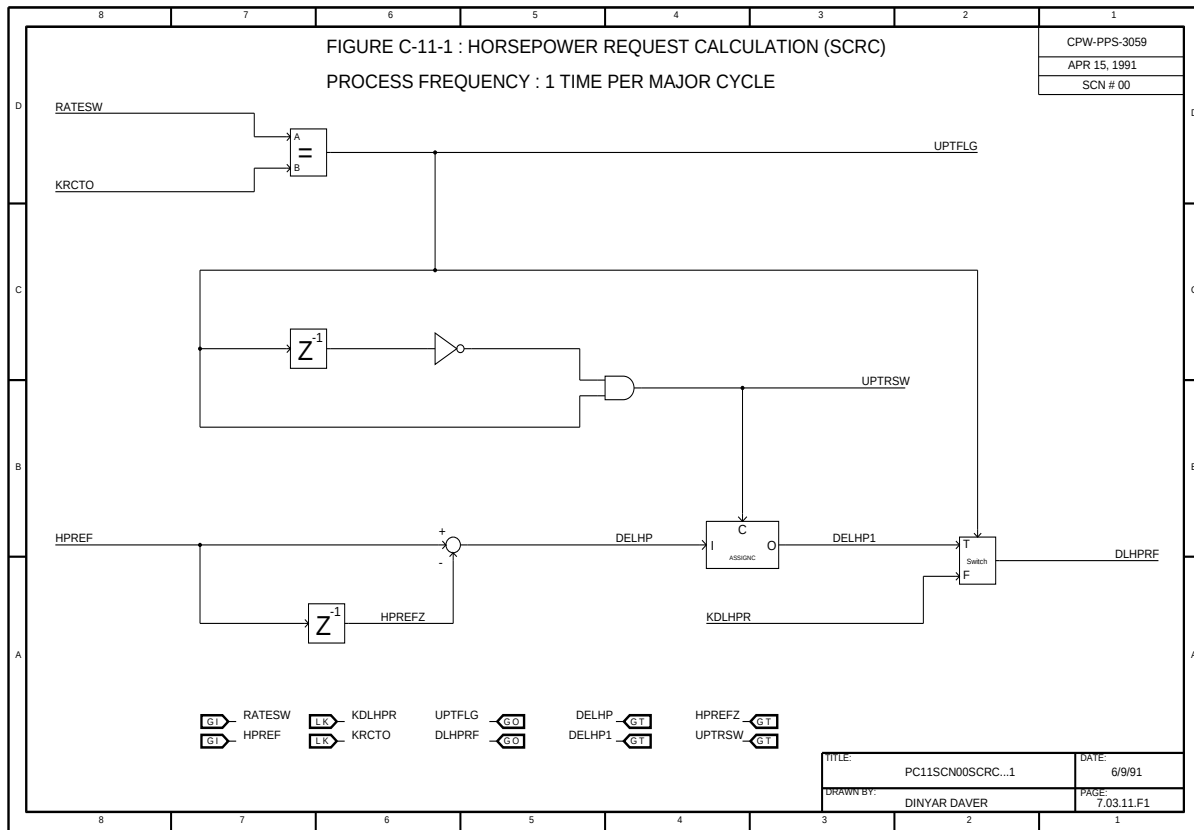
13.00 Figure 16: C08_SHPCL



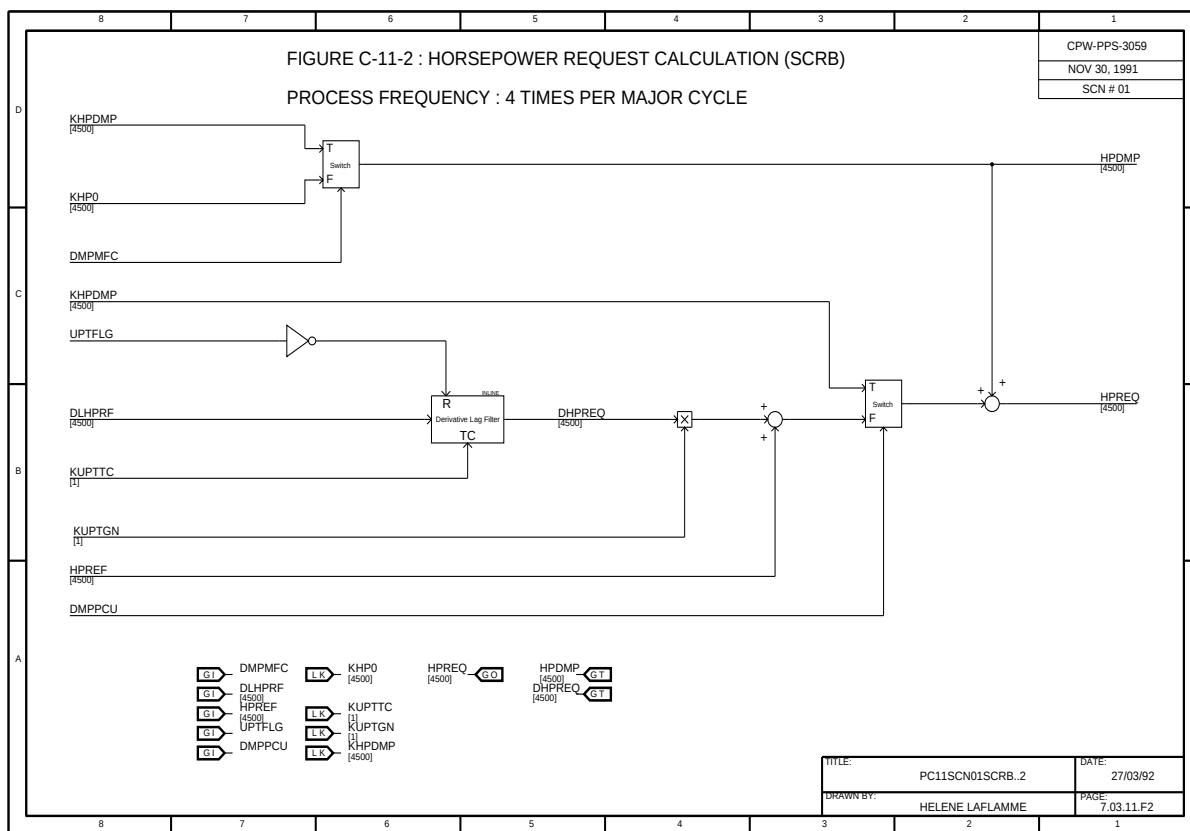
13.00 Figure 17: C10_ATOIGN



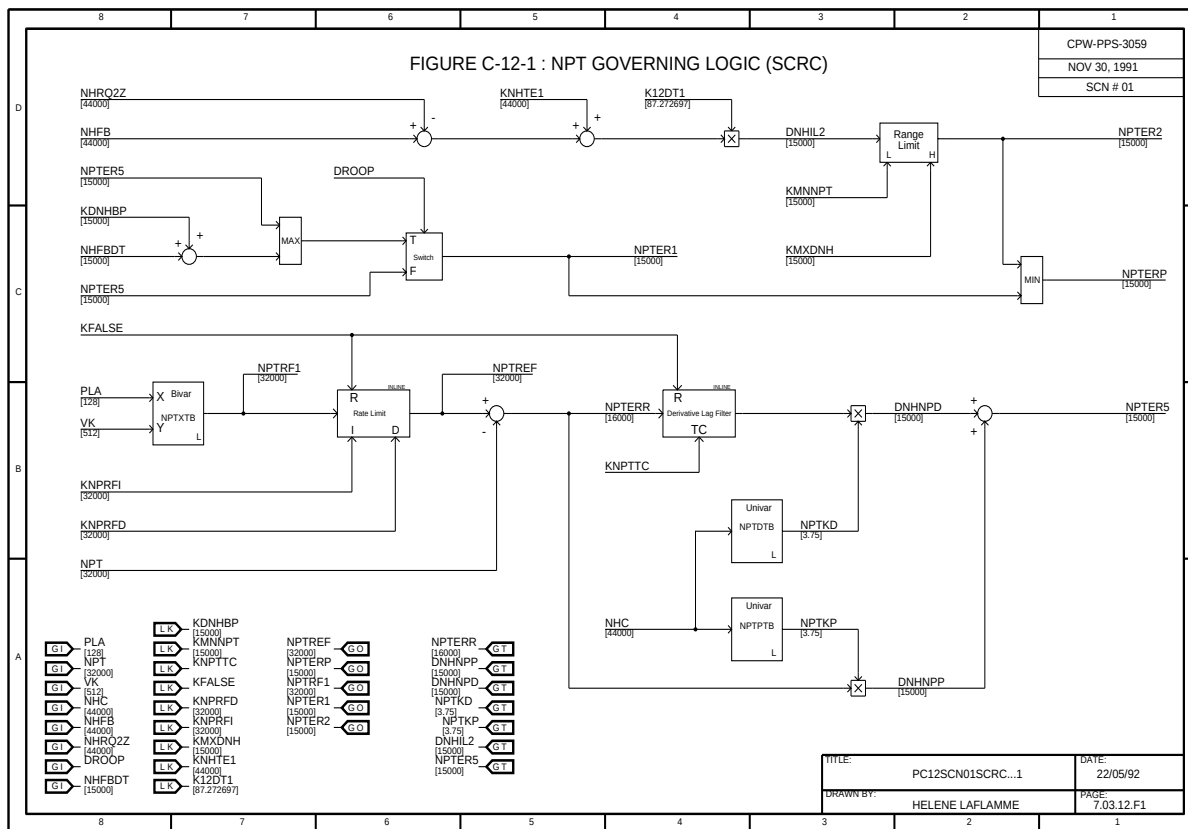
13.00 Figure 18: C11_SHPREQ_1



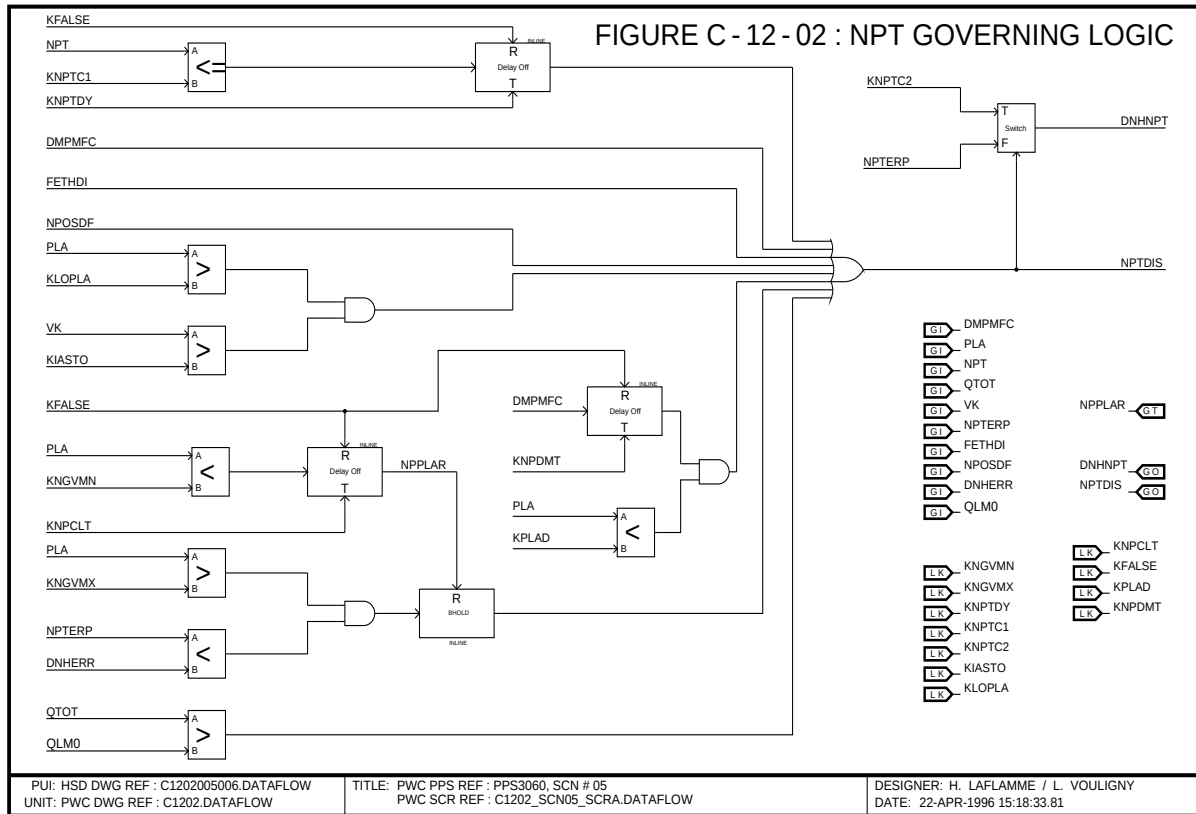
13.00 Figure 19: C11_SHPREQ_2



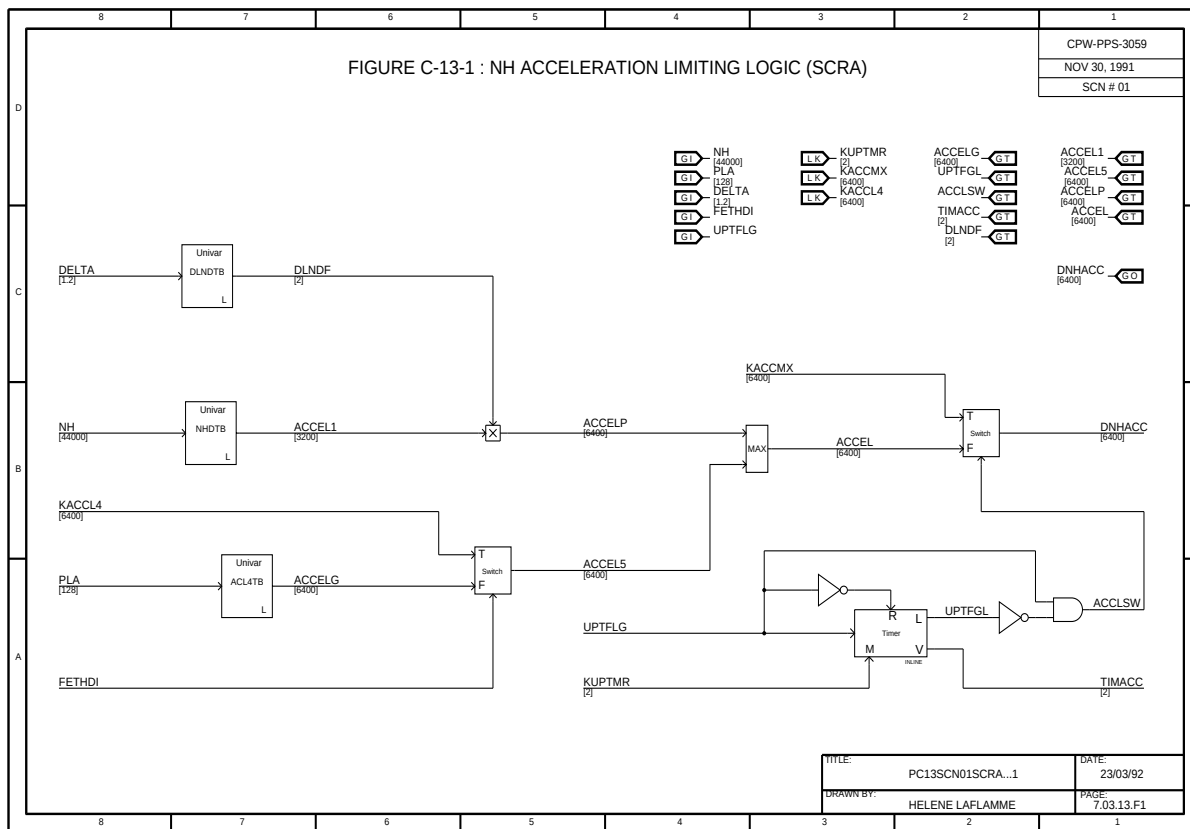
13.00 Figure 20: C12_NPTGOV_1



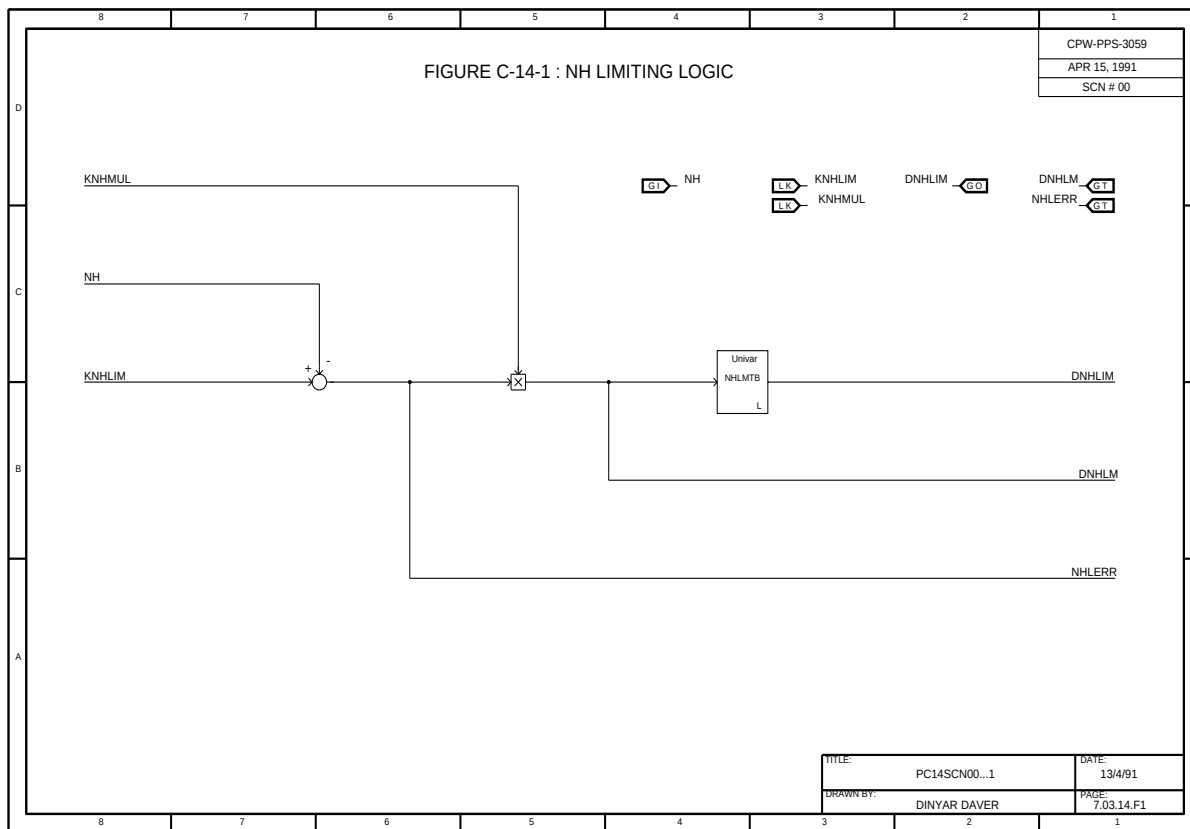
13.00 Figure 21: C12_NPTGOV_2



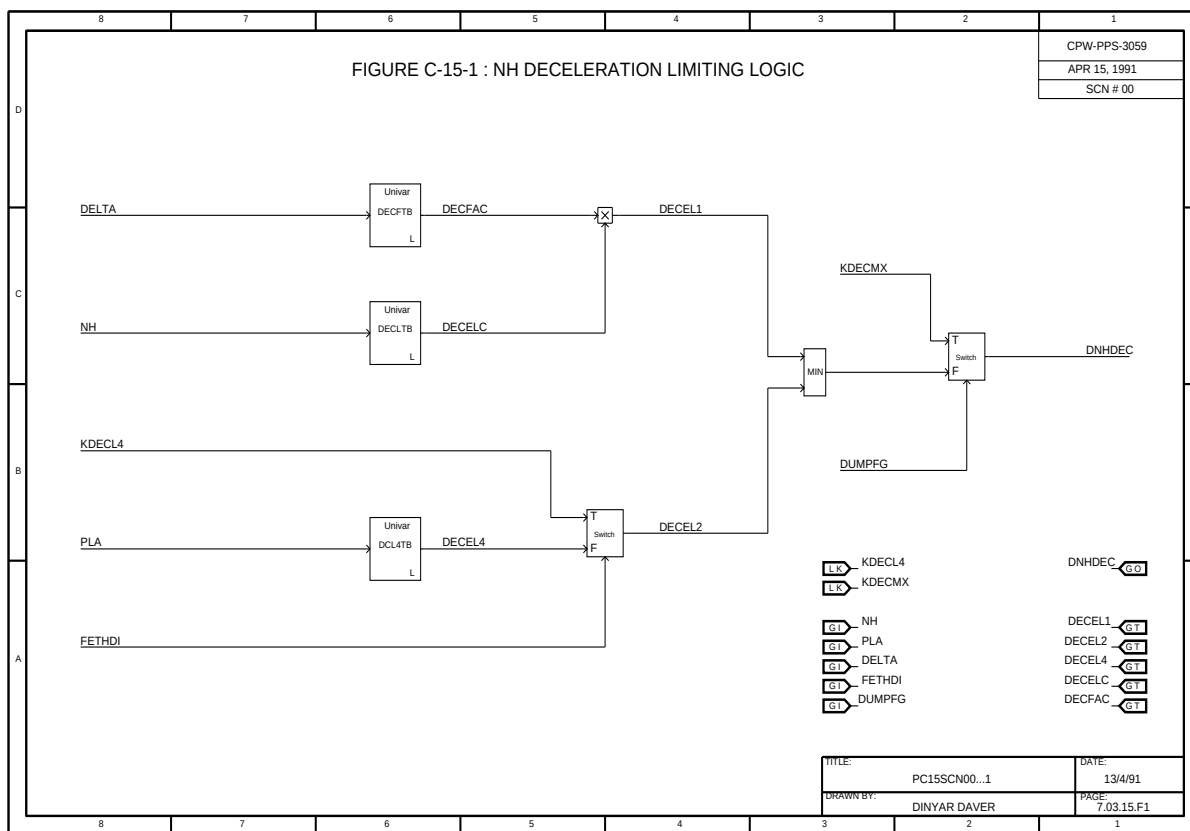
13.00 Figure 22: C13_NHACCELLIM



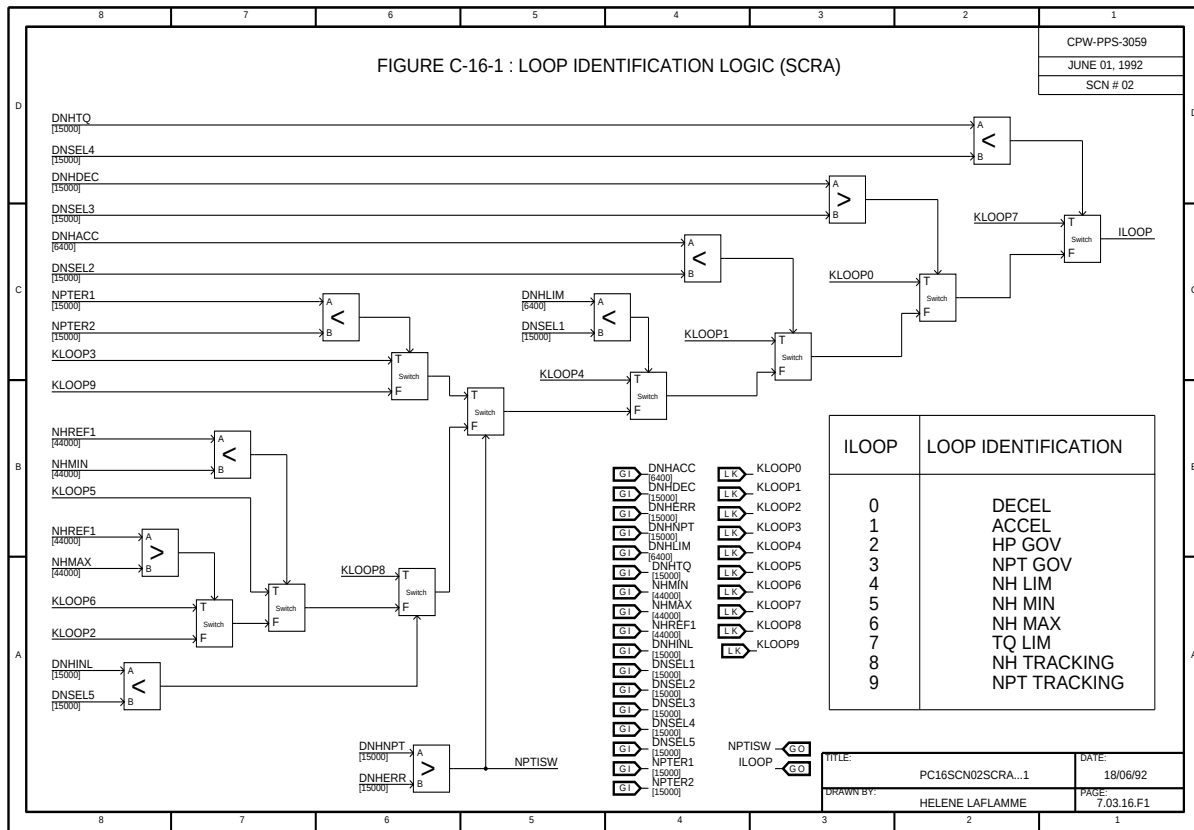
13.00 Figure 23: C14 _NHLIM



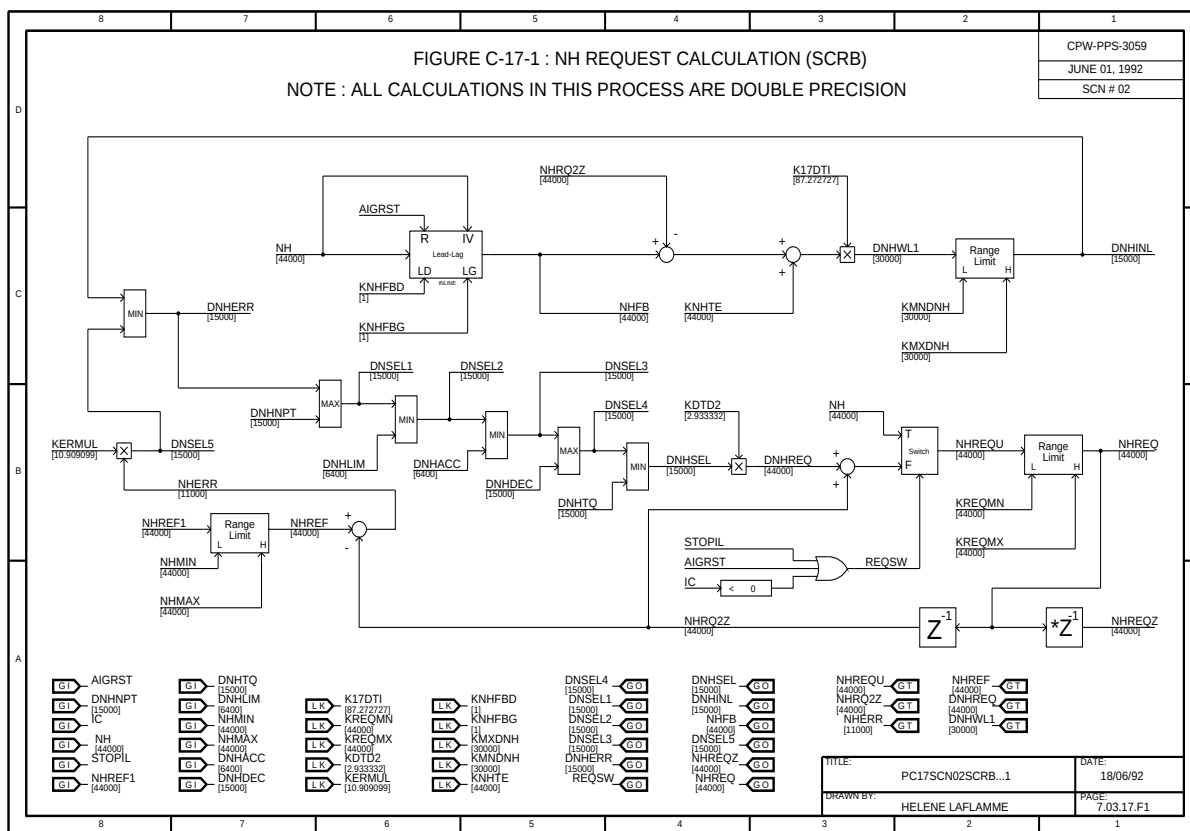
13.00 Figure 24: C15_NHDECELLIM



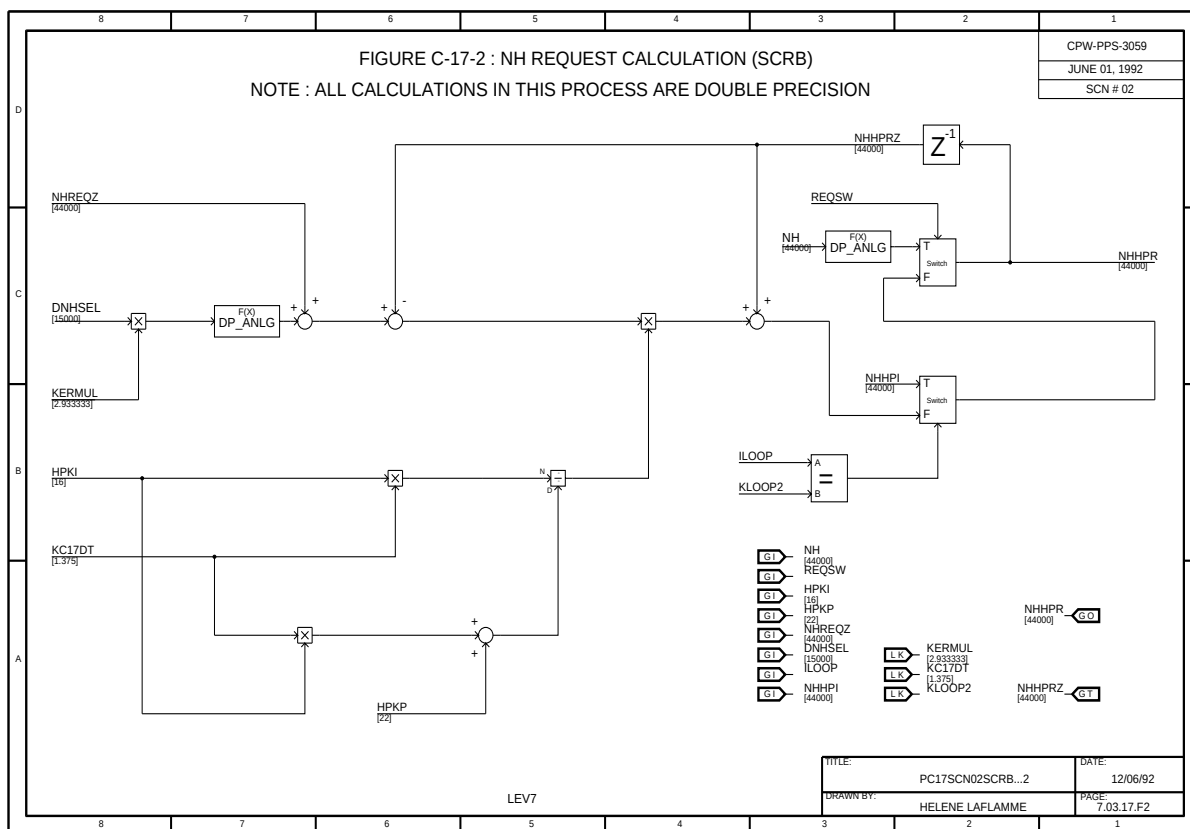
13.00 Figure 25: C16_LOOPSEL



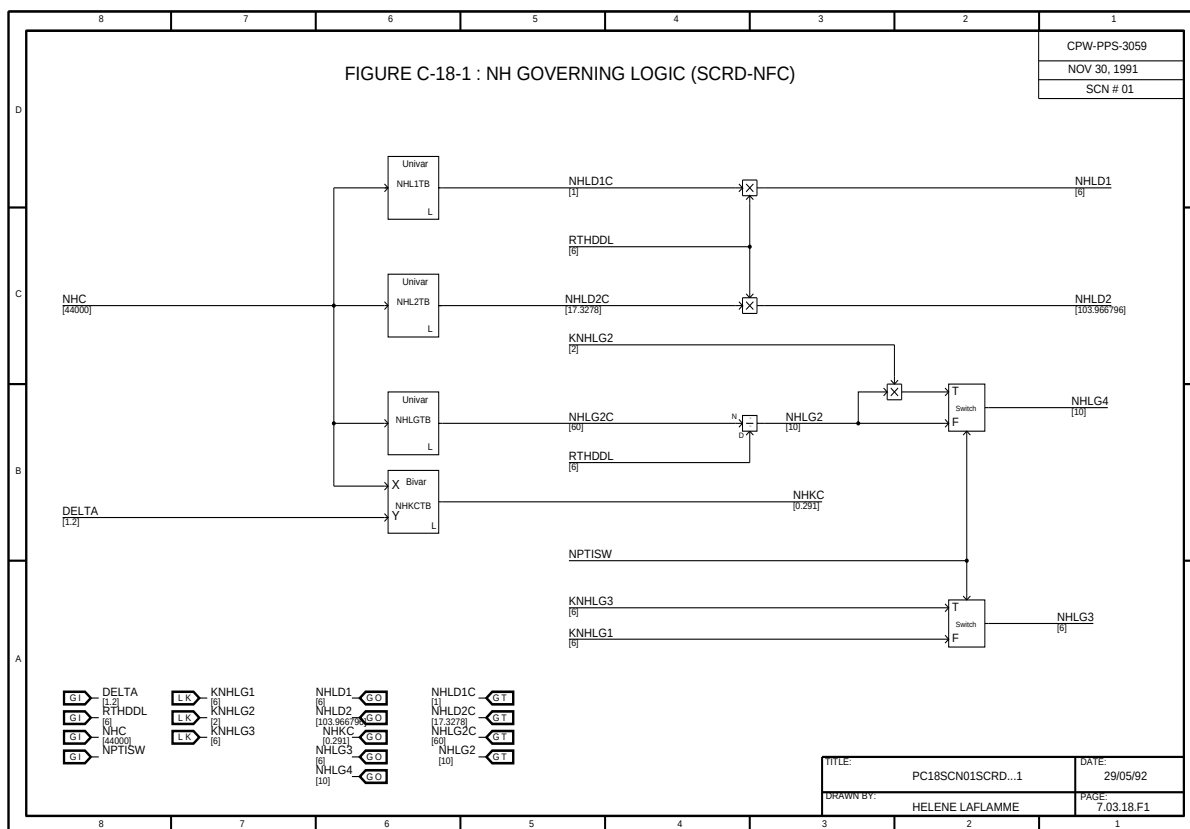
13.00 Figure 26: C17_NHREQUEST_1



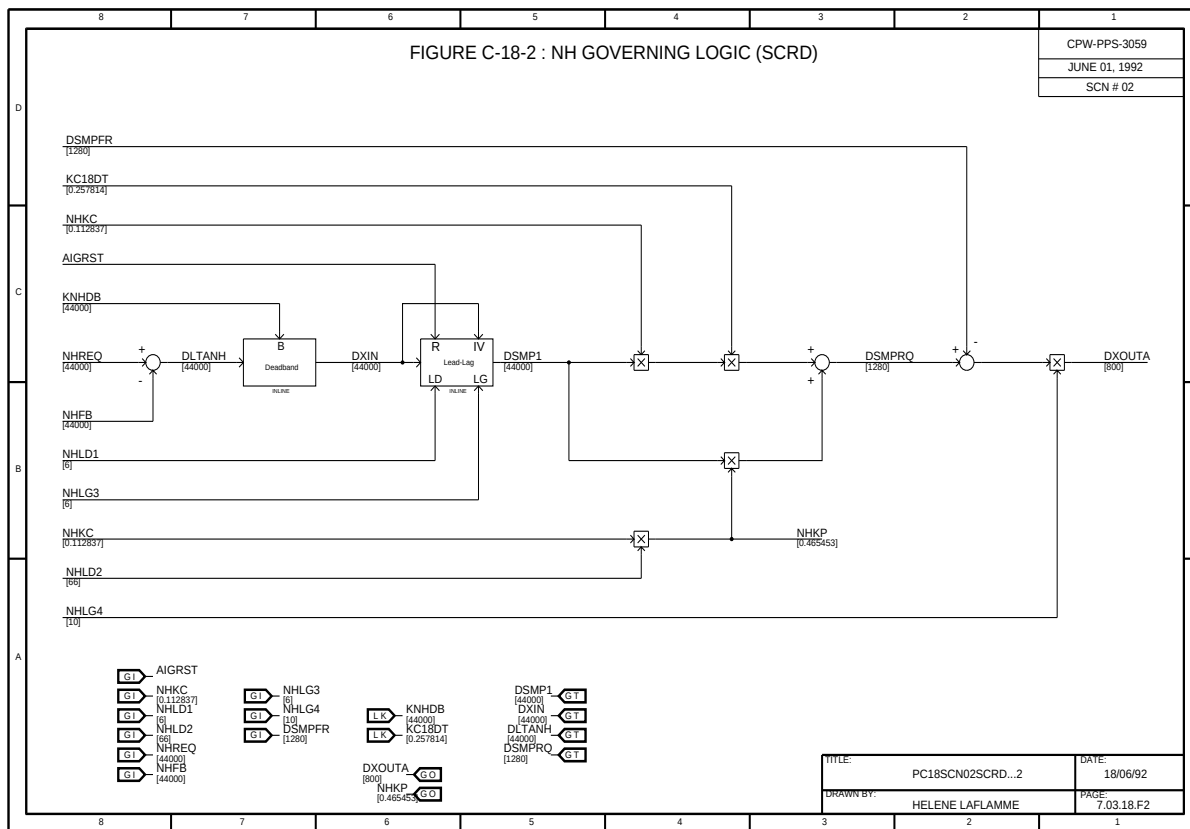
13.00 Figure 27: C17_NHREQUEST_2



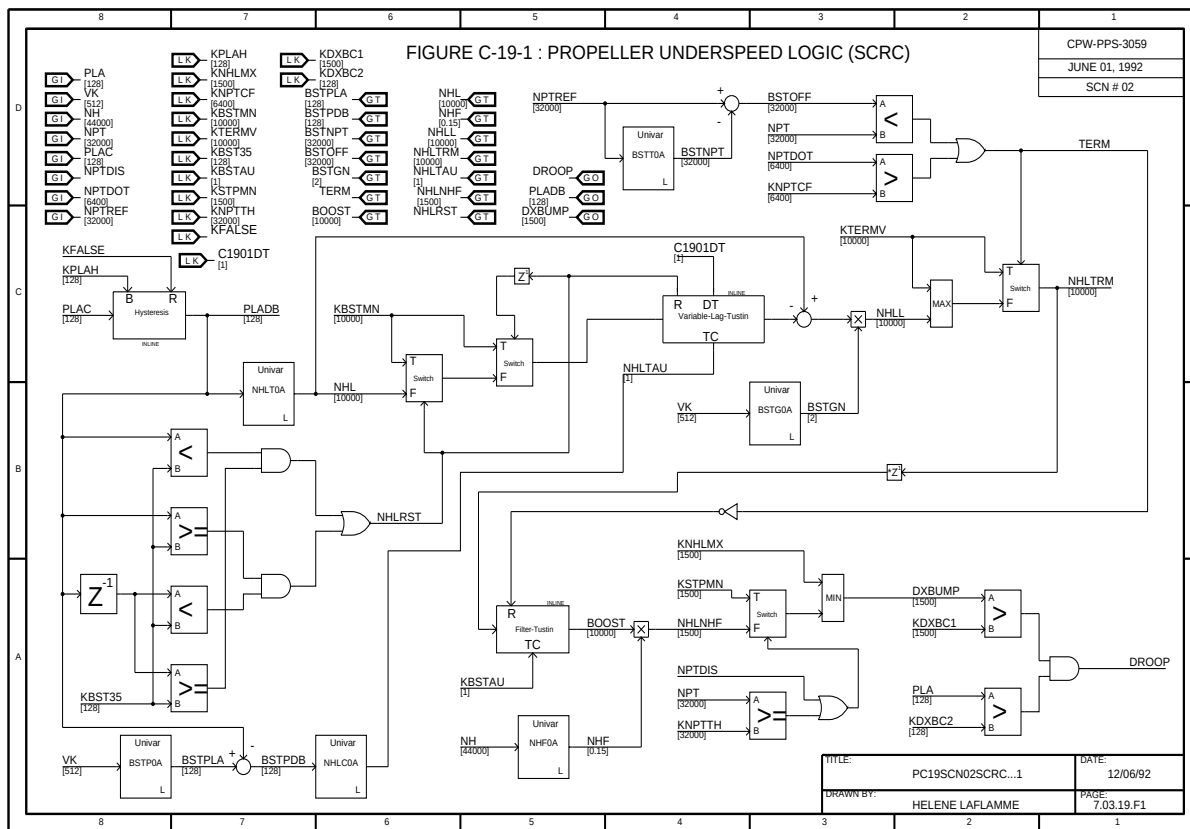
13.00 Figure 28: C18_NHGOV_1



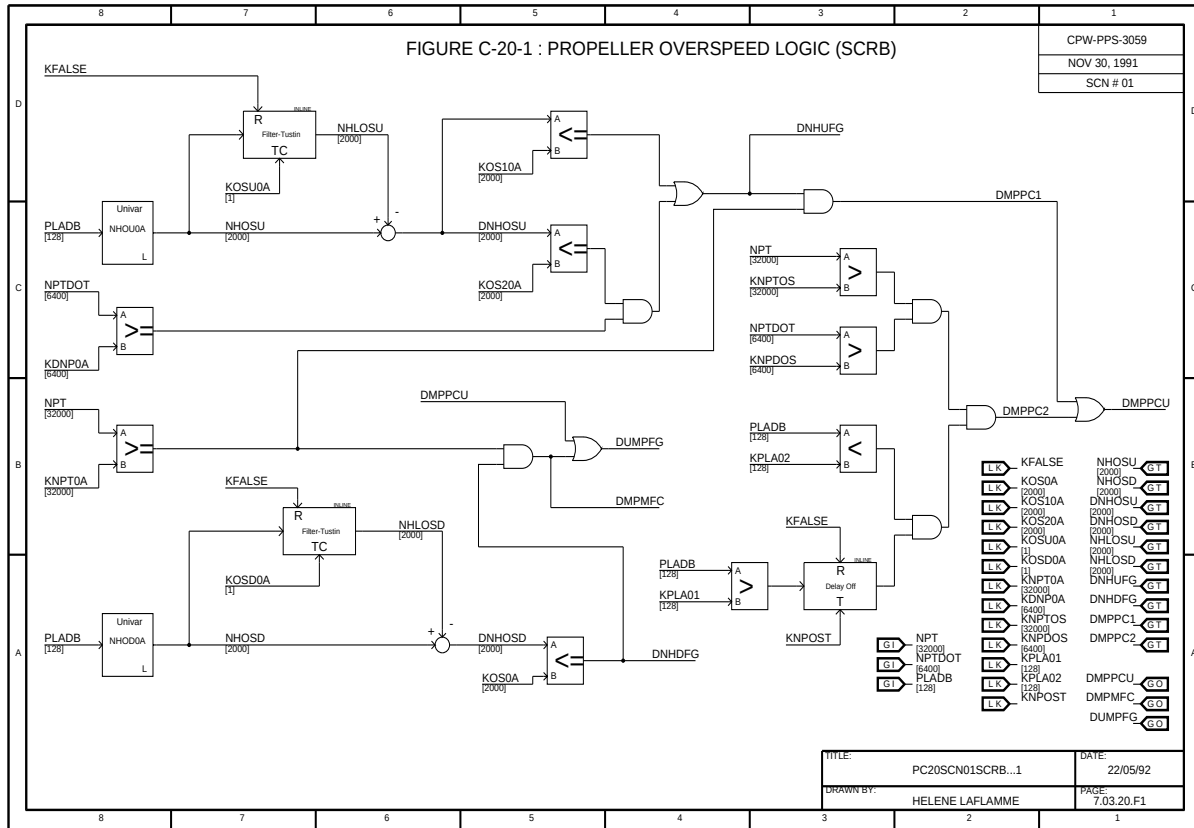
13.00 Figure 29: C18_NHGOV_2



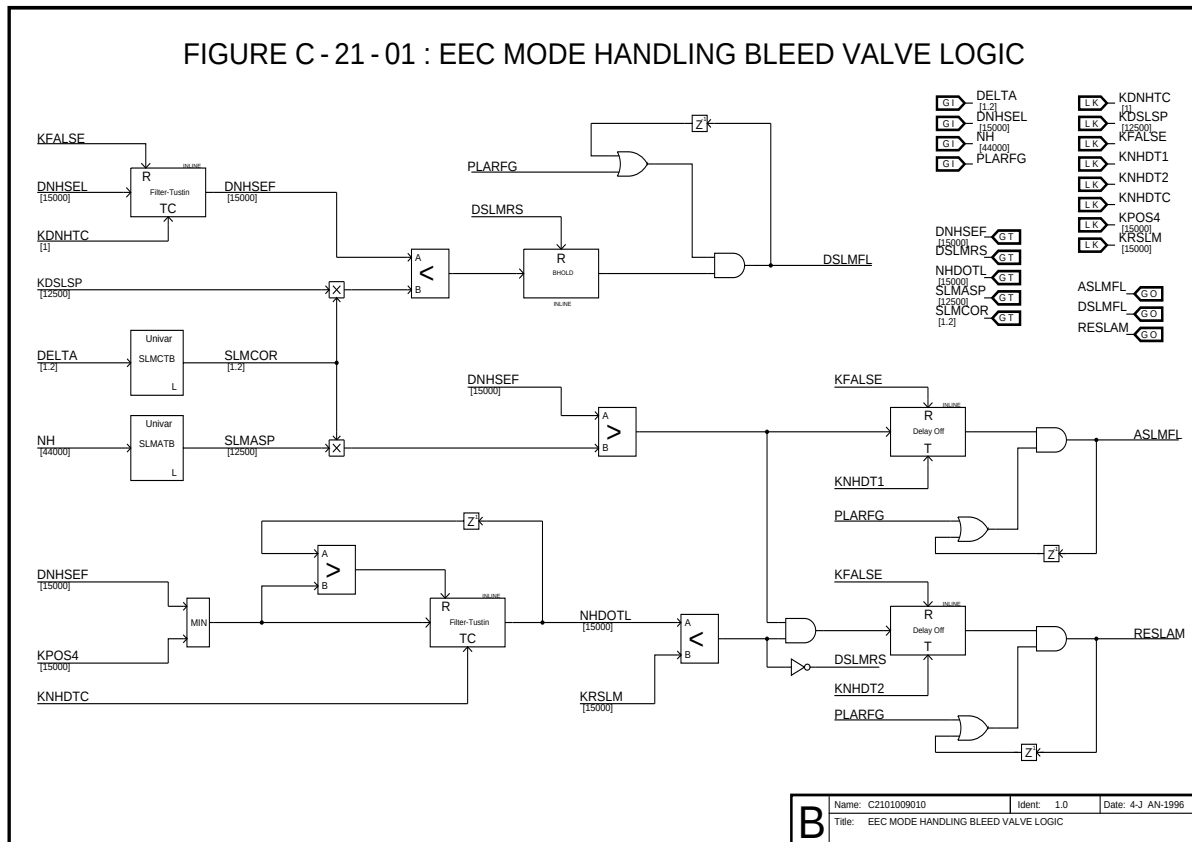
13.00 Figure 30: C19_UNDERSPEED



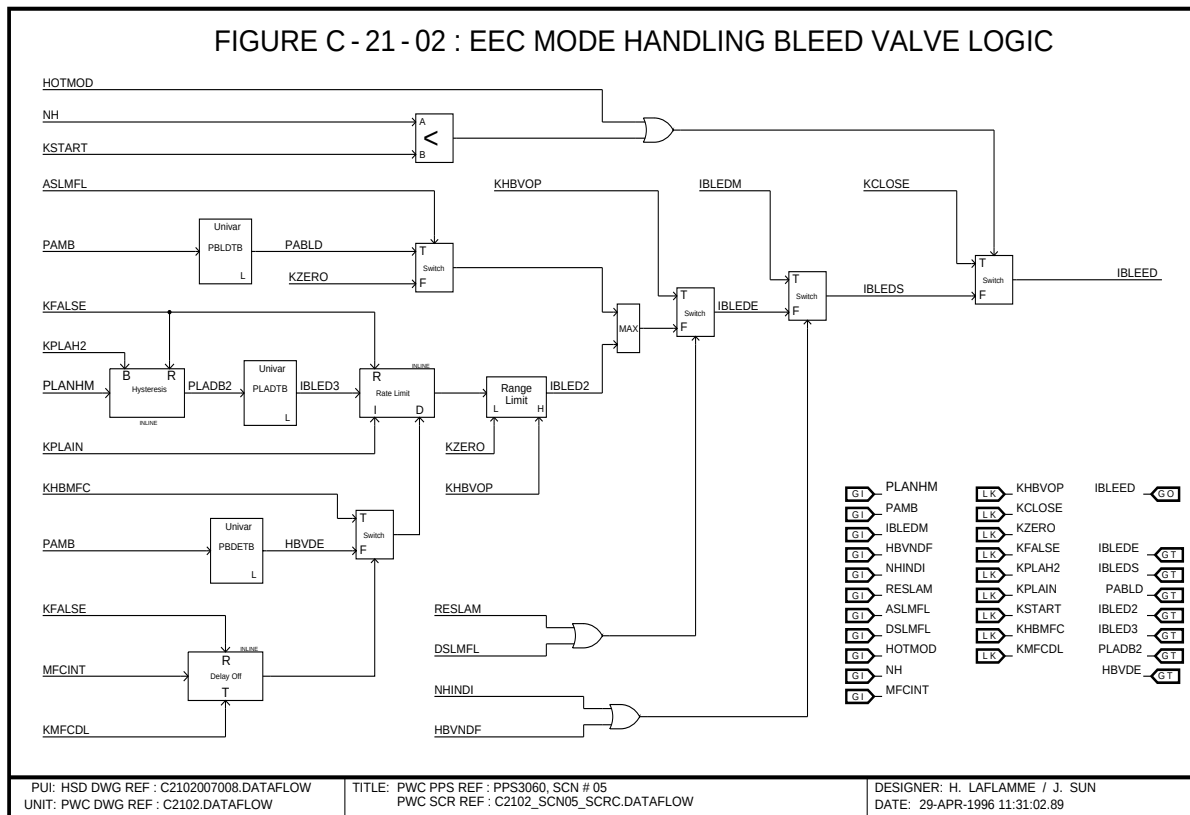
13.00 Figure 31: C20_OVSPEED



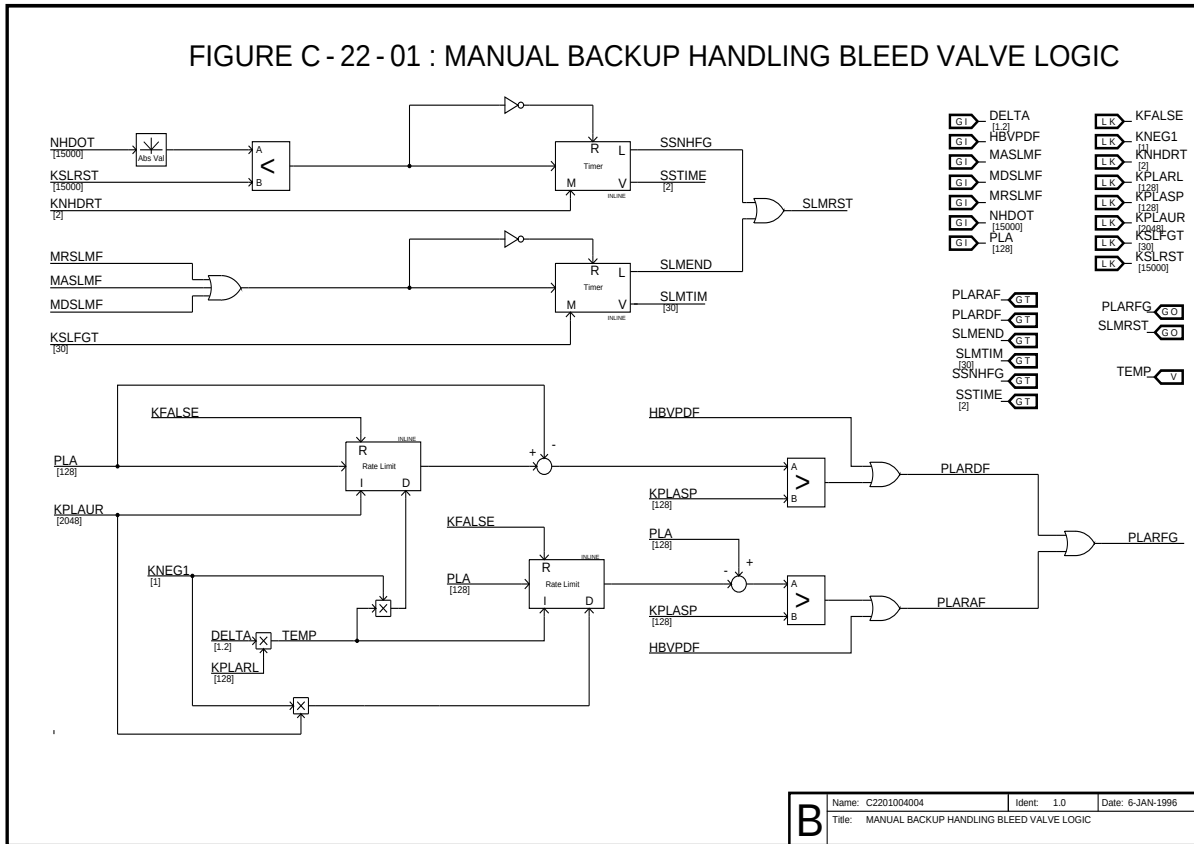
13.00 Figure 32: C21_HBOVEM_1



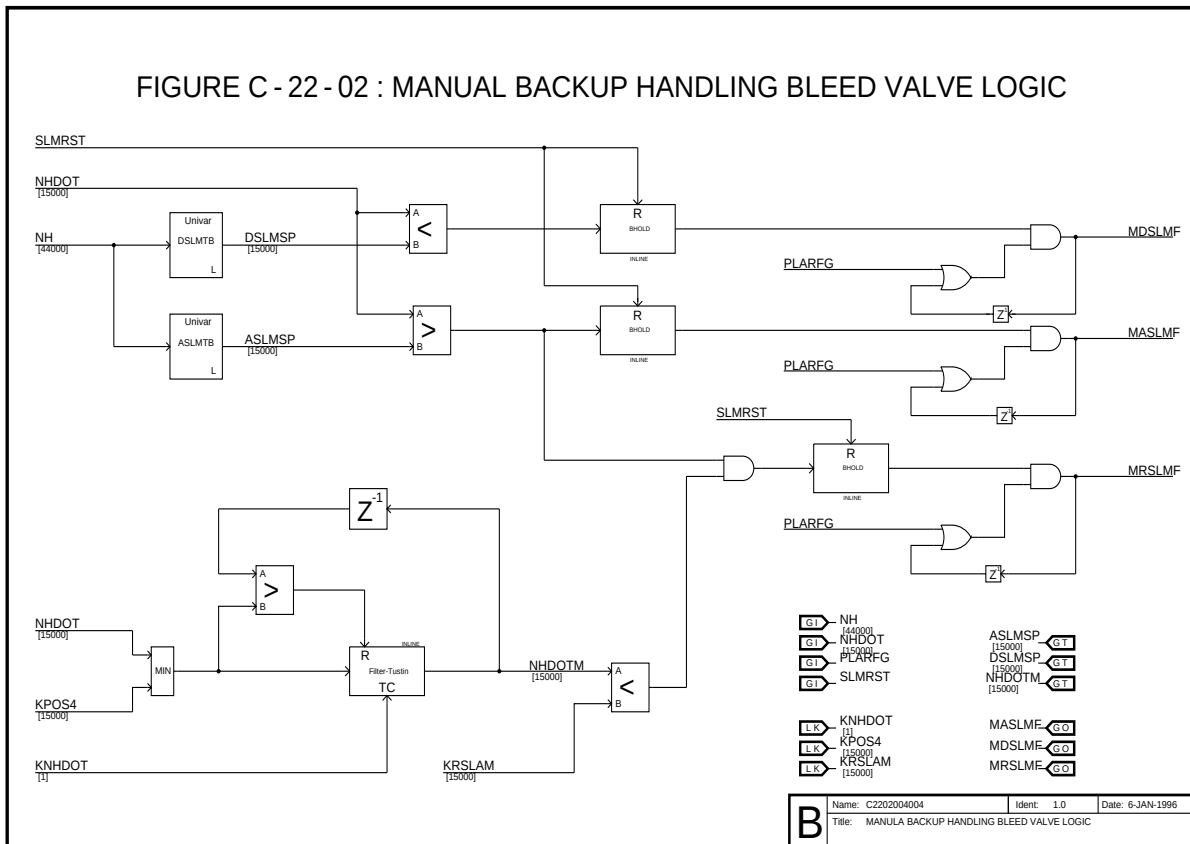
13.00 Figure 33: C21_HBOVEM_2



13.00 Figure 34: C22_HBOVMM_1



13.00 Figure 35: C22_HBOVMM_2



13.00 Figure 36: C22_HBOVMM_3

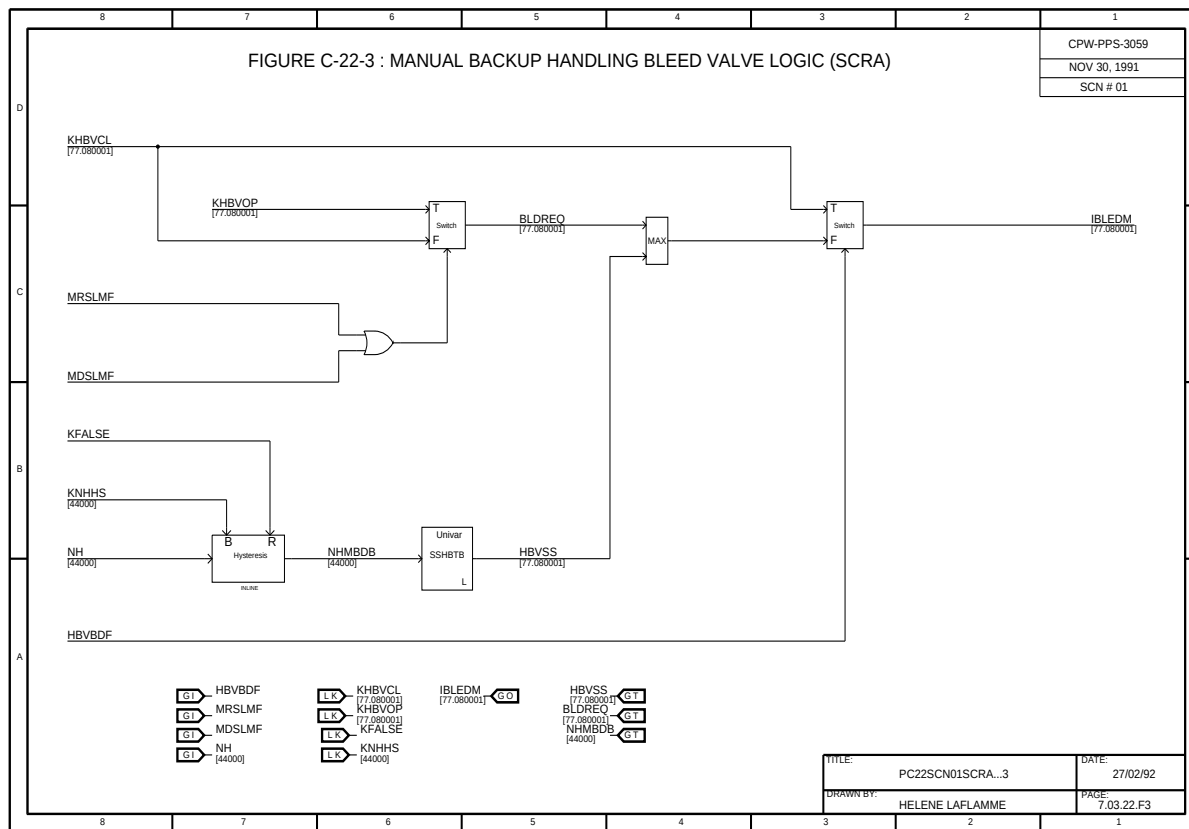
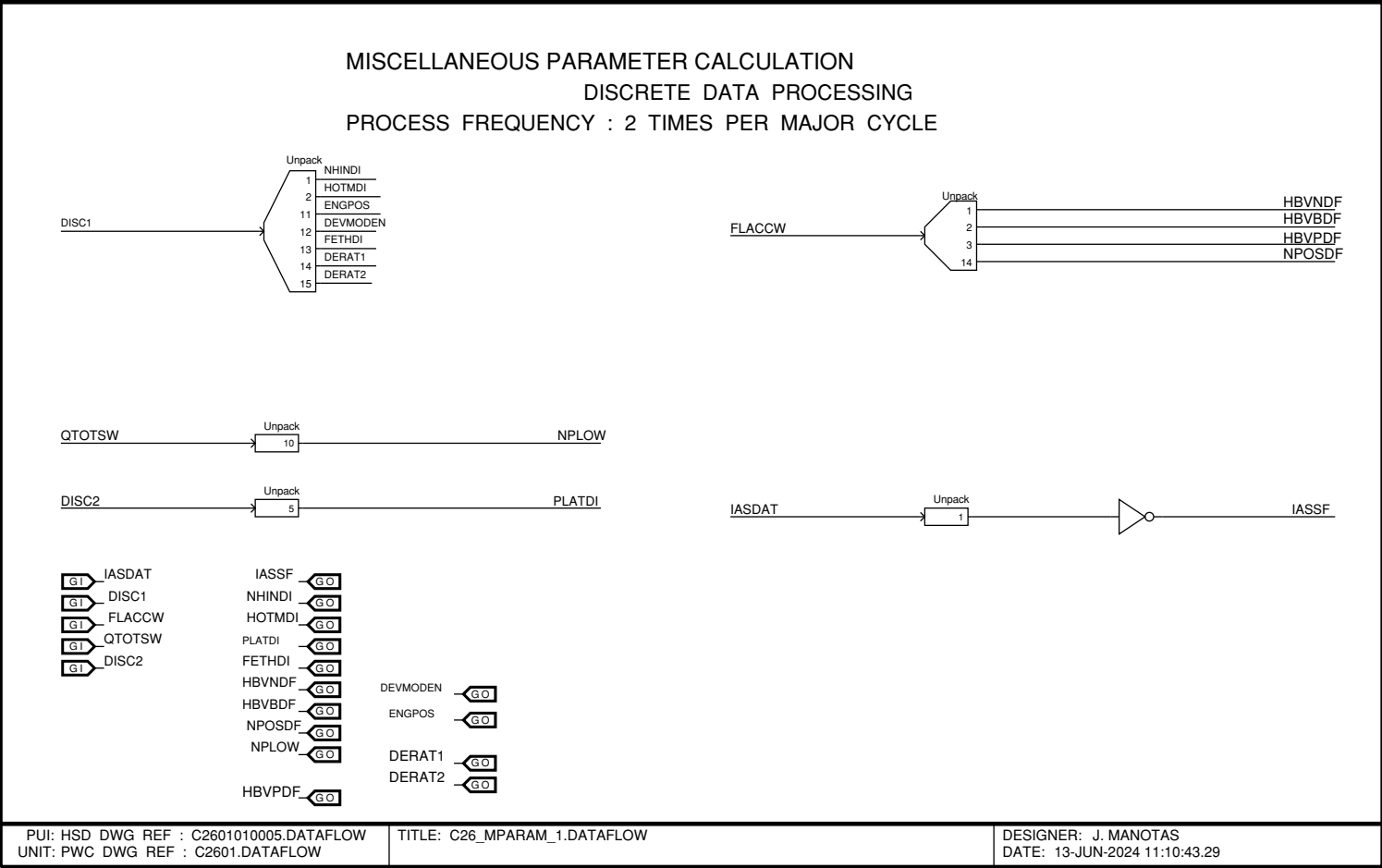


FIGURE C-24-01-FIGA : ENGINE OVERTORQUE TEST

FIGURE C-24-01-FIGB : ENGINE OVERTORQUE TEST

FIGURE C-24-01-FIGC : ENGINE OVERTORQUE TEST

13.00 Figure 38: C26_MPARAM_1



QTOTSW

Unpack

10

NFLOW

DISC2

Unpack

5

PLATDI

IASDAT

Unpack

1

AND

IASSF

IASDAT

DISC1

FLACCW

QTOTSW

DISC2

IASSF

NHINDI

HOTMDI

PLATDI

FETHDI

HBVNDF

HBVBDF

NPOSDF

NPLOW

HBVPDF

DEVMODEN

ENGPOS

DERAT1

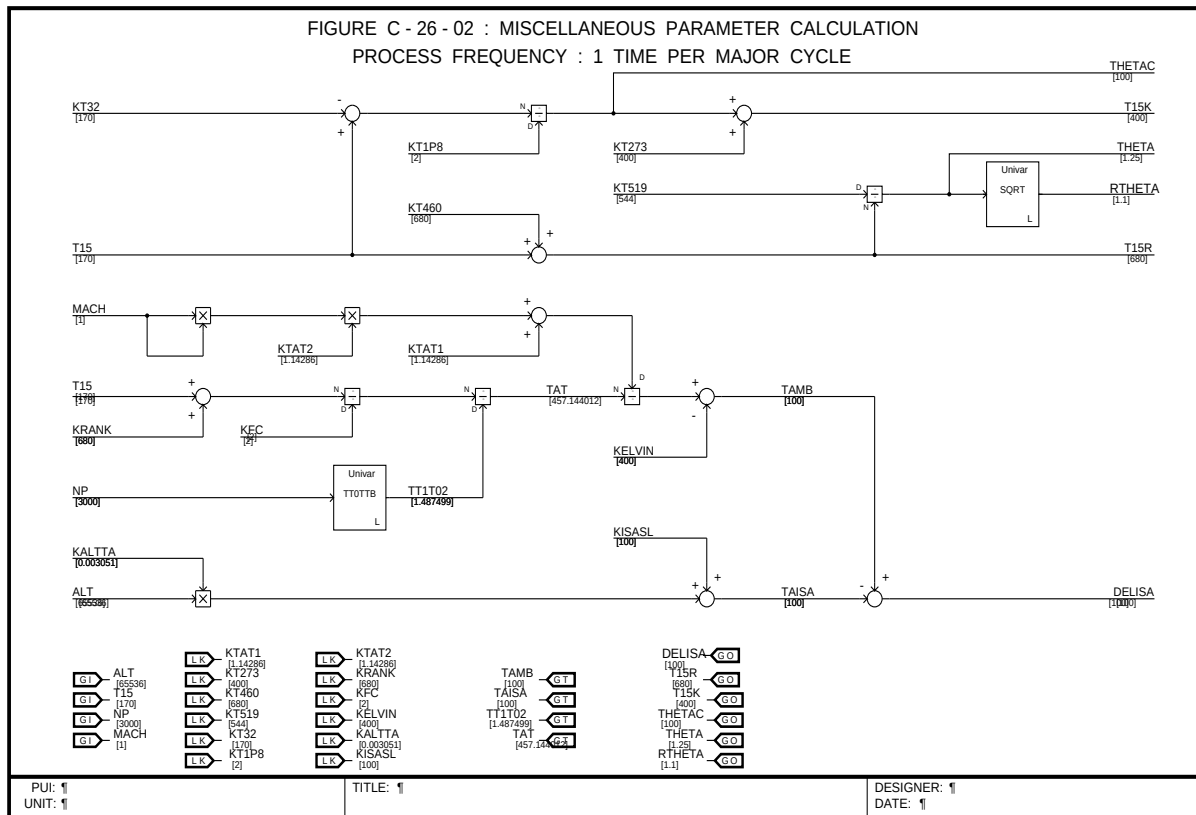
DERAT2

PUI: HSD DWG REF : C2601010005.DATAFLOW
UNIT: PWC DWG REF : C2601.DATAFLOW

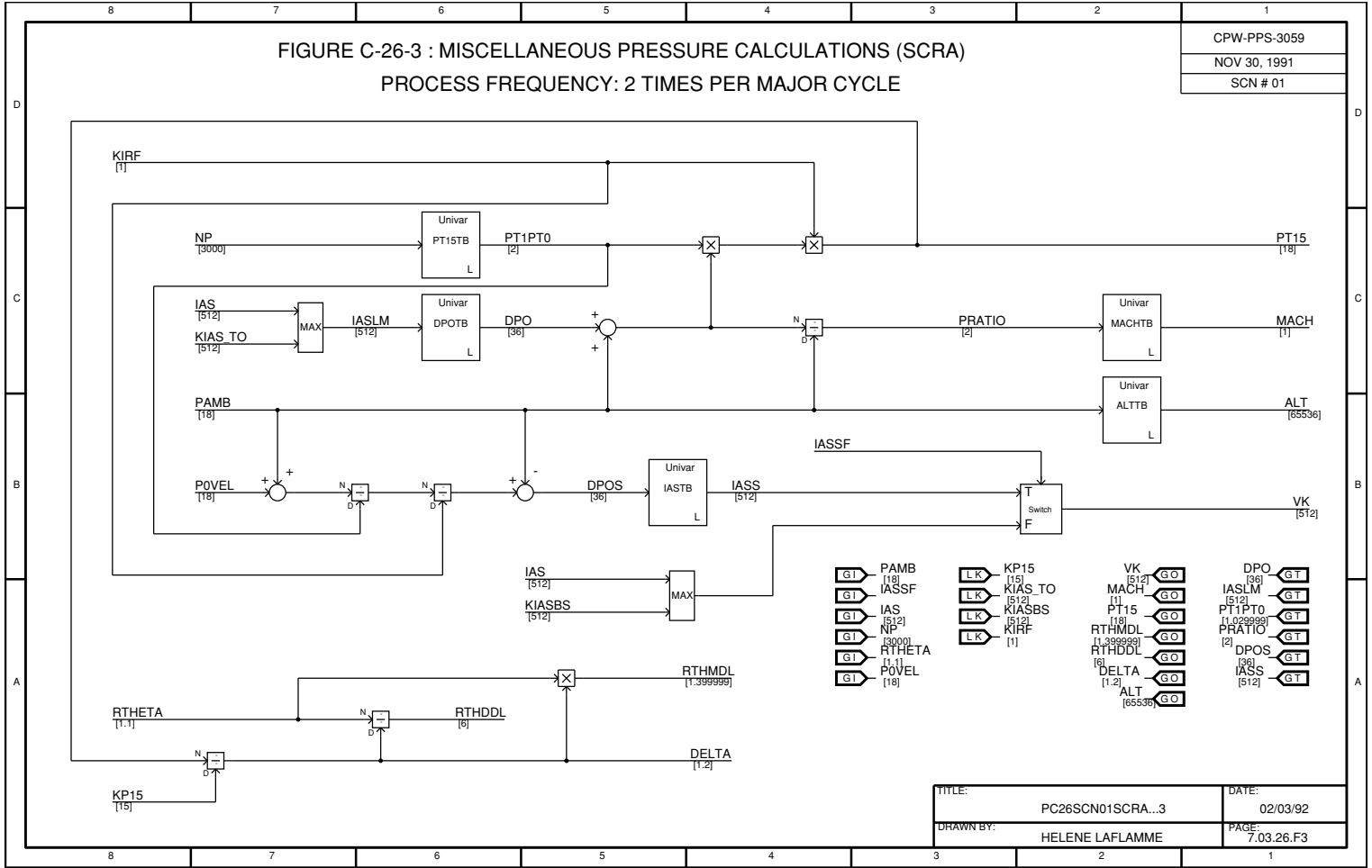
TITLE: C26_MPARAM_1.DATAFLOW

DESIGNER: J. MANOTAS
DATE: 13-JUN-2024 11:10:43.29

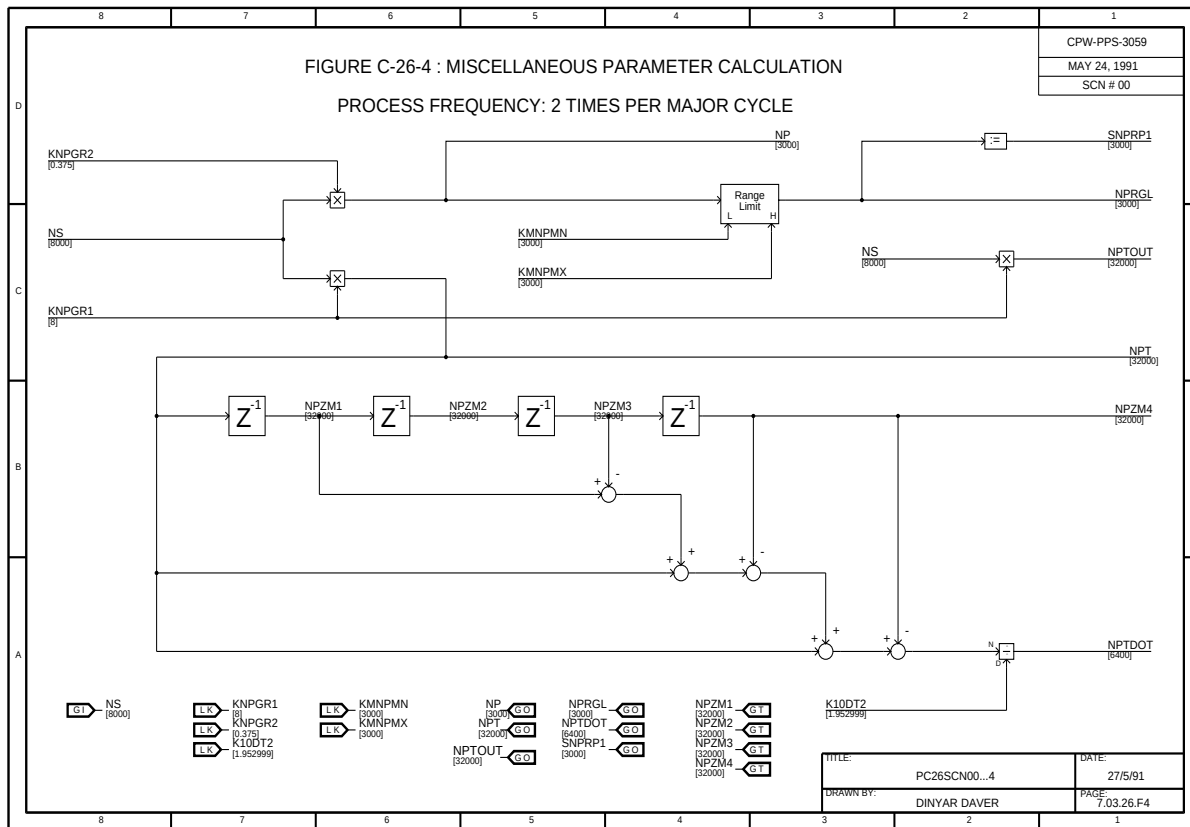
13.00 Figure 39: C26_MPARAM_2



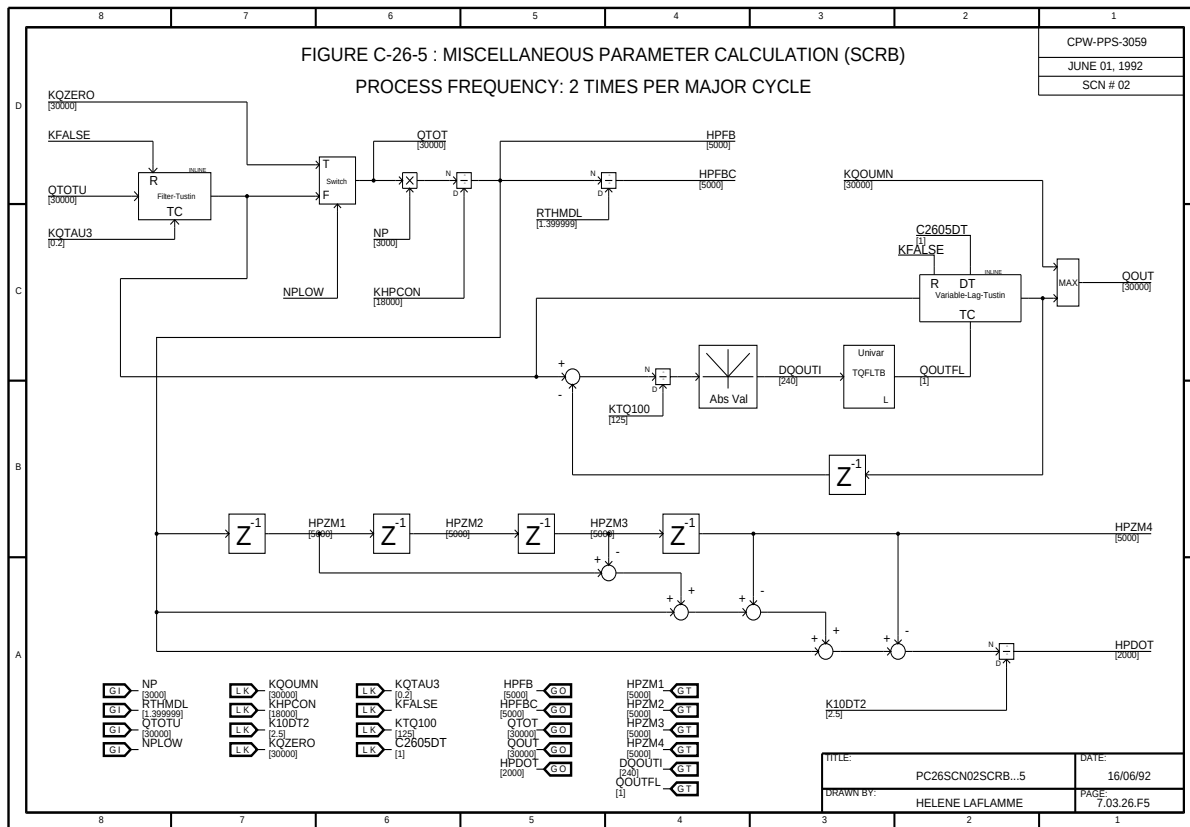
13.00 Figure 40: C26_MPARAM_3



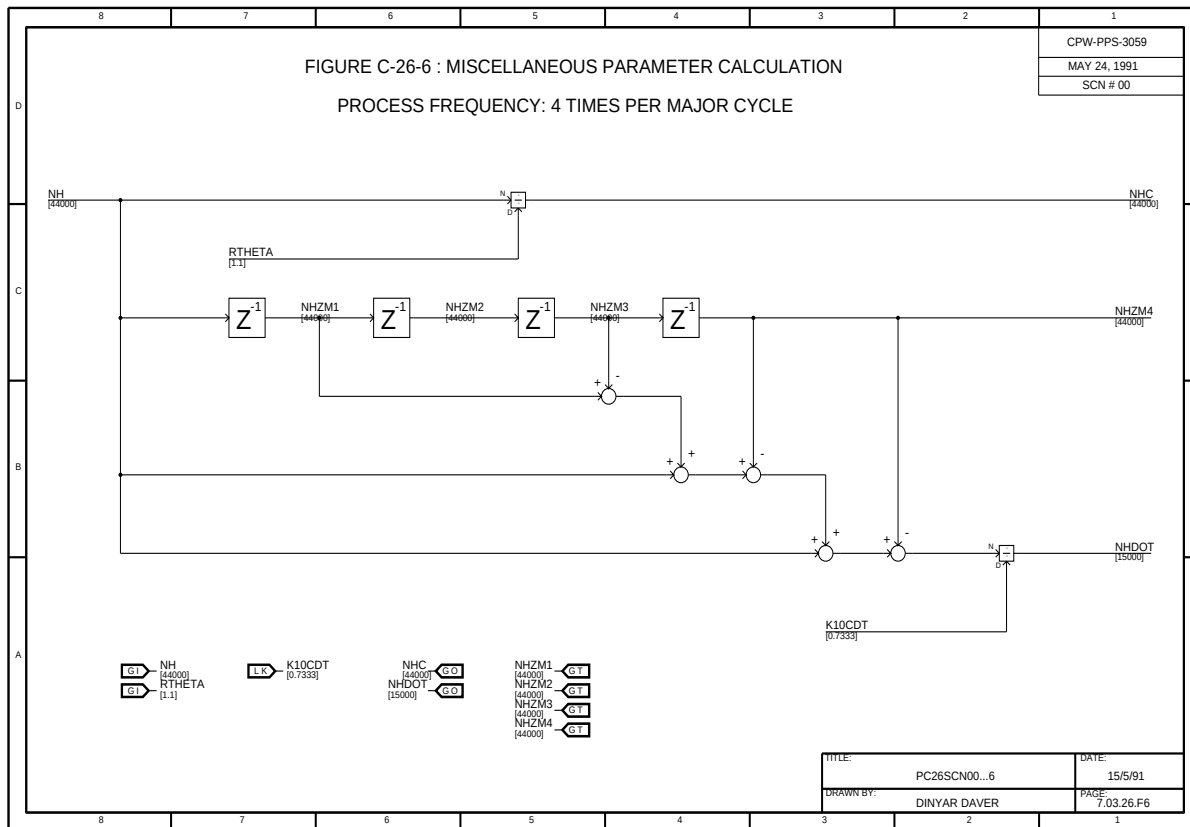
13.00 Figure 41: C26_MPARAM_4



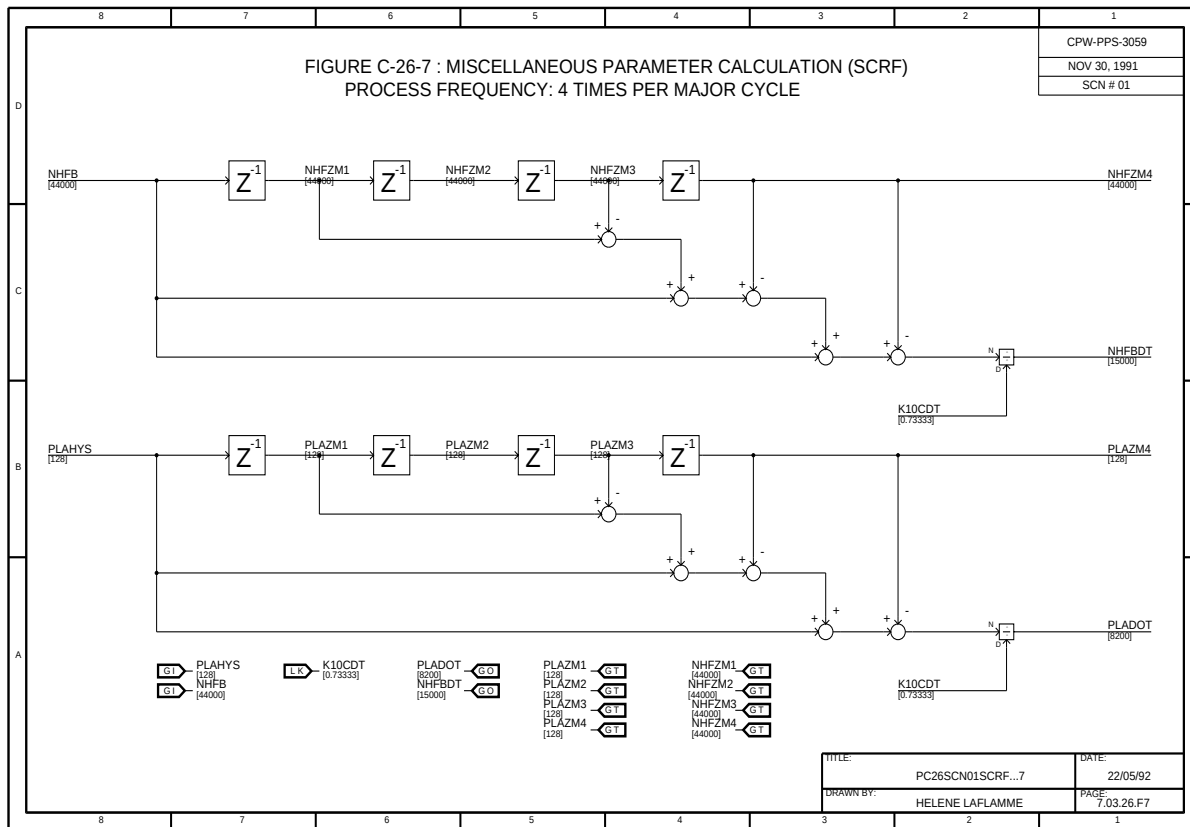
13.00 Figure 42: C26_MPARAM_5



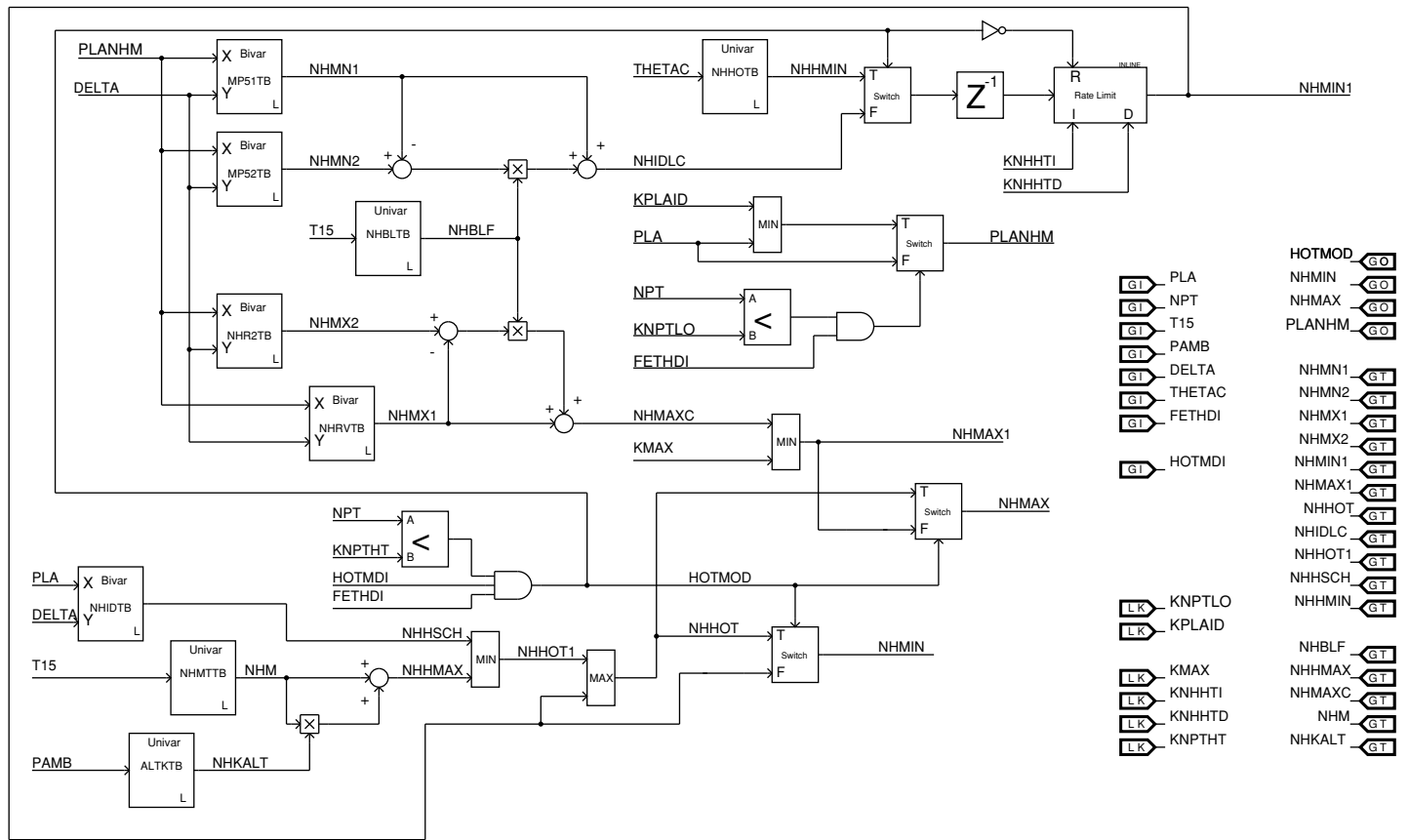
13.00 Figure 43: C26_MPARAM_6



13.00 Figure 44: C26_MPARAM_7



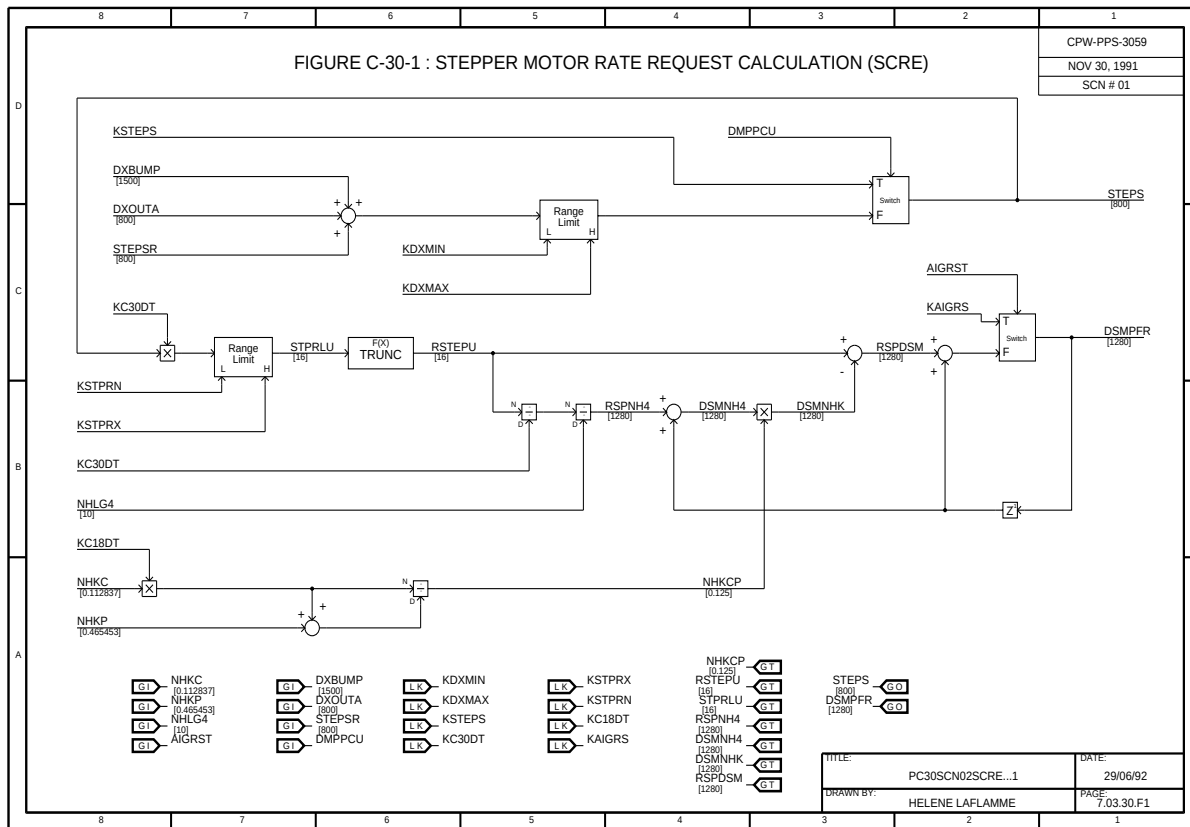
13.00 Figure 45: C27_NHMMX_1



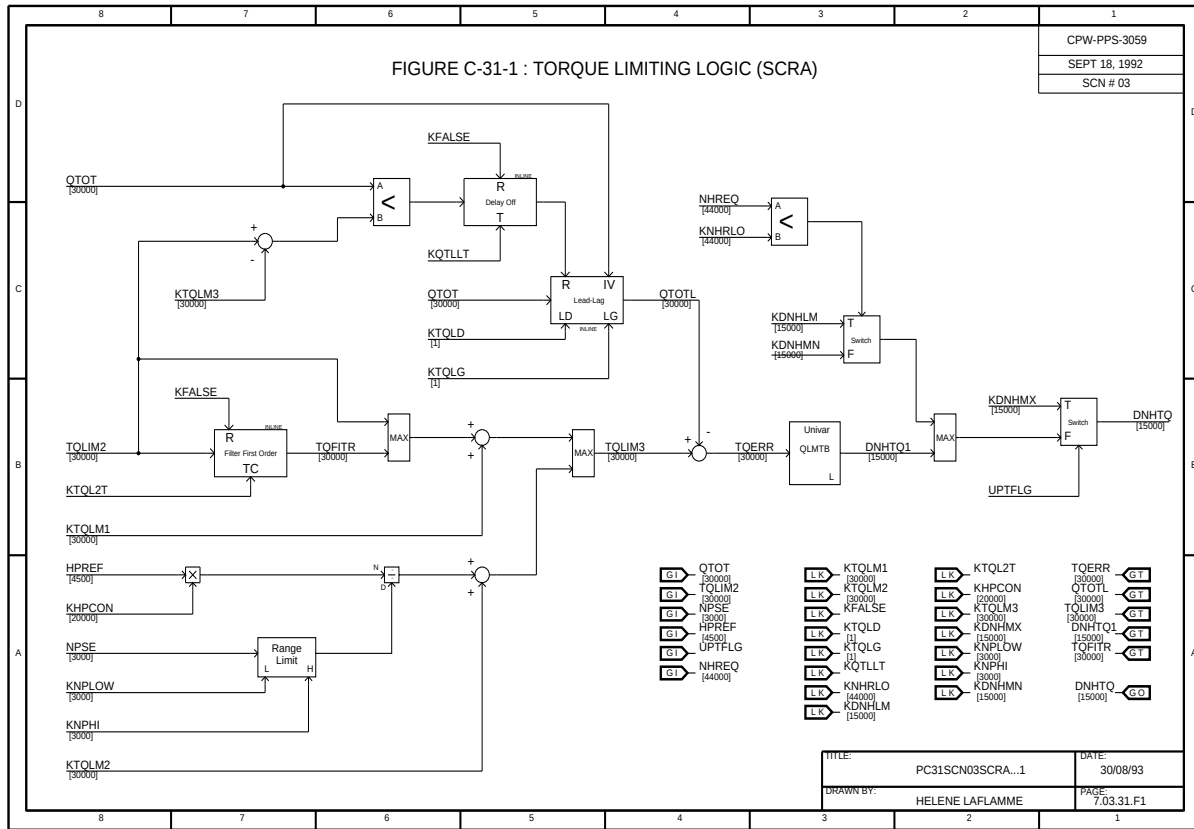
GO	PLA	GO	HOTMOD
GO	NPT	GO	NHMIN
GO	T15	GO	NHMAX
GO	PAMB	GO	PLANHM
GO	DELTA	GT	NHMIN1
GO	THETAC	GT	NHMIN2
GO	FETHDI	GT	NHMX1
GO		GT	NHMX2
		GT	NHMIN1
		GT	NHMAX1
		GT	NHHOT
		GT	NHIDLC
		GT	NHHOT1
		GT	NHHSCH
		GT	NHHMIN
LK	KNPTLO	GT	NHBLF
LK	KPLAID	GT	NHMAX
LK	KMAX	GT	NHMAXC
LK	KNHHTI	GT	NH
LK	KNHHTD	GT	NHKALT
LK	KNPTHT		

B	Name: C27_NHMMX_1	Ident:	Date: 1
	Title: NH Minimum / Maximum Calculation		20-JUN-2024 14:38:42.93

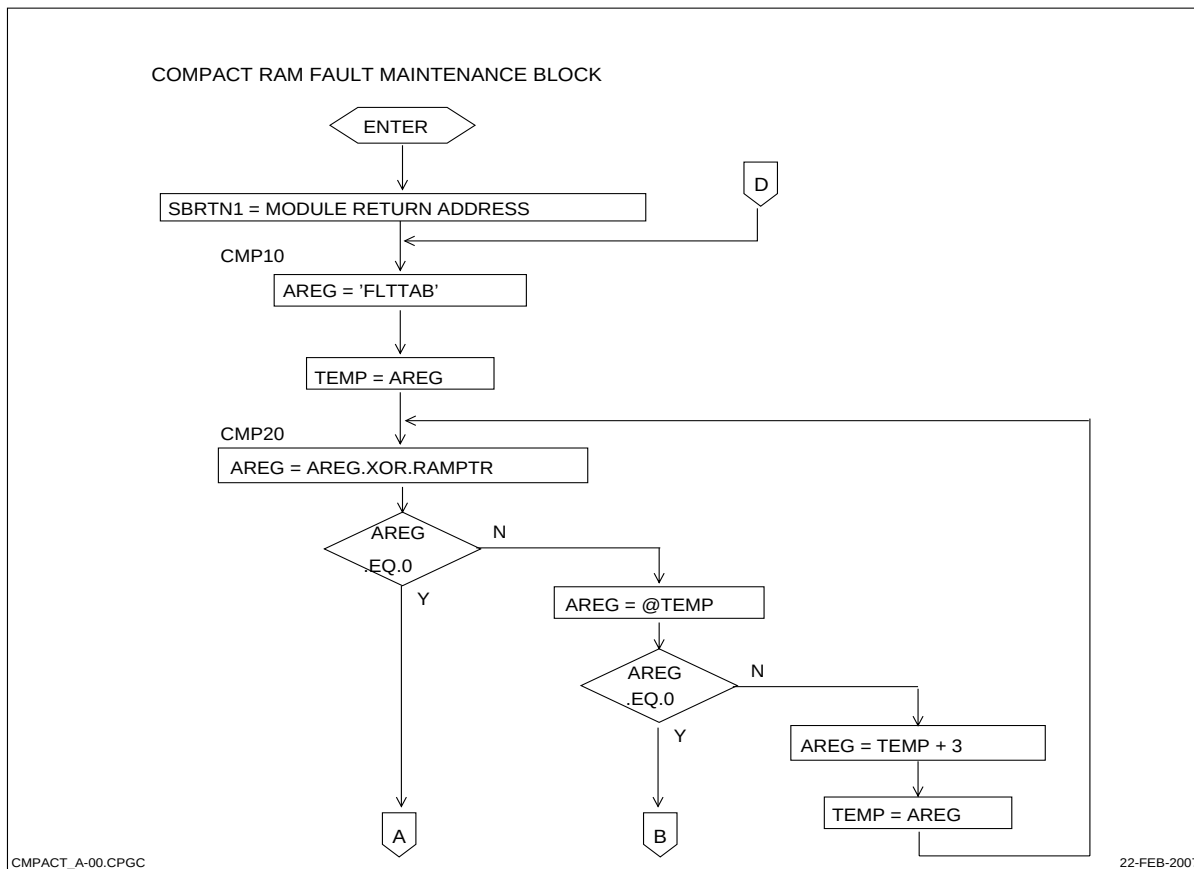
13.00 Figure 47: C30_SMRTREQ



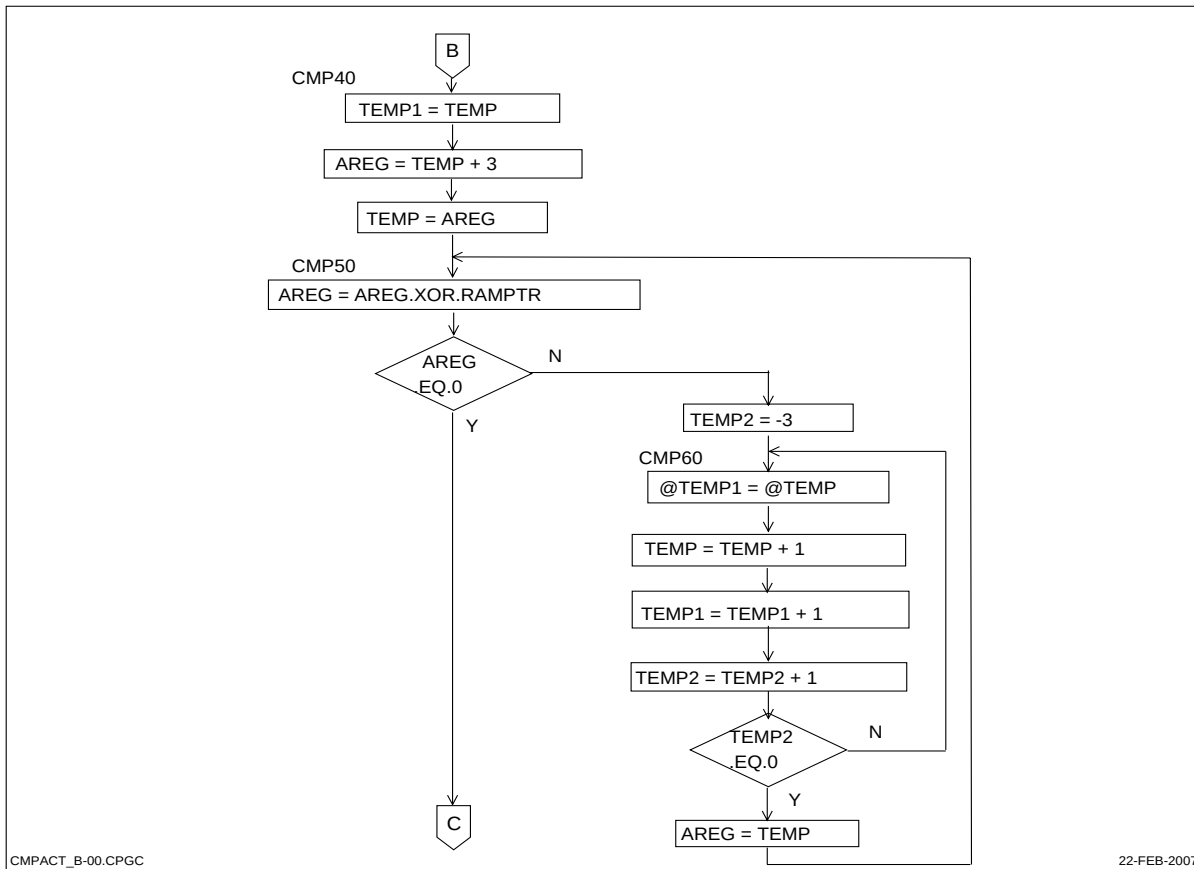
13.00 Figure 48: C31_TQLIM



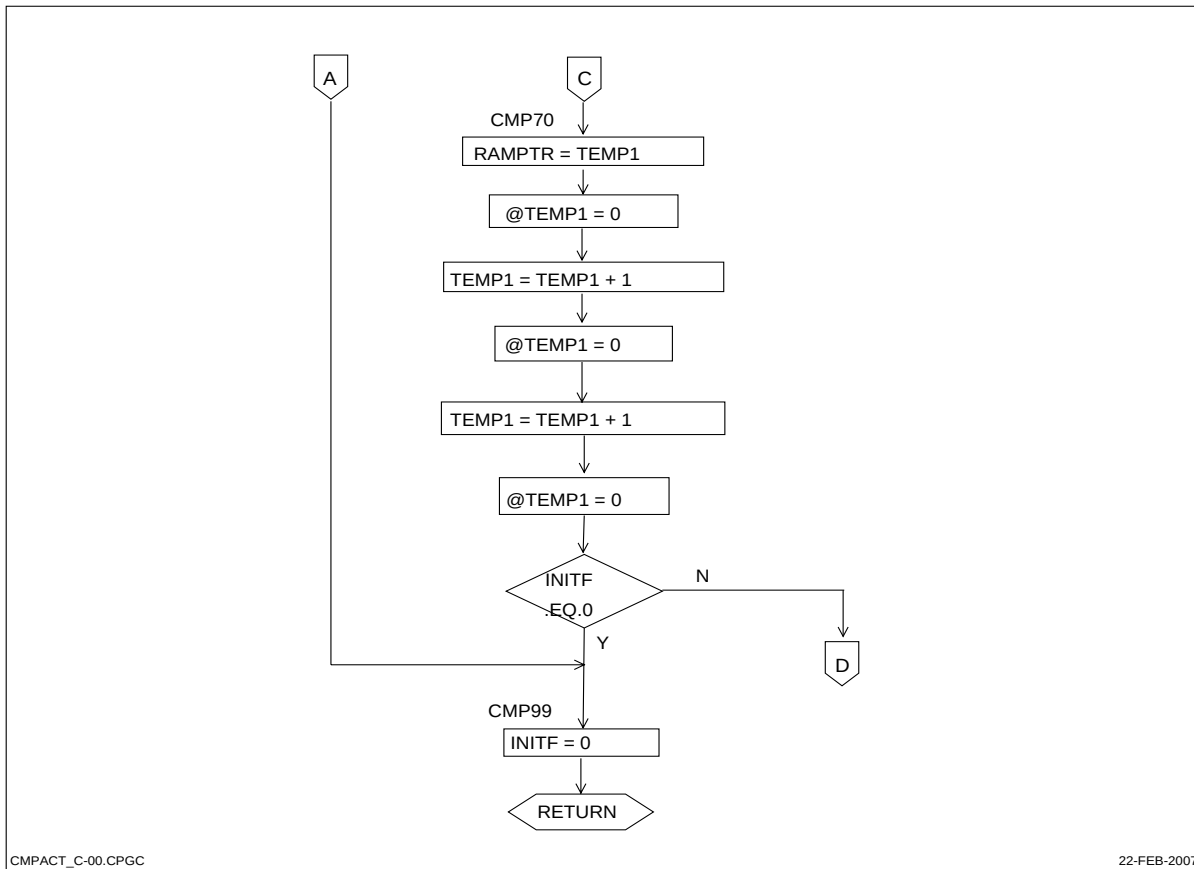
13.00 Figure 49: CMPACT_A



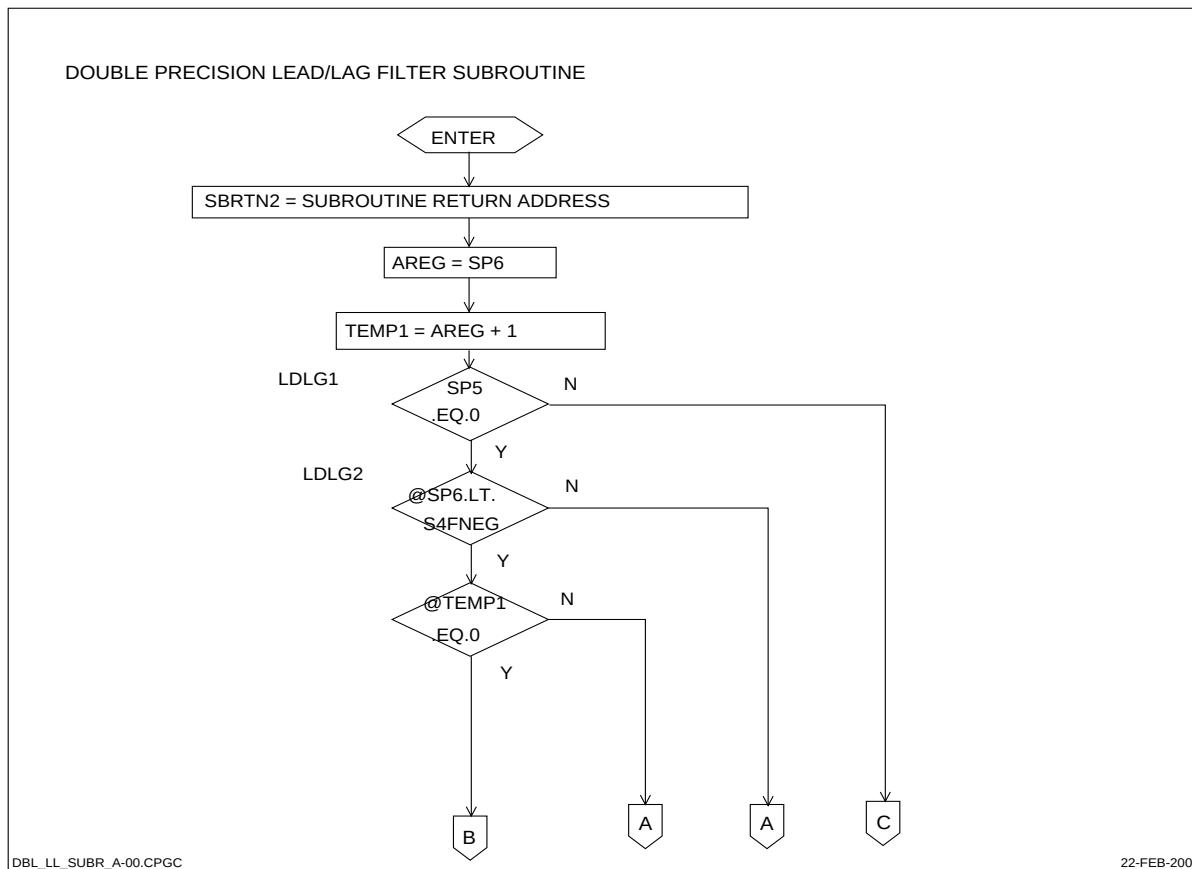
13.00 Figure 50: CMPACT_B



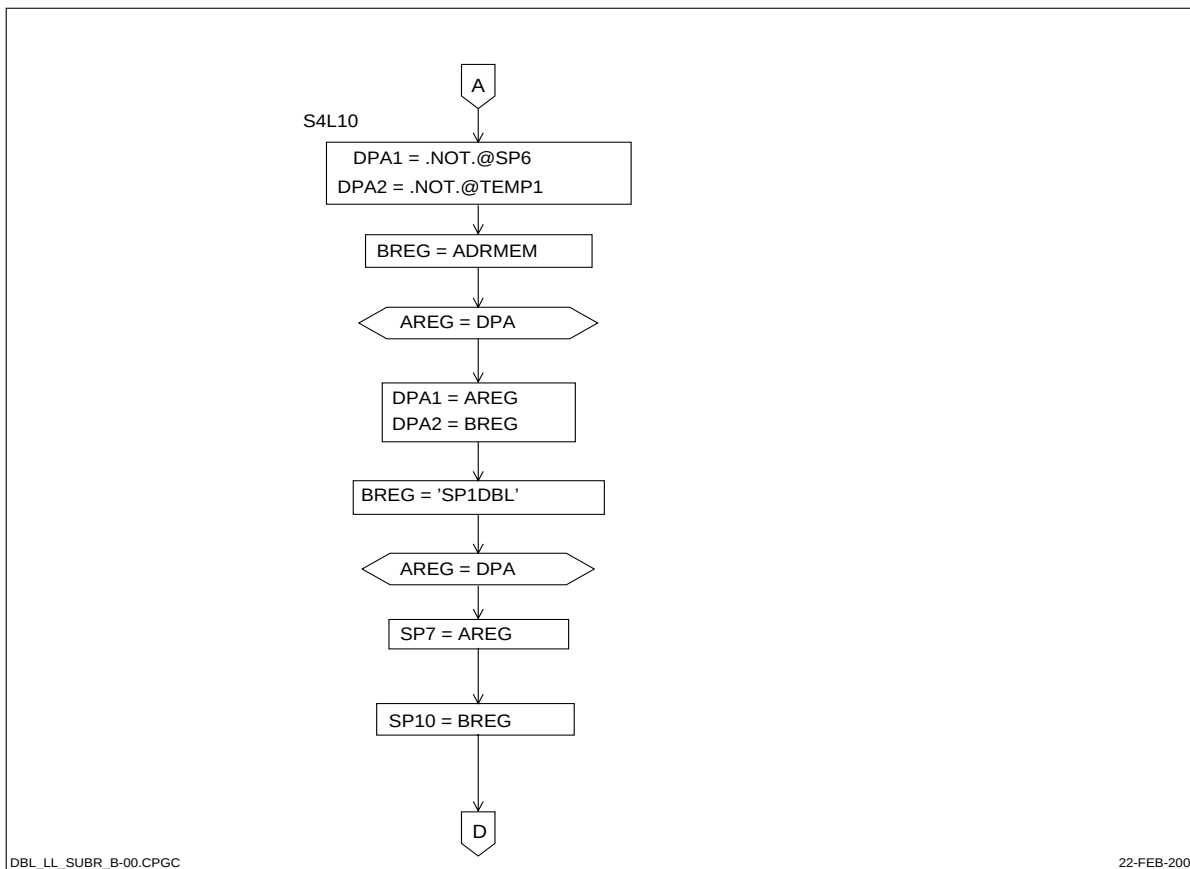
13.00 Figure 51: CMPACT_C



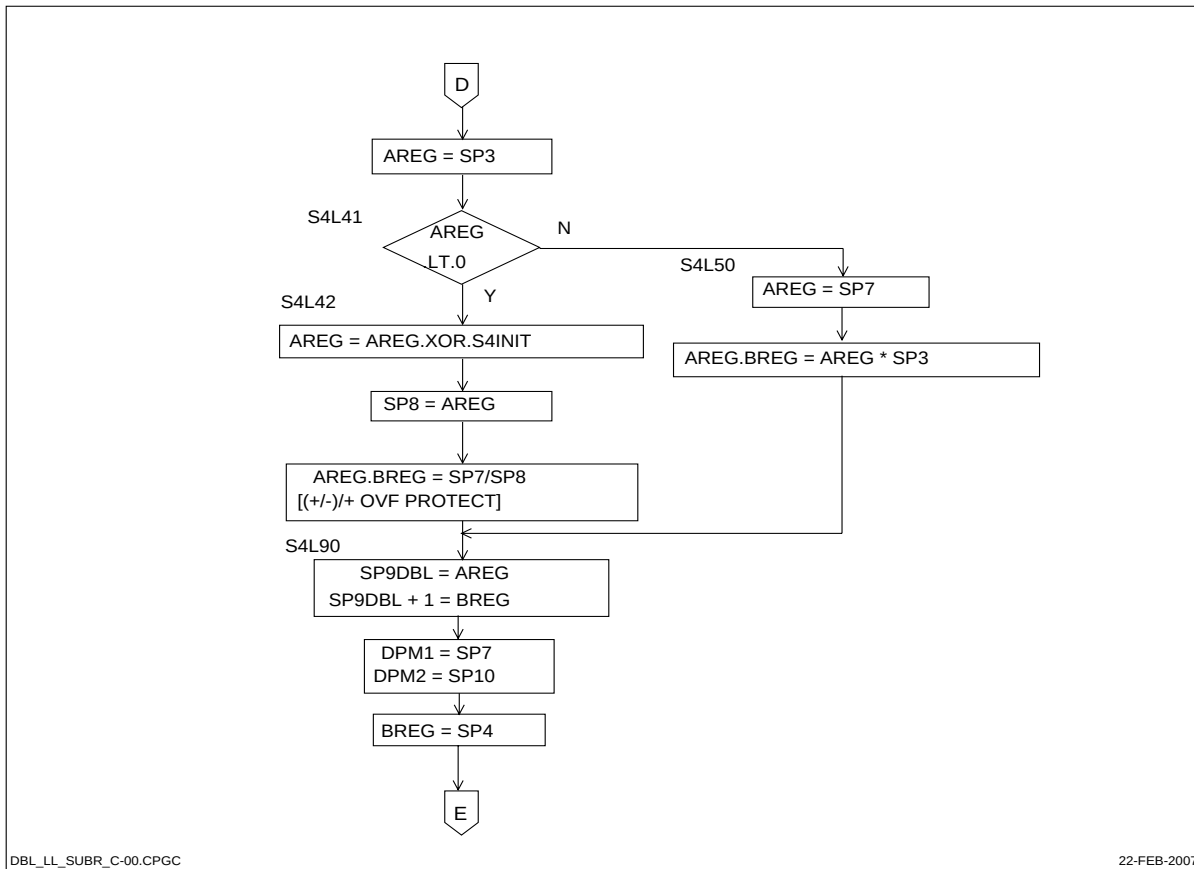
13.00 Figure 52: DBL_LL_SUBR_A



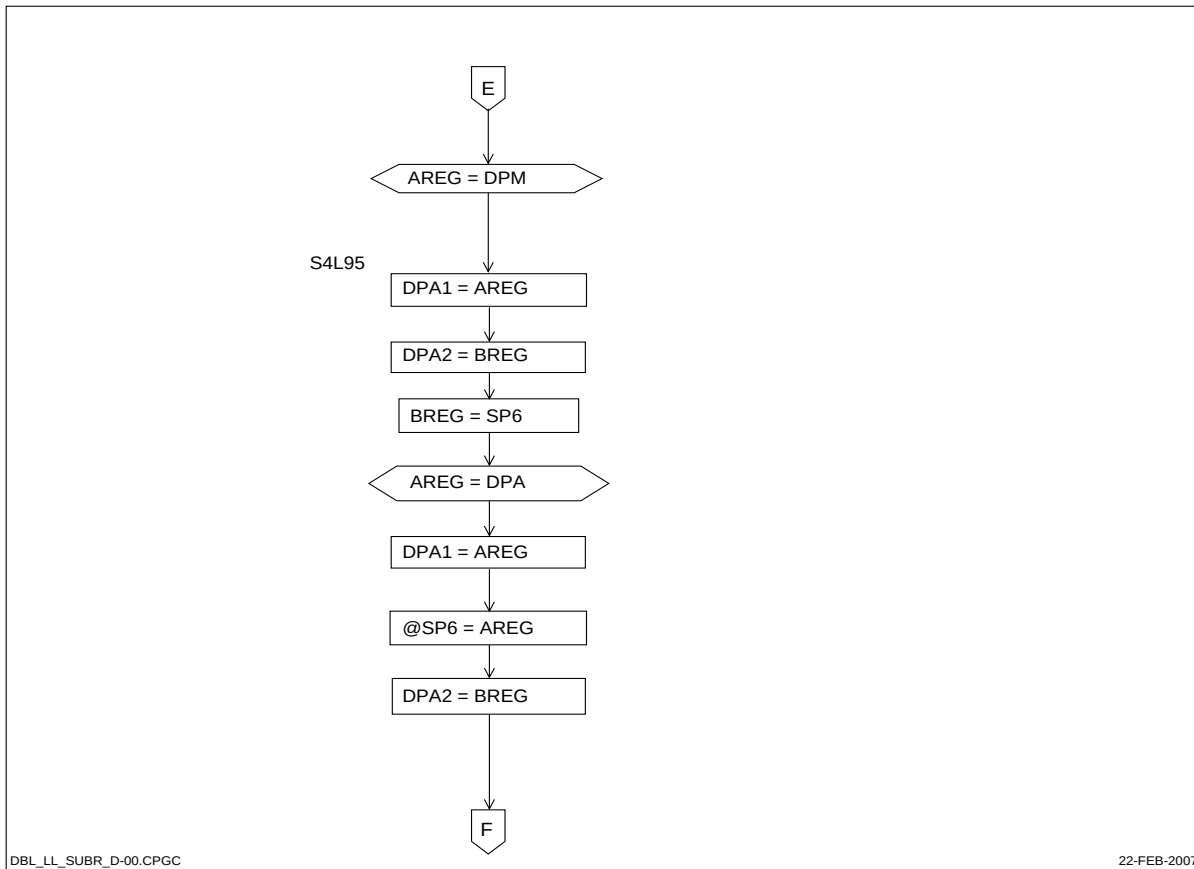
13.00 Figure 53: DBL_LL_SUBR_B



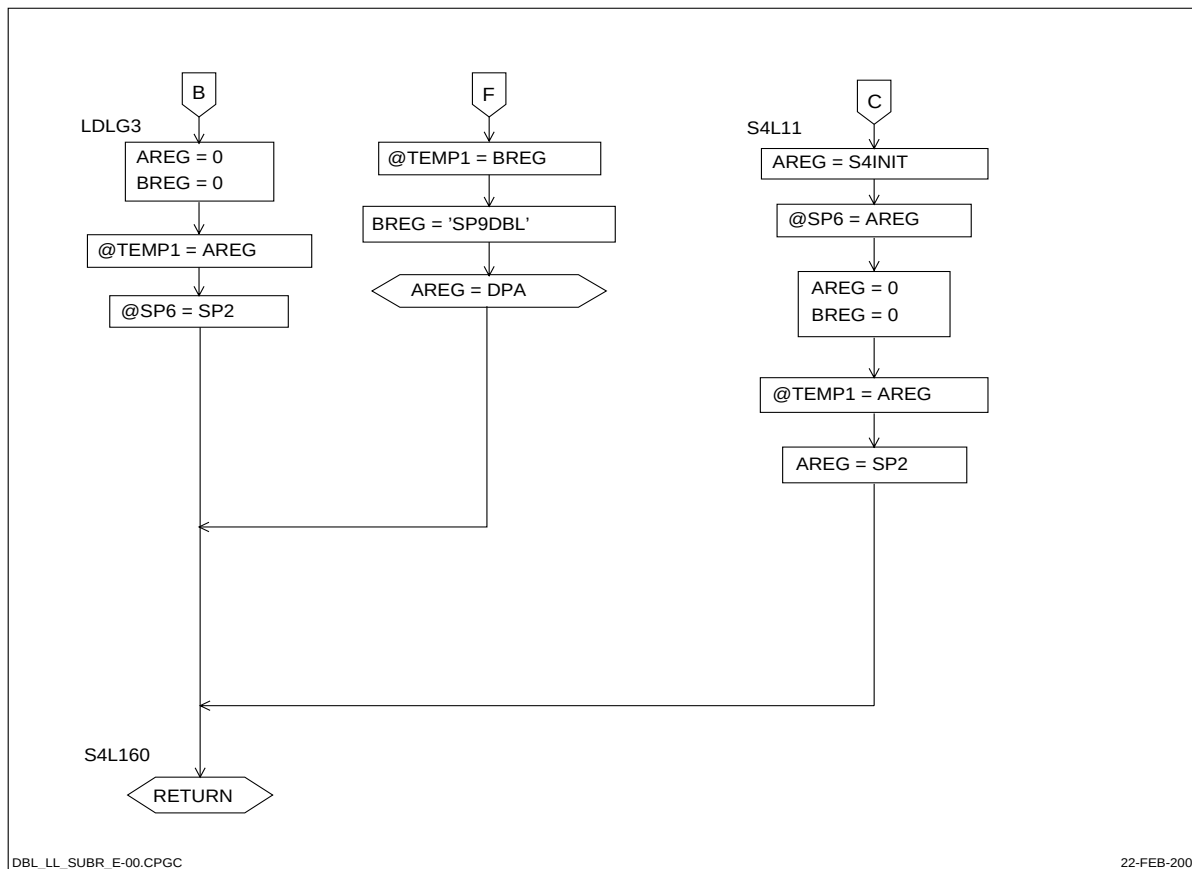
13.00 Figure 54: DBL_LL_SUBR_C



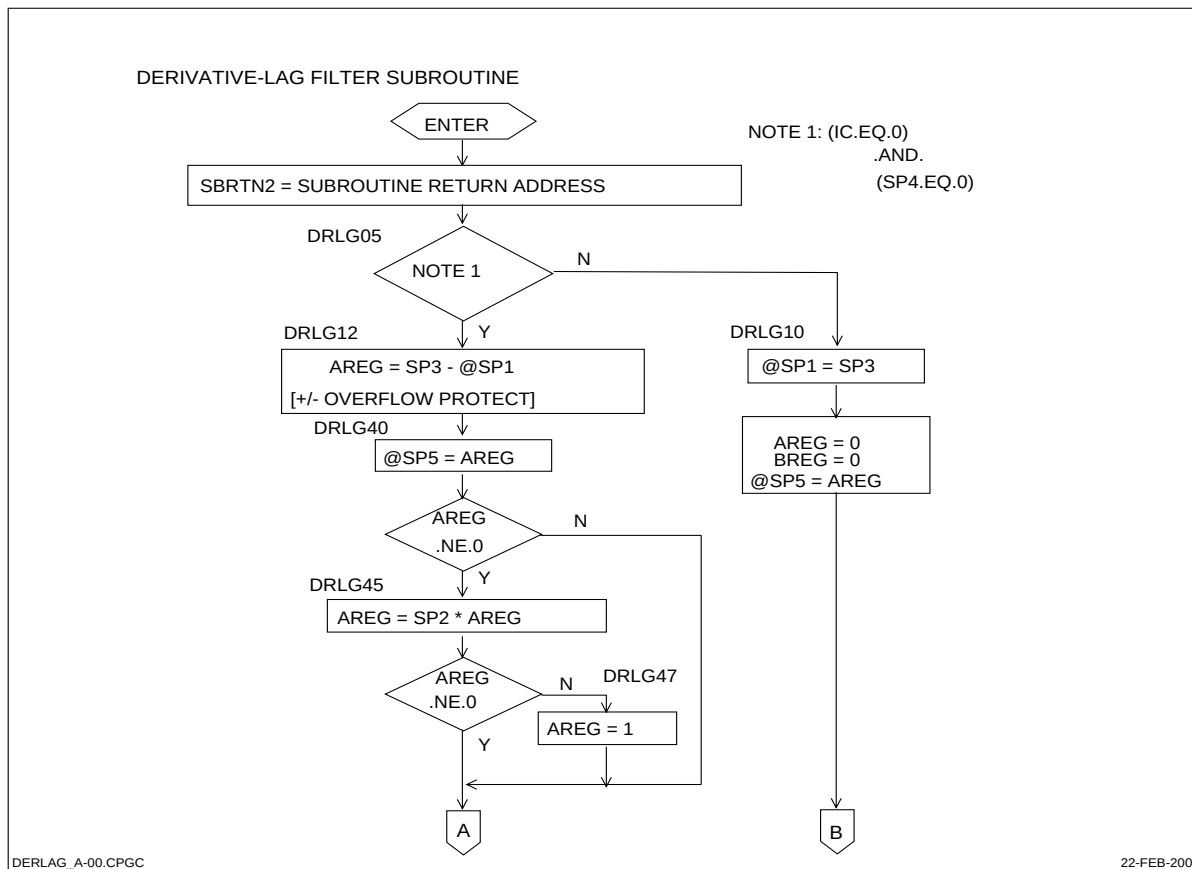
13.00 Figure 55: DBL_LL_SUBR_D



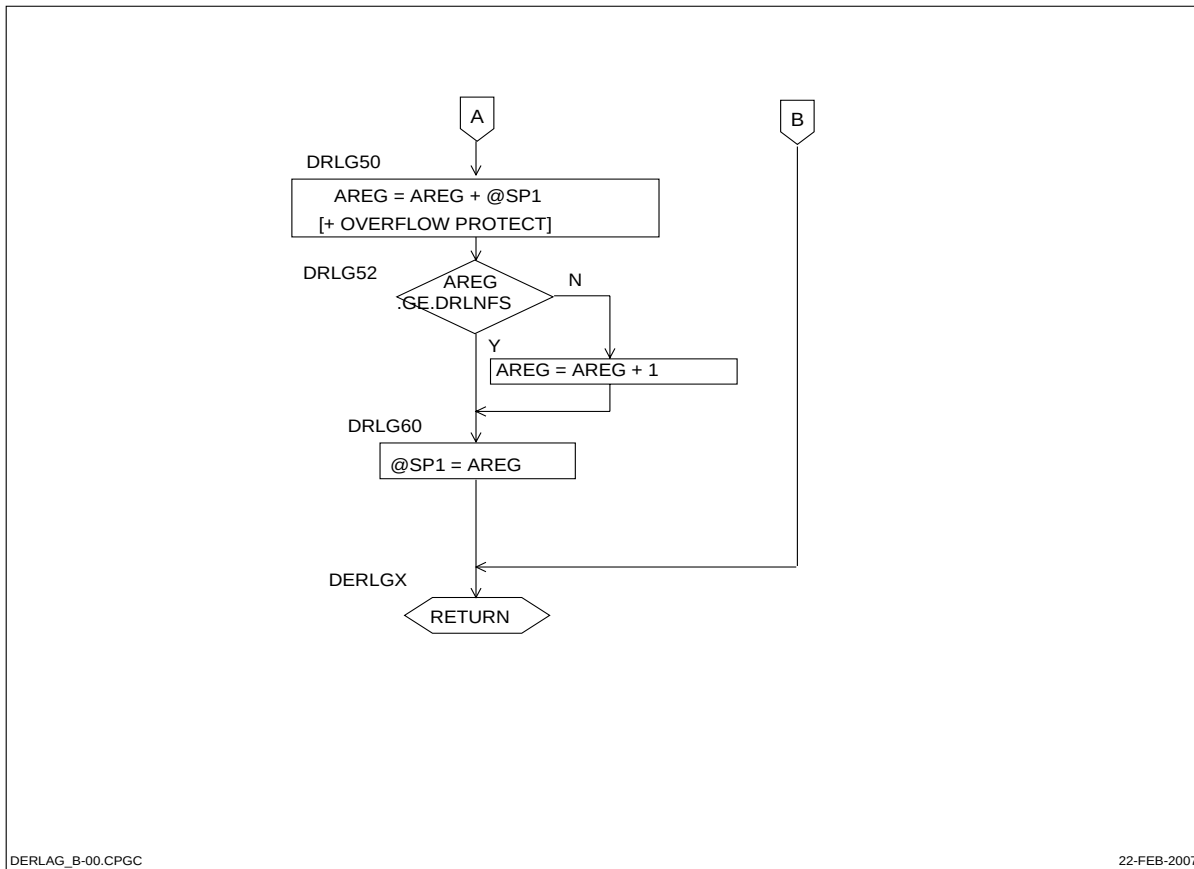
13.00 Figure 56: DBL_LL_SUBR_E



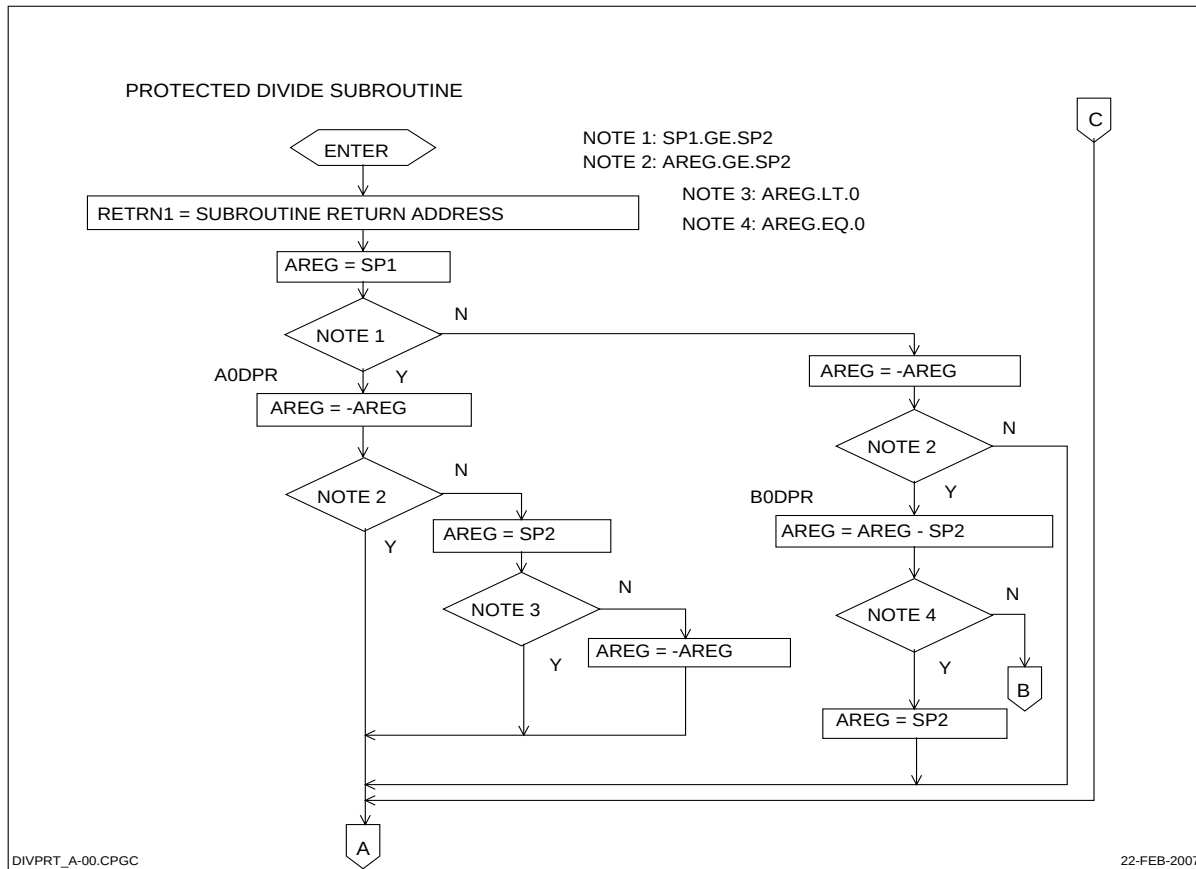
13.00 Figure 58: DERLAG_A



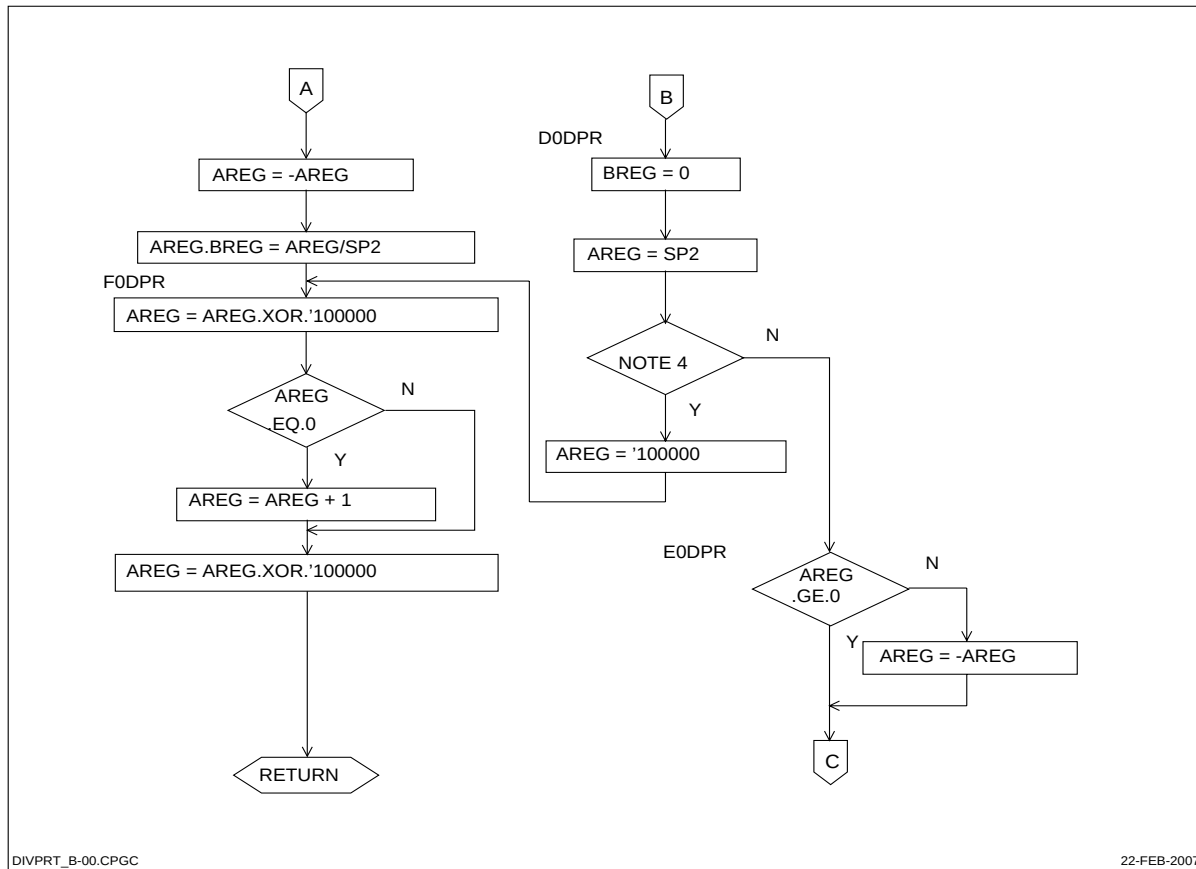
13.00 Figure 59: DERLAG_B



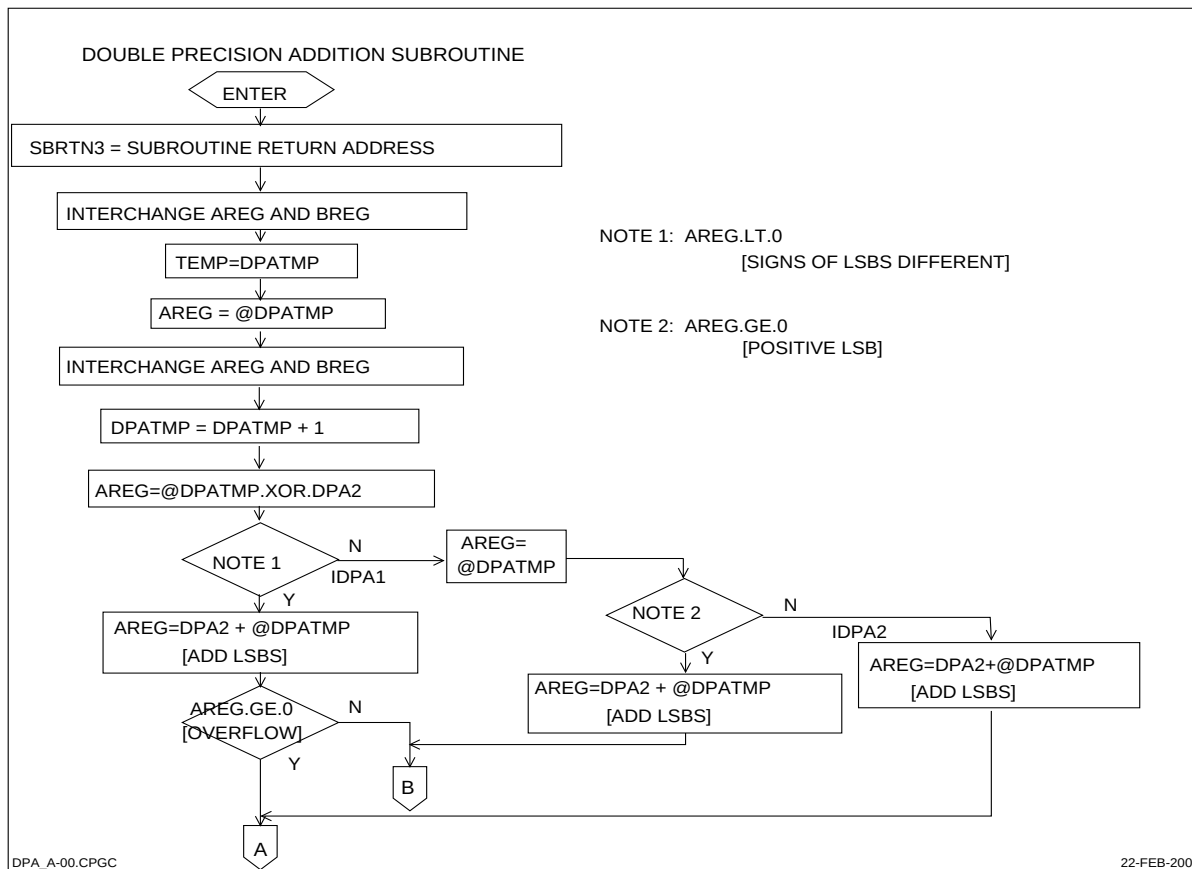
13.00 Figure 60: DIVPRT_A



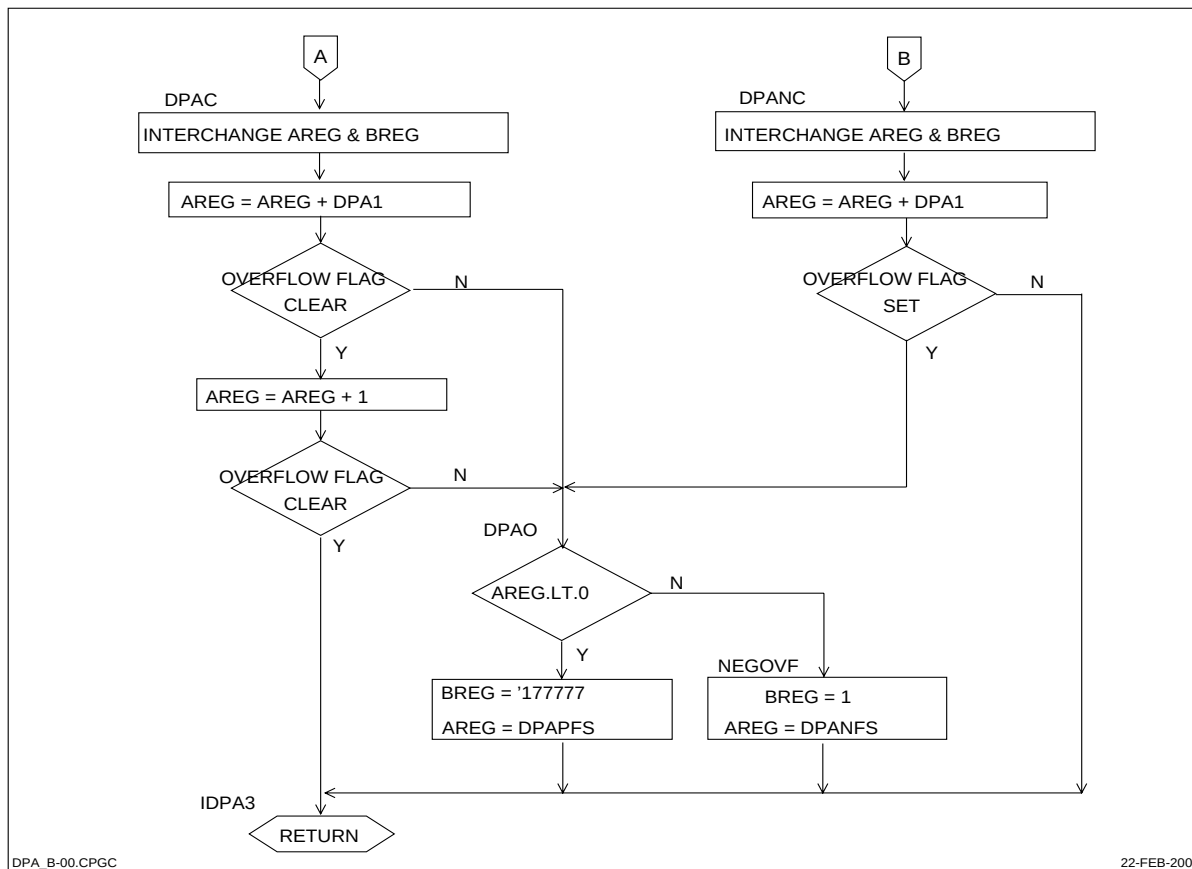
13.00 Figure 61: DIVPRT _B



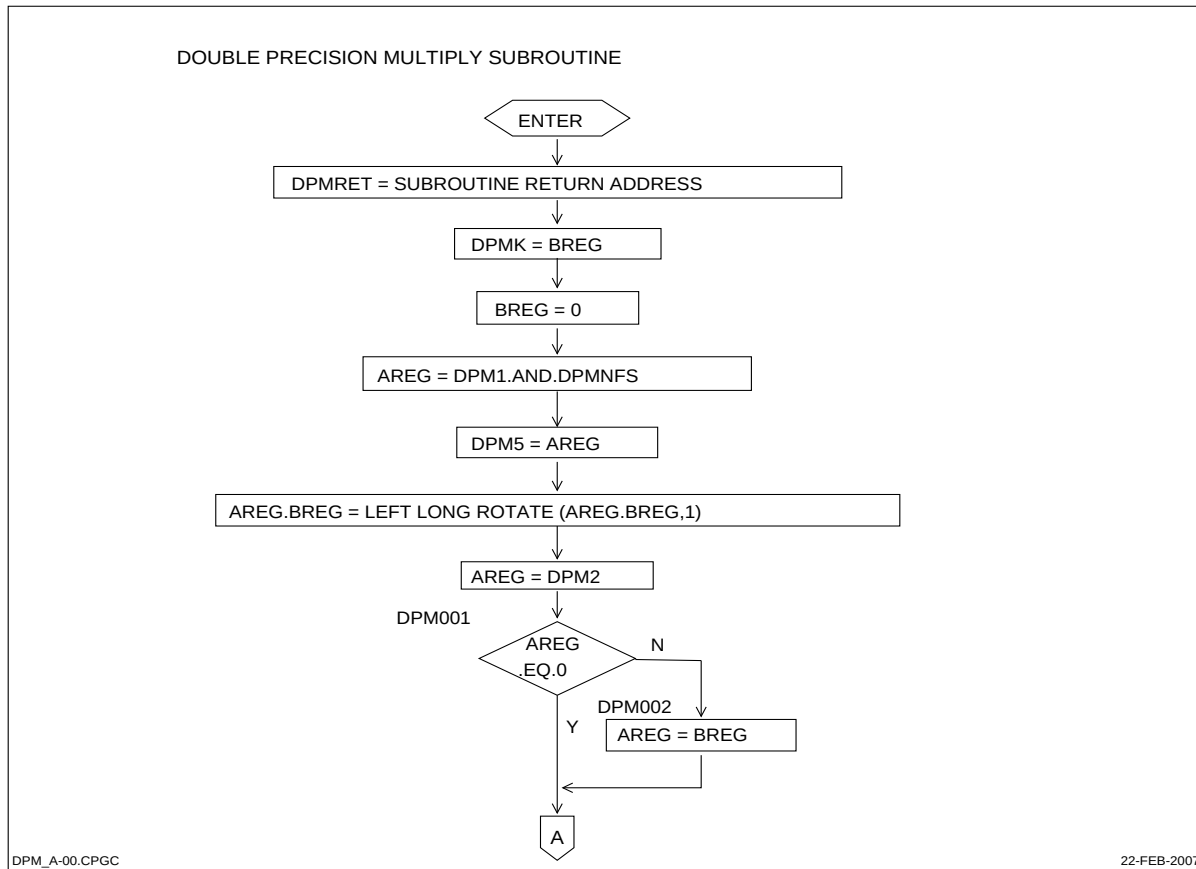
13.00 Figure 62: DPA_A



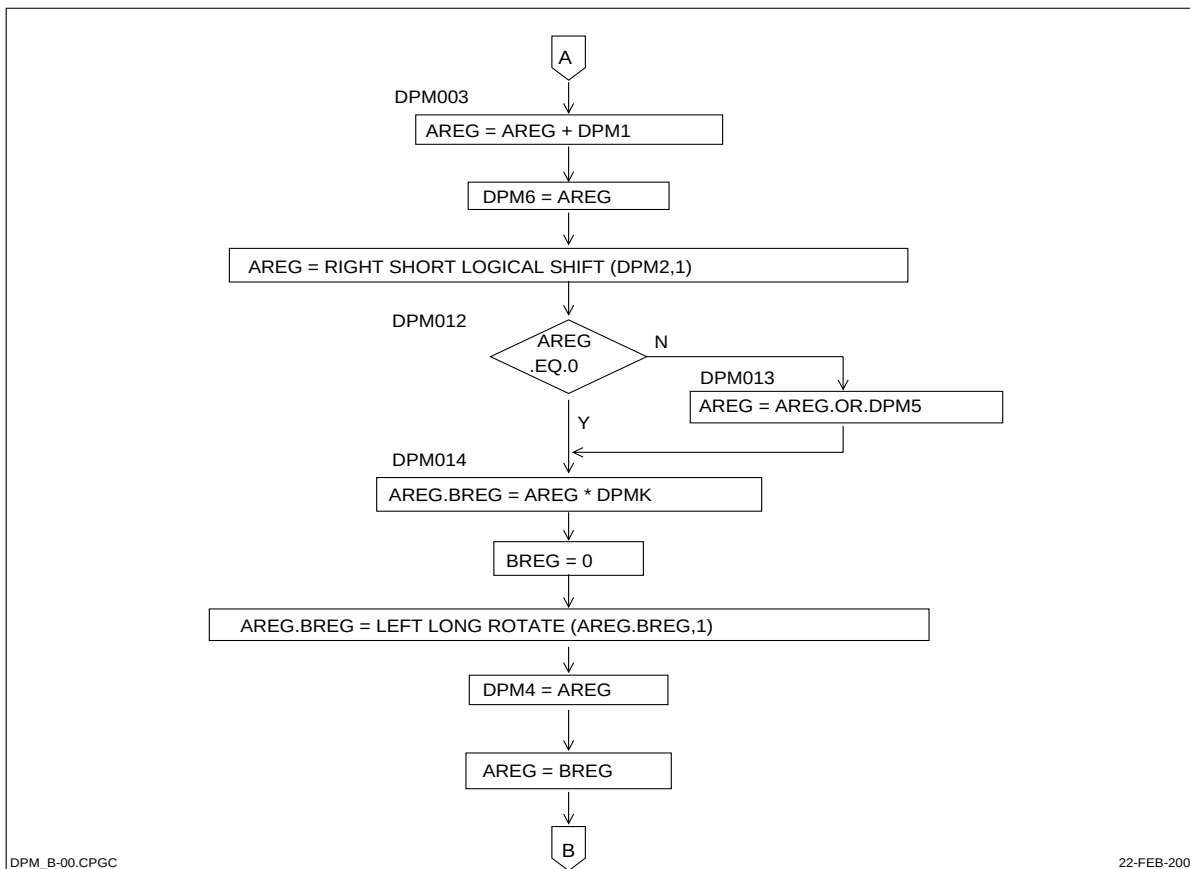
13.00 Figure 63: DPA_B



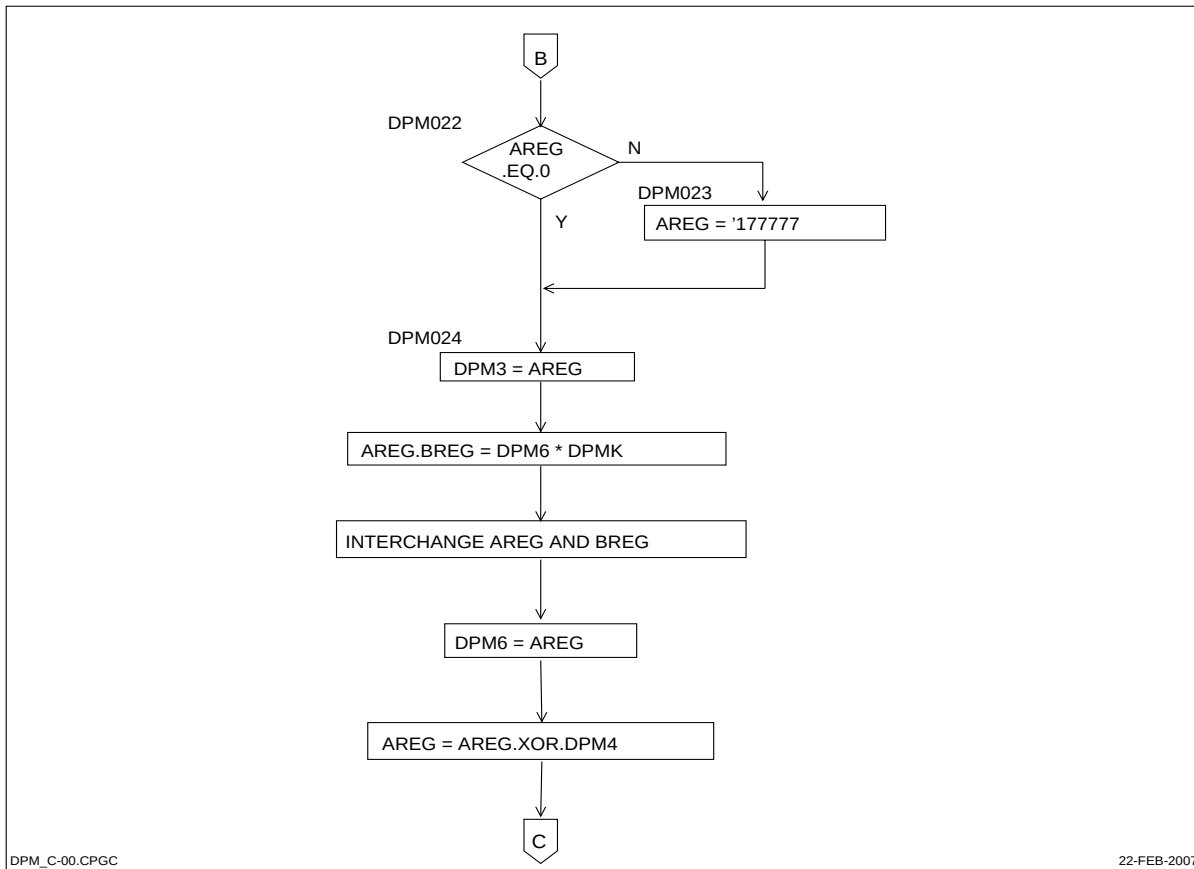
13.00 Figure 64: DPM_A



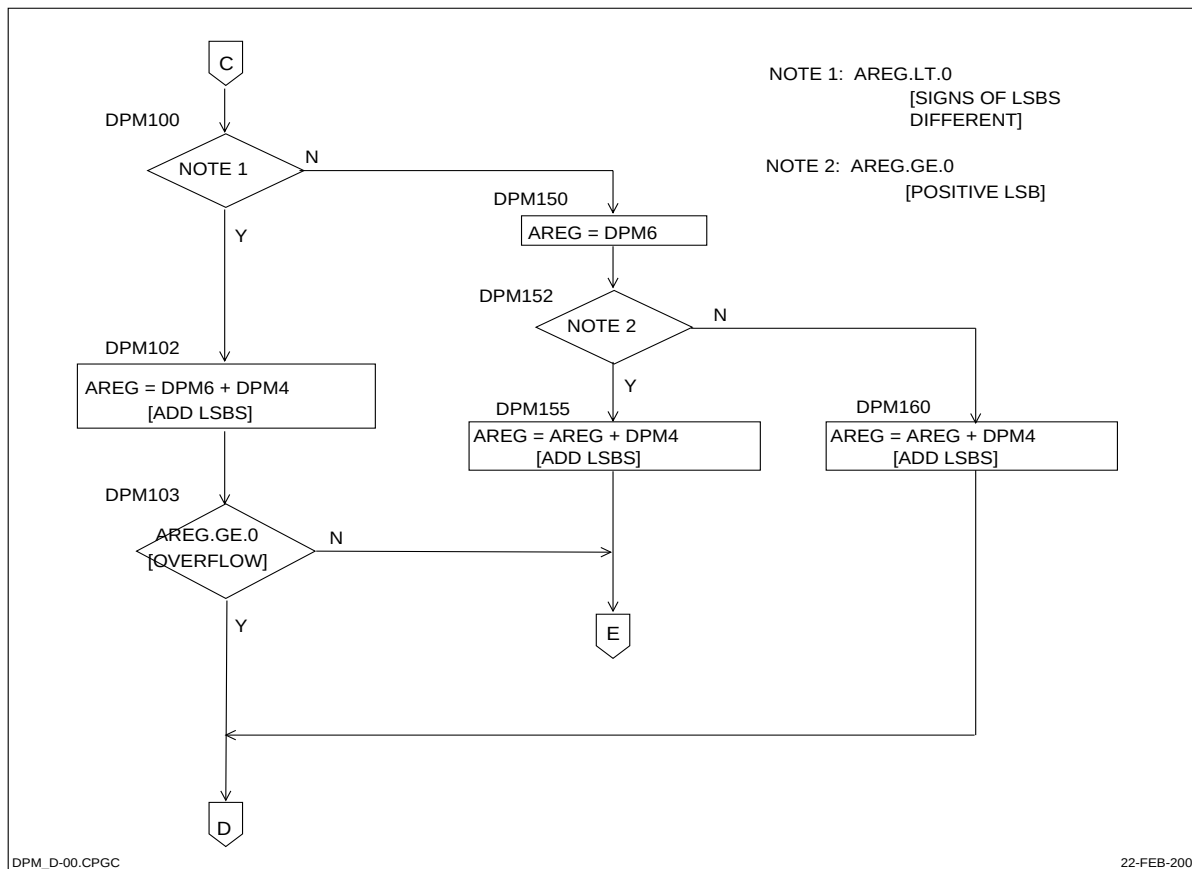
13.00 Figure 65: DPM_B



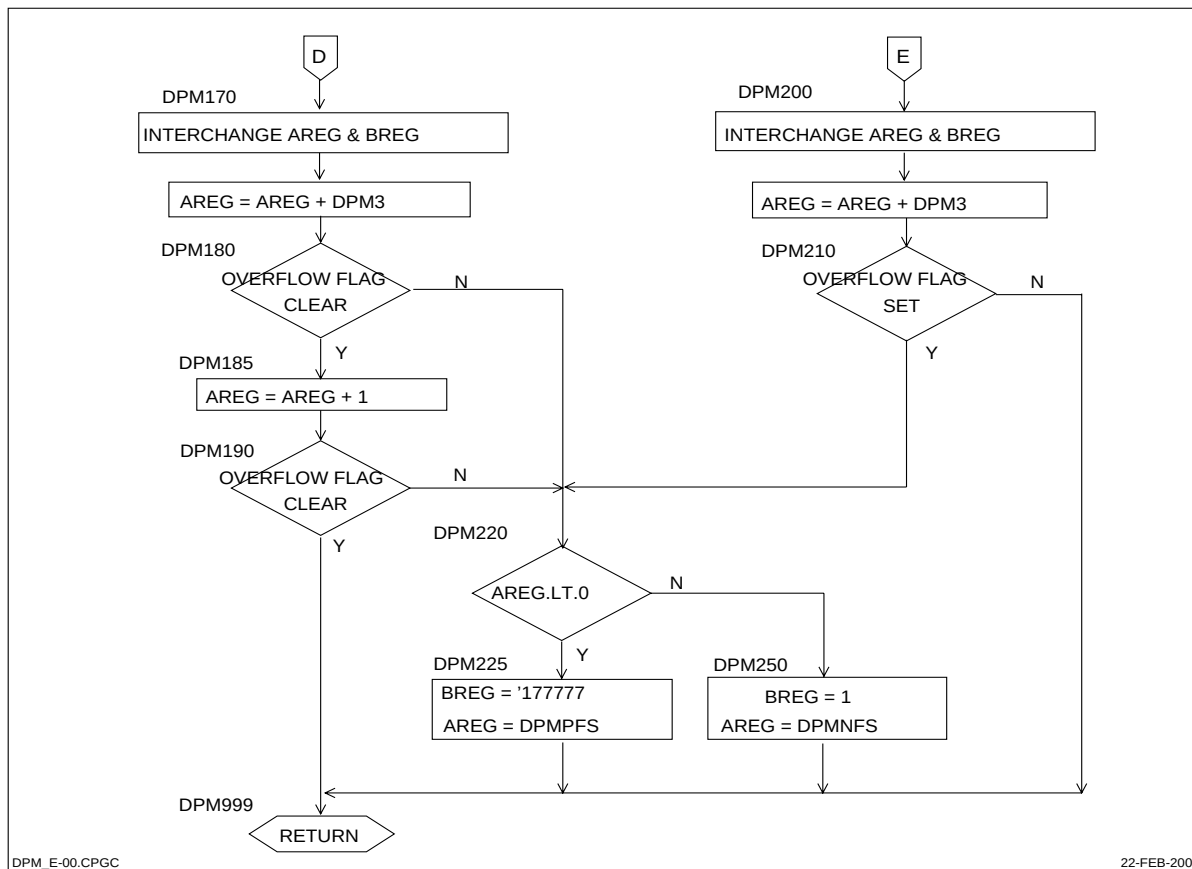
13.00 Figure 66: DPM_C



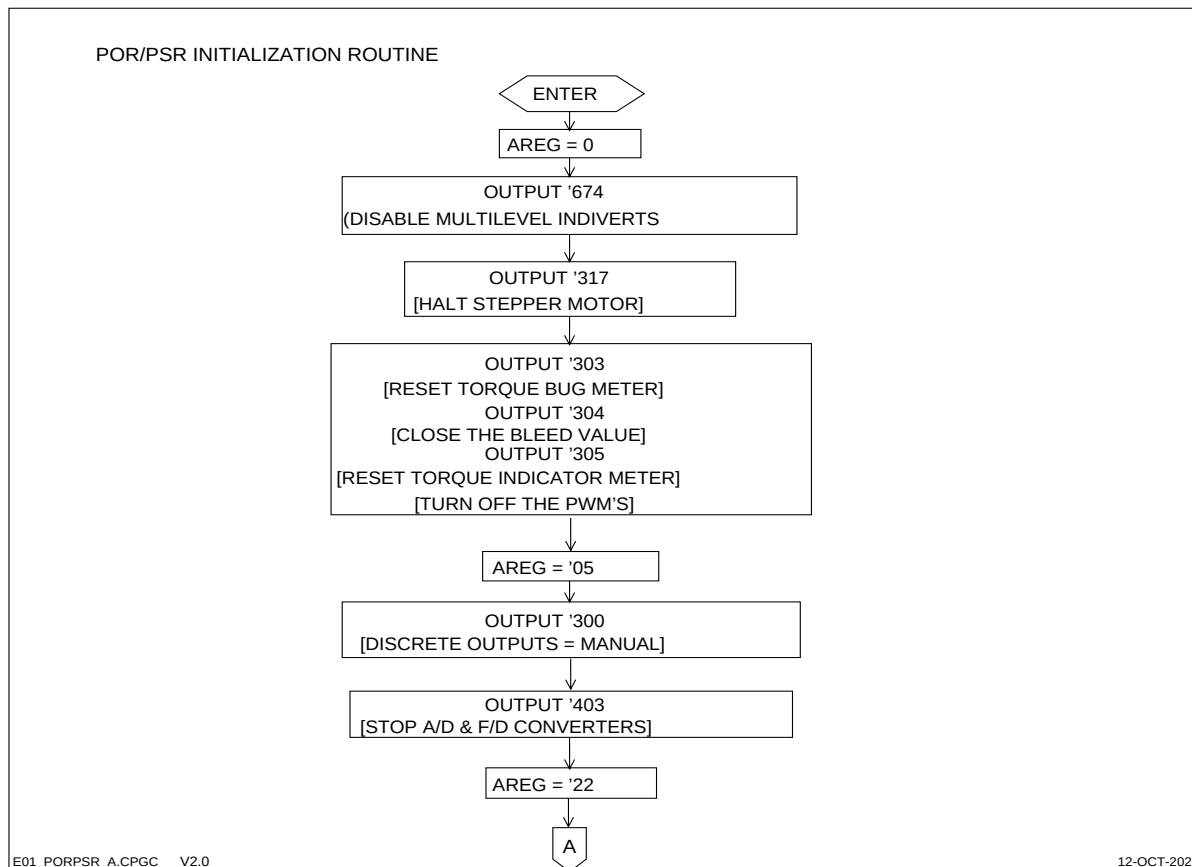
13.00 Figure 67: DPM _D



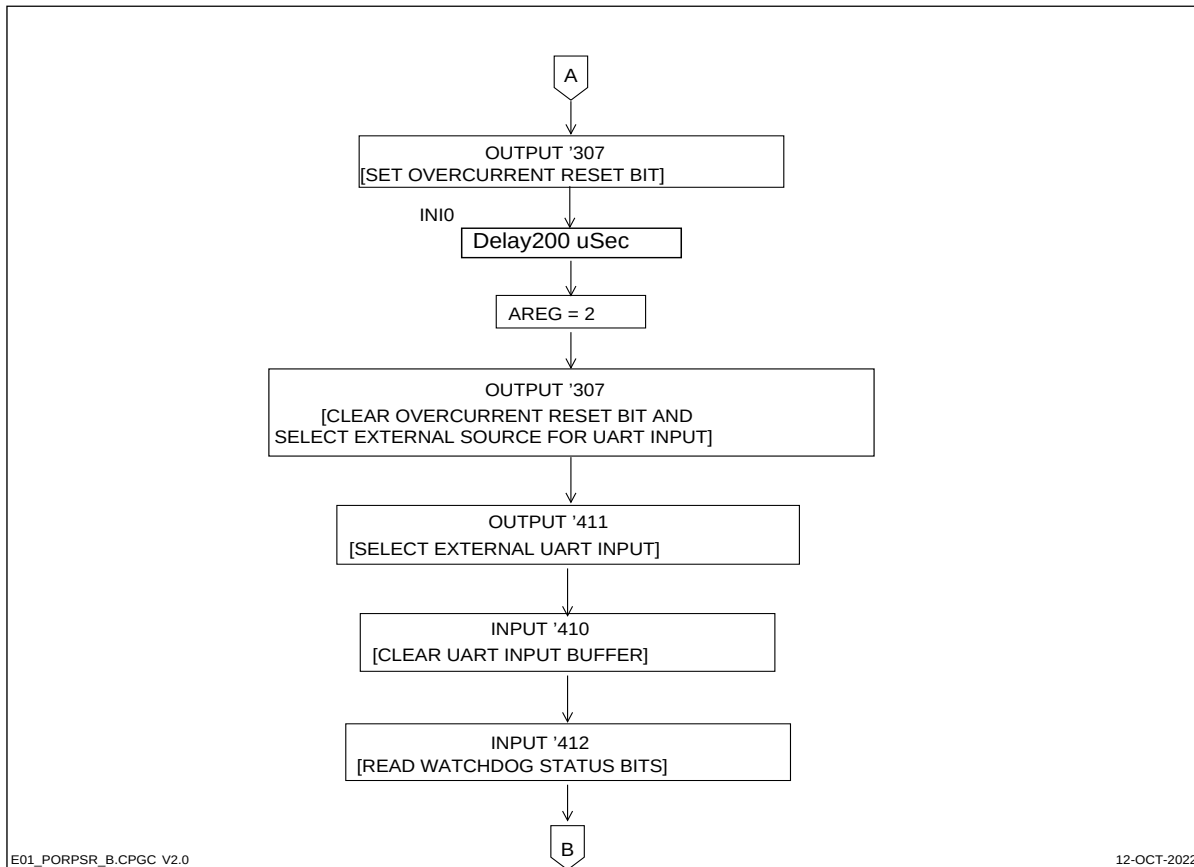
13.00 Figure 68: DPM_E



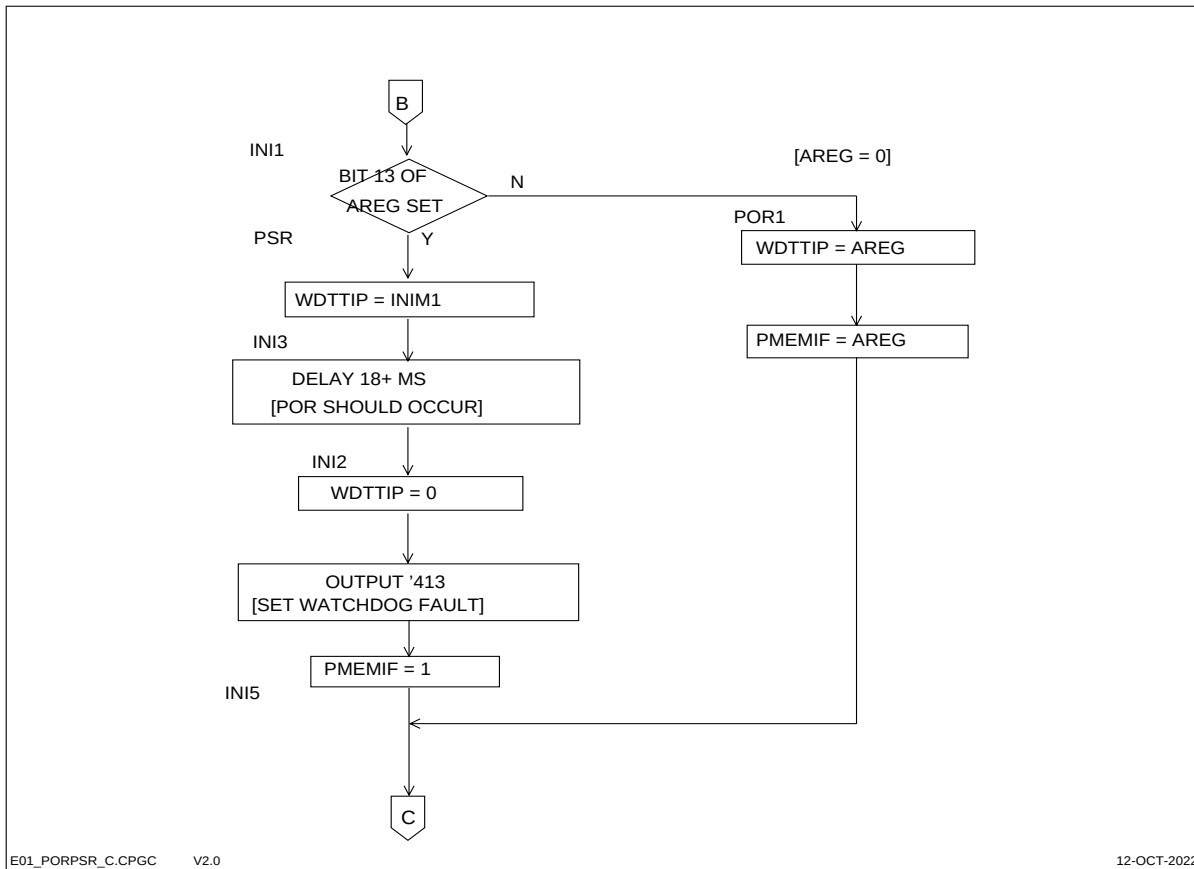
13.00 Figure 69: E01_PORPSR_A



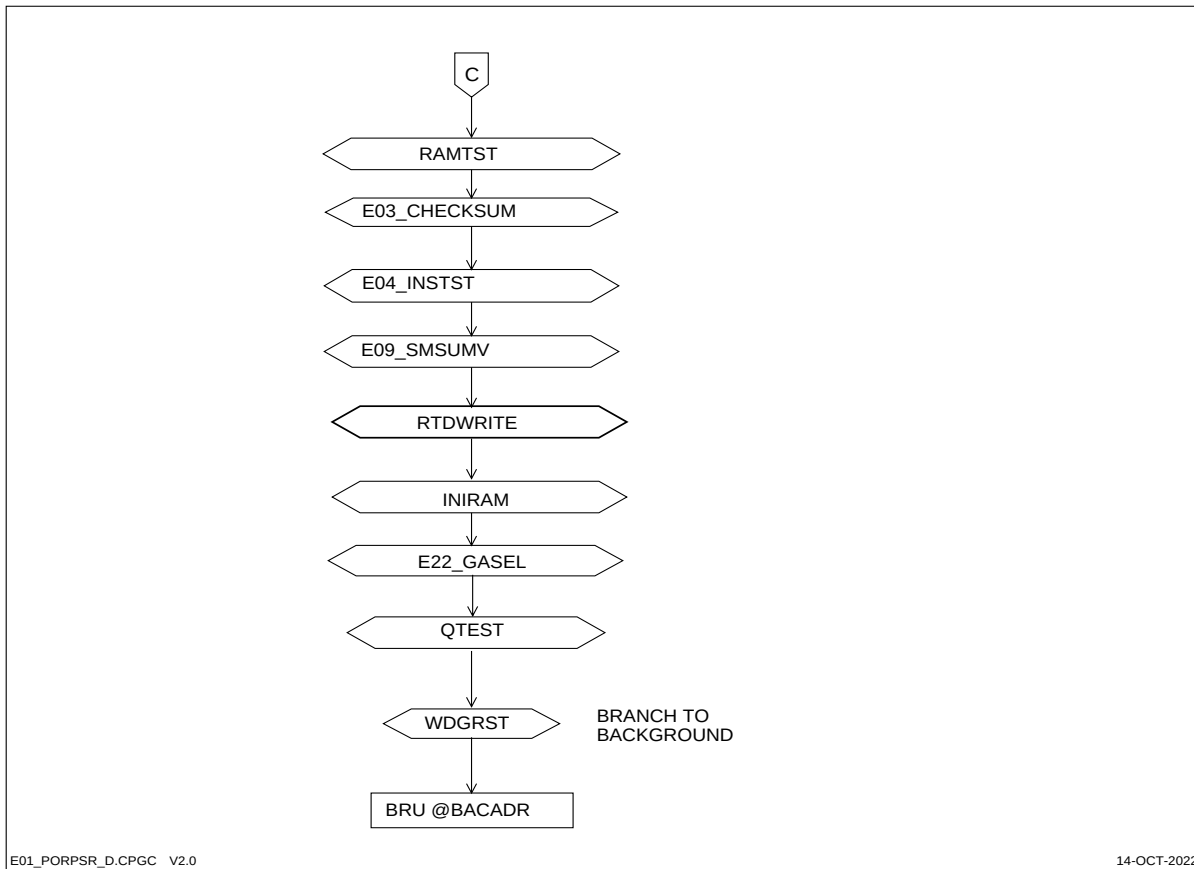
13.00 Figure 70: E01_PORPSR_B



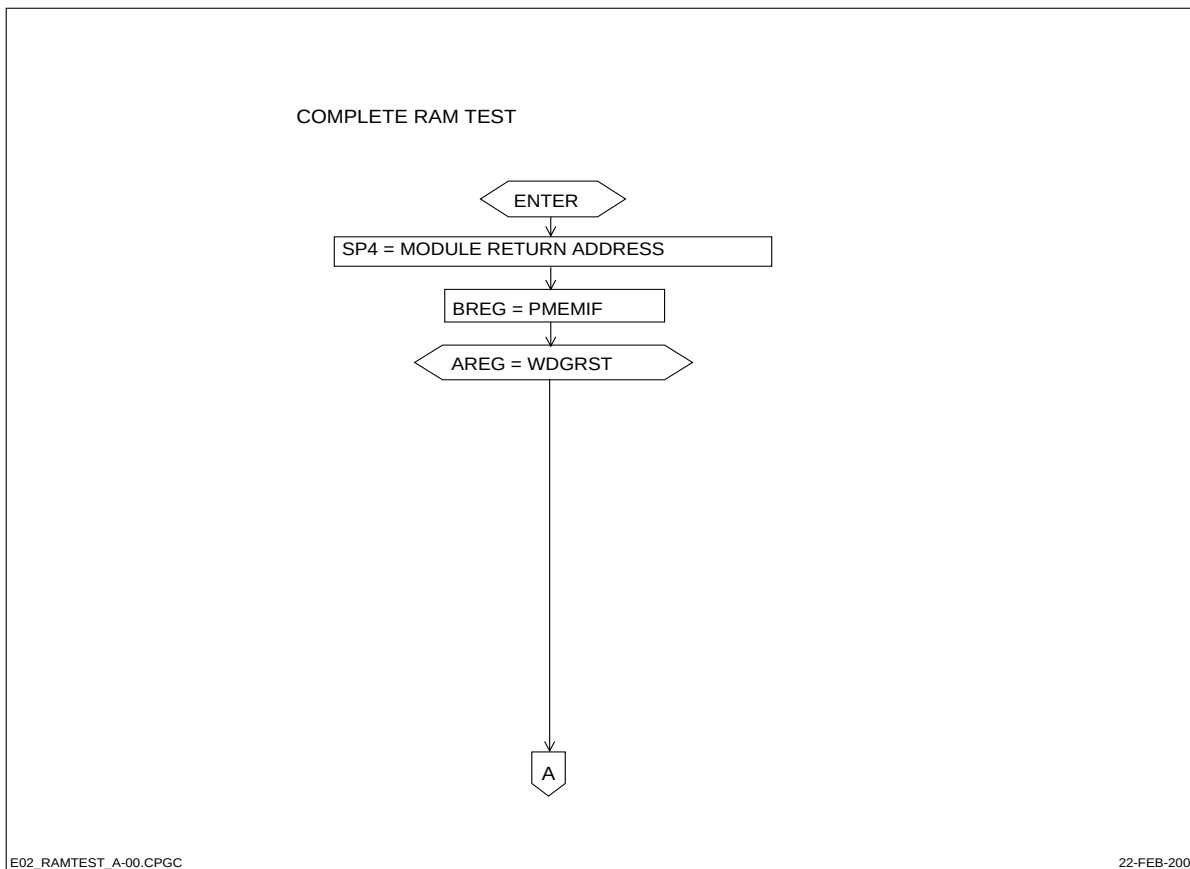
13.00 Figure 71: E01_PORPSR_C



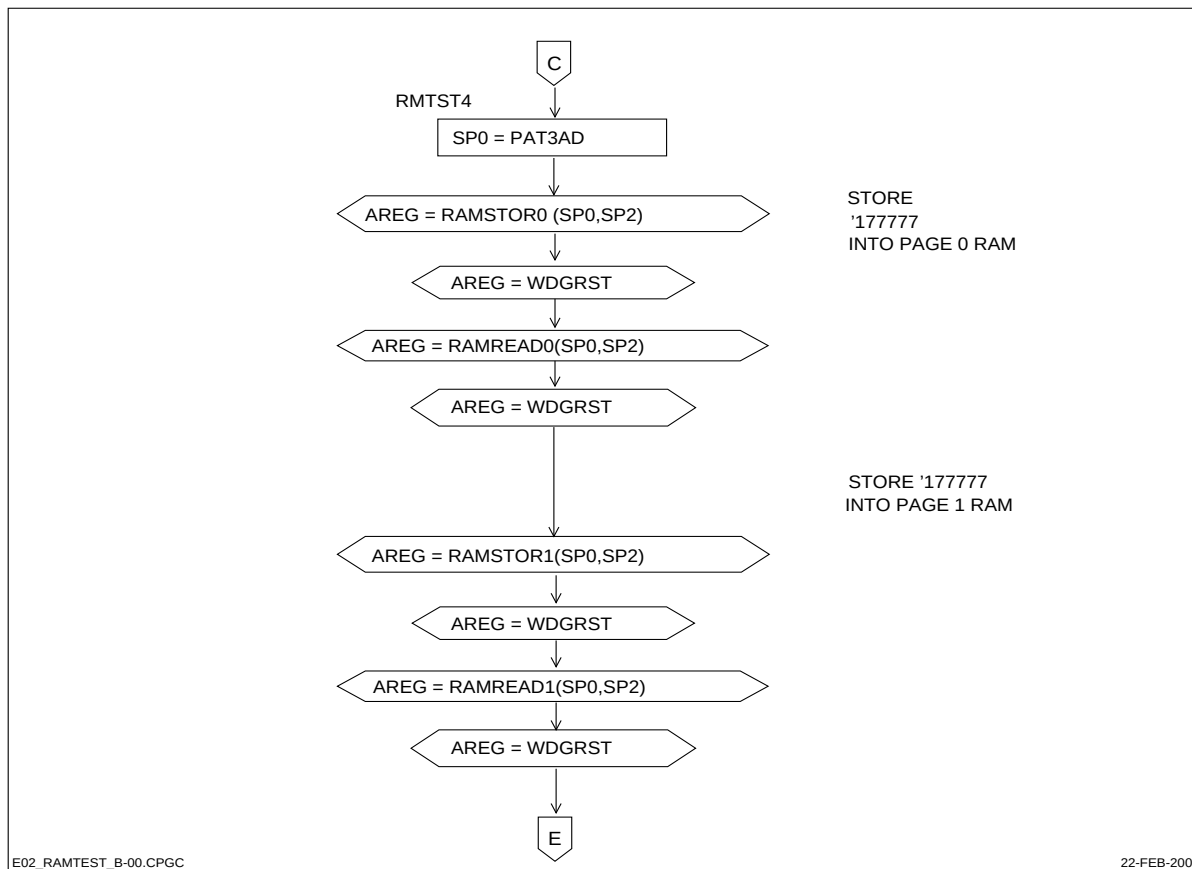
13.00 Figure 72: E01 _PORPSR _D



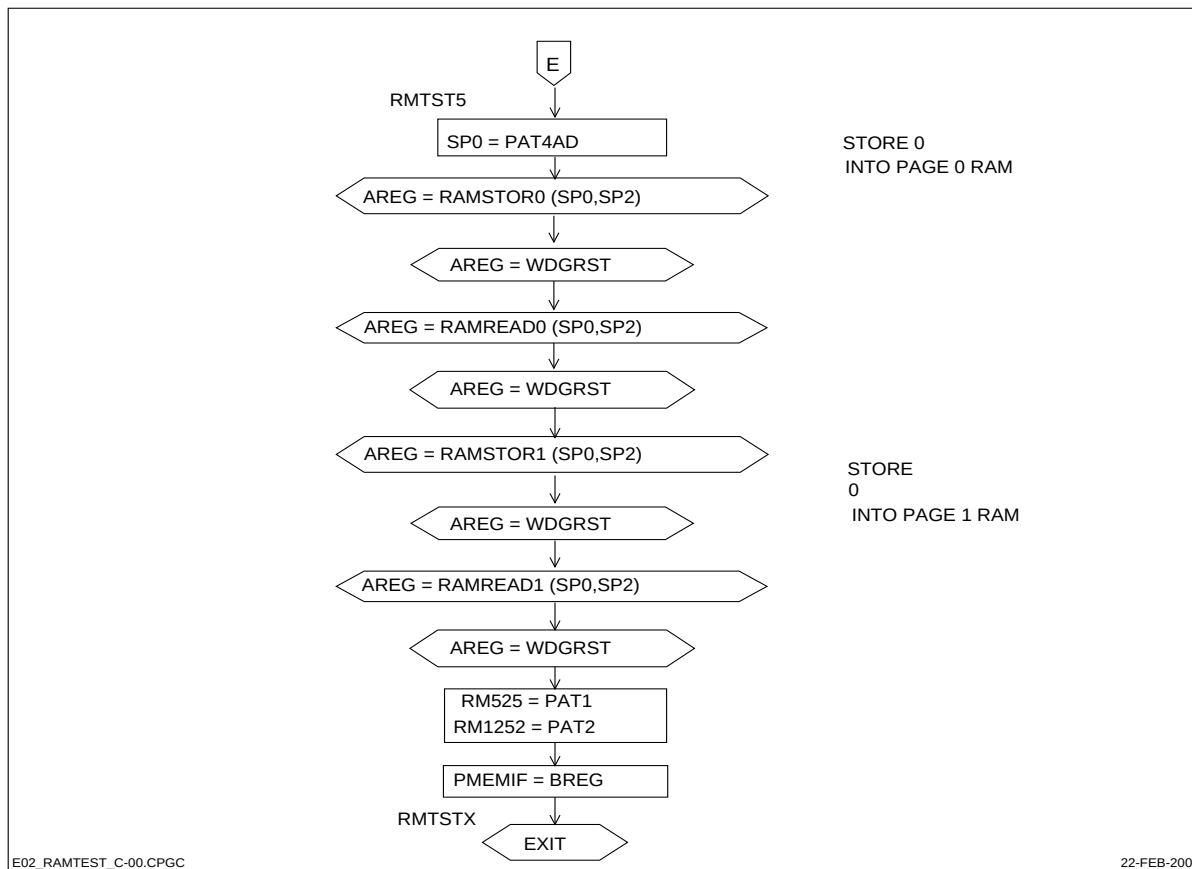
13.00 Figure 73: E02_RAMTEST_A



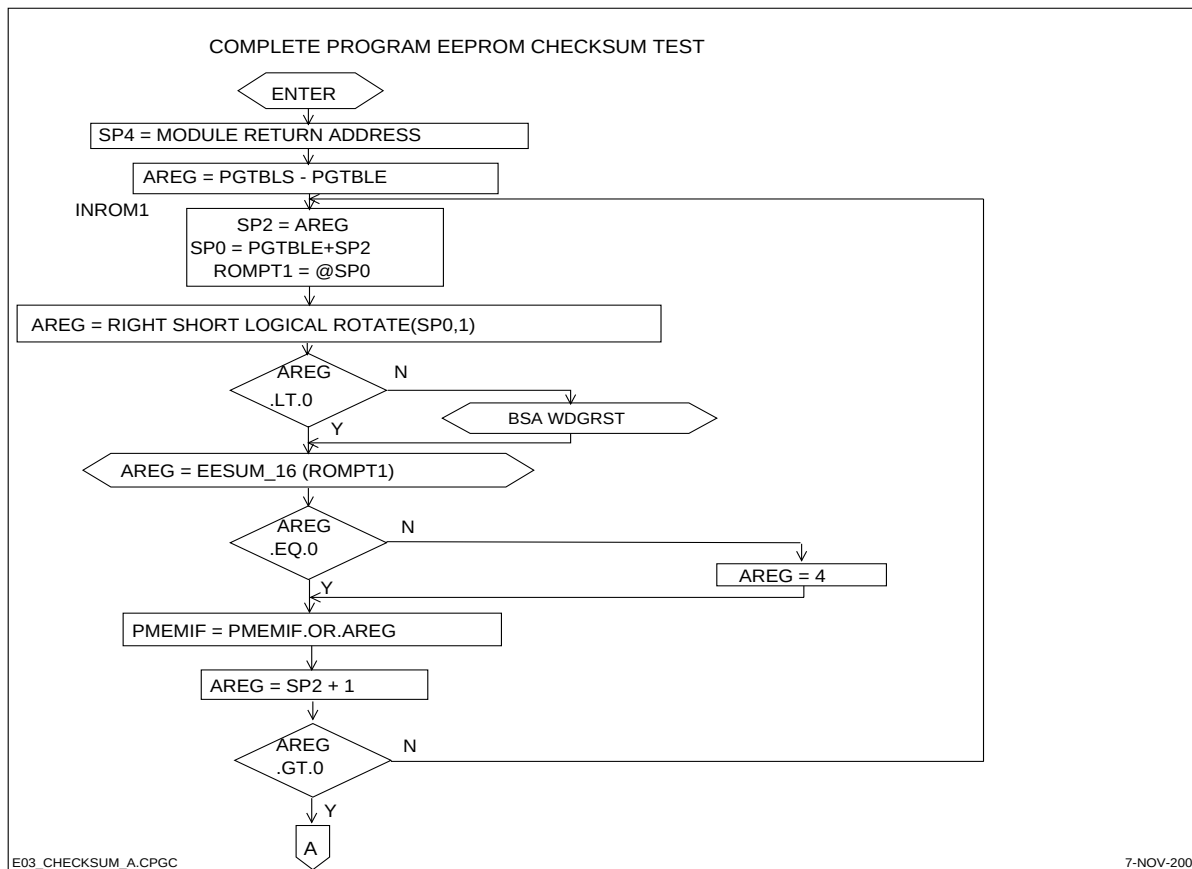
13.00 Figure 74: E02_RAMTEST_B



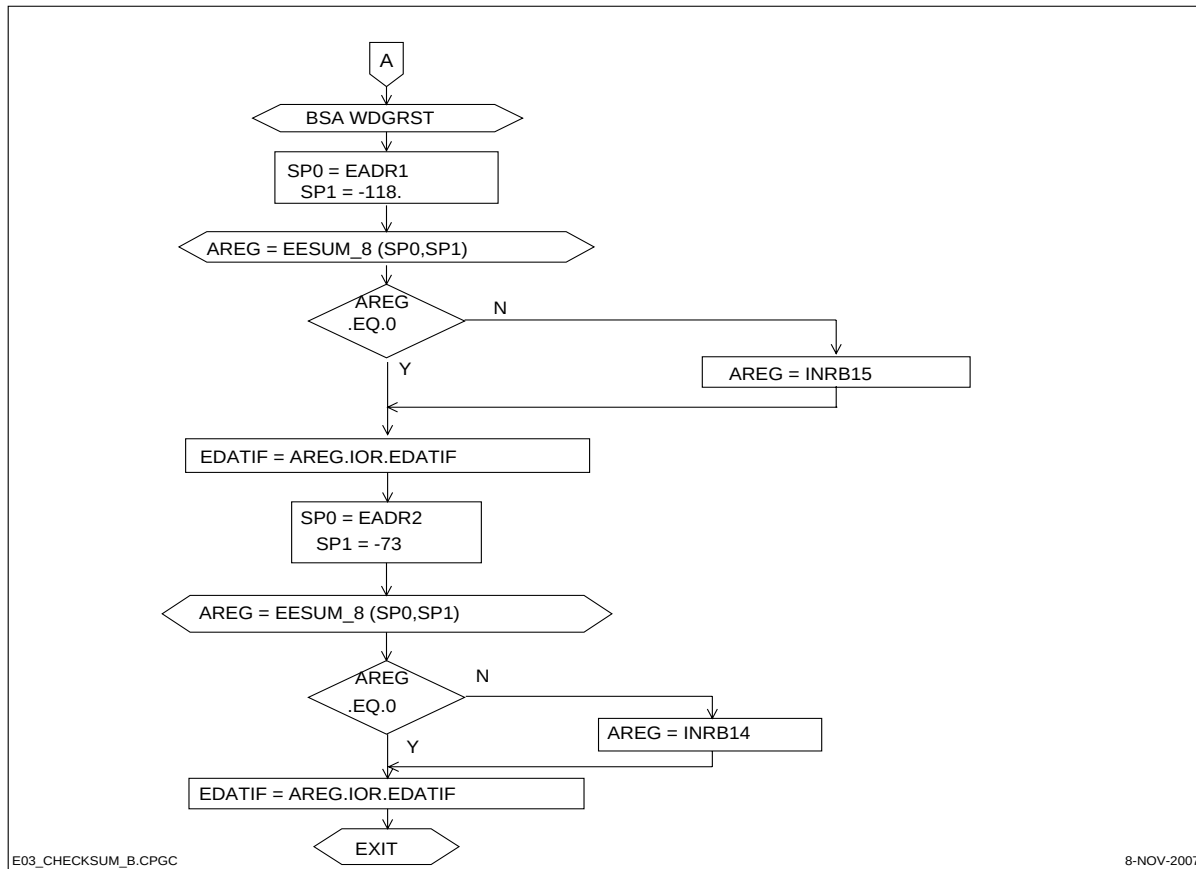
13.00 Figure 75: E02_RAMTEST_C



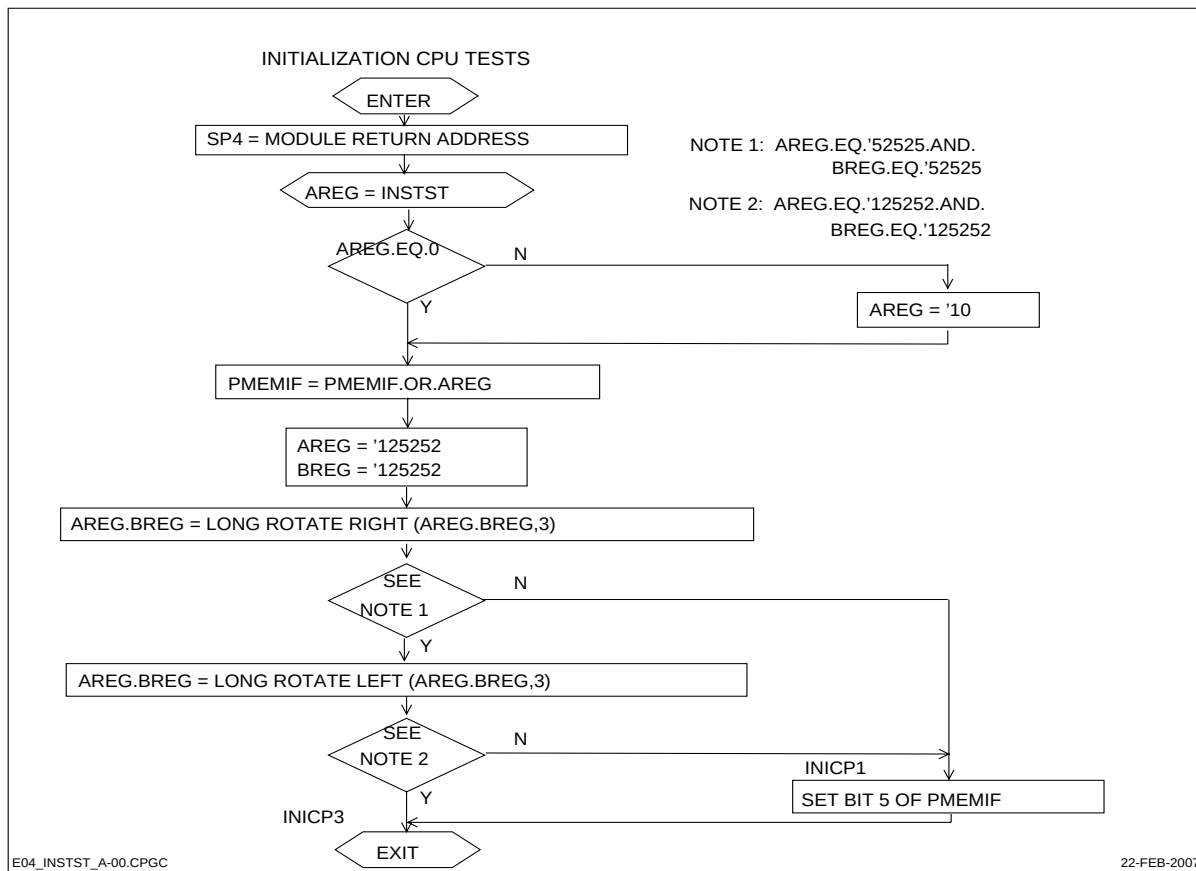
13.00 Figure 76: E03_CHECKSUM_A



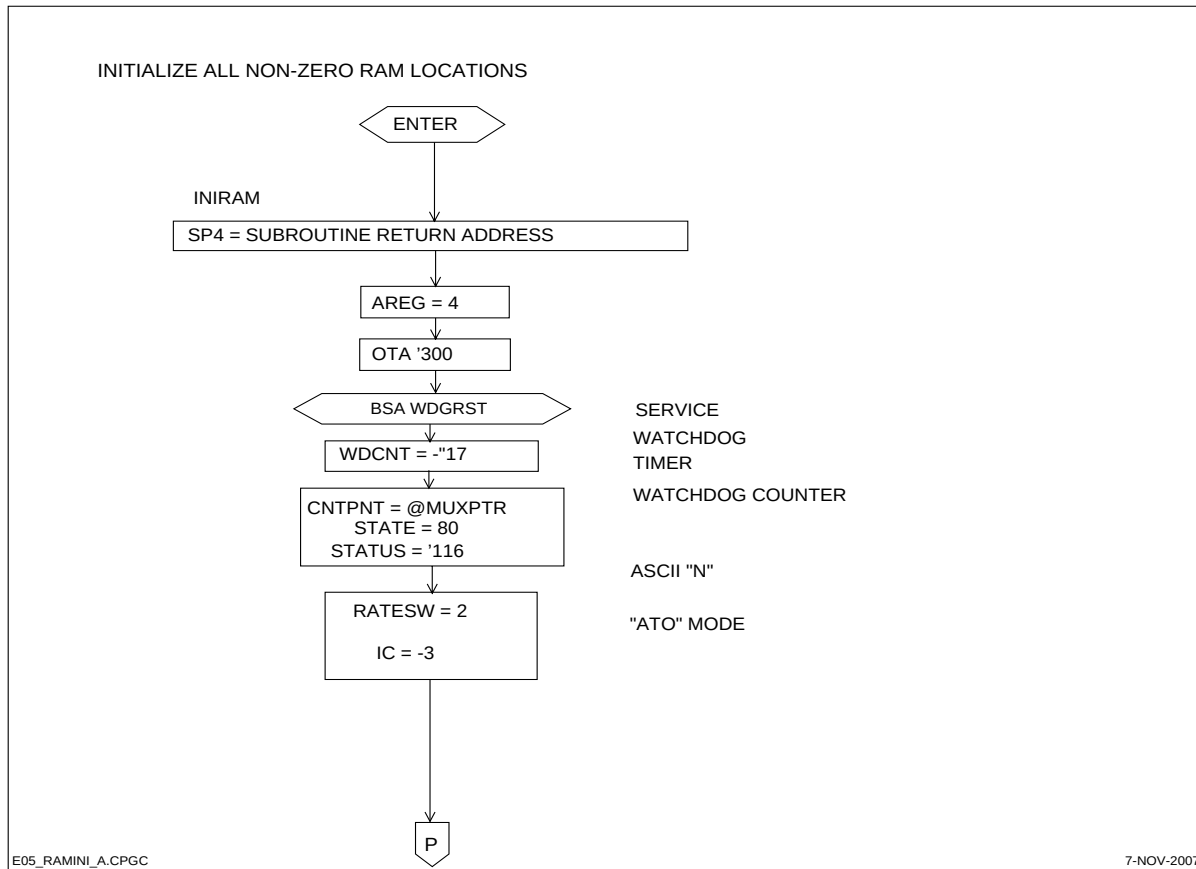
13.00 Figure 77: E03_CHECKSUM_B



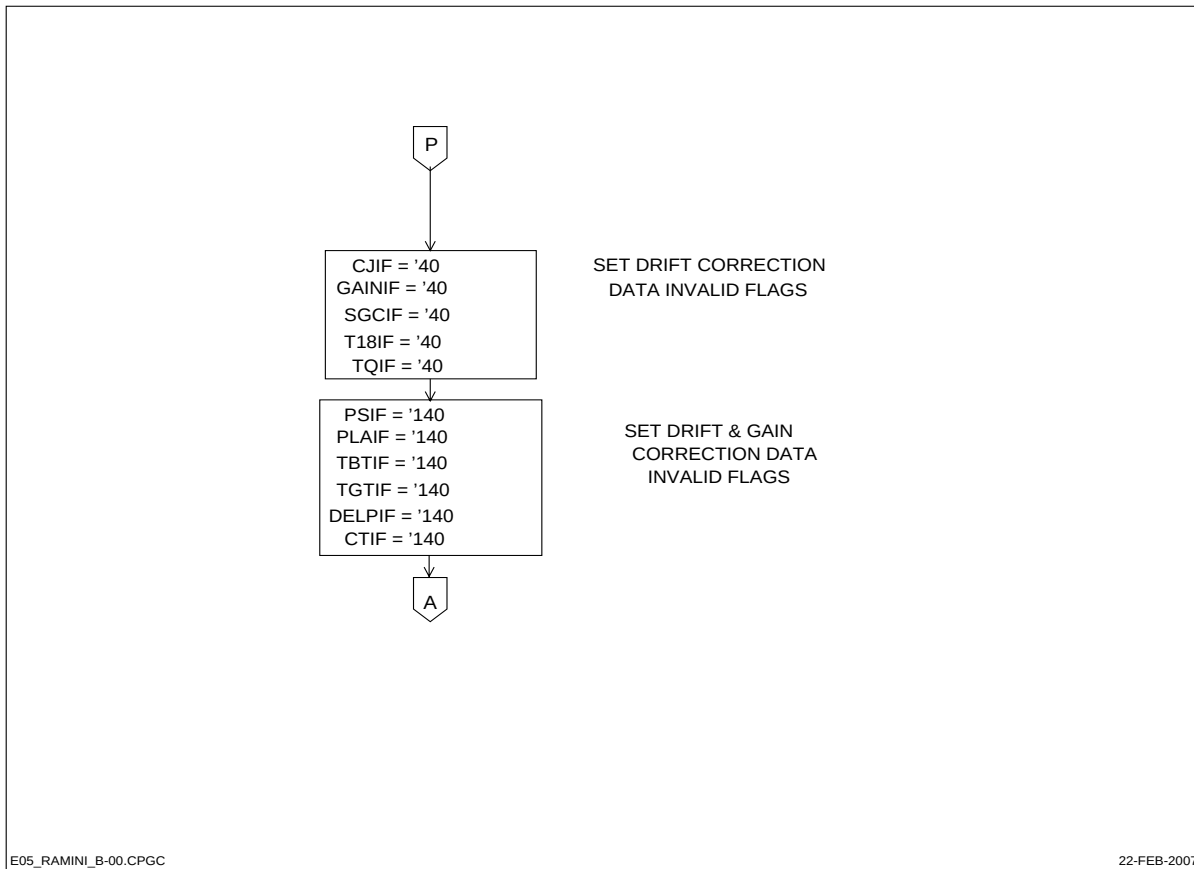
13.00 Figure 78: E04_INSTST_A



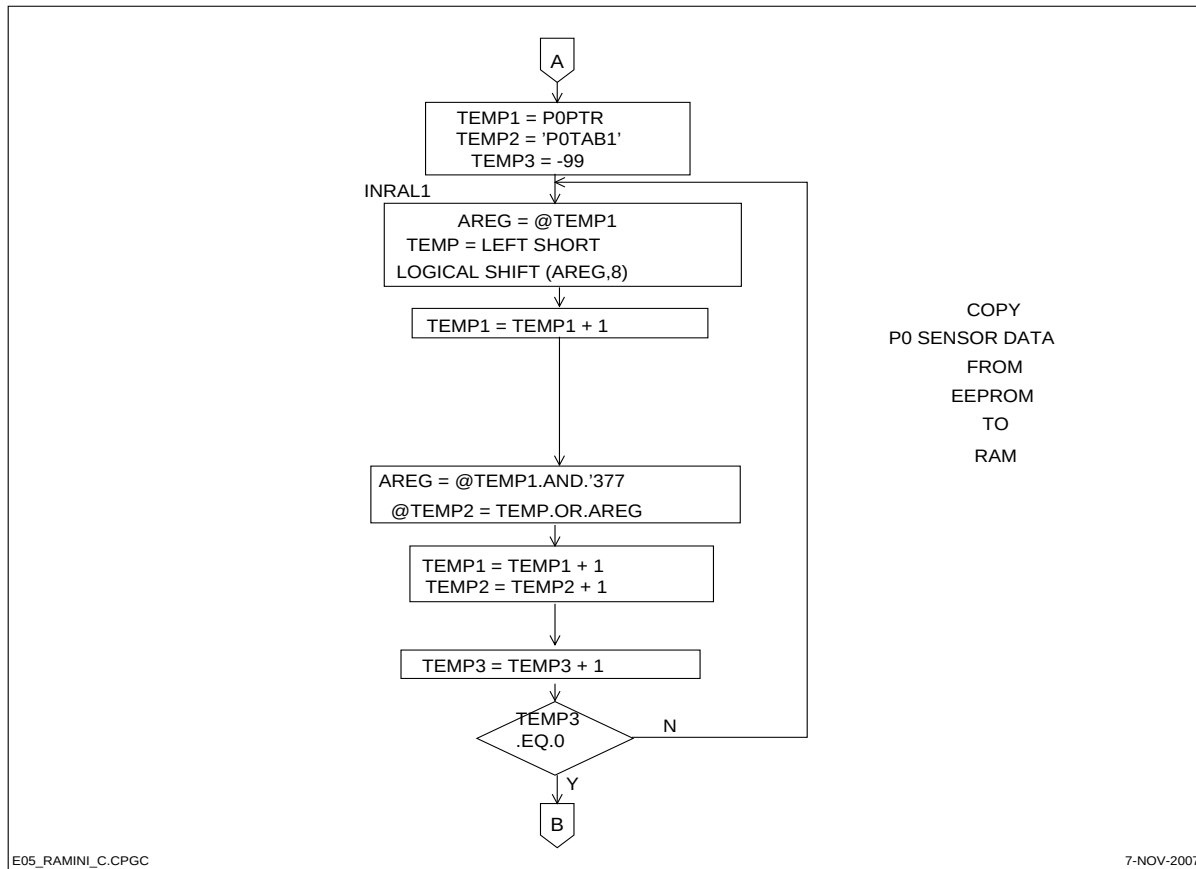
13.00 Figure 79: E05_RAMINI_A



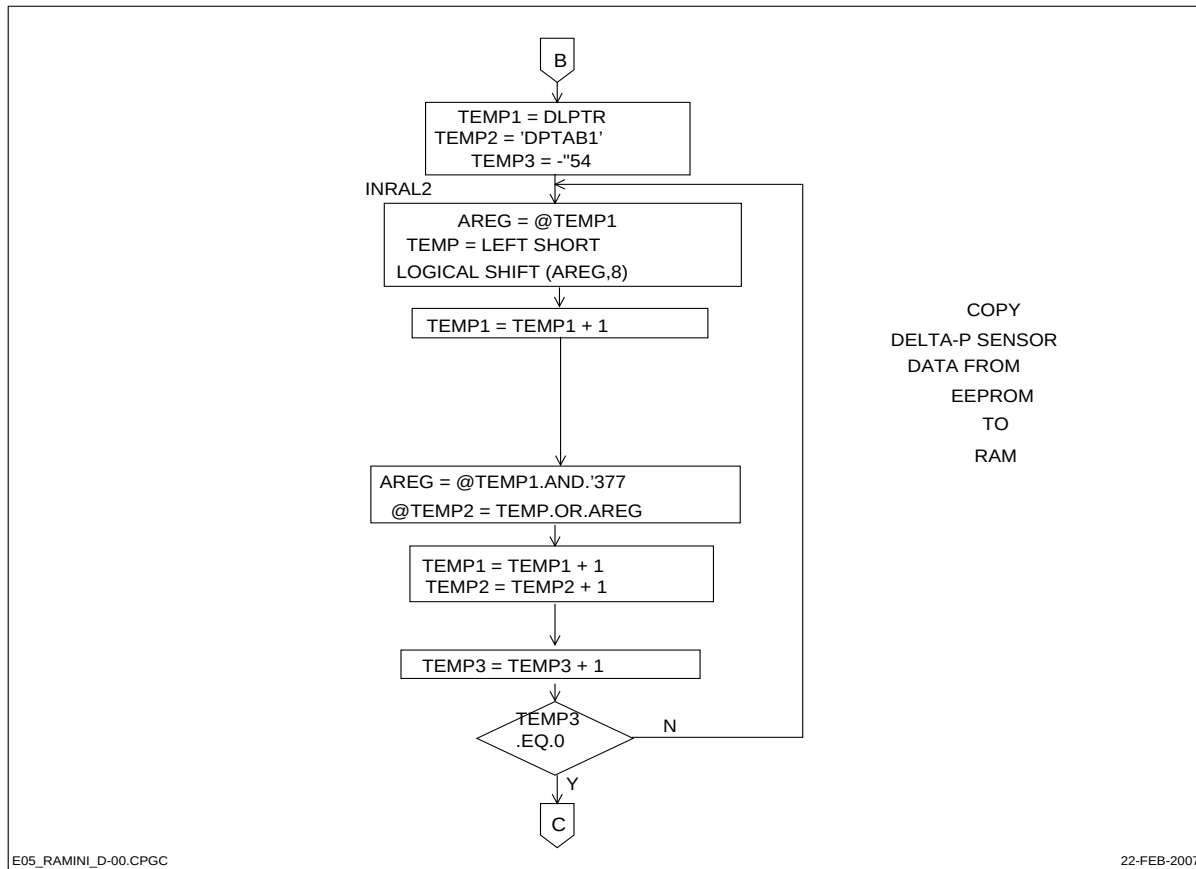
13.00 Figure 80: E05_RAMINI_B



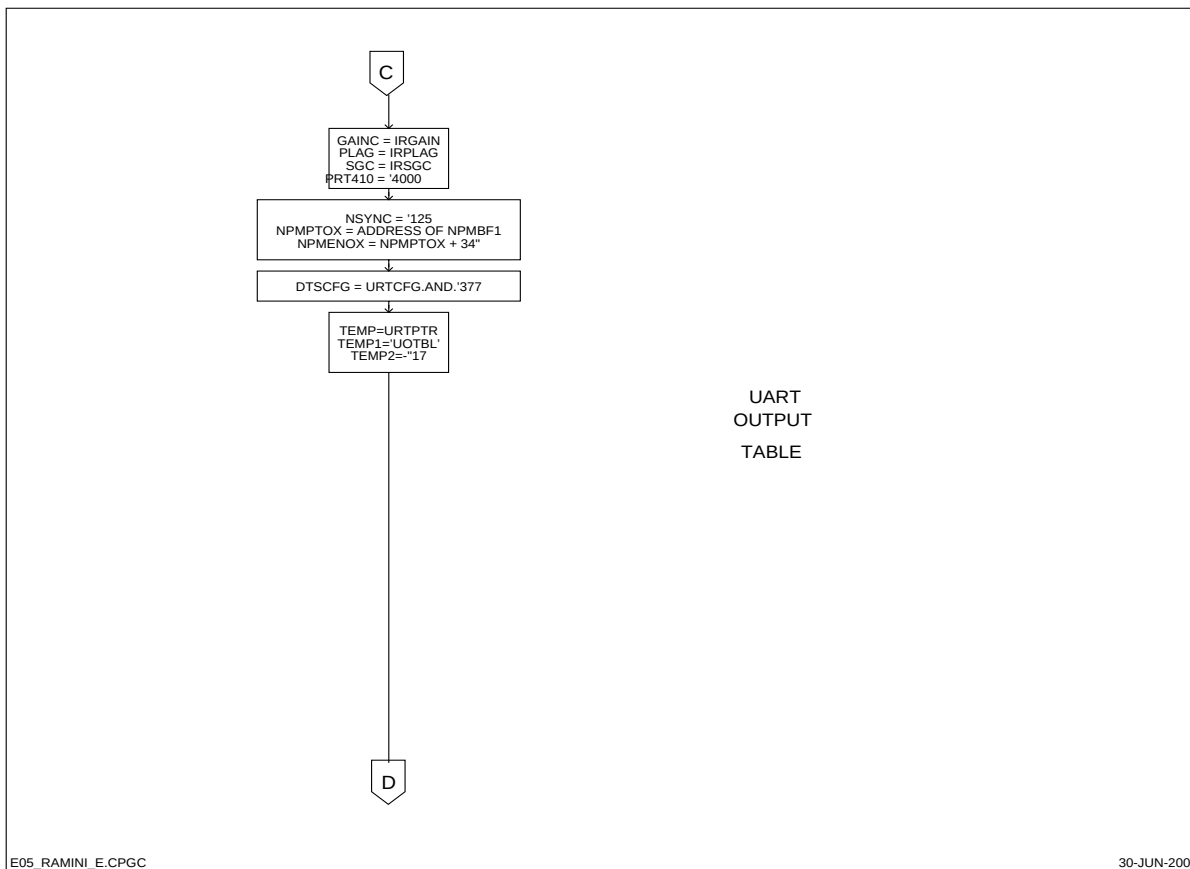
13.00 Figure 81: E05_RAMINI_C



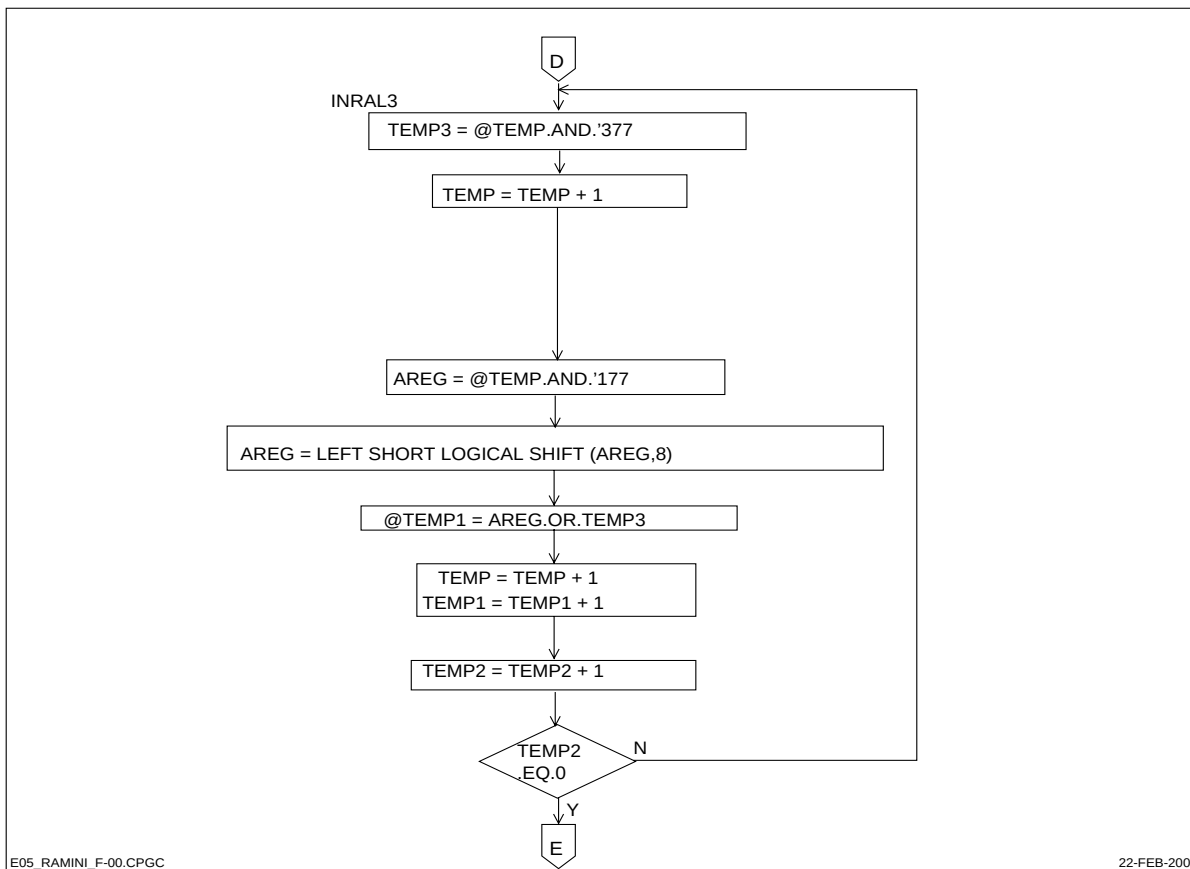
13.00 Figure 82: E05_RAMINI_D



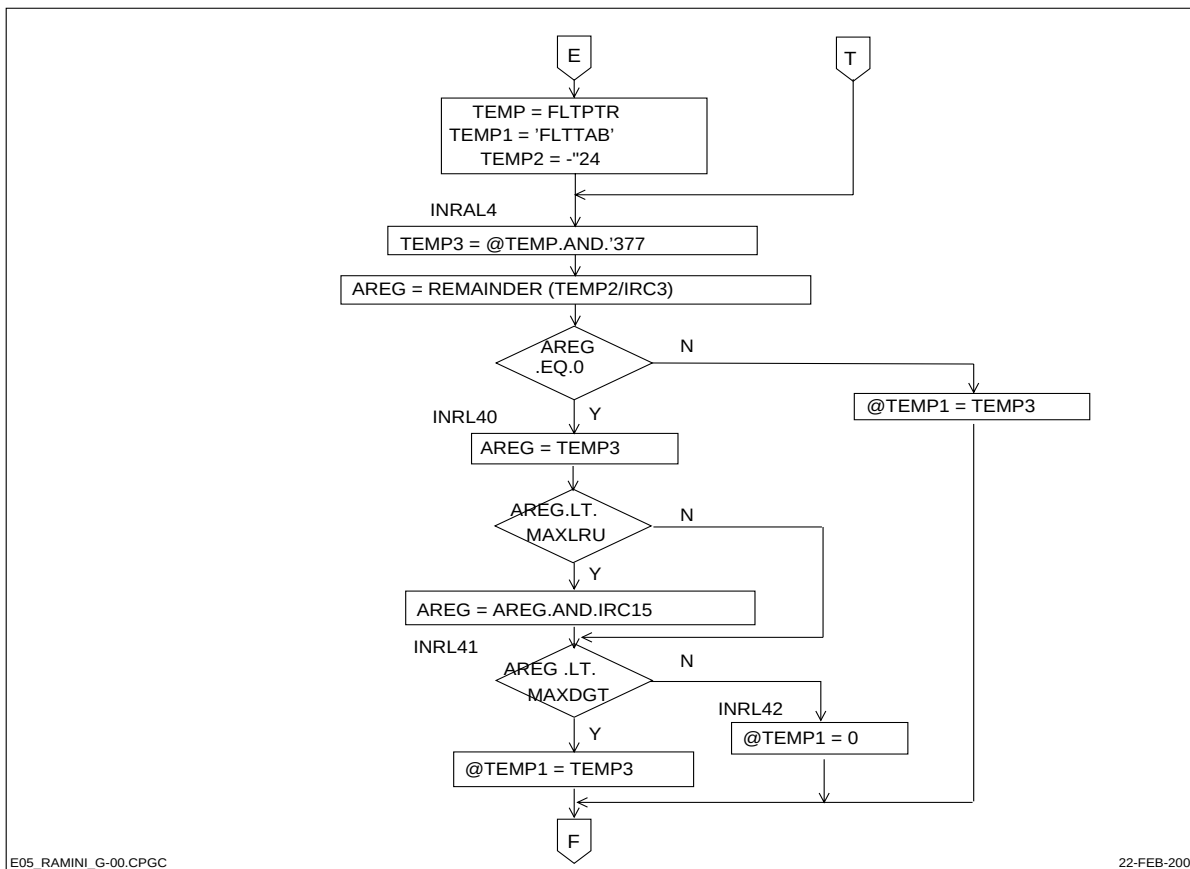
13.00 Figure 83: E05_RAMINI_E



13.00 Figure 84: E05_RAMINI_F



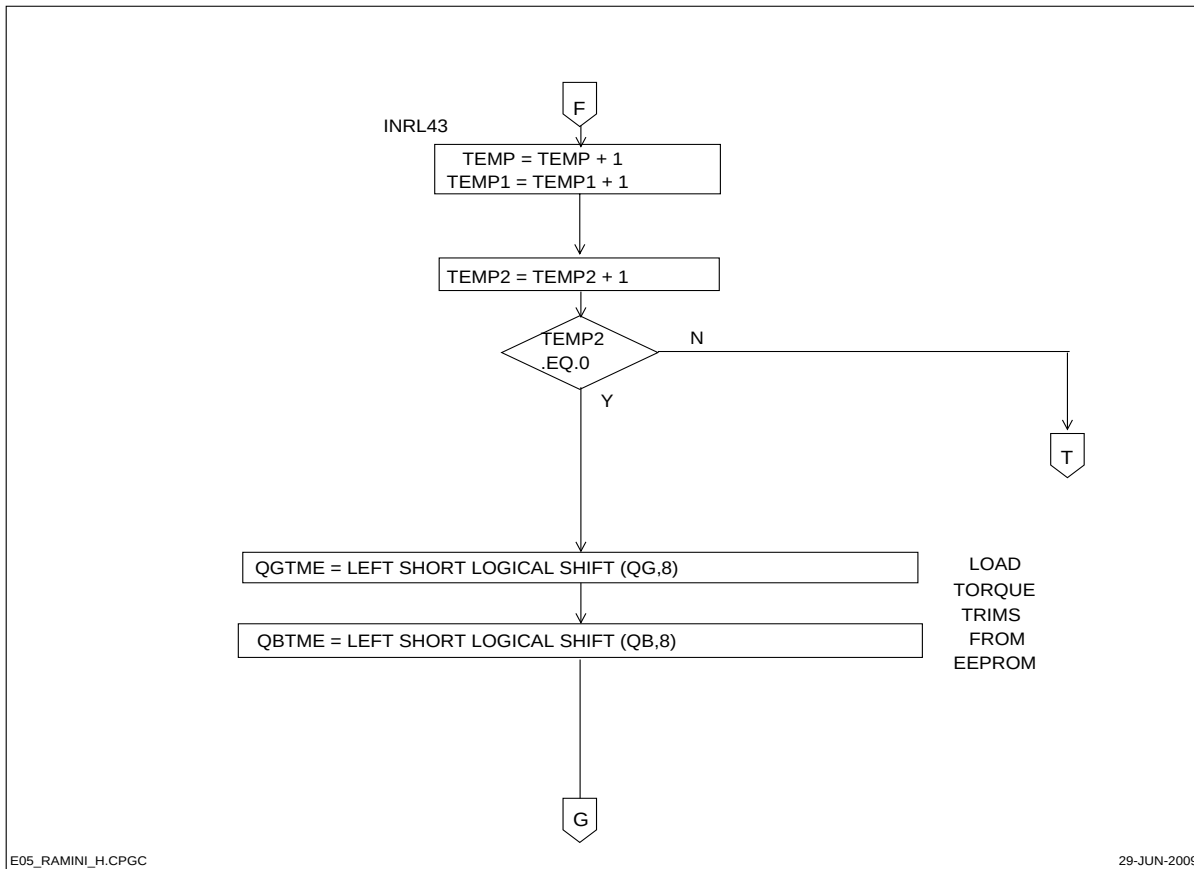
13.00 Figure 85: E05_RAMINI _G



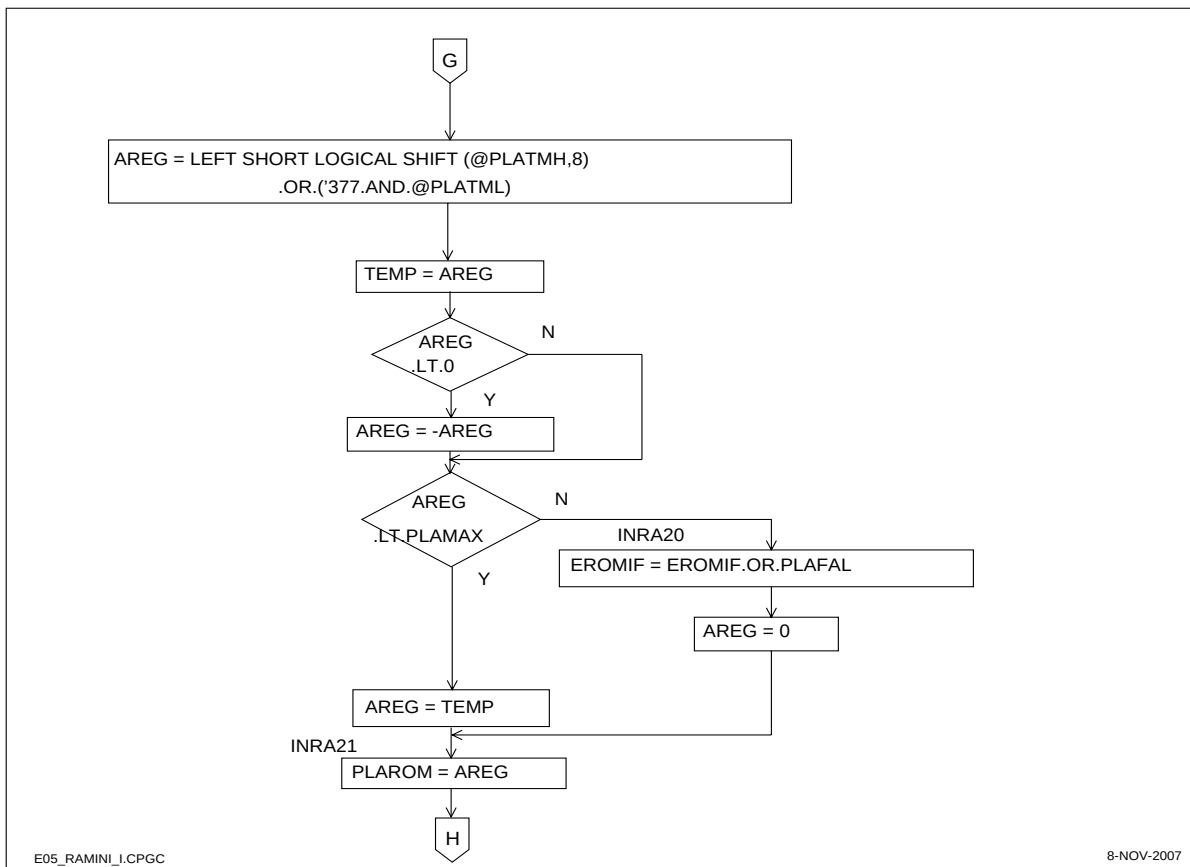
E05_RAMINI_G-00.CPGC

22-FEB-2007

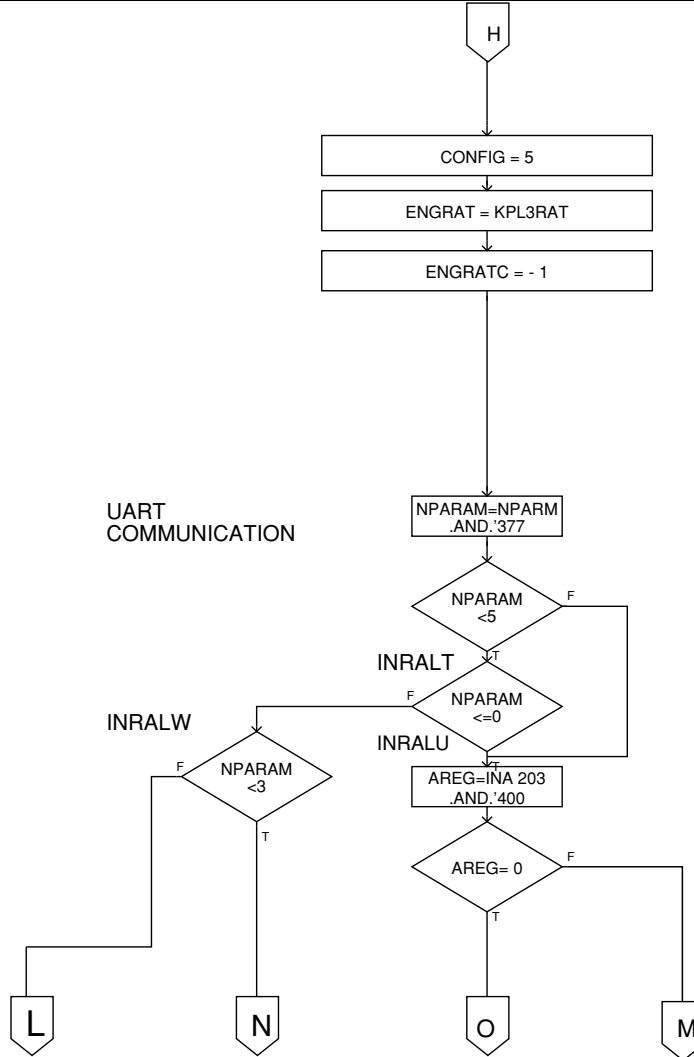
13.00 Figure 86: E05_RAMINI_H



13.00 Figure 87: E05_RAMINI_I



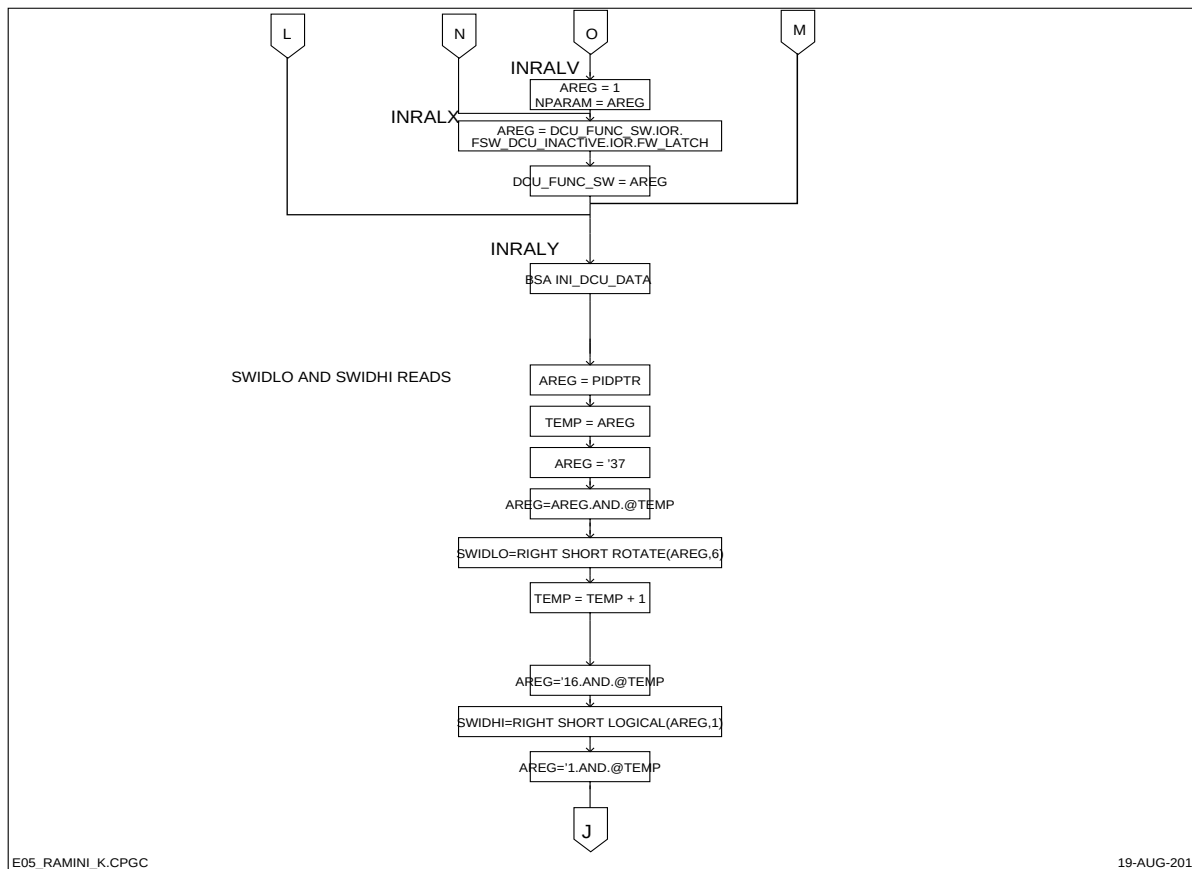
13.00 Figure 88: E05_RAMINI_J



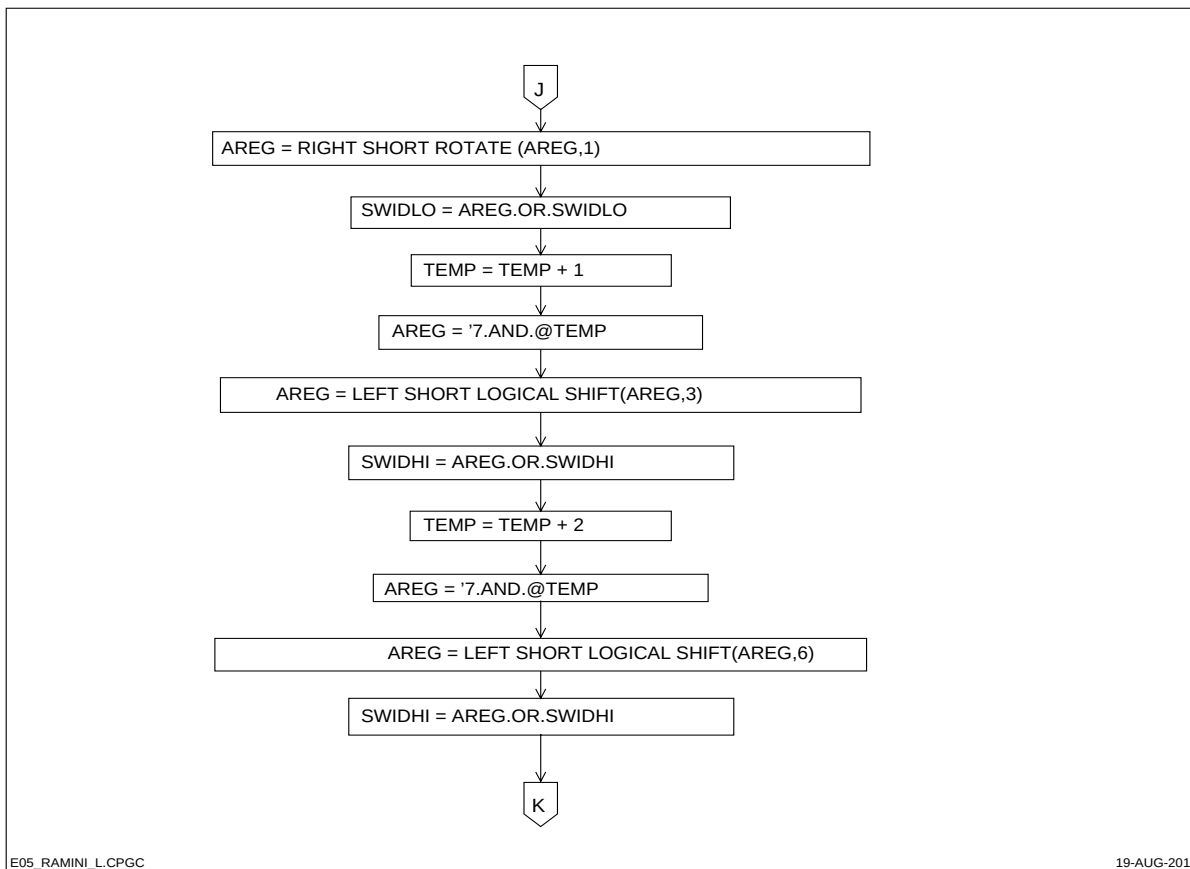
E05_RAMINI_J.CPGC

15-MAY-2024

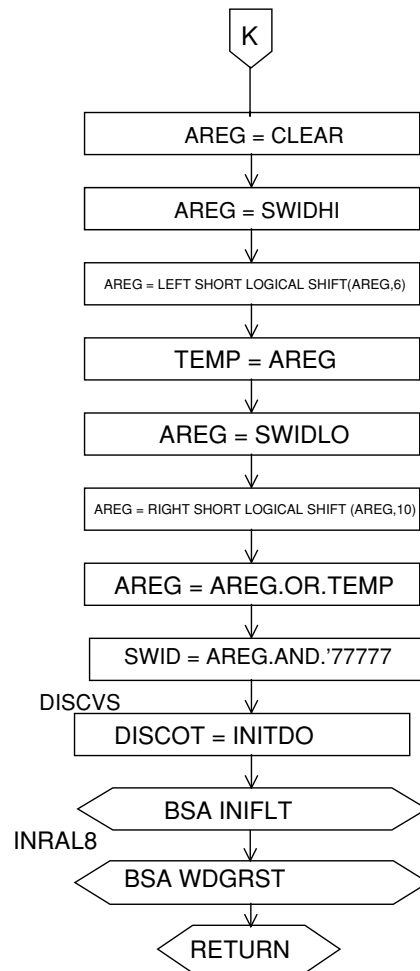
13.00 Figure 89: E05_RAMINI_K



13.00 Figure 90: E05_RAMINI_L



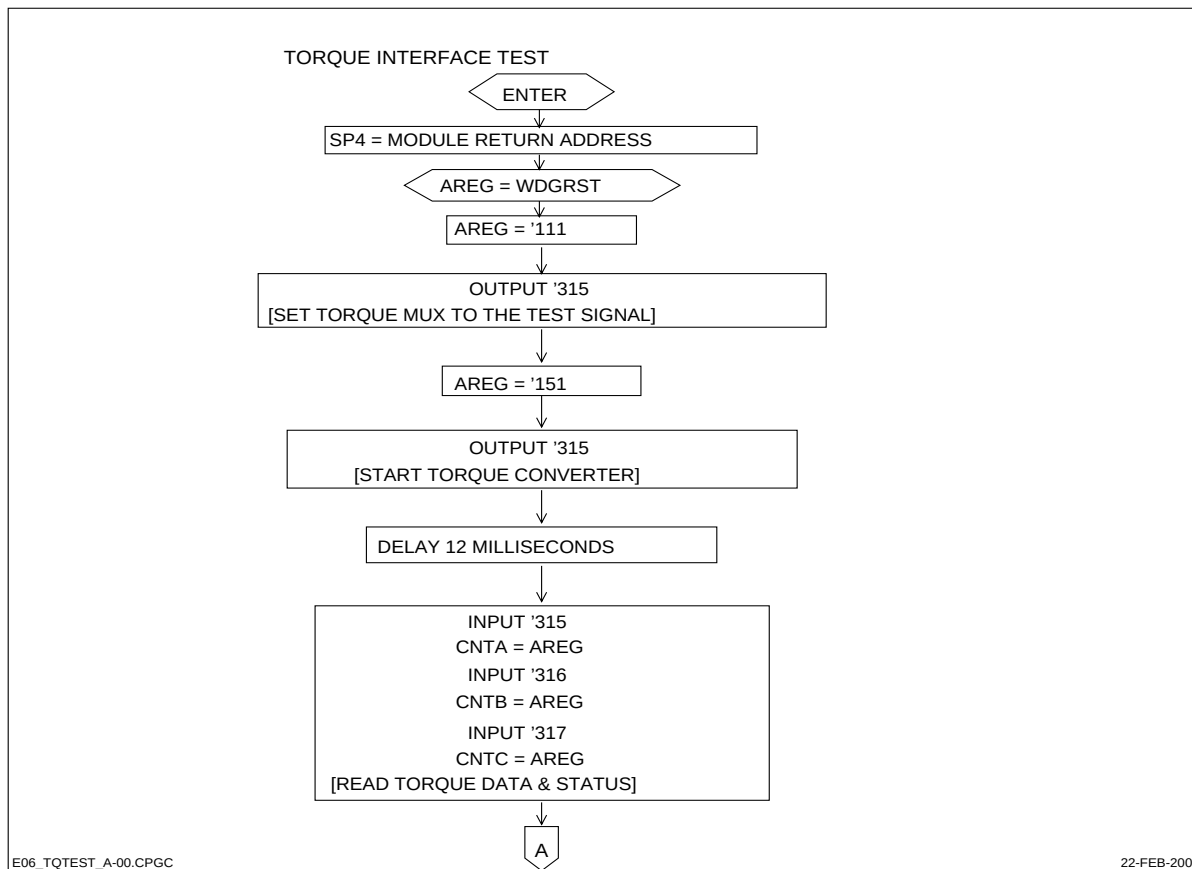
13.00 Figure 91: E05_RAMINI_M



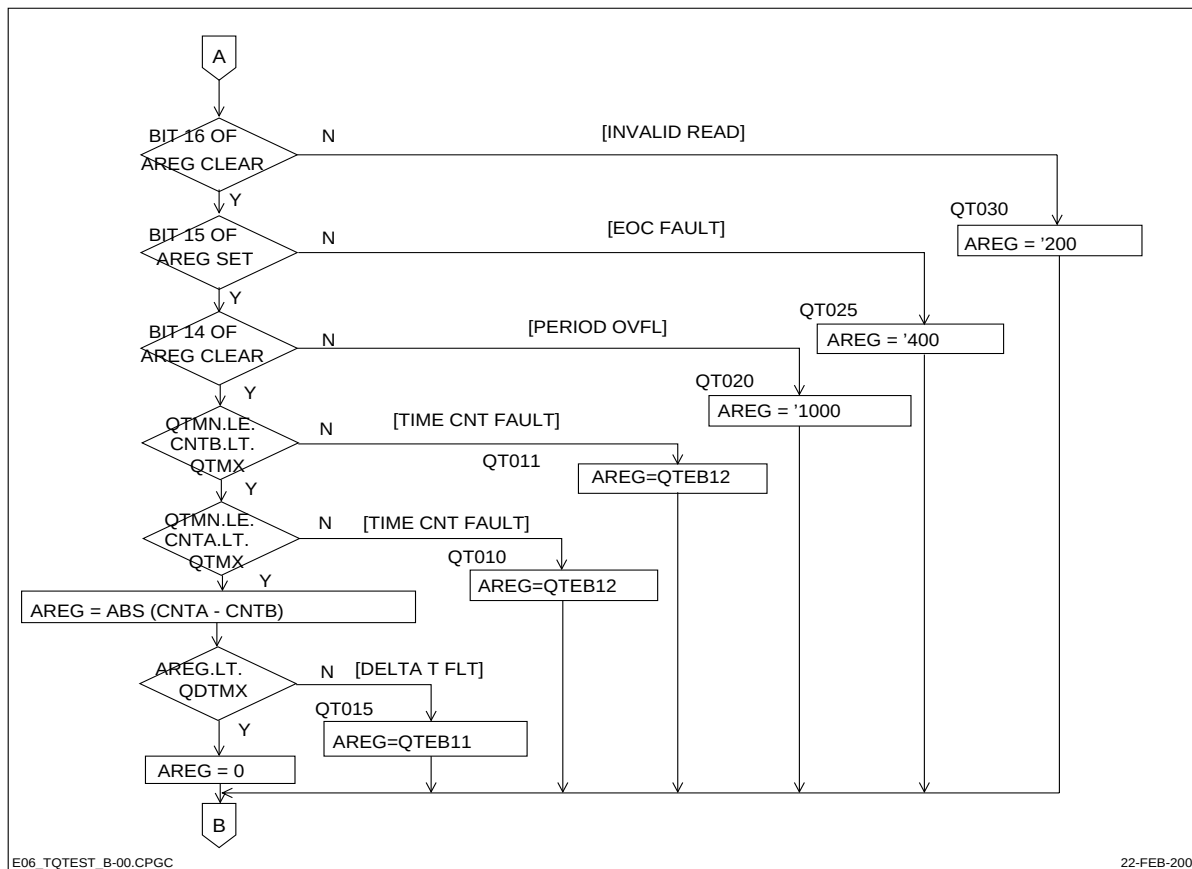
E05_RAMINI_M.CPGC

24-JUL-2024

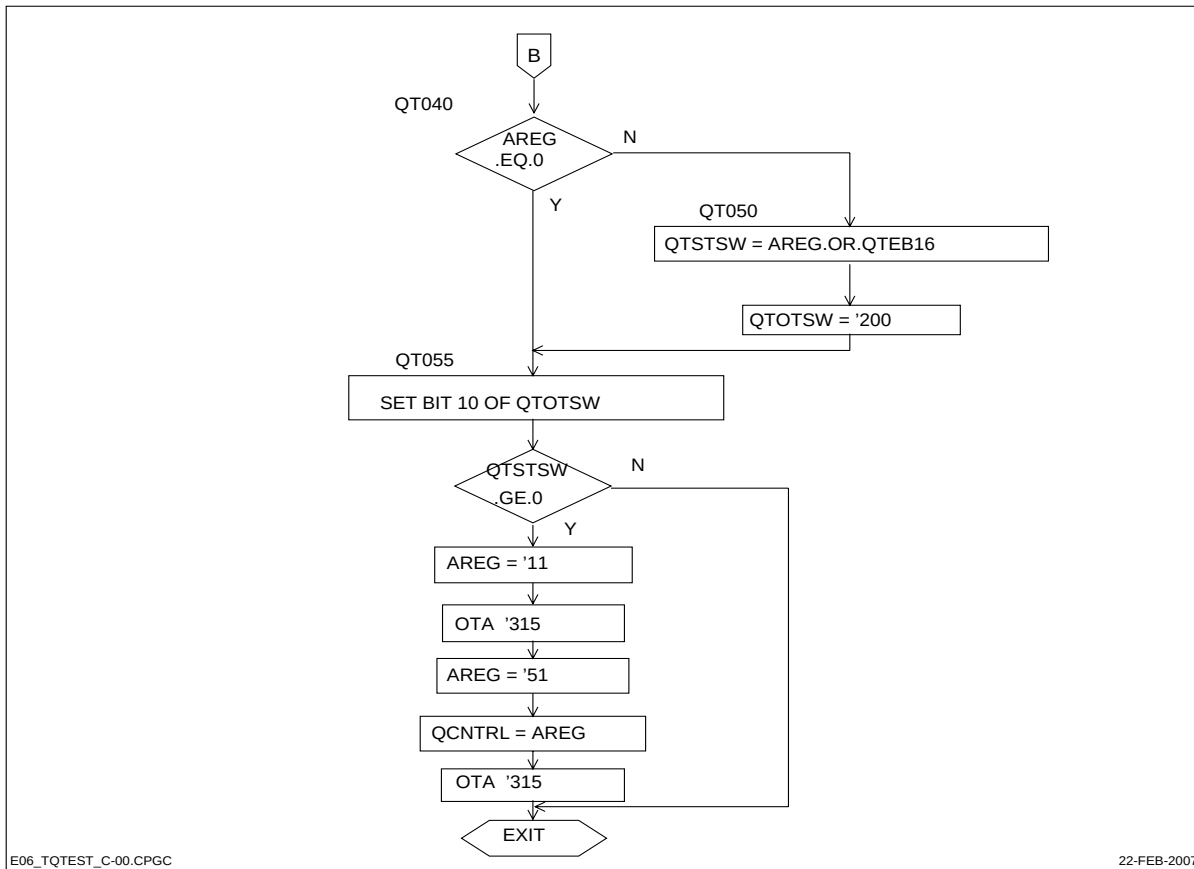
13.00 Figure 92: E06_TQTEST_A



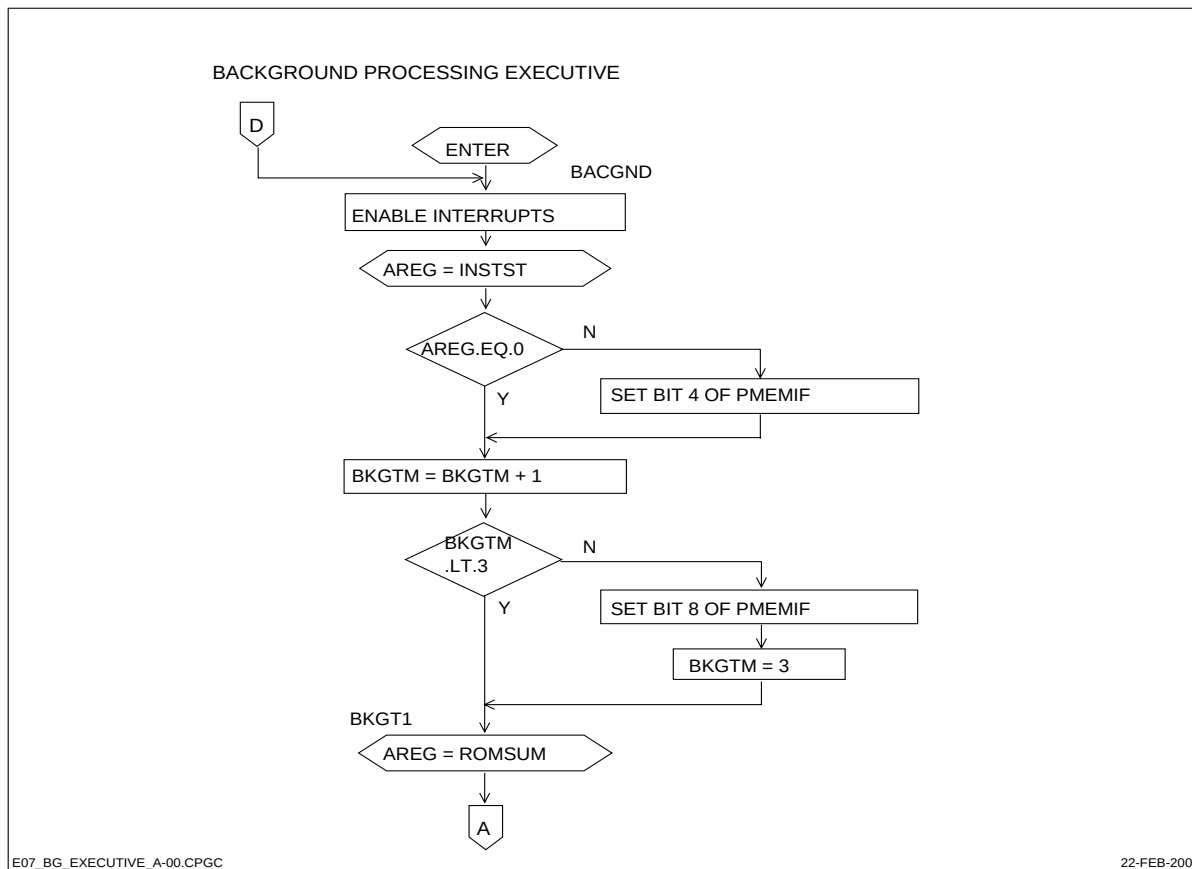
13.00 Figure 93: E06_TQTEST_B



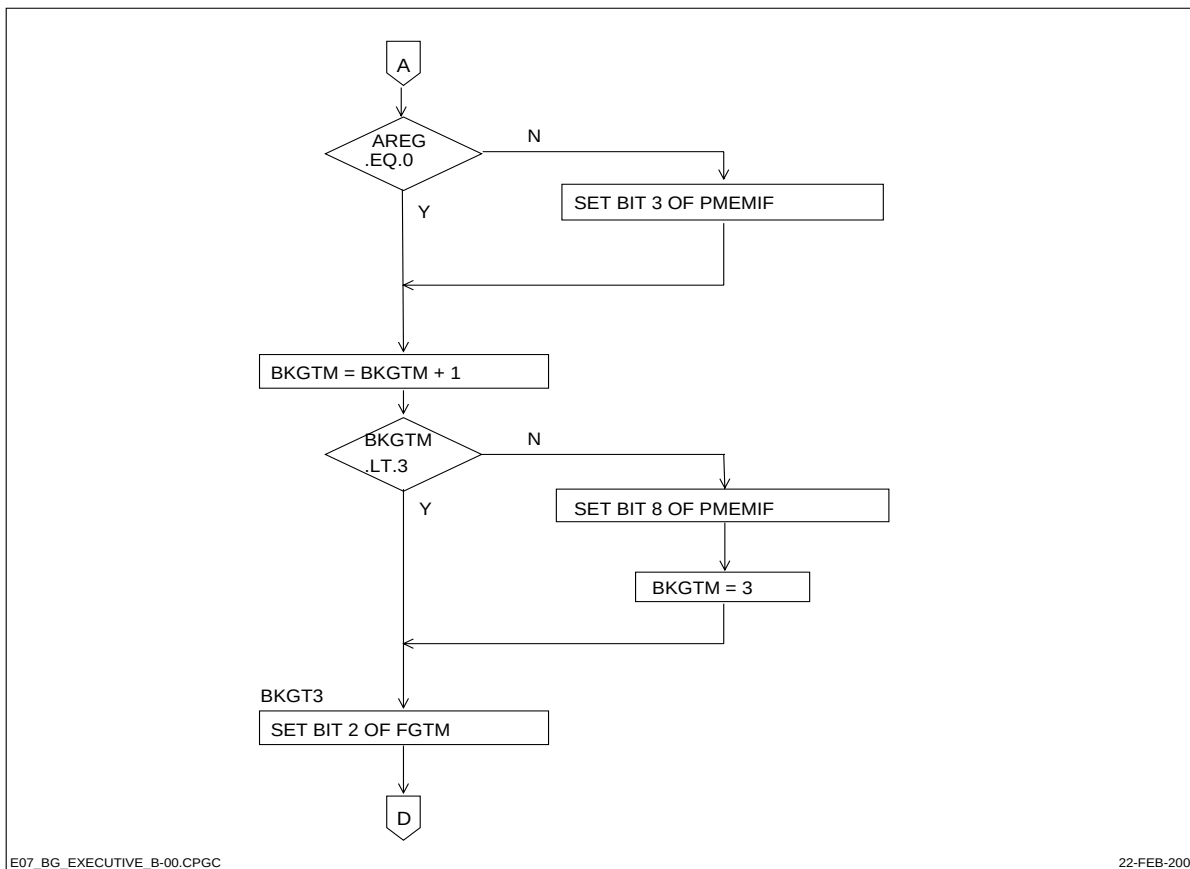
13.00 Figure 94: E06_TQTEST_C



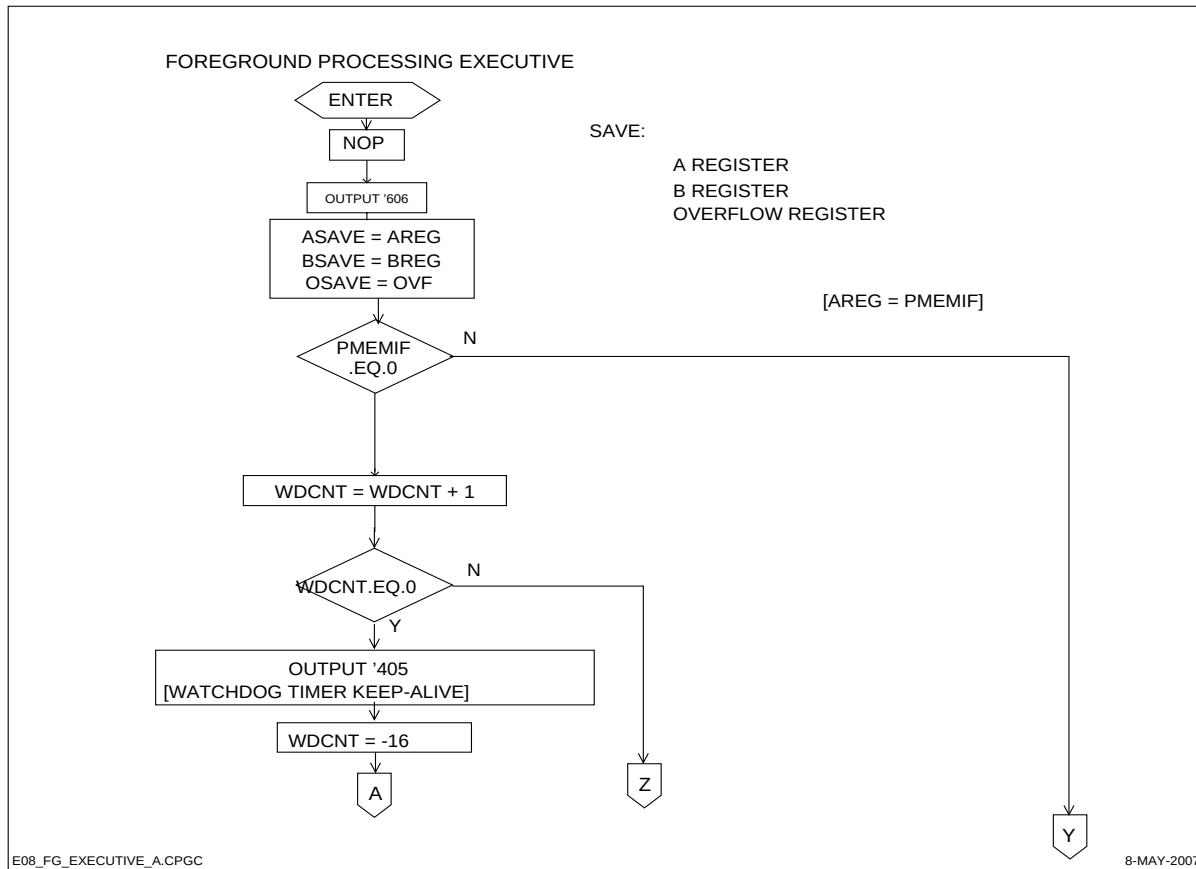
13.00 Figure 95: E07_BG_EXECUTIVE_A



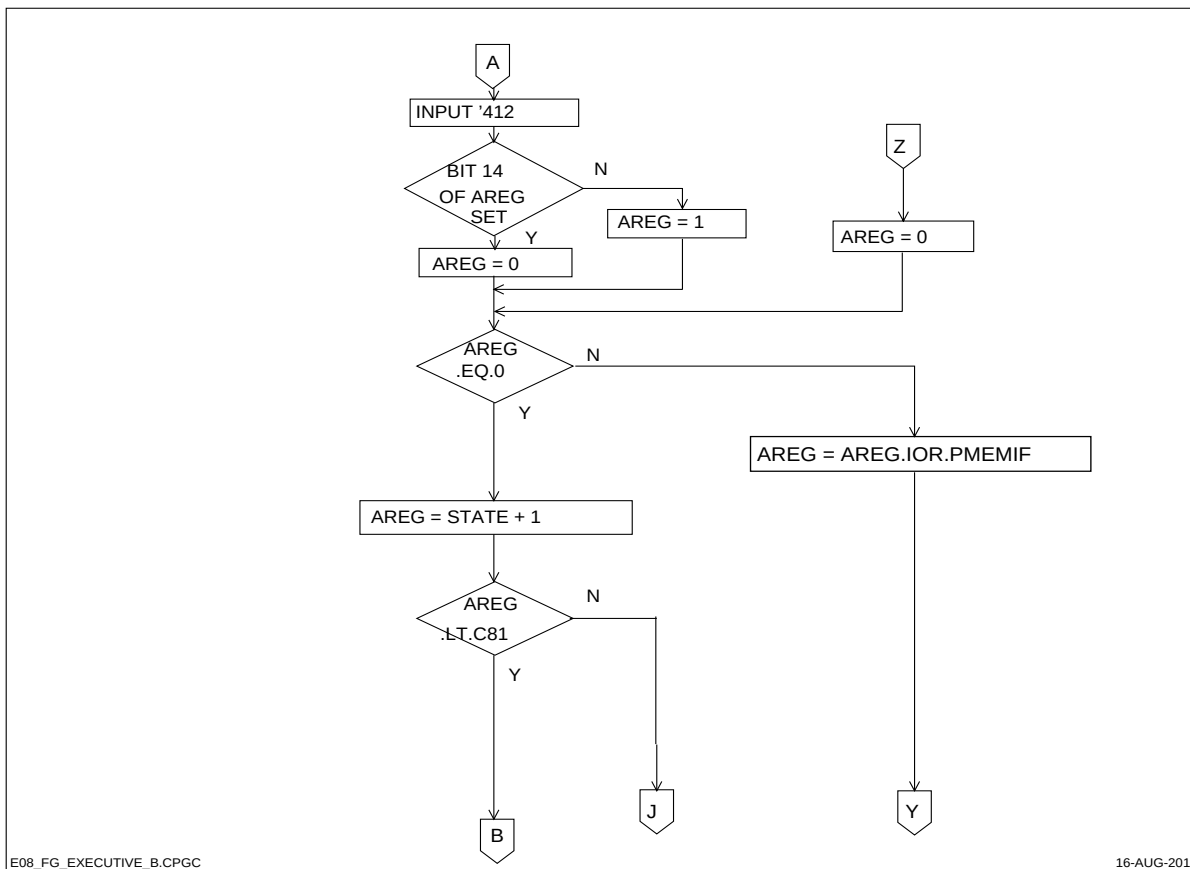
13.00 Figure 96: E07_BG_EXECUTIVE_B



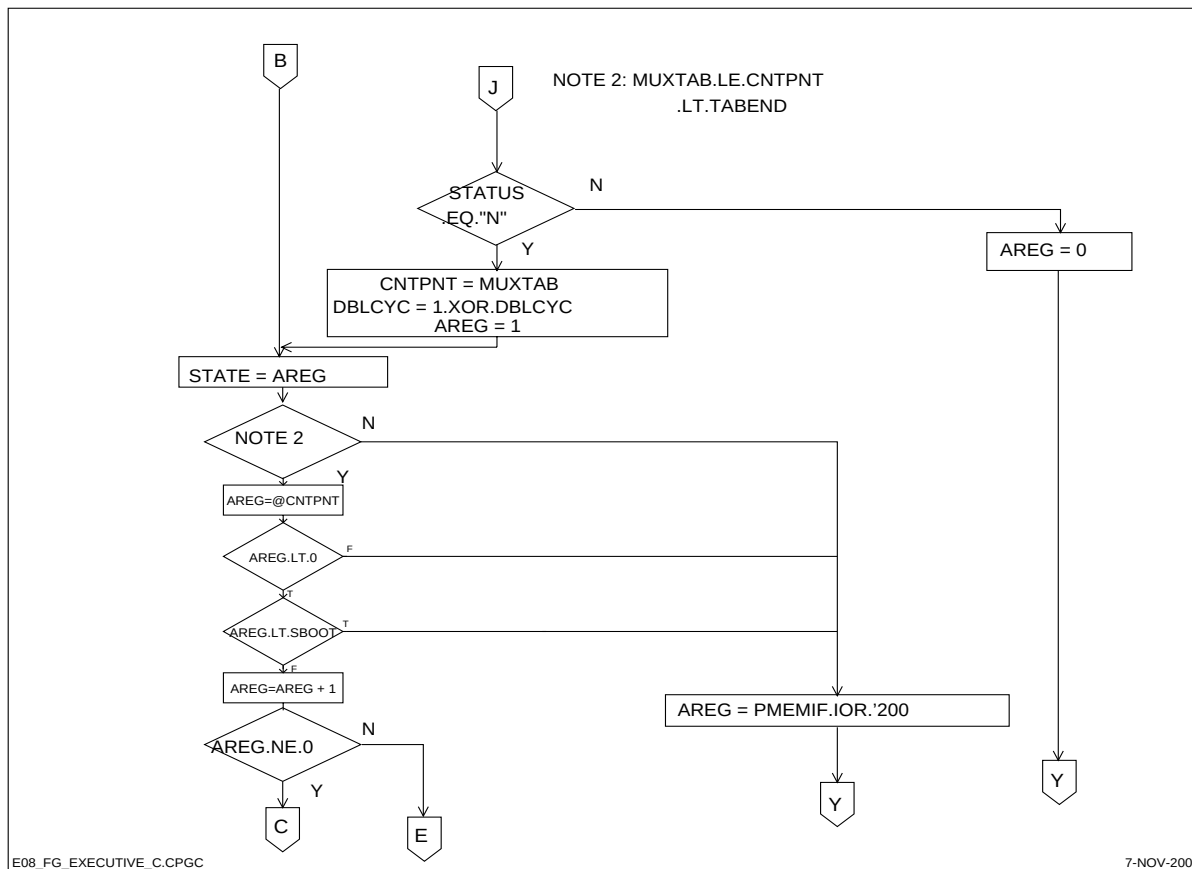
13.00 Figure 97: E08_FG_EXECUTIVE_A



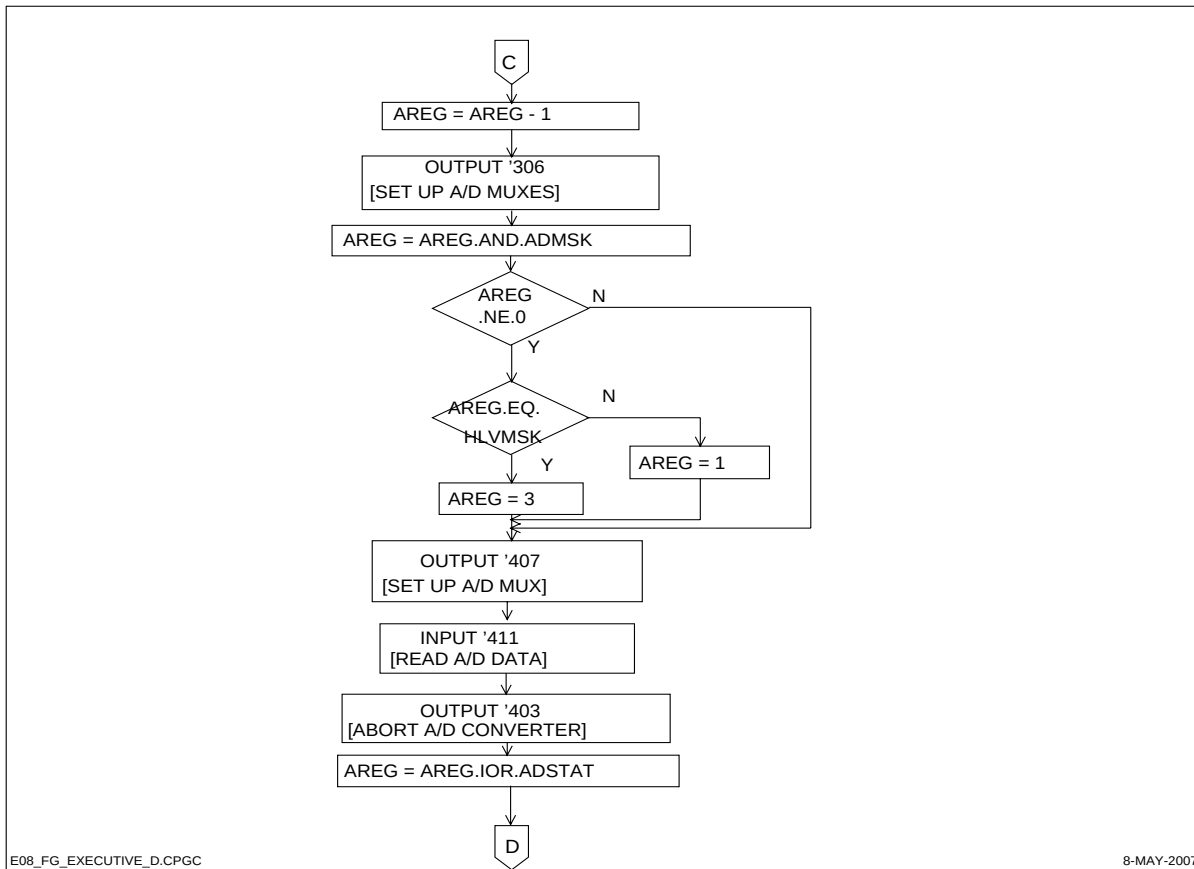
13.00 Figure 98: E08_FG_EXECUTIVE_B



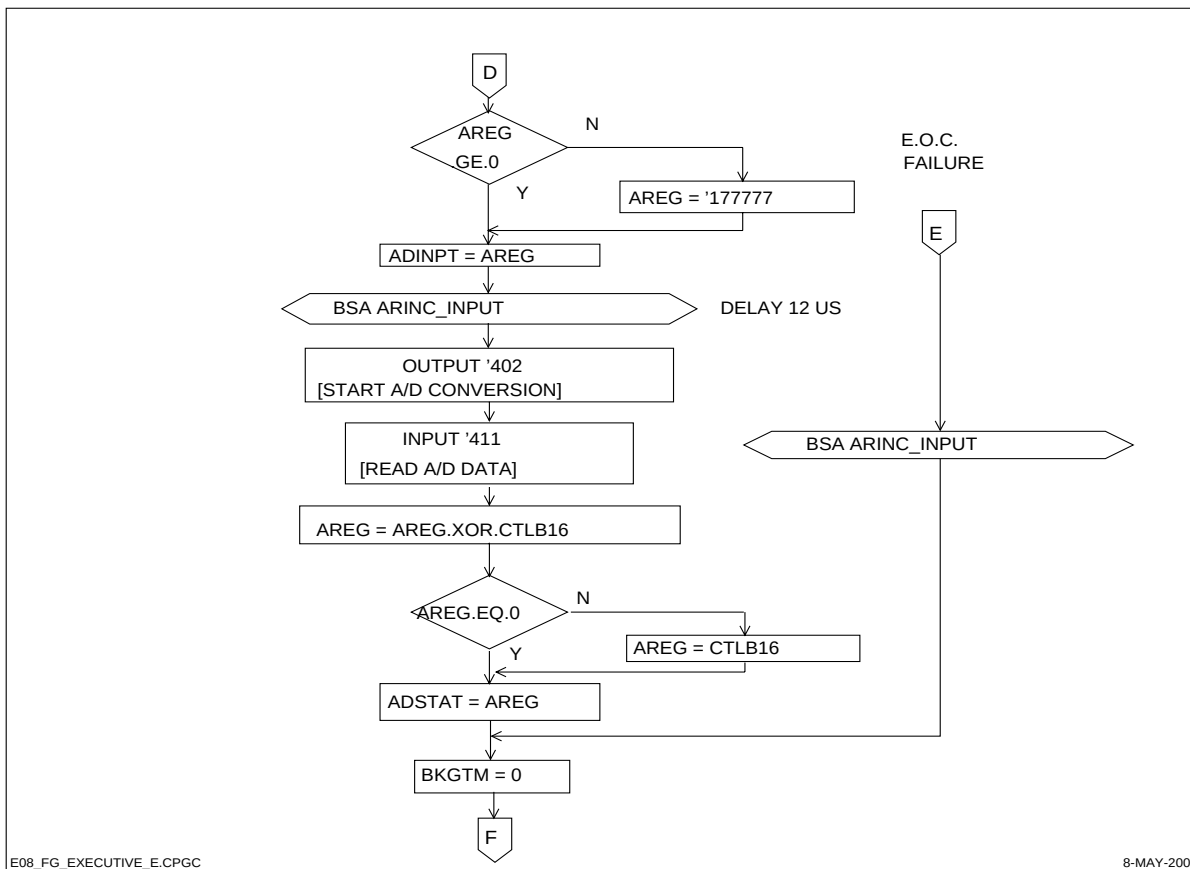
13.00 Figure 99: E08_FG_EXECUTIVE_C



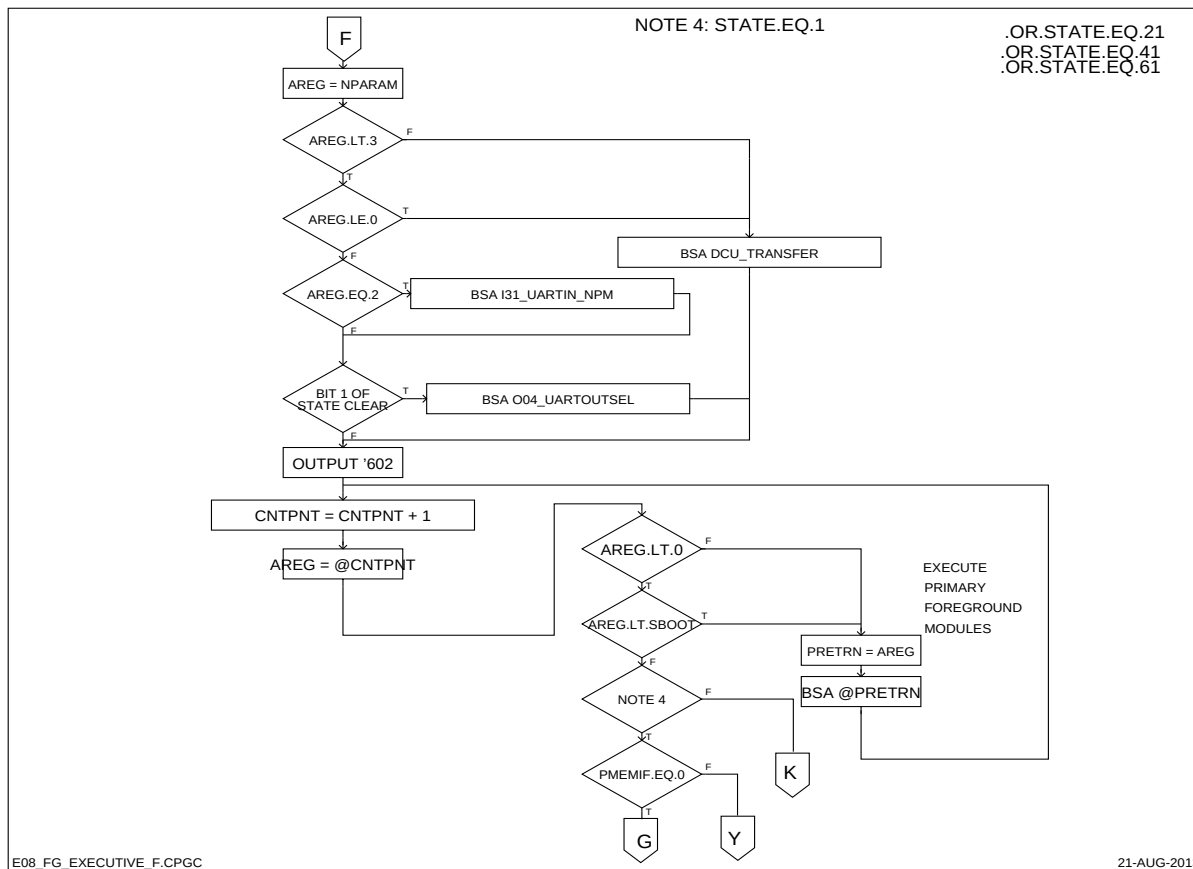
13.00 Figure 100: E08_FG_EXECUTIVE_D



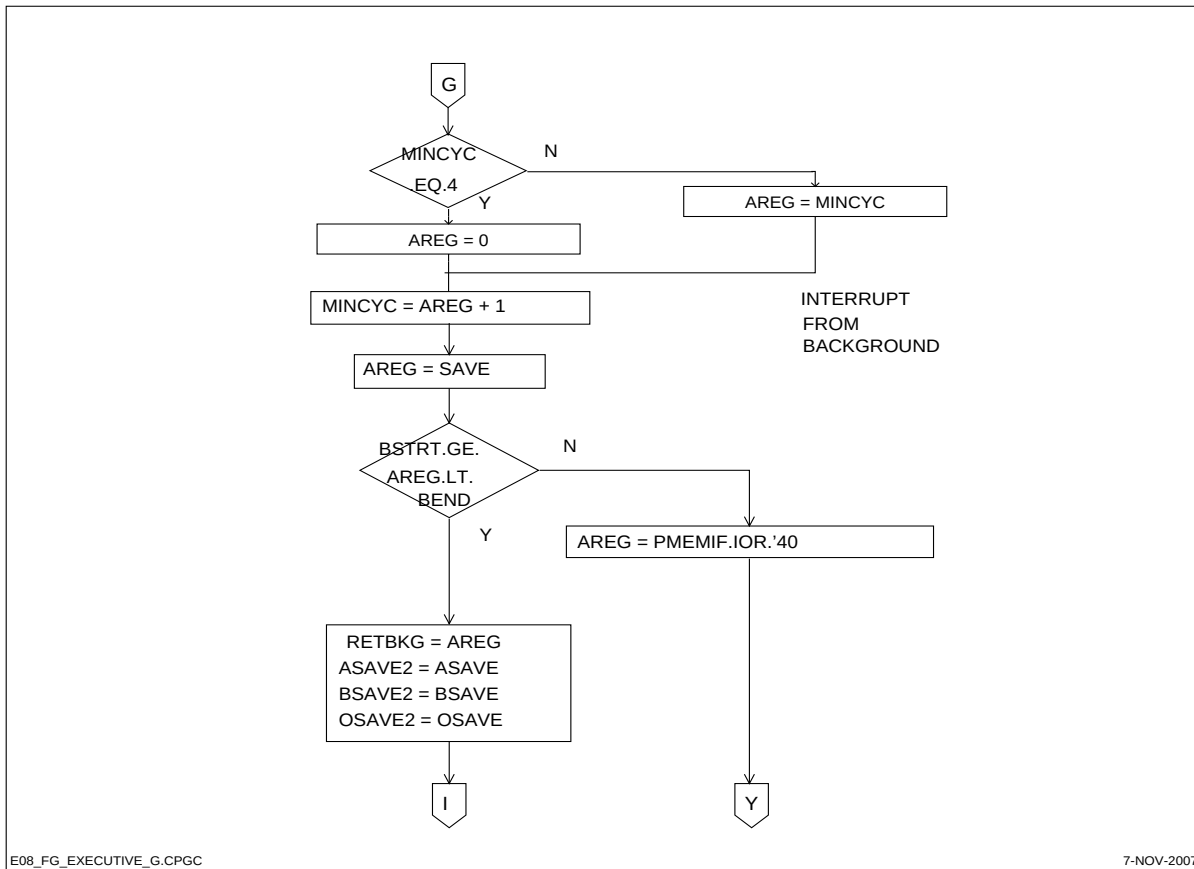
13.00 Figure 101: E08_FG_EXECUTIVE_E



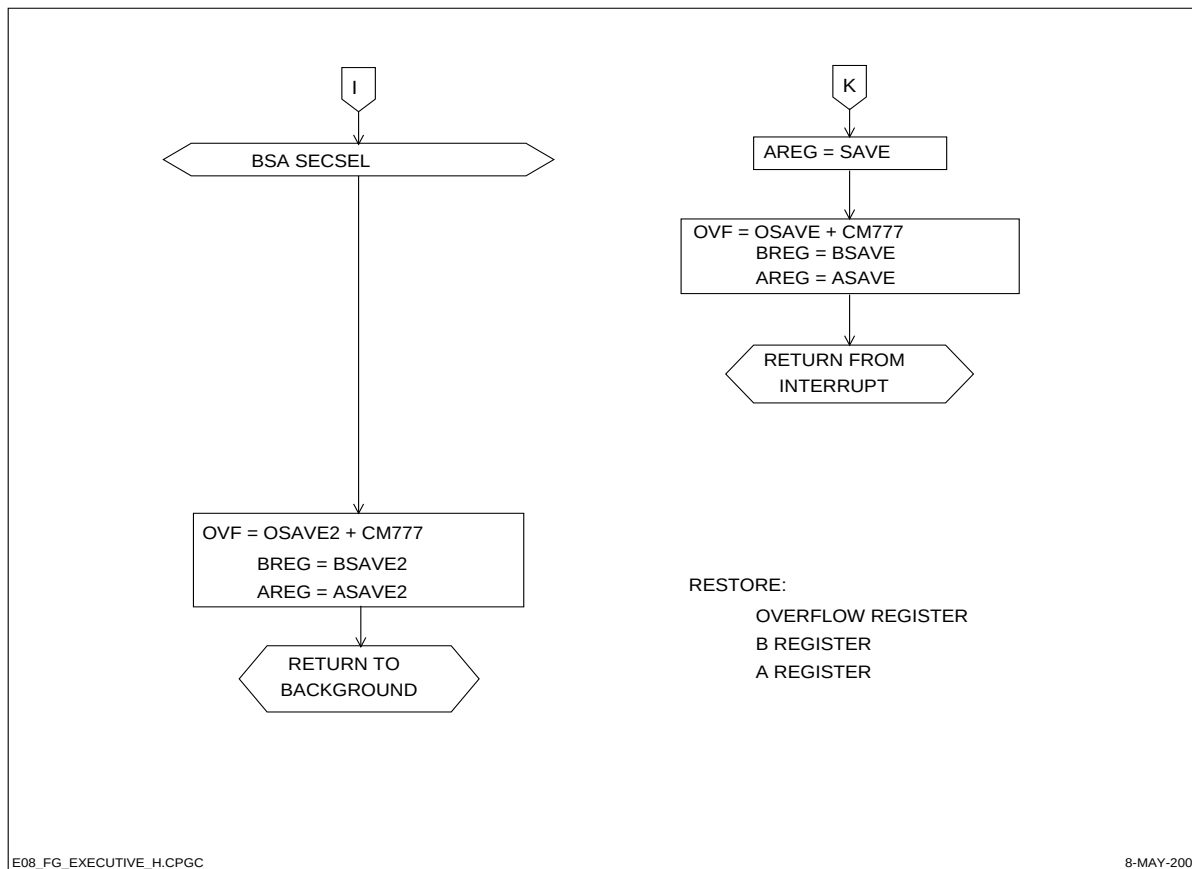
13.00 Figure 102: E08_FG_EXECUTIVE_F



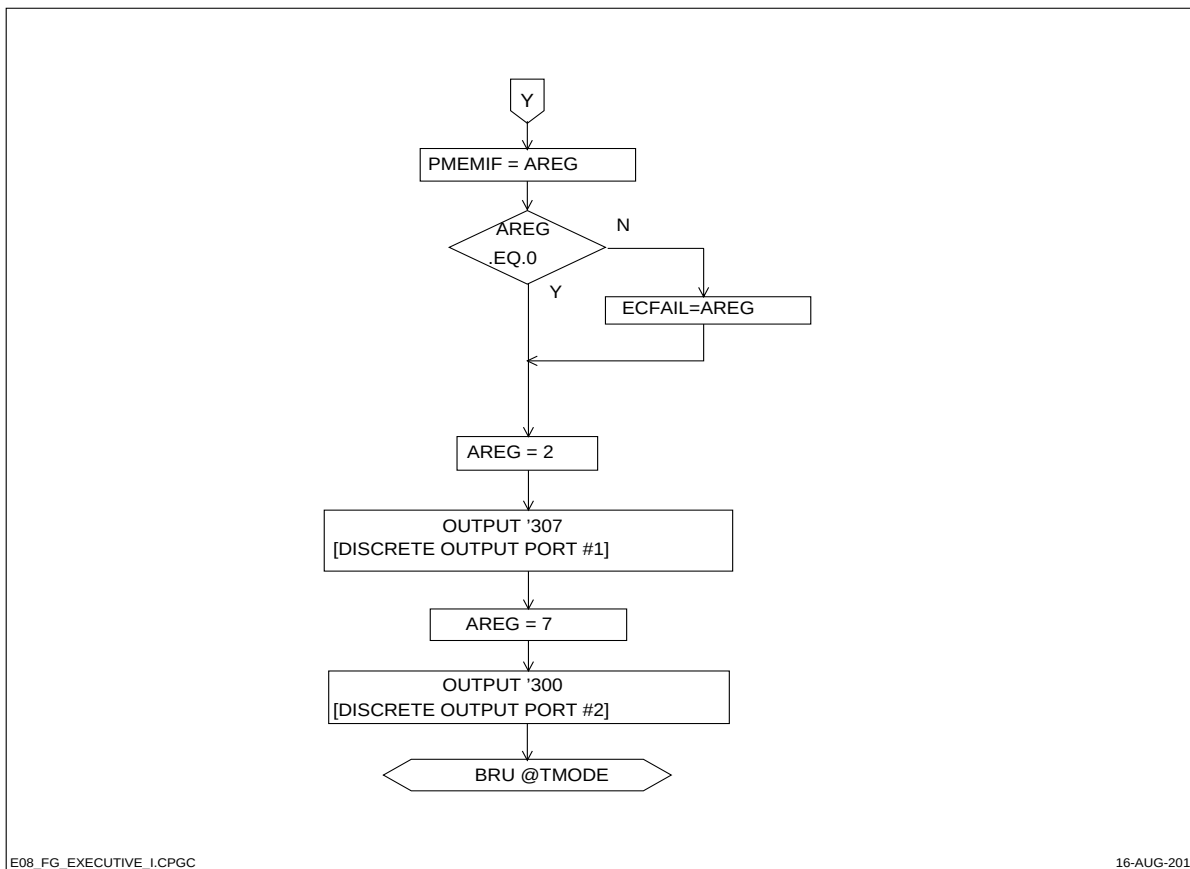
13.00 Figure 103: E08_FG_EXECUTIVE_G



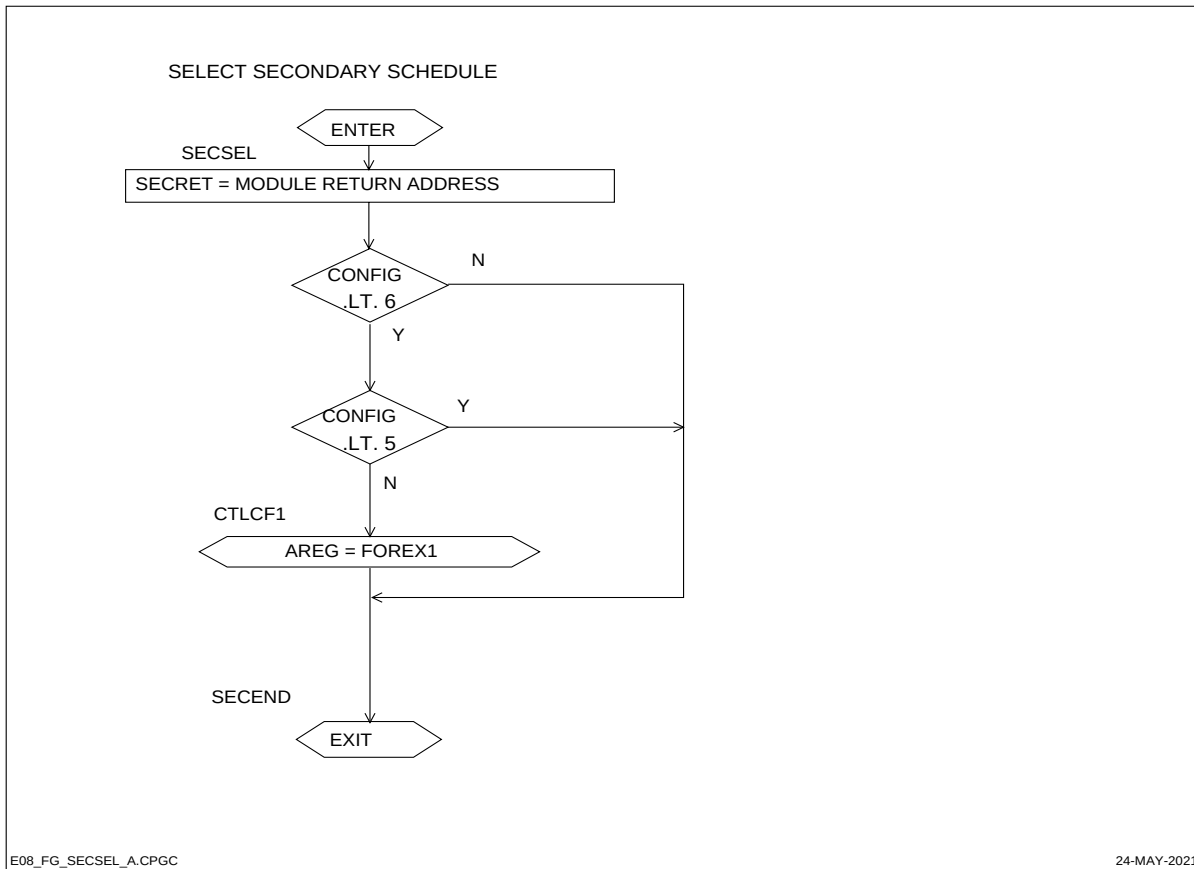
13.00 Figure 104: E08_FG_EXECUTIVE_H



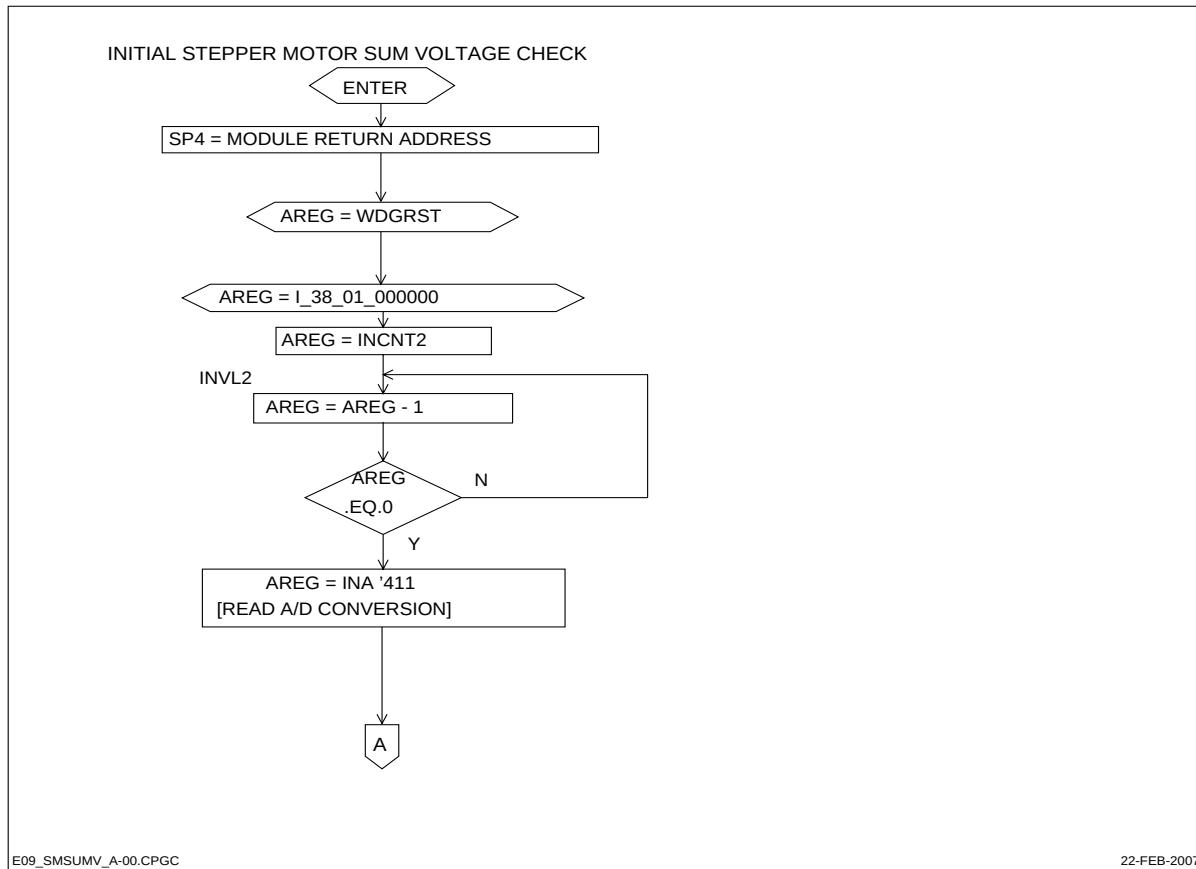
13.00 Figure 105: E08_FG_EXECUTIVE_I



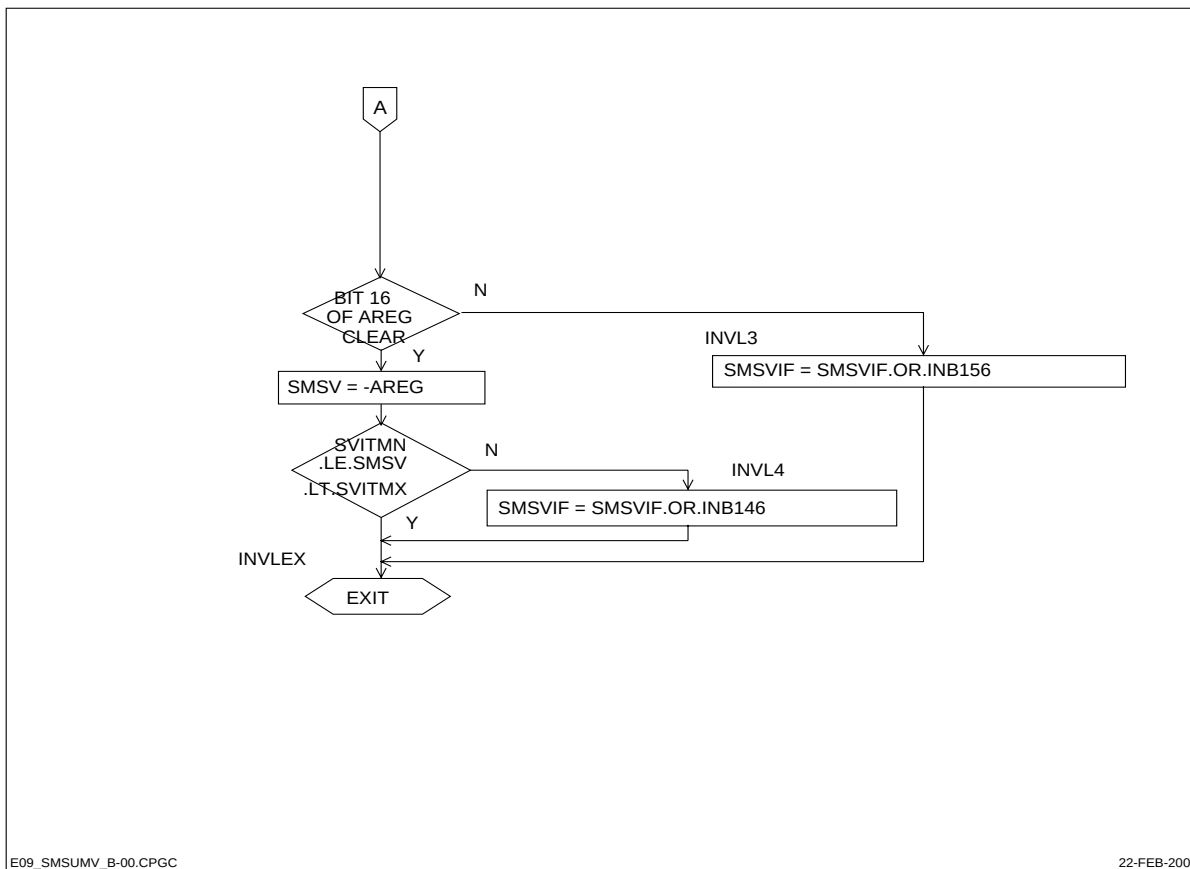
13.00 Figure 106: E08_FG_SECSEL_A



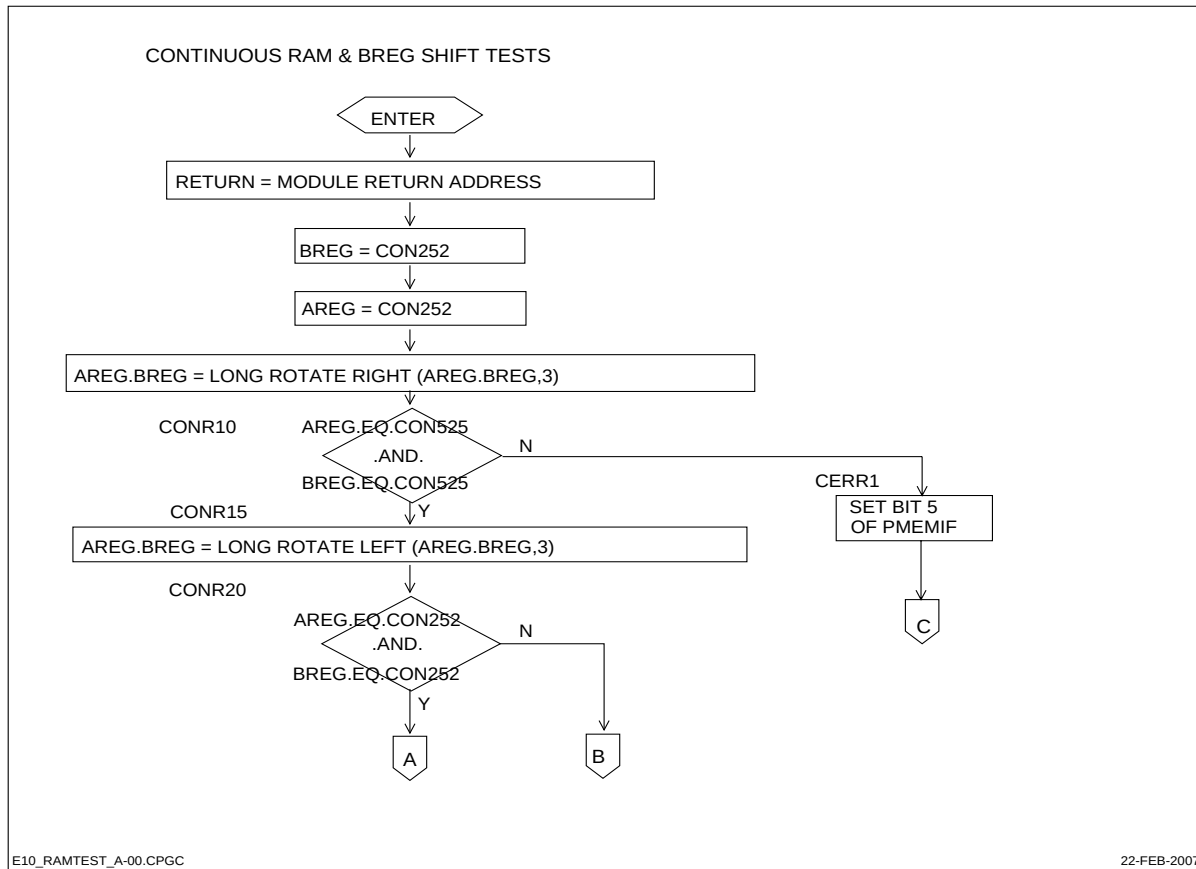
13.00 Figure 107: E09_SMSUMV_A



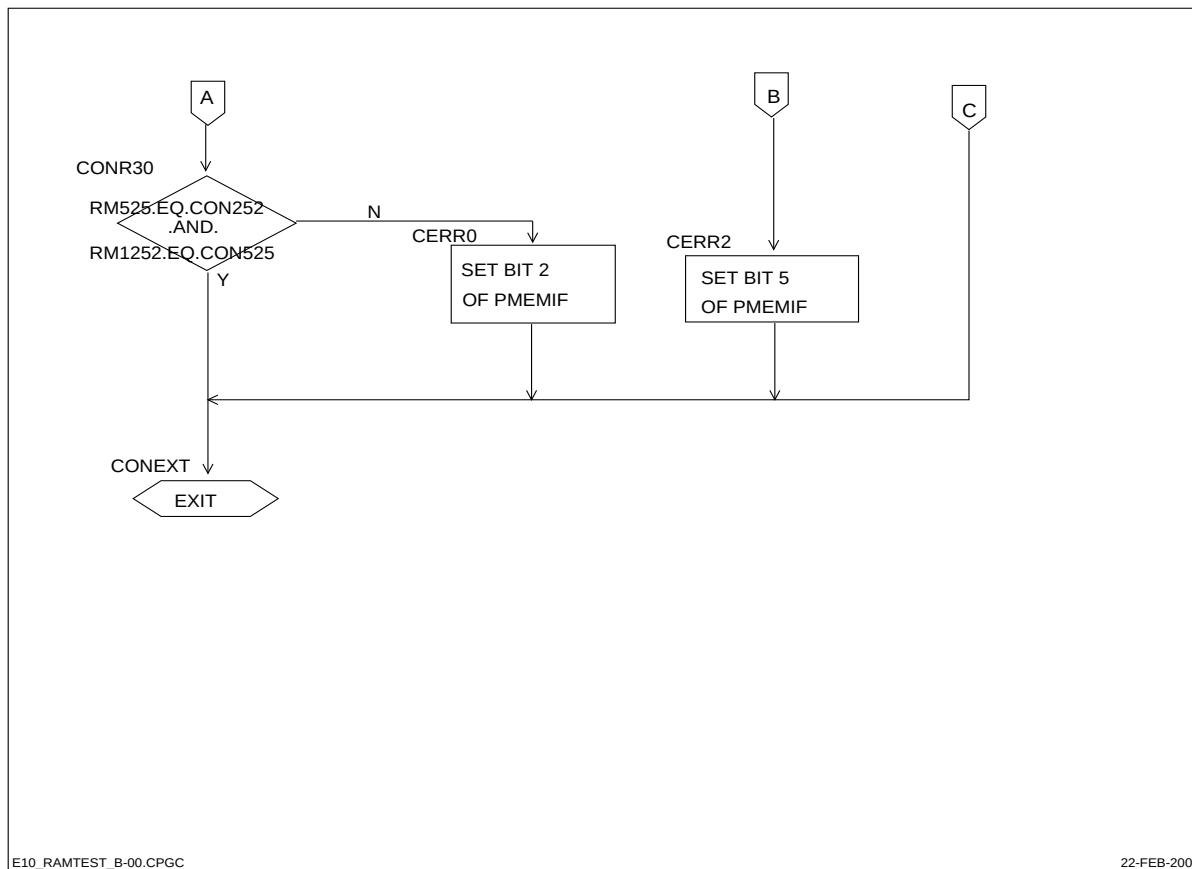
13.00 Figure 108: E09_SMSUMV_B



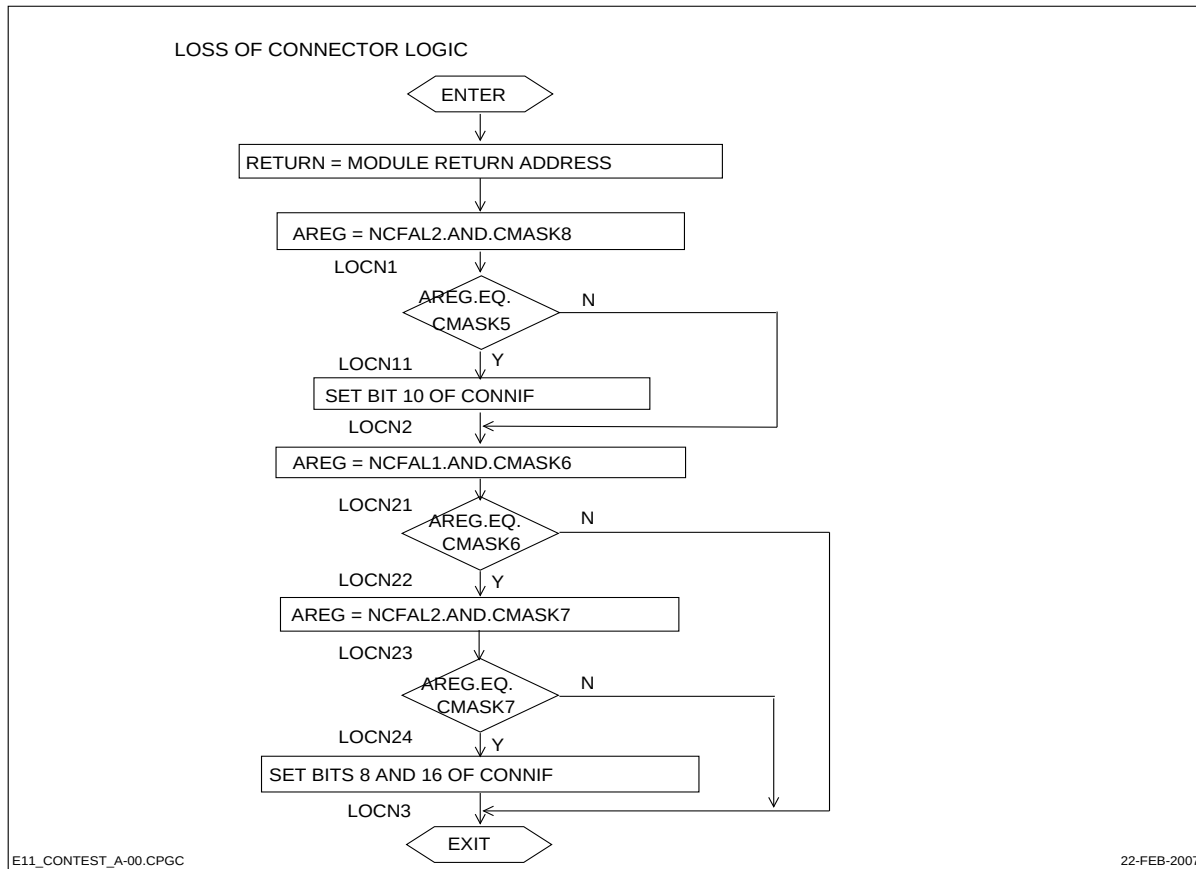
13.00 Figure 109: E10_RAMTEST_A



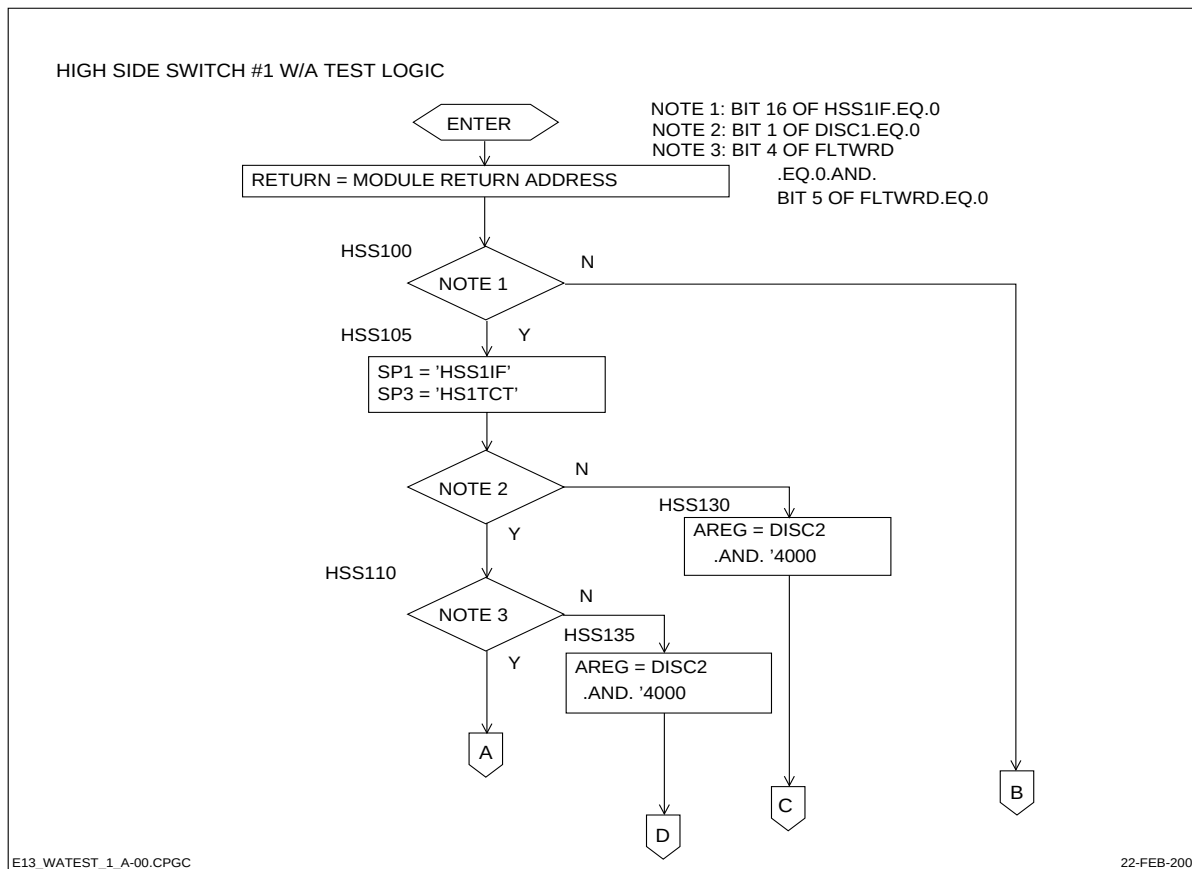
13.00 Figure 110: E10_RAMTEST_B



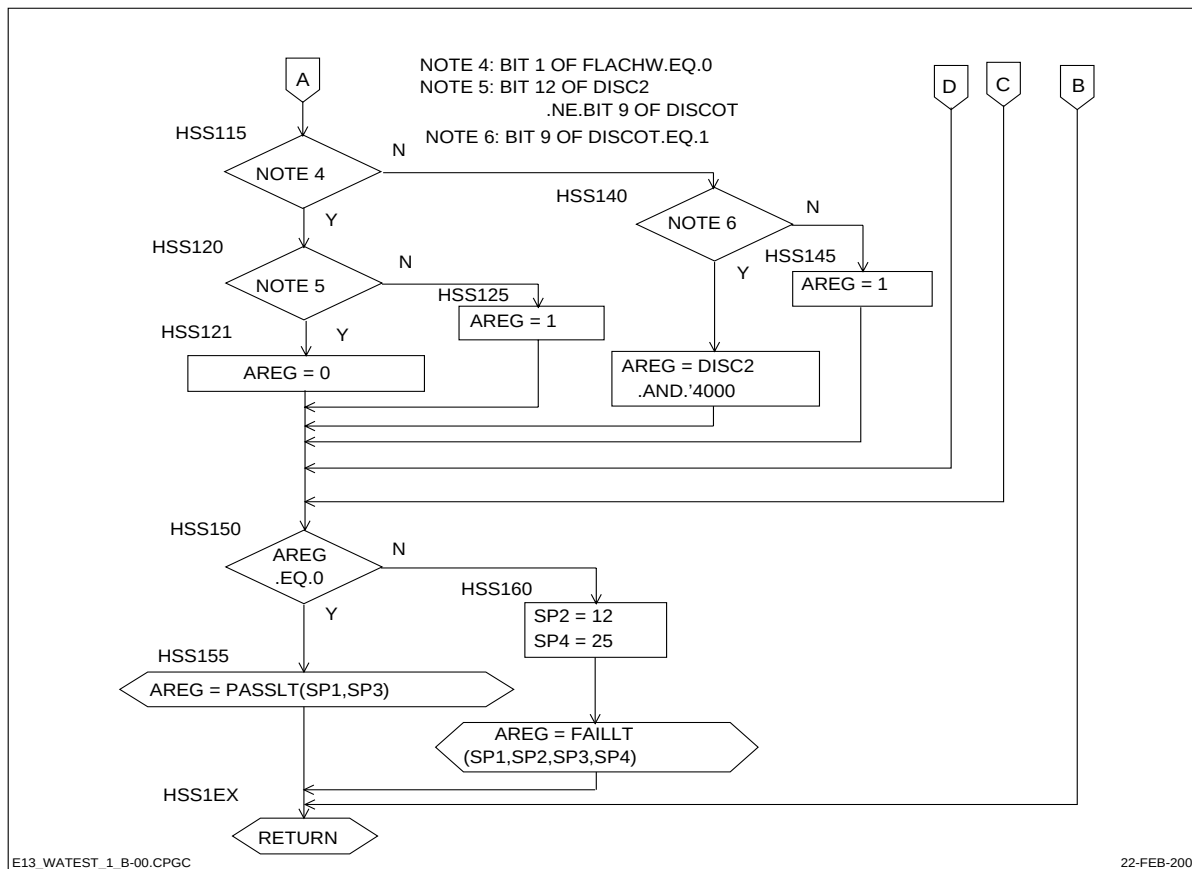
13.00 Figure 111: E11 _CONTEST _A



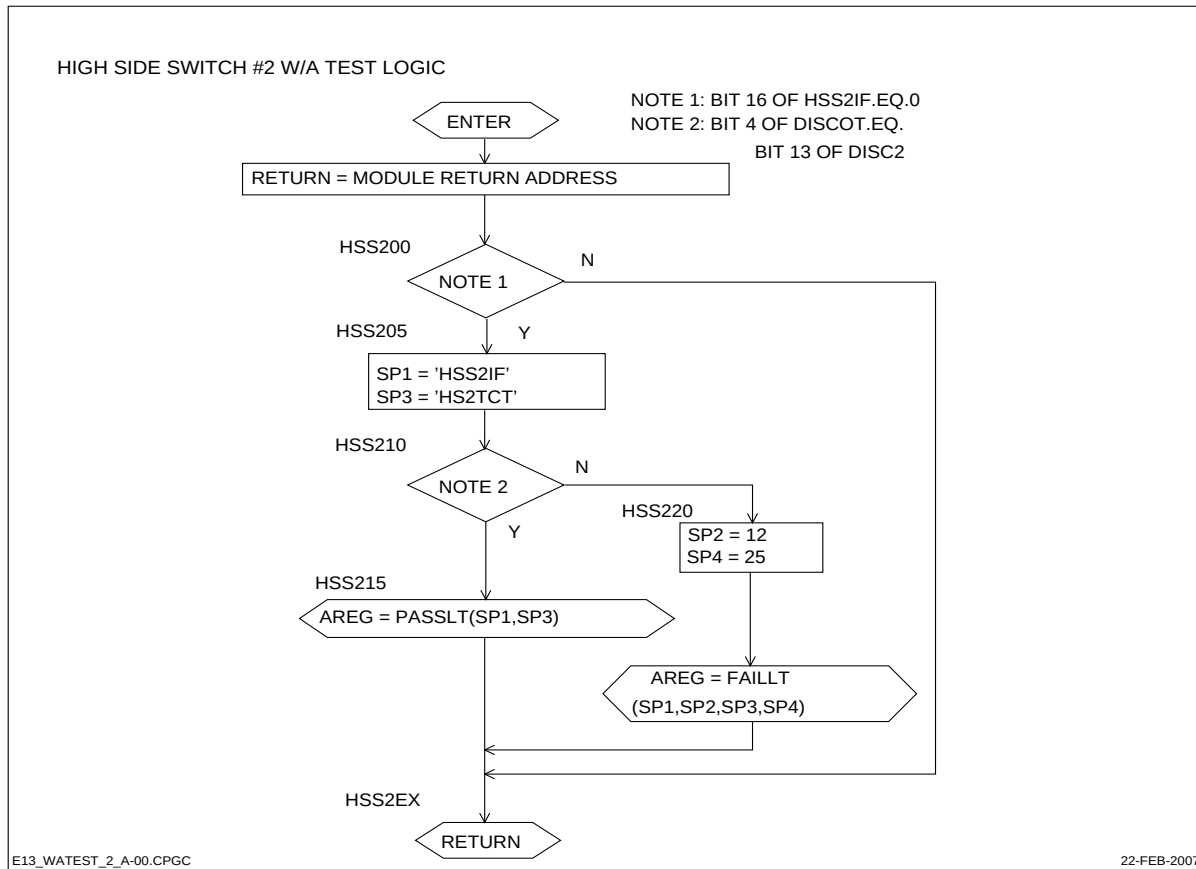
13.00 Figure 112: E13_WATEST_1_A



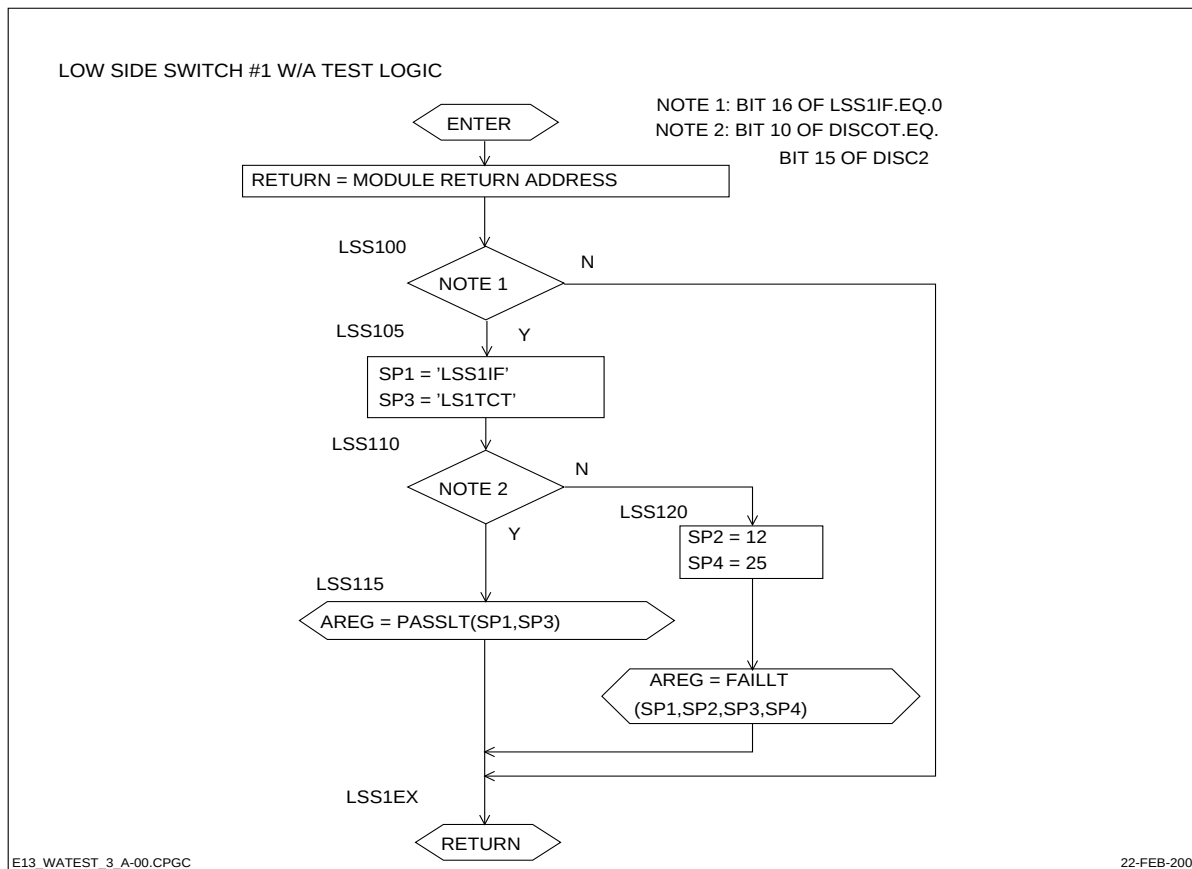
13.00 Figure 113: E13_WATEST_1_B



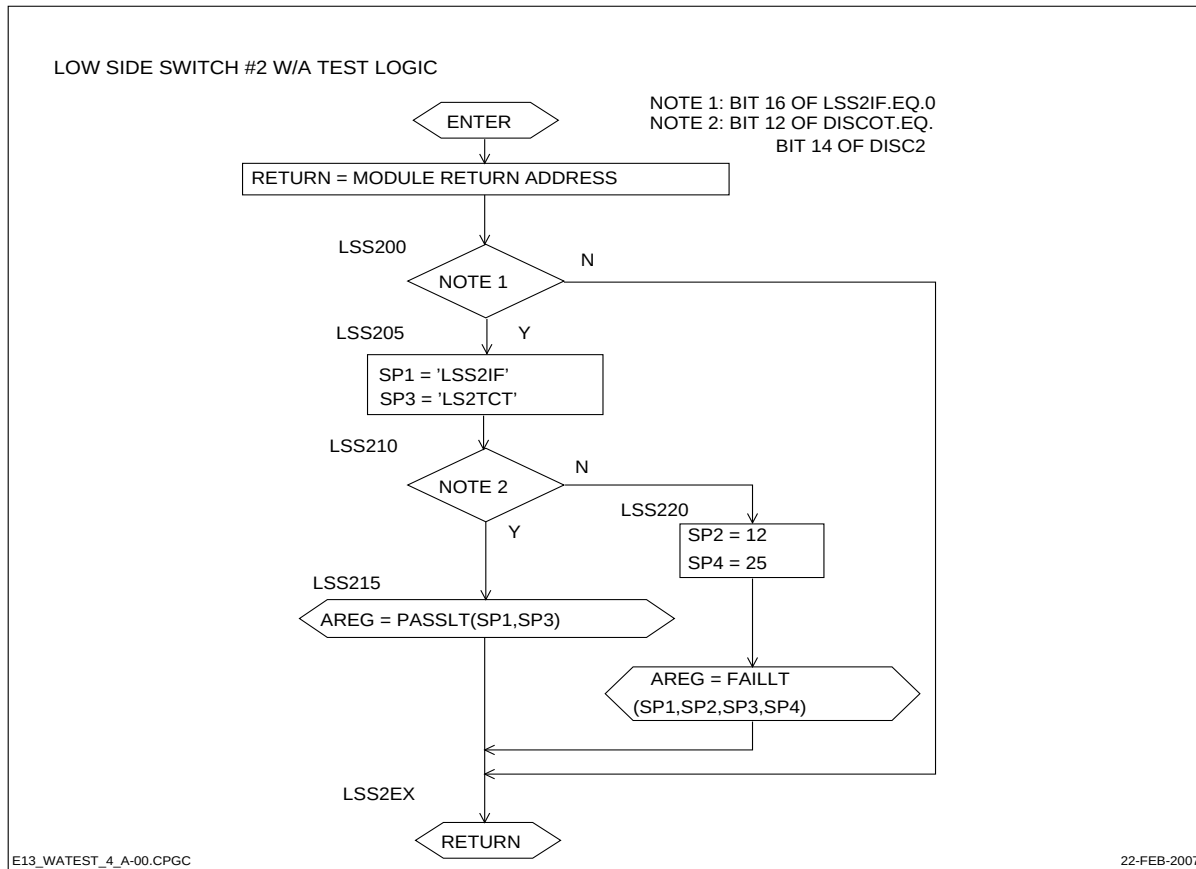
13.00 Figure 114: E13_WATEST_2_A



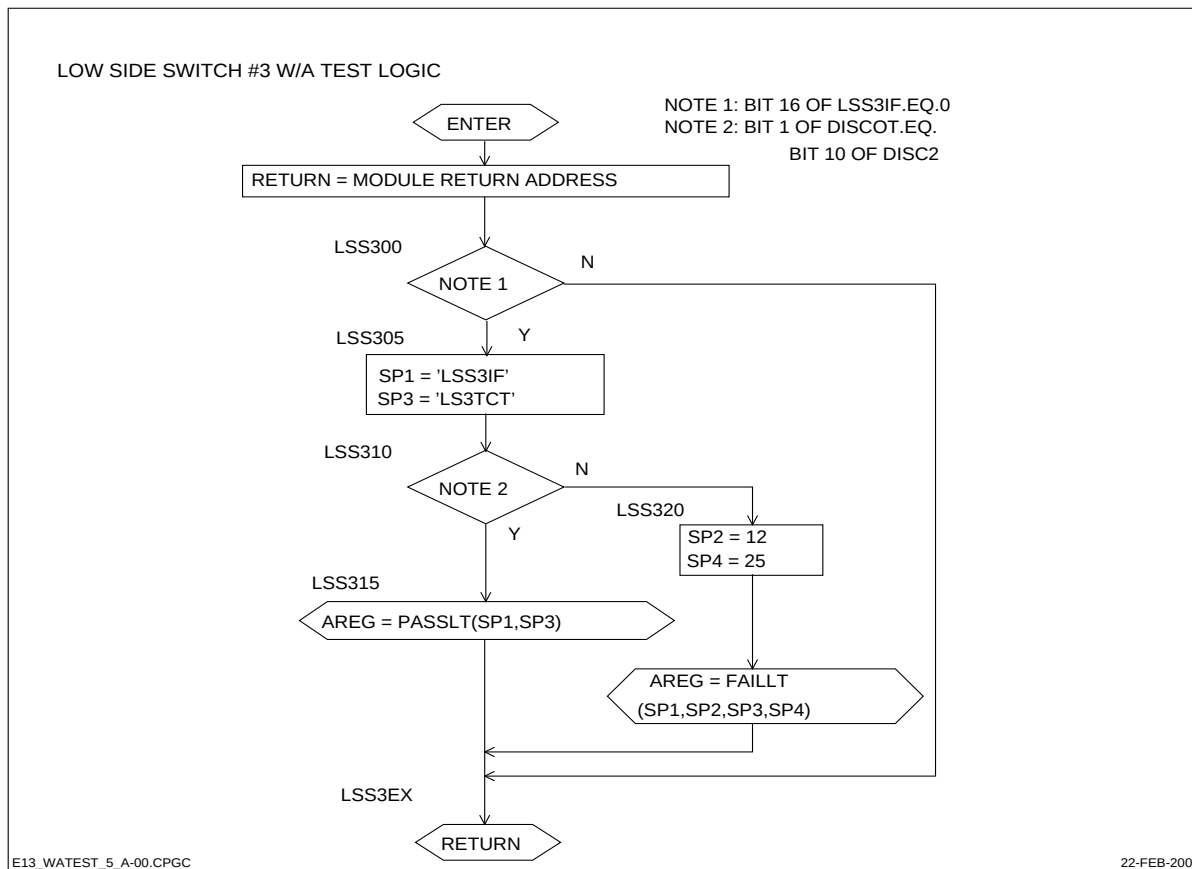
13.00 Figure 115: E13_WATEST_3_A



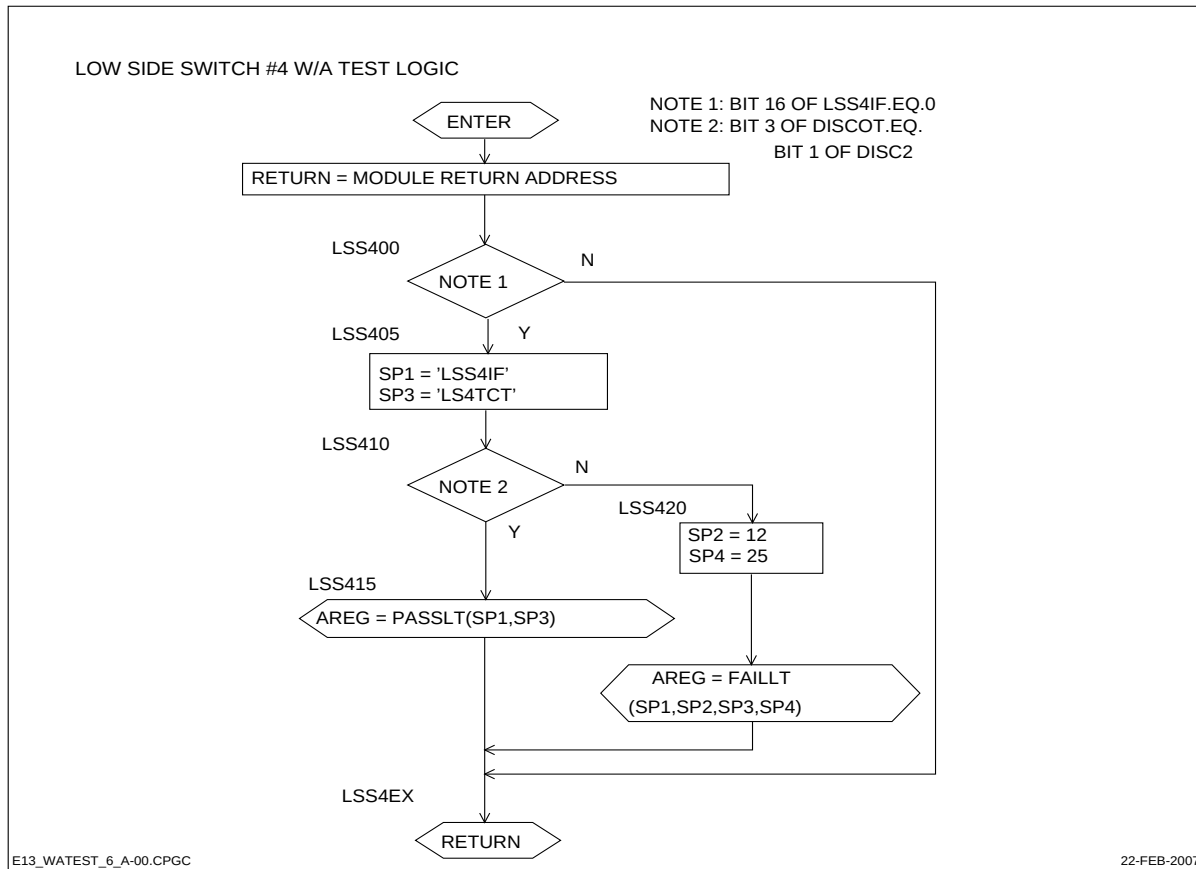
13.00 Figure 116: E13_WATEST_4_A



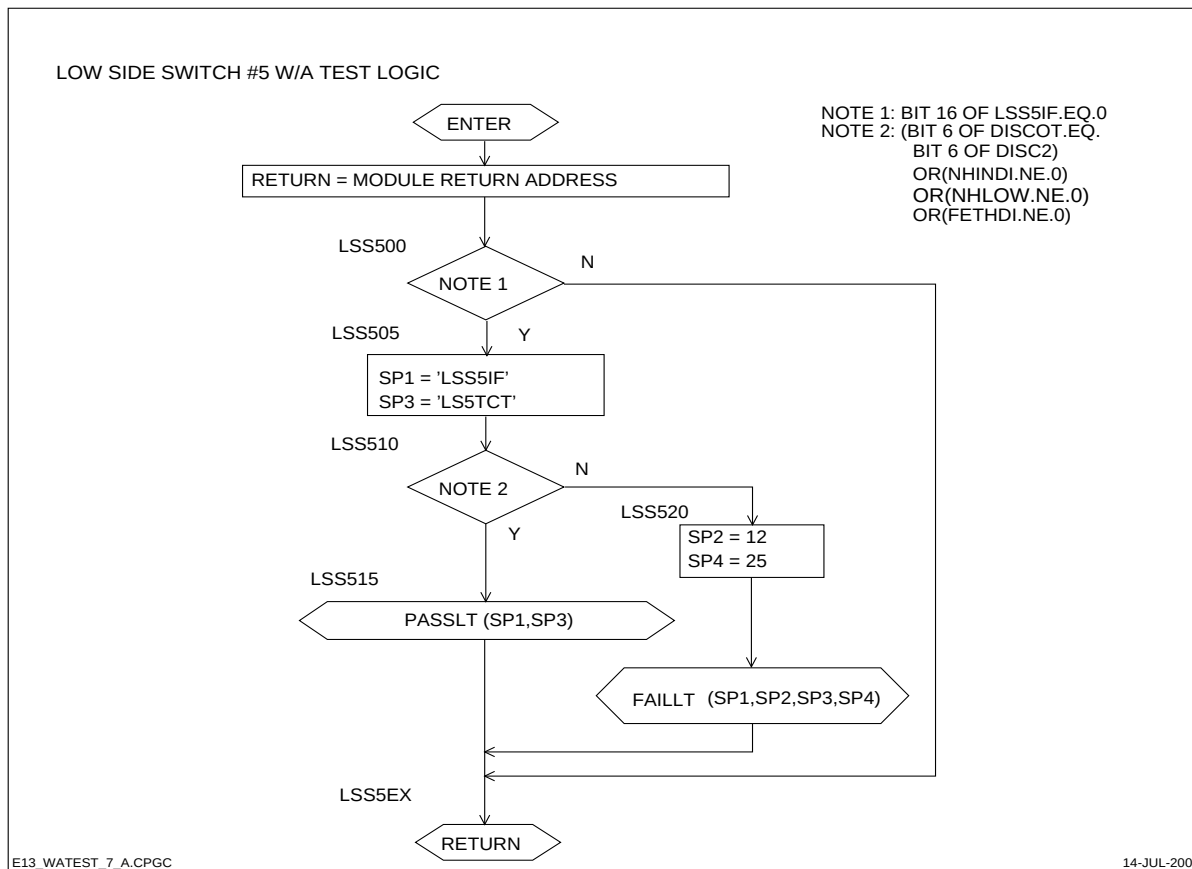
13.00 Figure 117: E13_WATEST_5_A



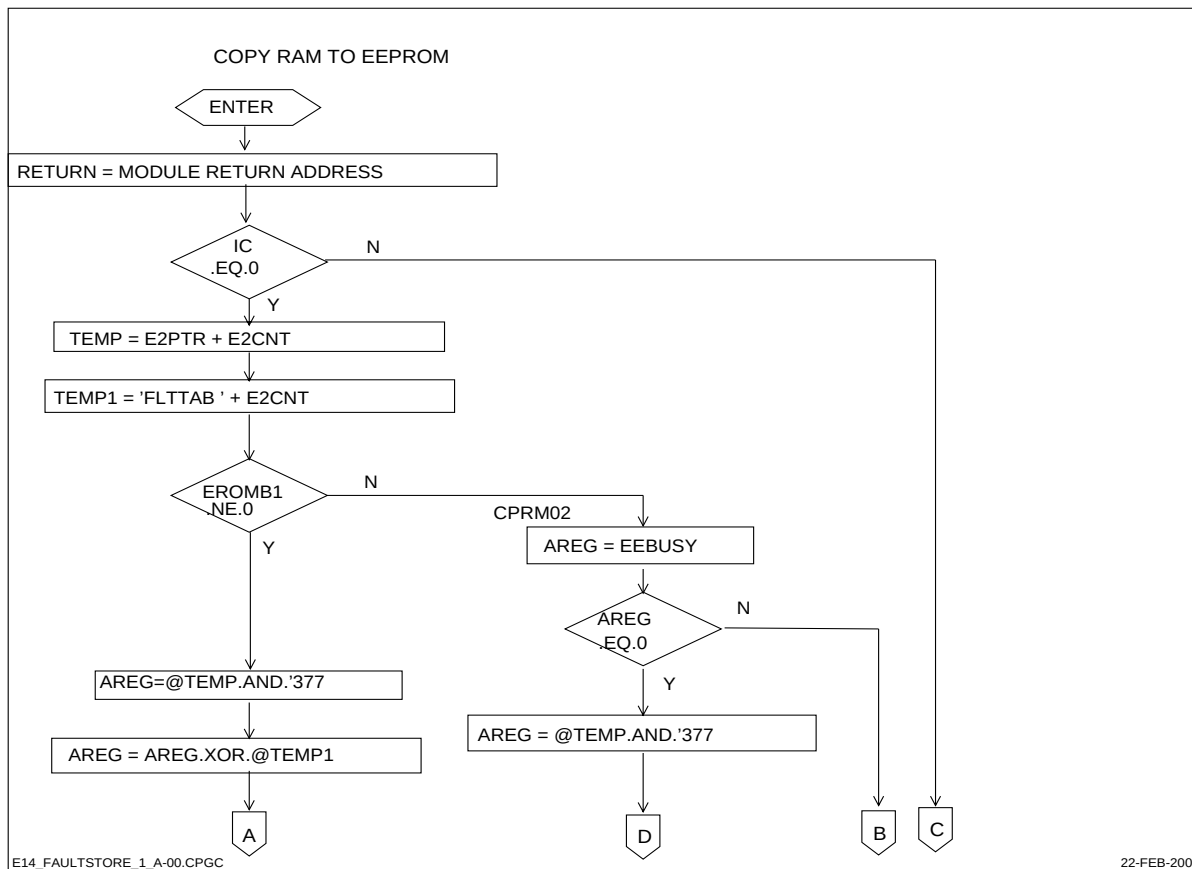
13.00 Figure 118: E13_WATEST_6_A



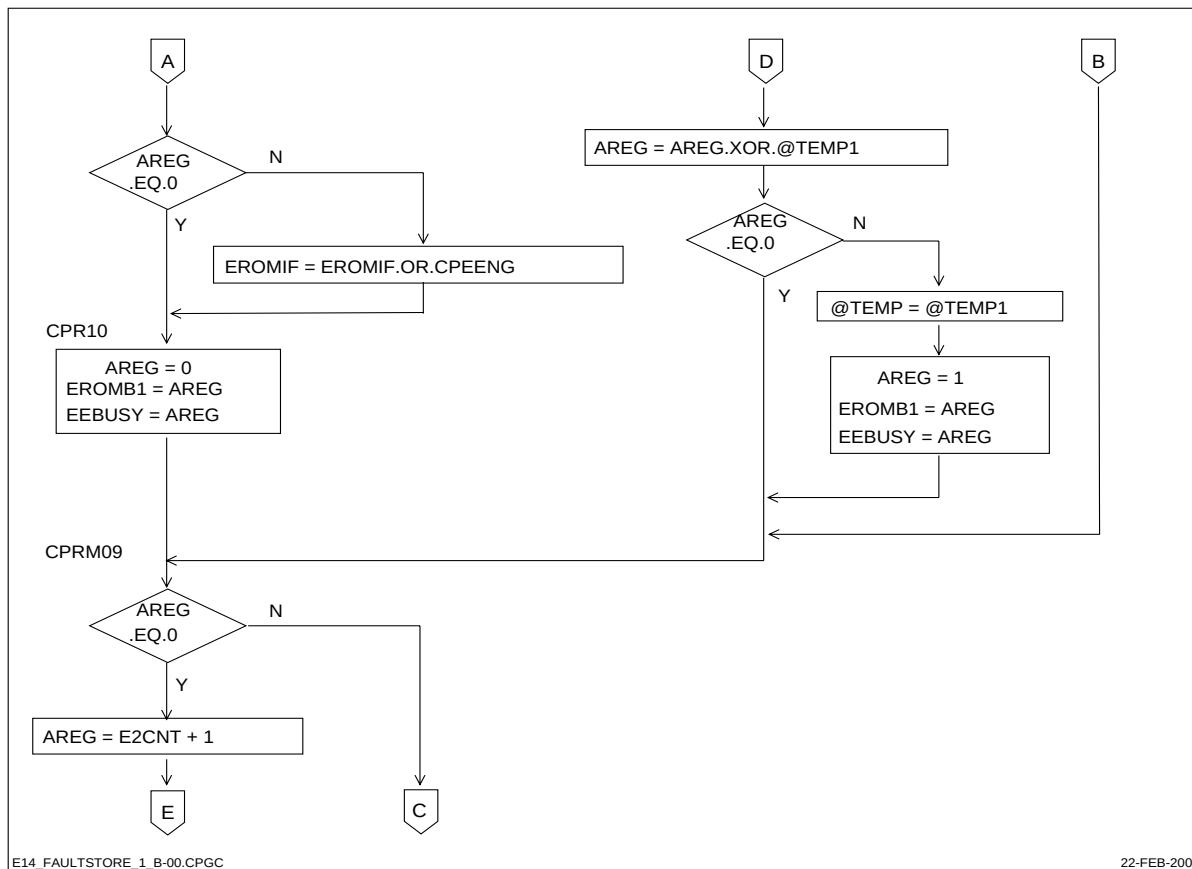
13.00 Figure 119: E13_WATEST_7_A



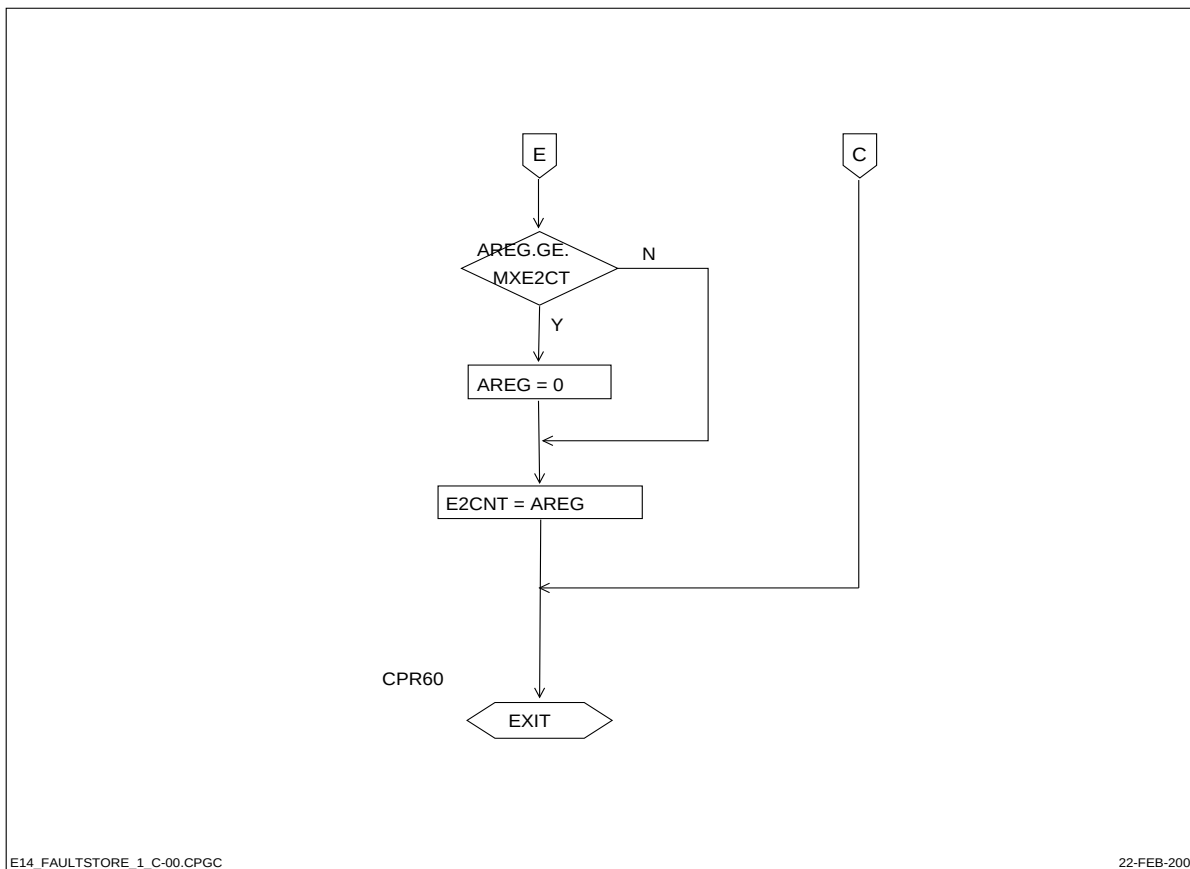
13.00 Figure 120: E14_FAULTSTORE_1_A



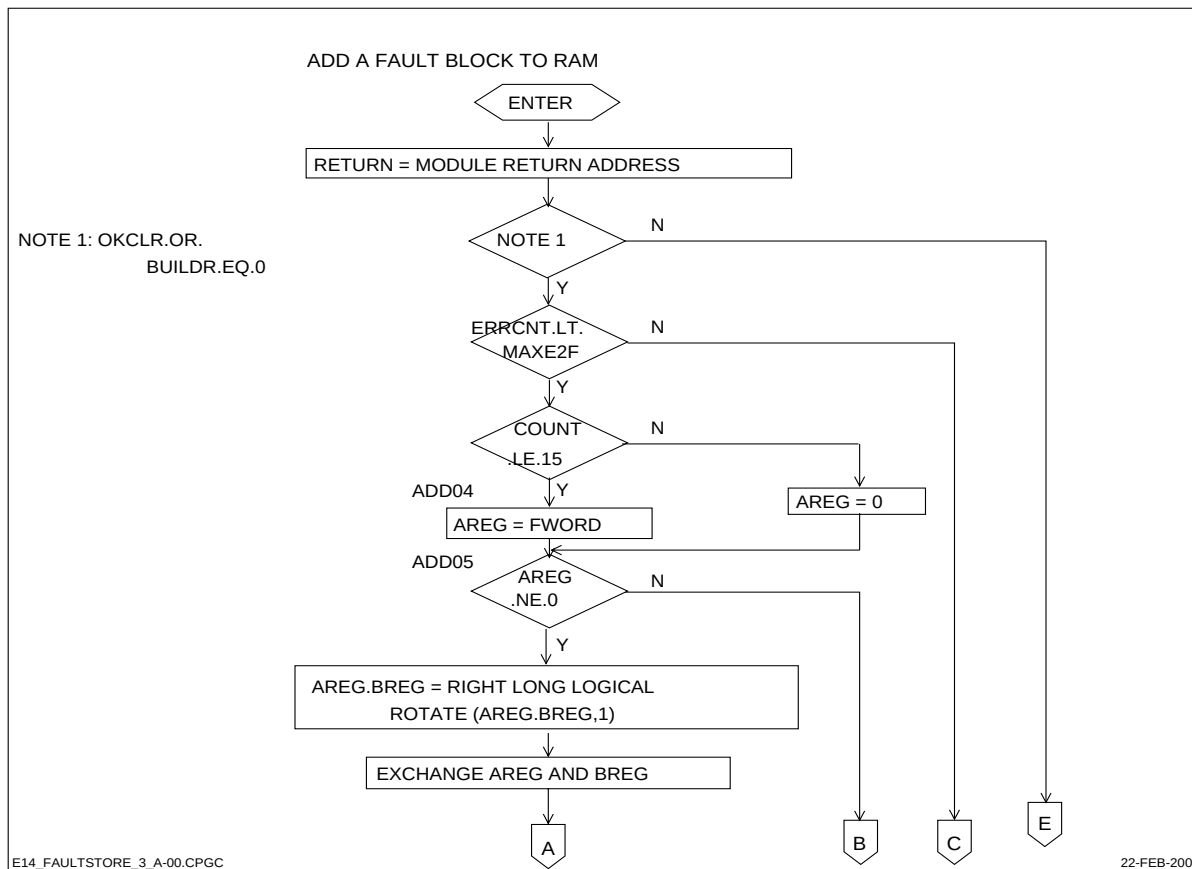
13.00 Figure 121: E14_FAULTSTORE_1_B



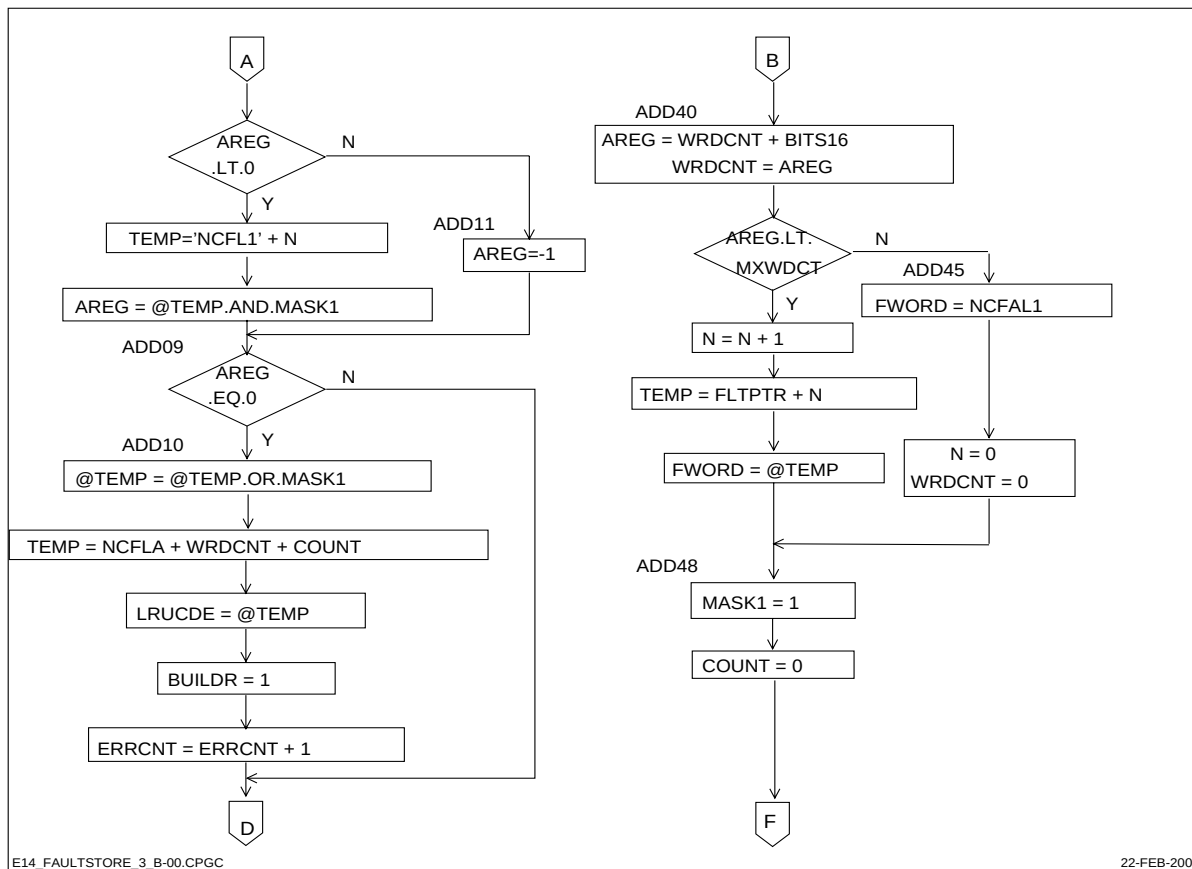
13.00 Figure 122: E14_FAULTSTORE_1_C



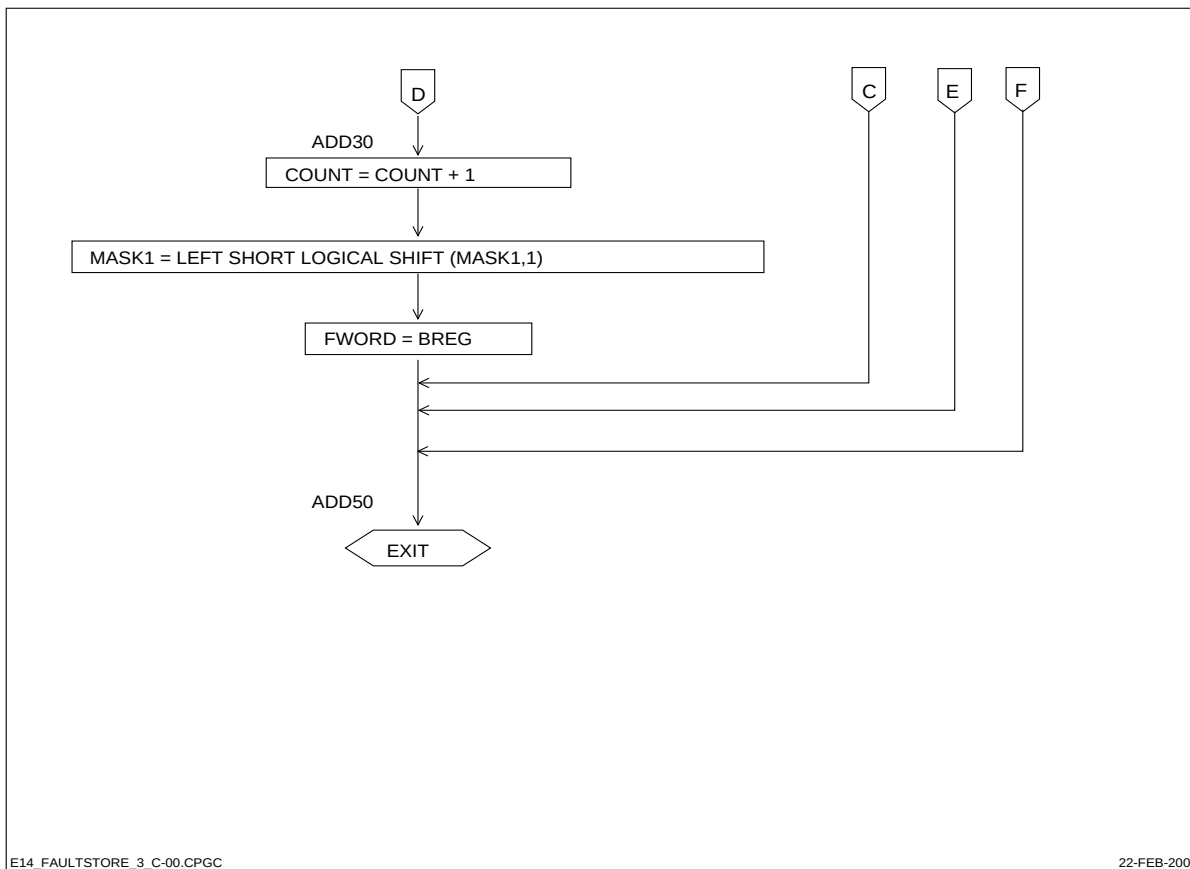
13.00 Figure 123: E14_FAULTSTORE_3_A



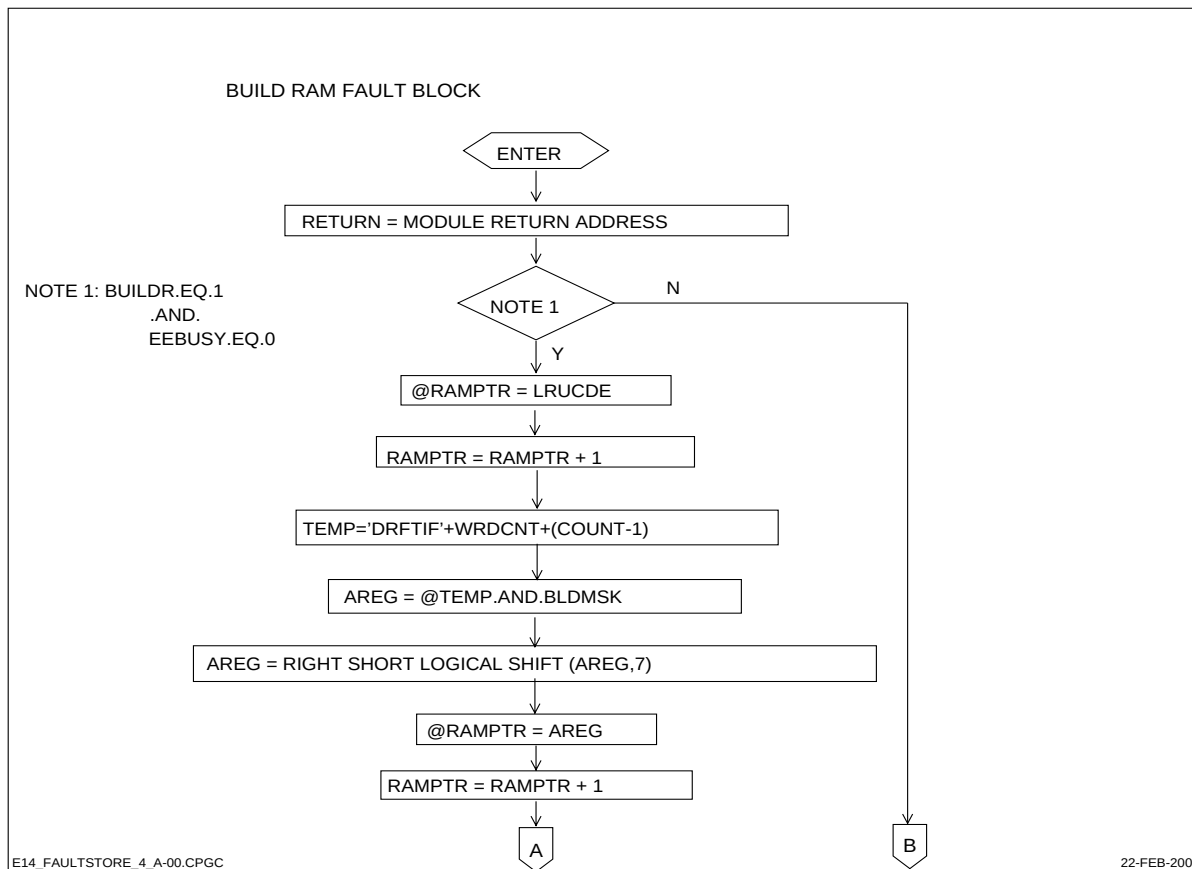
13.00 Figure 124: E14_FAULTSTORE_3_B



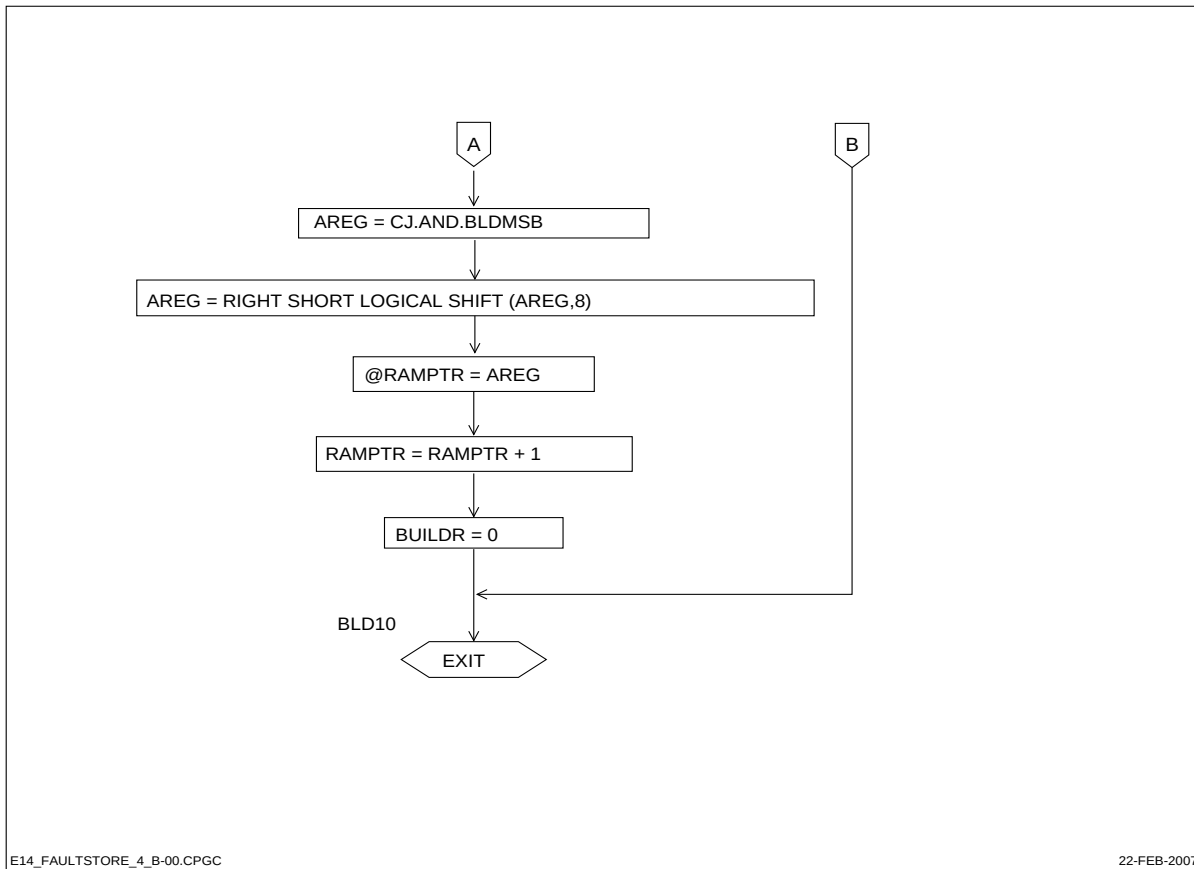
13.00 Figure 125: E14_FAULTSTORE_3_C



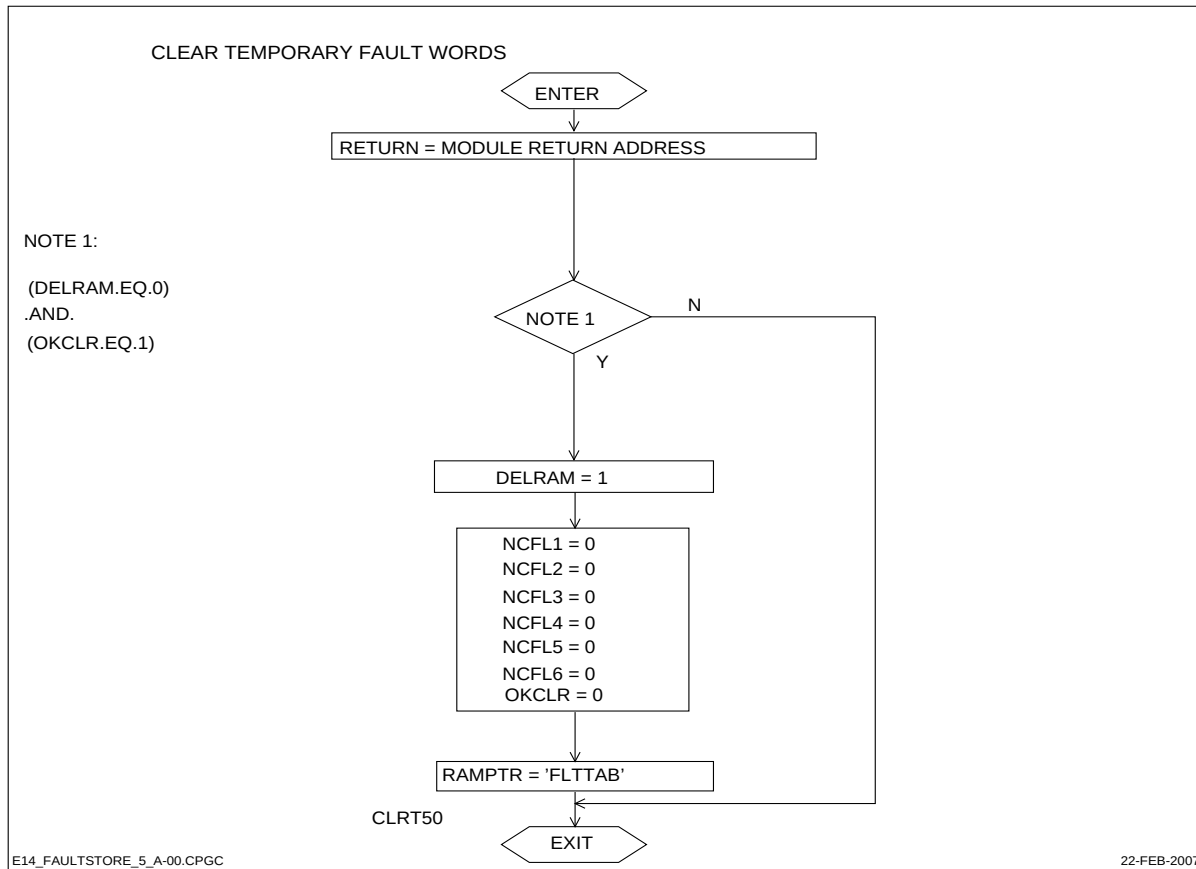
13.00 Figure 126: E14_FAULTSTORE_4_A



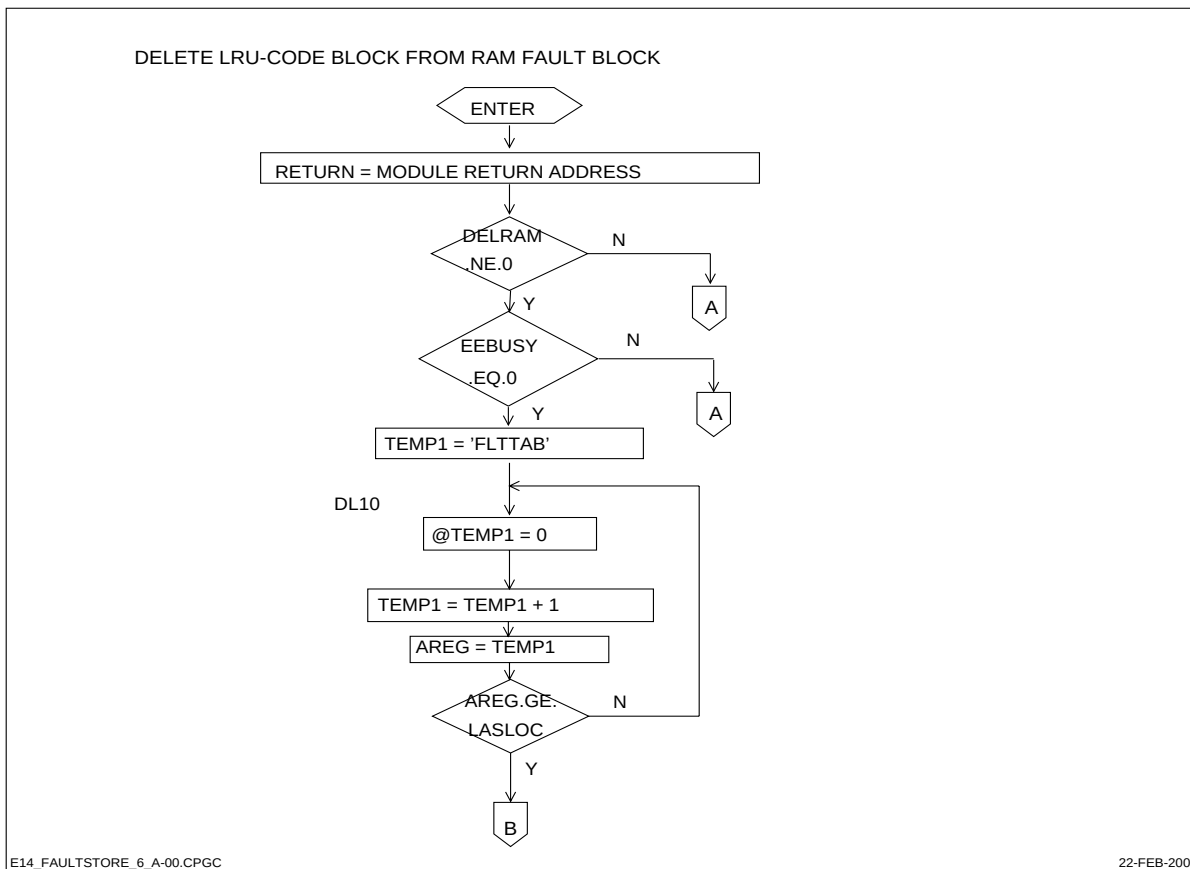
13.00 Figure 127: E14_FAULTSTORE_4_B



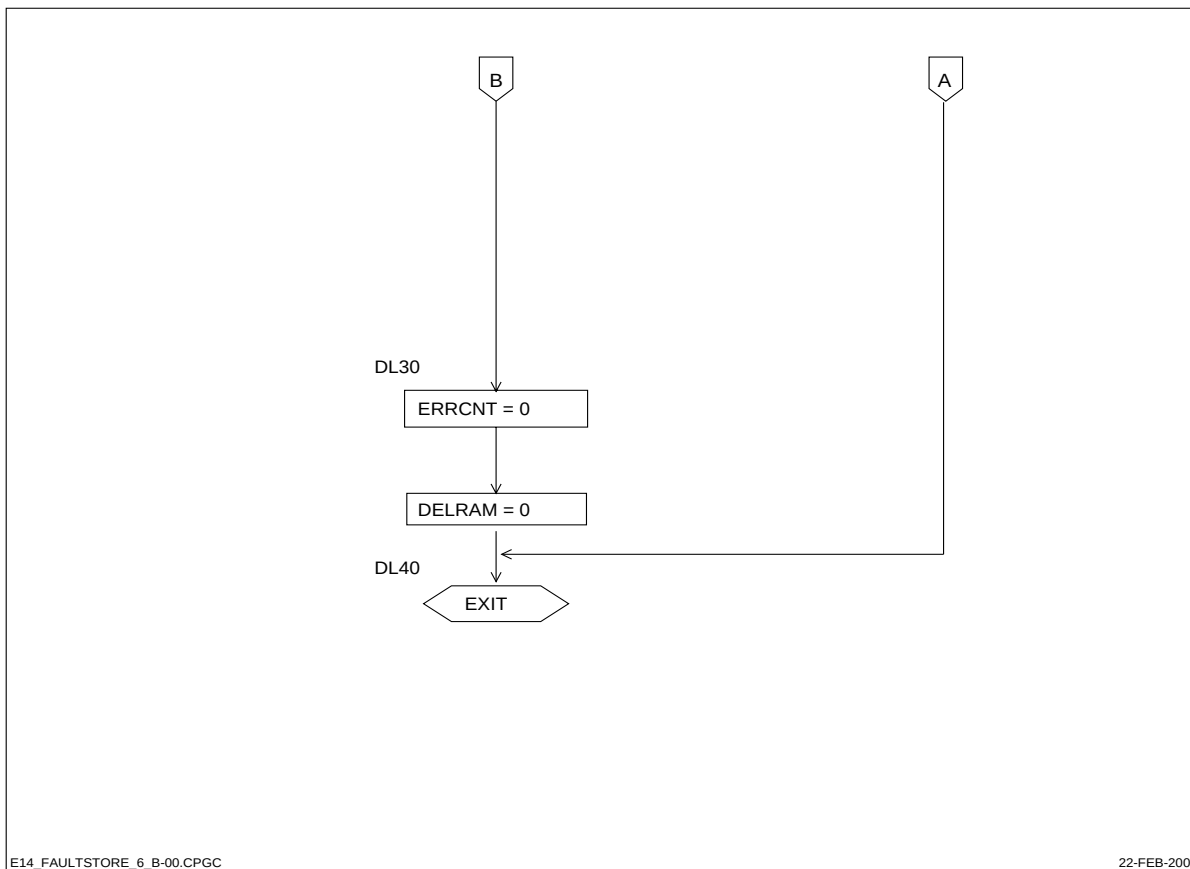
13.00 Figure 128: E14_FAULTSTORE_5_A



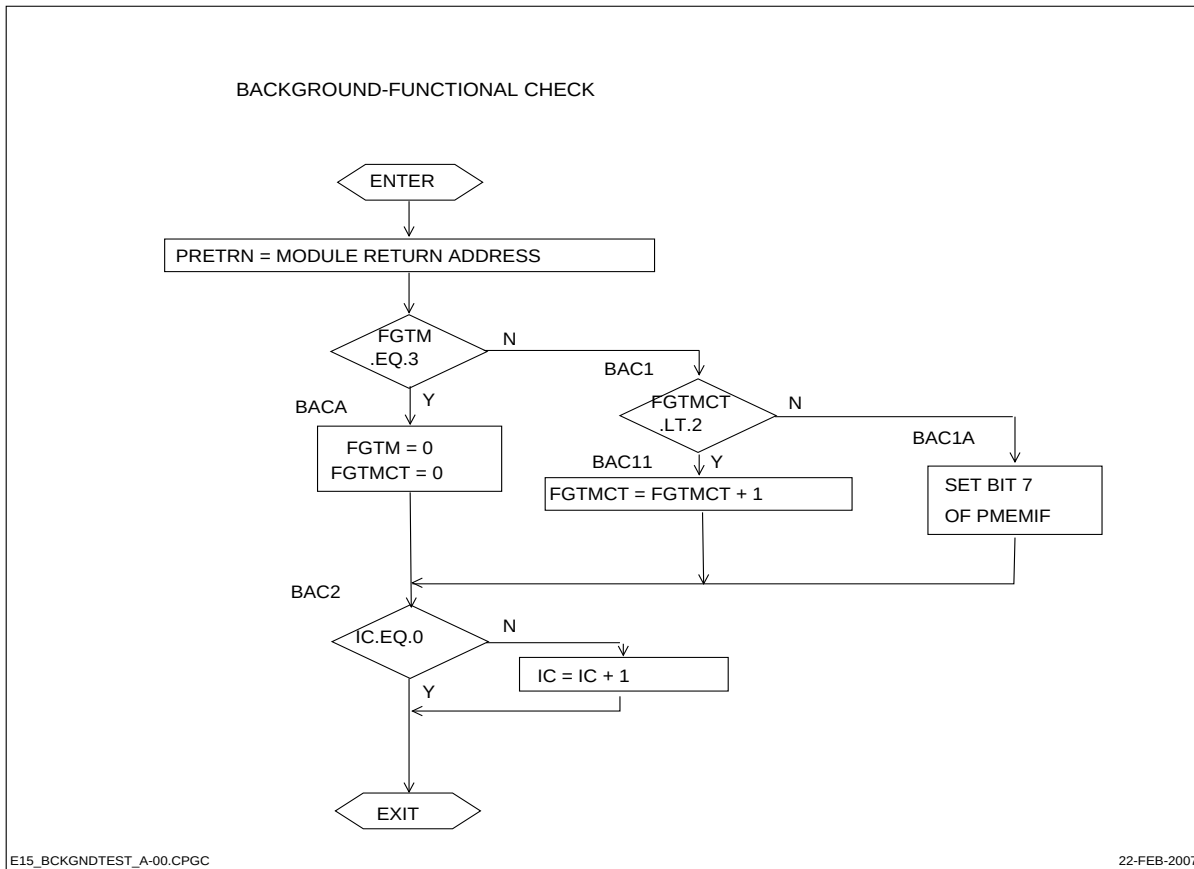
13.00 Figure 129: E14_FAULTSTORE_6_A



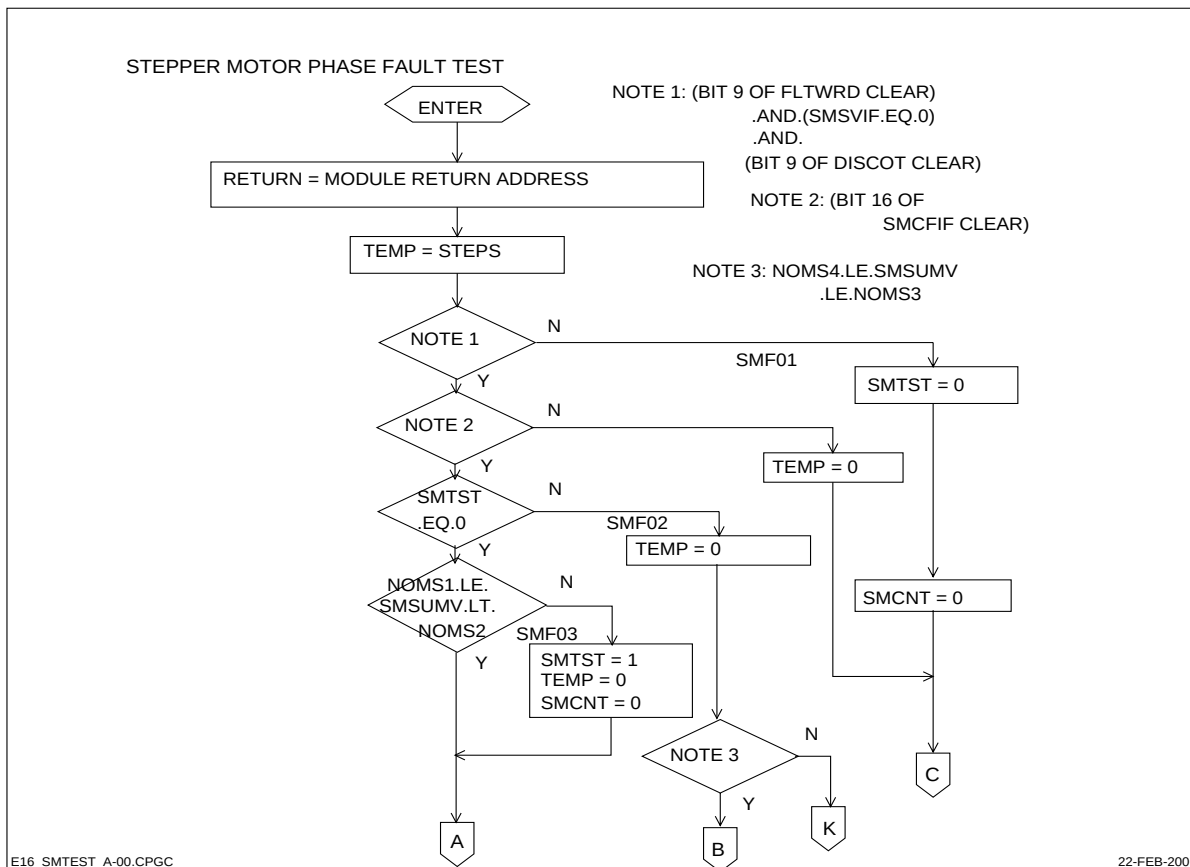
13.00 Figure 130: E14_FAULTSTORE_6_B



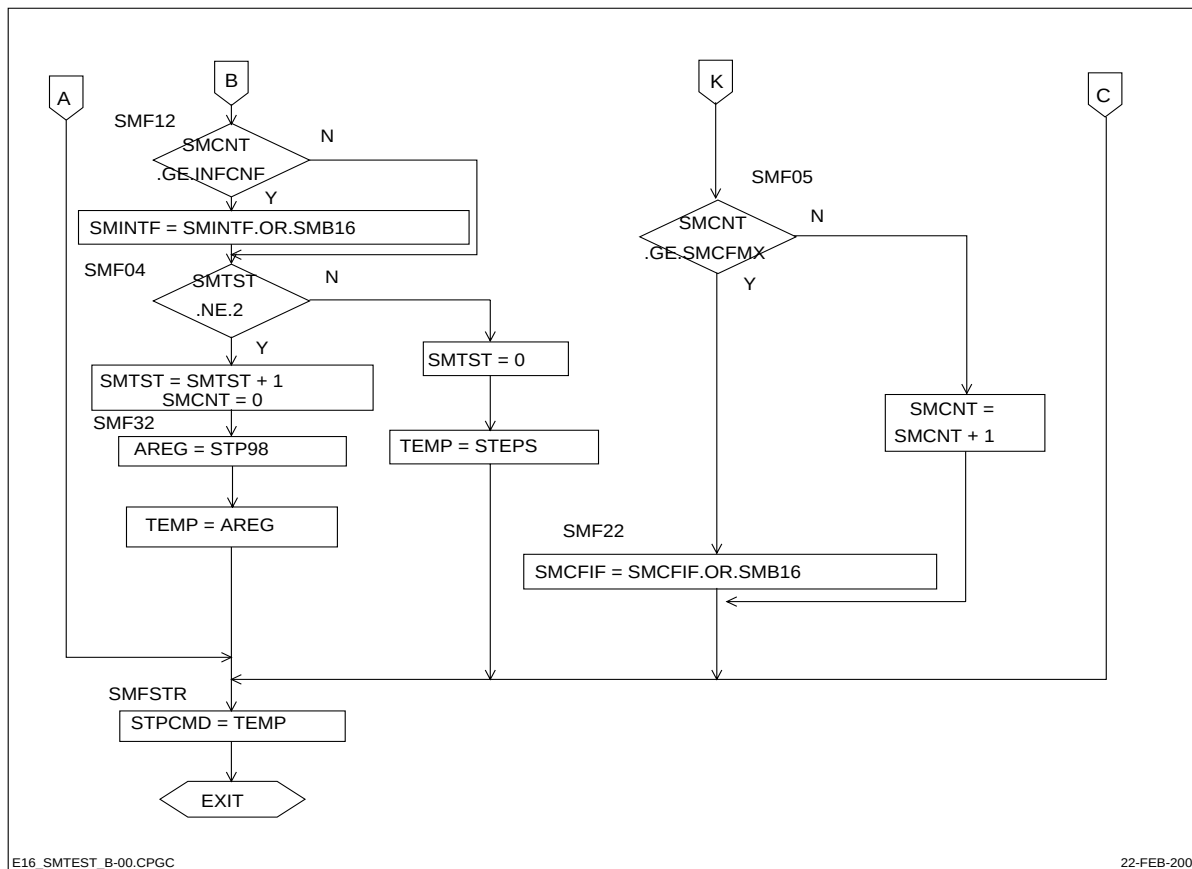
13.00 Figure 131: E15_BCKGNDTEST_A



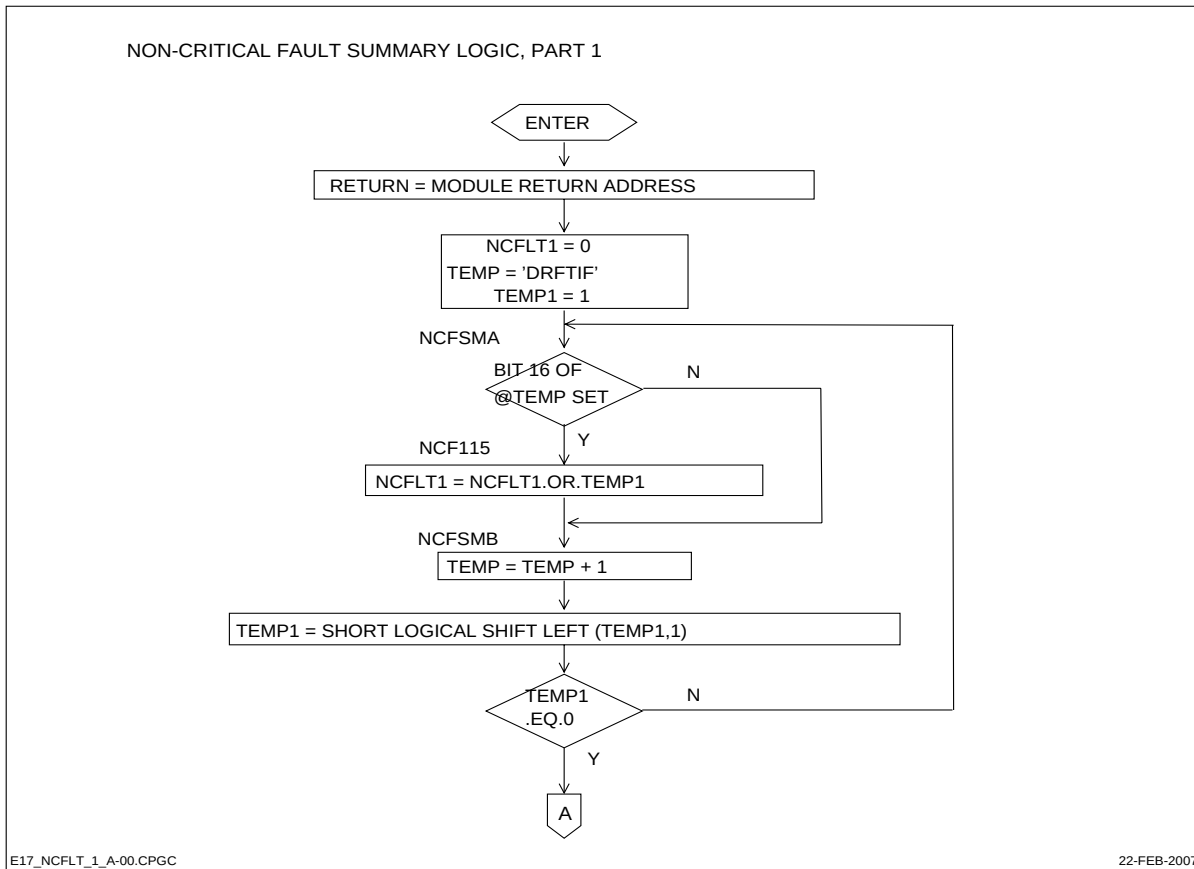
13.00 Figure 132: E16_SMTEST_A



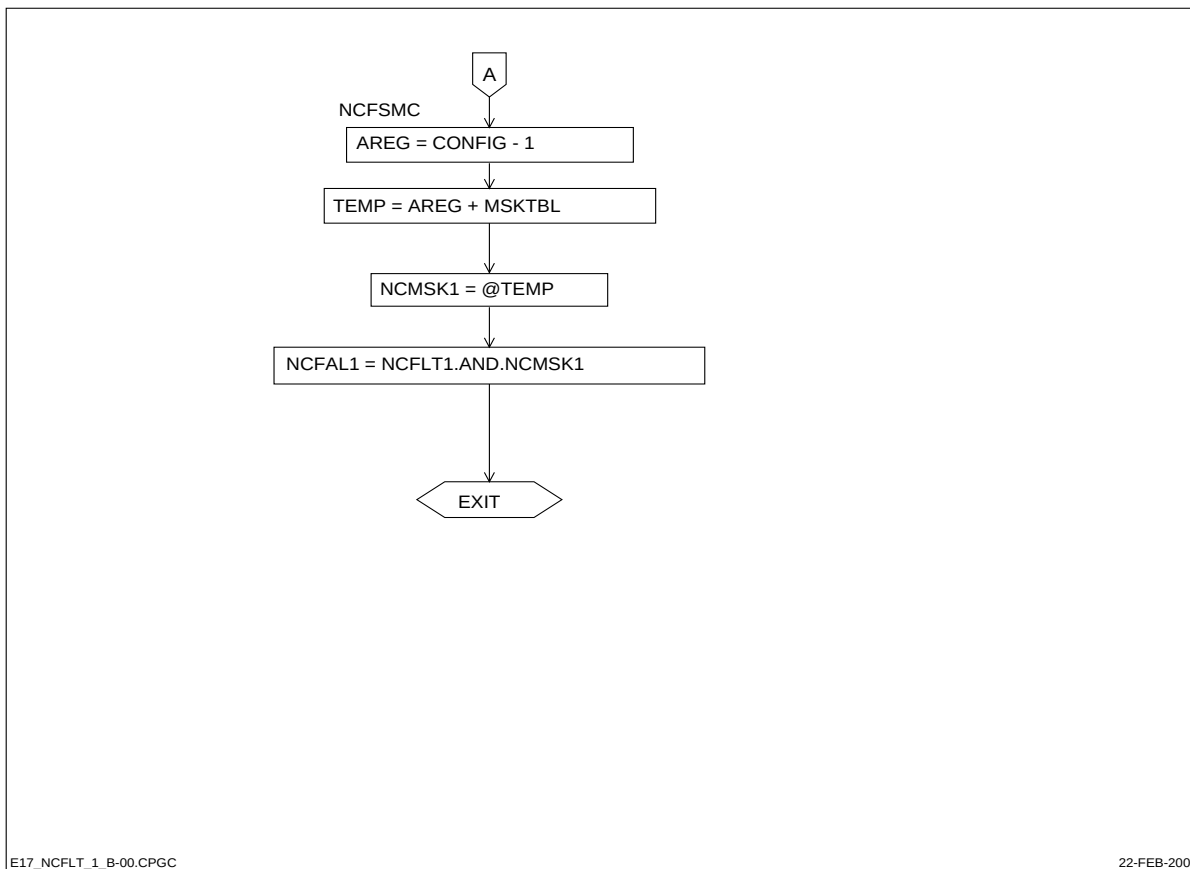
13.00 Figure 133: E16_SMTEST_B



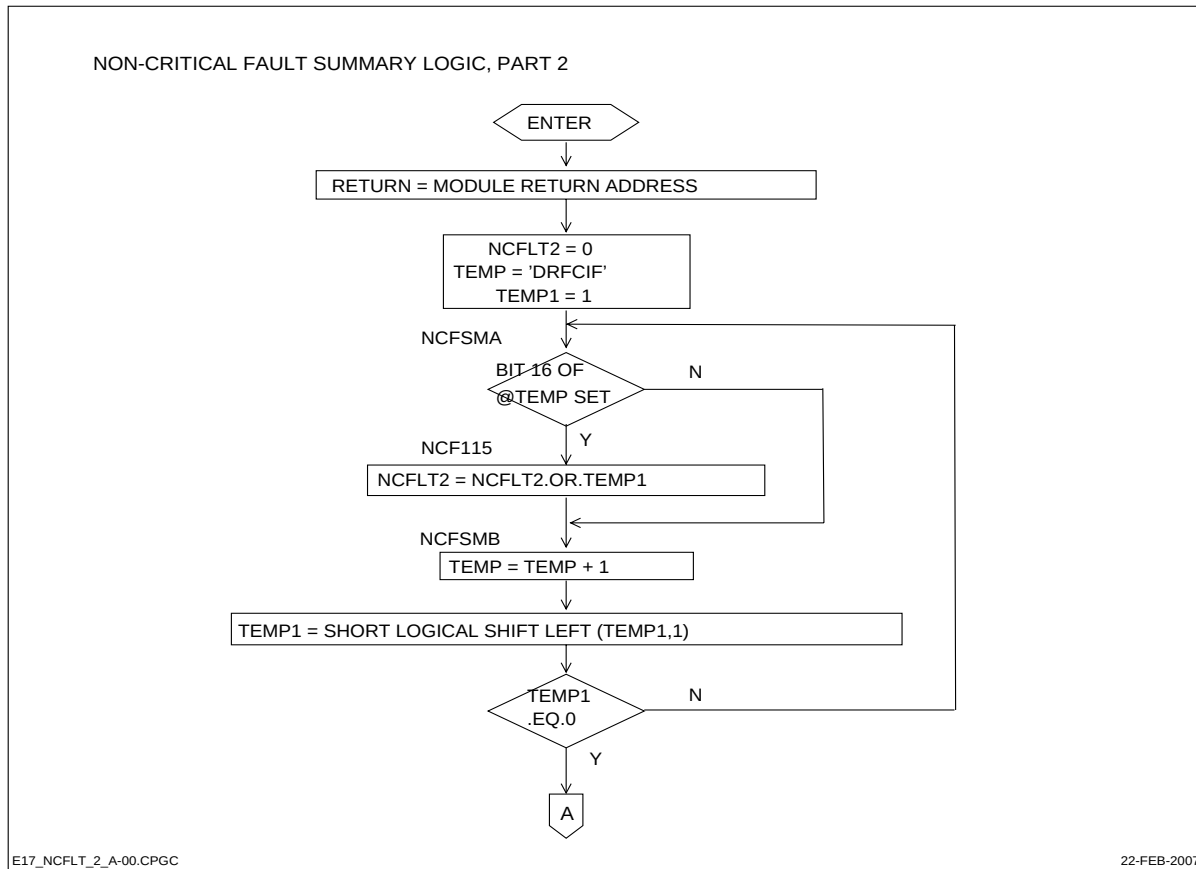
13.00 Figure 134: E17_NCFLT_1_A



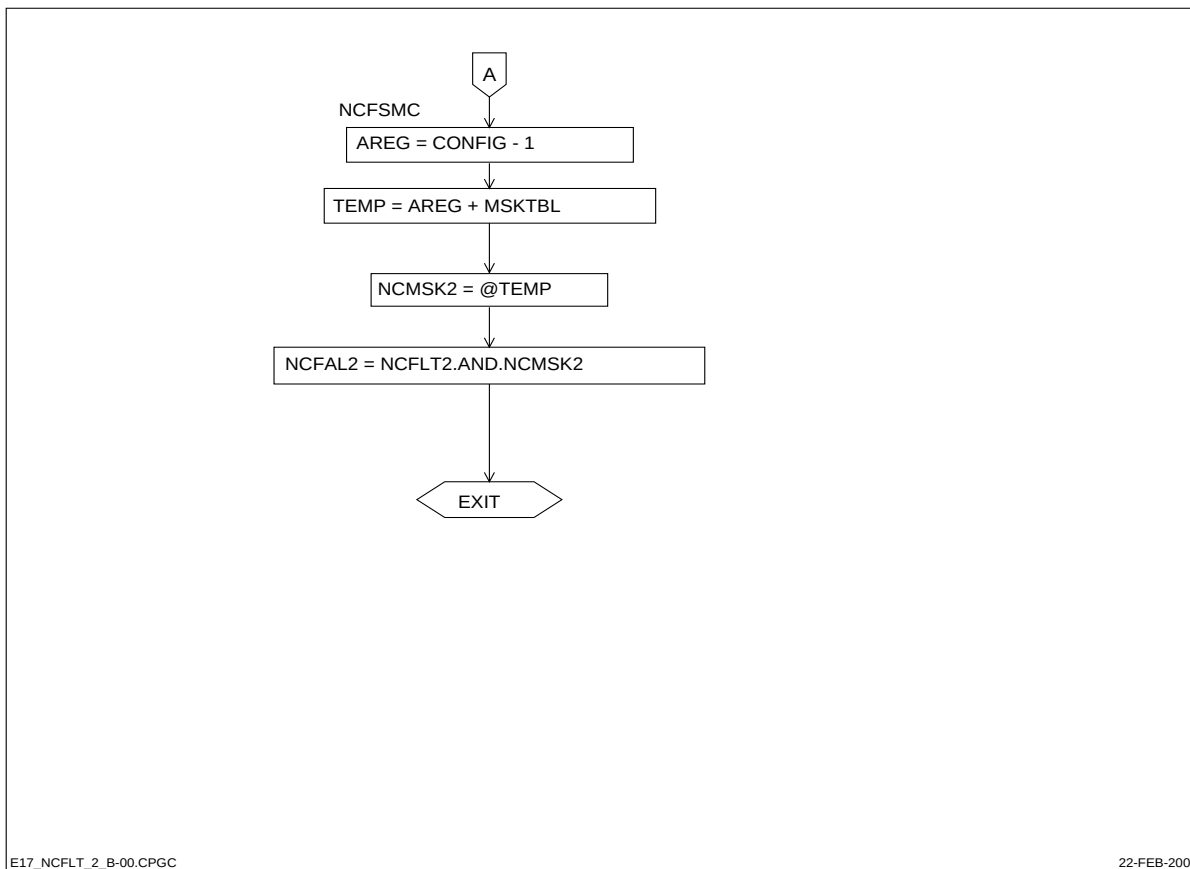
13.00 Figure 135: E17_NCFLT_1_B



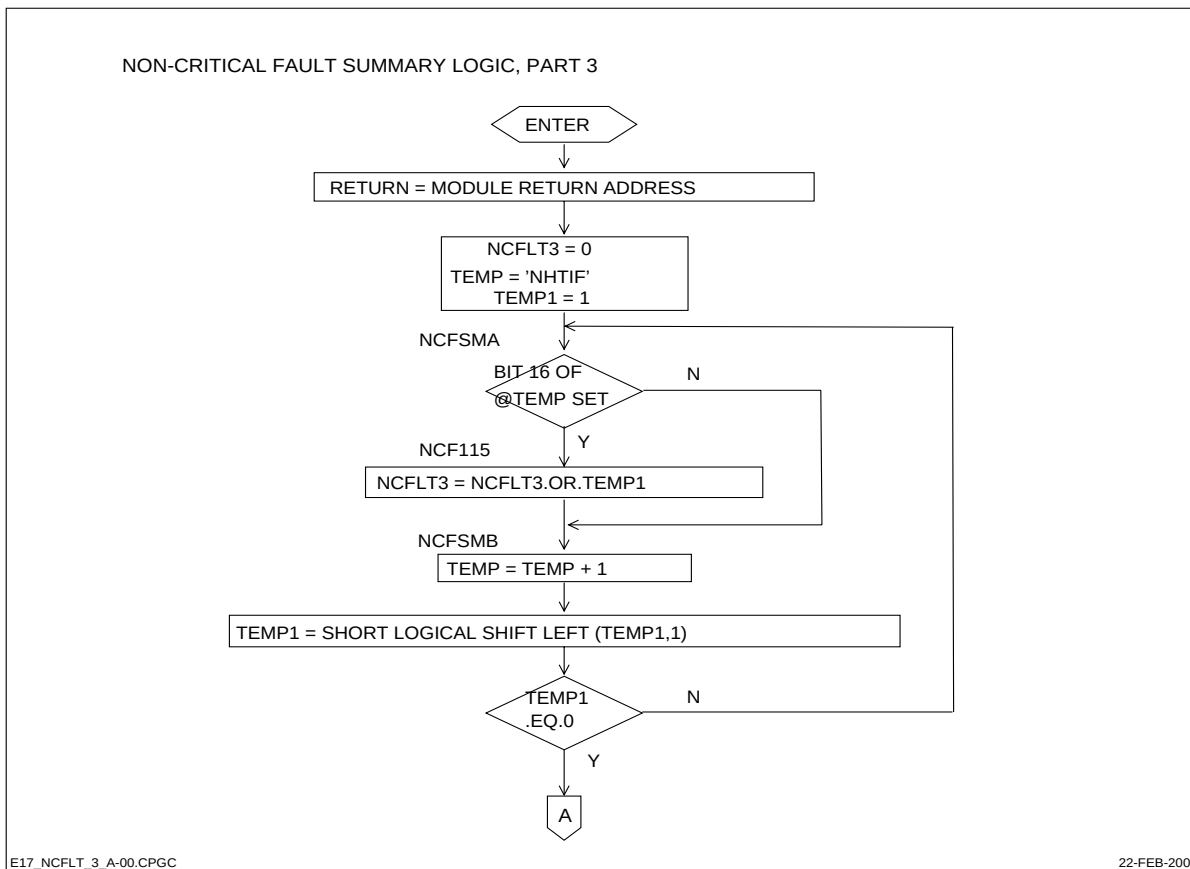
13.00 Figure 136: E17_NCFLT_2_A



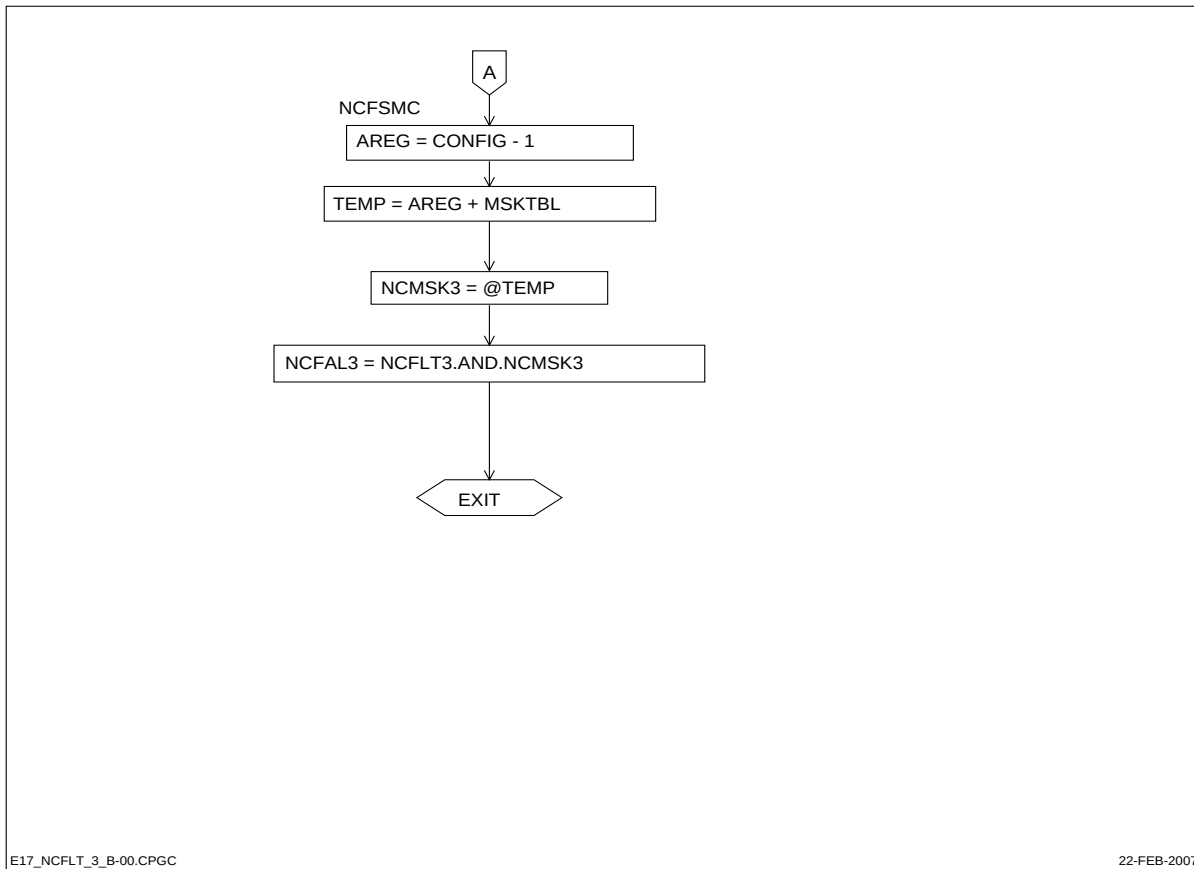
13.00 Figure 137: E17_NCFLT_2_B



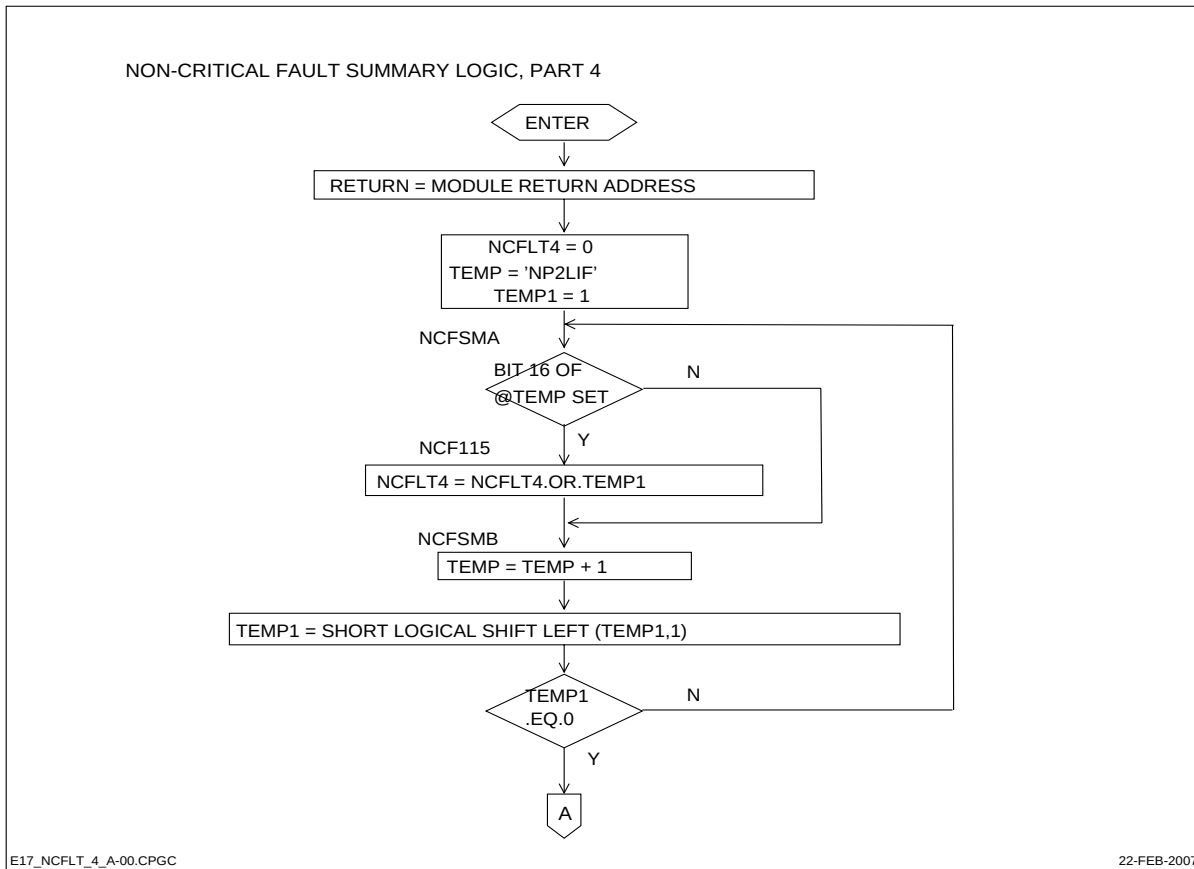
13.00 Figure 138: E17_NCFLT_3_A



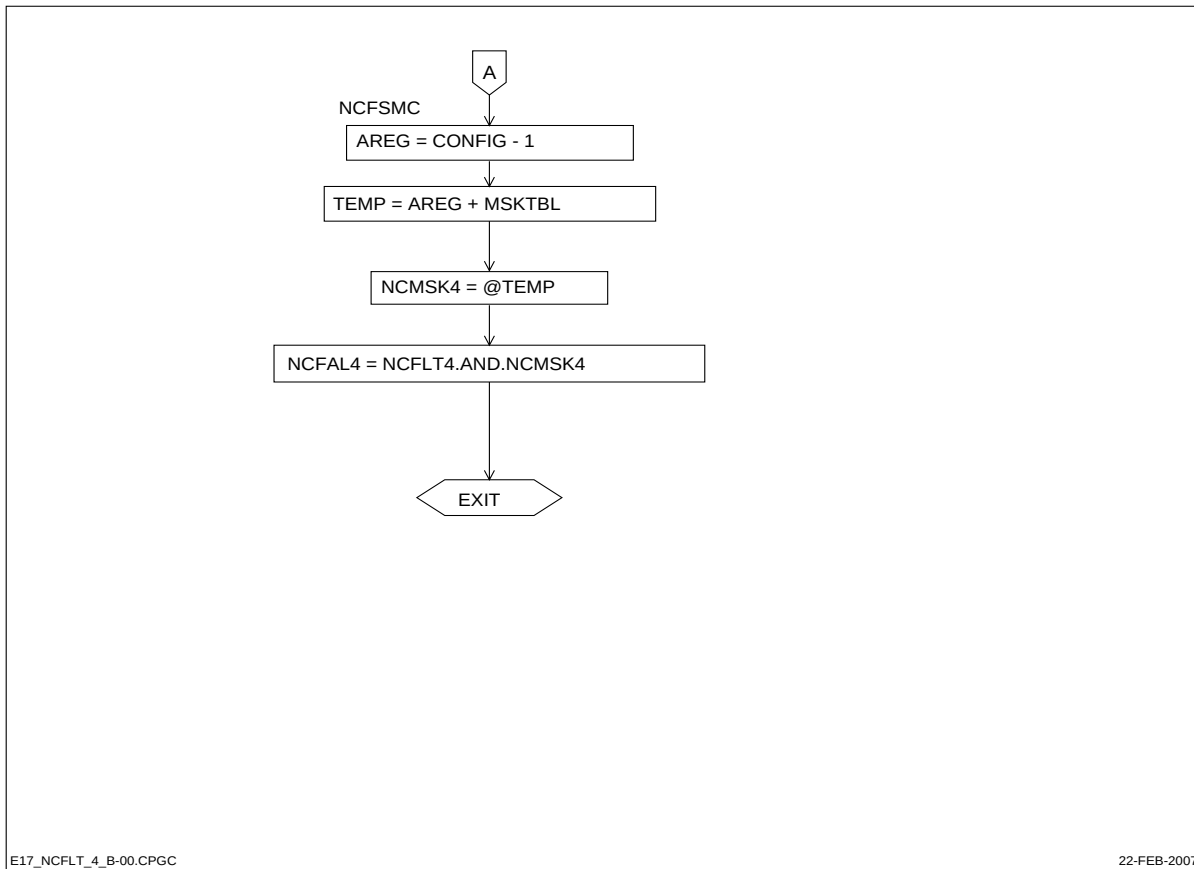
13.00 Figure 139: E17_NCFLT_3_B



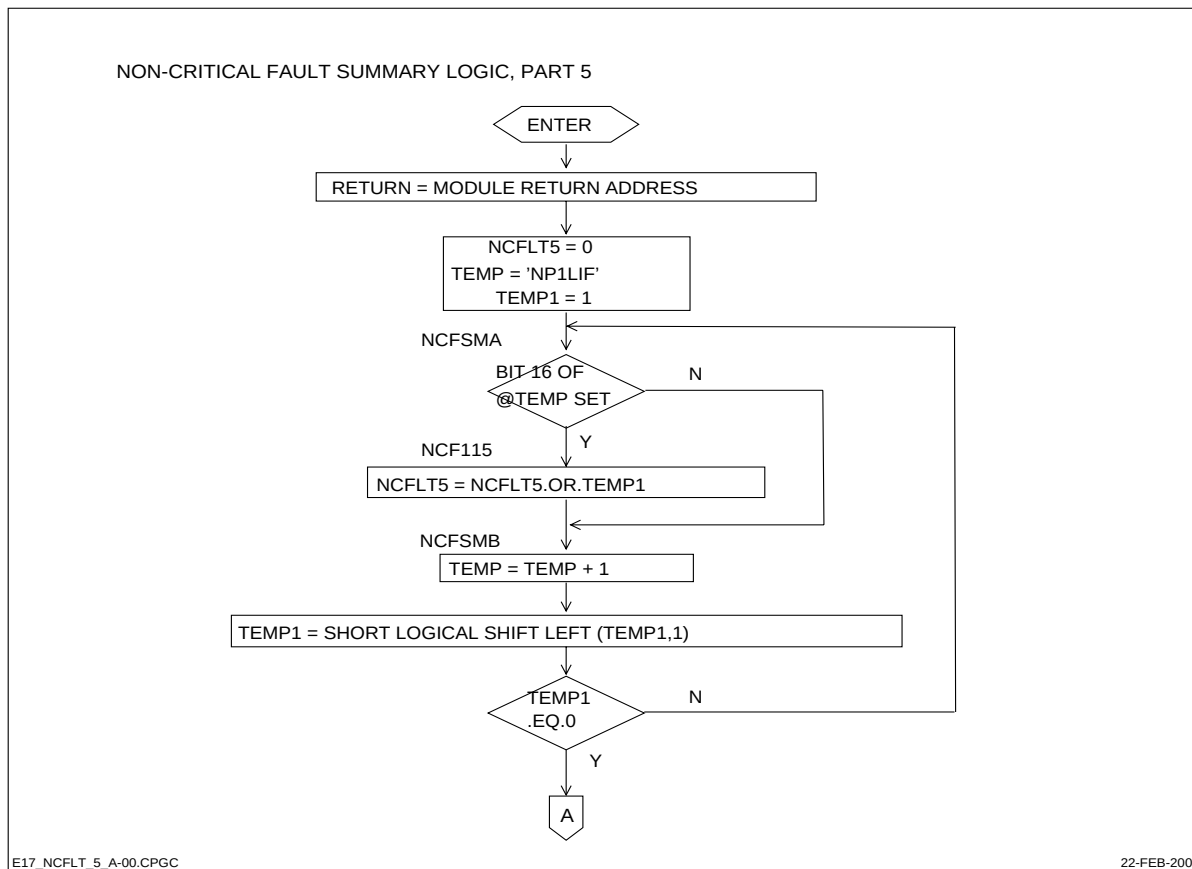
13.00 Figure 140: E17_NCFLT_4_A



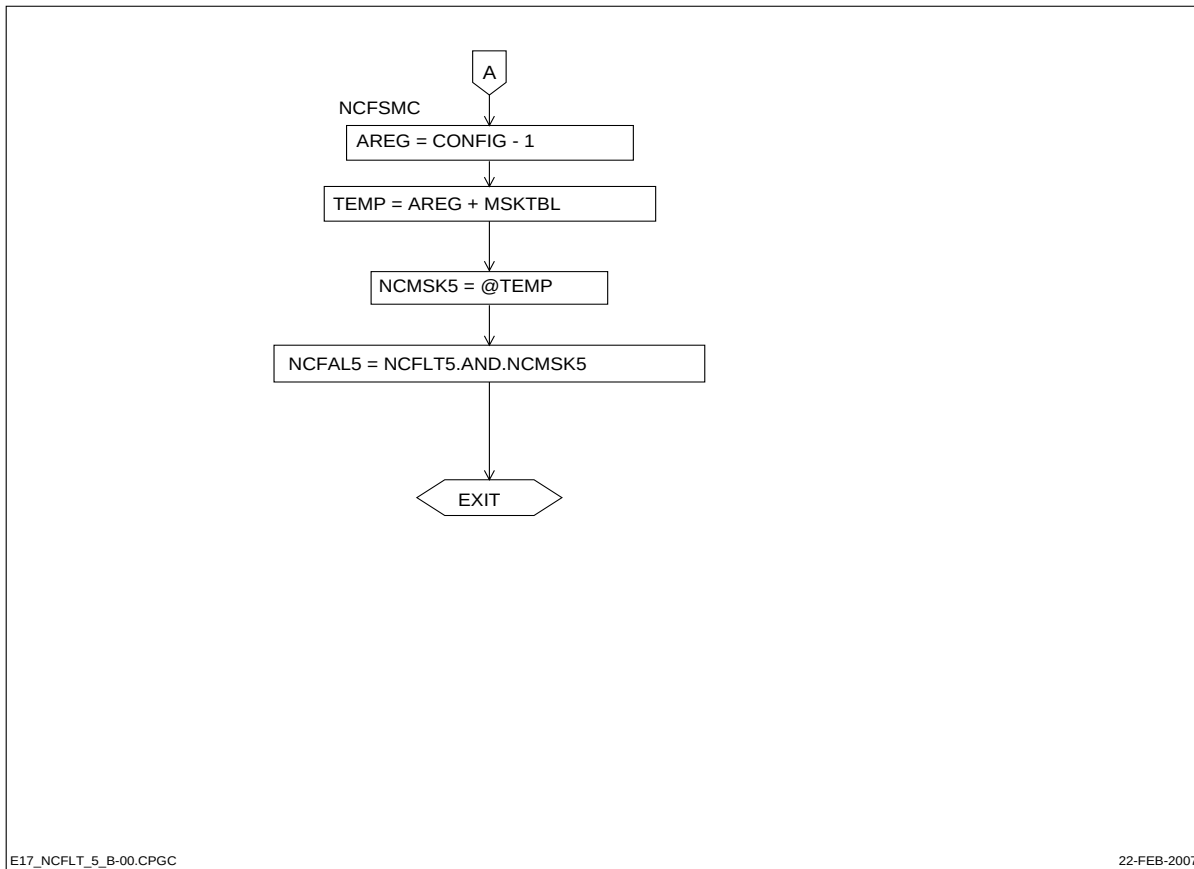
13.00 Figure 141: E17_NCFLT_4_B



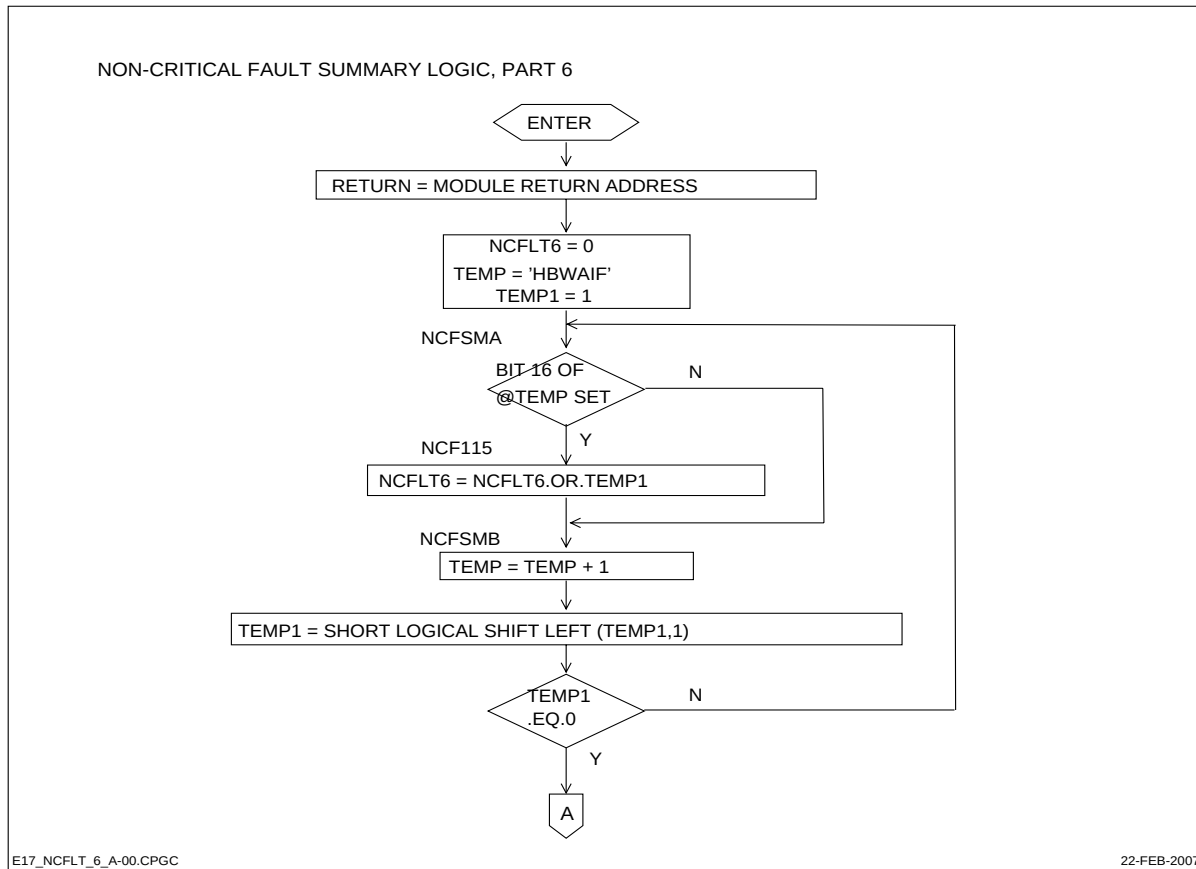
13.00 Figure 142: E17_NCFLT_5_A



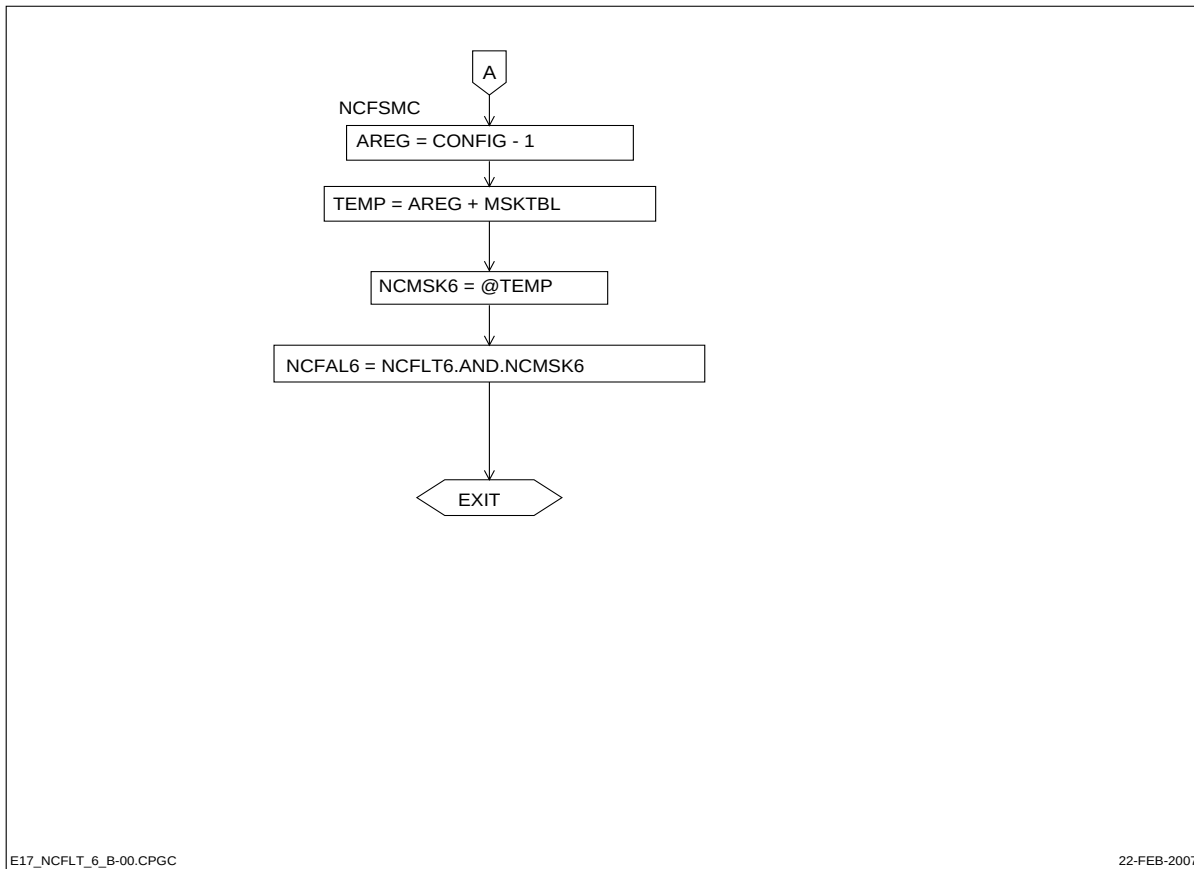
13.00 Figure 143: E17_NCFLT_5_B



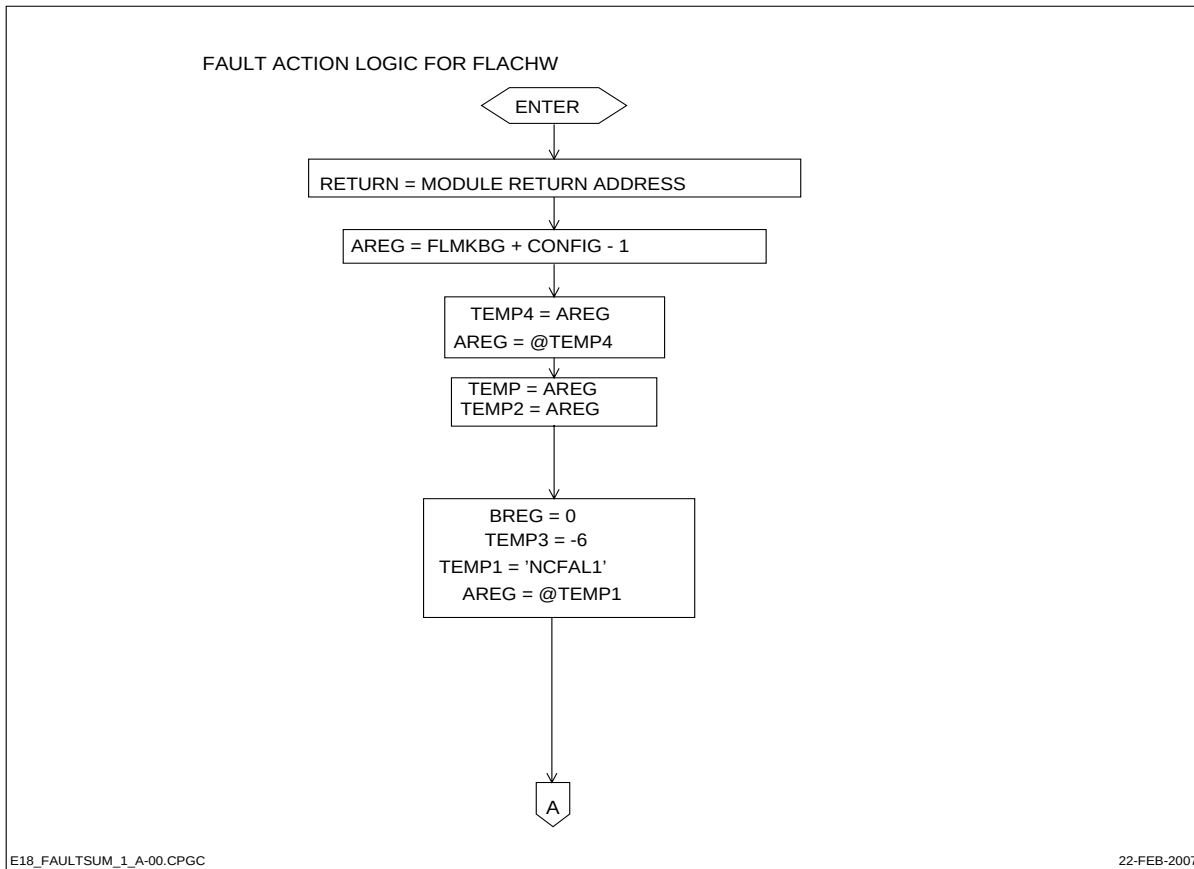
13.00 Figure 144: E17_NCFLT_6_A



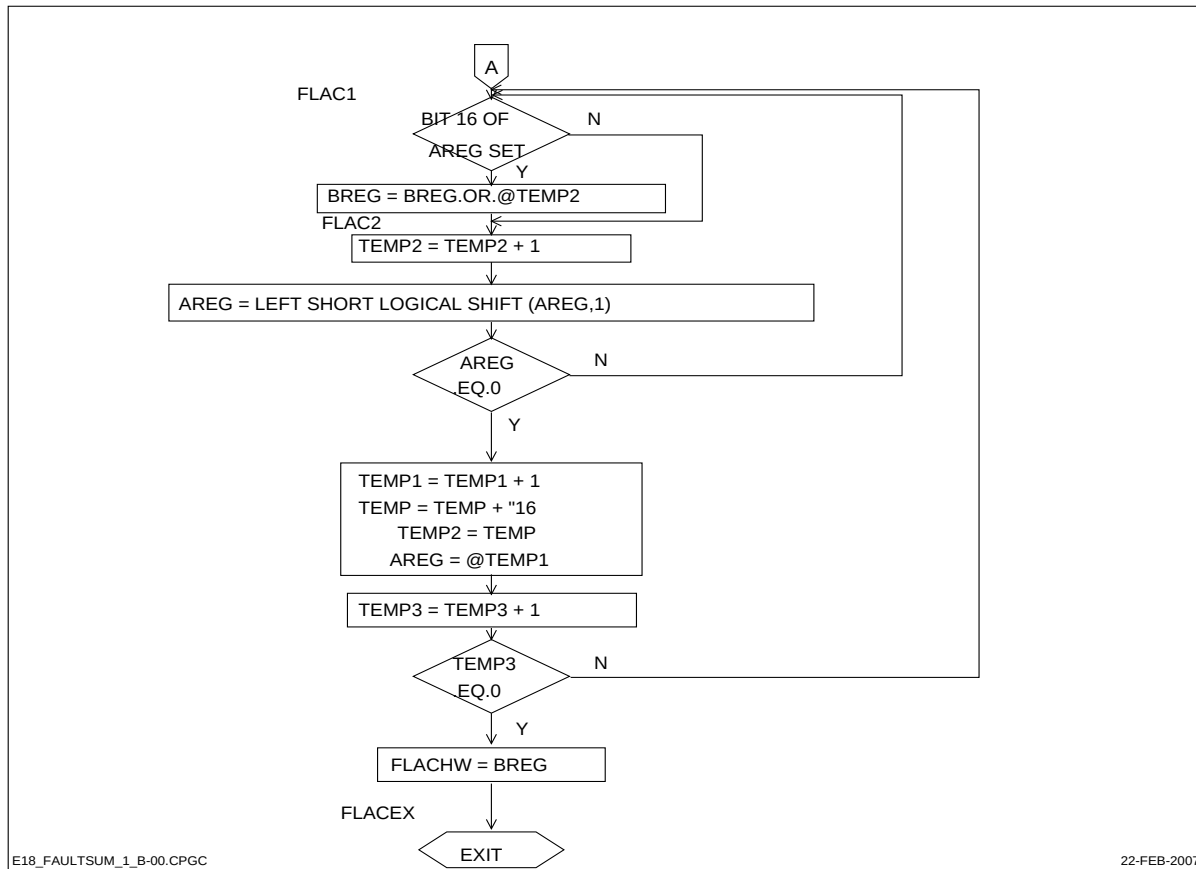
13.00 Figure 145: E17_NCFLT_6_B



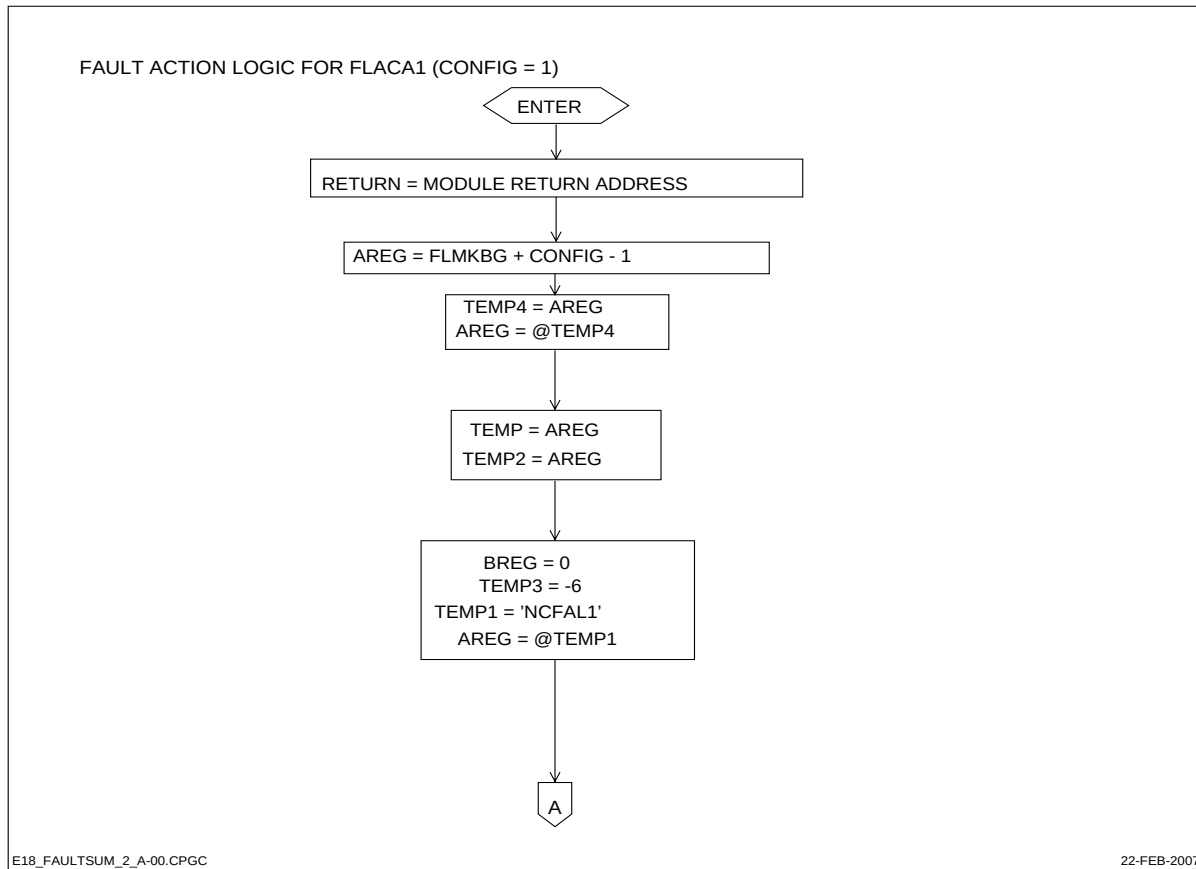
13.00 Figure 146: E18_FAULTSUM_1_A



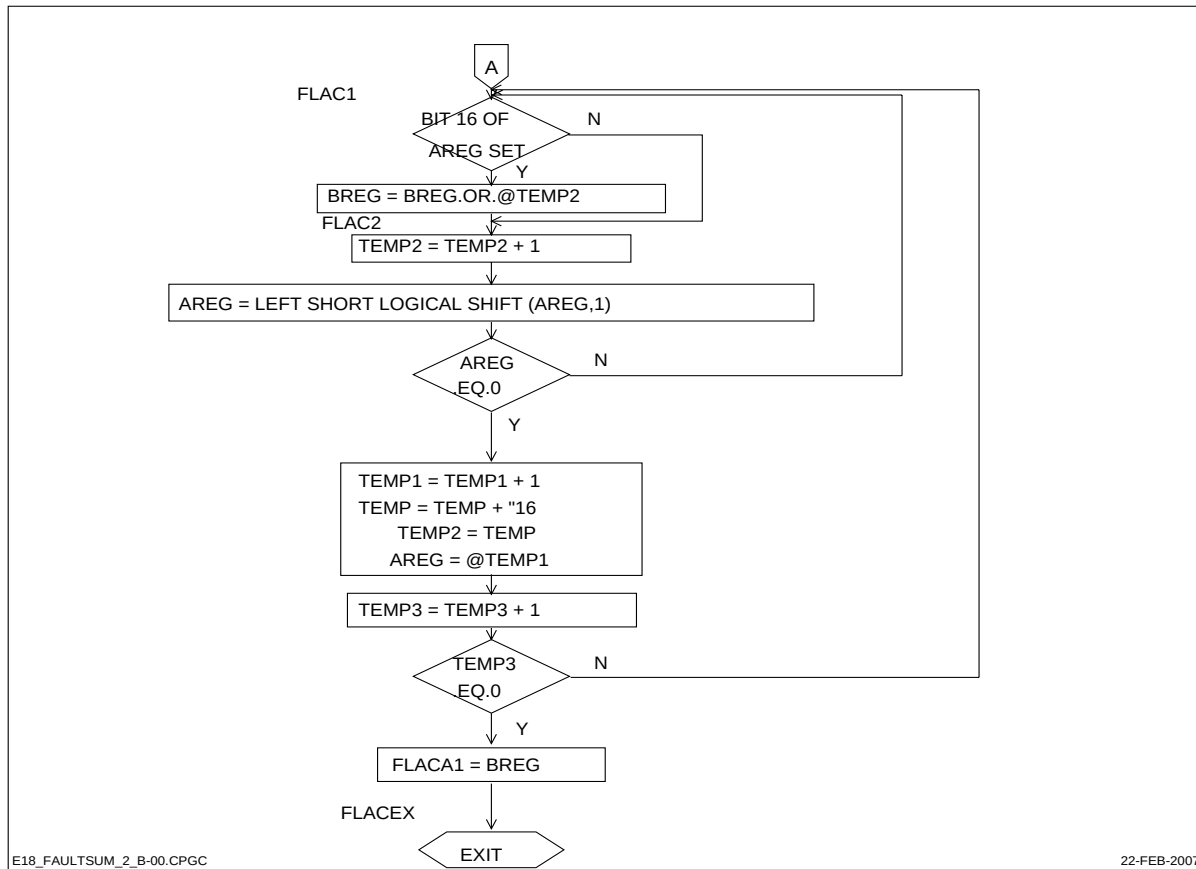
13.00 Figure 147: E18_FAULTSUM_1_B



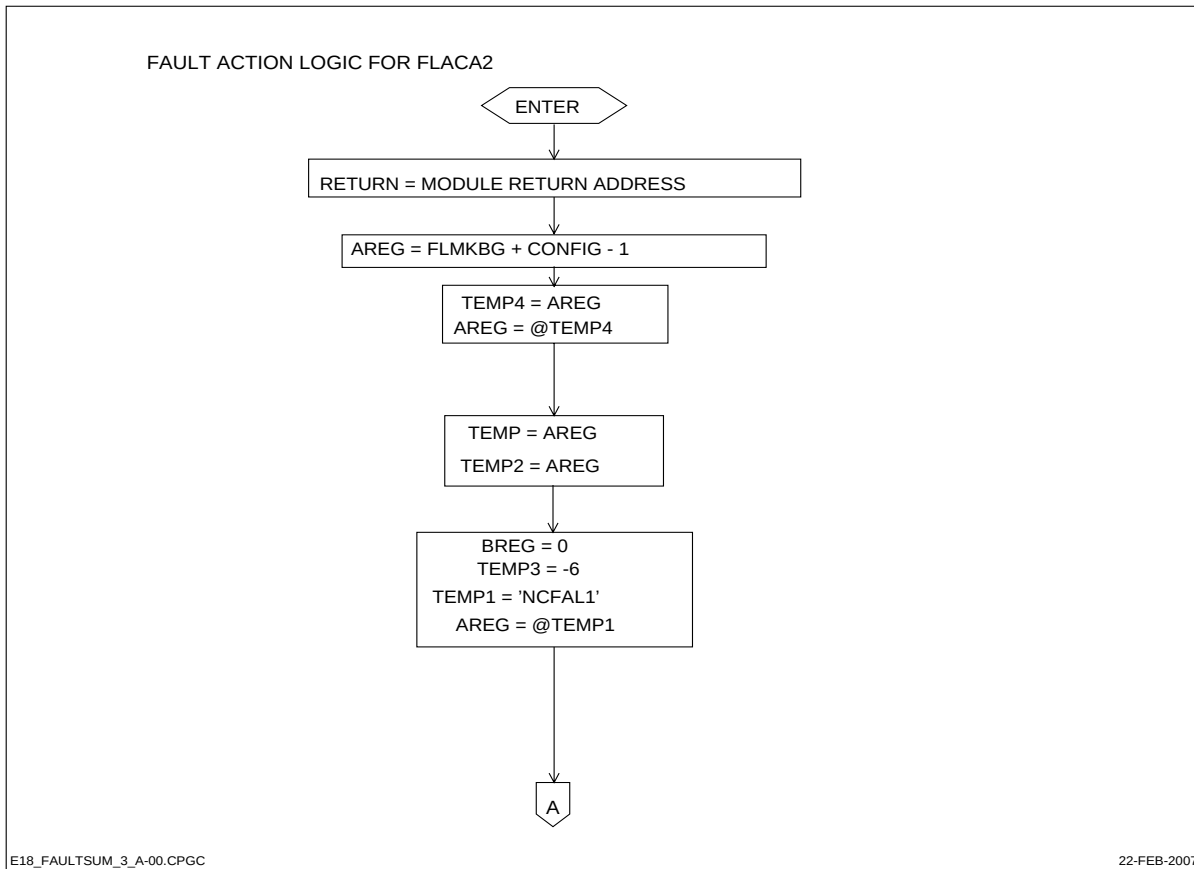
13.00 Figure 148: E18_FAULTSUM_2_A



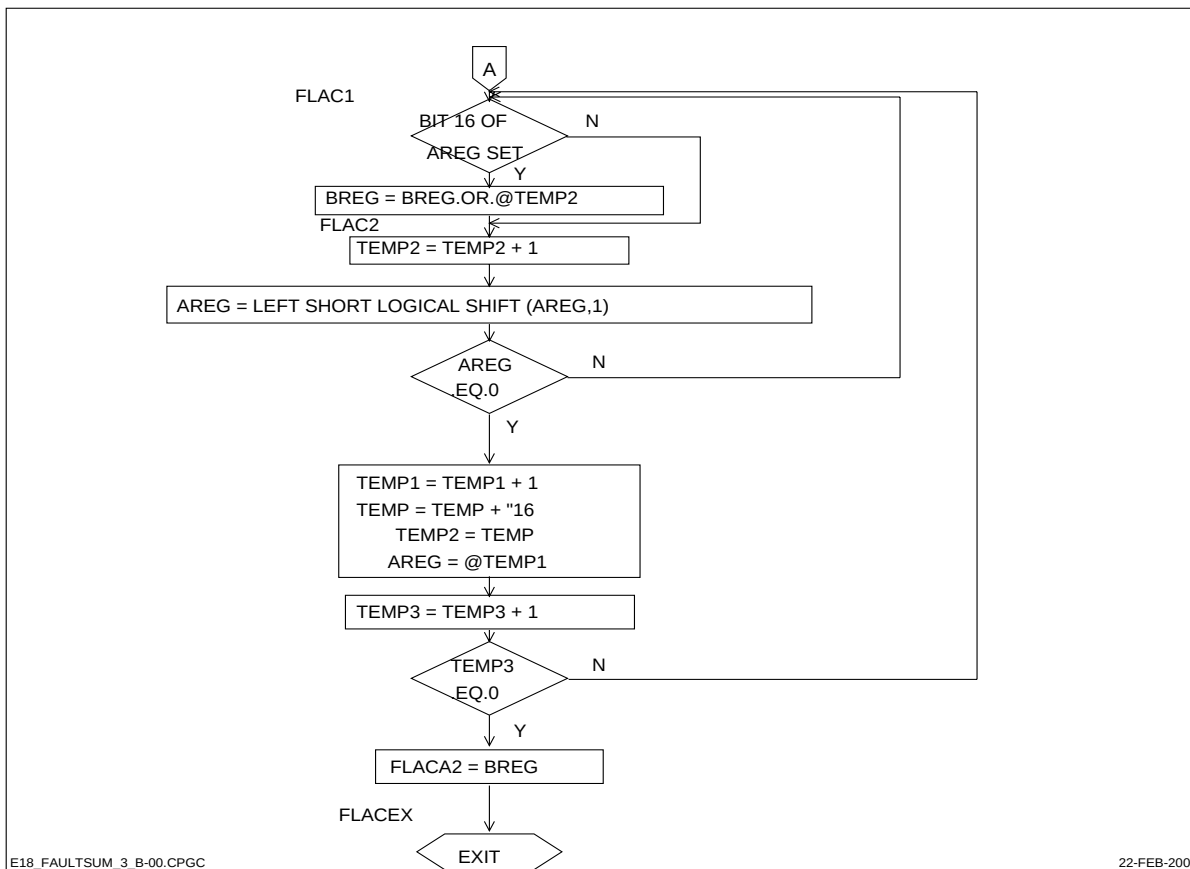
13.00 Figure 149: E18_FAULTSUM_2_B



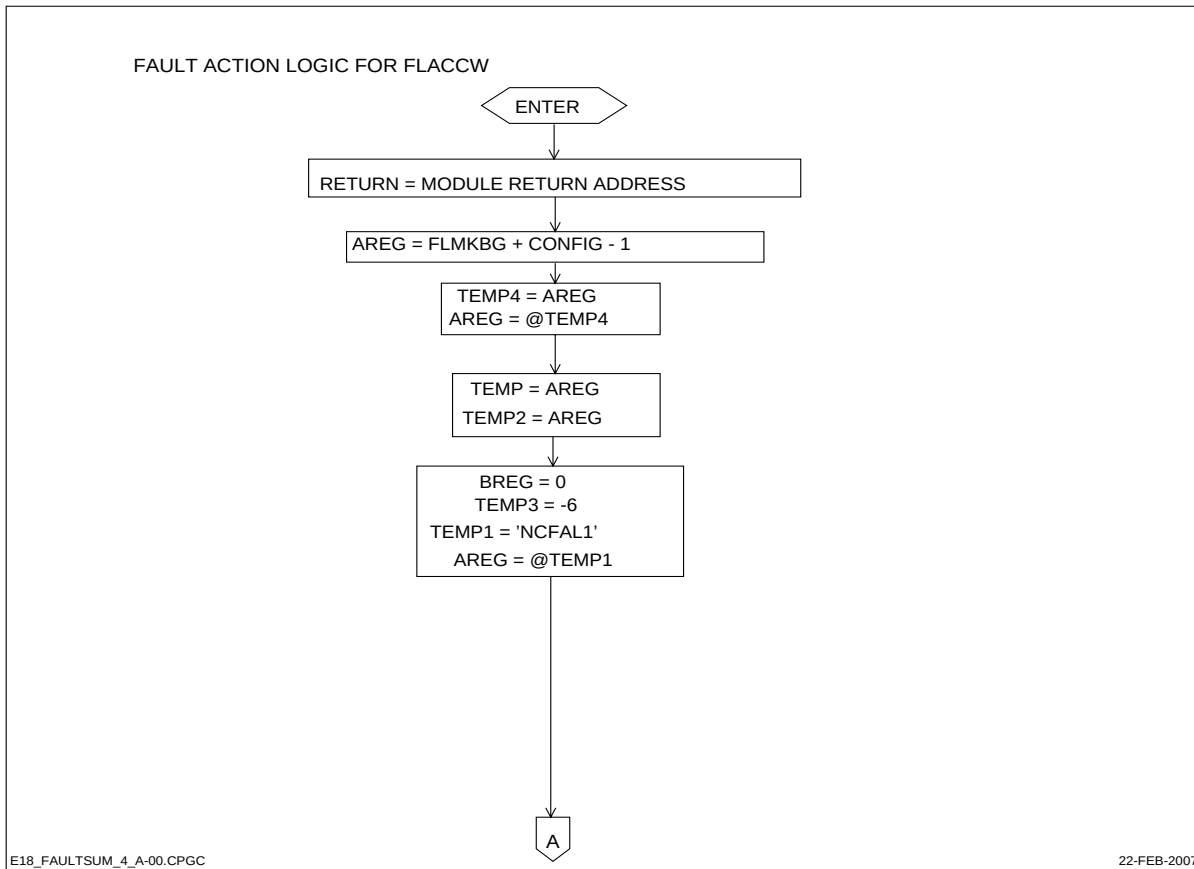
13.00 Figure 150: E18_FAULTSUM_3_A



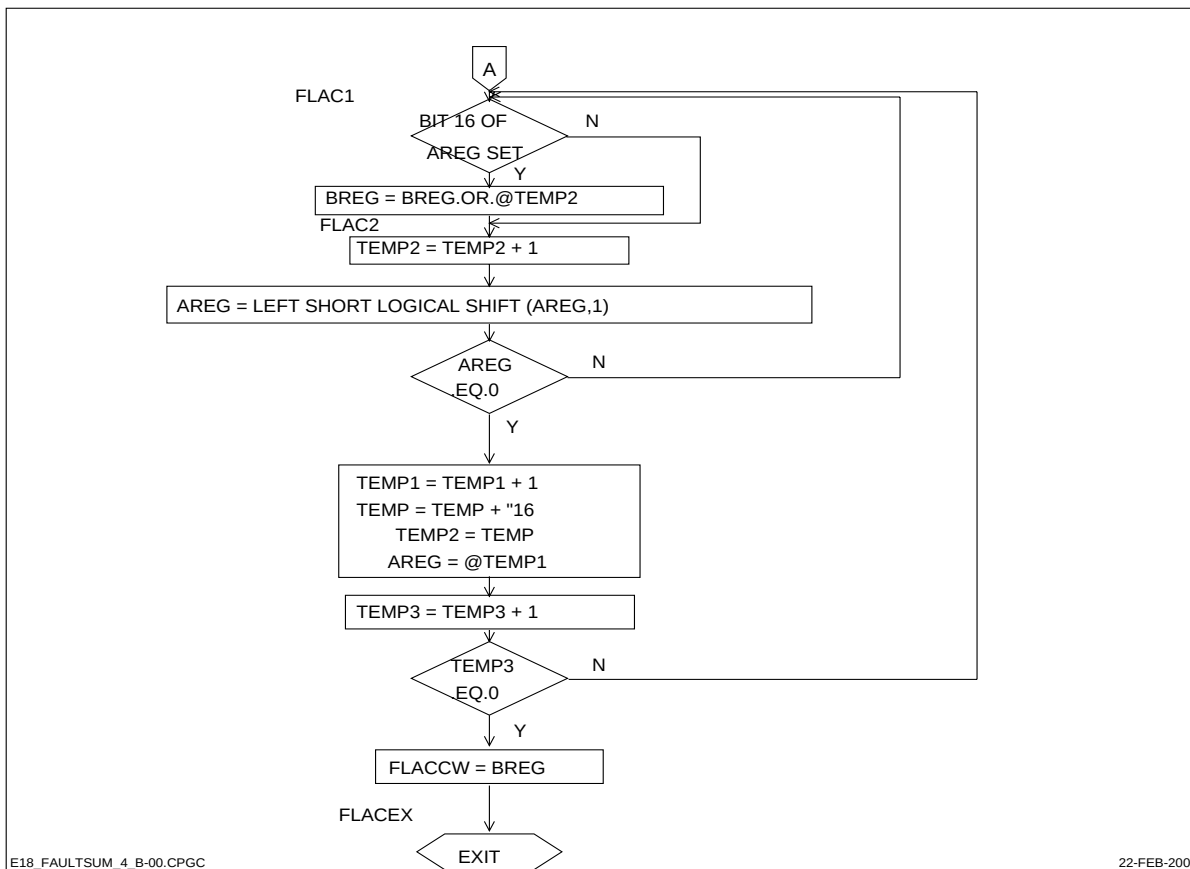
13.00 Figure 151: E18_FAULTSUM_3_B



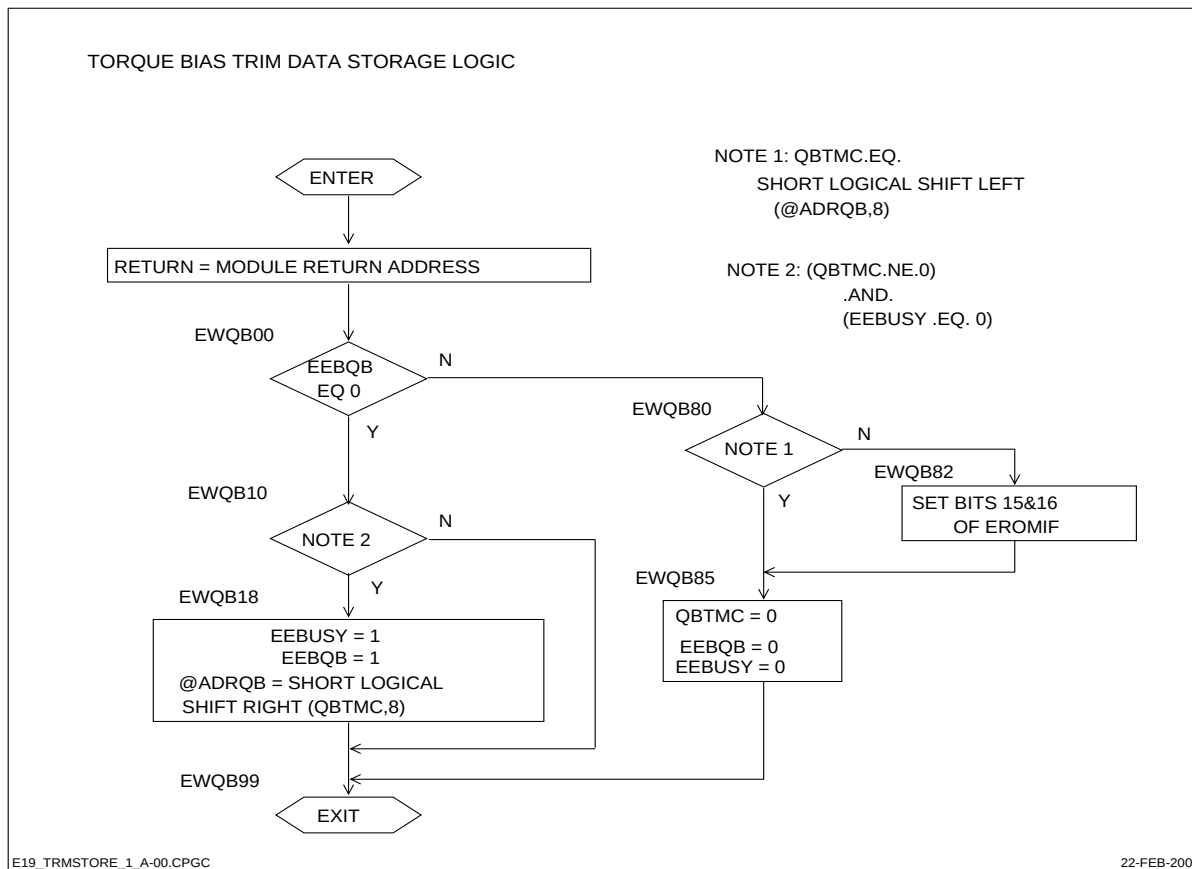
13.00 Figure 152: E18_FAULTSUM_4_A



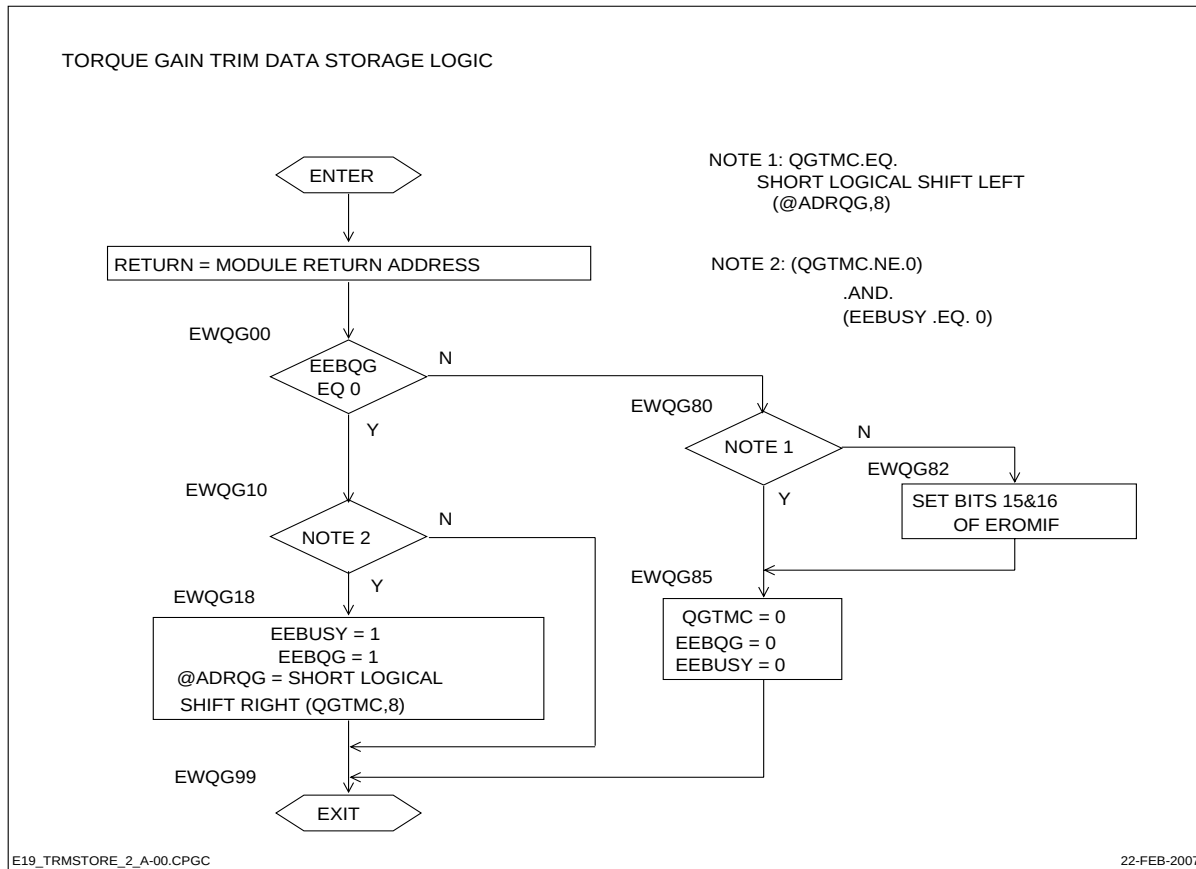
13.00 Figure 153: E18_FAULTSUM_4_B



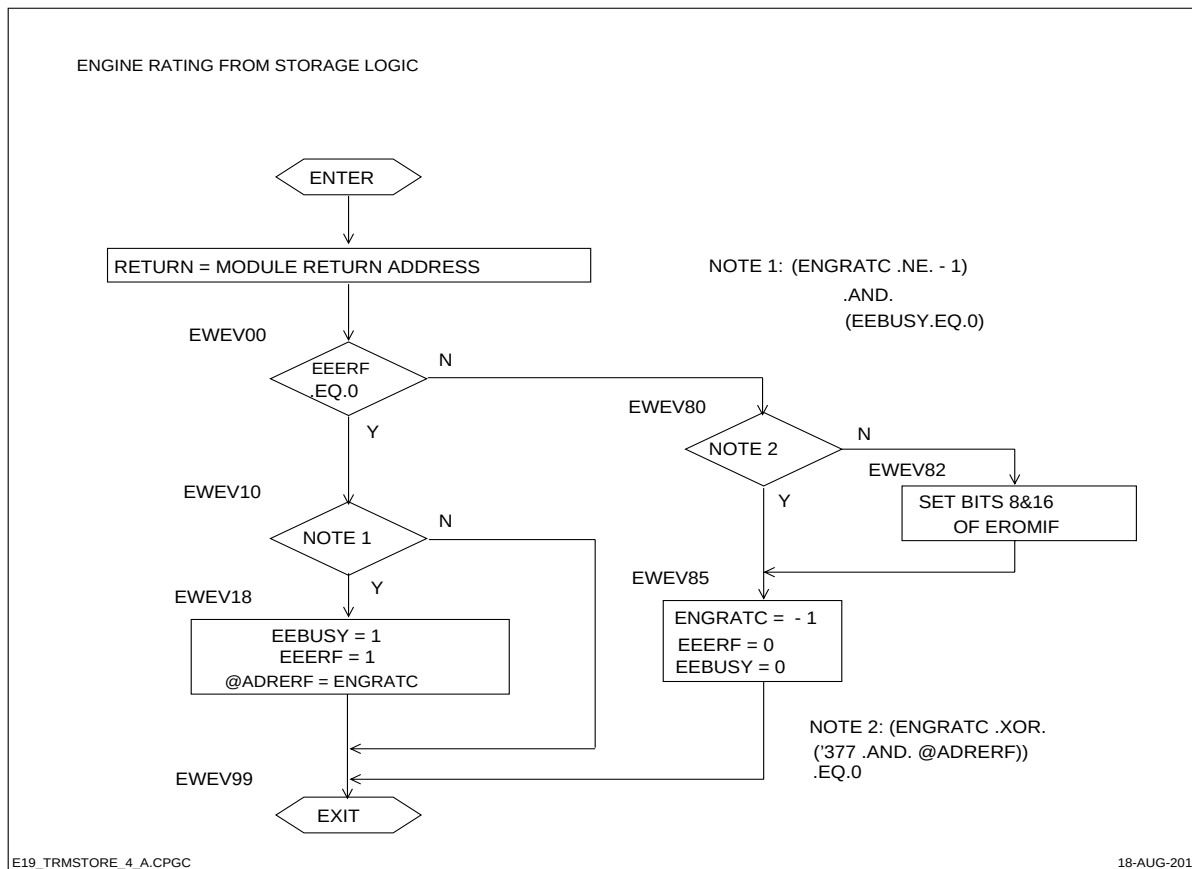
13.00 Figure 154: E19_TRMSTORE_1_A



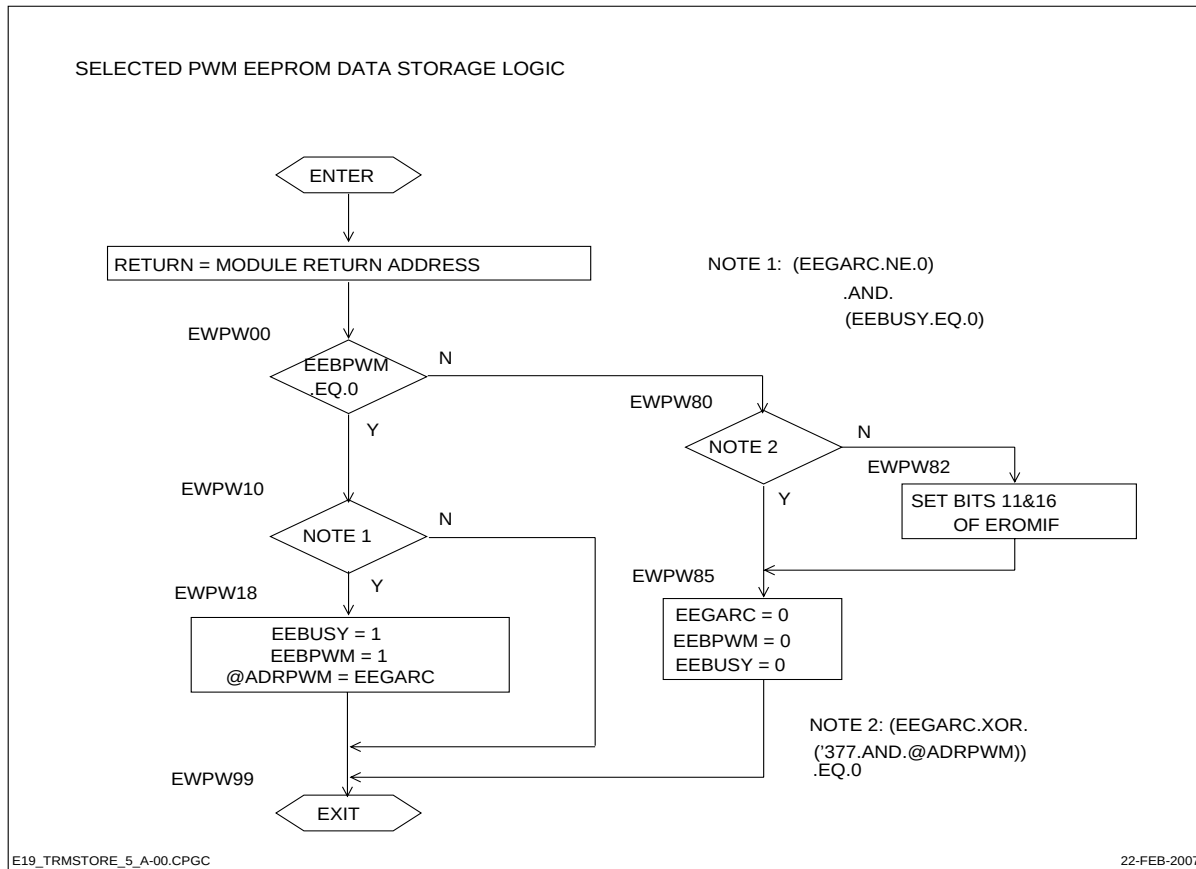
13.00 Figure 155: E19_TRMSTORE_2_A



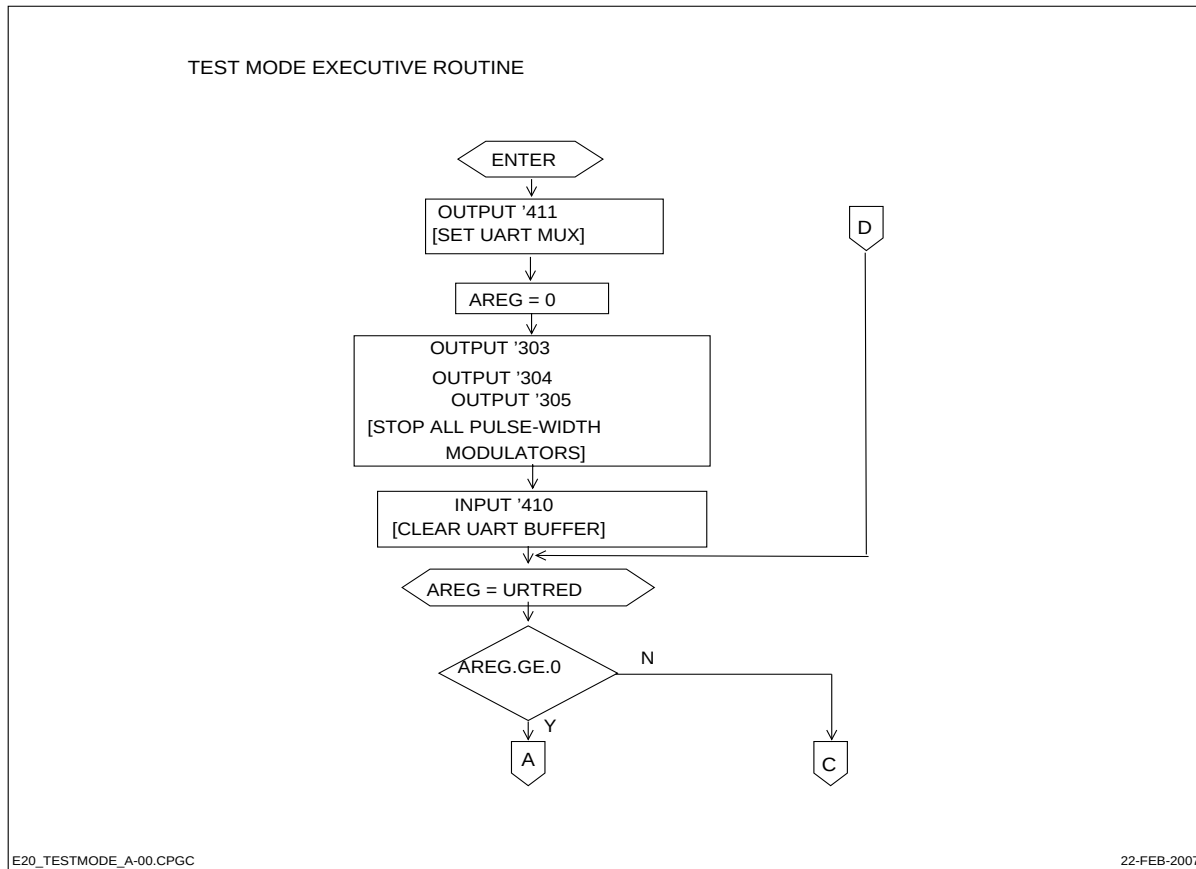
13.00 Figure 156: E19_TRMSTORE_4_A



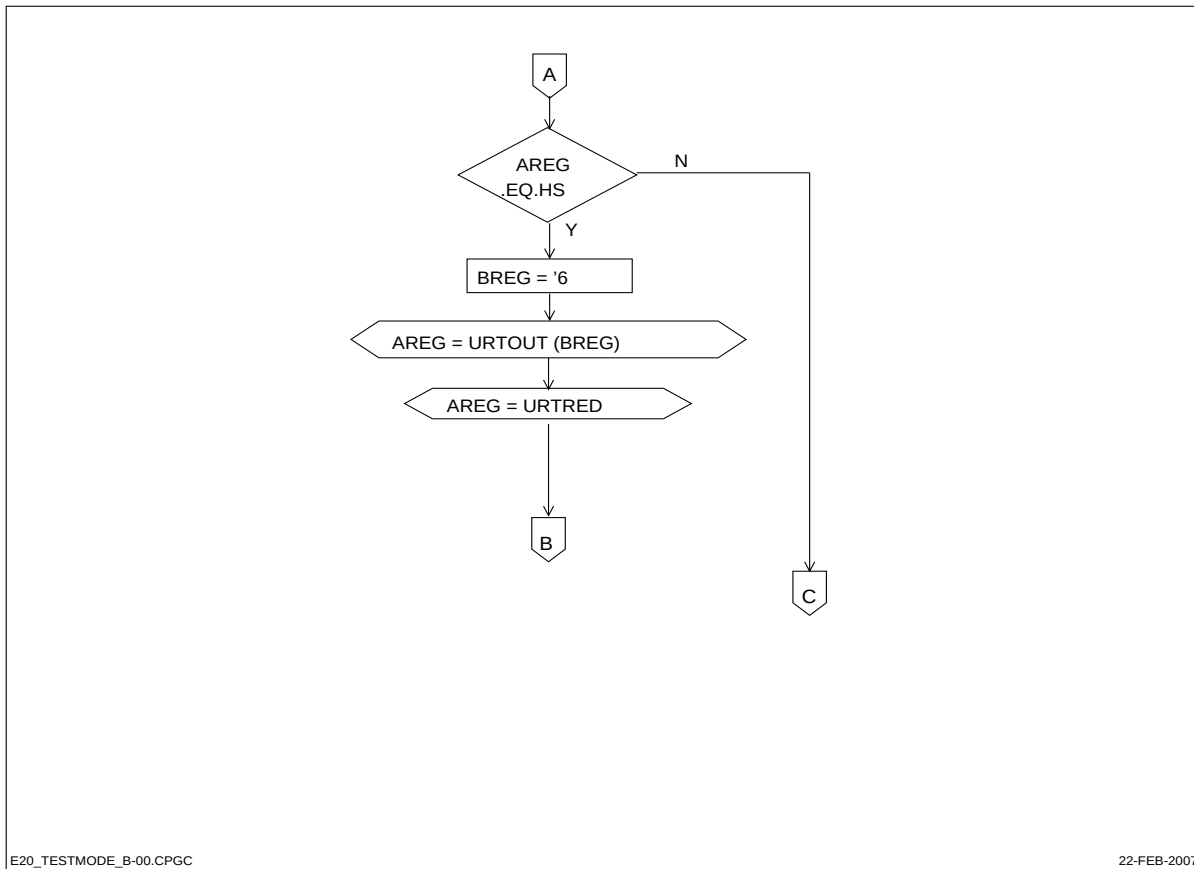
13.00 Figure 157: E19_TRMSTORE_5_A



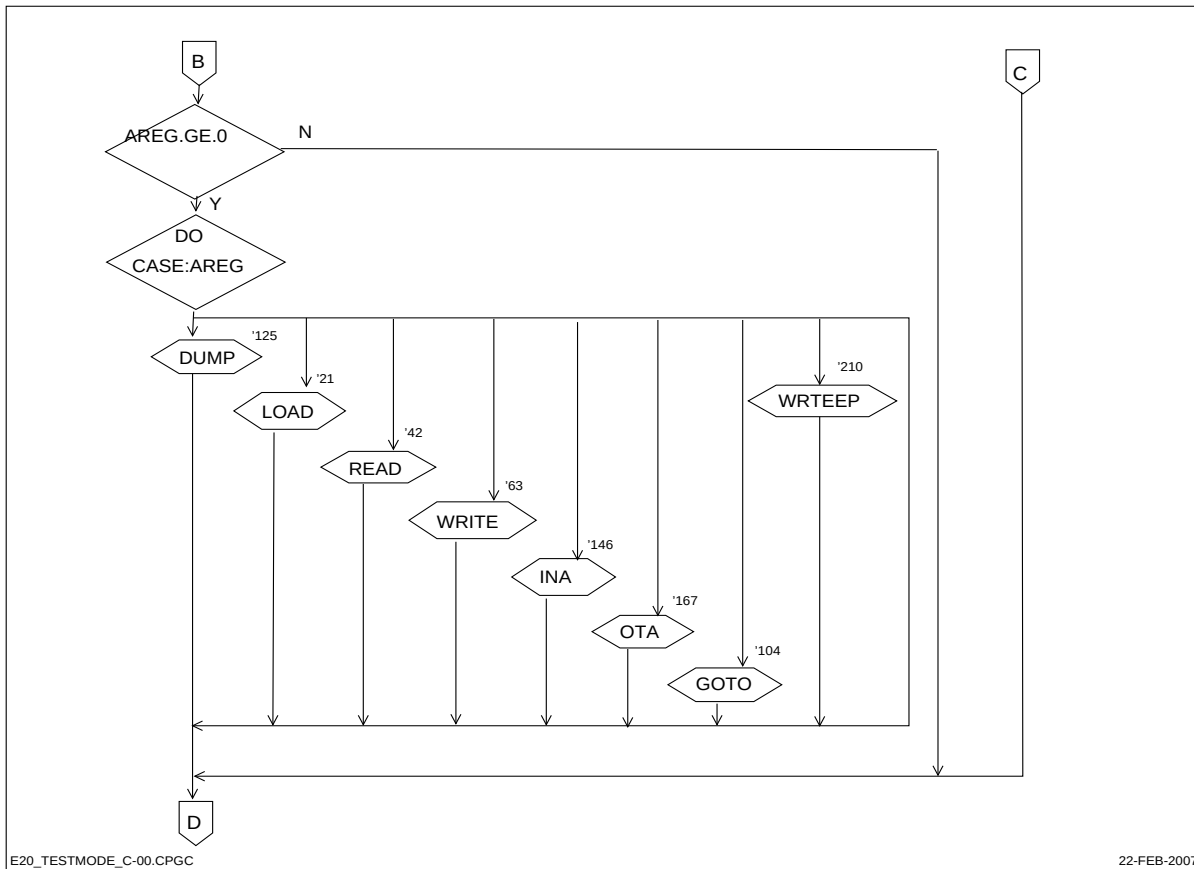
13.00 Figure 158: E20_TESTMODE _A



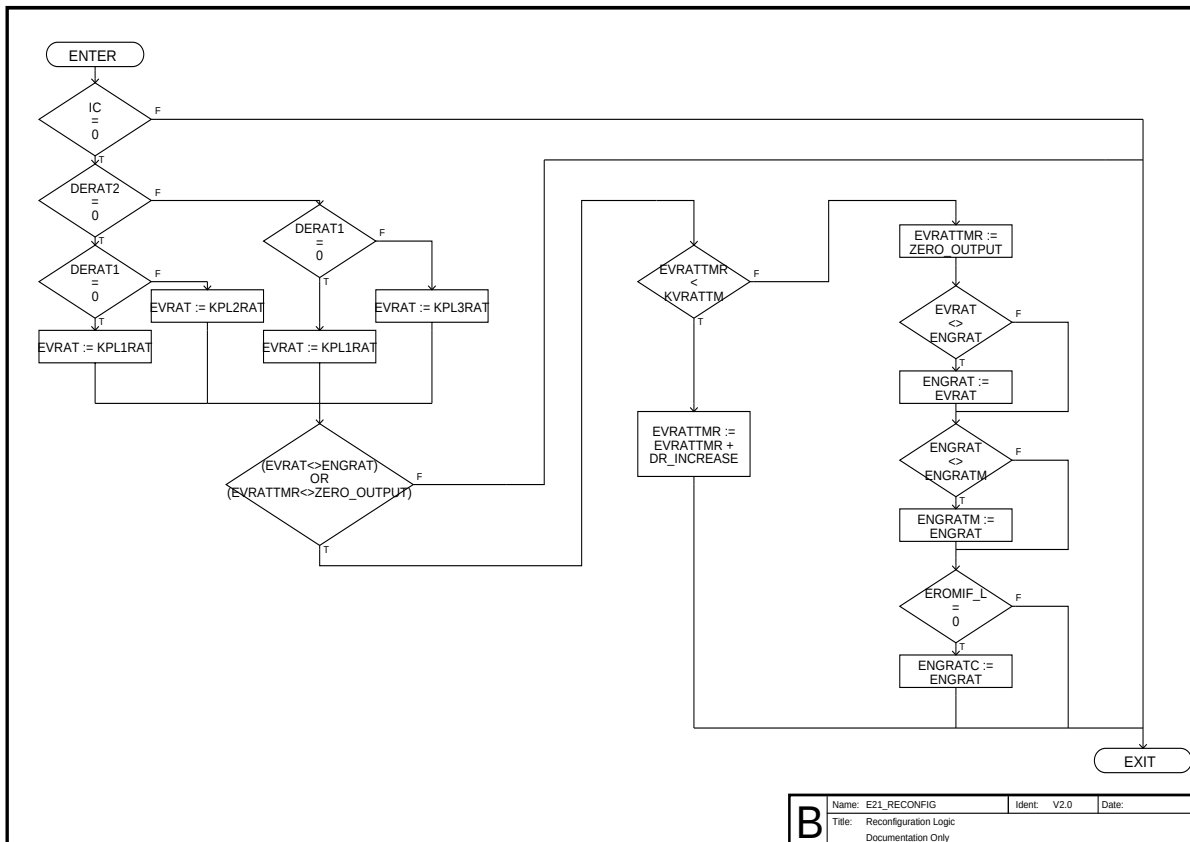
13.00 Figure 159: E20_TESTMODE _B



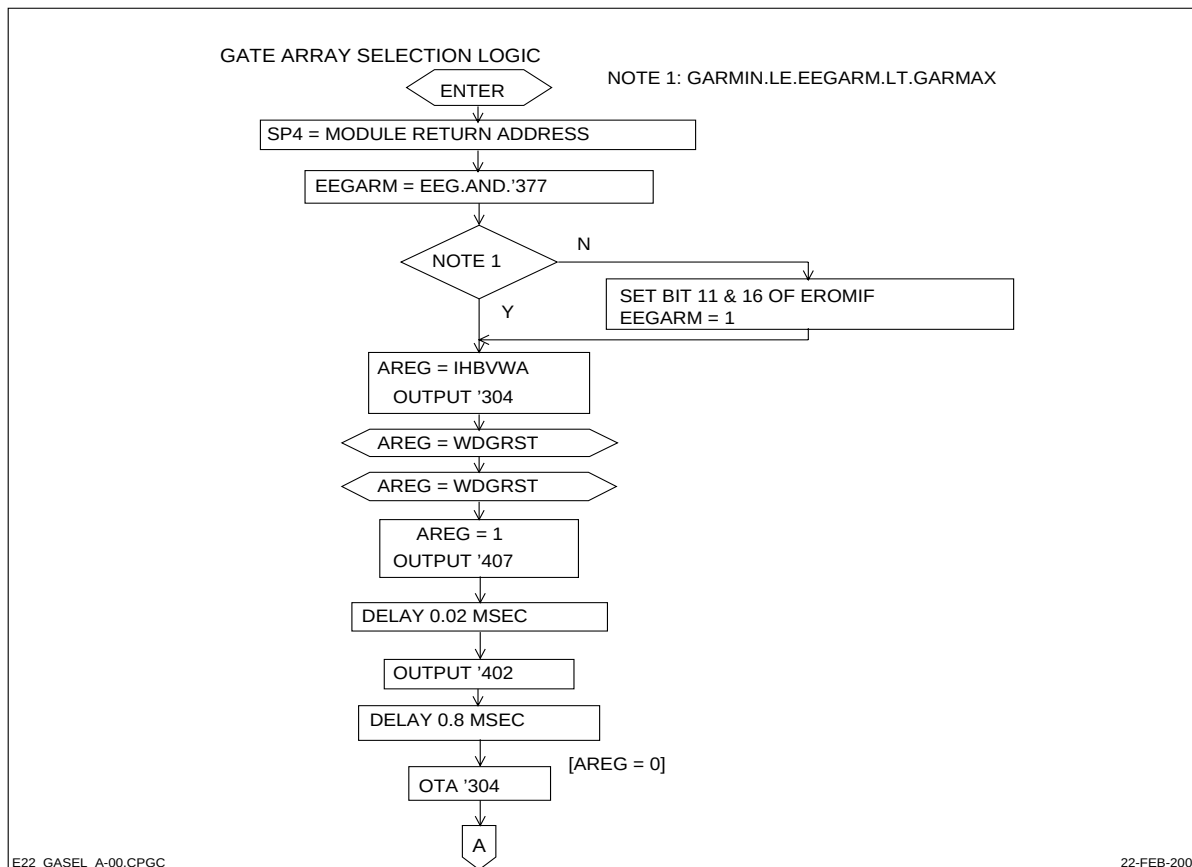
13.00 Figure 160: E20_TESTMODE _C



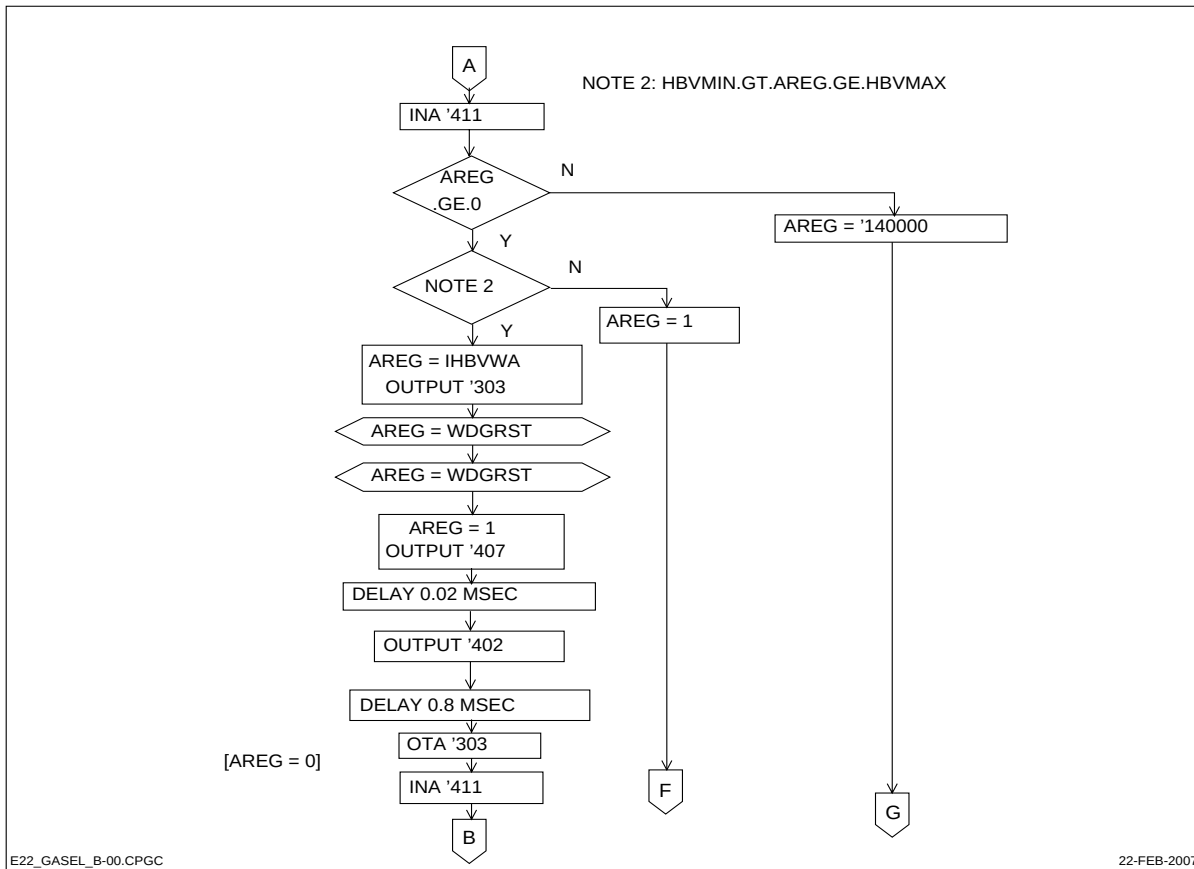
13.00 Figure 161: E21_RECONFIG



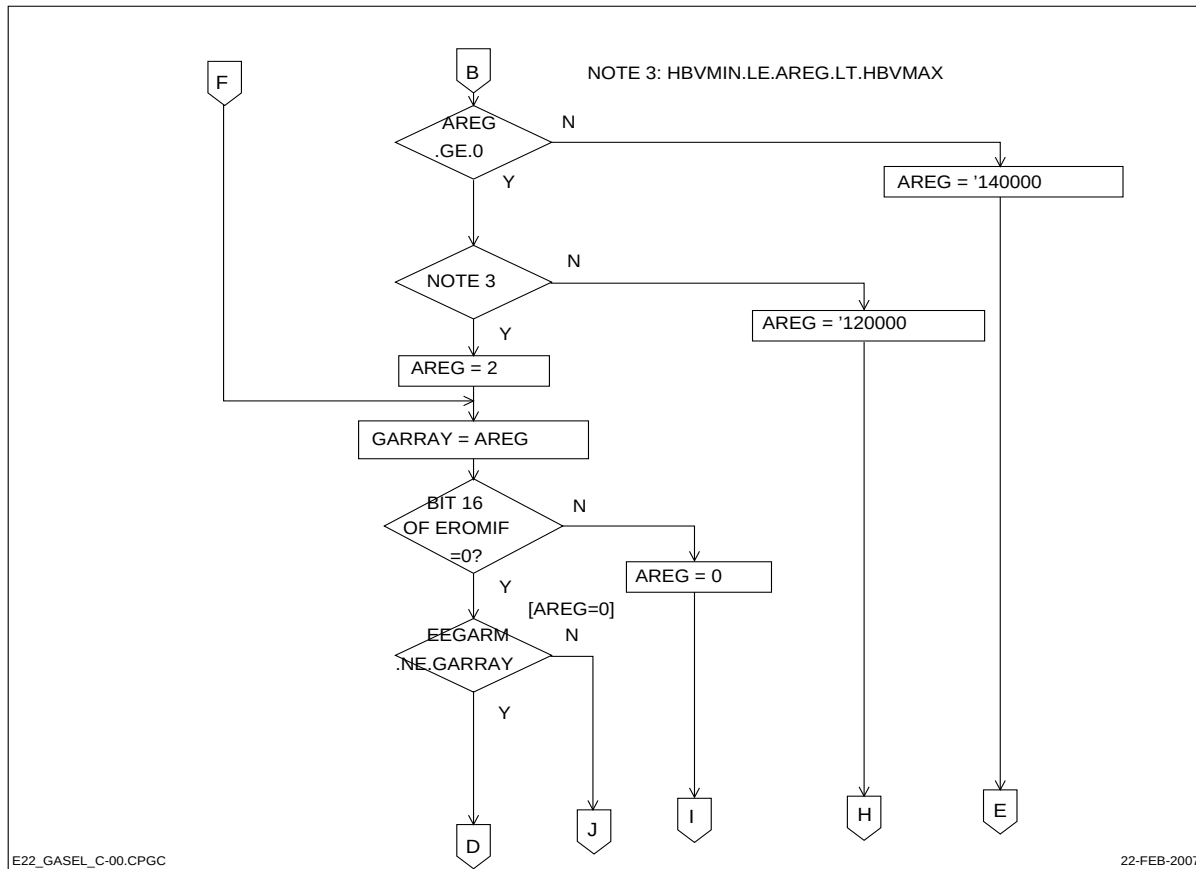
13.00 Figure 162: E22_GASEL_A



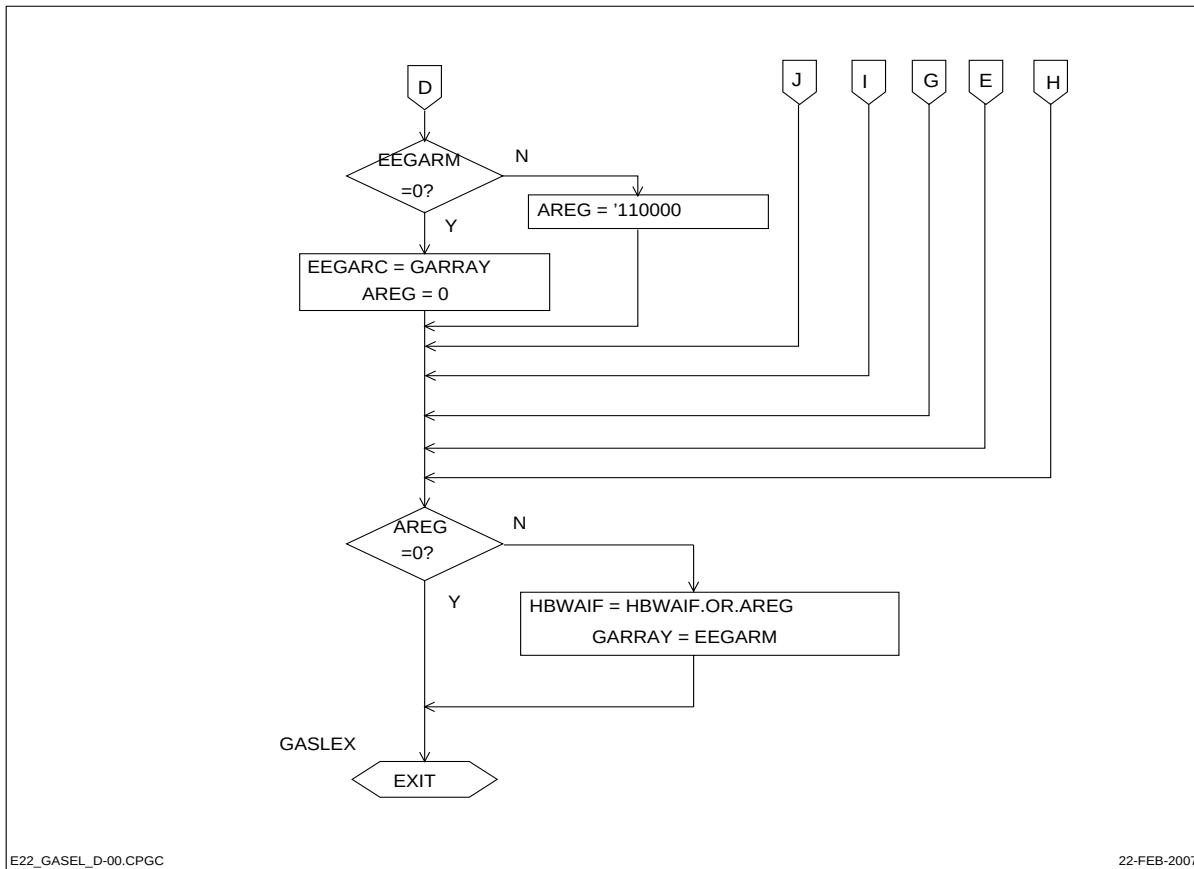
13.00 Figure 163: E22_GASEL_B



13.00 Figure 164: E22_GASEL_C



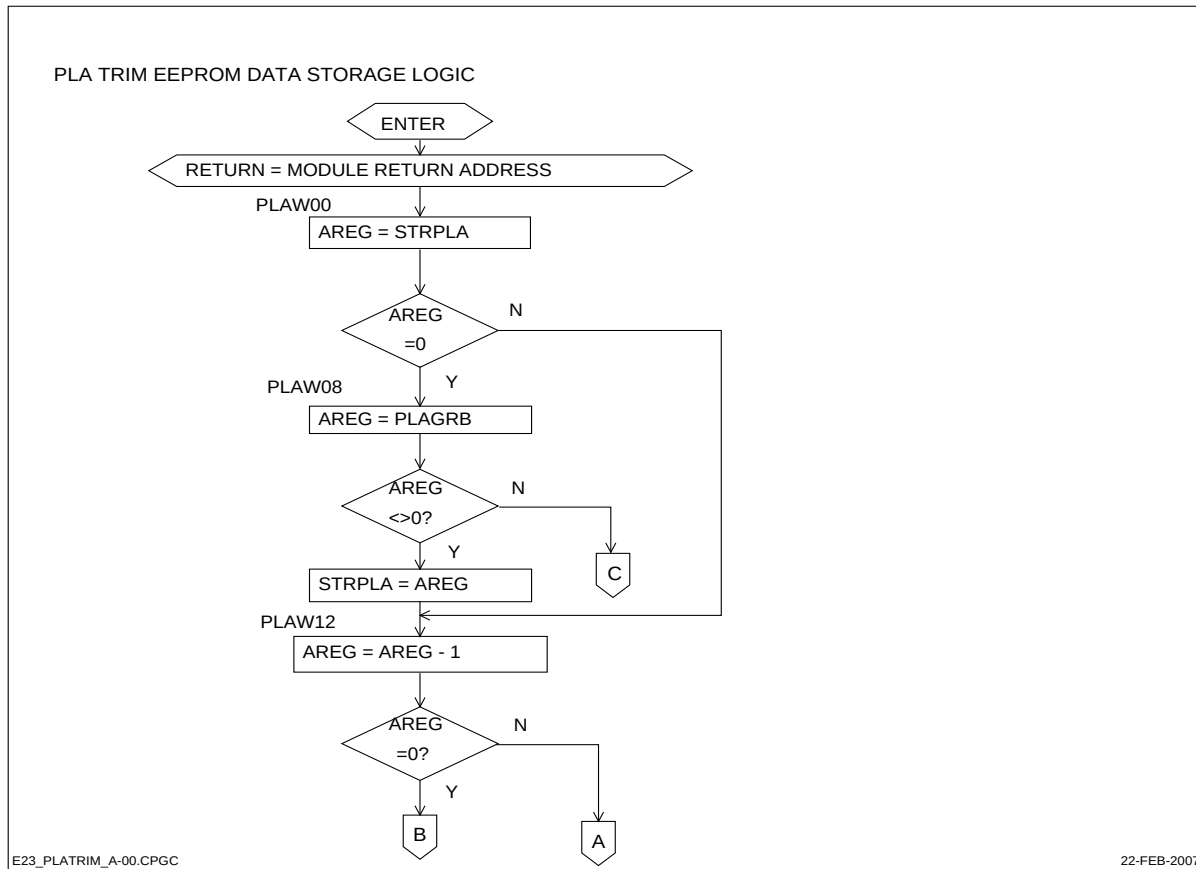
13.00 Figure 165: E22_GASEL_D



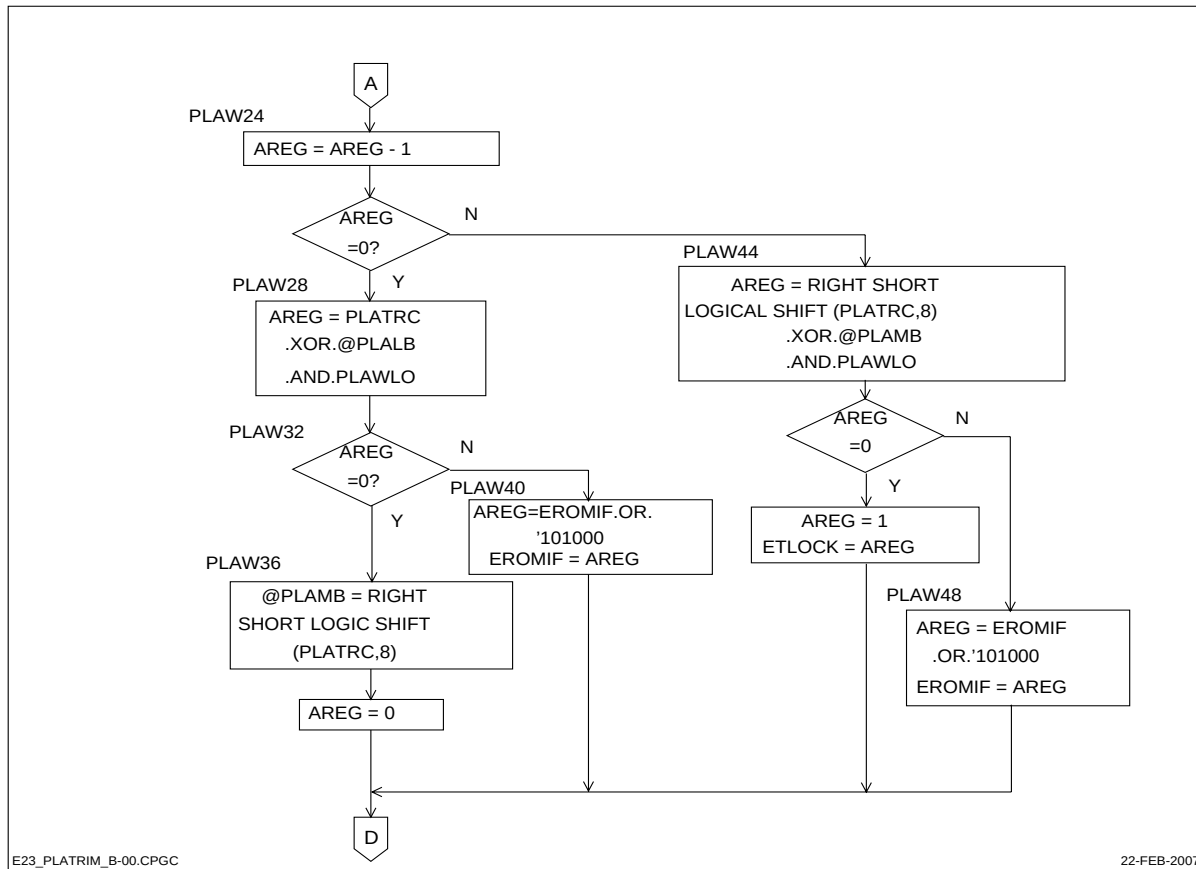
E22_GASEL_D-00.CPGC

22-FEB-2007

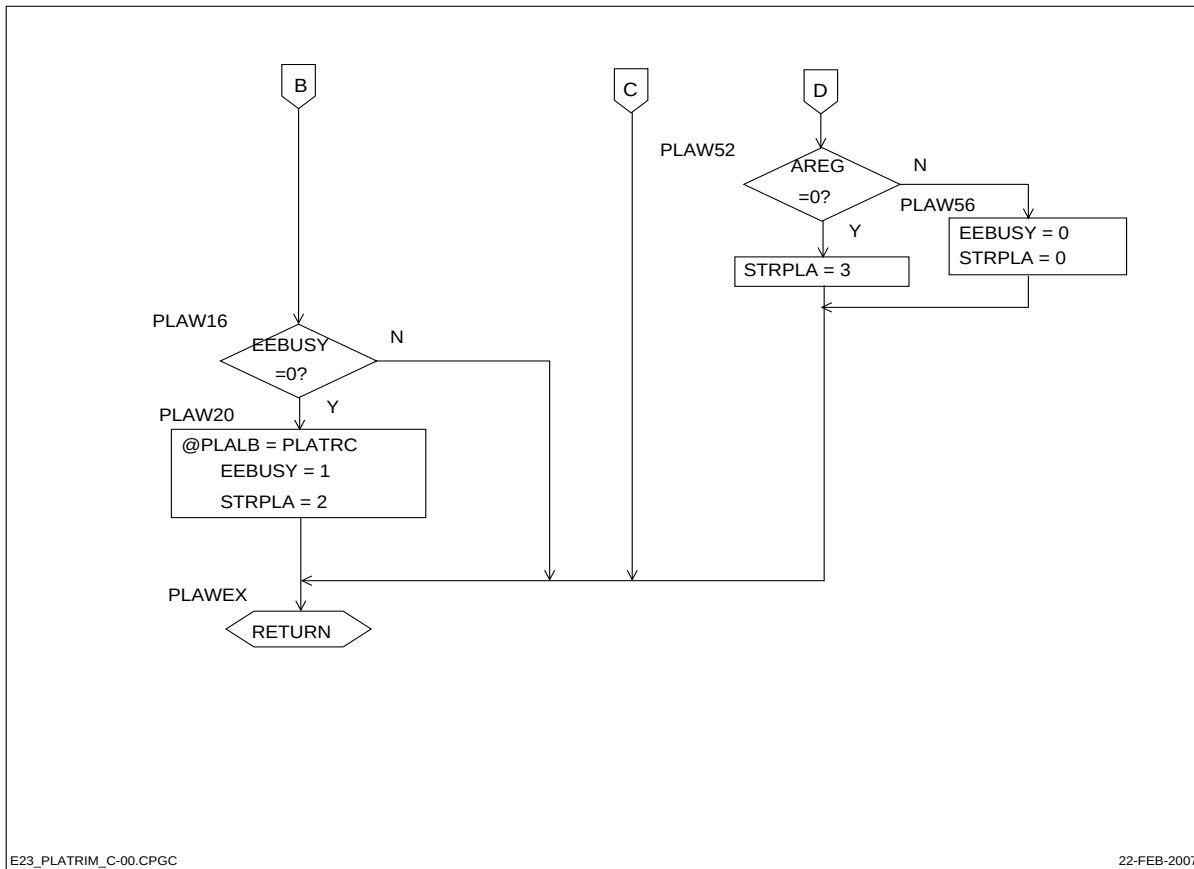
13.00 Figure 166: E23_PLATRIM_A



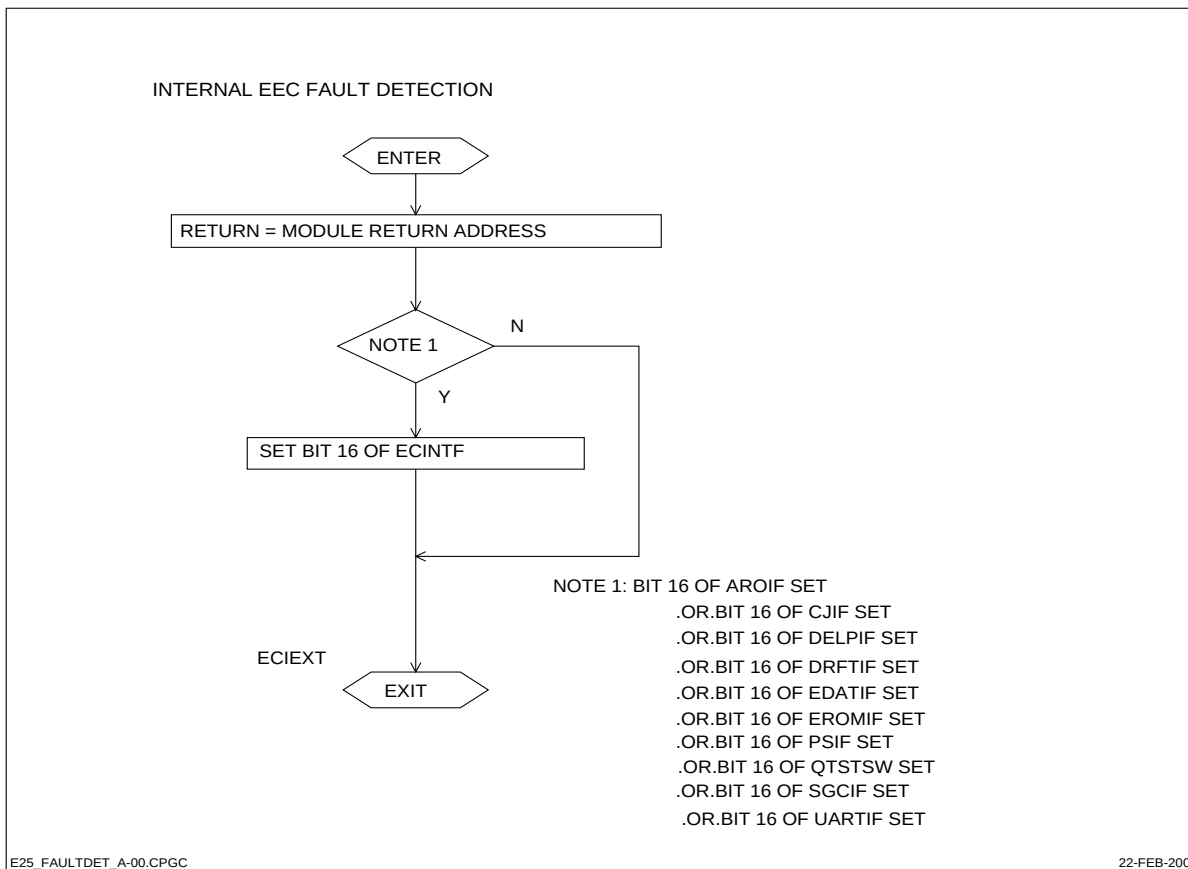
13.00 Figure 167: E23_PLATRIM_B



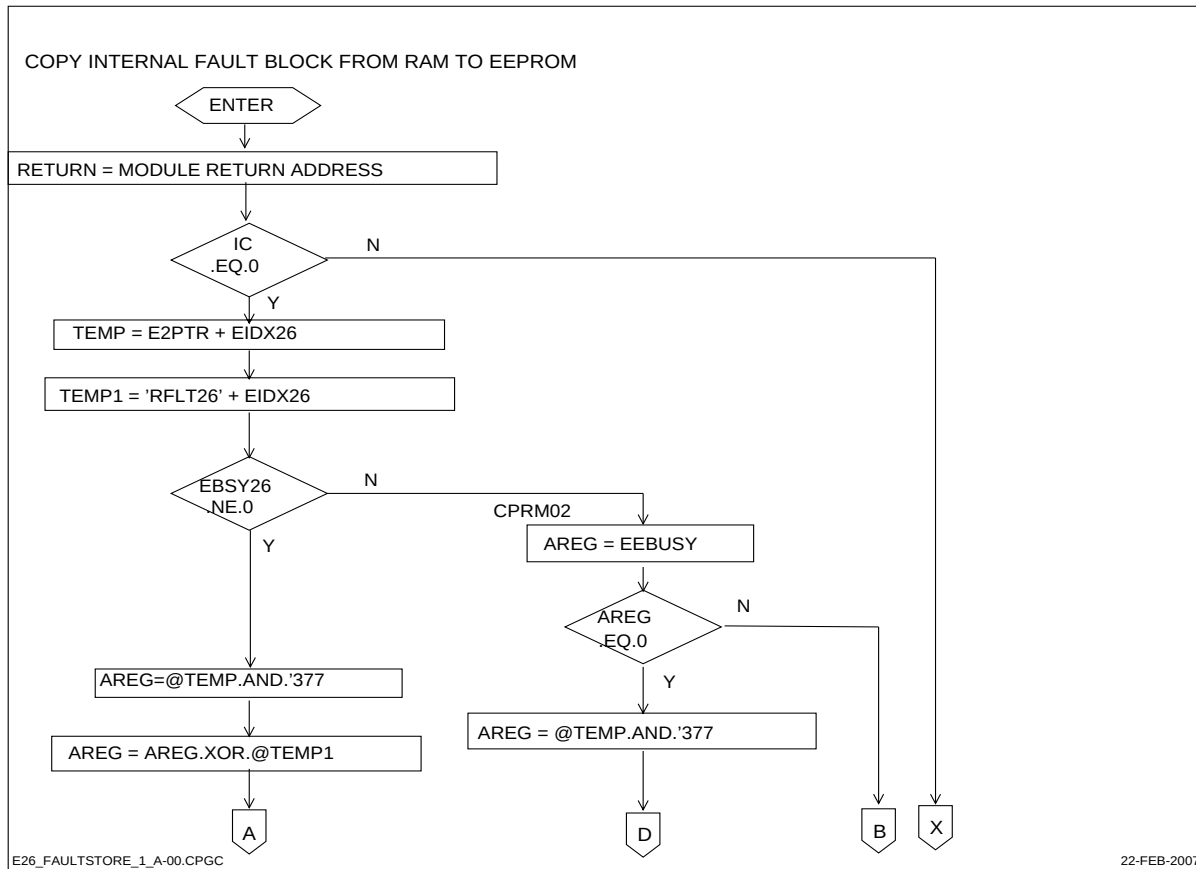
13.00 Figure 168: E23_PLATRIM_C



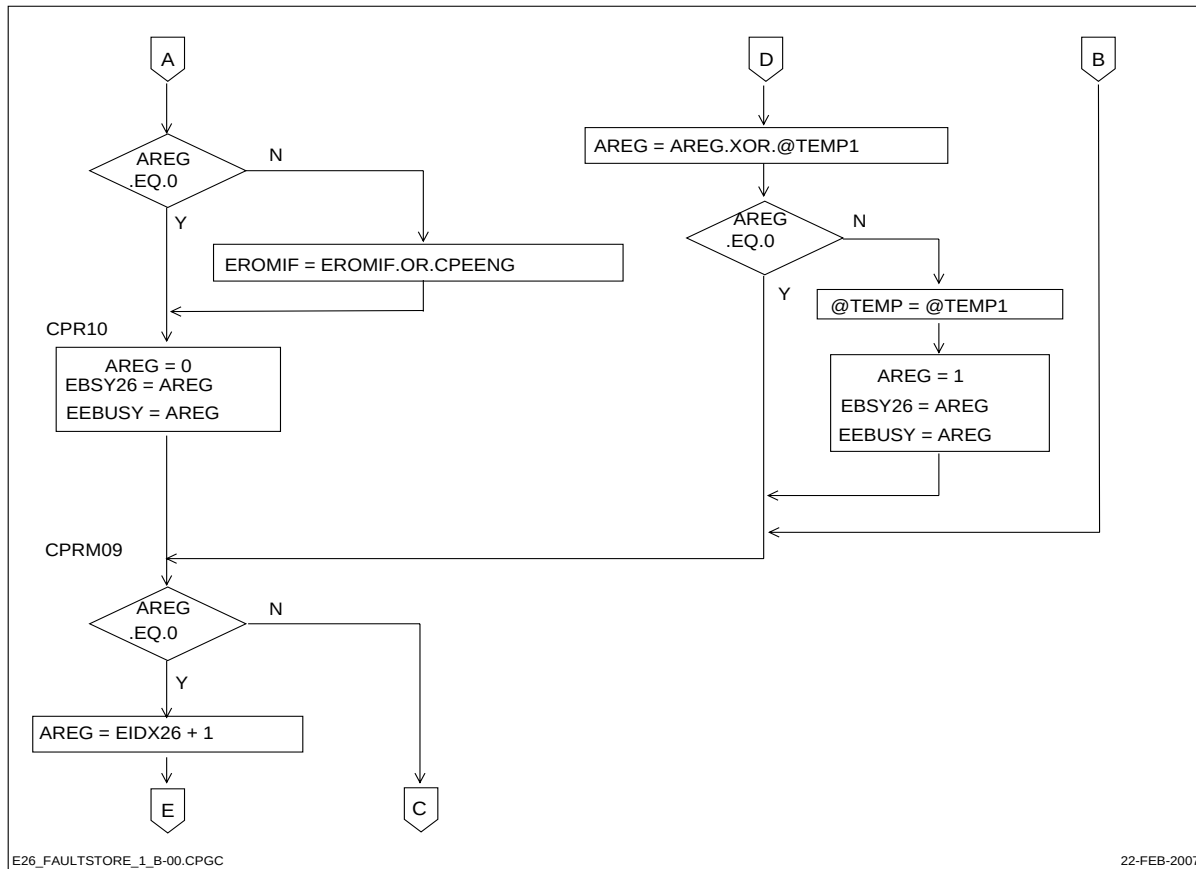
13.00 Figure 169: E25_FAULTDET_A



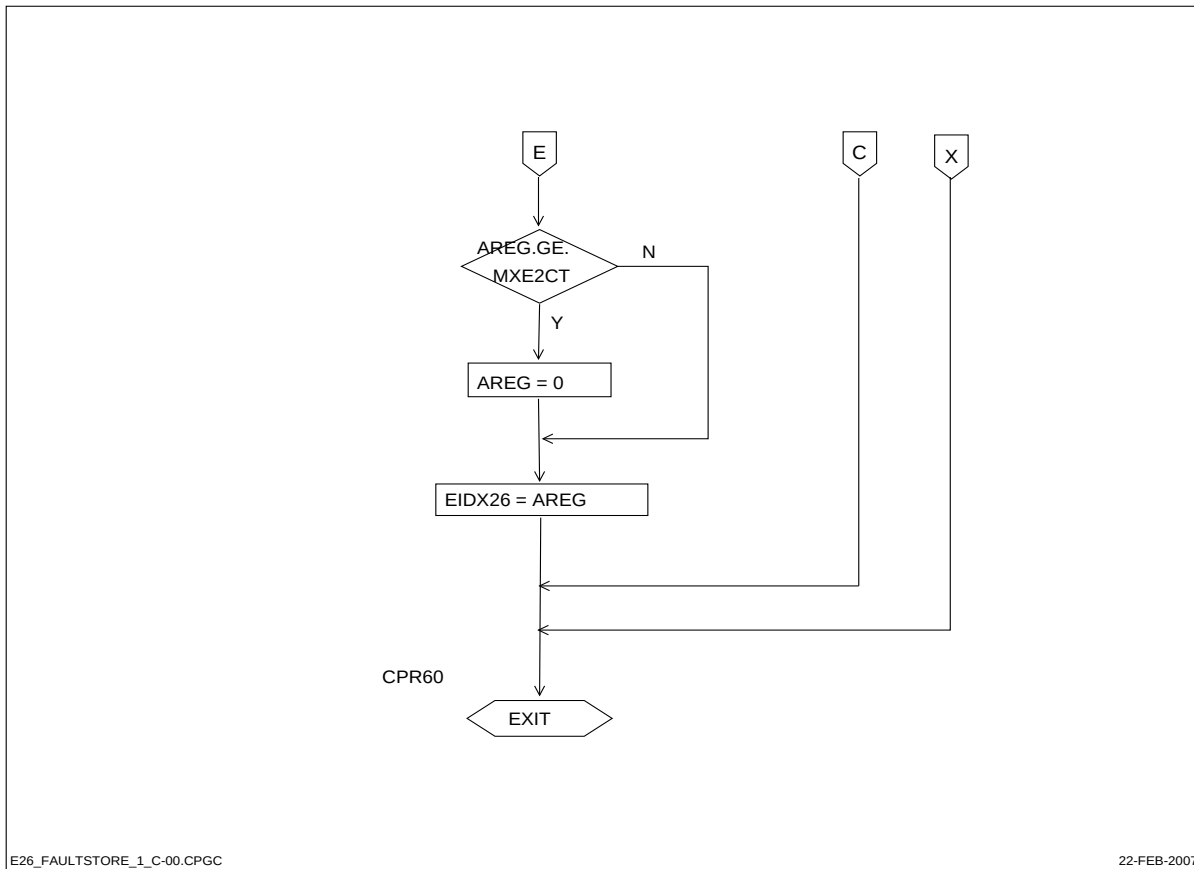
13.00 Figure 170: E26_FAULTSTORE_1_A



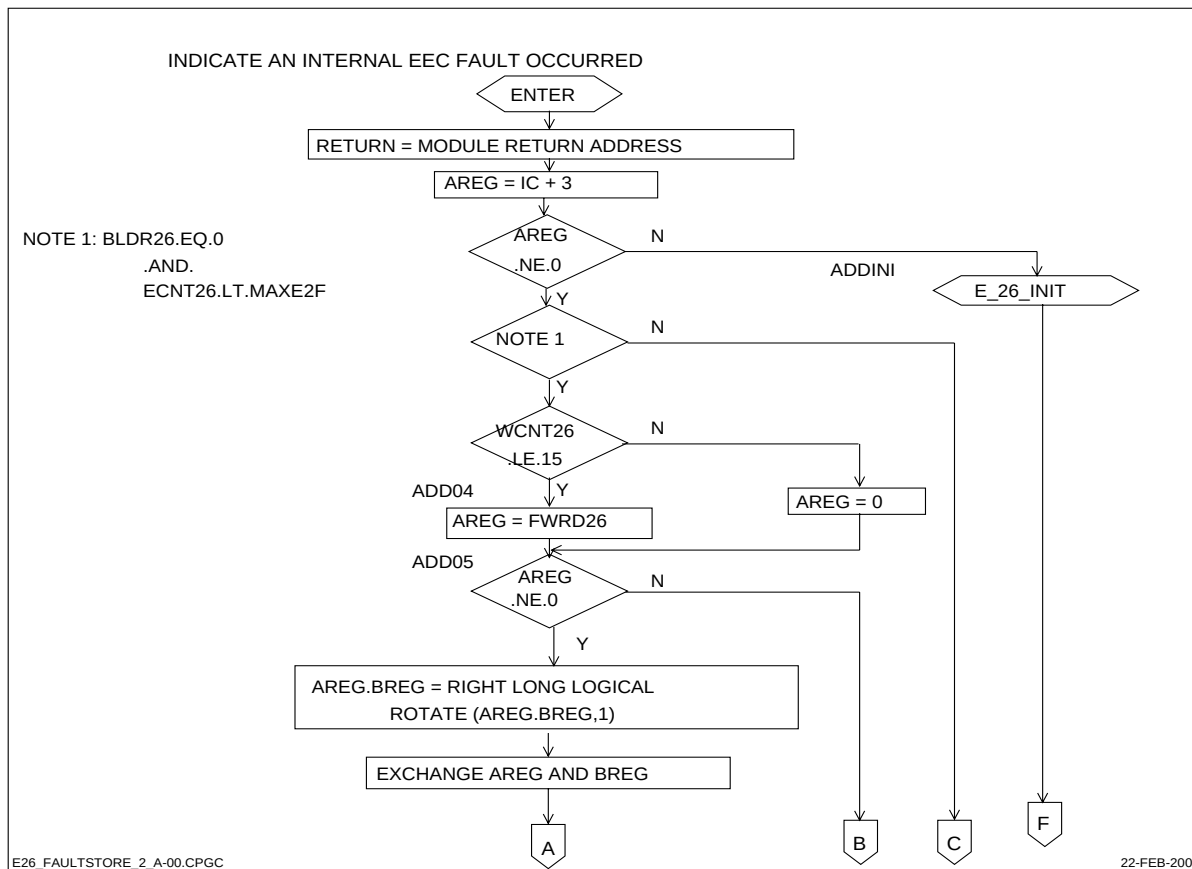
13.00 Figure 171: E26_FAULTSTORE_1_B



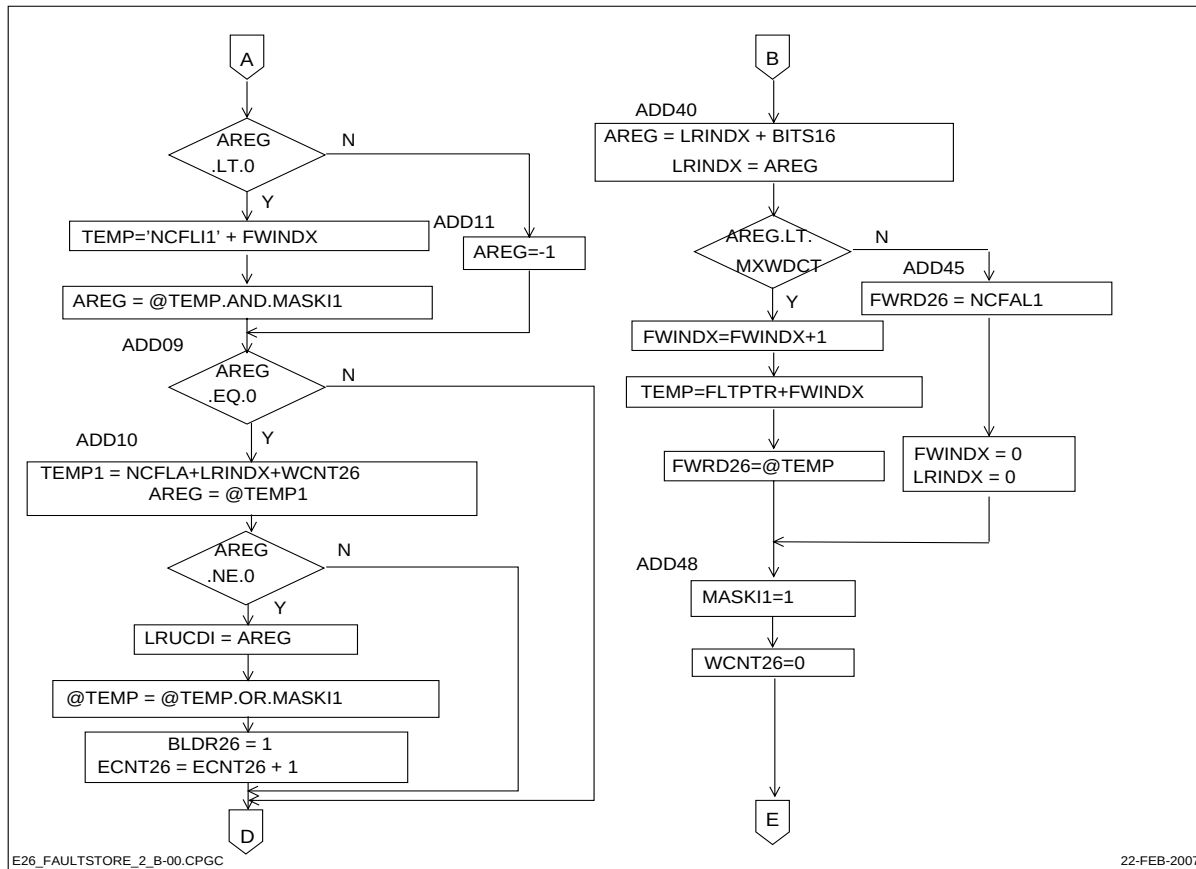
13.00 Figure 172: E26_FAULTSTORE_1_C



13.00 Figure 173: E26_FAULTSTORE_2_A



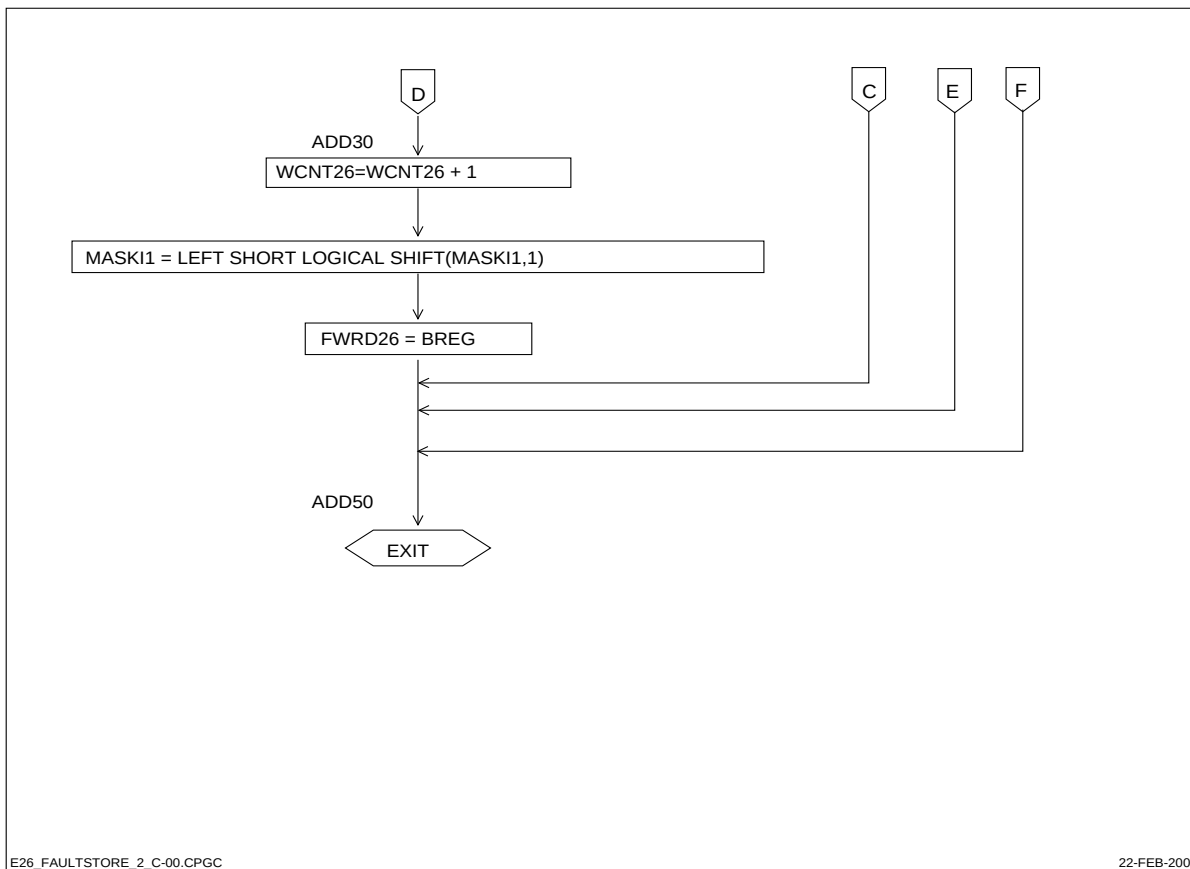
13.00 Figure 174: E26_FAULTSTORE_2_B



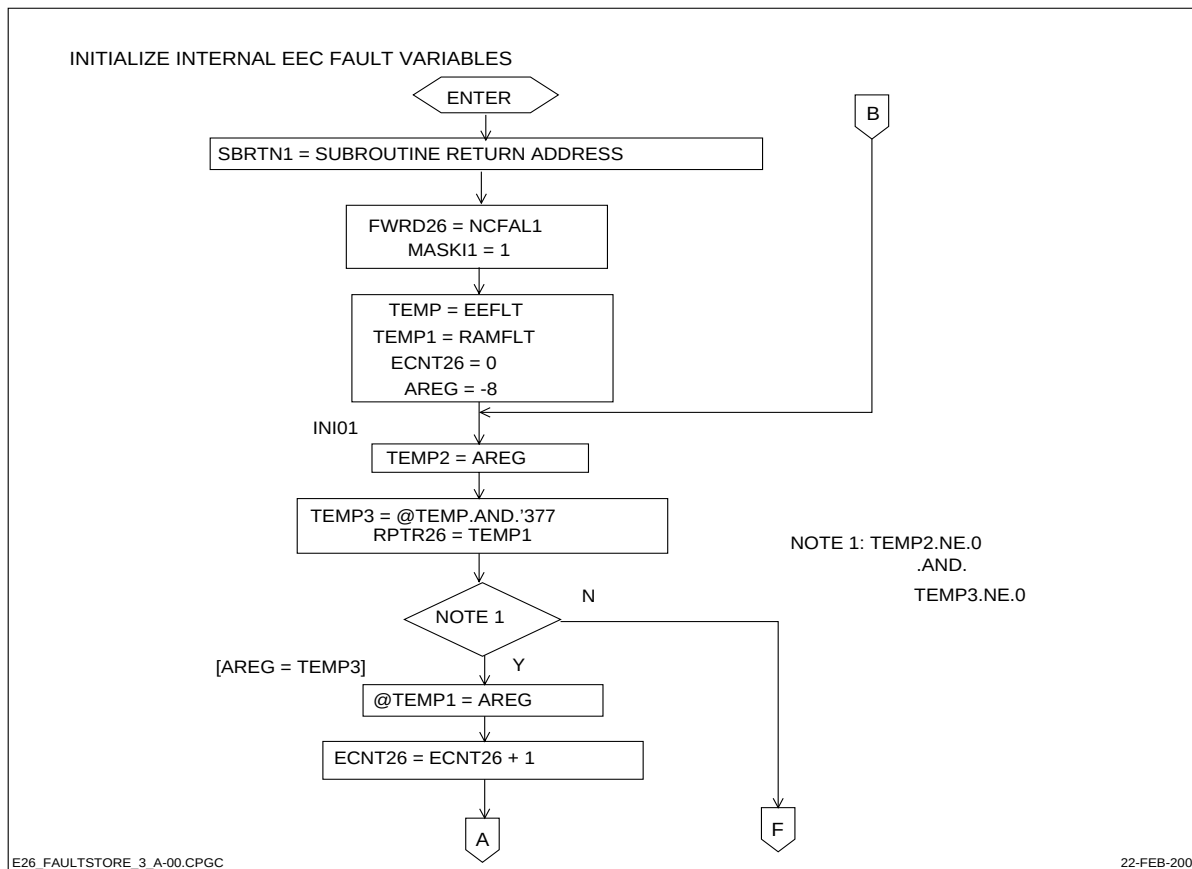
E26_FAULTSTORE_2_B-00.CPGC

22-FEB-2007

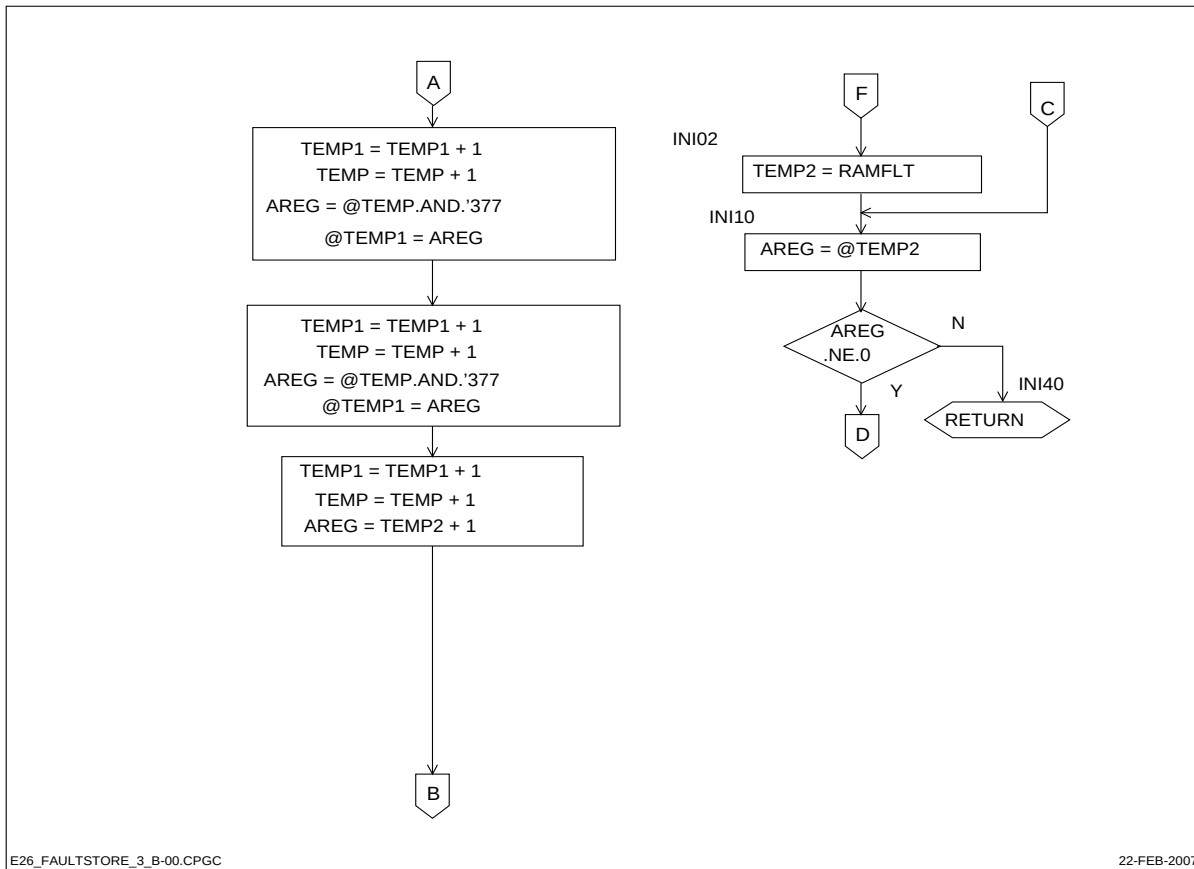
13.00 Figure 175: E26_FAULTSTORE_2_C



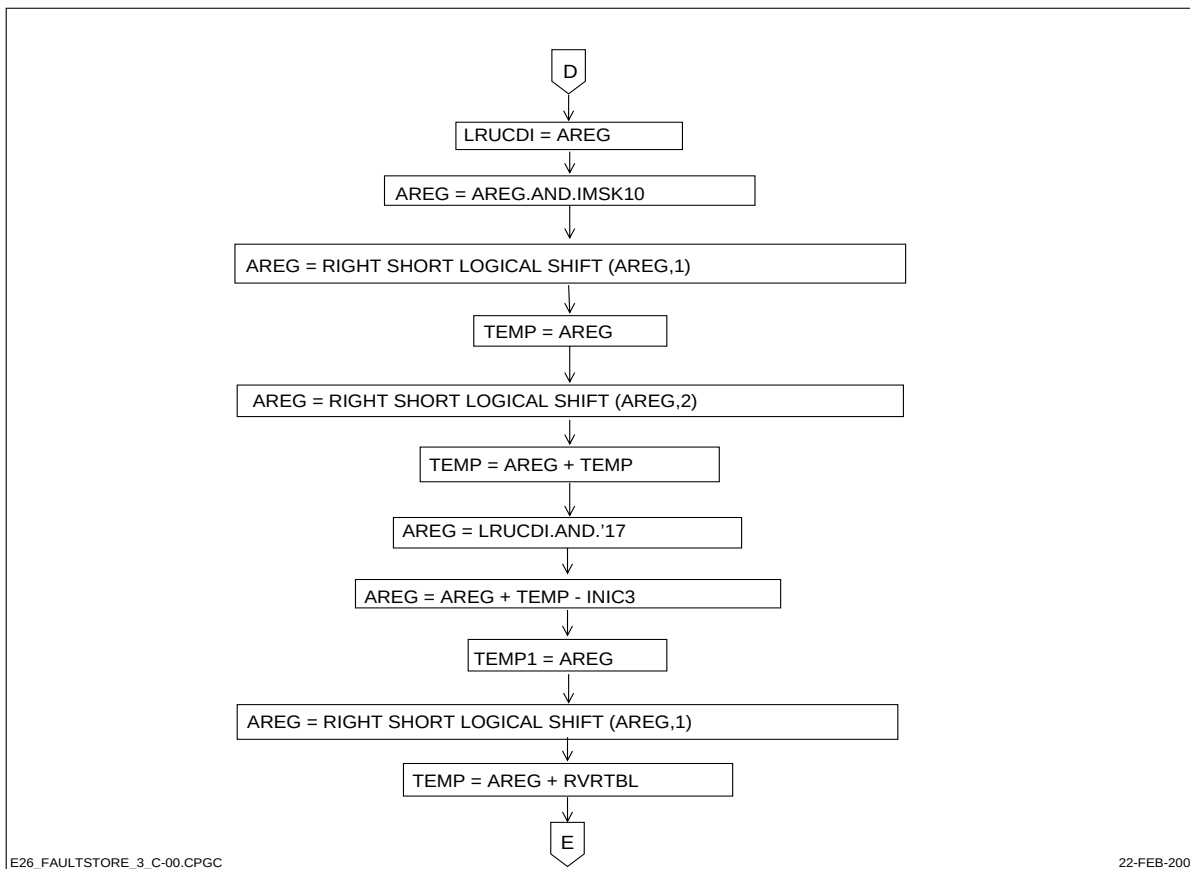
13.00 Figure 176: E26_FAULTSTORE_3_A



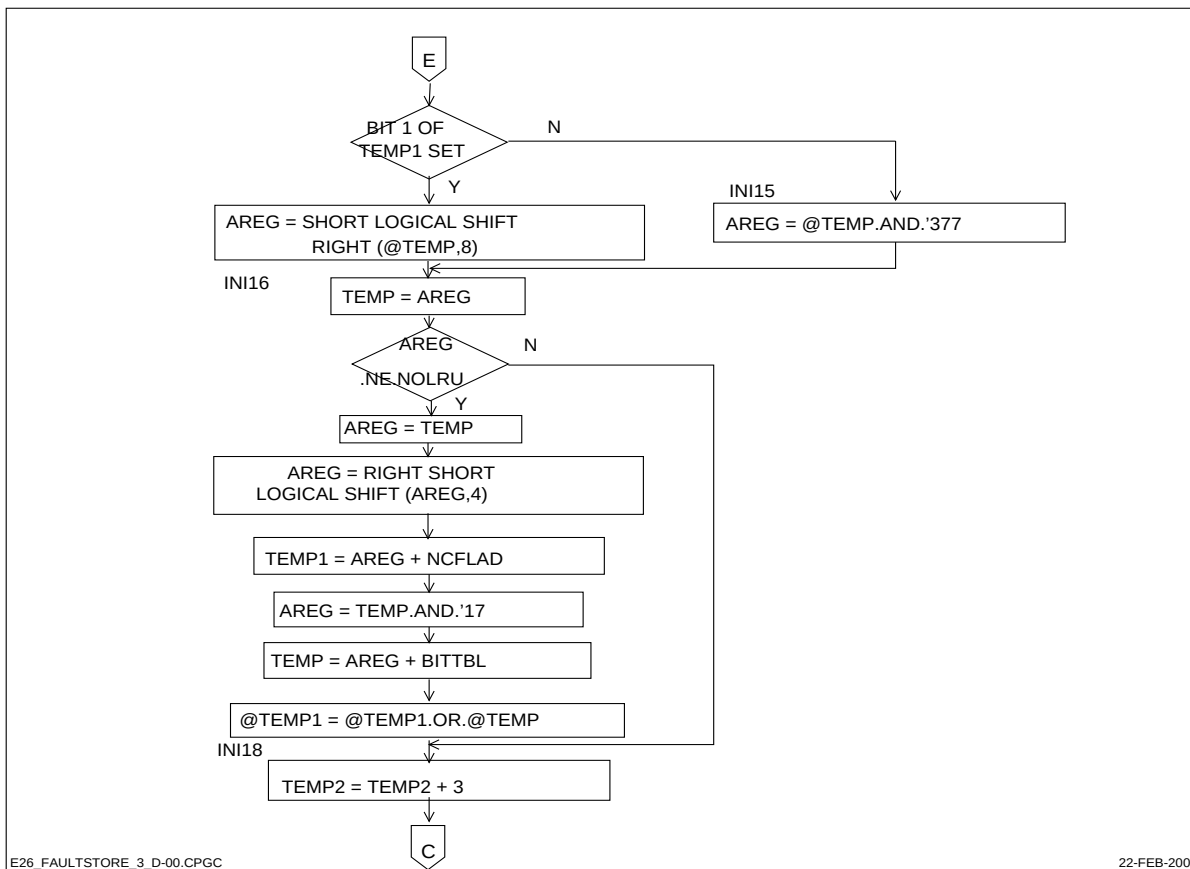
13.00 Figure 177: E26_FAULTSTORE_3_B



13.00 Figure 178: E26_FAULTSTORE_3_C



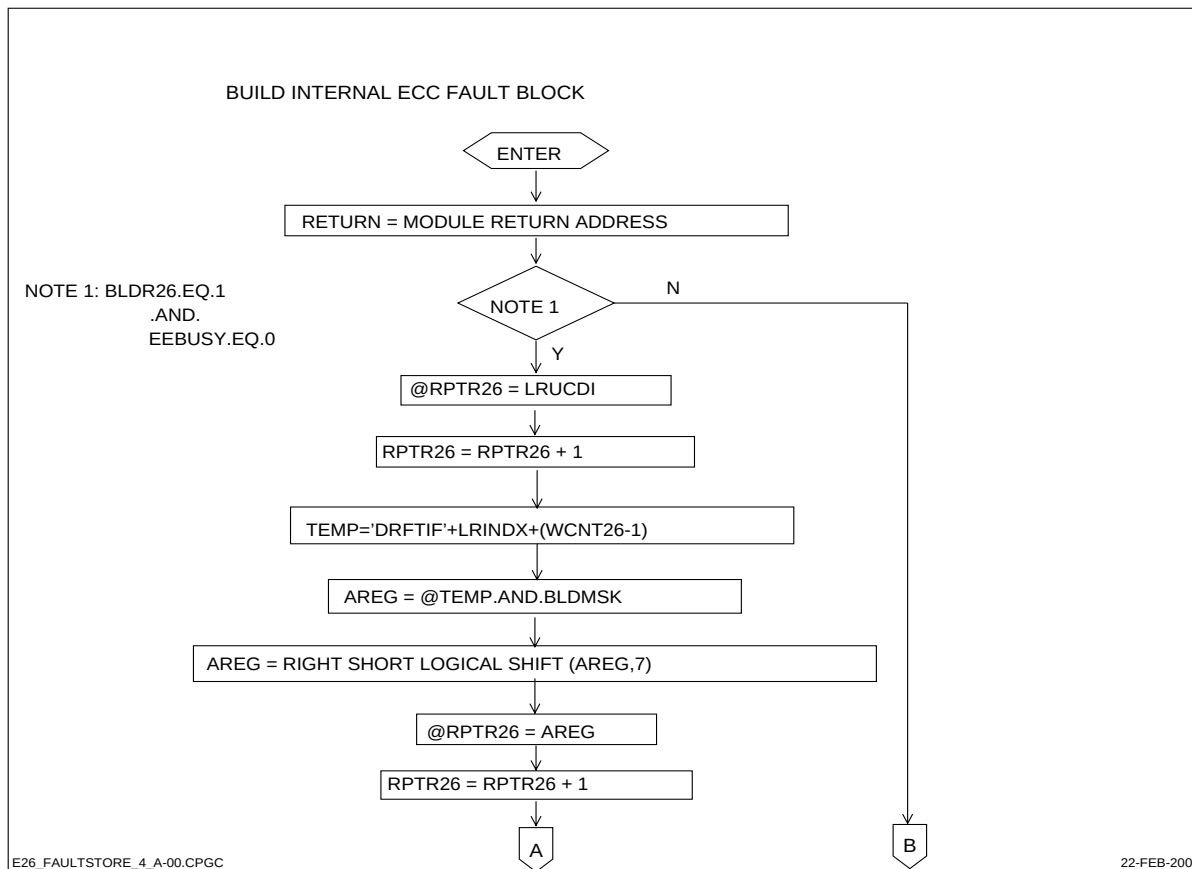
13.00 Figure 179: E26_FAULTSTORE_3_D



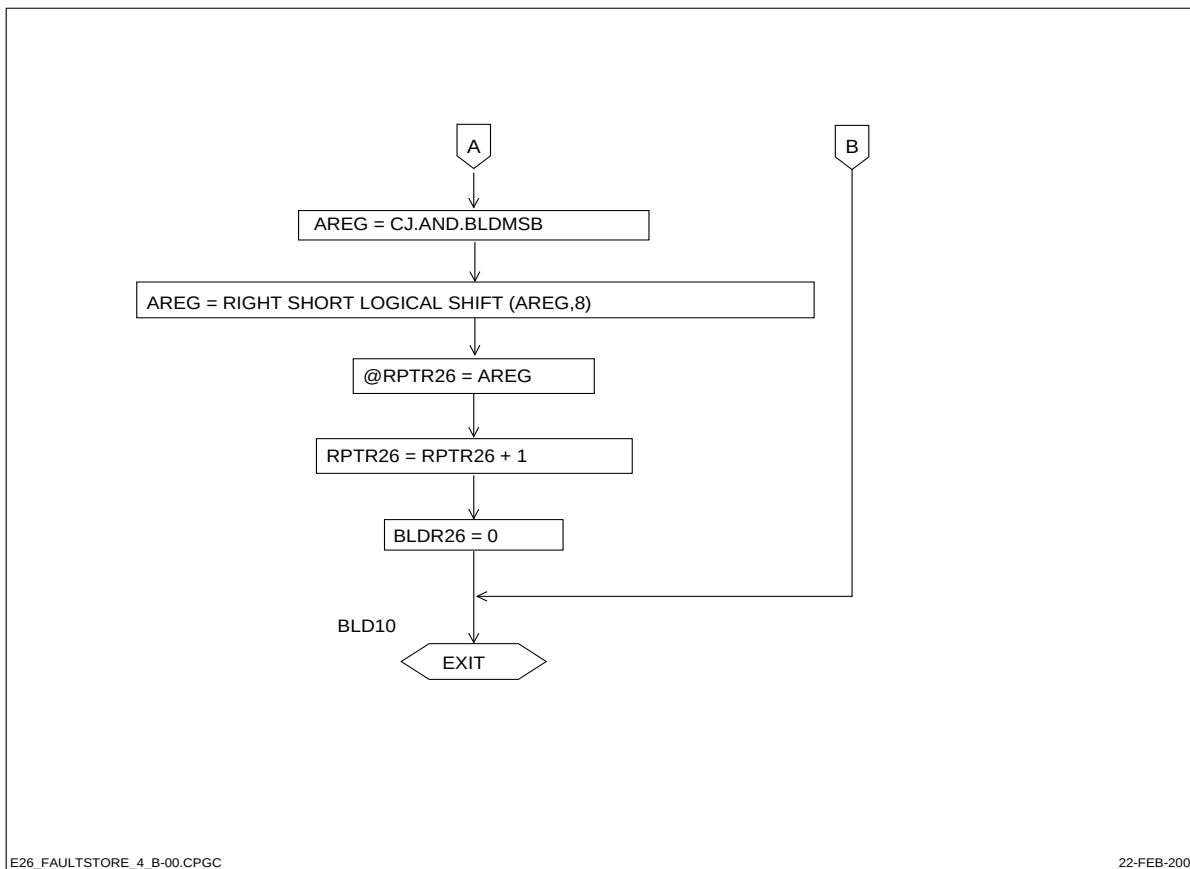
E26_FAULTSTORE_3_D-00.CPGC

22-FEB-2007

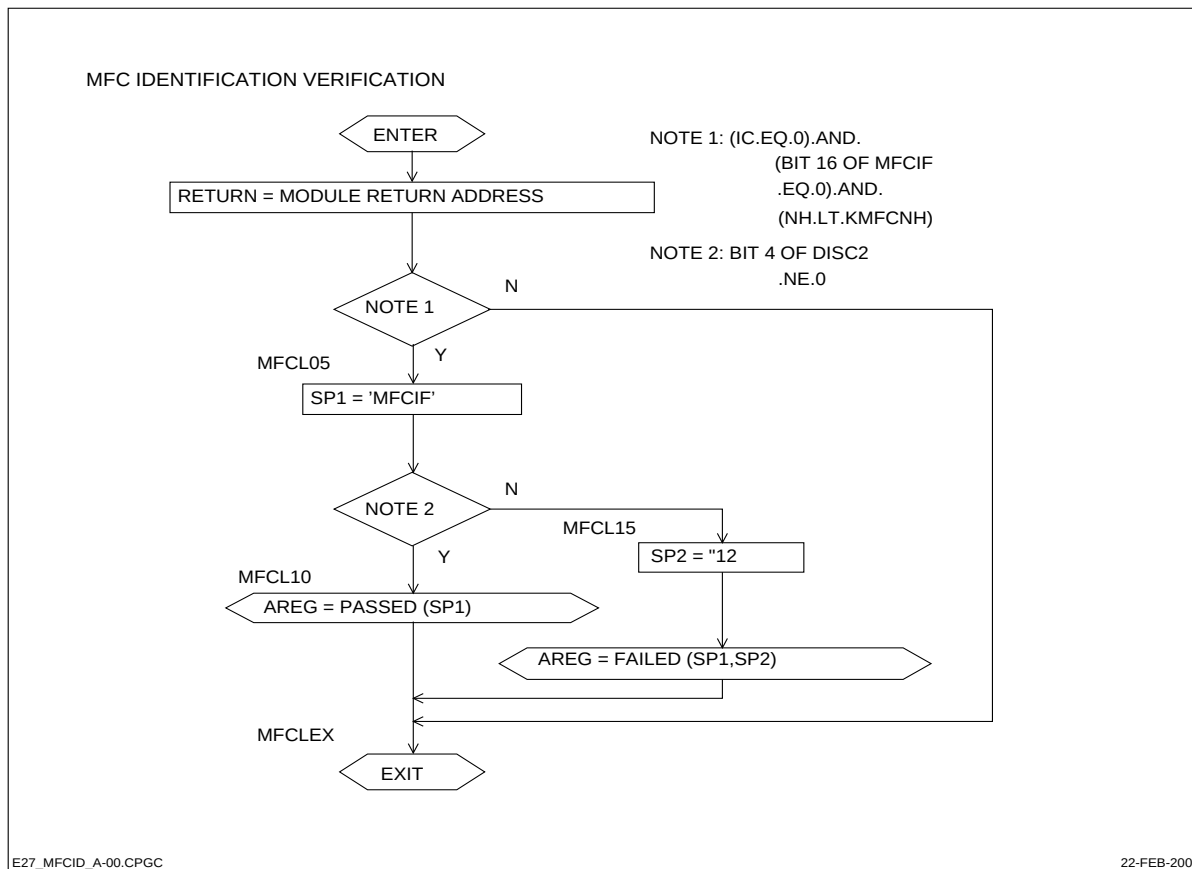
13.00 Figure 180: E26_FAULTSTORE_4_A



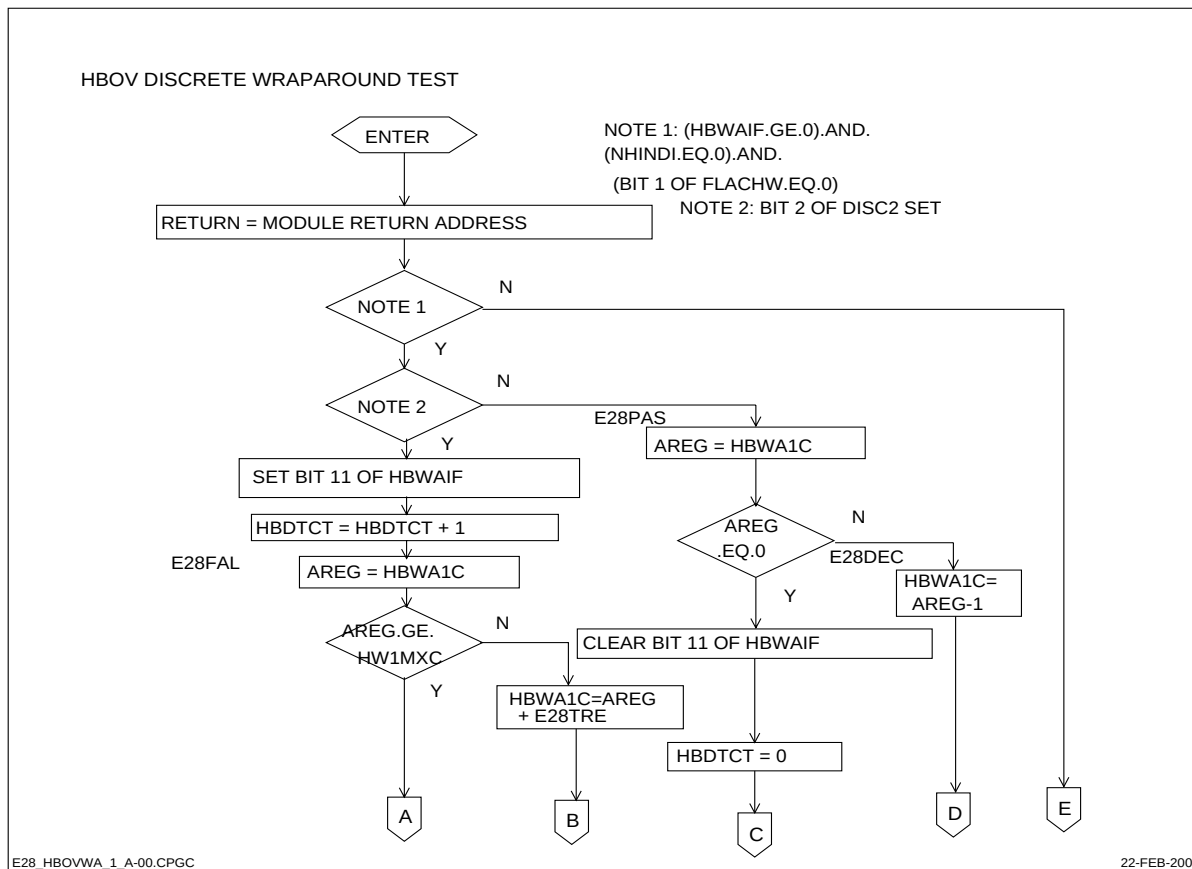
13.00 Figure 181: E26_FAULTSTORE_4_B



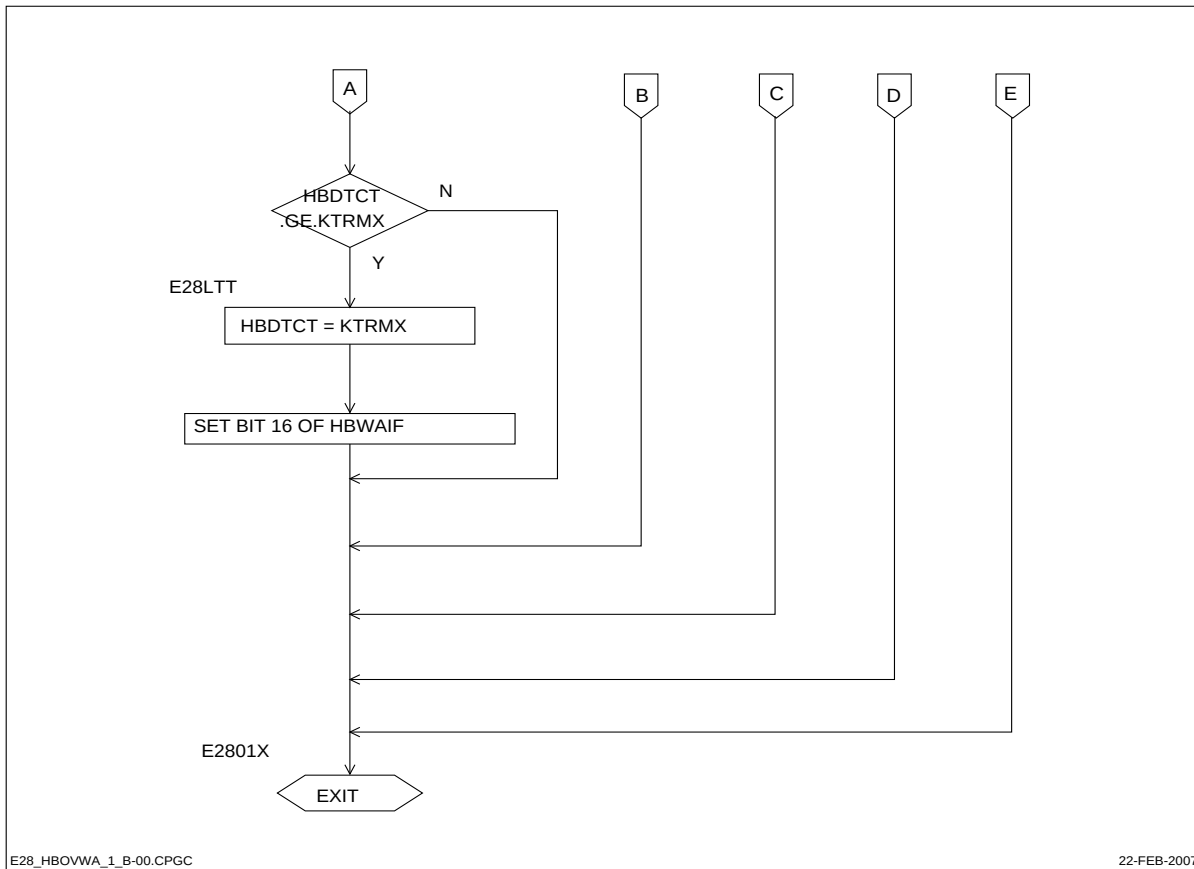
13.00 Figure 182: E27_MFCID_A



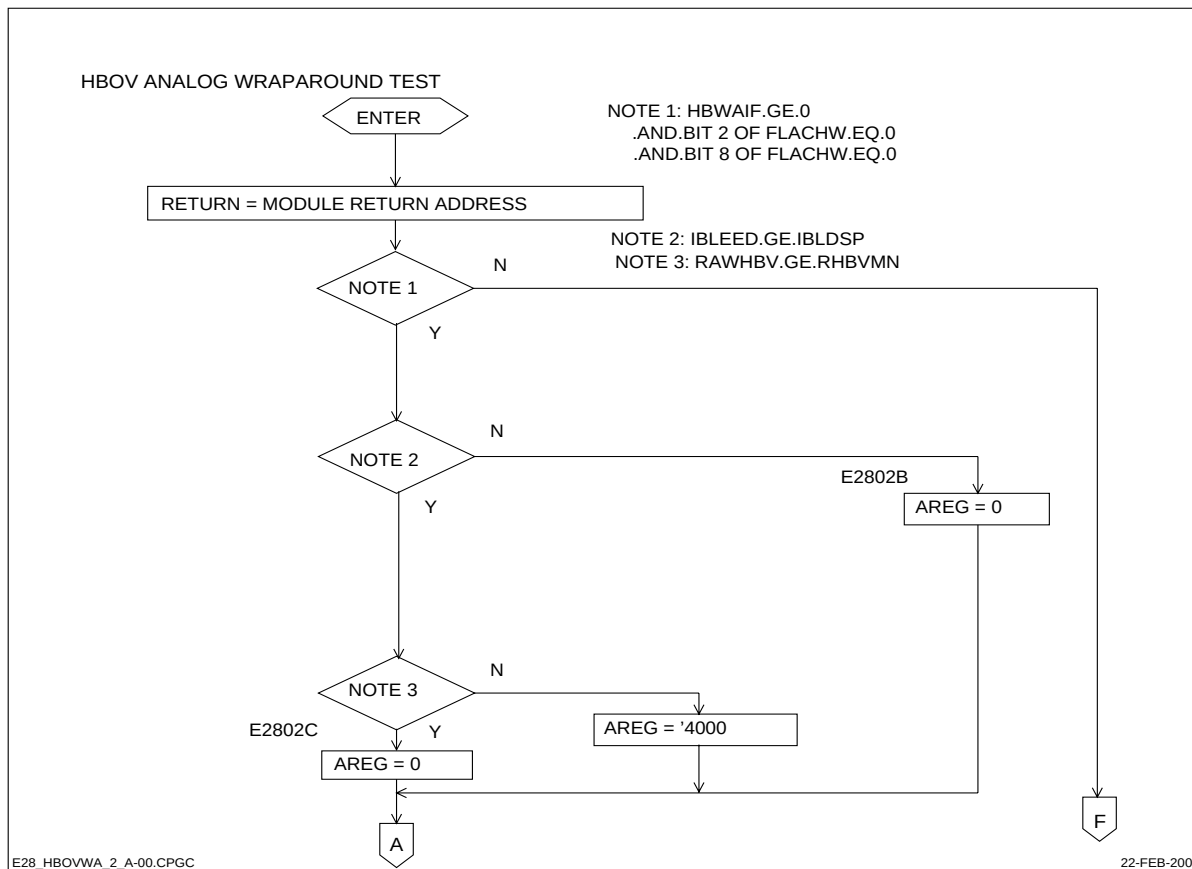
13.00 Figure 183: E28_HBOVWA_1_A



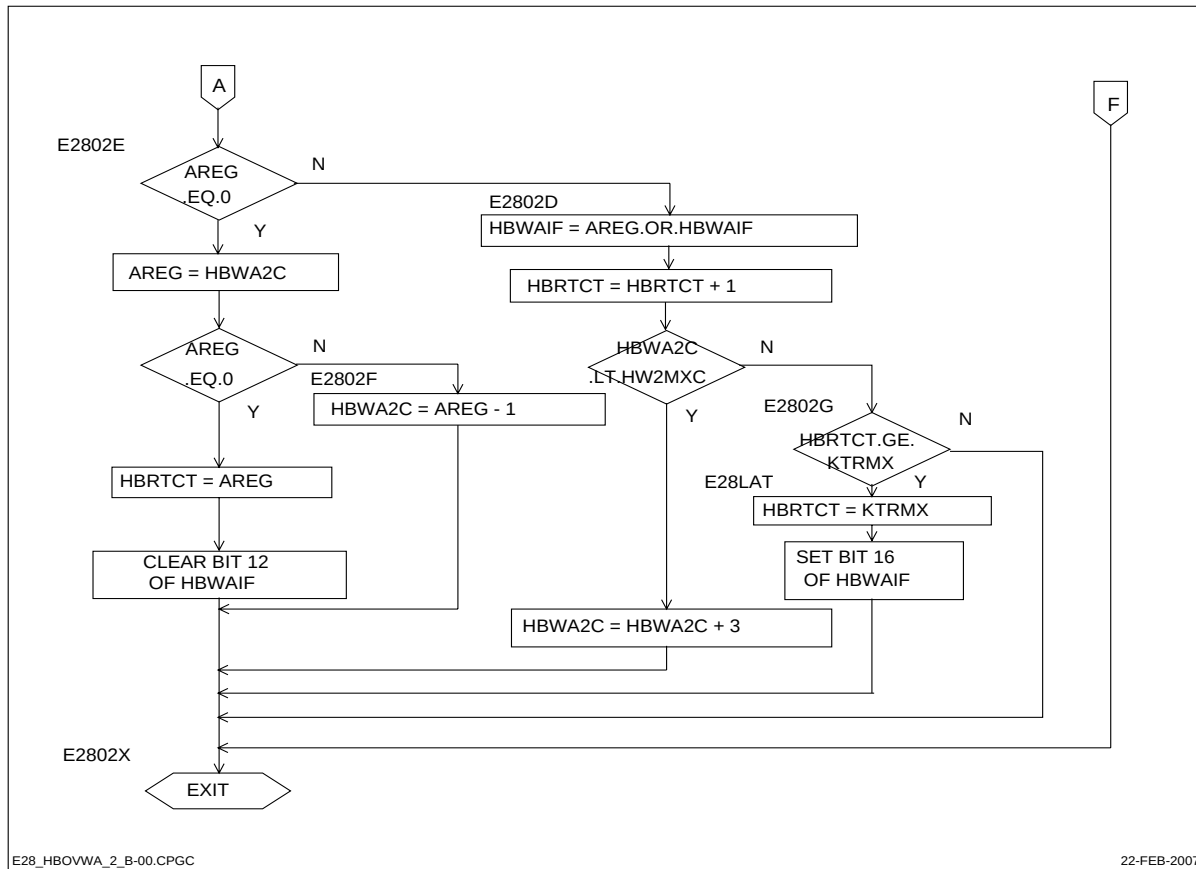
13.00 Figure 184: E28_HBOVWA_1_B



13.00 Figure 185: E28_HBOVWA_2_A



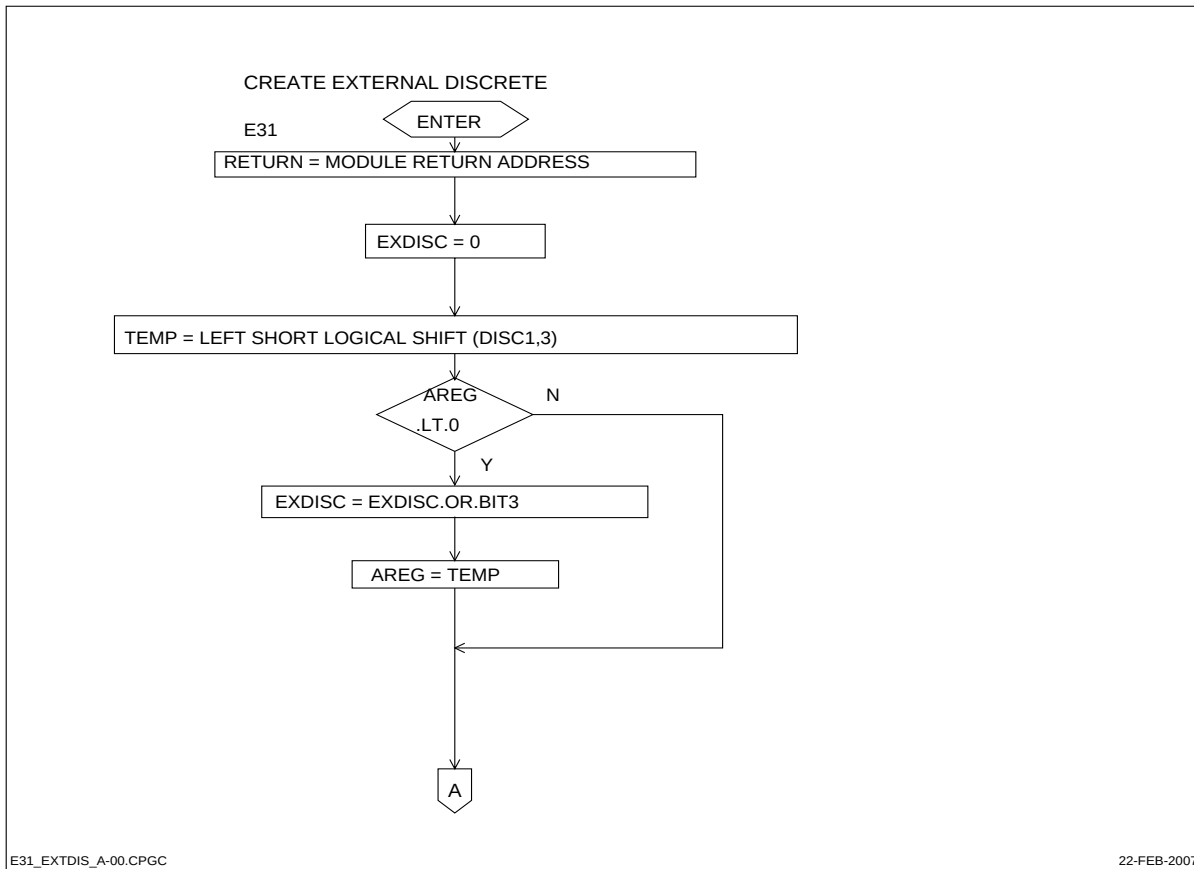
13.00 Figure 186: E28_HBOVWA_2_B



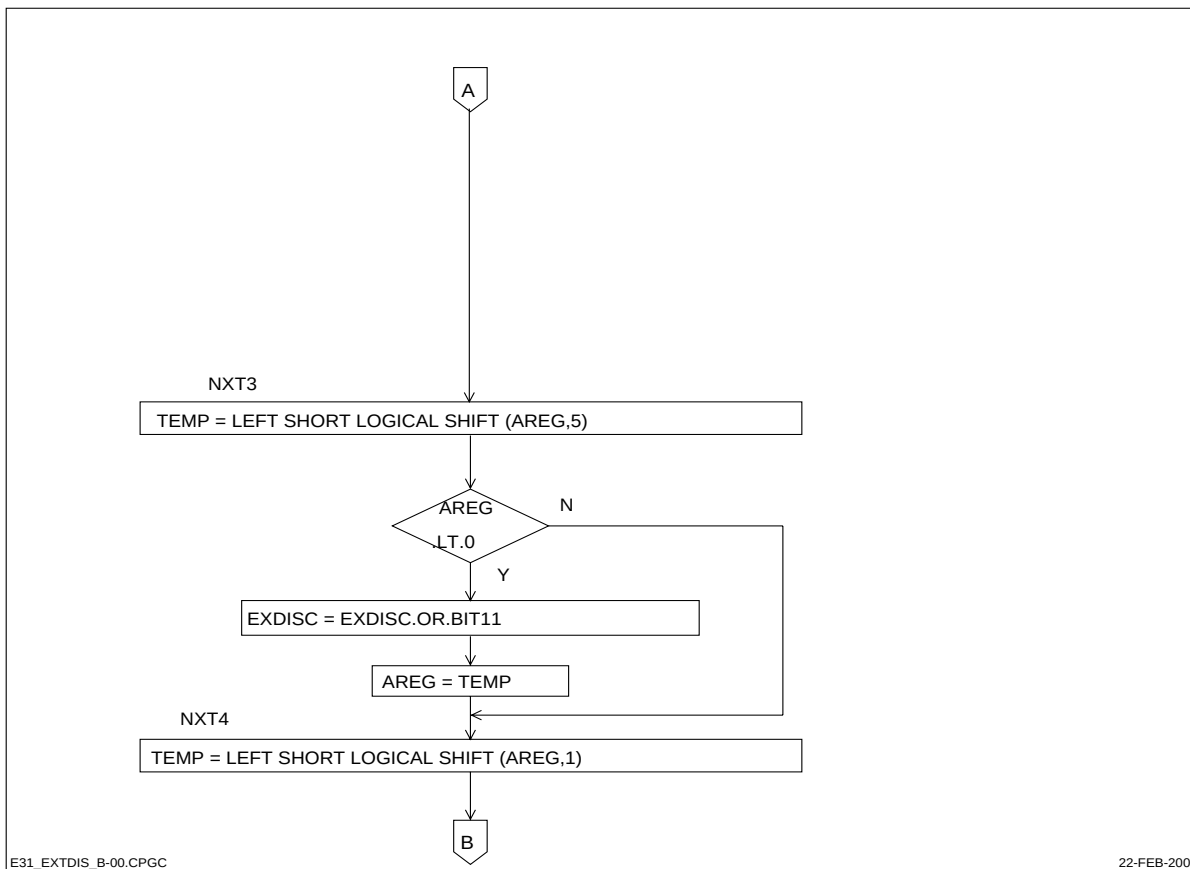
E28_HBOVWA_2_B-00.CPGC

22-FEB-2007

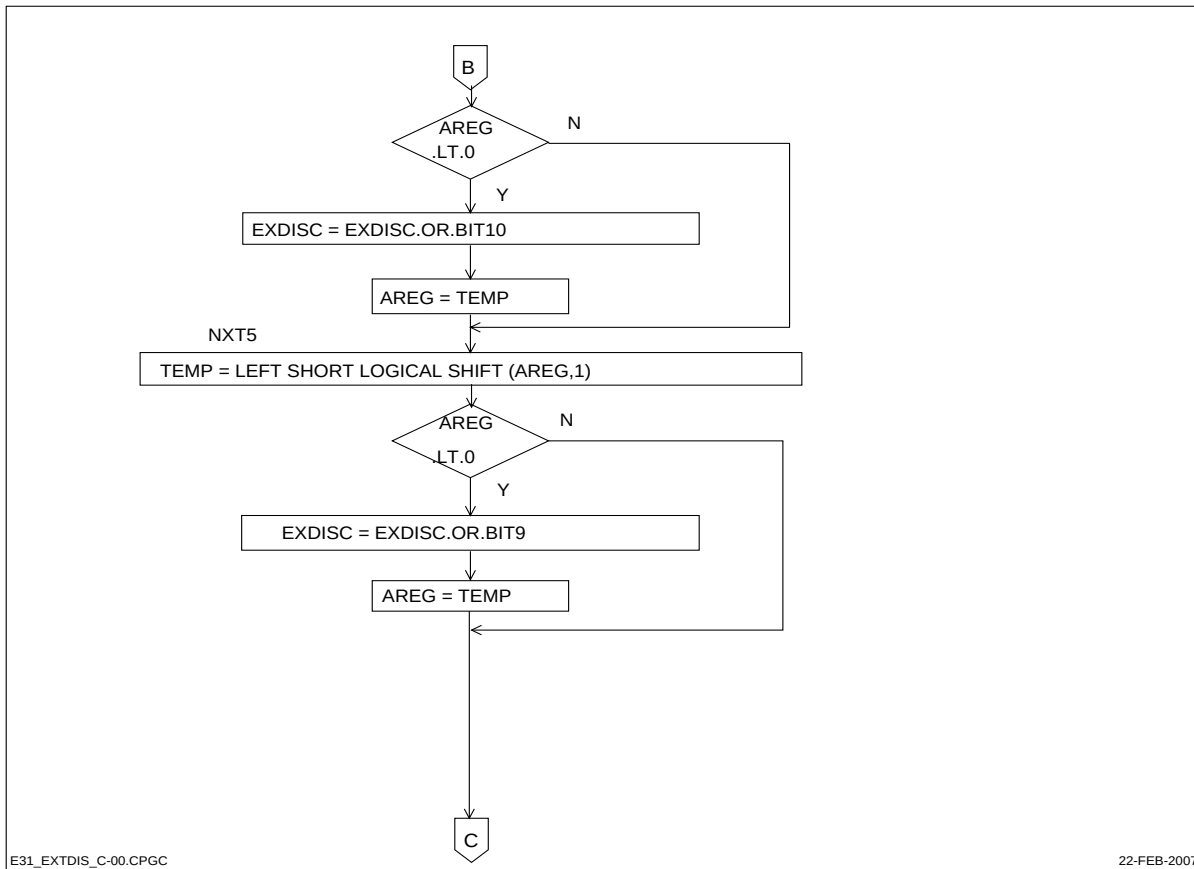
13.00 Figure 187: E31_EXTDIS_A



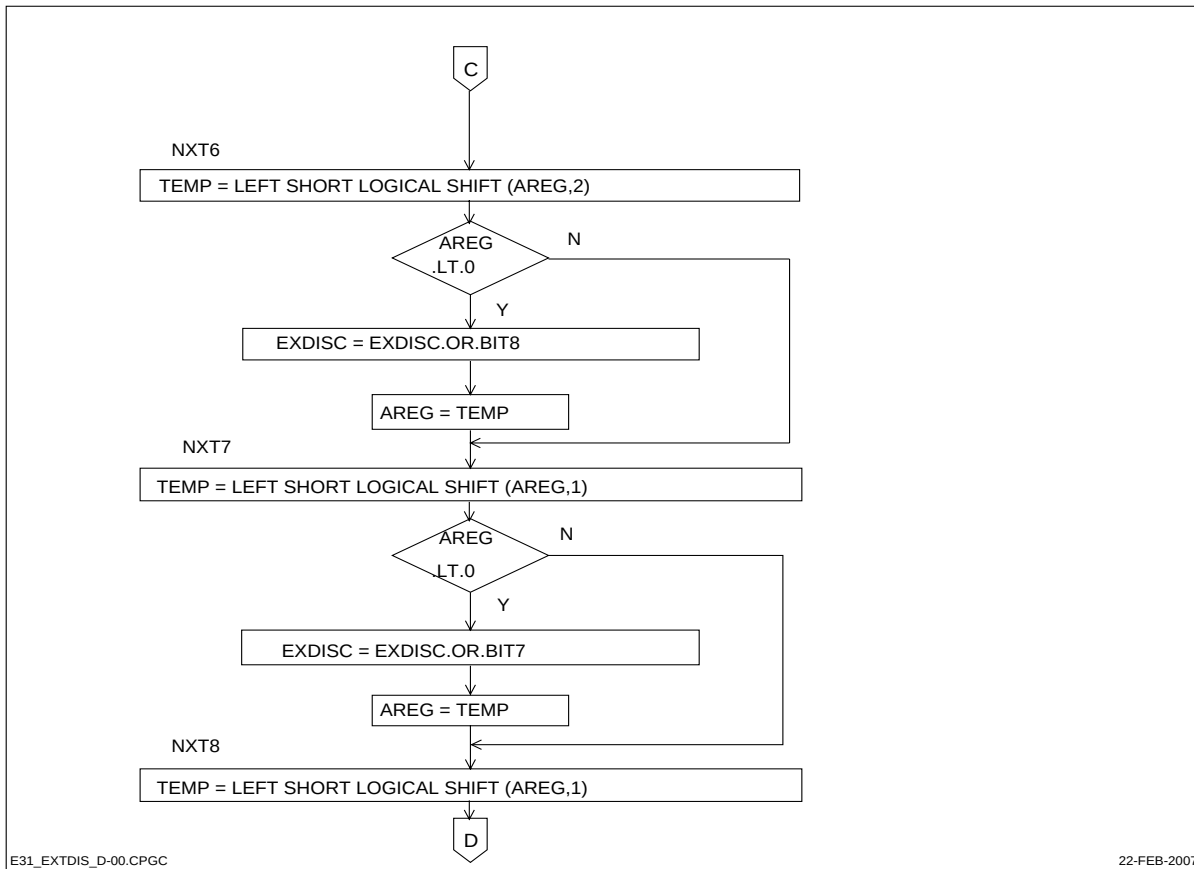
13.00 Figure 188: E31_EXTDIS_B



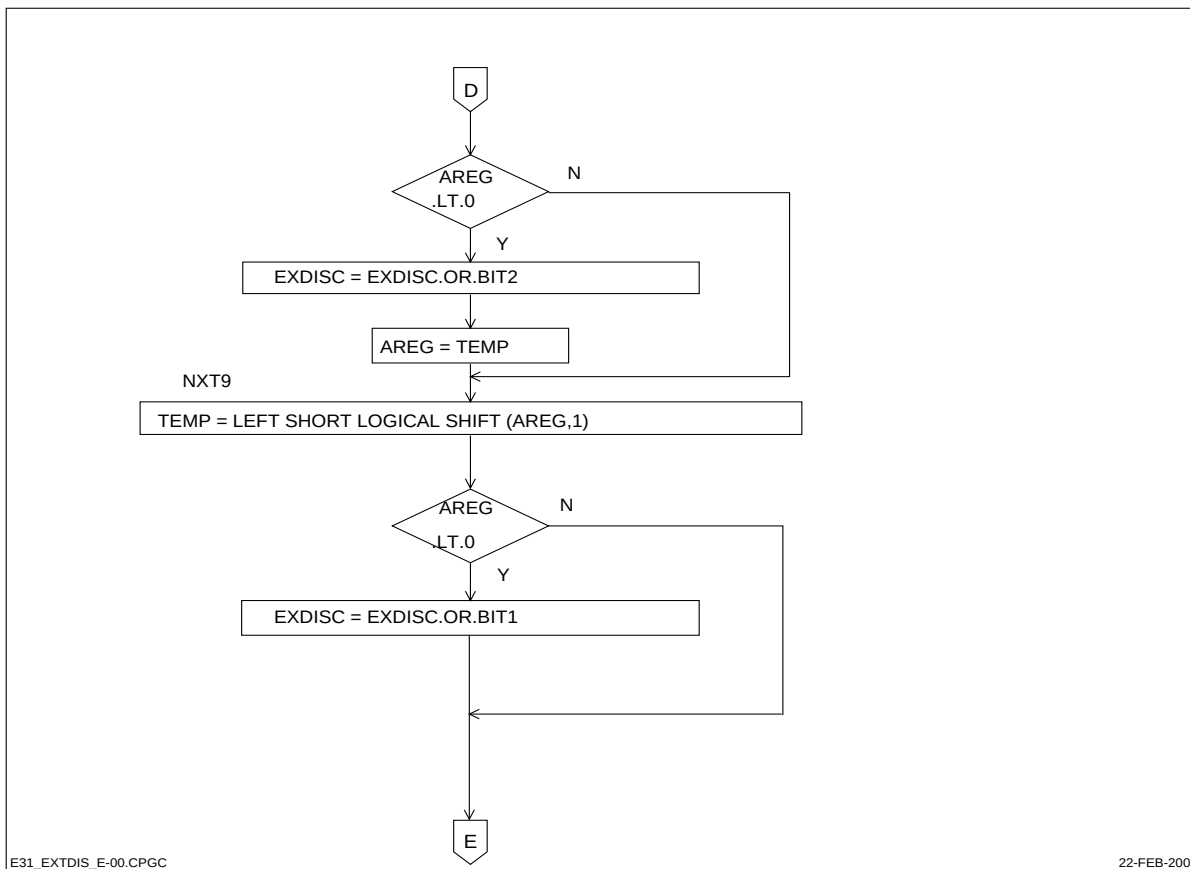
13.00 Figure 189: E31_EXTDIS_C



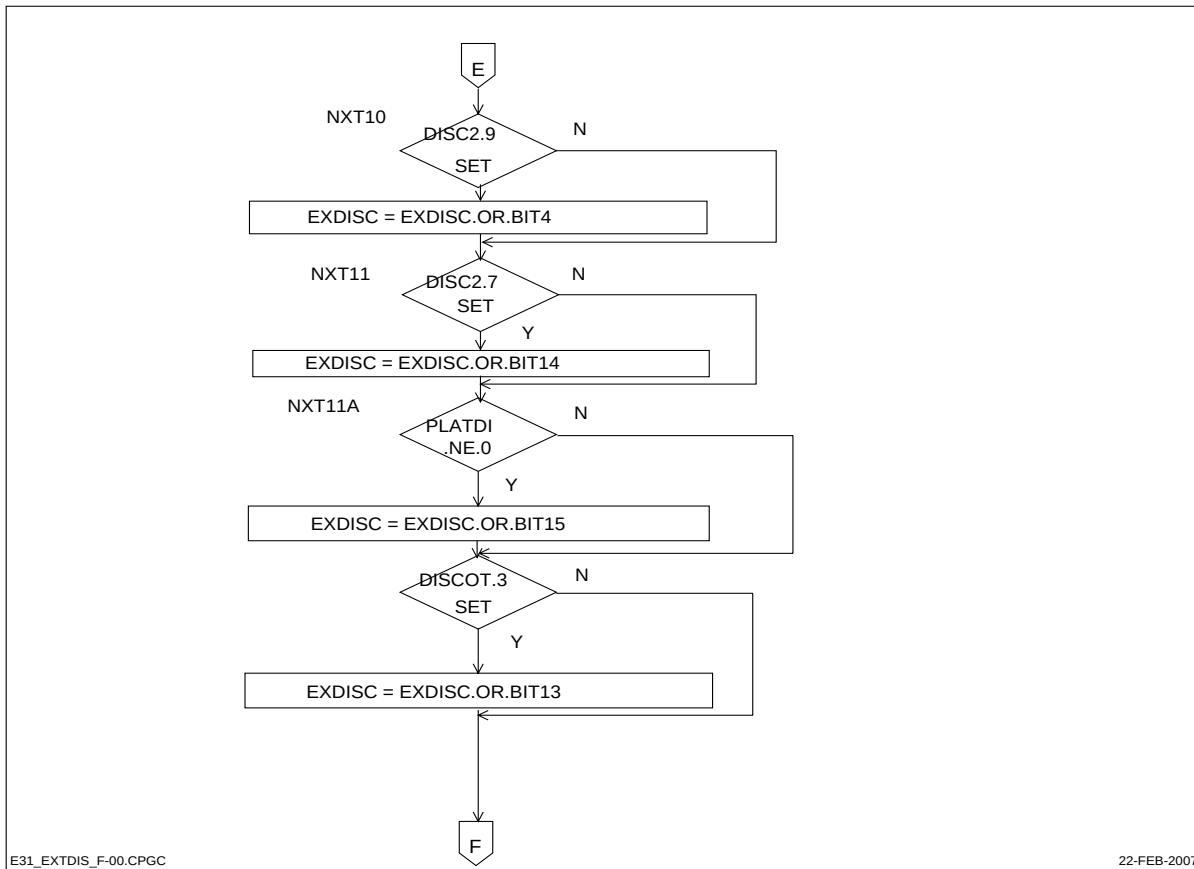
13.00 Figure 190: E31_EXTDIS_D



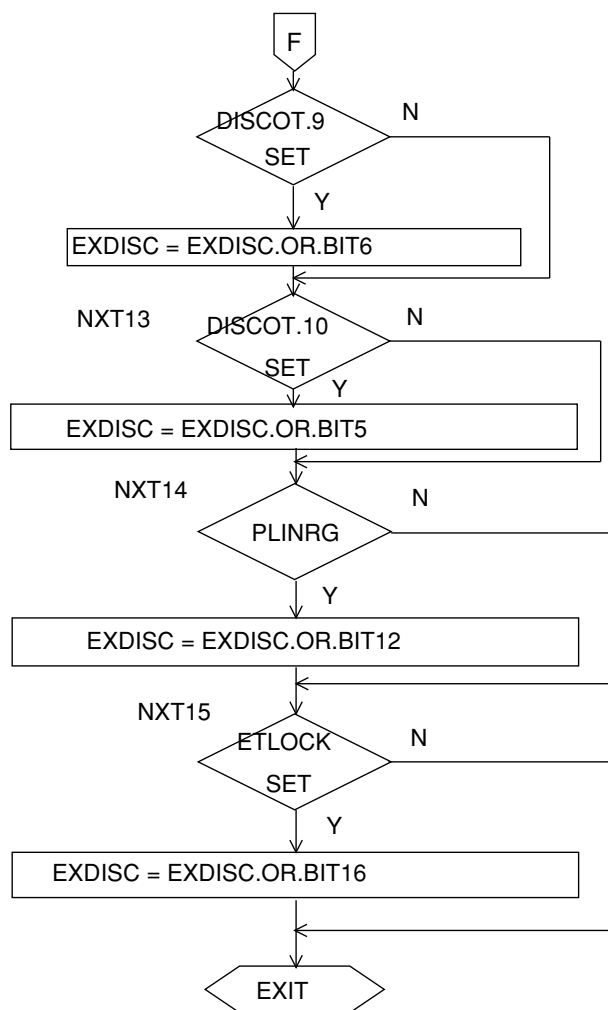
13.00 Figure 191: E31_EXTDIS_E



13.00 Figure 192: E31_EXTDIS_F



13.00 Figure 193: E31 _EXTDIS _G

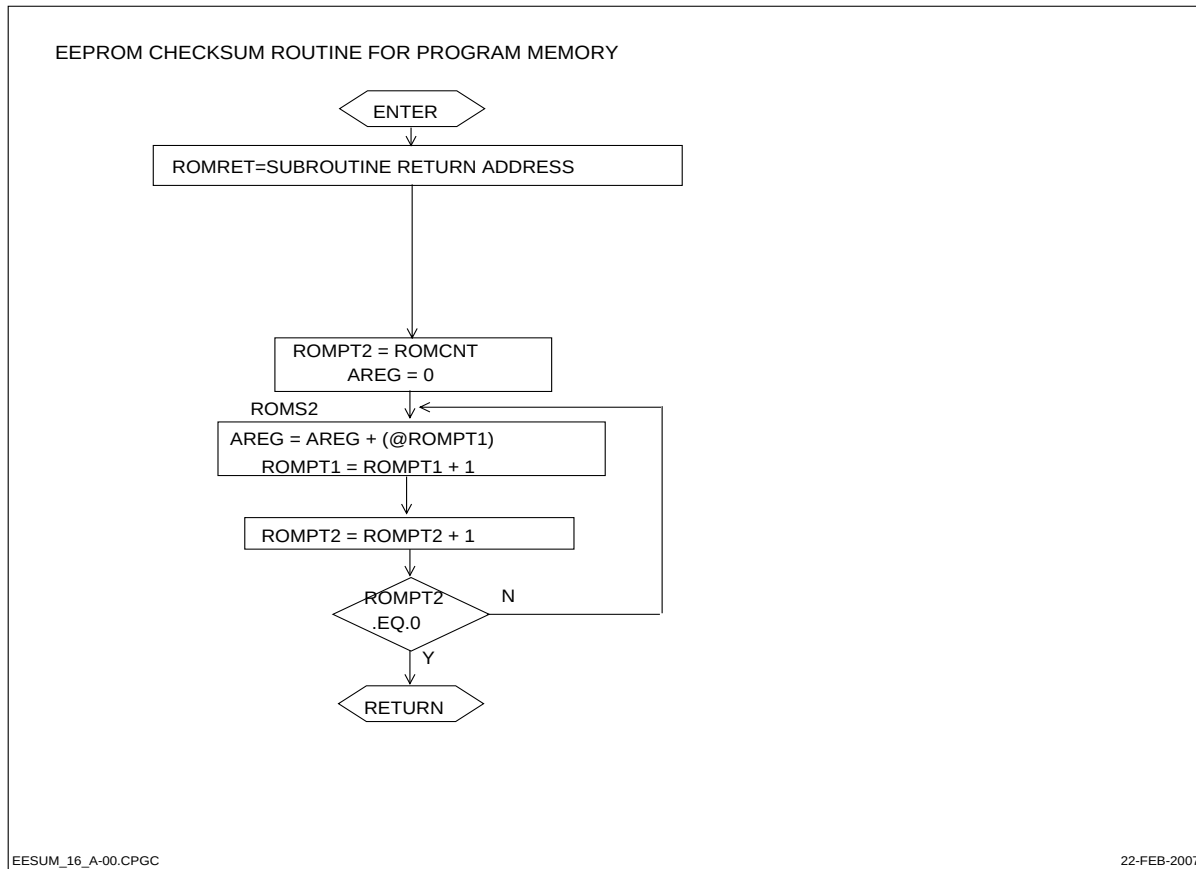


E31_EXTDIS_G.CPGC

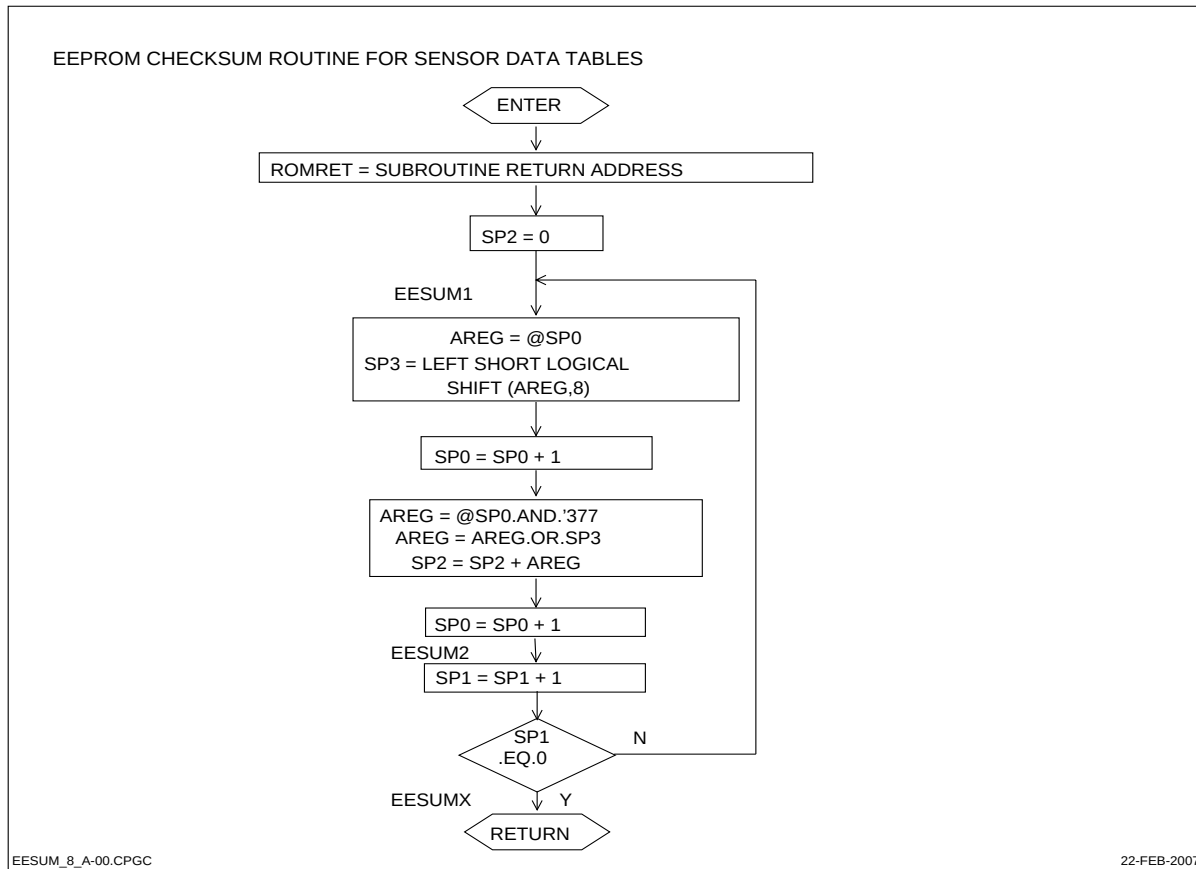
EXTE31

21-MAY-2024

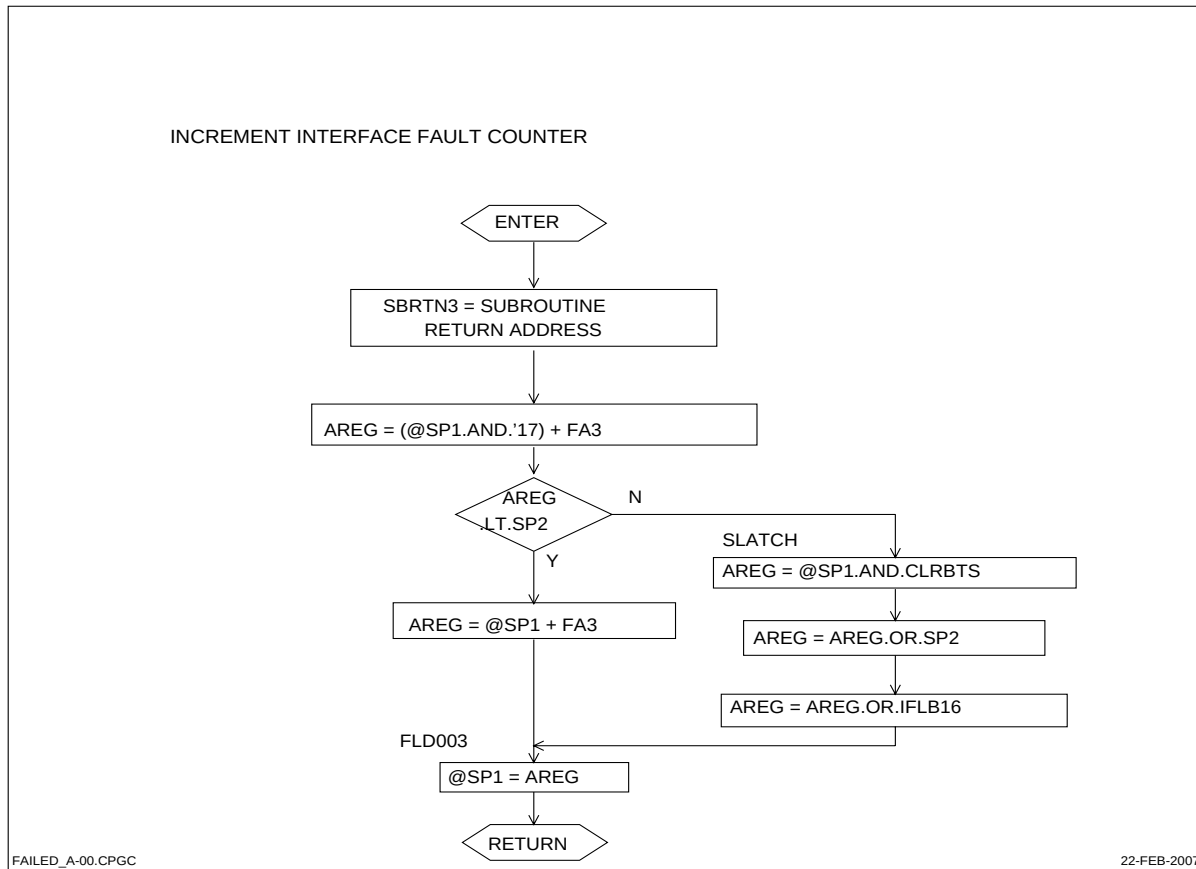
13.00 Figure 194: EESUM_16_A



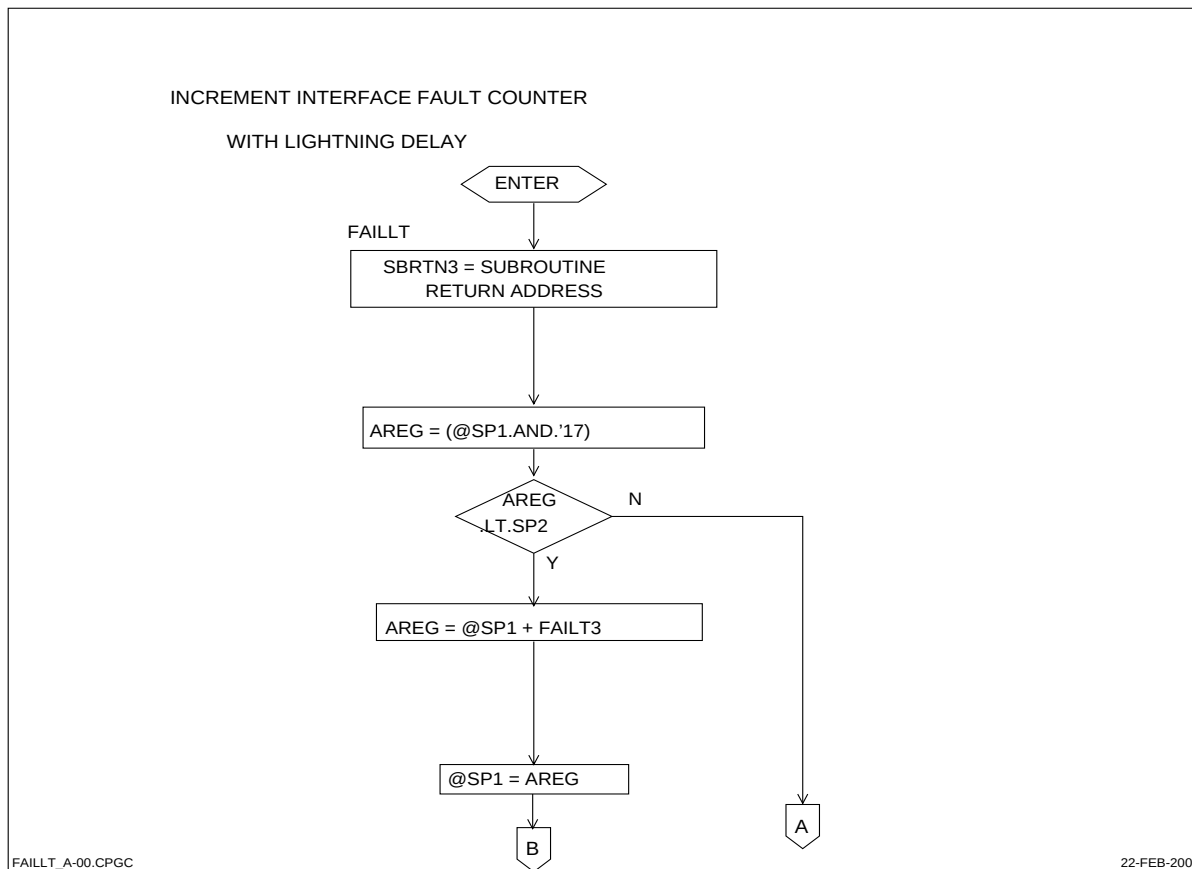
13.00 Figure 195: EESUM_8_A



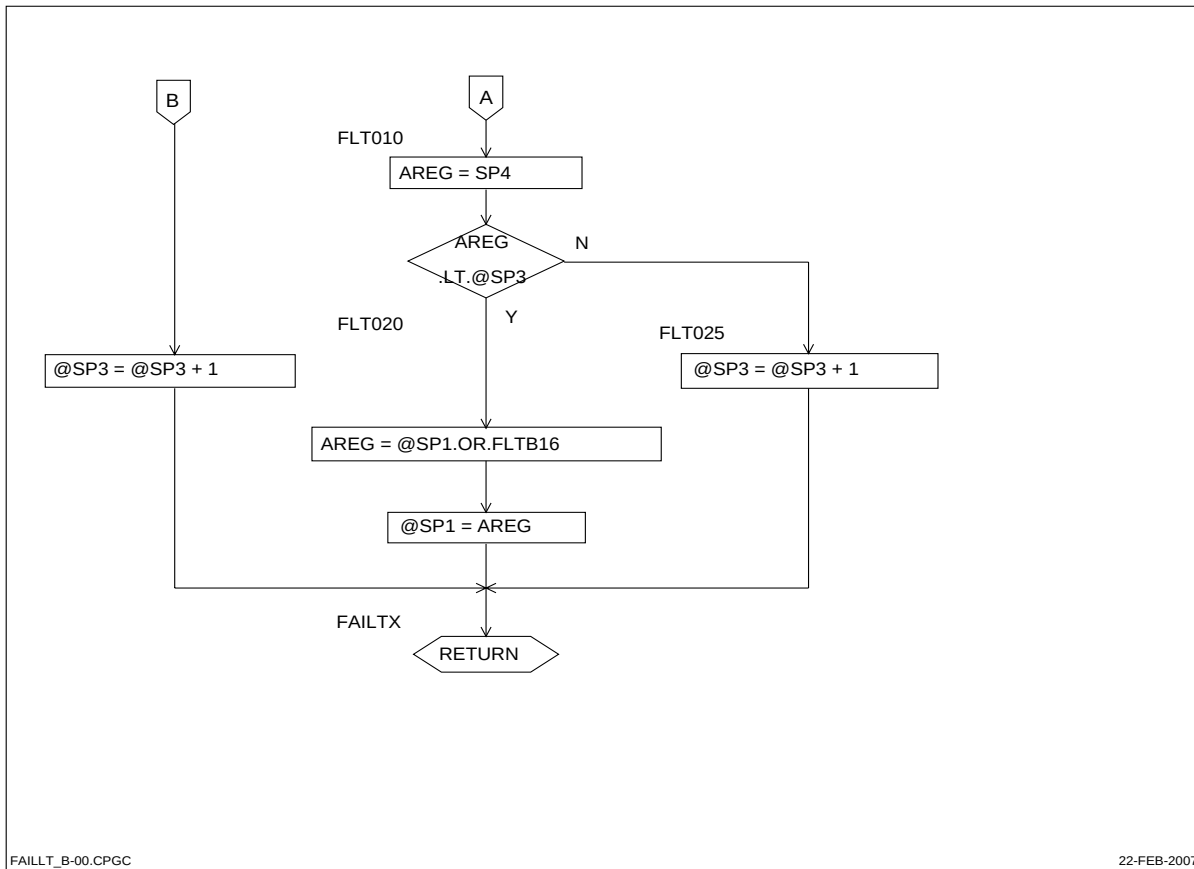
13.00 Figure 196: FAILED_A



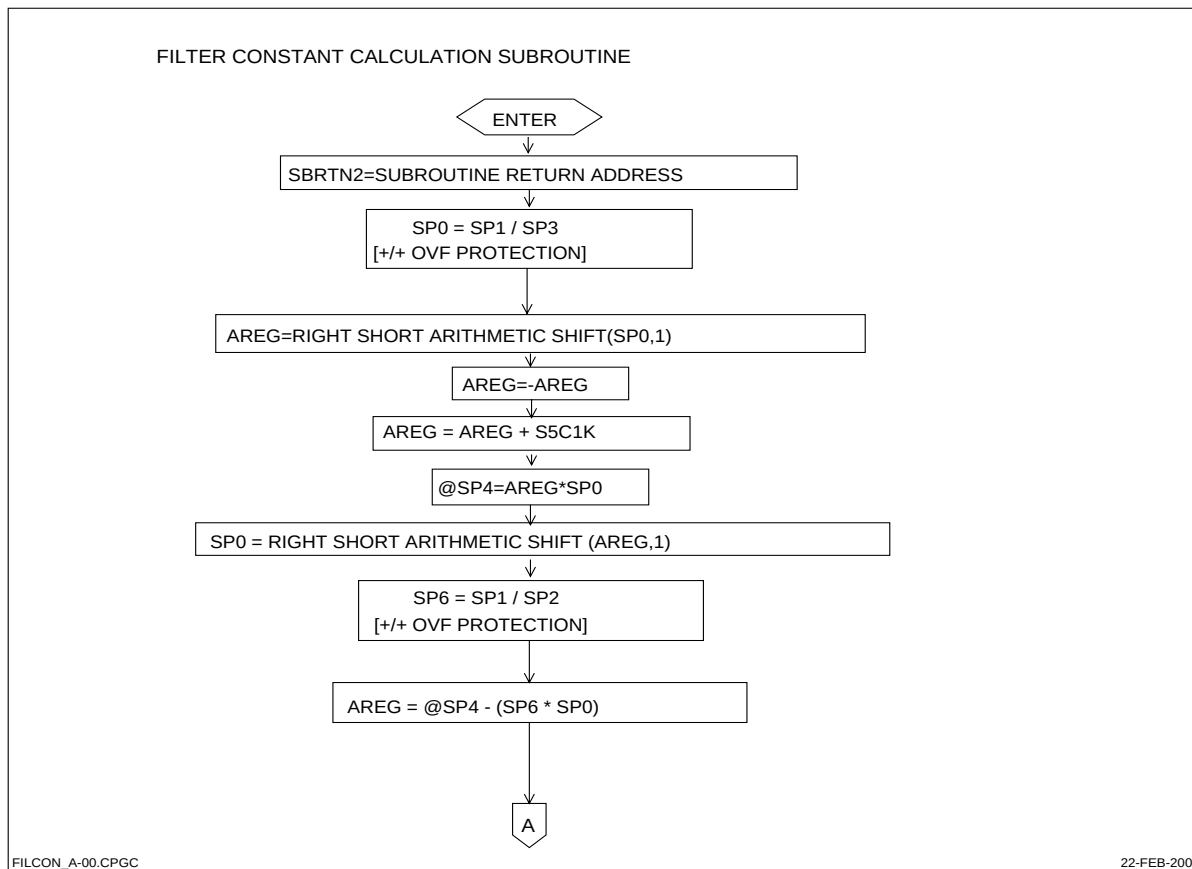
13.00 Figure 197: FAILLT _A



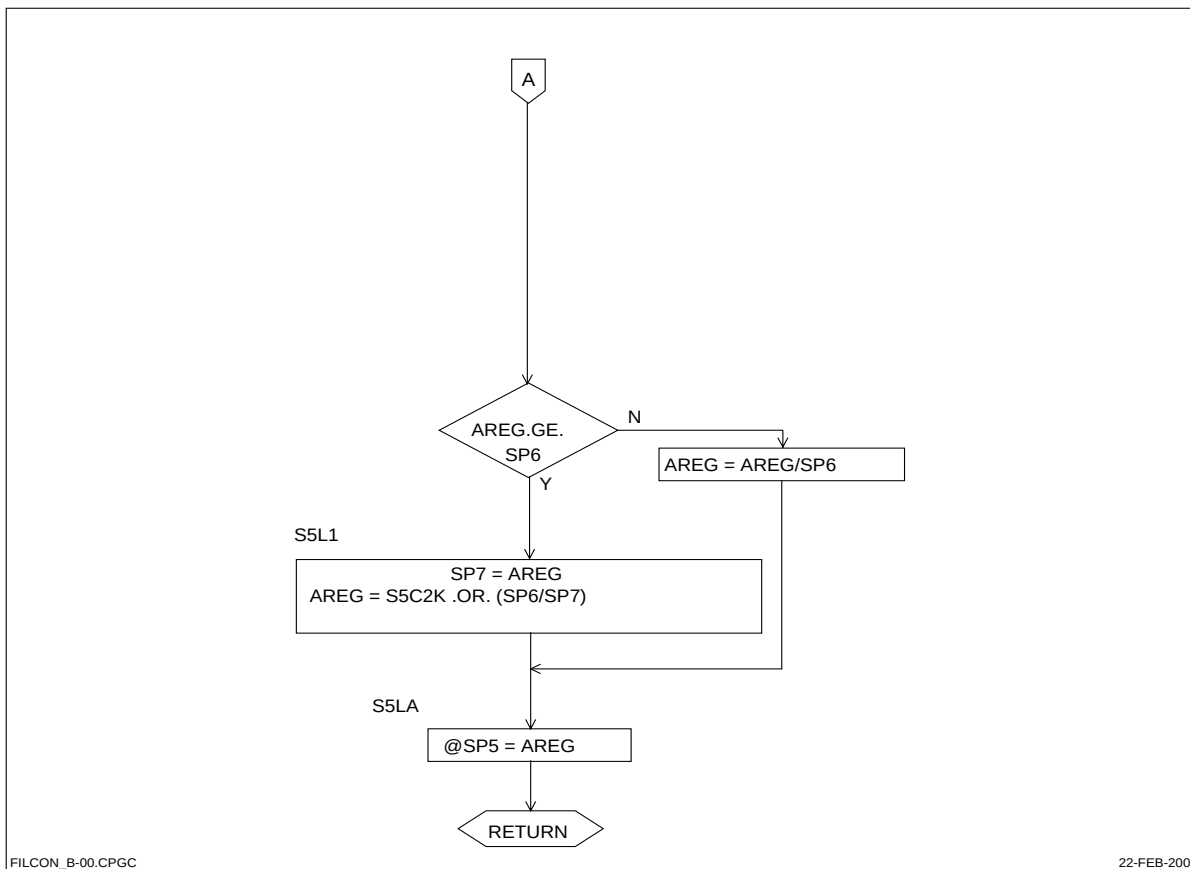
13.00 Figure 198: FAILLT _B



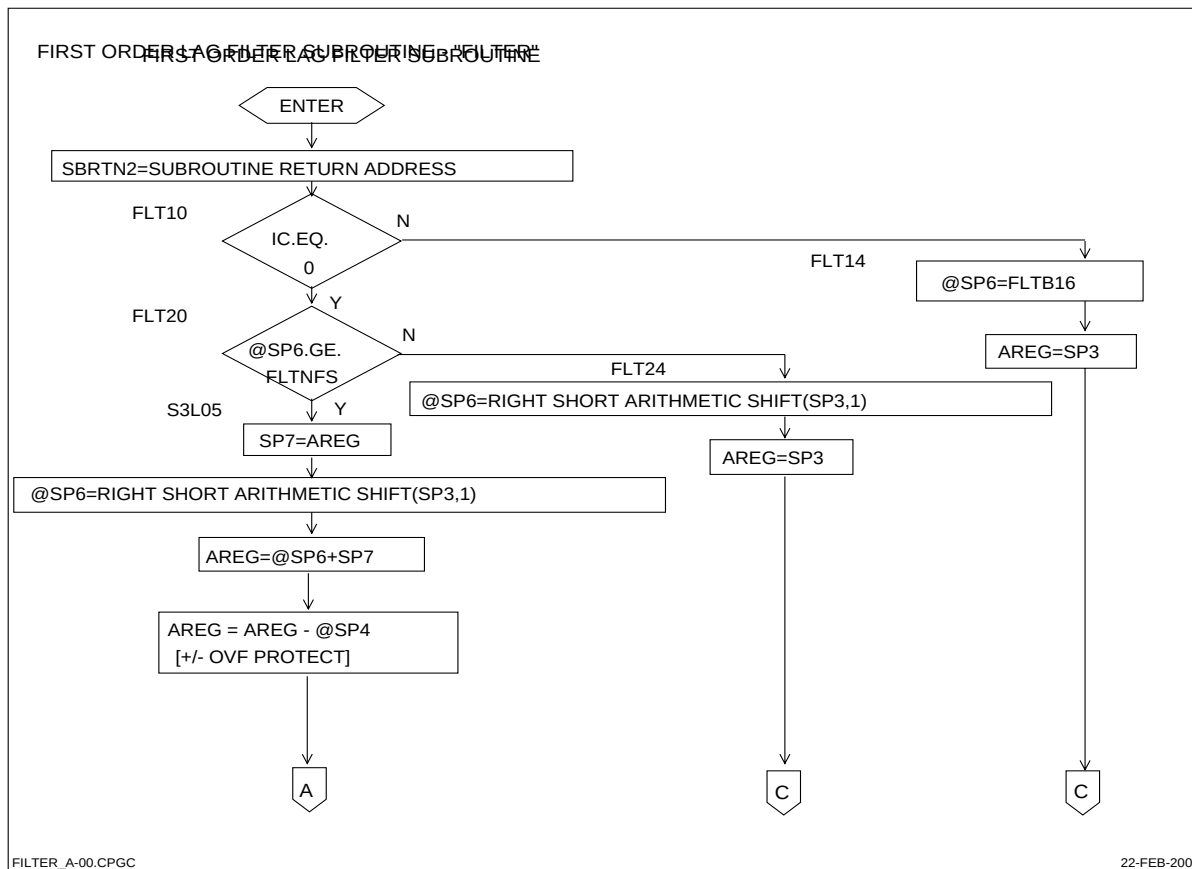
13.00 Figure 199: FILCON_A



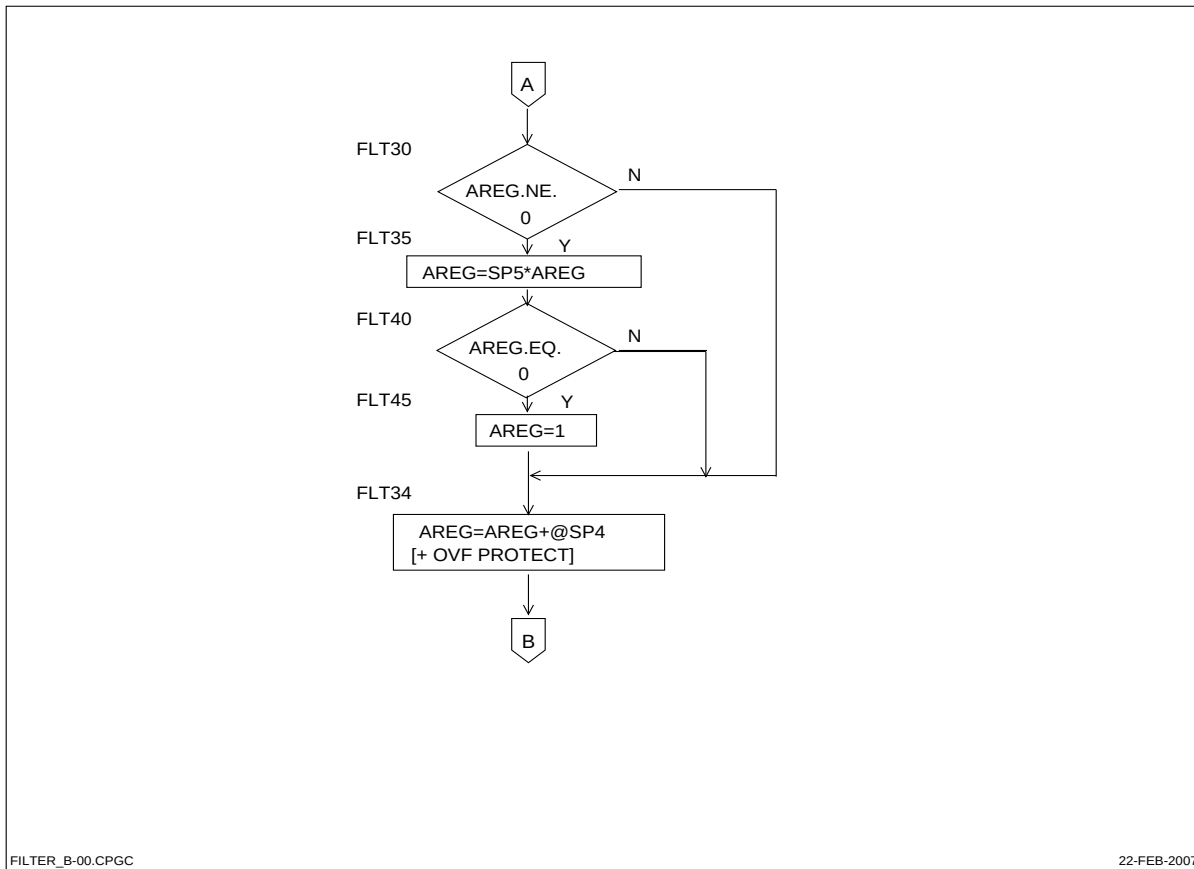
13.00 Figure 200: FILCON _B



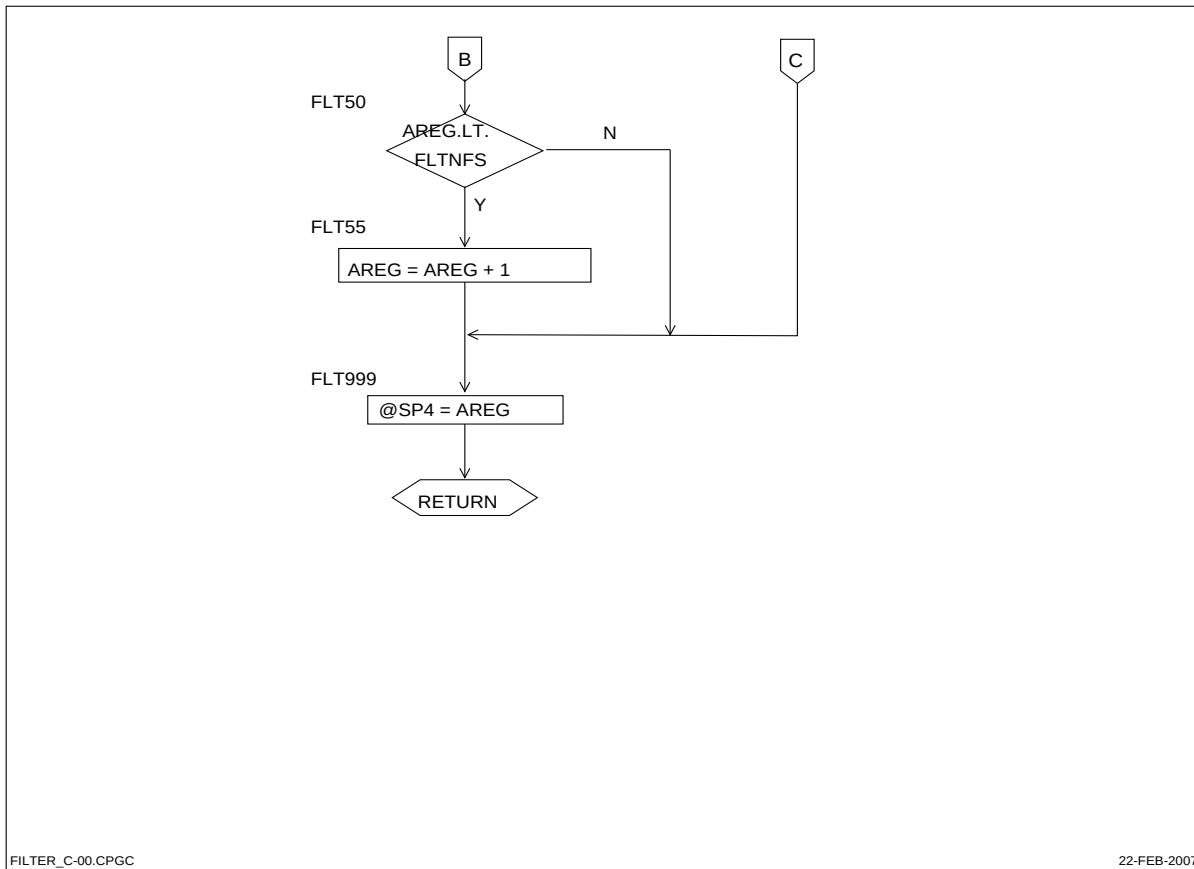
13.00 Figure 201: FILTER_A



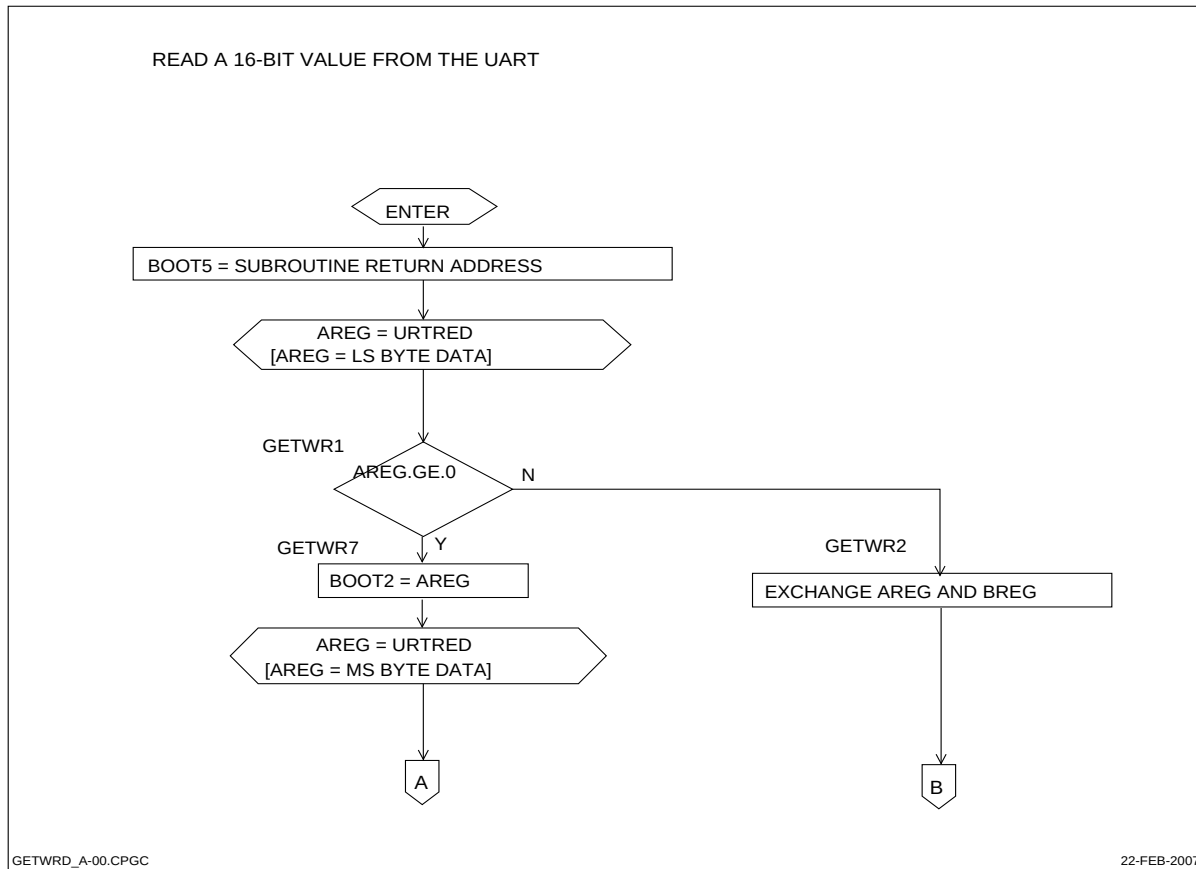
13.00 Figure 202: FILTER _B



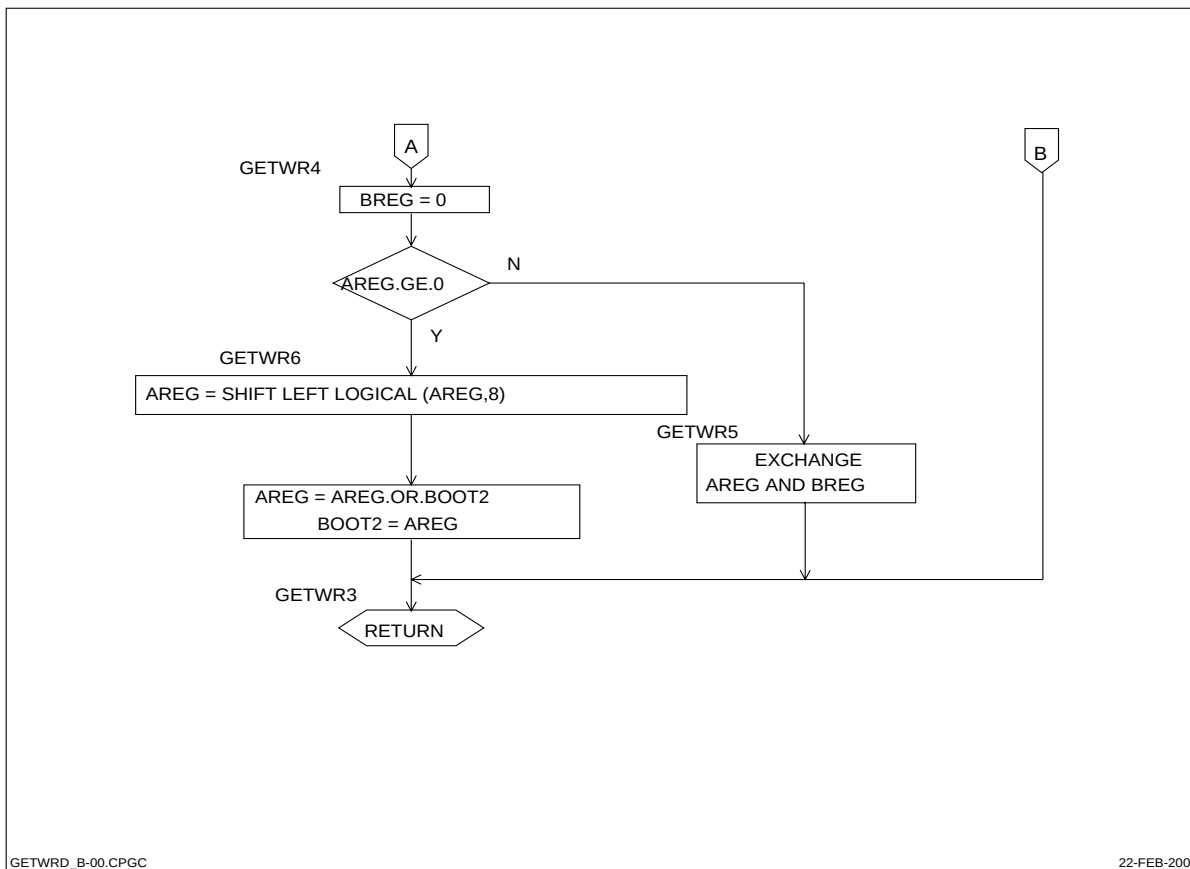
13.00 Figure 203: FILTER _C



13.00 Figure 204: GETWRD_A



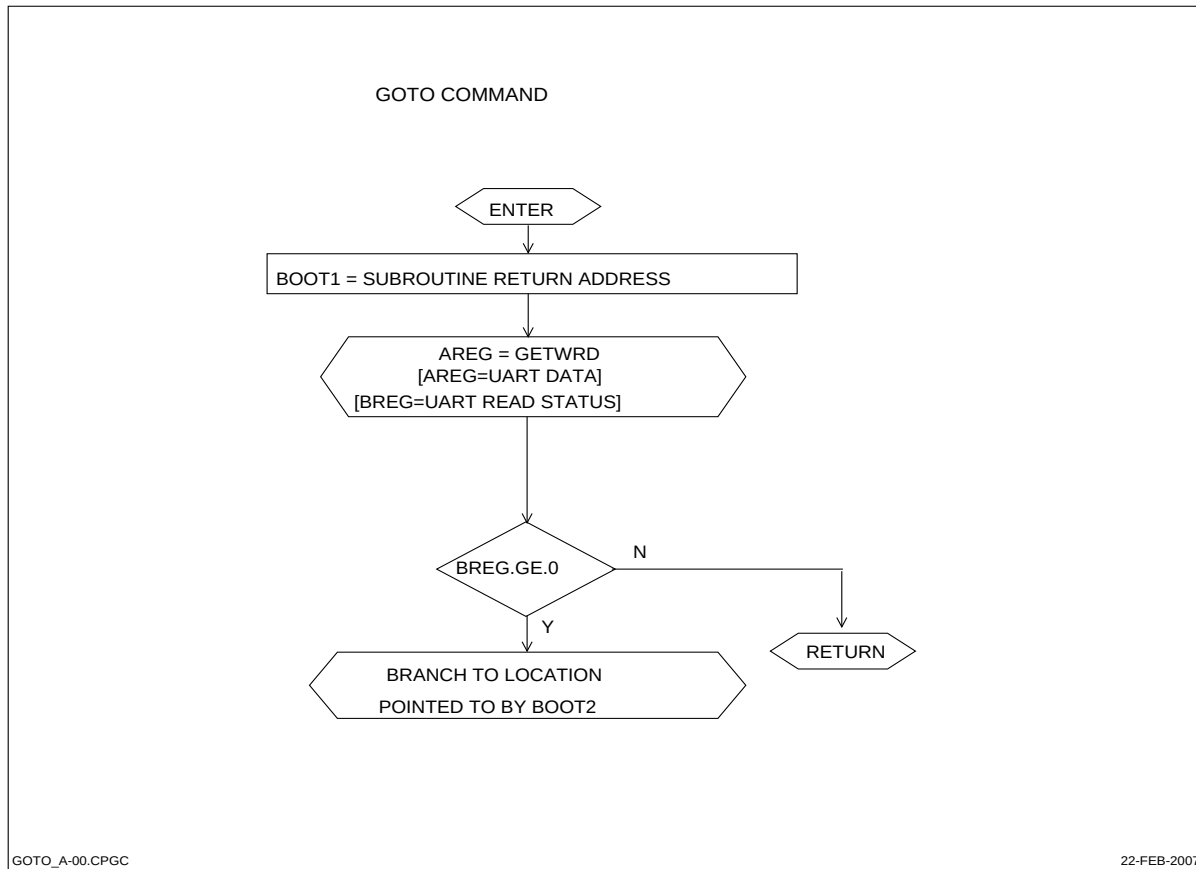
13.00 Figure 205: GETWRD _B



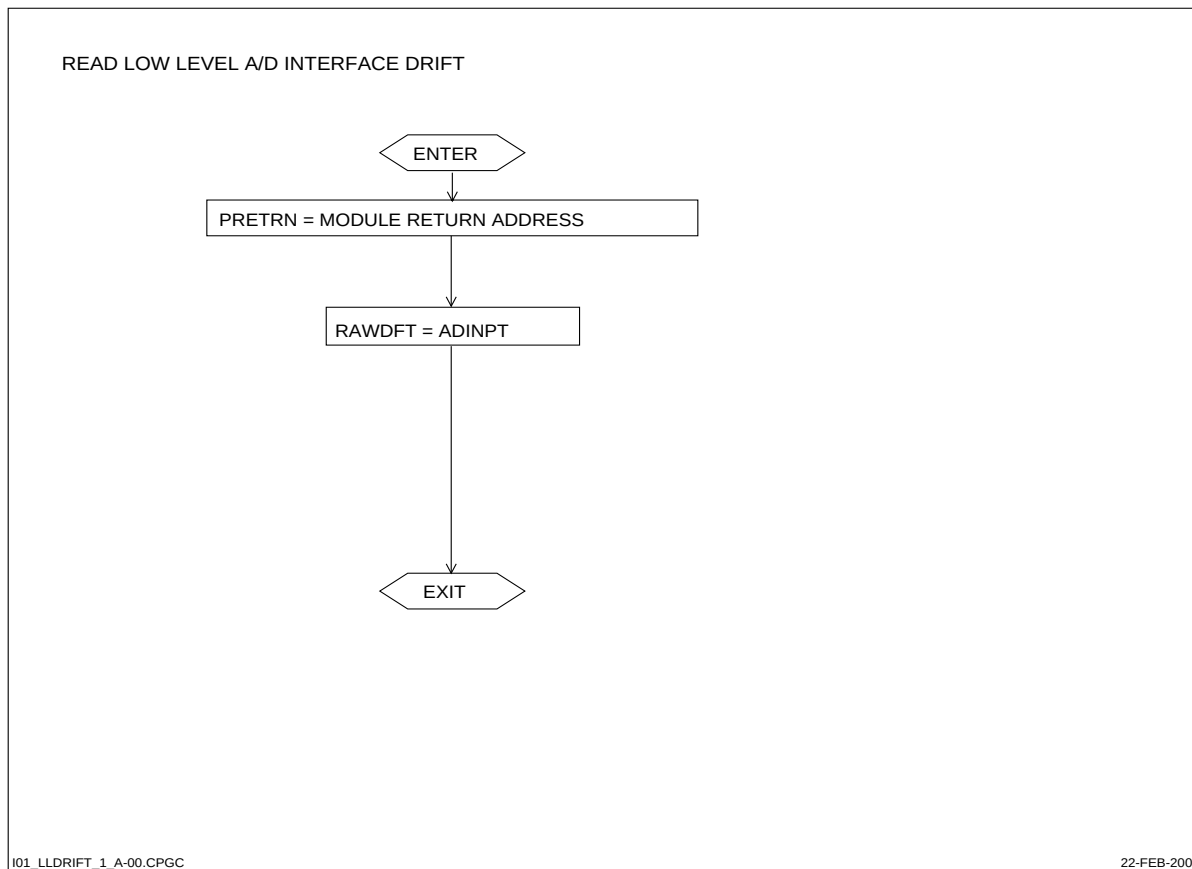
GETWRD_B-00.CPGC

22-FEB-2007

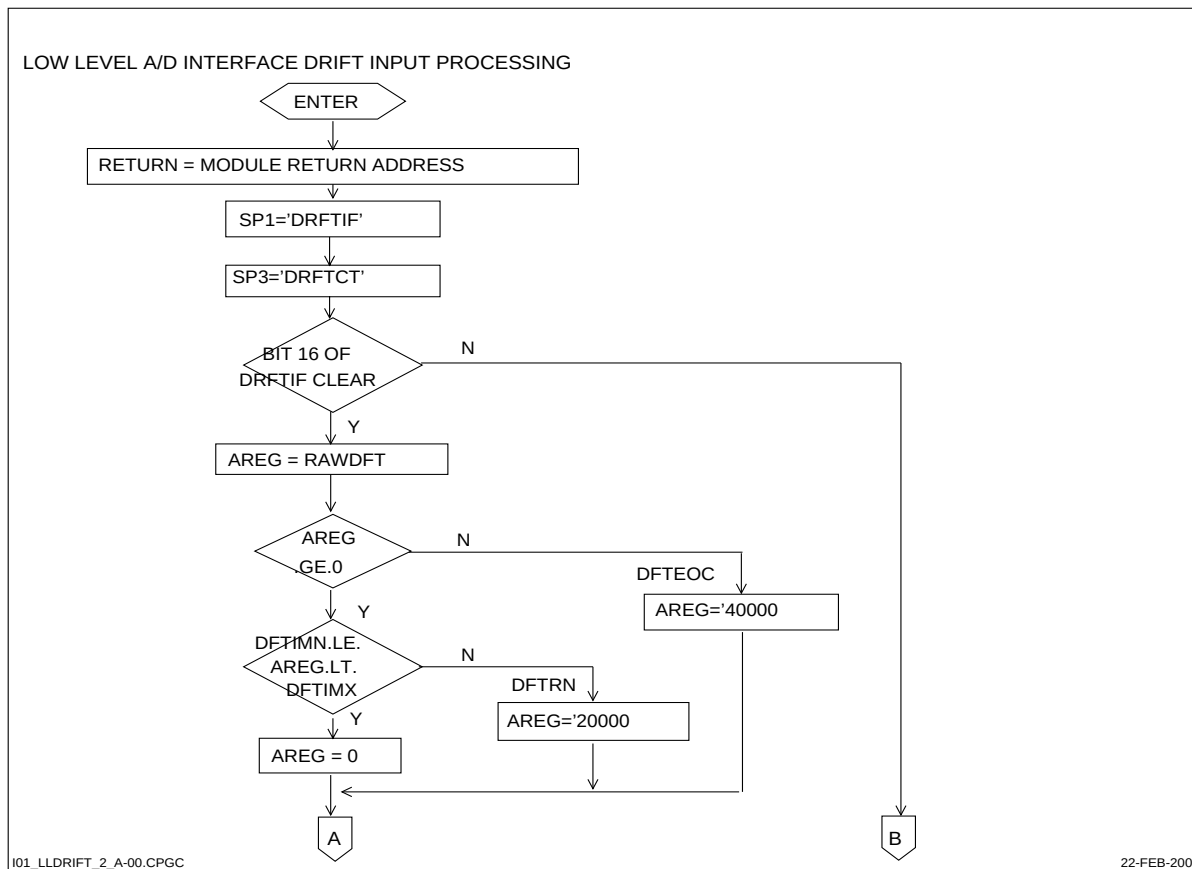
13.00 Figure 206: GOTO _A



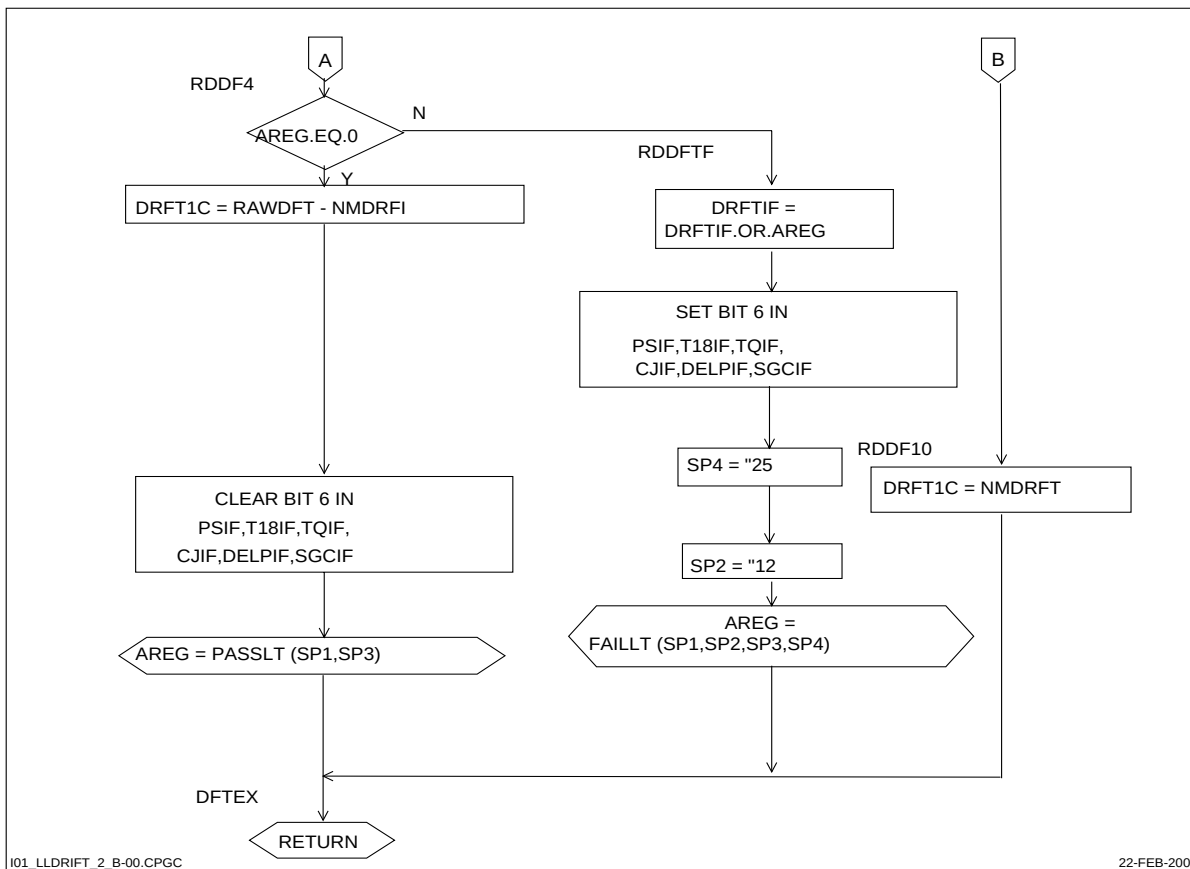
13.00 Figure 207: I01 _LLDRIFT _1 _A



13.00 Figure 208: I01_LLDRIFT_2_A



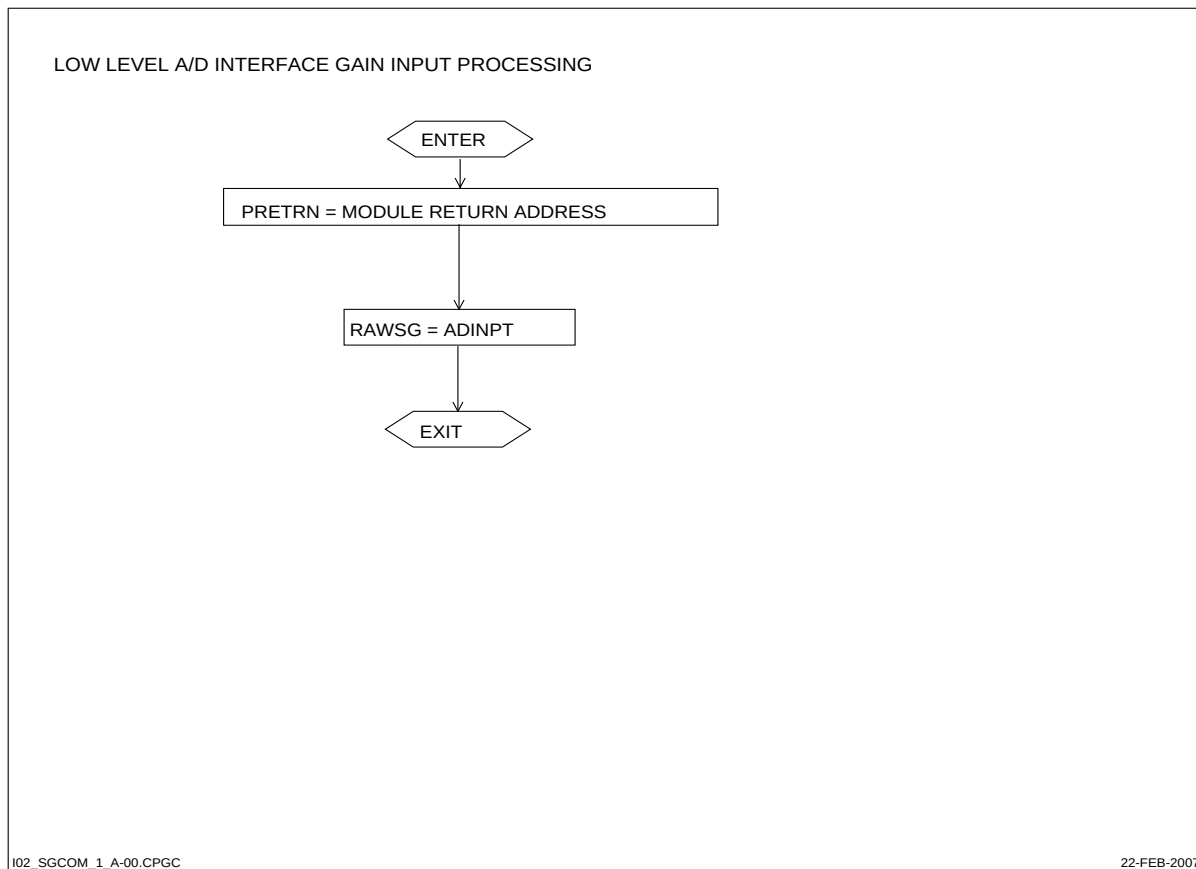
13.00 Figure 209: I01_LLDRIFT_2_B



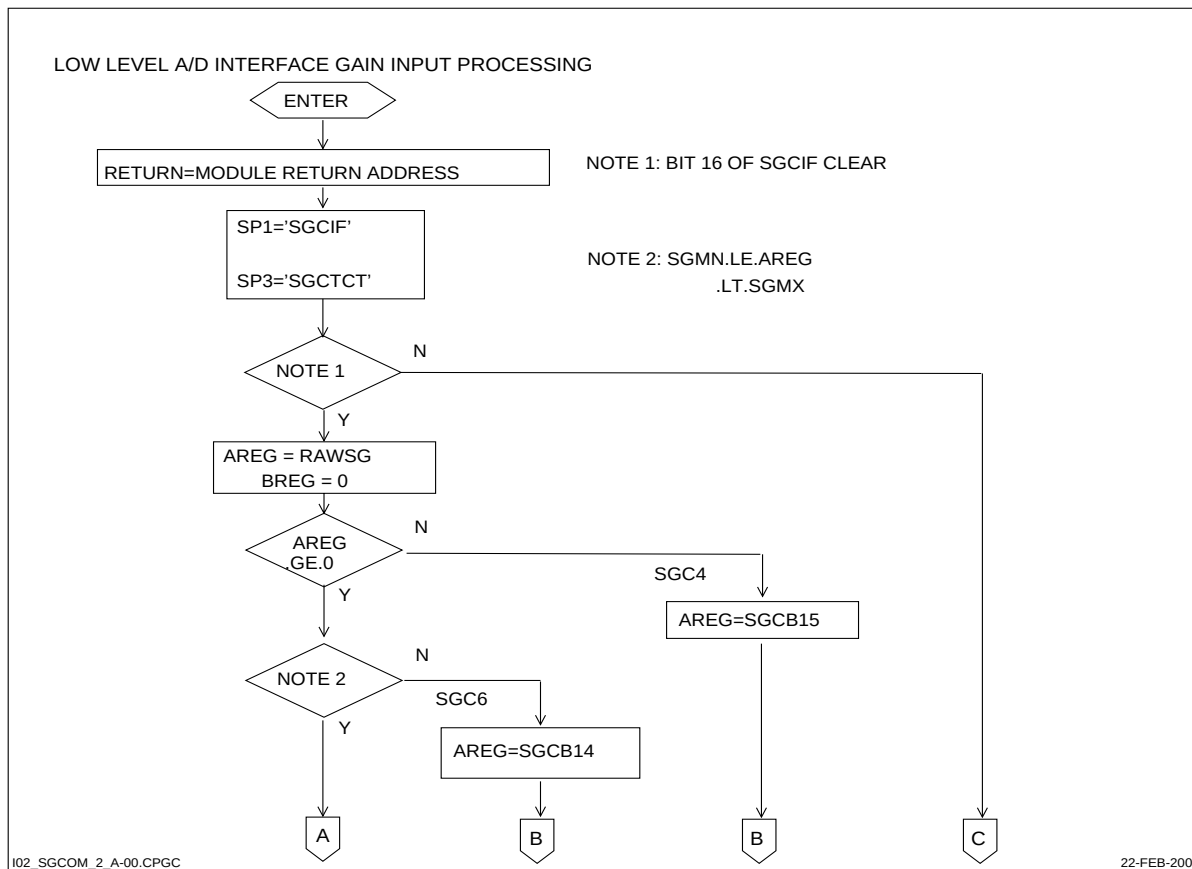
I01_LLDRIFT_2_B-00.CPGC

22-FEB-2007

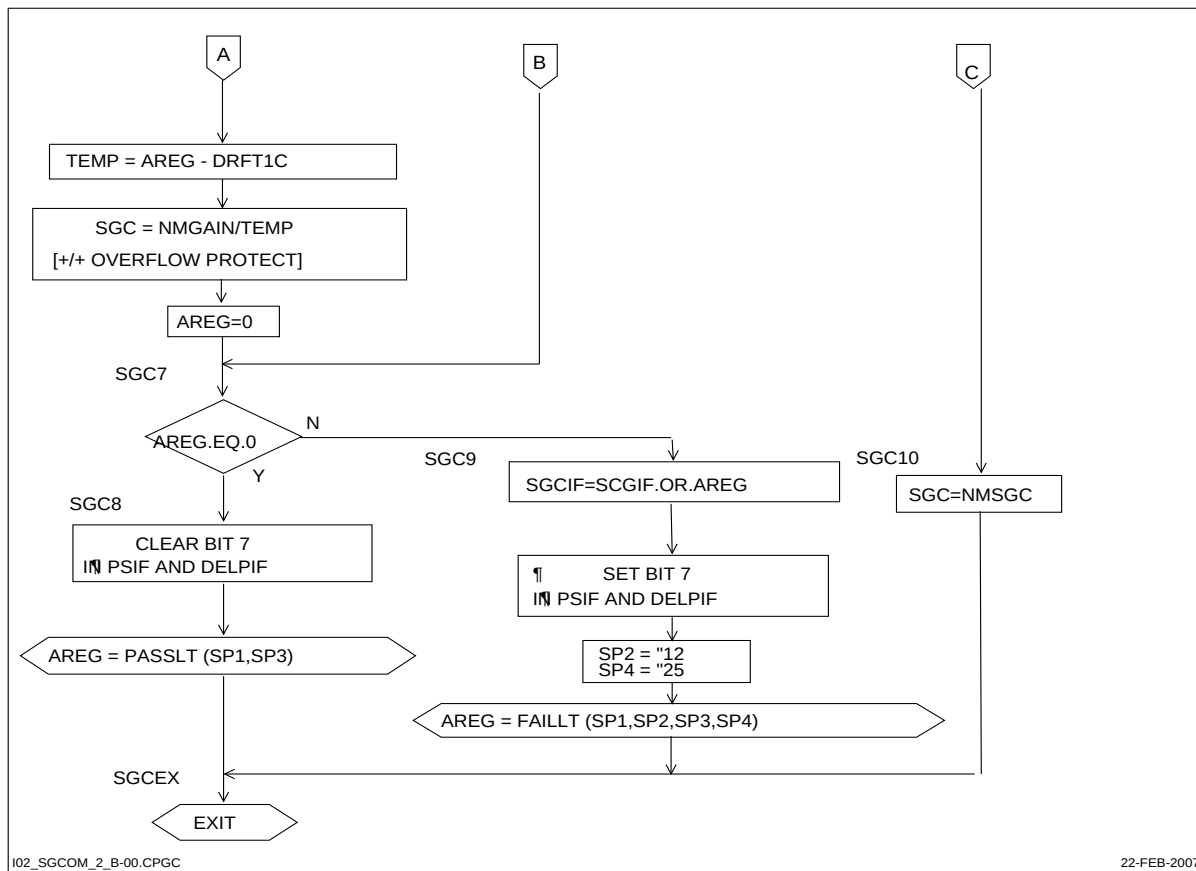
13.00 Figure 210: I02_SGCOM_1_A



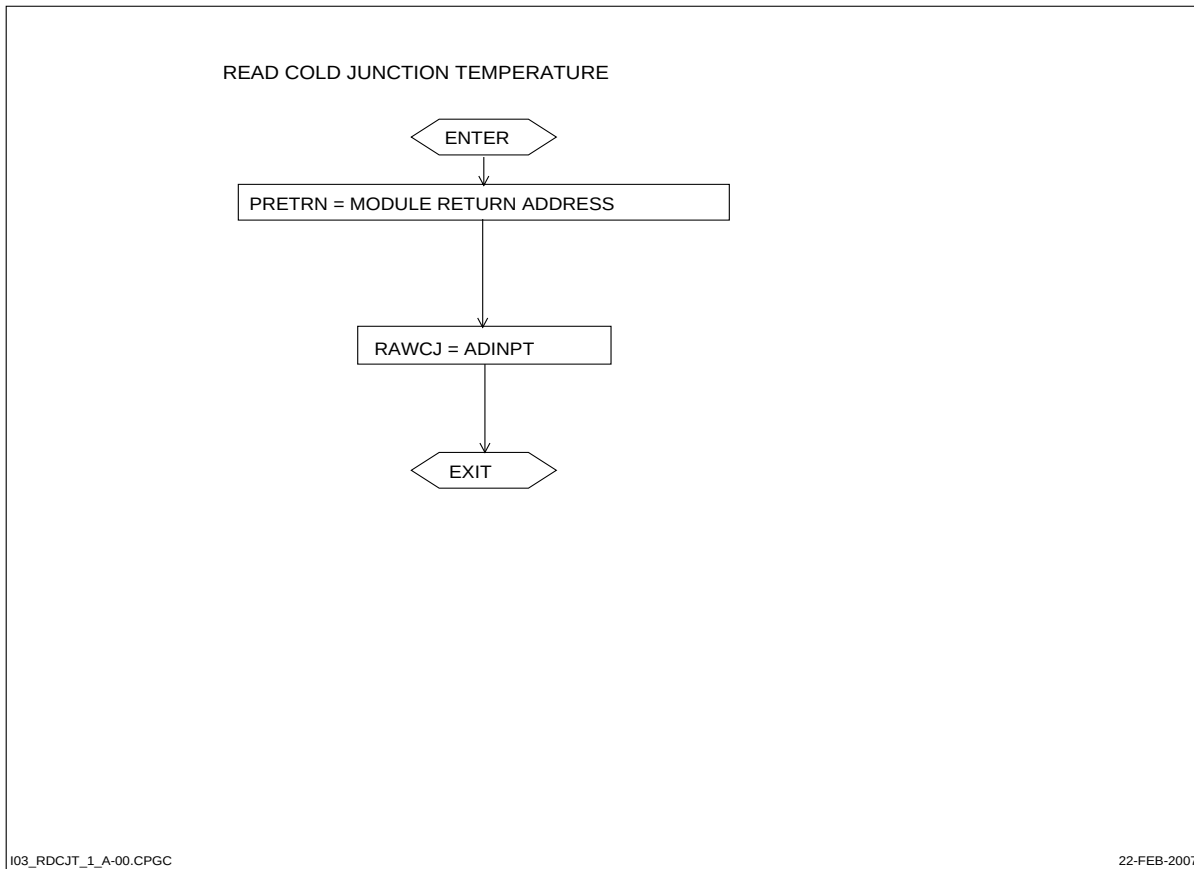
13.00 Figure 211: I02_SGCOM_2_A



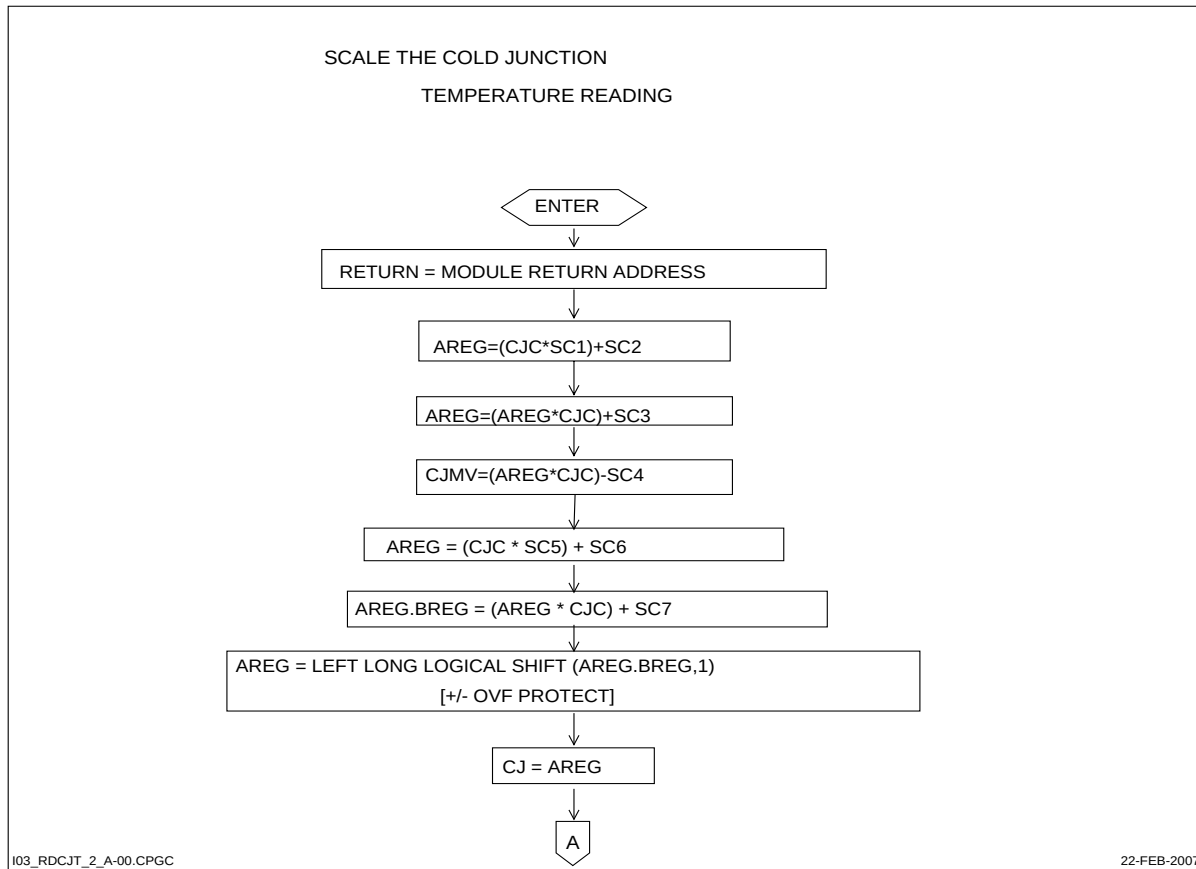
13.00 Figure 212: I02_SGCOM_2_B



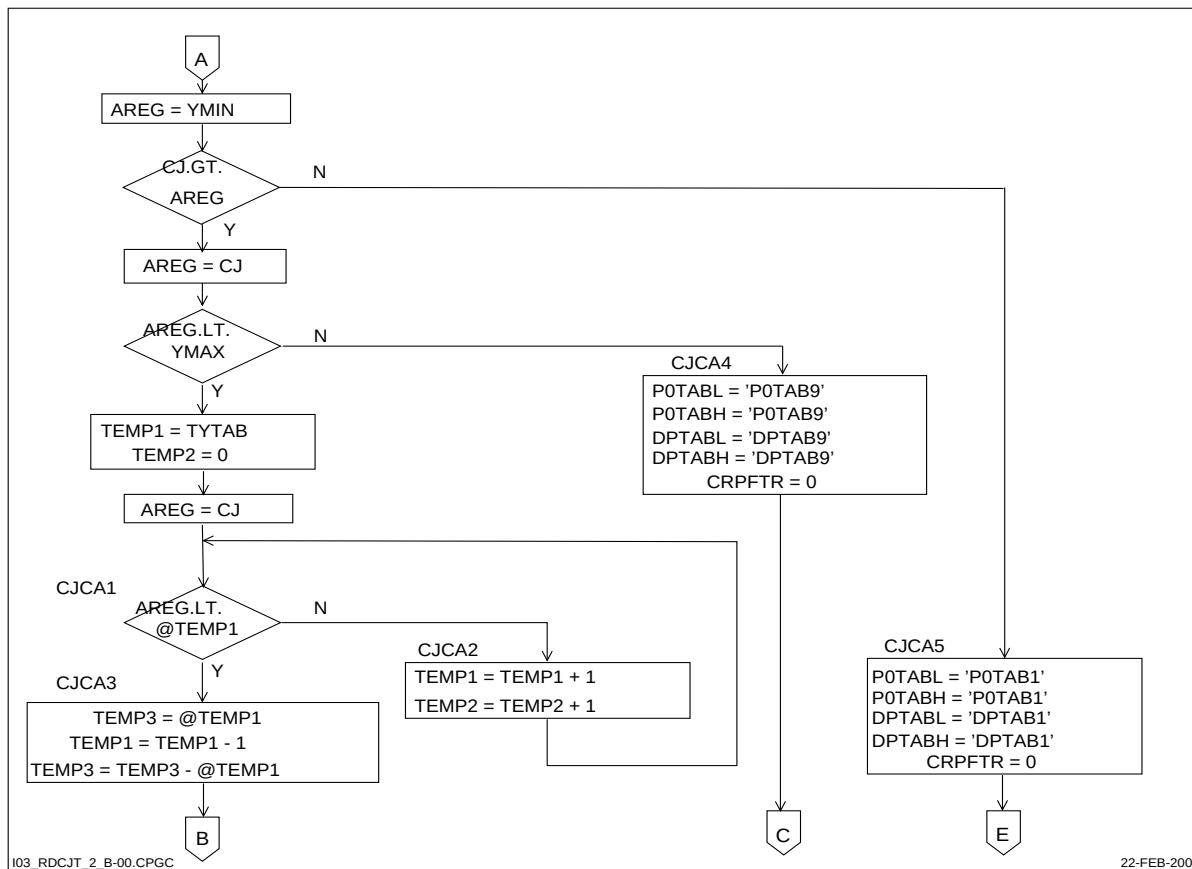
13.00 Figure 213: I03_RDCJT_1_A



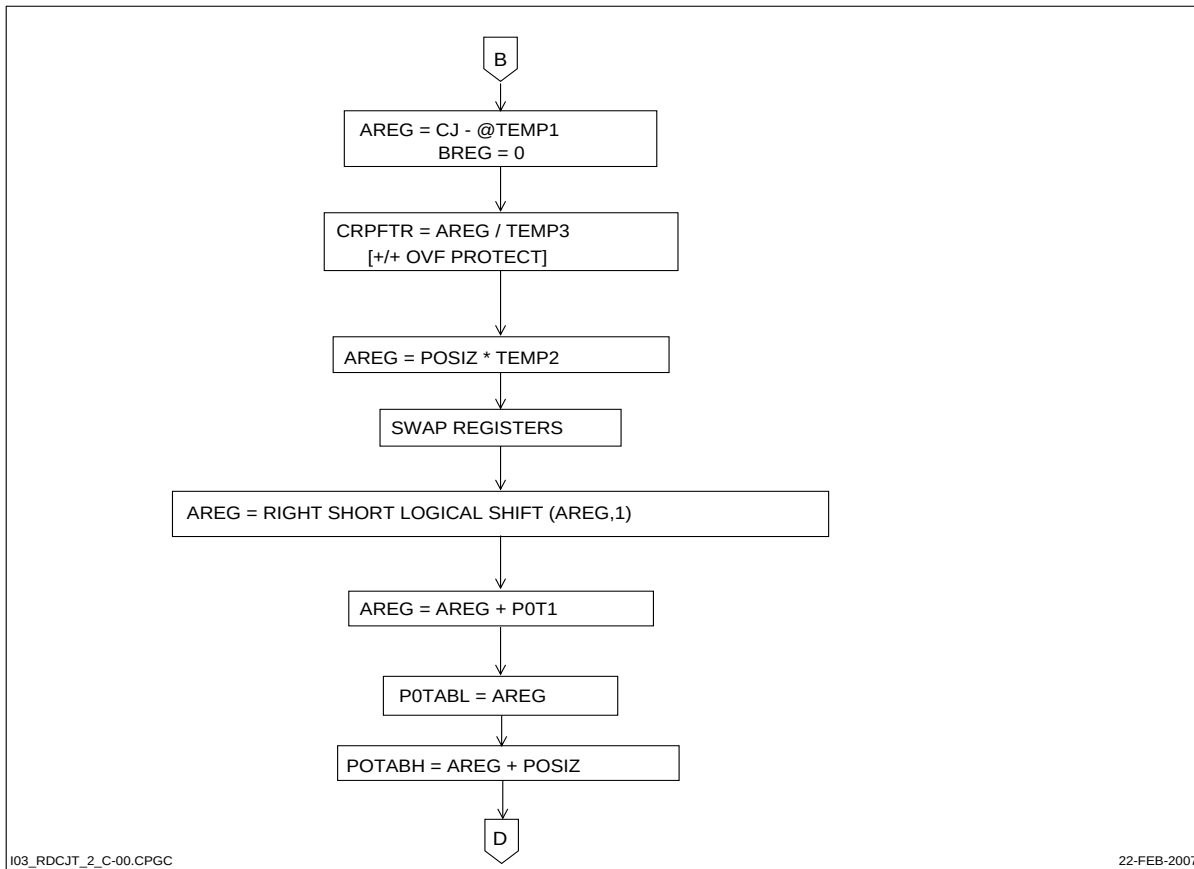
13.00 Figure 214: I03_RDCJT_2_A



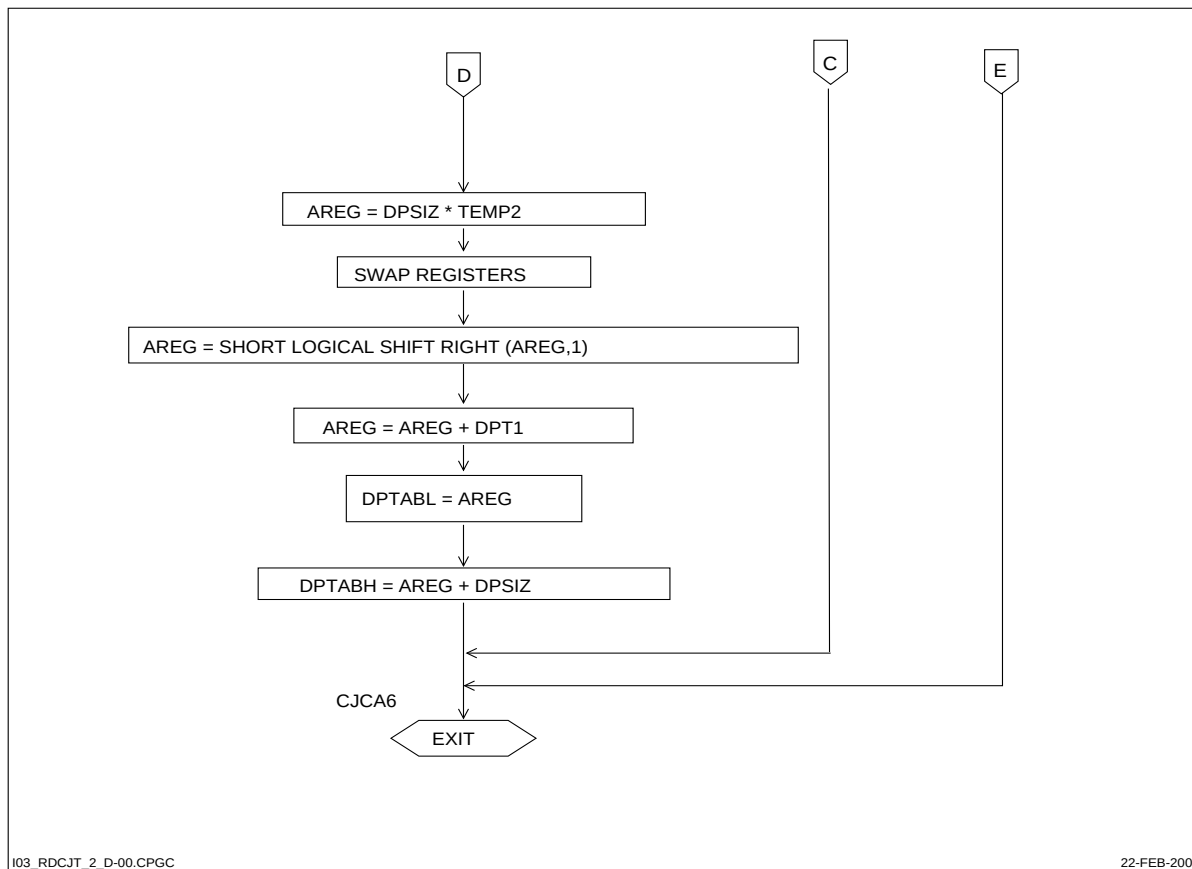
13.00 Figure 215: I03_RDCJT_2_B



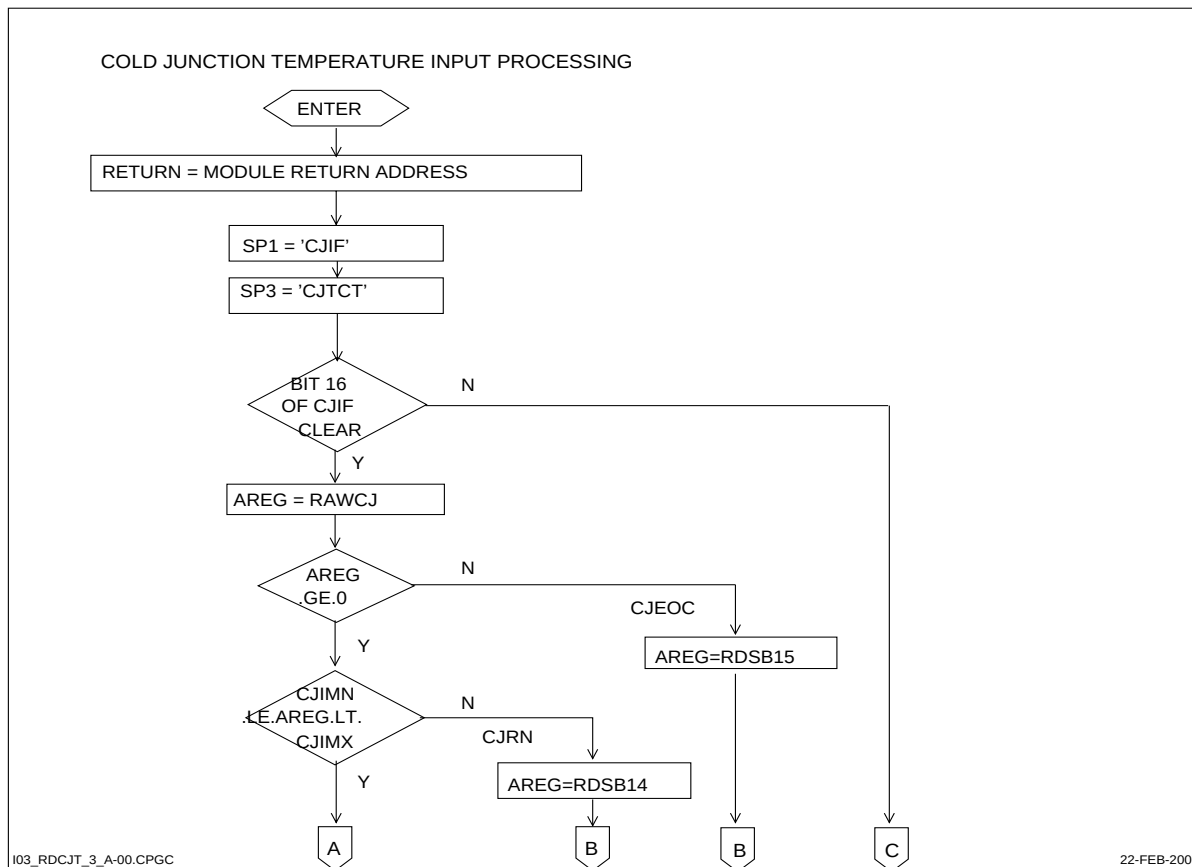
13.00 Figure 216: I03_RDCJT_2_C



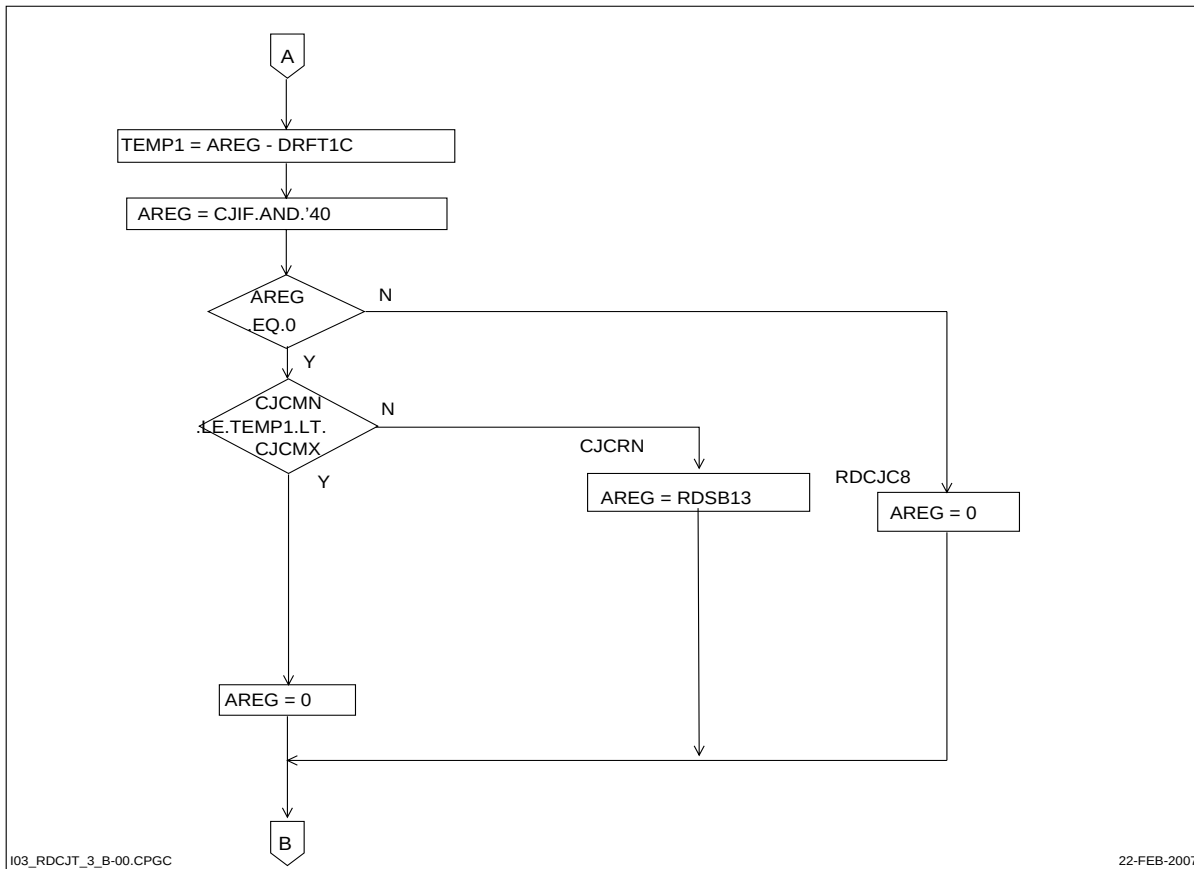
13.00 Figure 217: I03_RDCJT_2_D



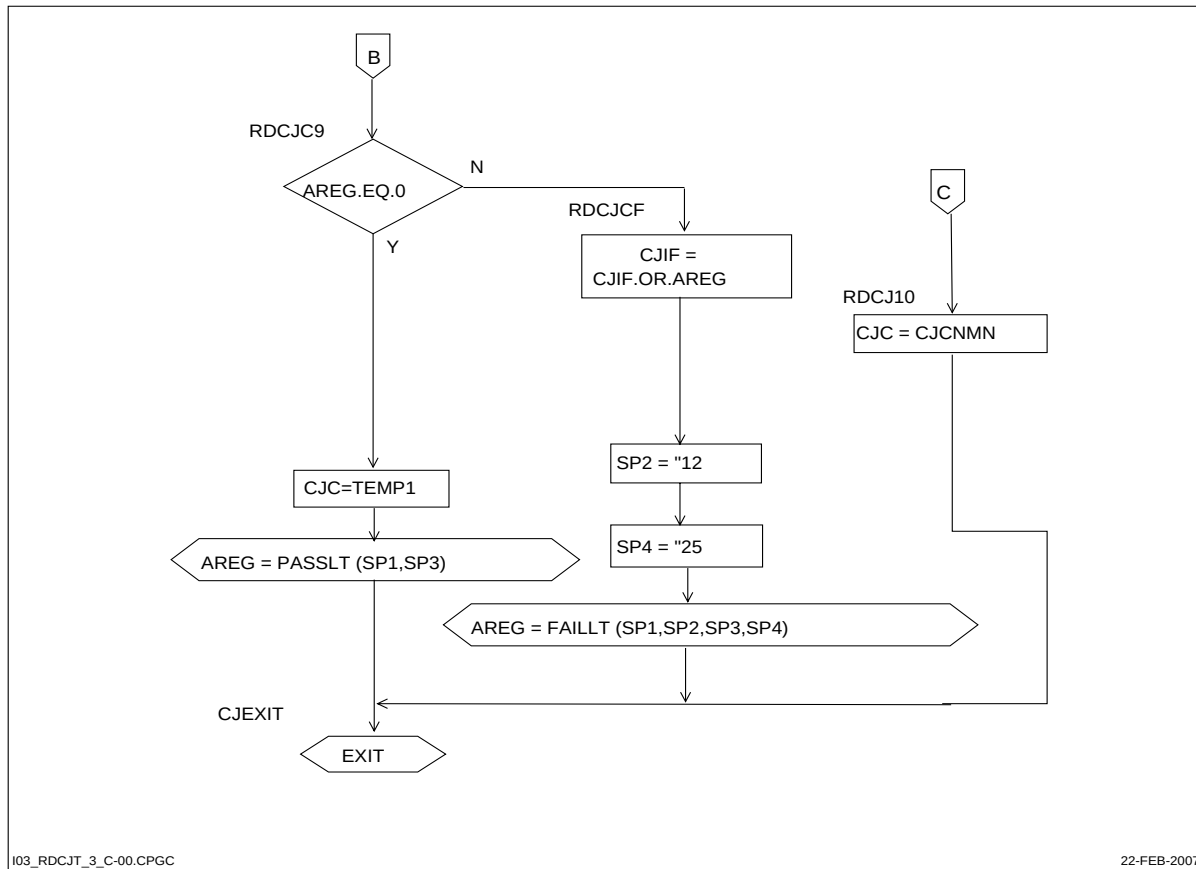
13.00 Figure 218: I03_RDCJT_3_A



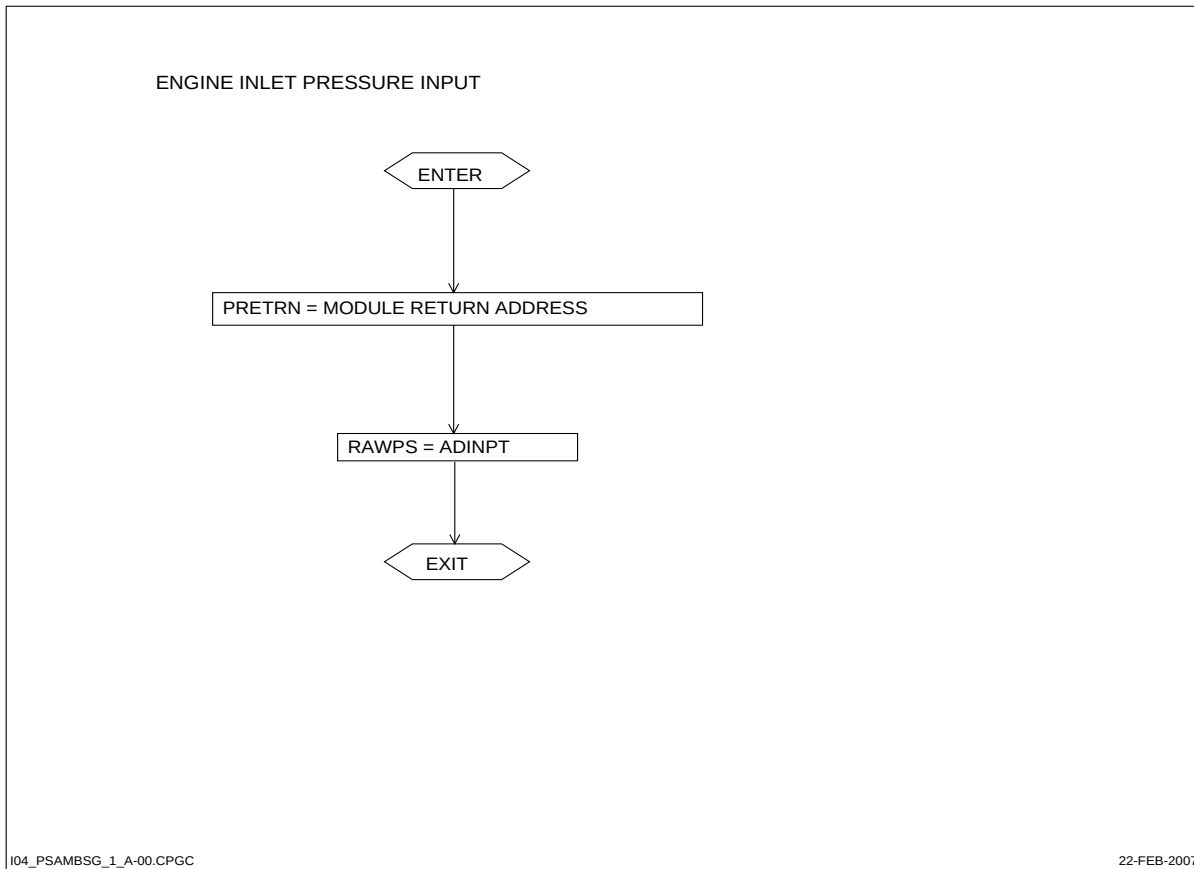
13.00 Figure 219: I03_RDCJT_3_B



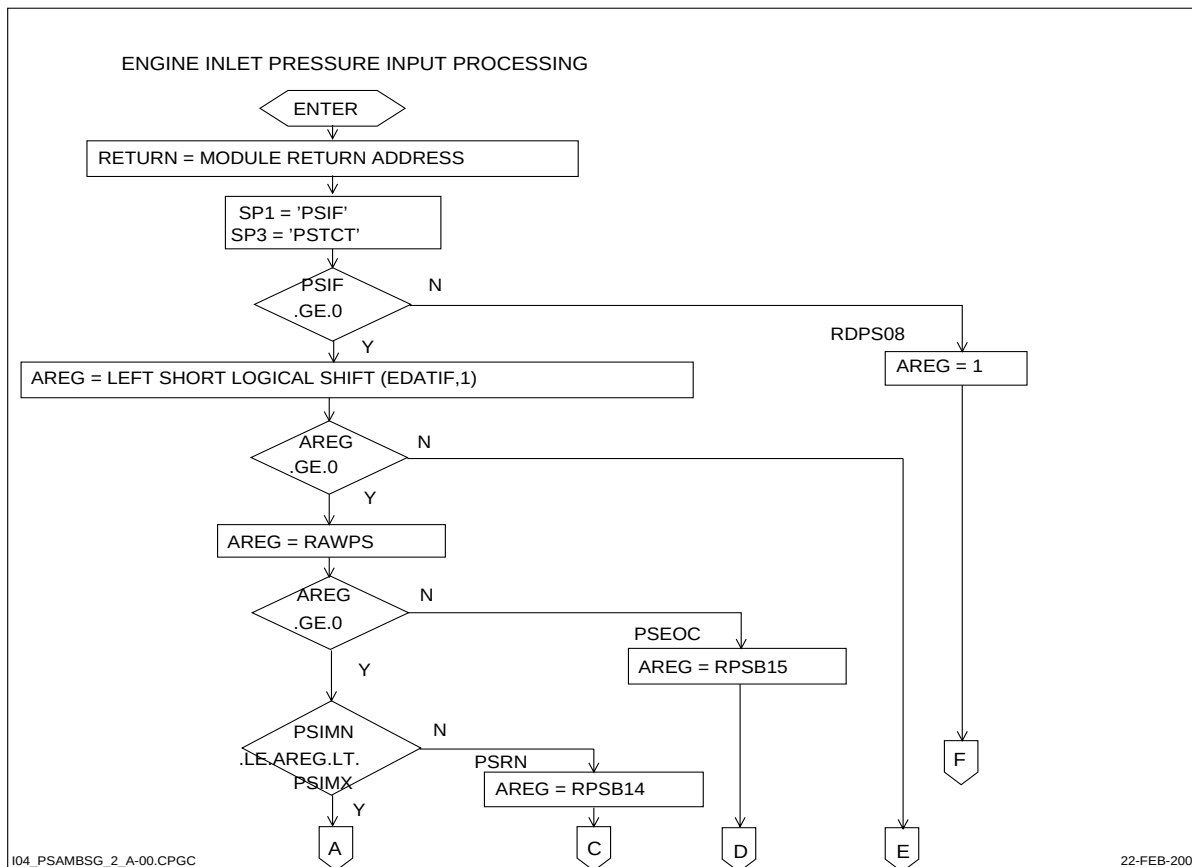
13.00 Figure 220: I03_RDCJT_3_C



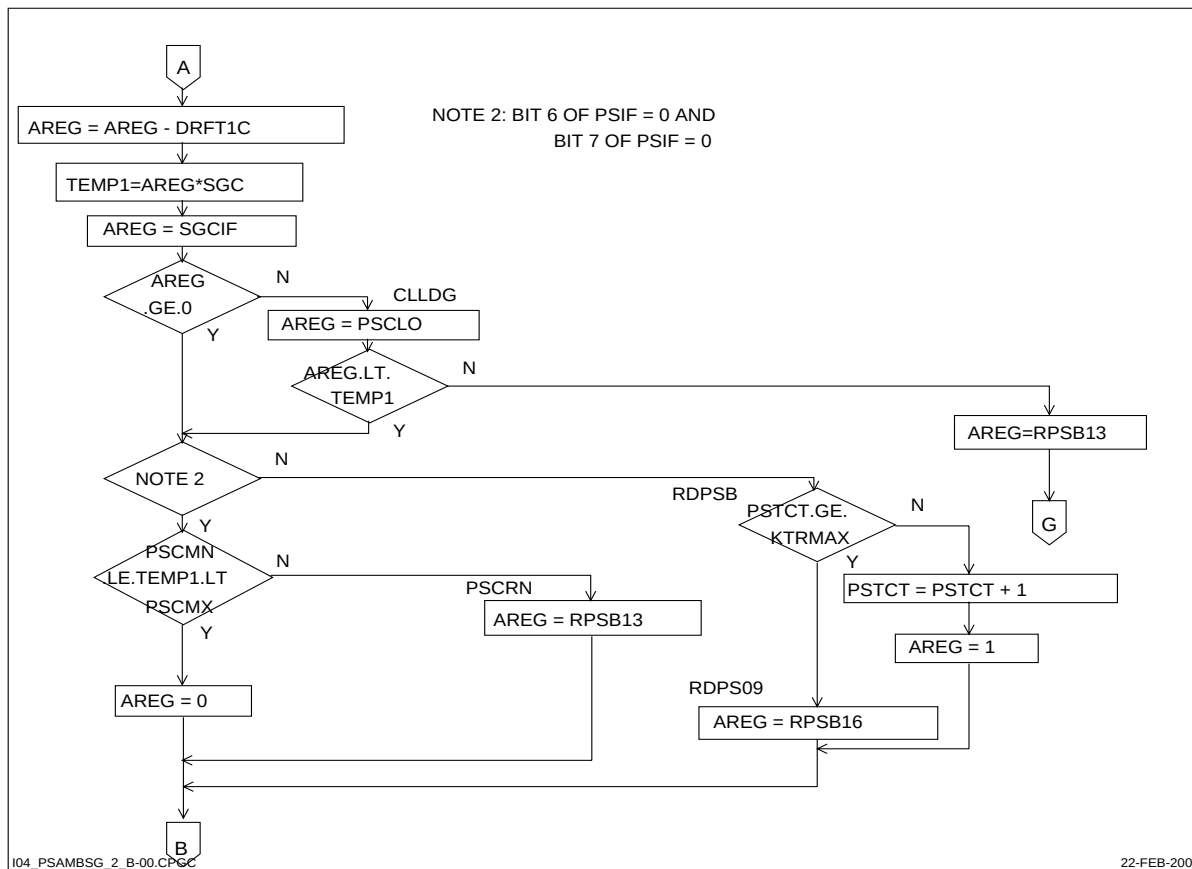
13.00 Figure 221: I04_PSAMBSG_1_A



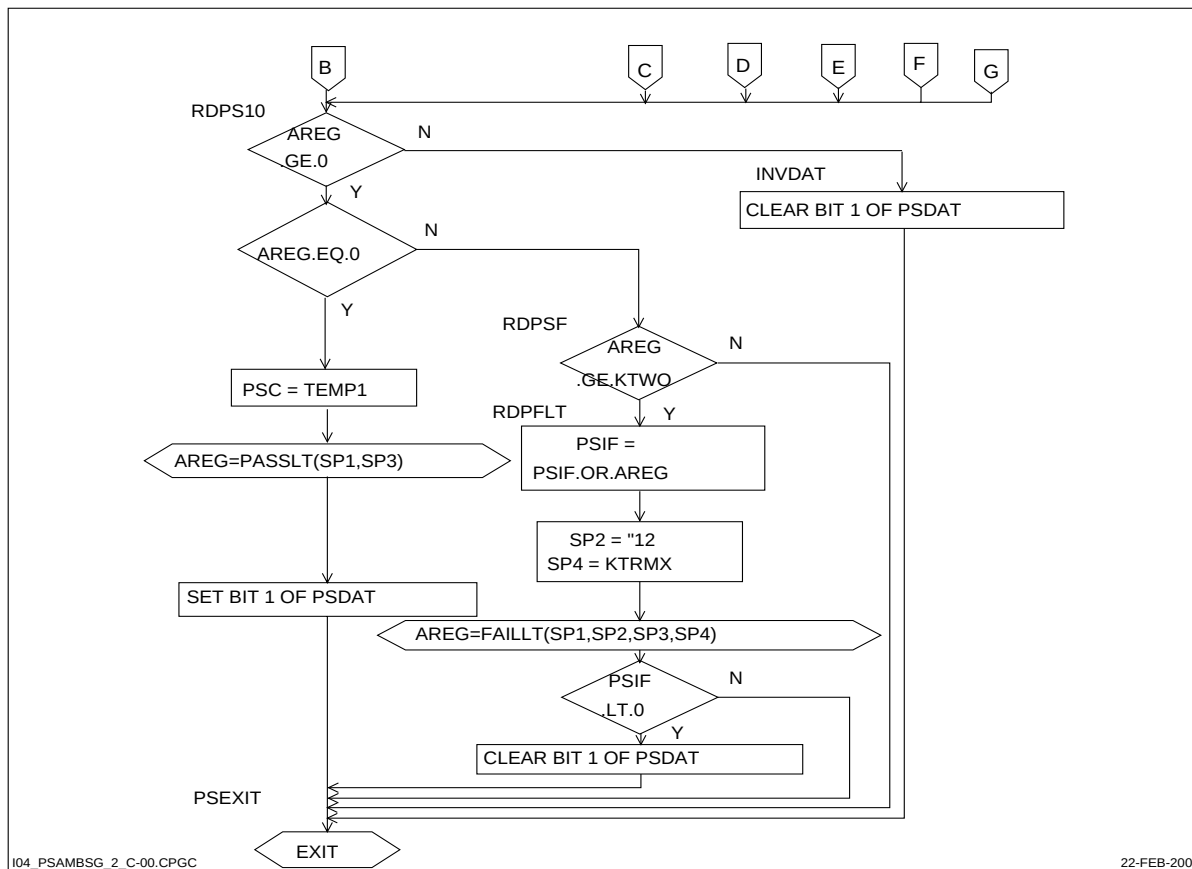
13.00 Figure 222: I04_PSAMBSG_2_A



13.00 Figure 223: I04_PSAMBSG_2_B



13.00 Figure 224: I04_PSAMBSG_2_C

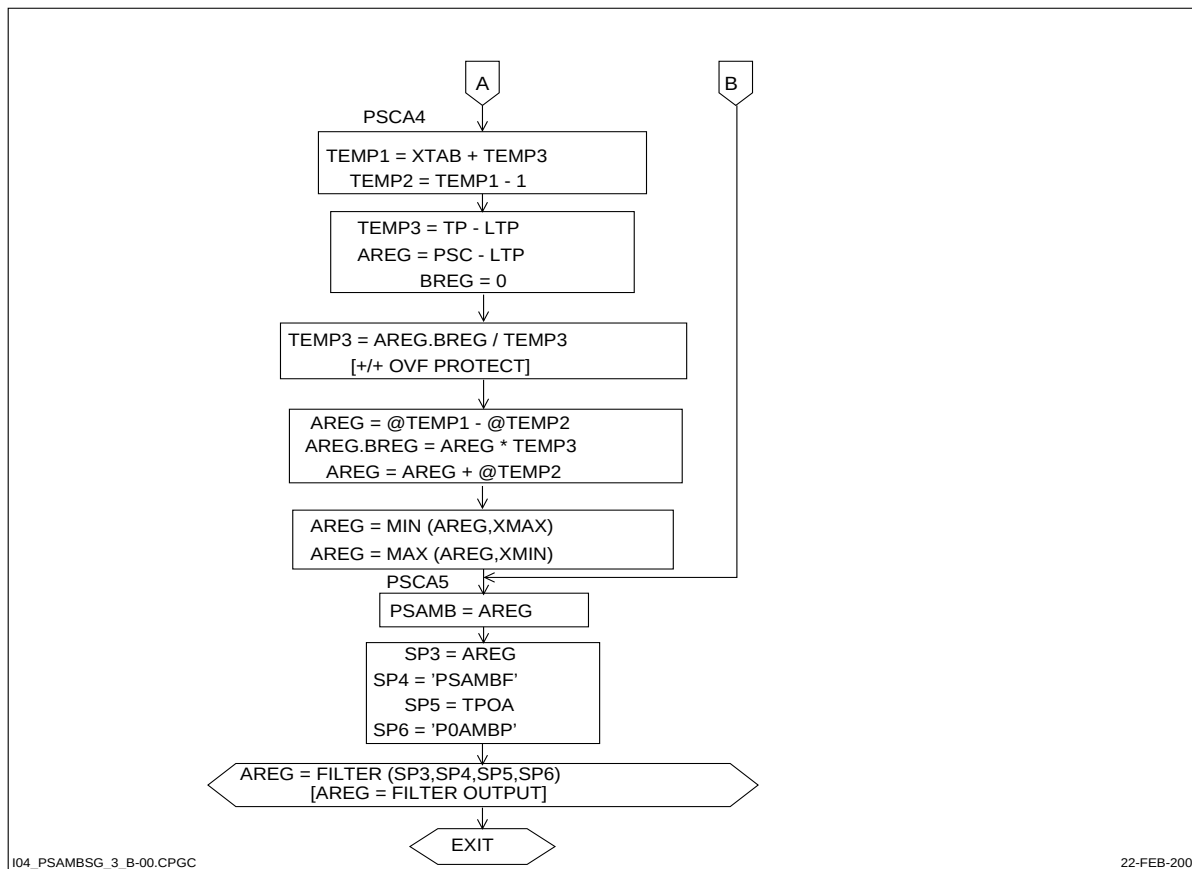


```

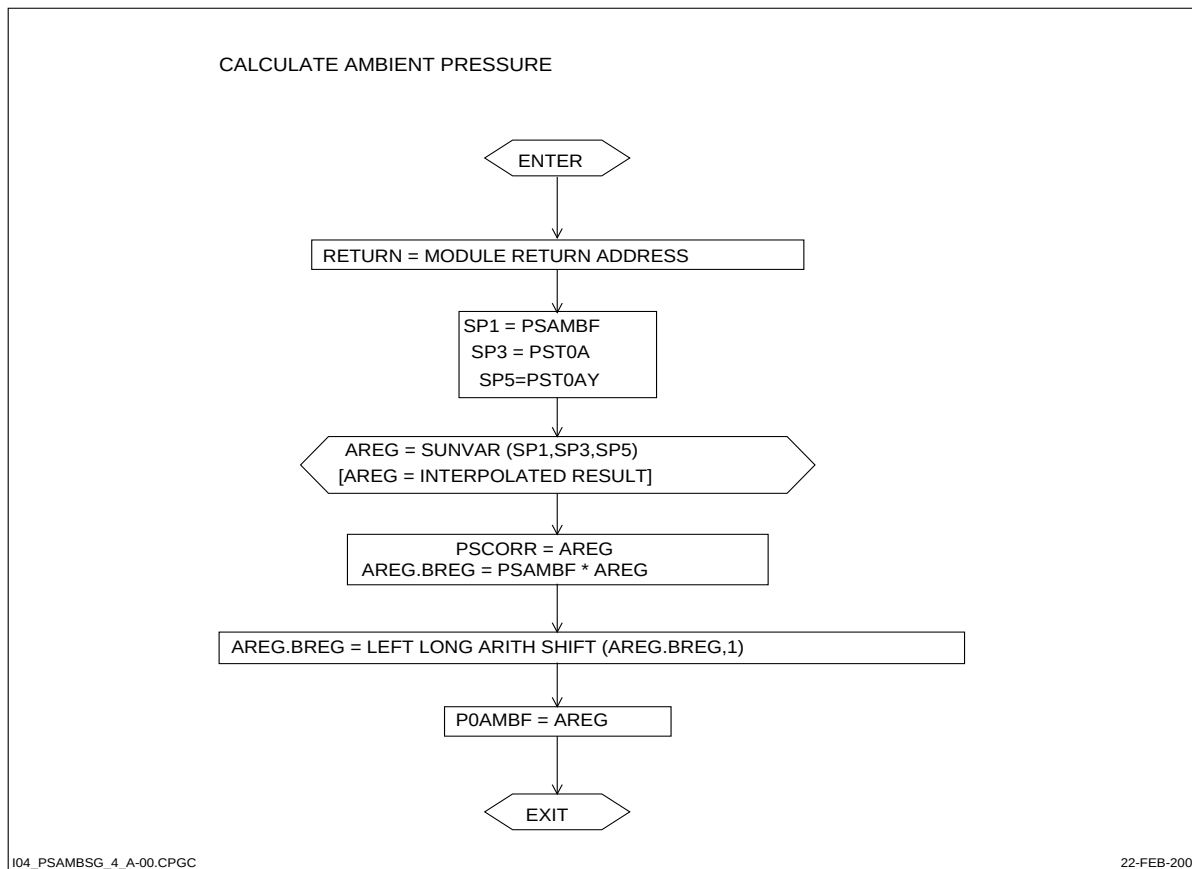
graph TD
    ENTER{{ENTER}} --> RETURN[RETURN = MODULE RETURN ADDRESS]
    RETURN --> BIT1{BIT 1  
OF PSDAT  
SET}
    BIT1 -- N --> AREG_PSDEF[AREG = PSDEF]
    AREG_PSDEF --> B1{{B}}
    BIT1 -- Y --> PSCA0[PSCA0]
    PSCA0 --> TEMP_INIT[TEMP1 = P0TABL  
TEMP2 = P0TABH  
TEMP3 = 0]
    TEMP_INIT --> PSCA1[PSCA1]
    PSCA1 --> AREG_CALC[AREG = @TEMP2 - @TEMP1  
AREG.BREG = AREG * CRPFTR  
AREG = AREG + @TEMP1]
    AREG_CALC --> AREG_GE_PSC{AREG.GE.  
PSC}
    AREG_GE_PSC -- N --> LTP_CALC[LTP = AREG  
TEMP1 = TEMP1 + 1  
TEMP2 = TEMP2 + 1  
AREG = TEMP3 + 1]
    AREG_GE_PSC -- Y --> PSCA3[PSCA3]
    PSCA3 --> TP_ASSIGN[TP = AREG]
    TP_ASSIGN --> TEMP3_NE0{TEMP3  
.NE.0}
    TEMP3_NE0 -- N --> AREG_XMIN[AREG = XMIN]
    AREG_XMIN --> B2{{B}}
    TEMP3_NE0 -- Y --> A{{A}}
    LTP_CALC --> AREG_GE_POSIZE{AREG.GE.  
POSIZE}
    AREG_GE_POSIZE -- N --> TEMP3_ASSIGN[TEMP3 = AREG]
    TEMP3_ASSIGN --> B3{{B}}
    AREG_GE_POSIZE -- Y --> PSCA2[PSCA2]
    PSCA2 --> AREG_XMAX[AREG = XMAX]
    AREG_XMAX --> B4{{B}}

```

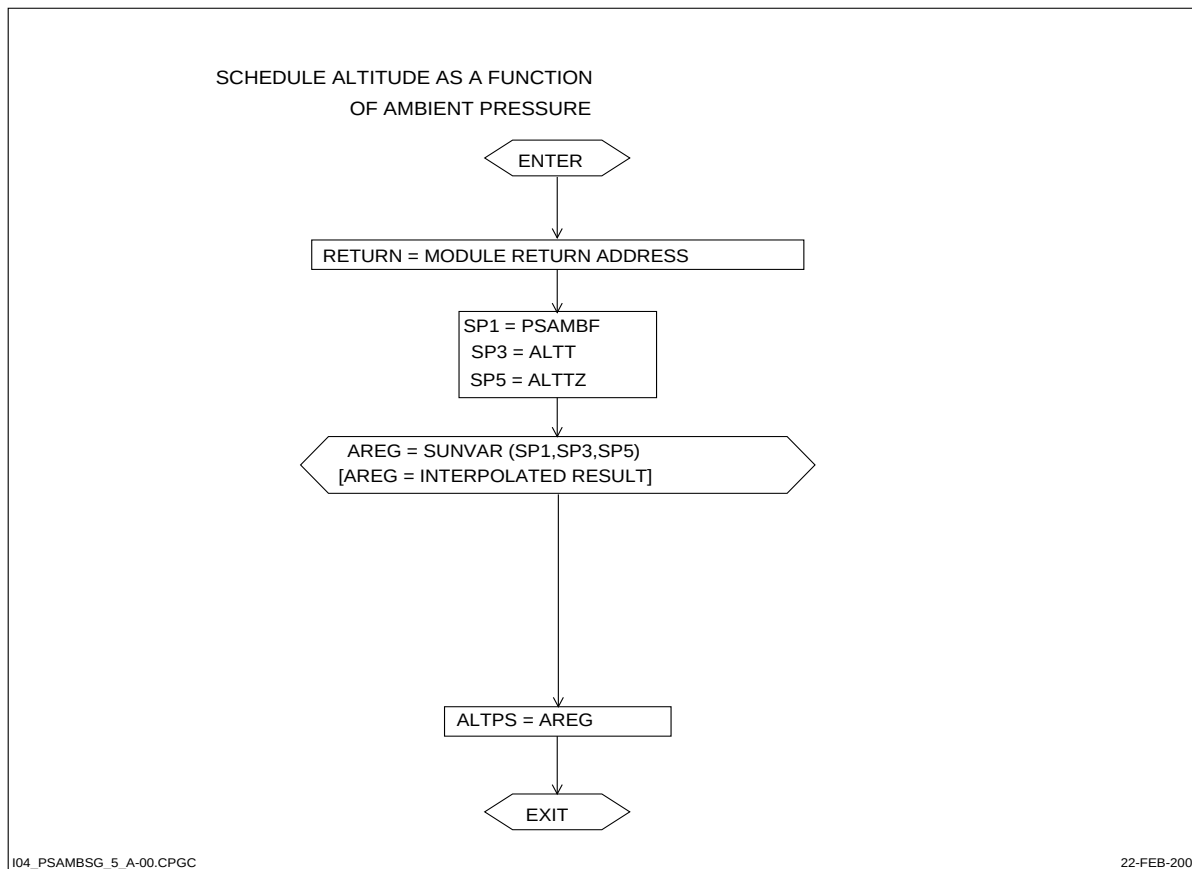
13.00 Figure 226: I04_PSAMBSG_3_B



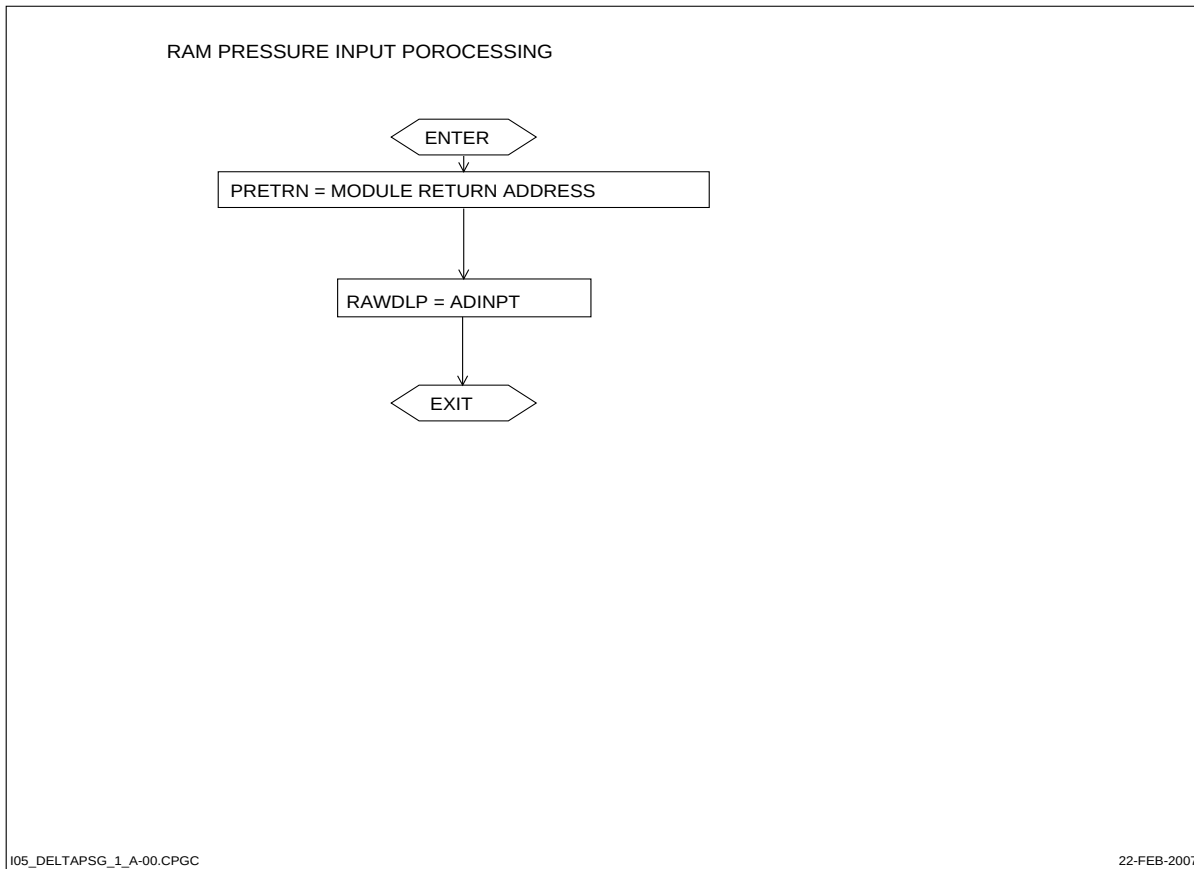
13.00 Figure 227: I04_PSAMBSG_4_A



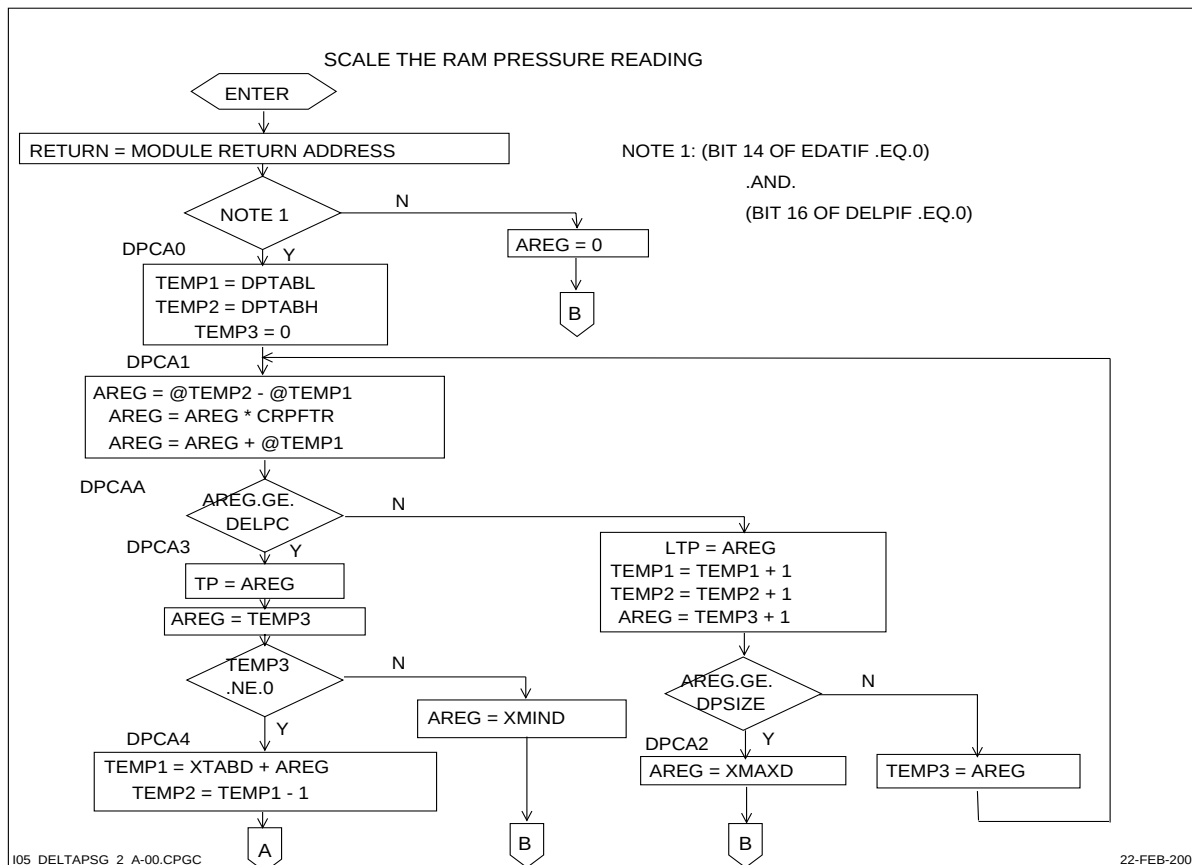
13.00 Figure 228: I04_PSAMBSG_5_A



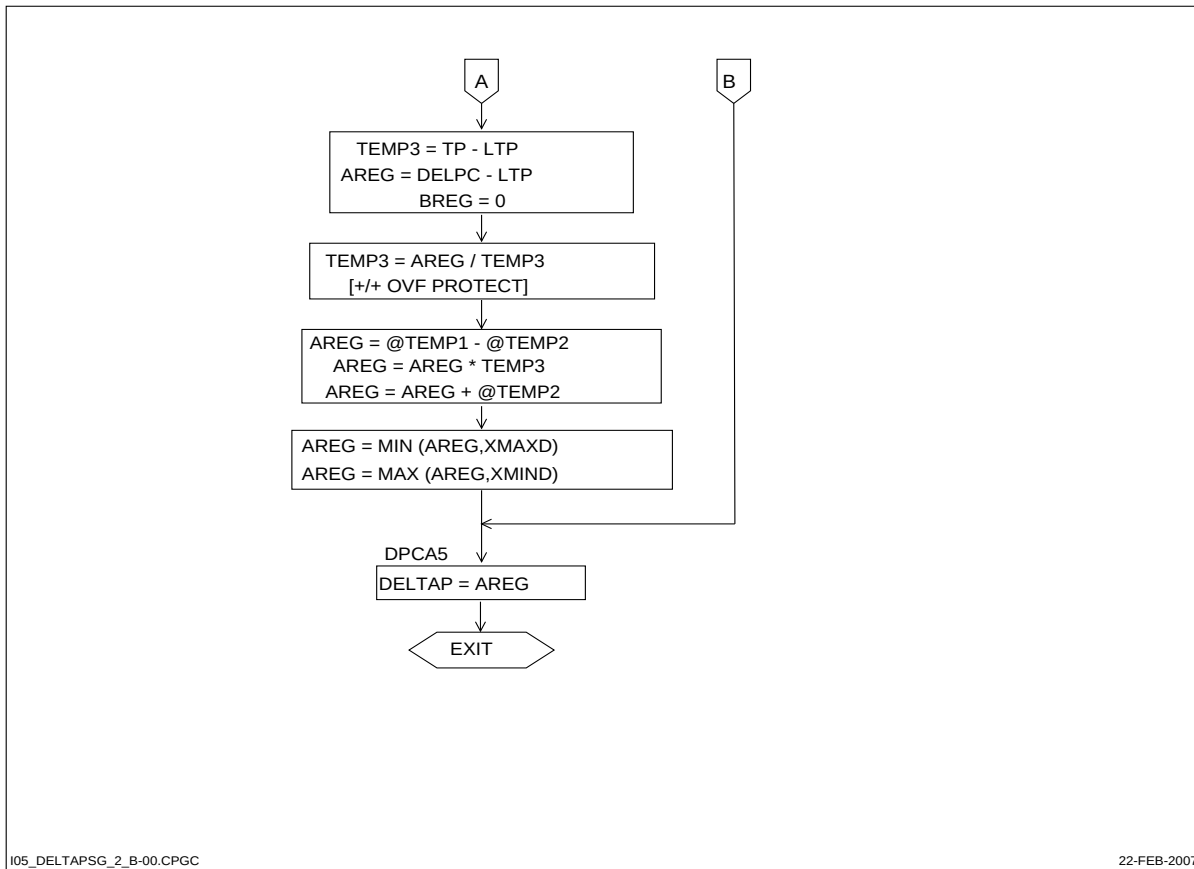
13.00 Figure 229: I05 _DELTAPSG _1 _A



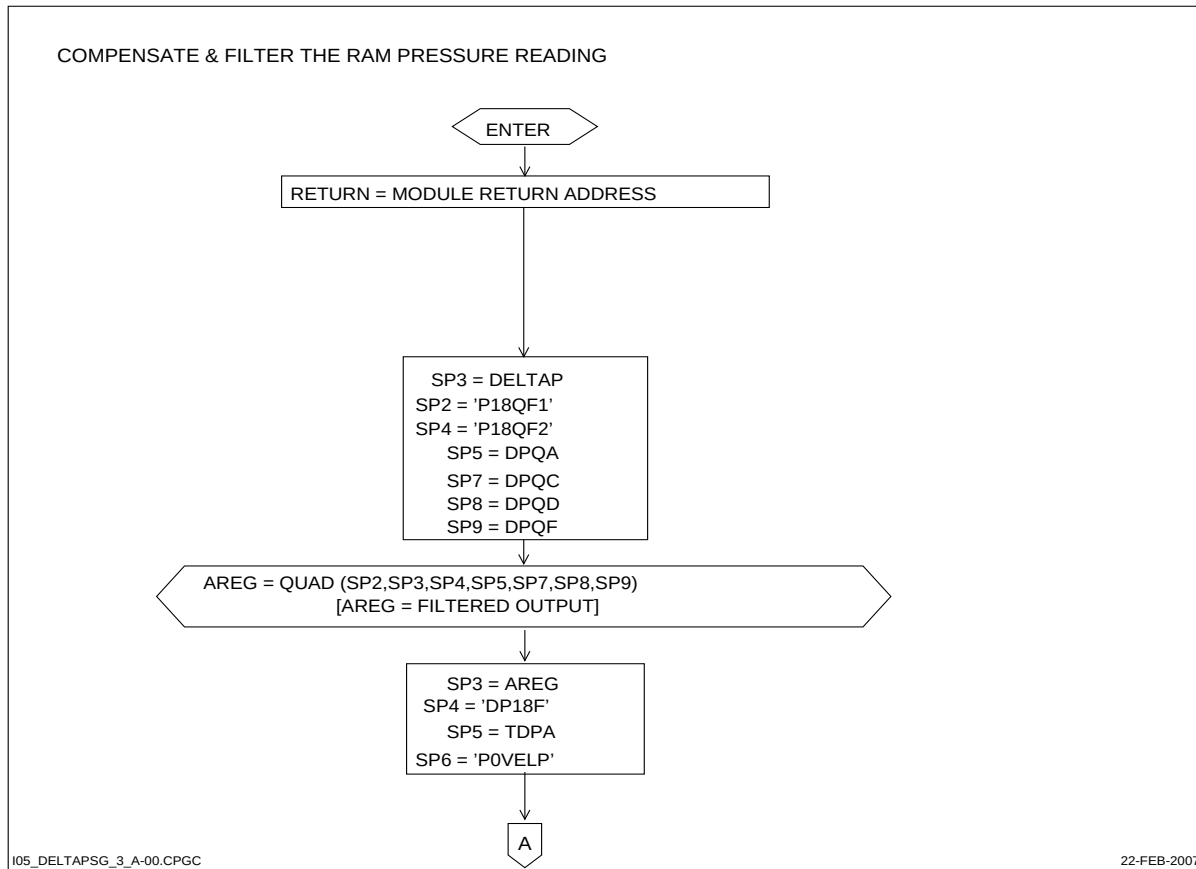
13.00 Figure 230: I05_DELTAPSG_2_A



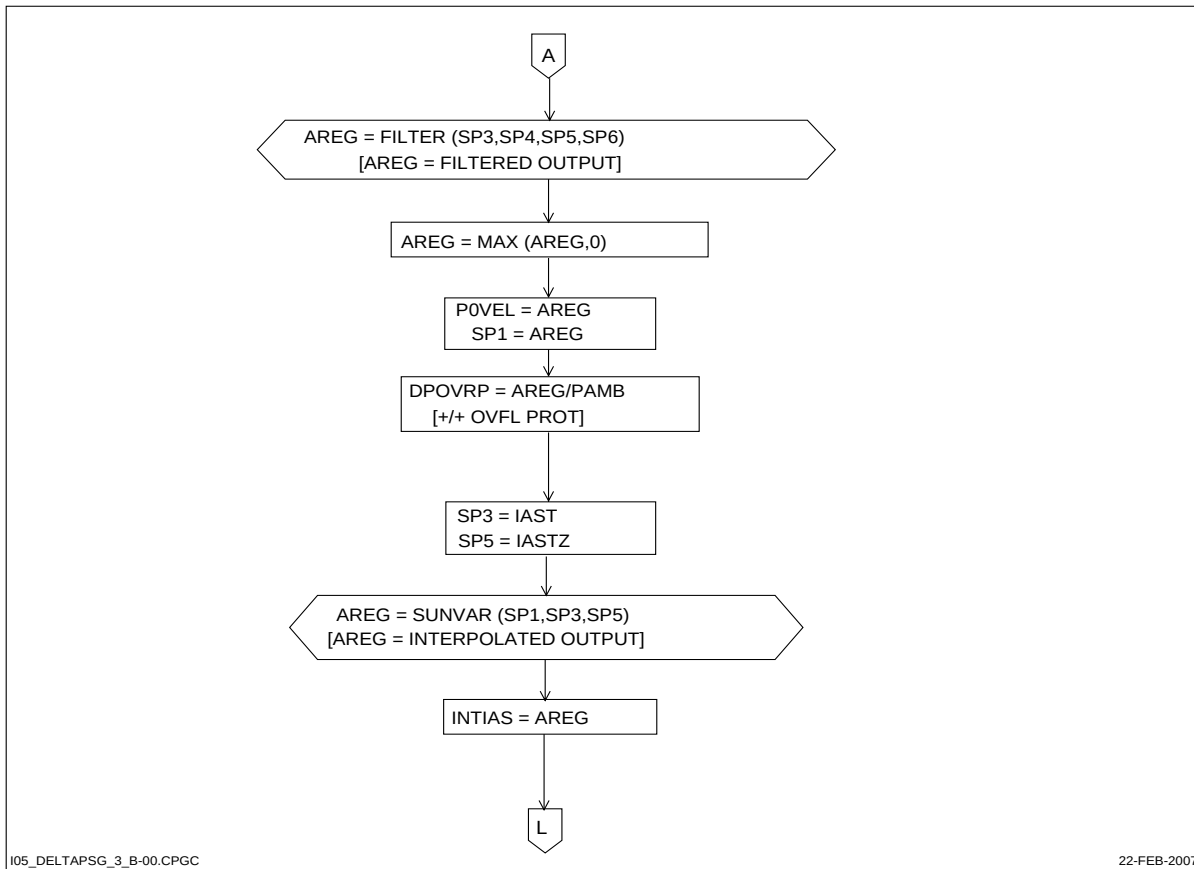
13.00 Figure 231: I05 _DELTAPSG _2 _B



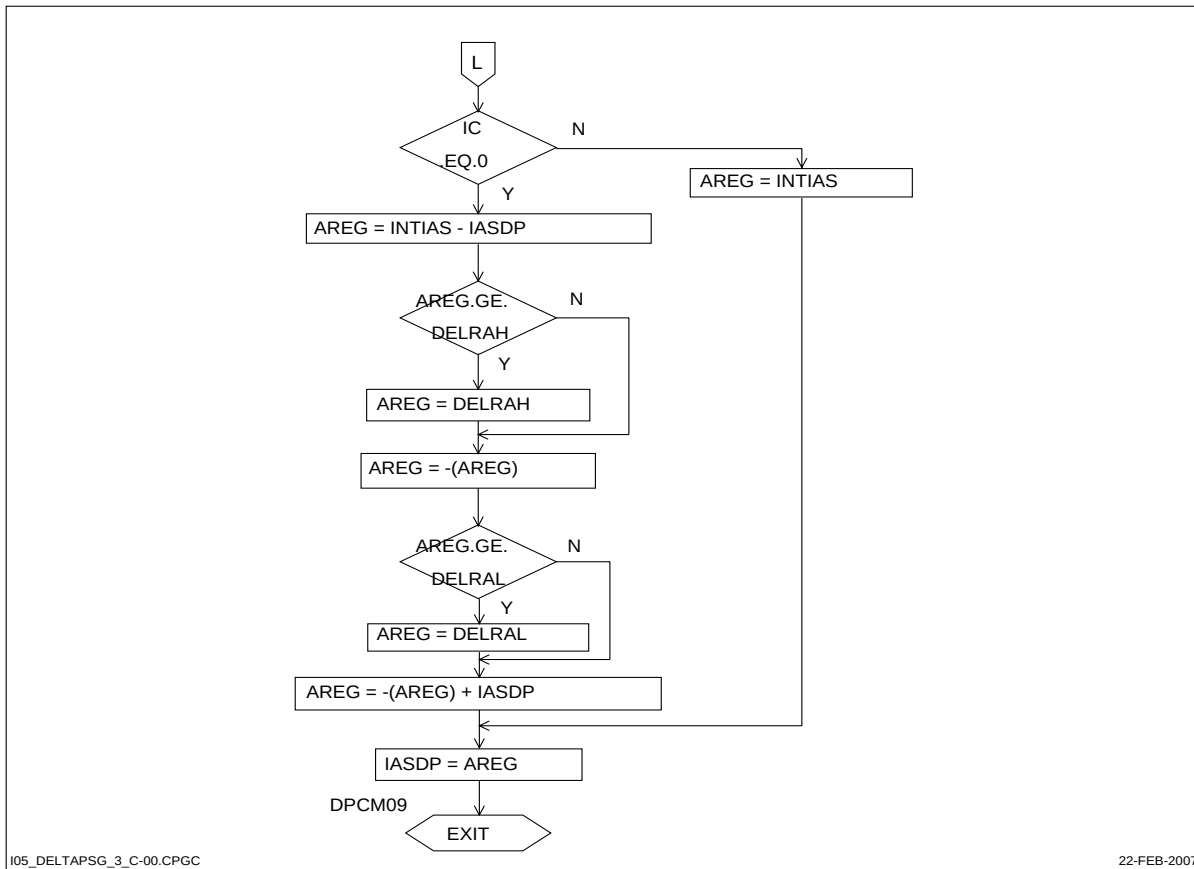
13.00 Figure 232: I05 _DELTAPSG _3 _A



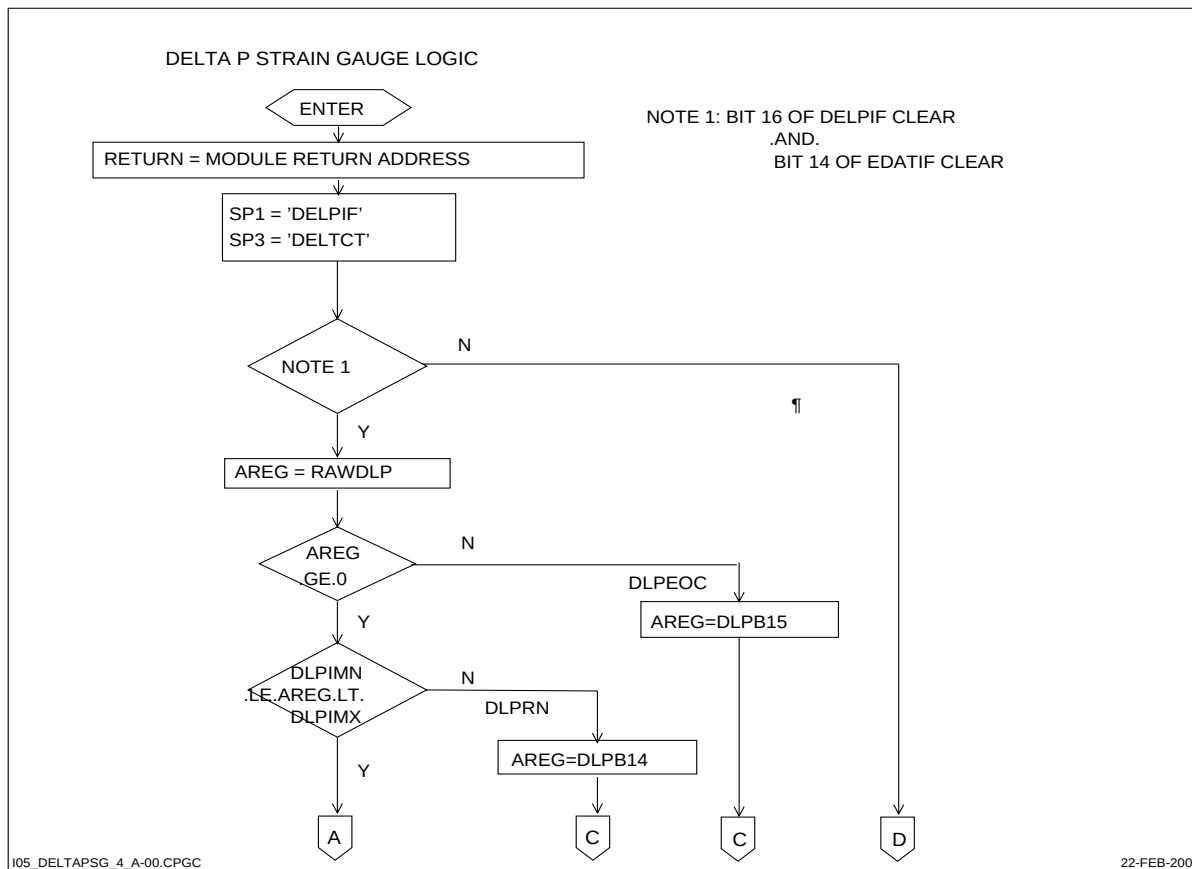
13.00 Figure 233: I05 _DELTAPSG _3 _B



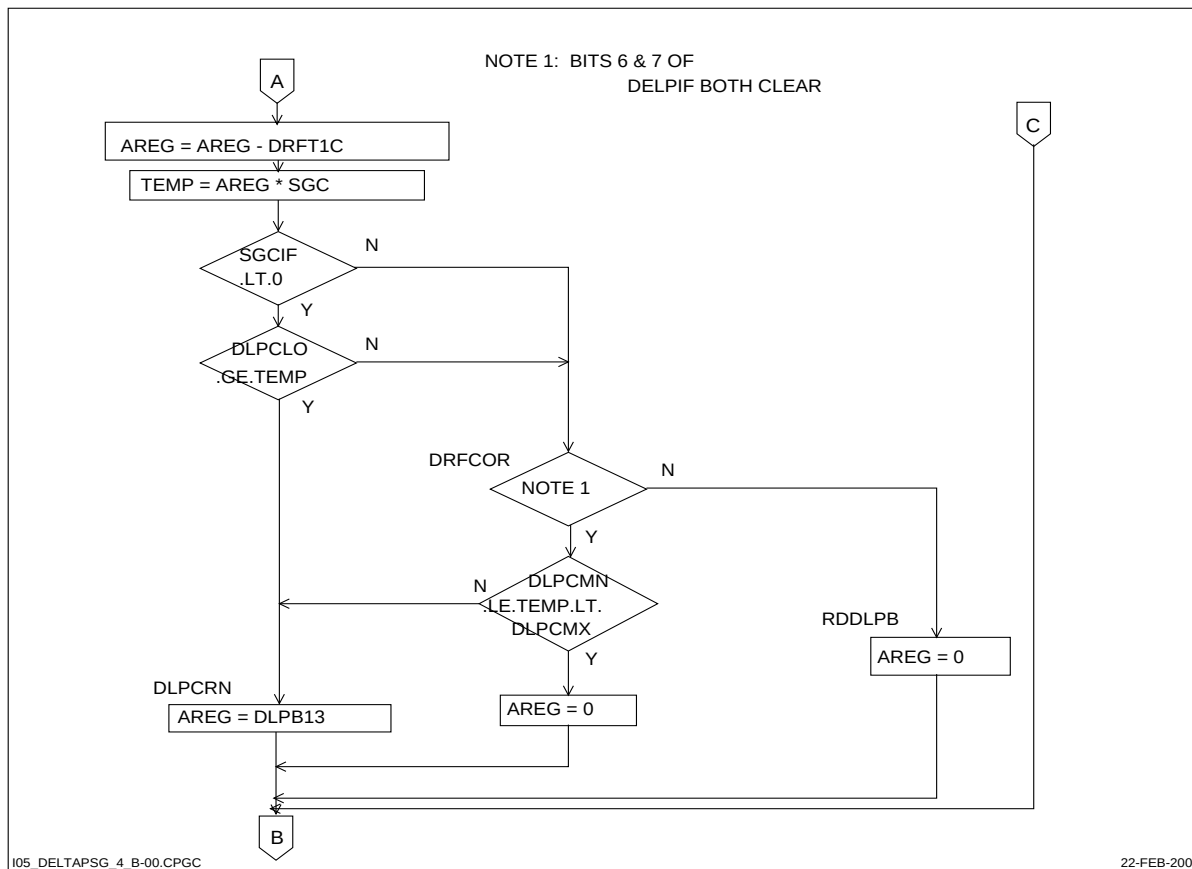
13.00 Figure 234: I05 _DELTAPSG _3 _C



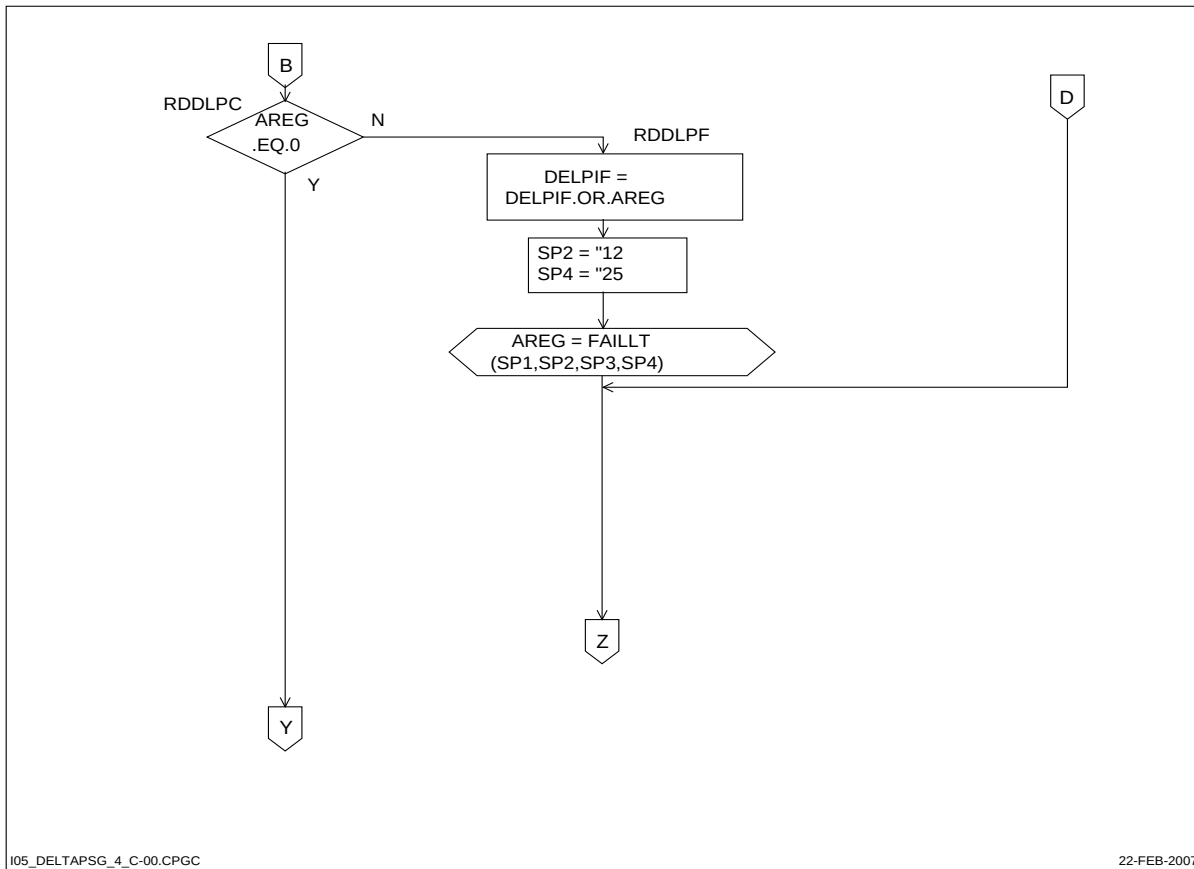
13.00 Figure 235: I05 _DELTAPSG_4_A



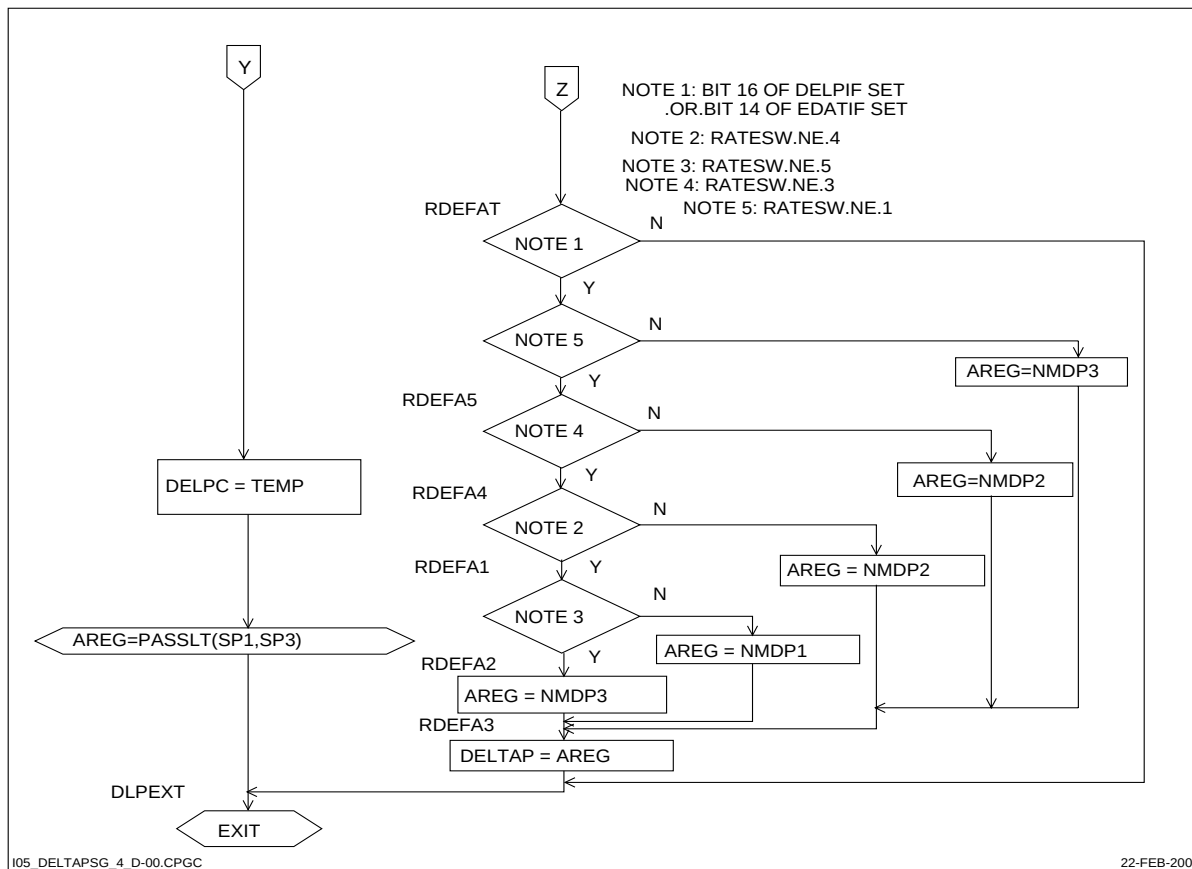
13.00 Figure 236: I05_DELTAPSG_4_B



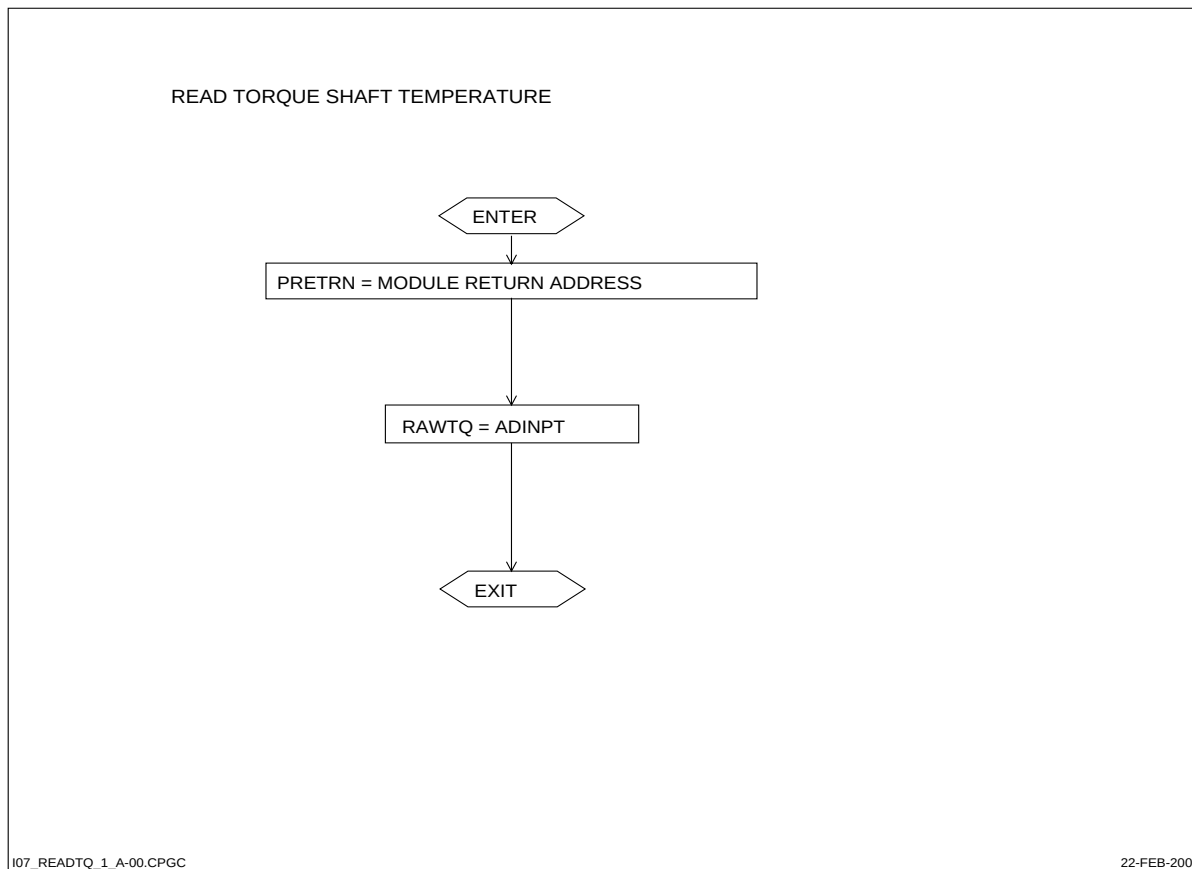
13.00 Figure 237: I05 _DELTAPSG _4 _C



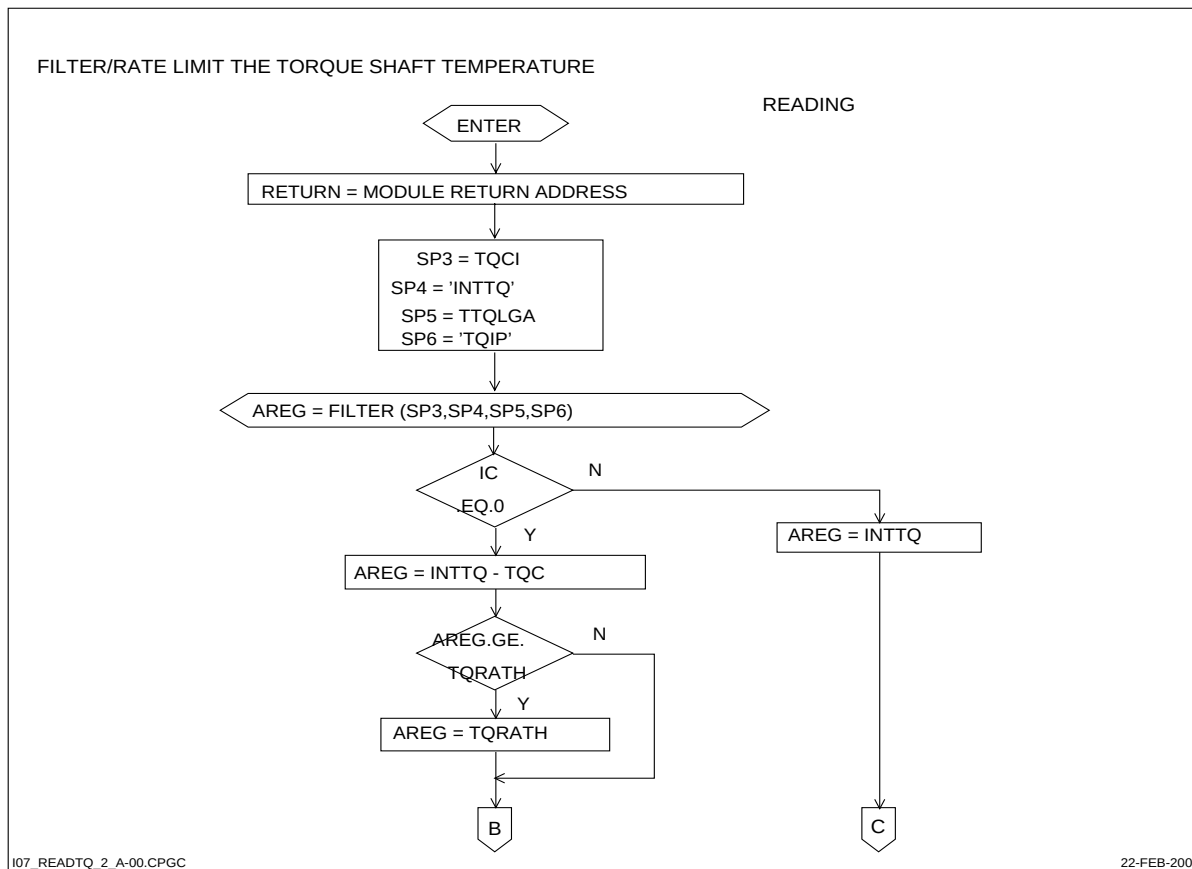
13.00 Figure 238: I05_DELTAPSG_4_D



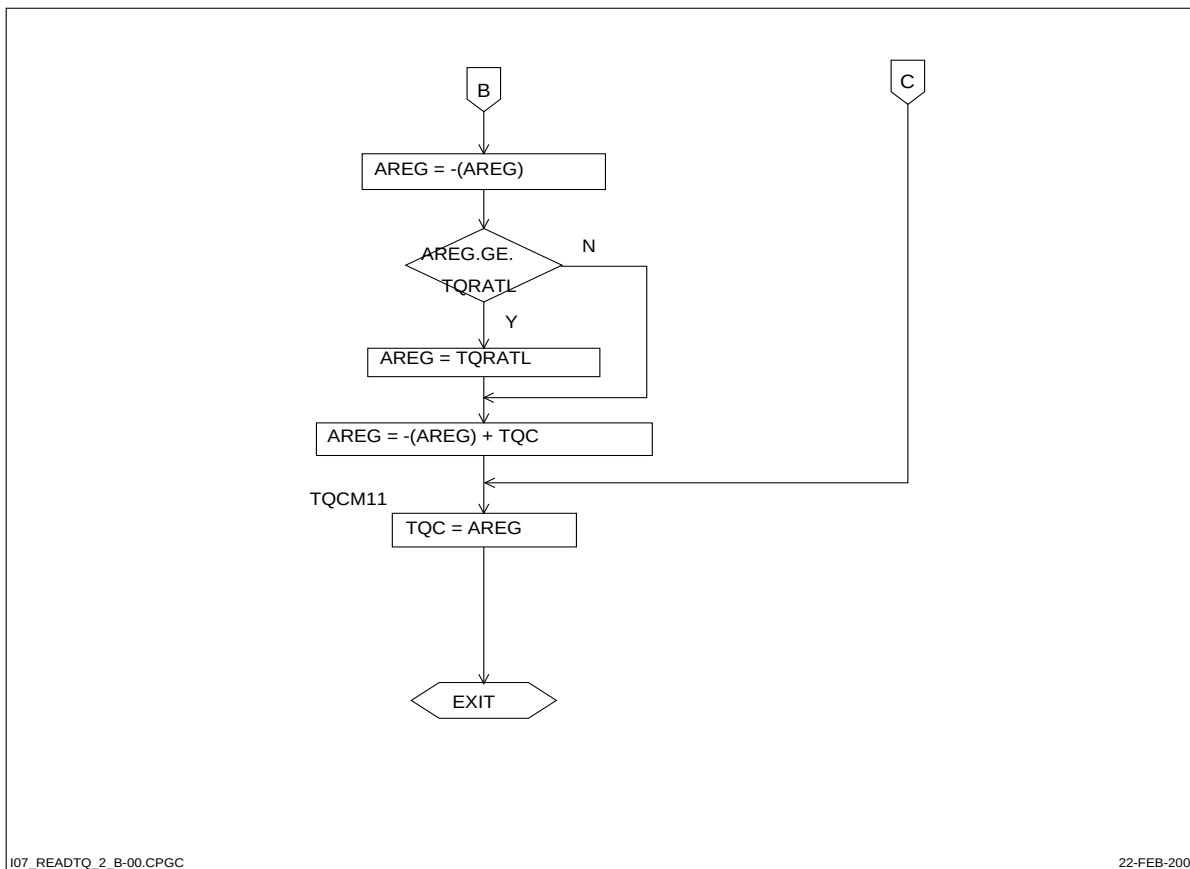
13.00 Figure 239: I07_READTQ_1_A



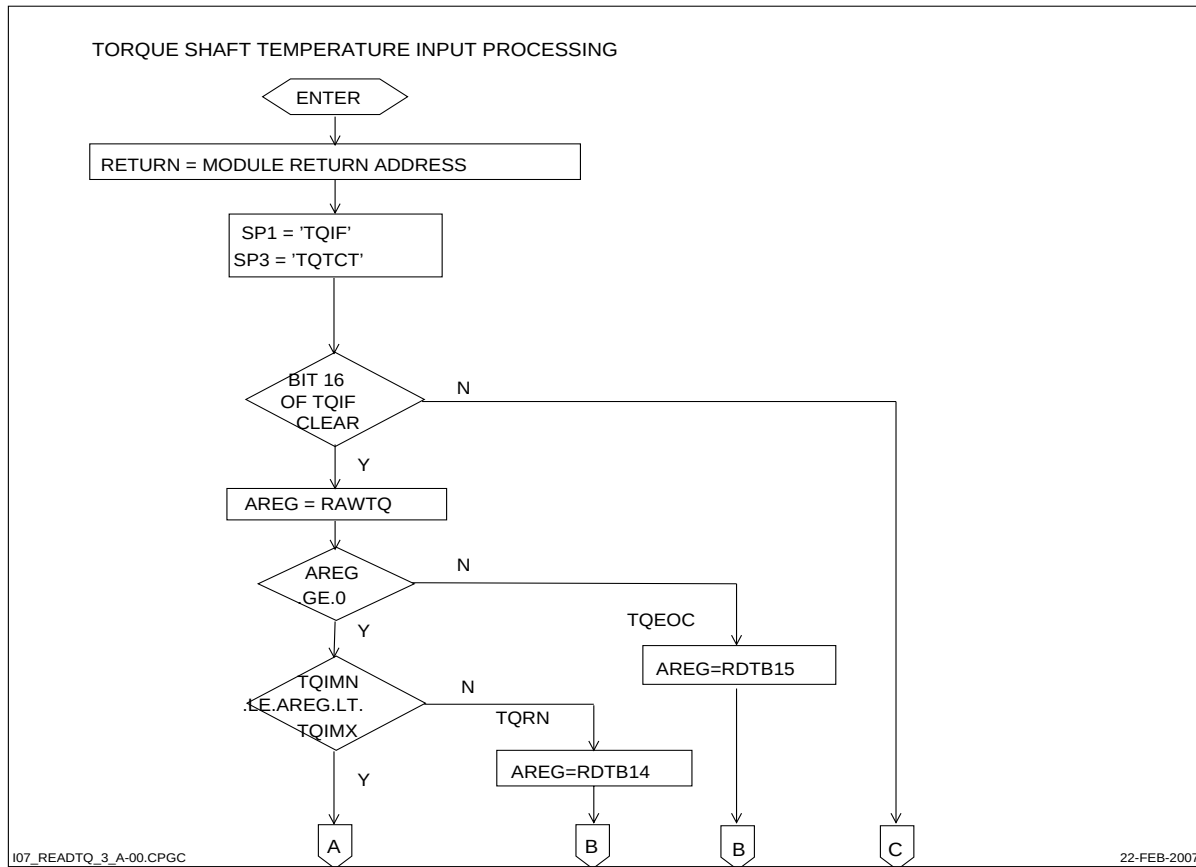
13.00 Figure 240: I07_READTQ_2_A



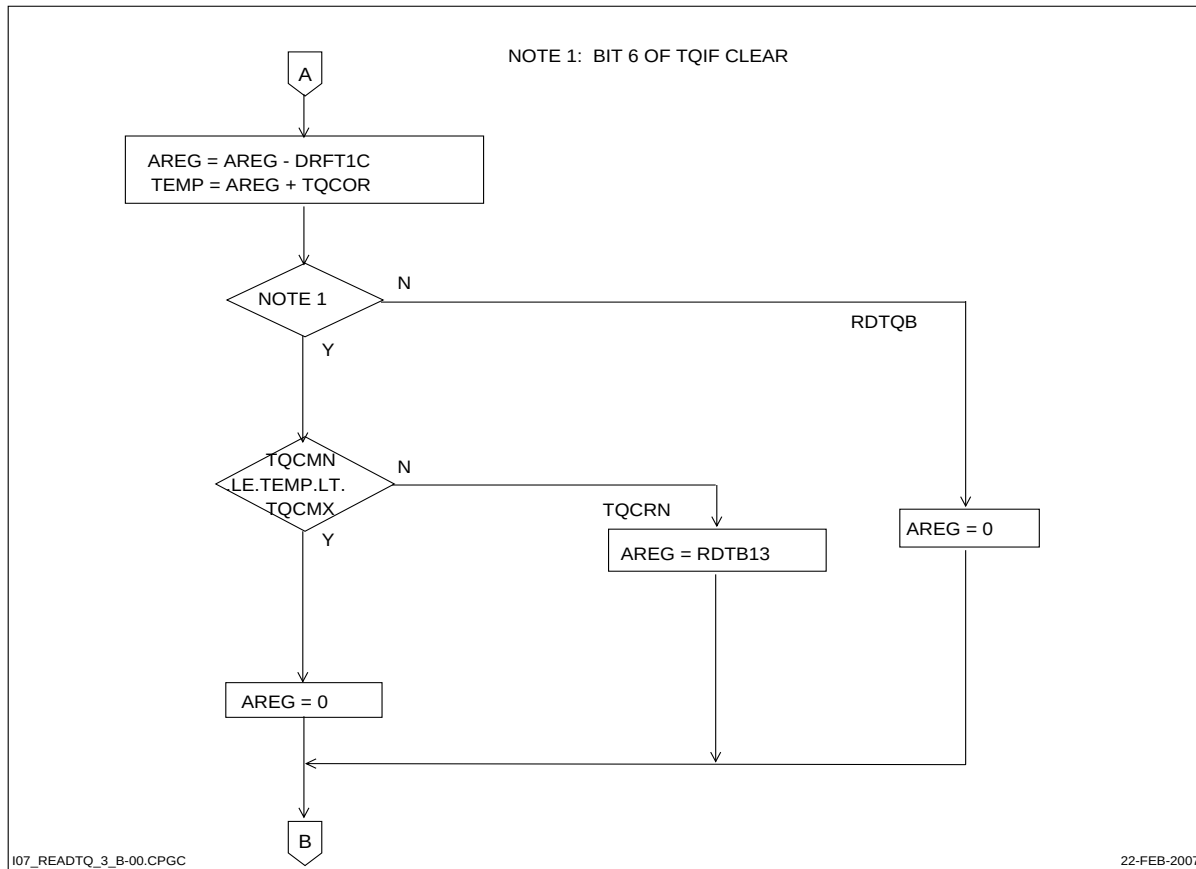
13.00 Figure 241: I07_READTQ_2_B



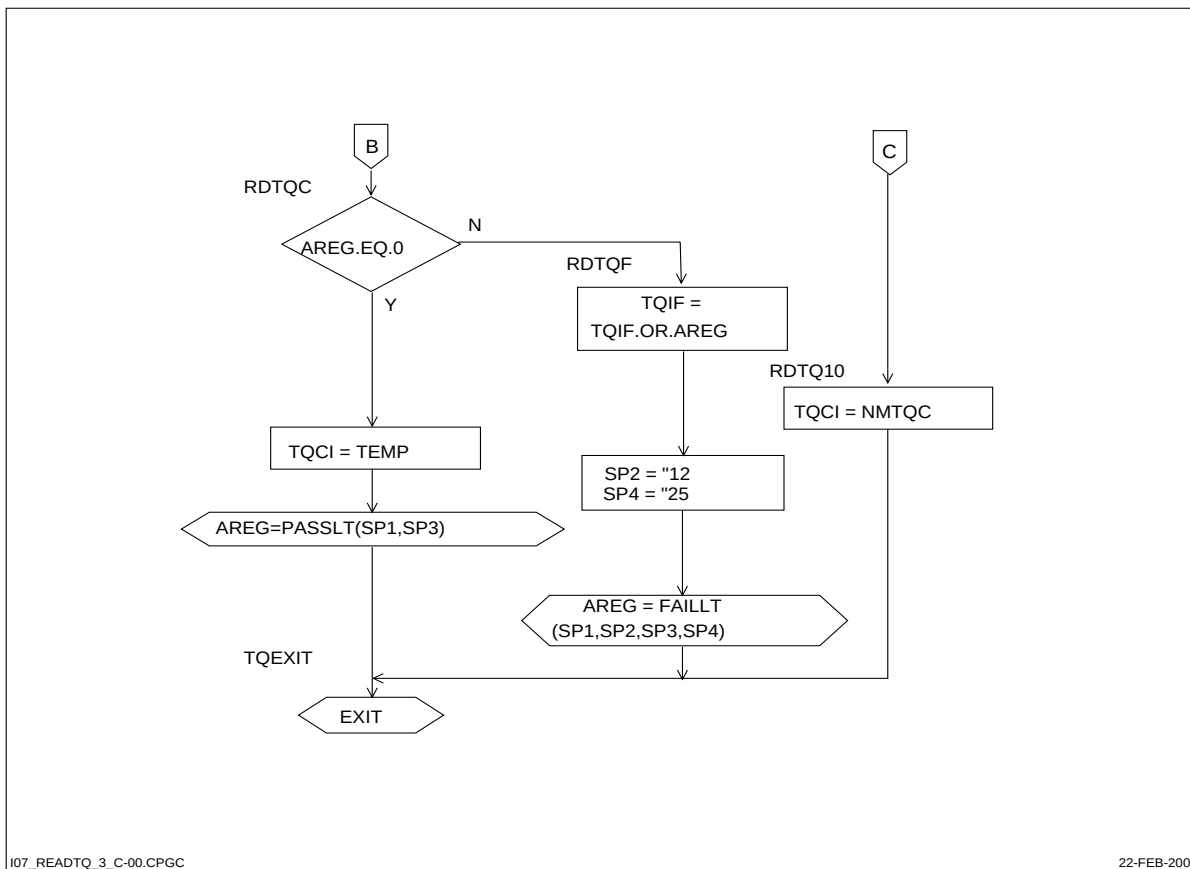
13.00 Figure 242: I07_READTQ_3_A



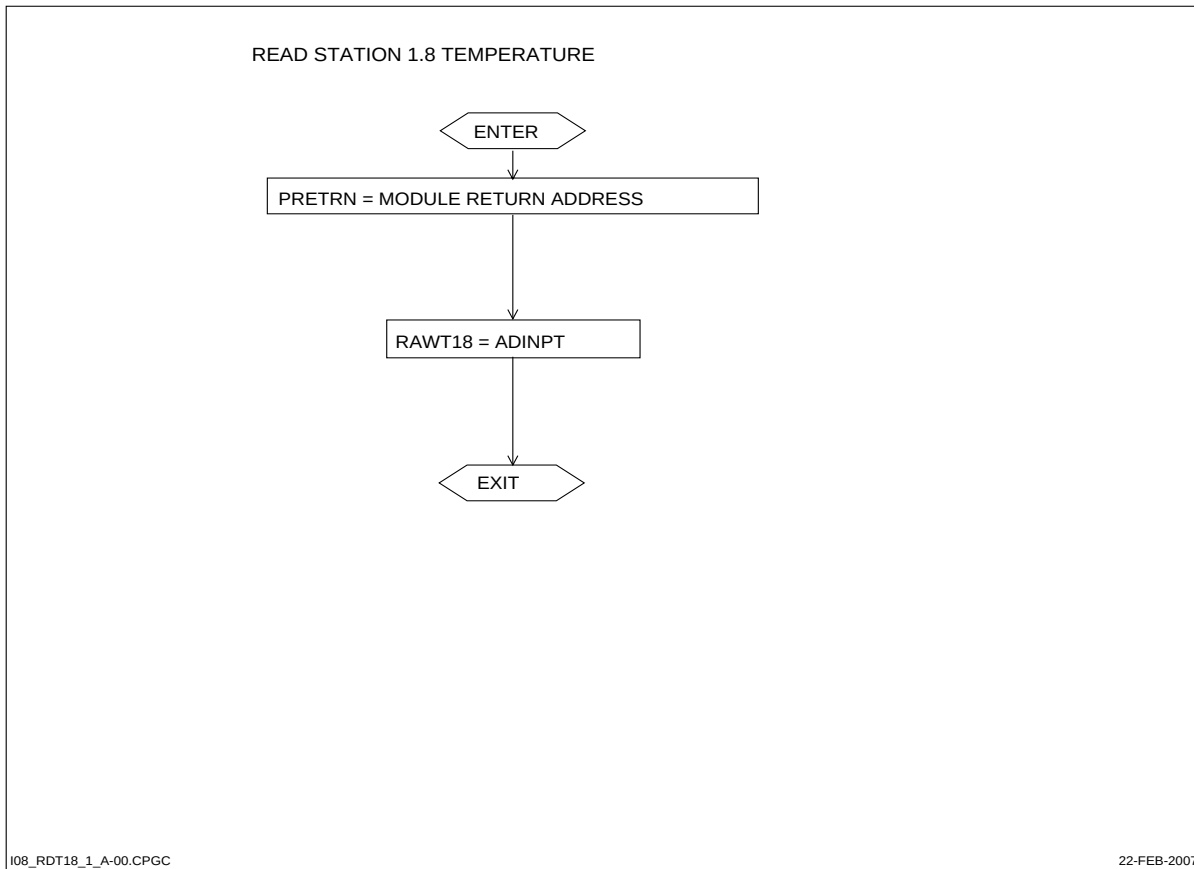
13.00 Figure 243: I07_READTQ_3_B



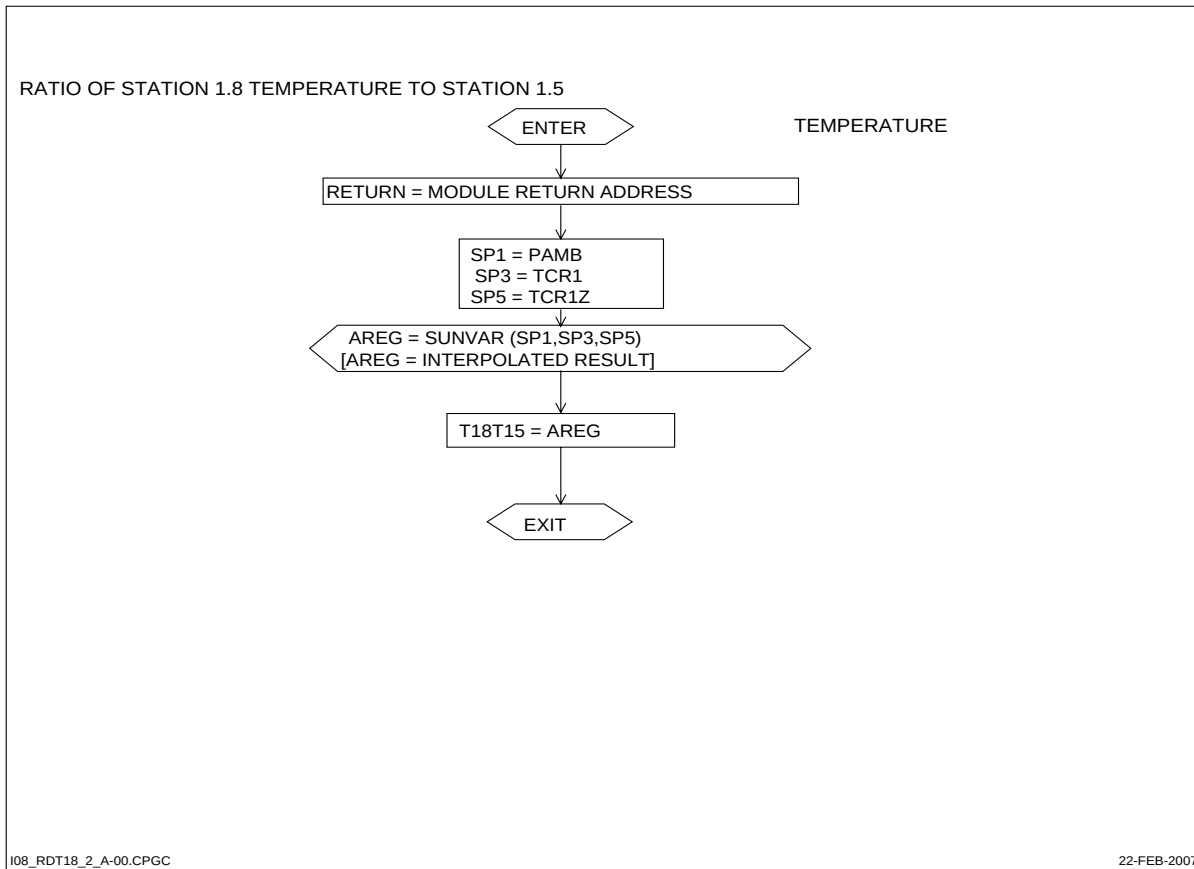
13.00 Figure 244: I07_READTQ_3_C



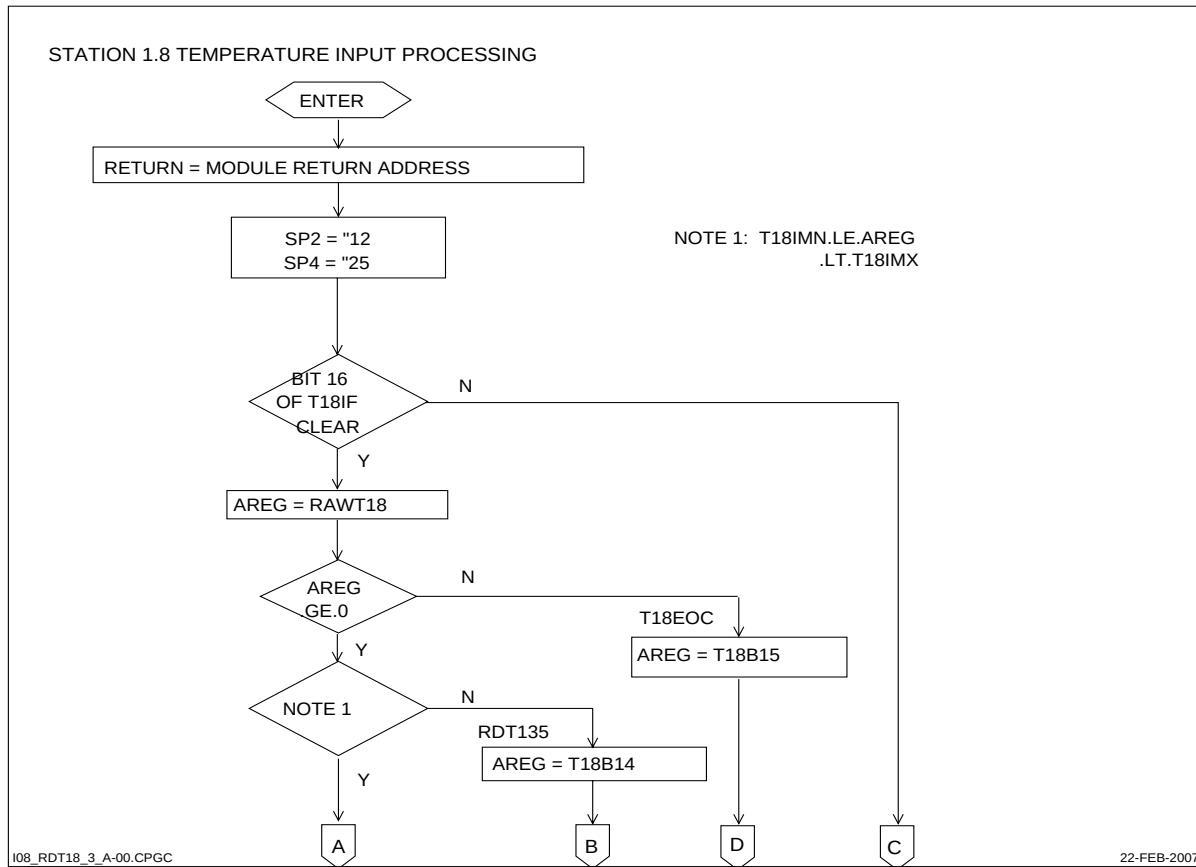
13.00 Figure 245: I08 _RDT18 _1 _A



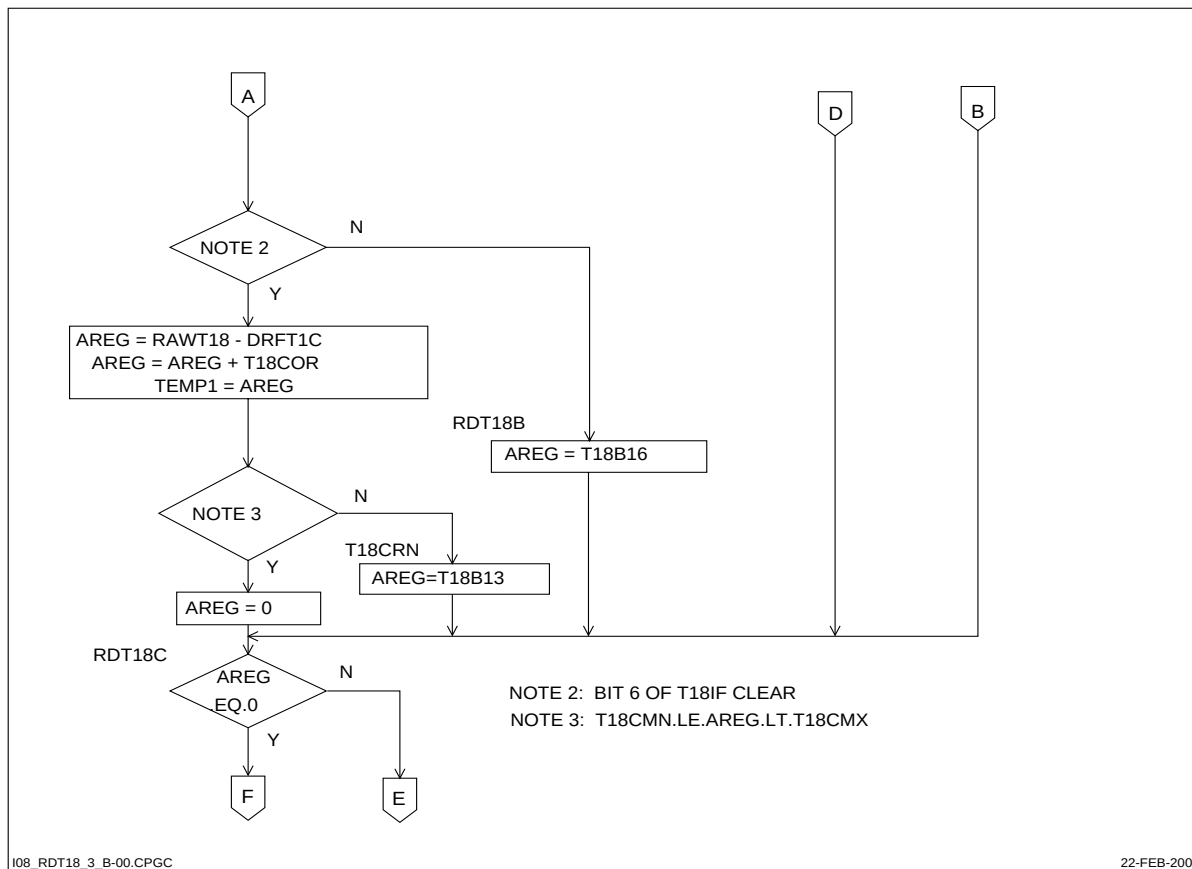
13.00 Figure 246: I08 _RDT18 _2 _A



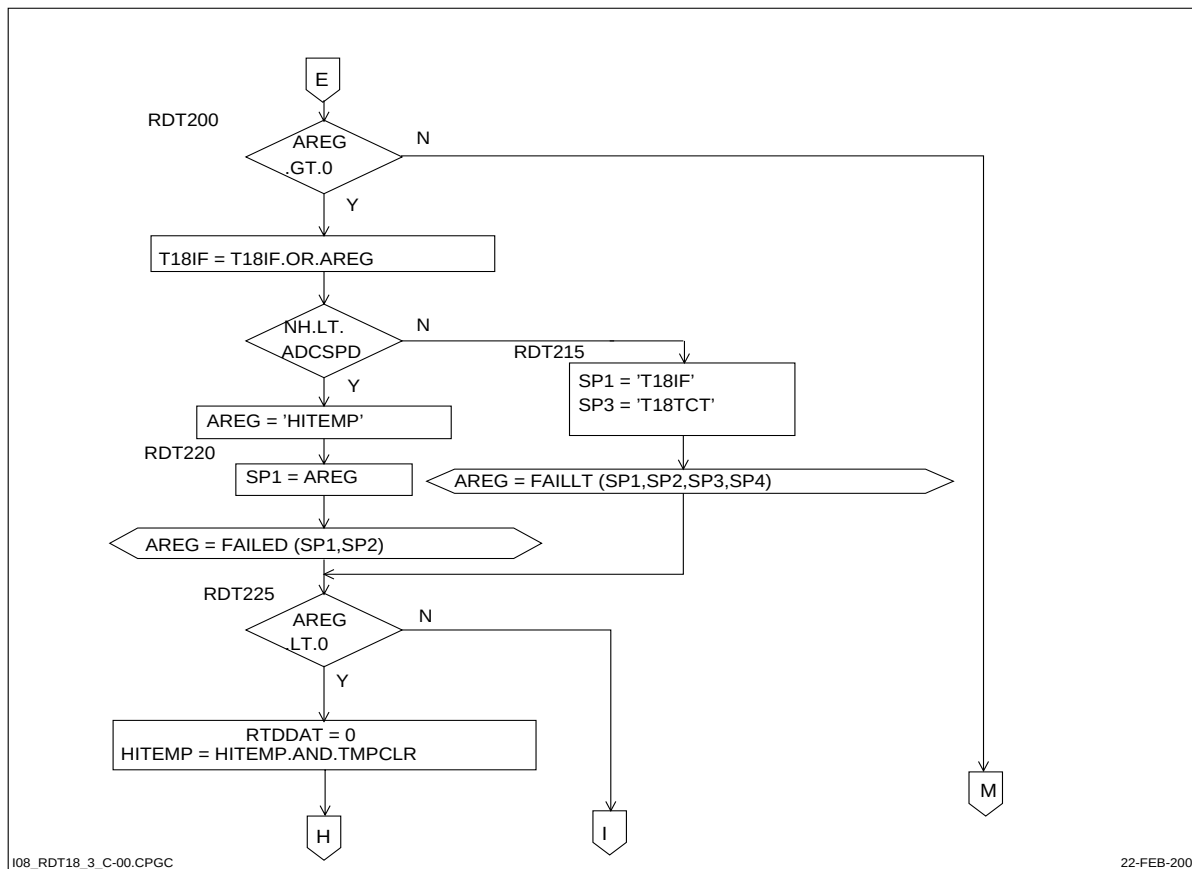
13.00 Figure 247: I08_RDT18_3_A



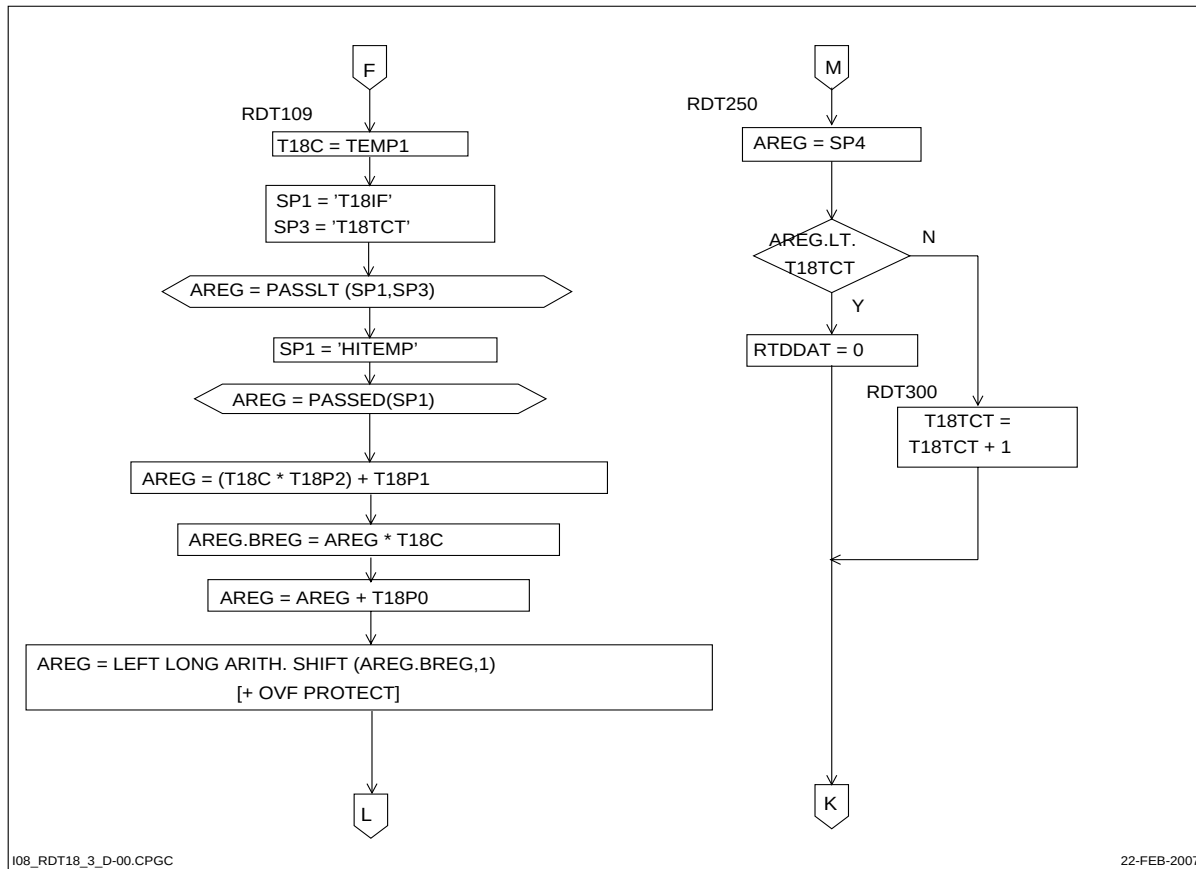
13.00 Figure 248: I08_RDT18_3_B



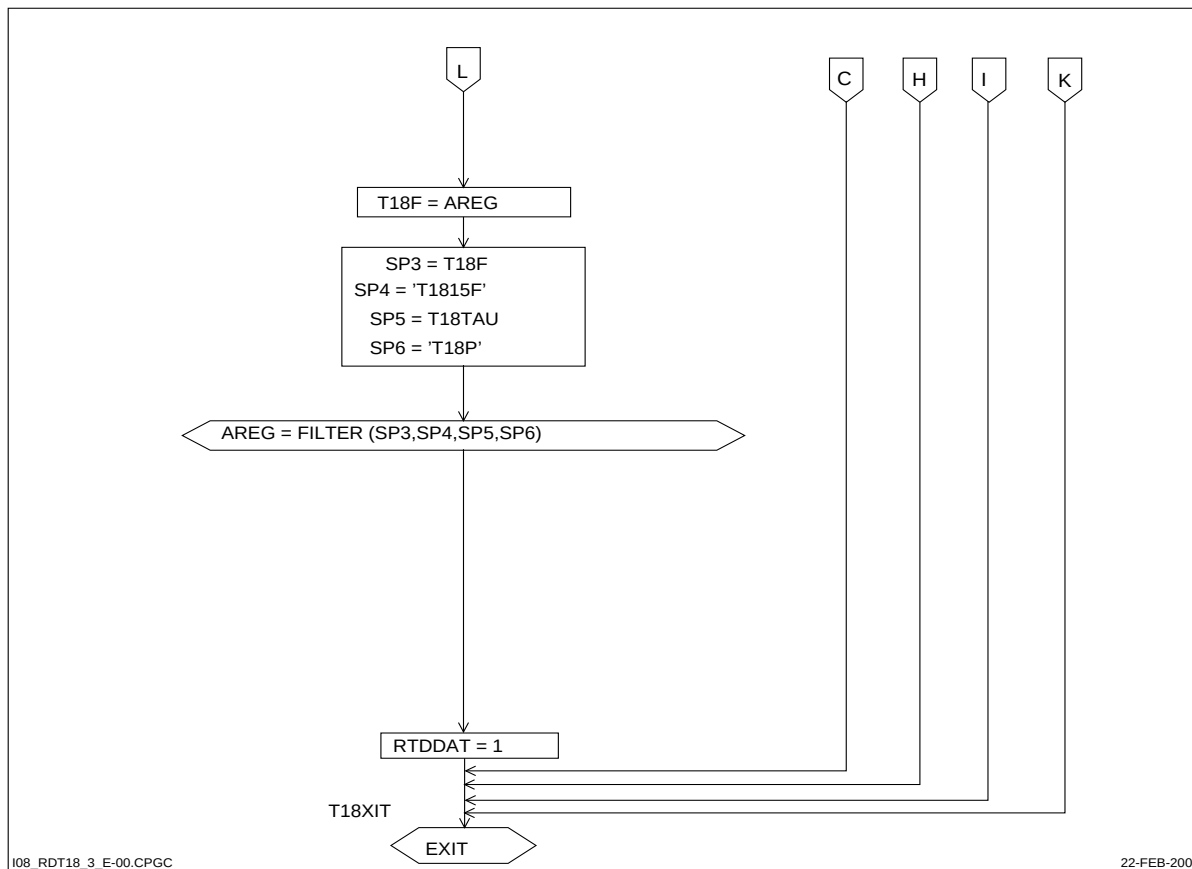
13.00 Figure 249: I08_RDT18_3_C



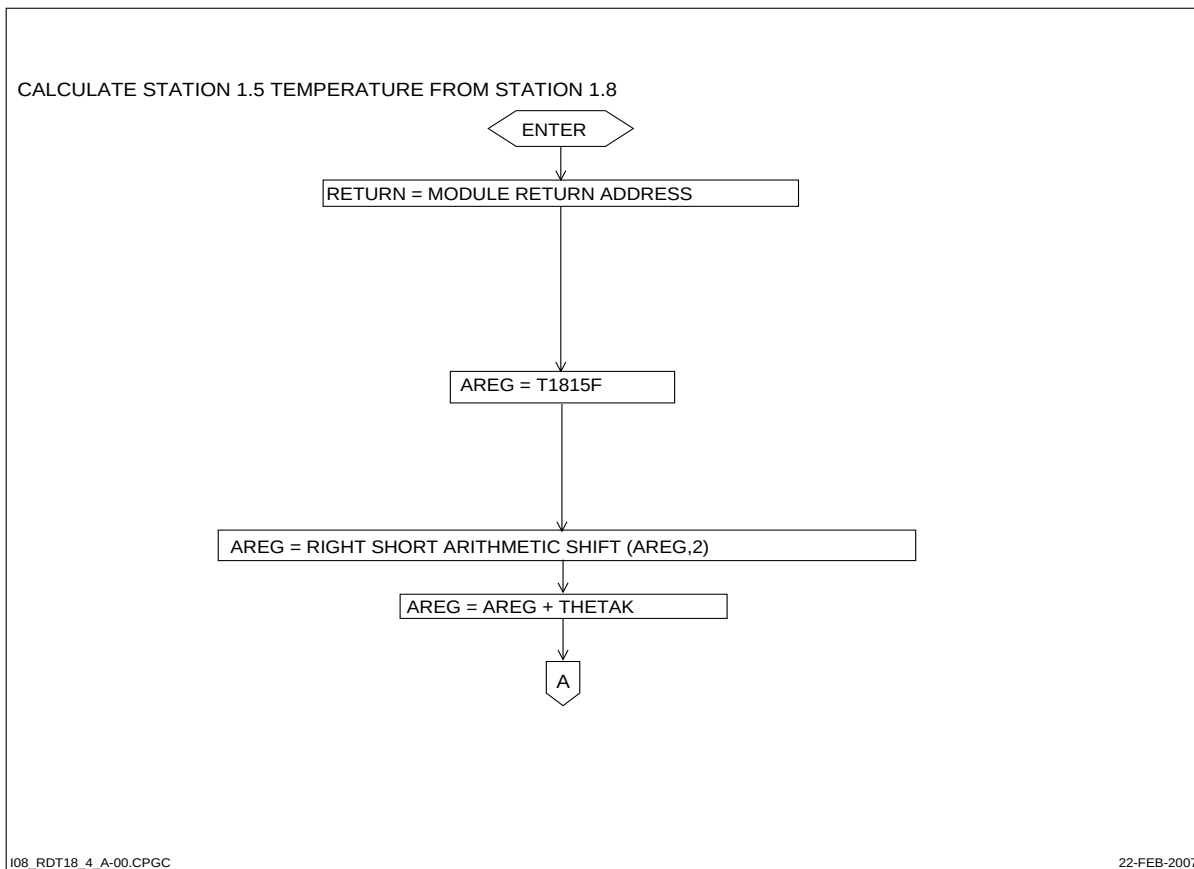
13.00 Figure 250: I08_RDT18_3_D



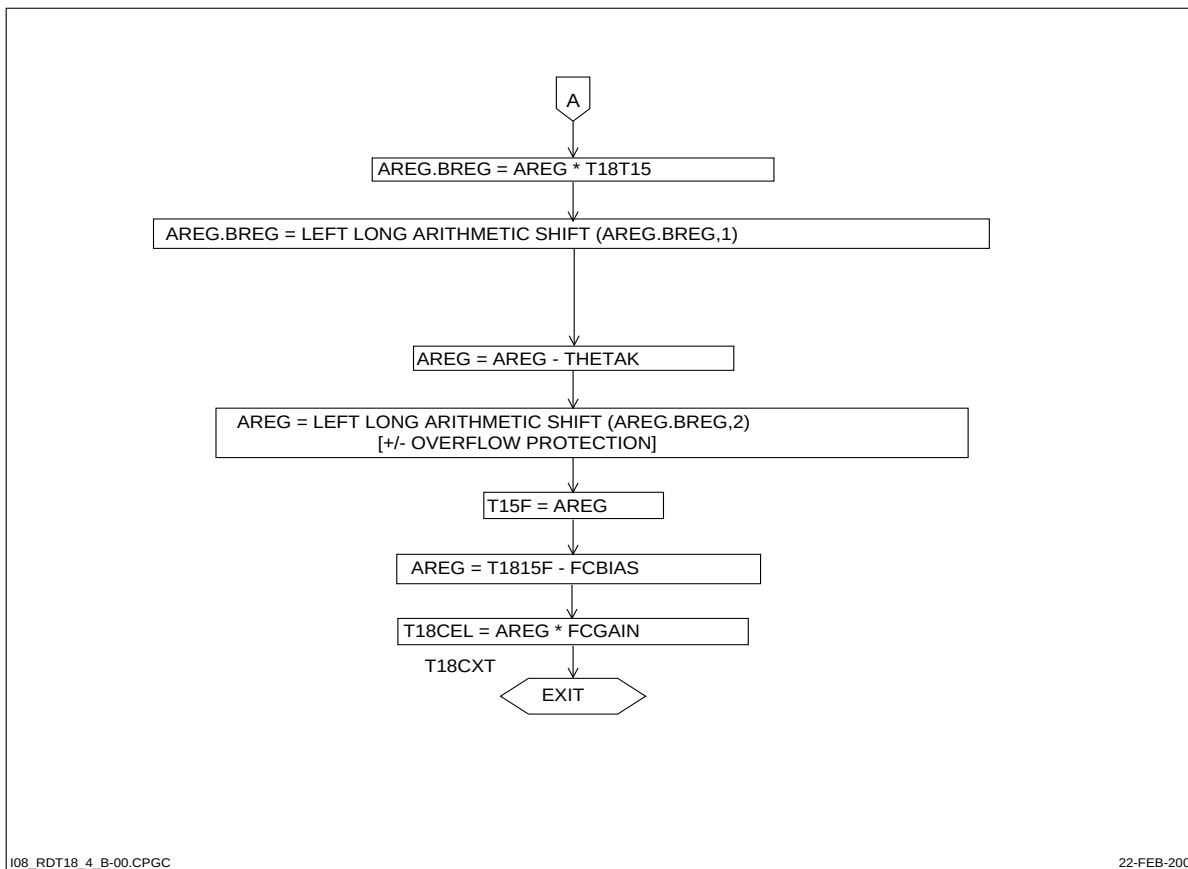
13.00 Figure 251: I08 _RDT18 _3 _E



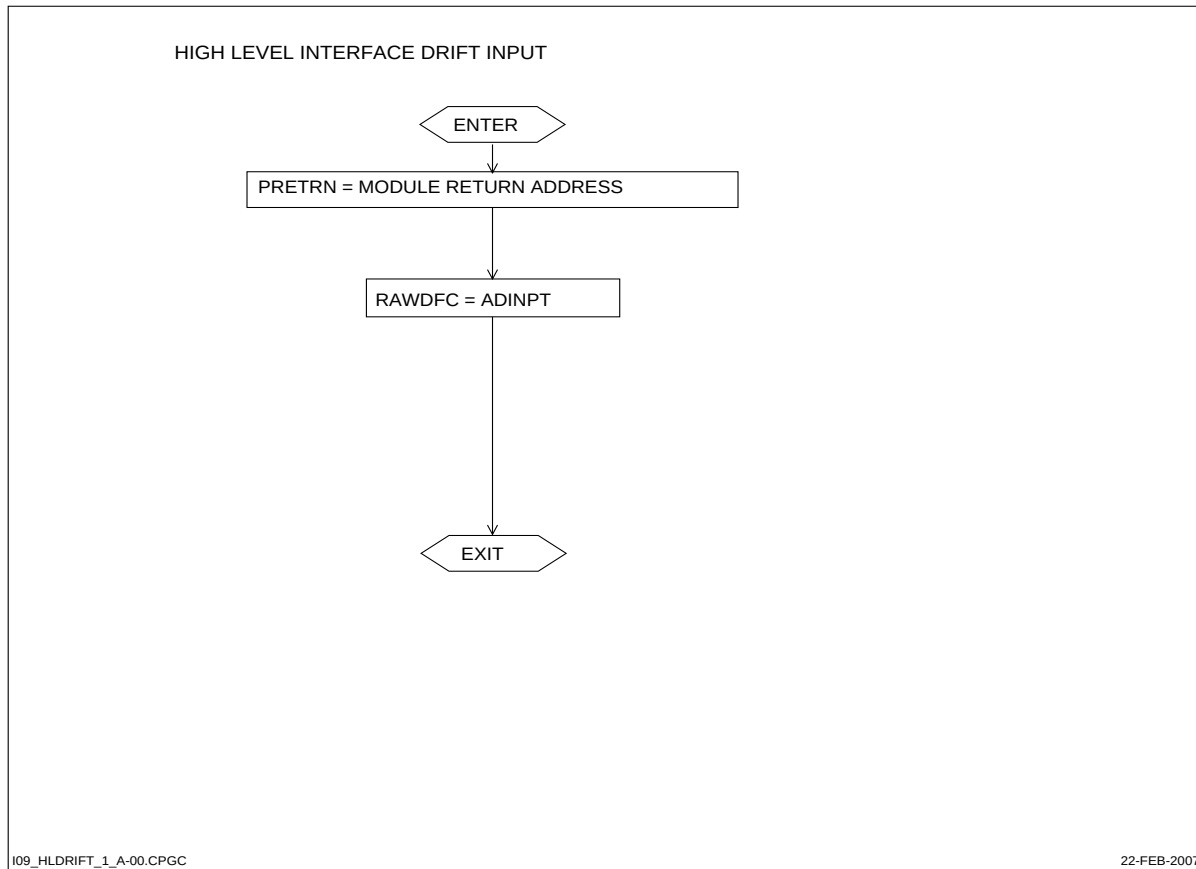
13.00 Figure 252: I08 _RDT18 _4 _A



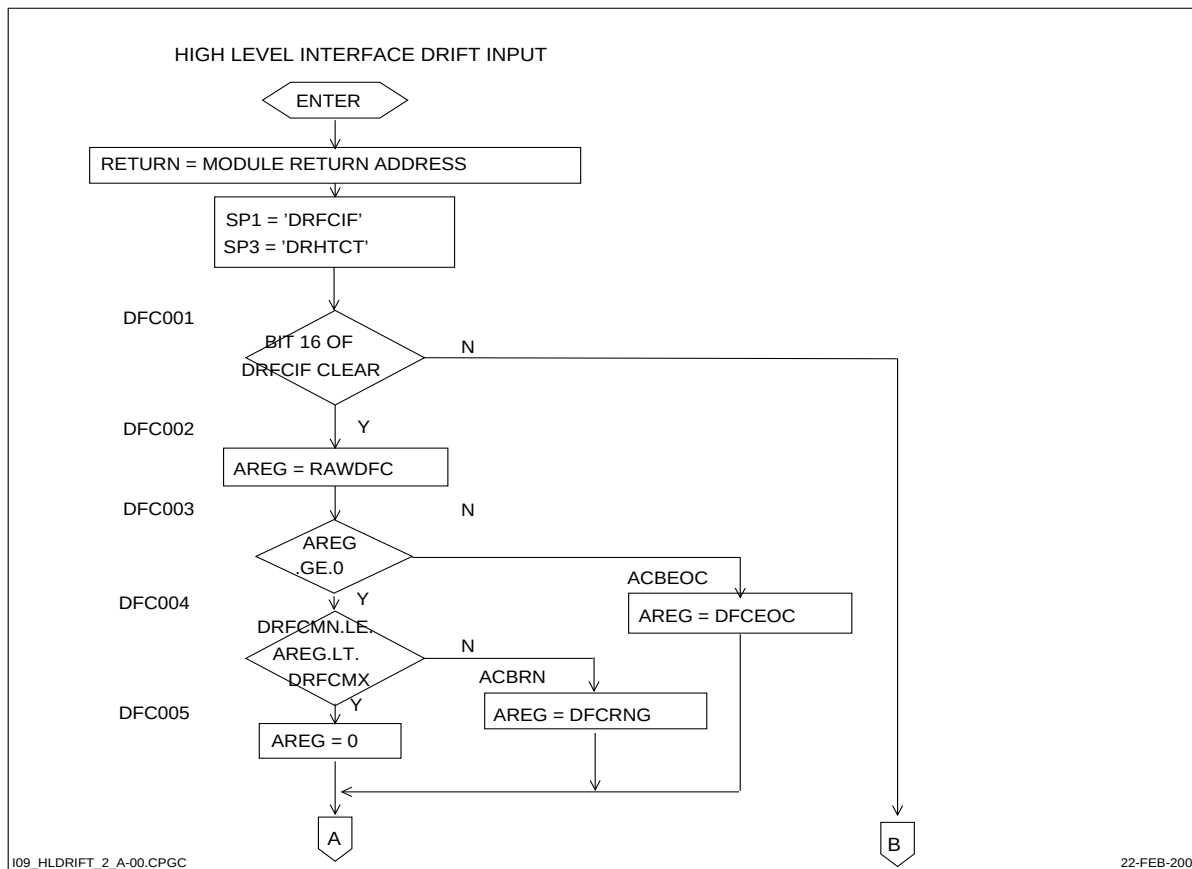
13.00 Figure 253: I08_RDT18_4_B



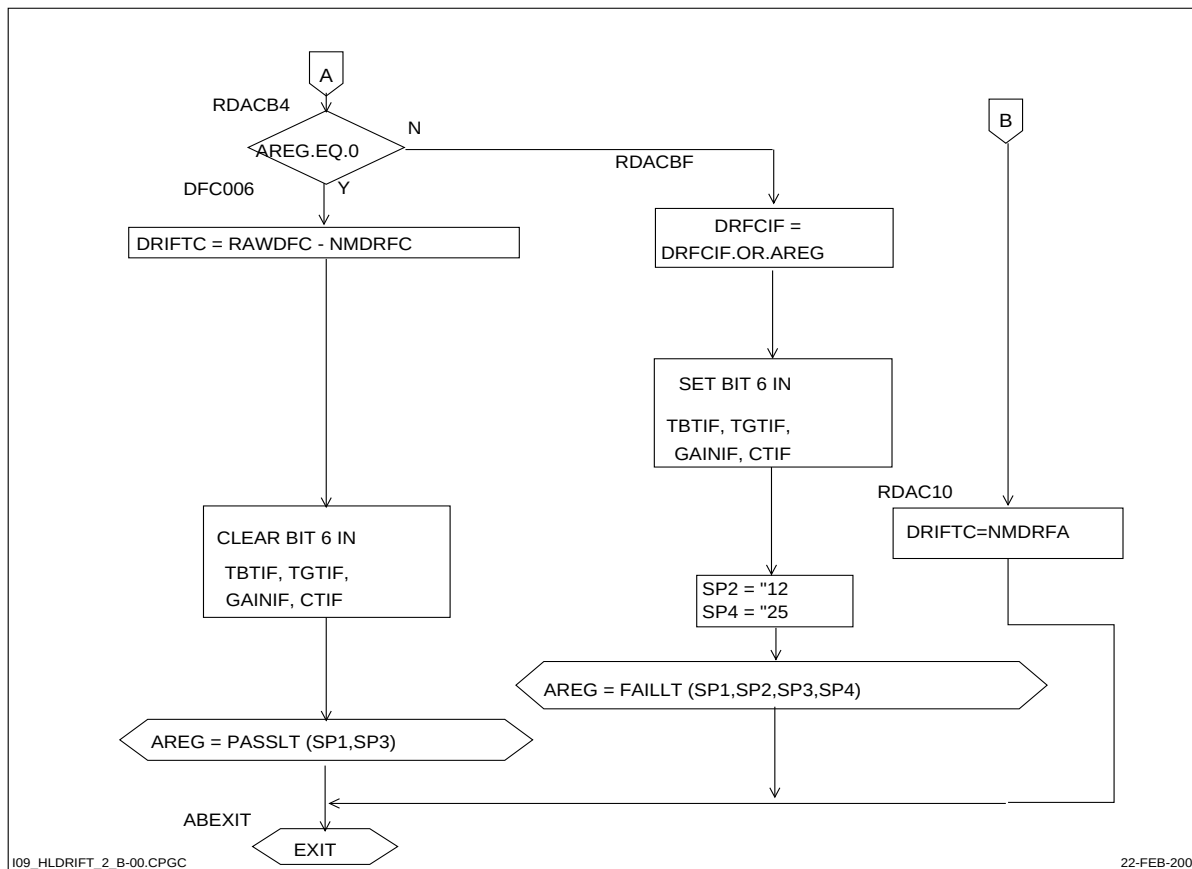
13.00 Figure 254: I09_HLDRIFT_1_A



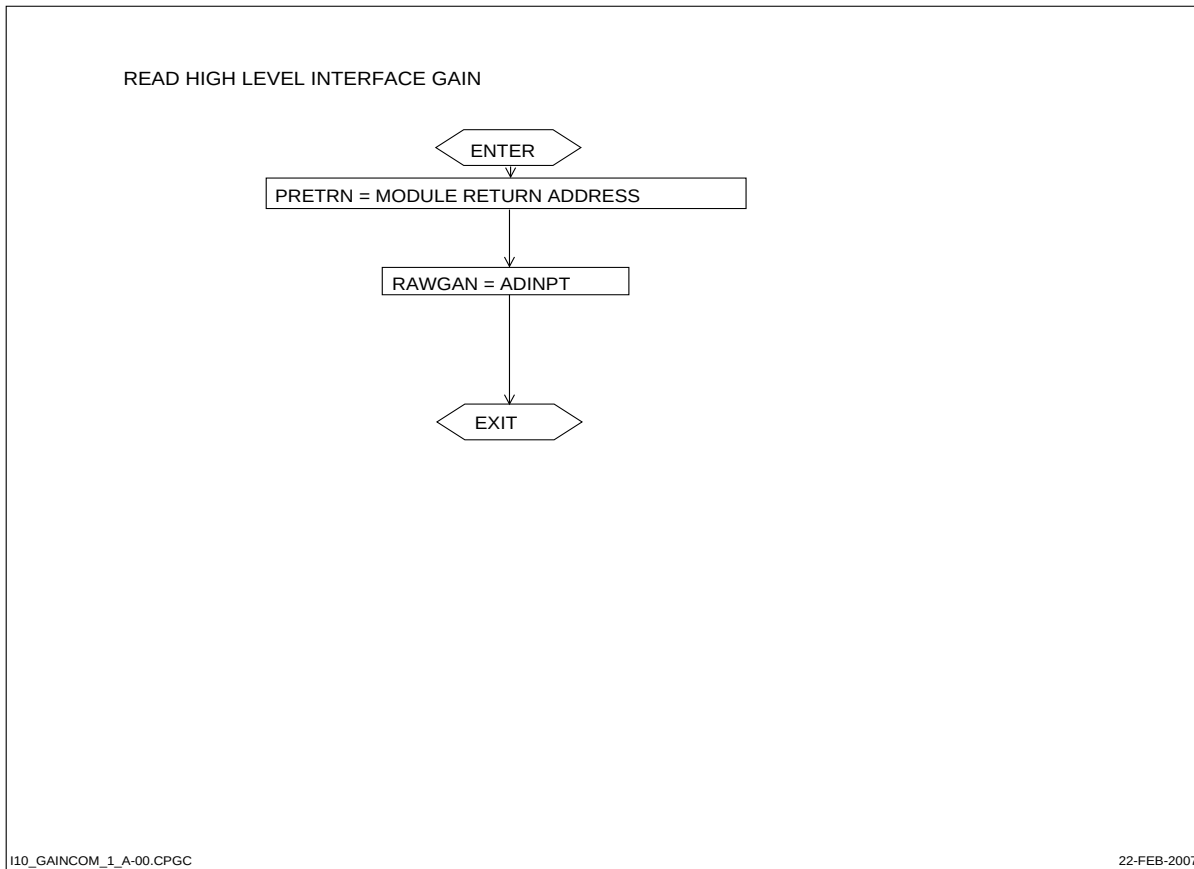
13.00 Figure 255: I09_HLDRIFT_2_A



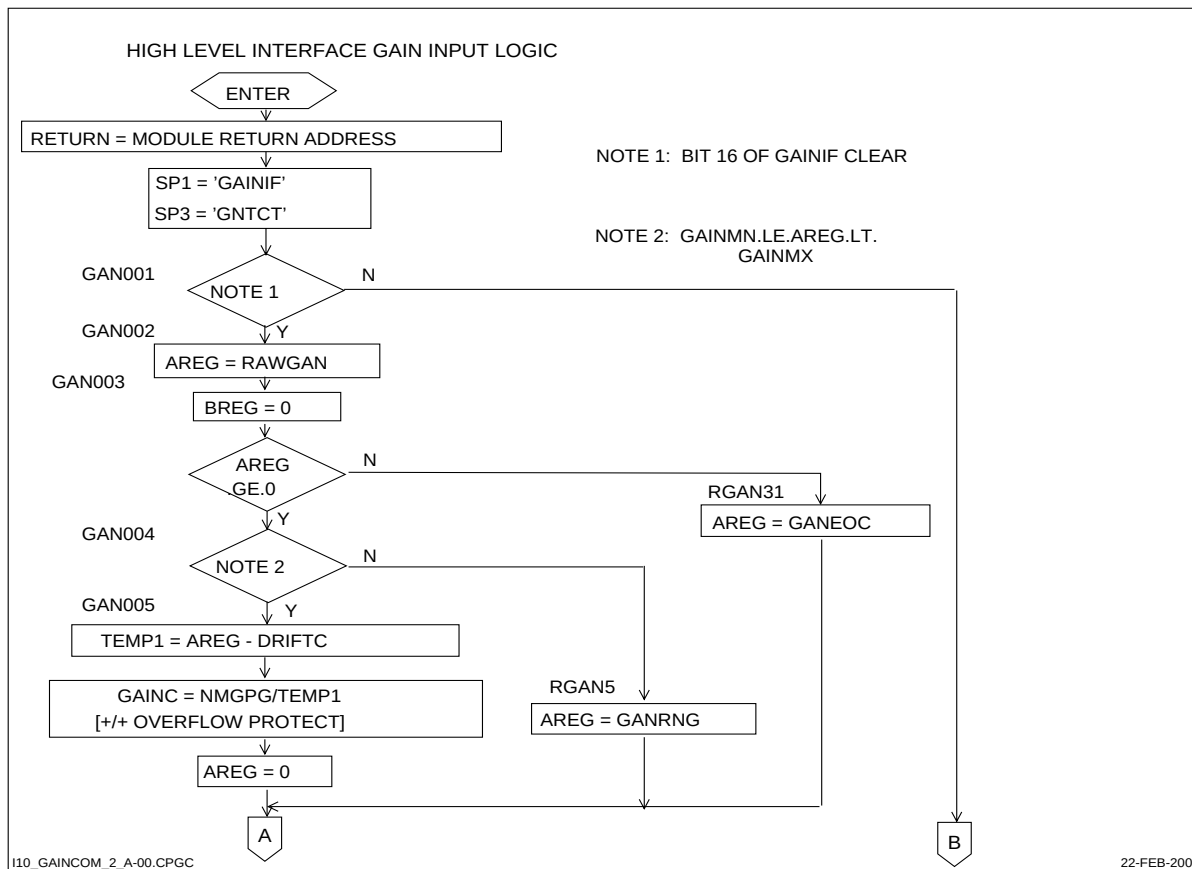
13.00 Figure 256: I09_HLDRIFT_2_B



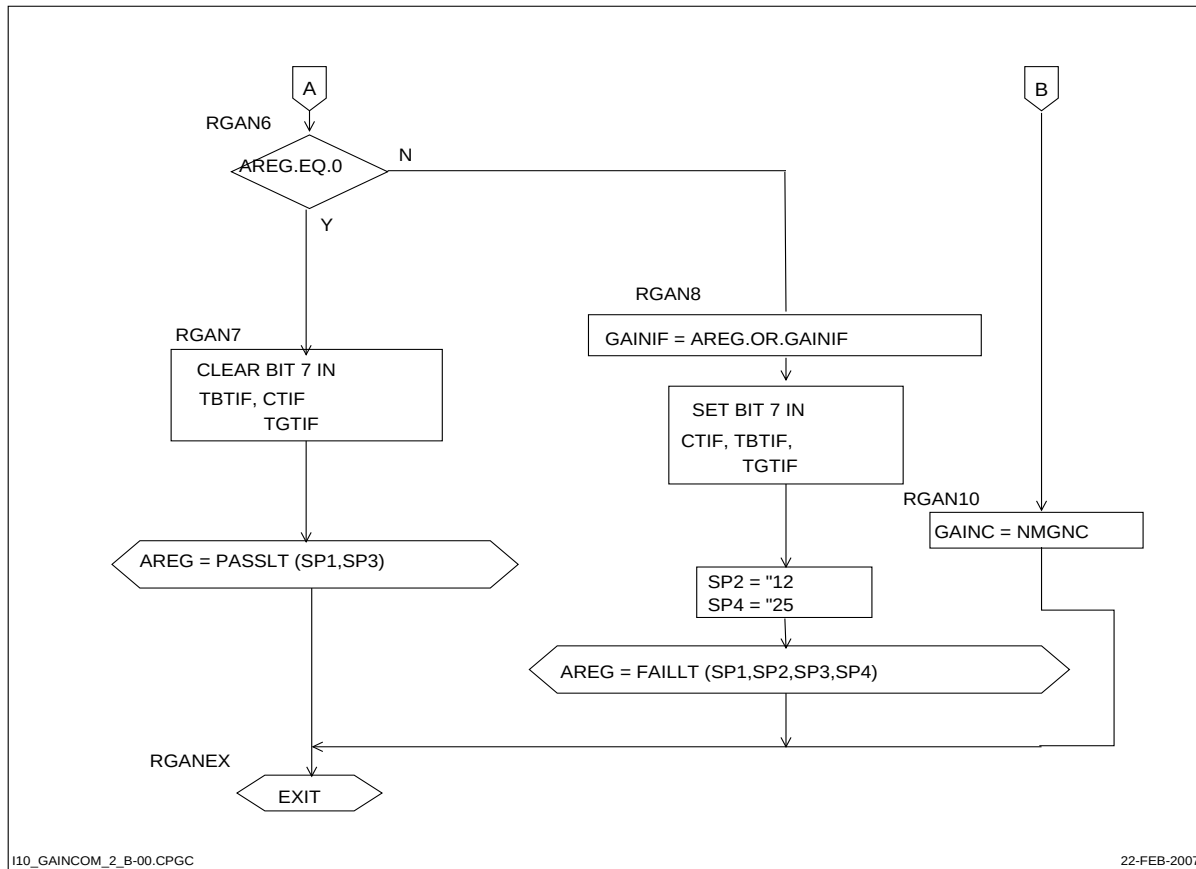
13.00 Figure 257: I10 _GAINCOM _1 _A



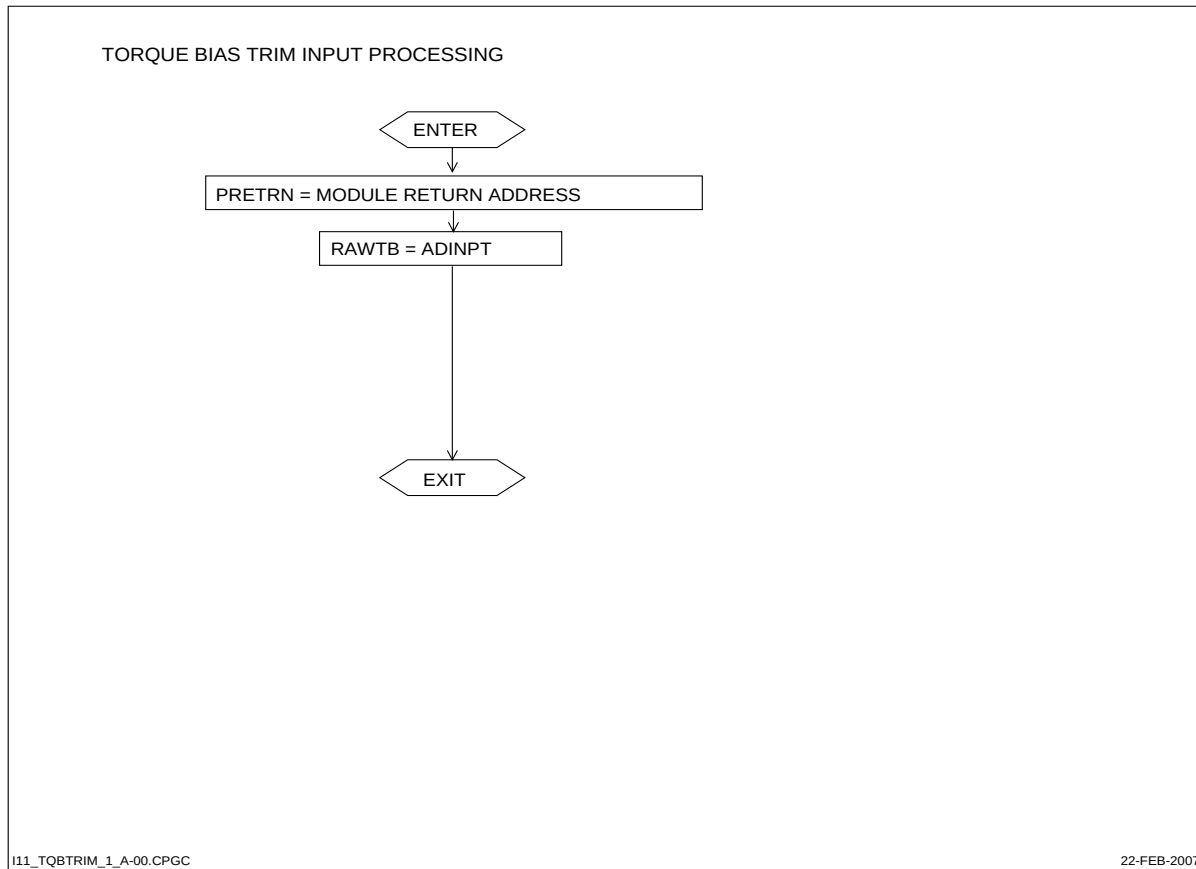
13.00 Figure 258: I10_GAINCOM_2_A



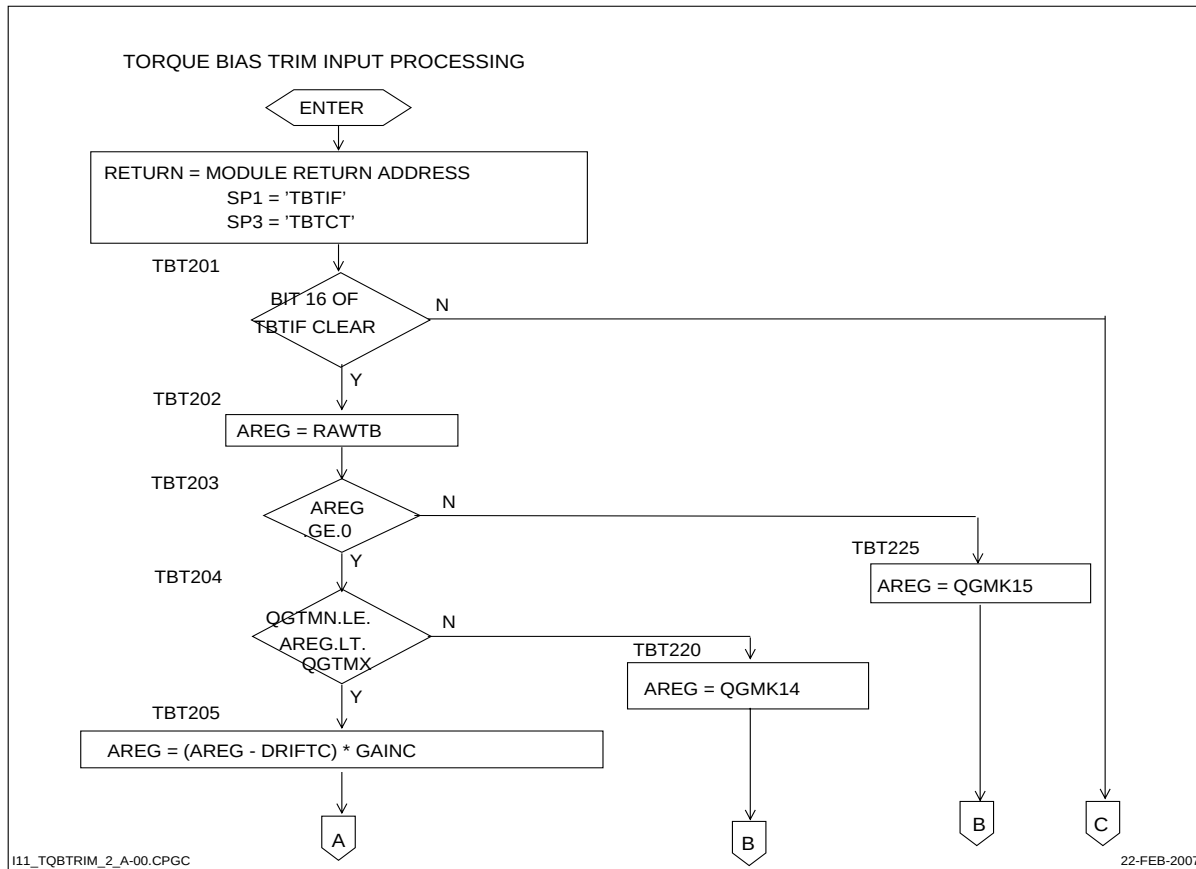
13.00 Figure 259: I10_GAINCOM_2_B



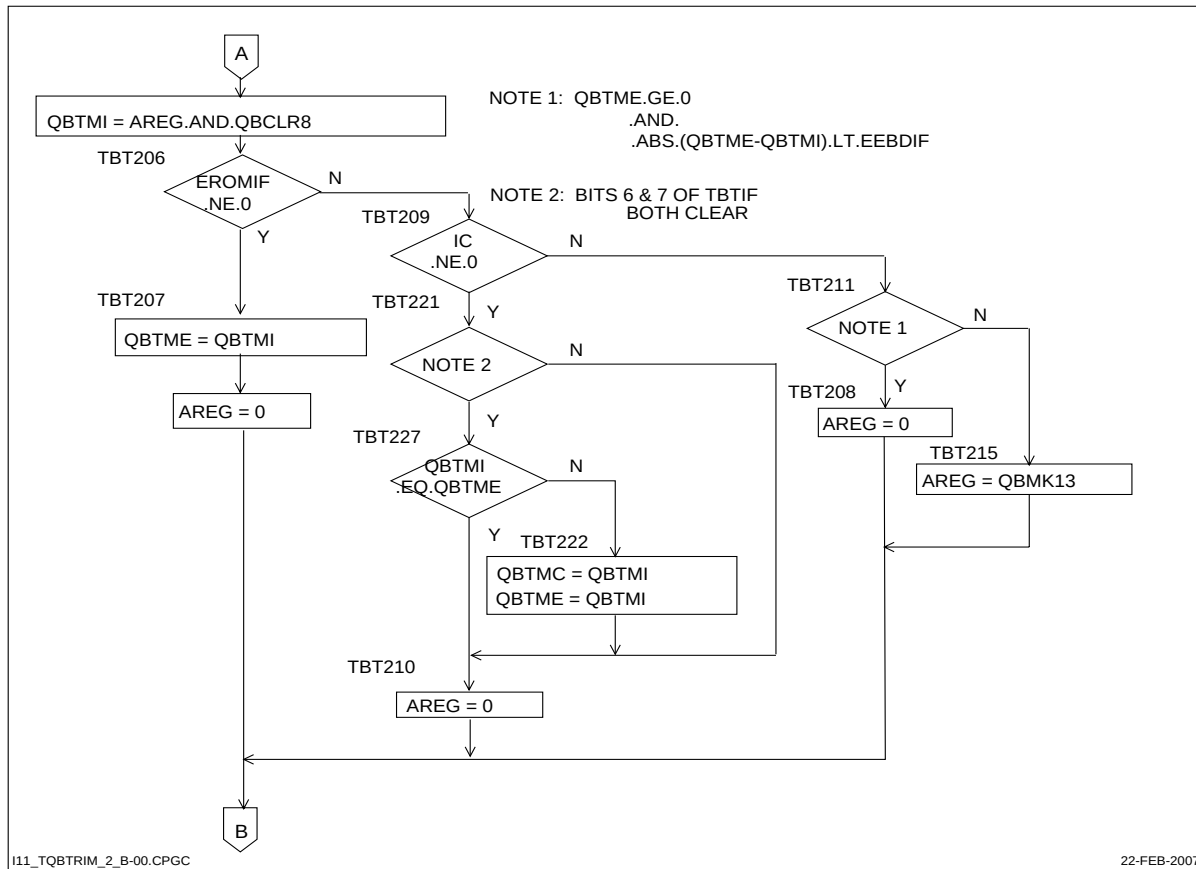
13.00 Figure 260: I11 _TQBTRIM _1 _A



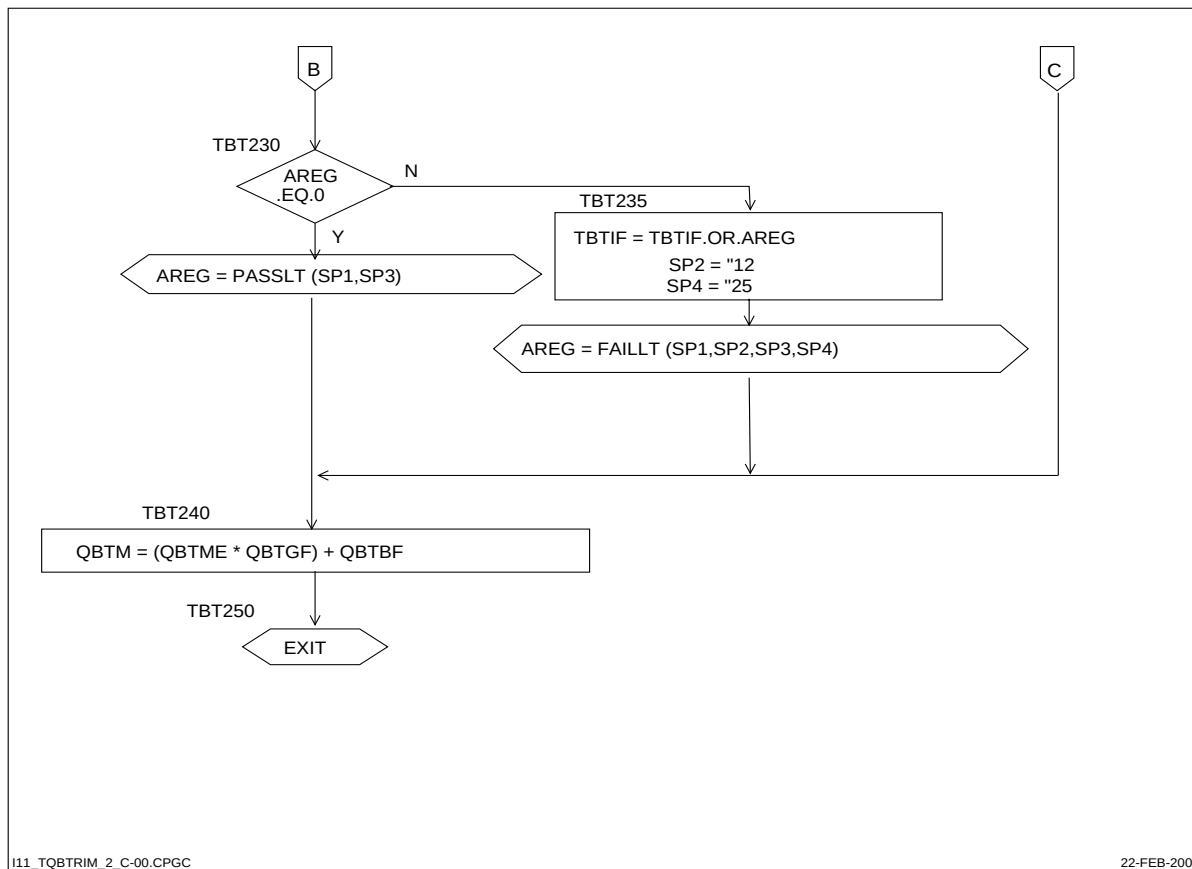
13.00 Figure 261: I11_TQBTRIM_2_A



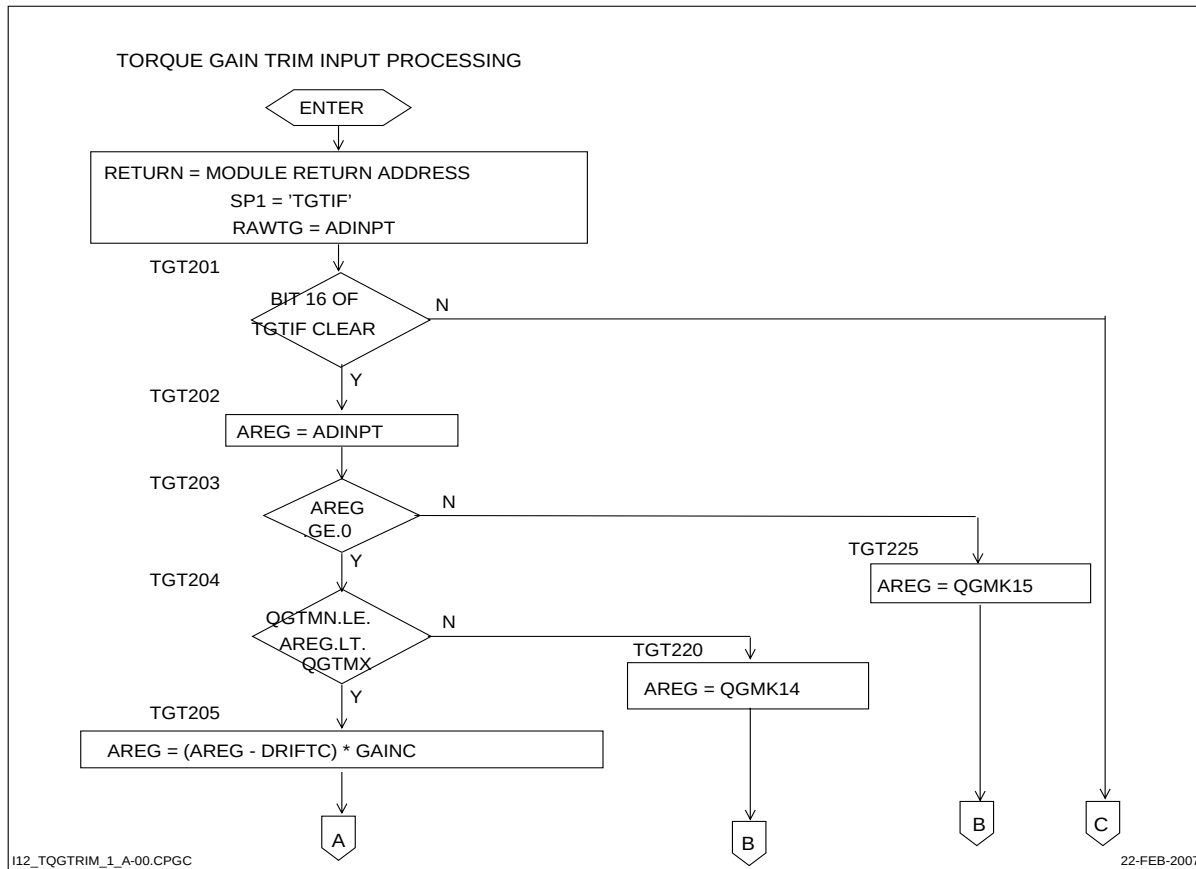
13.00 Figure 262: I11_TQBTRIM_2_B



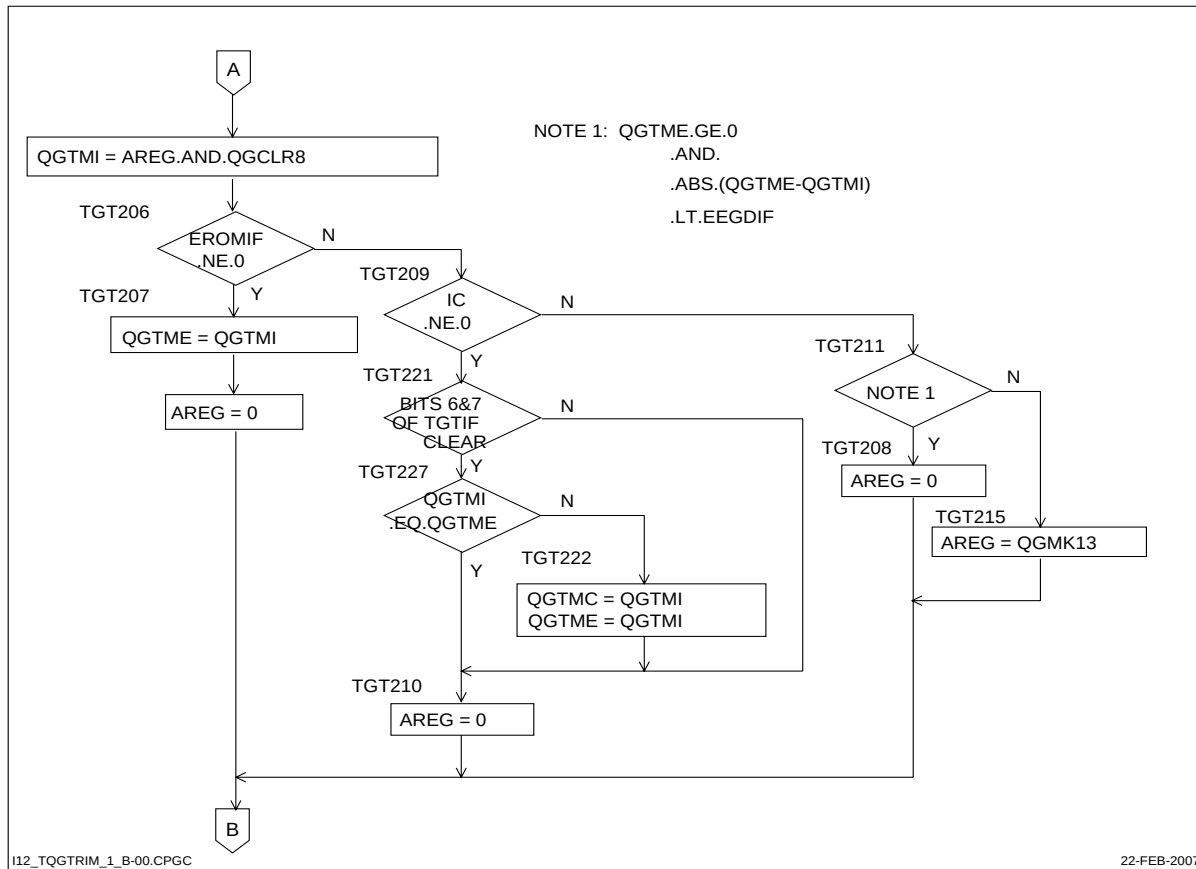
13.00 Figure 263: I11_TQBTRIM_2_C



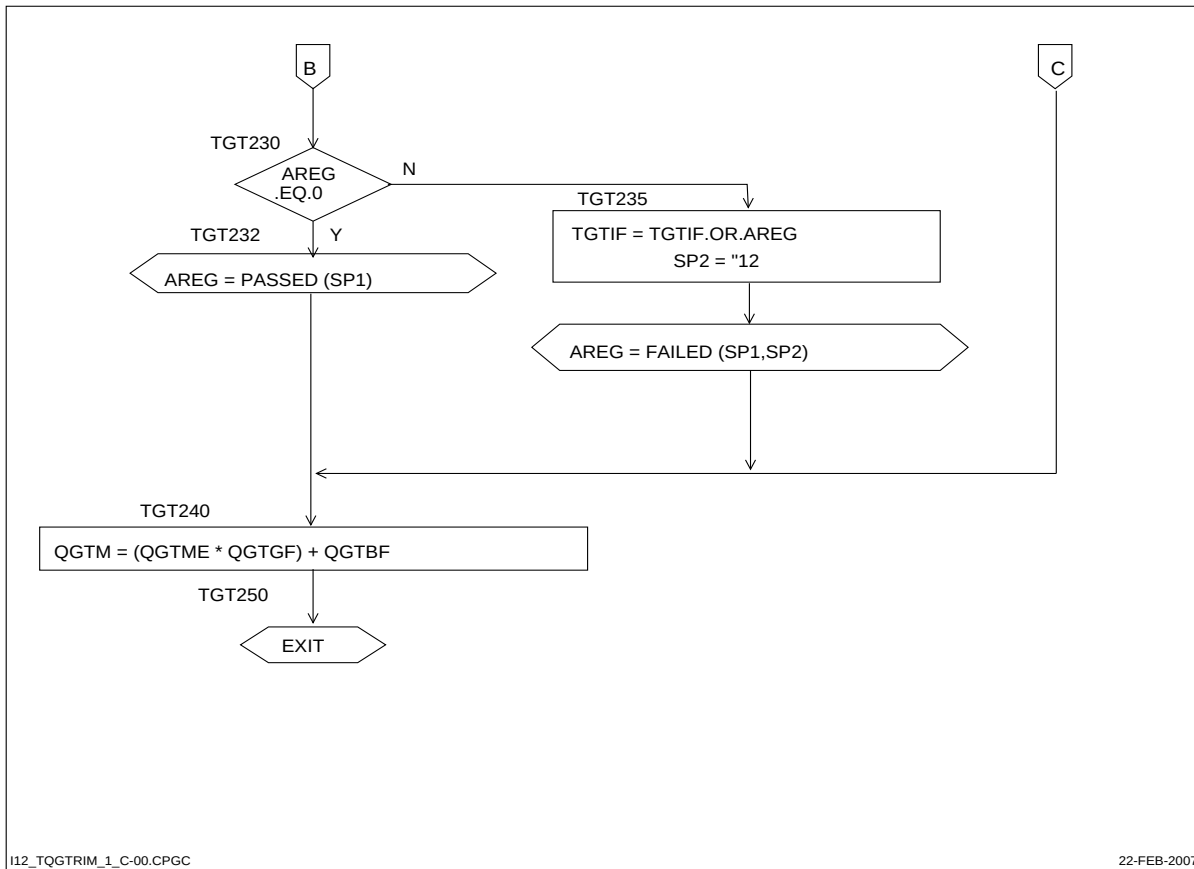
13.00 Figure 264: I12_TQGTRIM_1_A



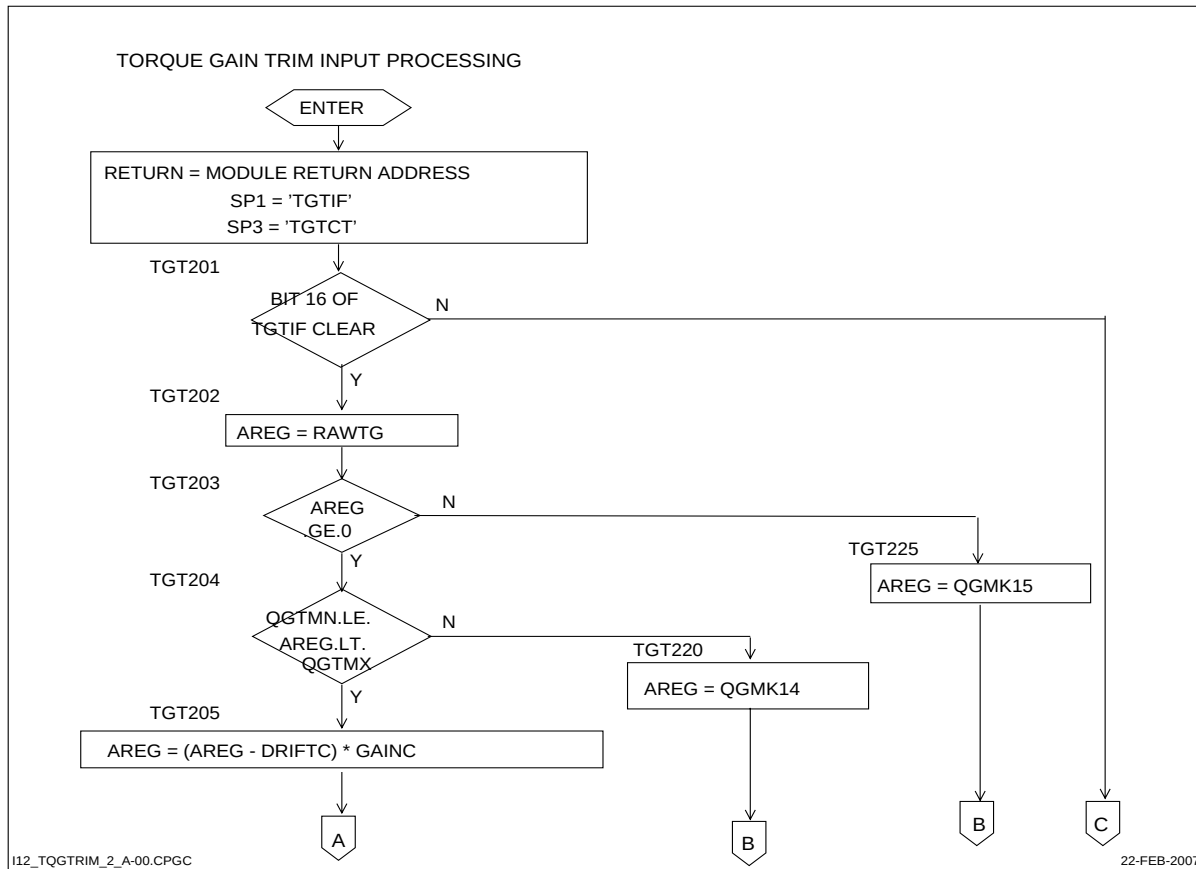
13.00 Figure 265: I12_TQGTRIM_1_B



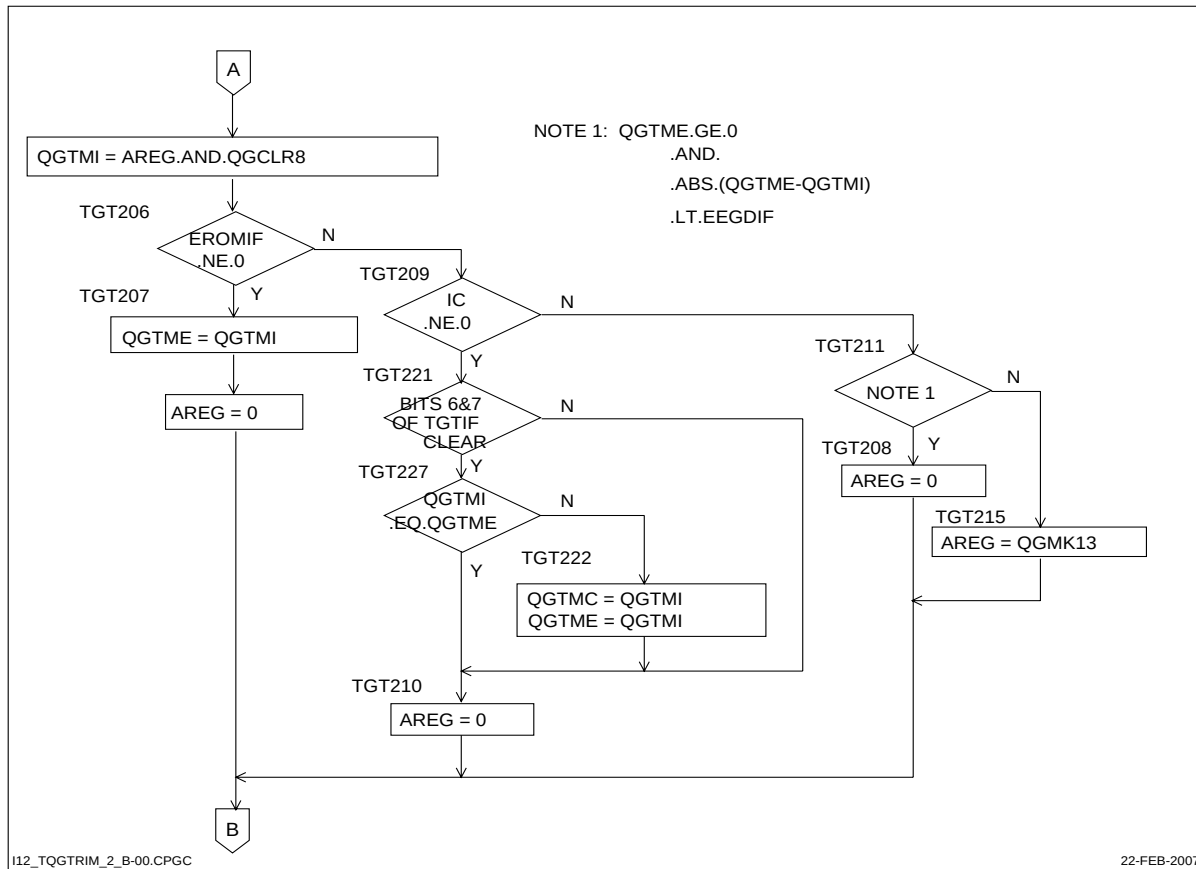
13.00 Figure 266: I12_TQGTRIM_1_C



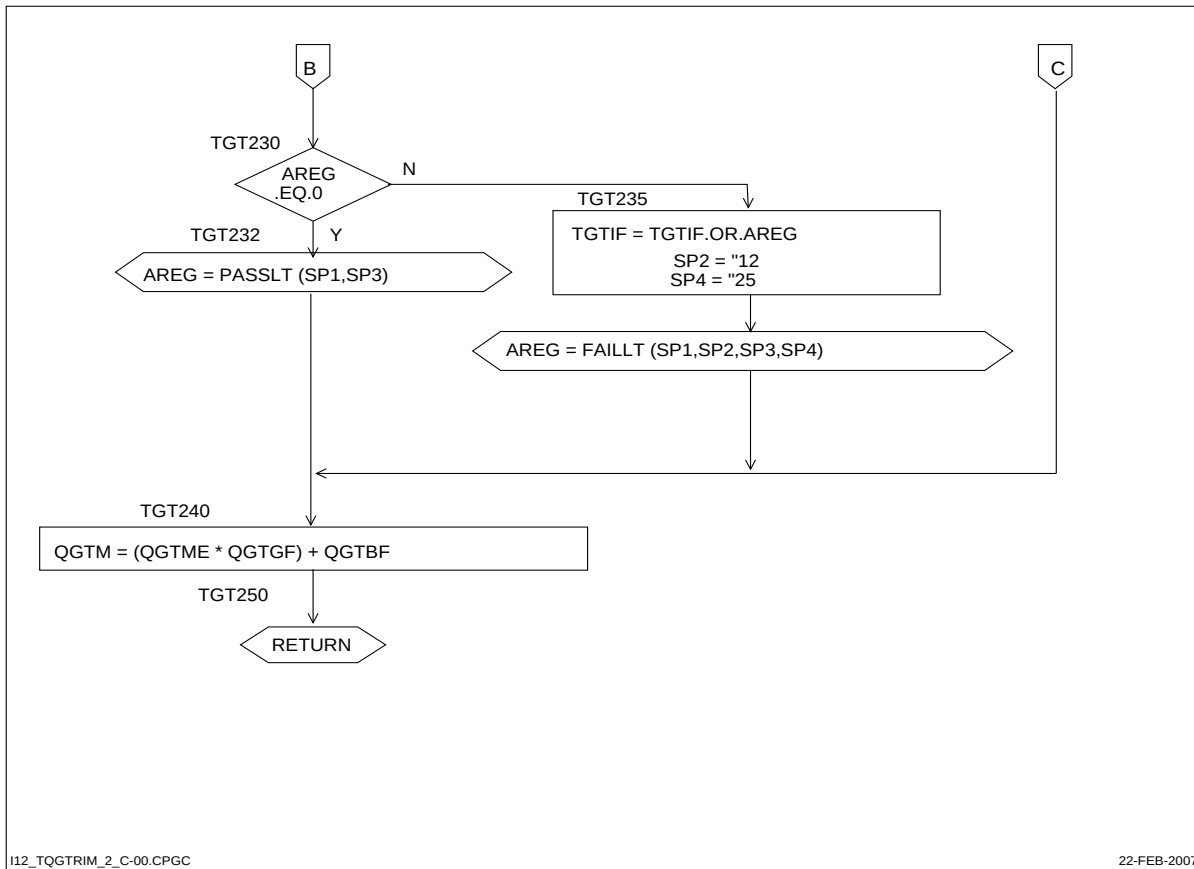
13.00 Figure 267: I12_TQGTRIM_2_A



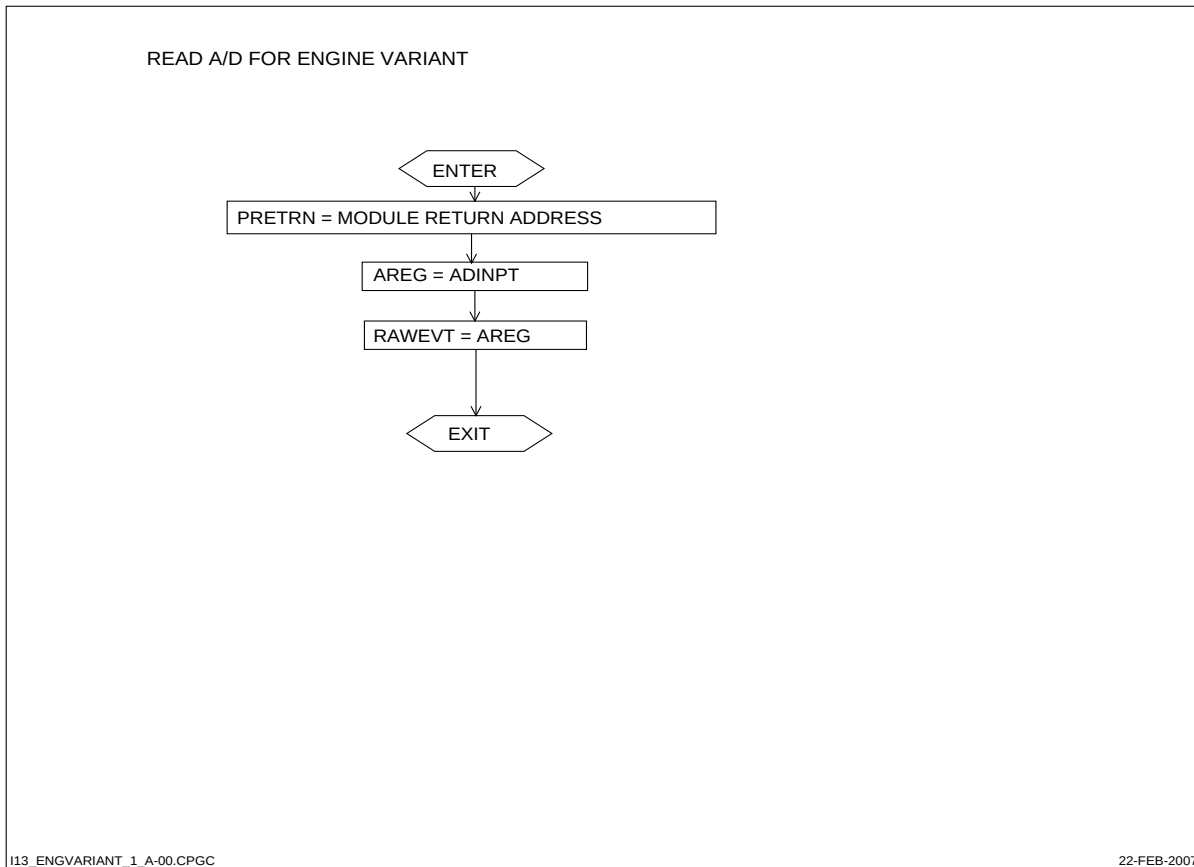
13.00 Figure 268: I12_TQGTRIM_2_B



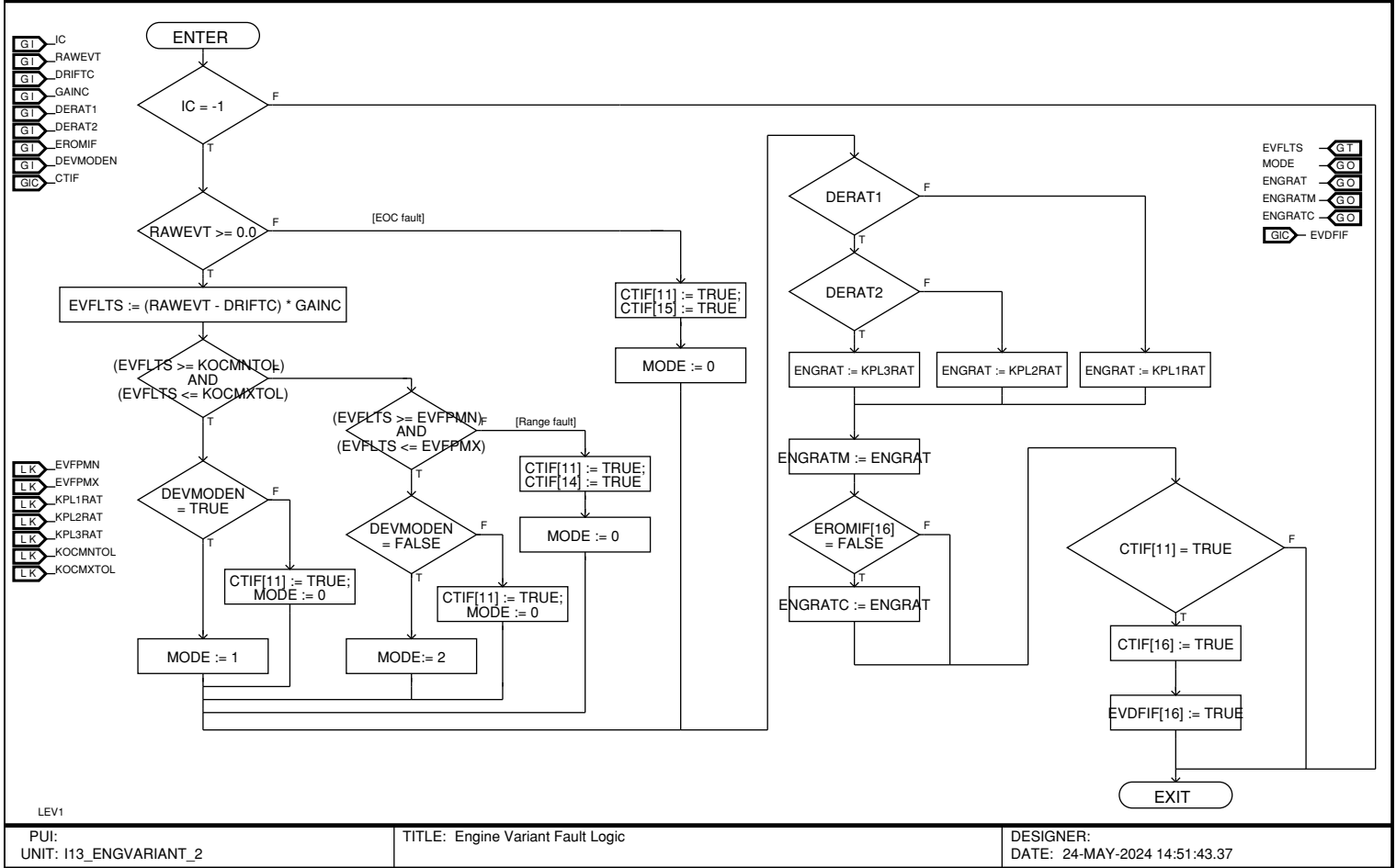
13.00 Figure 269: I12_TQGTRIM_2_C



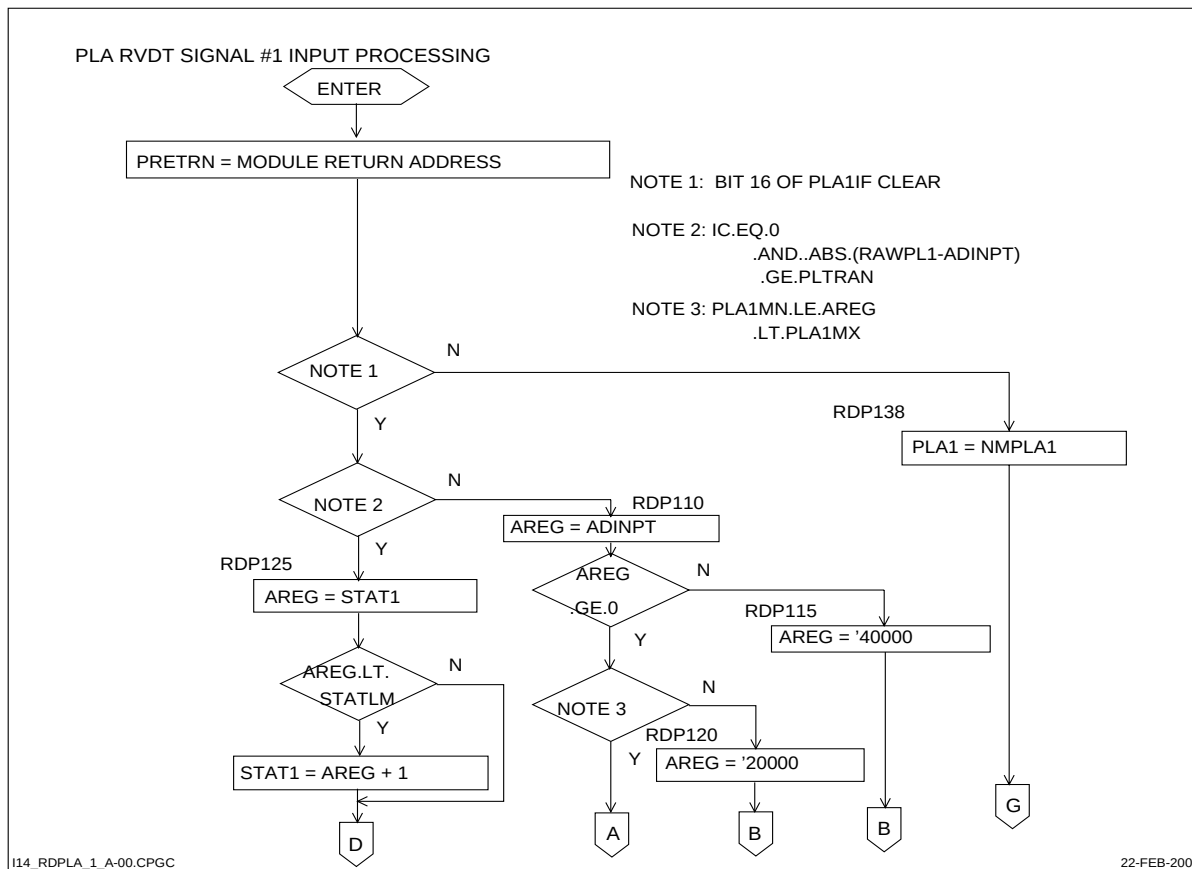
13.00 Figure 270: I13 _ENG VARIANT _1 _A



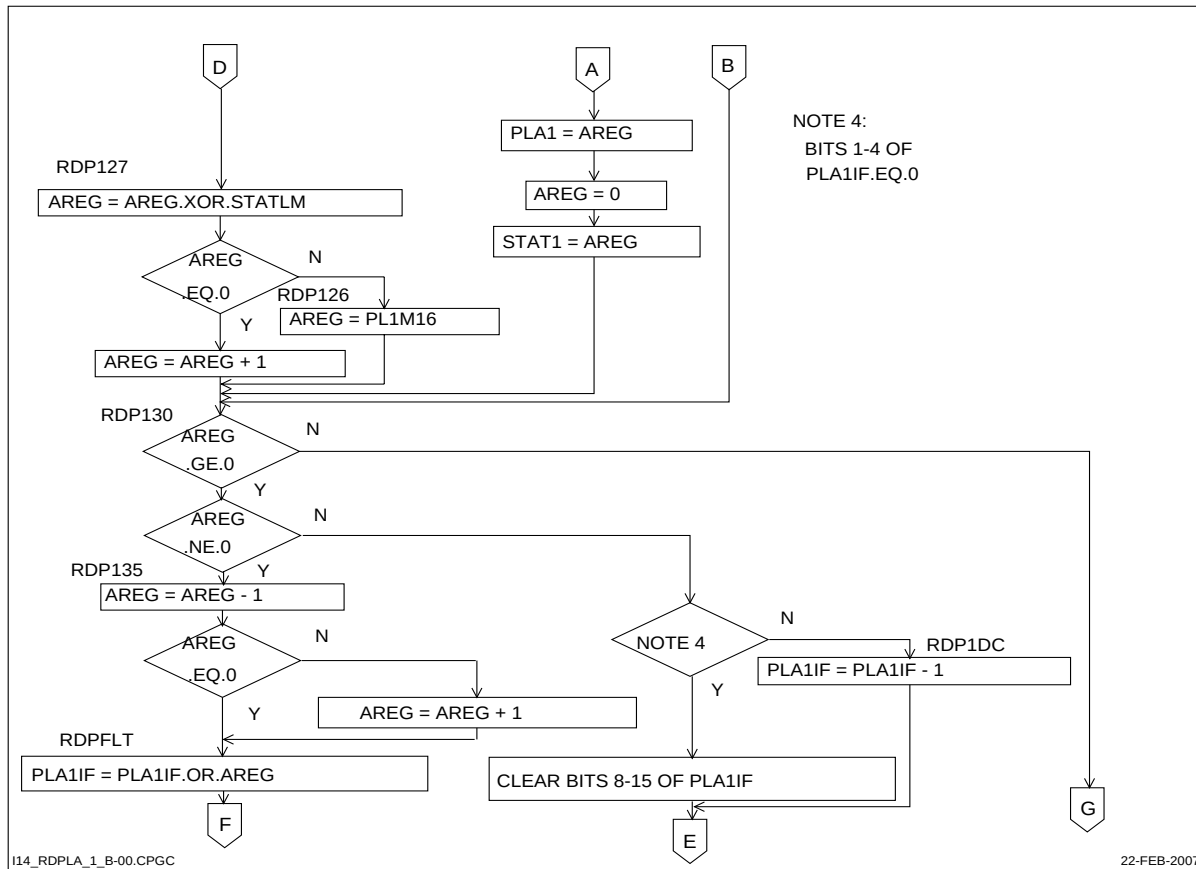
13.00 Figure 271: I13_ENGVARIANT_2



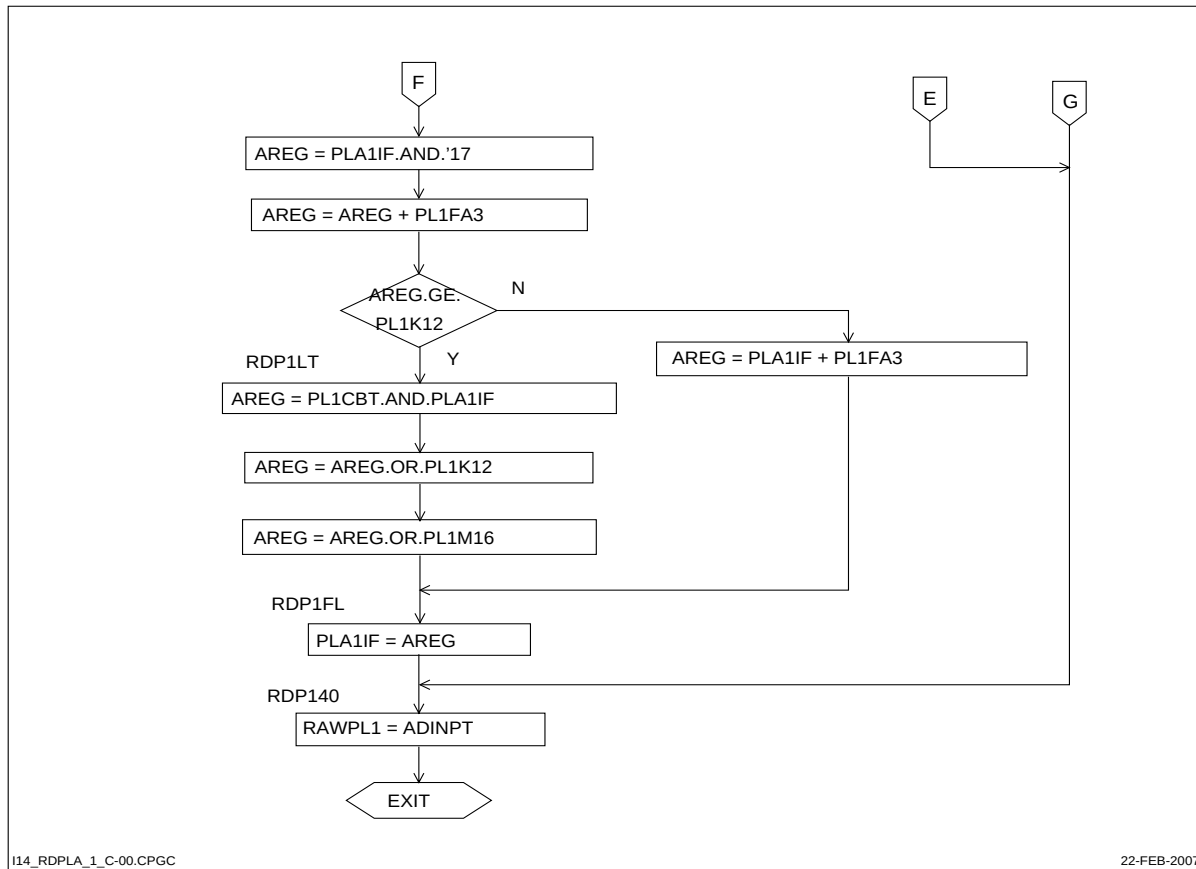
13.00 Figure 272: I14_RDPLA_1_A



13.00 Figure 273: I14_RDPLA_1_B



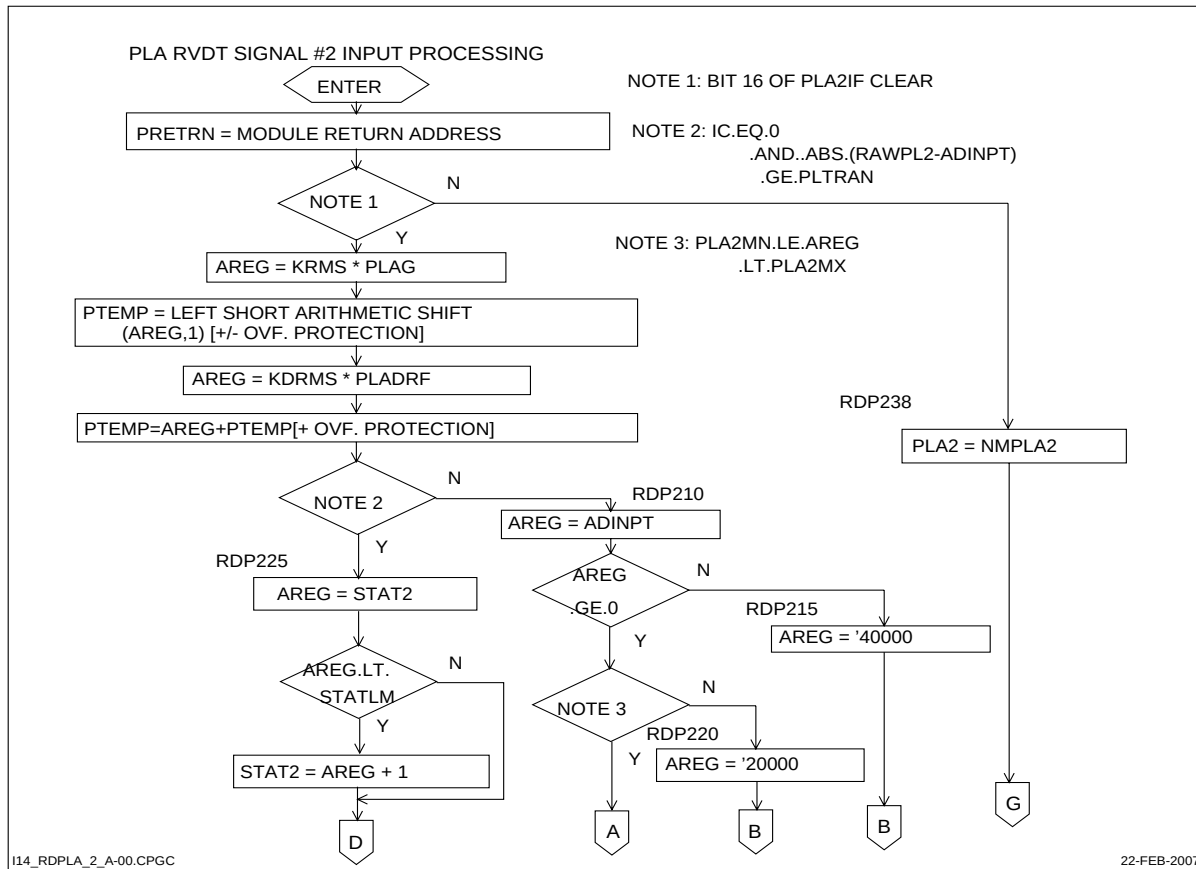
13.00 Figure 274: I14_RDPLA_1_C



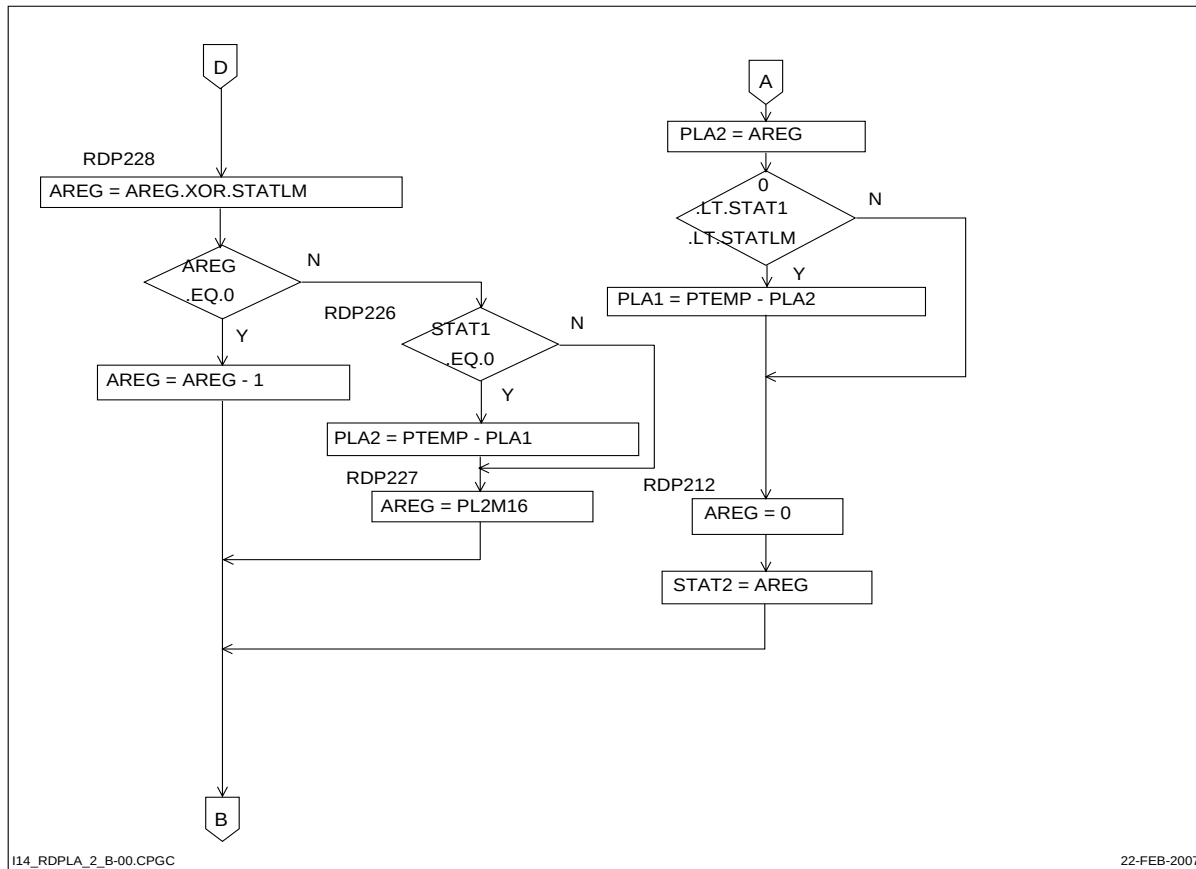
I14_RDPLA_1_C-00.CPGC

22-FEB-2007

13.00 Figure 275: I14_RDPLA_2_A



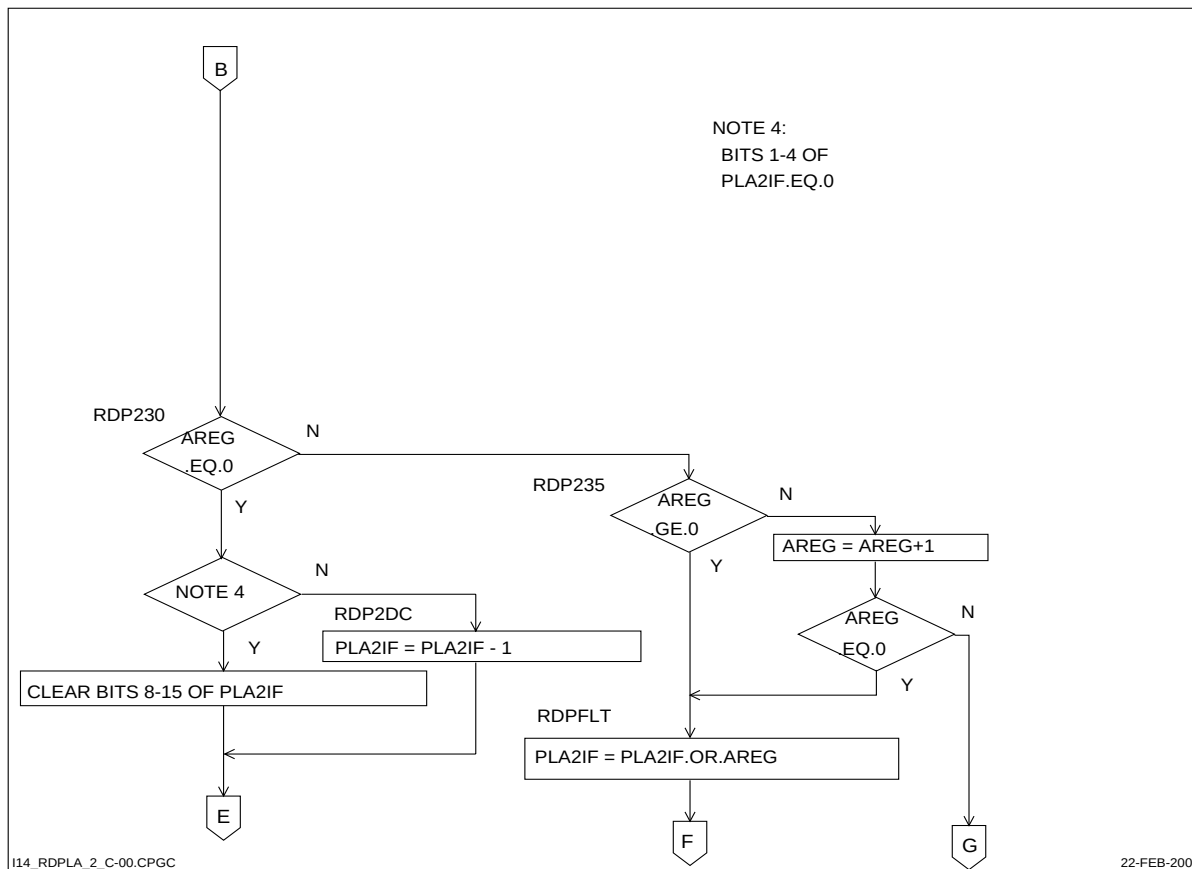
13.00 Figure 276: I14_RDPLA_2_B



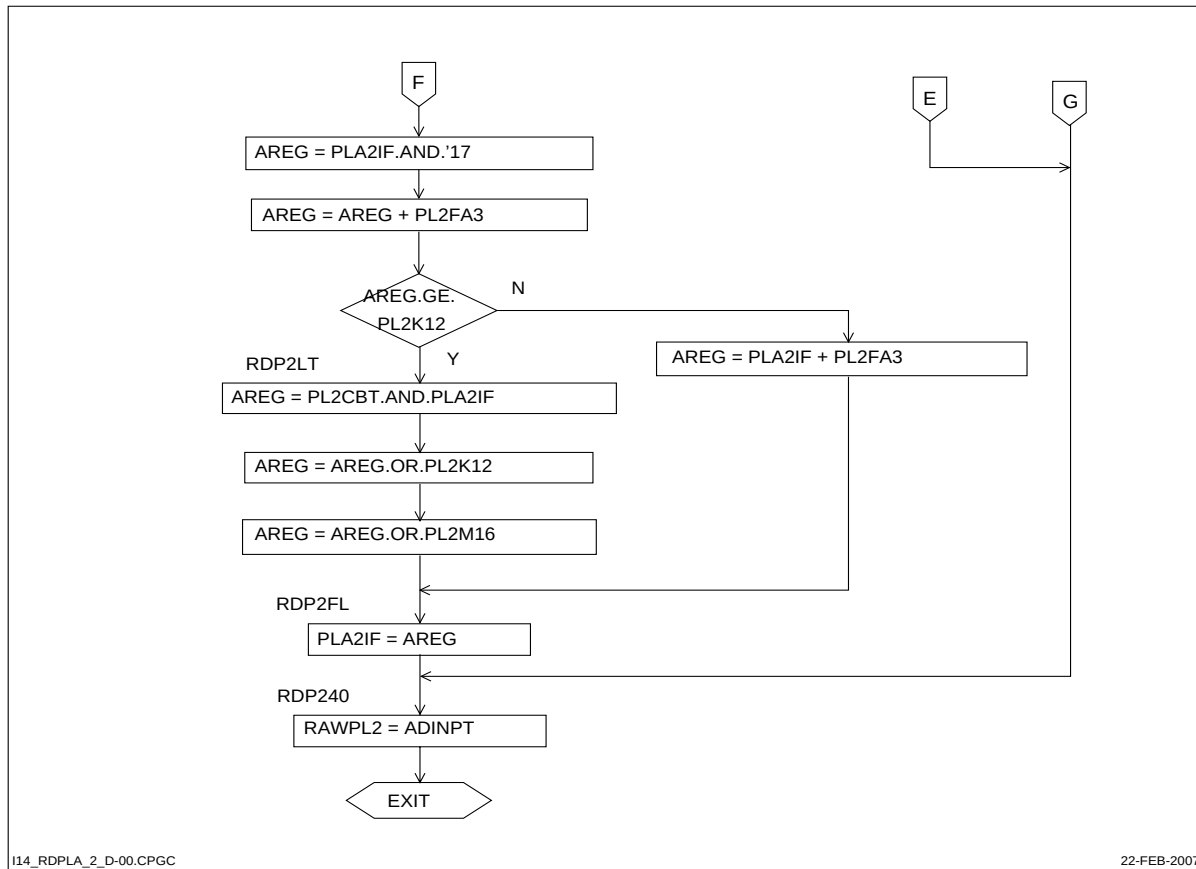
I14_RDPLA_2_B-00.CPGC

22-FEB-2007

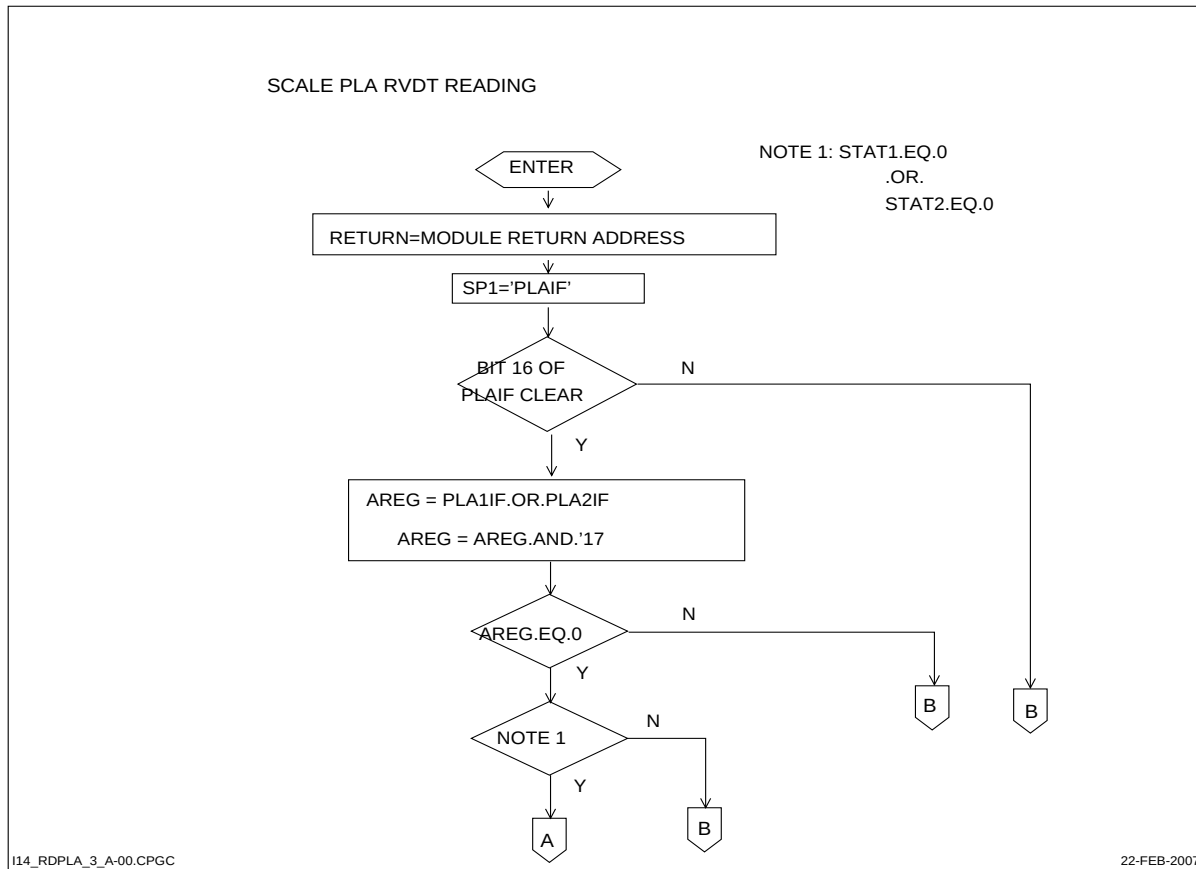
13.00 Figure 277: I14_RDPLA_2_C



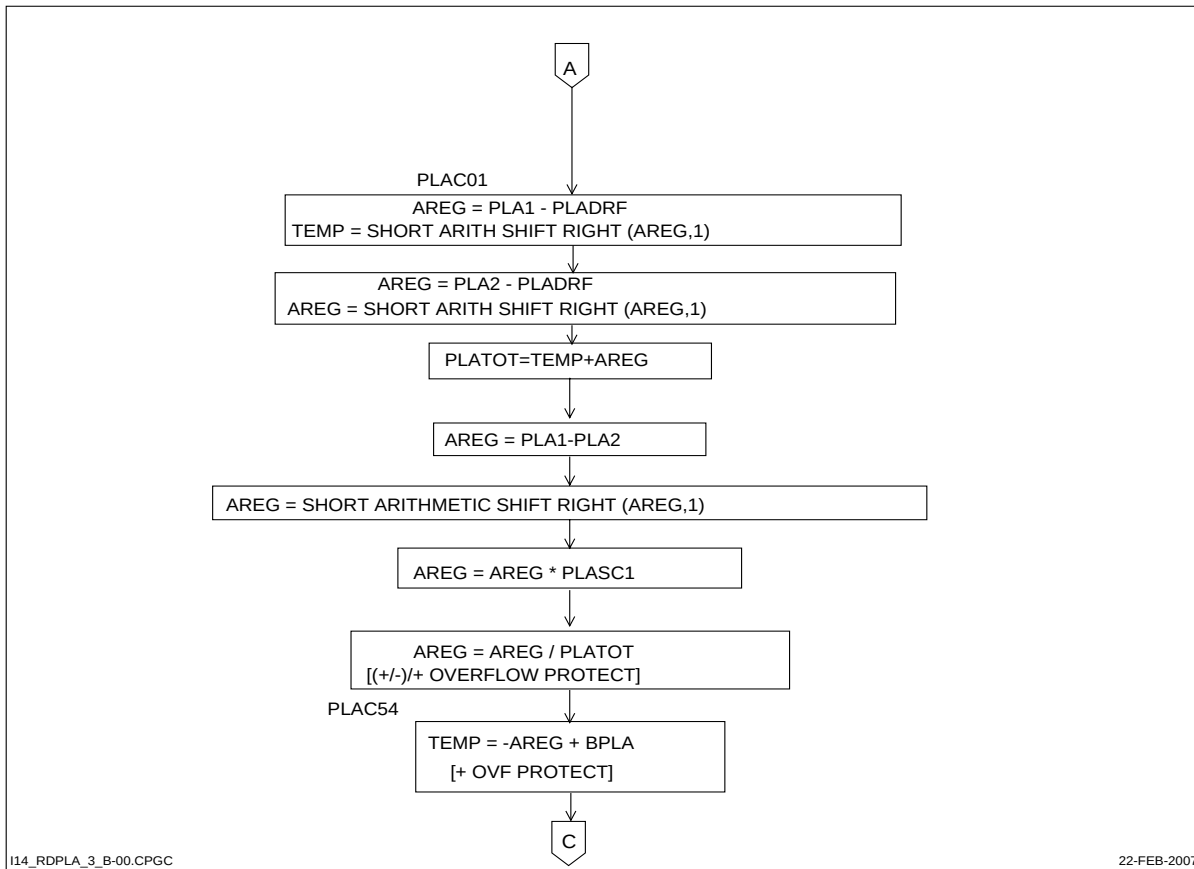
13.00 Figure 278: I14_RDPLA_2_D



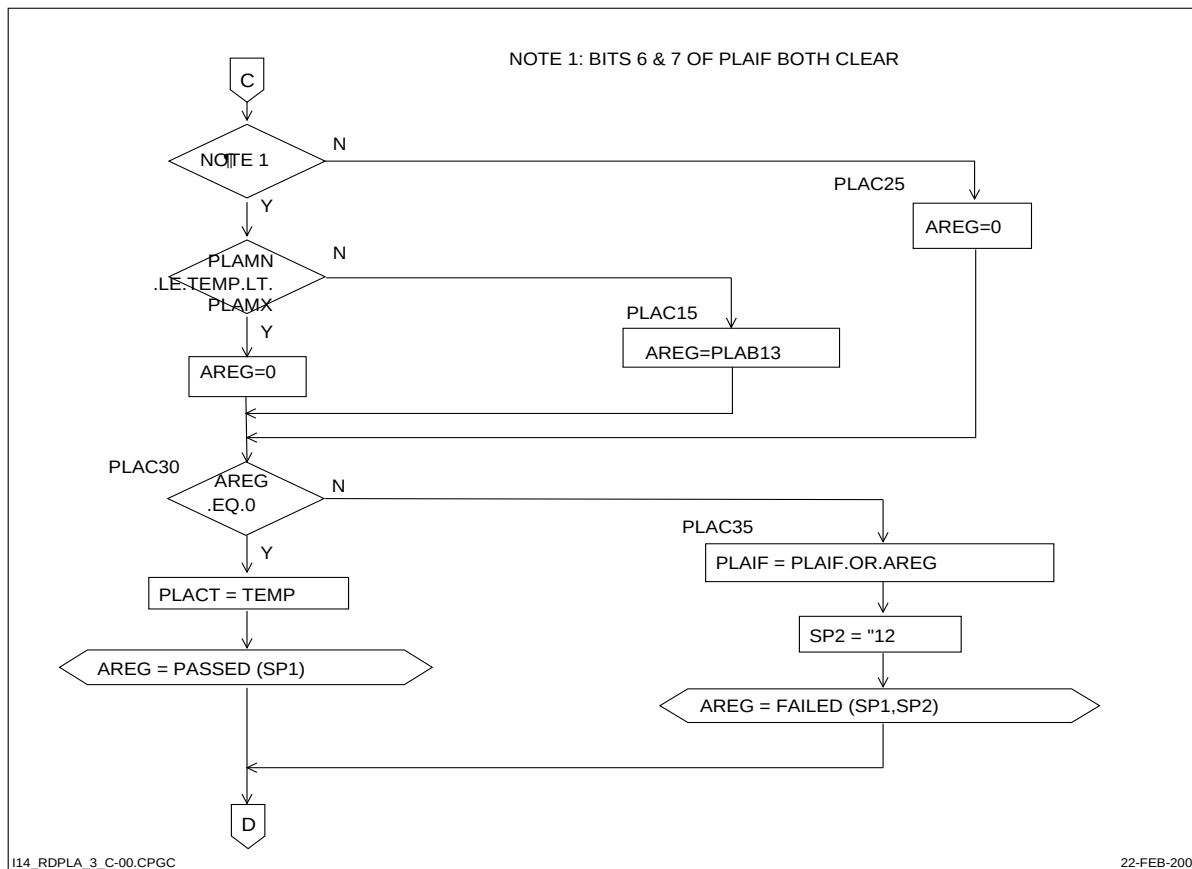
13.00 Figure 279: I14_RDPLA_3_A



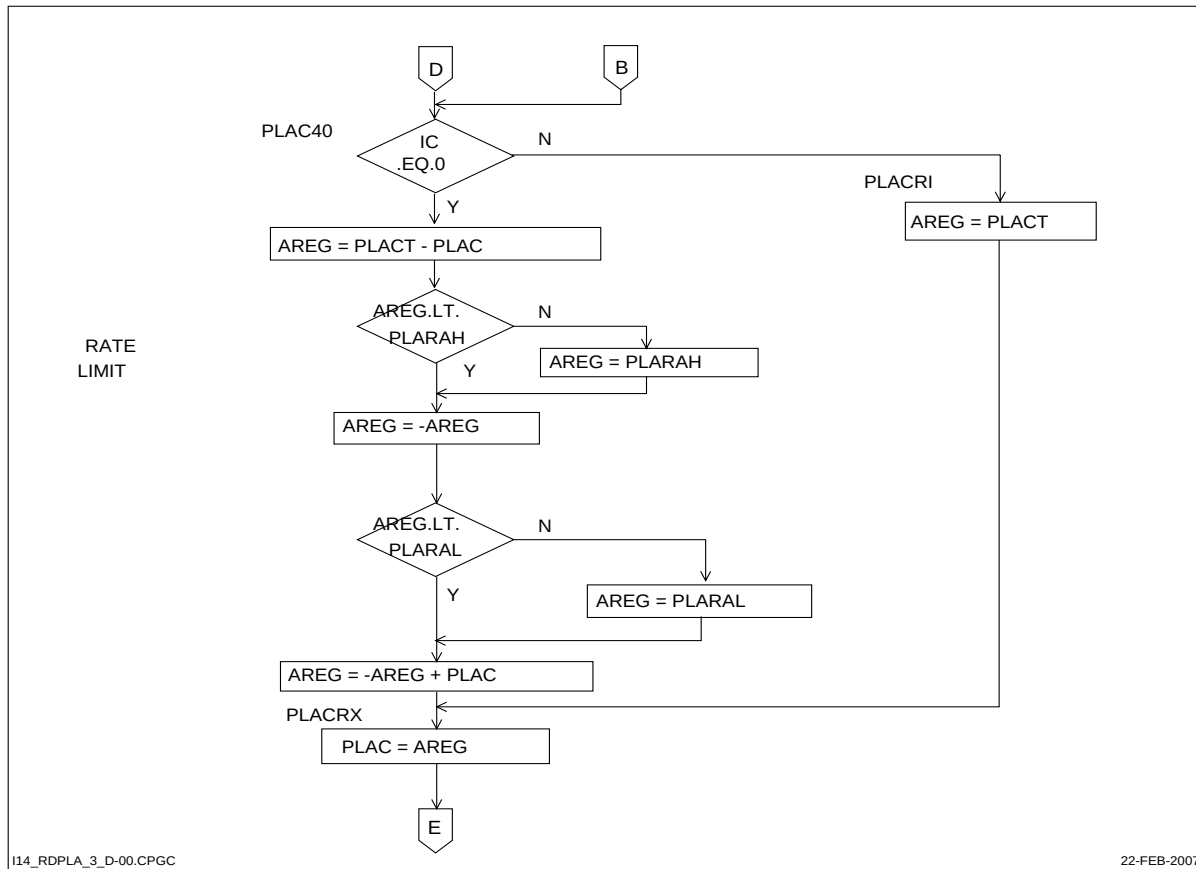
13.00 Figure 280: I14_RDPLA_3_B



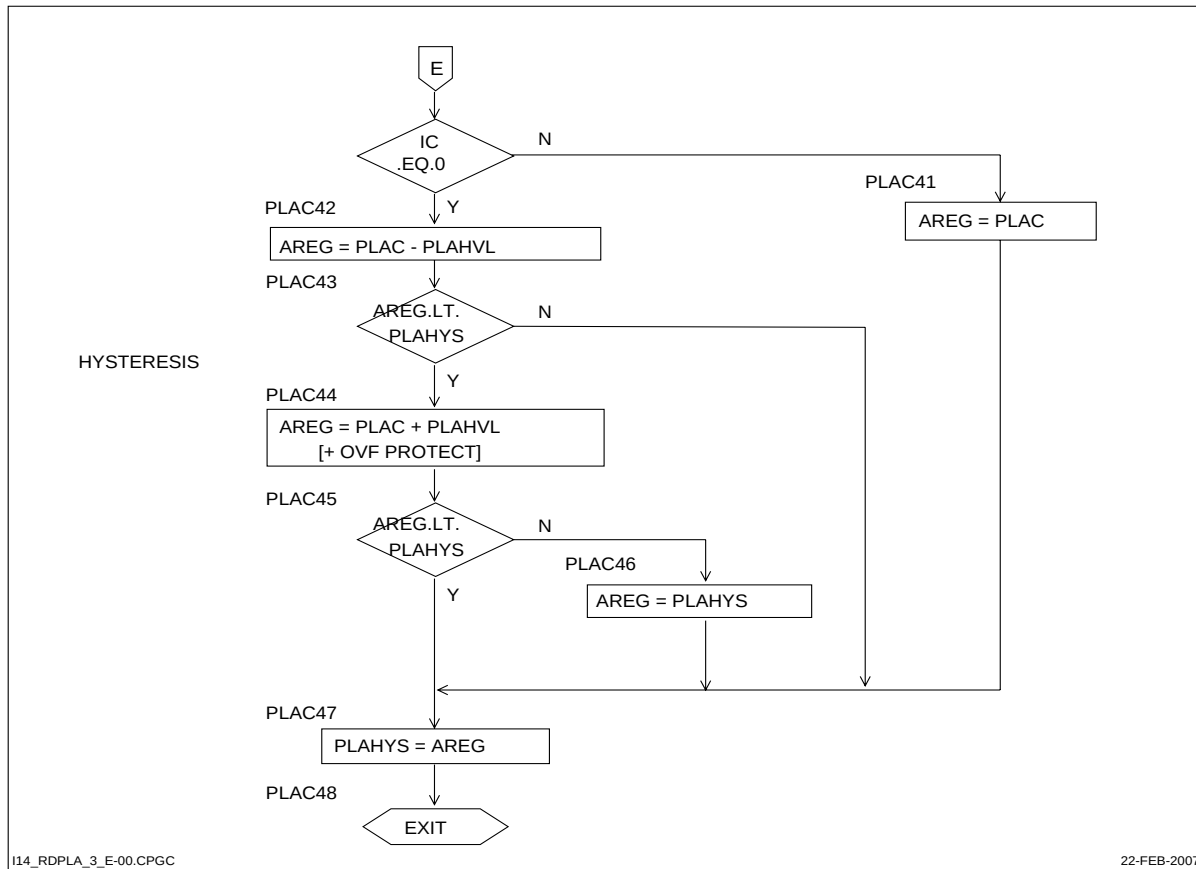
13.00 Figure 281: I14_RDPLA_3_C



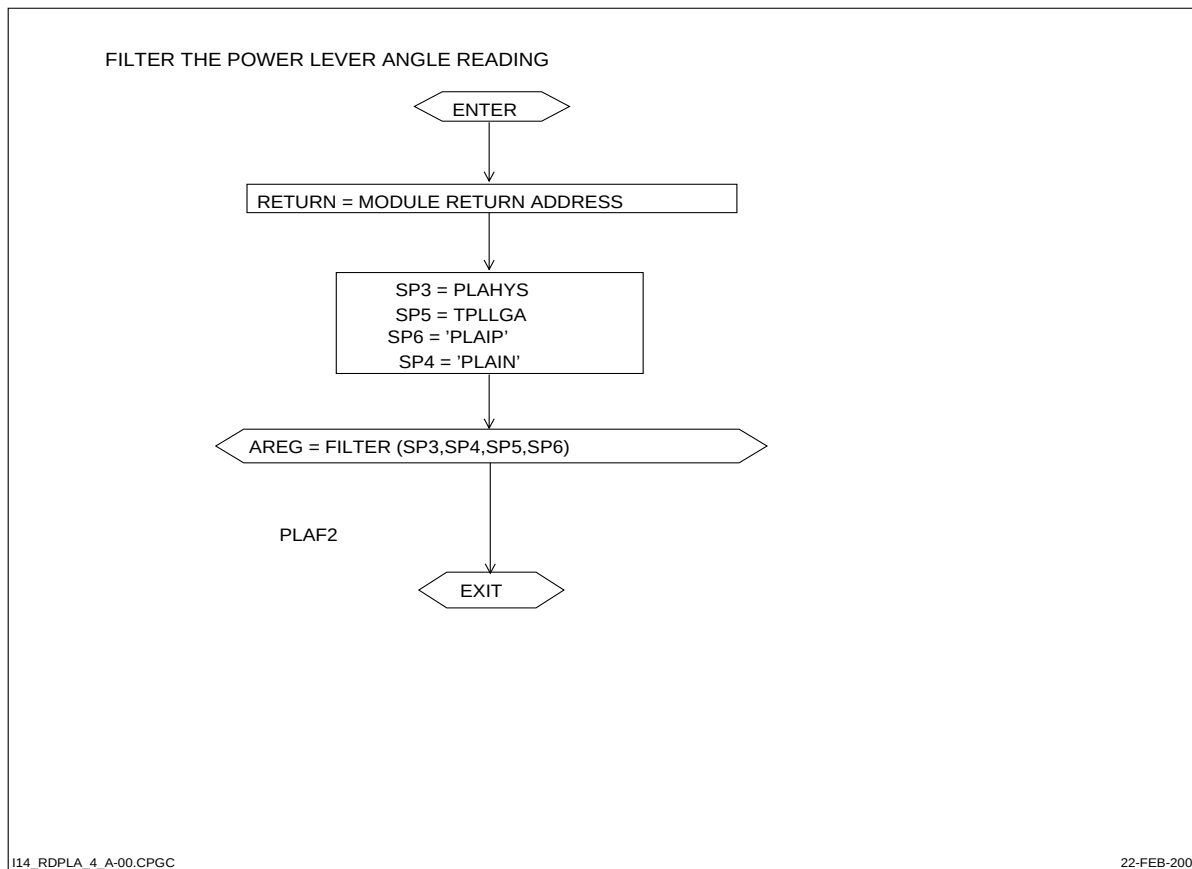
13.00 Figure 282: I14_RDPLA_3_D



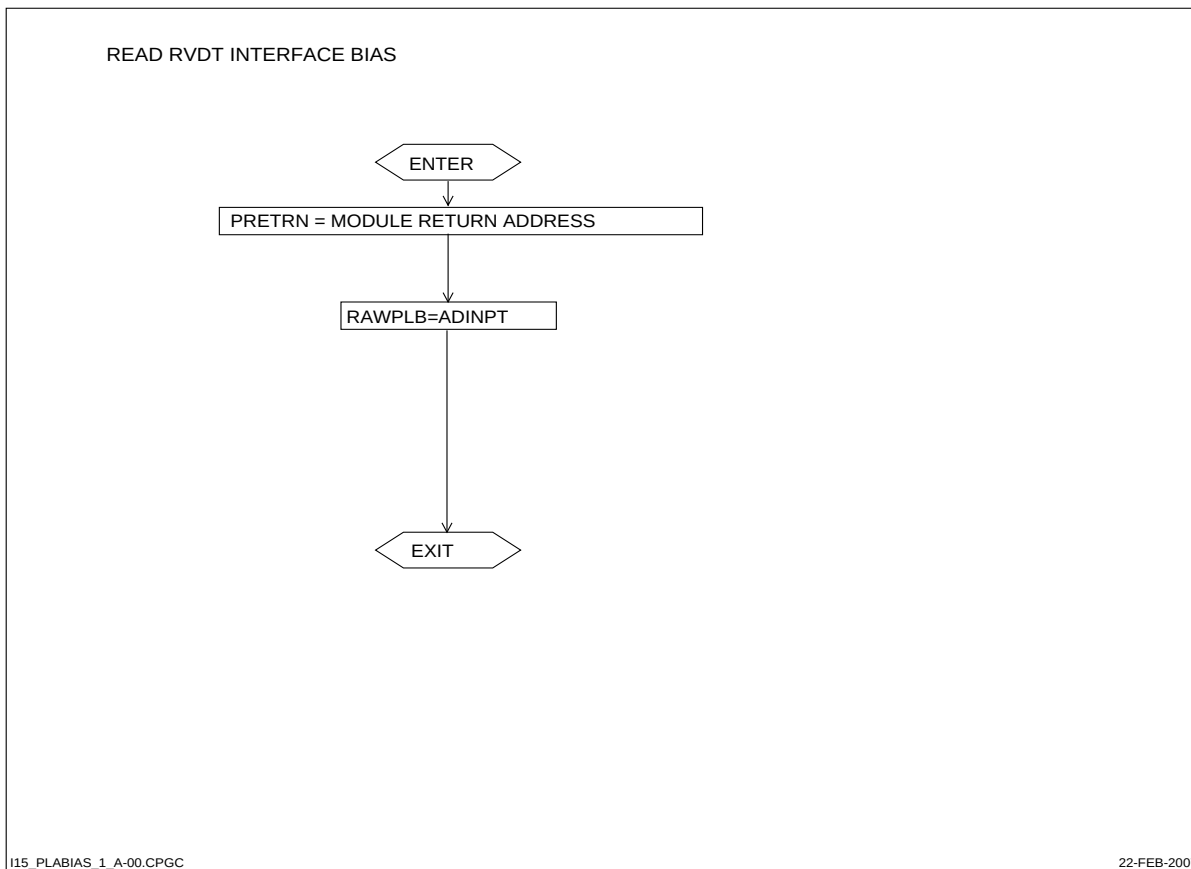
13.00 Figure 283: I14_RDPLA_3_E



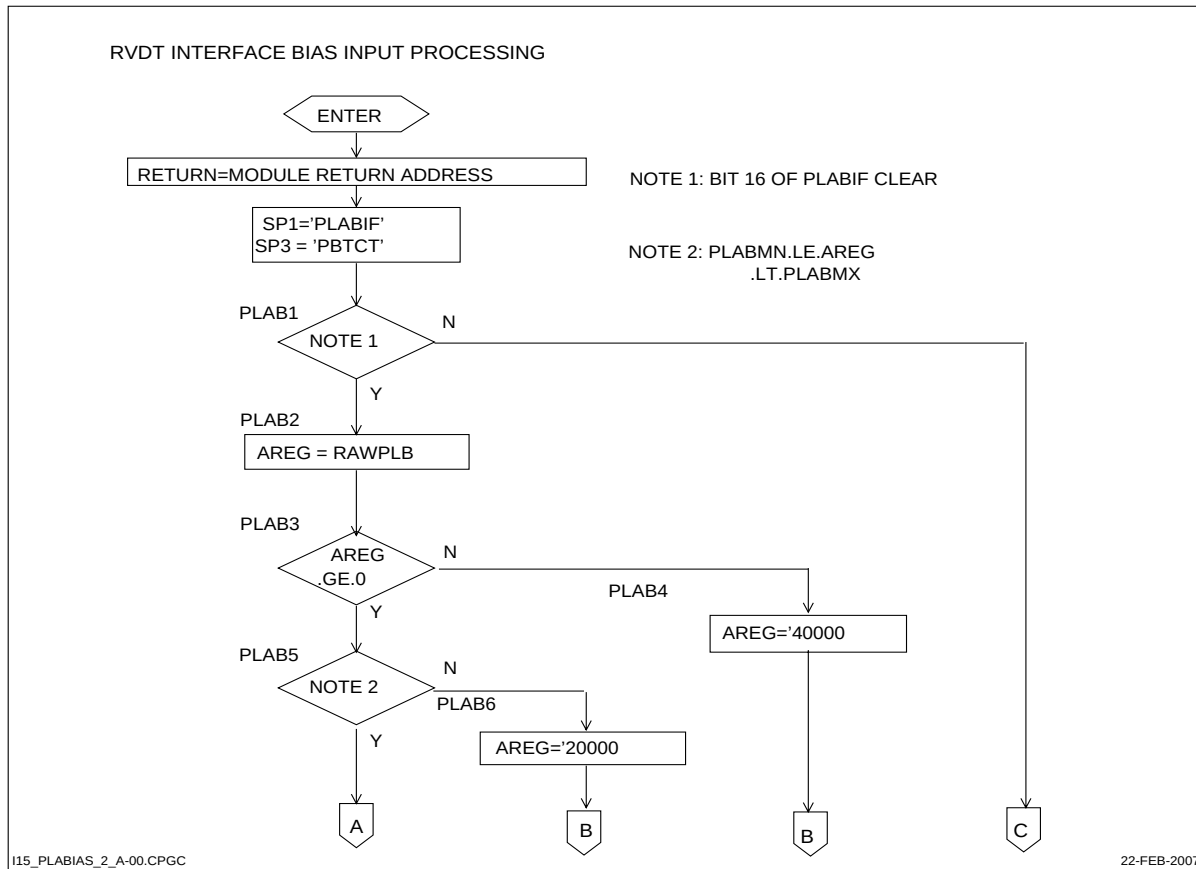
13.00 Figure 284: I14_RDPLA_4_A



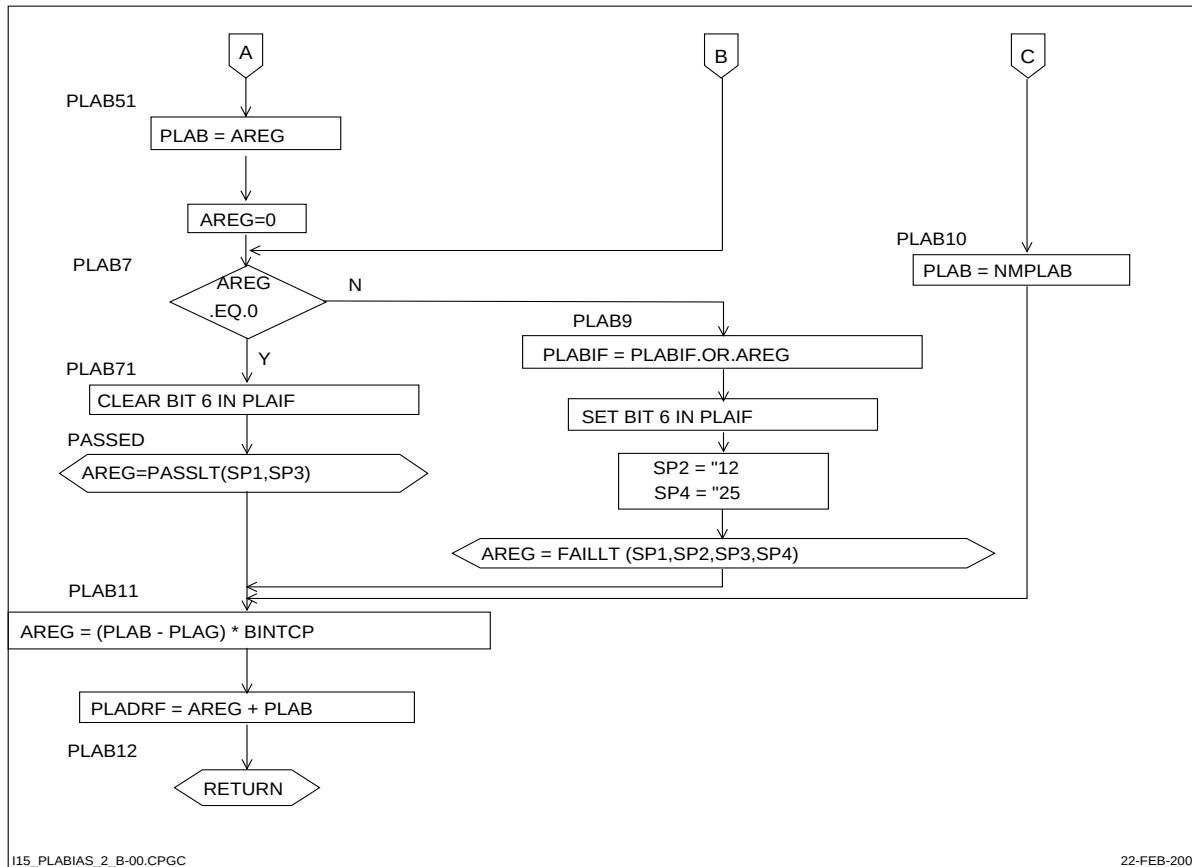
13.00 Figure 285: I15 _PLABIAS _1 _A



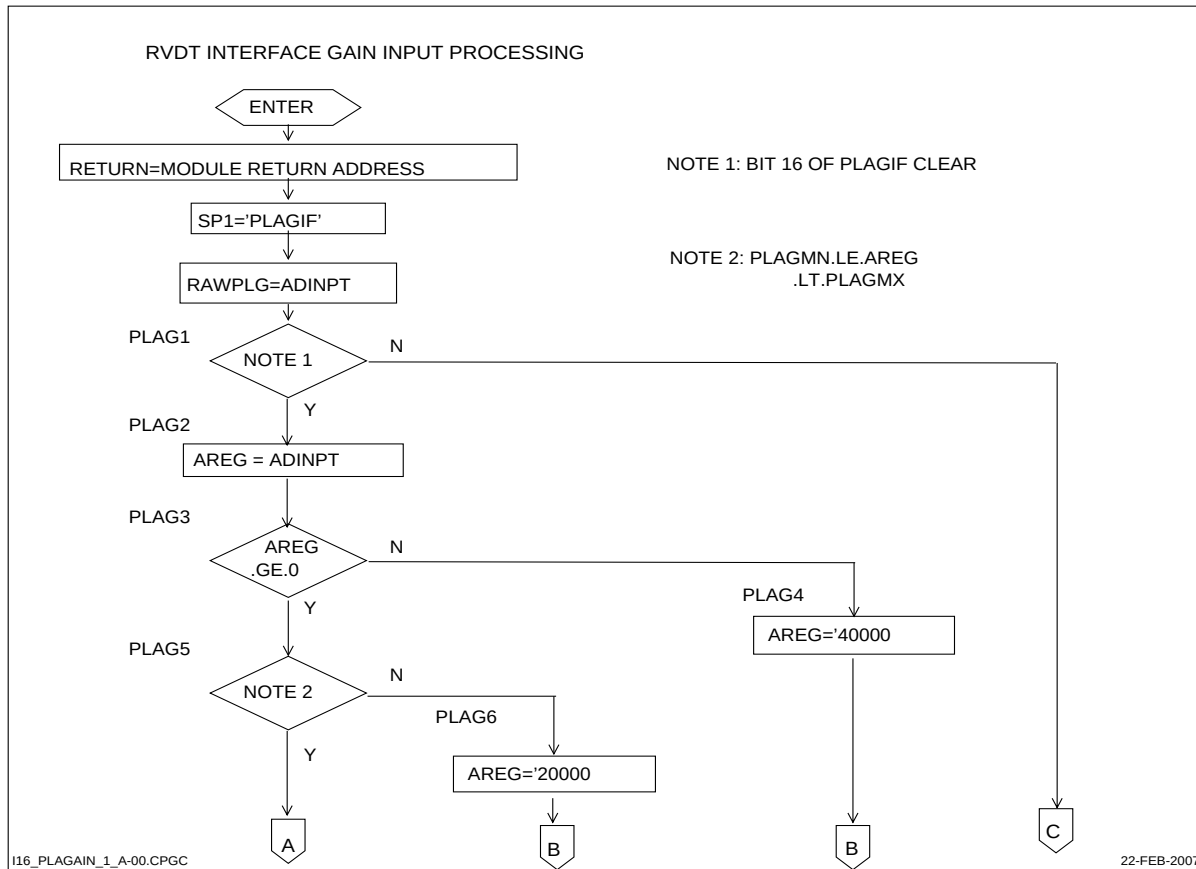
13.00 Figure 286: I15_PLABIAS_2_A



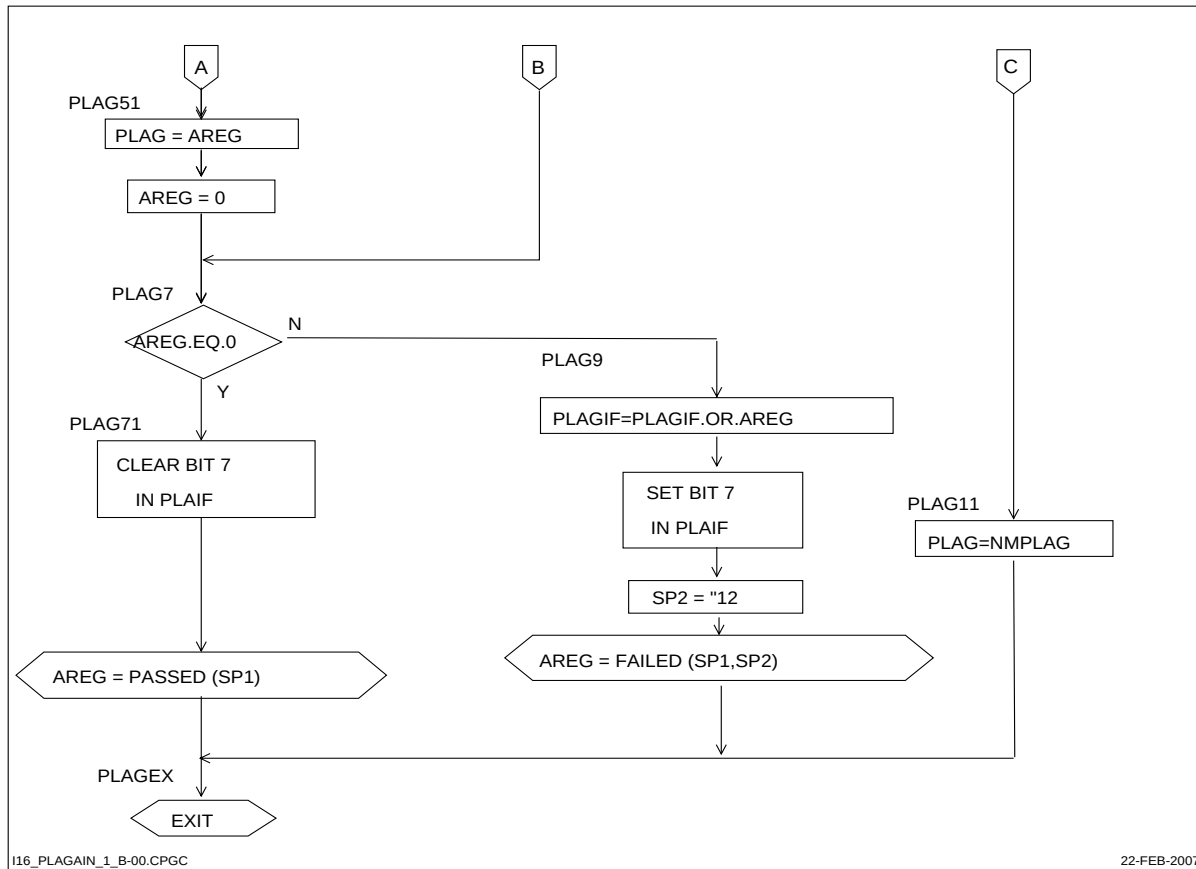
13.00 Figure 287: I15_PLABIAS_2_B



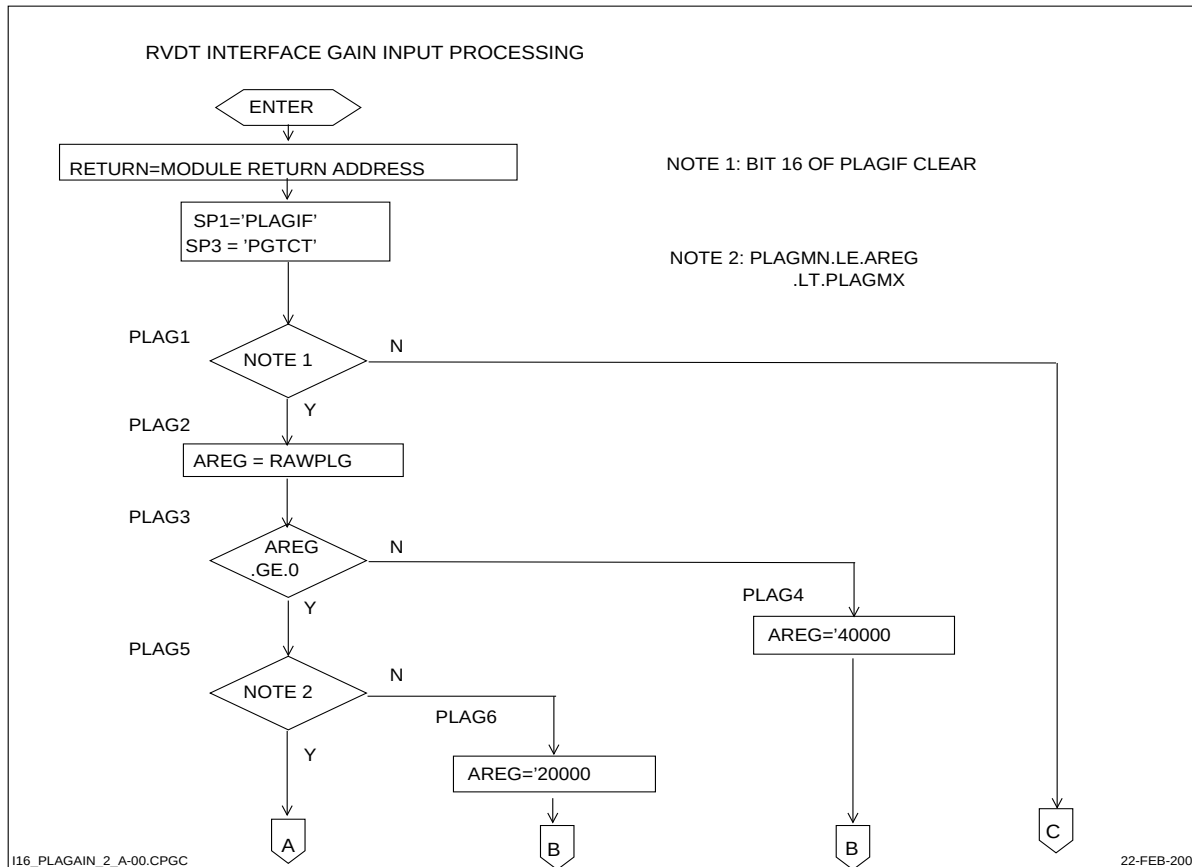
13.00 Figure 288: I16_PLAGAIN_1_A



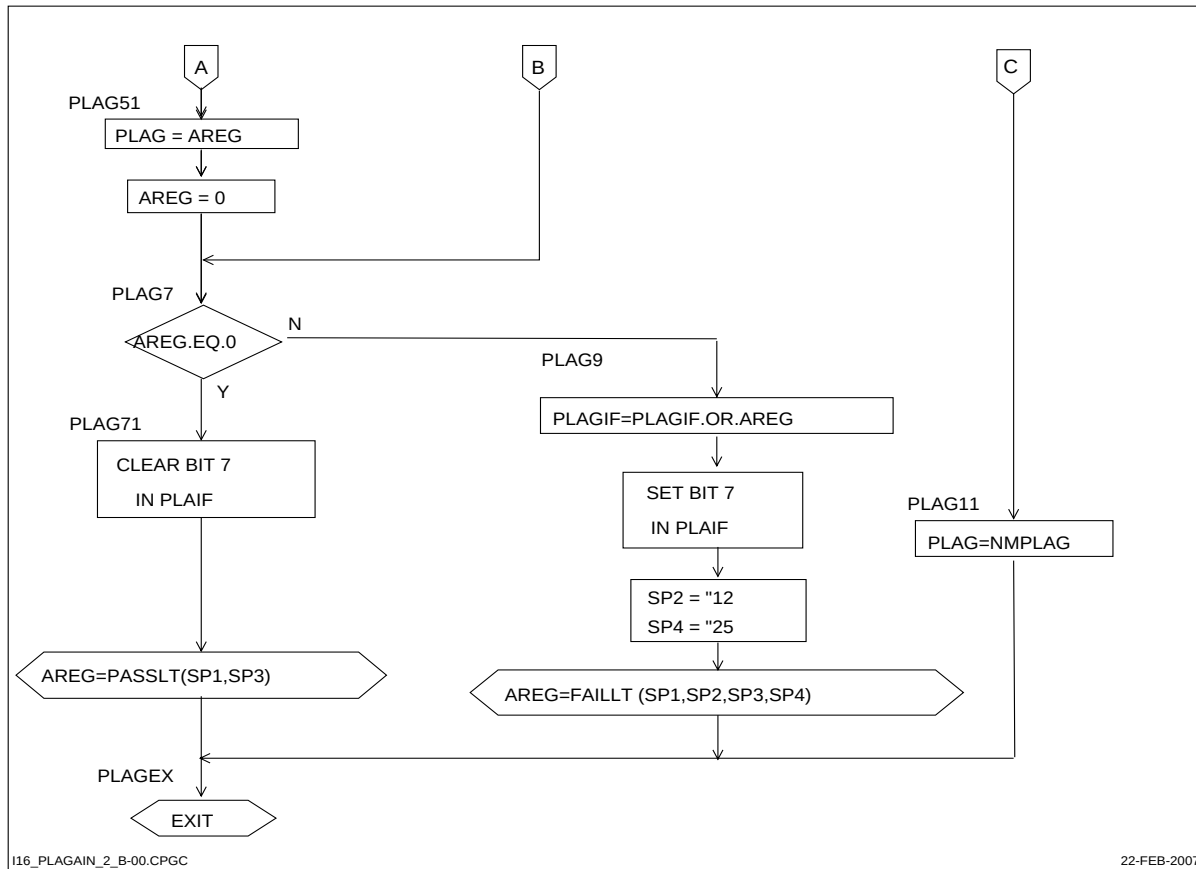
13.00 Figure 289: I16_PLAGAIN_1_B



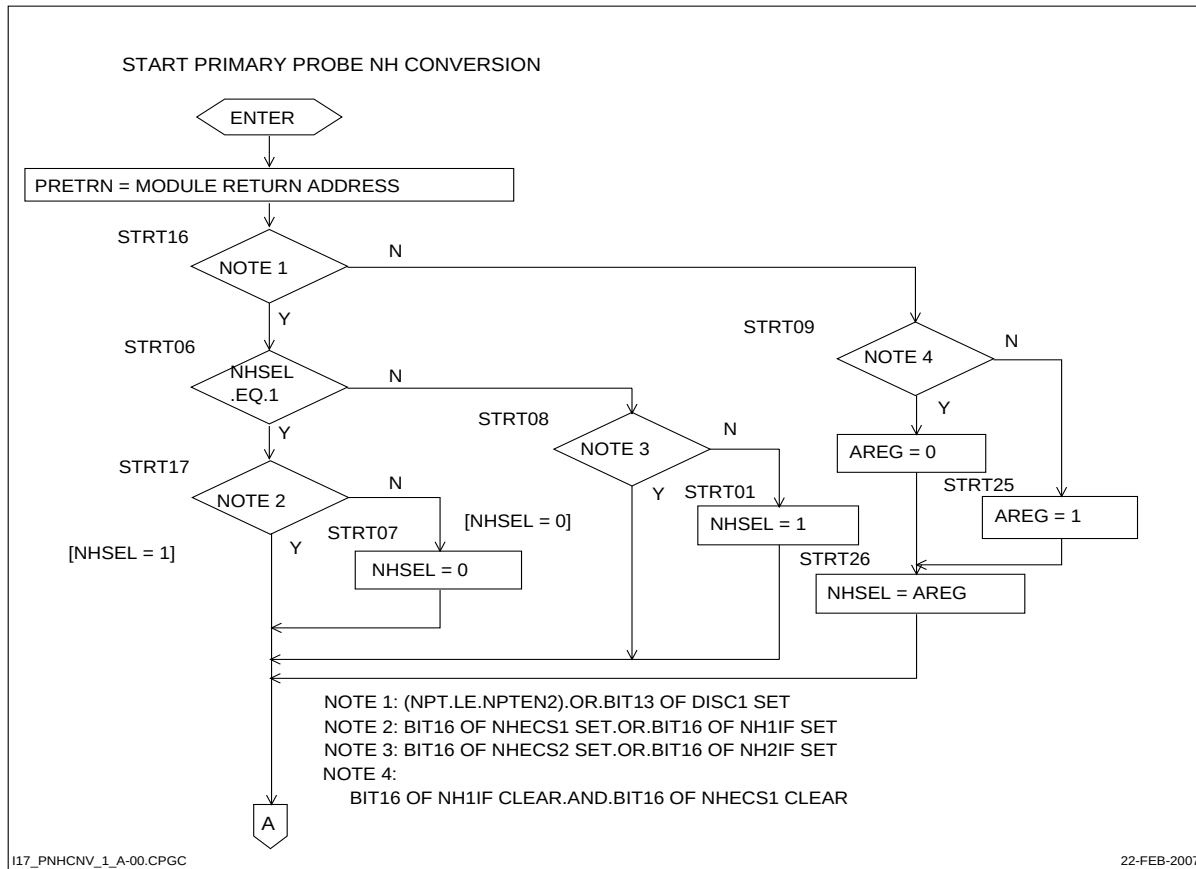
13.00 Figure 290: I16_PLAGAIN_2_A



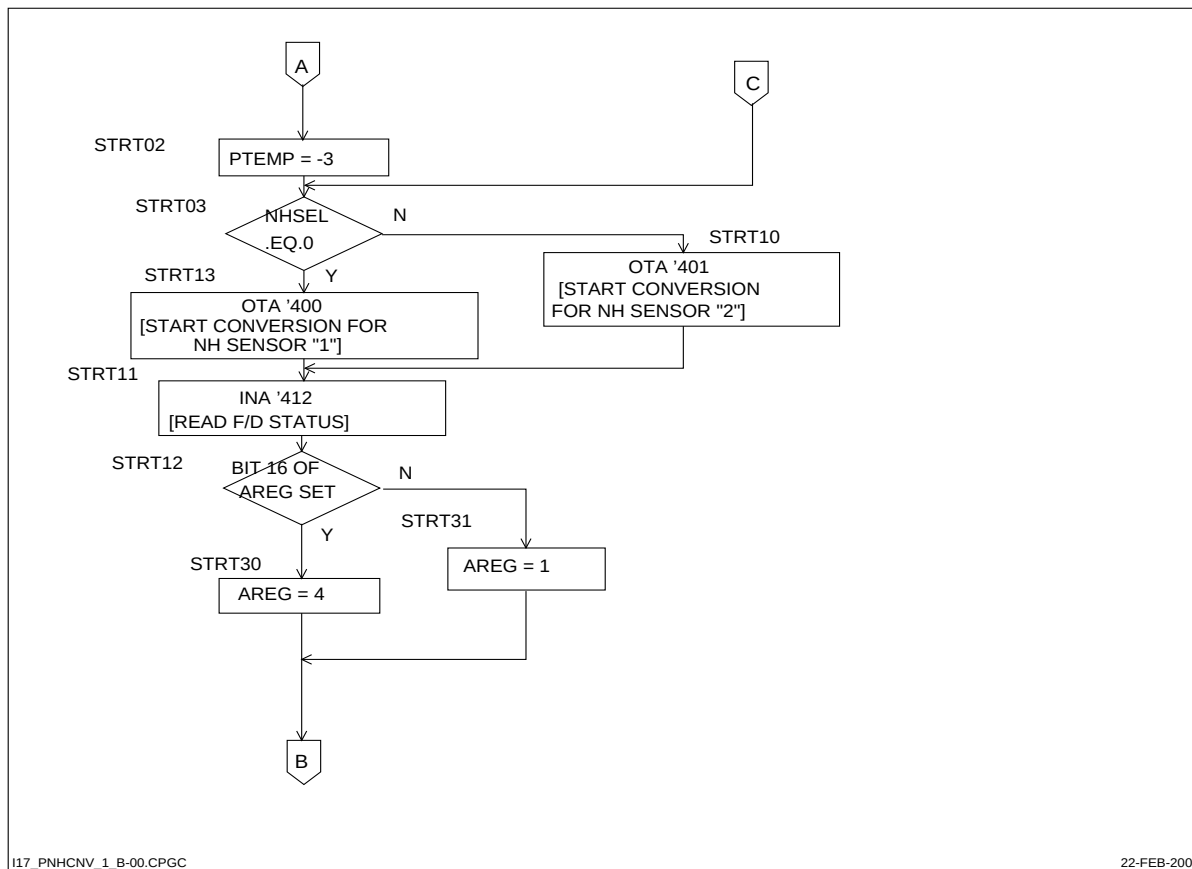
13.00 Figure 291: I16_PLAGAIN_2_B



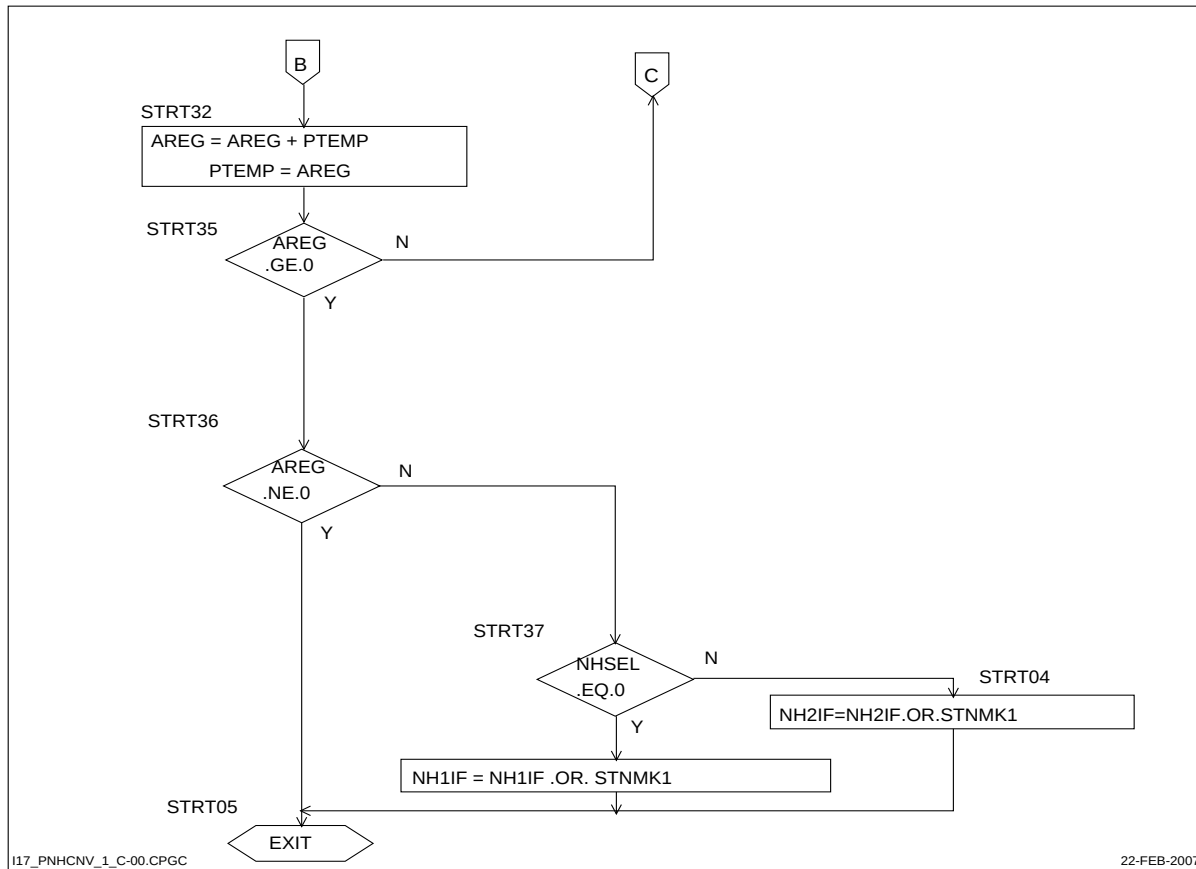
13.00 Figure 292: I17_PNHCNV_1_A



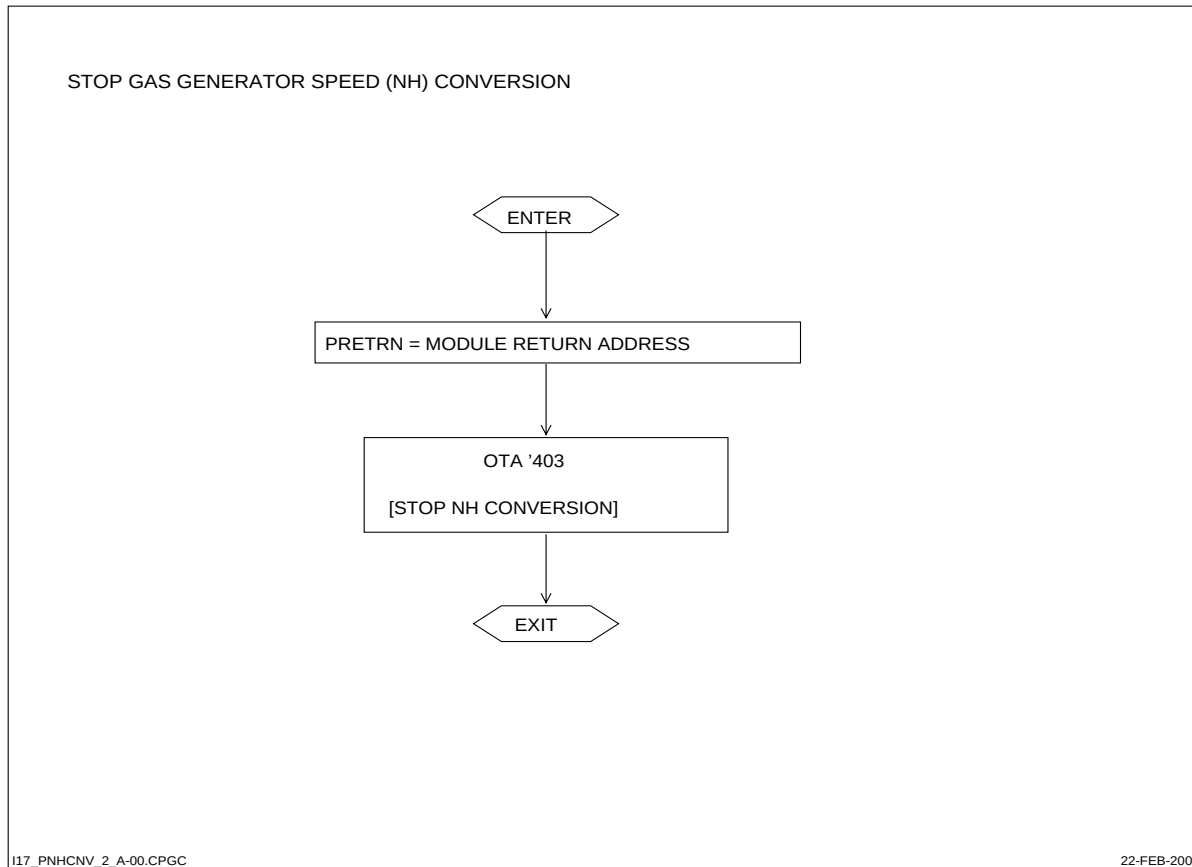
13.00 Figure 293: I17_PNHCNV_1_B



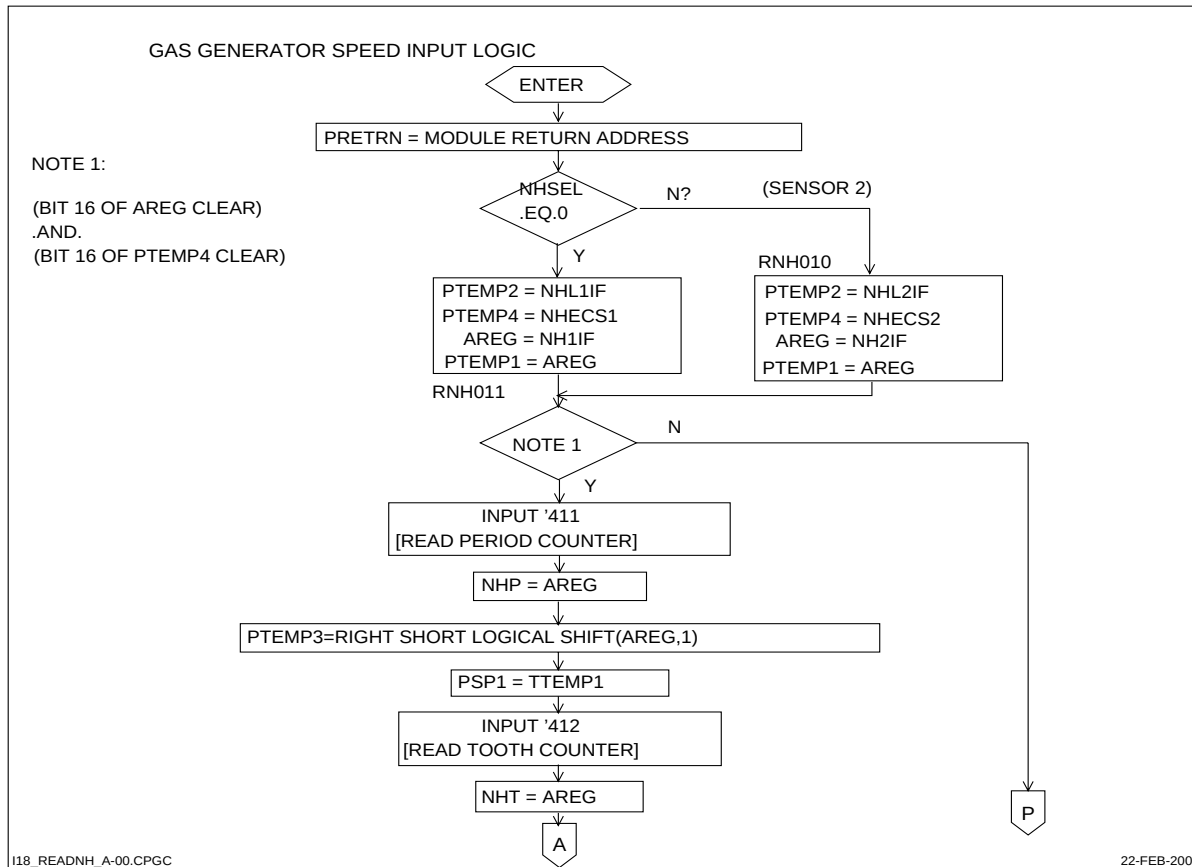
13.00 Figure 294: I17_PNHCNV_1_C



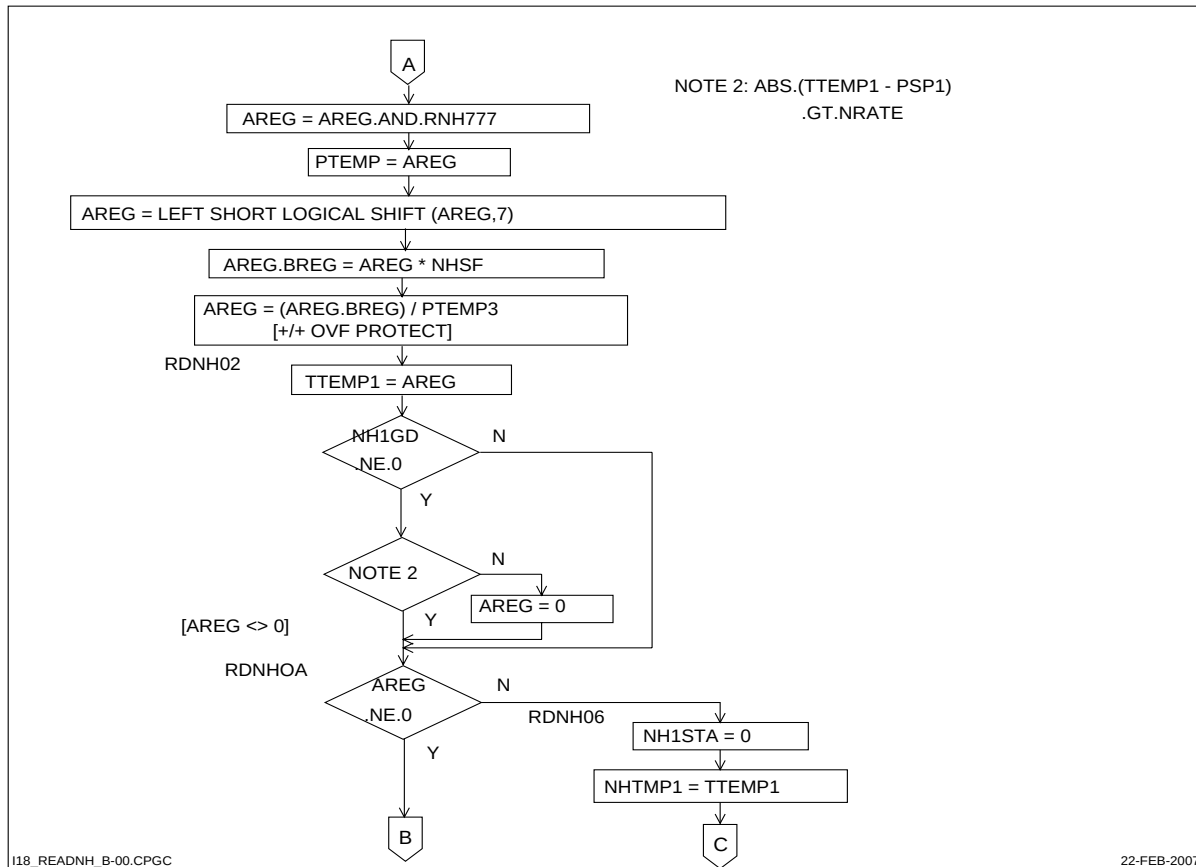
13.00 Figure 295: I17_PNHCVN_2_A



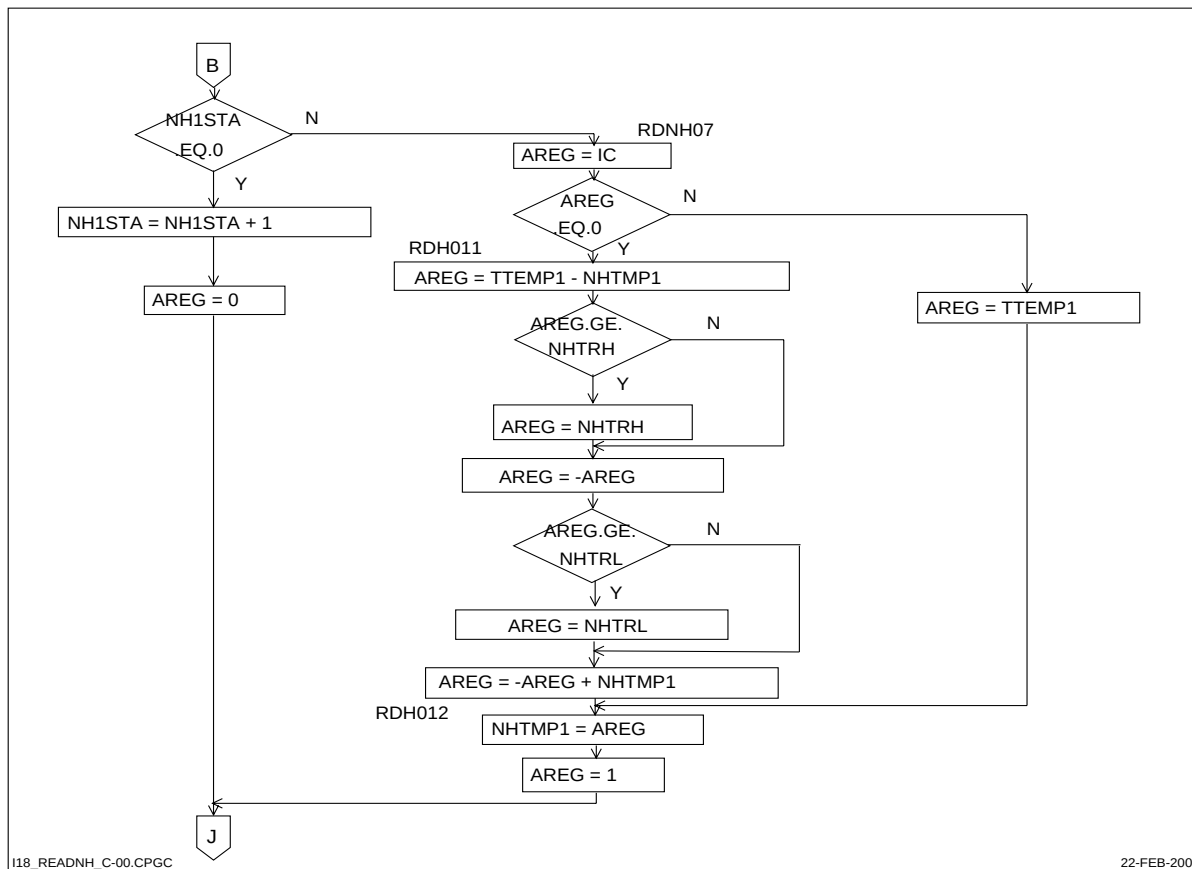
13.00 Figure 296: I18_READNH_A



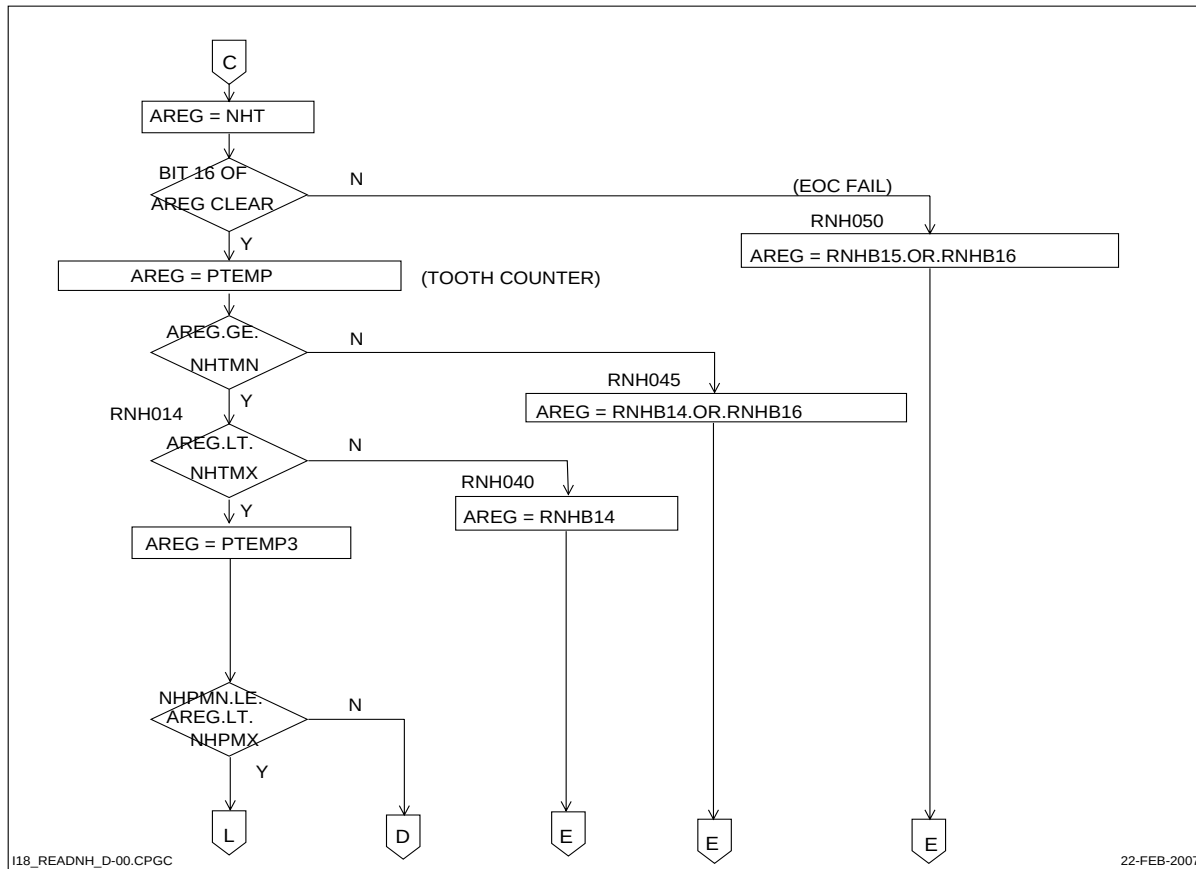
13.00 Figure 297: I18_READNH_B



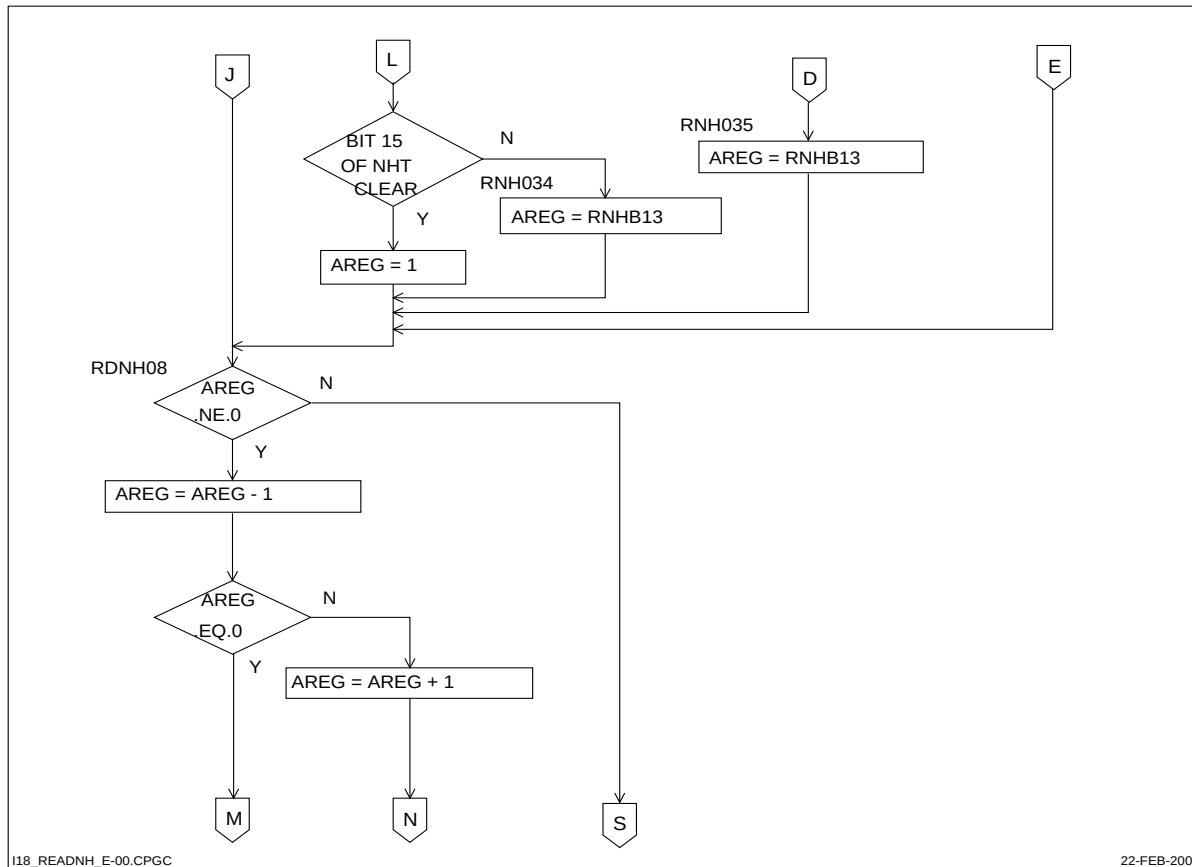
13.00 Figure 298: I18_READNH_C



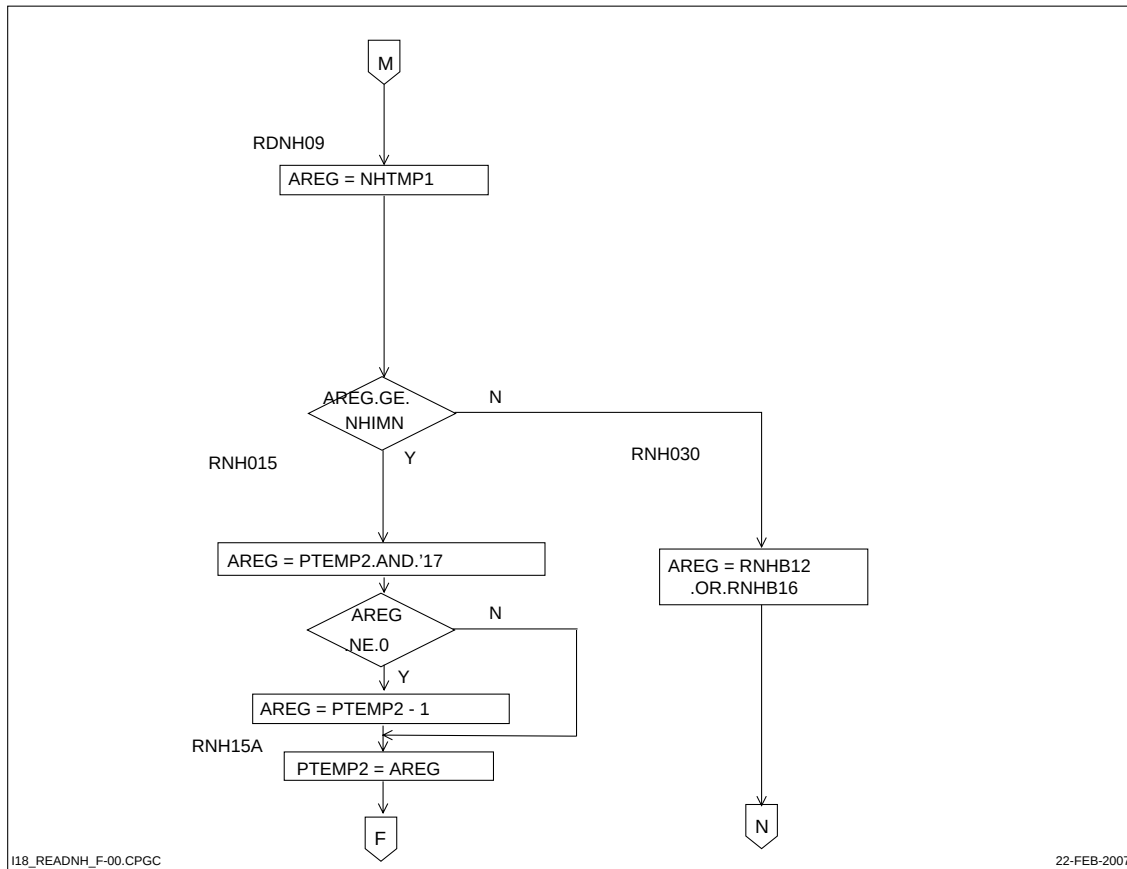
13.00 Figure 299: I18_READNH_D



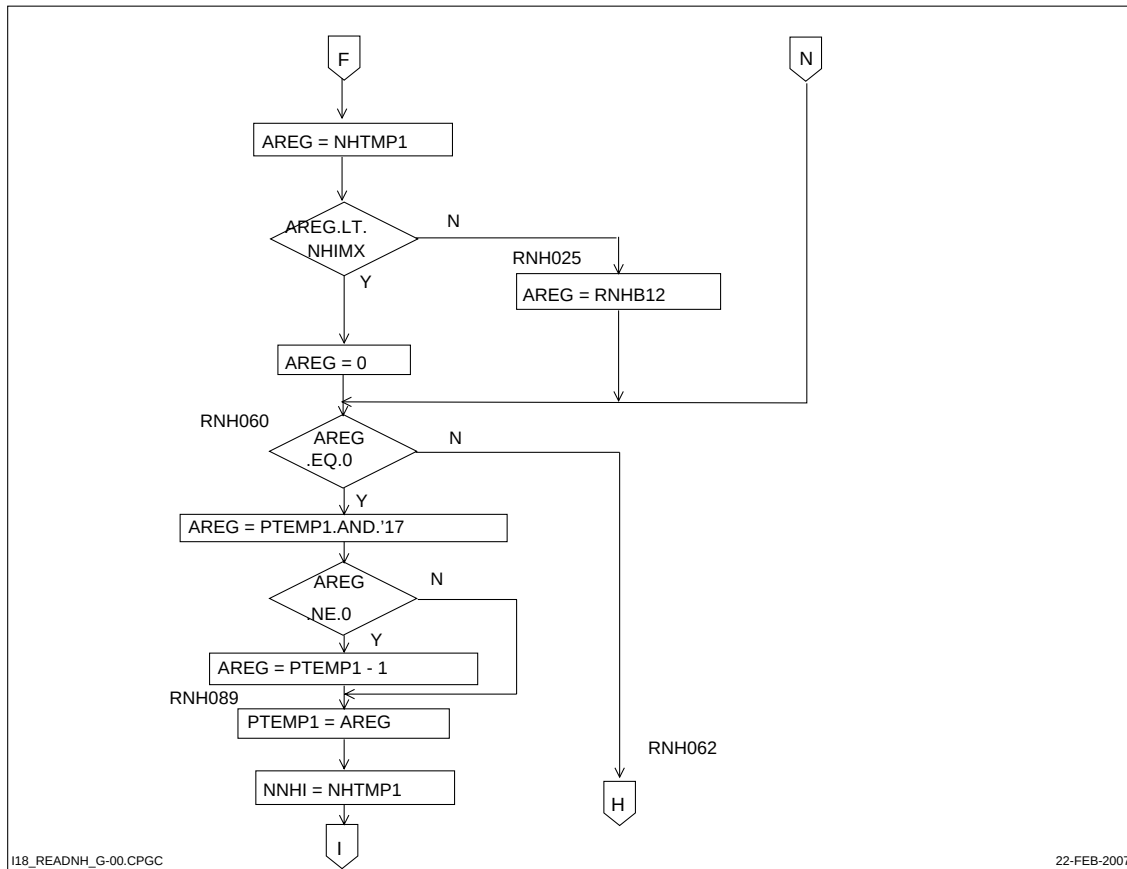
13.00 Figure 300: I18_READNH_E



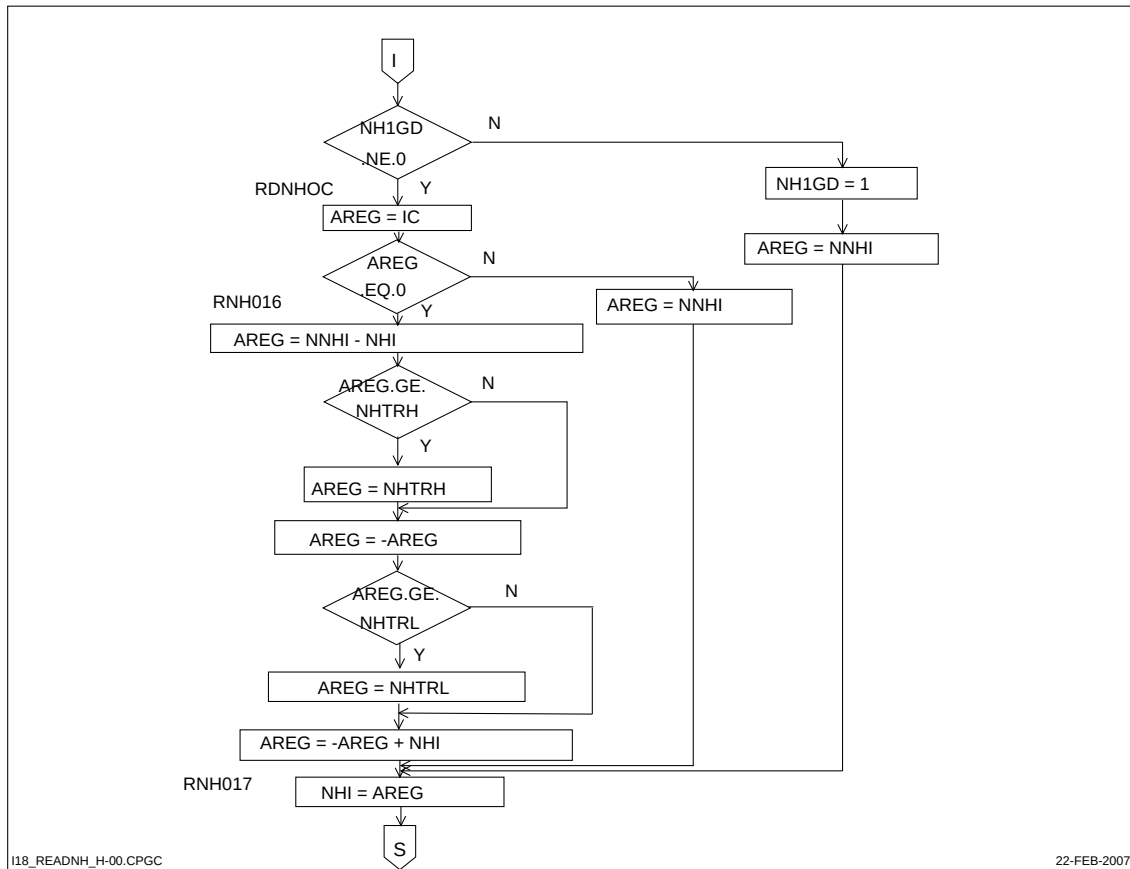
13.00 Figure 301: I18_READNH_F



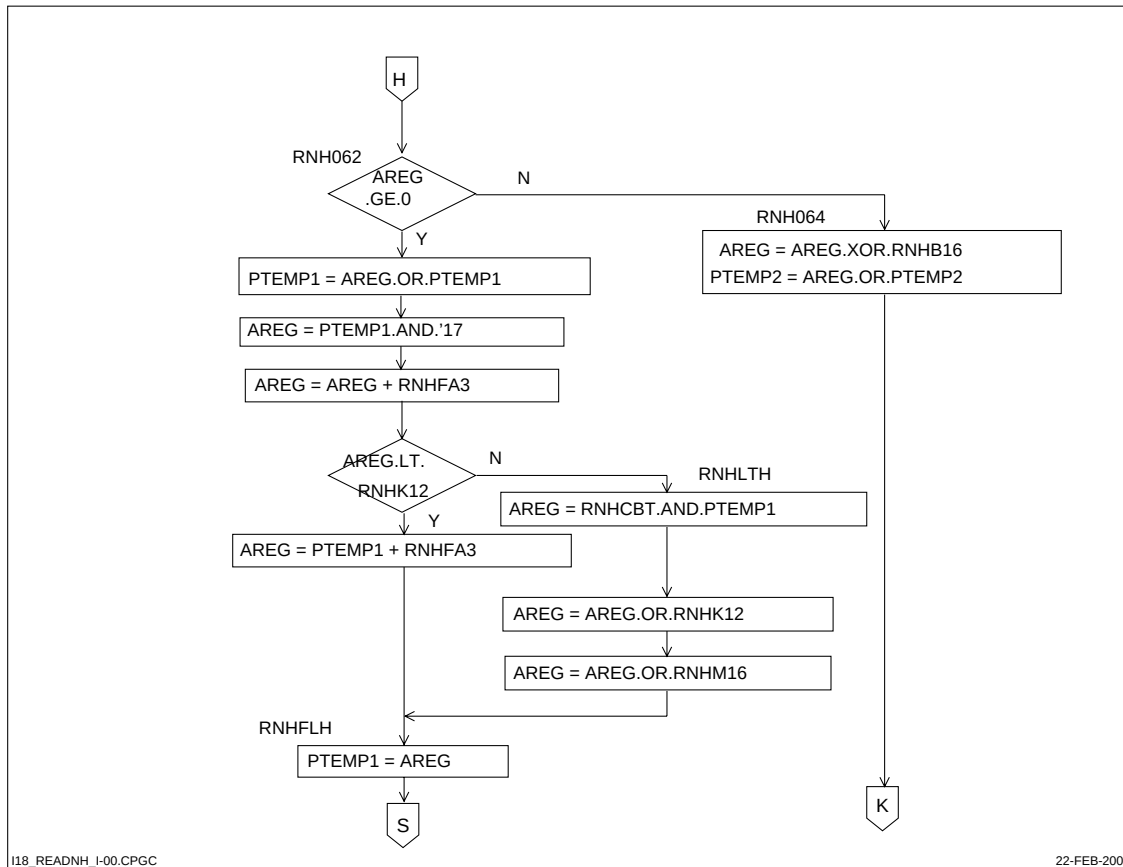
13.00 Figure 302: I18_READNH_G



13.00 Figure 303: I18_READNH_H



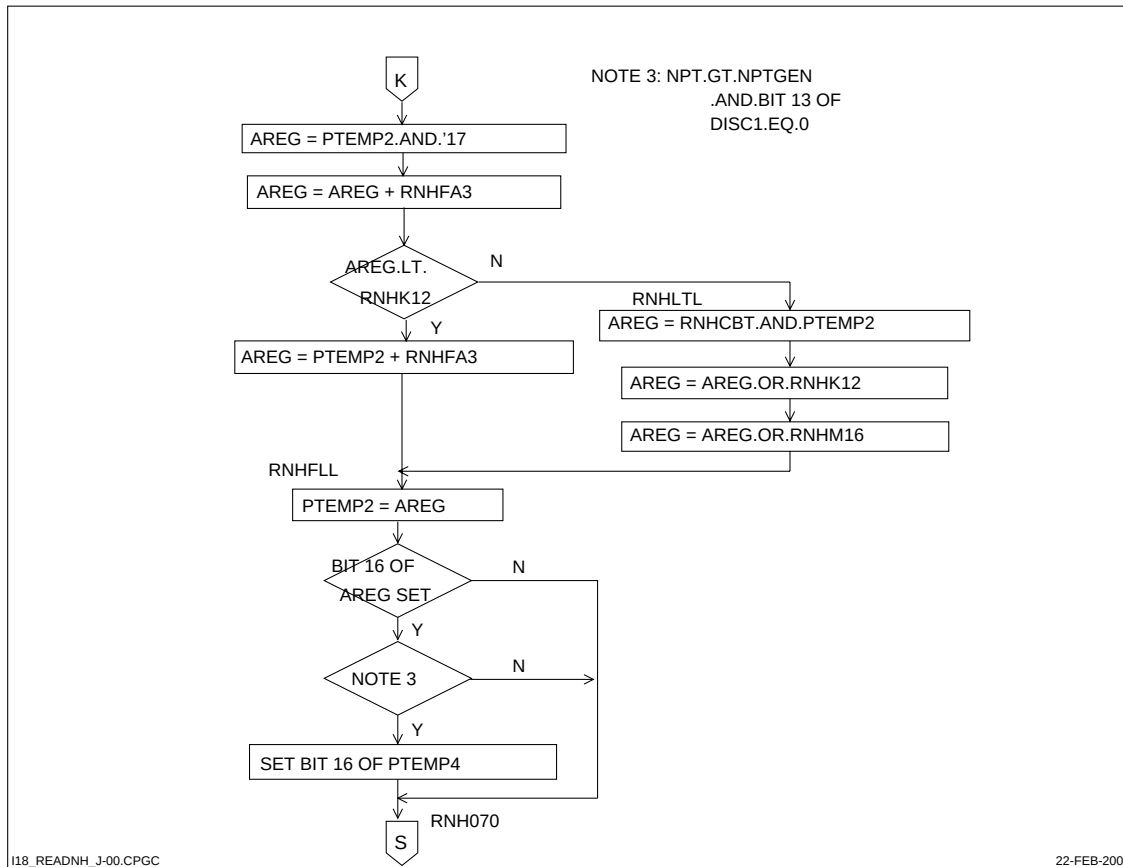
13.00 Figure 304: I18_READNH_I



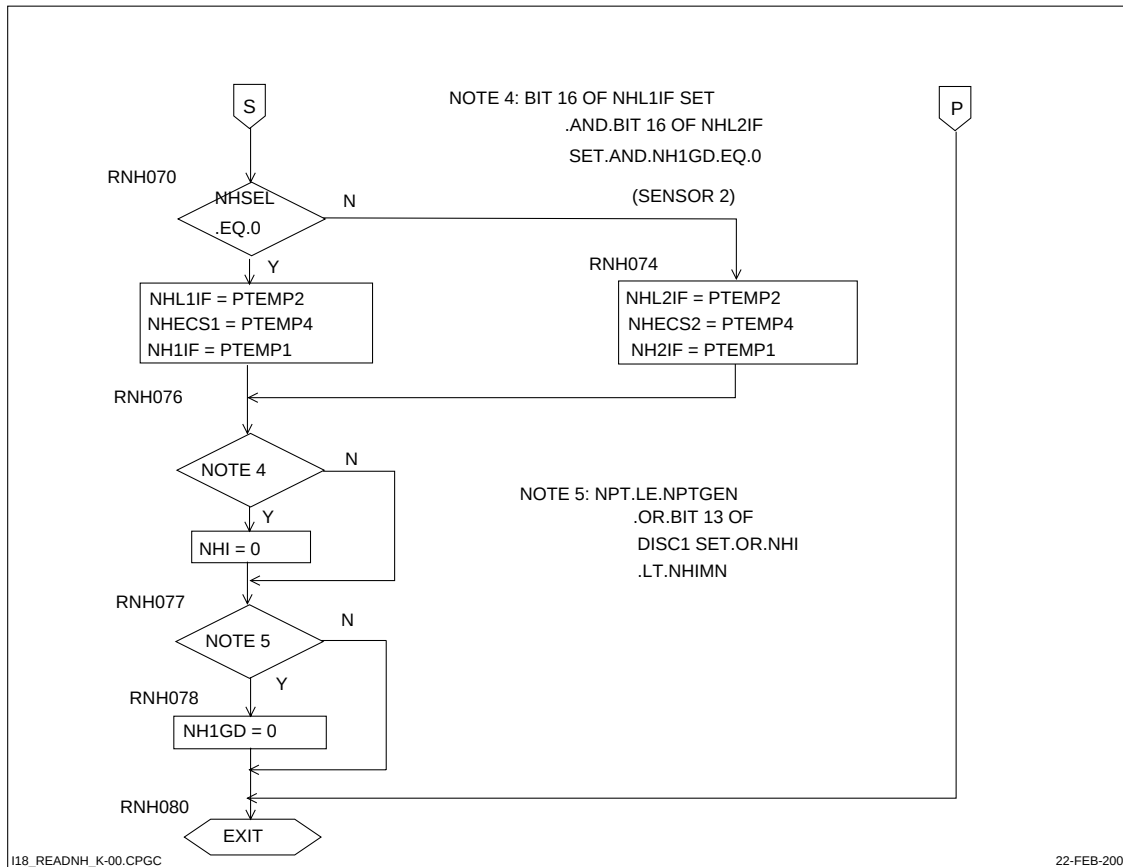
I18_READNH_I-00.CPGC

22-FEB-2007

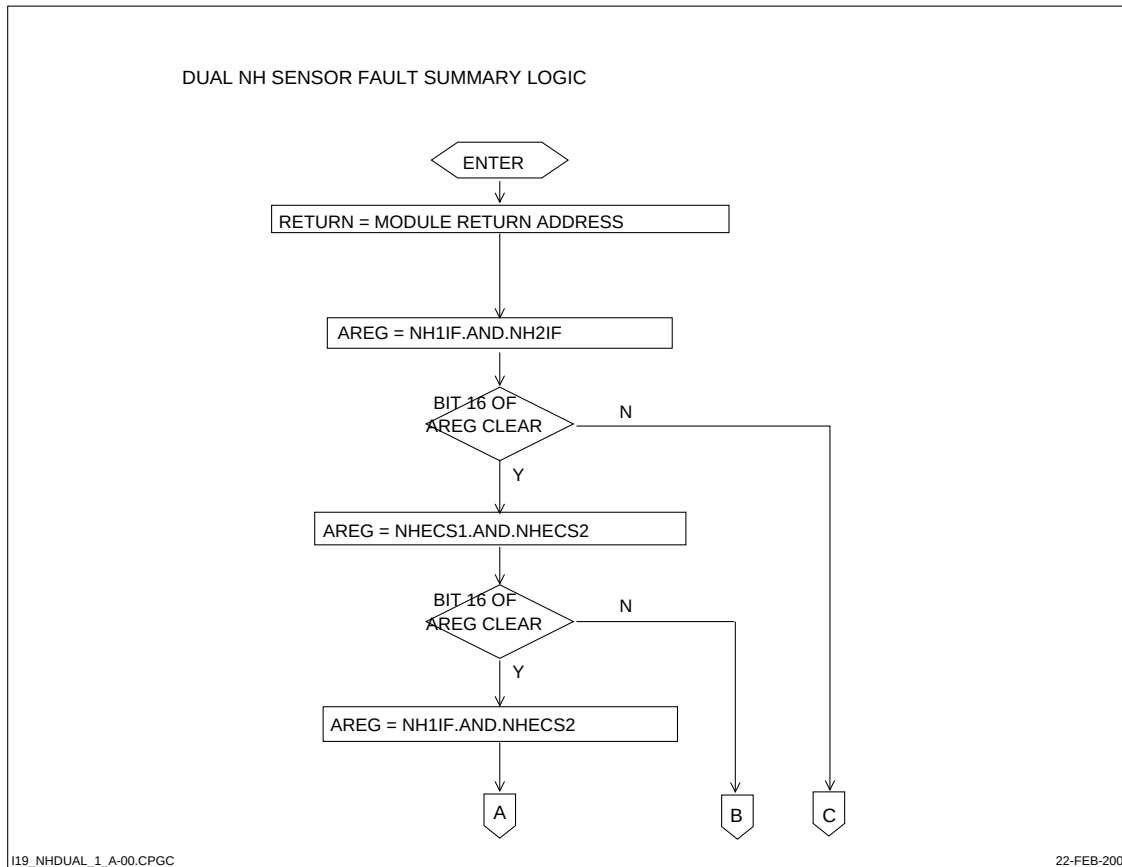
13.00 Figure 305: I18_READNH_J



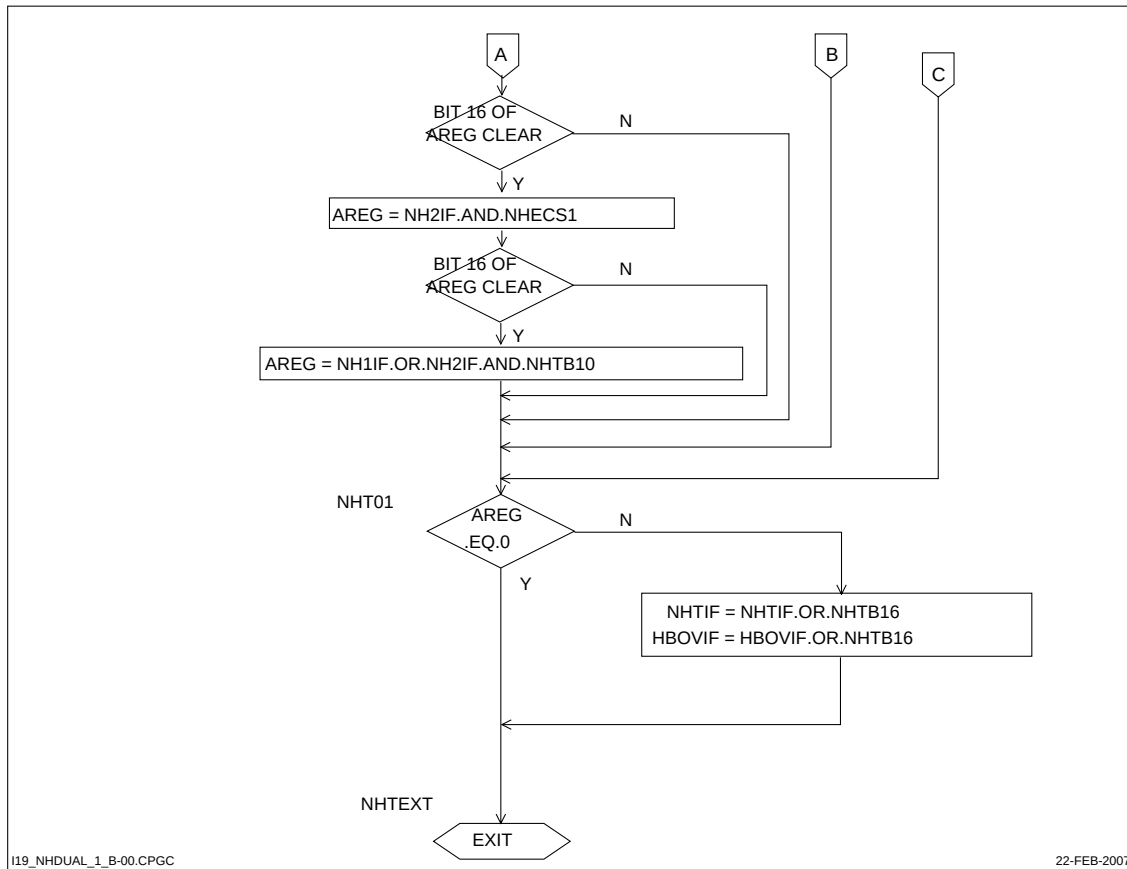
13.00 Figure 306: I18_READNH_K



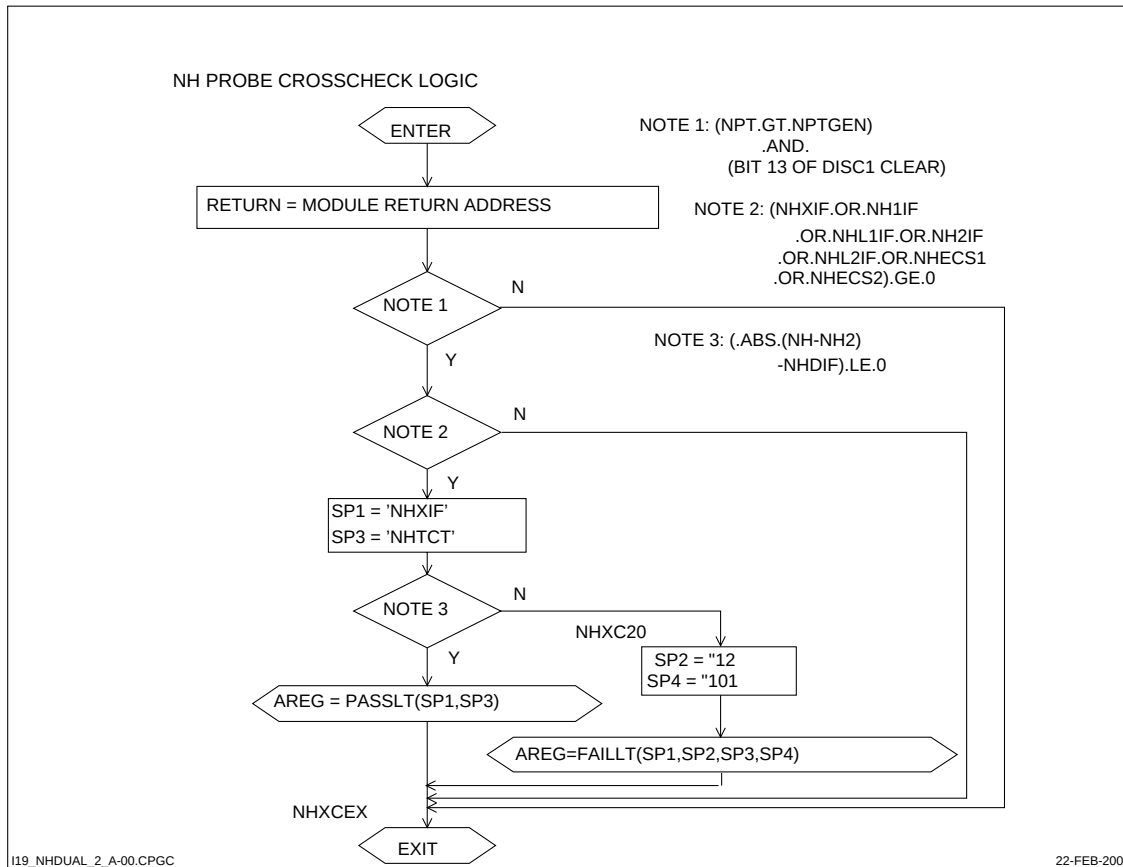
13.00 Figure 307: I19 _NHDUAL _1 _A



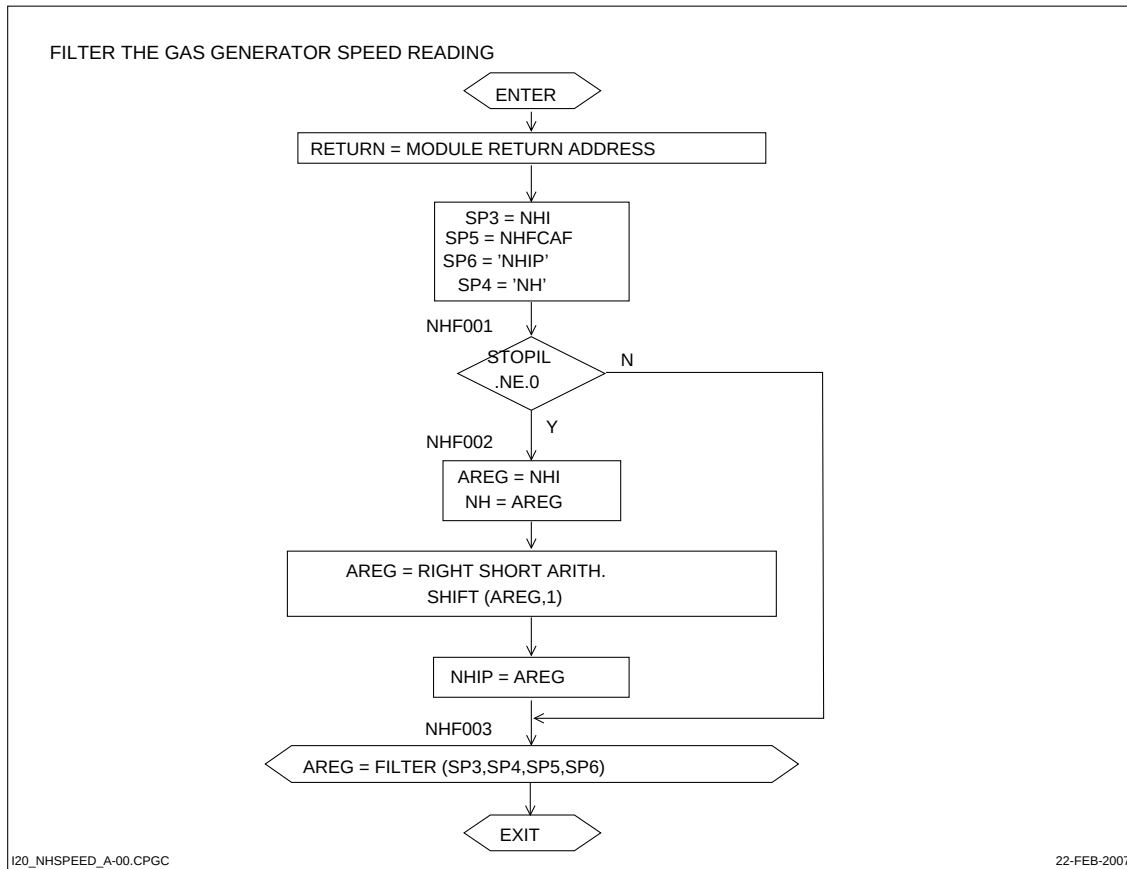
13.00 Figure 308: I19 _NHDUAL_1_B



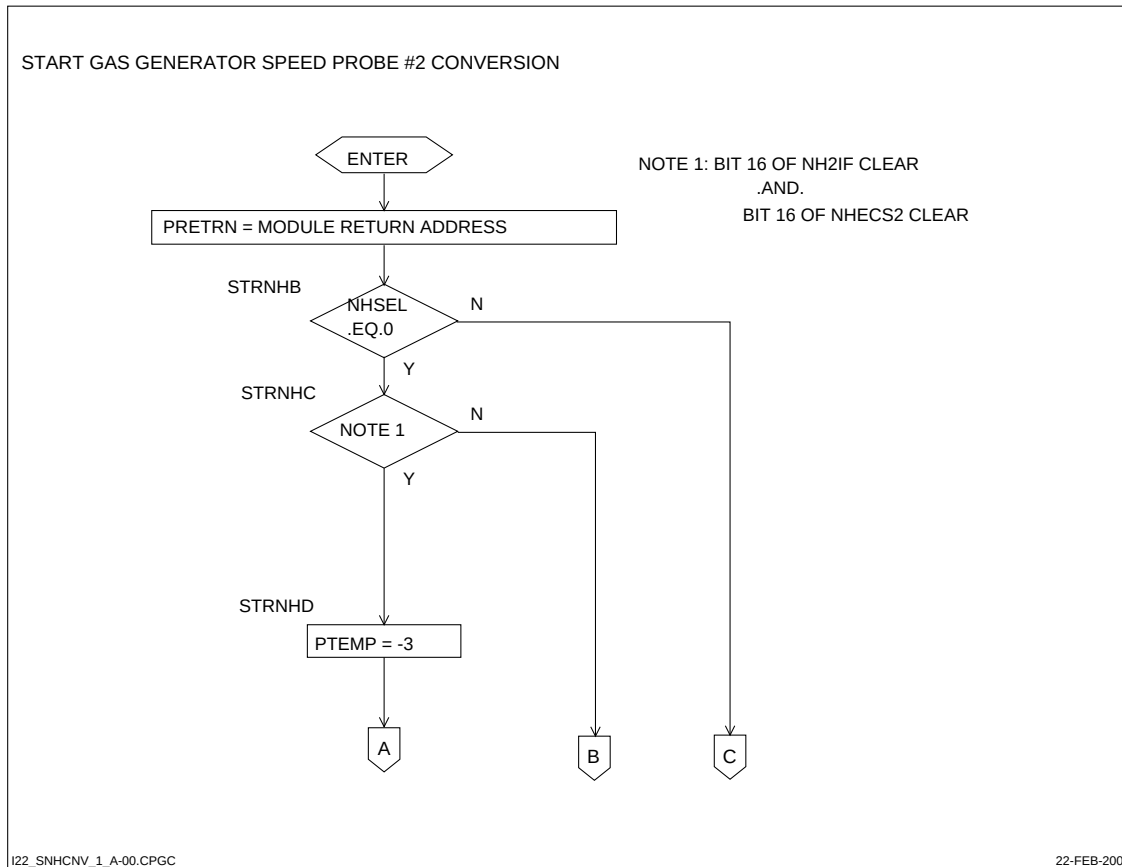
13.00 Figure 309: I19_NHDUAL_2_A



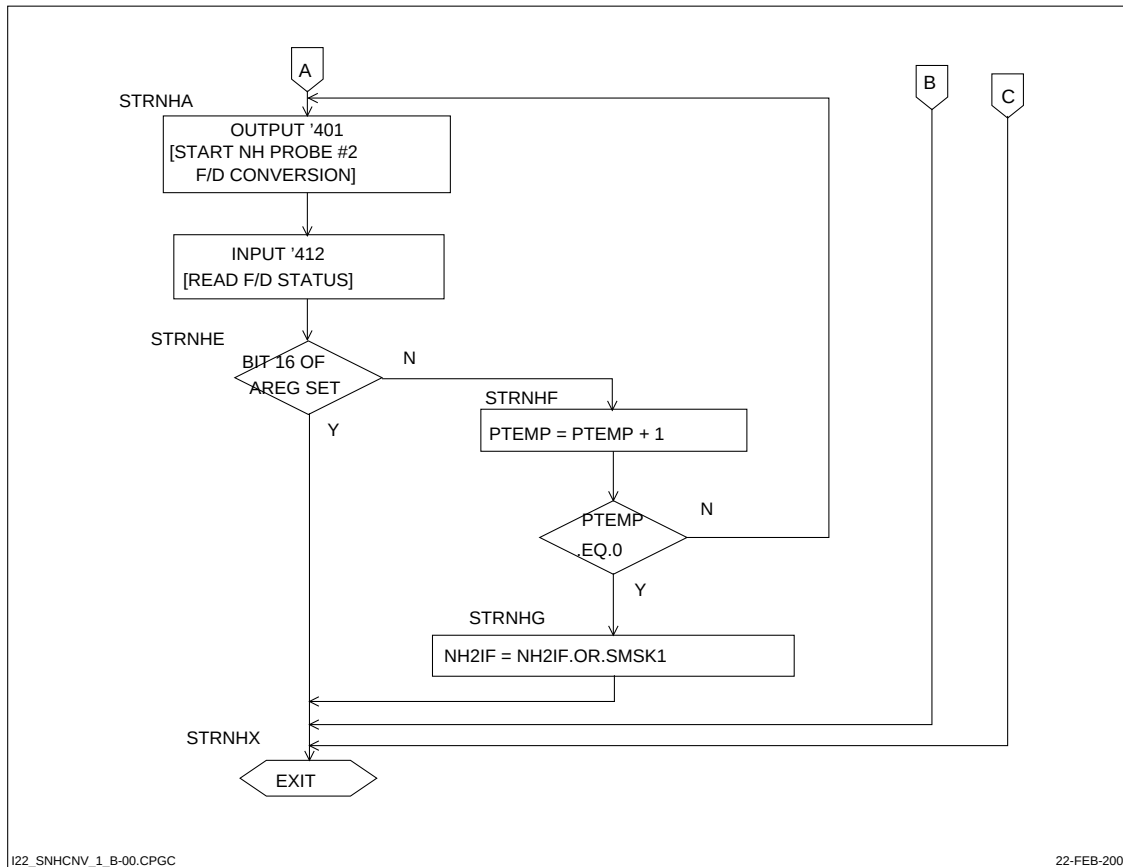
13.00 Figure 310: I20 _NHSPEED _A



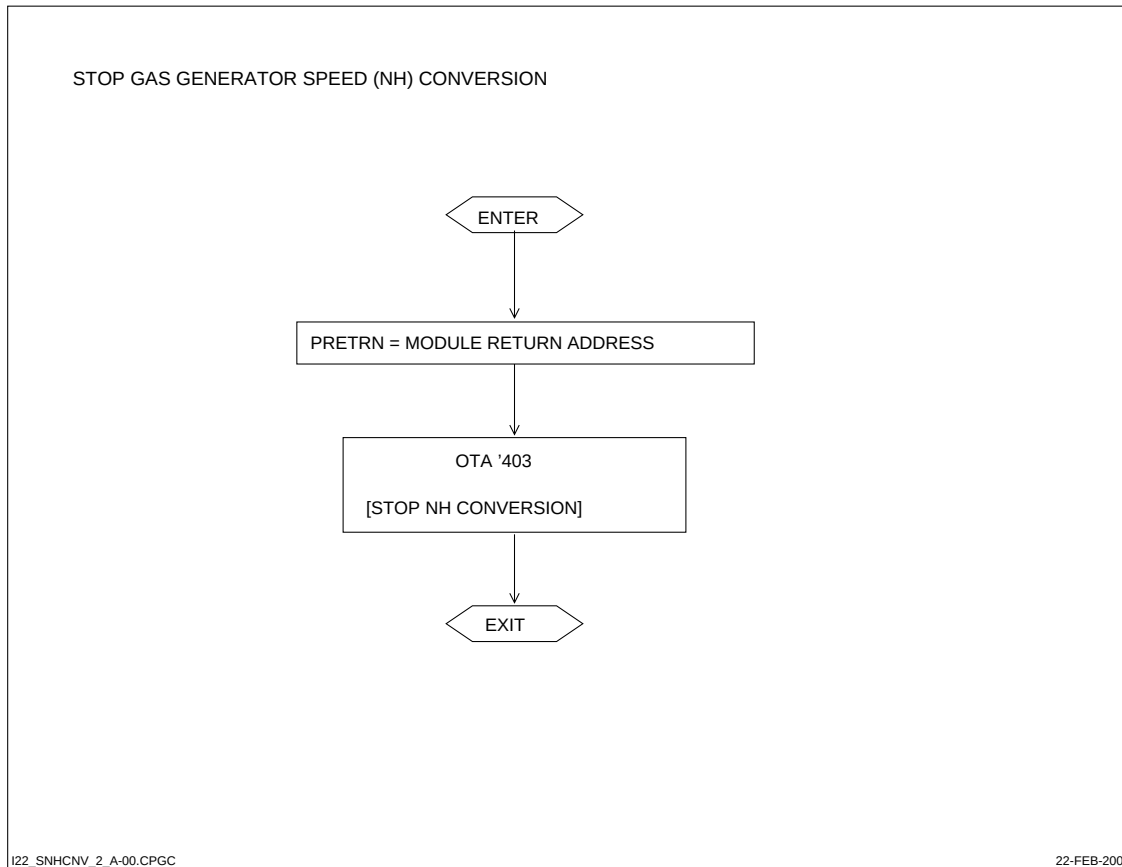
13.00 Figure 311: I22 _SNHCNV _1 _A



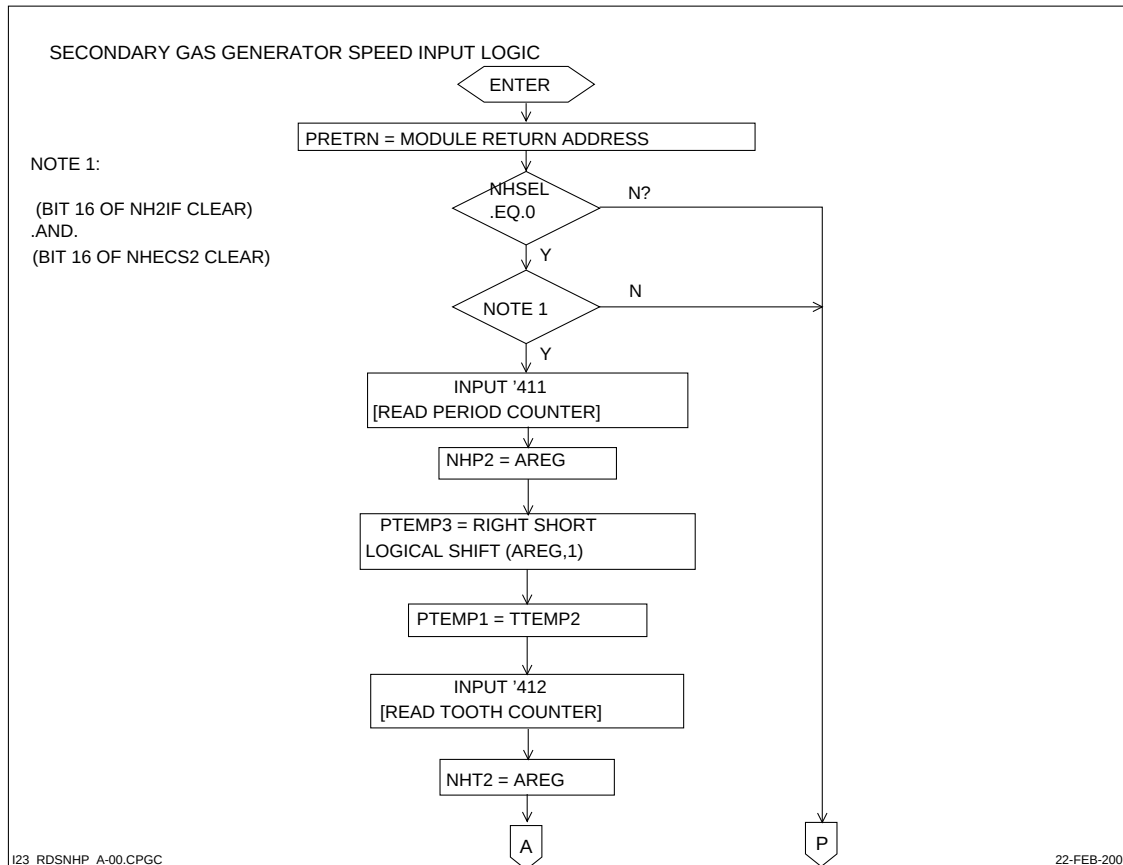
13.00 Figure 312: I22 _SNHCNV _1 _B



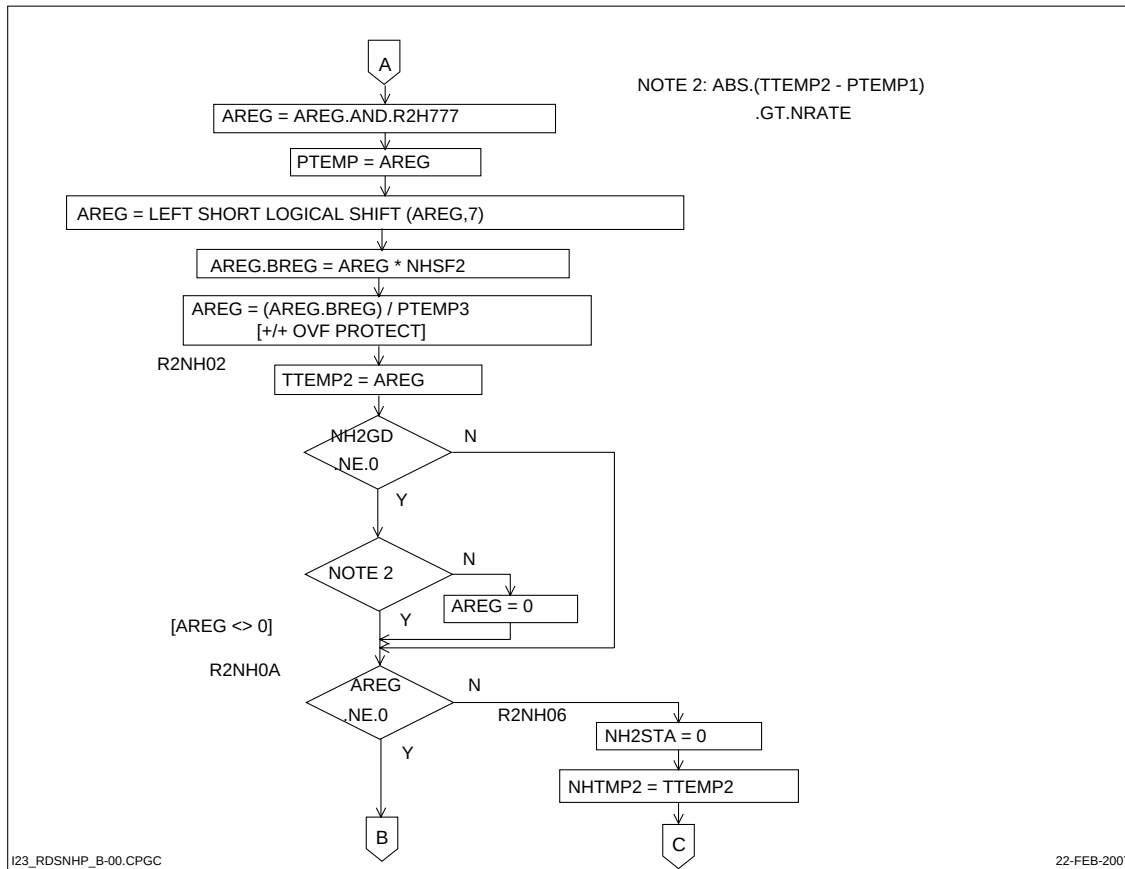
13.00 Figure 313: I22 _SNHCNV _2 _A



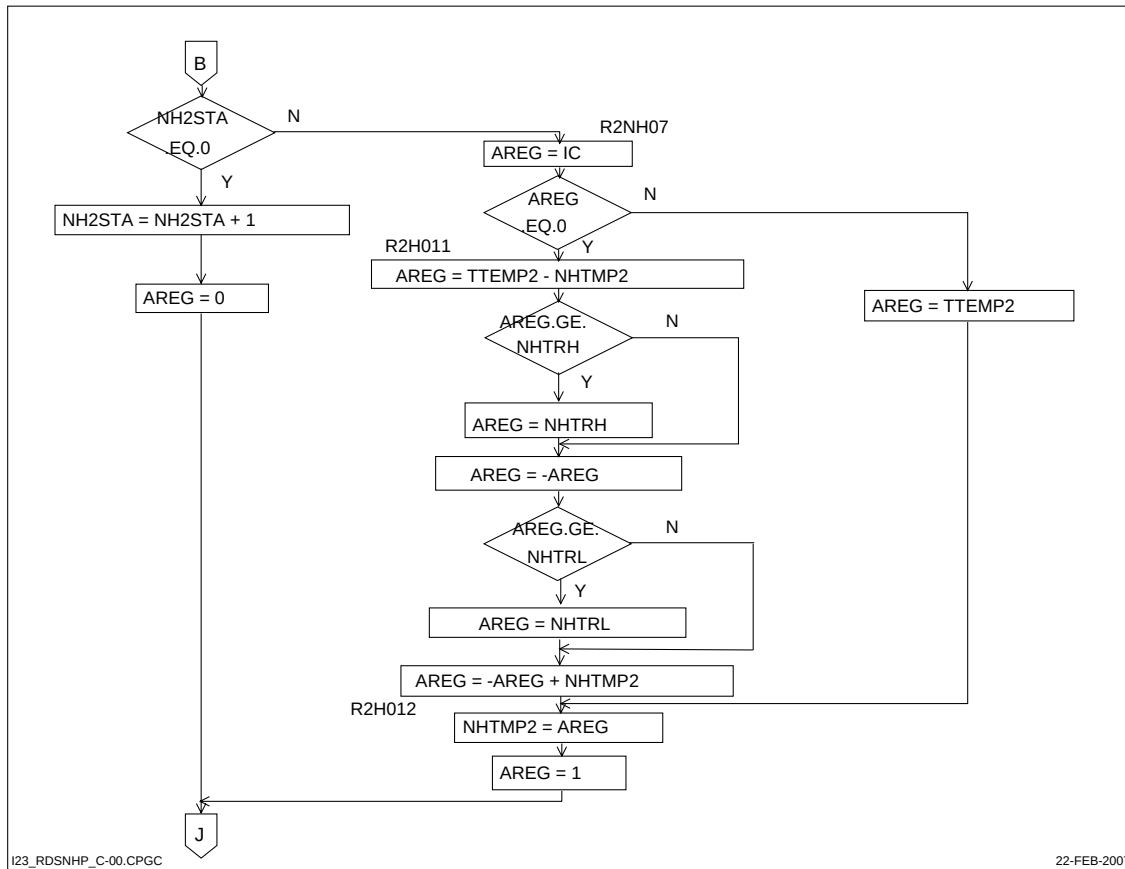
13.00 Figure 314: I23 _RDSNHP _A



13.00 Figure 315: I23_RDSNHP_B



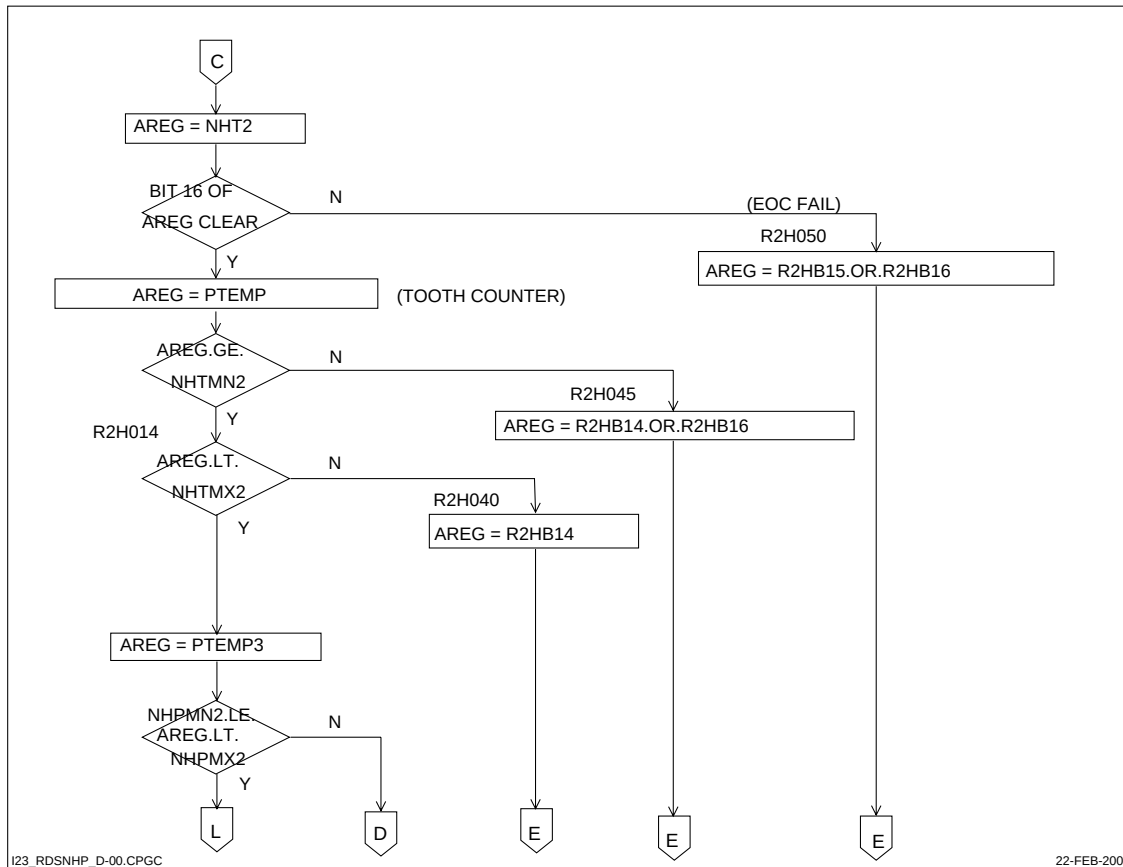
13.00 Figure 316: I23_RDSNHP_C



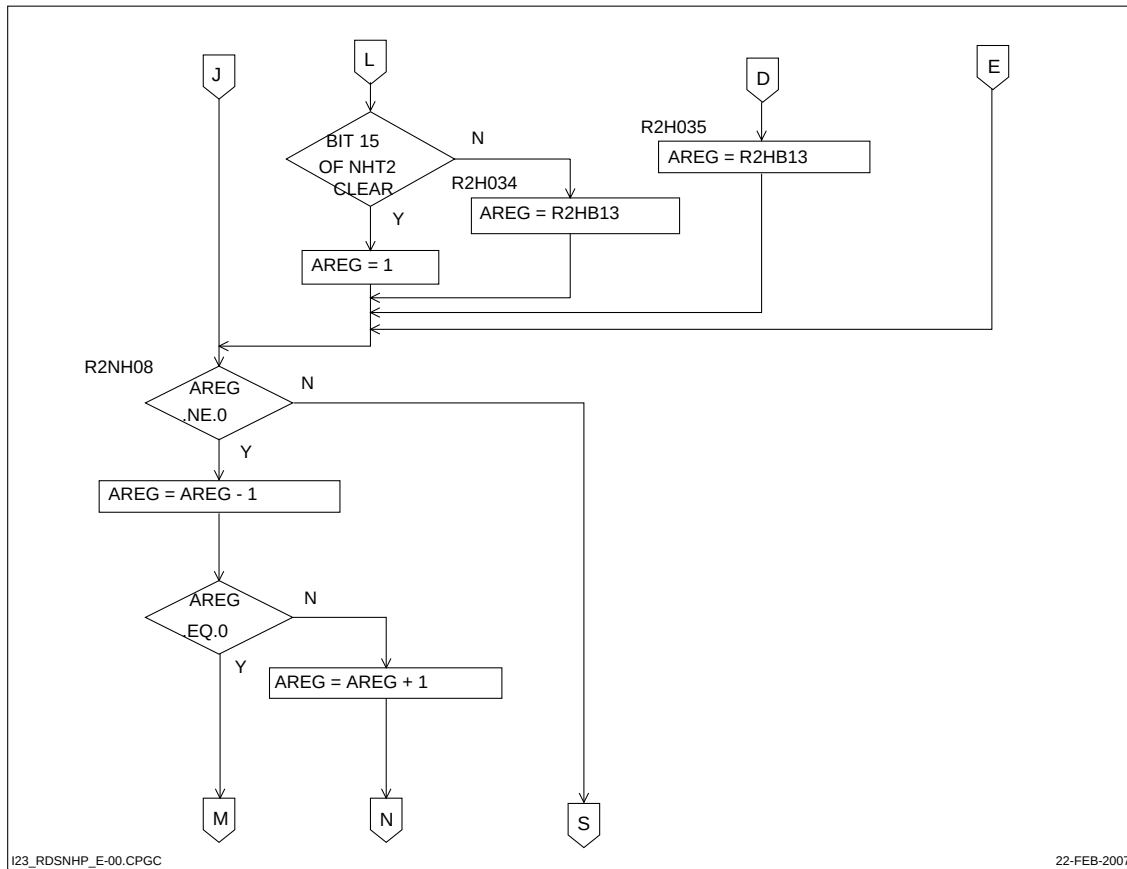
I23_RDSNHP_C-00.CPGC

22-FEB-2007

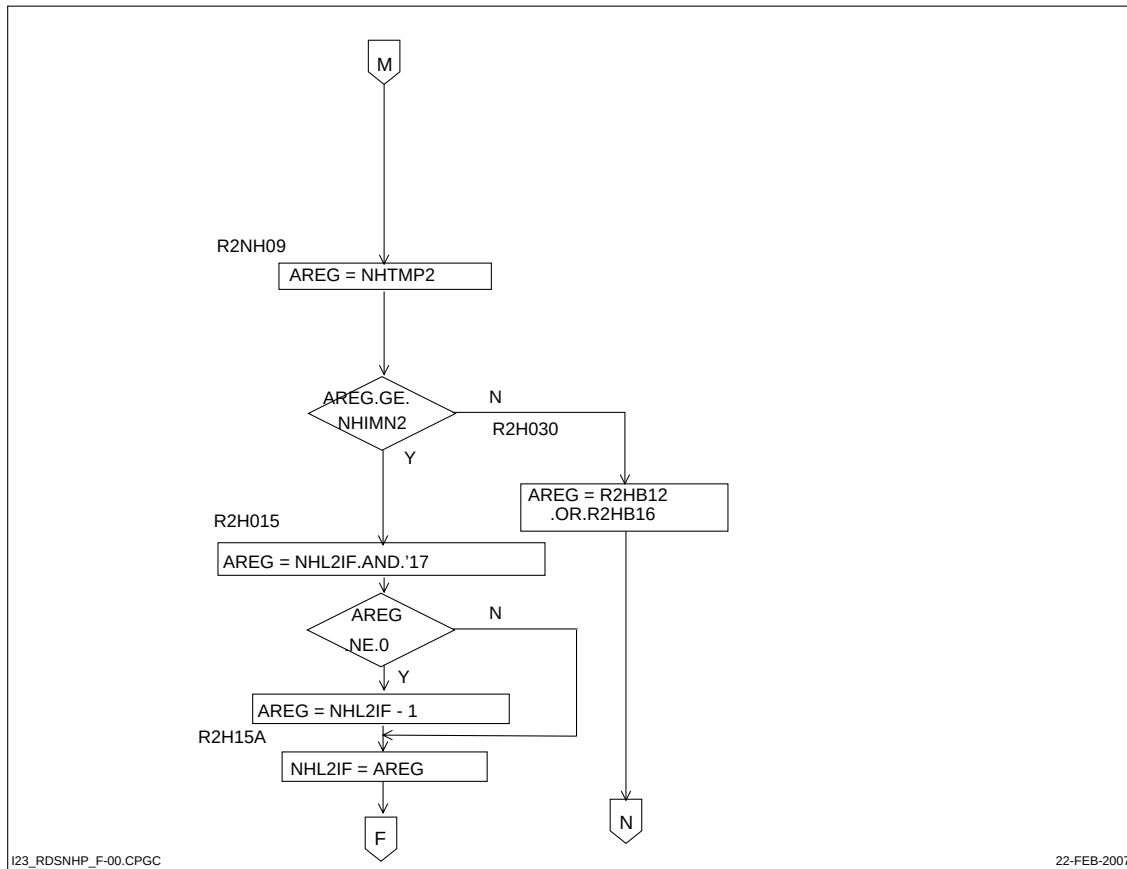
13.00 Figure 317: I23_RDSNHP_D



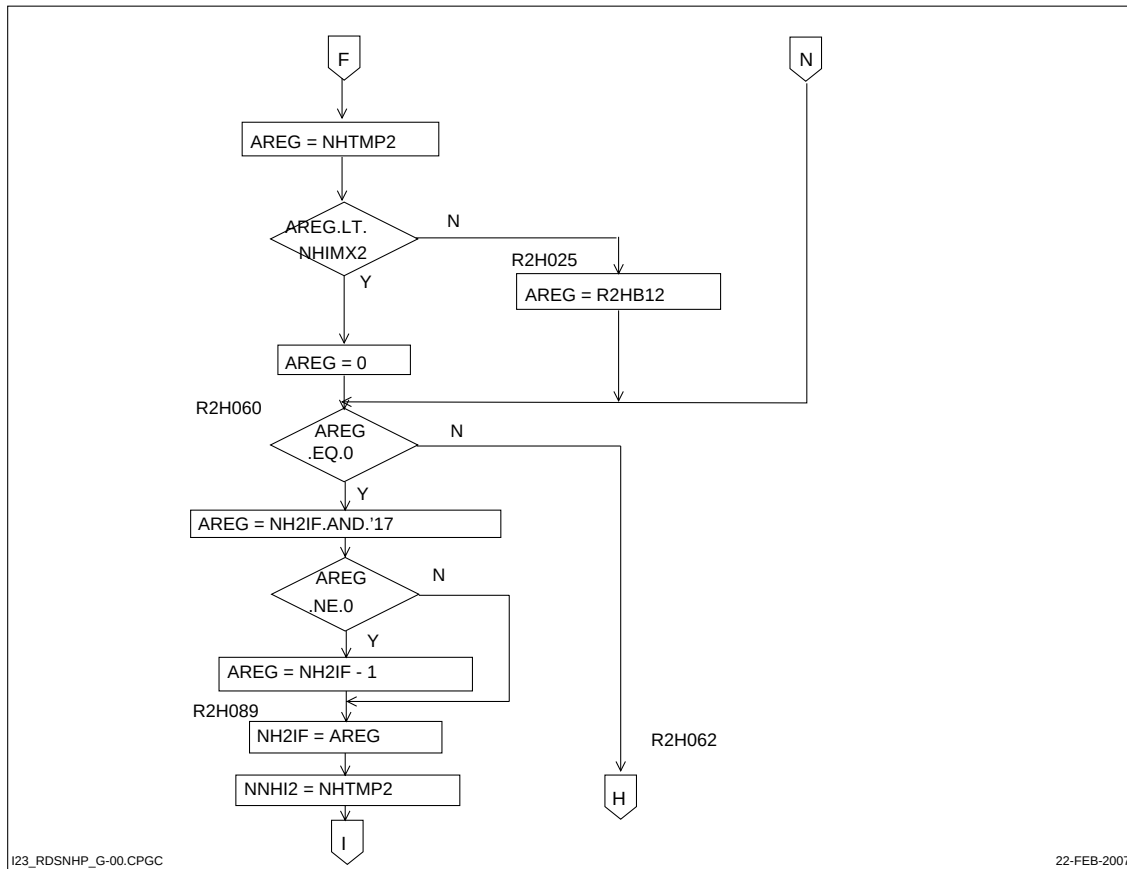
13.00 Figure 318: I23_RDSNHP_E



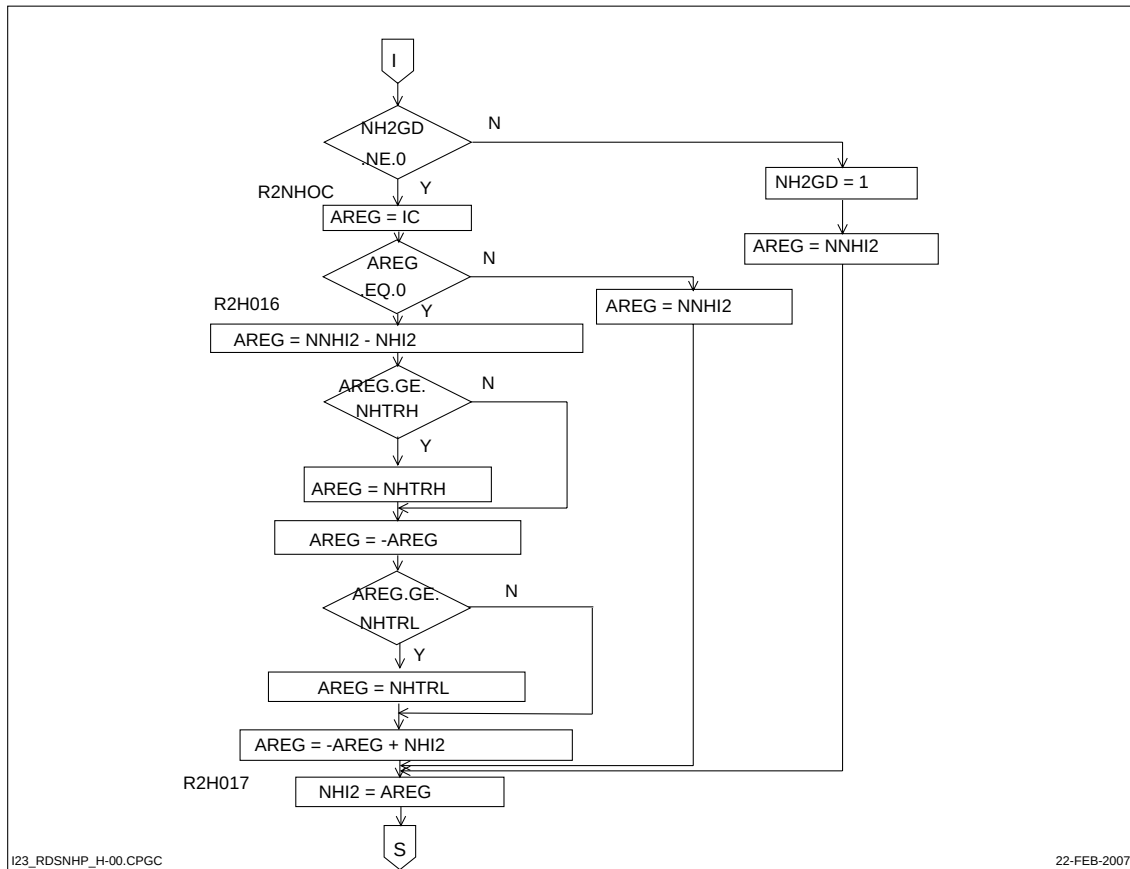
13.00 Figure 319: I23_RDSNHP_F



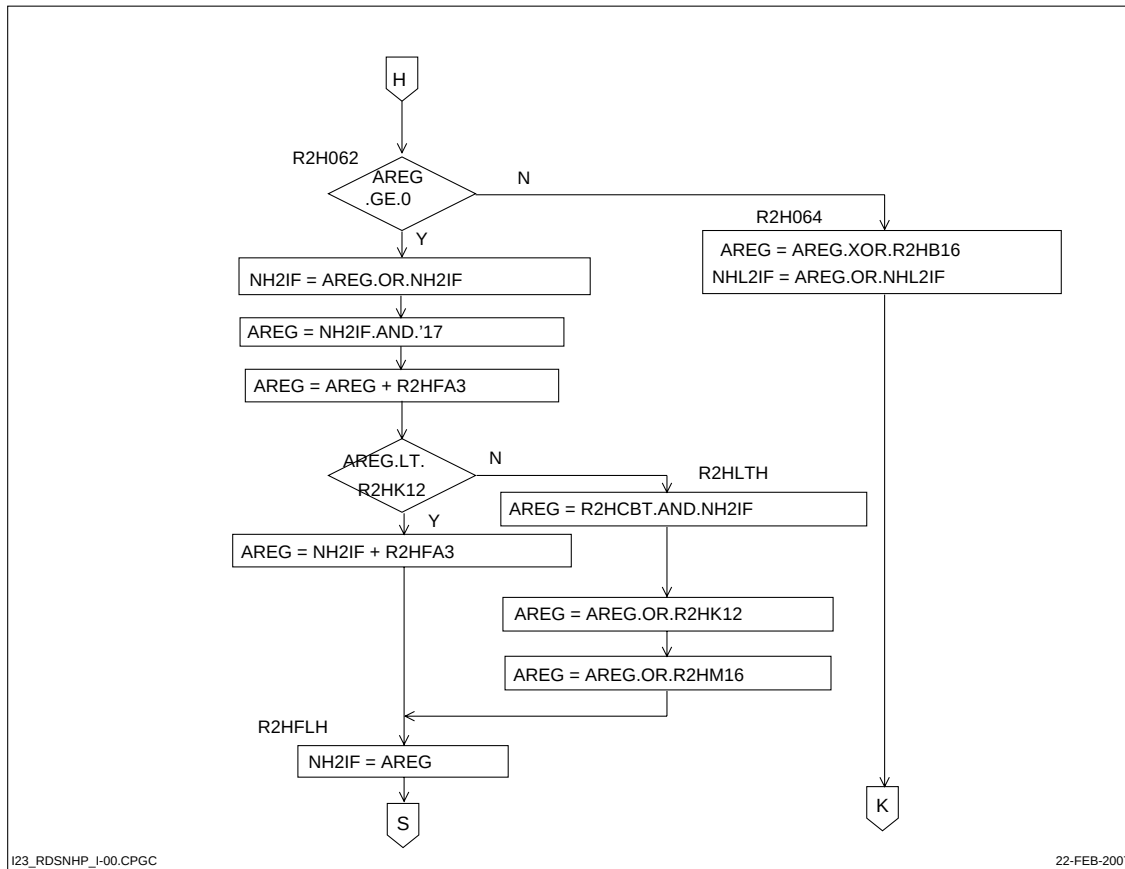
13.00 Figure 320: I23_RDSNHP_G



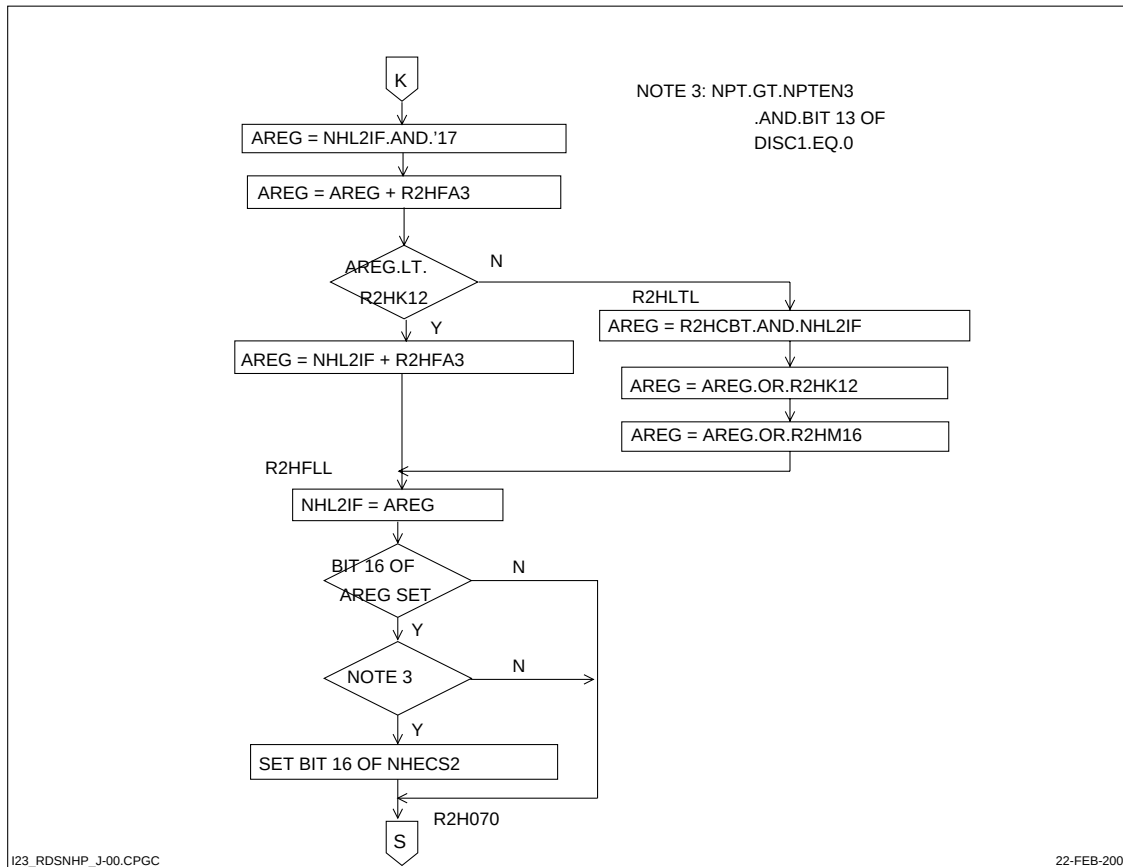
13.00 Figure 321: I23_RDSNHP_H



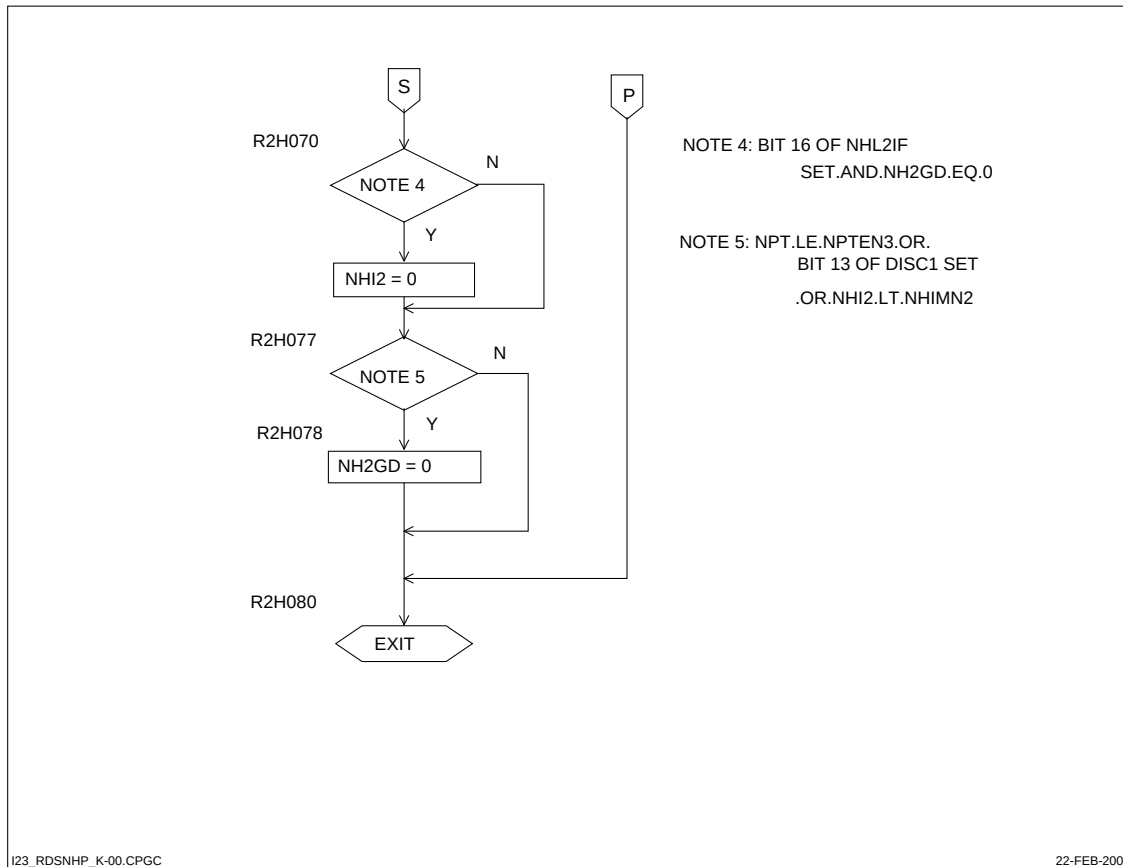
13.00 Figure 322: I23 _RDSNHP _I



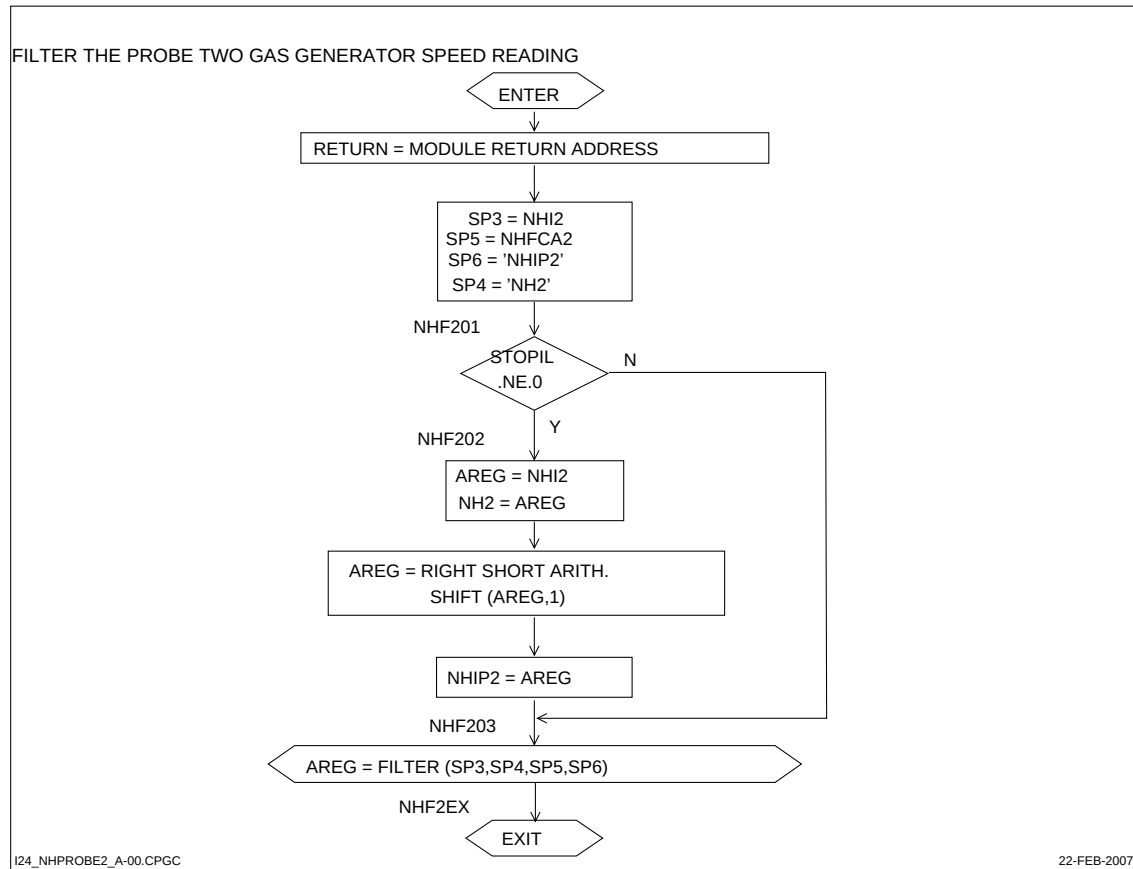
13.00 Figure 323: I23_RDSNHP_J



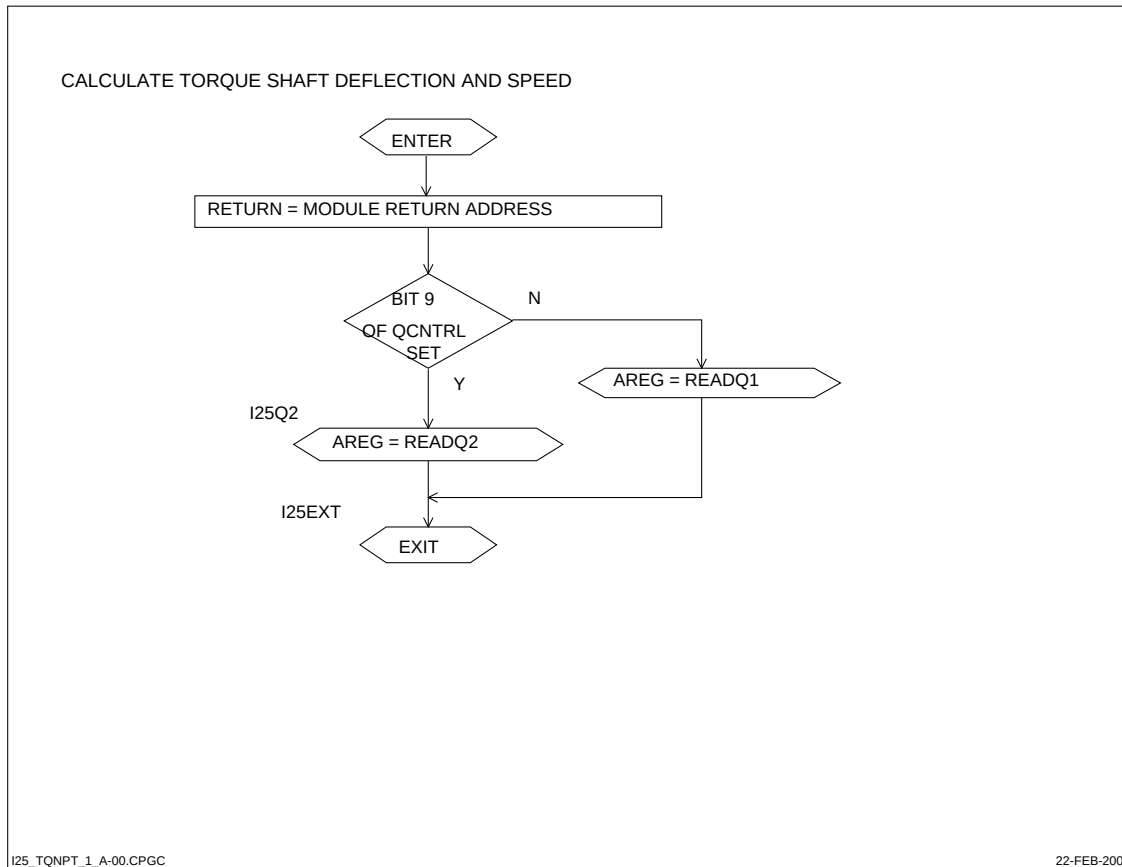
13.00 Figure 324: I23 _RDSNHP _K



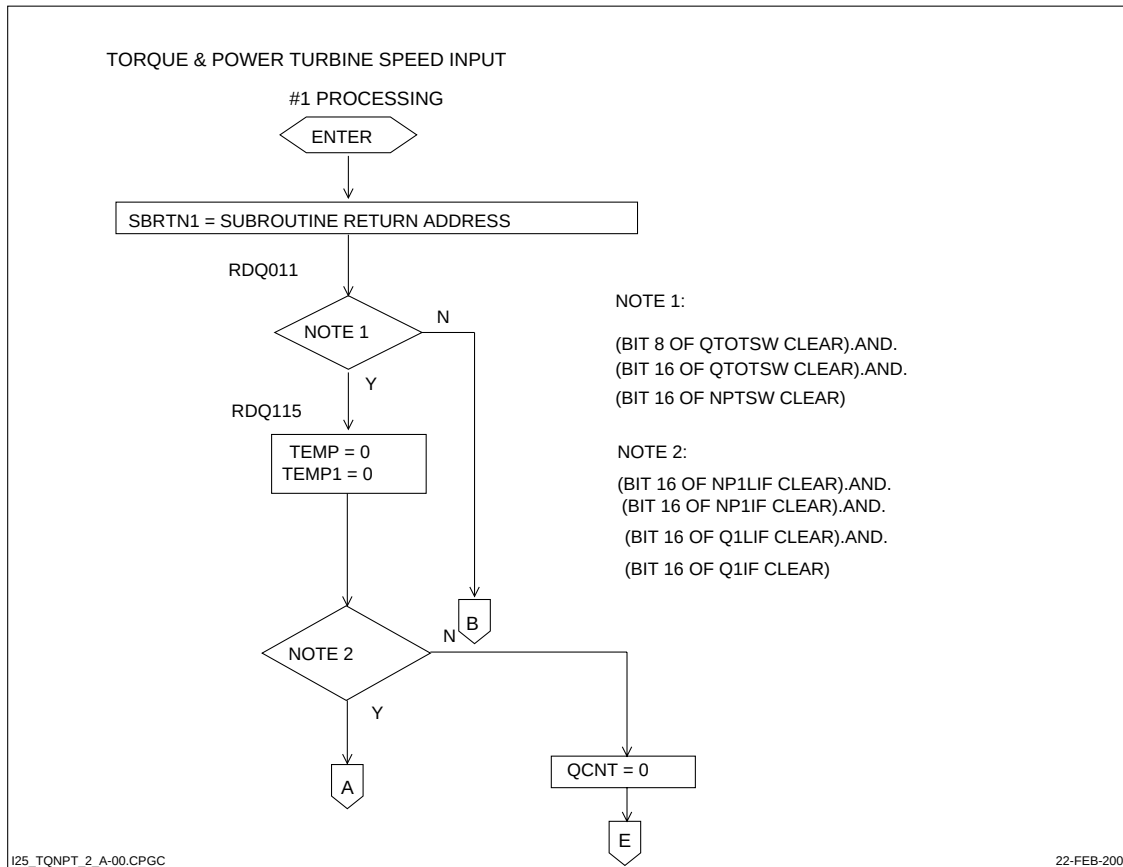
13.00 Figure 325: I24 _NHPRBE2 _A



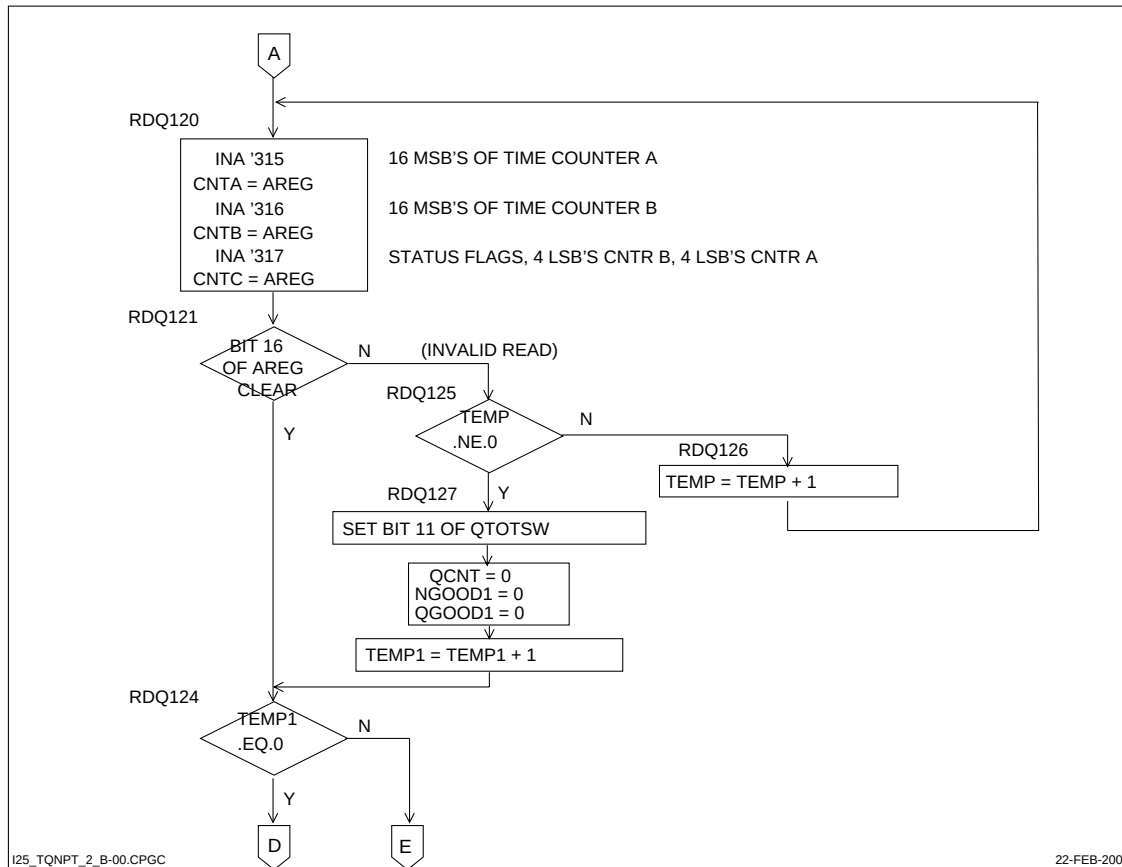
13.00 Figure 326: I25 _TQNPT _1 _A



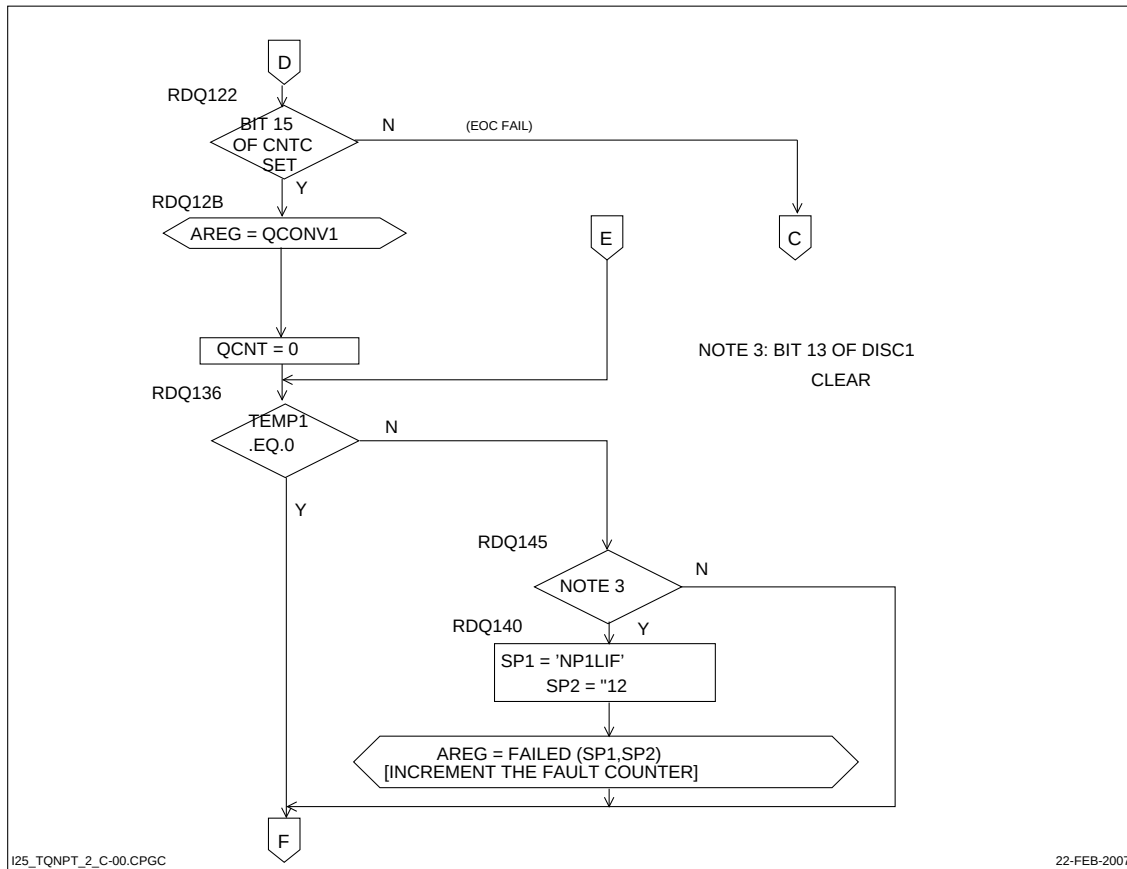
13.00 Figure 327: I25 _TQNPT _2 _A



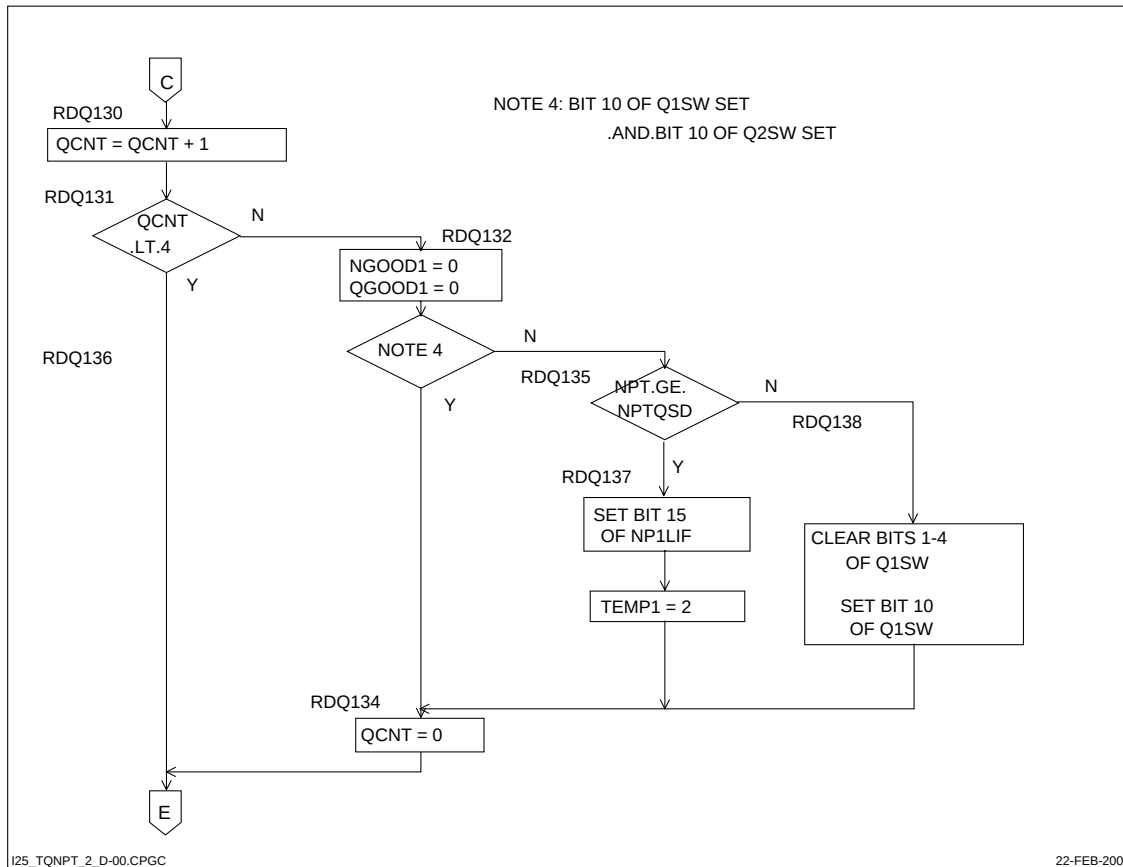
13.00 Figure 328: I25_TQNPT_2_B



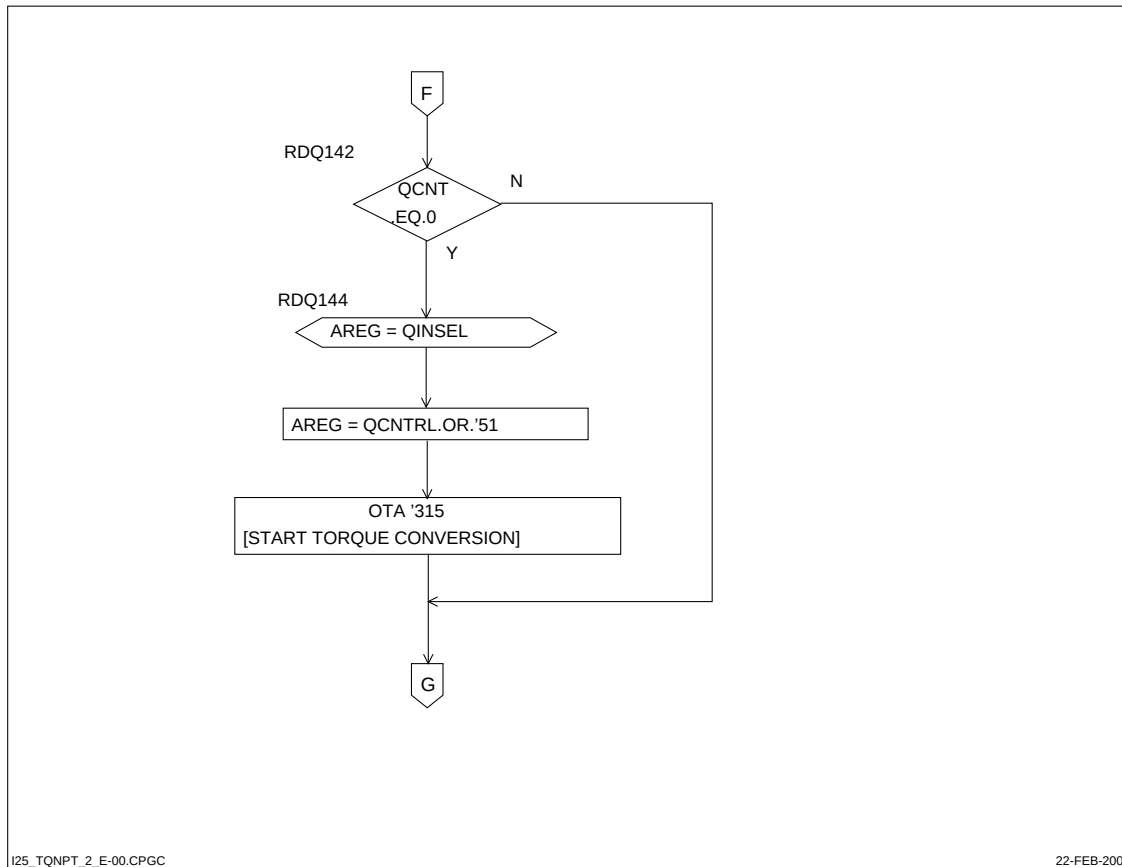
13.00 Figure 329: I25_TQNPT_2_C



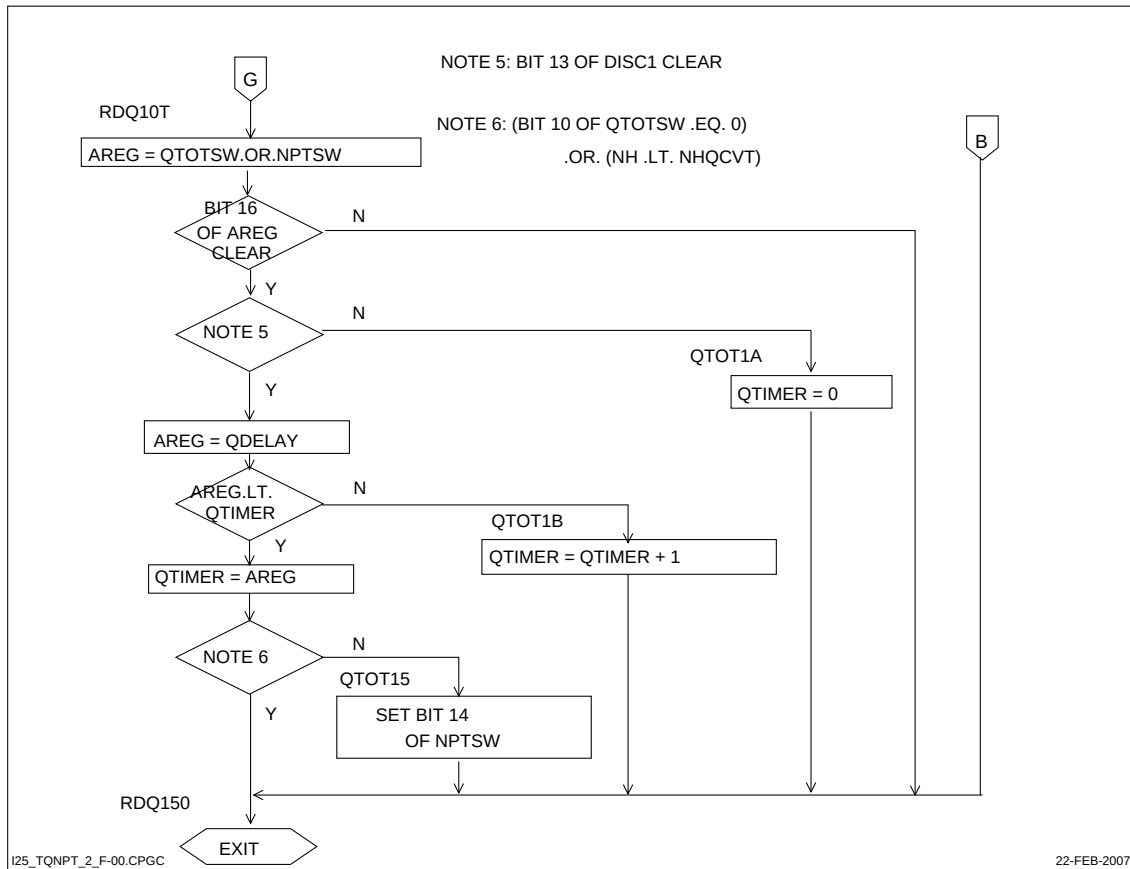
13.00 Figure 330: I25_TQNPT_2_D



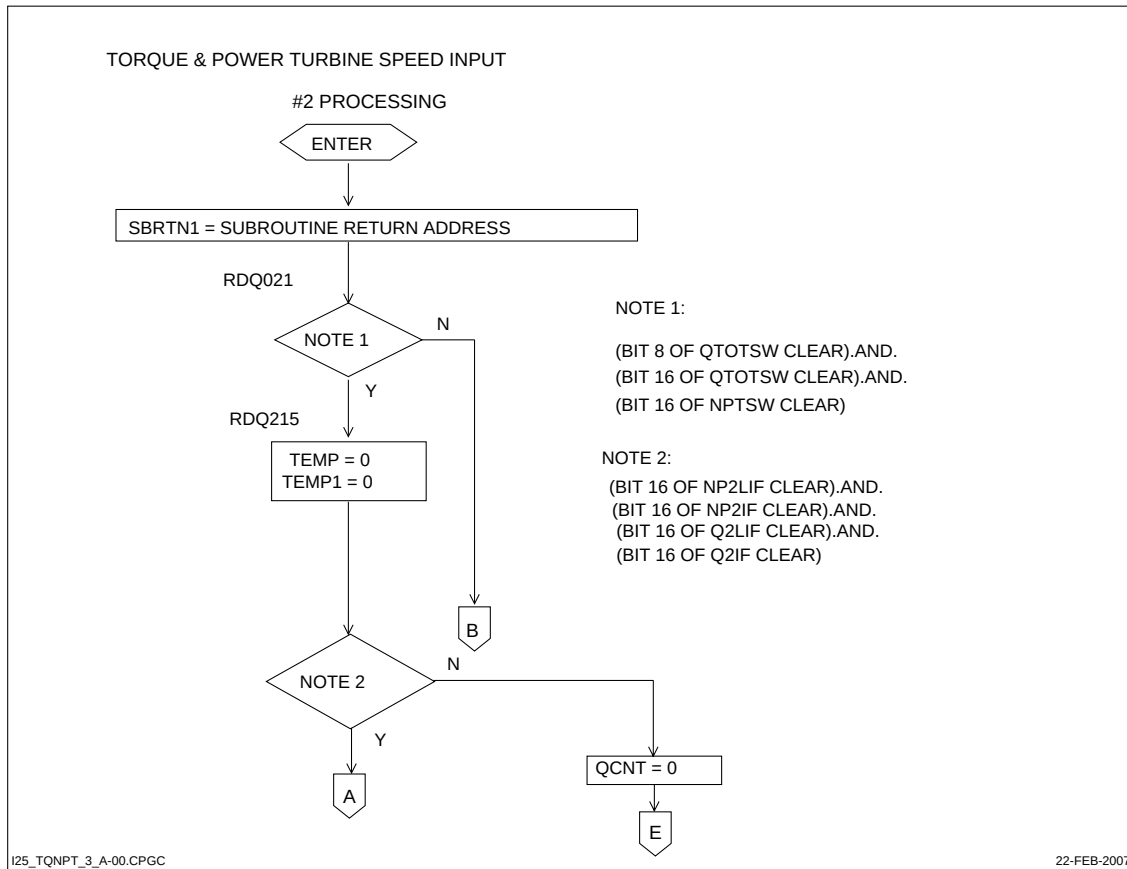
13.00 Figure 331: I25 _TQNPT _2 _E



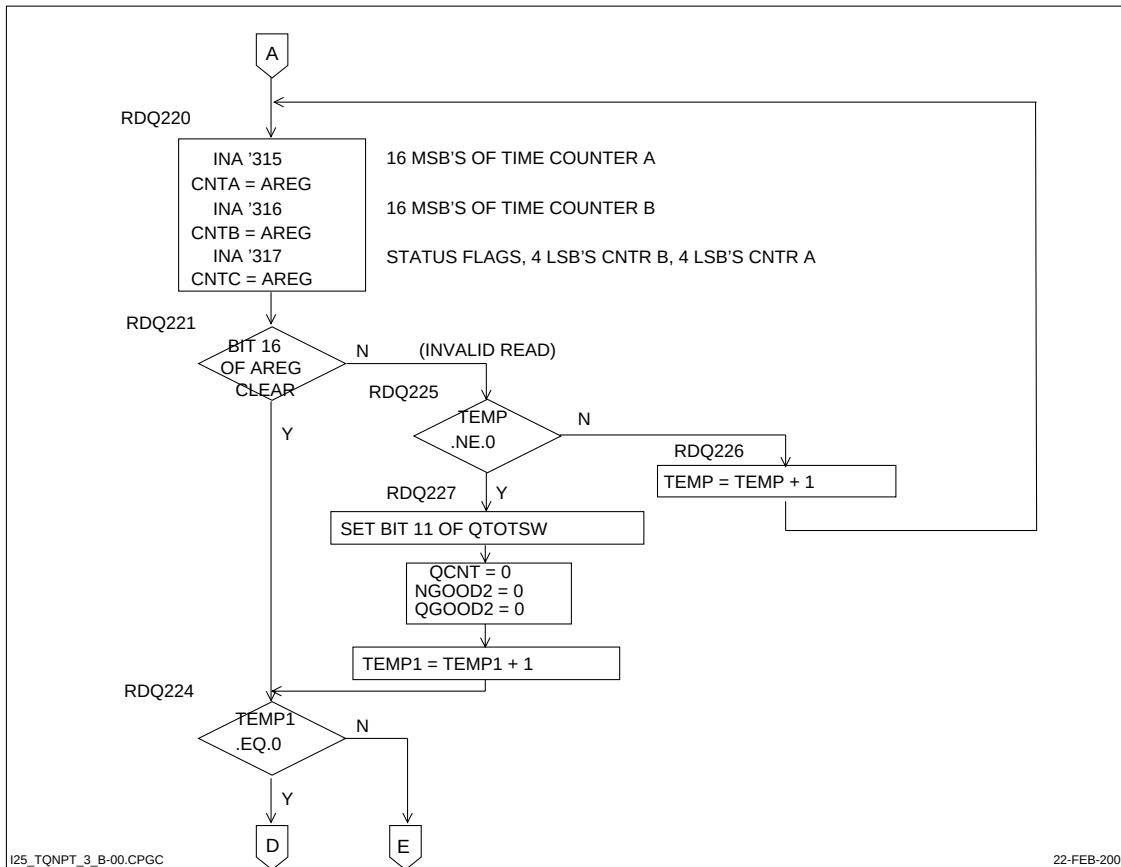
13.00 Figure 332: I25_TQNPT_2_F



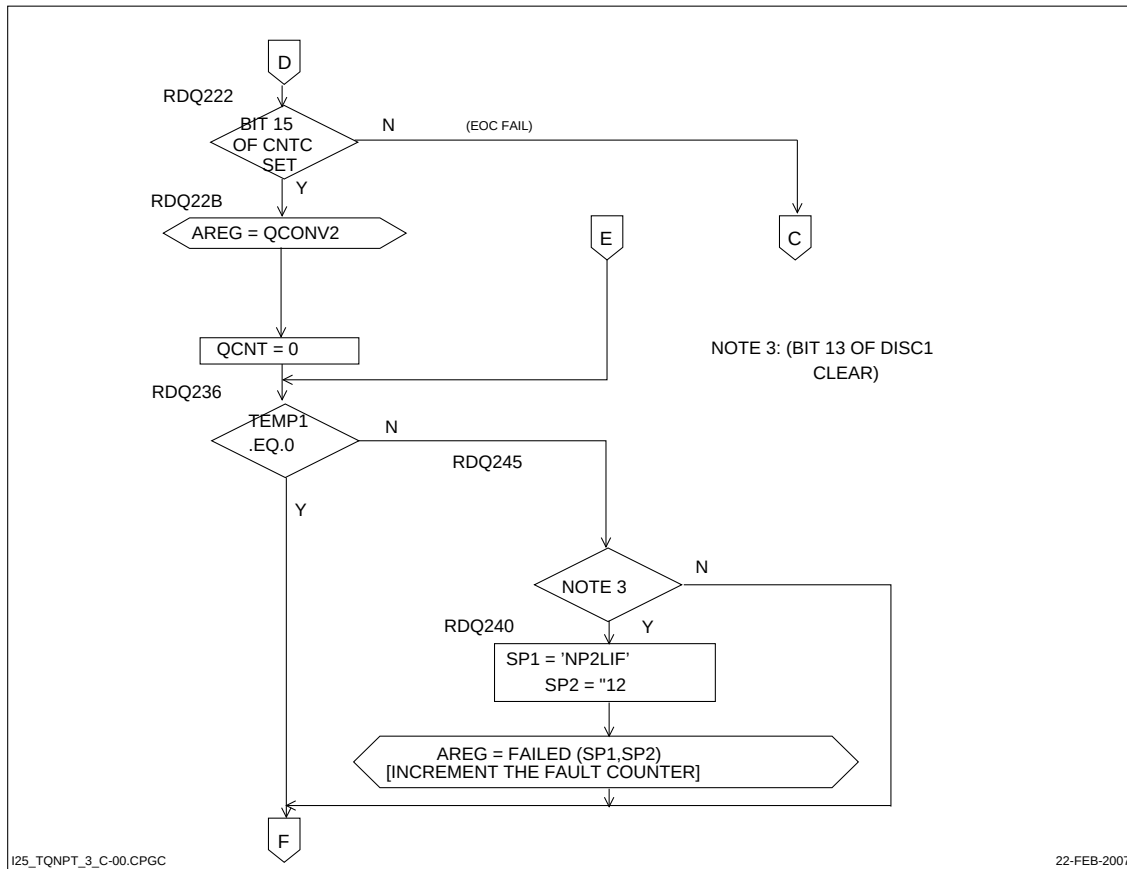
13.00 Figure 333: I25_TQNPT_3_A



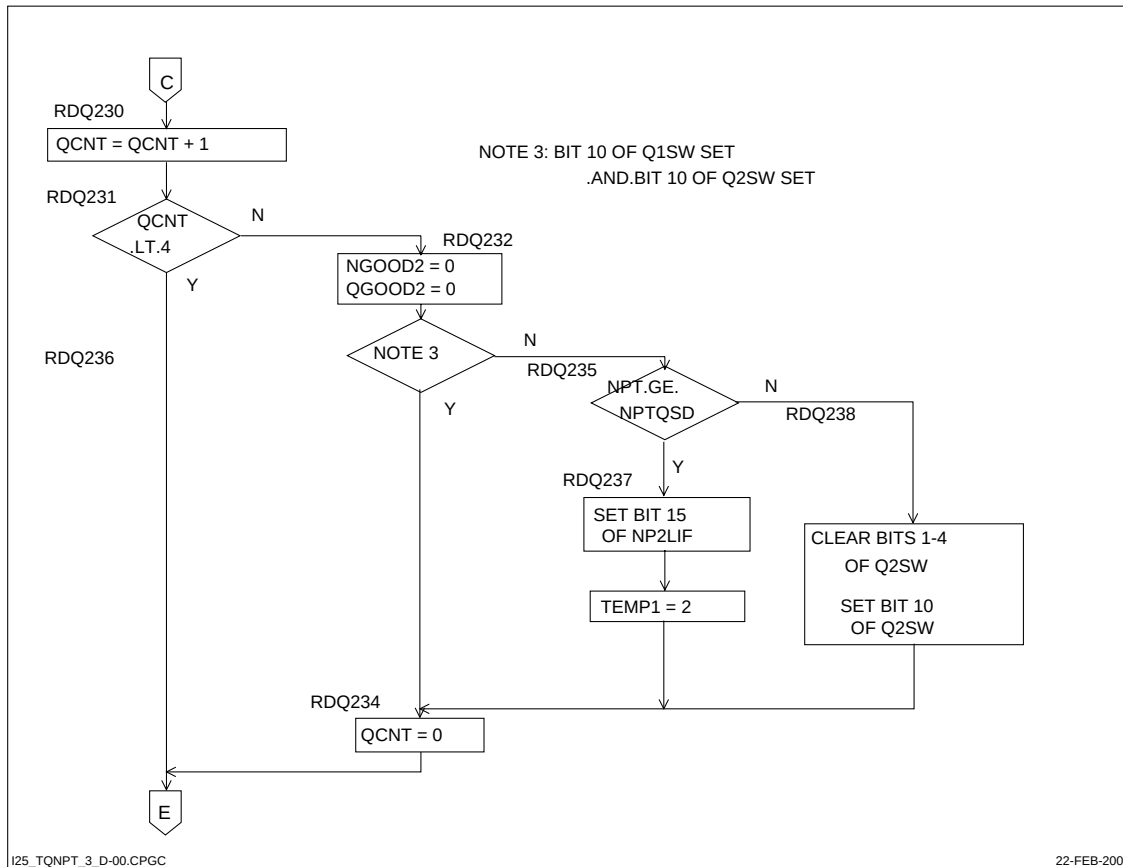
13.00 Figure 334: I25 _TQNPT_3_B



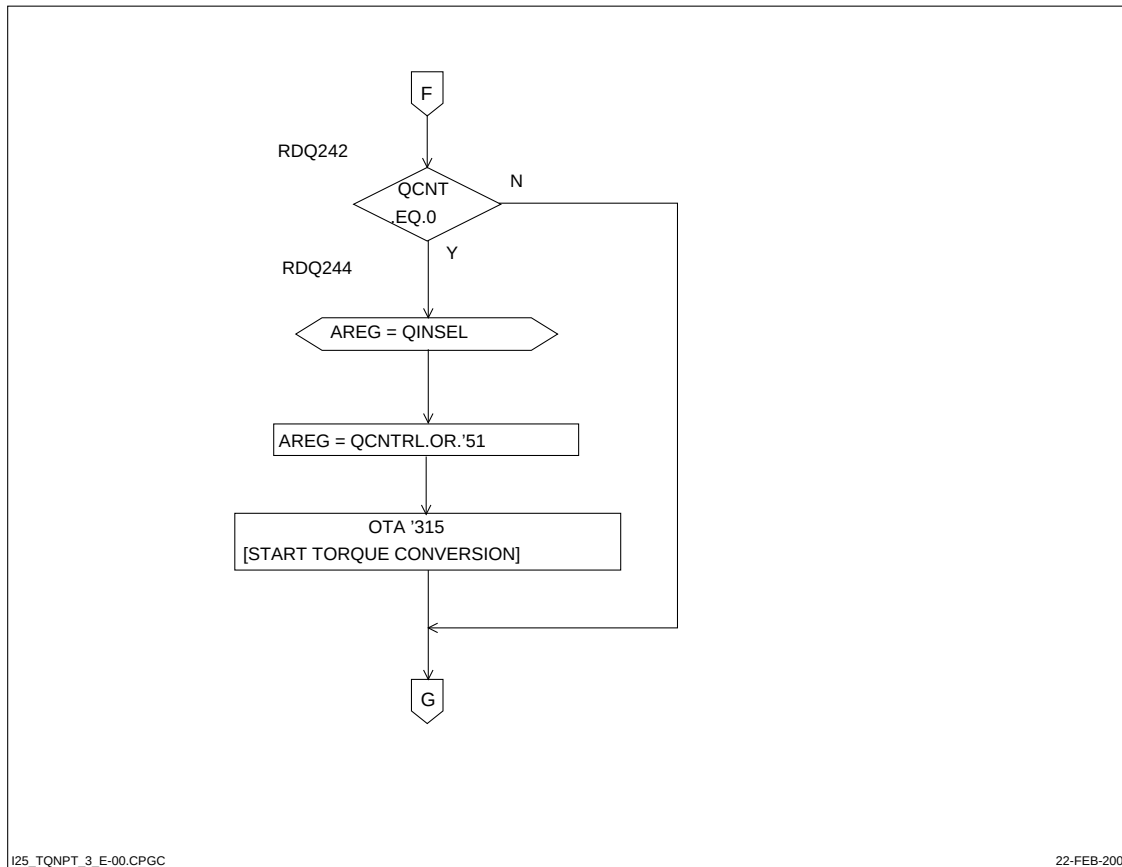
13.00 Figure 335: I25_TQNPT_3_C



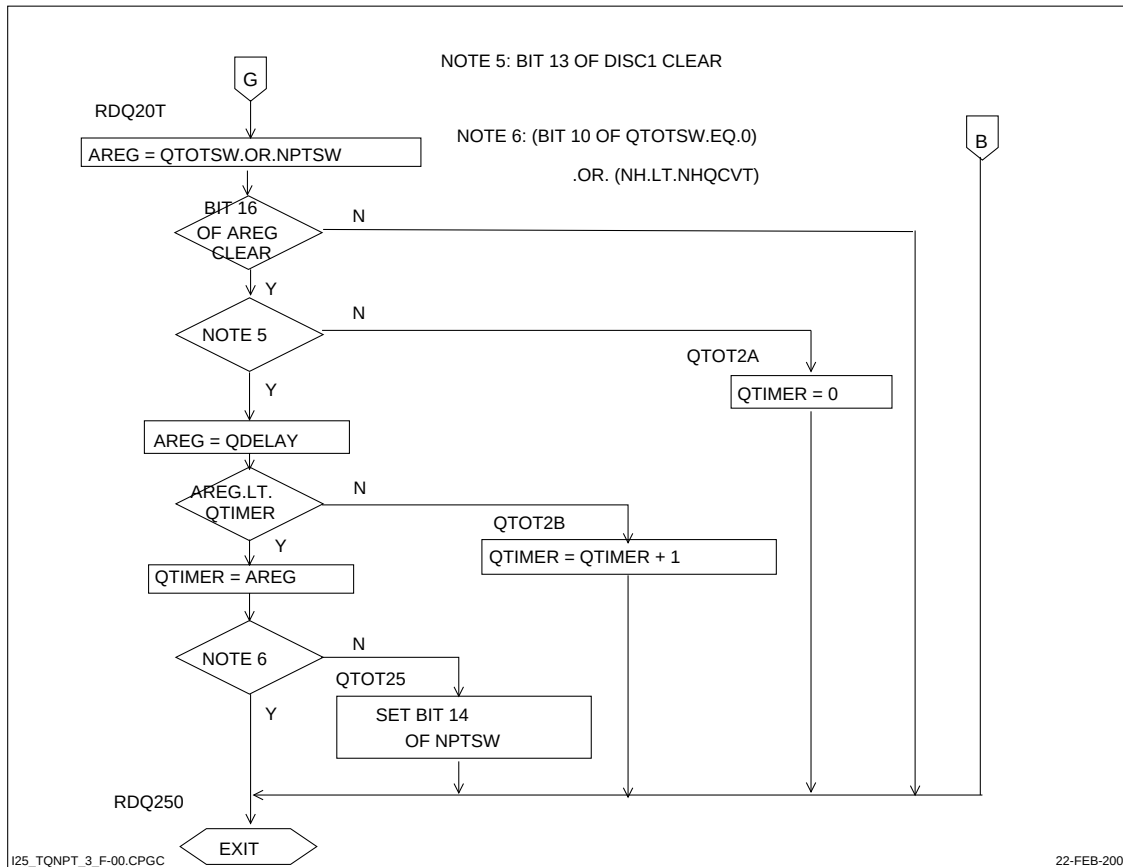
13.00 Figure 336: I25_TQNPT_3_D



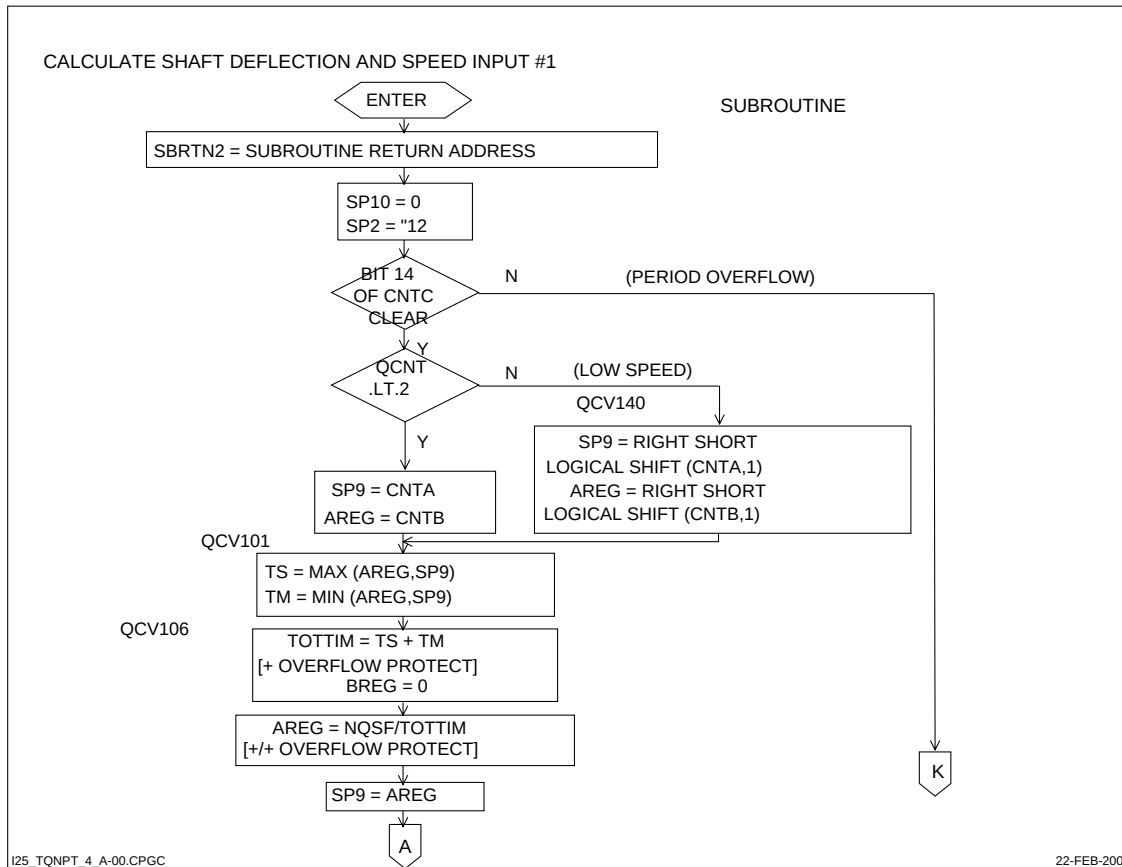
13.00 Figure 337: I25 _TQNPT _3 _E



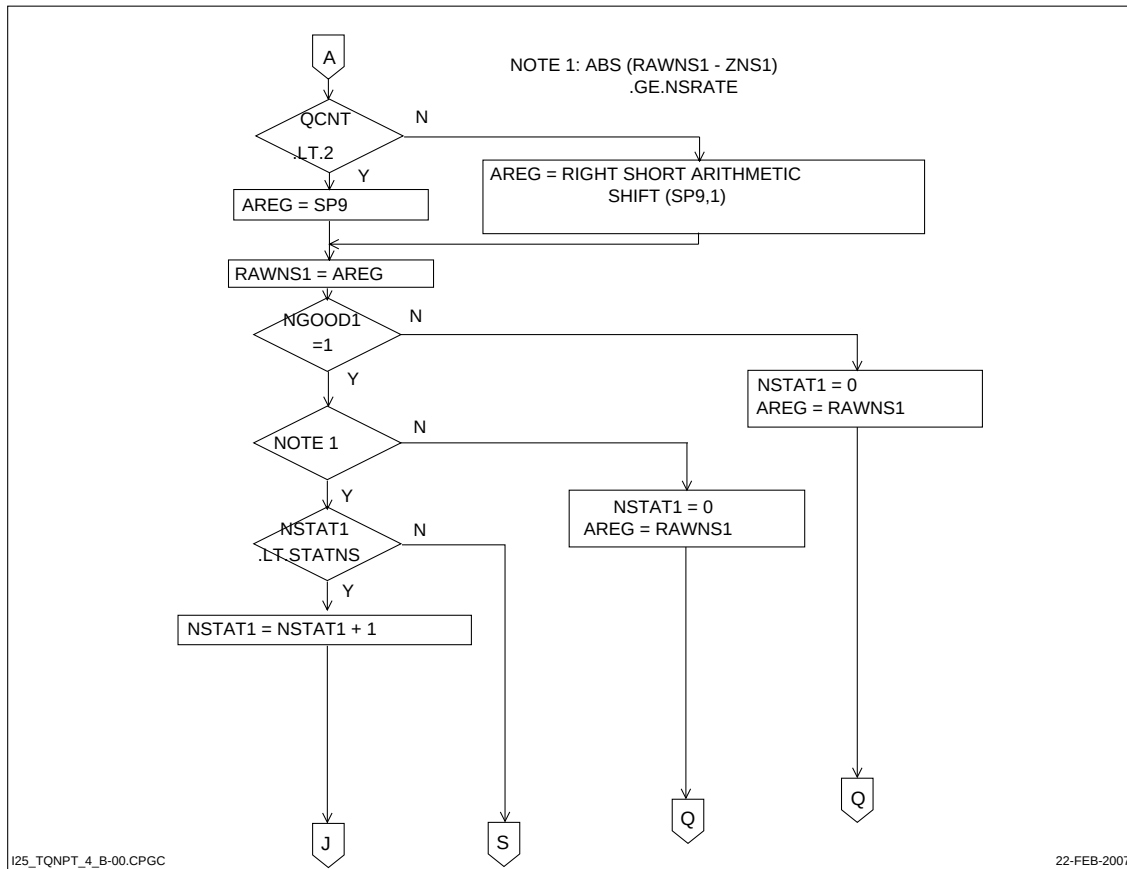
13.00 Figure 338: I25_TQNPT_3_F



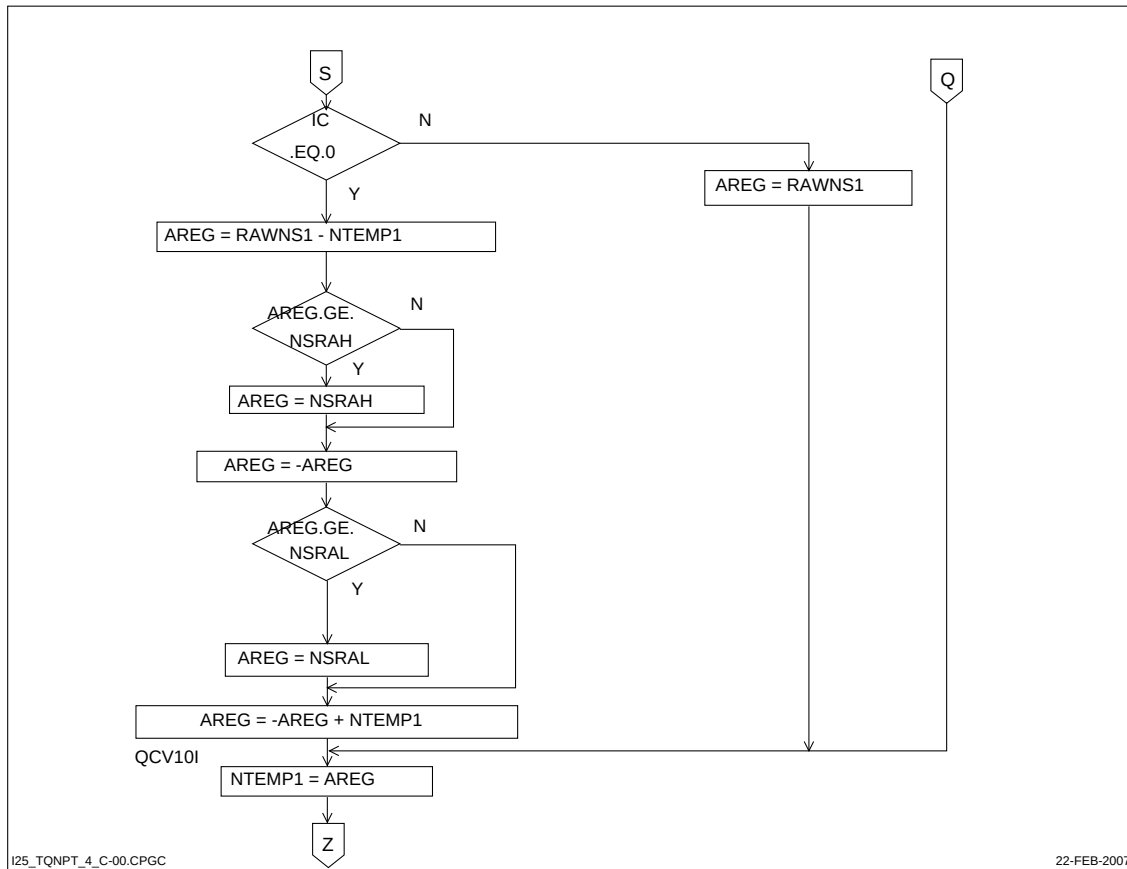
13.00 Figure 339: I25 _TQNPT _4 _A



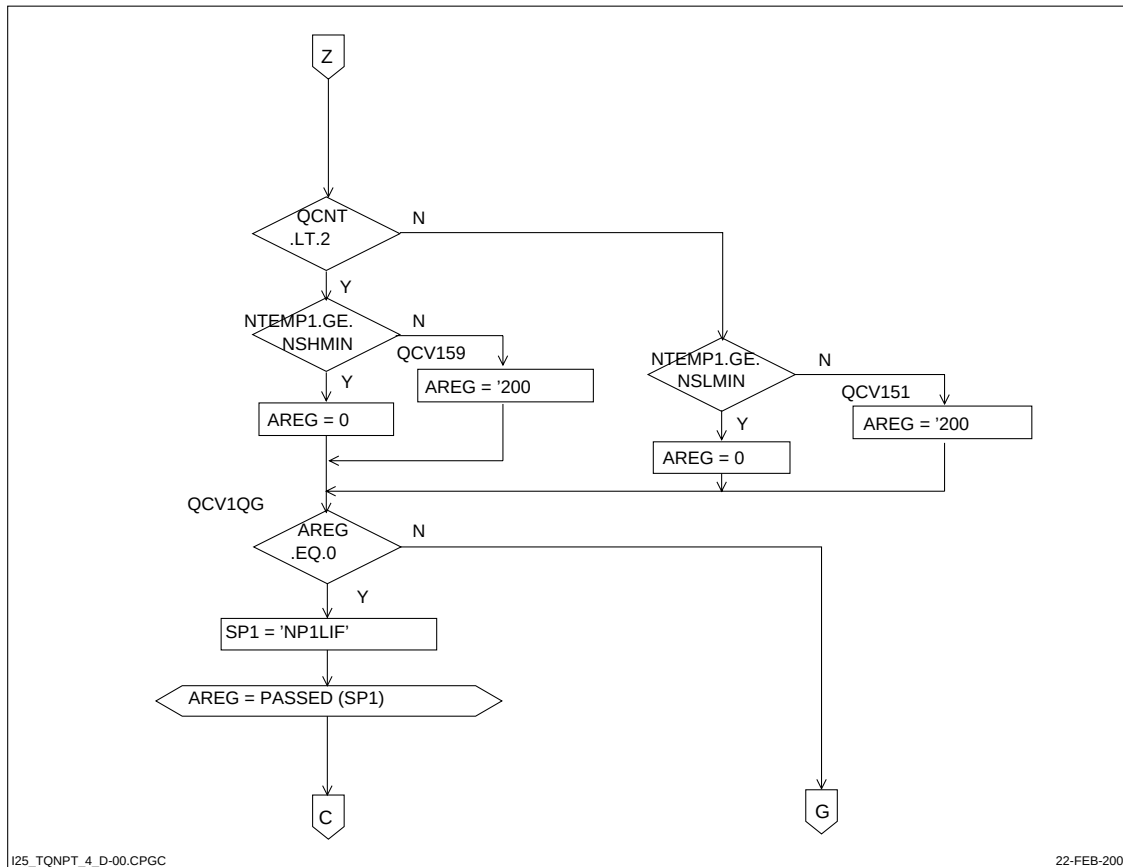
13.00 Figure 340: I25 _TQNPT _4 _B



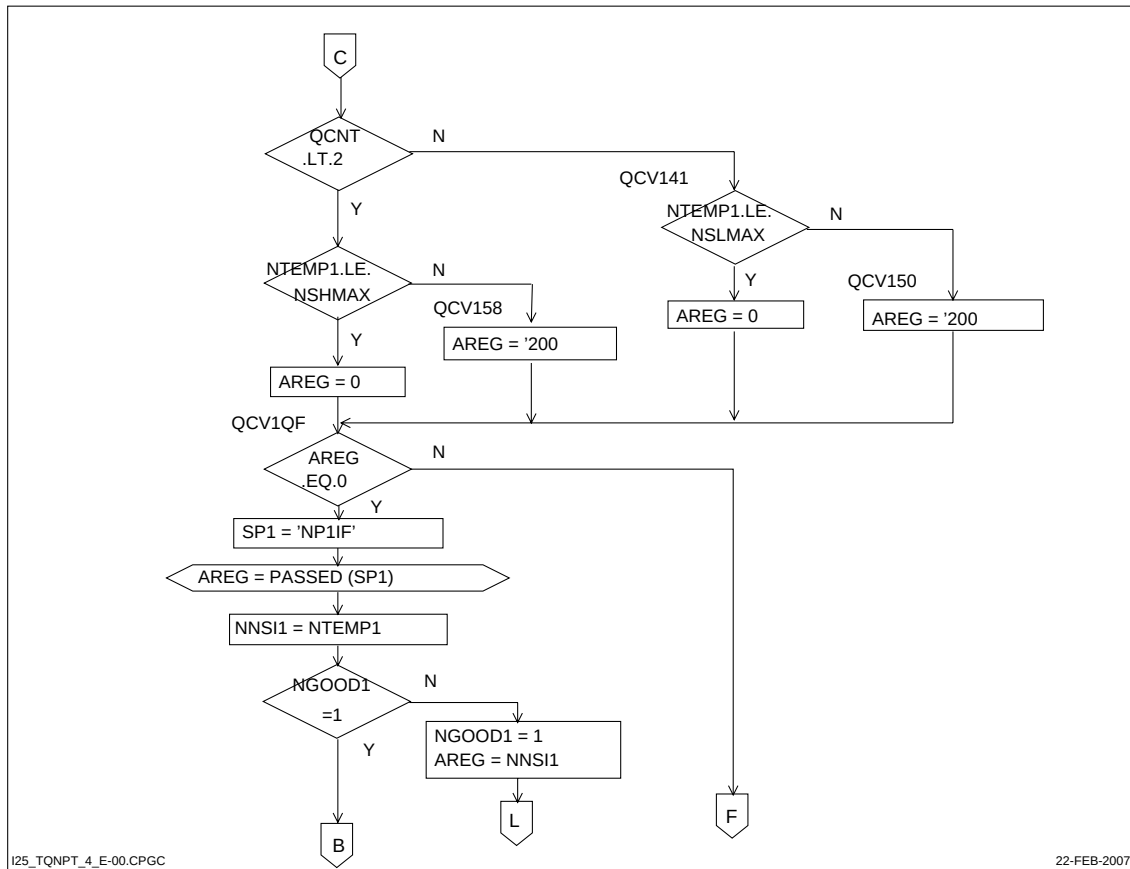
13.00 Figure 341: I25 _TQNPT _4 _C



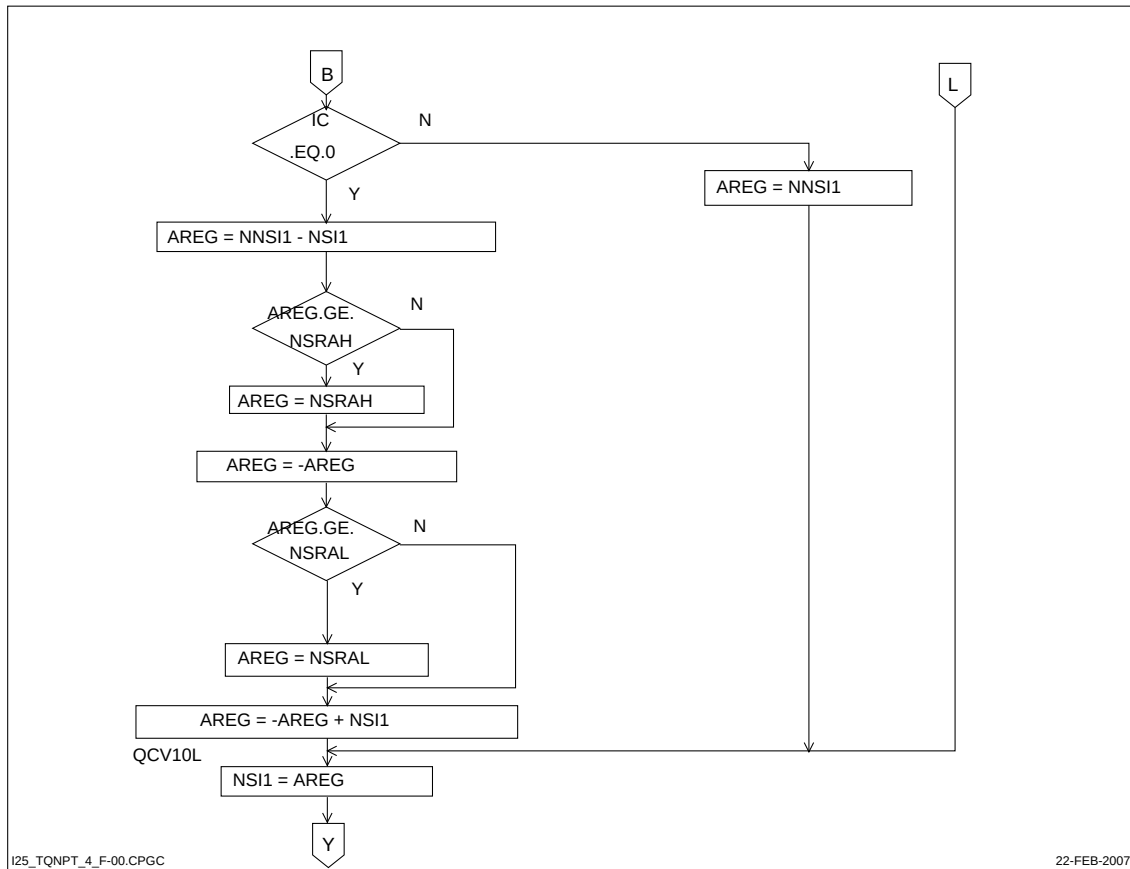
13.00 Figure 342: I25 _TQNPT _4 _D



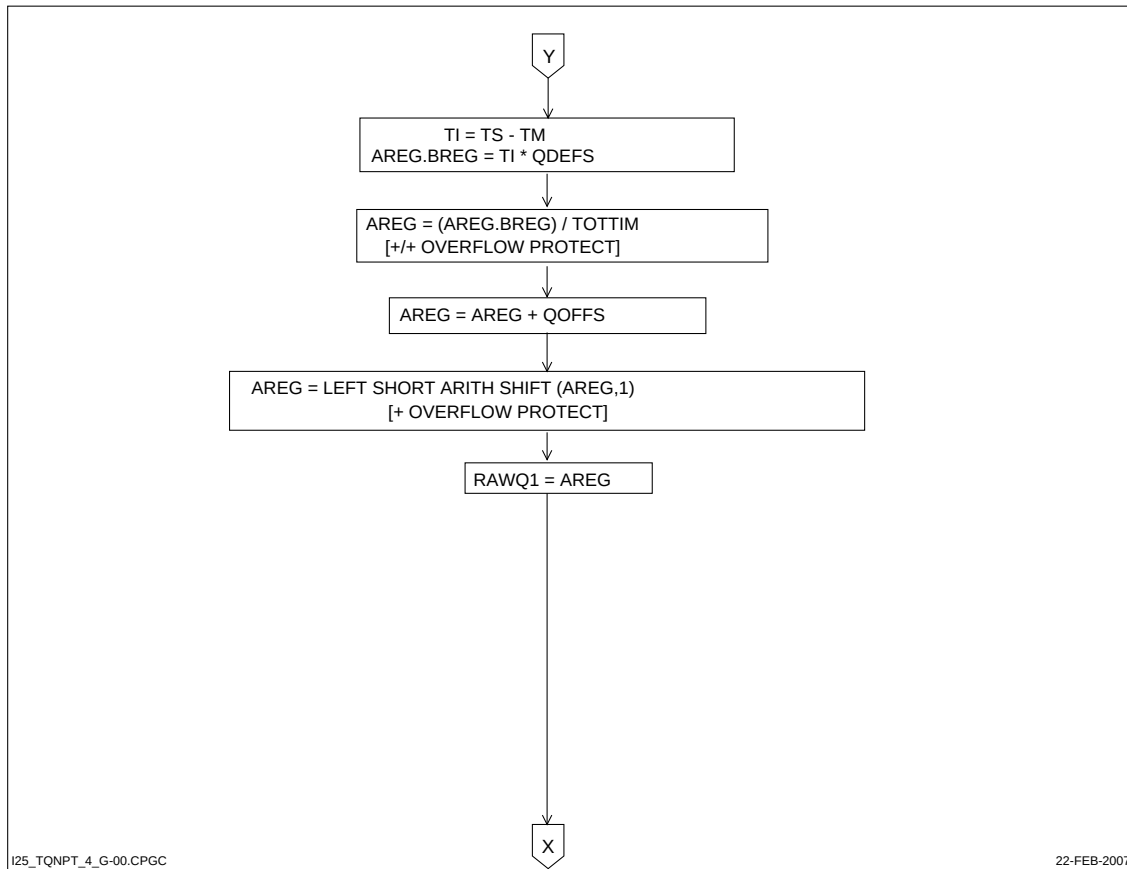
13.00 Figure 343: I25 _TQNPT _4 _E



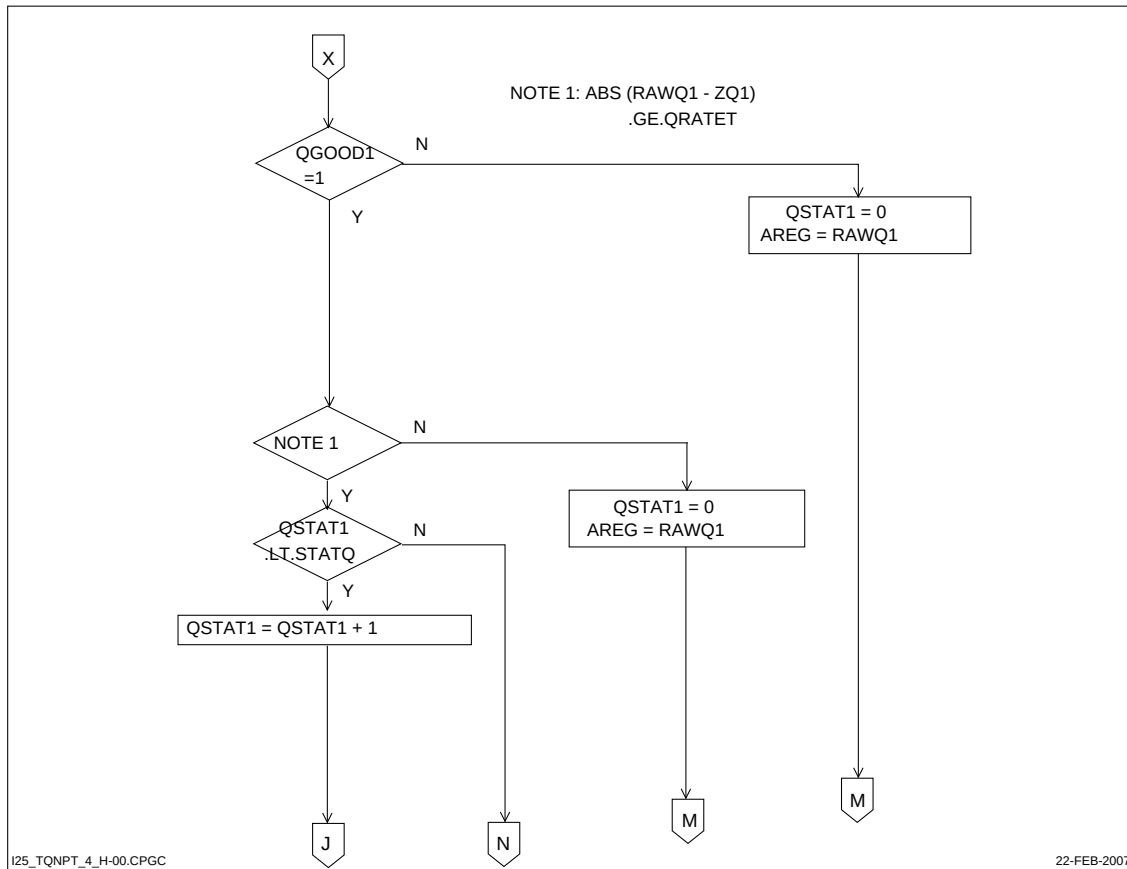
13.00 Figure 344: I25 _TQNPT _4 _F



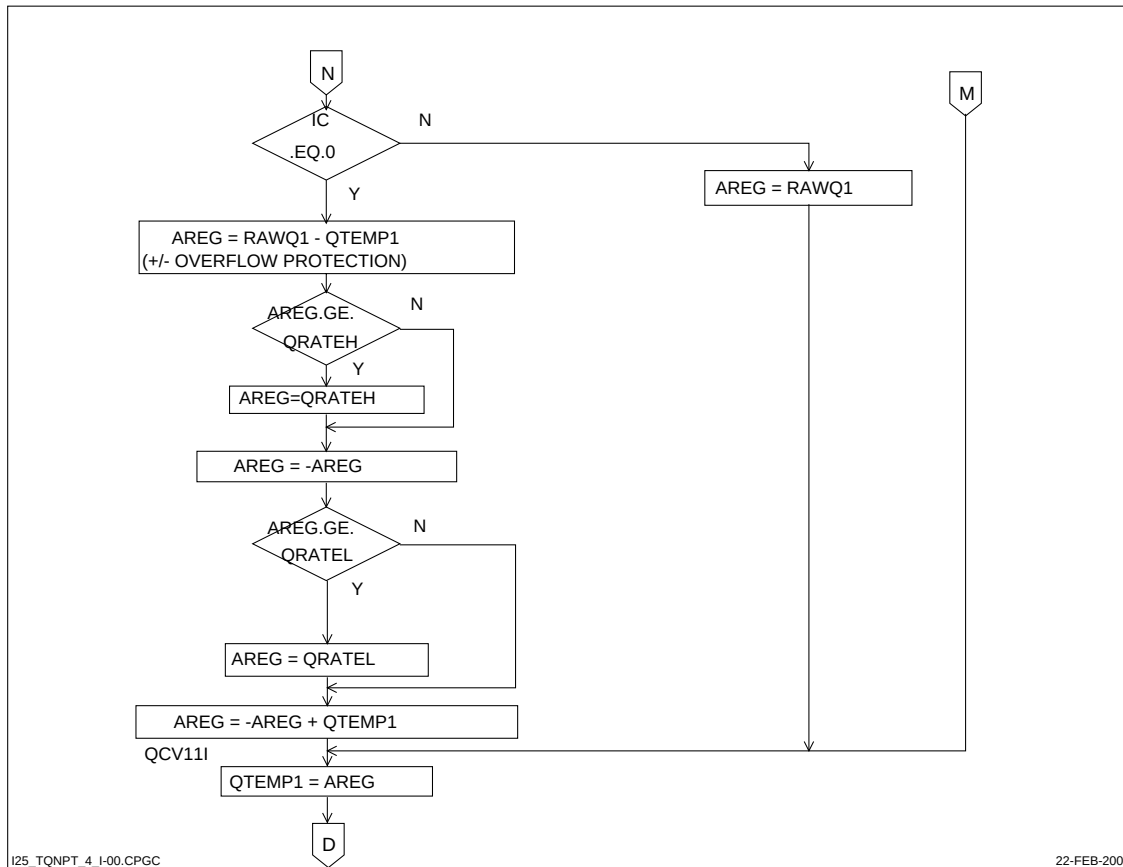
13.00 Figure 345: I25 _TQNPT _4 _G



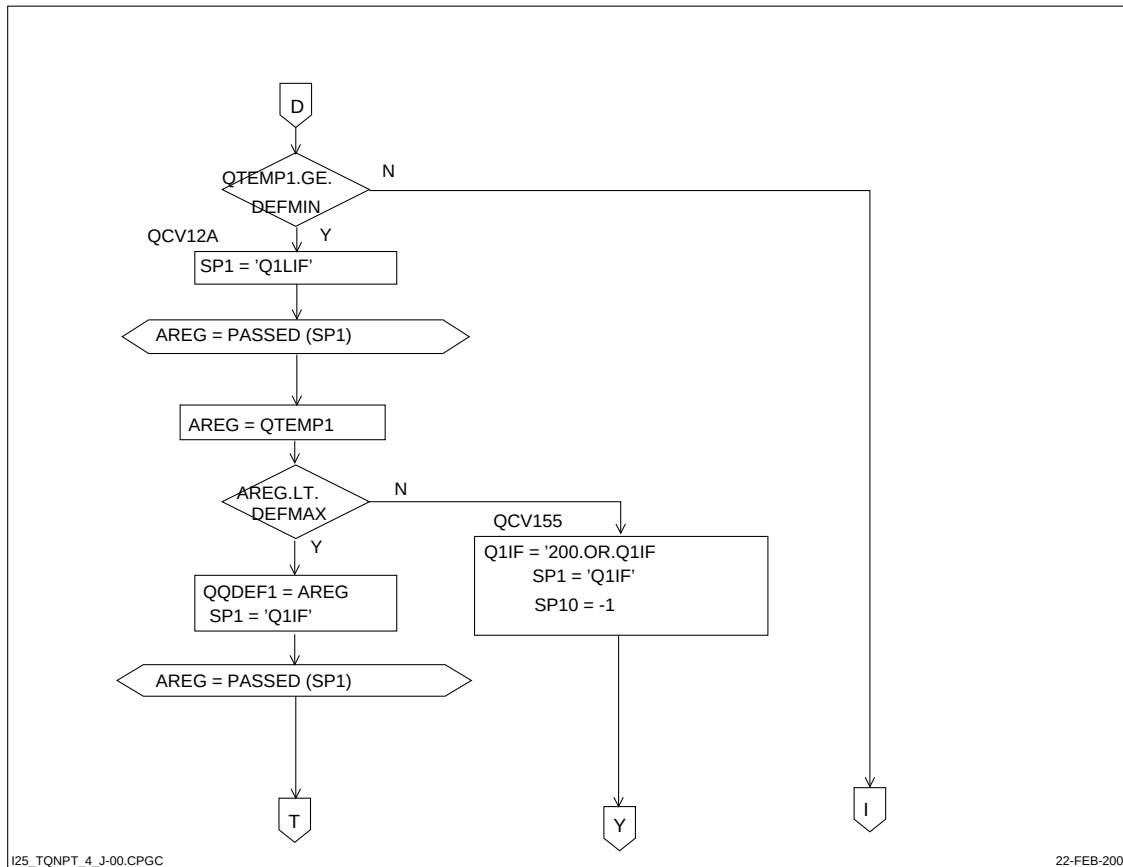
13.00 Figure 346: I25 _TQNPT _4 _H



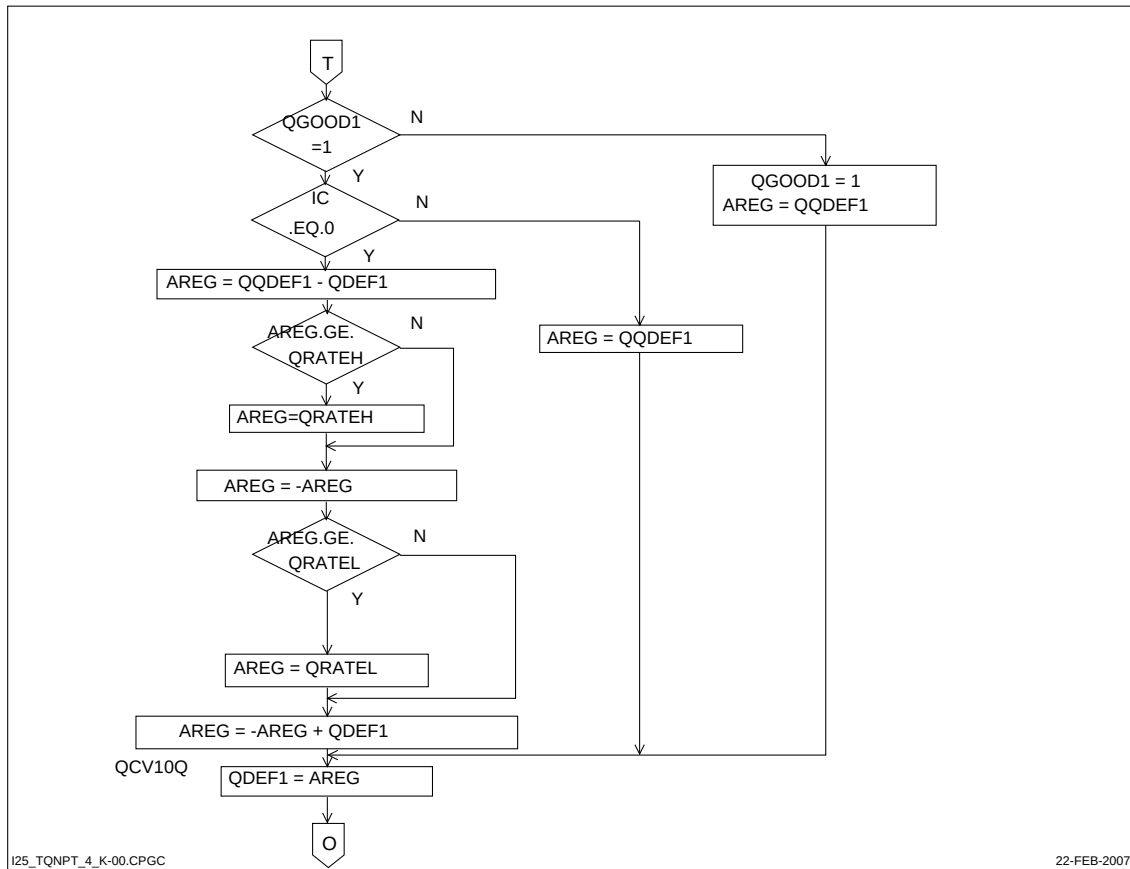
13.00 Figure 347: I25_TQNPT_4_I



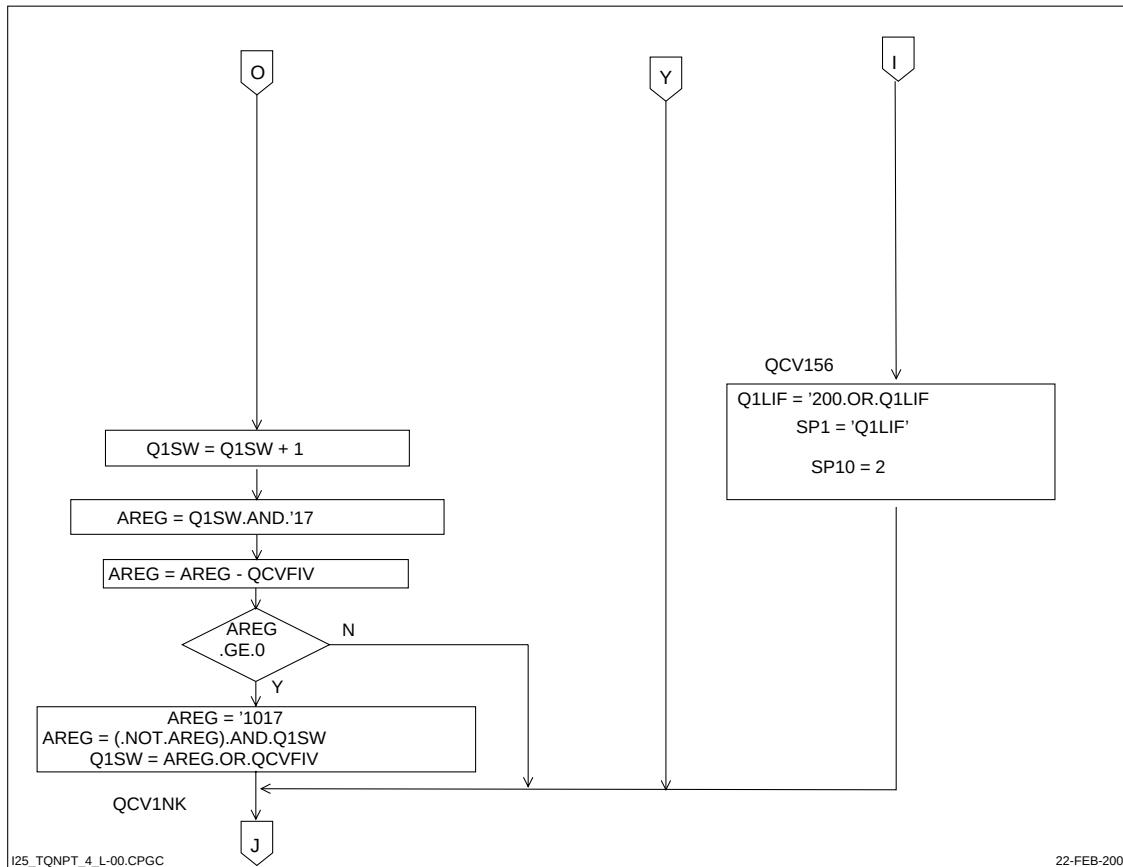
13.00 Figure 348: I25 _TQNPT _4 _J



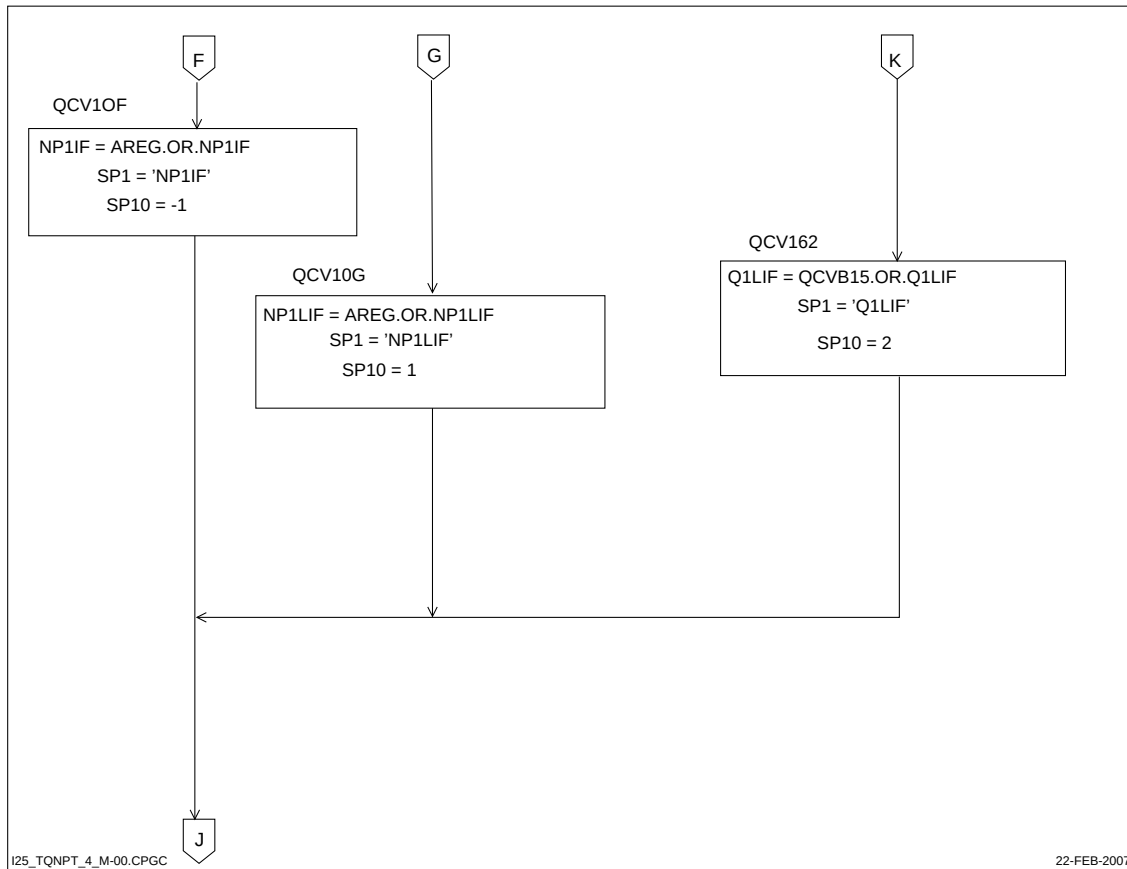
13.00 Figure 349: I25_TQNPT_4_K



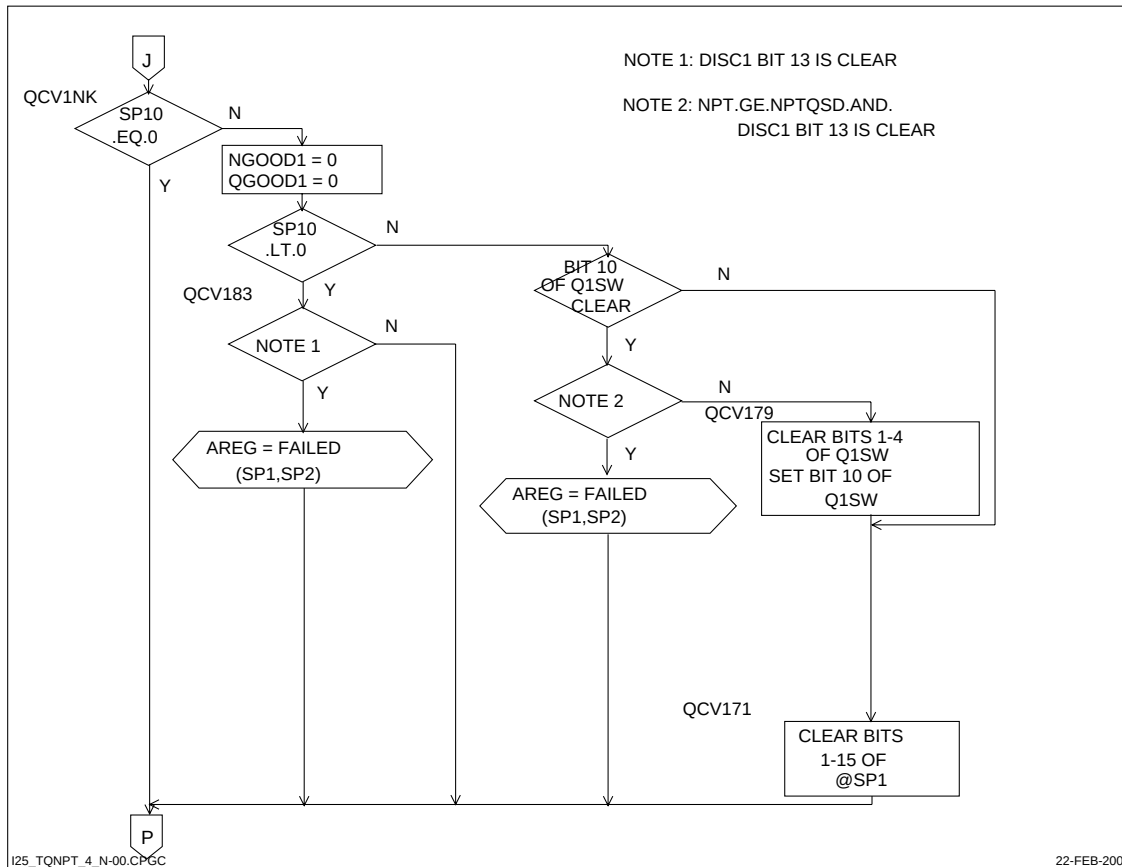
13.00 Figure 350: I25_TQNPT_4_L



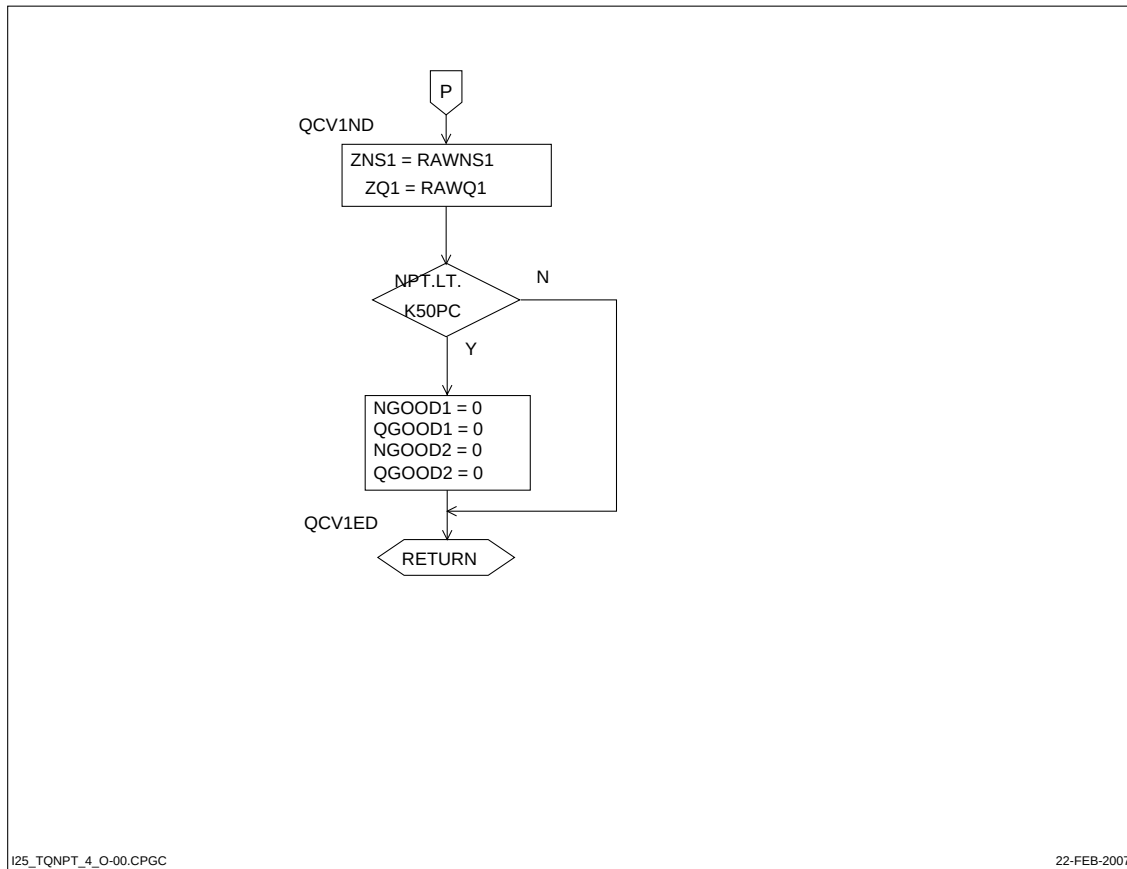
13.00 Figure 351: I25 _TQNPT _4 _M



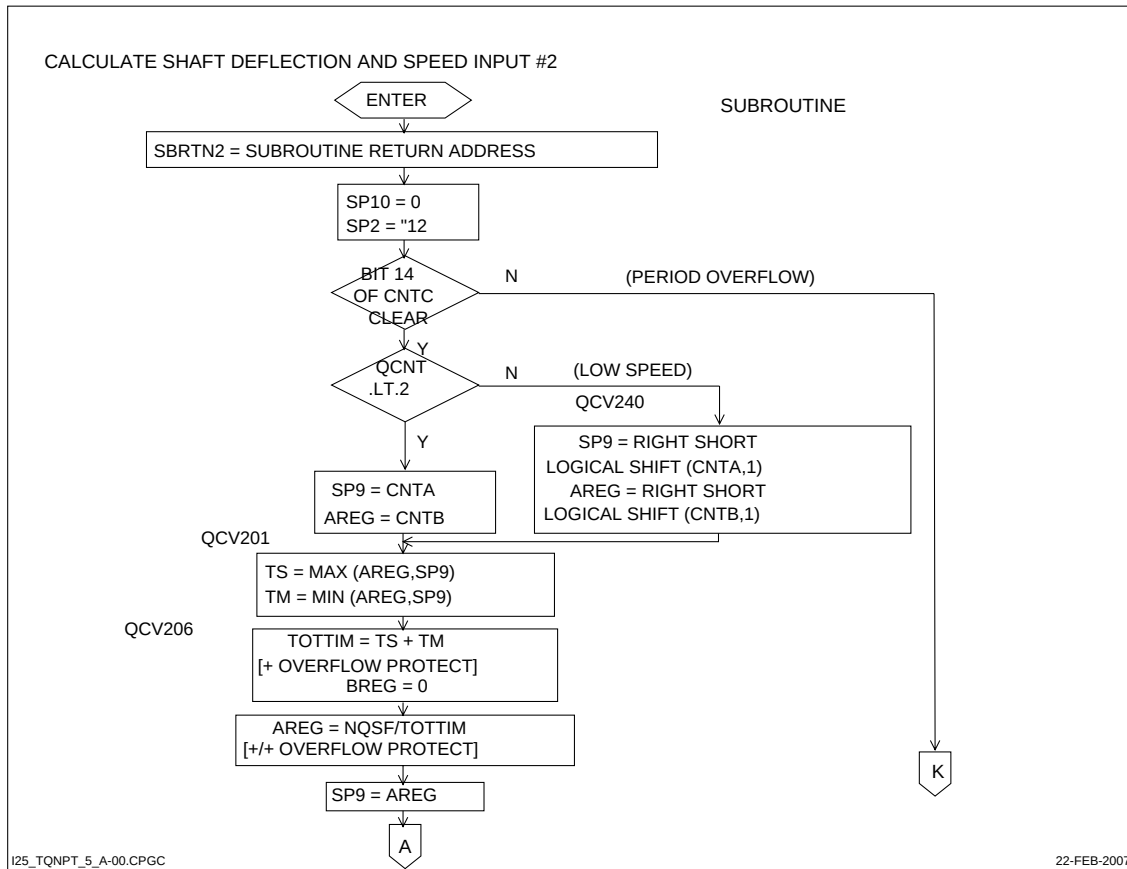
13.00 Figure 352: I25_TQNPT_4_N



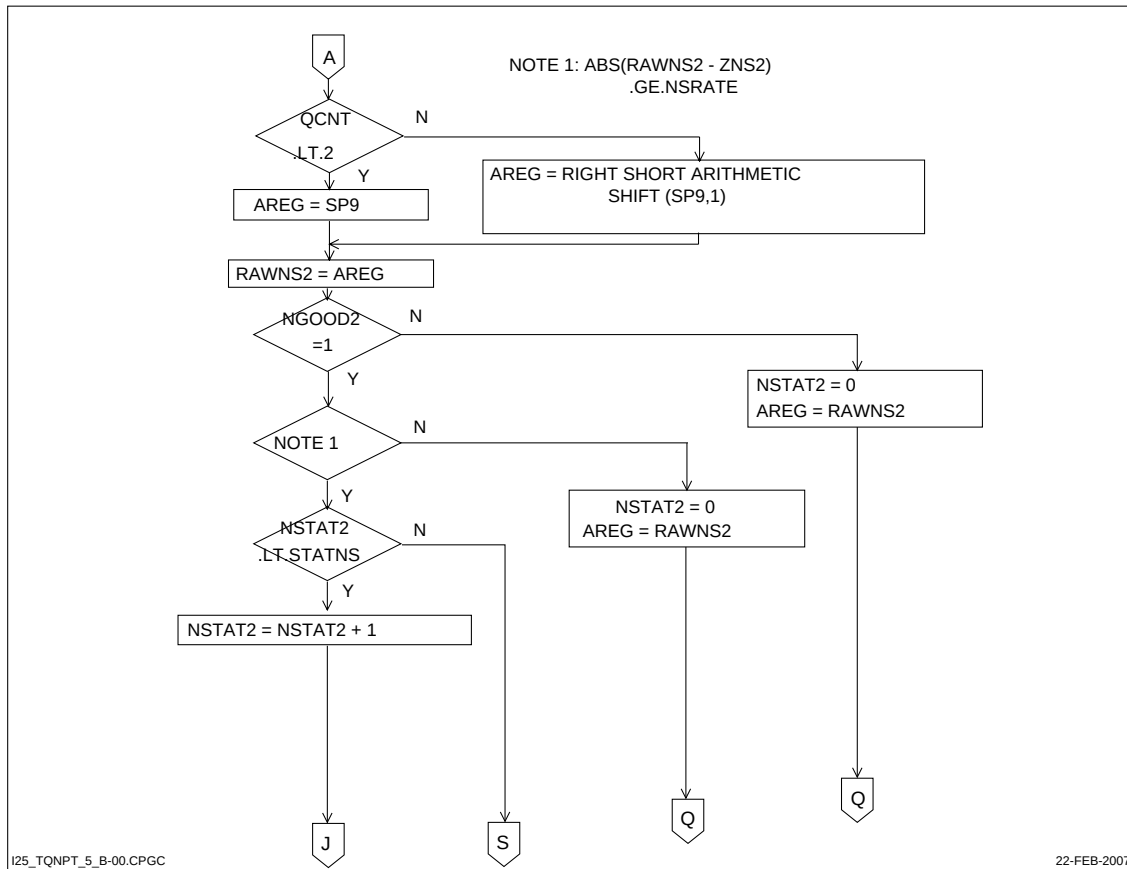
13.00 Figure 353: I25 _TQNPT _4 _O



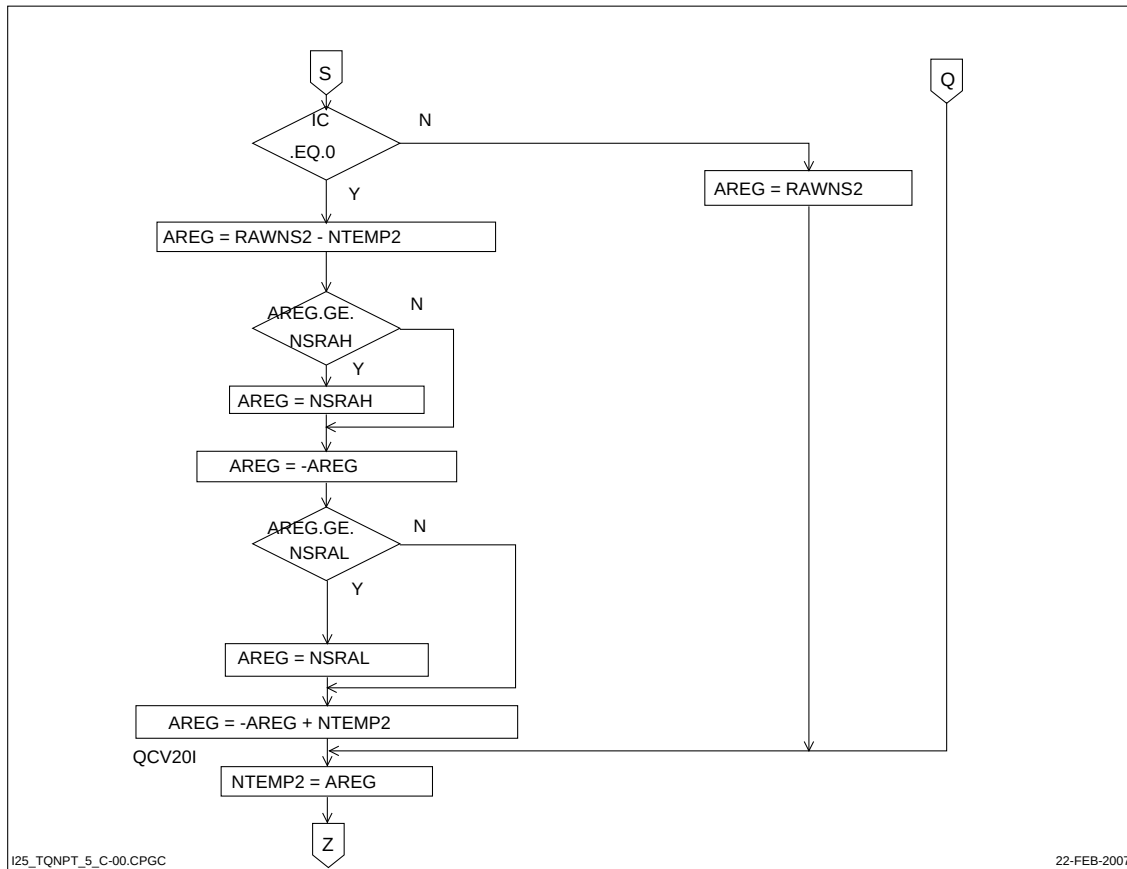
13.00 Figure 354: I25 _TQNPT _5 _A



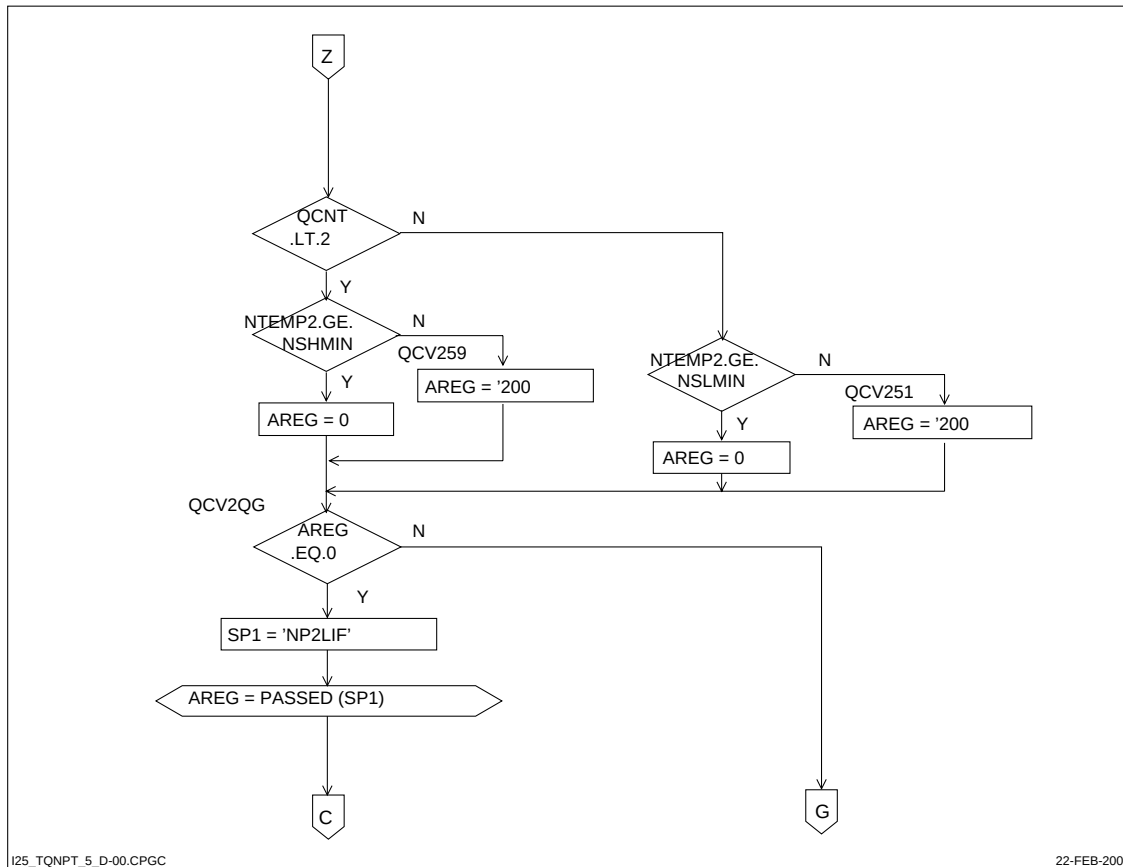
13.00 Figure 355: I25 _TQNPT _5 _B



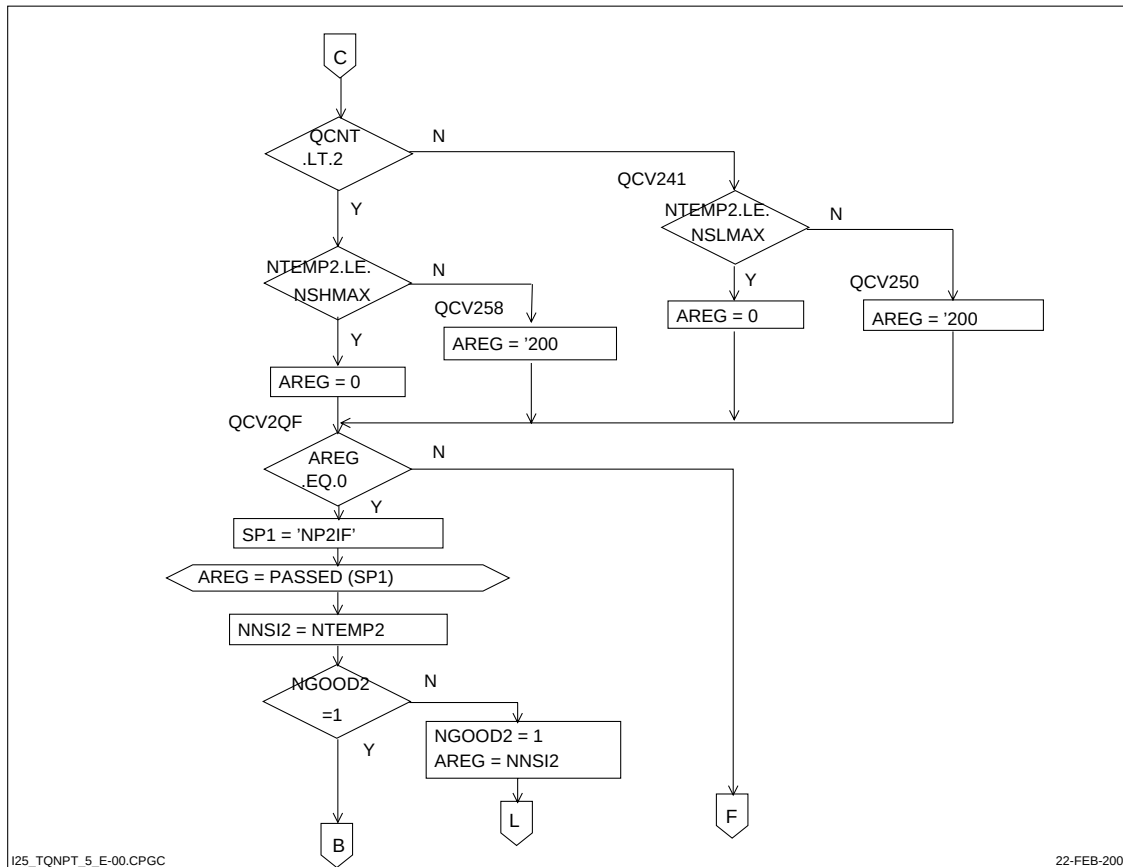
13.00 Figure 356: I25 _TQNPT _5 _C



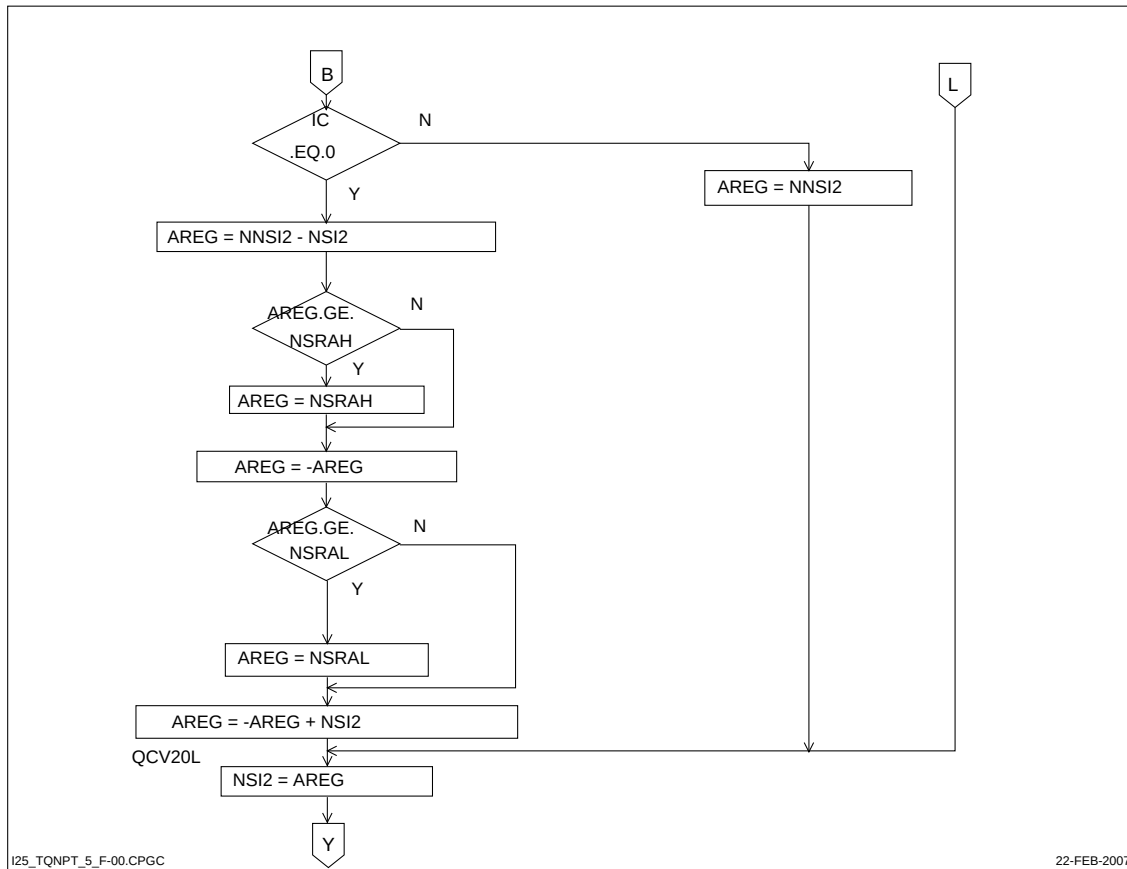
13.00 Figure 357: I25 _TQNPT _5 _D



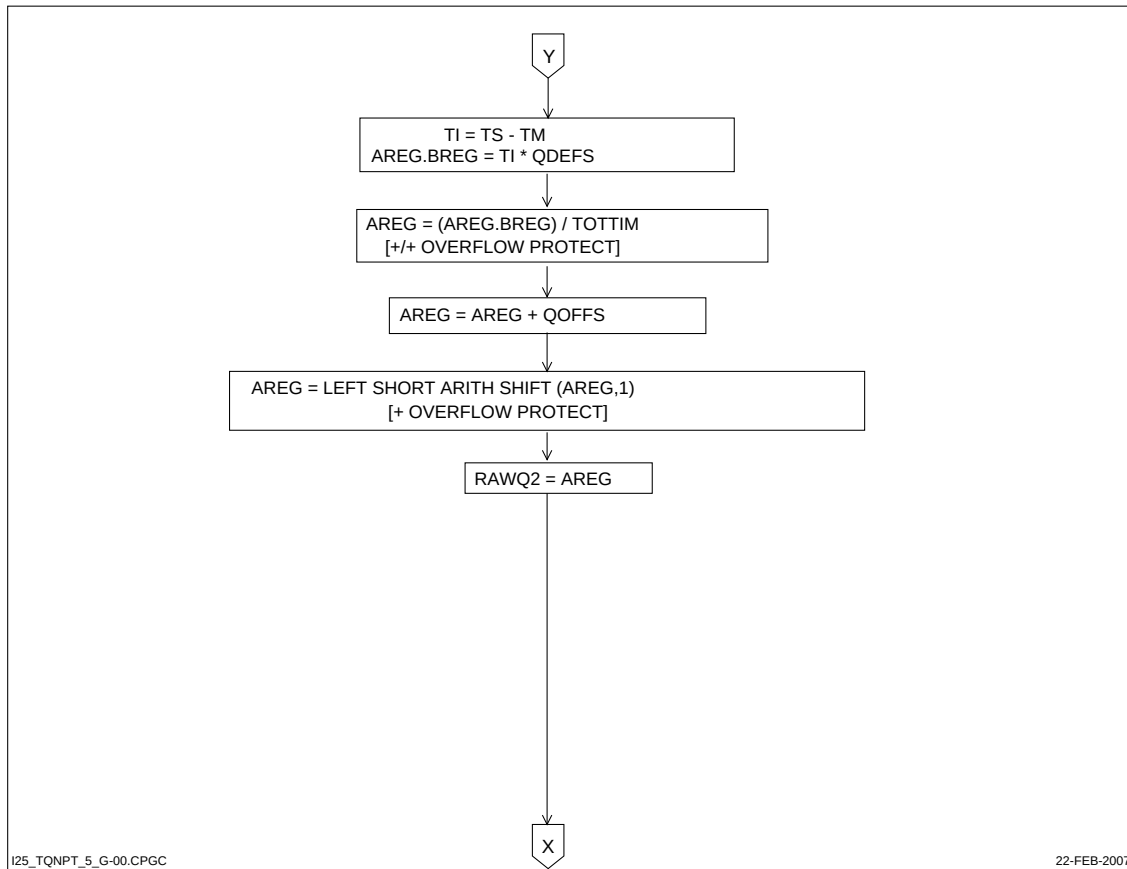
13.00 Figure 358: I25_TQNPT_5_E



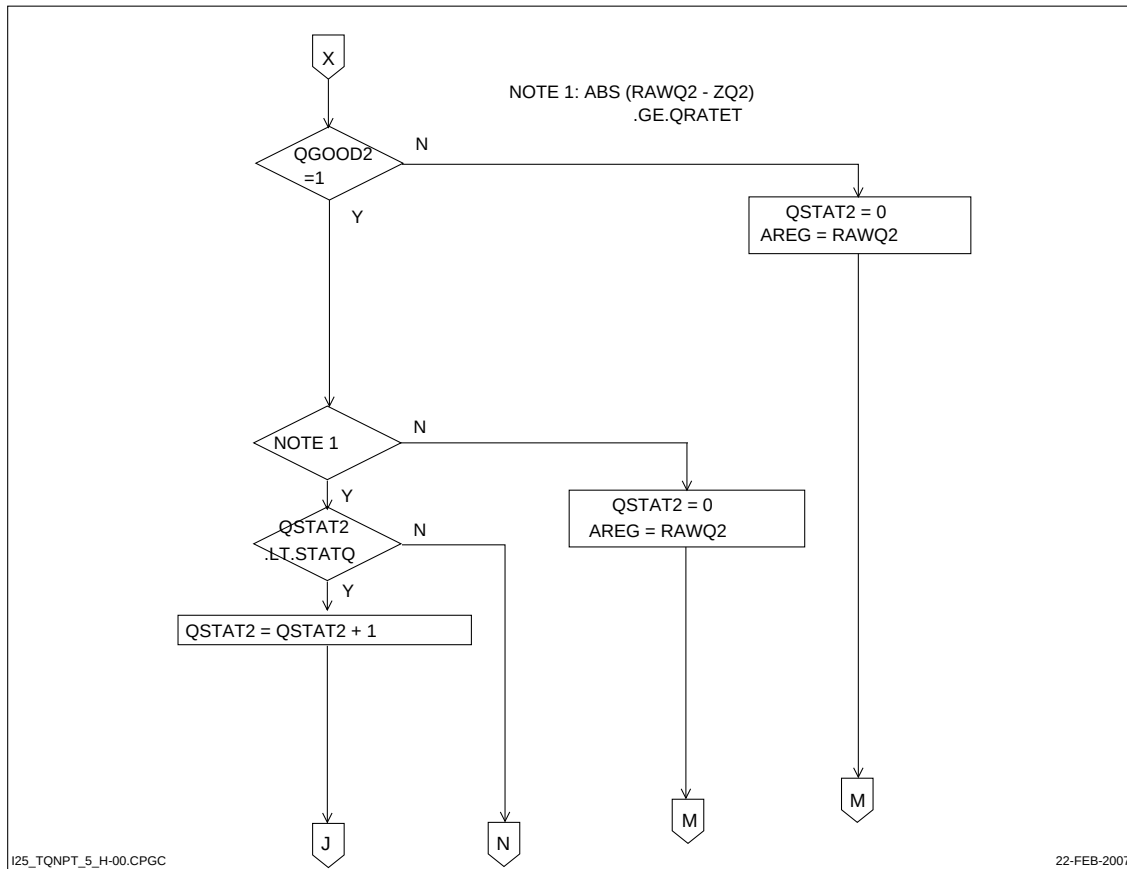
13.00 Figure 359: I25_TQNPT_5_F



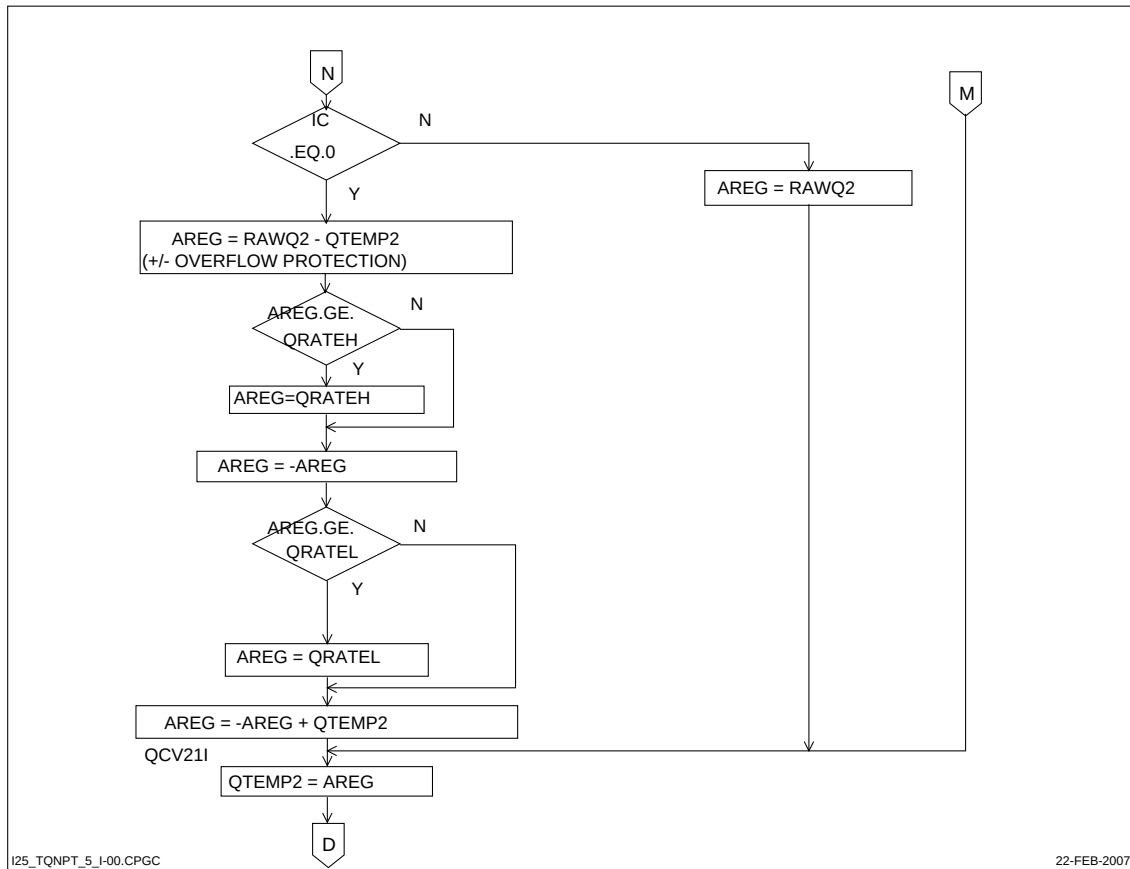
13.00 Figure 360: I25 _TQNPT _5 _G



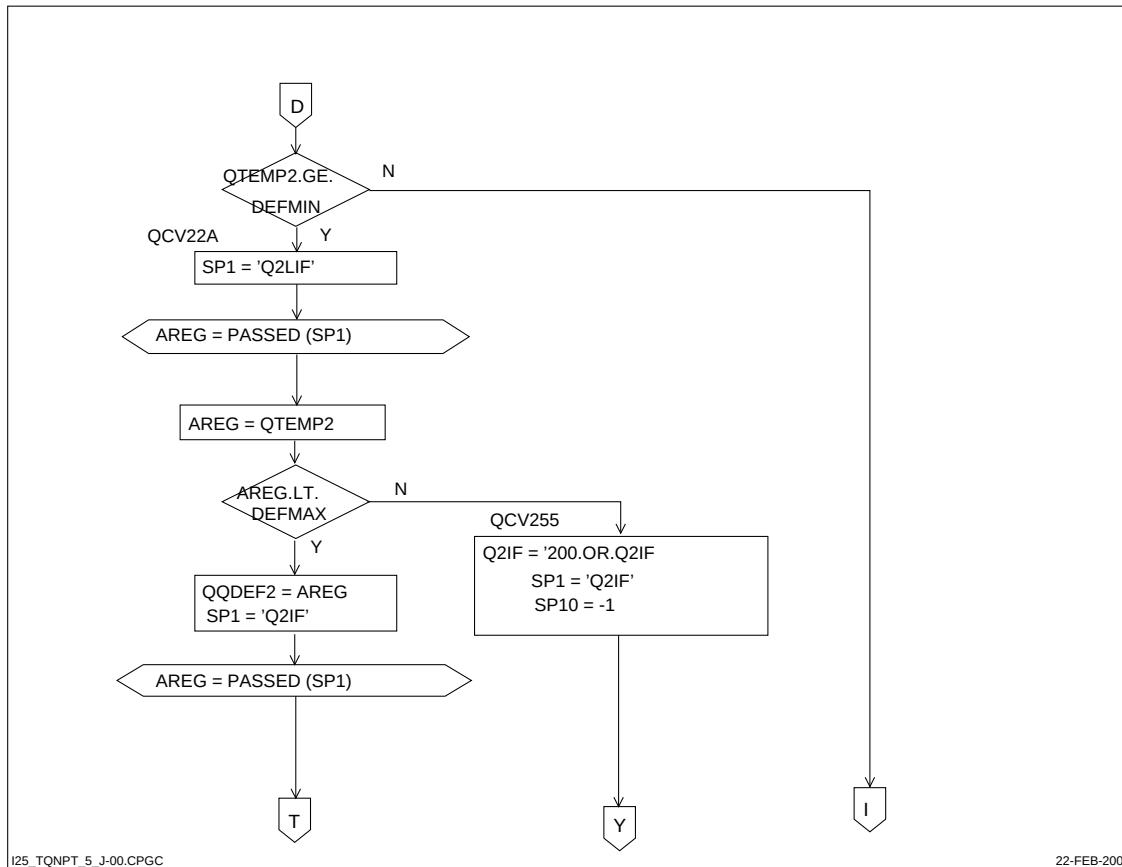
13.00 Figure 361: I25 _TQNPT _5 _H



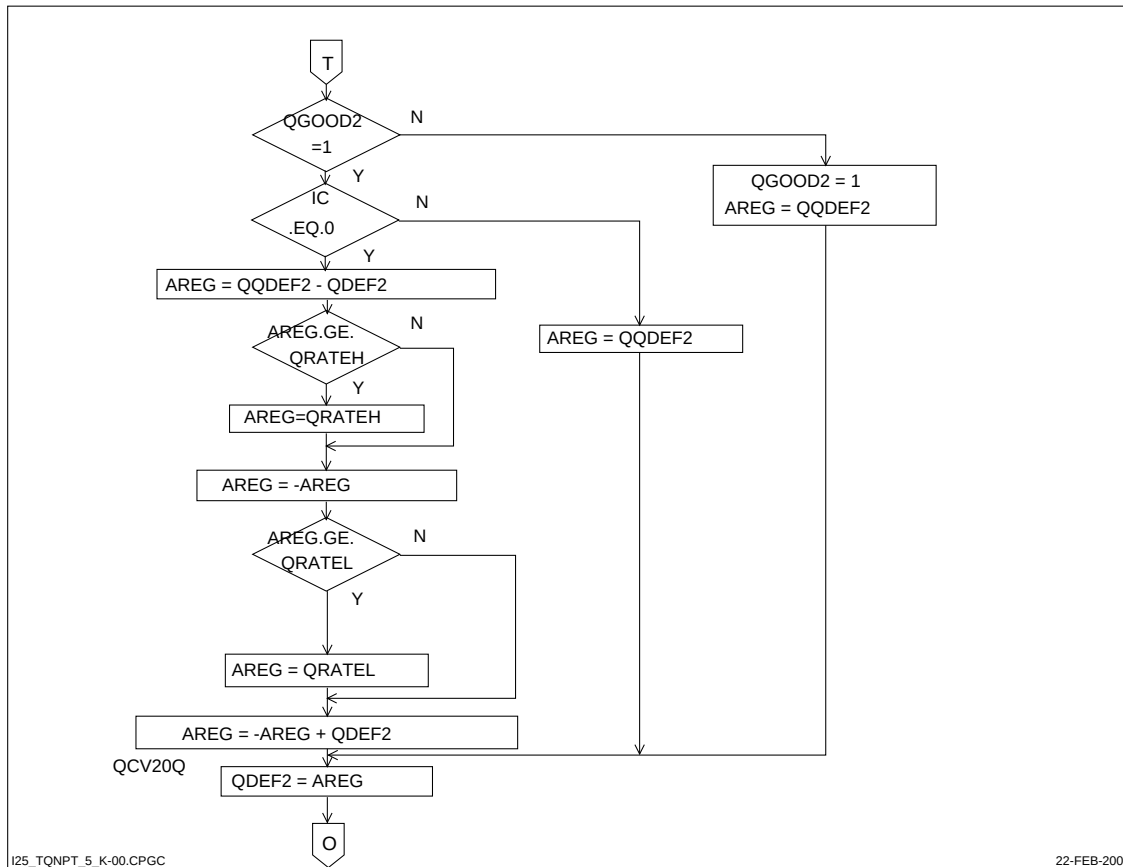
13.00 Figure 362: I25_TQNPT_5_I



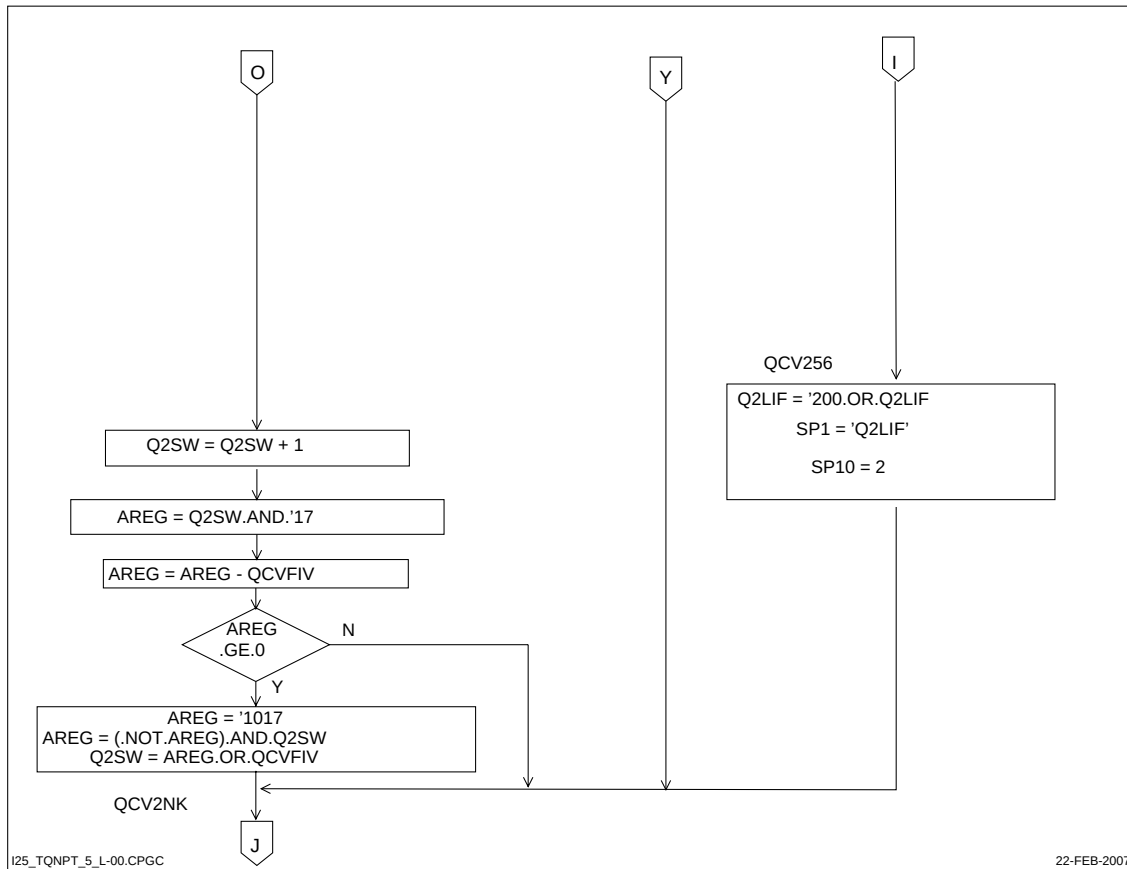
13.00 Figure 363: I25 _TQNPT _5 _J



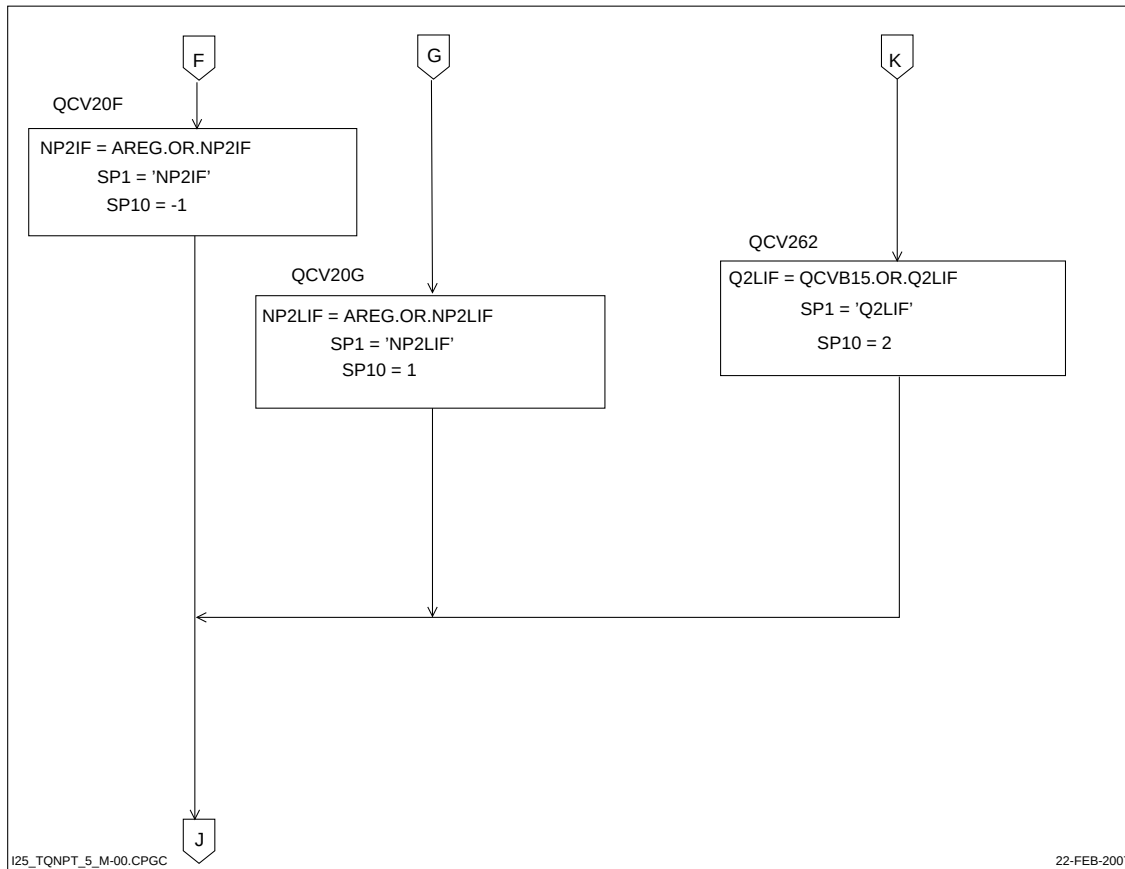
13.00 Figure 364: I25_TQNPT_5_K



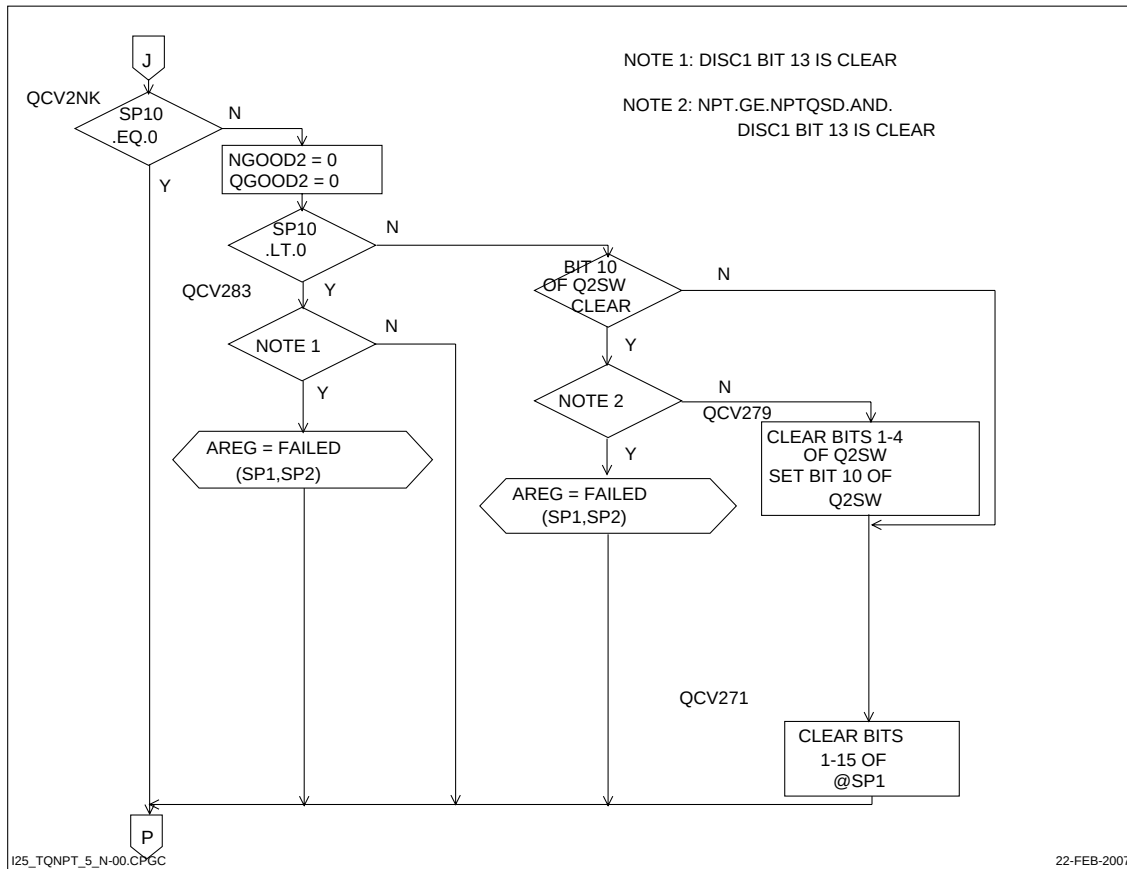
13.00 Figure 365: I25_TQNPT_5_L



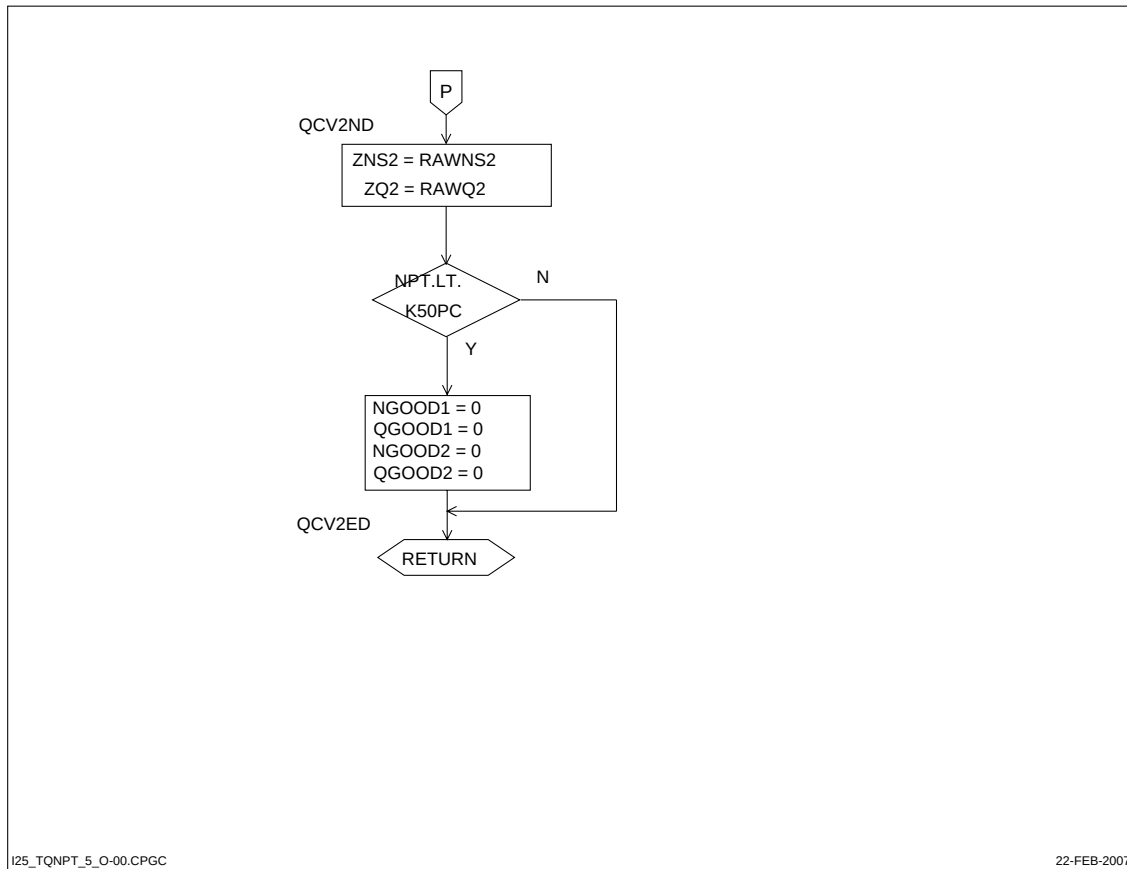
13.00 Figure 366: I25_TQNPT_5_M



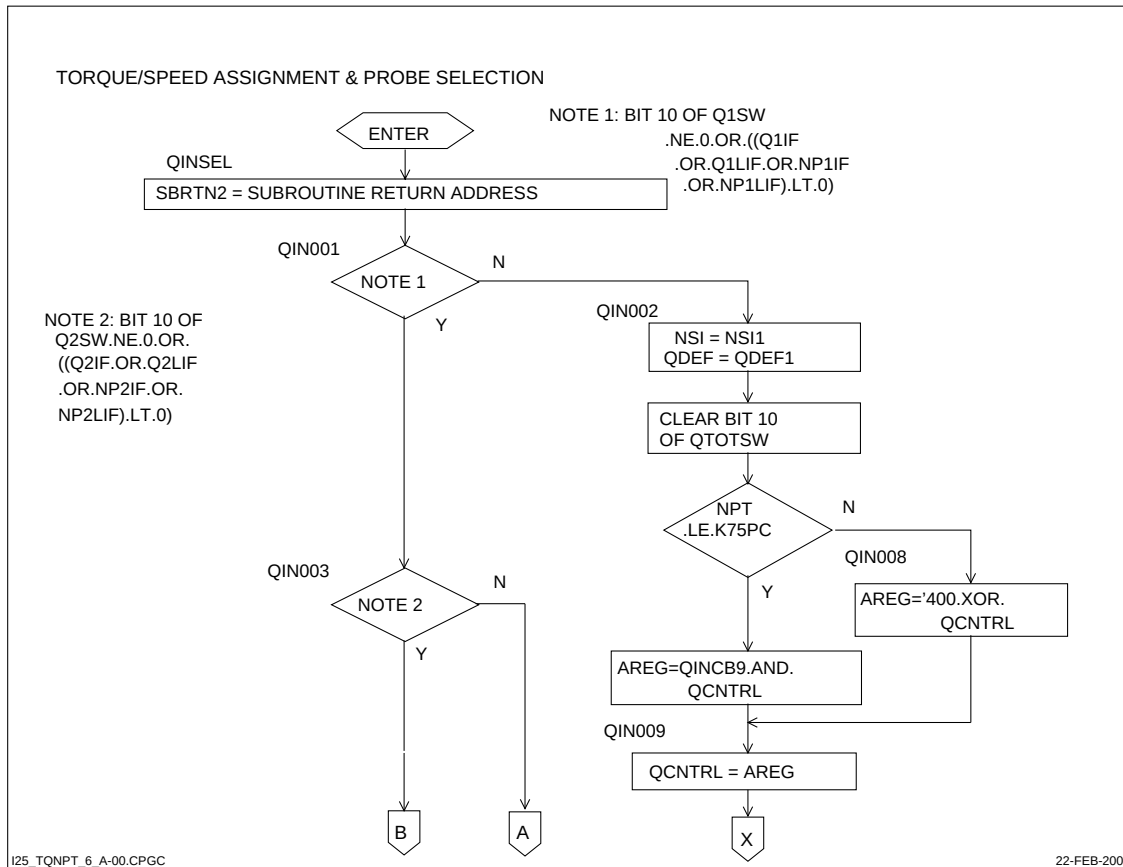
13.00 Figure 367: I25_TQNPT_5_N



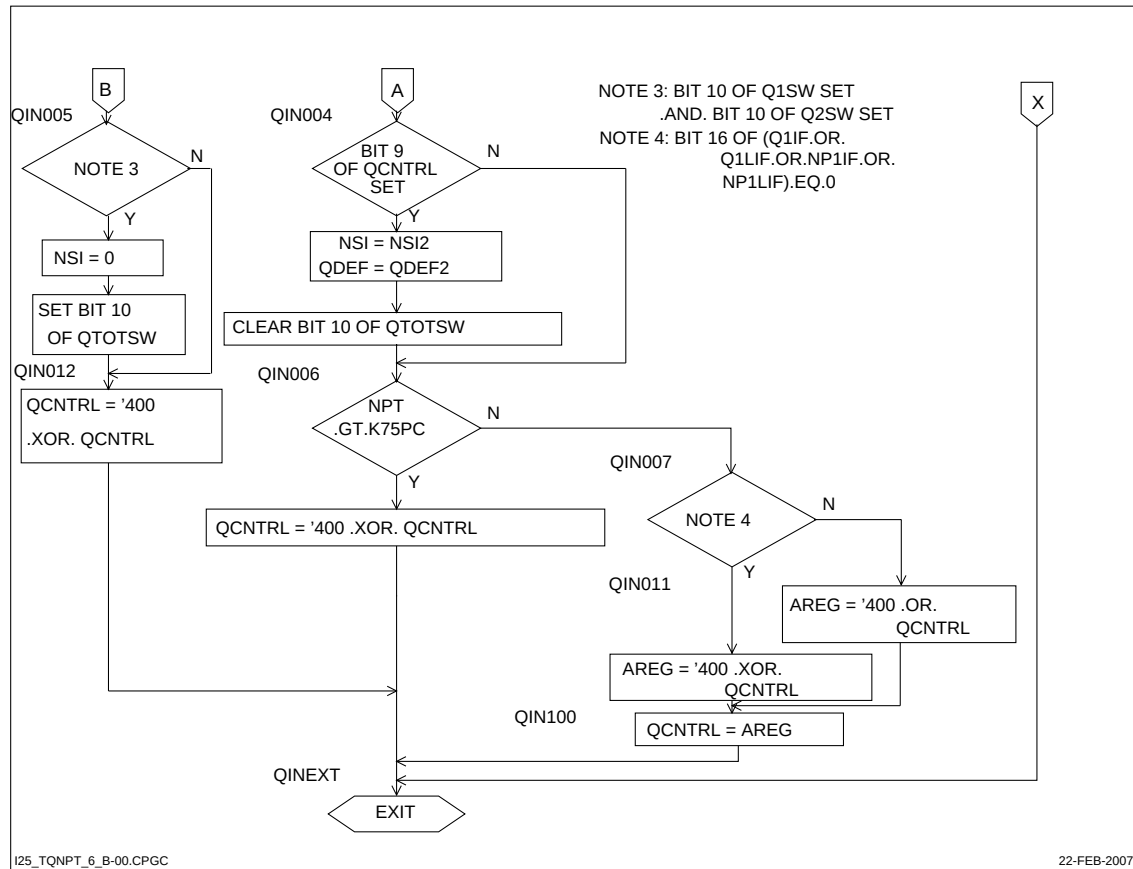
13.00 Figure 368: I25 _TQNPT _5 _O



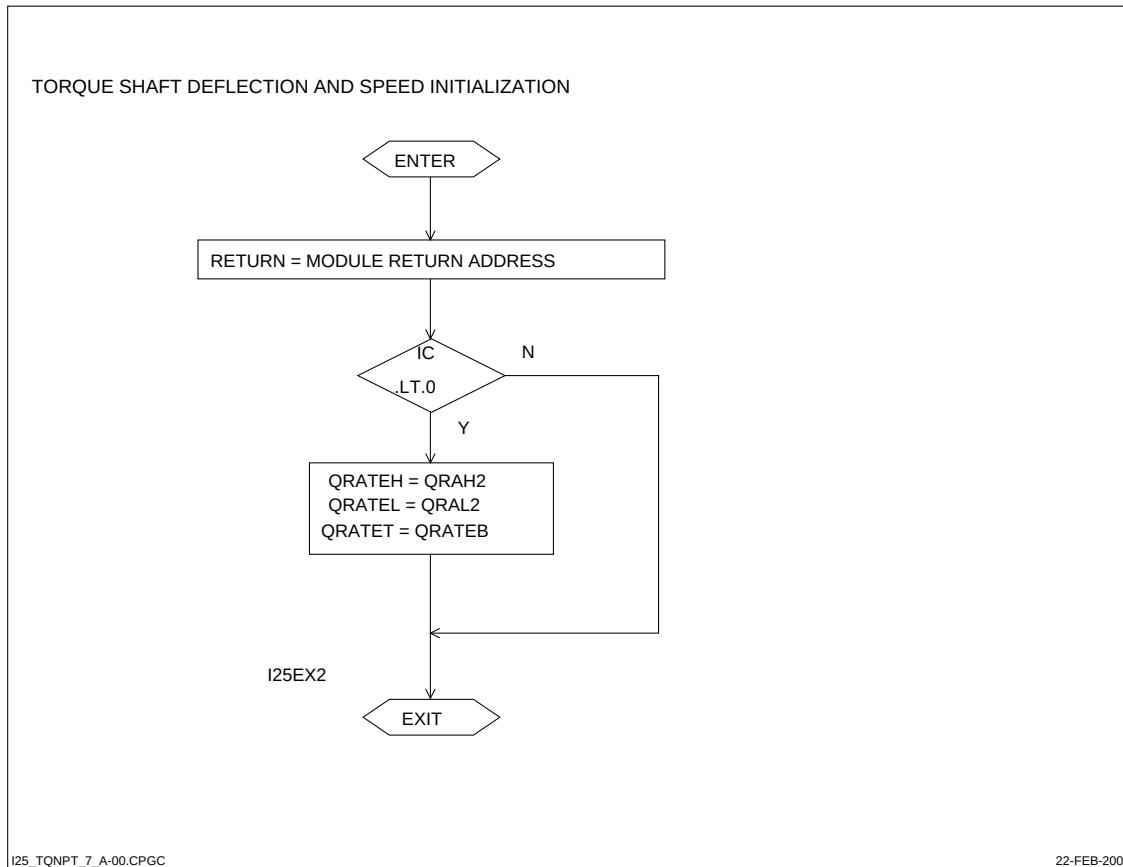
13.00 Figure 369: I25 _TQNPT _6 _A



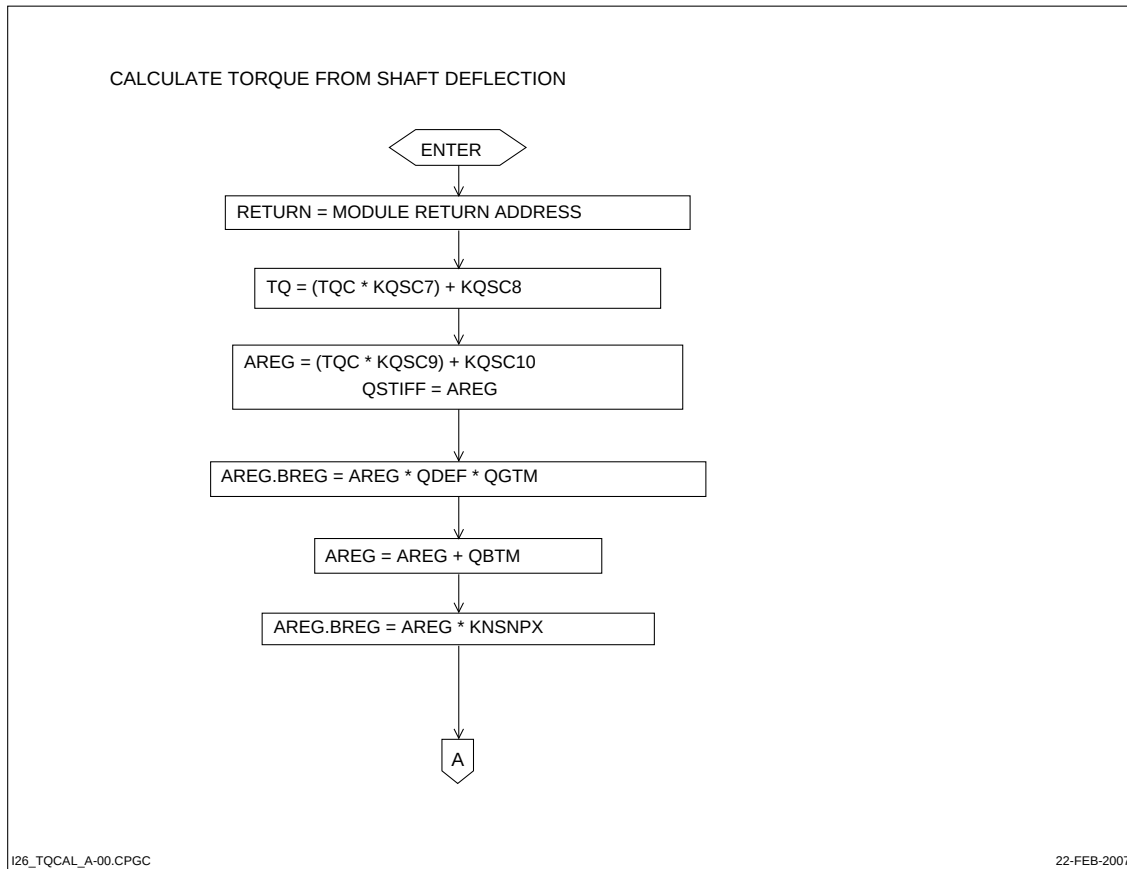
13.00 Figure 370: I25_TQNPT_6_B



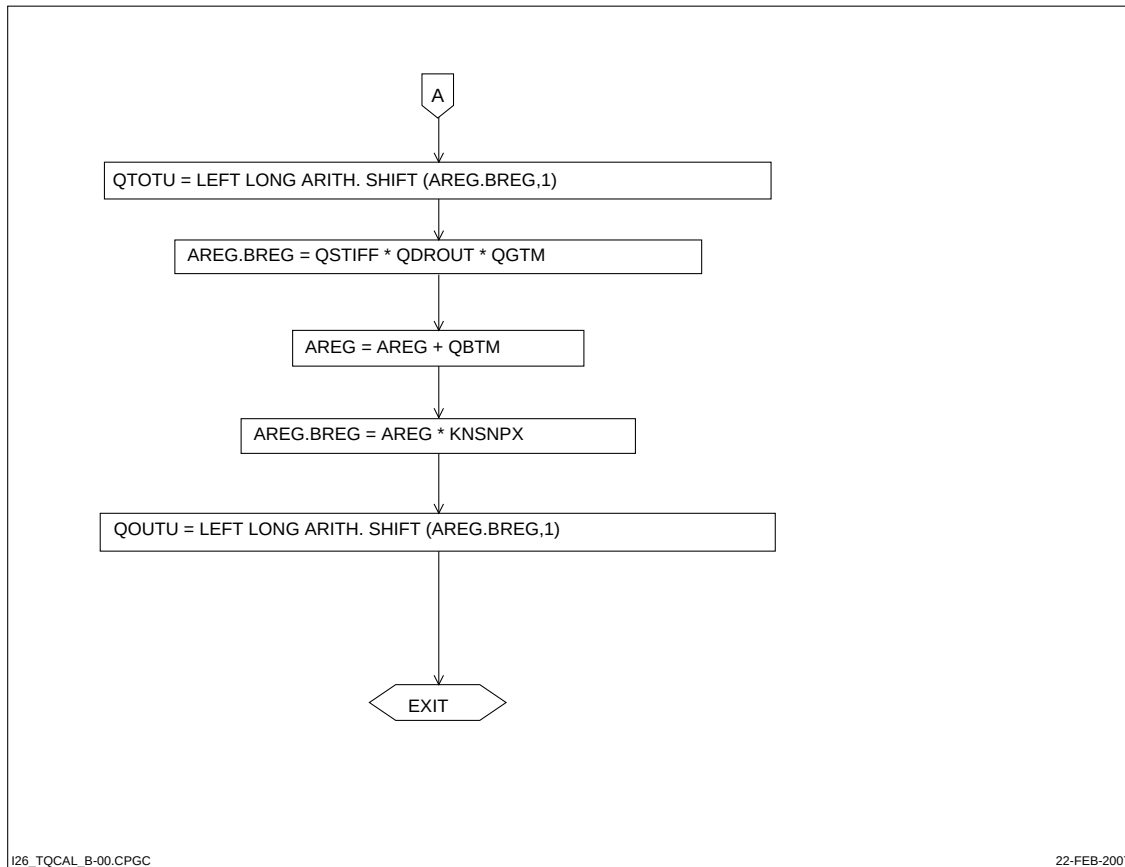
13.00 Figure 371: I25 _TQNPT _7 _A



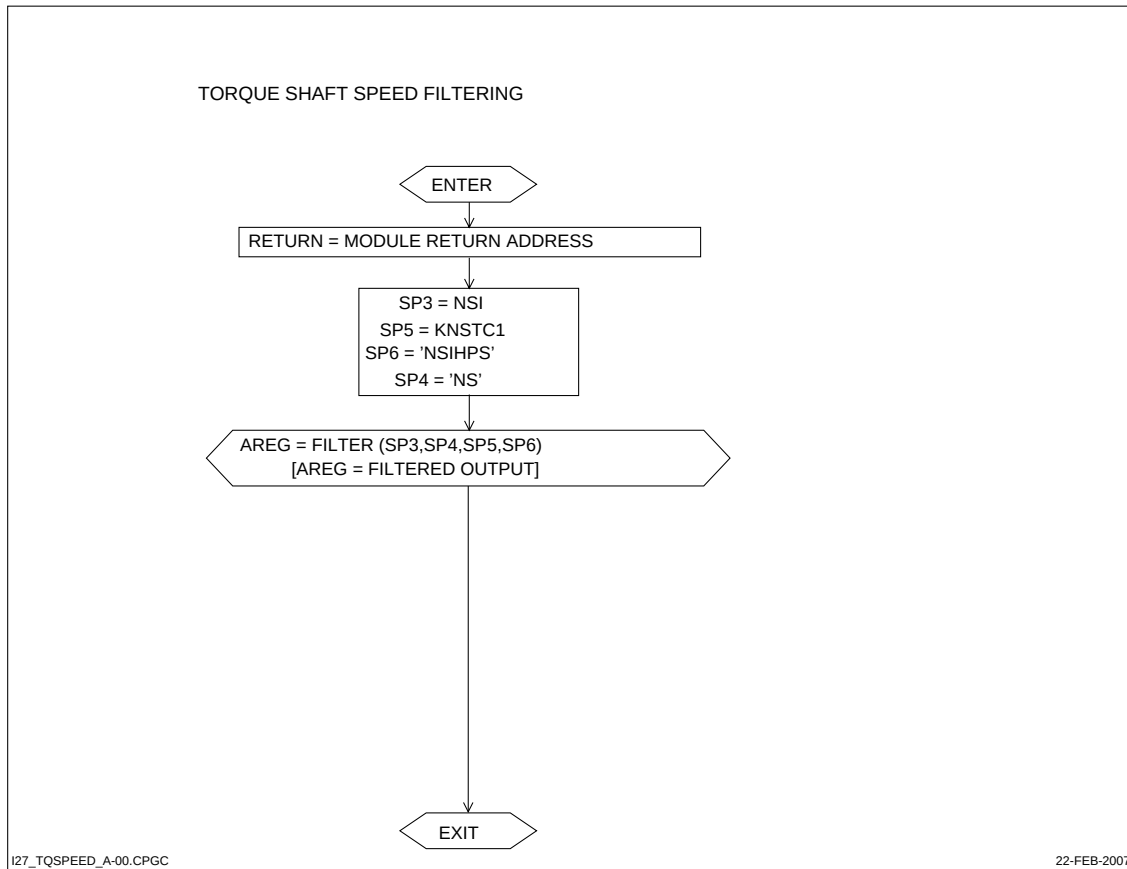
13.00 Figure 372: I26 _TQCAL _A



13.00 Figure 373: I26 _TQCAL _B

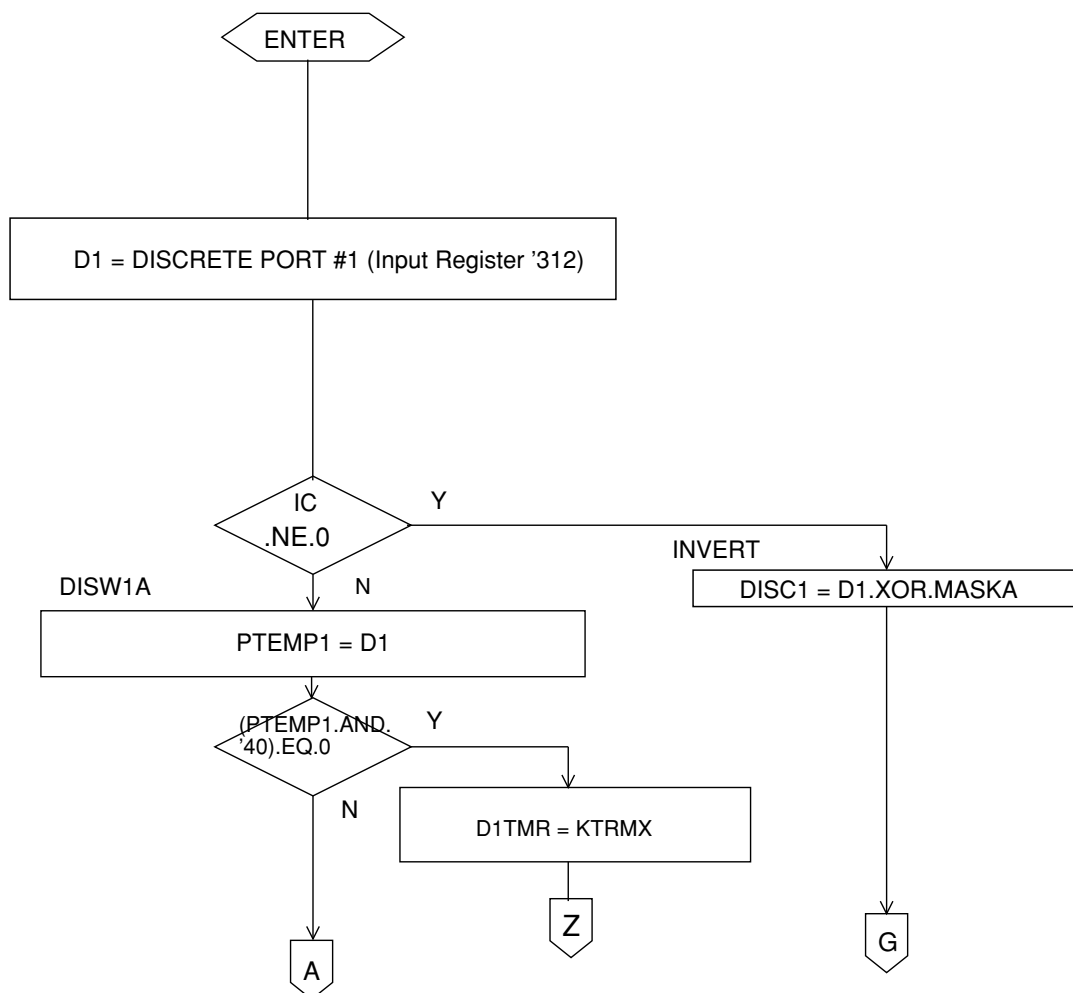


13.00 Figure 374: I27 _TQSPEED _A



13.00 Figure 375: I28_DISINW1_A

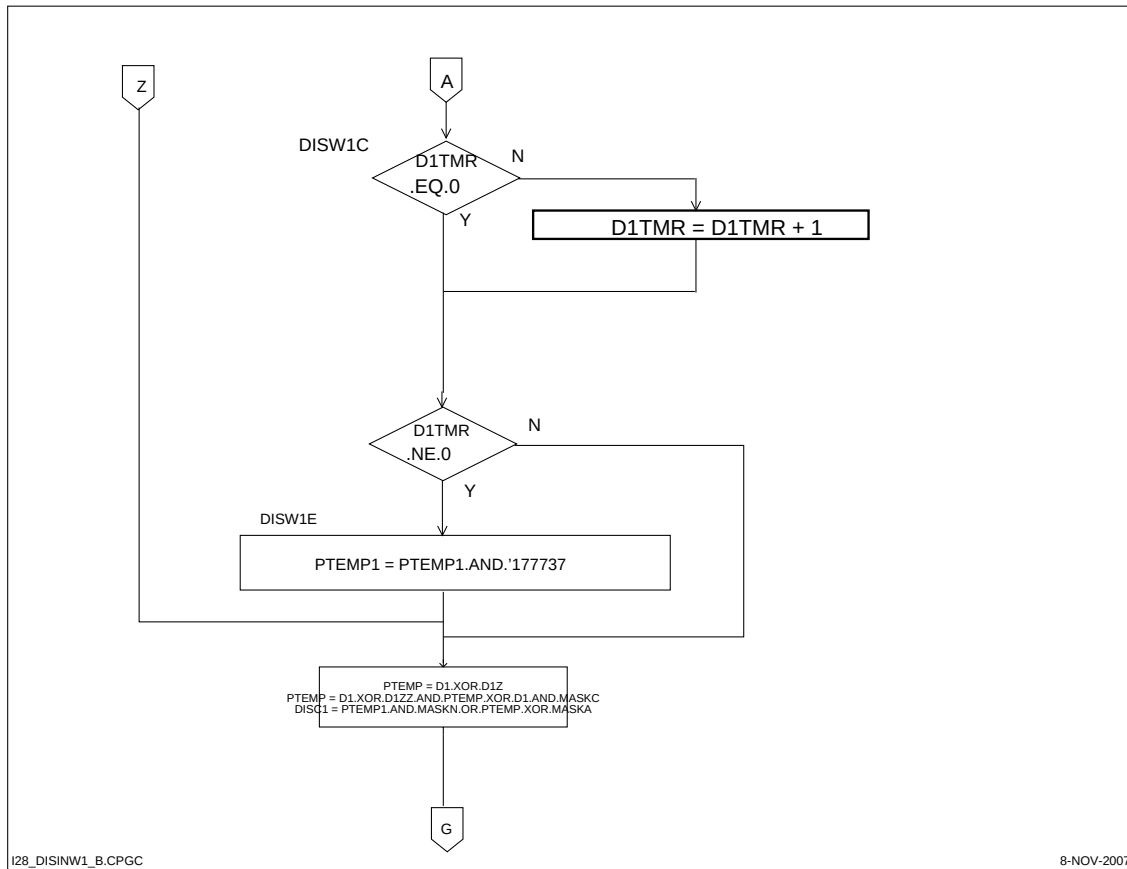
DISCRETE INPUT WORD #1 LOGIC



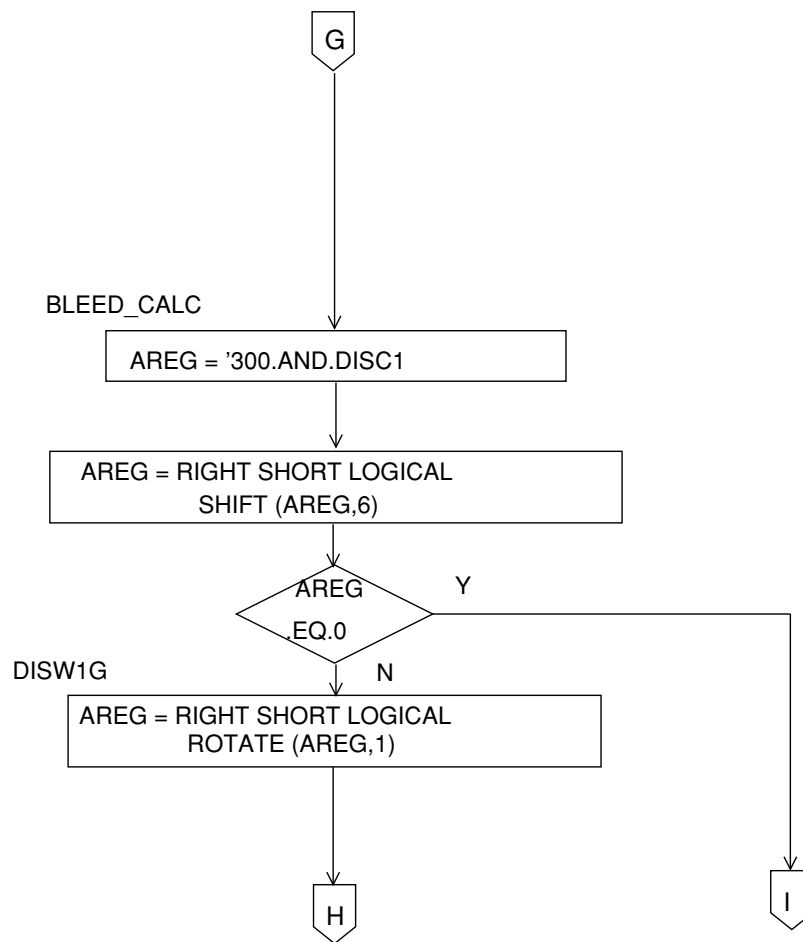
I28_DISINW1_A.CPGC

3-JAN-2025

13.00 Figure 376: I28 _DISINW1 _B



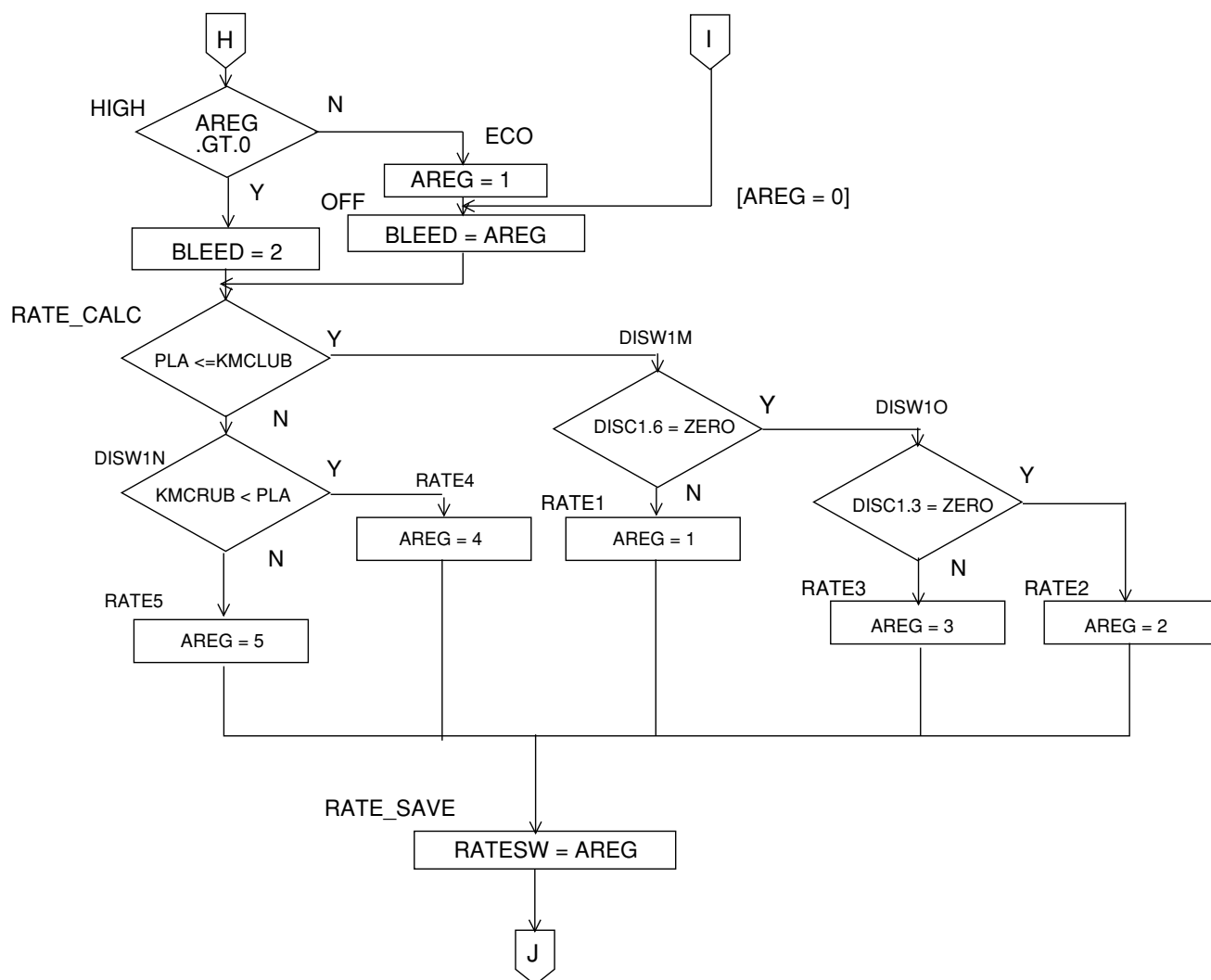
13.00 Figure 377: I28_DISINW1 _C



I28_DISINW1 _C.CPGC

6-JAN-2025

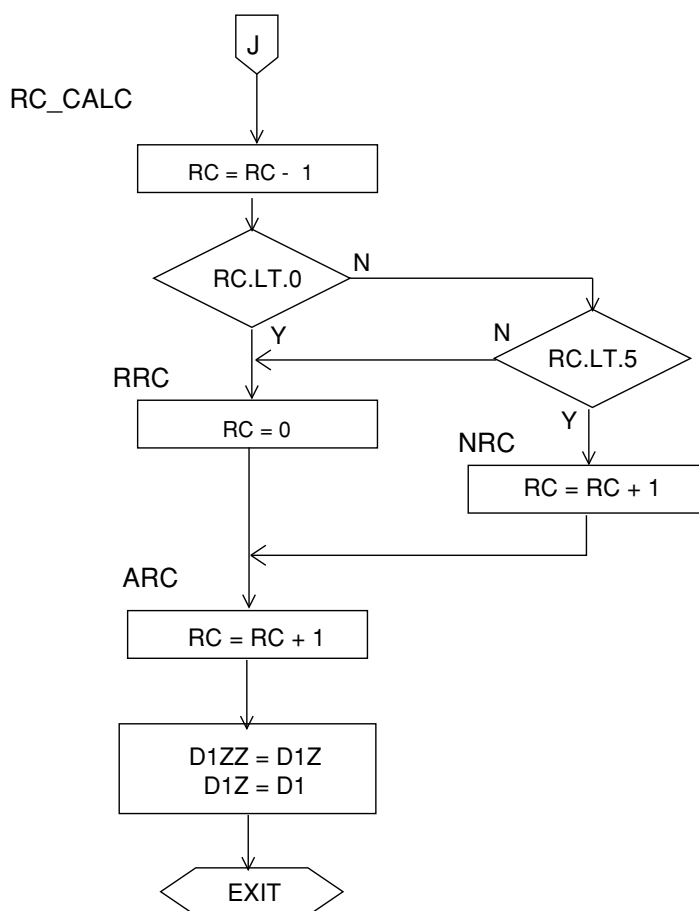
13.00 Figure 378: I28_DISINW1_D



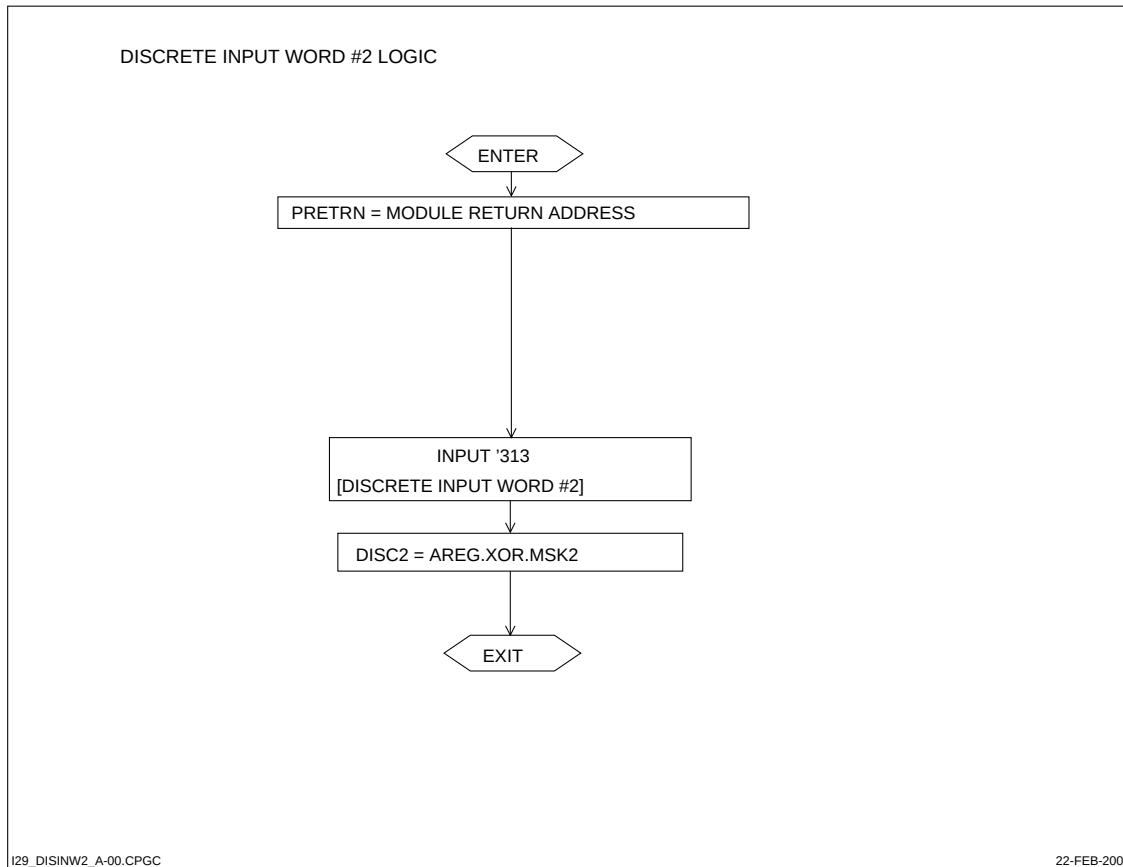
I28_DISINW1_D.CPGC

12-FEB-2025

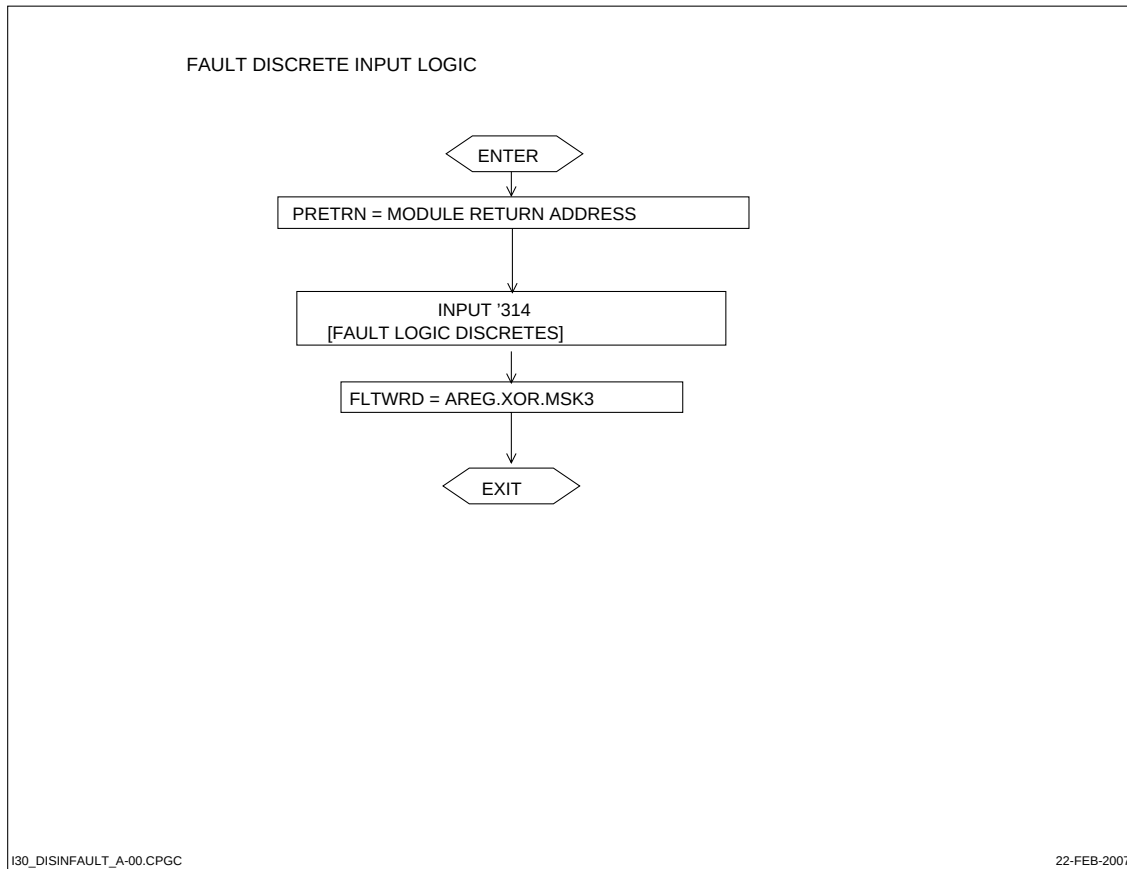
13.00 Figure 379: I28 _DISINW1 _E



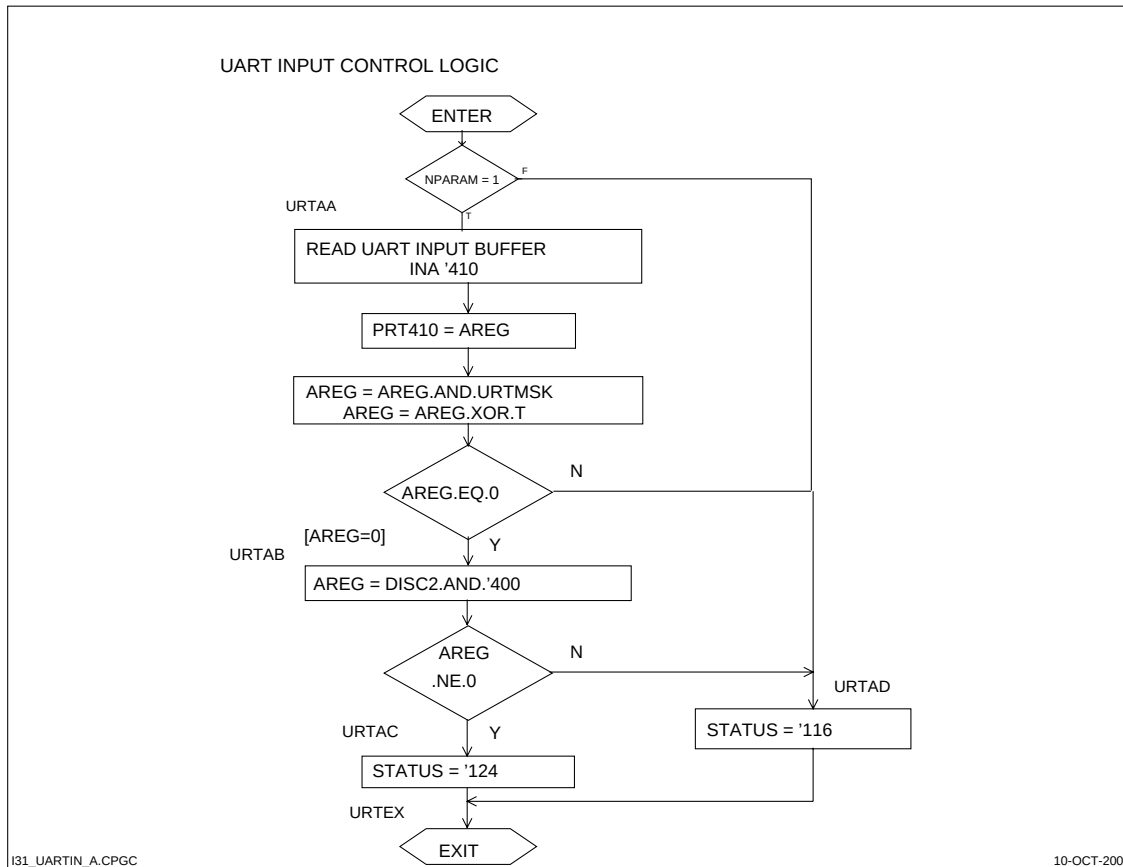
13.00 Figure 380: I29 _DISINW2 _A



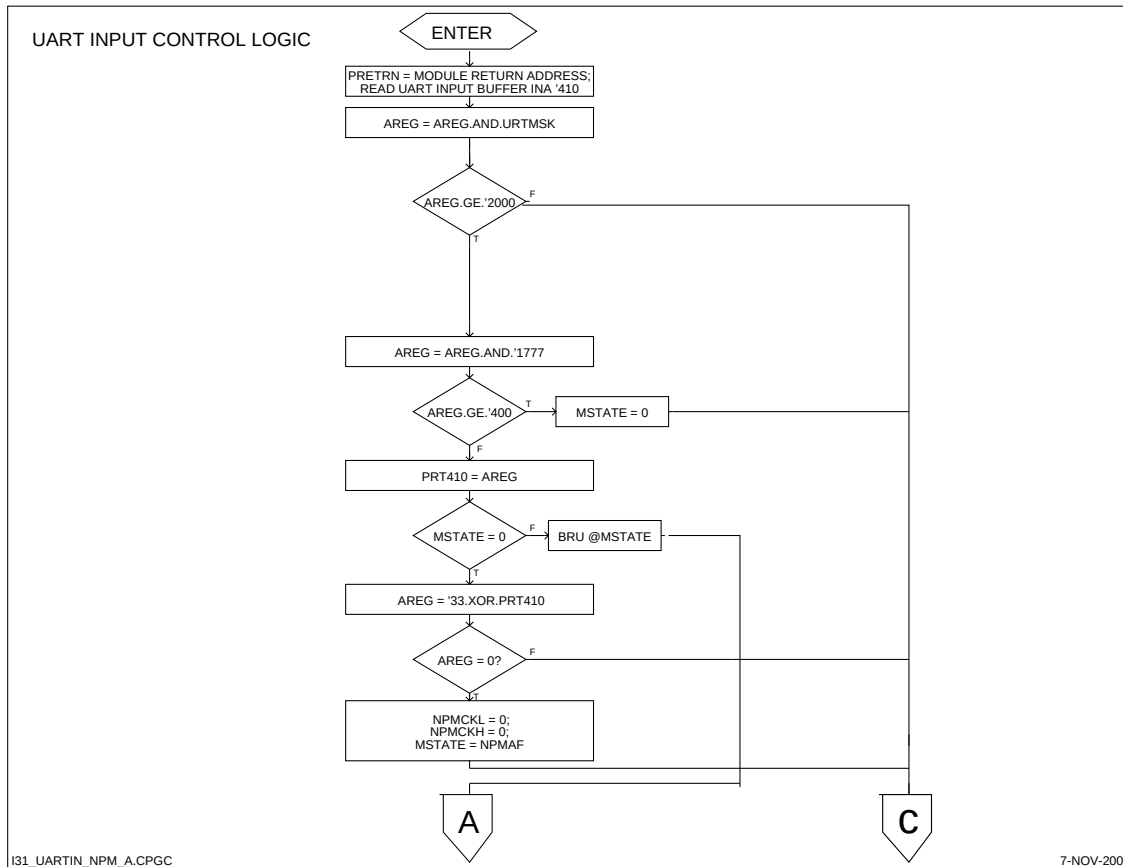
13.00 Figure 381: I30 _DISINFALT _A



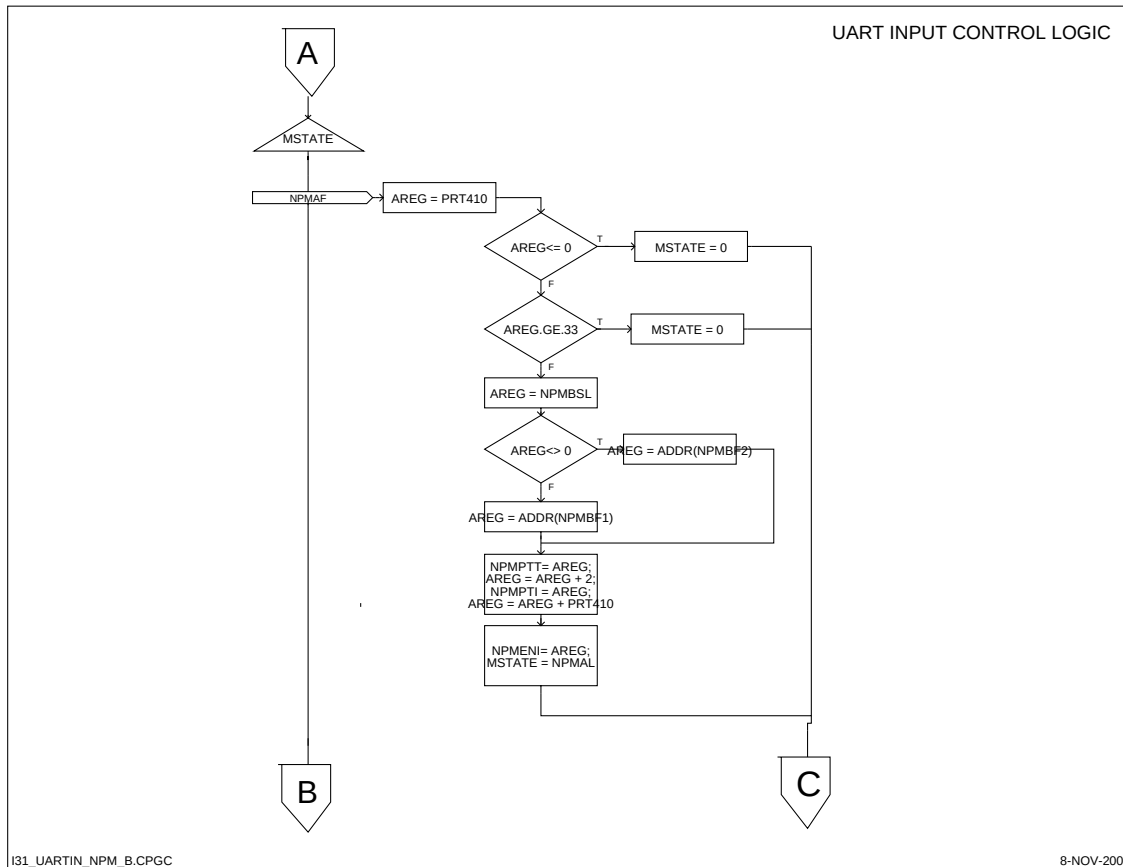
13.00 Figure 382: I31 _UARTIN _A



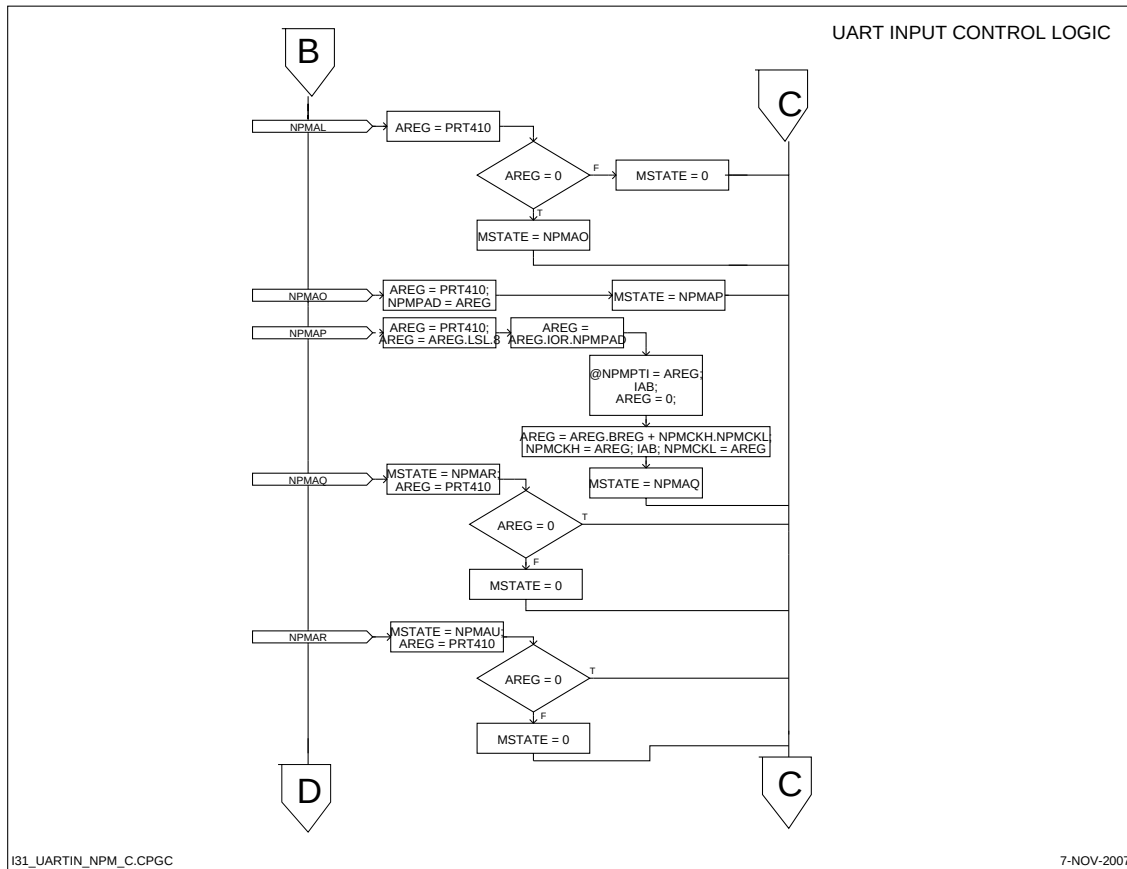
13.00 Figure 383: I31_UARTIN_NPM_A



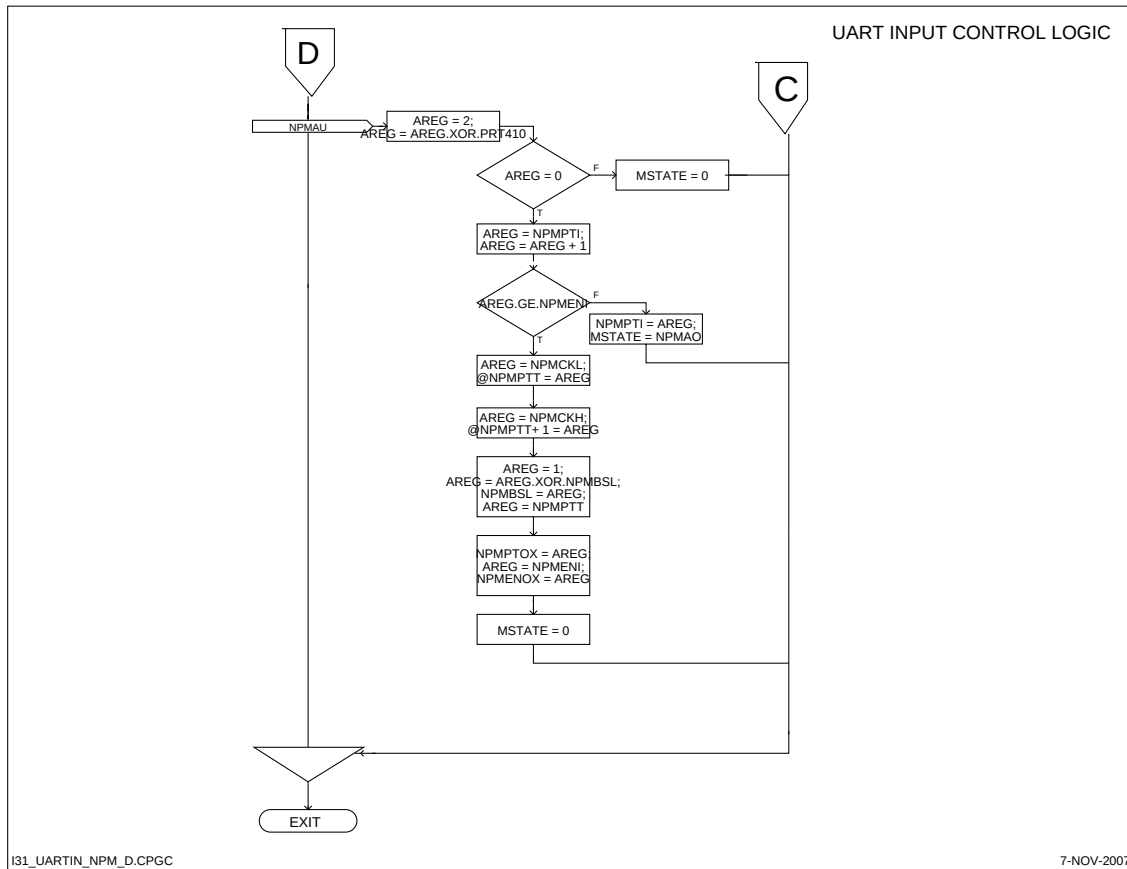
13.00 Figure 384: I31_UARTIN_NPM_B



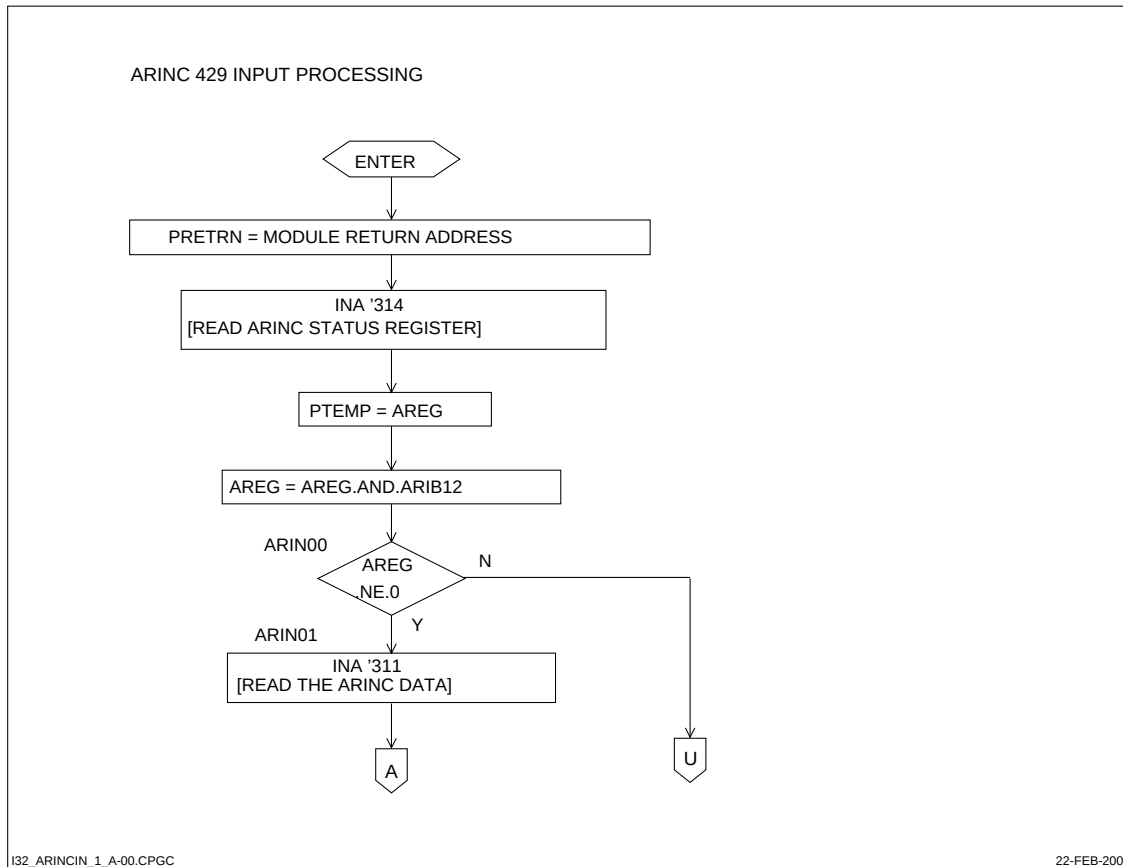
13.00 Figure 385: I31_UARTIN_NPM_C



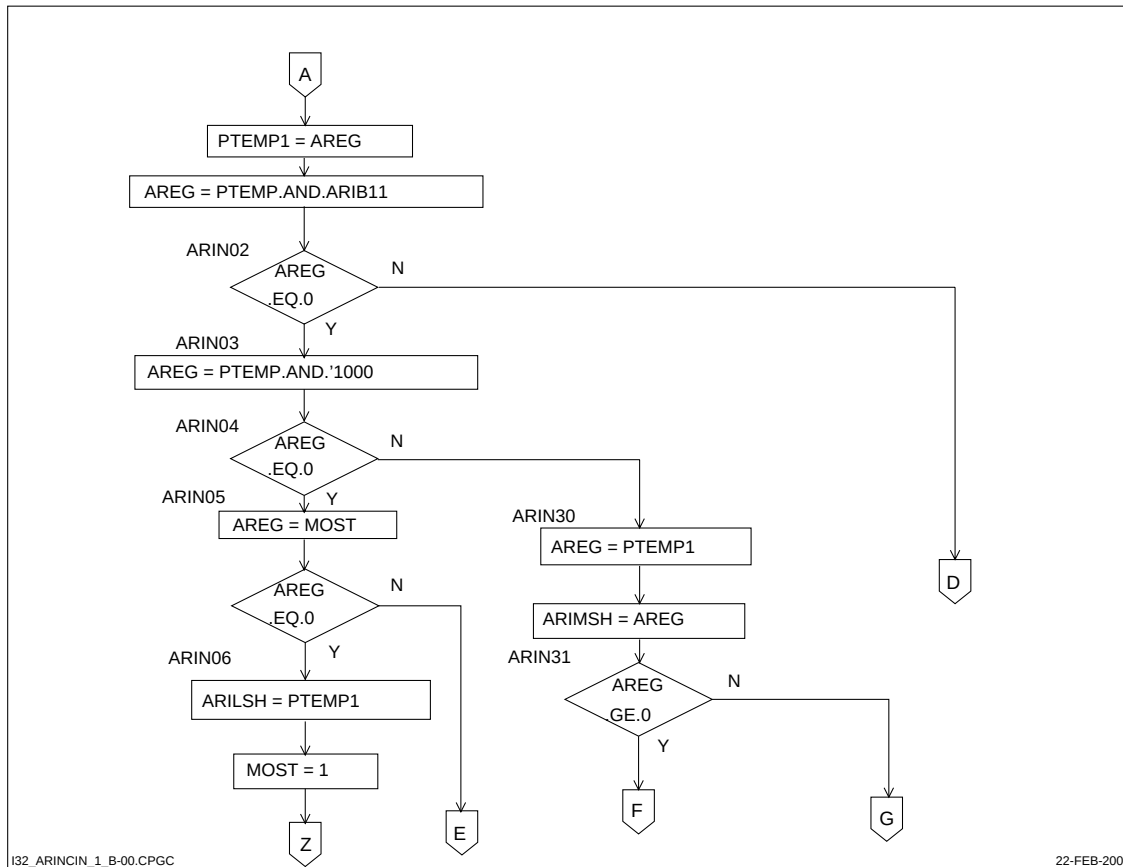
13.00 Figure 386: I31_UARTIN_NPM_D



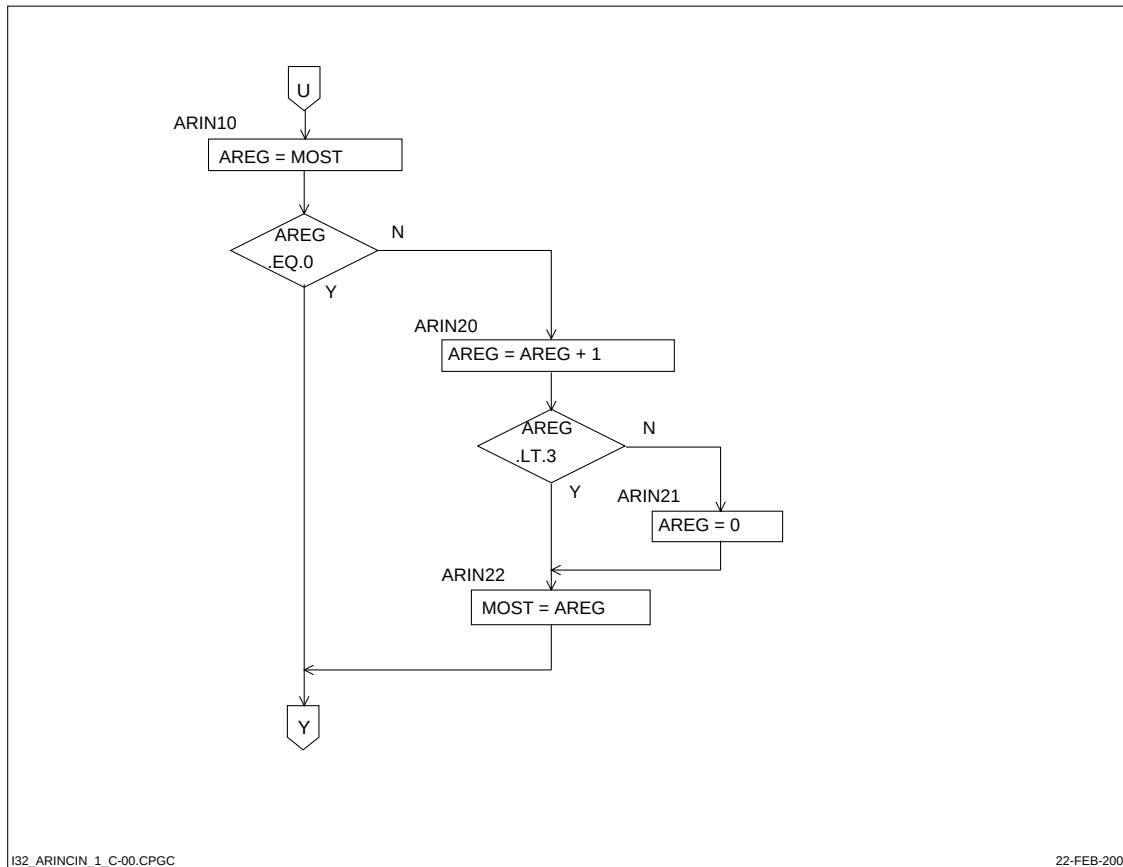
13.00 Figure 387: I32_ARINCIN_1_A



13.00 Figure 388: I32_ARINCIN_1_B



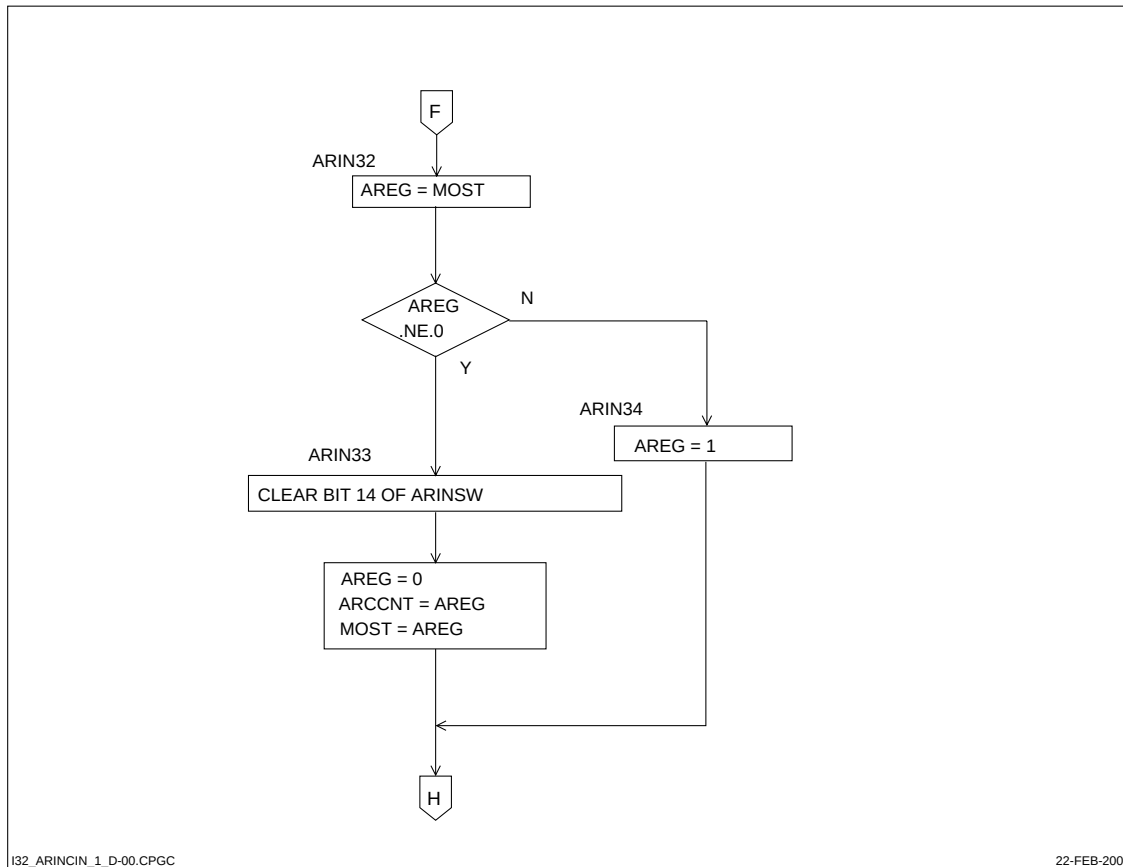
13.00 Figure 389: I32_ARINCIN_1_C



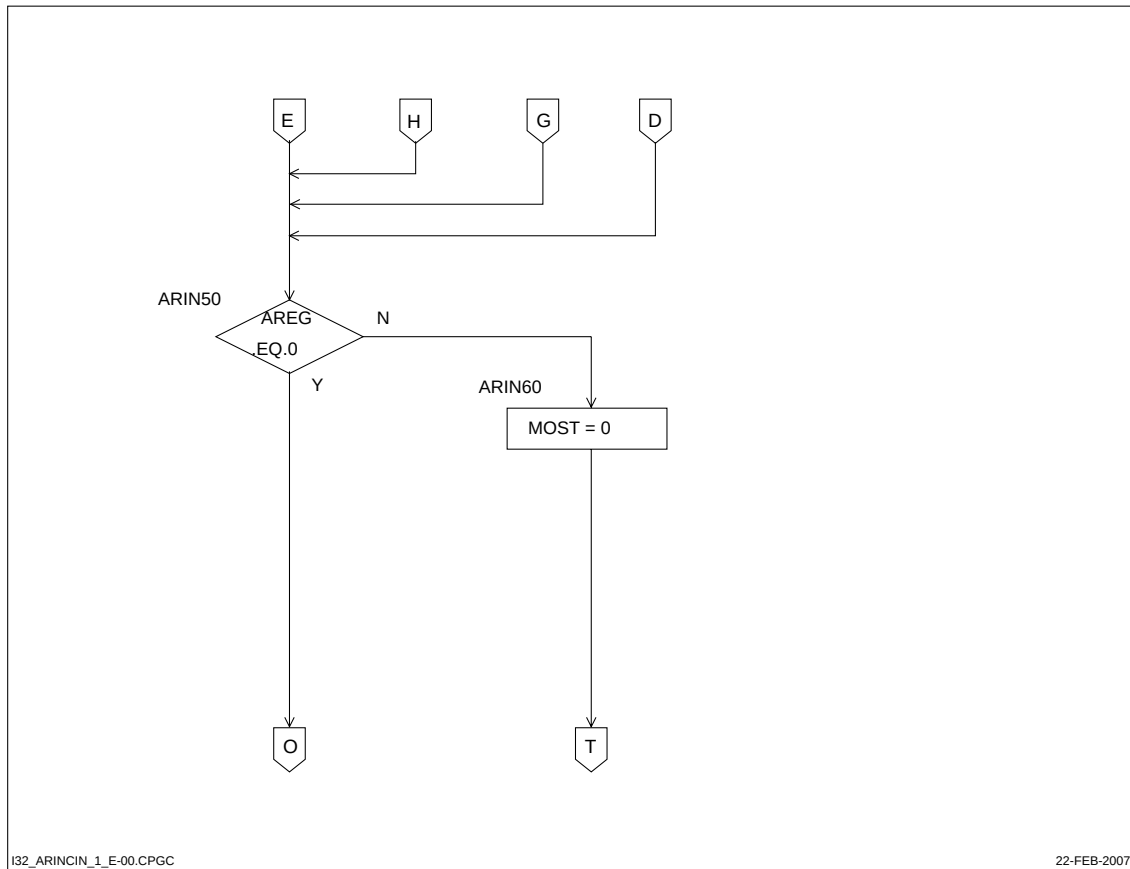
I32_ARINCIN_1_C-00.CPGC

22-FEB-2007

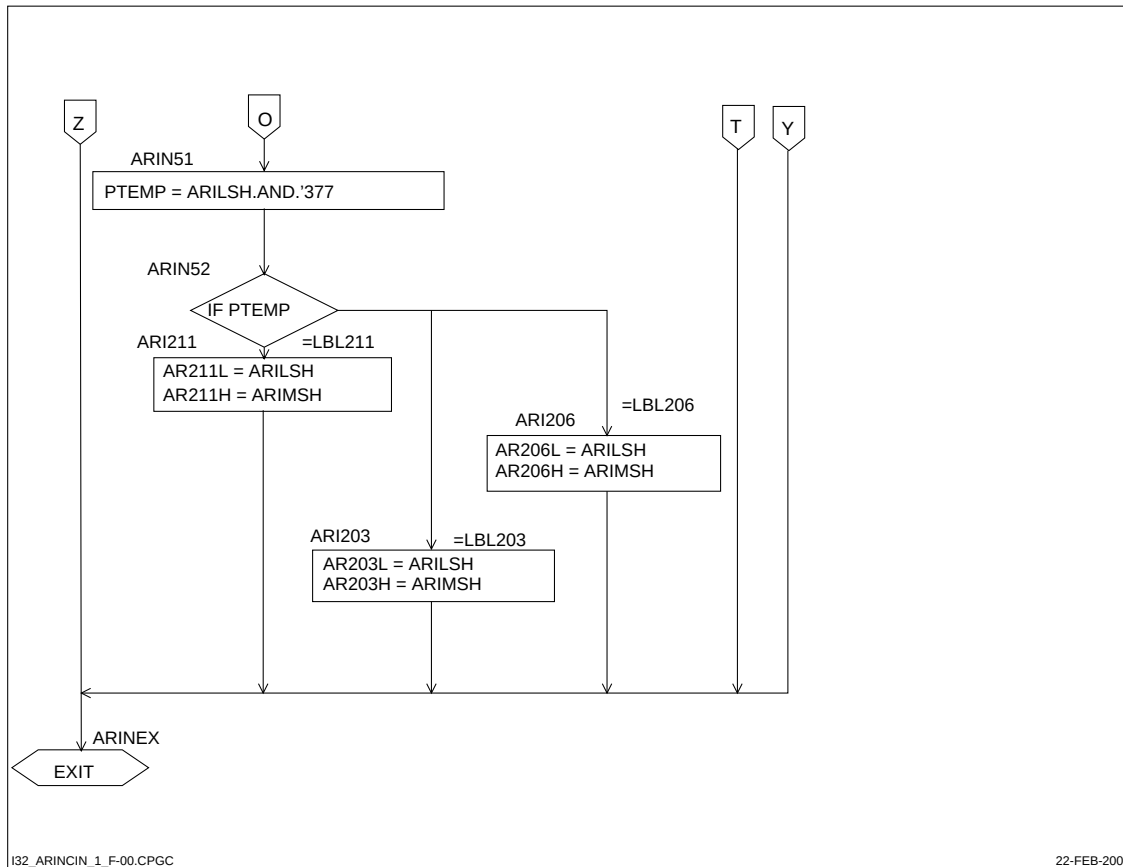
13.00 Figure 390: I32_ARINCIN_1_D



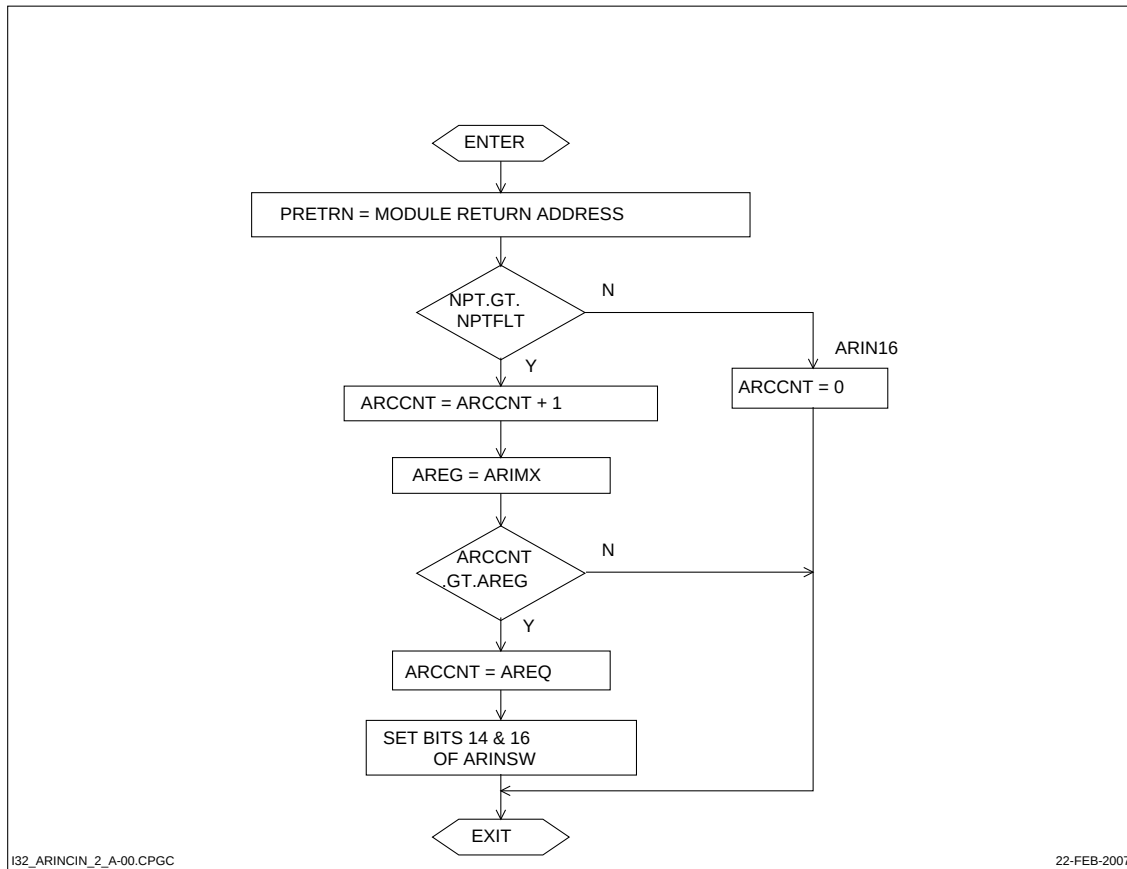
13.00 Figure 391: I32_ARINCIN_1_E



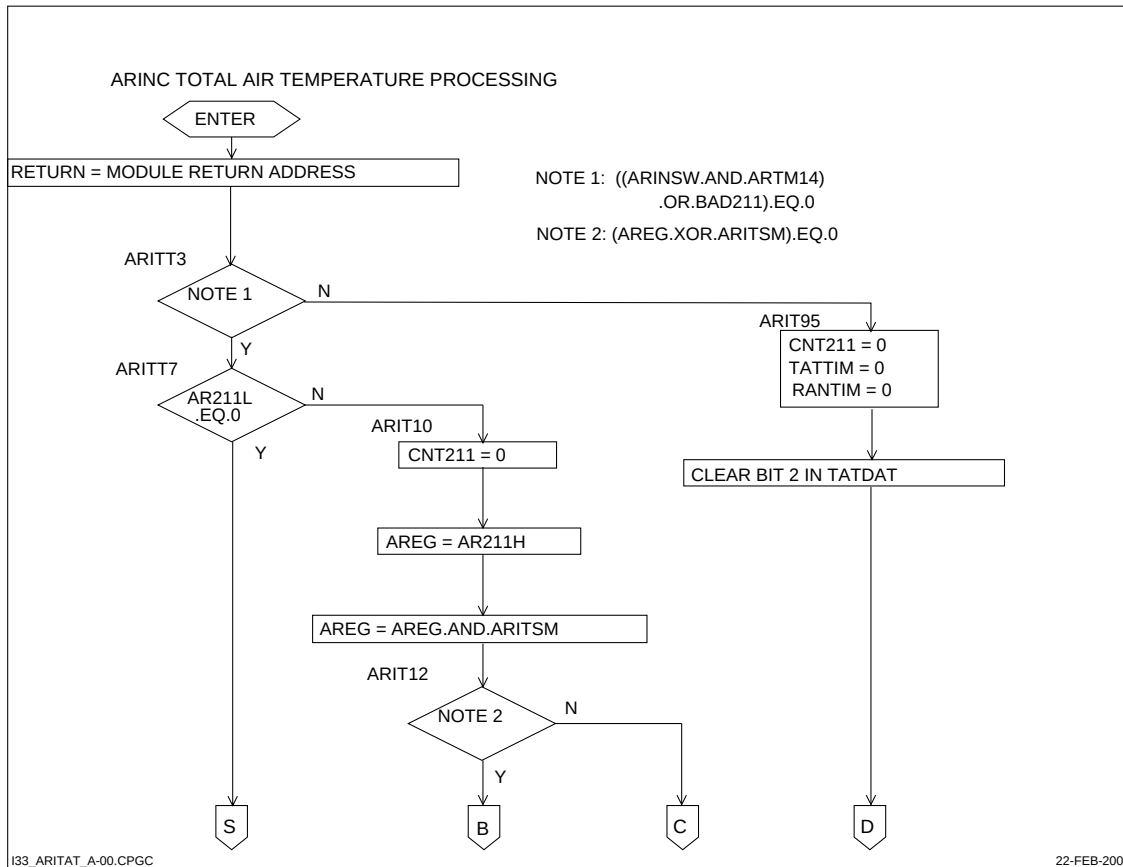
13.00 Figure 392: I32_ARINCIN_1_F



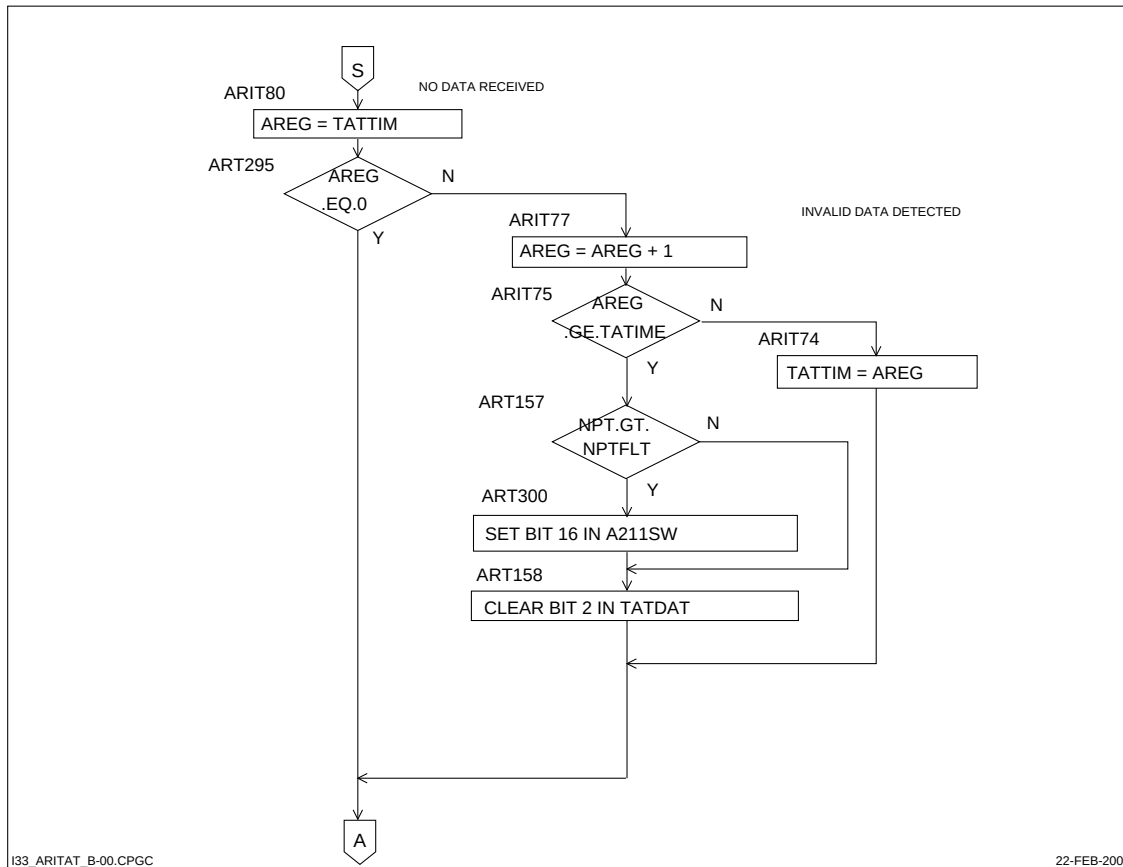
13.00 Figure 393: I32_ARINCIN_2_A



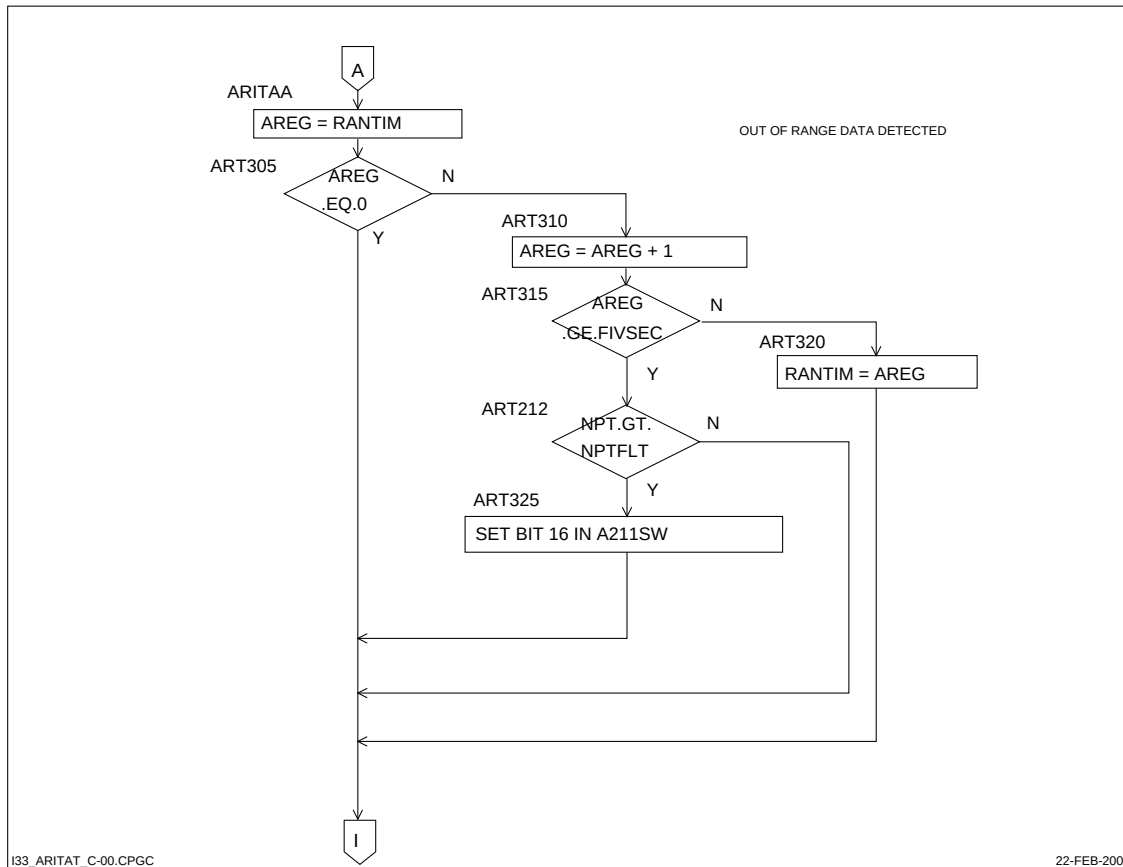
13.00 Figure 394: I33_ARITAT_A



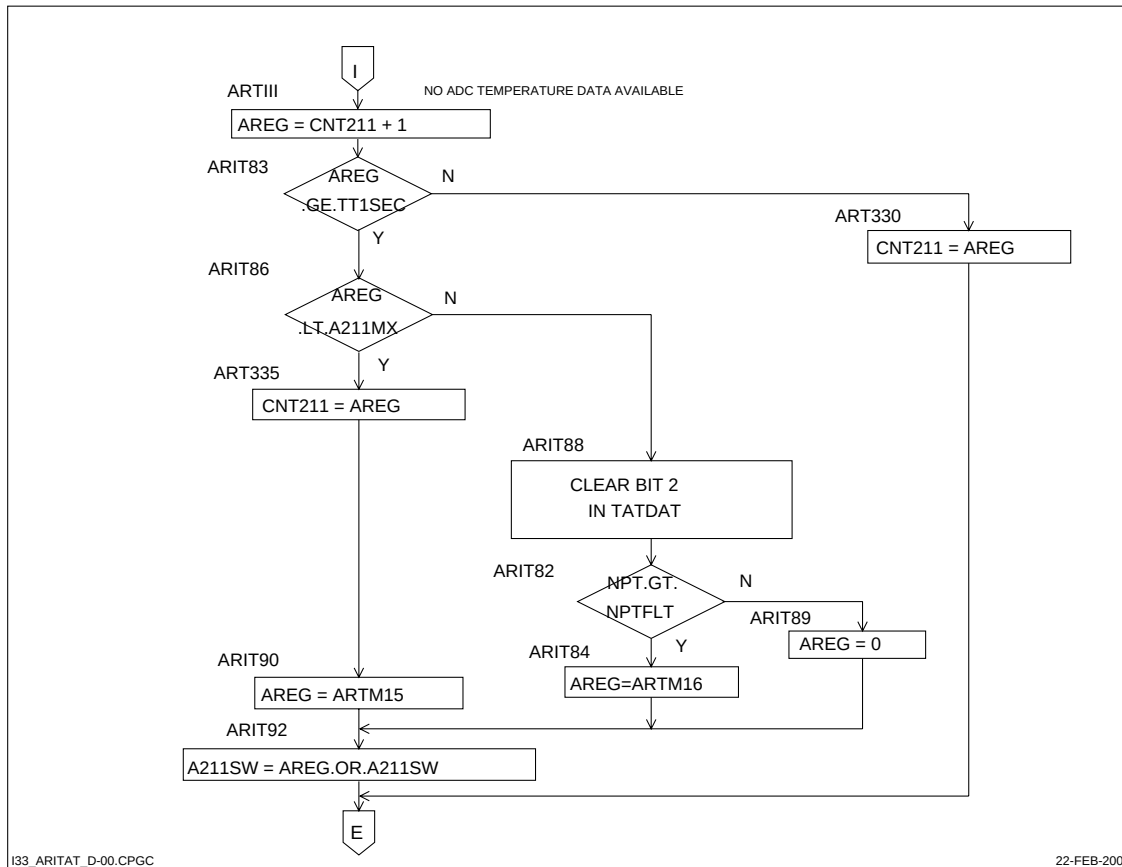
13.00 Figure 395: I33_ARITAT_B



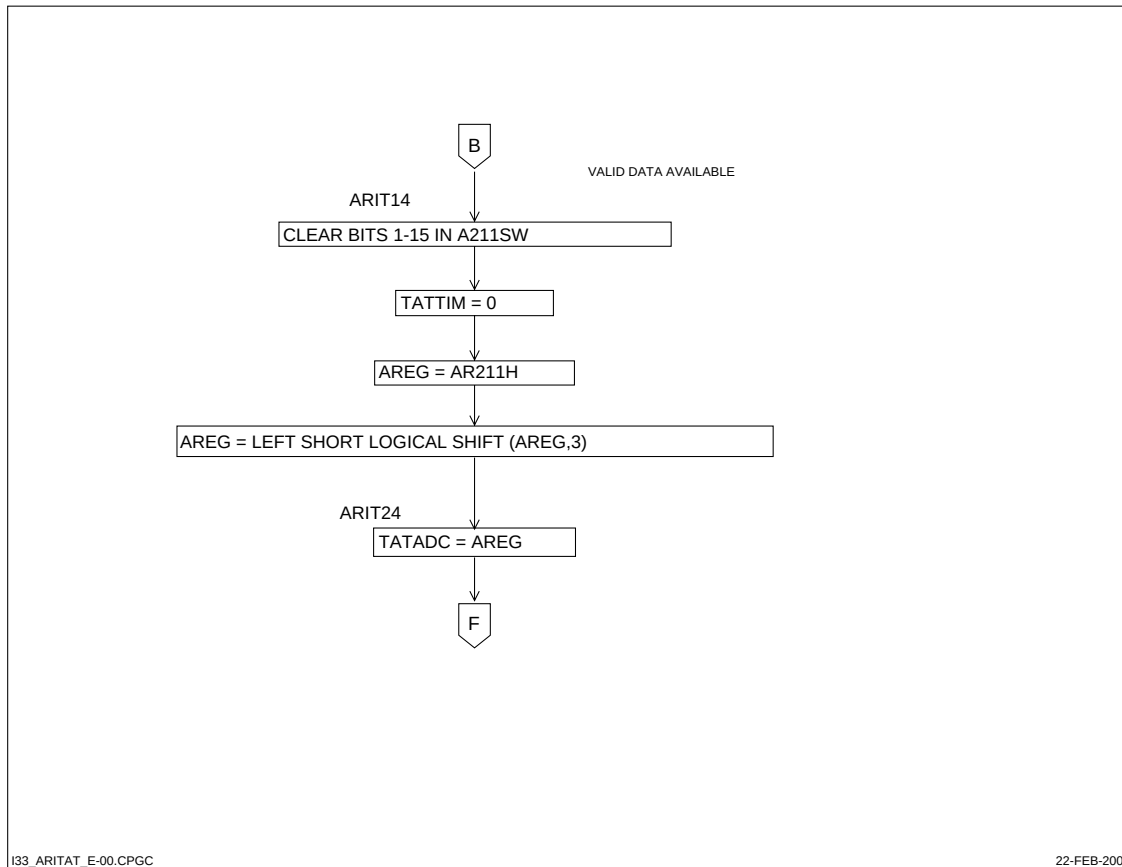
13.00 Figure 396: I33_ARITAT_C



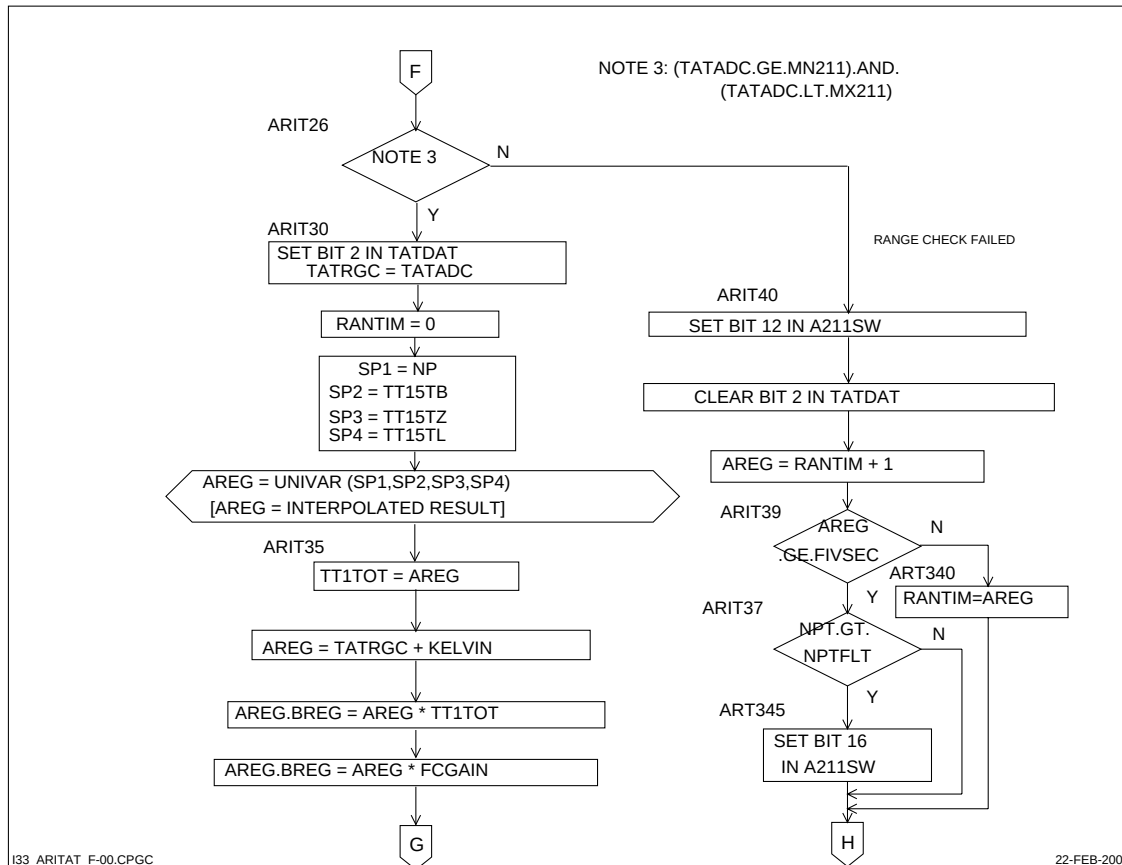
13.00 Figure 397: I33_ARITAT_D



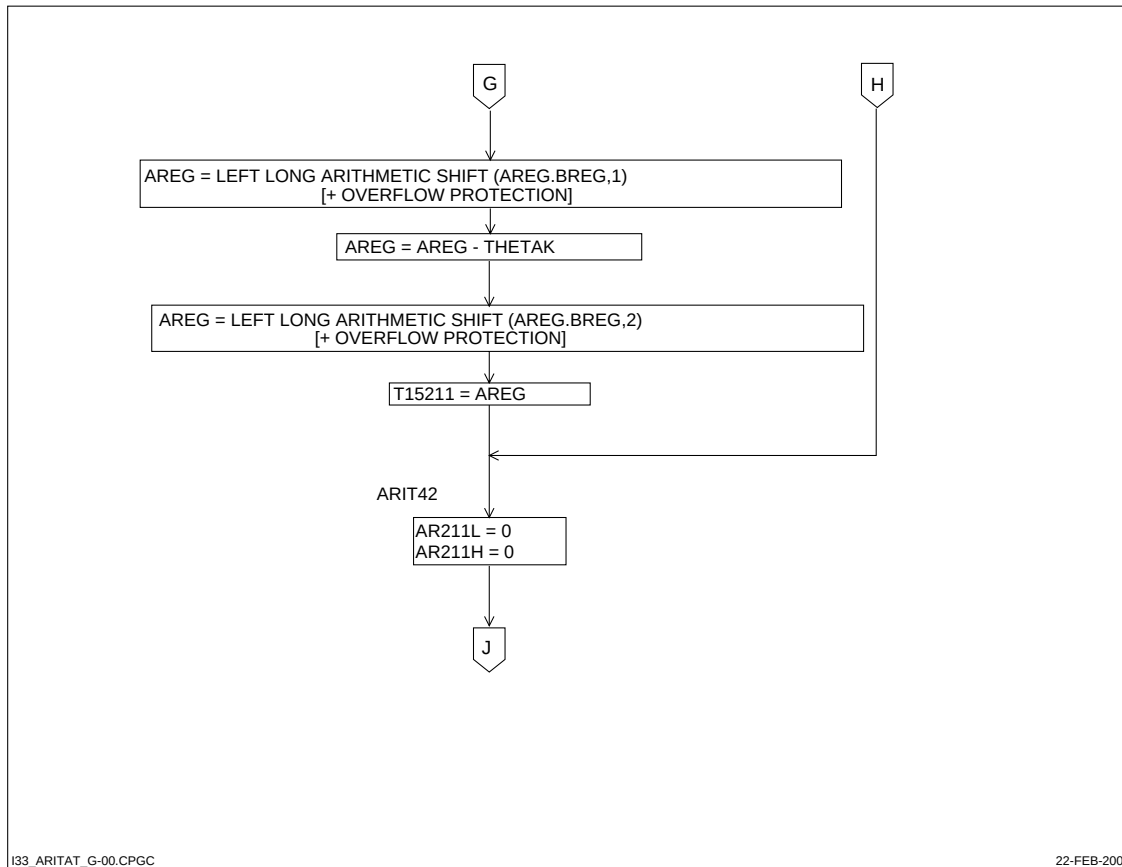
13.00 Figure 398: I33_ARITAT_E



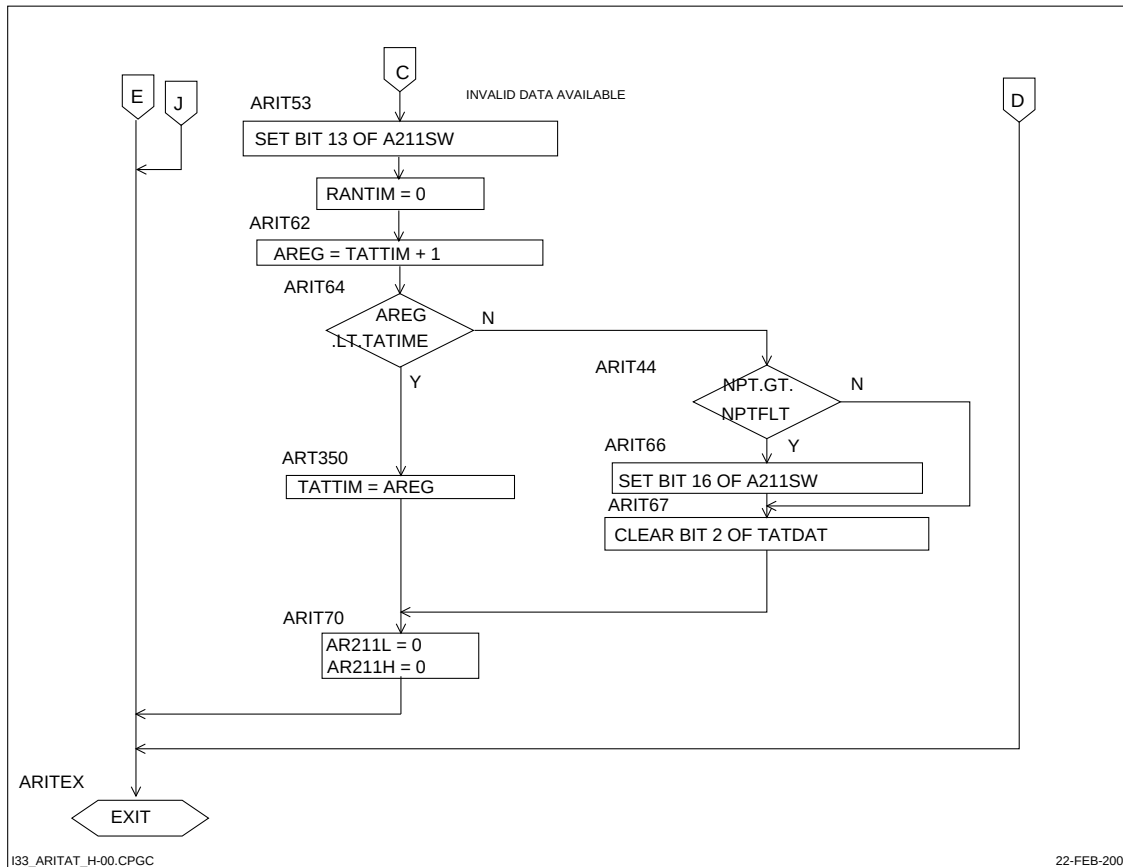
13.00 Figure 399: I33_ARITAT_F



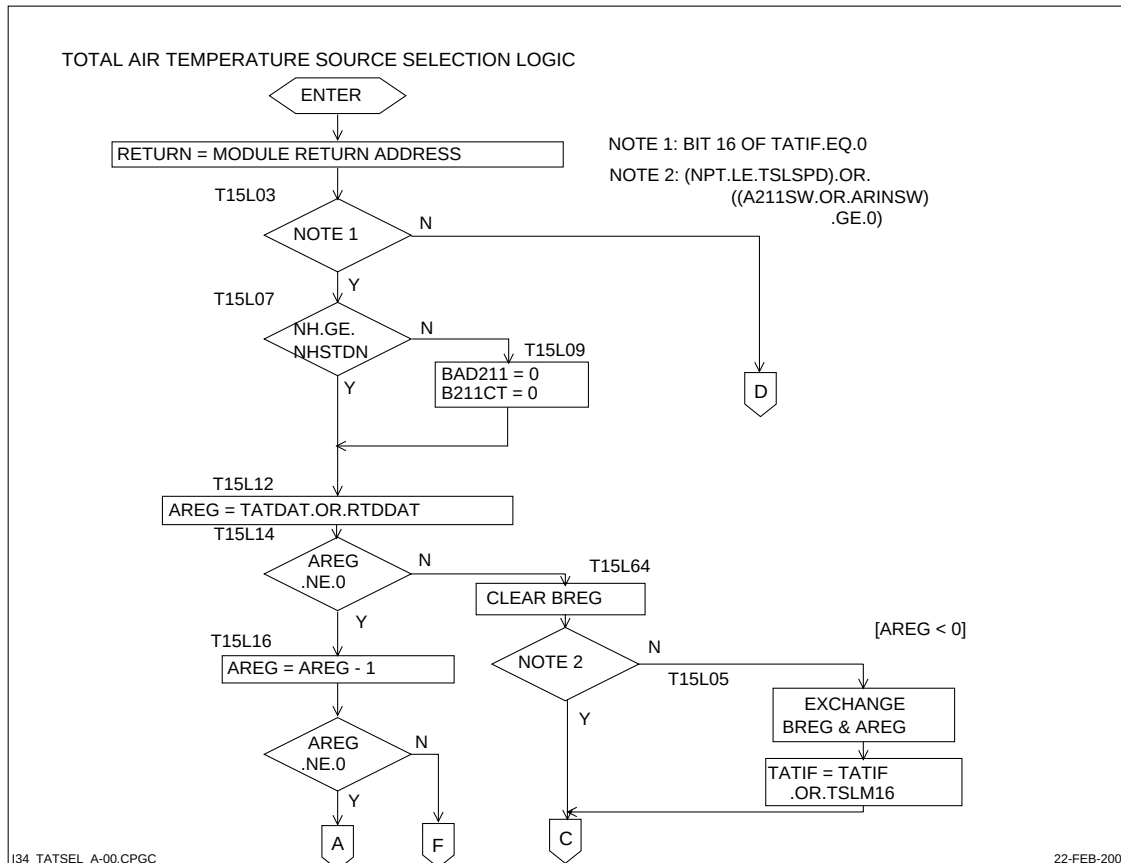
13.00 Figure 410: I33 _ARITAT _G



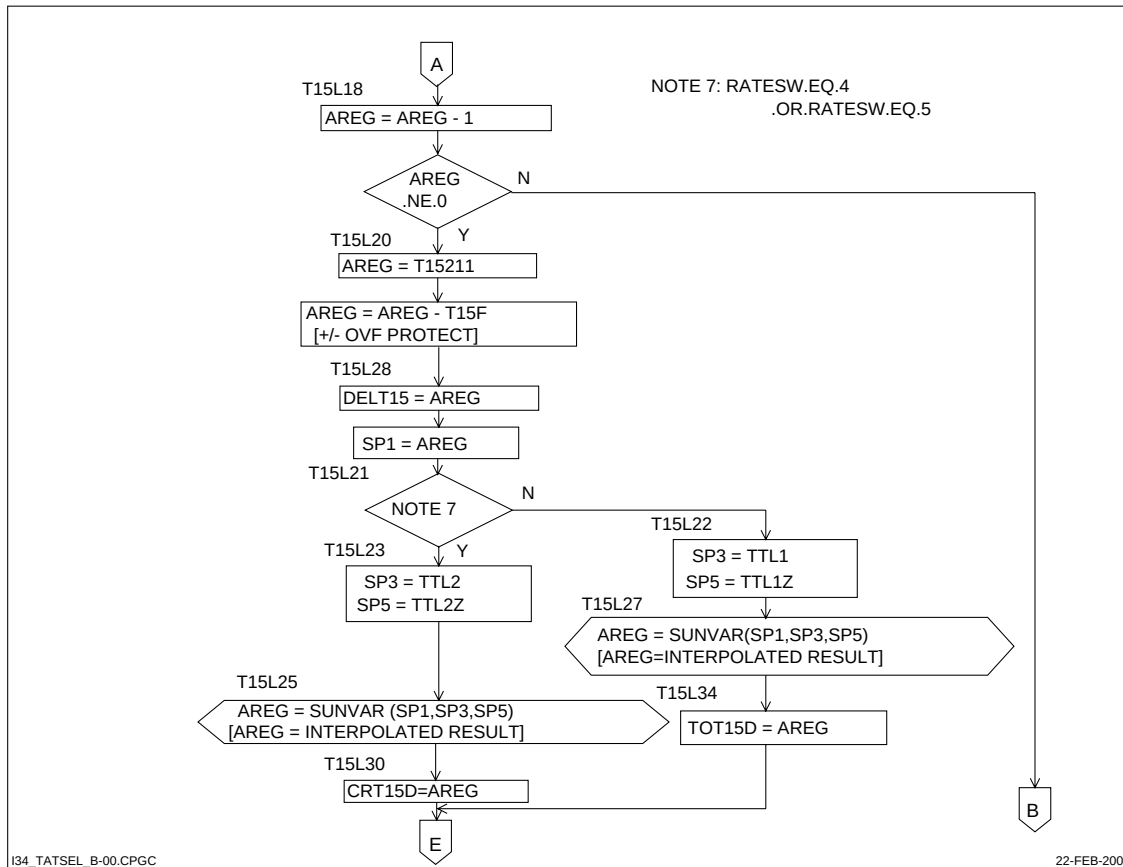
13.00 Figure 411: I33_ARITAT_H



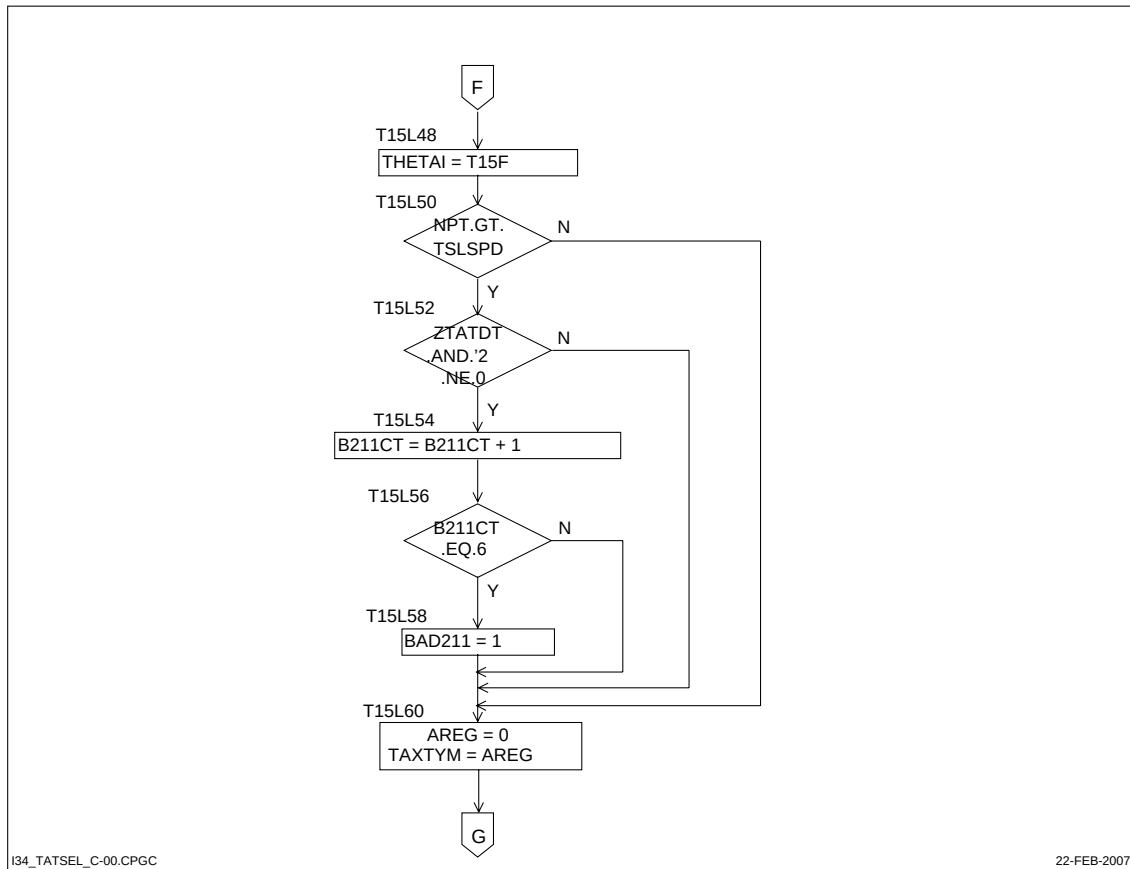
13.00 Figure 412: I34 _TATSEL _A



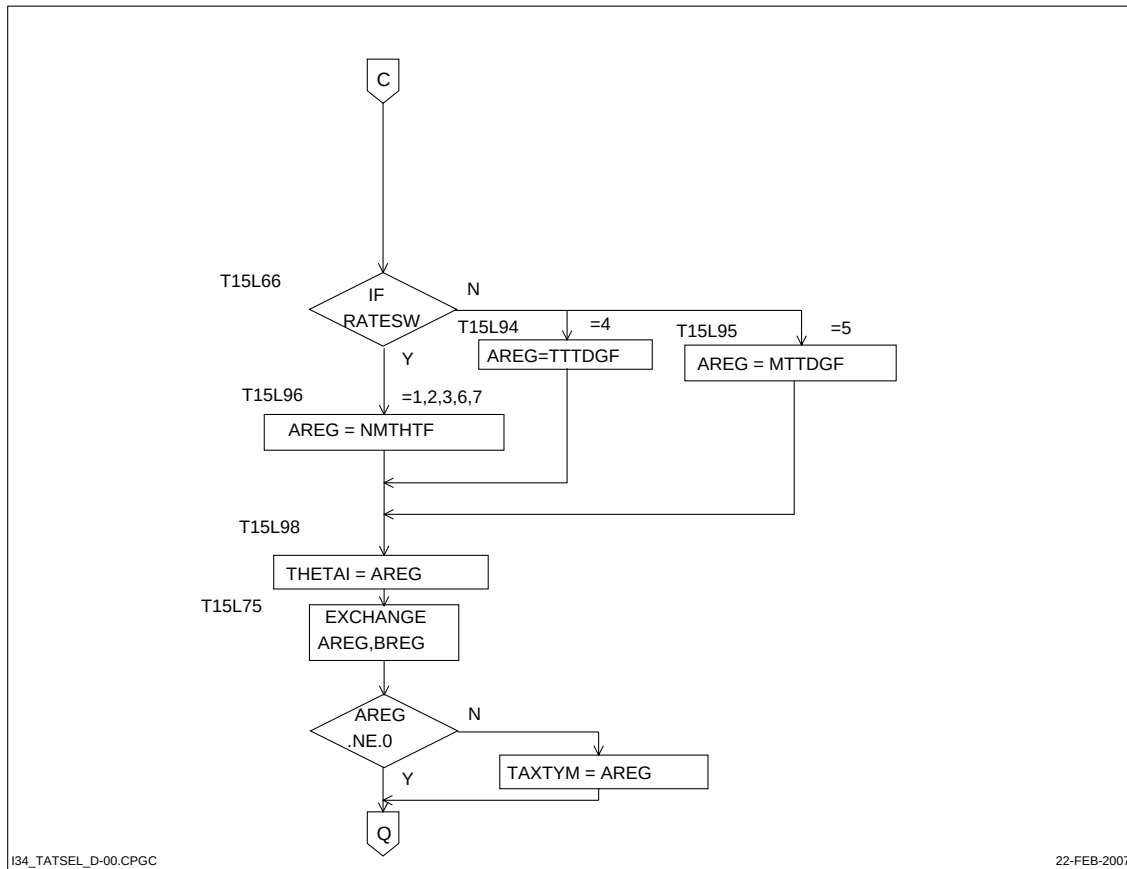
13.00 Figure 413: I34 _TATSEL _B



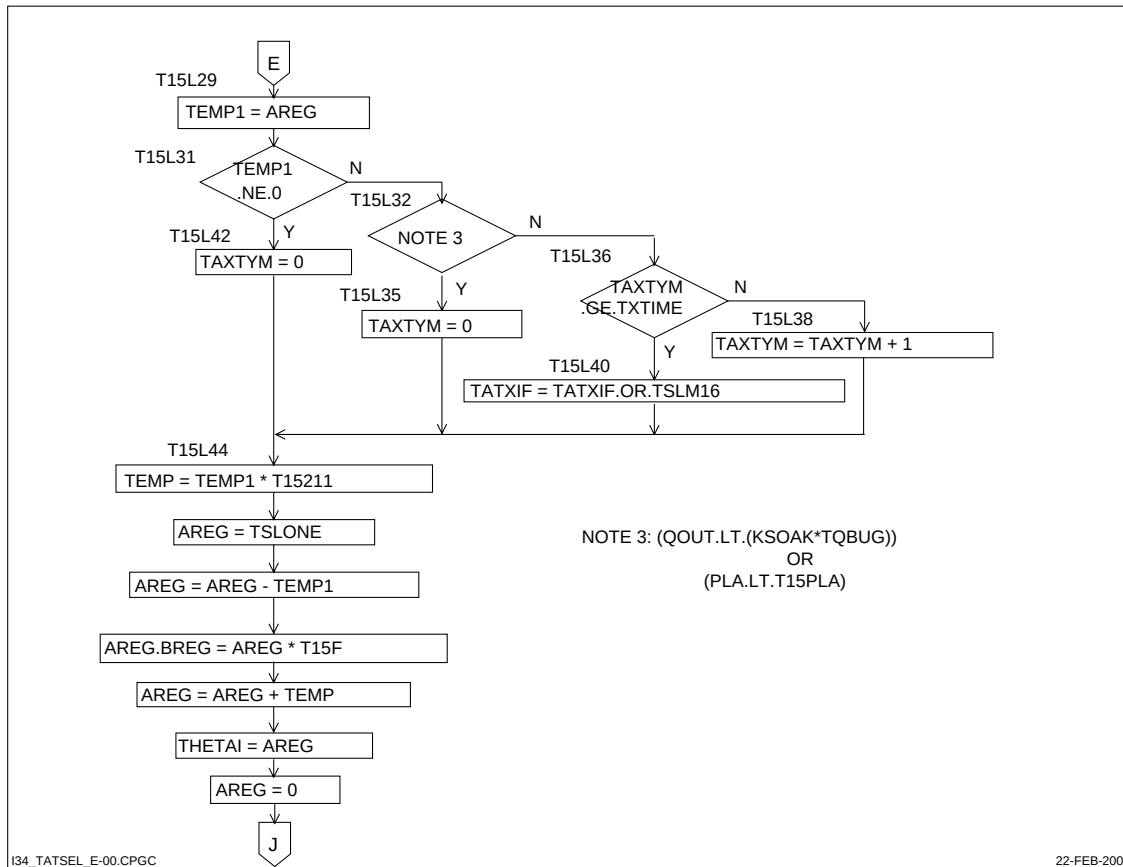
13.00 Figure 414: I34 _TATSEL _C



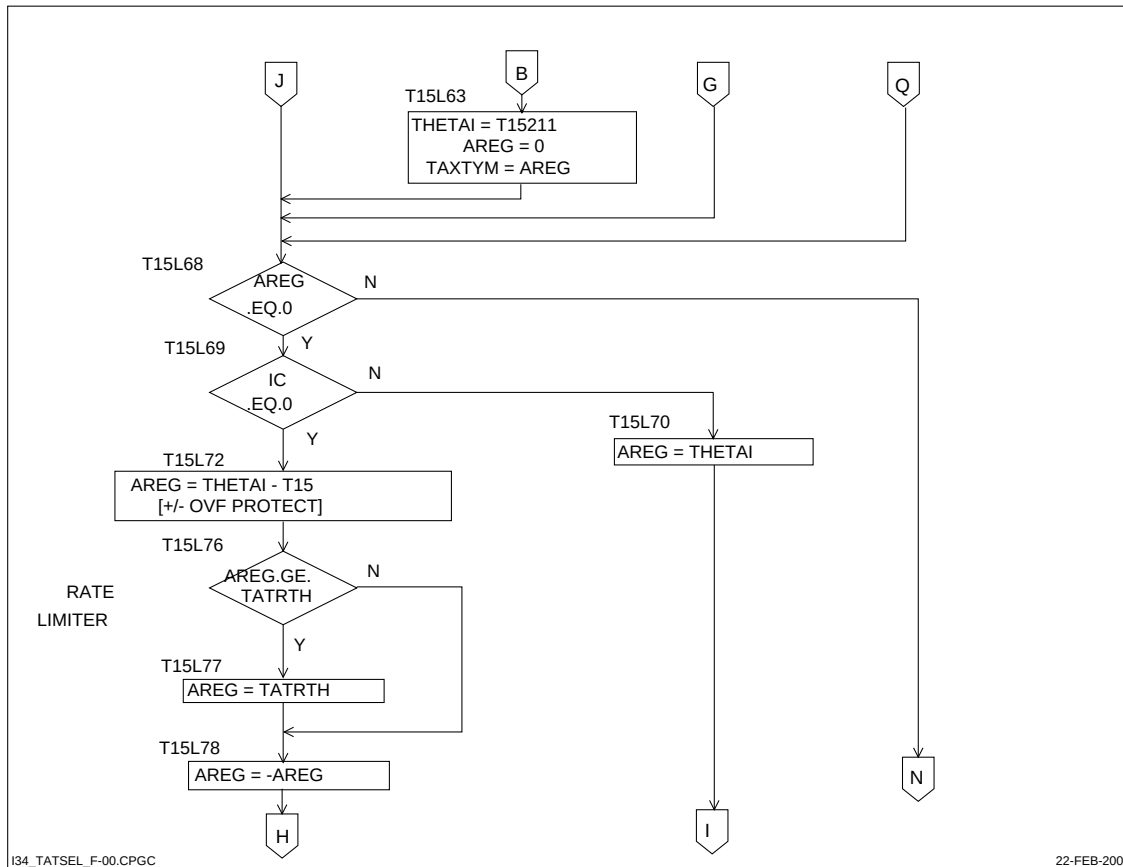
13.00 Figure 415: I34 _TATSEL _D



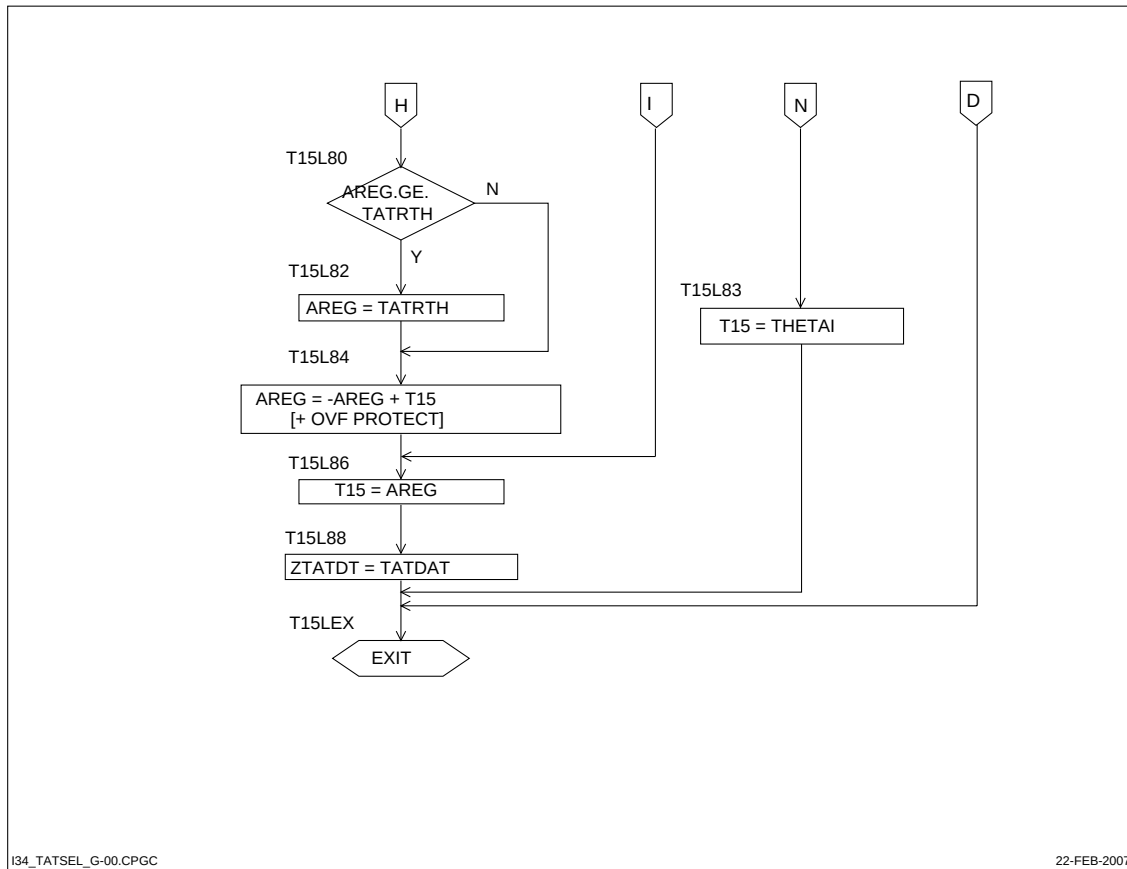
13.00 Figure 416: I34_TATSEL_E



13.00 Figure 417: I34_TATSEL_F



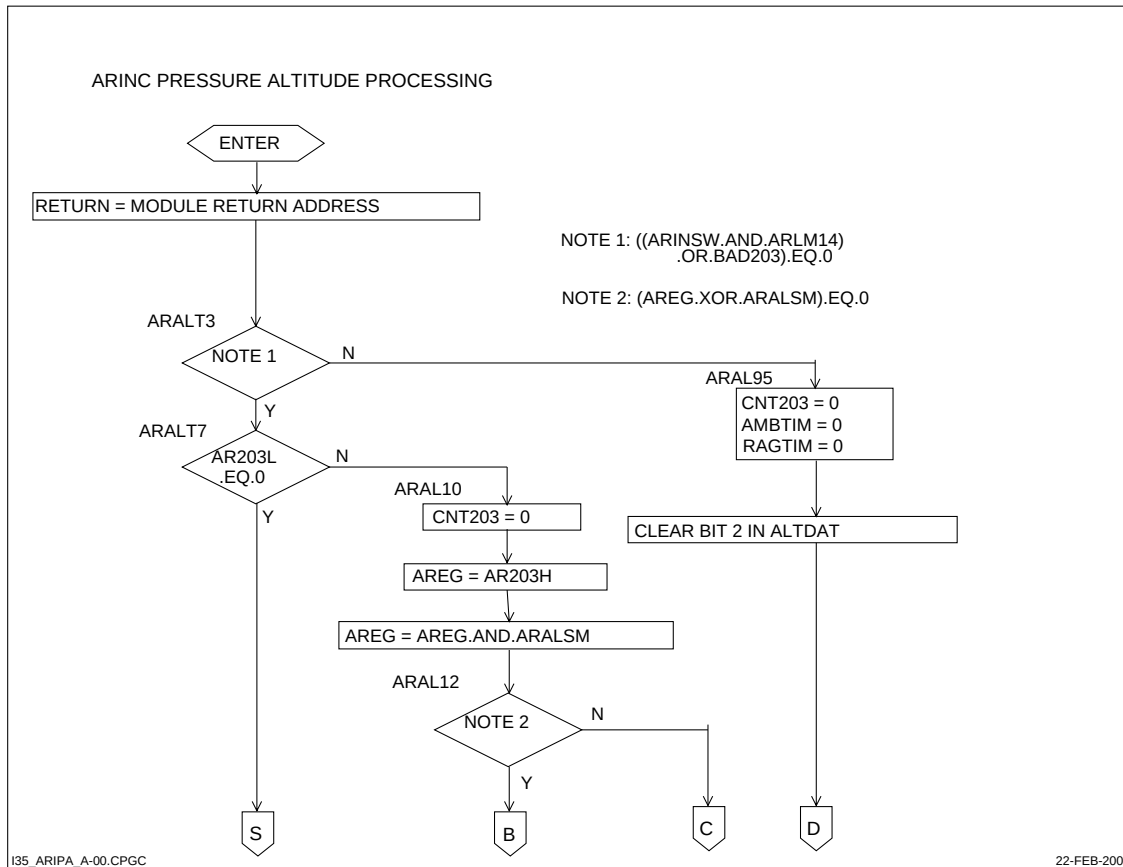
13.00 Figure 418: I34 _TATSEL _G



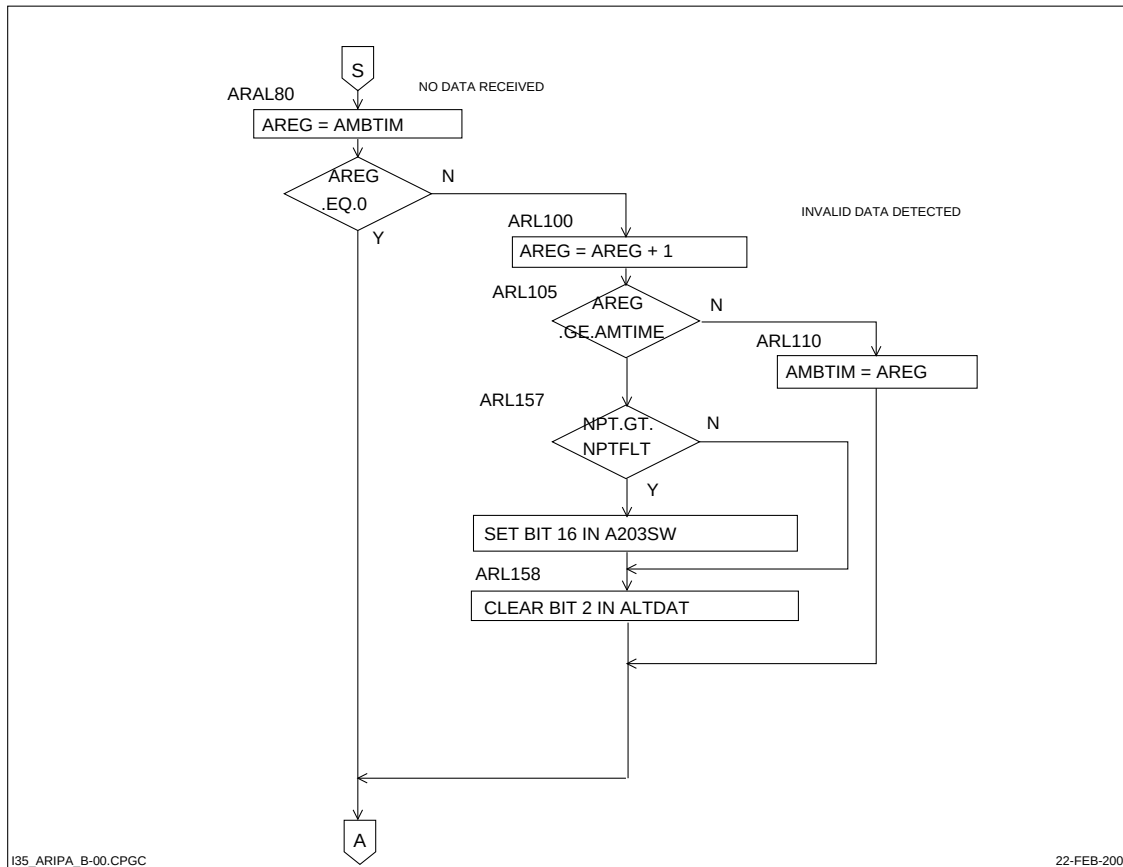
I34 _TATSEL _G-00.CPGC

22-FEB-2007

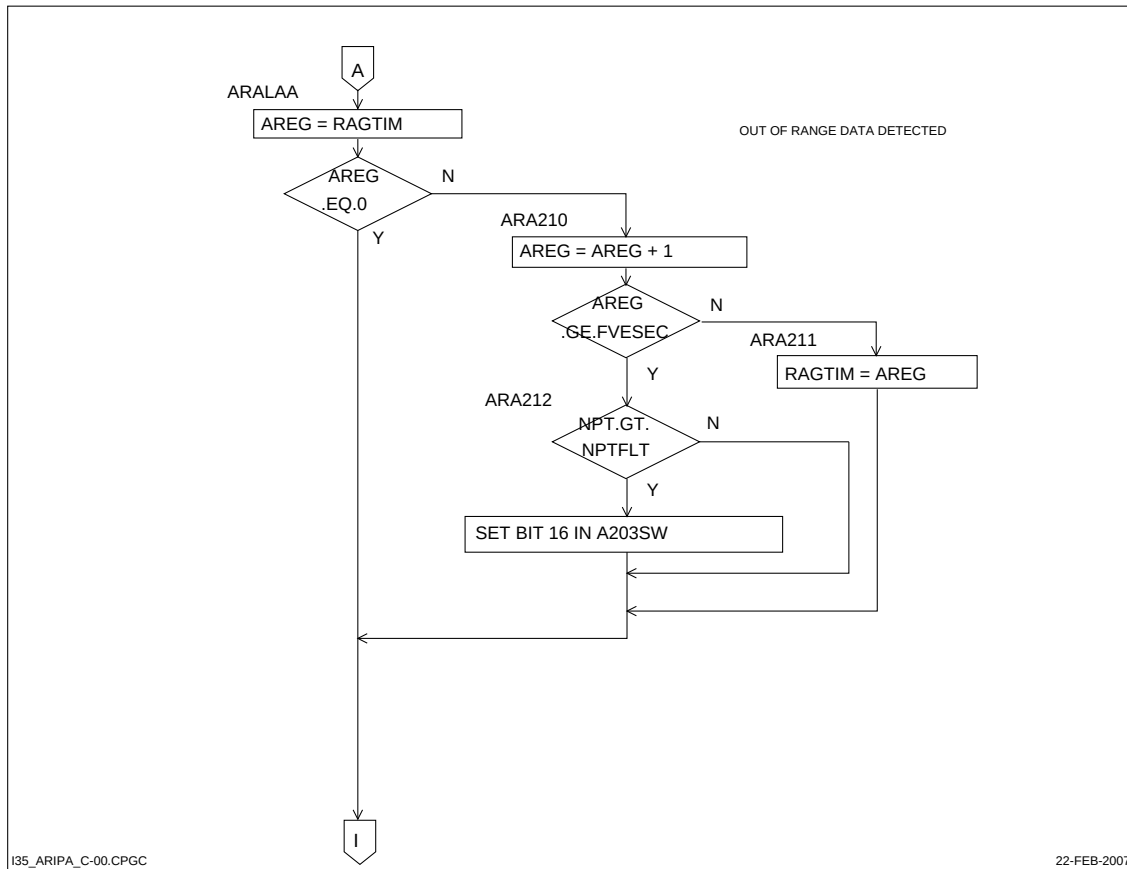
13.00 Figure 419: I35_ARIPA_A



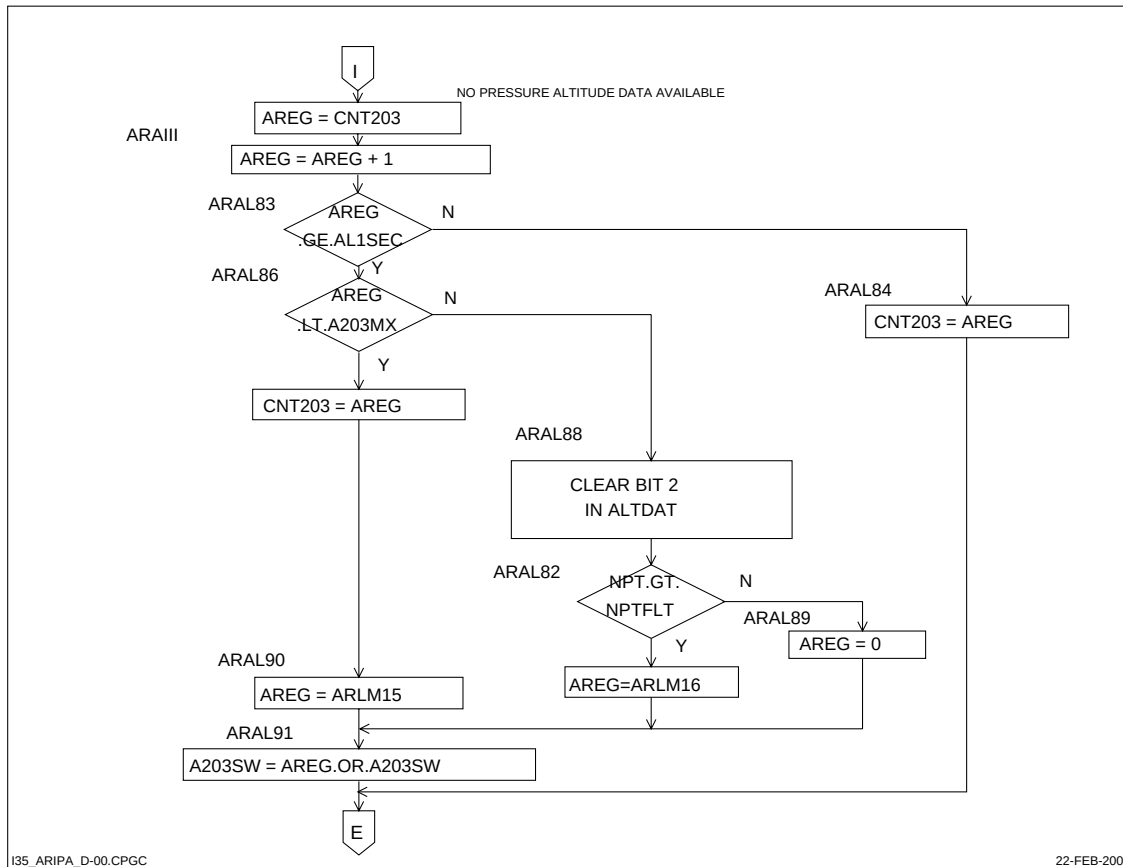
13.00 Figure 420: I35_ARIPA_B



13.00 Figure 421: I35_ARIPA_C



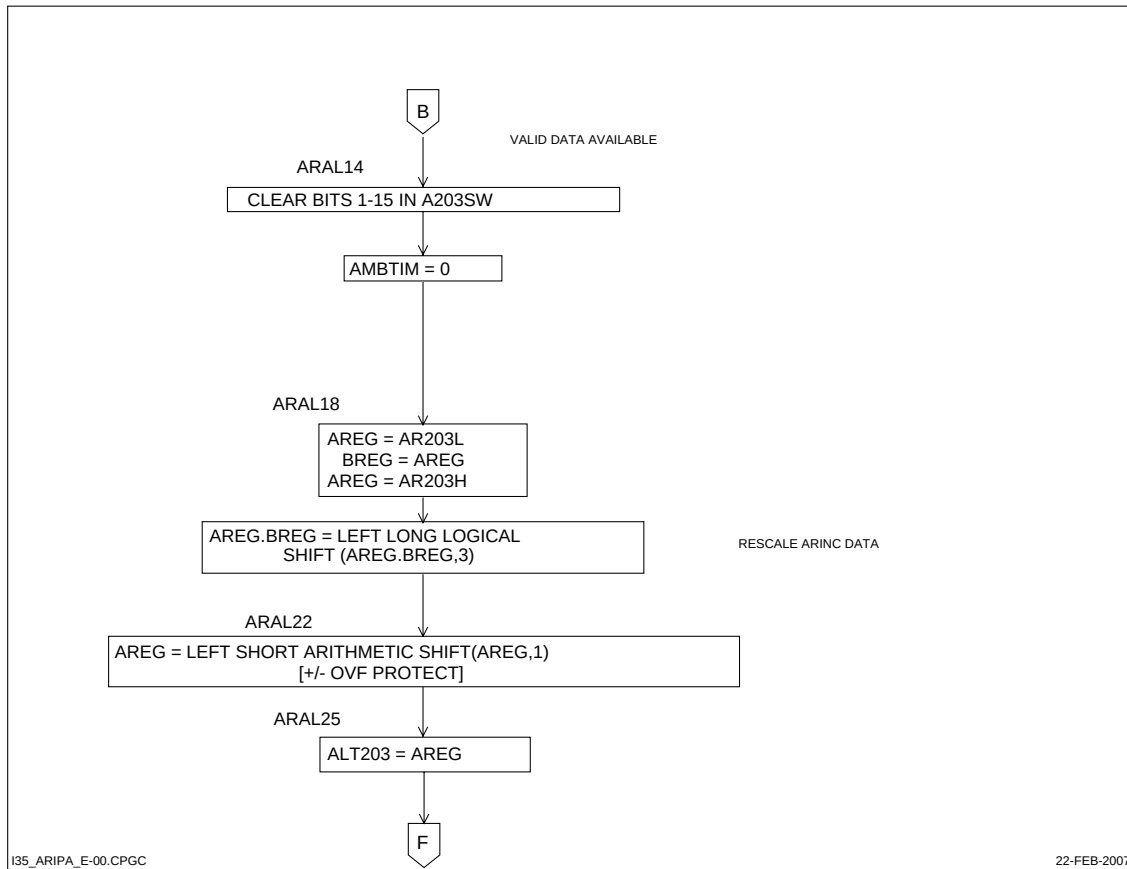
13.00 Figure 422: I35_ARIPA_D



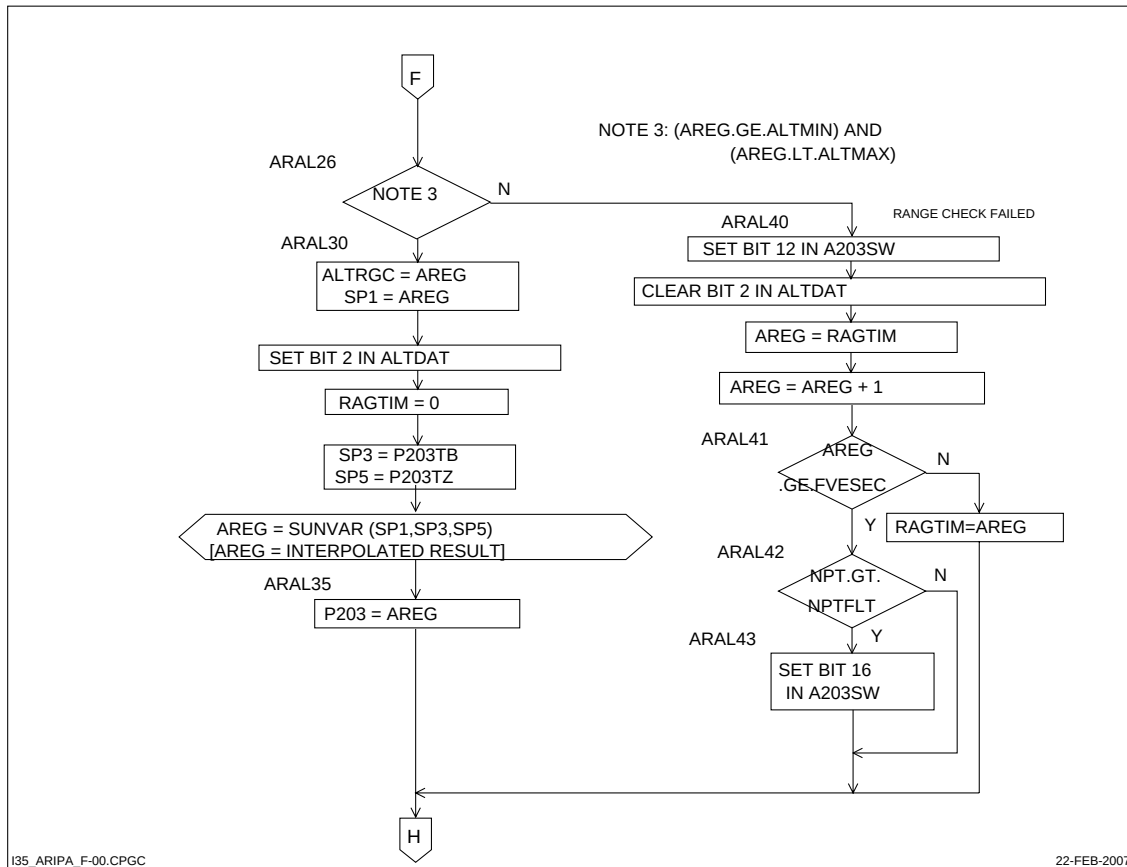
I35_ARIPA_D-00.CPGC

22-FEB-2007

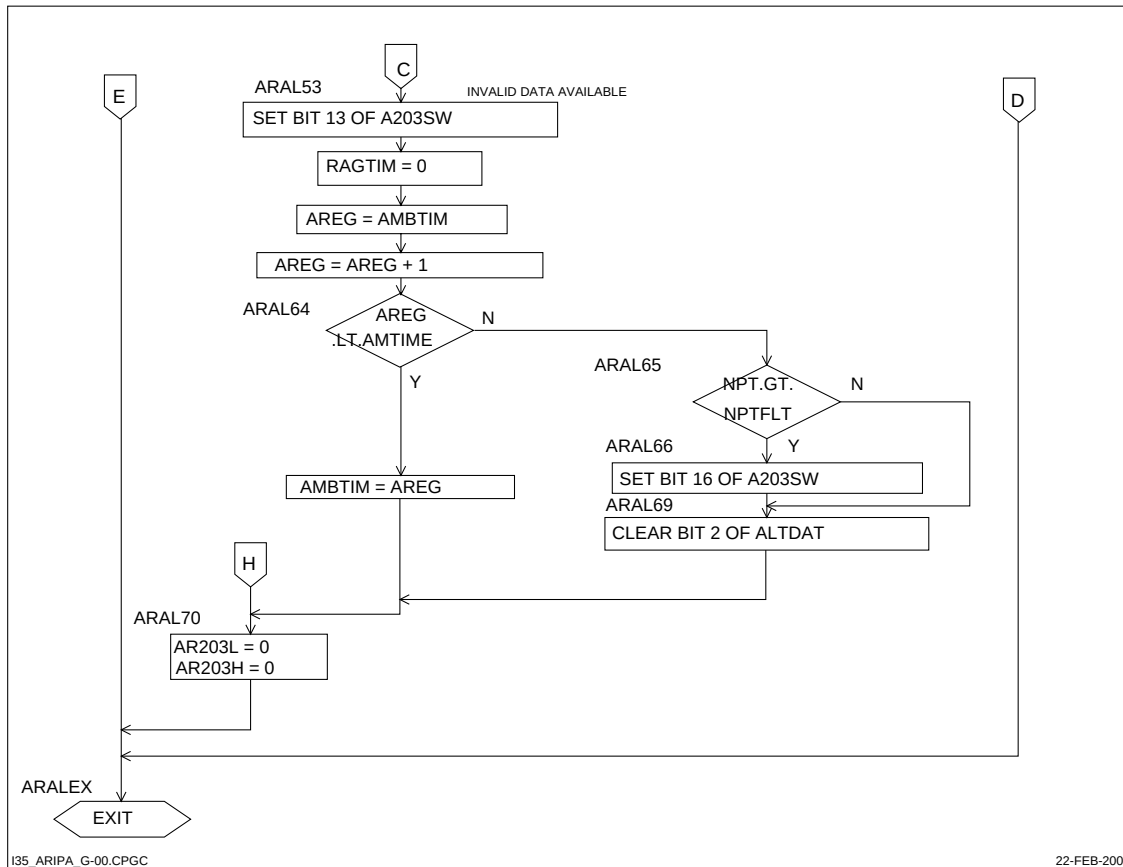
13.00 Figure 423: I35 _ARIPA _E



13.00 Figure 424: I35_ARIPA_F



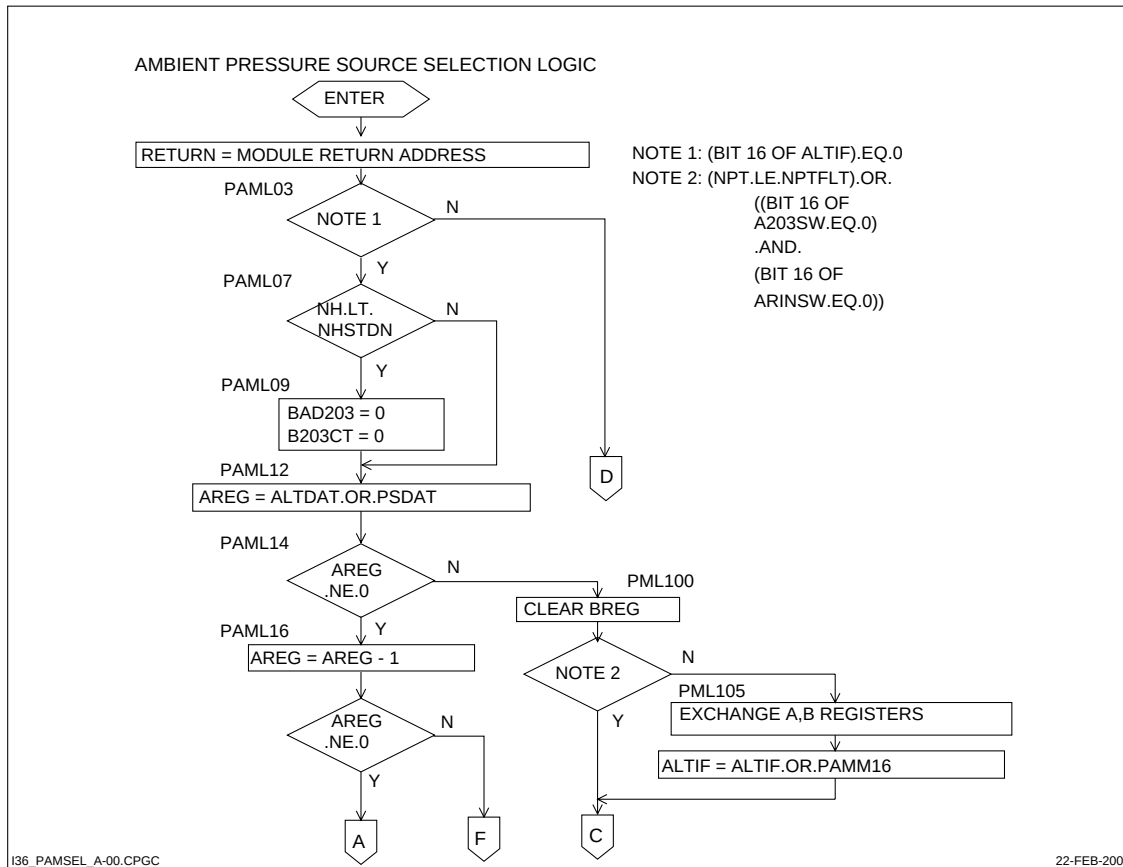
13.00 Figure 425: I35_ARIPA_G



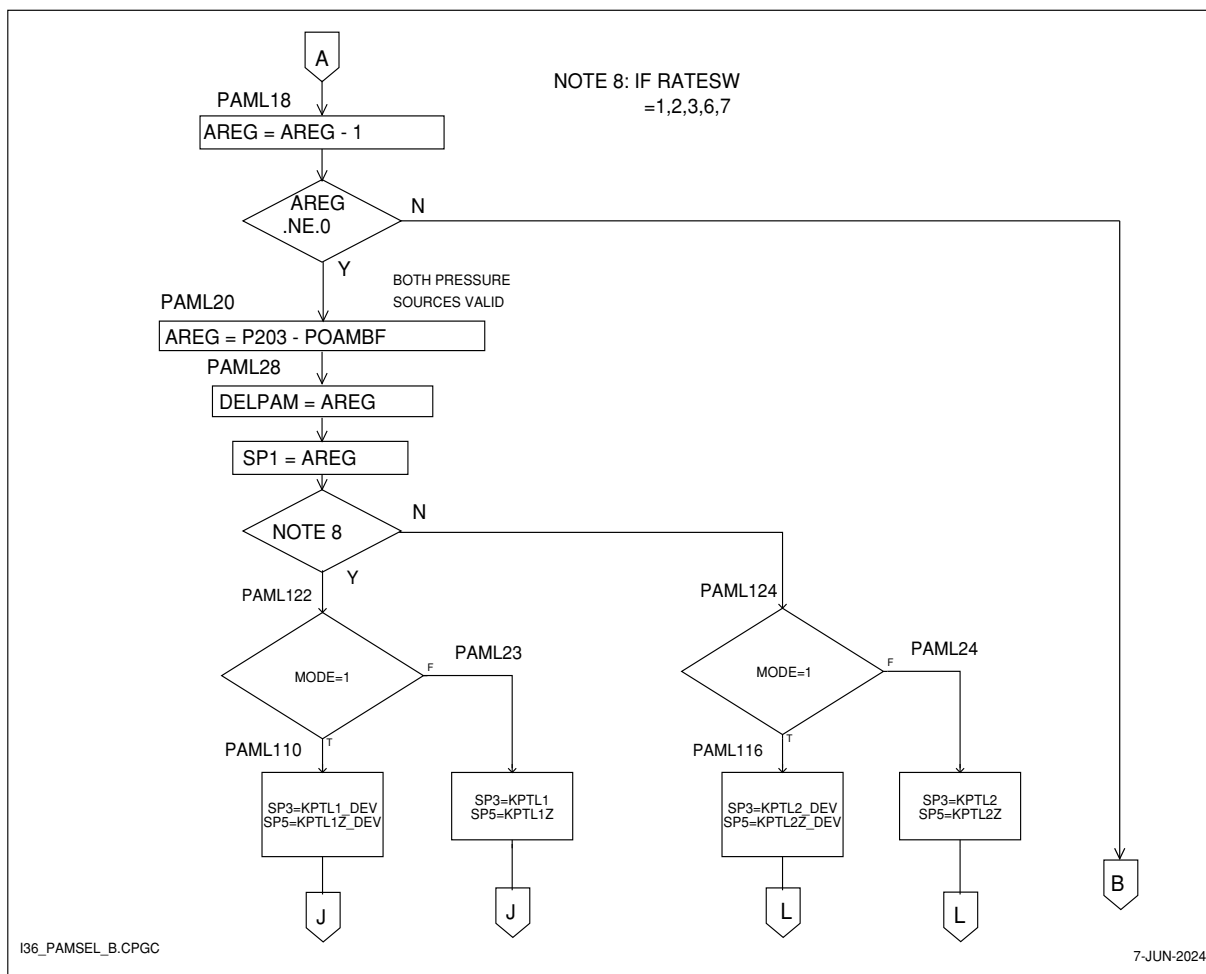
I35_ARIPA_G-00.CPGC

22-FEB-2007

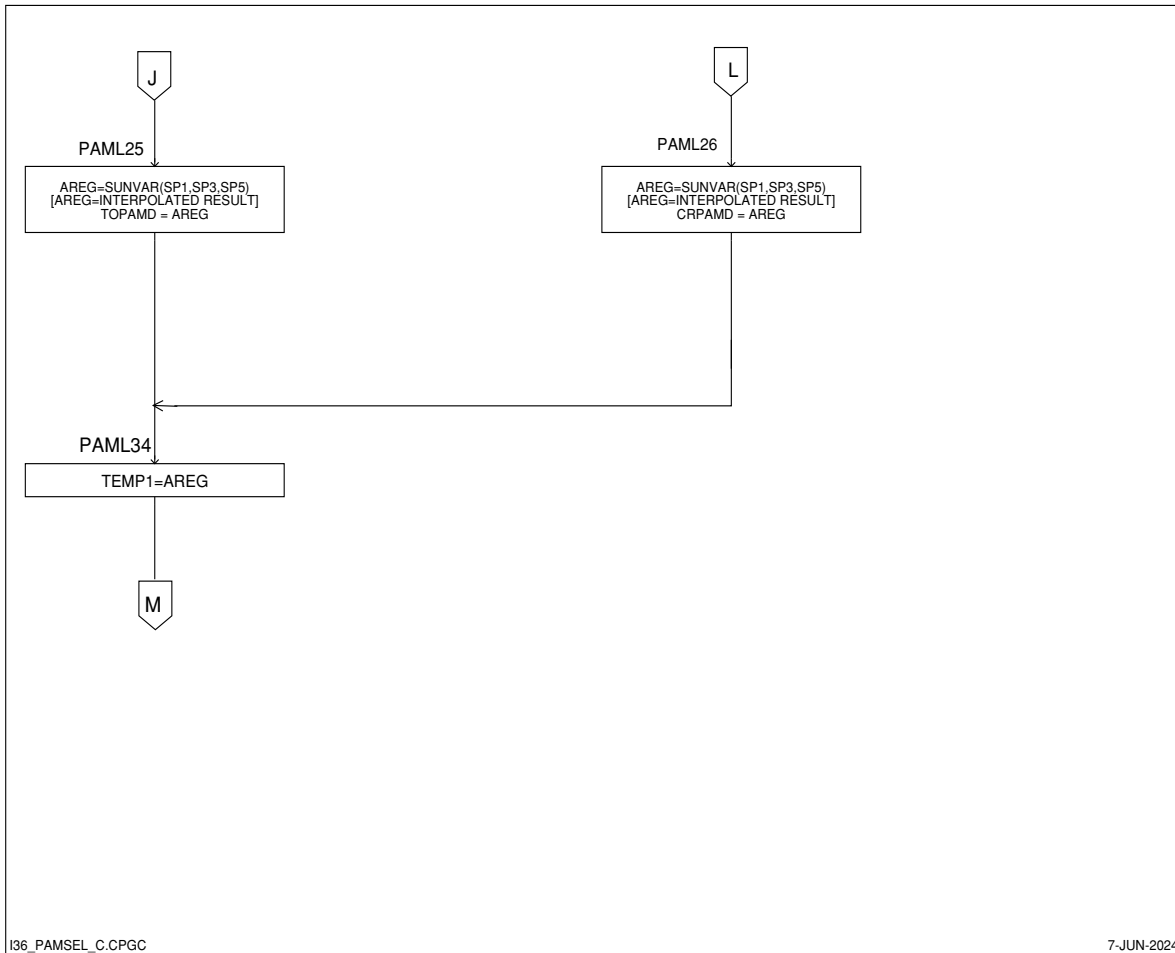
13.00 Figure 426: I36 _PAMSEL _A



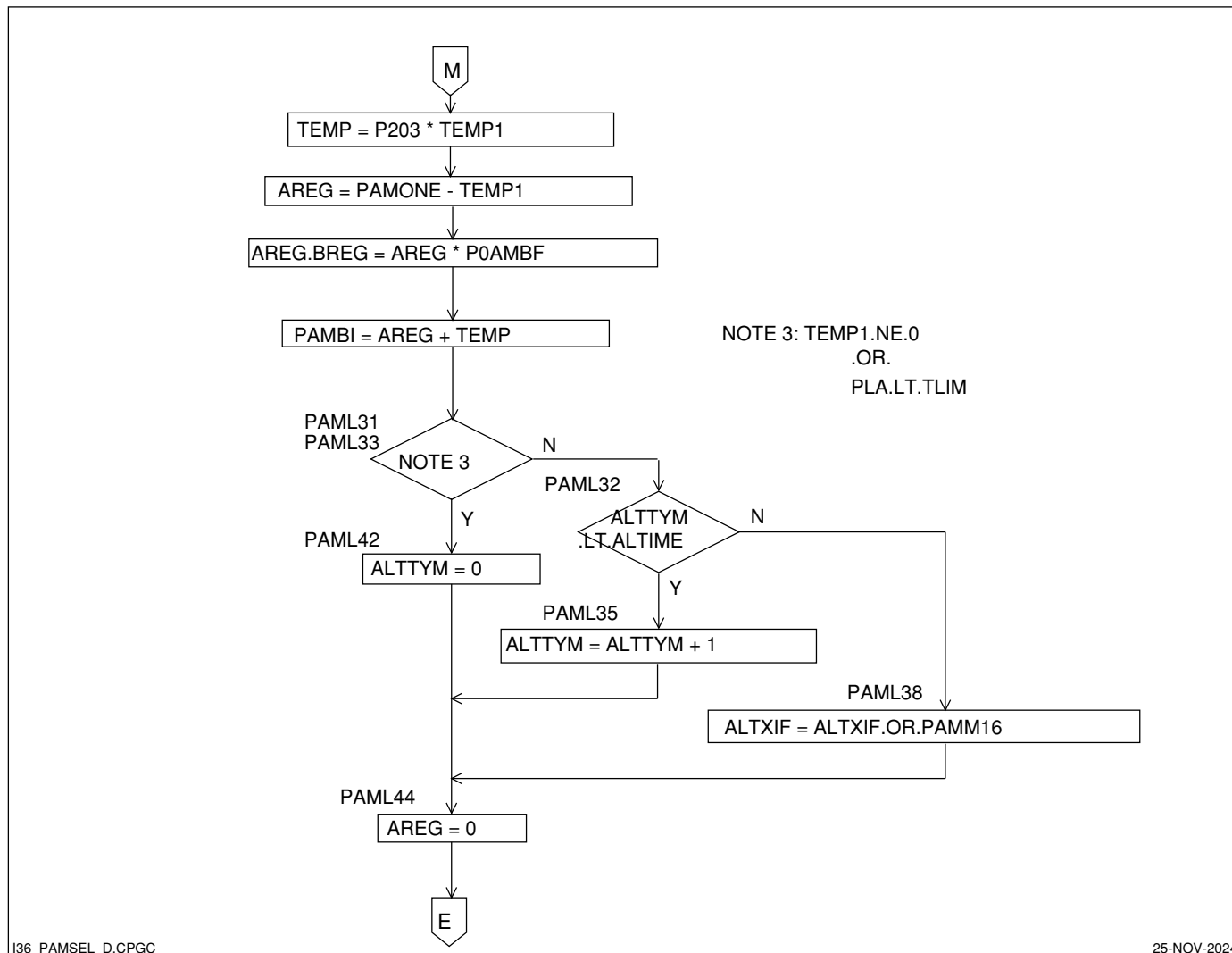
13.00 Figure 427: I36_PAMSEL_B



13.00 Figure 428: I36 _PAMSEL _C



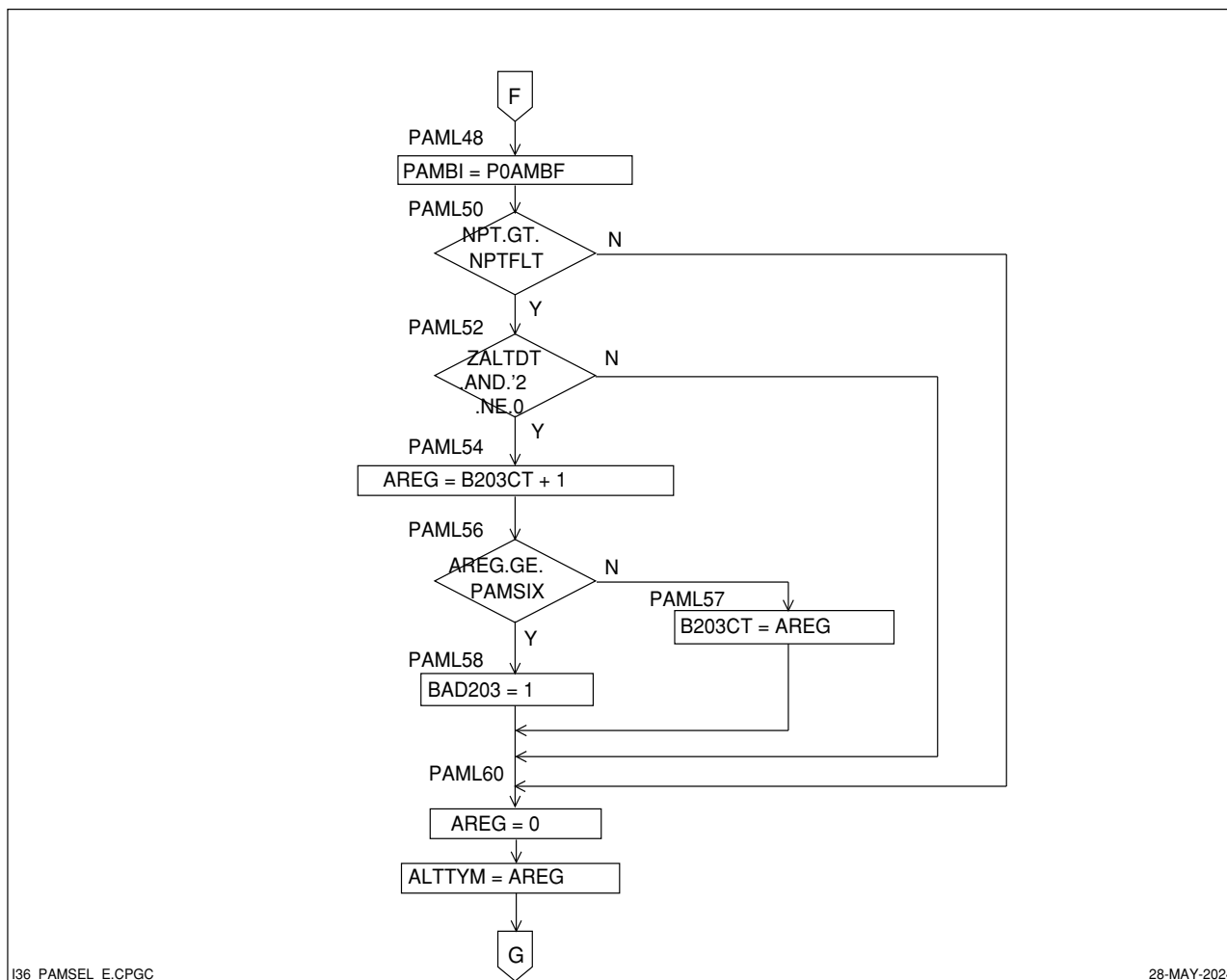
13.00 Figure 429: I36 _PAMSEL _D



I36 _PAMSEL _D.CPGC

25-NOV-2024

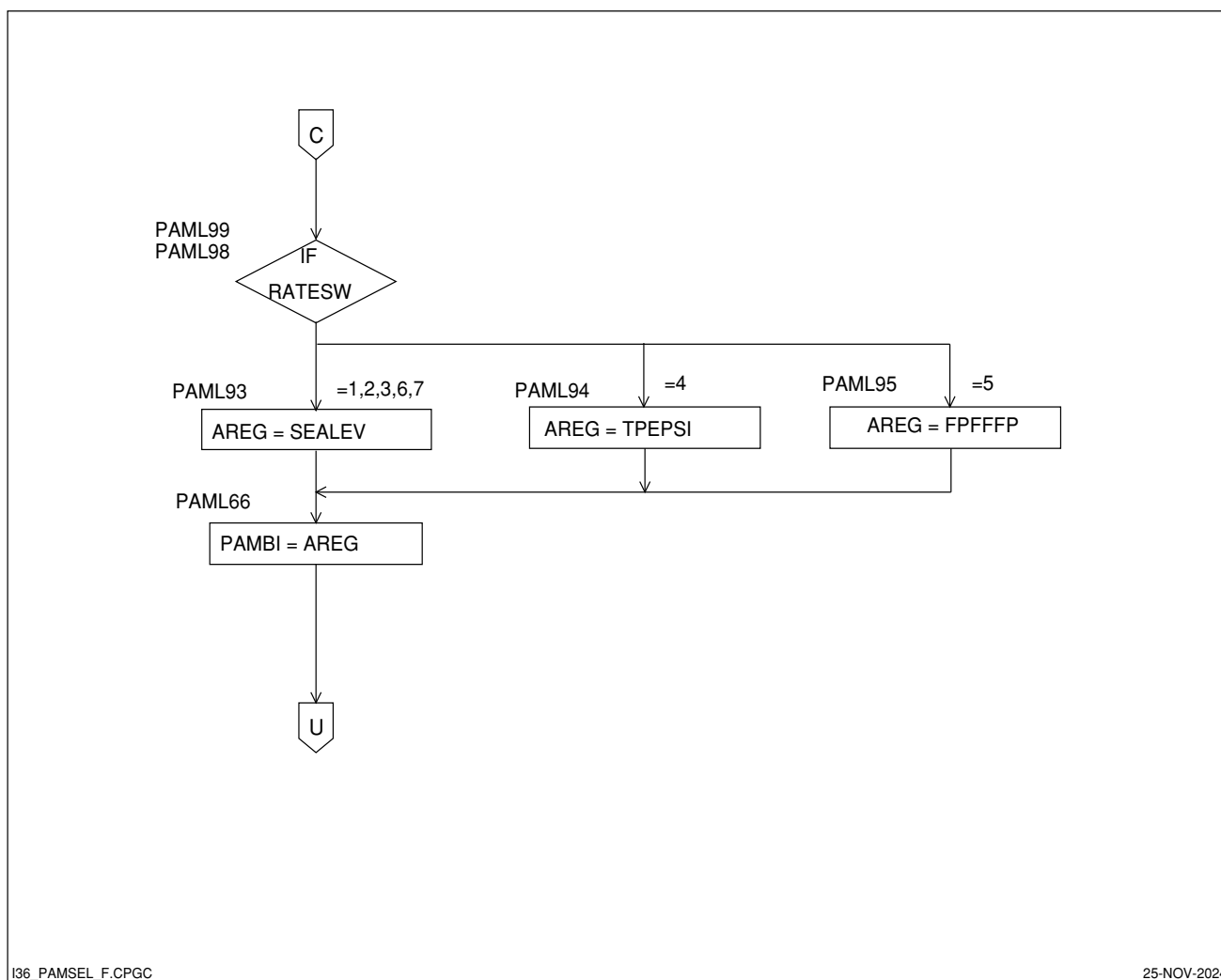
13.00 Figure 430: I36 _PAMSEL _E



I36 _PAMSEL _E.CPGC

28-MAY-2024

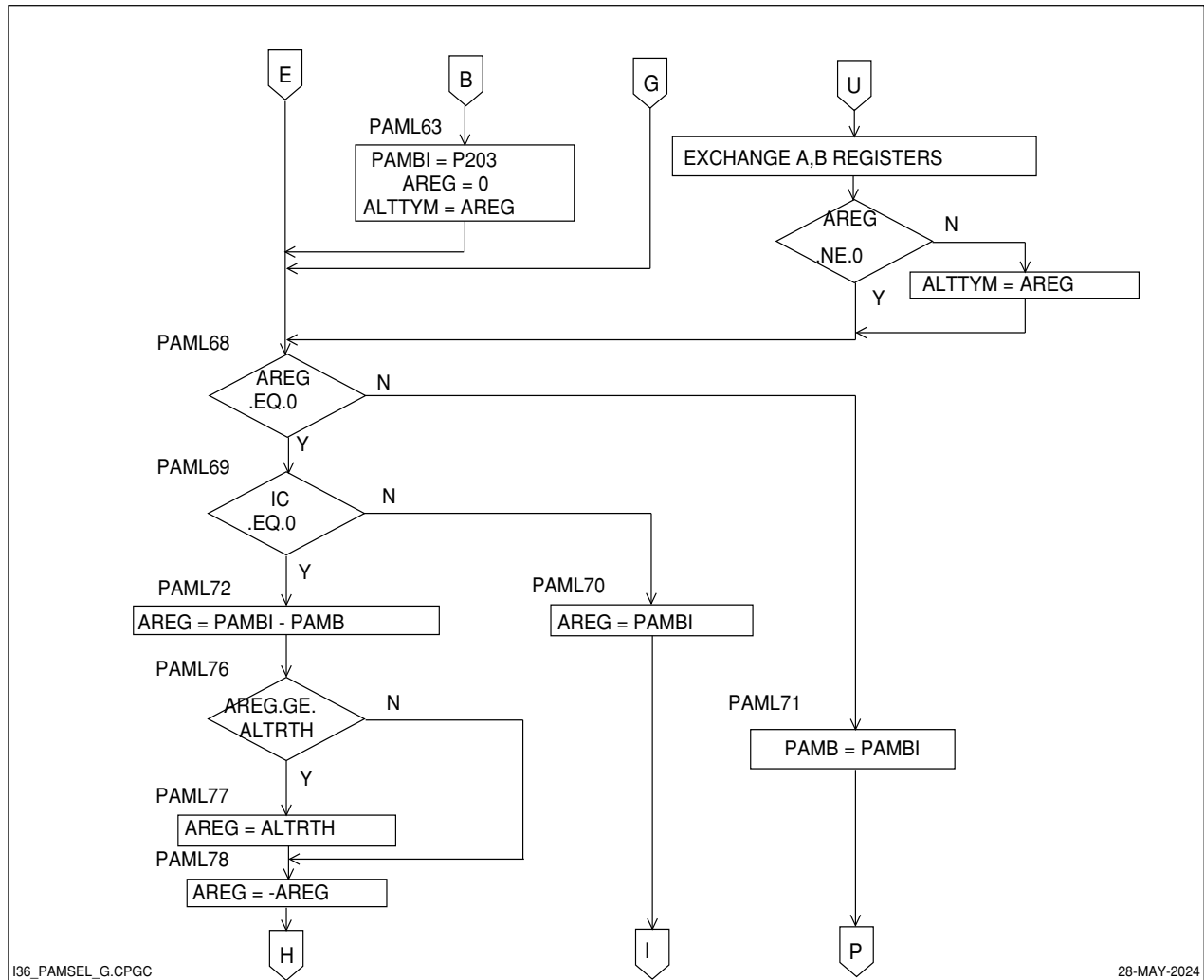
13.00 Figure 431: I36_PAMSEL_F



I36_PAMSEL_F.CPGC

25-NOV-2024

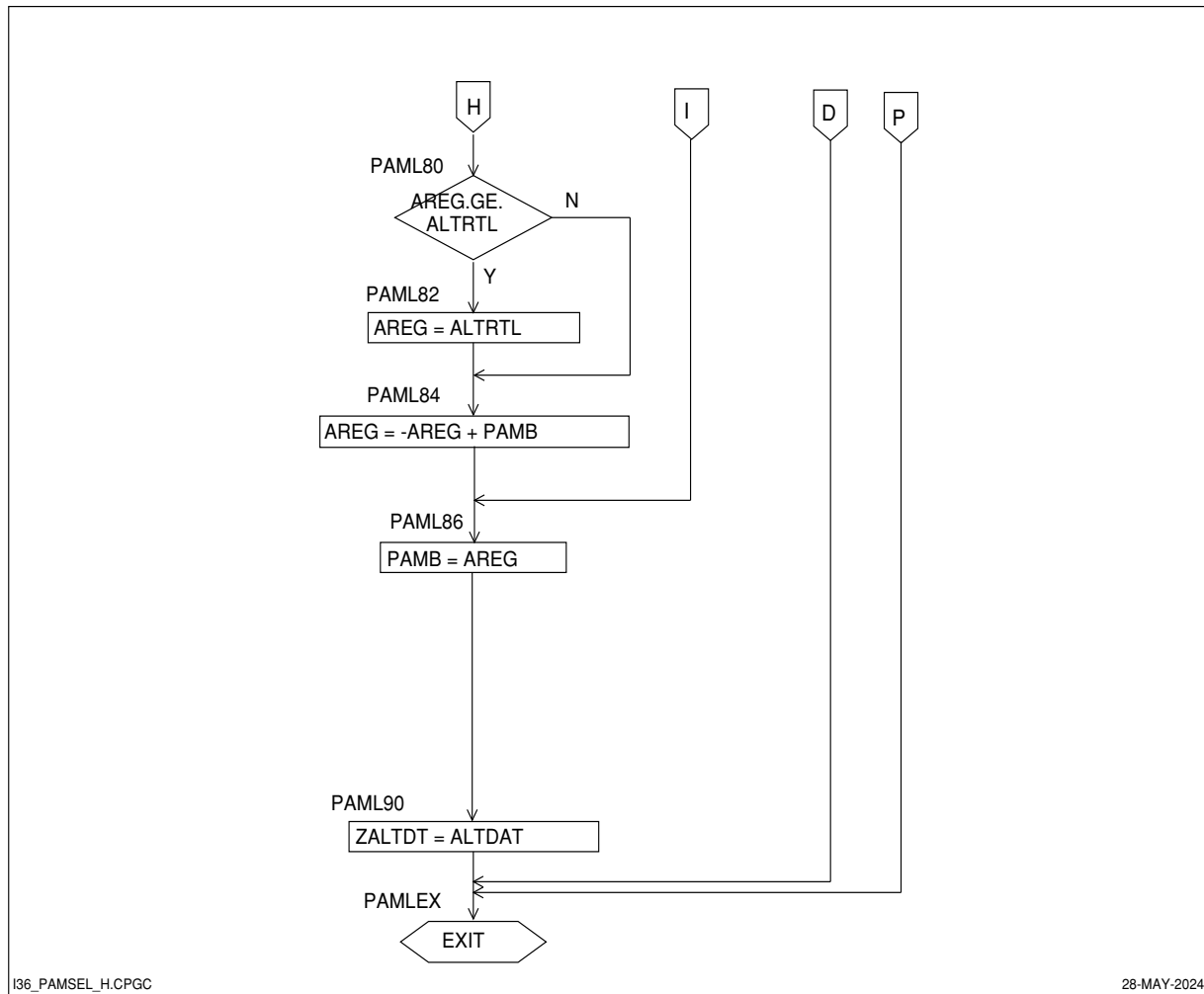
13.00 Figure 432: I36 _PAMSEL _G



I36 _PAMSEL _G.CPGC

28-MAY-2024

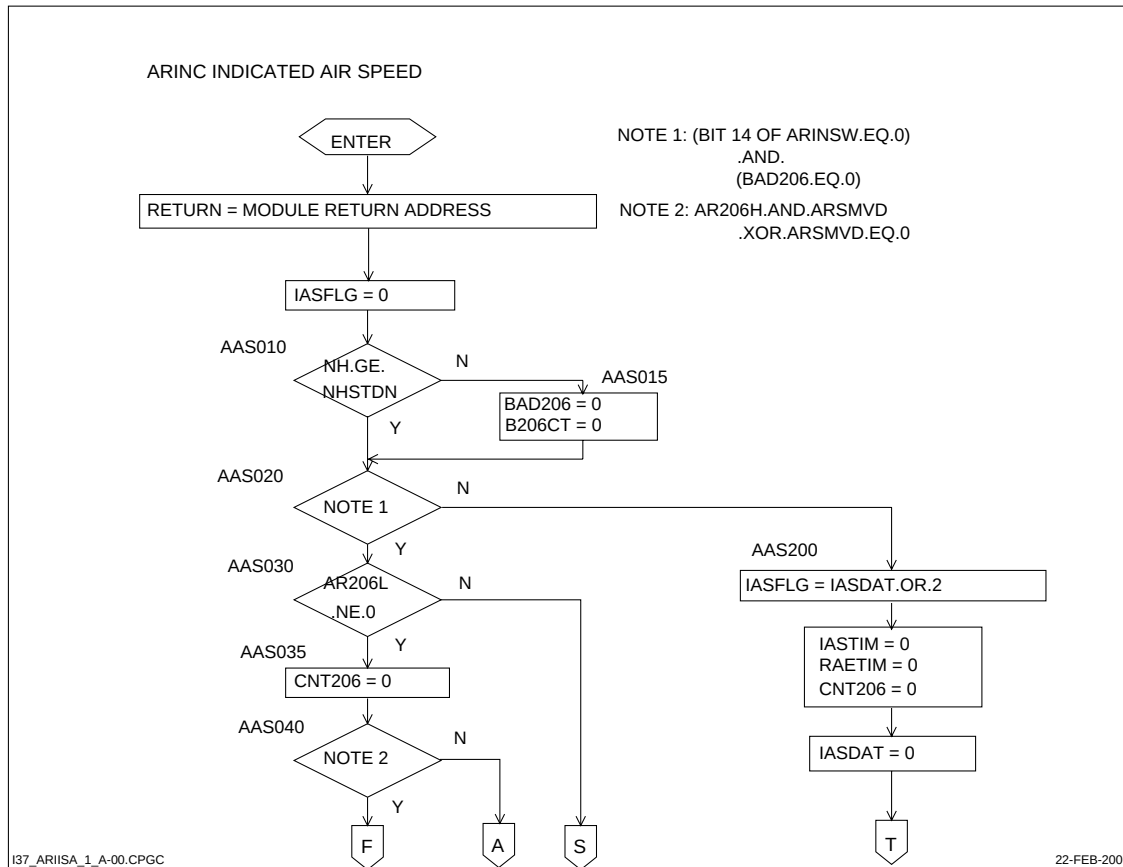
13.00 Figure 433: I36 _PAMSEL _H



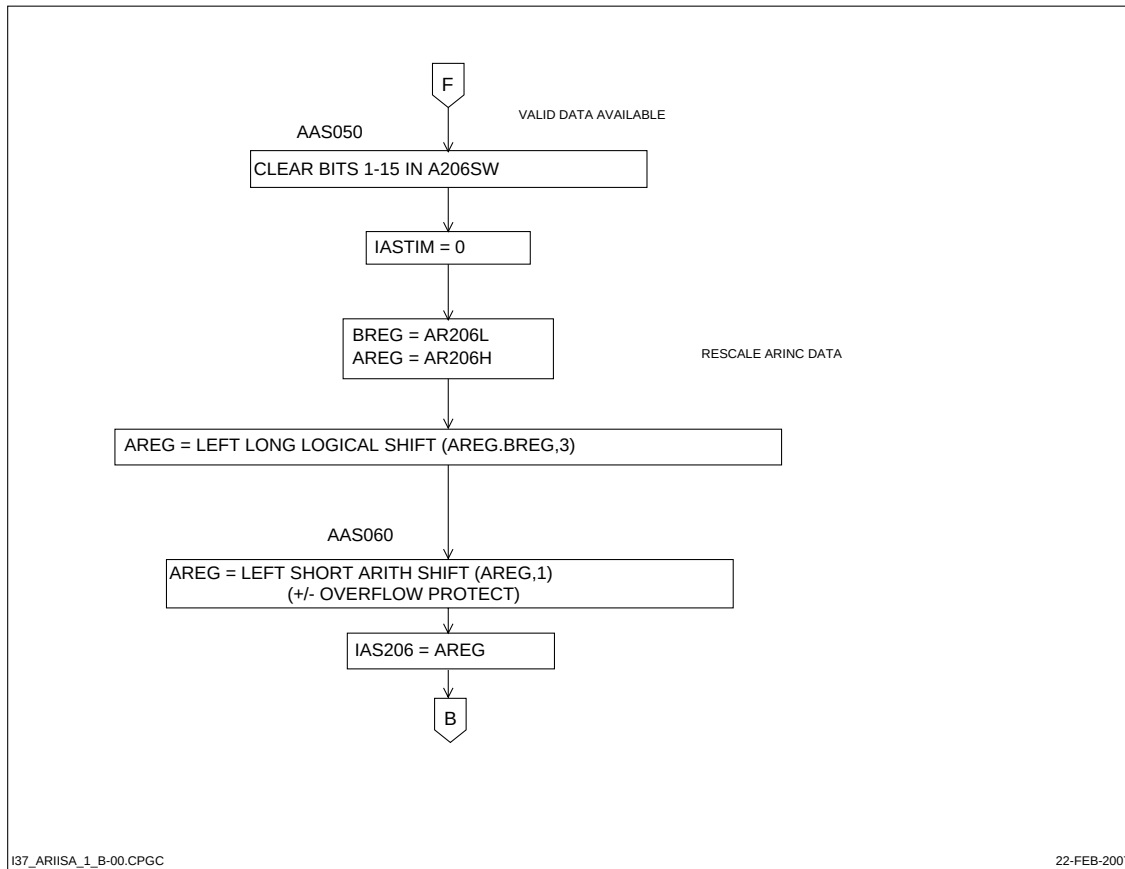
I36_PAMSEL_H.CPGC

28-MAY-2024

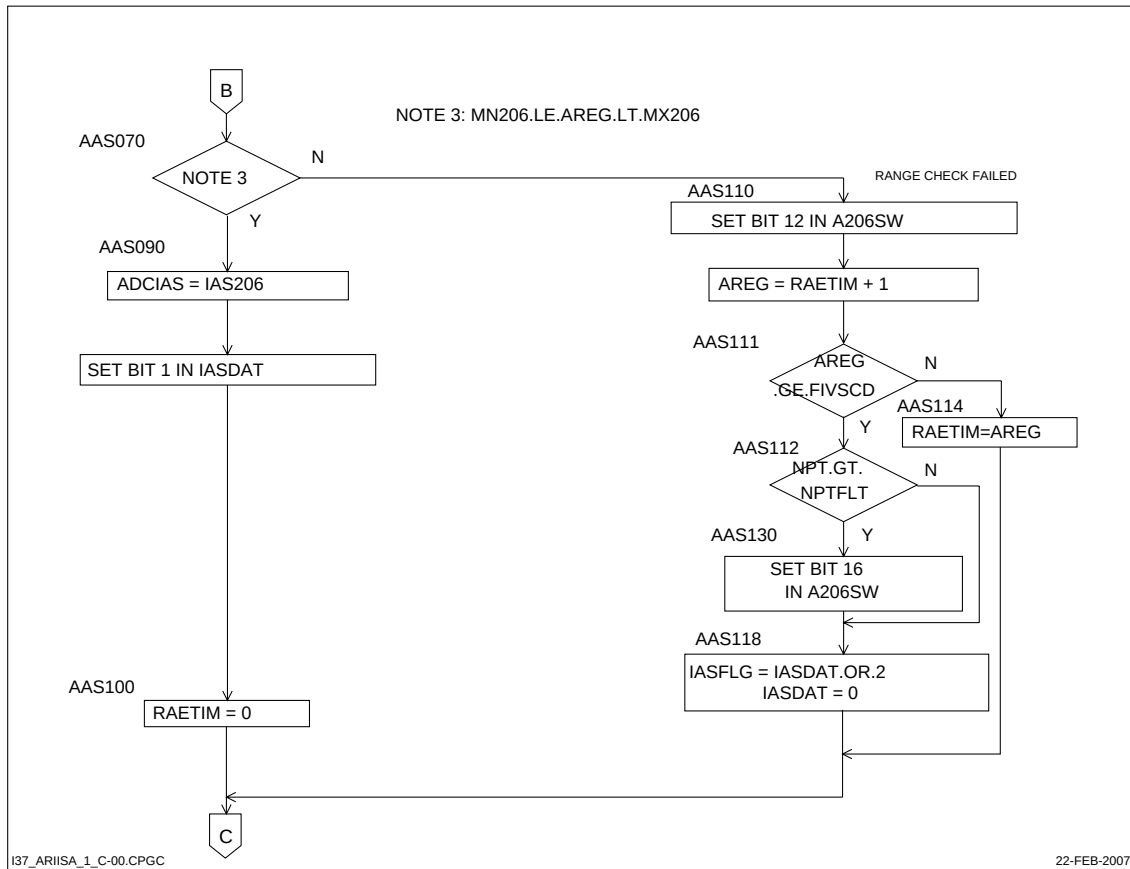
13.00 Figure 434: I37 _ARIISA_1_A



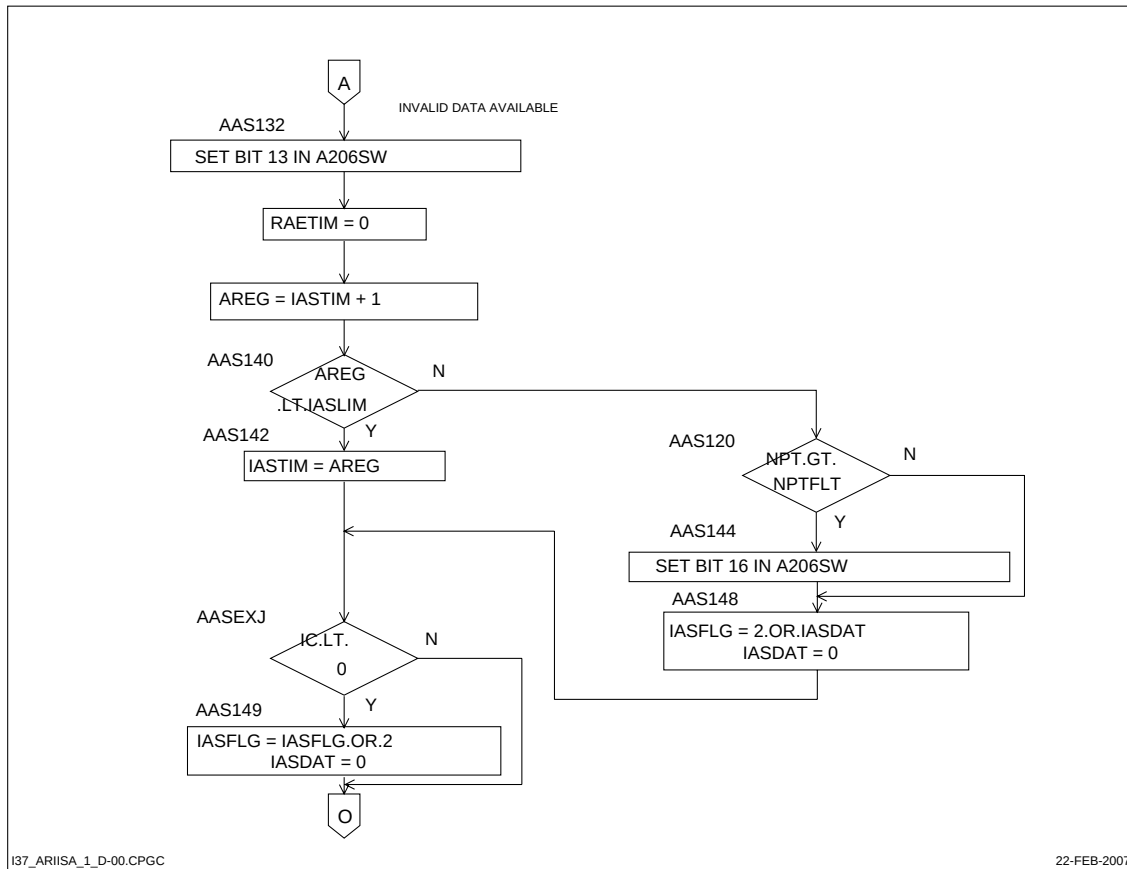
13.00 Figure 435: I37 _ARIISA _1 _B



13.00 Figure 436: I37 _ARIISA _1 _C



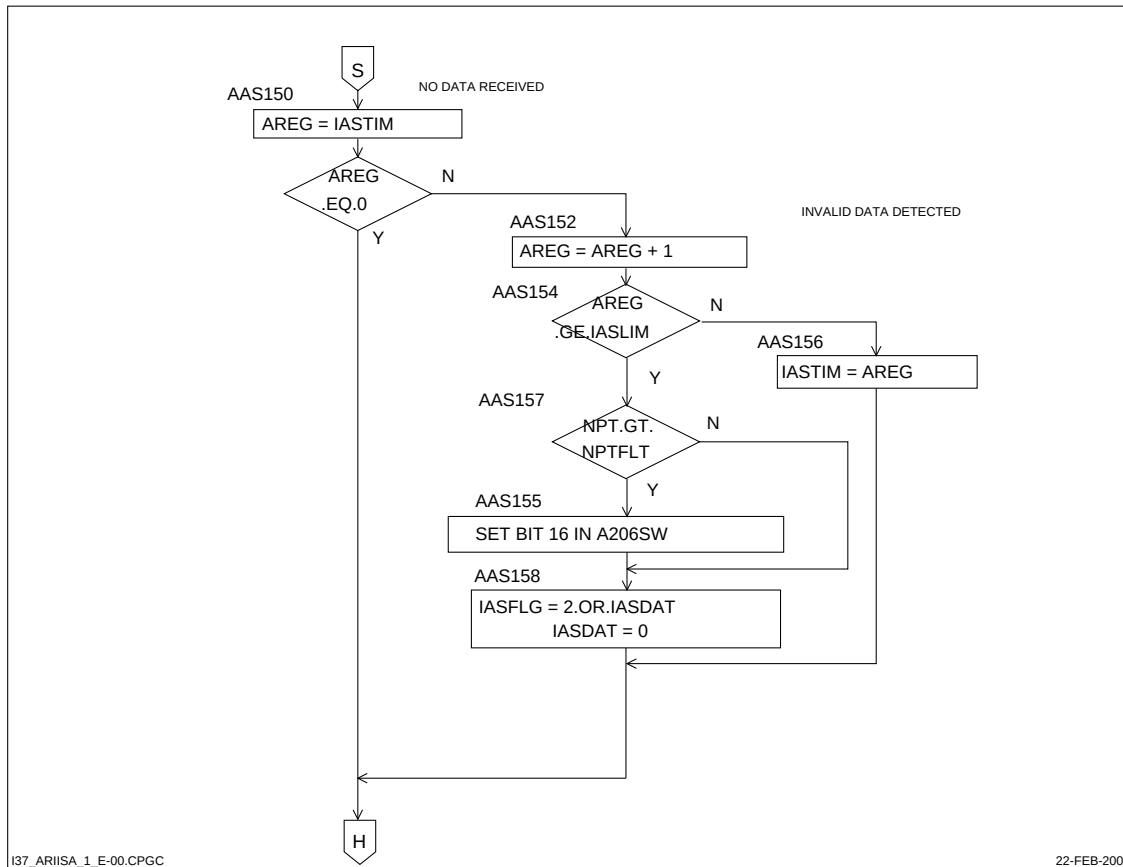
13.00 Figure 437: I37 _ARIISA_1_D



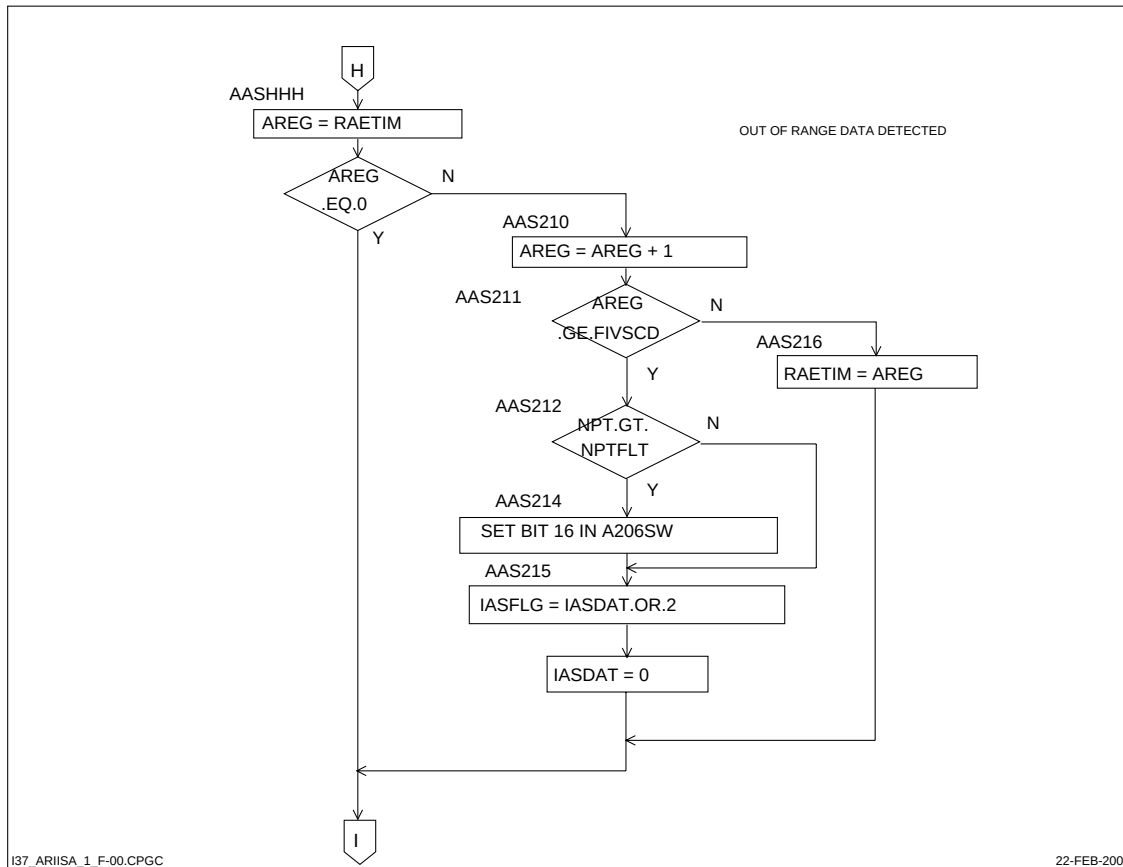
I37 _ARIISA_1_D-00.CPGC

22-FEB-2007

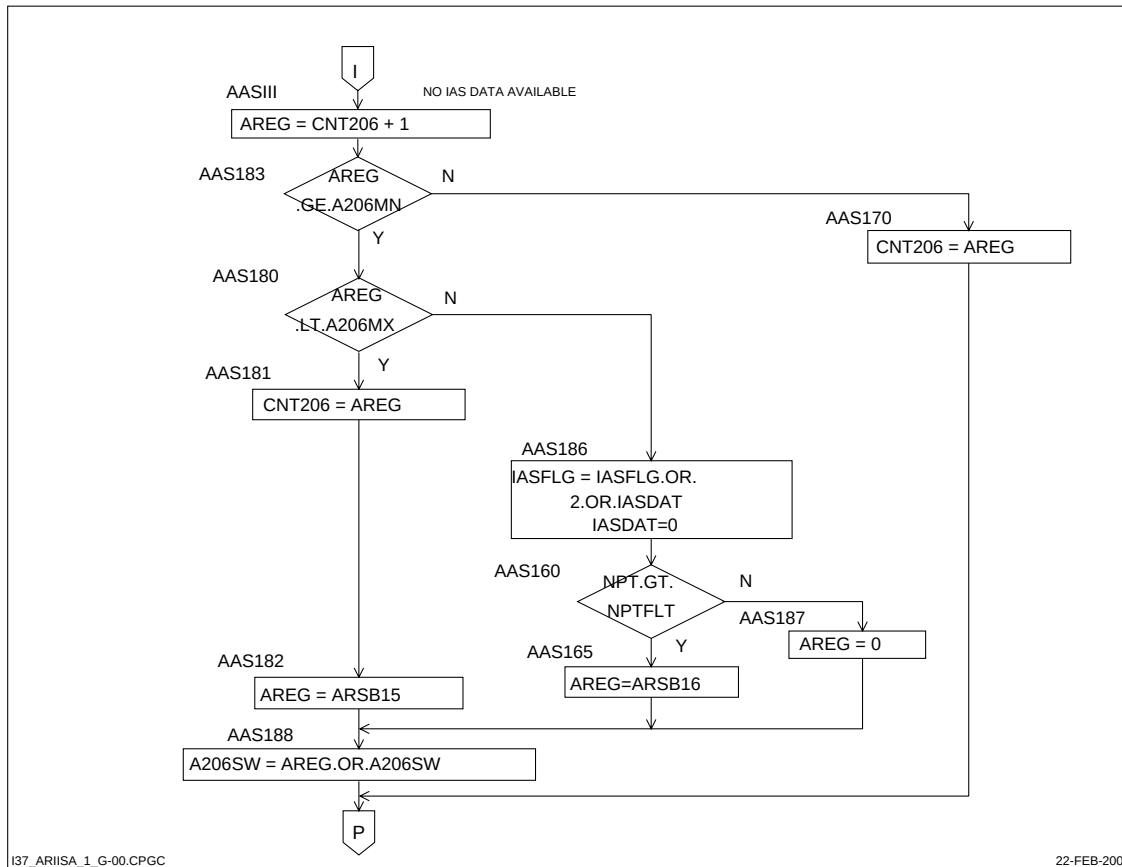
13.00 Figure 438: I37 _ARIISA _1 _E



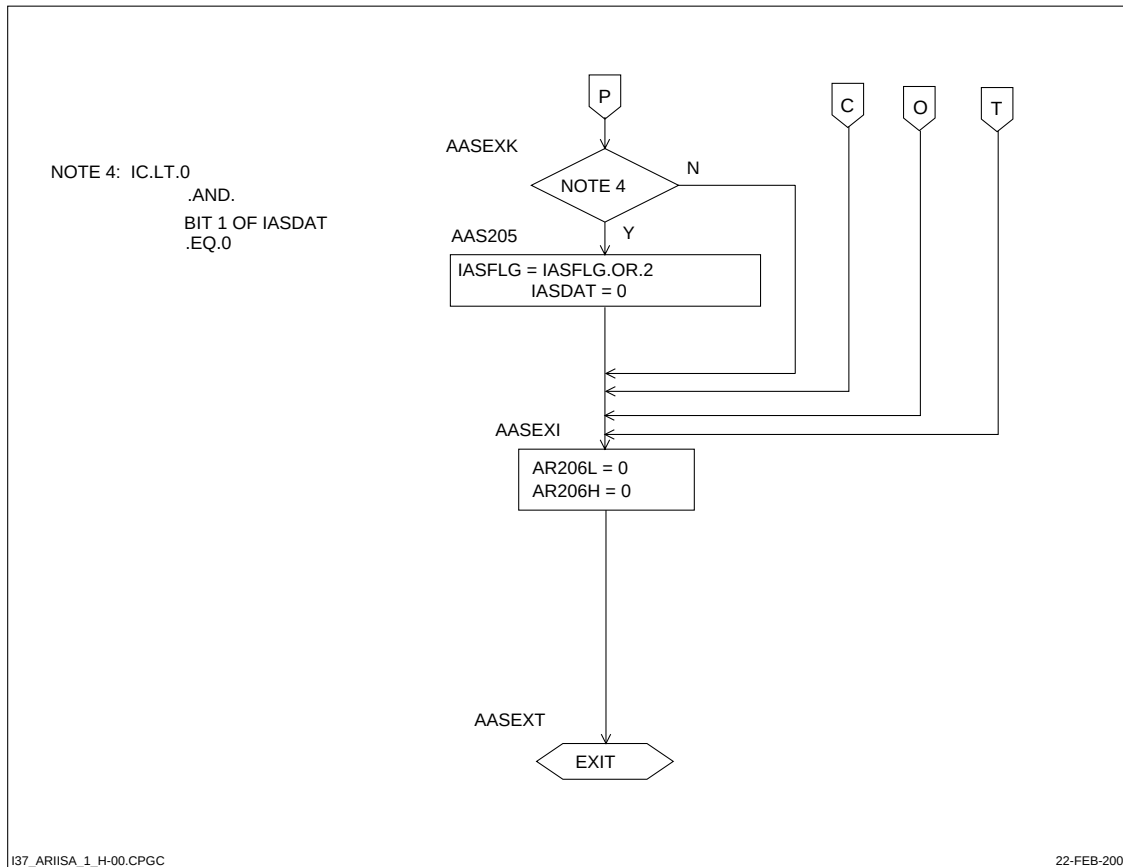
13.00 Figure 439: I37 _ARIISA _1 _F



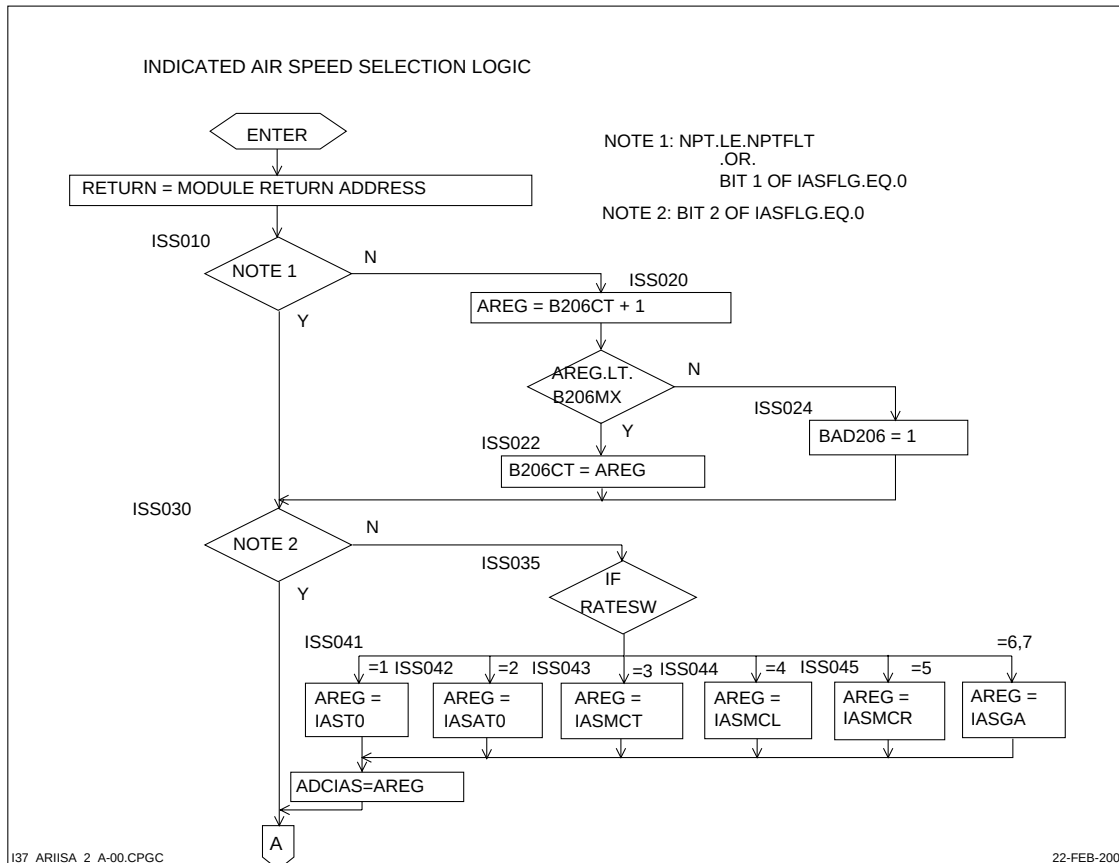
13.00 Figure 440: I37 _ARIISA_1 _G



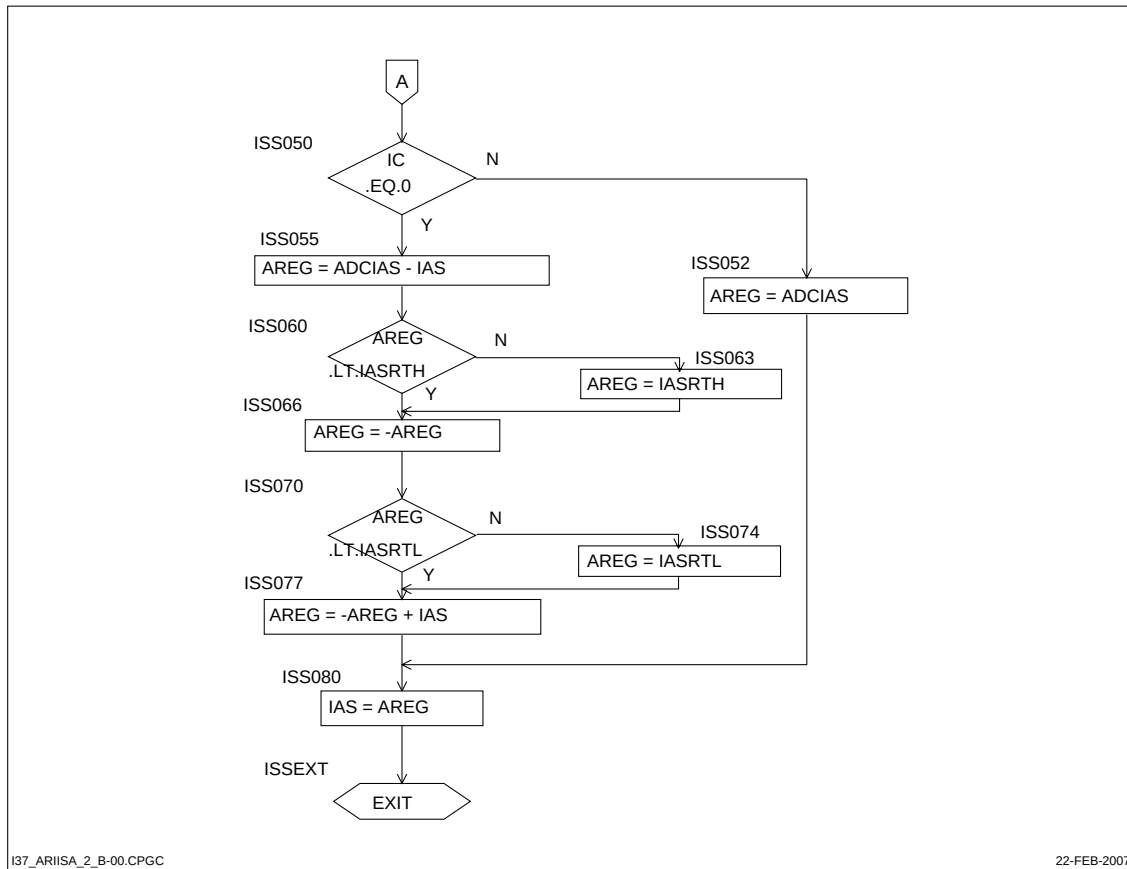
13.00 Figure 441: I37 _ARIISA _1 _H



13.00 Figure 442: I37 _ARIISA _2 _A



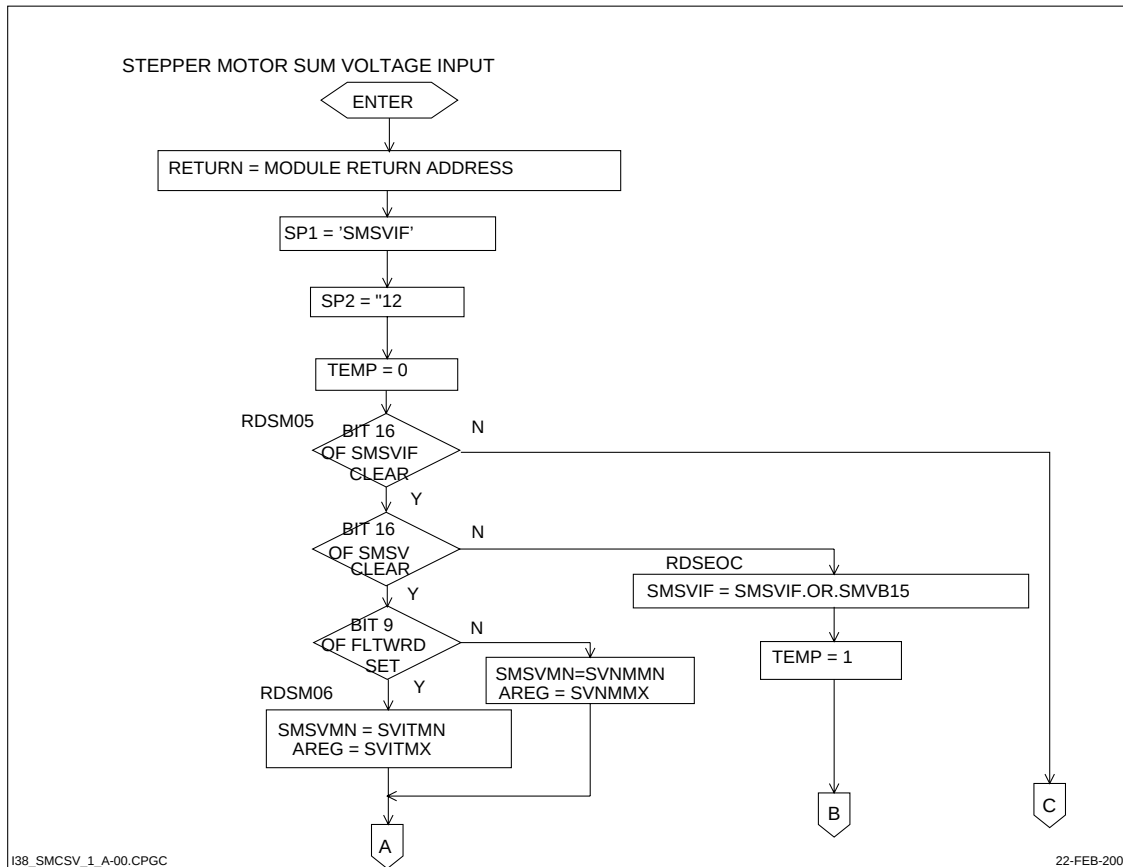
13.00 Figure 443: I37 _ARIISA _2 _B



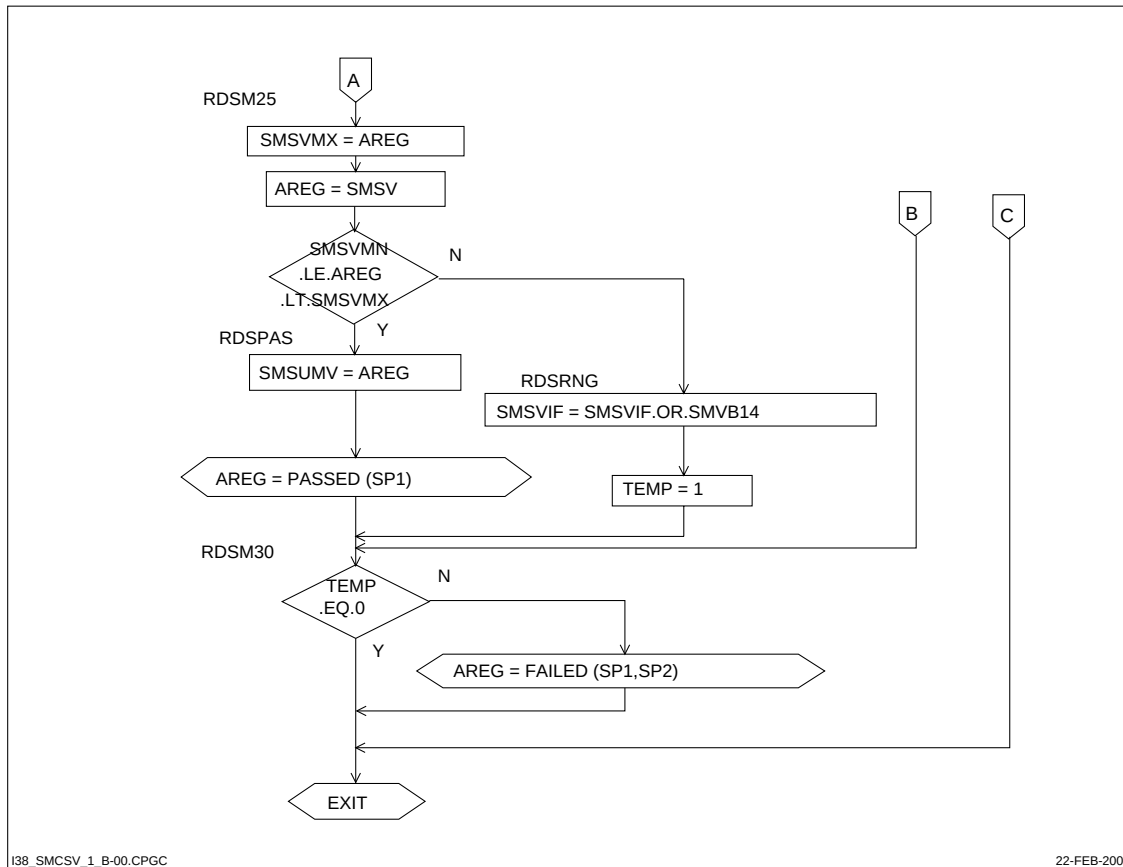
I37 _ARIISA _2 _B-00.CPGC

22-FEB-2007

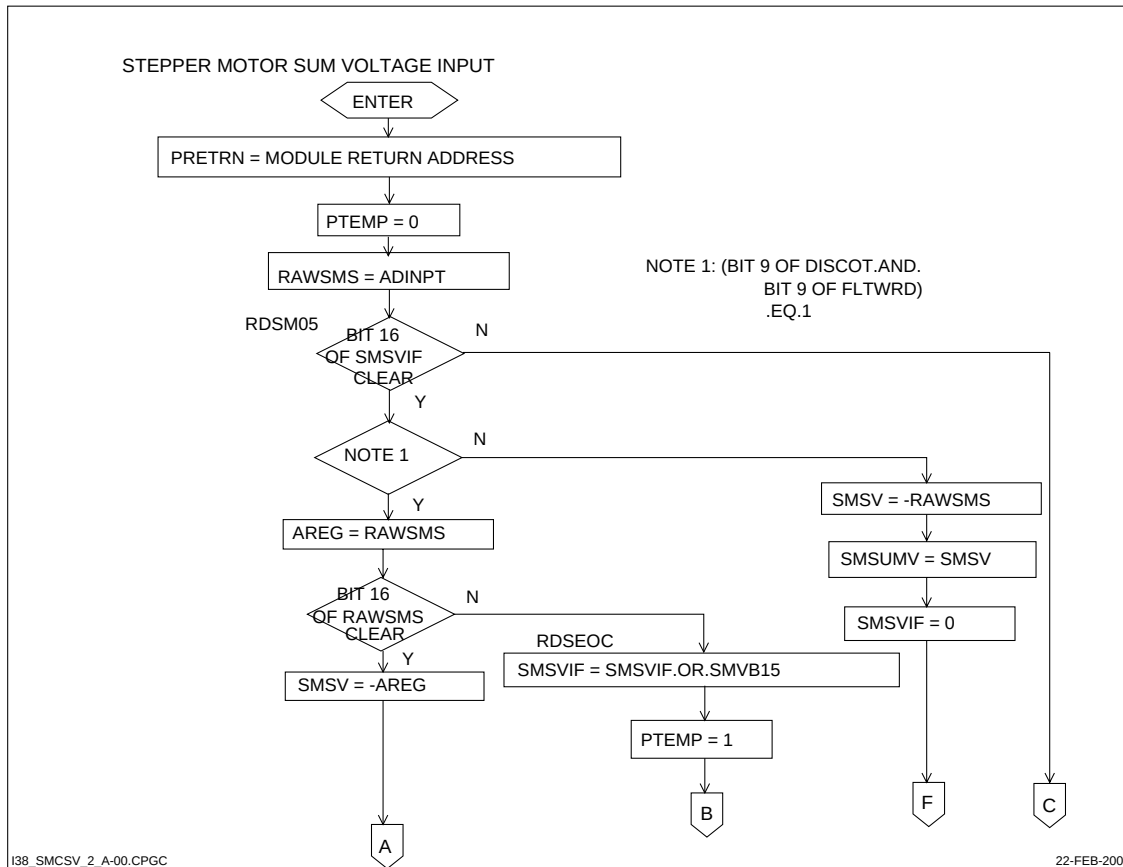
13.00 Figure 444: I38_SMCSV_1_A



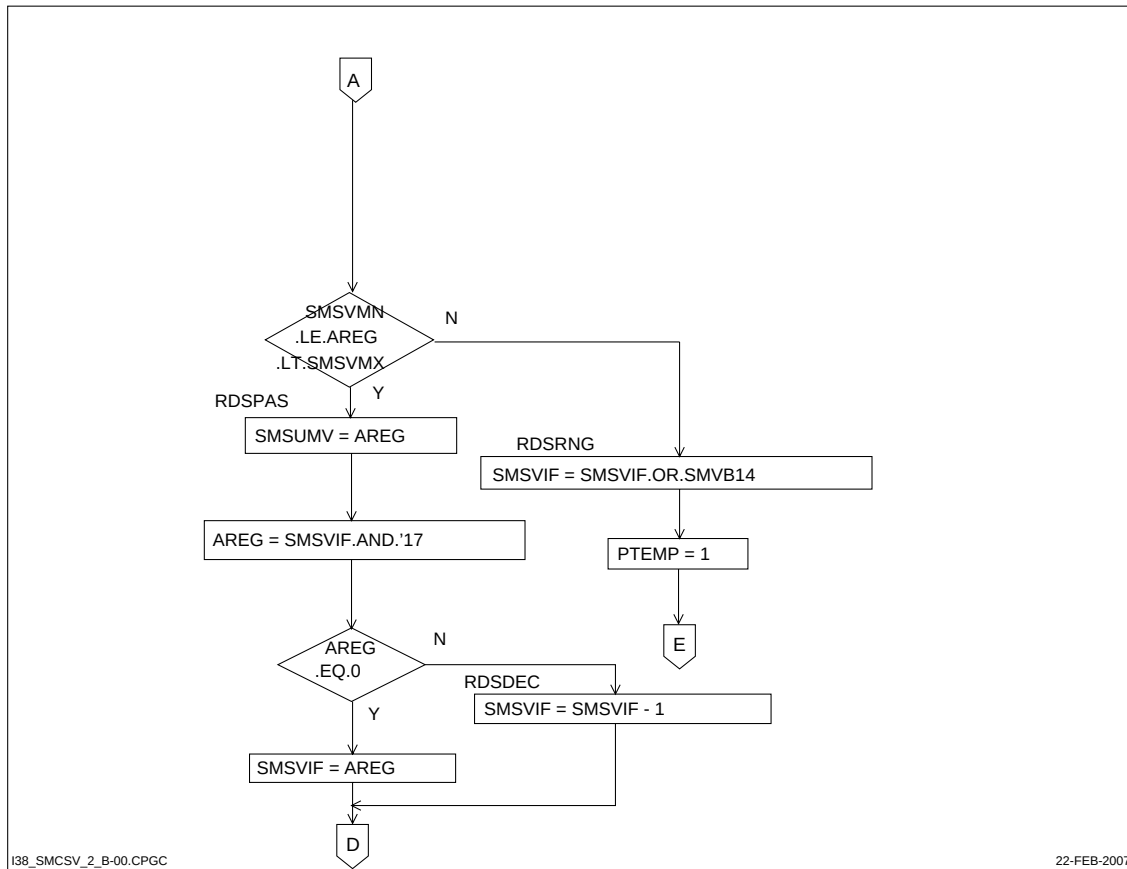
13.00 Figure 445: I38_SMCSV_1_B



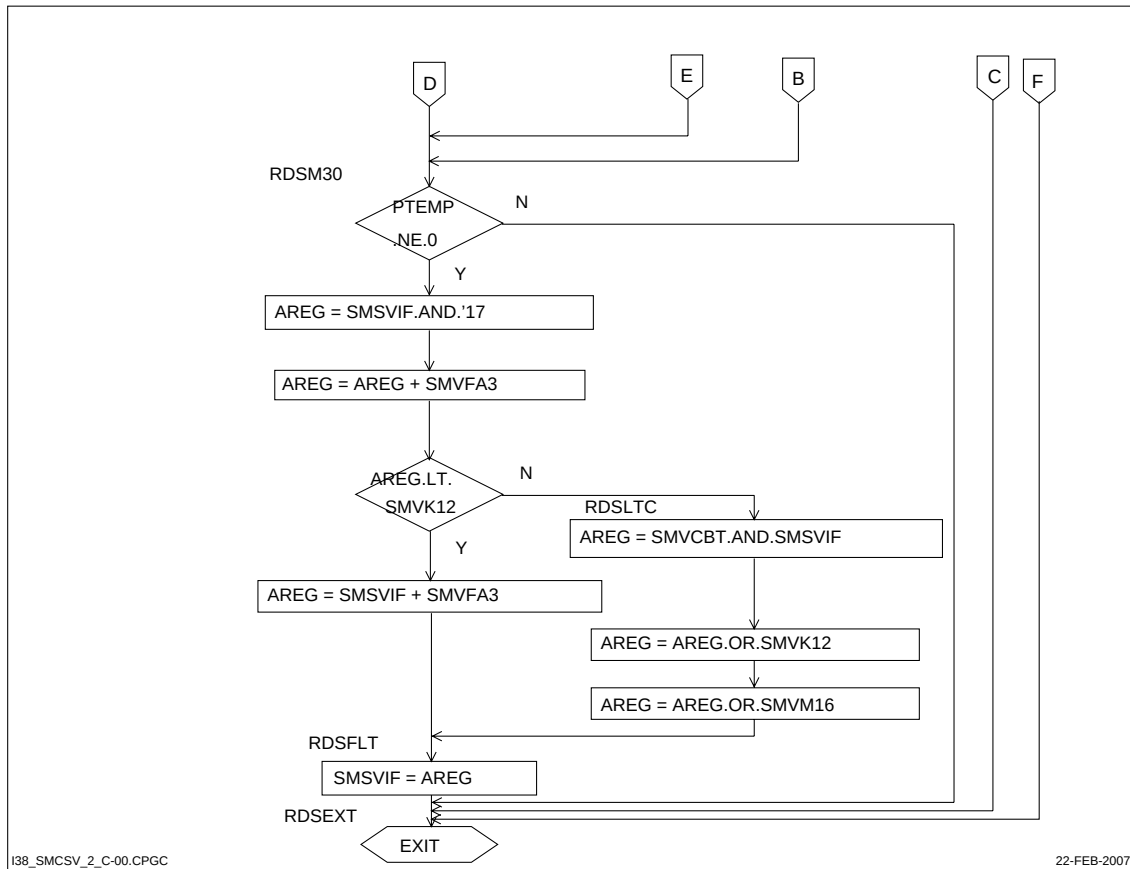
13.00 Figure 446: I38_SMCSV_2_A



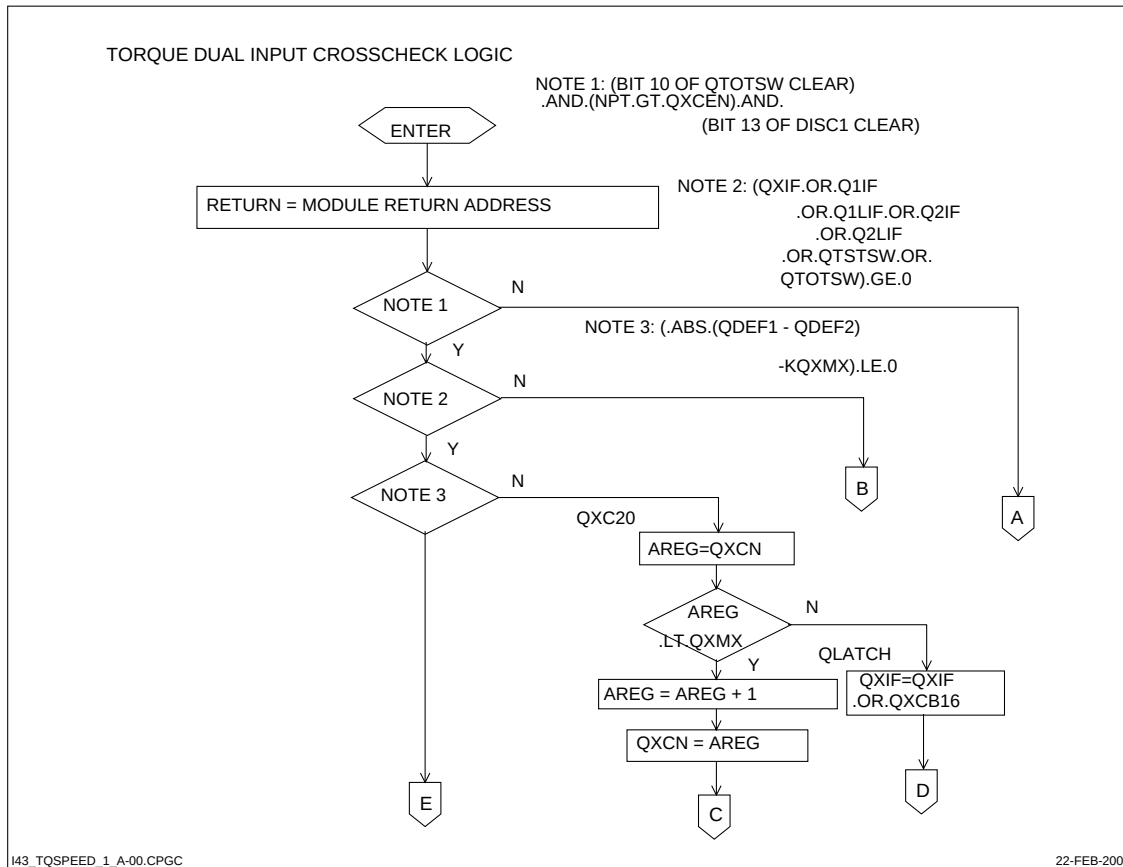
13.00 Figure 447: I38_SMCSV_2_B



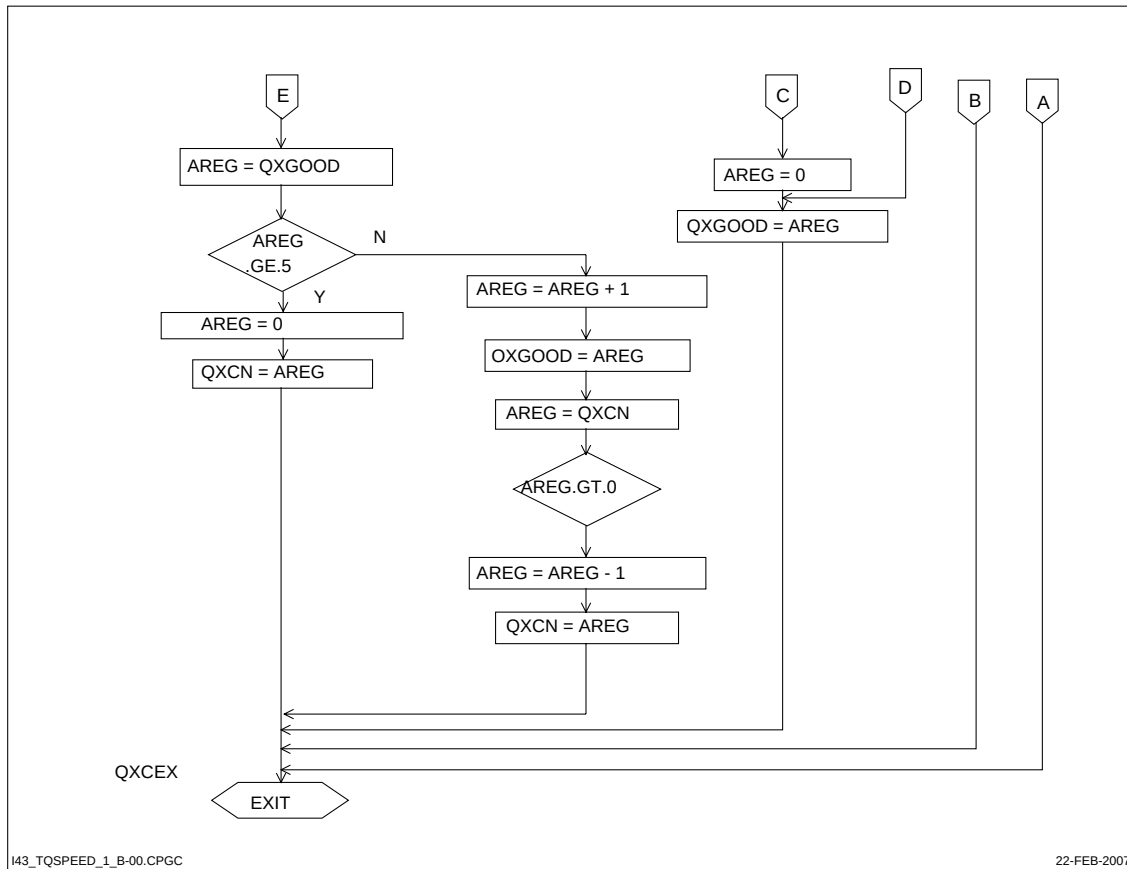
13.00 Figure 448: I38 _SMCSV _2 _C



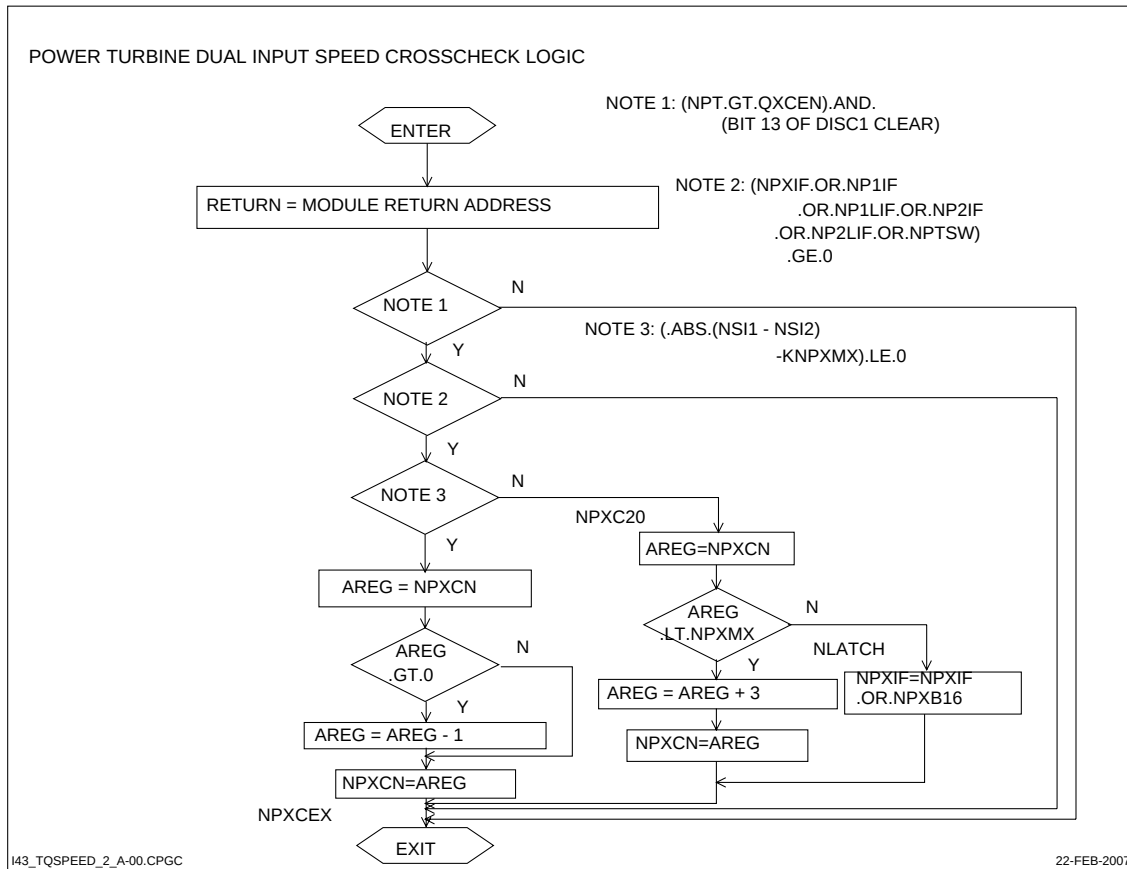
13.00 Figure 449: I43 _TQSPEED _1 _A



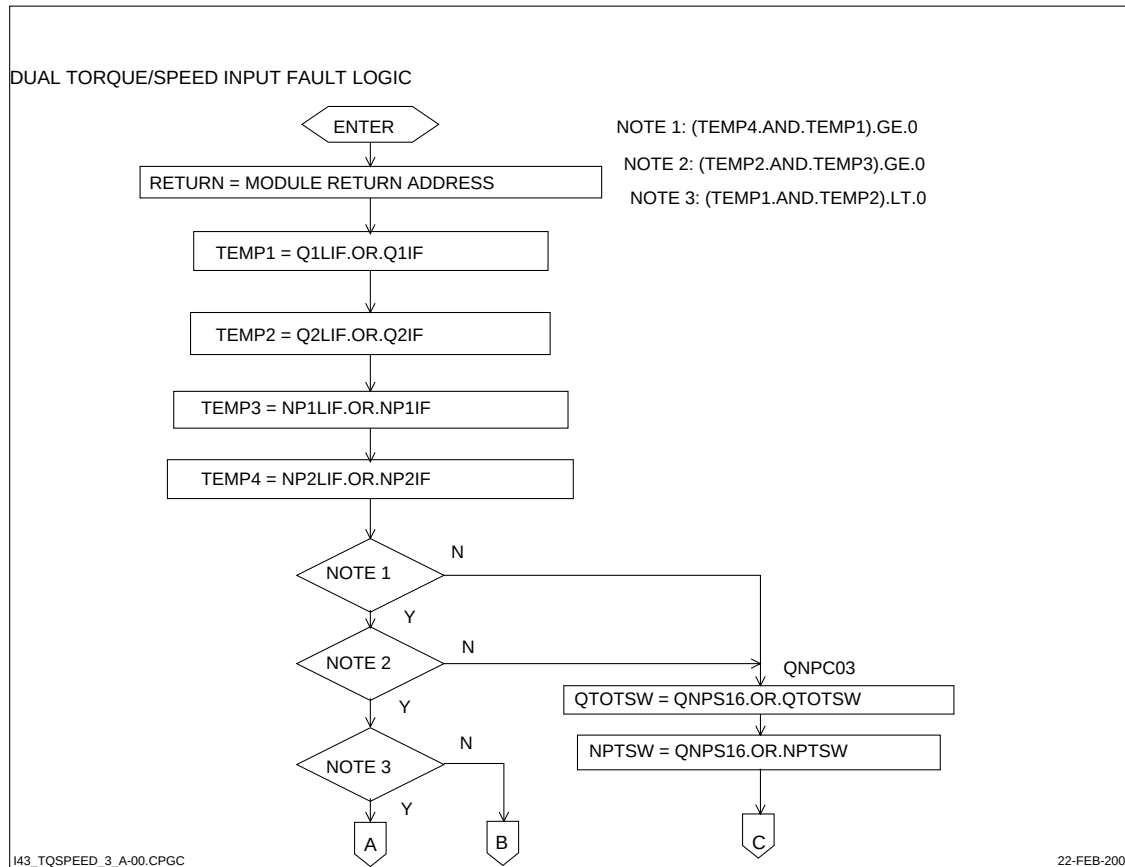
13.00 Figure 450: I43_TQSPEED_1_B



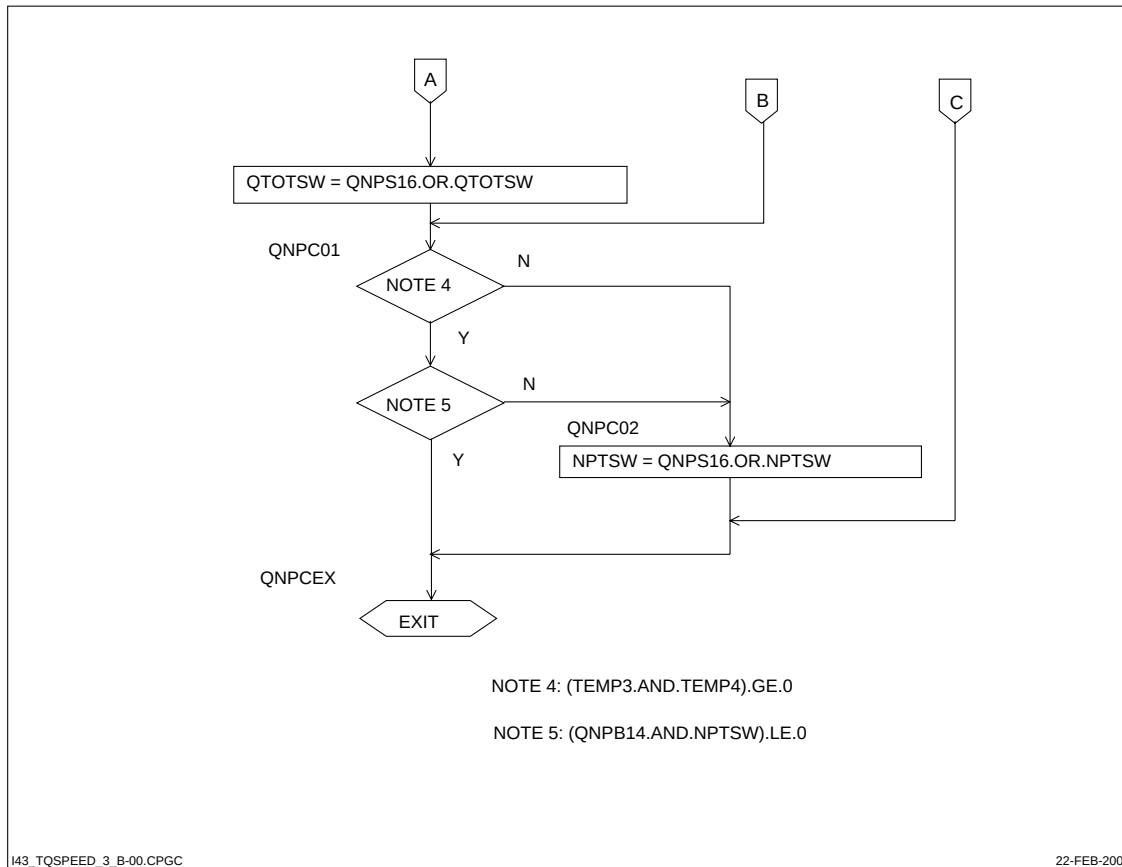
13.00 Figure 451: I43_TQSPEED_2_A



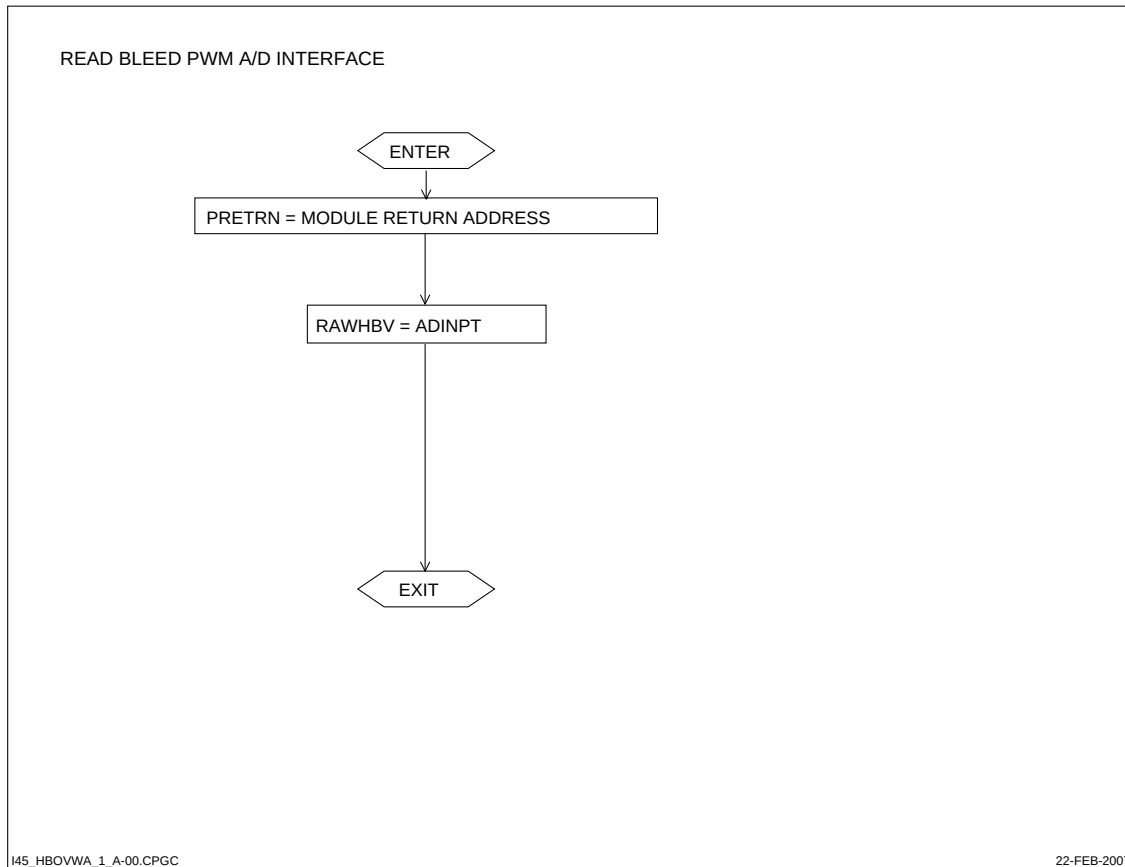
13.00 Figure 452: I43_TQSPEED_3_A



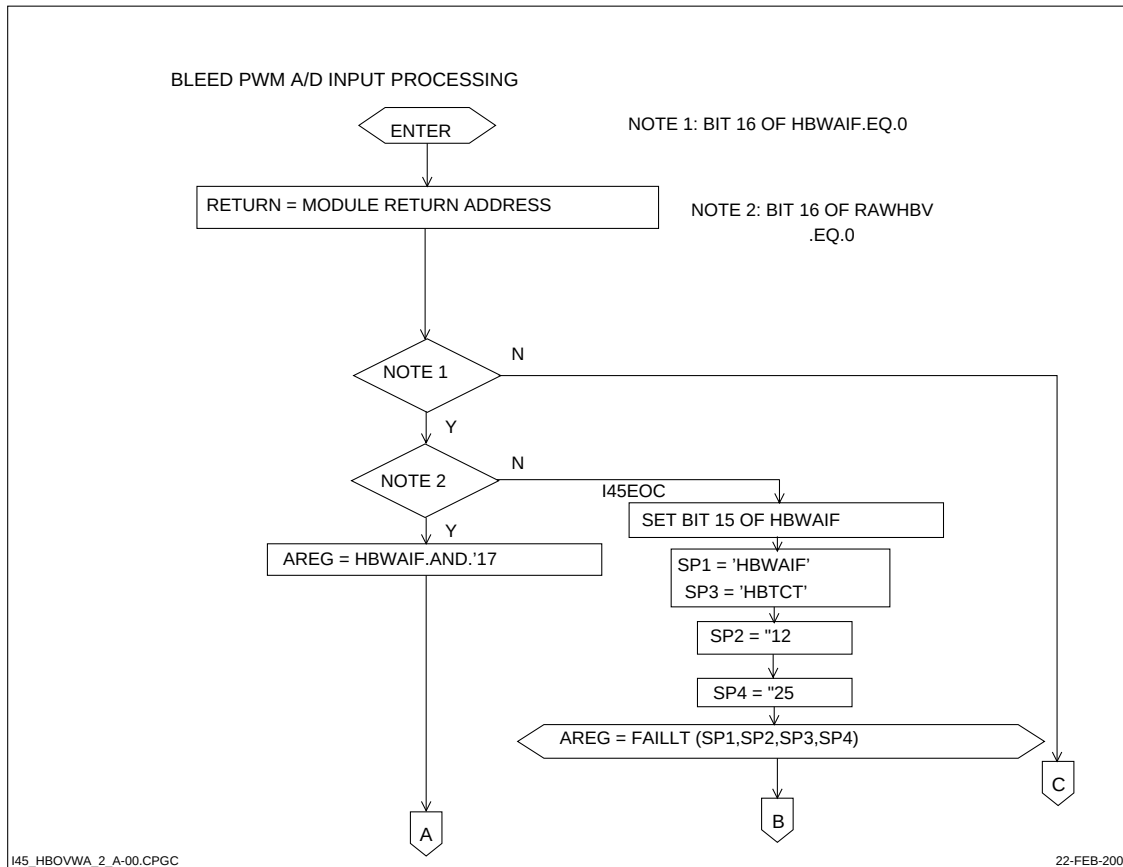
13.00 Figure 453: I43 _TQSPEED _3 _B



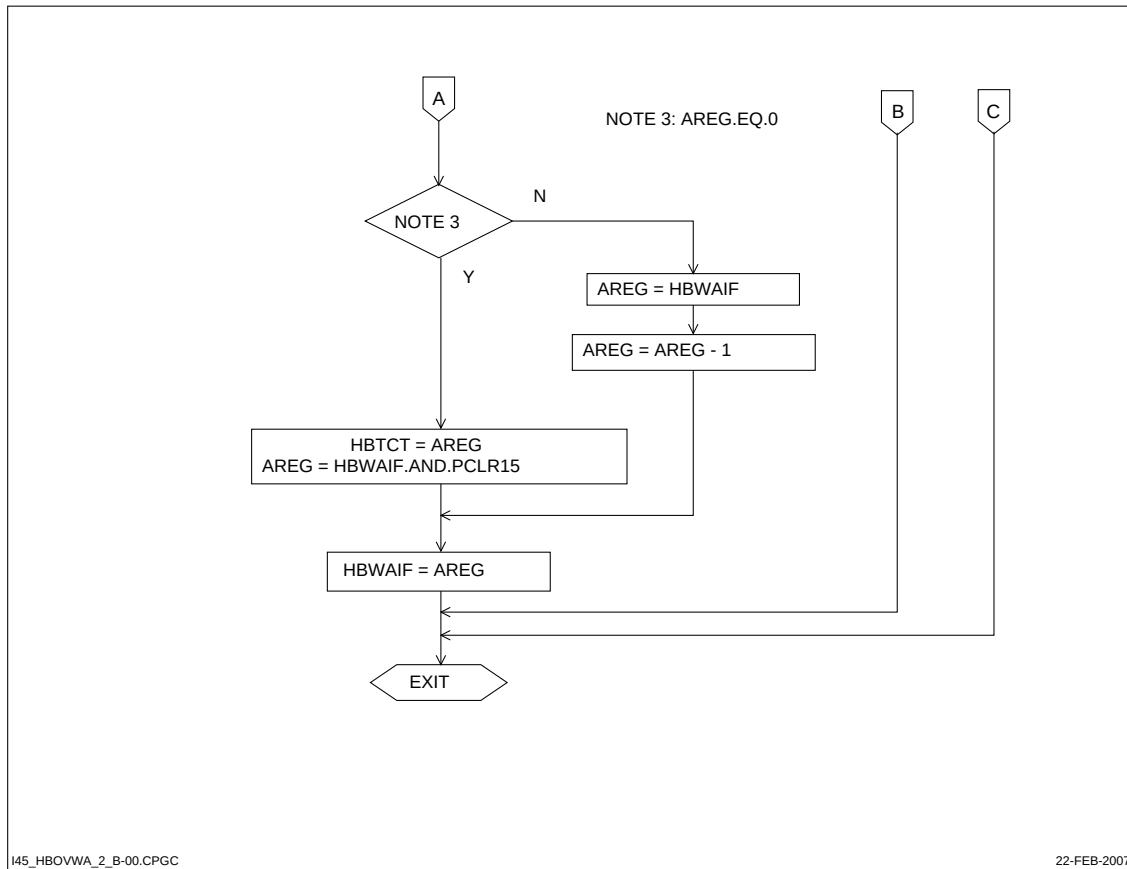
13.00 Figure 454: I45_HBOVWA_1_A



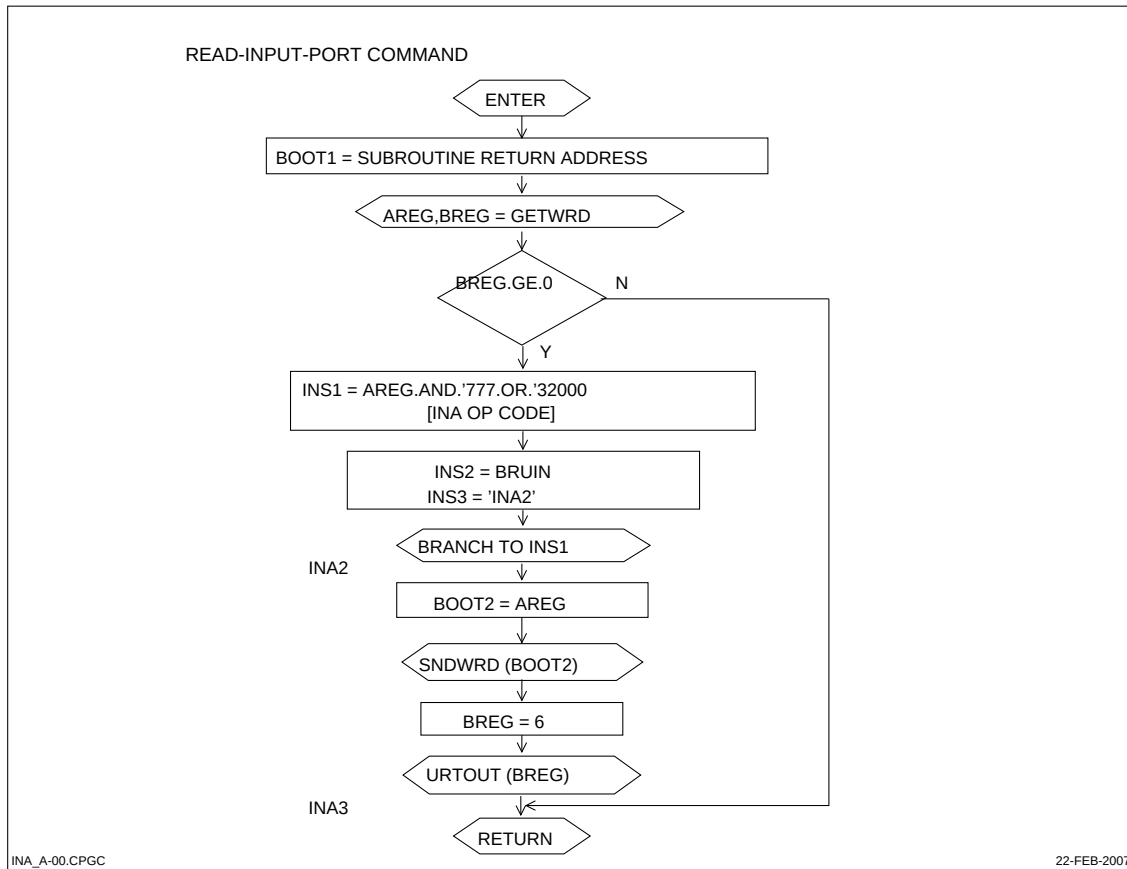
13.00 Figure 455: I45_HBOVWA_2_A



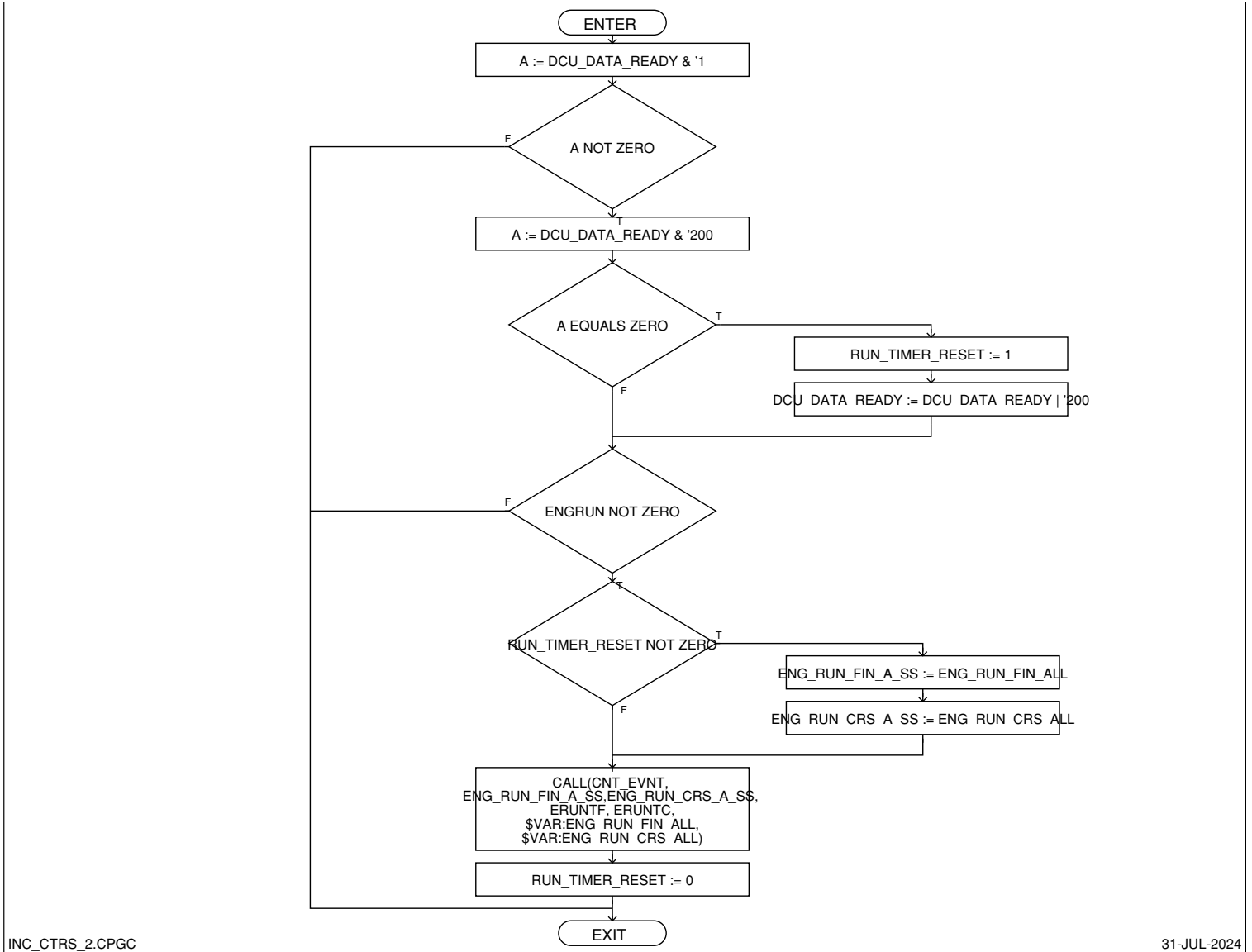
13.00 Figure 456: I45_HBOVWA_2_B



13.00 Figure 457: INA_A



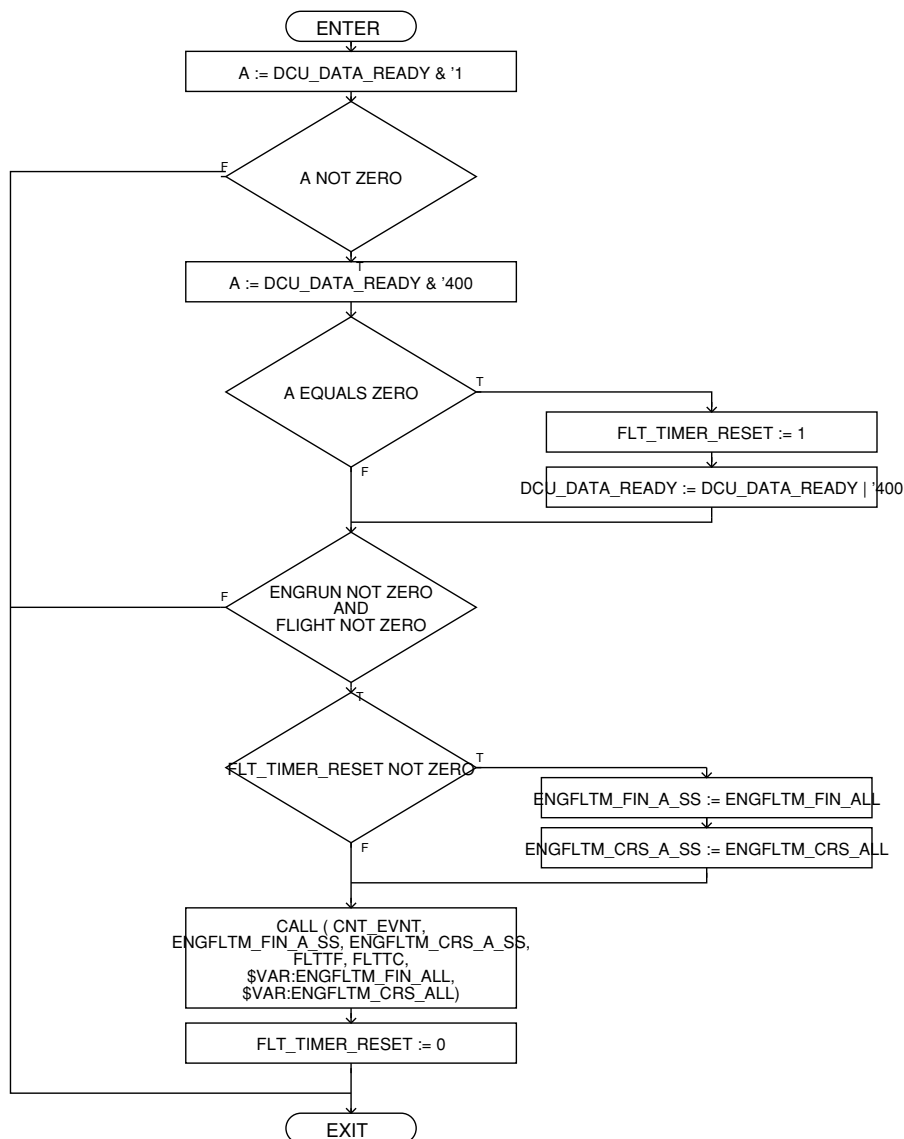
13.00 Figure 458: INC_CTRS_2



INC_CTRS_2.CPGC

31-JUL-2024

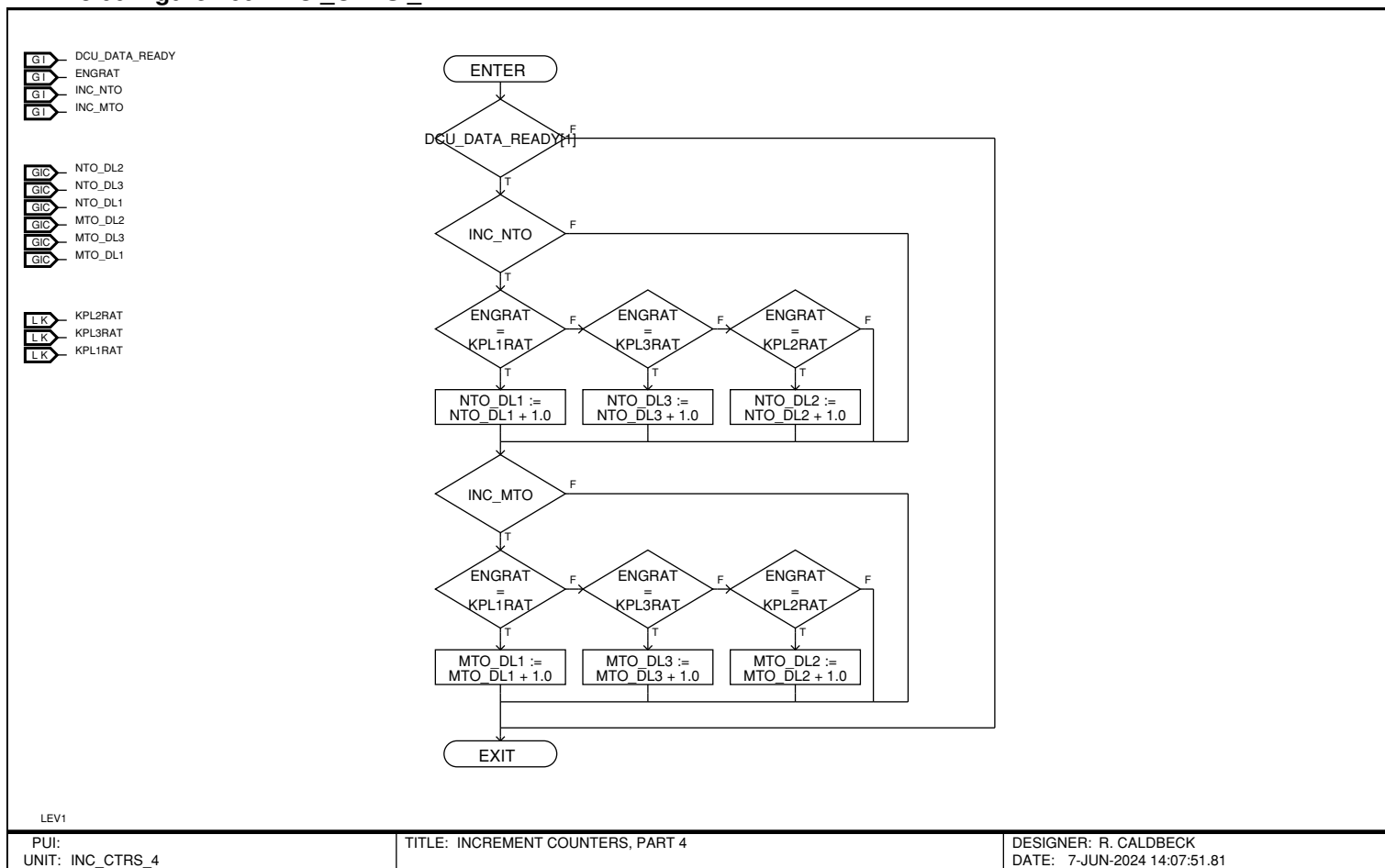
13.00 Figure 459: INC_CTRS_3



INC_CTRS_3.CPGC

31-JUL-2024

13.00 Figure 460: INC_CTRS_4



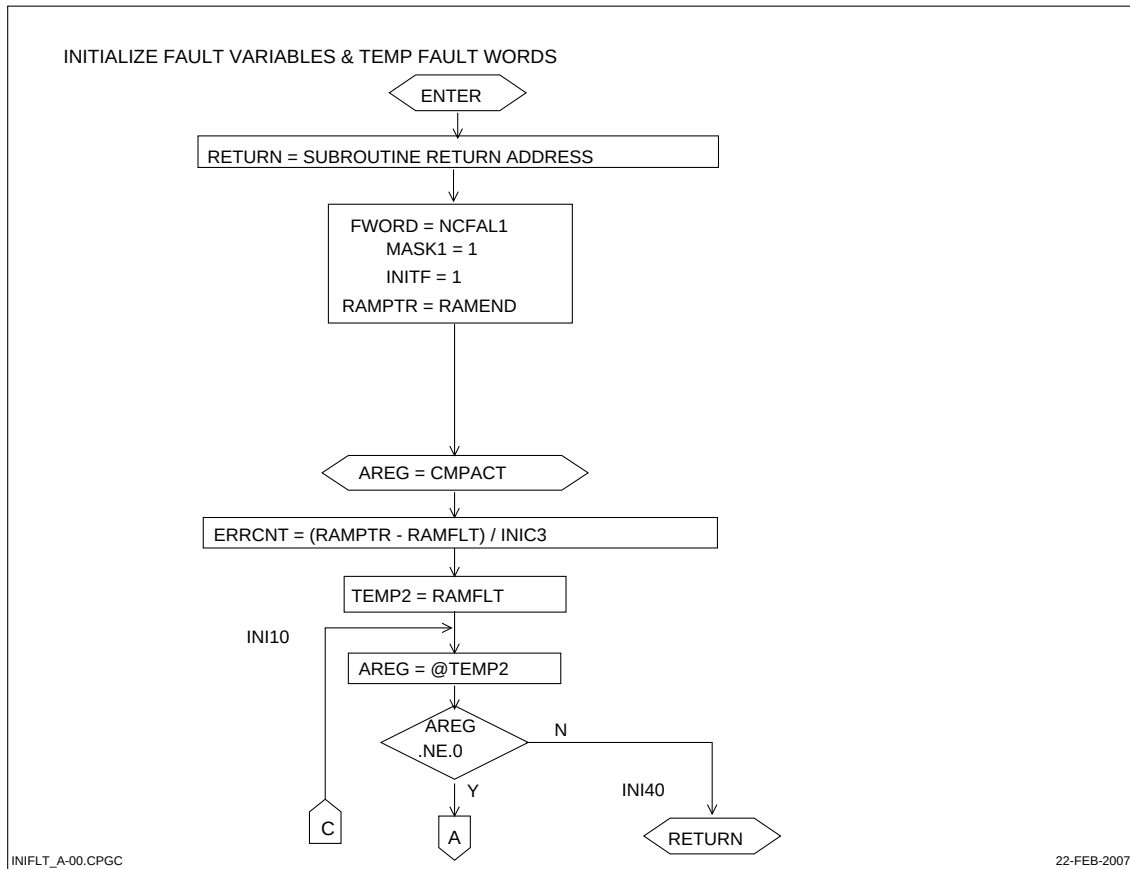
LEV1

PUI:
UNIT: INC_CTRS_4

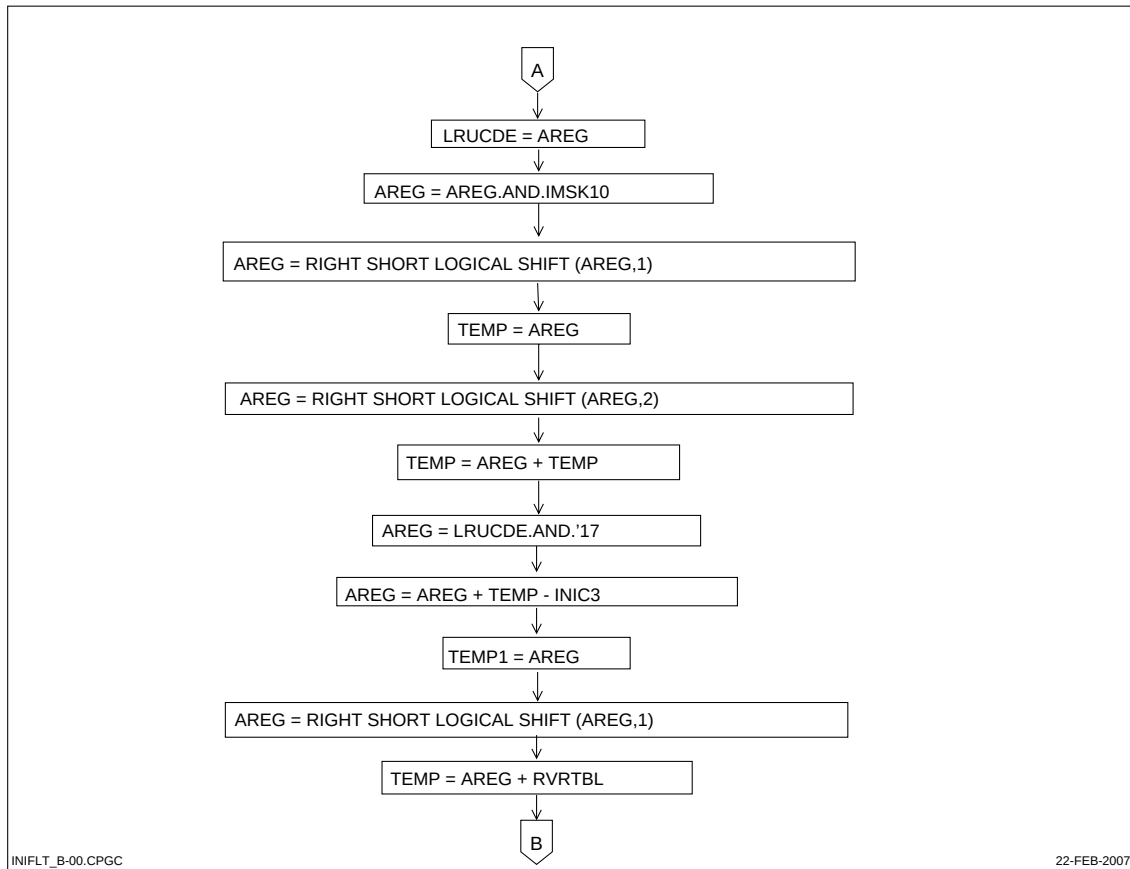
TITLE: INCREMENT COUNTERS, PART 4

DESIGNER: R. CALDBECK
DATE: 7-JUN-2024 14:07:51.81

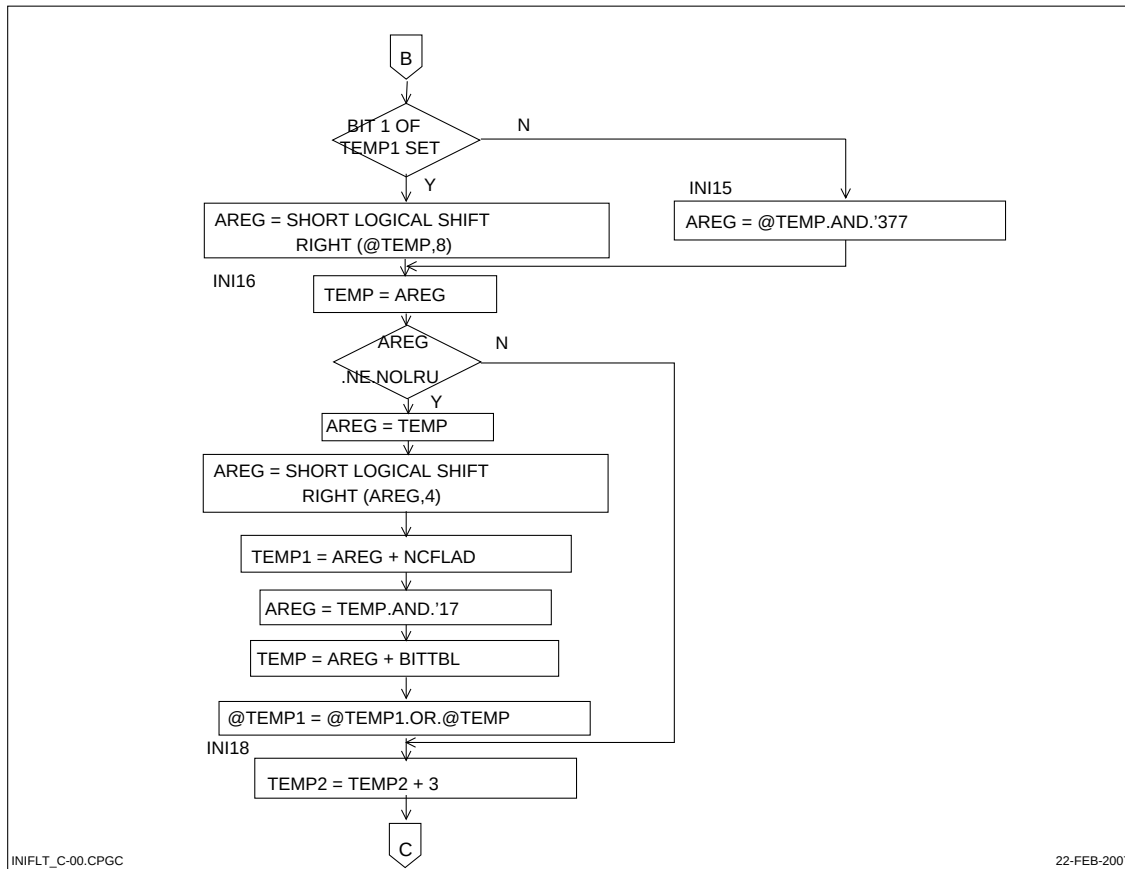
13.00 Figure 461: INIFLT_A



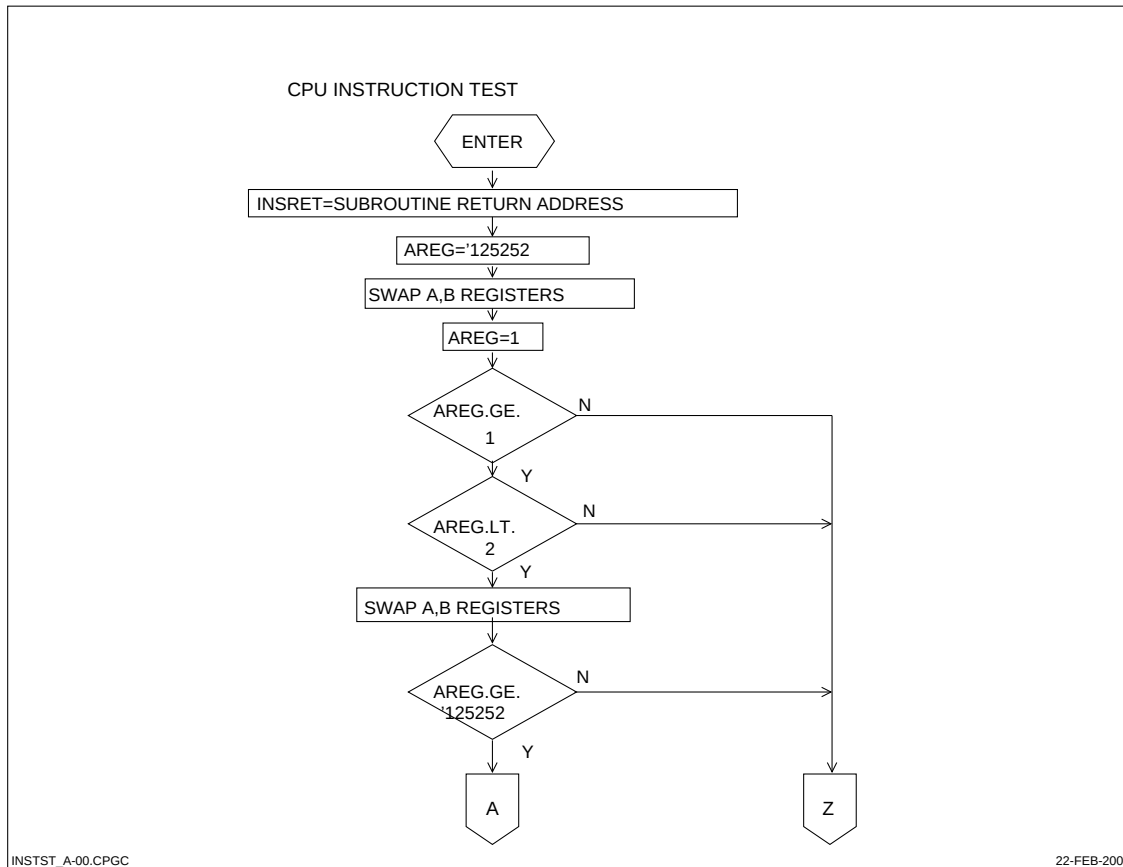
13.00 Figure 462: INIFLT _B



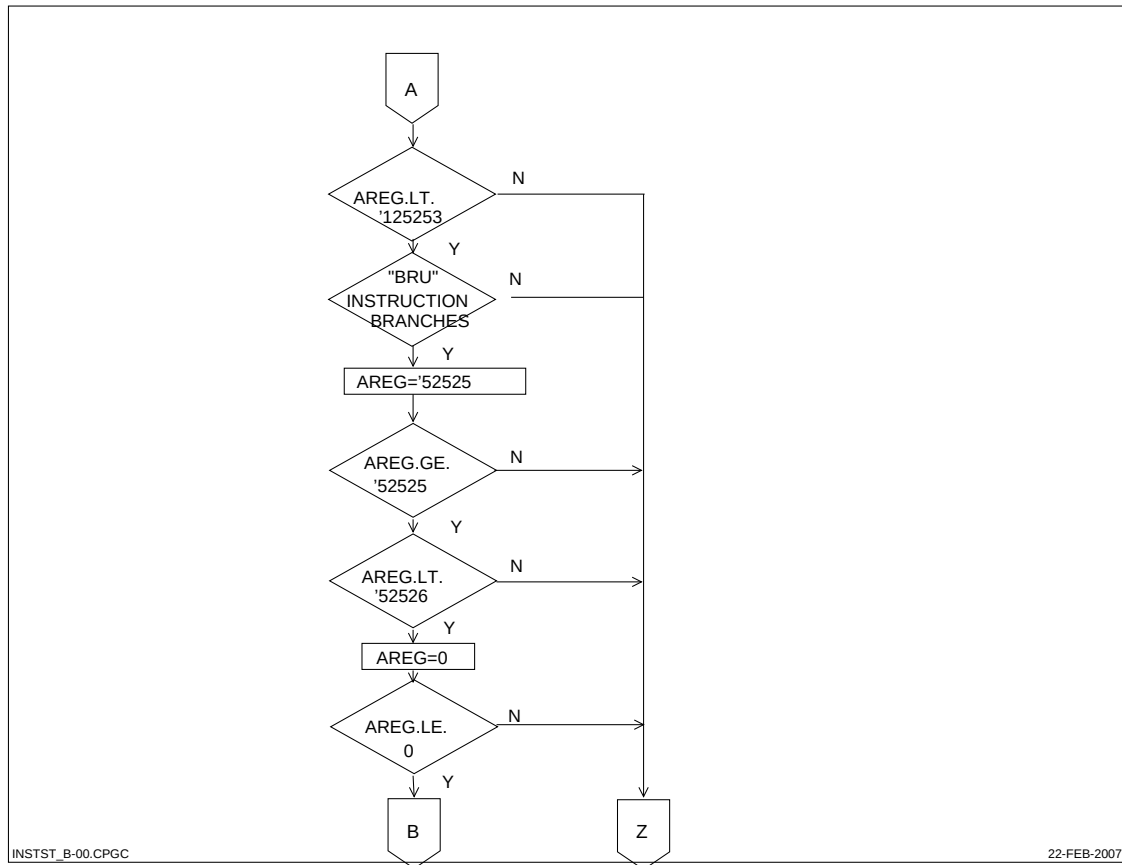
13.00 Figure 463: INIFLT_C



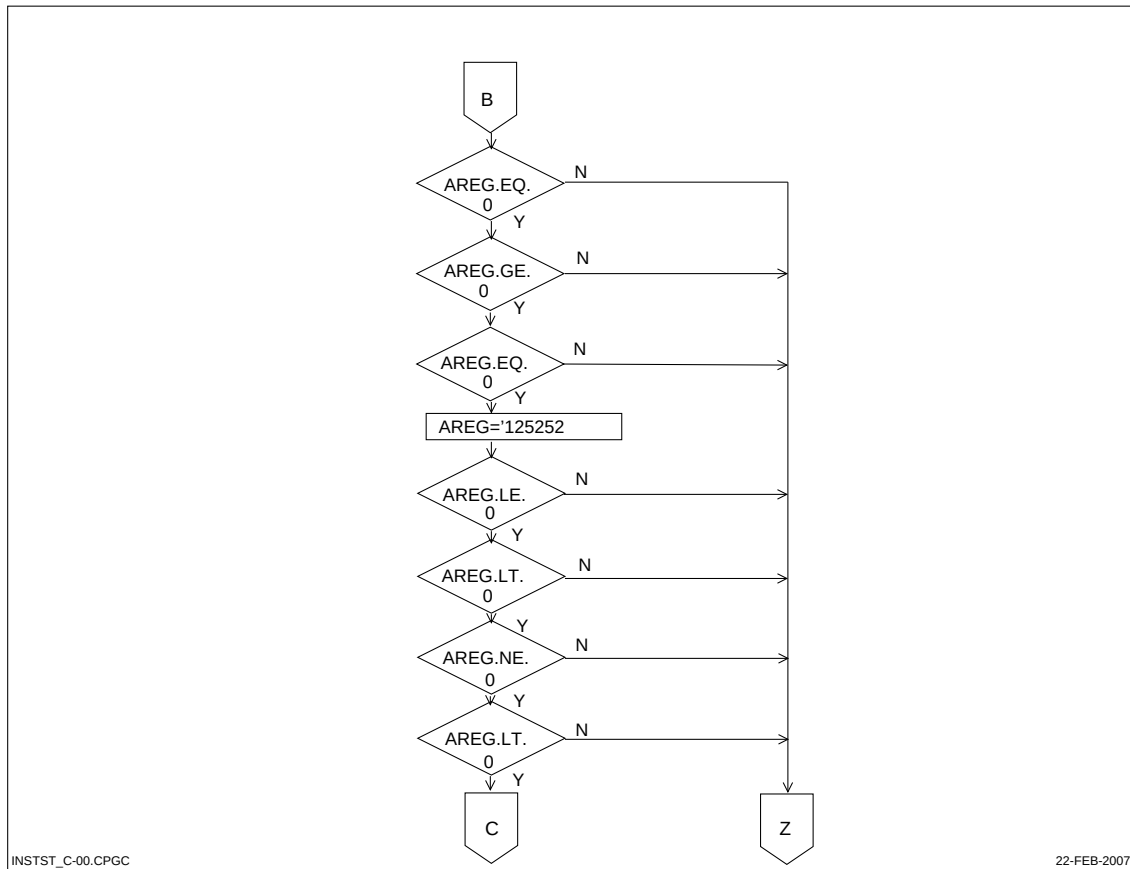
13.00 Figure 464: INSTST_A



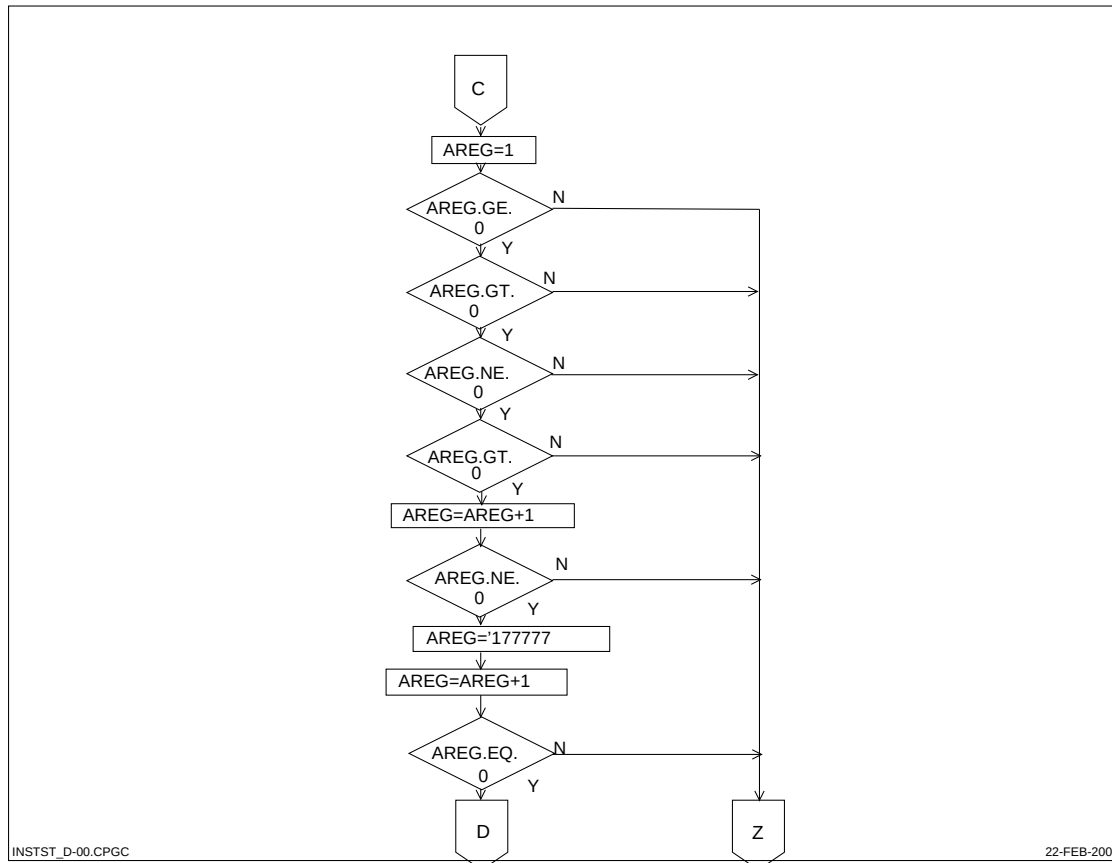
13.00 Figure 465: INSTST_B



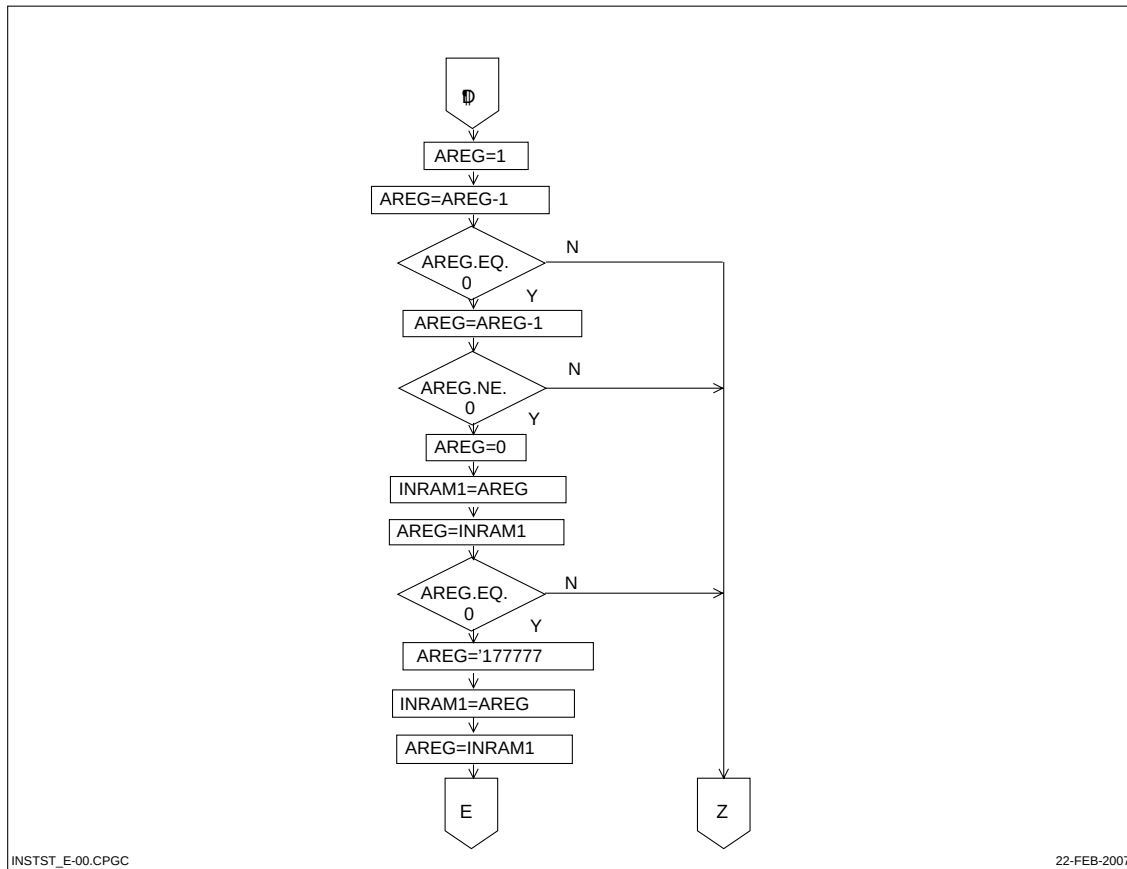
13.00 Figure 466: INSTST_C



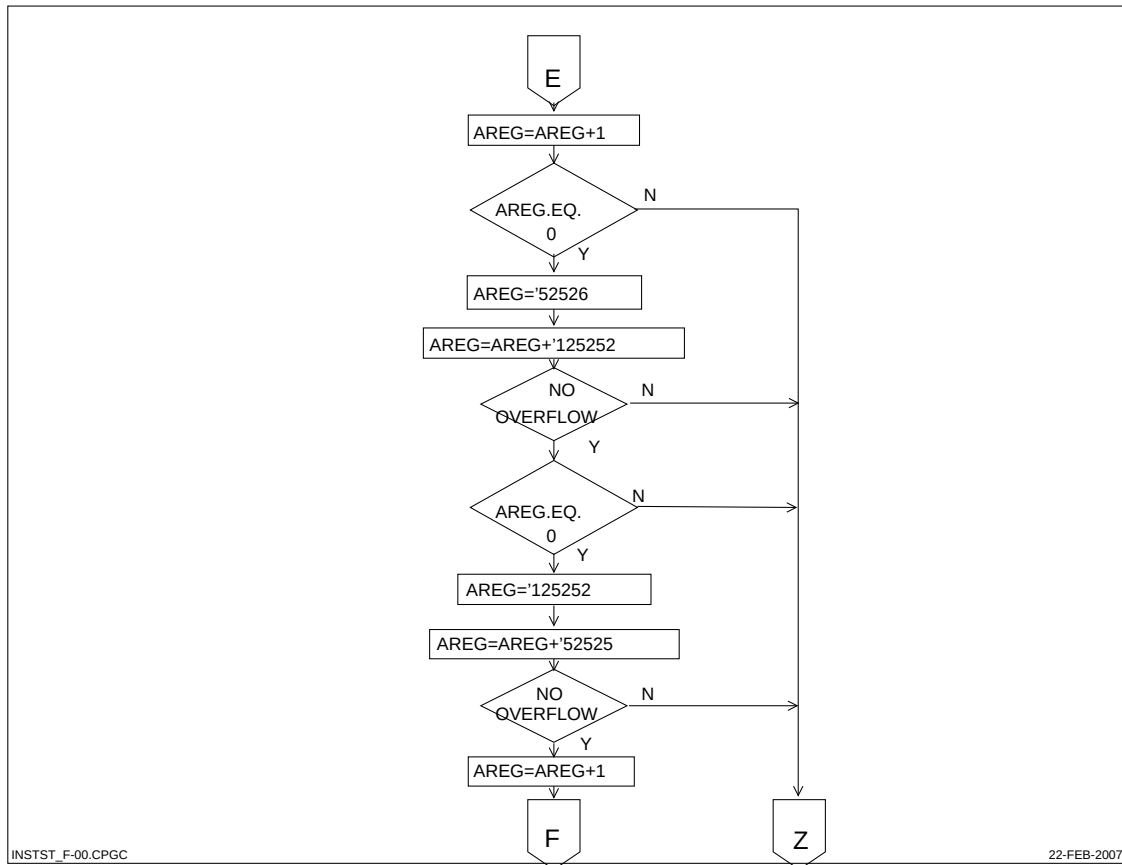
13.00 Figure 467: INSTST _D



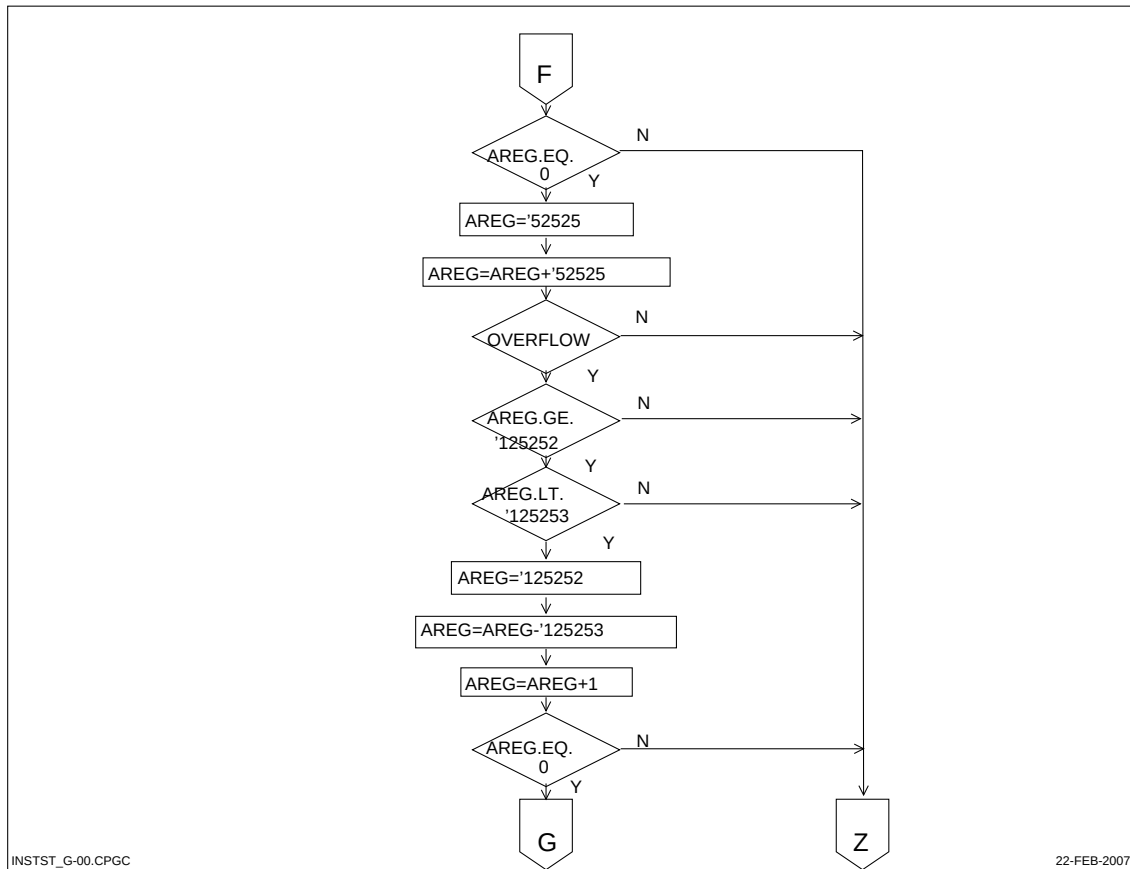
13.00 Figure 468: INSTST_E



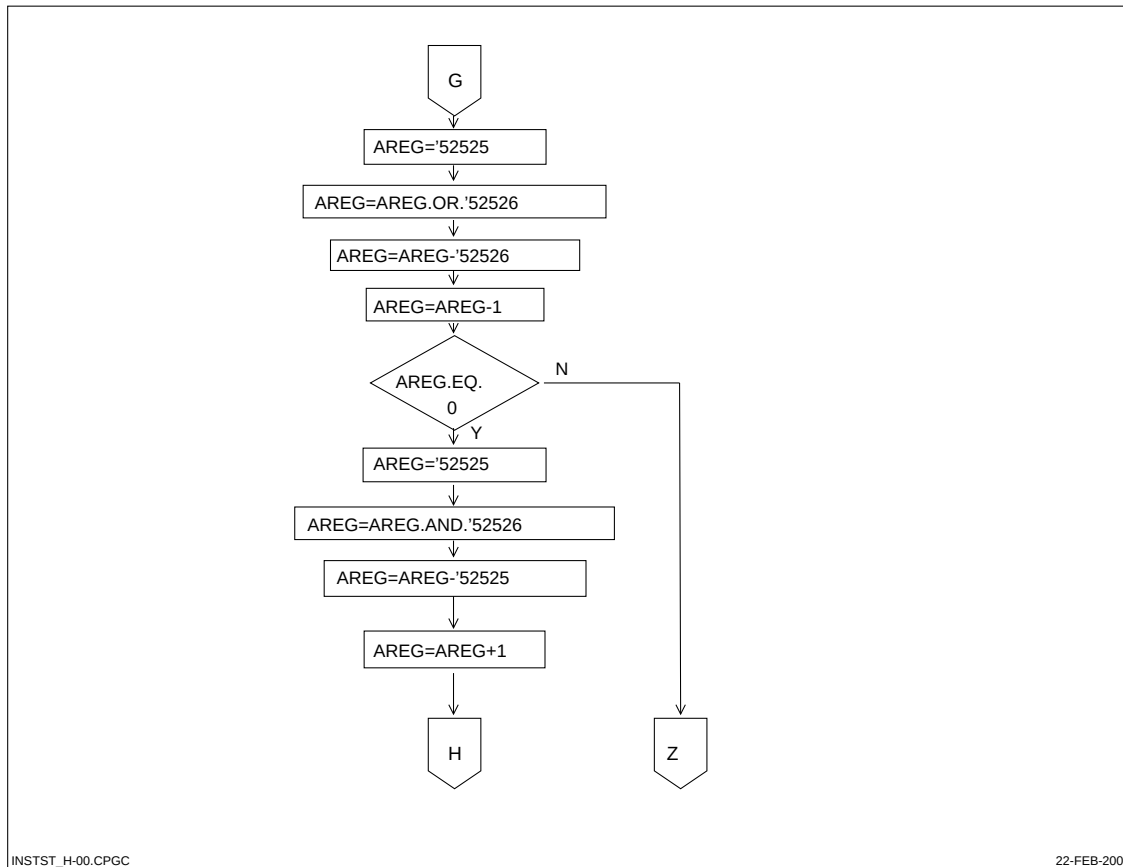
13.00 Figure 469: INSTST_F



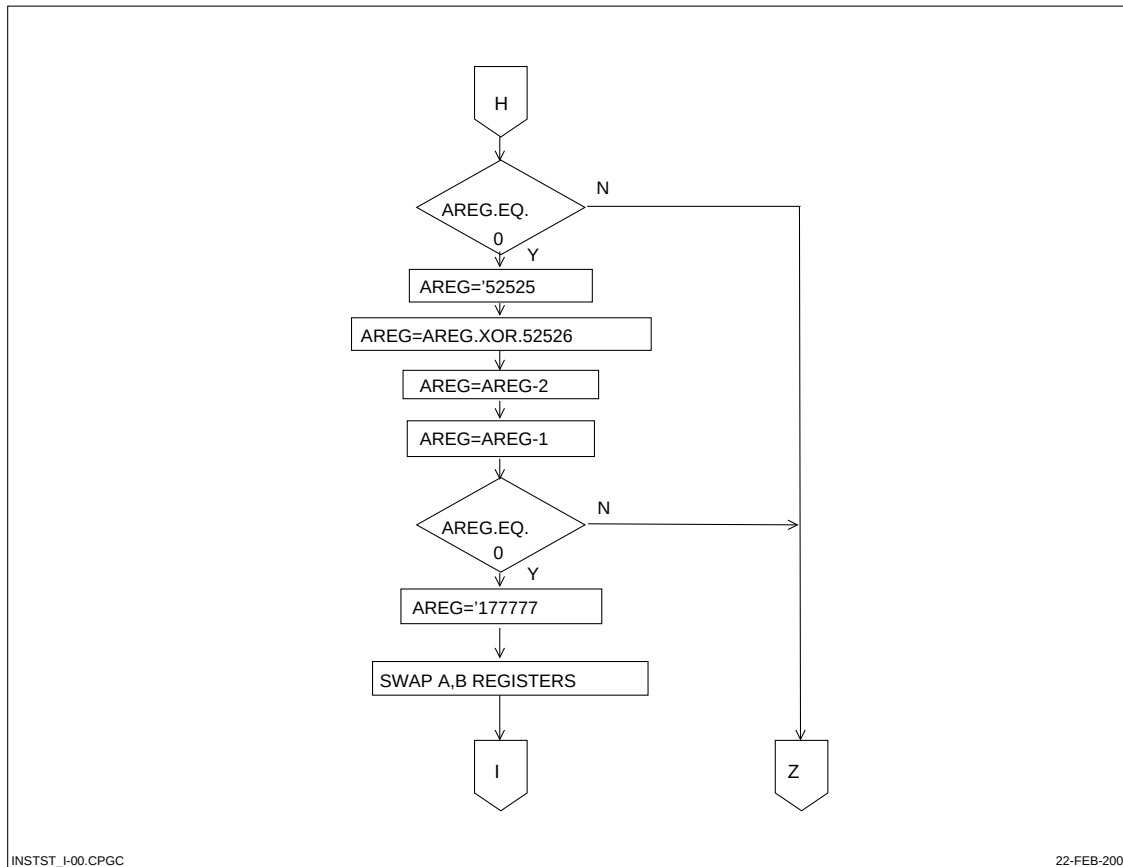
13.00 Figure 470: INSTST_G



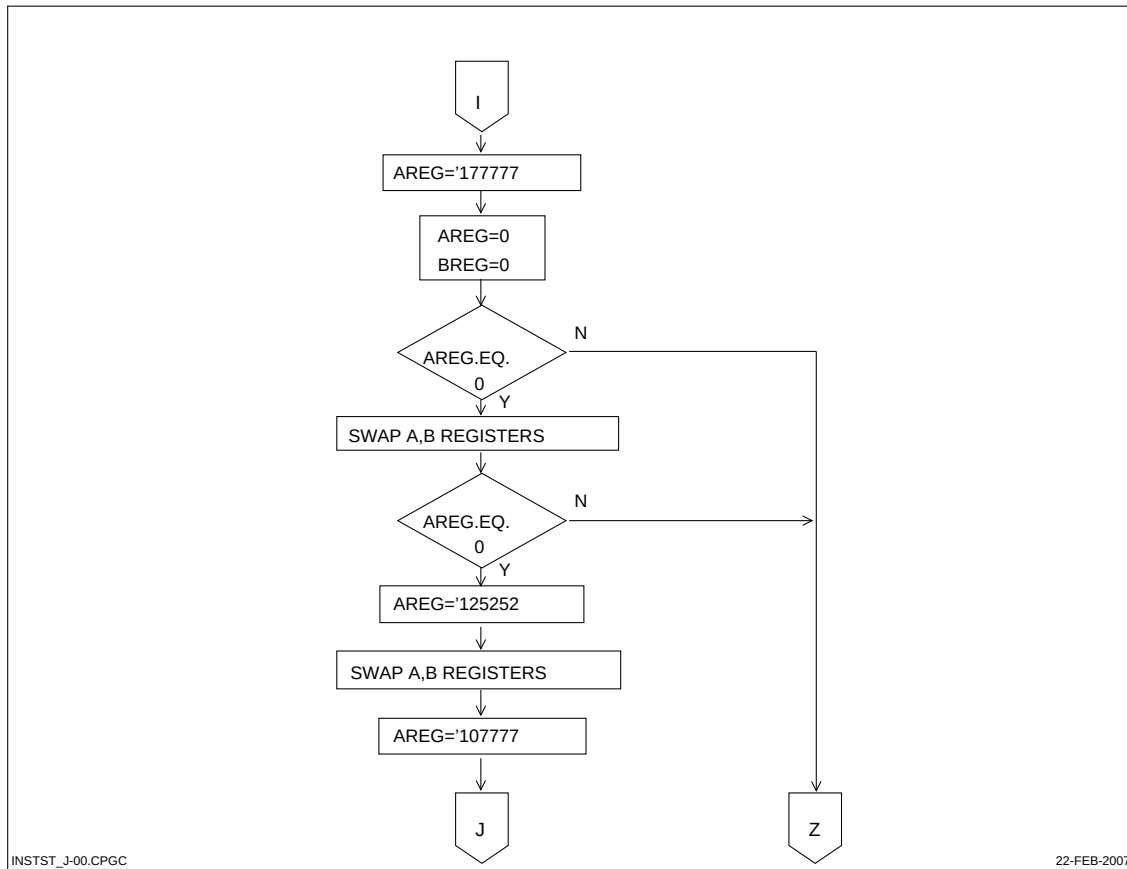
13.00 Figure 471: INSTST_H



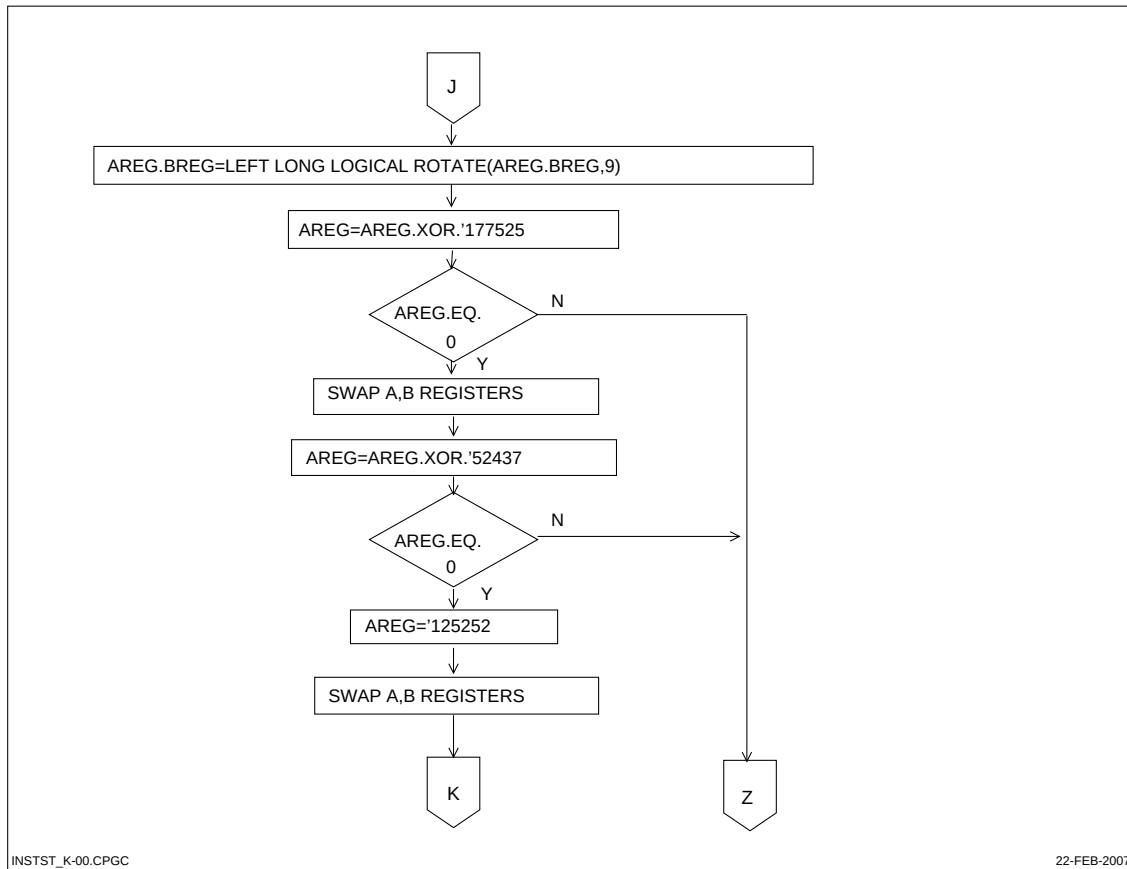
13.00 Figure 472: INSTST_I



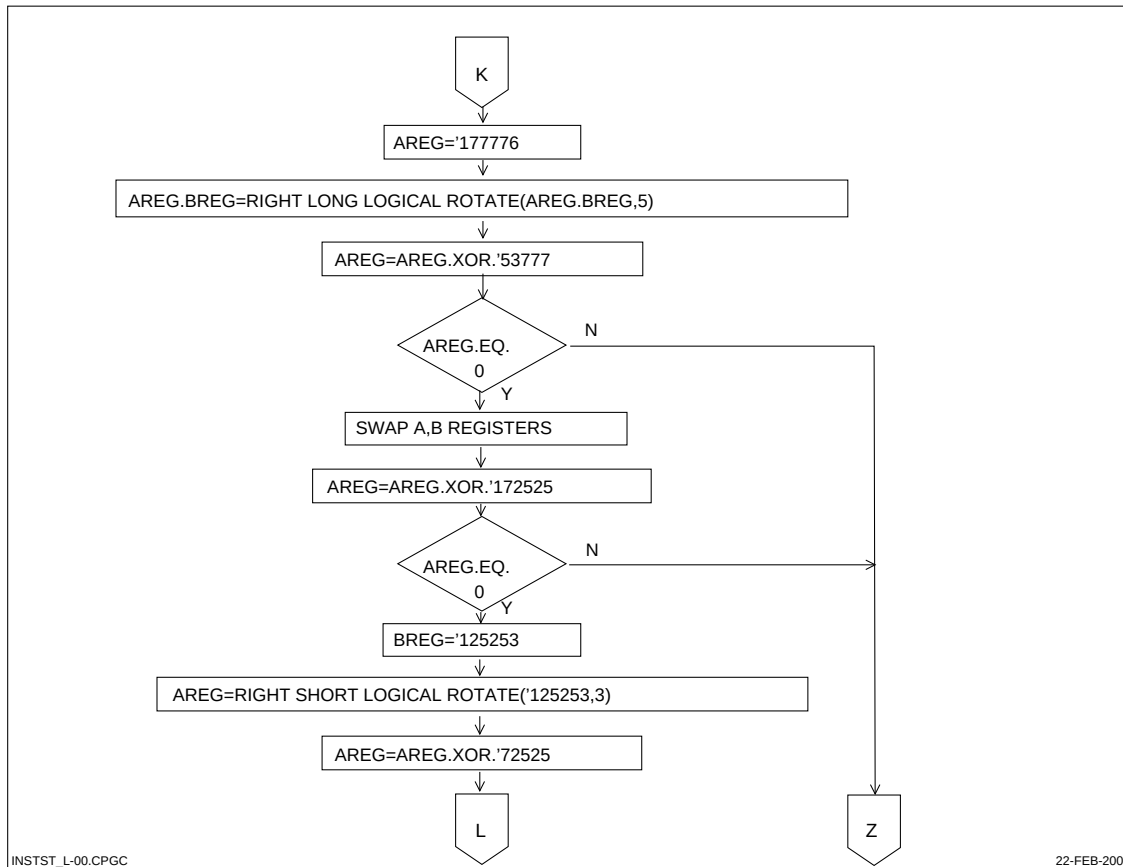
13.00 Figure 473: INSTST_J



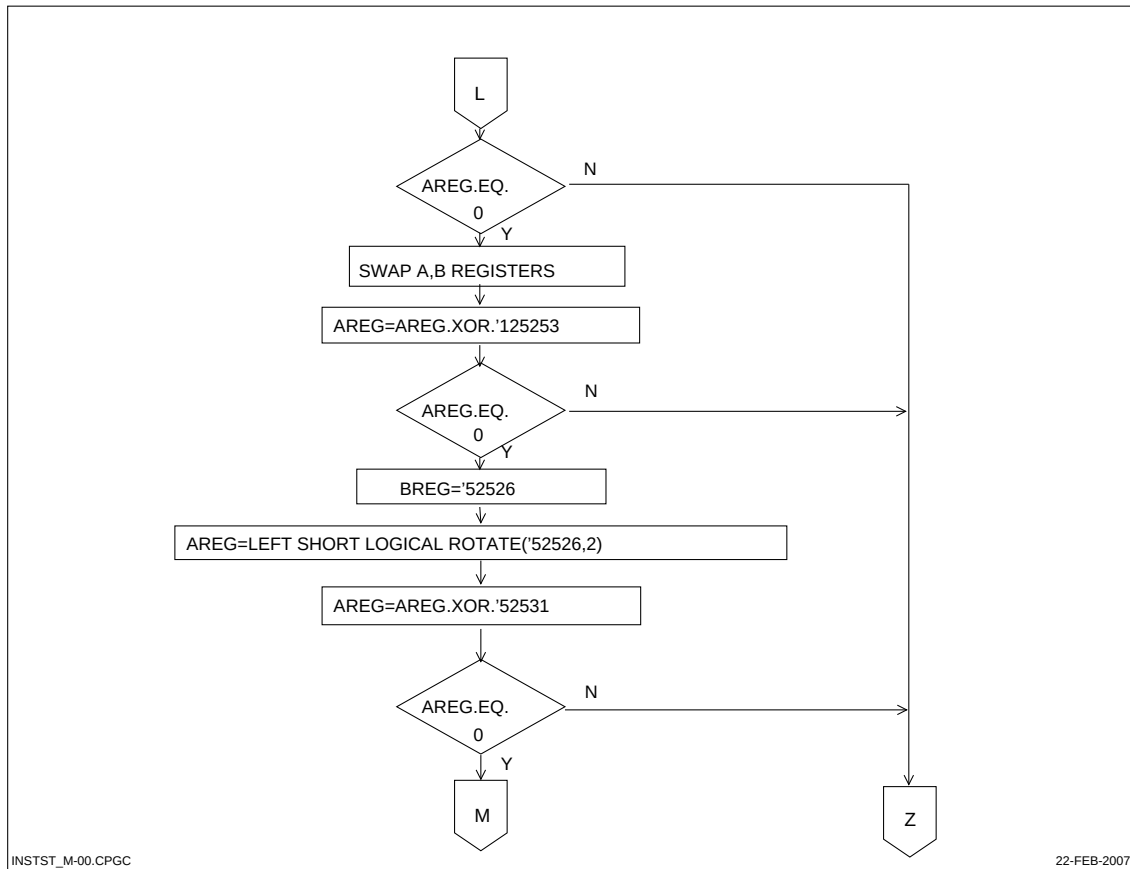
13.00 Figure 474: INSTST_K



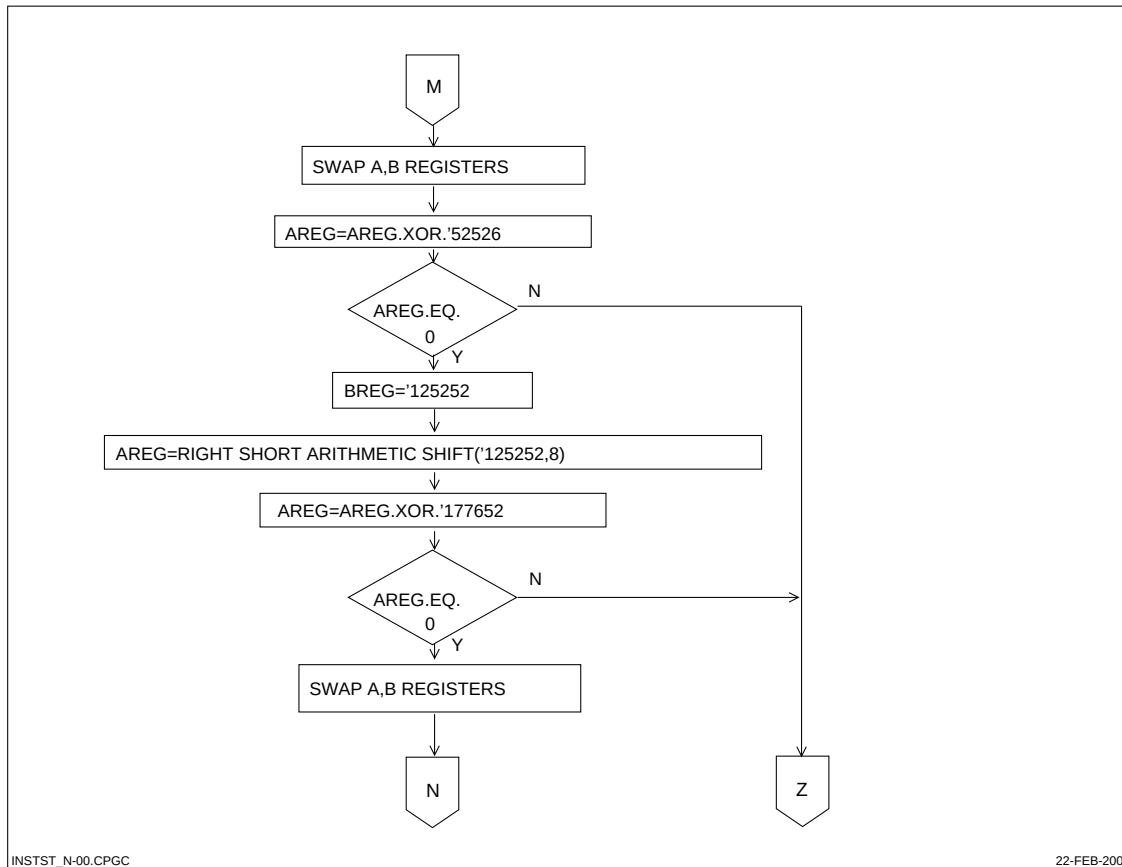
13.00 Figure 475: INSTST _L



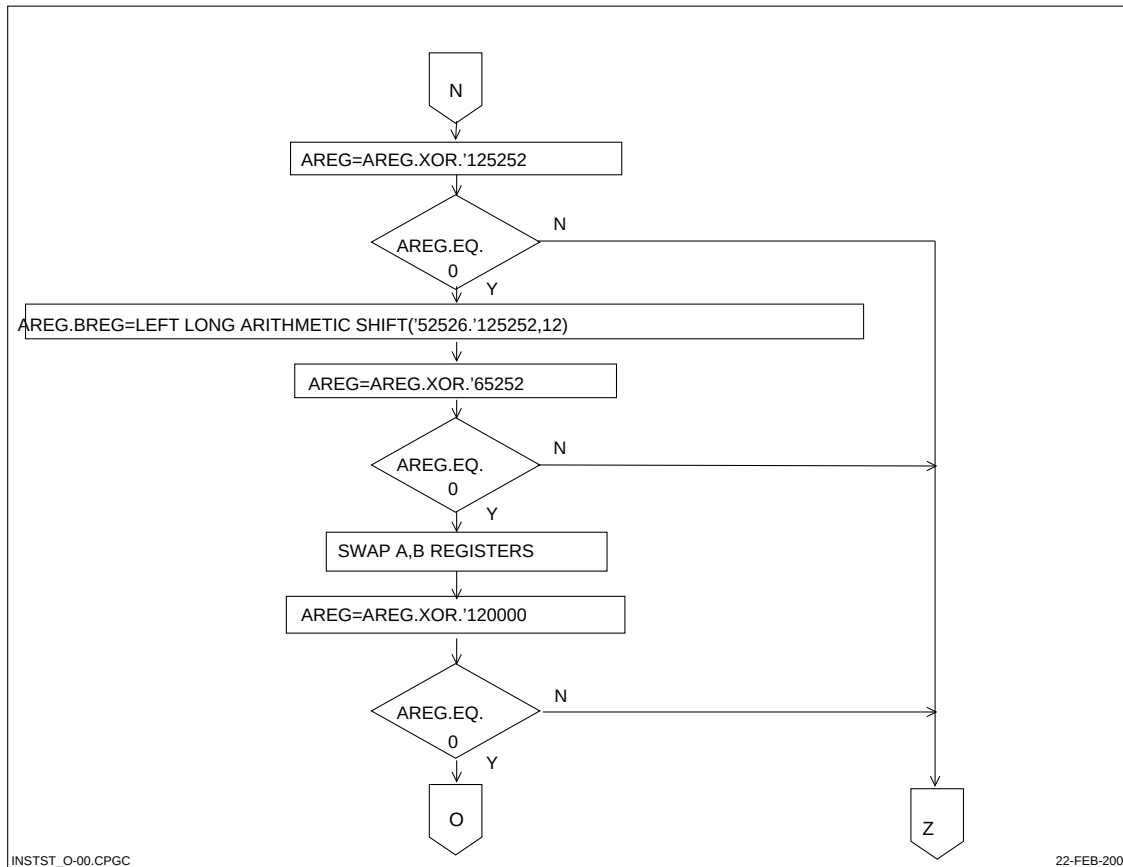
13.00 Figure 476: INSTST_M



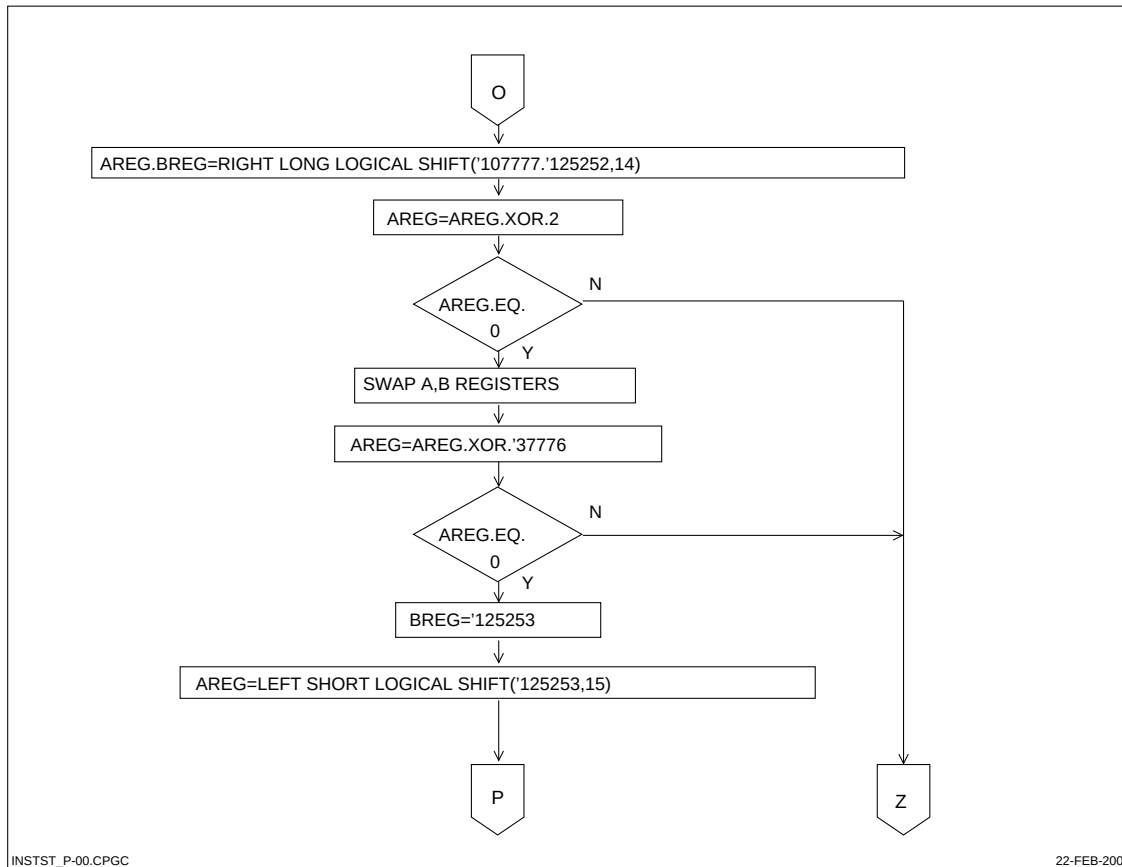
13.00 Figure 477: INSTST_N



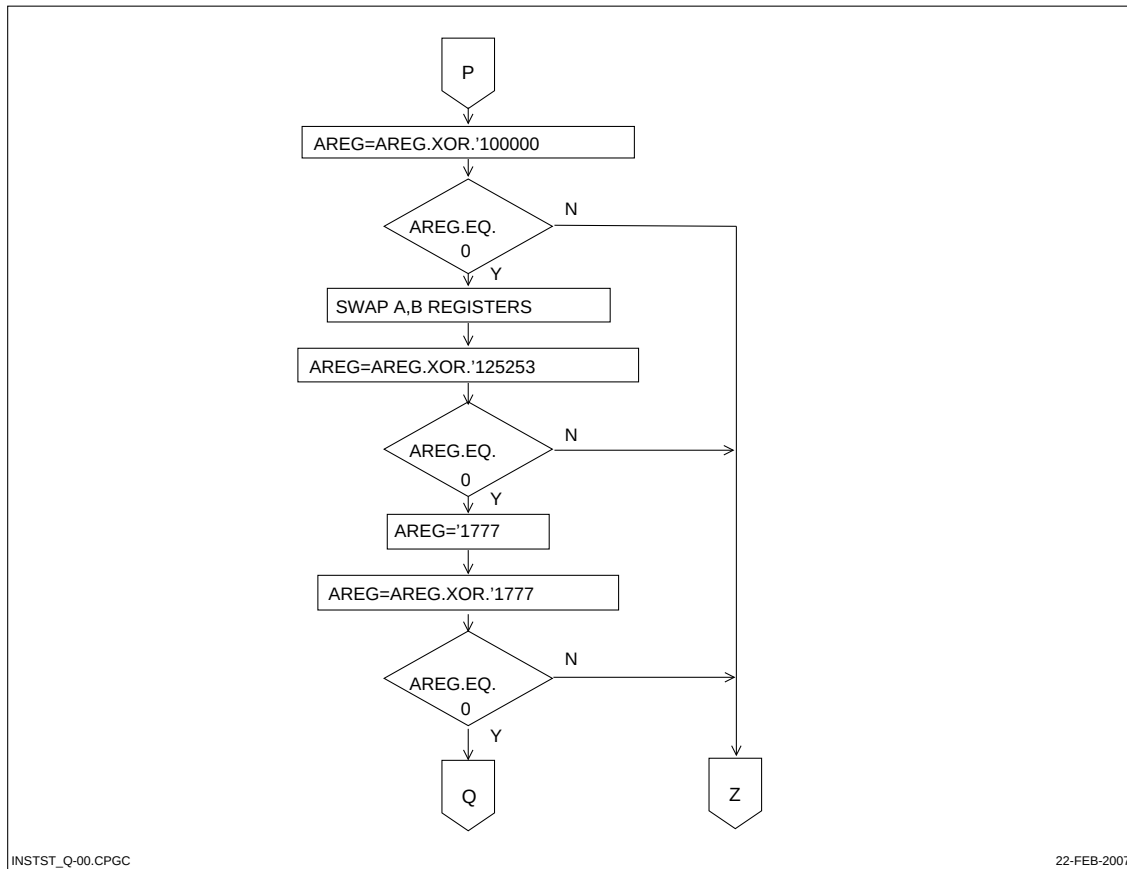
13.00 Figure 478: INSTST_O



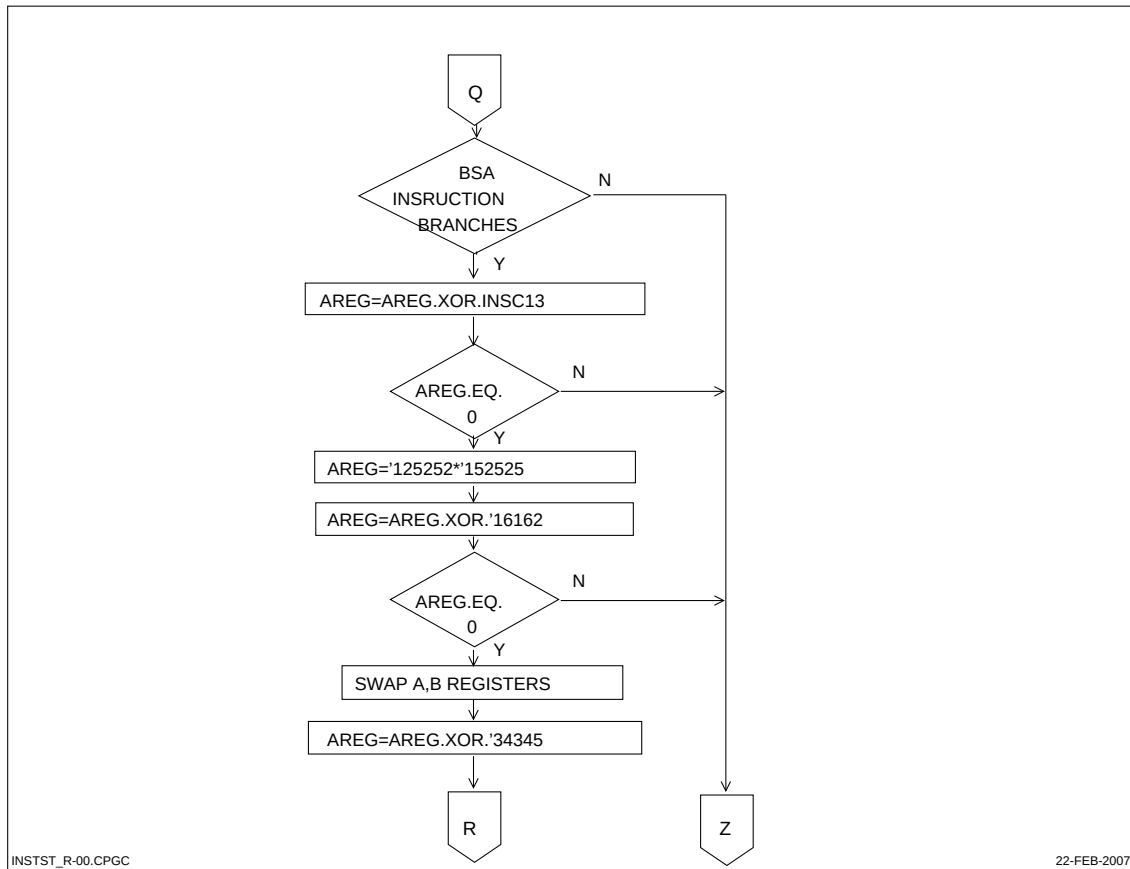
13.00 Figure 479: INSTST_P



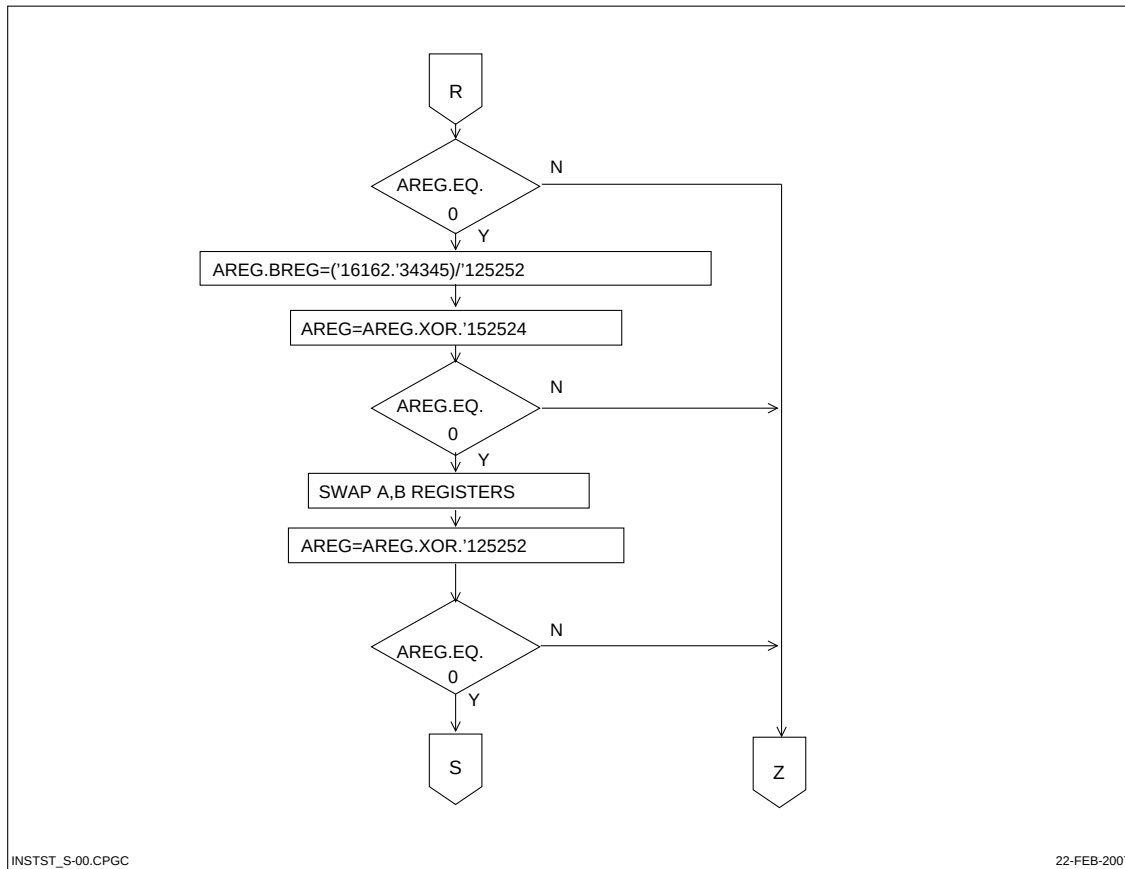
13.00 Figure 480: INSTST_Q



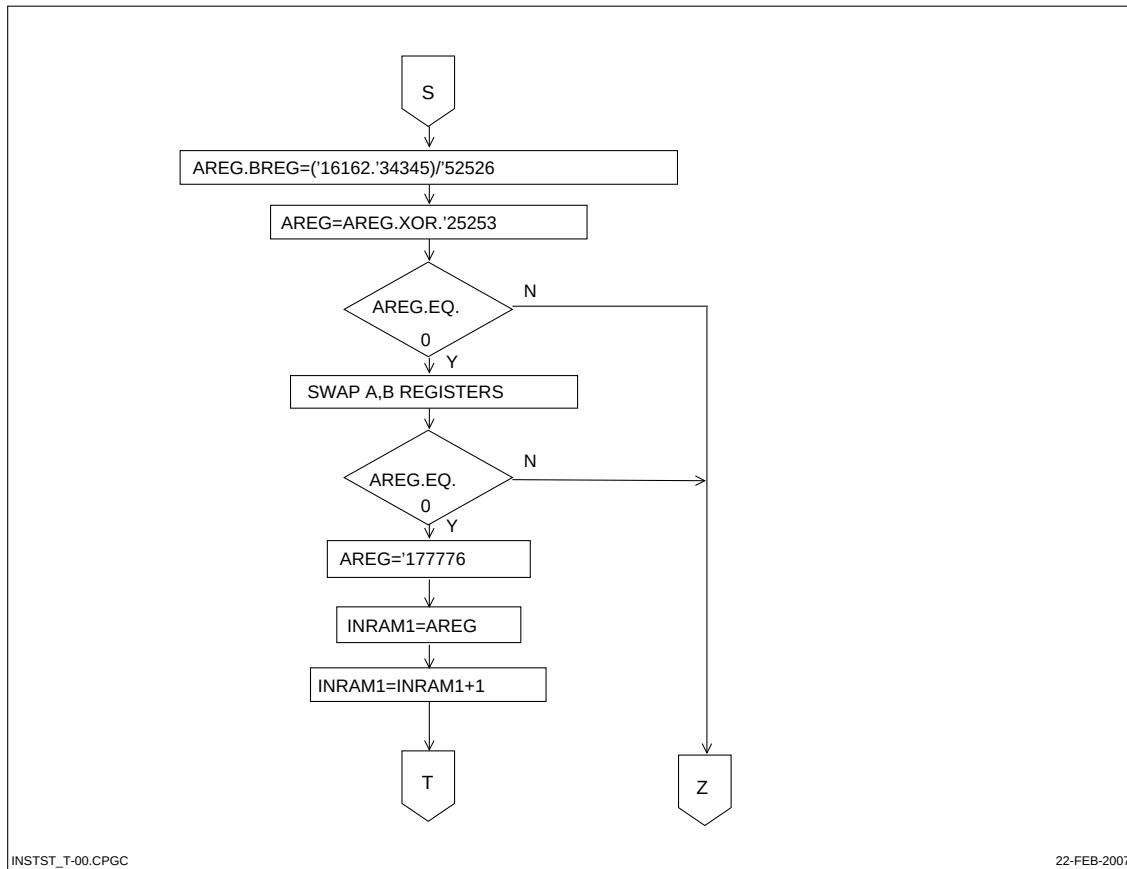
13.00 Figure 481: INSTST_R



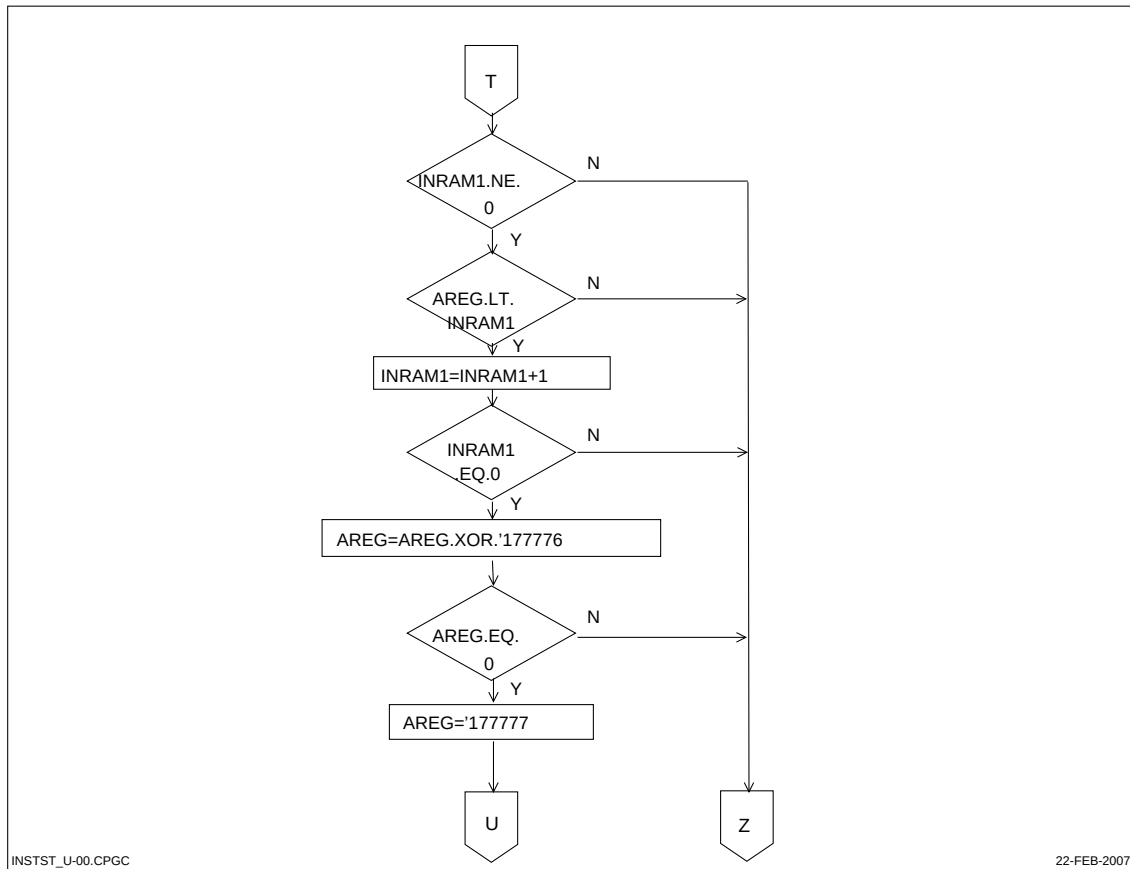
13.00 Figure 482: INSTST_S



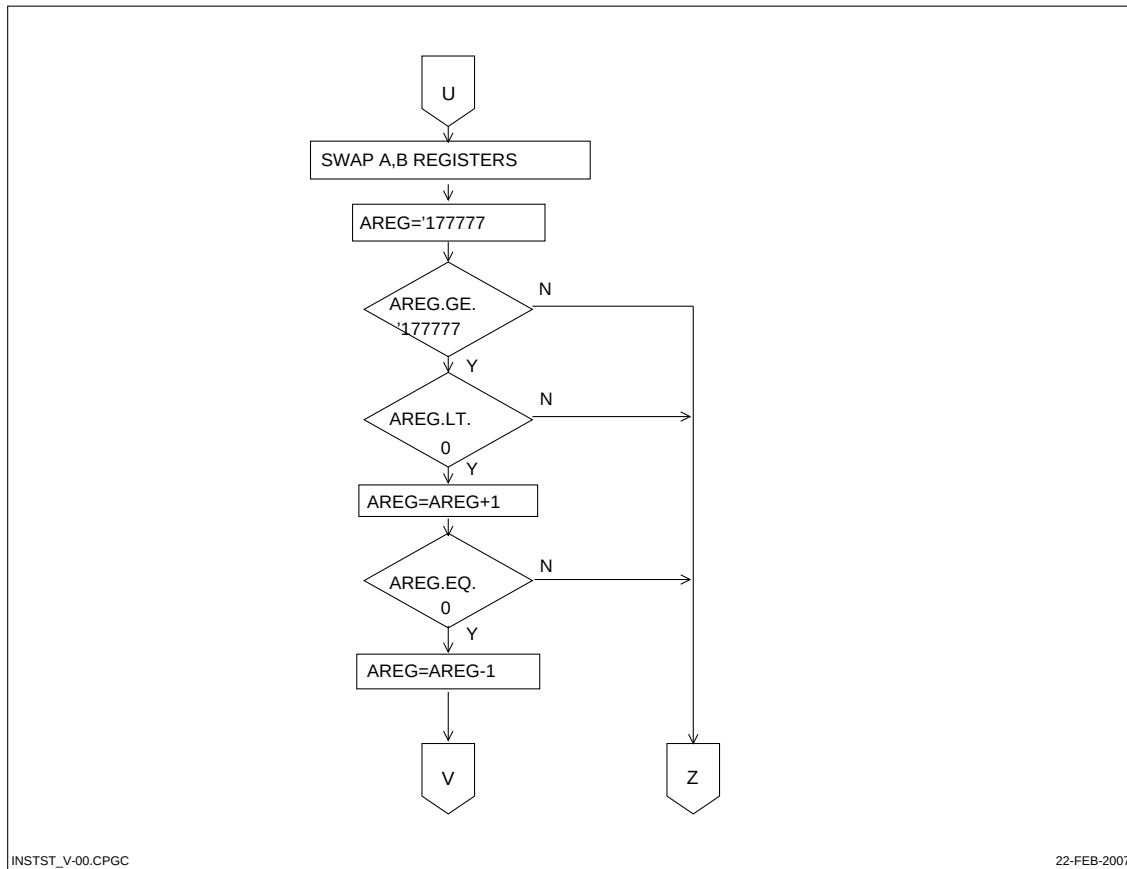
13.00 Figure 483: INSTST_T



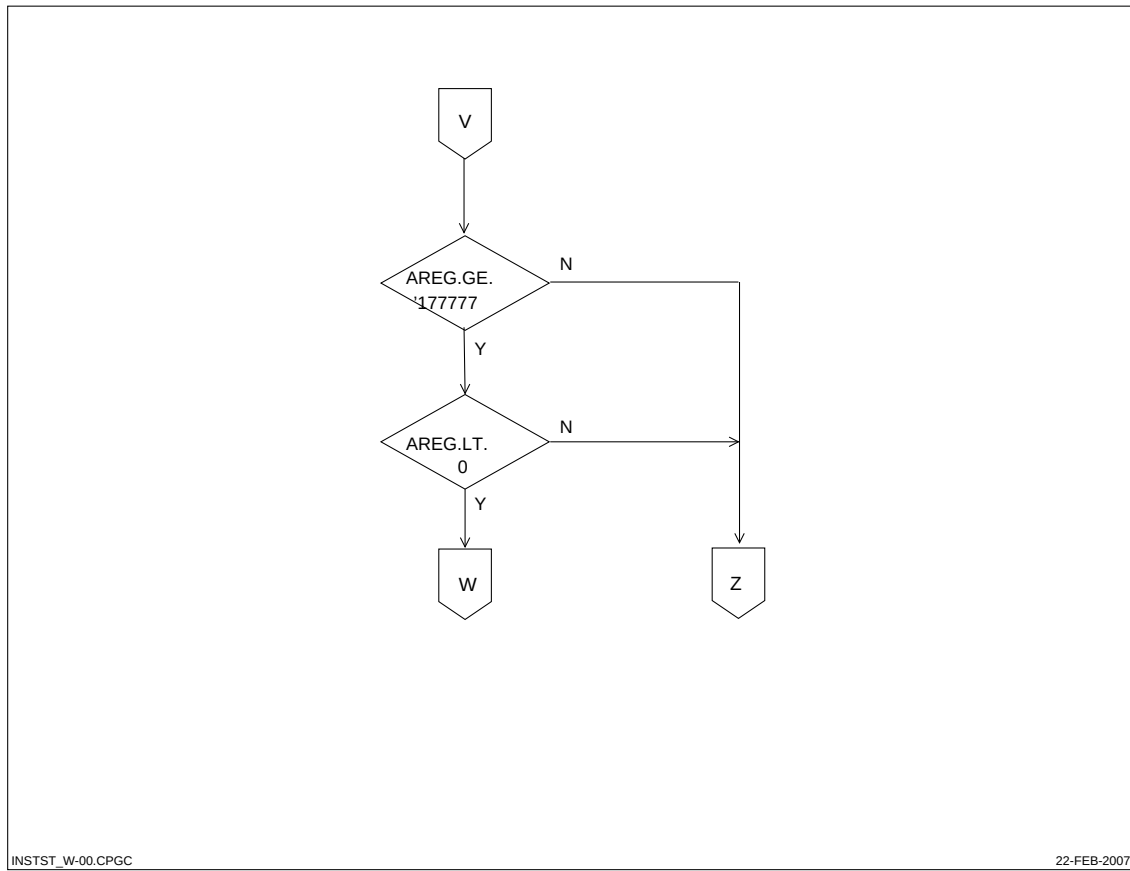
13.00 Figure 484: INSTST_U



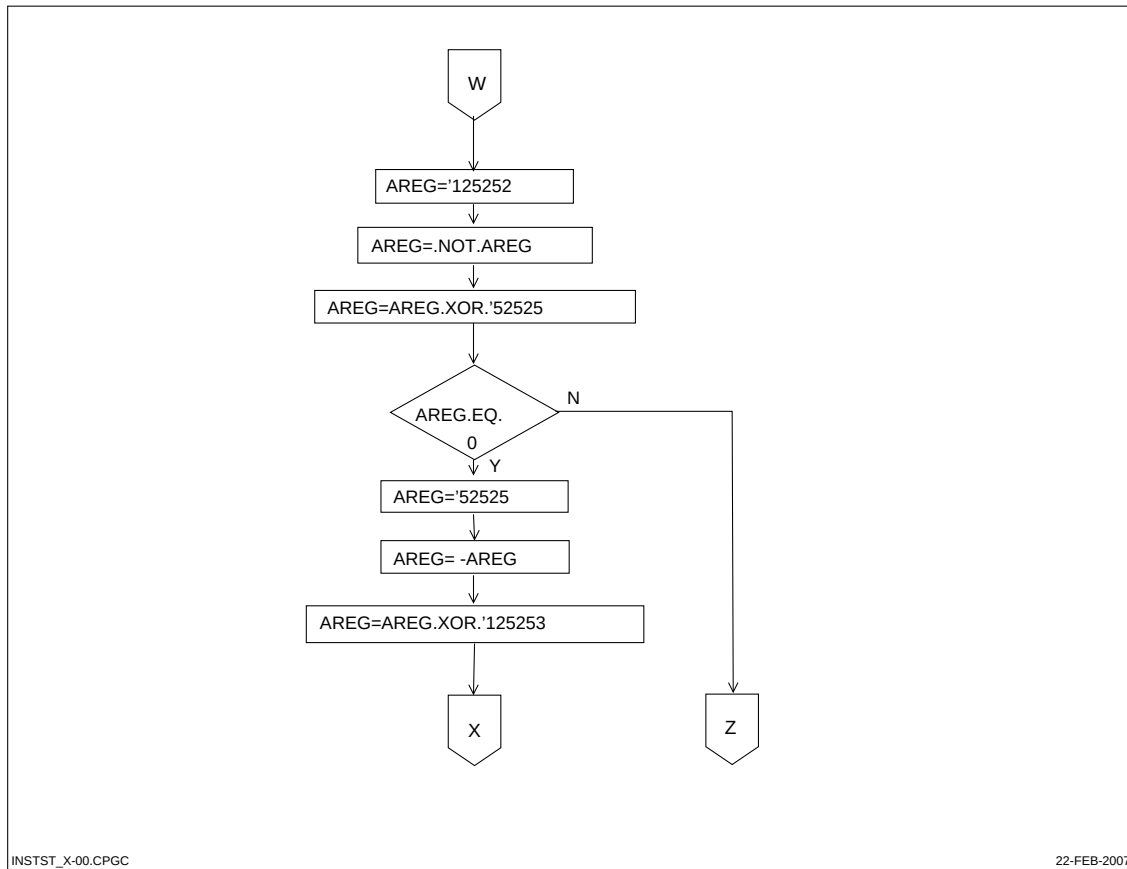
13.00 Figure 485: INSTST_V



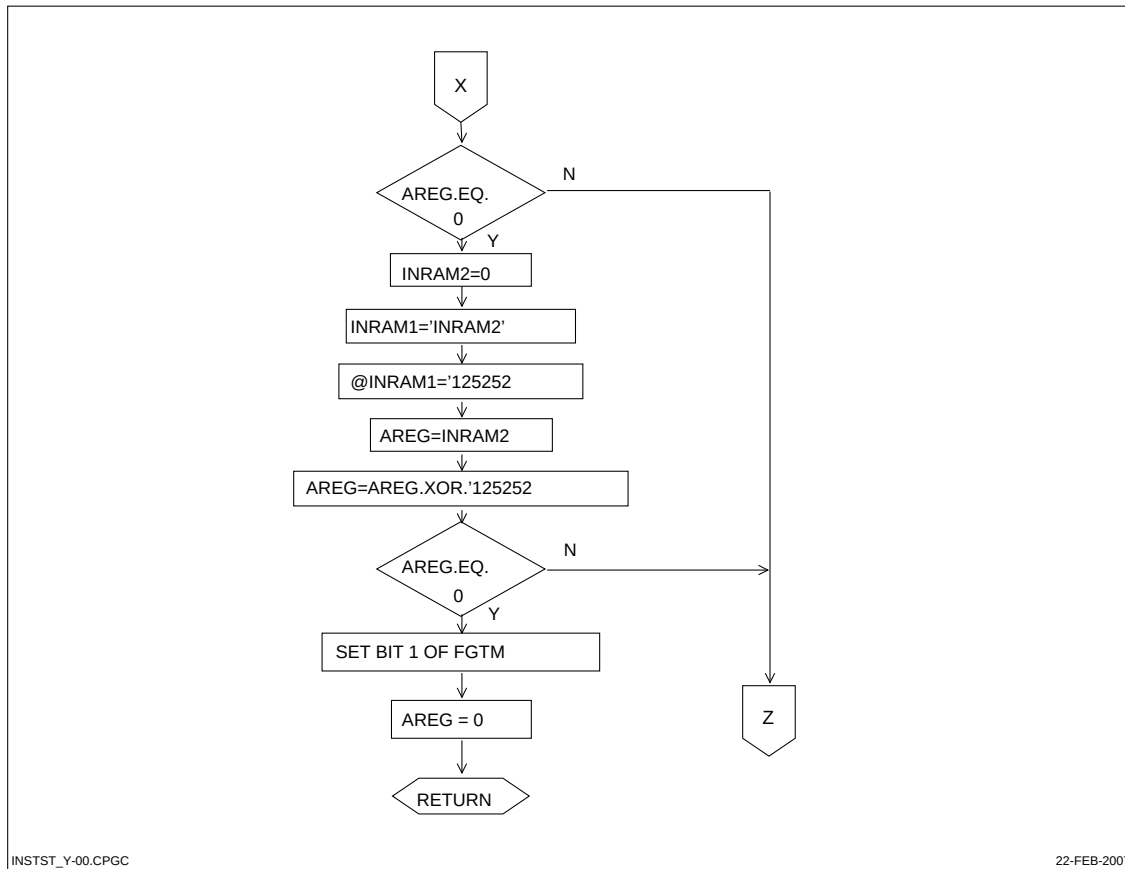
13.00 Figure 486: INSTST_W



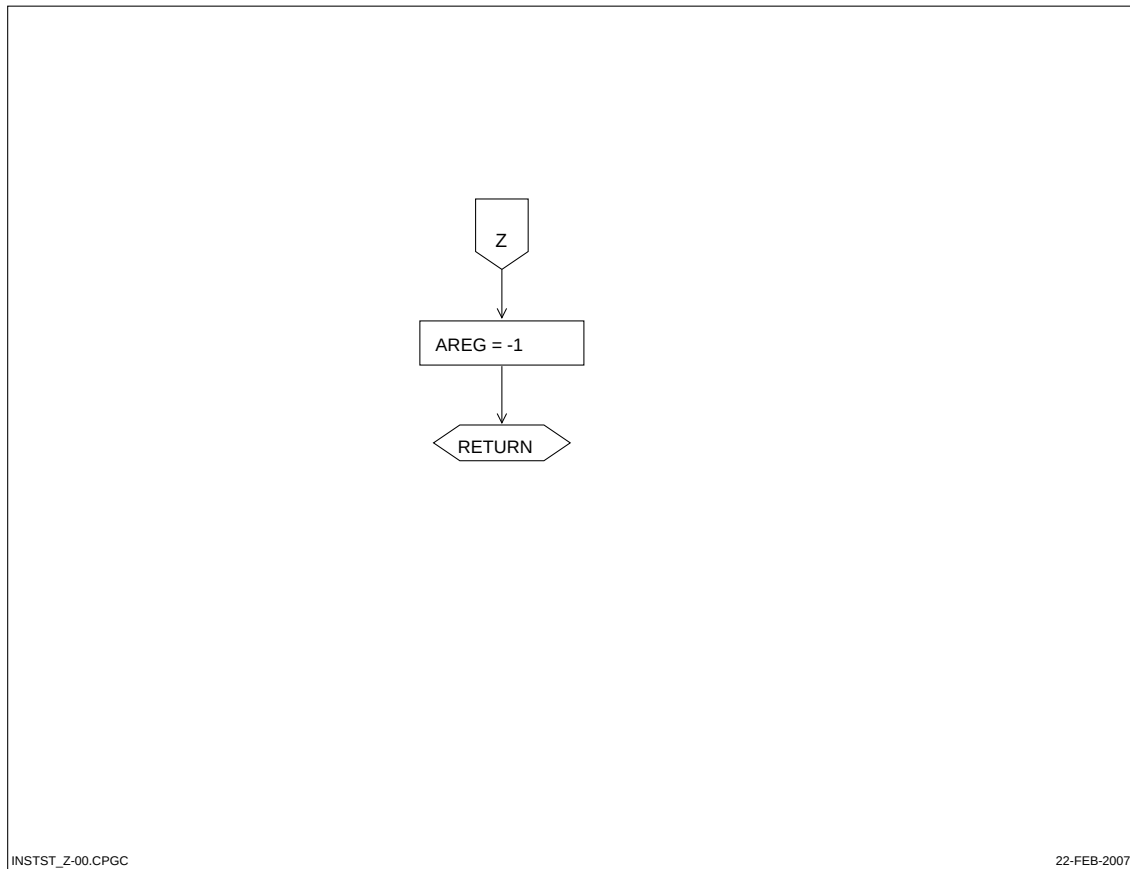
13.00 Figure 487: INSTST_X



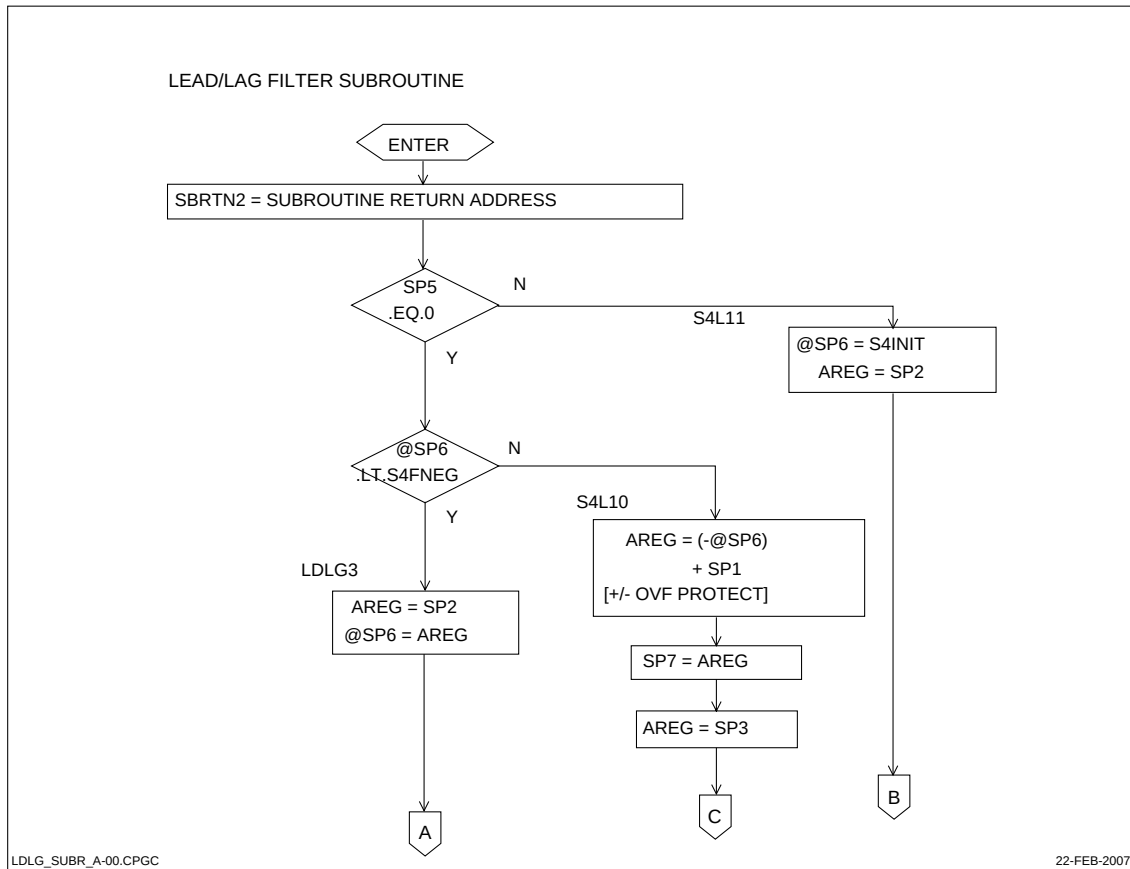
13.00 Figure 488: INSTST_Y



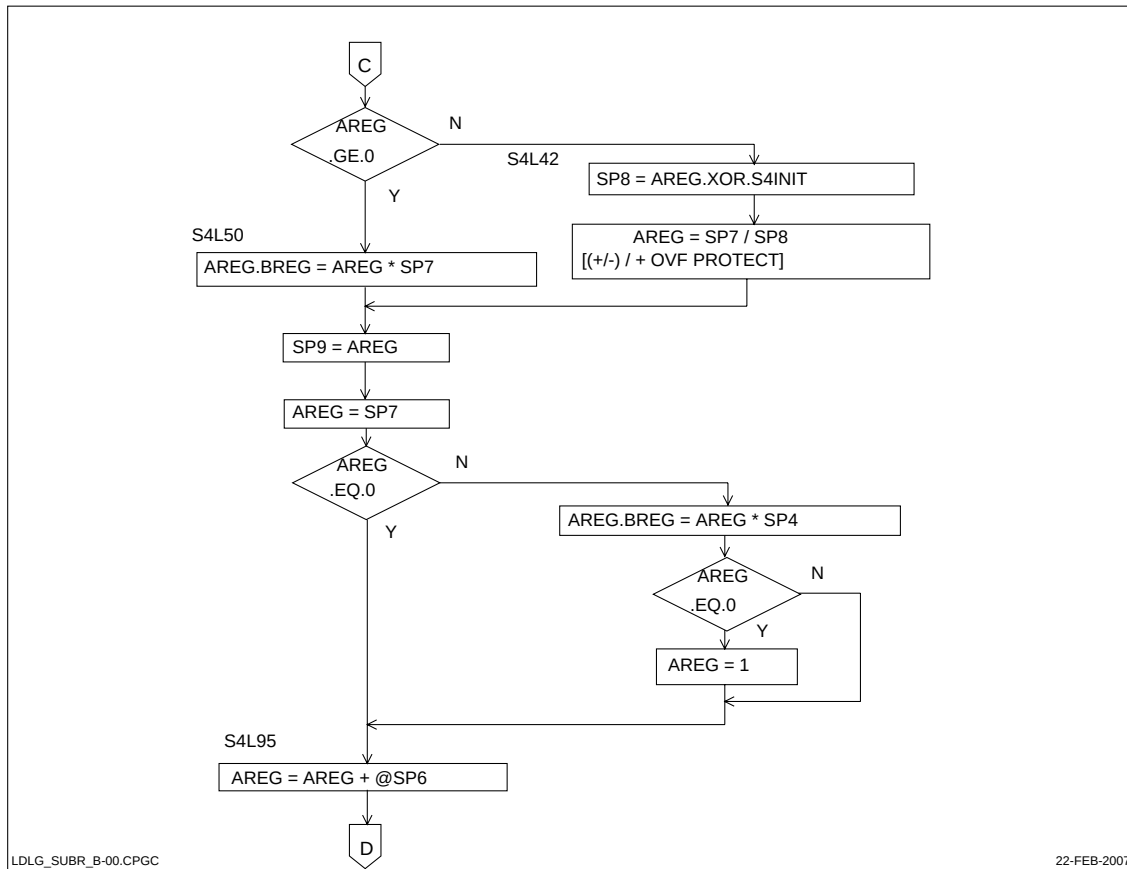
13.00 Figure 489: INSTST_Z



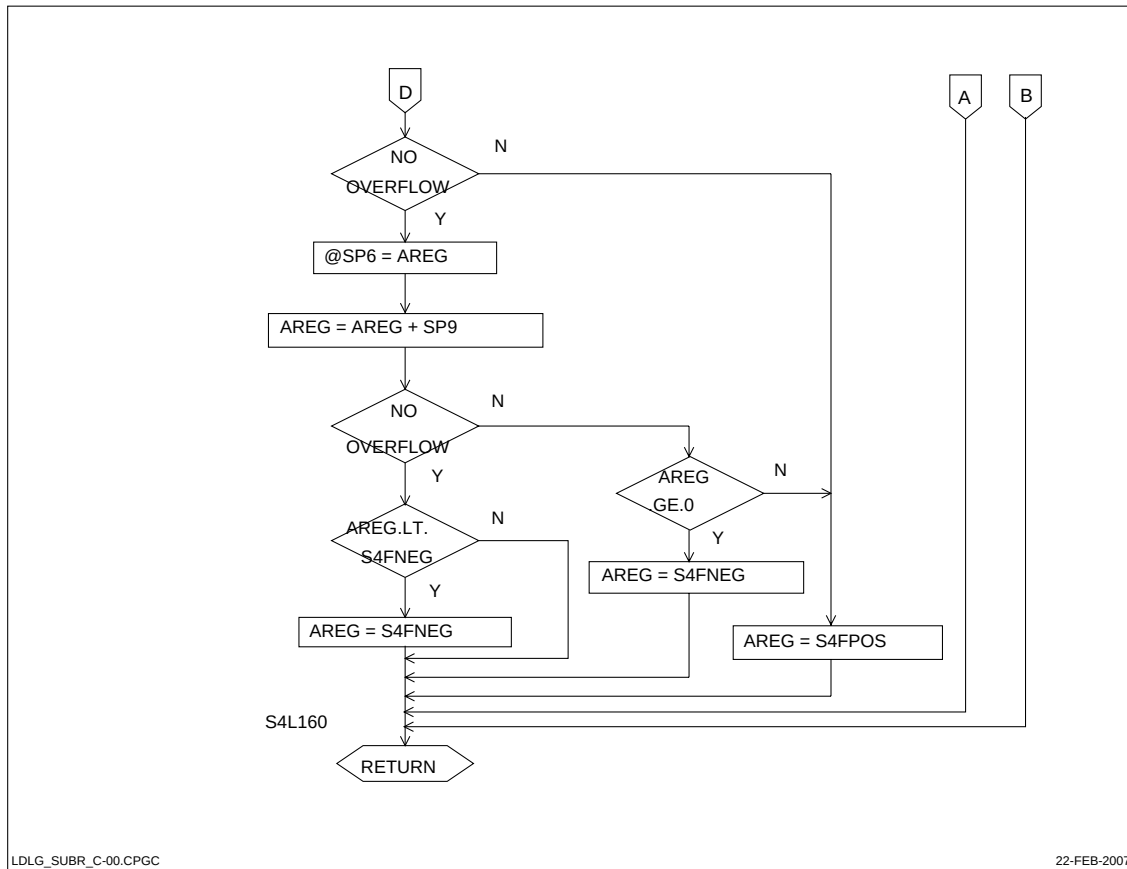
13.00 Figure 490: LDLG_SUBR_A



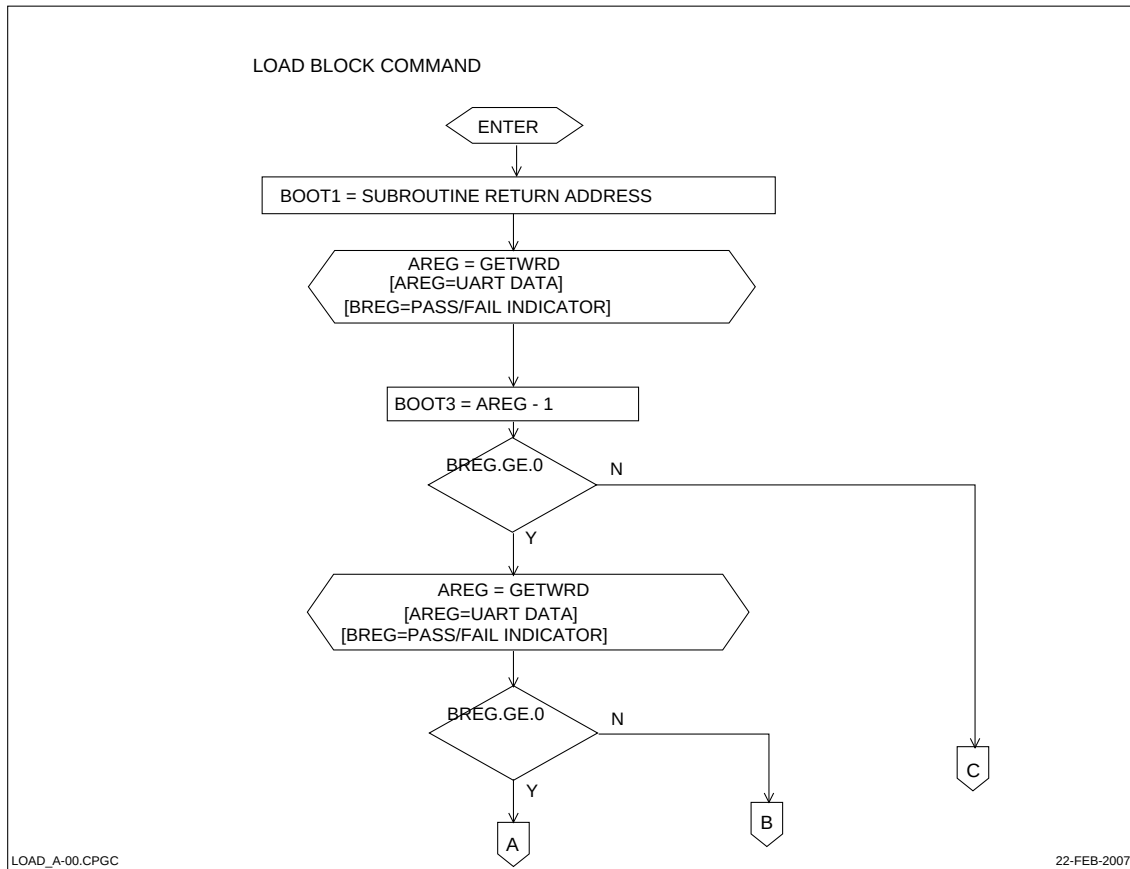
13.00 Figure 491: LDLG_SUBR_B



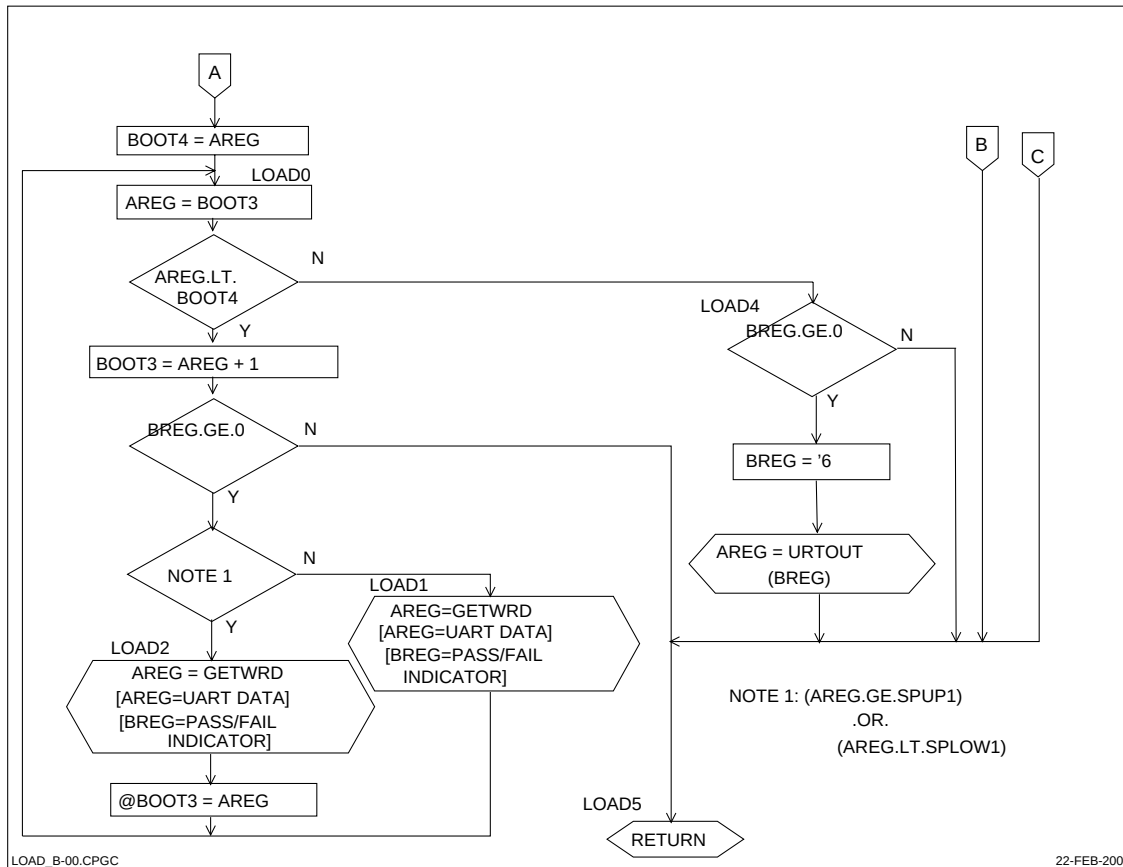
13.00 Figure 492: LDLG _SUBR _C



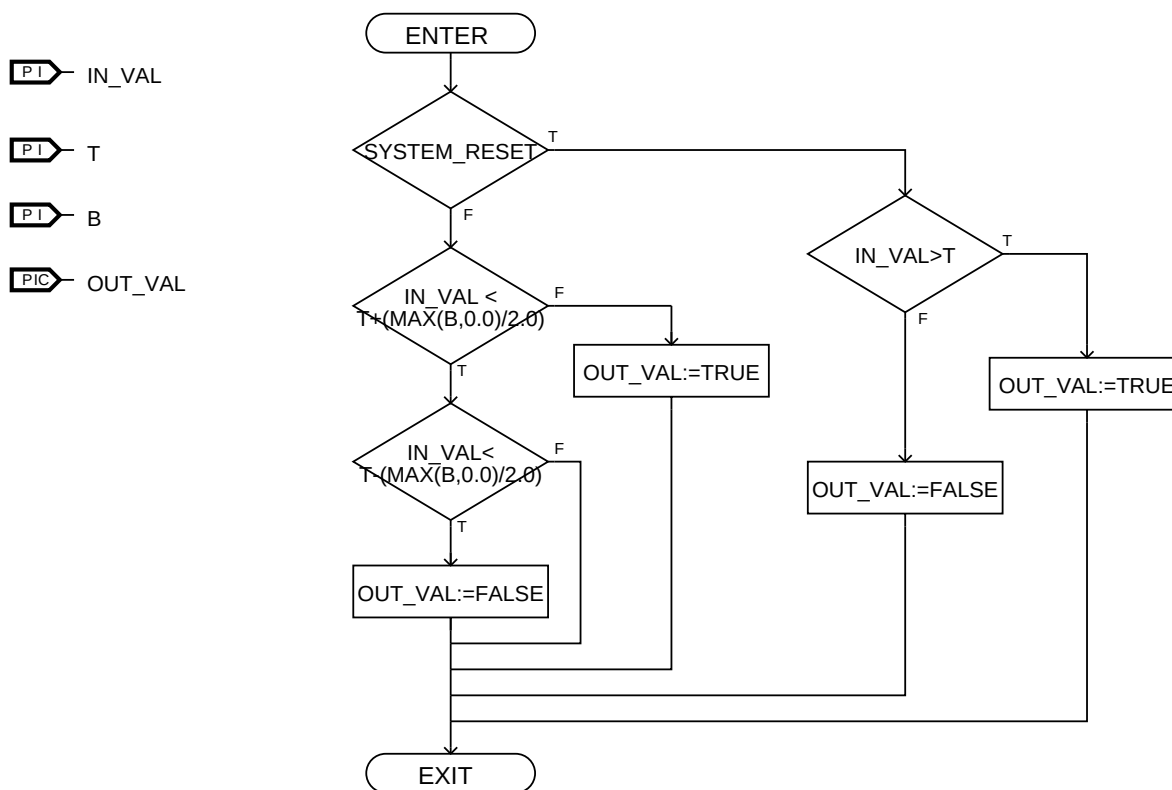
13.00 Figure 493: LOAD _A



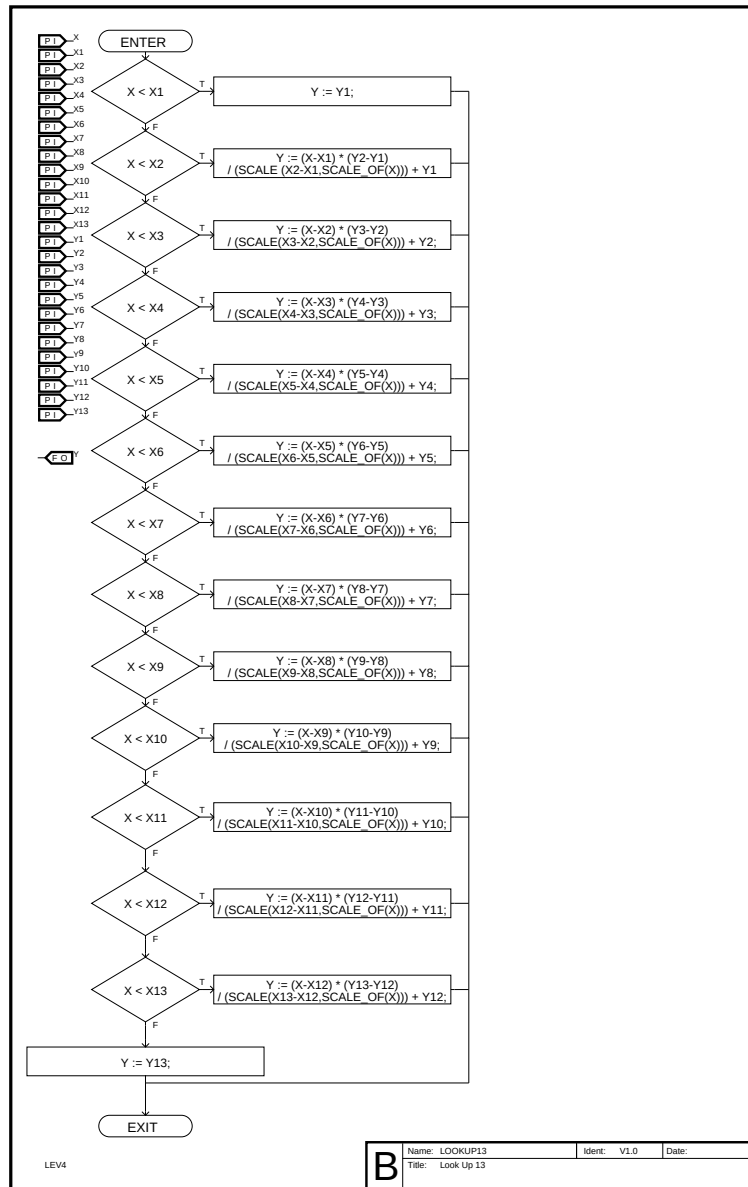
13.00 Figure 494: LOAD_B



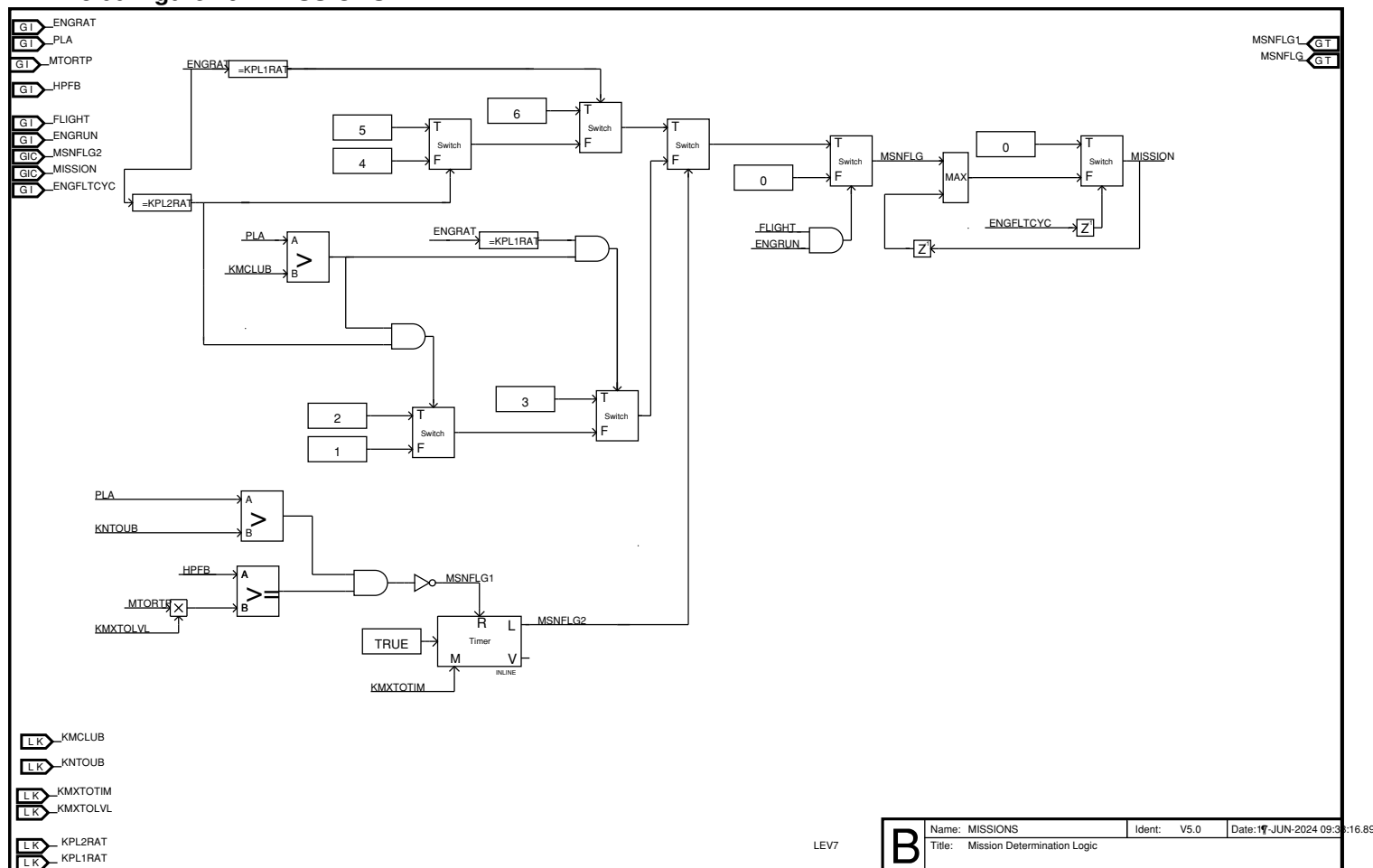
13.00 Figure 495: LOGHYS



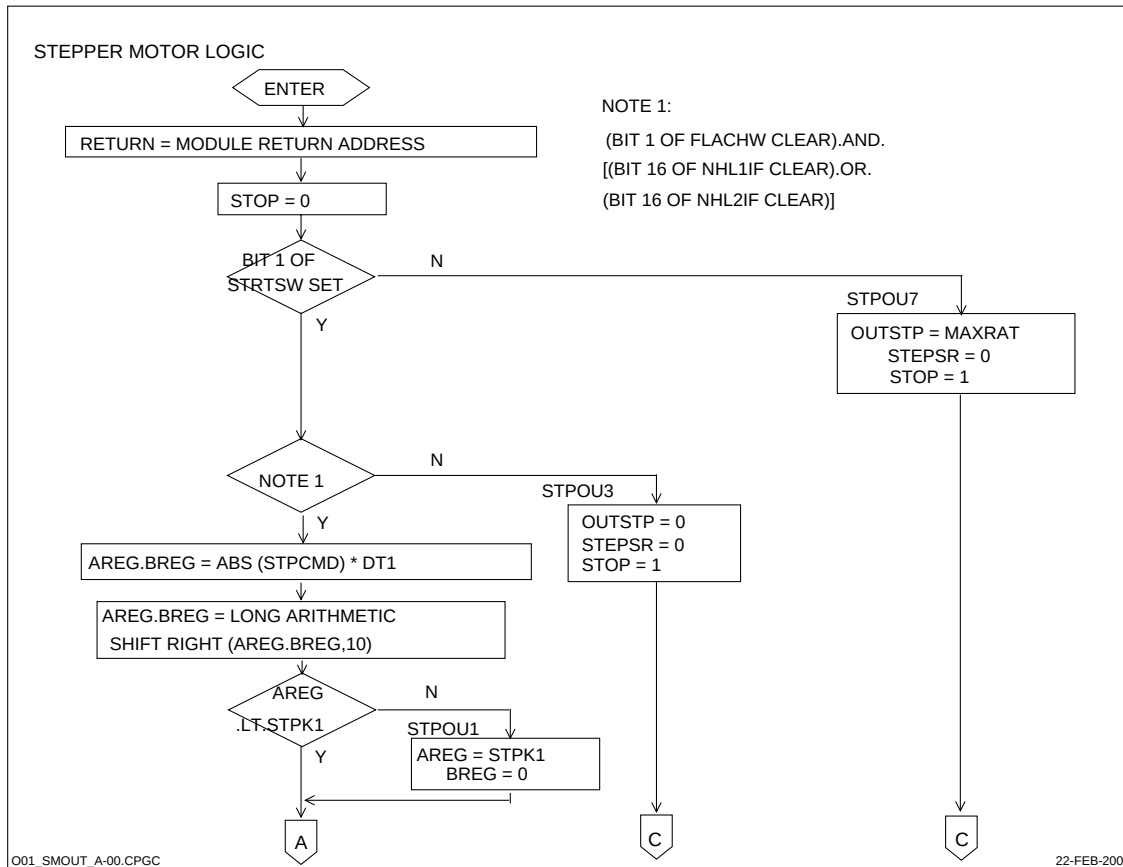
13.00 Figure 496: LOOKUP13



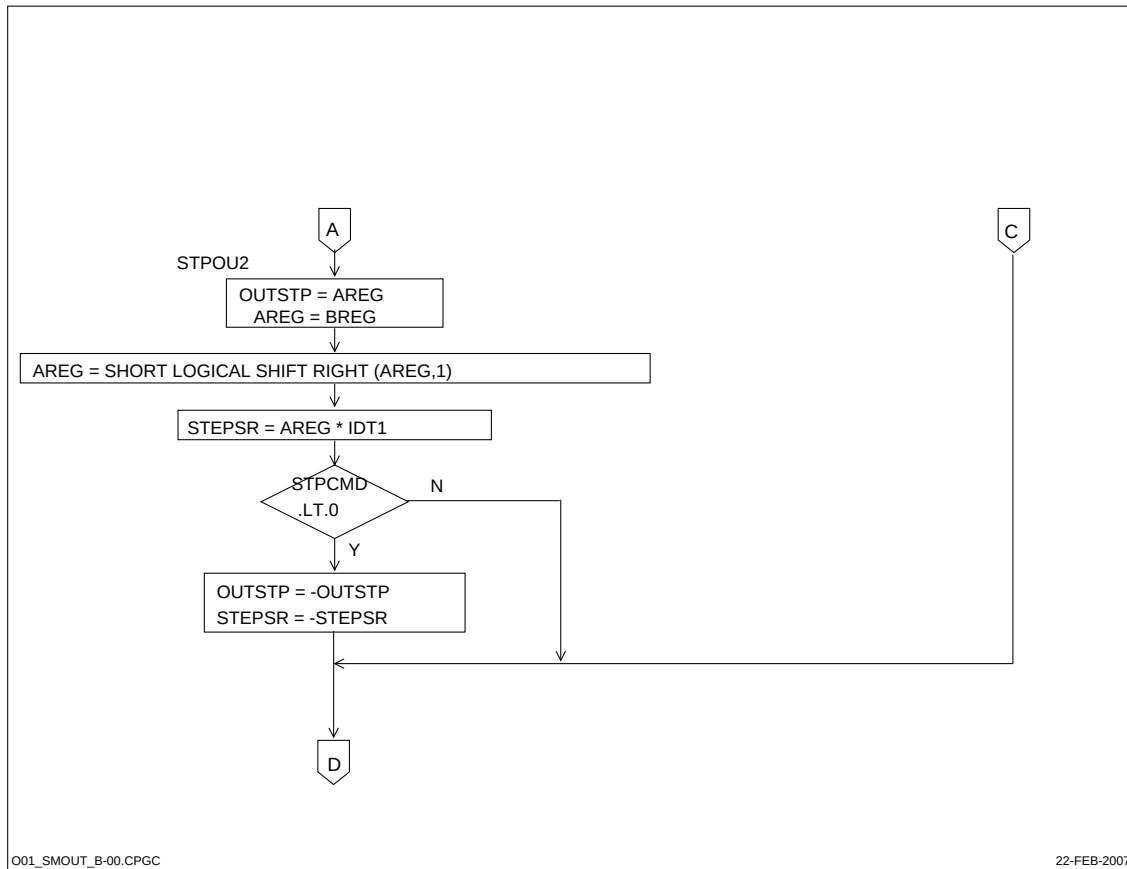
13.00 Figure 497: MISSIONS



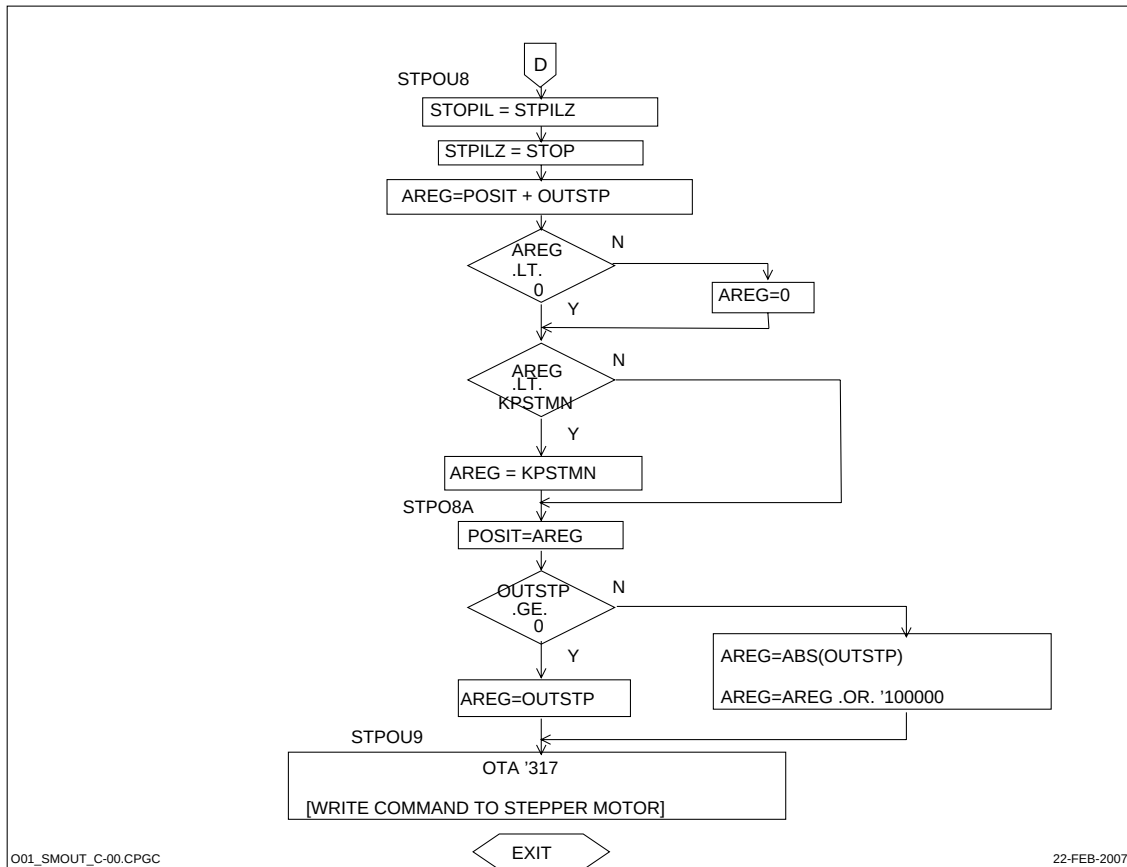
13.00 Figure 498: O01_SMOUT_A



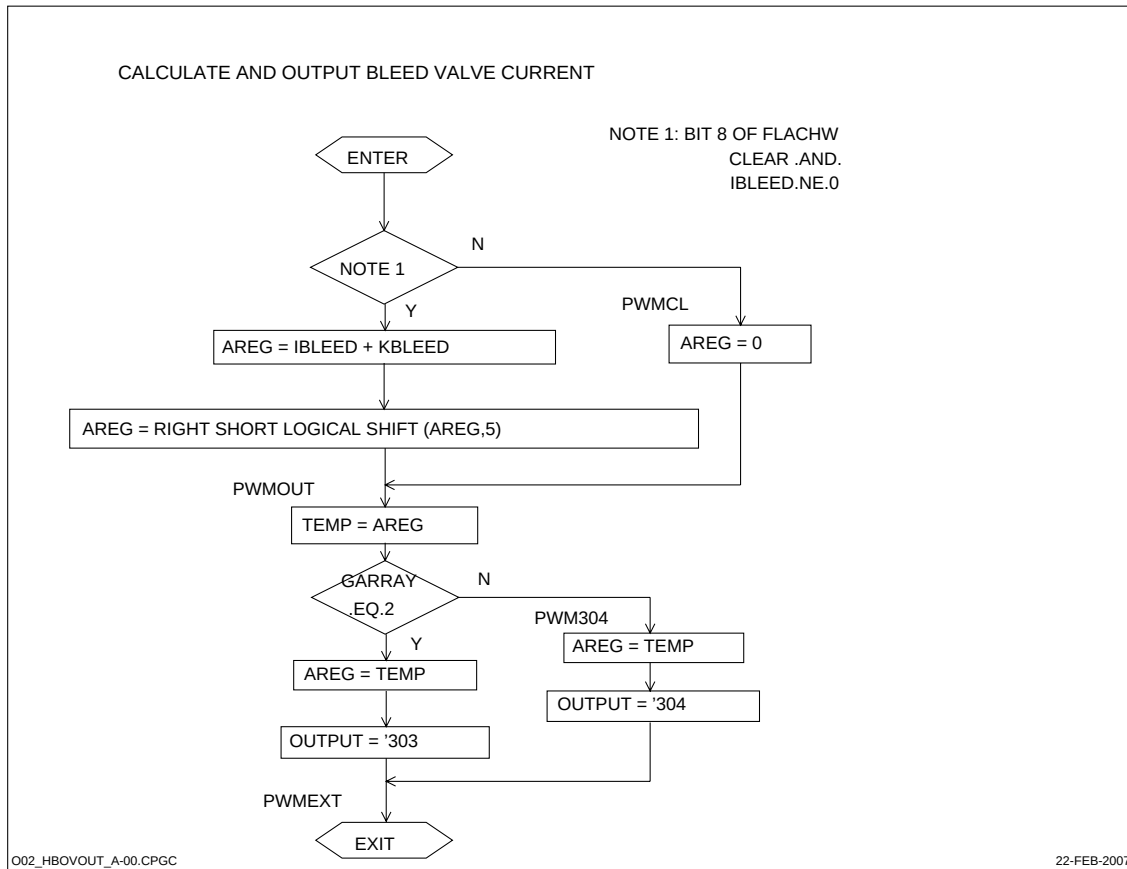
13.00 Figure 499: O01_SMOUT_B



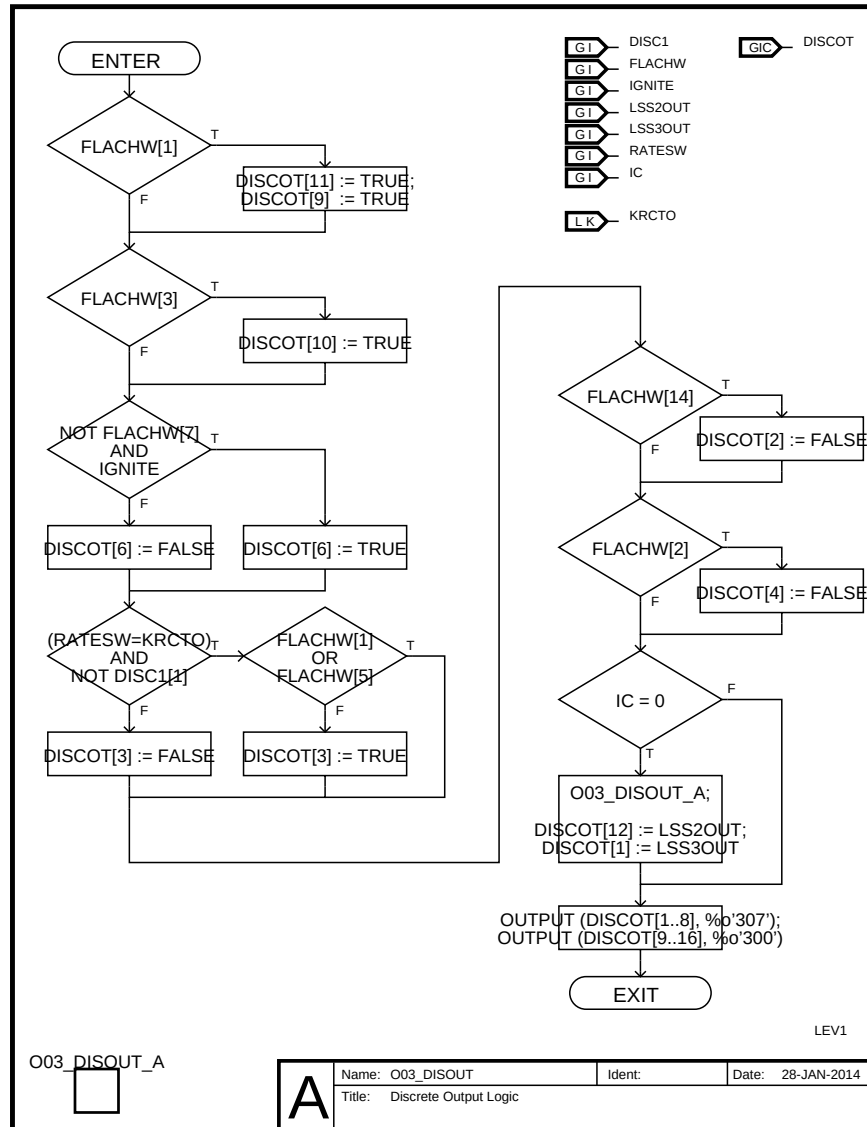
13.00 Figure 500: O01 _SMOUT _C



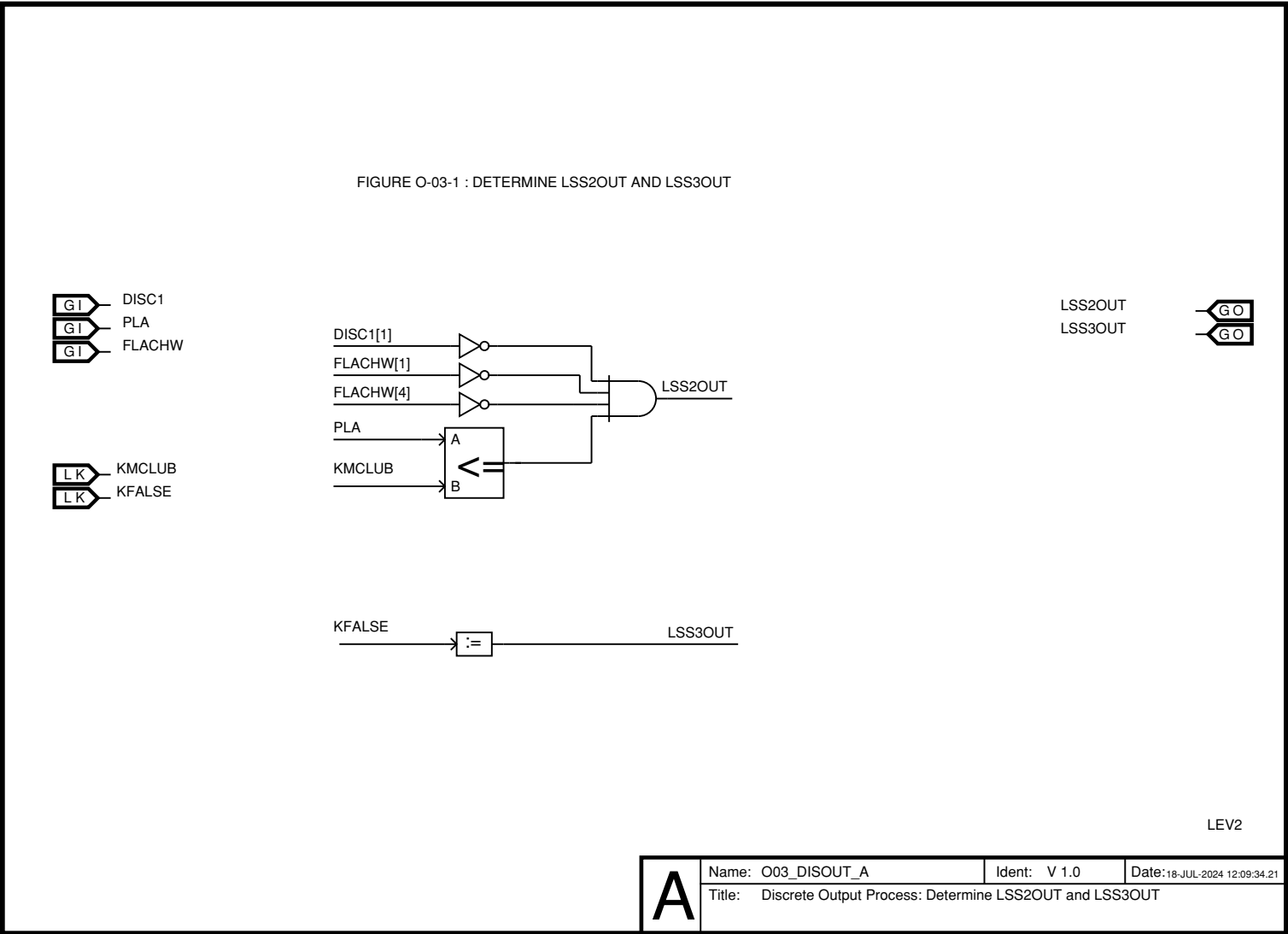
13.00 Figure 501: O02_HBOVOUT_A



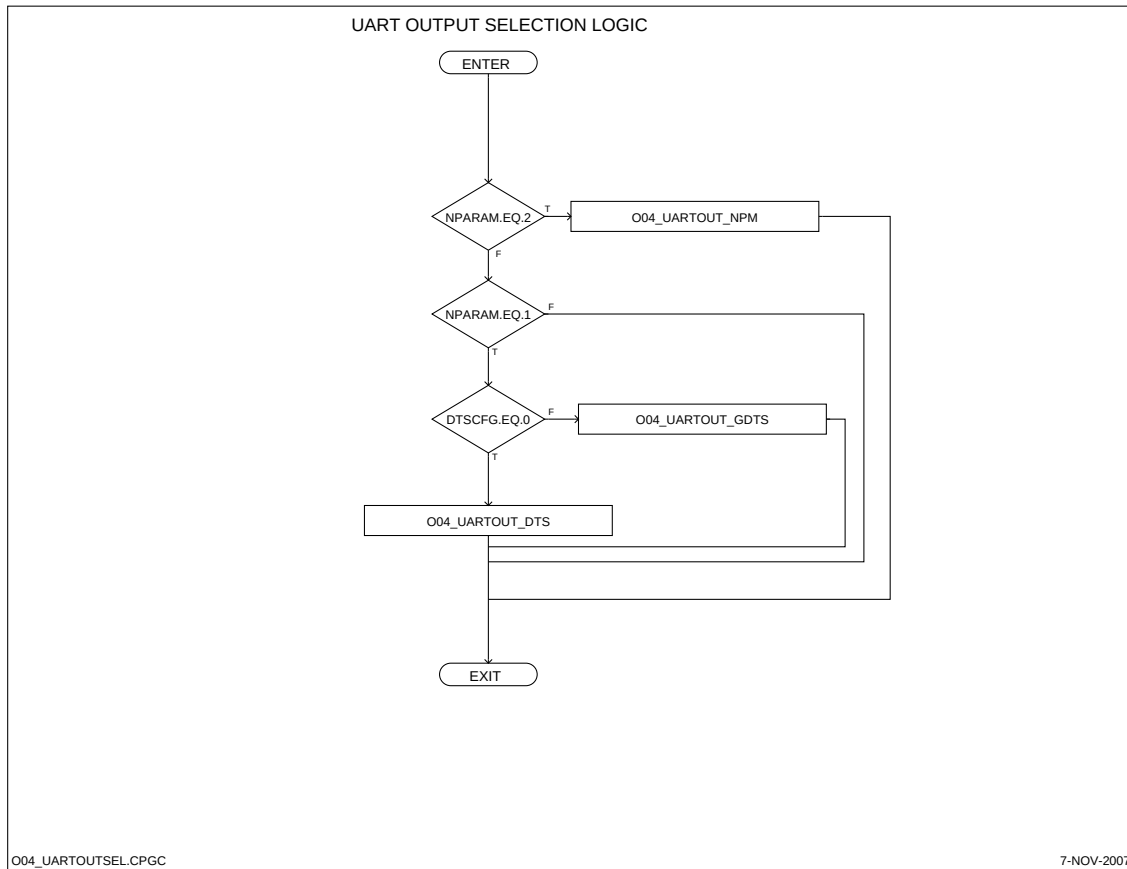
13.00 Figure 502: O03_DISOUT



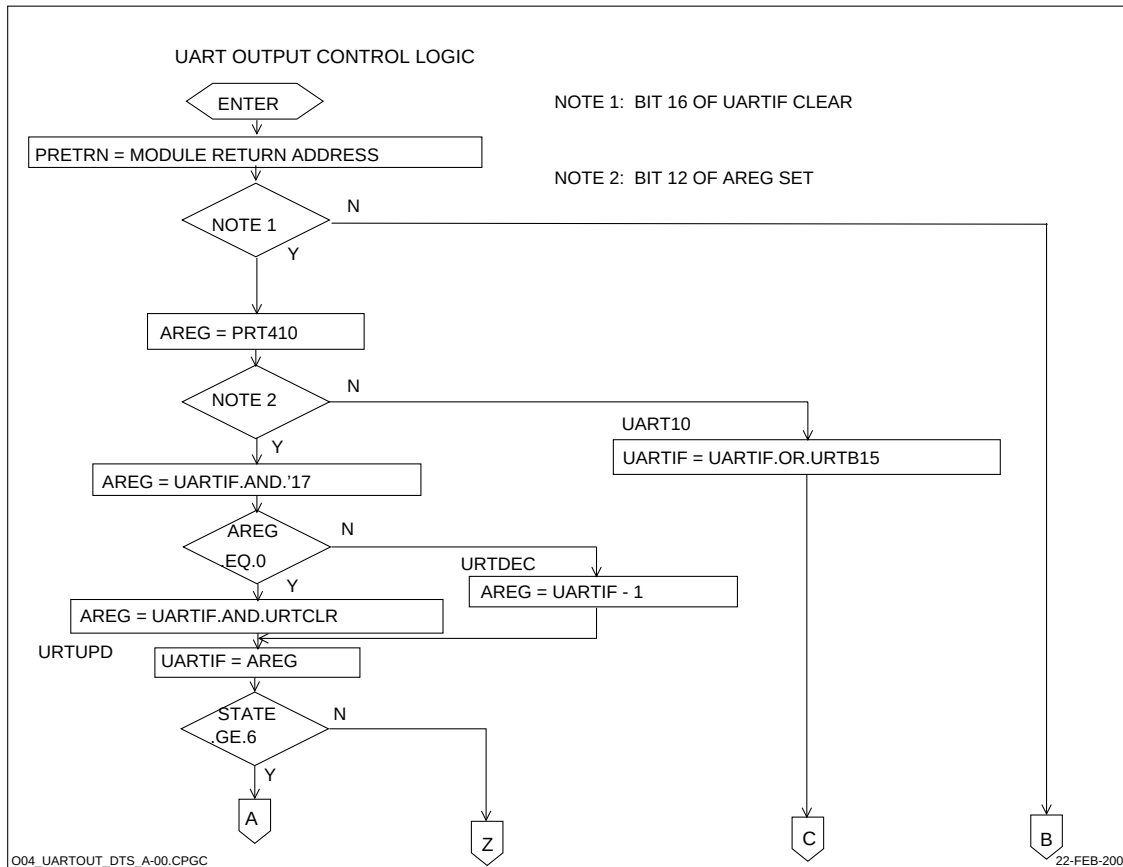
13.00 Figure 503: O03_DISOUT_A



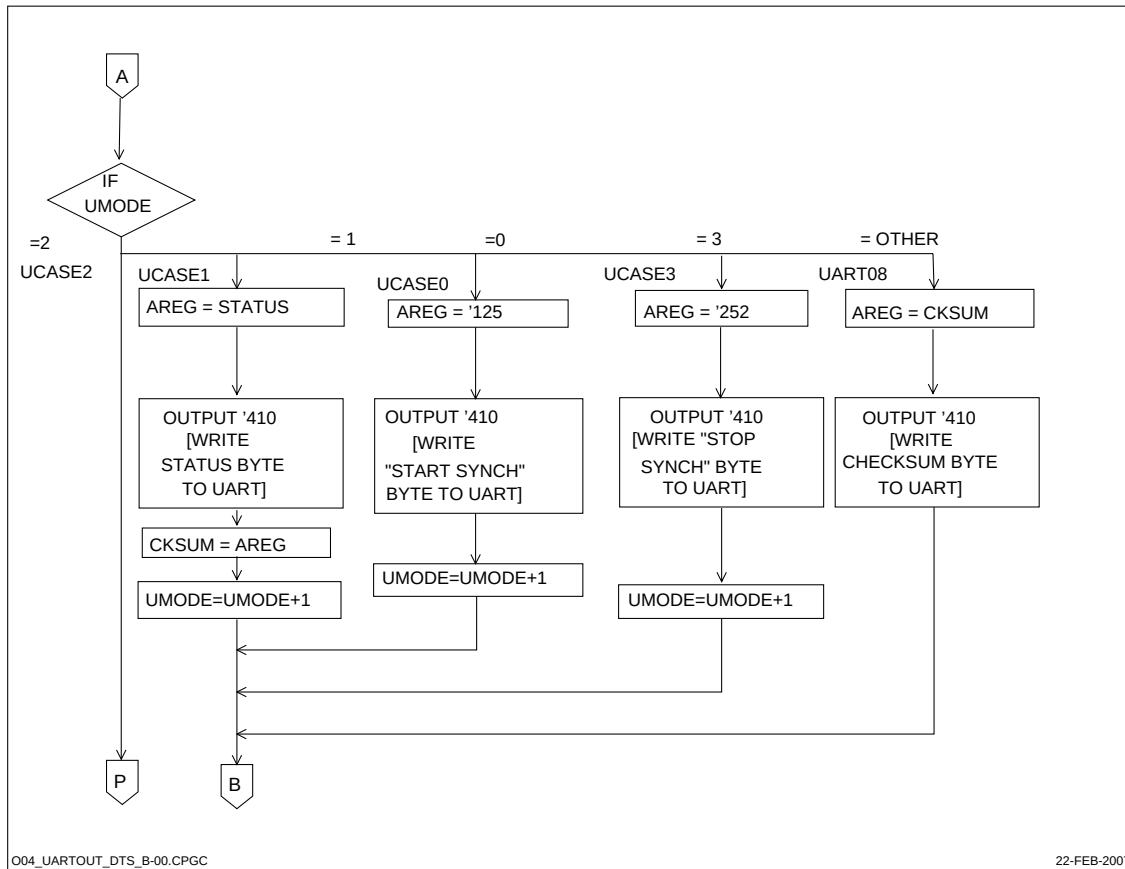
13.00 Figure 504: O04_UARTOUTSEL



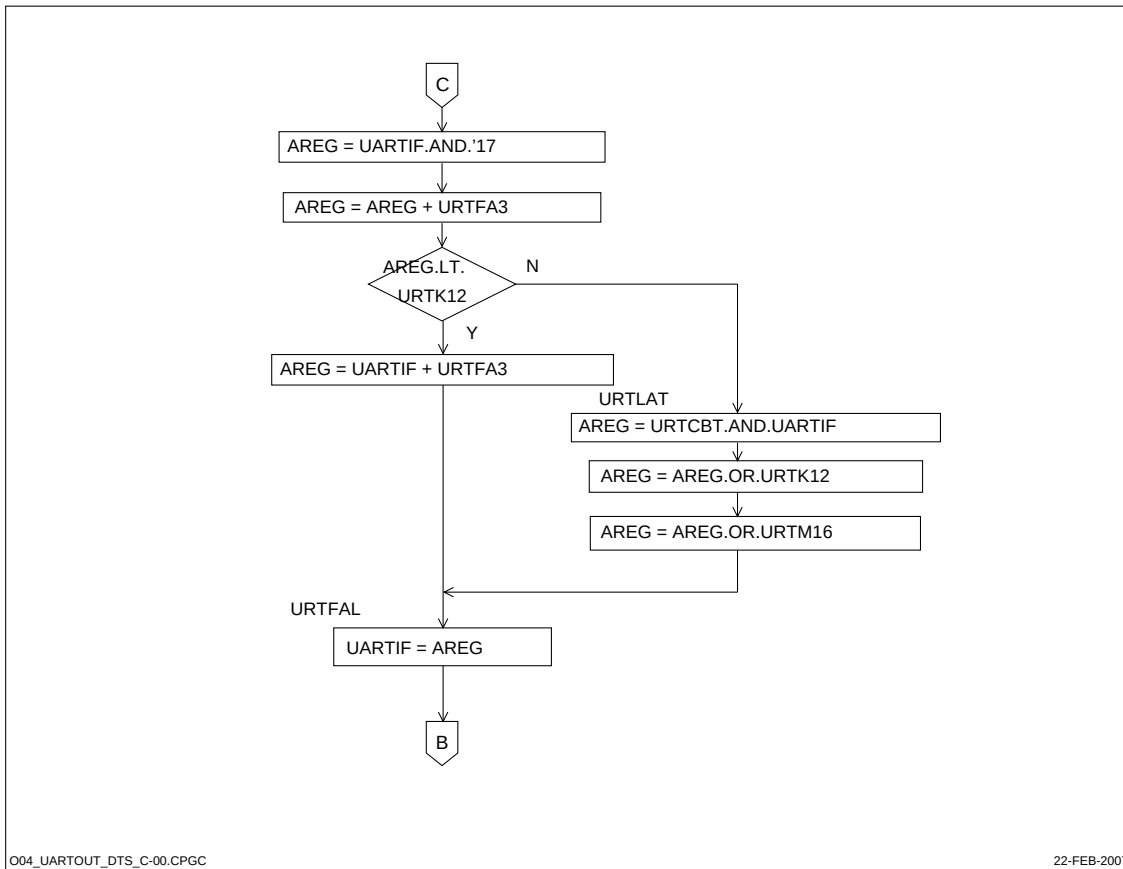
13.00 Figure 505: O04_UARTOUT_DTS_A



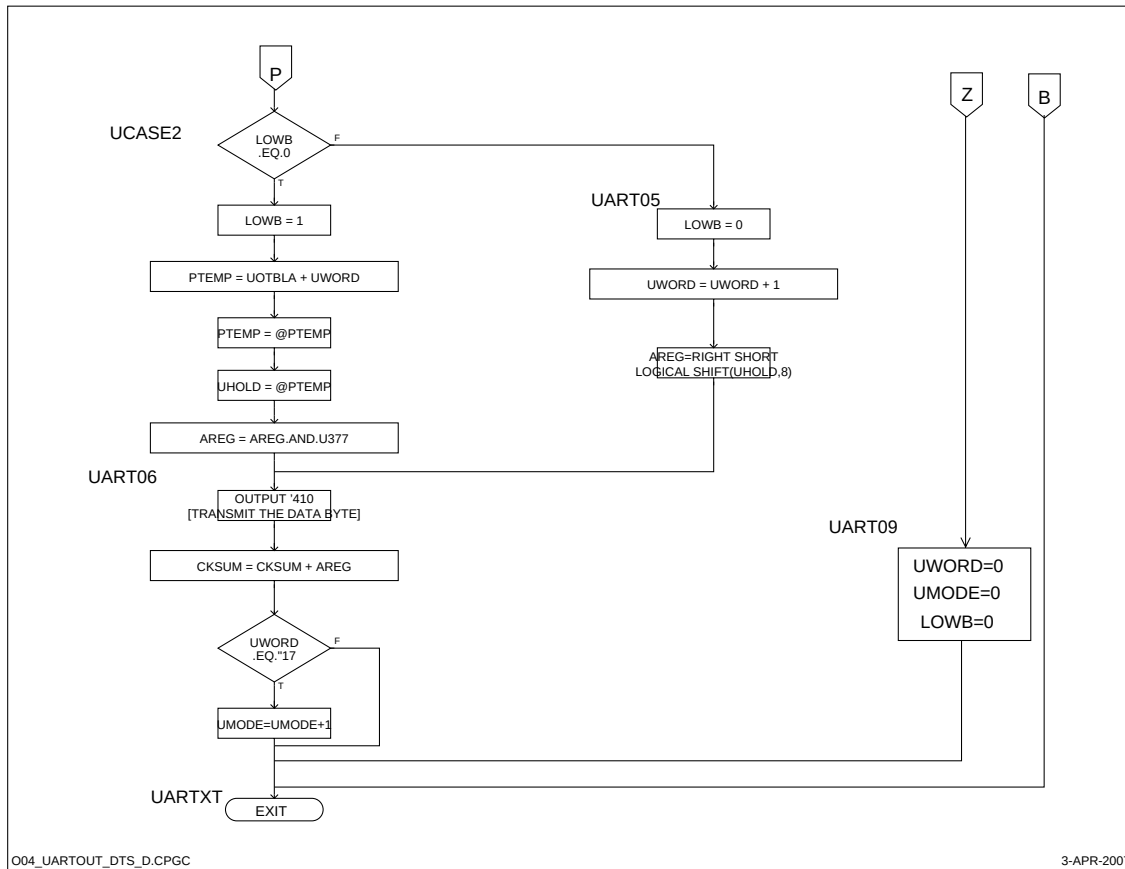
13.00 Figure 506: O04_UARTOUT_DTS_B



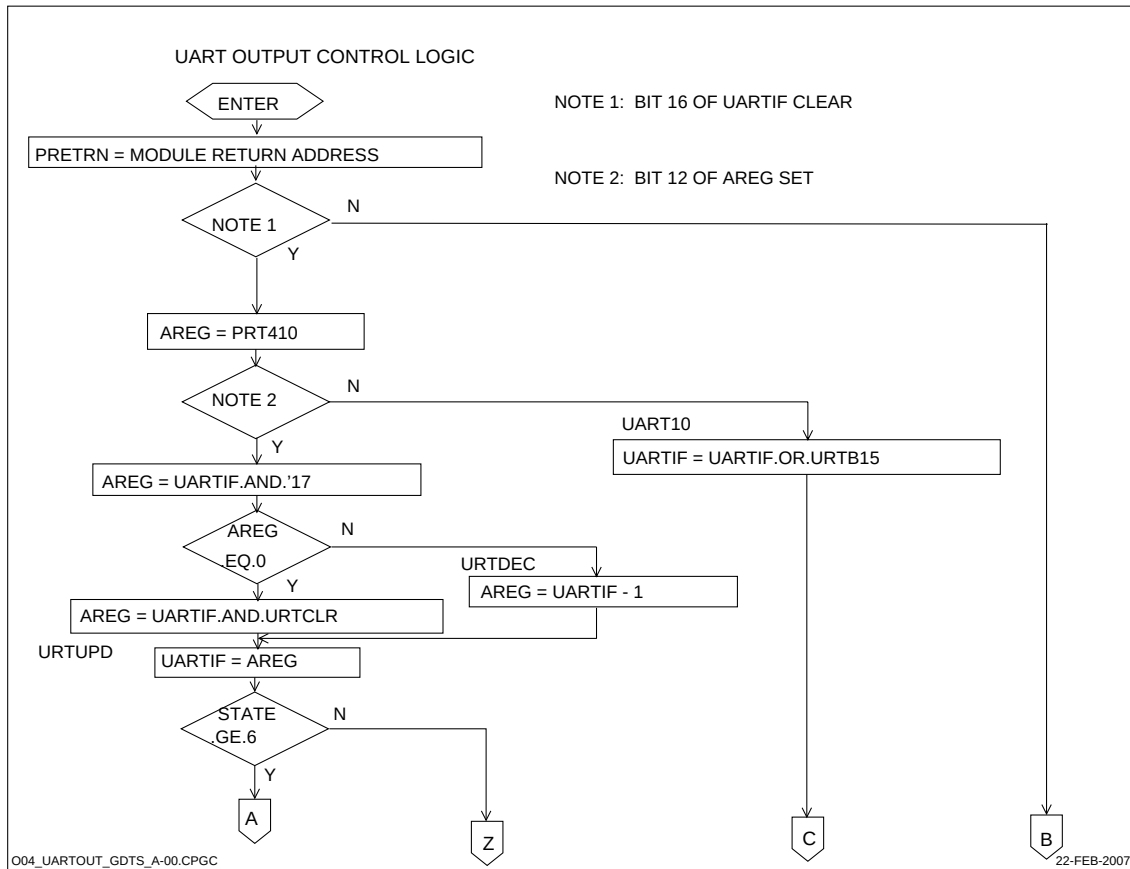
13.00 Figure 507: O04_UARTOUT_DTS_C



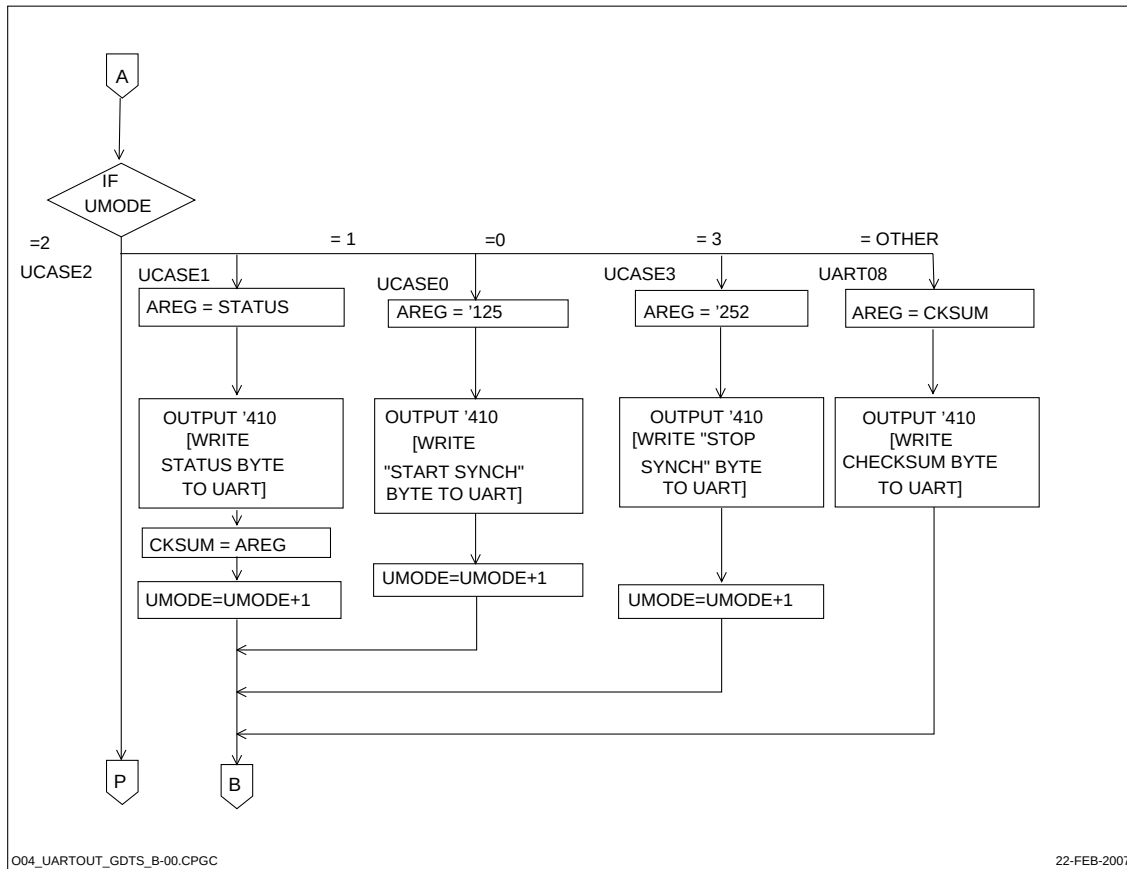
13.00 Figure 508: O04_UARTOUT_DTS_D



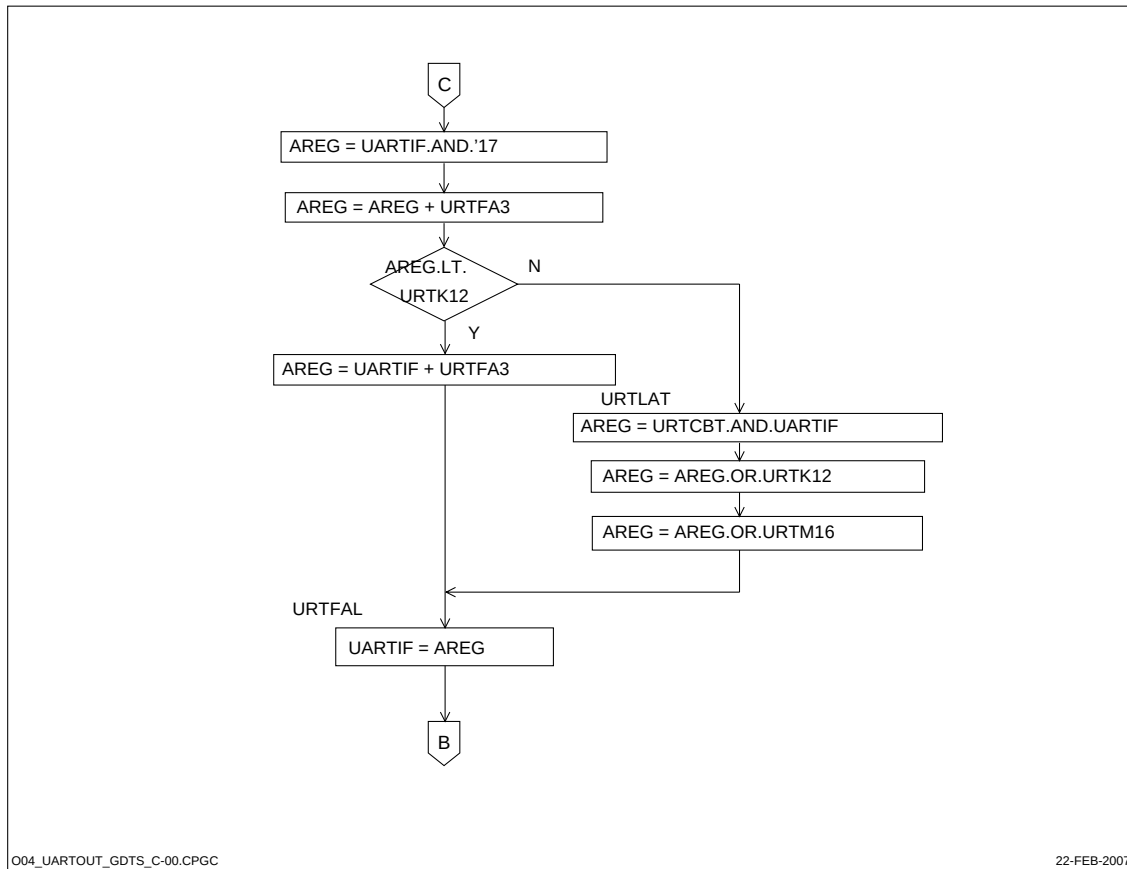
13.00 Figure 509: O04_UARTOUT_GDTS_A



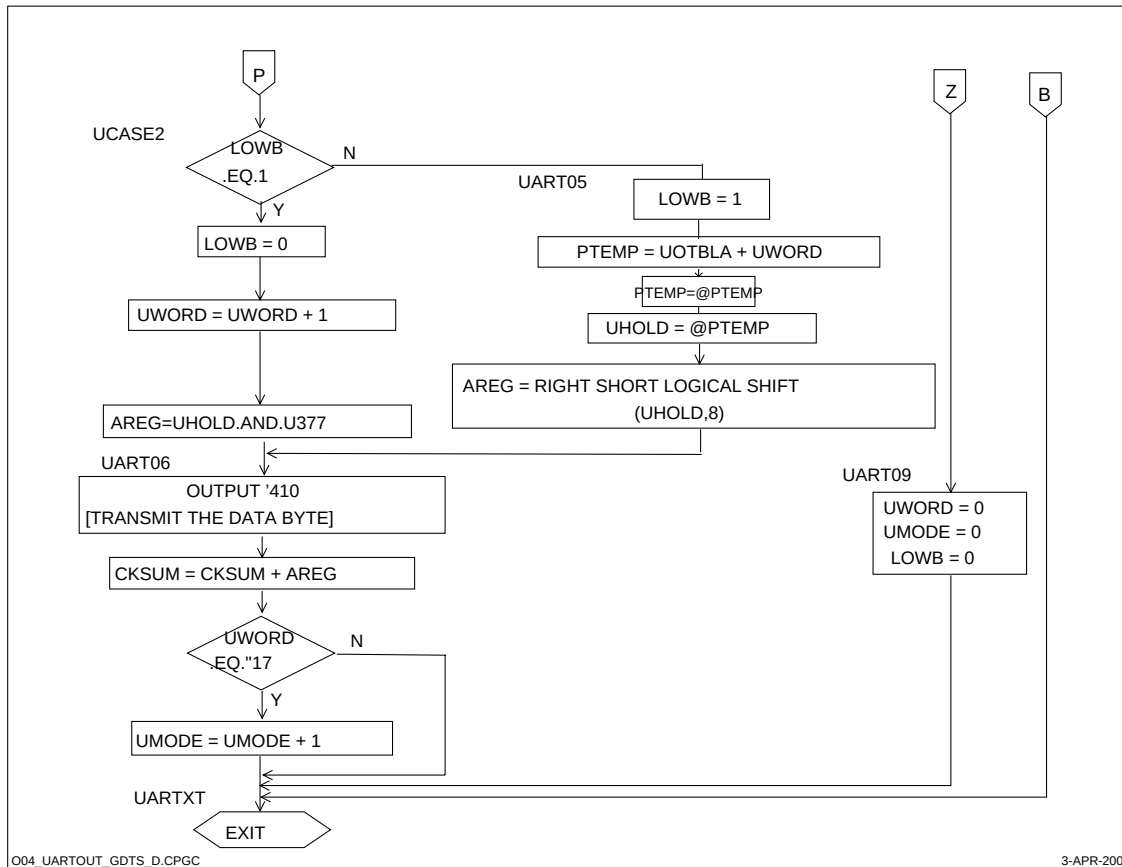
13.00 Figure 510: O04_UARTOUT_GDTS_B



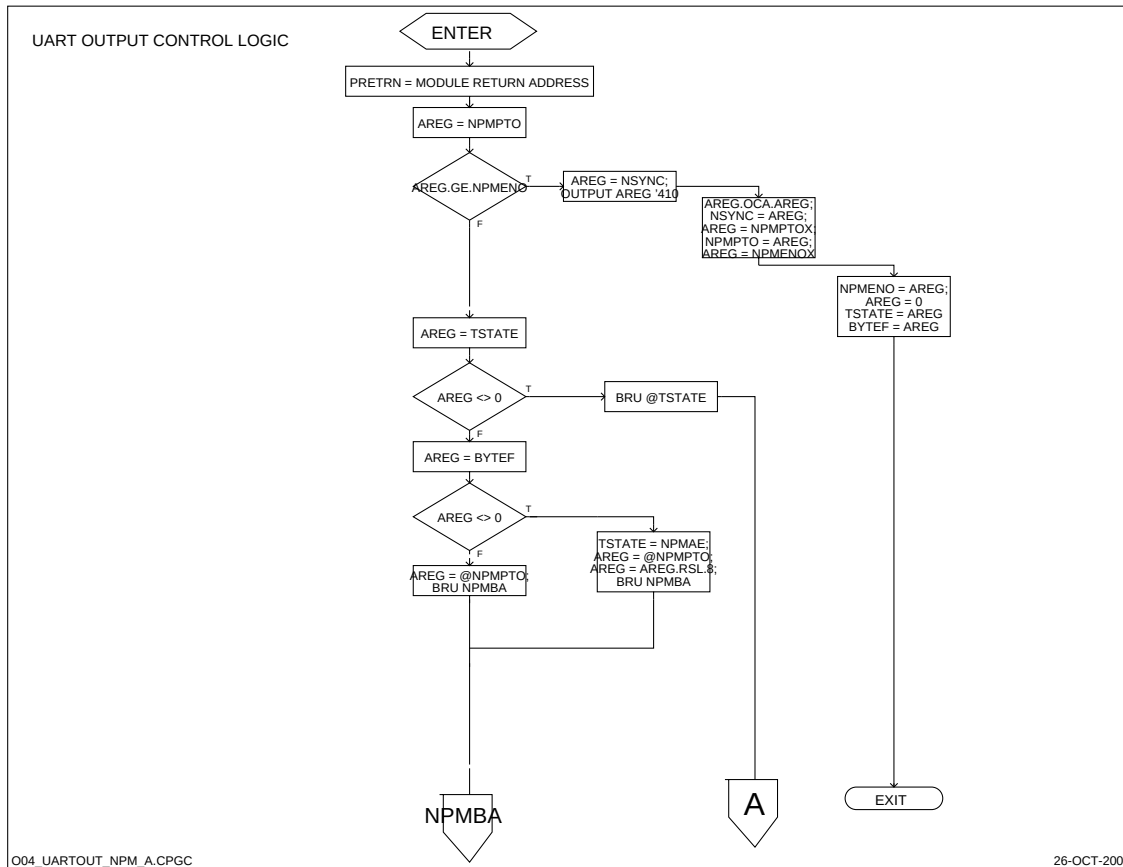
13.00 Figure 511: O04 _UARTOUT _GDTS _C



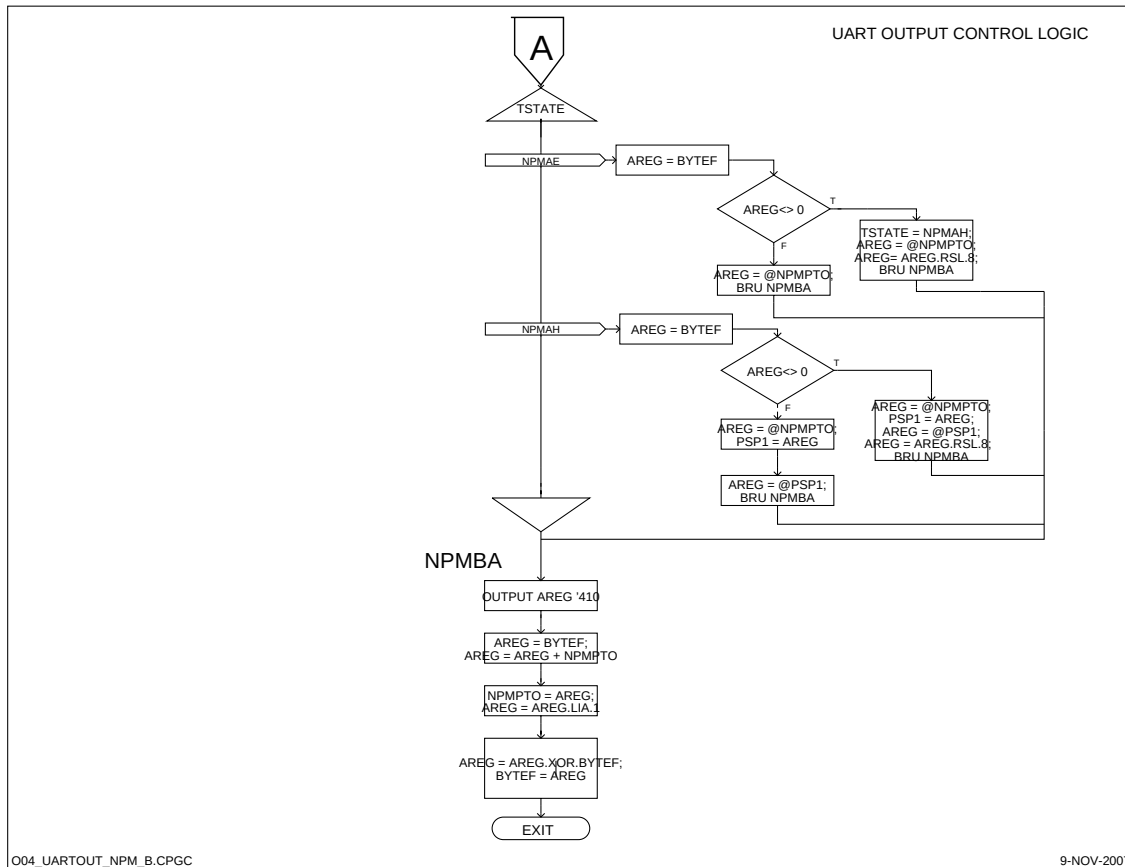
13.00 Figure 512: O04_UARTOUT_GDTS_D



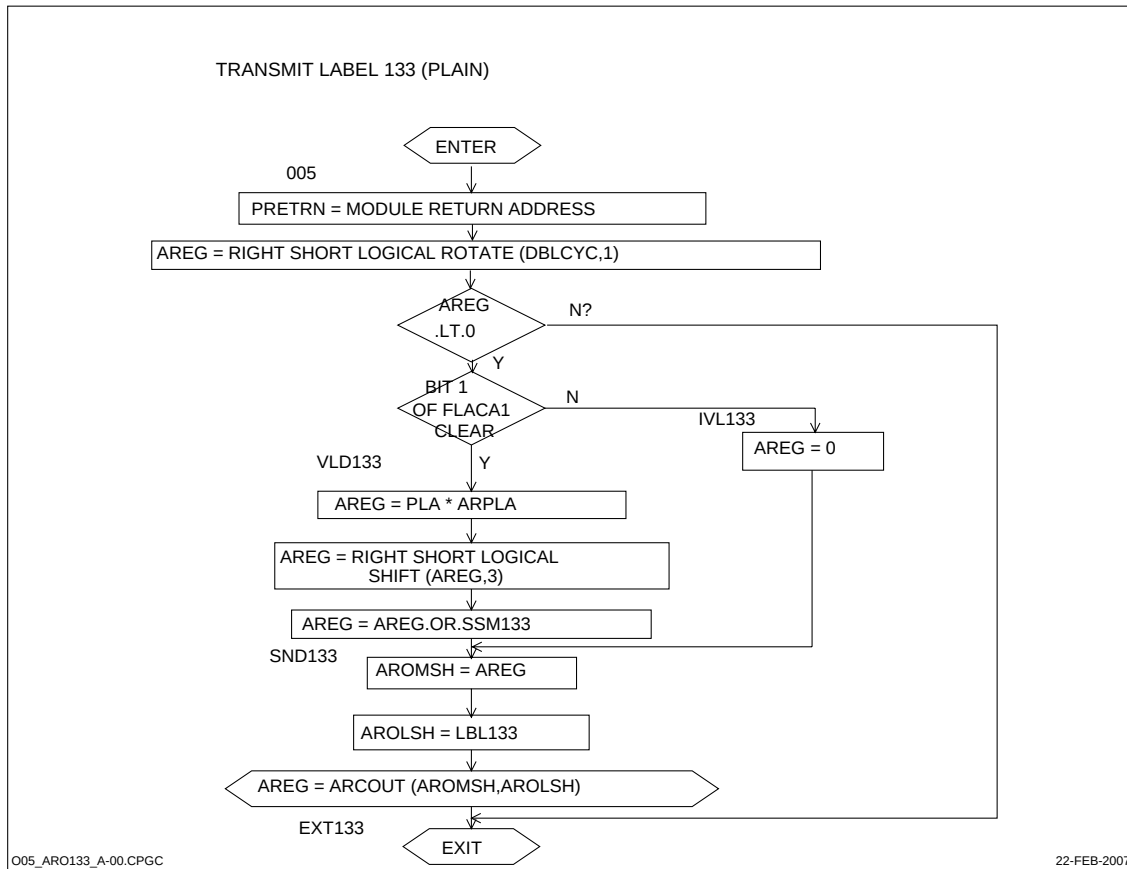
13.00 Figure 513: O04_UARTOUT_NPM_A



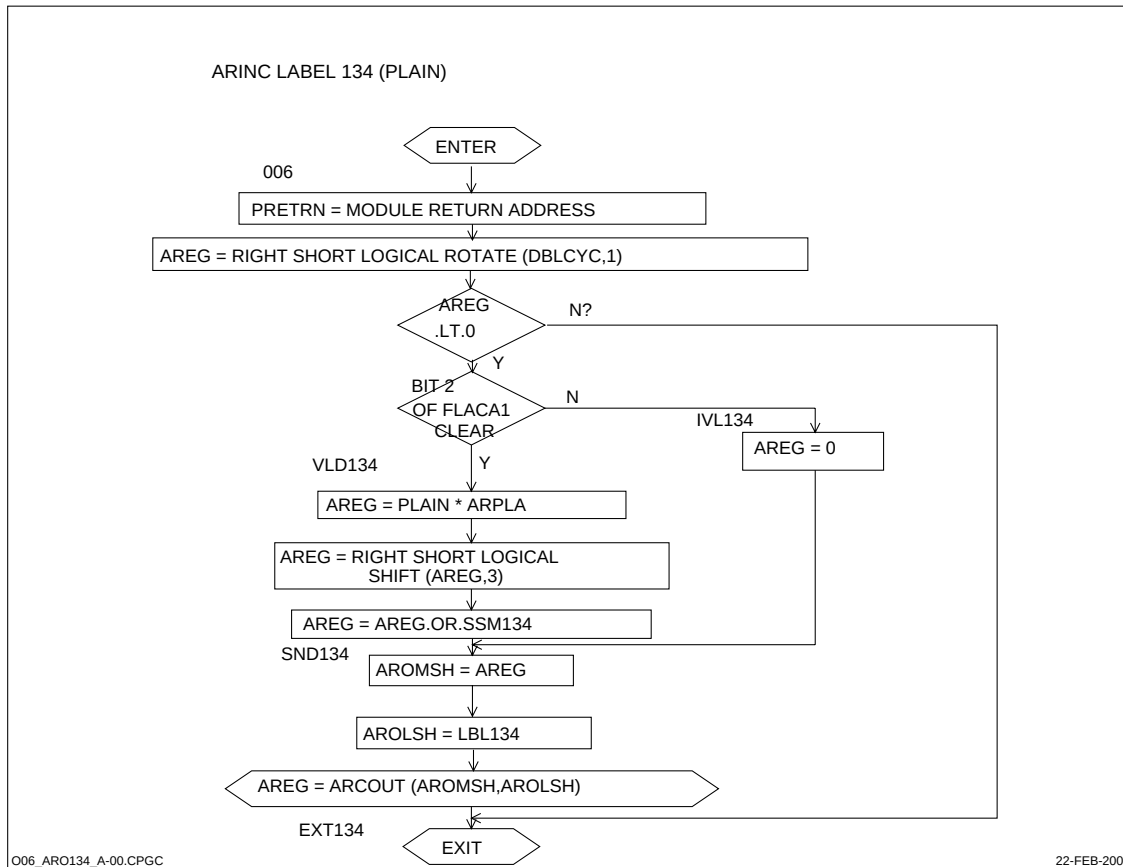
13.00 Figure 514: O04 _UARTOUT _NPM _B



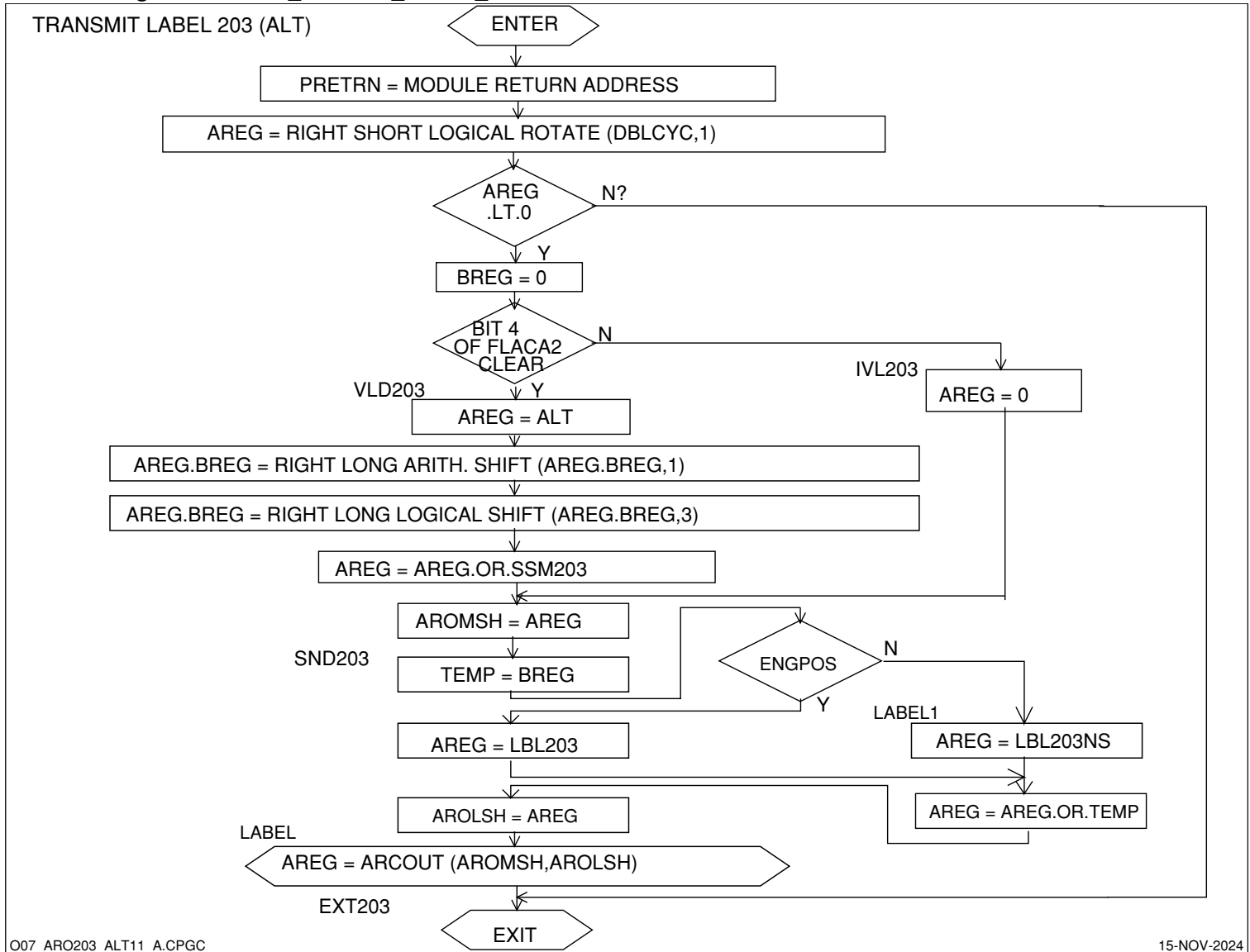
13.00 Figure 515: O05_ARO133_A



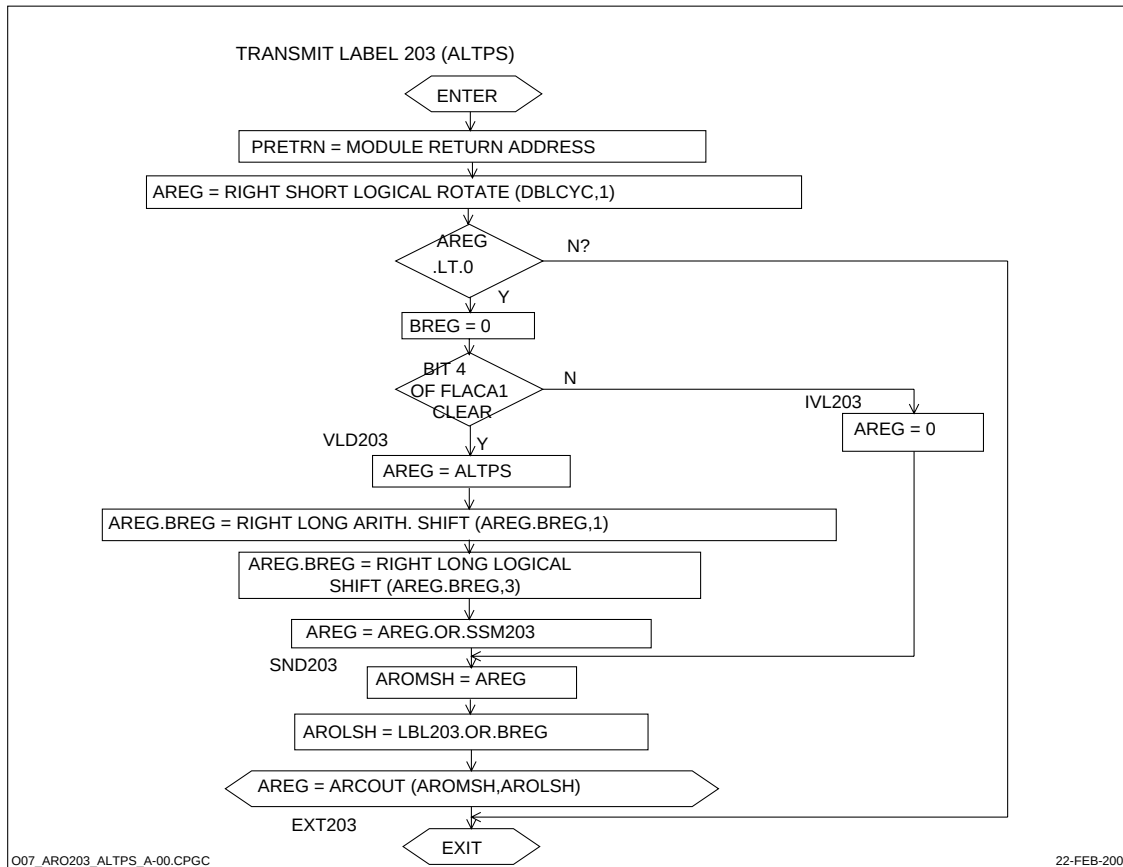
13.00 Figure 516: O06_ARO134_A



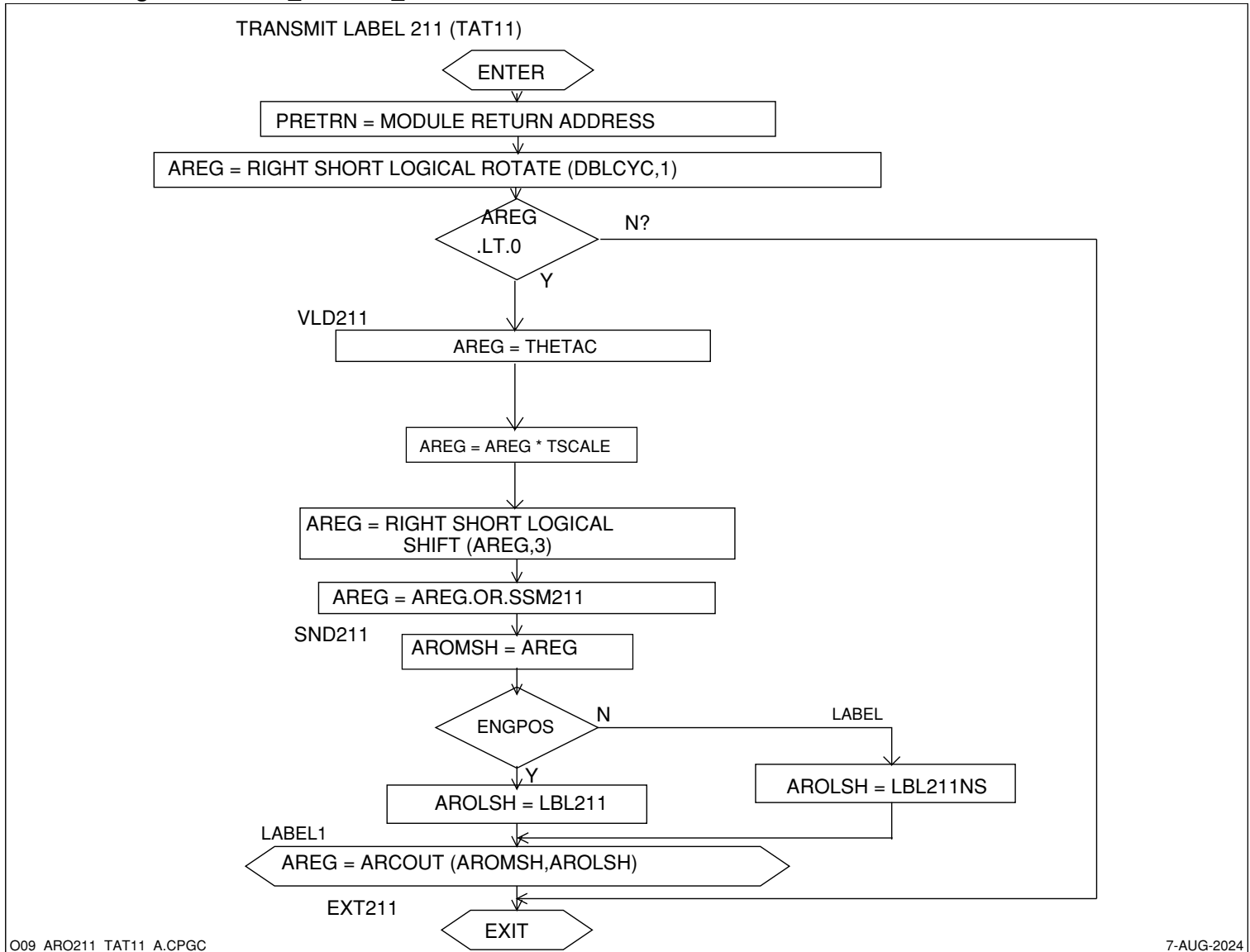
13.00 Figure 517: O07_ARO203_ALT11_A



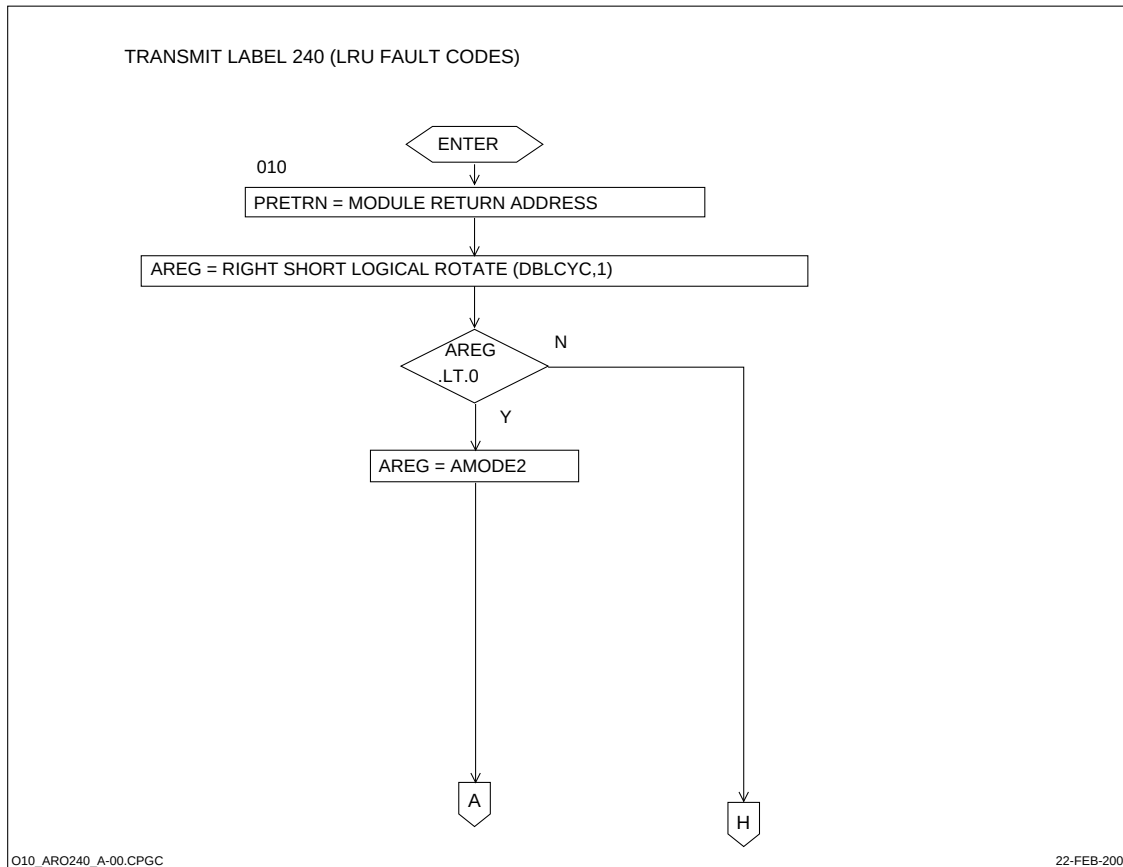
13.00 Figure 518: O07_ARO203_ALTPS_A



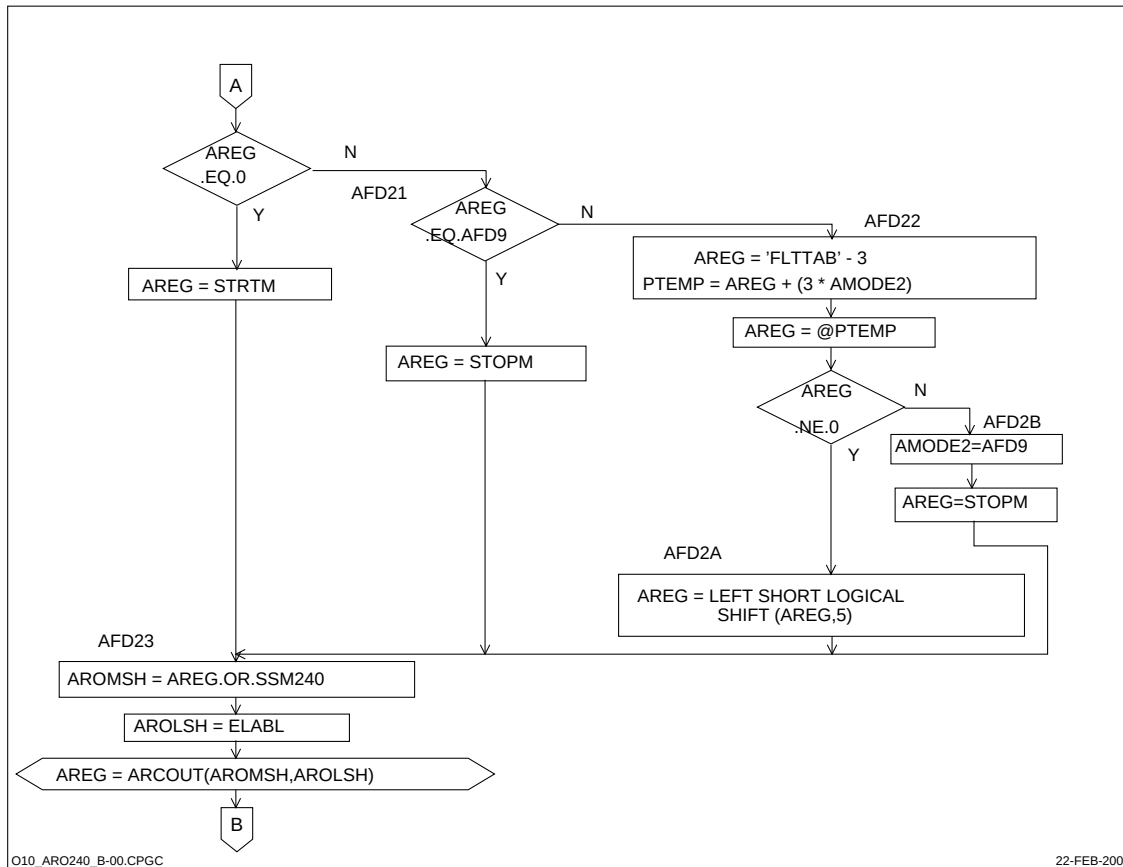
13.00 Figure 519: O09_ARO211_TAT11



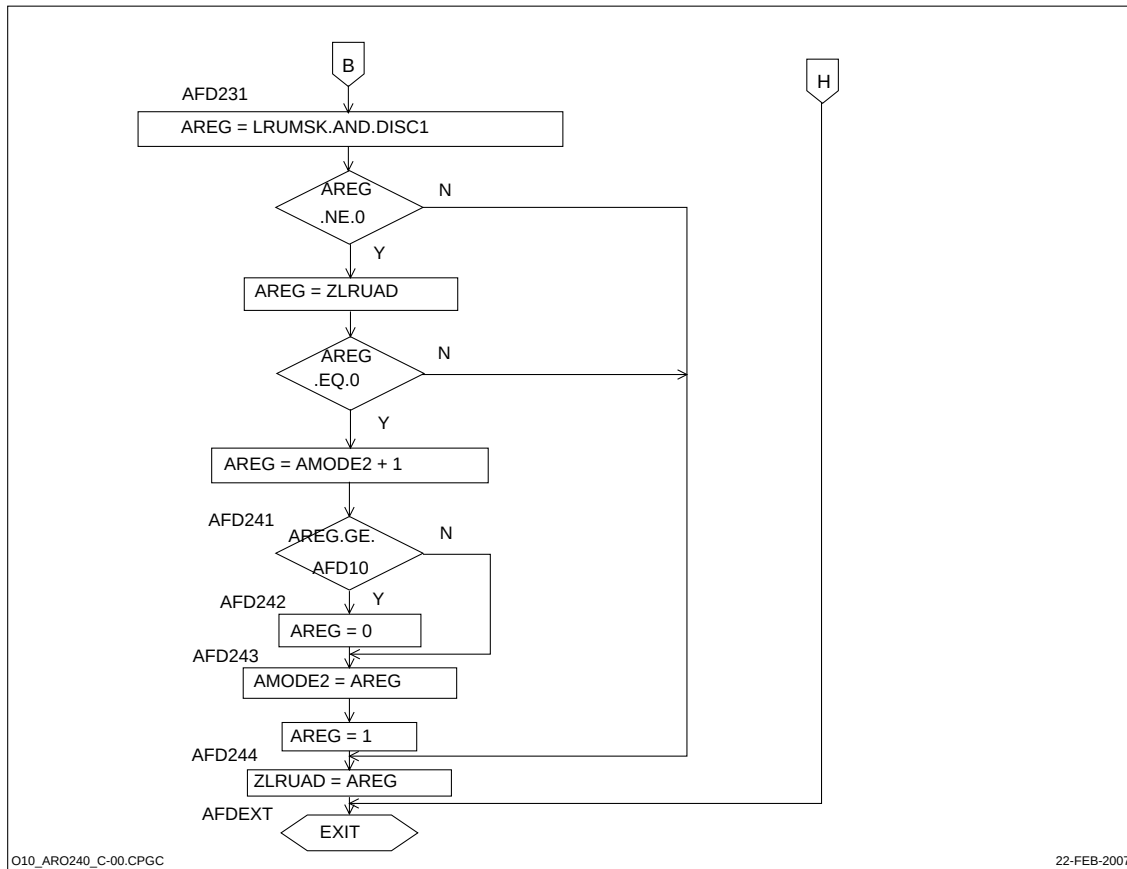
13.00 Figure 520: O10_ARO240_A



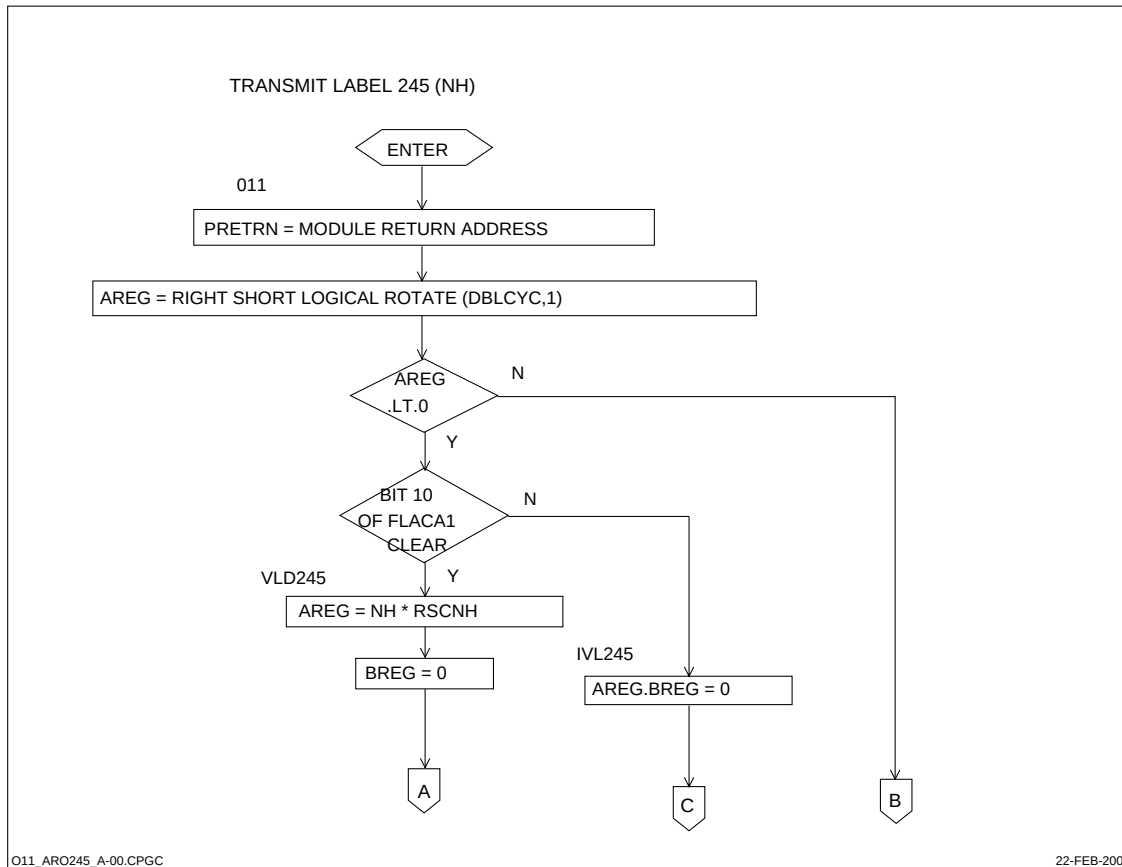
13.00 Figure 521: O10_ARO240_B



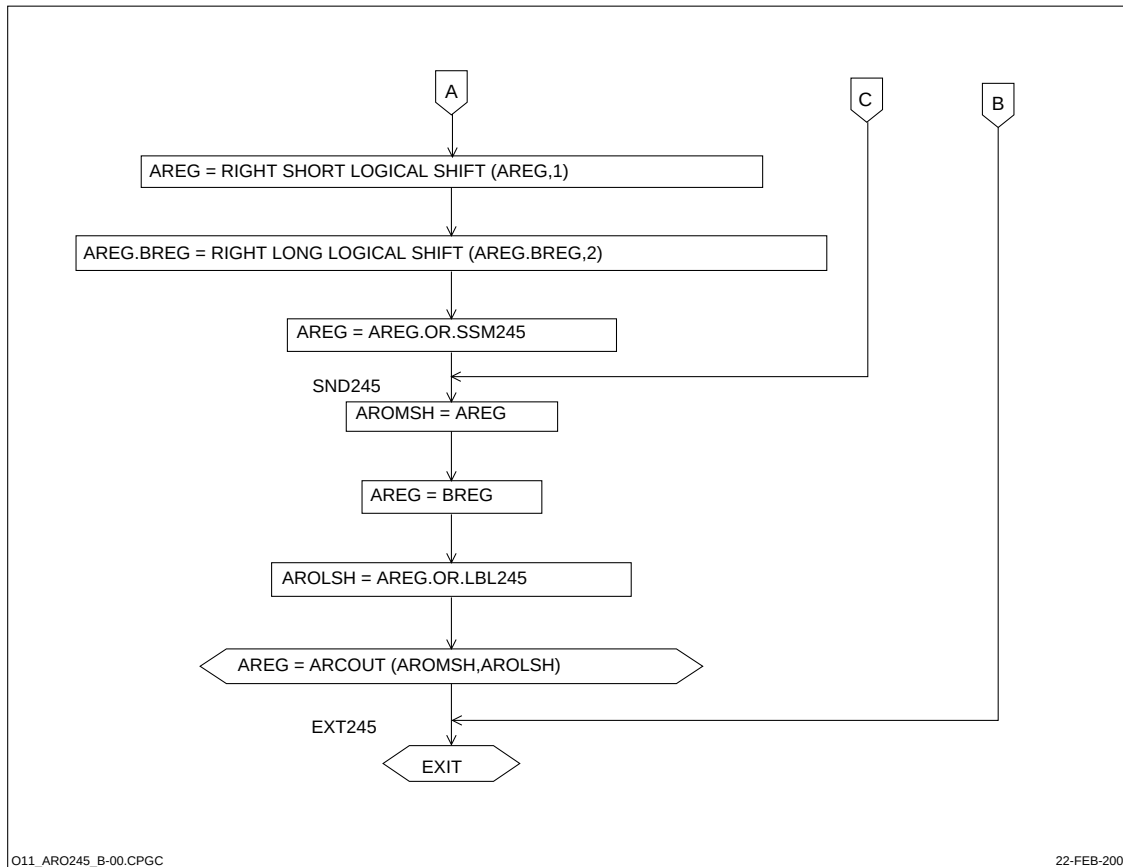
13.00 Figure 522: O10_ARO240_C



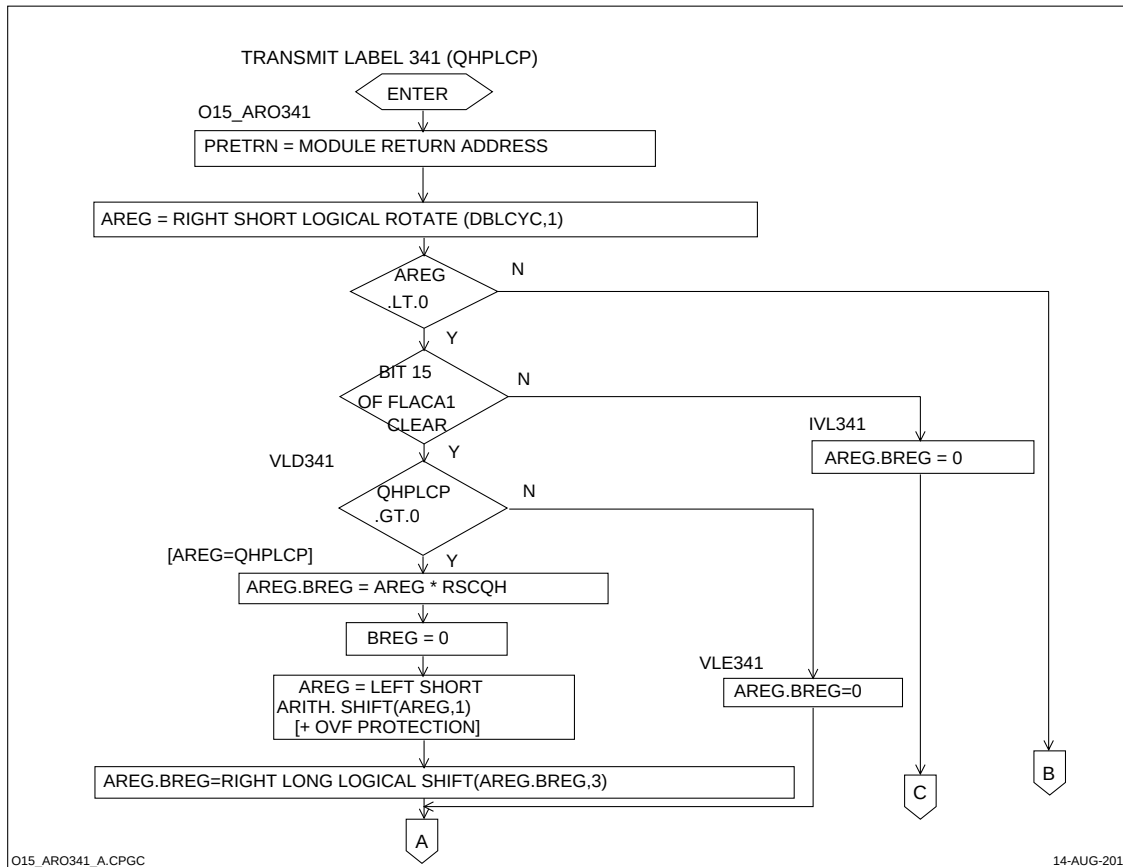
13.00 Figure 523: O11 _ARO245 _A



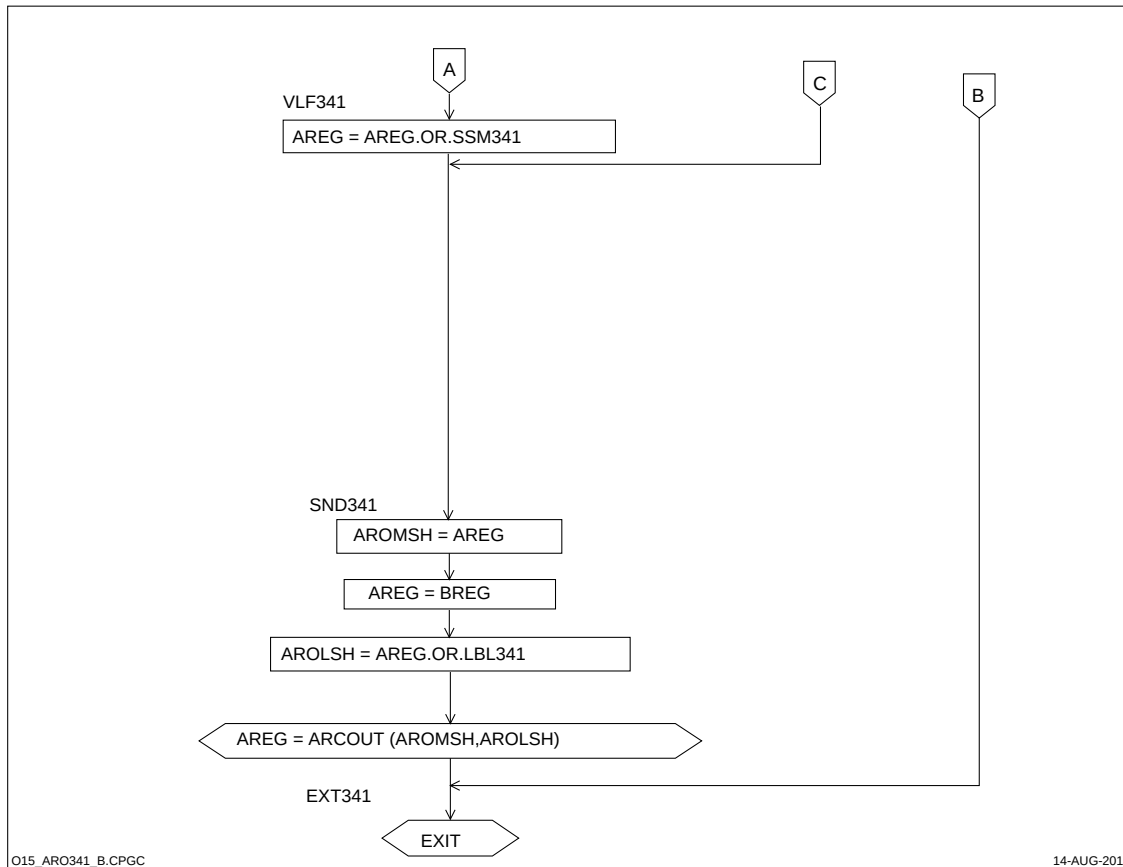
13.00 Figure 524: O11 _ARO245 _B



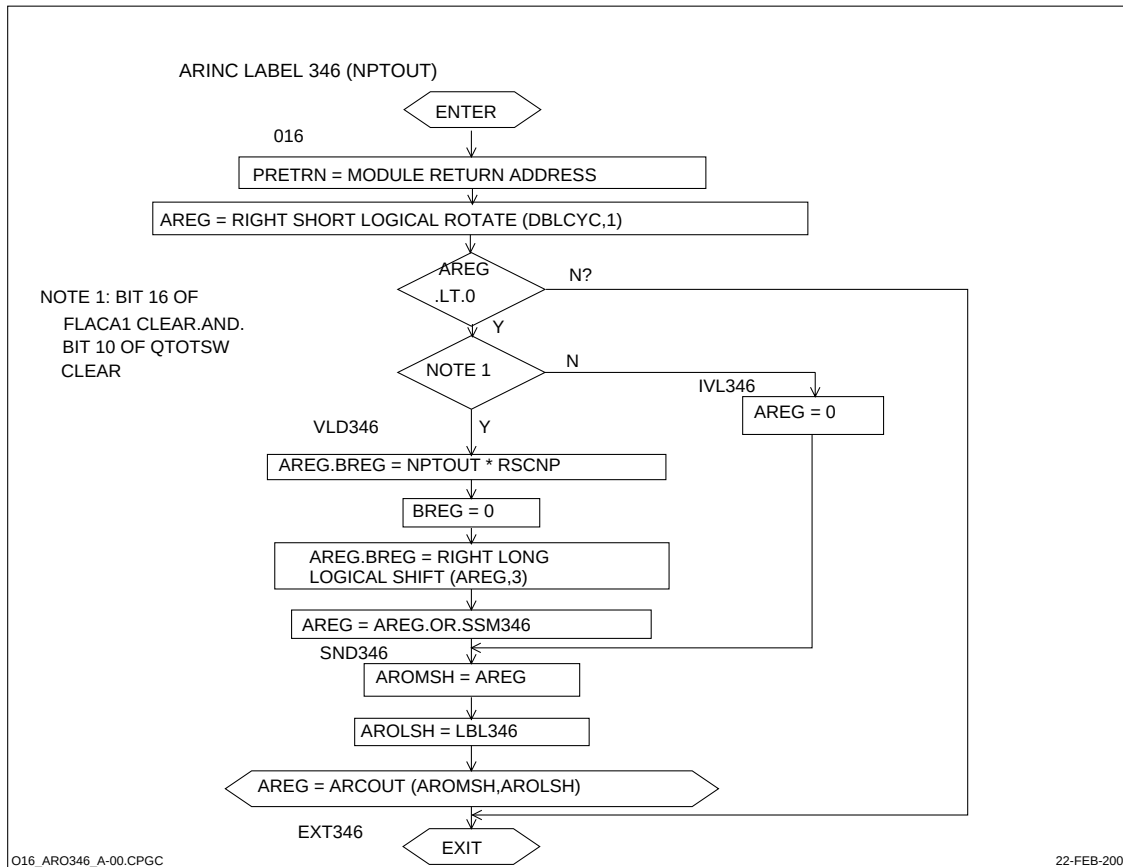
13.00 Figure 525: O15_ARO341_A



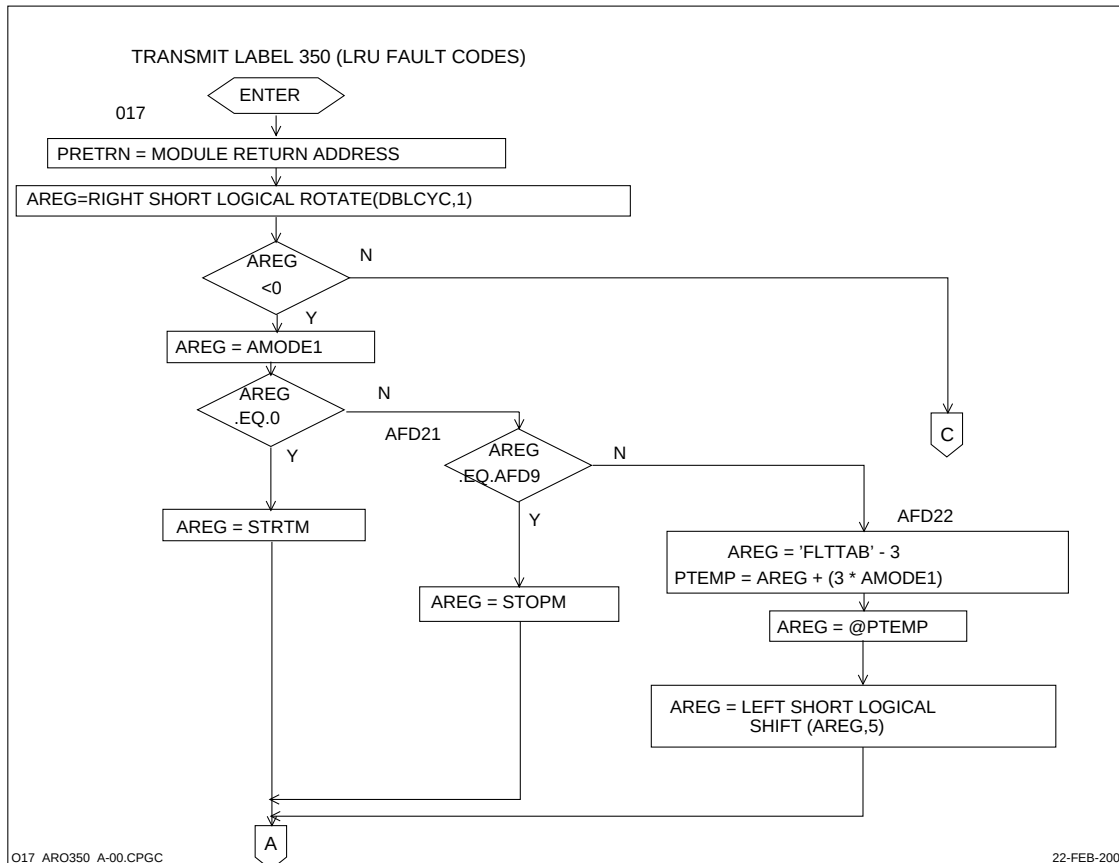
13.00 Figure 526: O15_ARO341_B



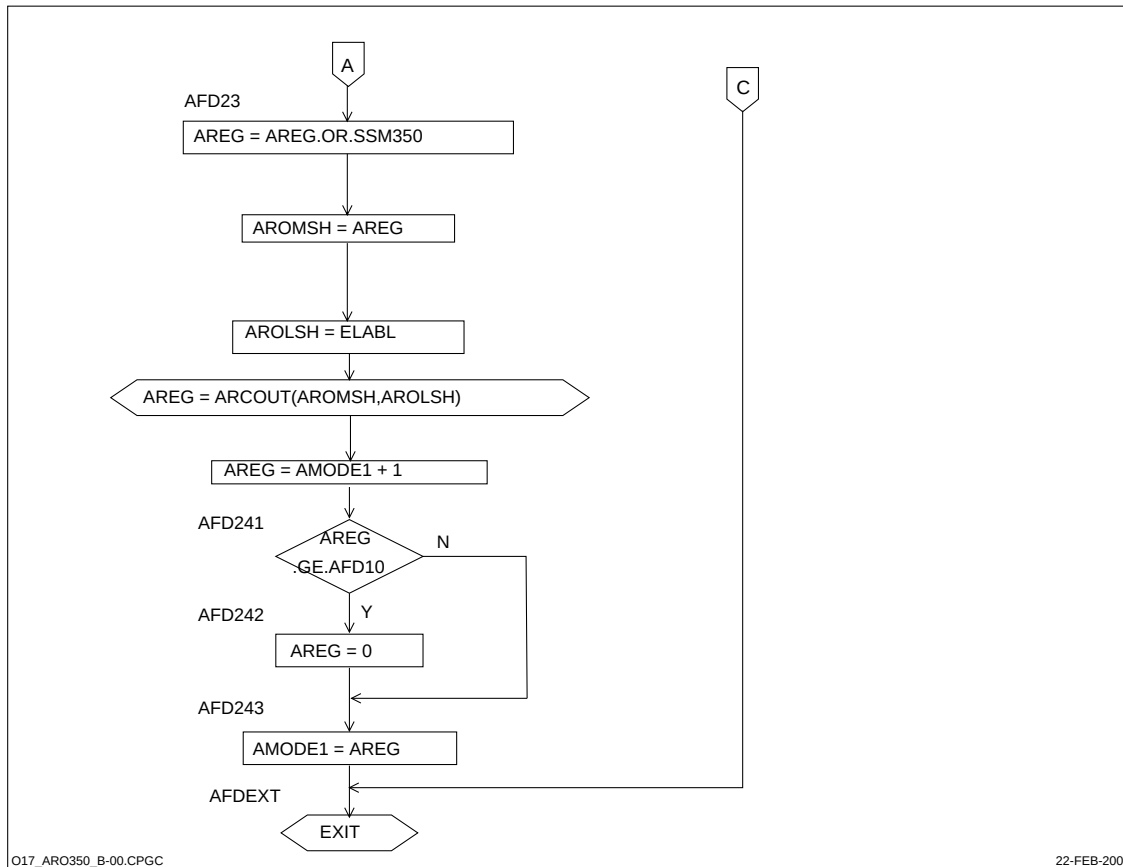
13.00 Figure 527: O16 _ARO346 _A



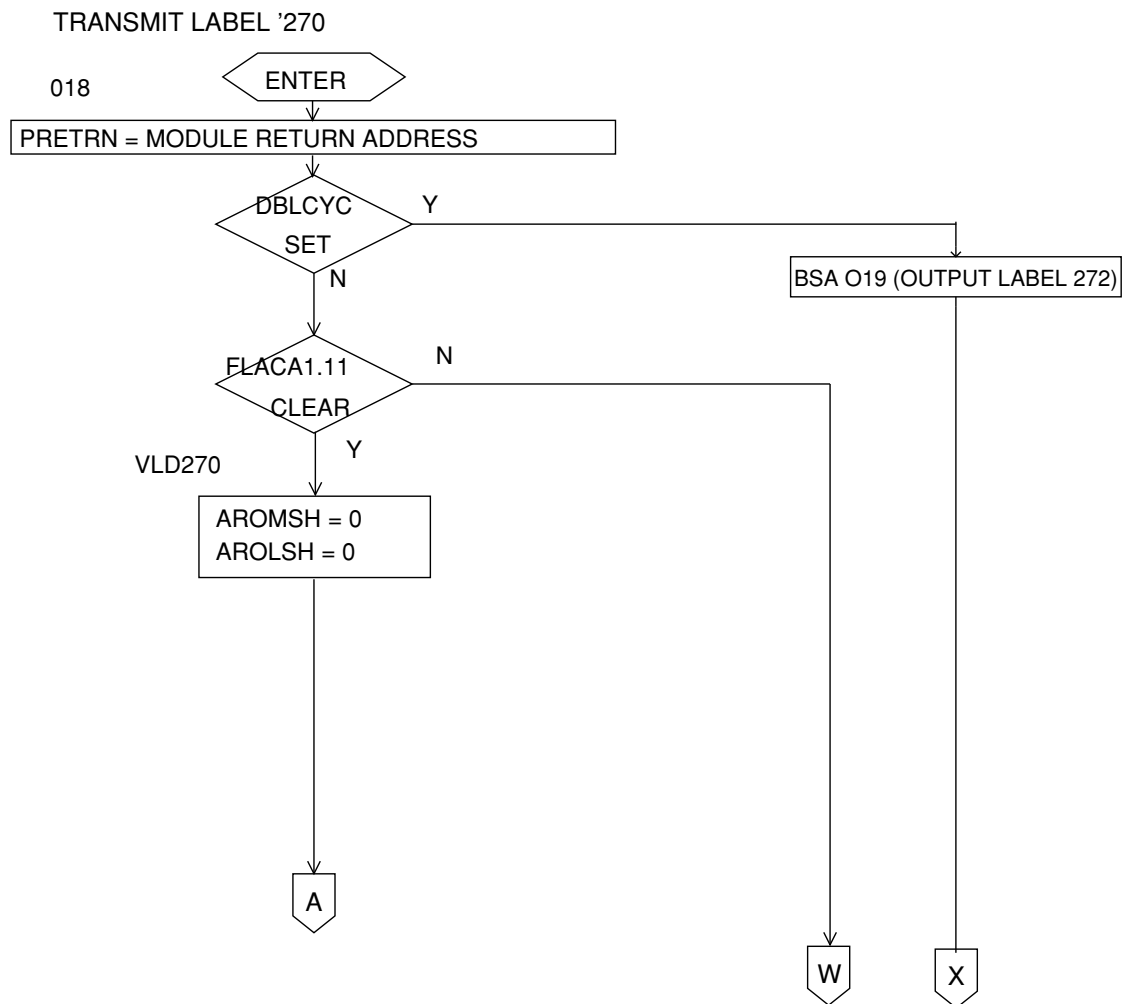
13.00 Figure 528: O17 _ARO350 _A



13.00 Figure 529: O17_ARO350_B



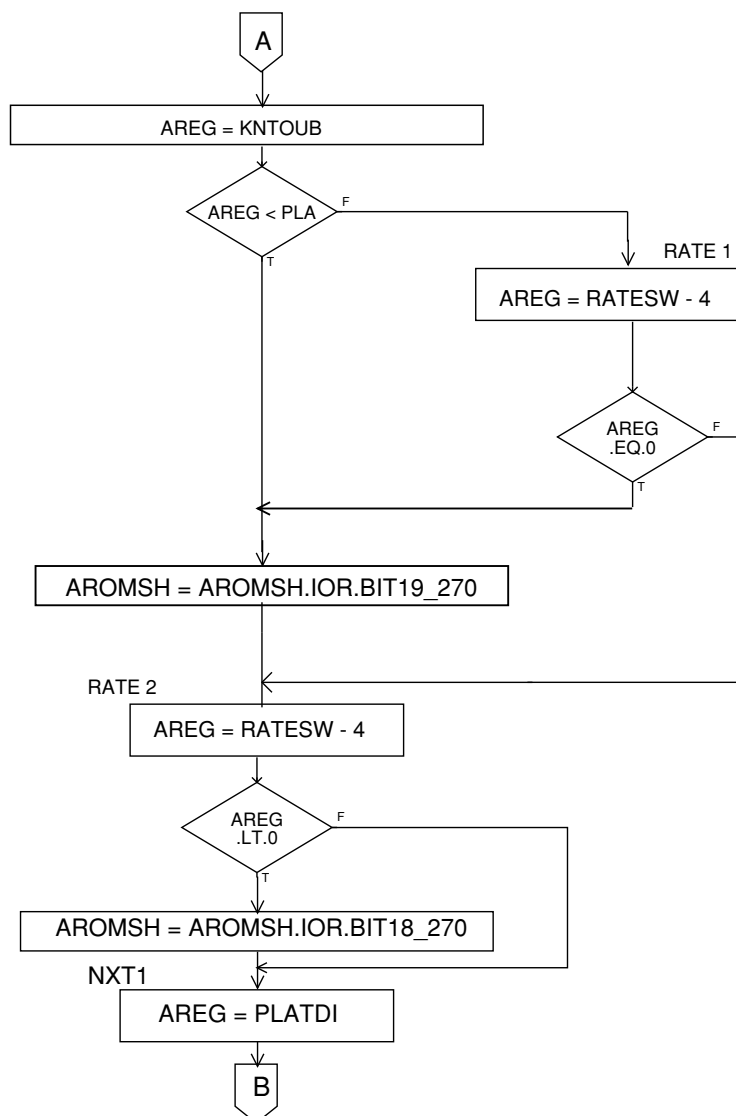
13.00 Figure 530: O18 _ARO270 _A



O18_ARO270_A.CPGC

30-JUL-2024

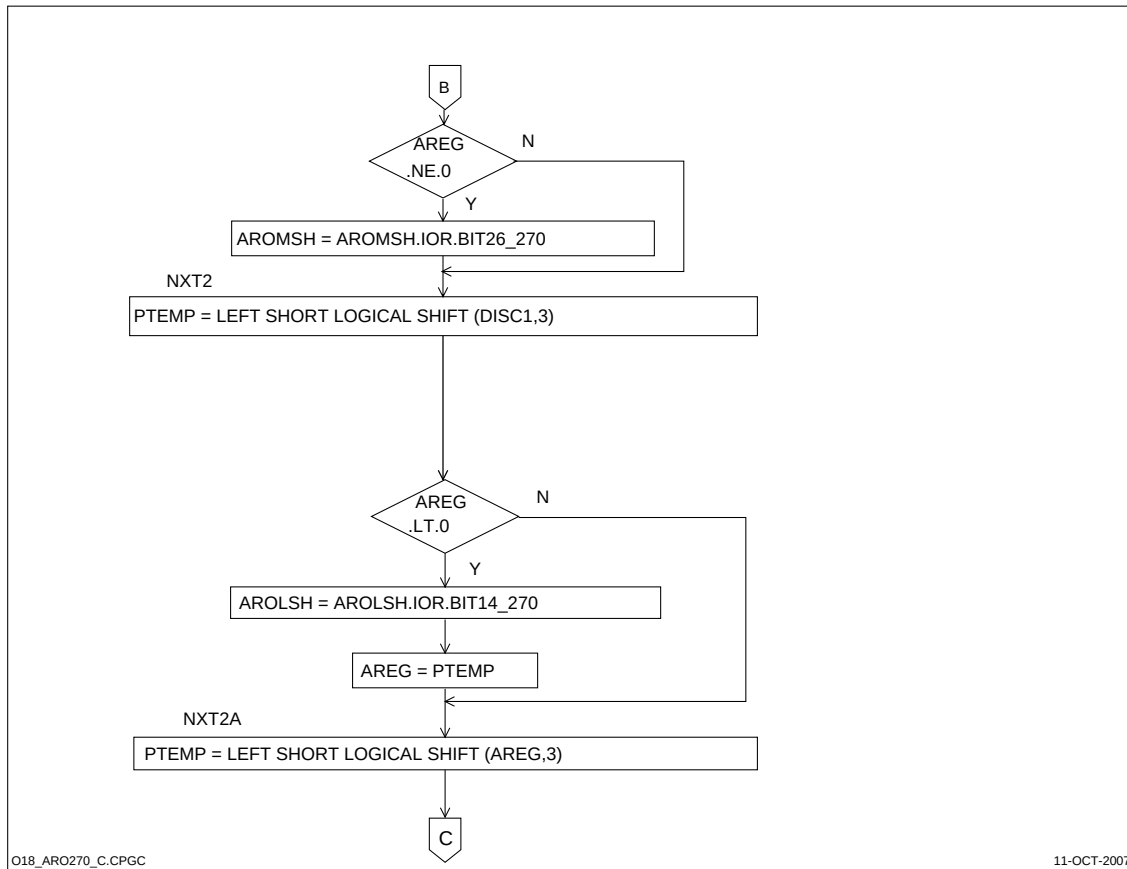
13.00 Figure 531: O18_ARO270_B



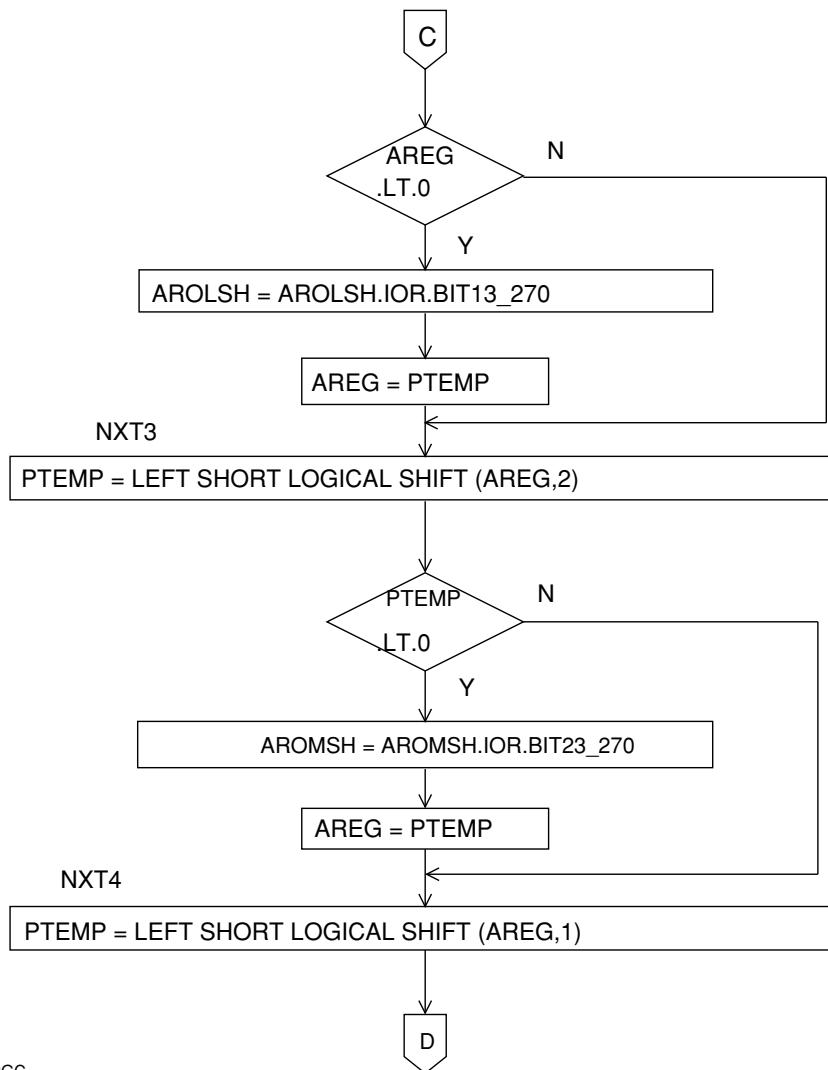
O18_ARO270_B.CPGC

5-AUG-2024

13.00 Figure 532: O18_ARO270_C



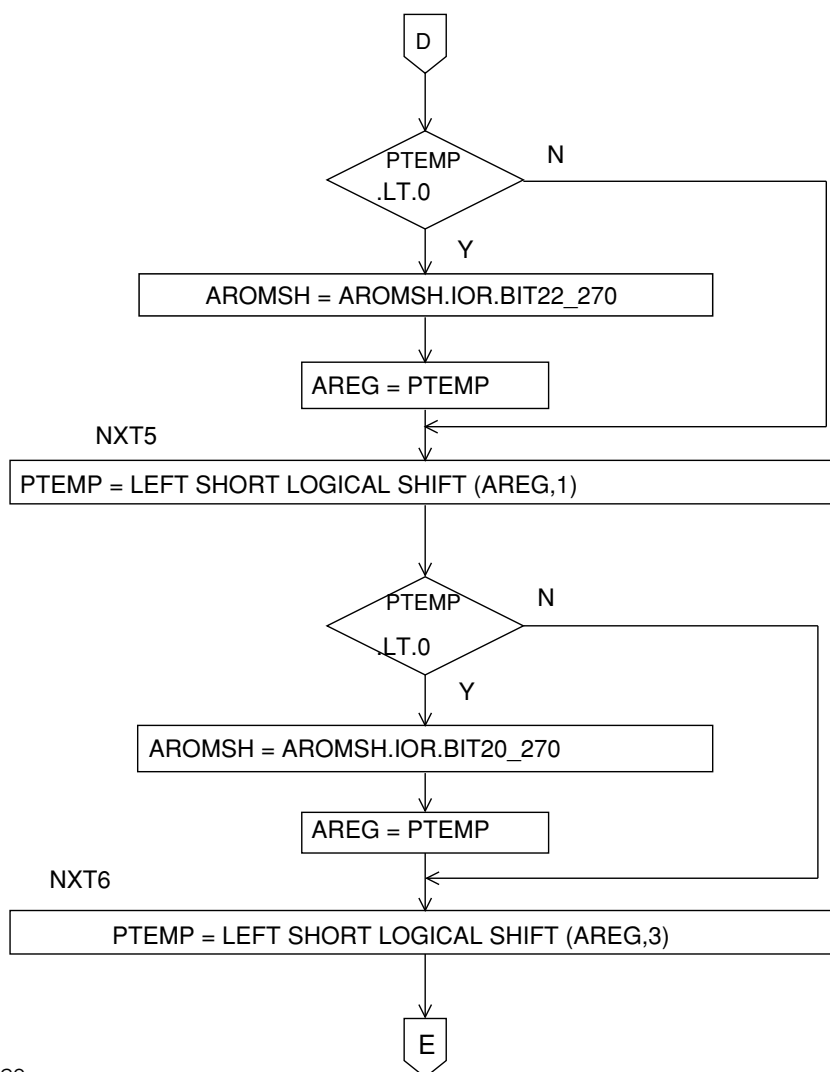
13.00 Figure 533: O18_ARO270_D



O18_ARO270_D.CPGC

7-AUG-2024

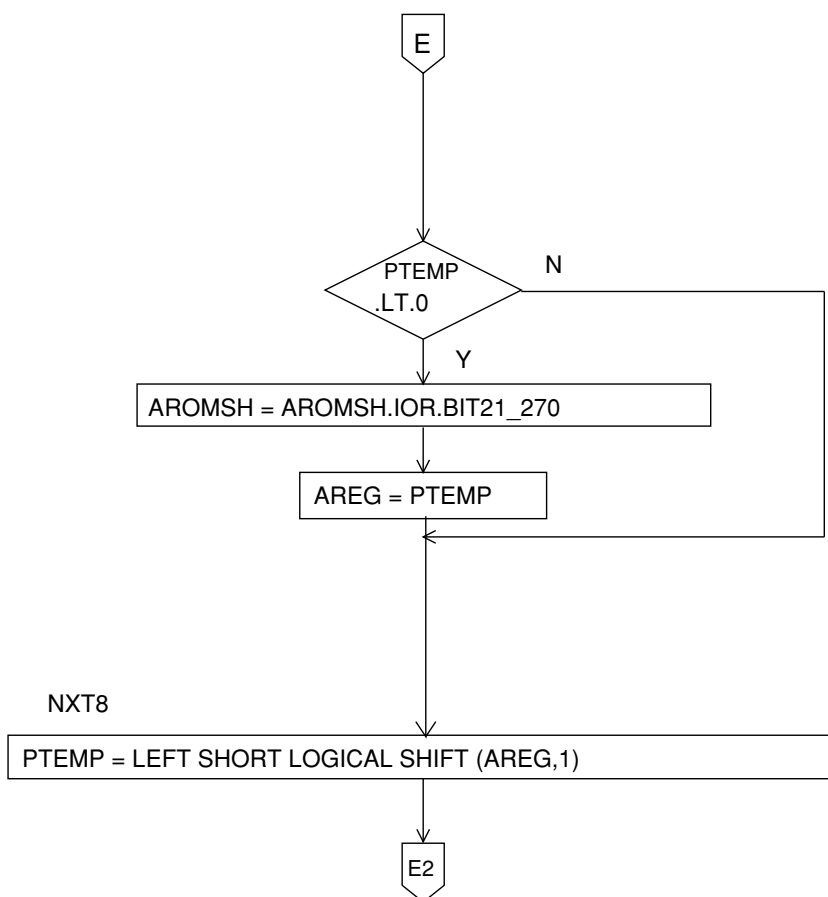
13.00 Figure 534: O18_ARO270_E



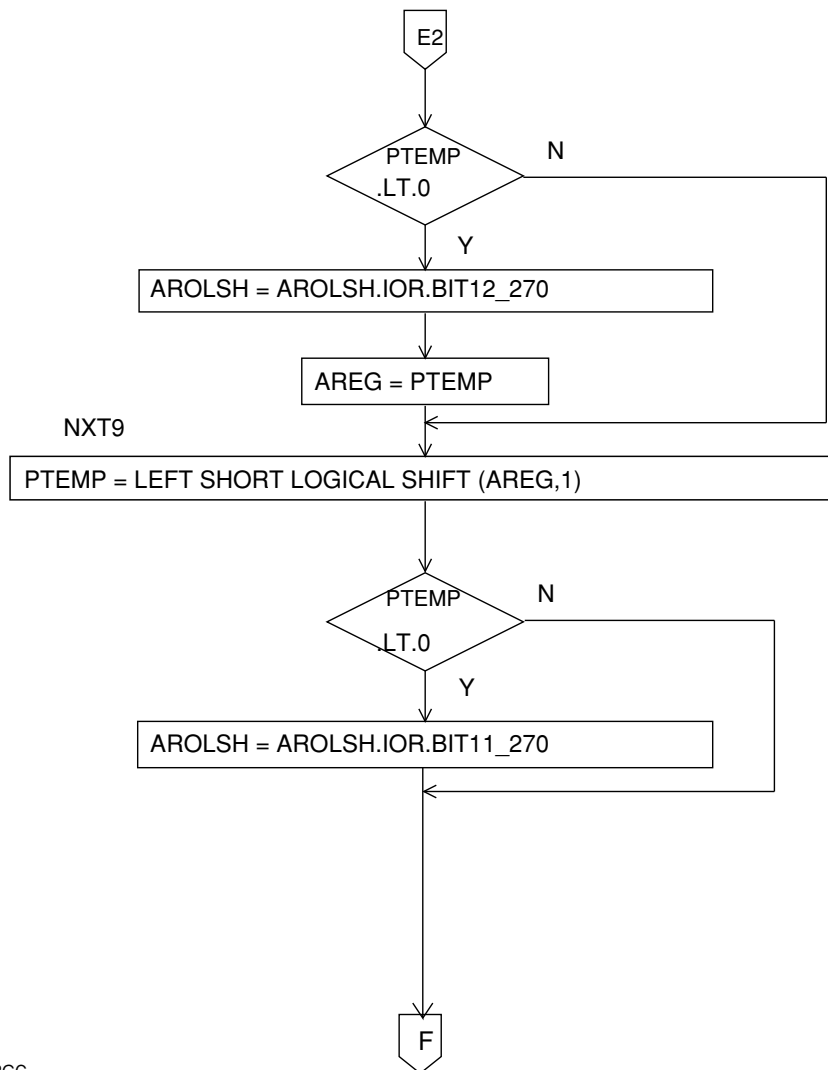
O18_ARO270_E.CPGC

7-AUG-2024

13.00 Figure 535: O18_ARO270_F



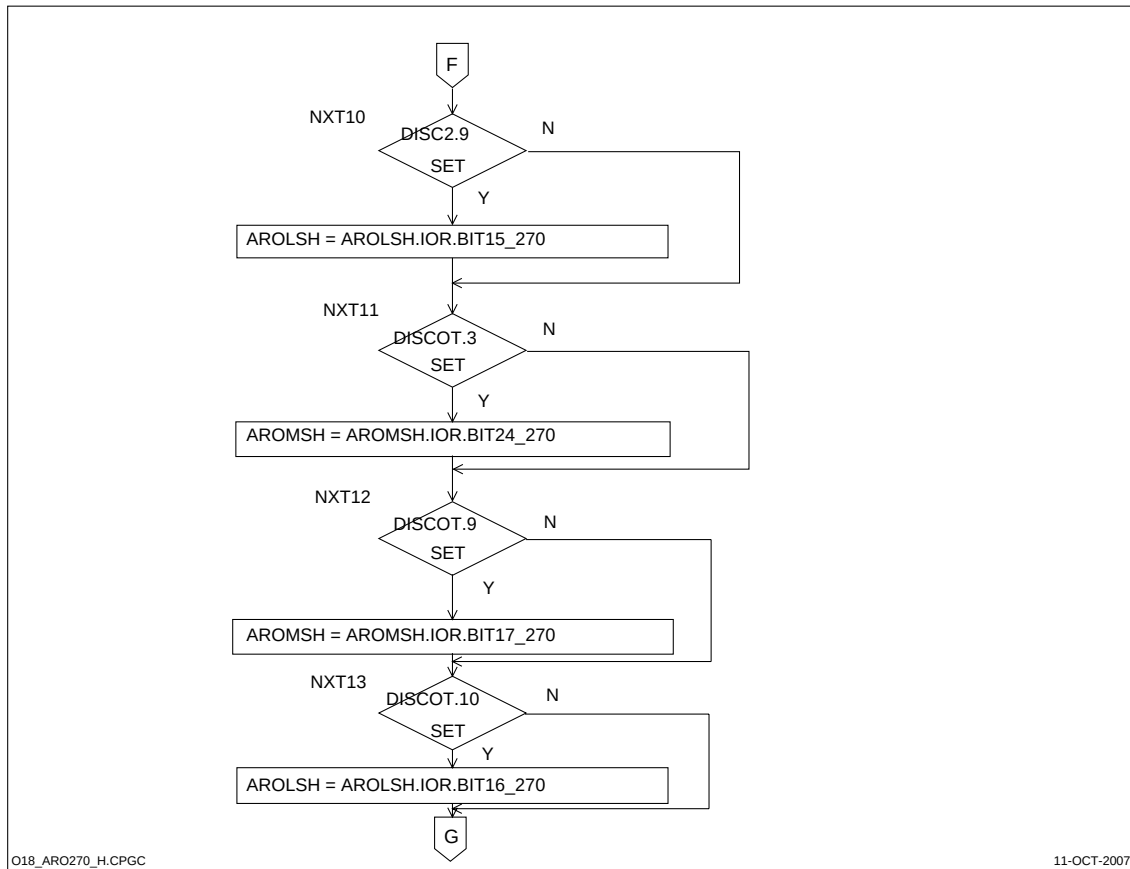
13.00 Figure 536: O18_ARO270_G



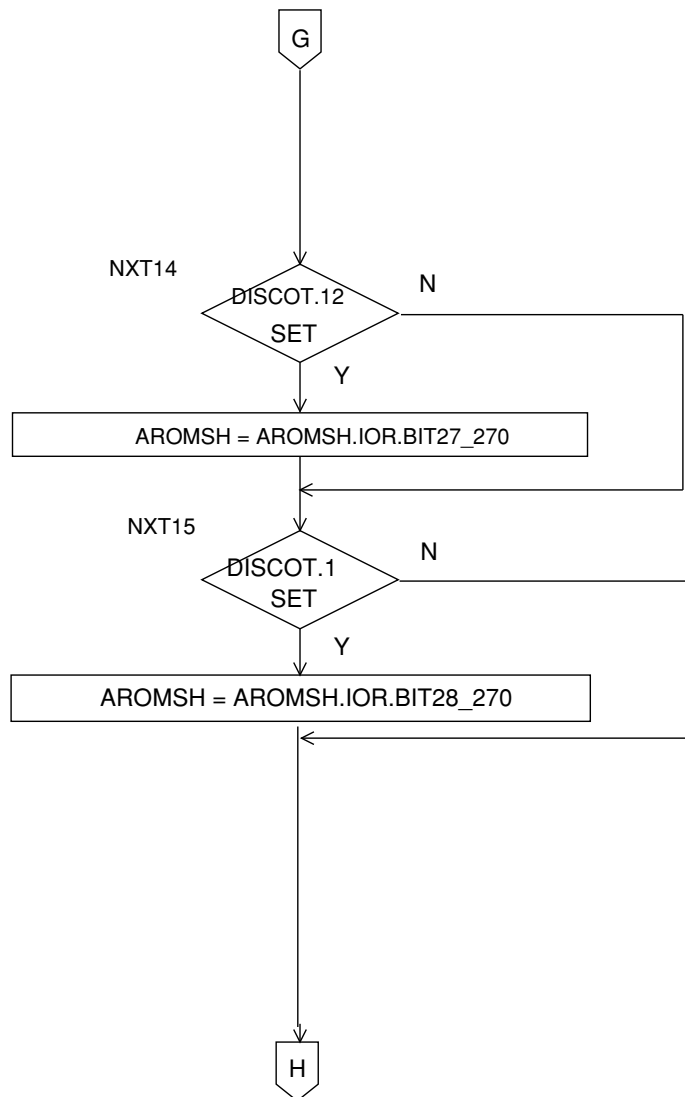
O18_ARO270_G.CPGC

7-AUG-2024

13.00 Figure 537: O18_ARO270_H



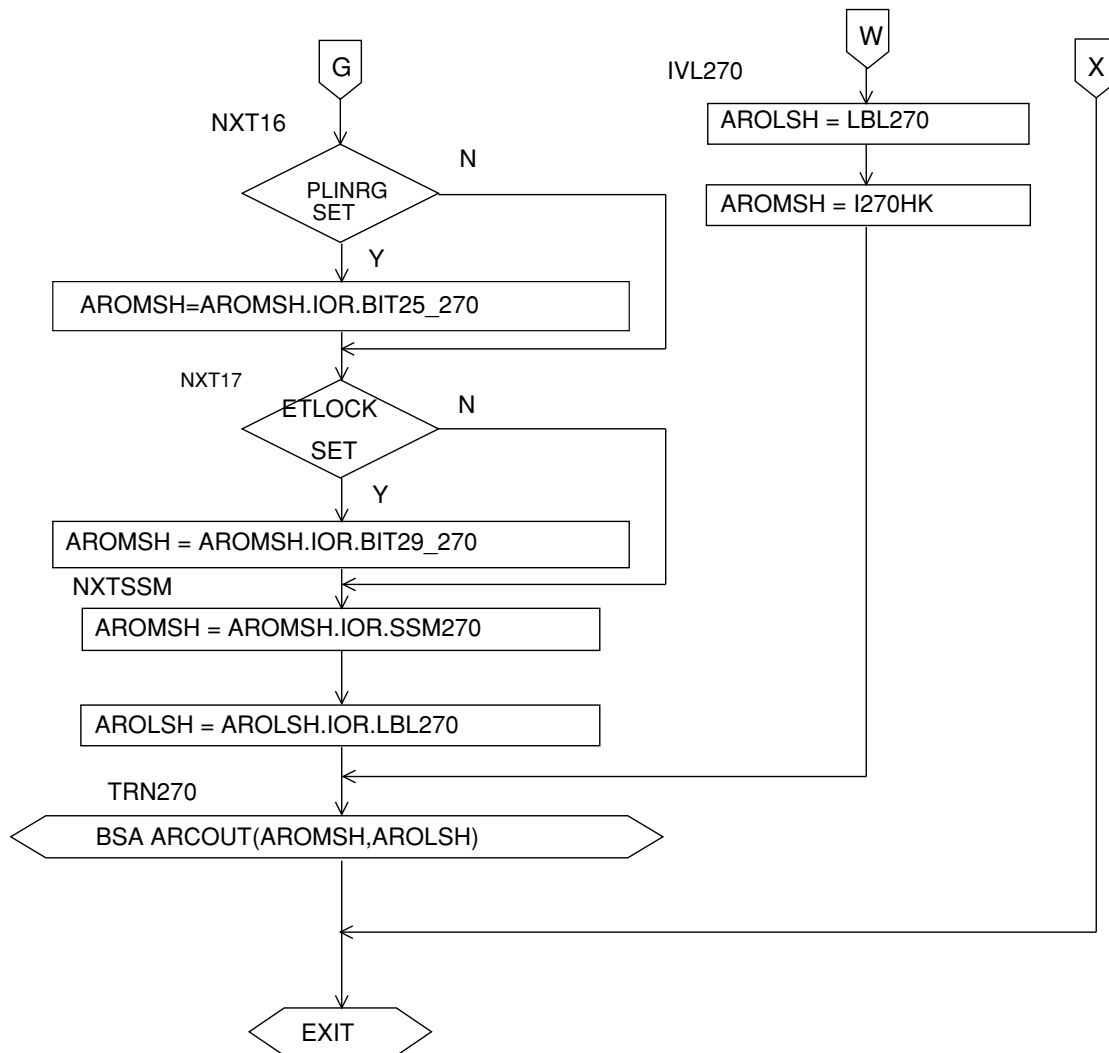
13.00 Figure 538: O18_ARO270_HA



O18_ARO270_HA.CPGC

23-MAY-2024

13.00 Figure 539: O18 _ARO270 _I

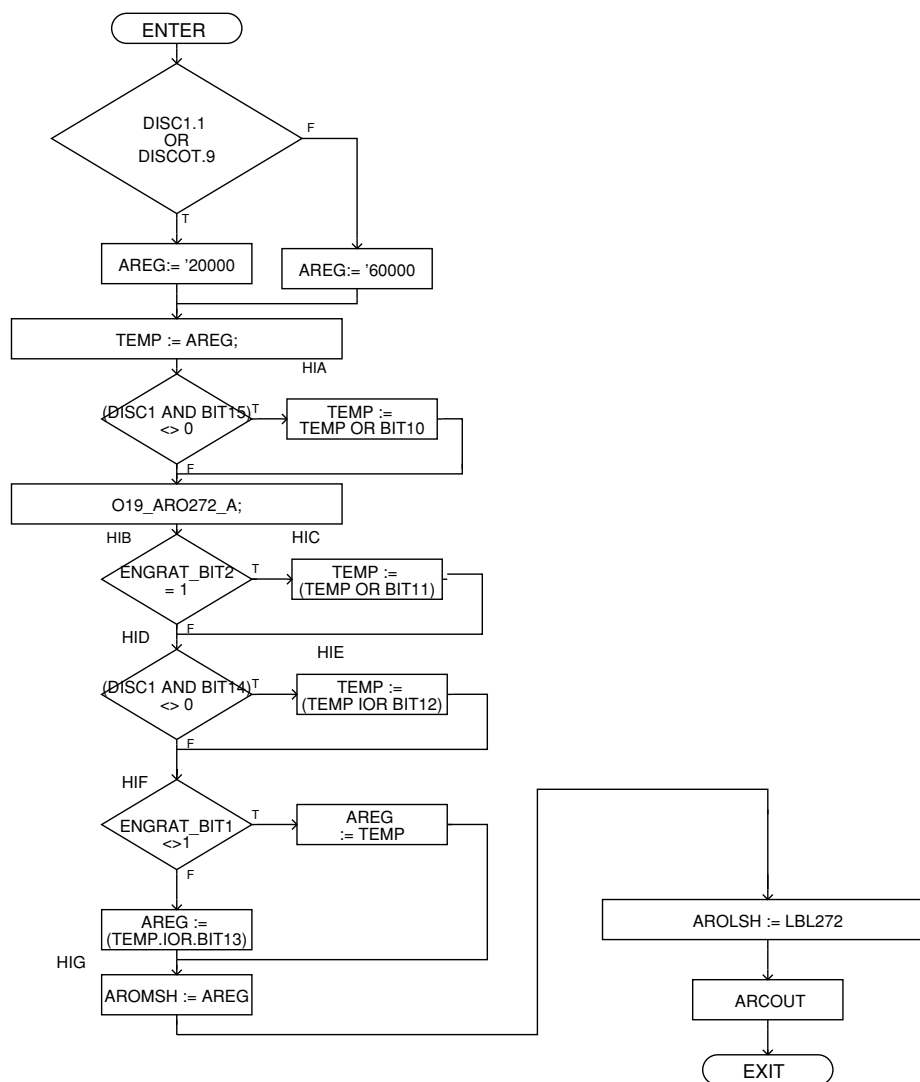


O18_ARO270_I.CPGC

30-JUL-2024

13.00 Figure 540: O19_ARO272

TRANSMIT LABEL 272

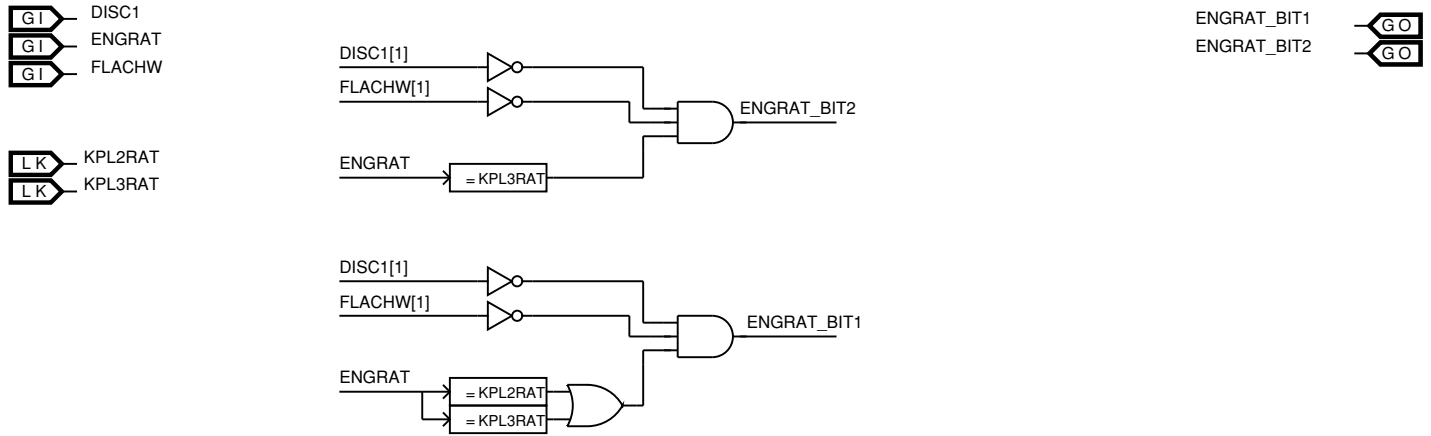


O19_ARO272.CPGC

7-AUG-2024

13.00 Figure 541: O19_ARO272_A

FIGURE O-19-2 : ARINC Label 272 Outputs Process: Determine ENGRAT_BIT1 and ENGRAT_BIT2

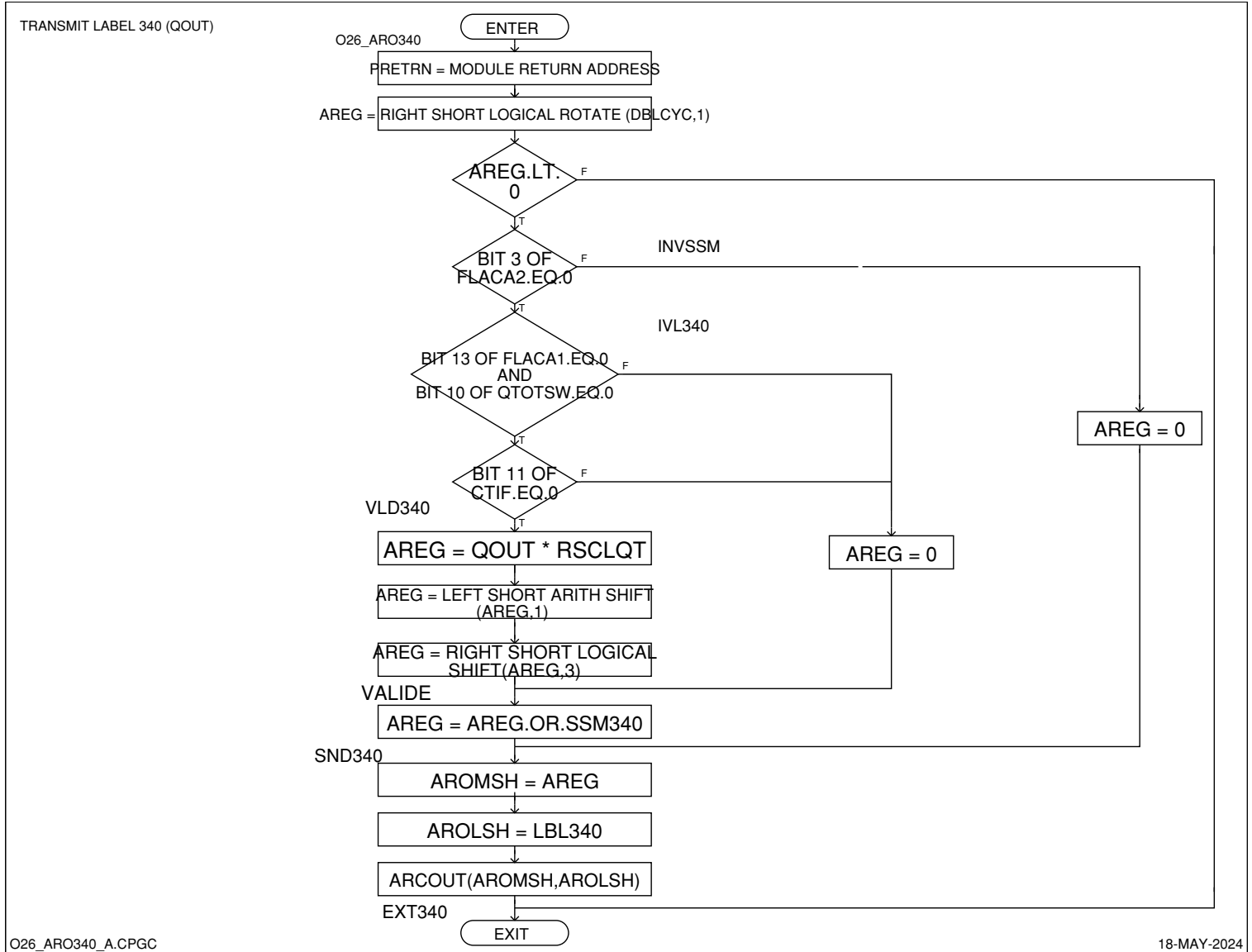


LEV6

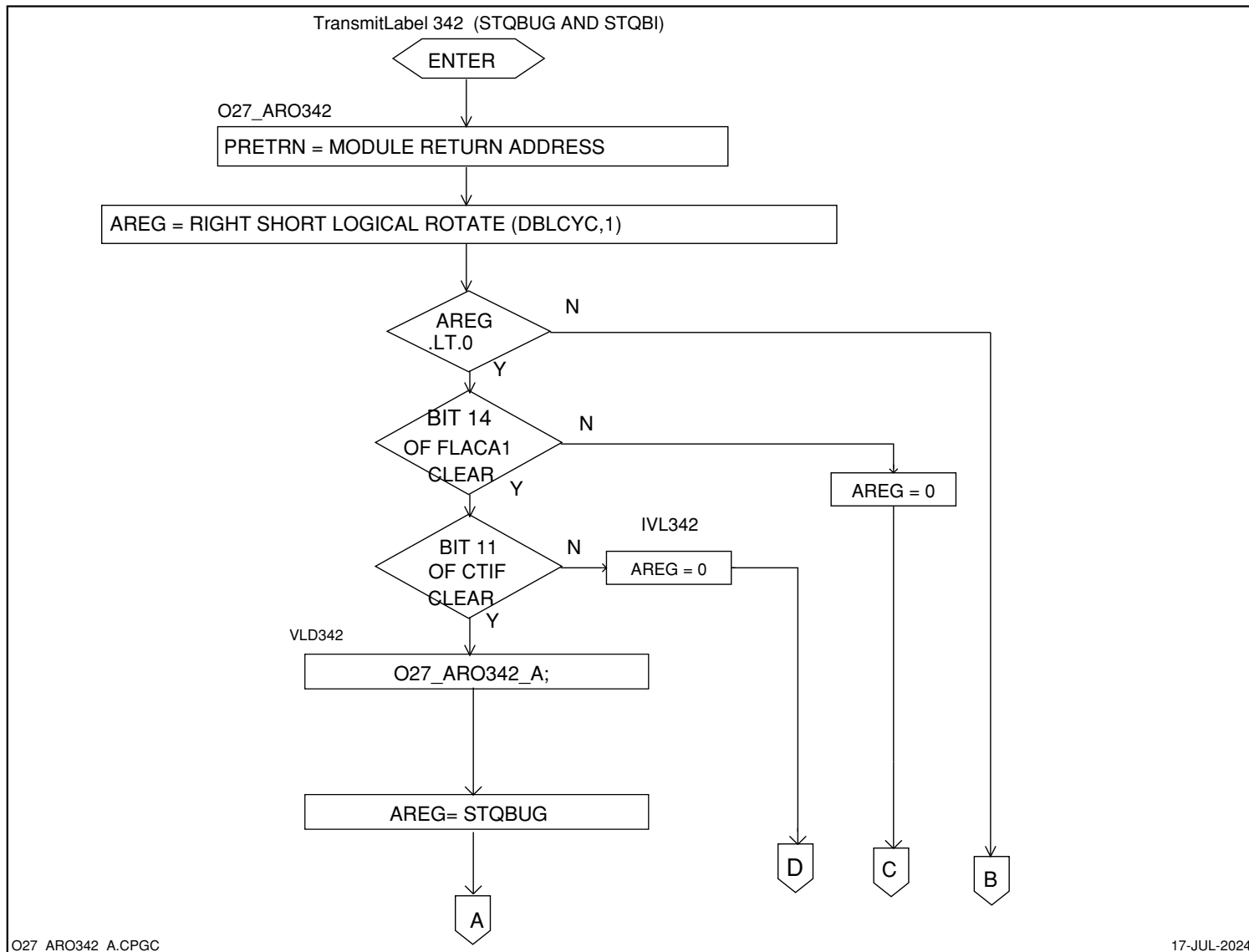
A

Name: O19_ARO272_A	Ident: V 1.0	Date: 6-JAN-2026 14:00:50.16
Title: ARINC Label 272 Output Process: Determine ENGRAT_BIT1 and ENGRAT_BIT2		

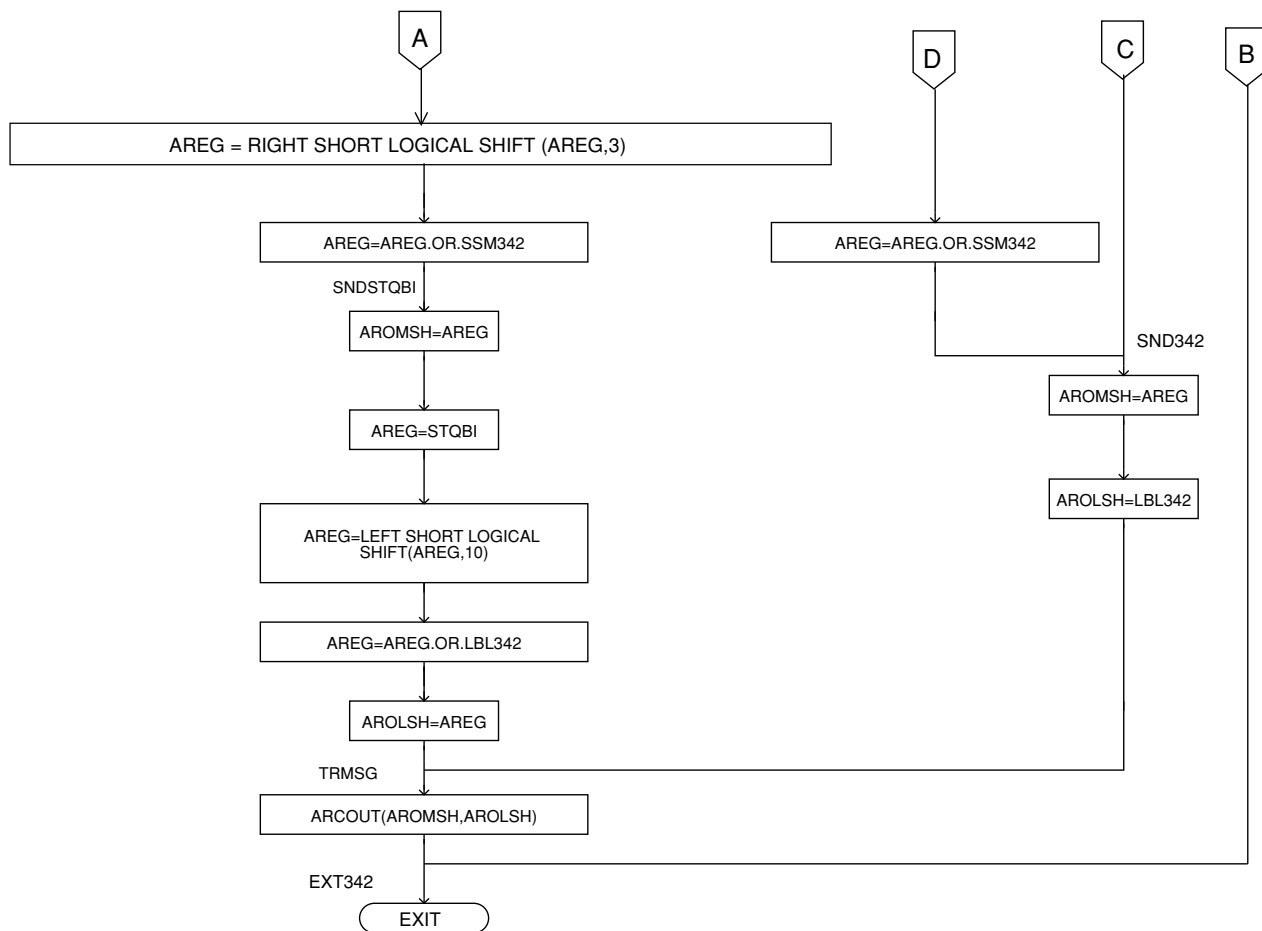
13.00 Figure 543: O26_ARO340_A



13.00 Figure 544: O27_ARO342_A



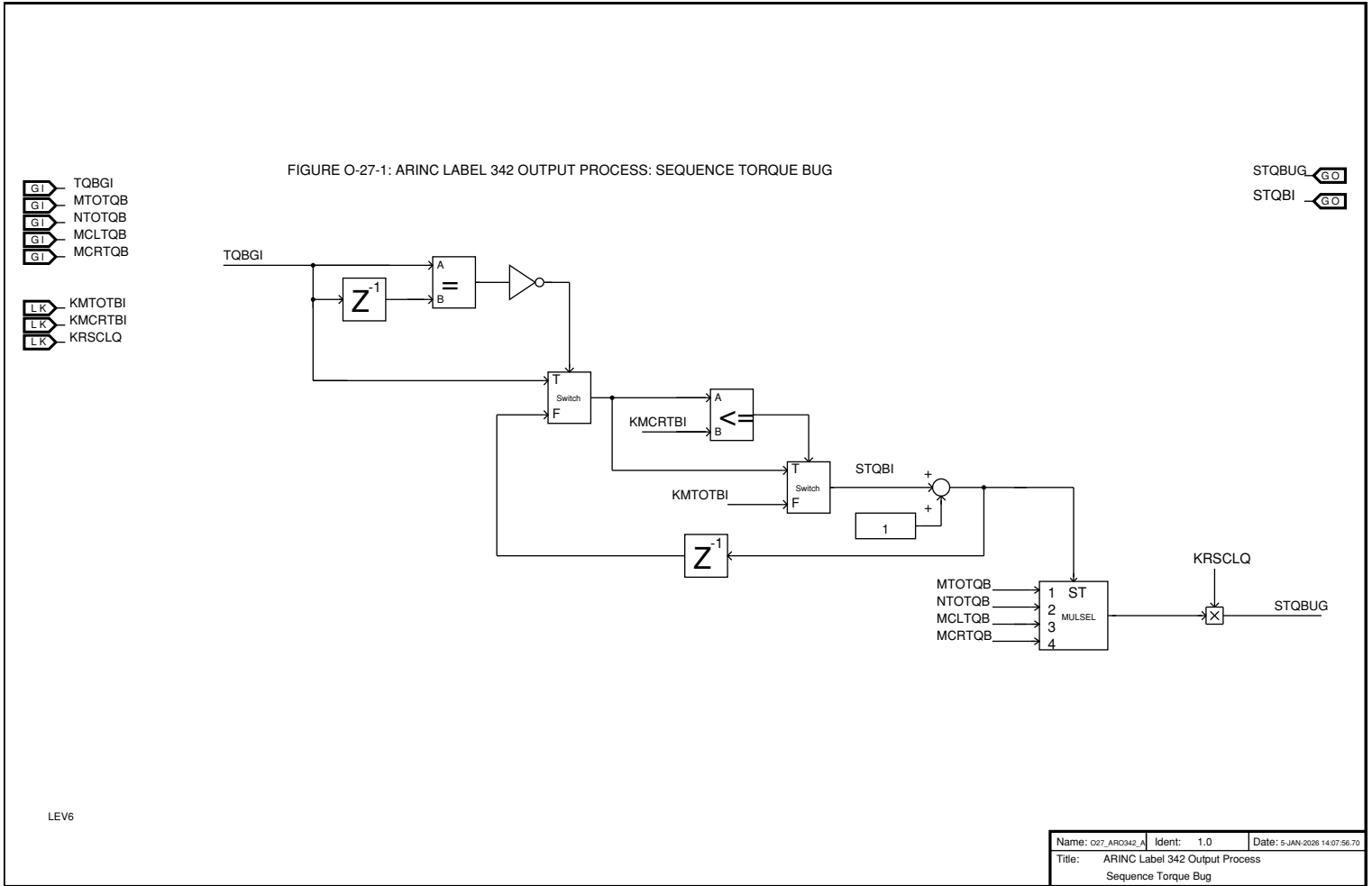
13.00 Figure 545: O27 _ARO342 _B



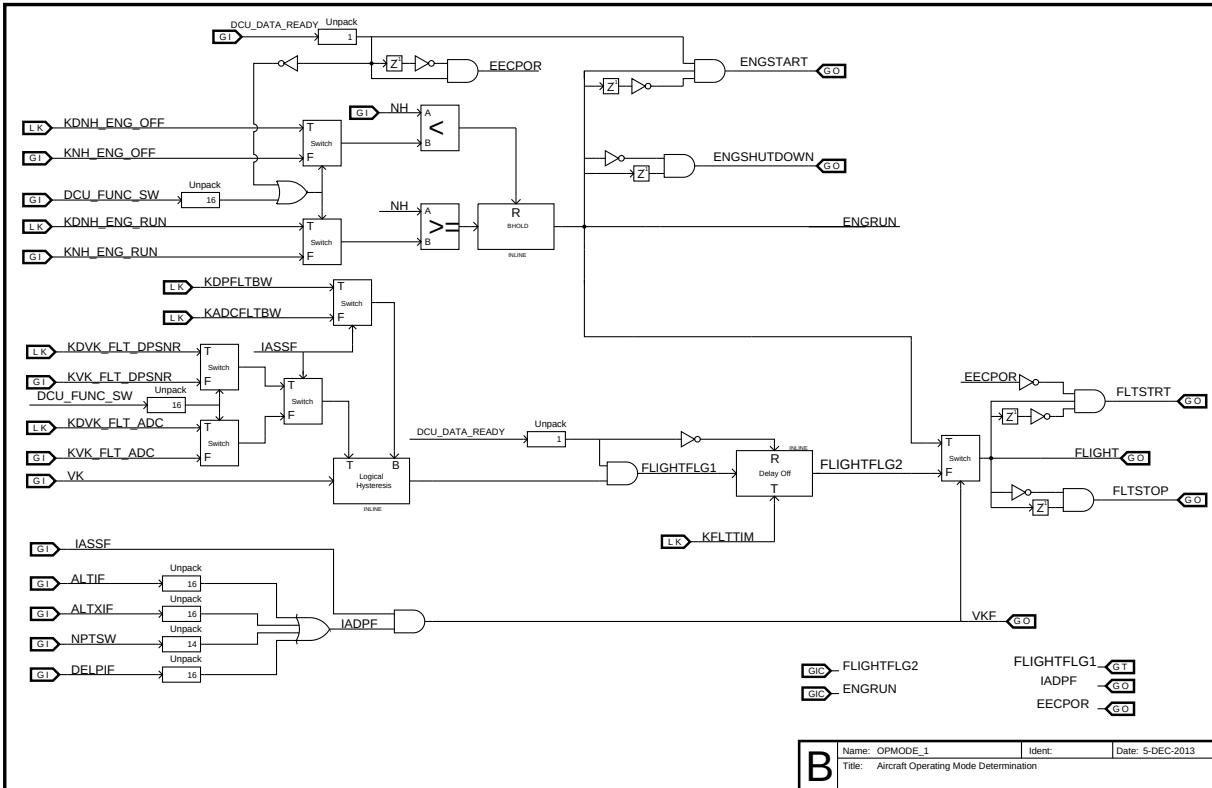
O27 _ARO342 _B.CPGC

17-JUL-2024

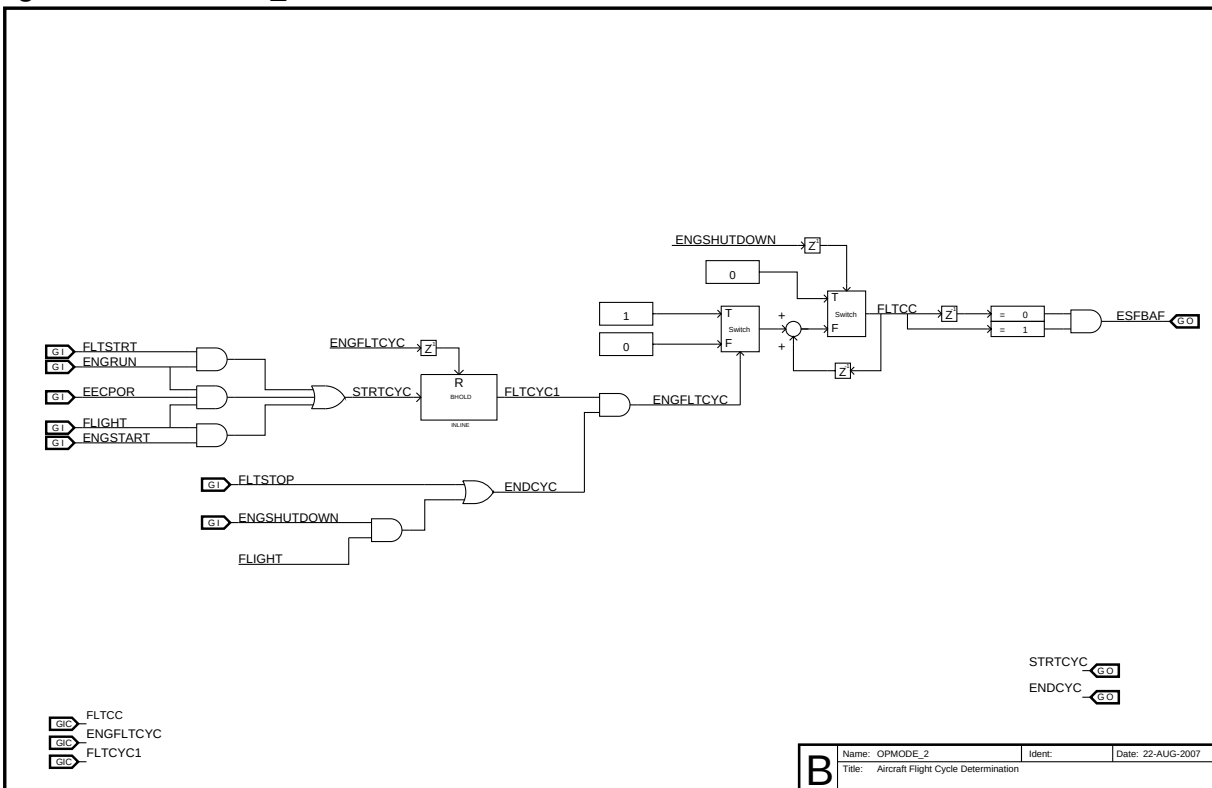
13.00 Figure 546: O27_ARO342_A_Subfunc



13.00 Figure 547: OPMODE _1

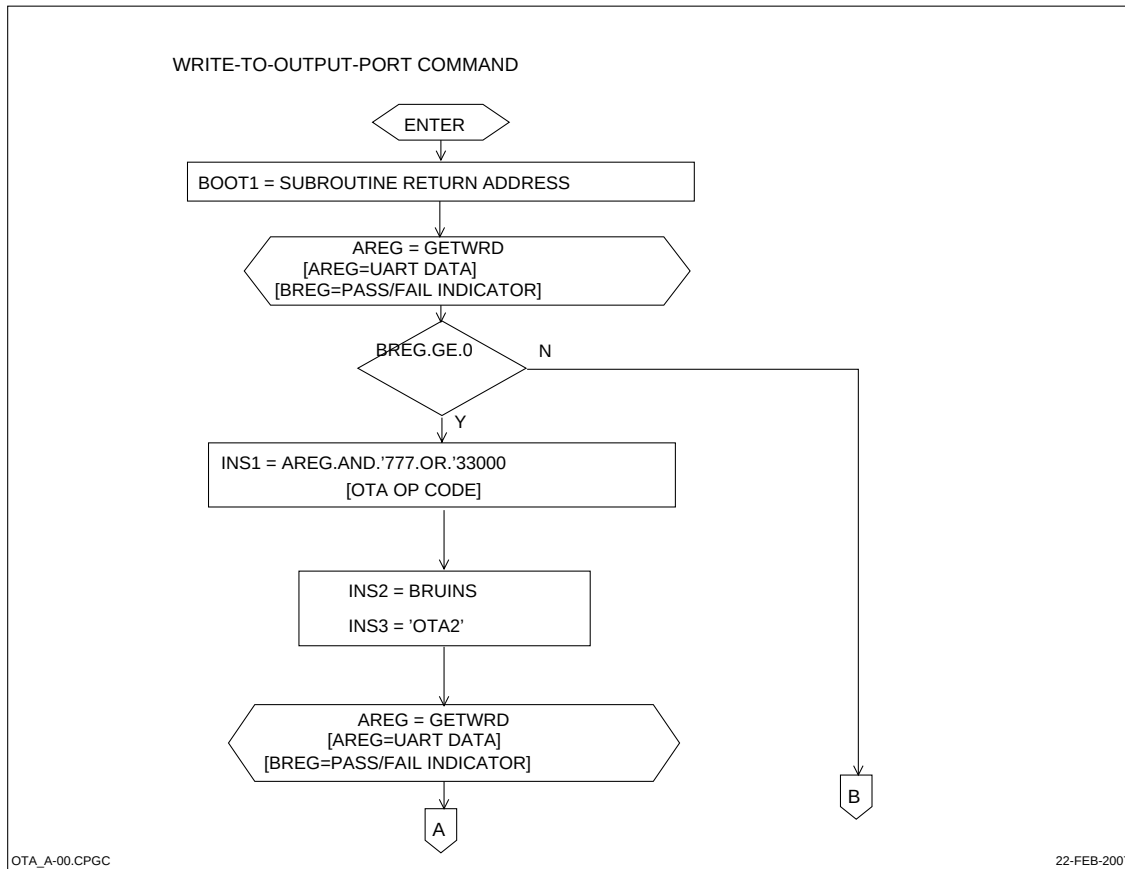


13.00 Figure 548: OPMODE _2

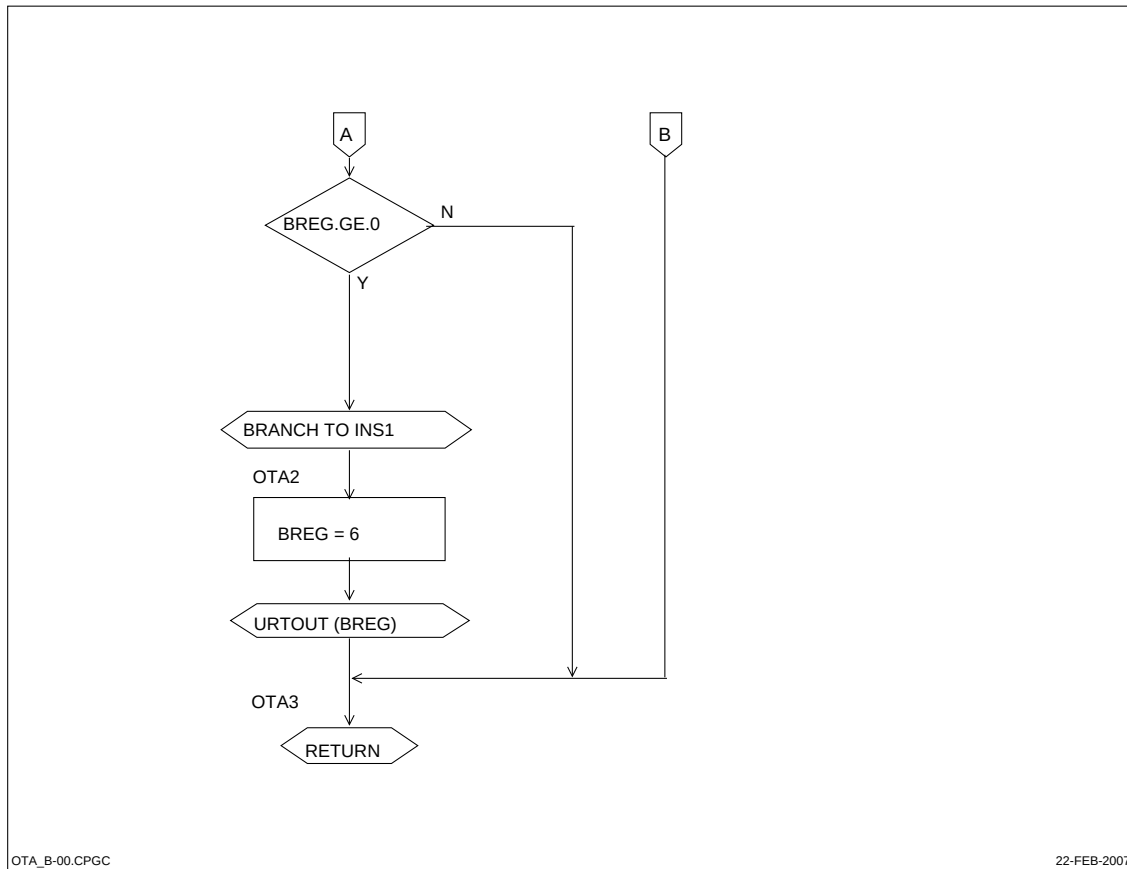


B	Name: OPMODE_3	Ident:	Date: ¶
	Title: Takeoff Determination		7-JUN-2024 14:27:11.87

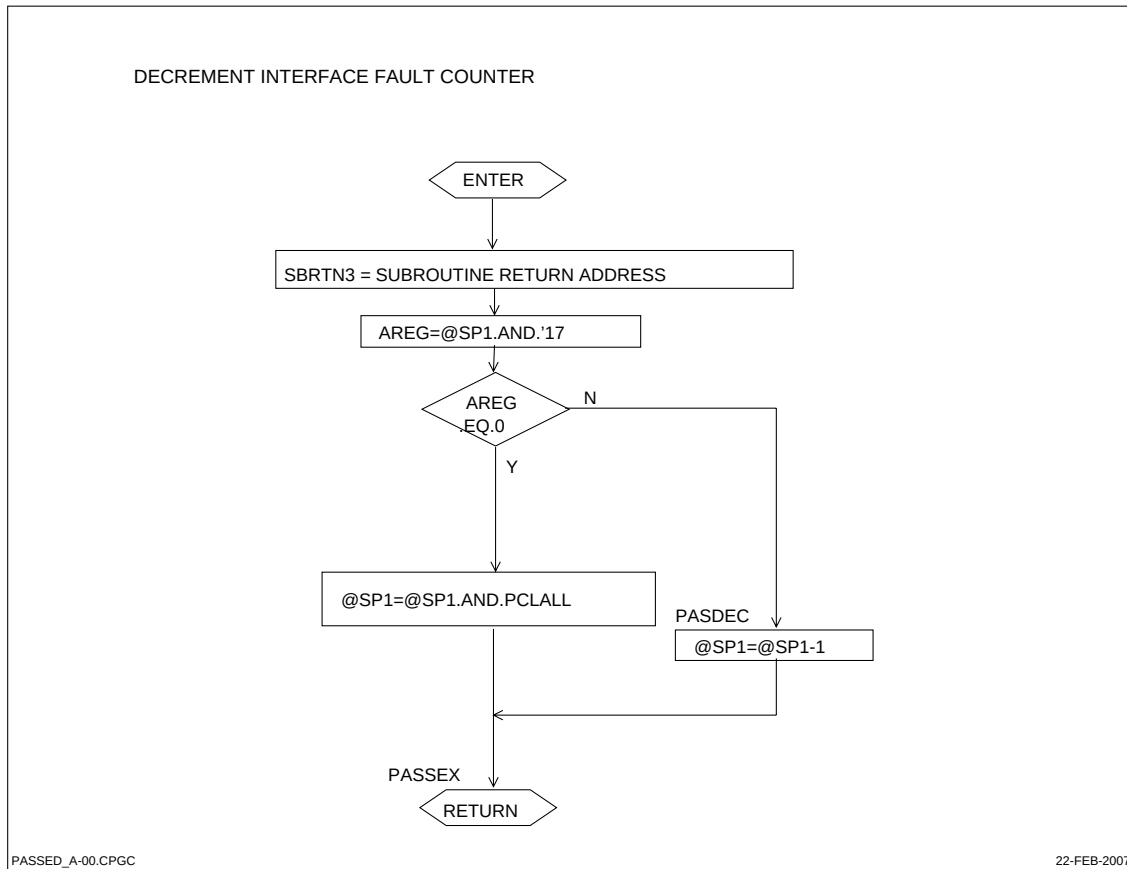
13.00 Figure 550: OTA_A



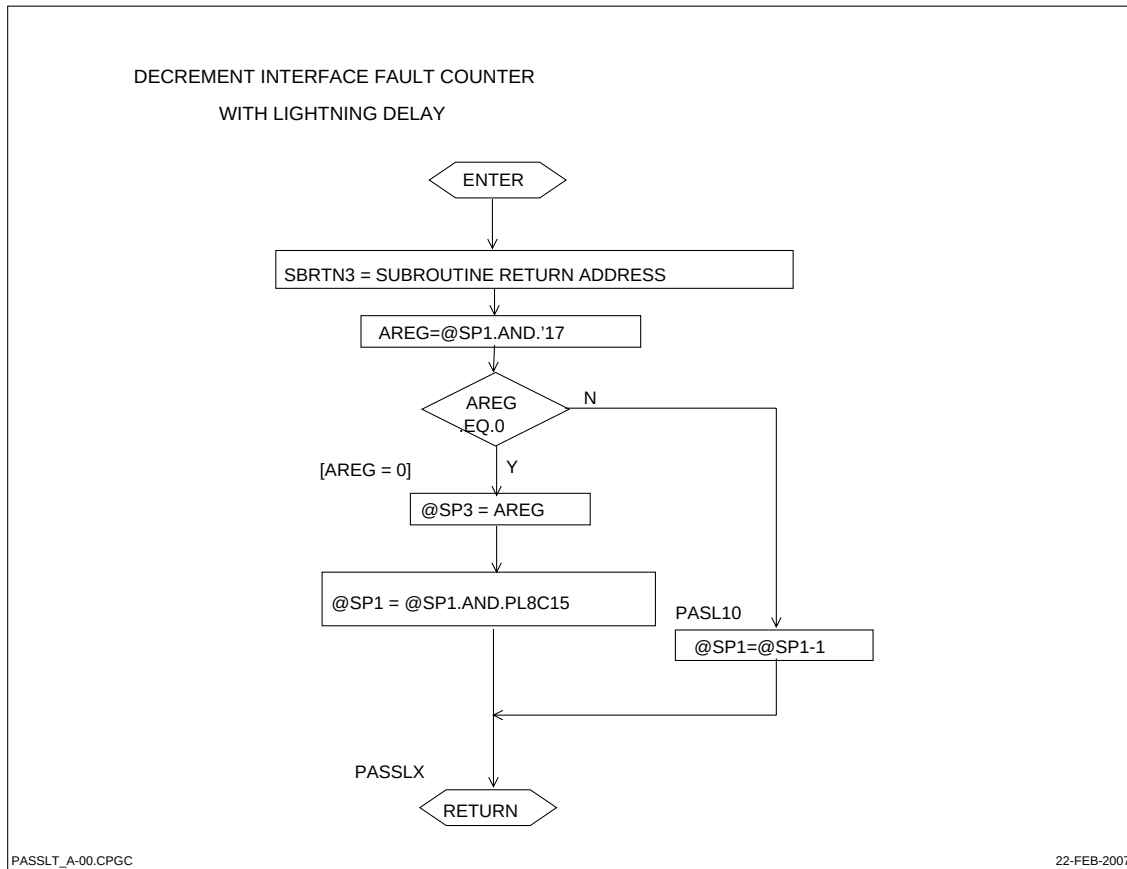
13.00 Figure 551: OTA_B



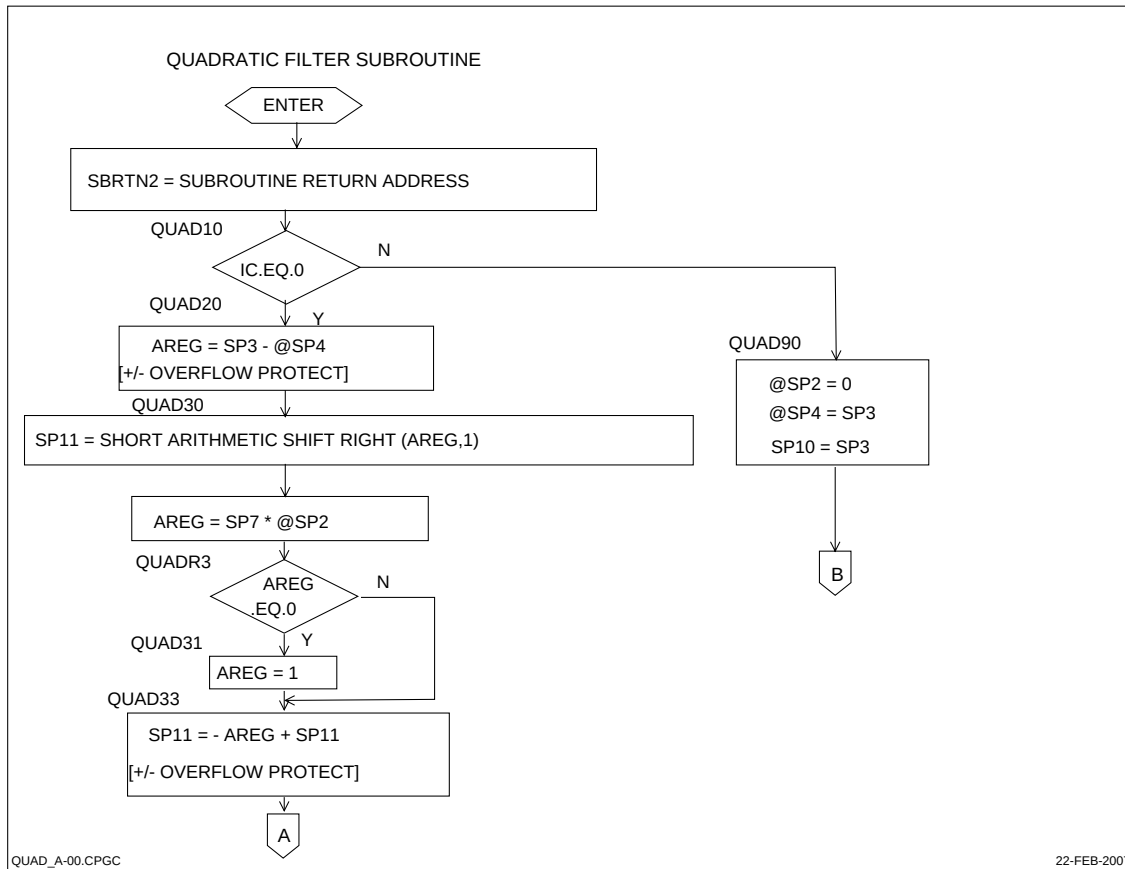
13.00 Figure 552: PASSED_A



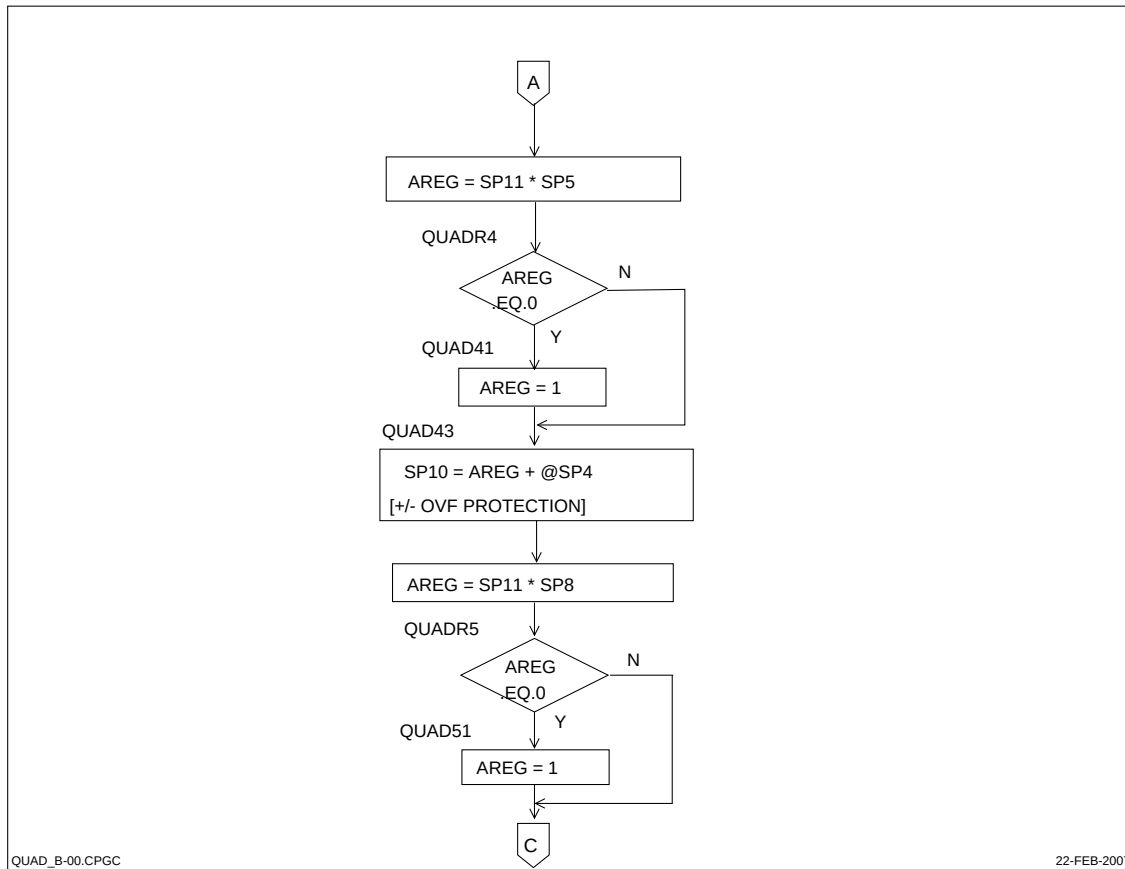
13.00 Figure 553: PASSLT_A



13.00 Figure 554: QUAD_A



13.00 Figure 555: QUAD _B

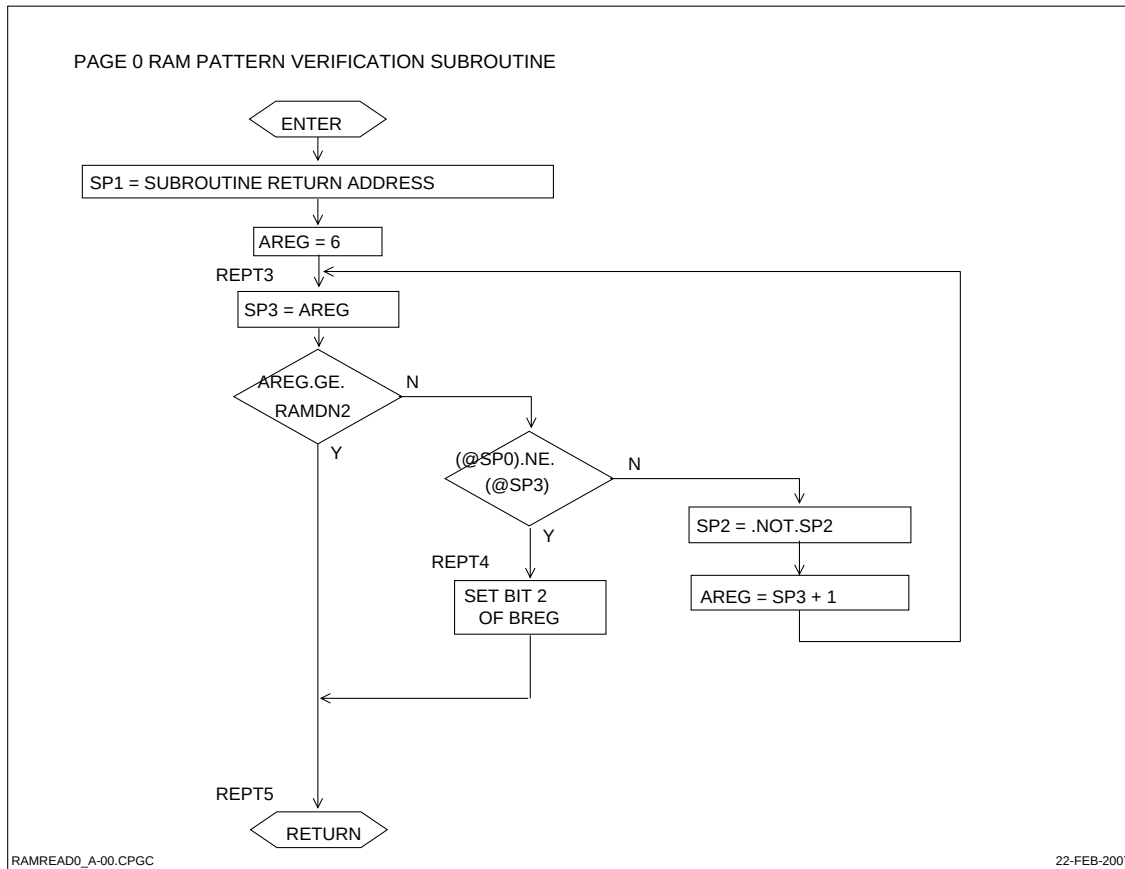


```

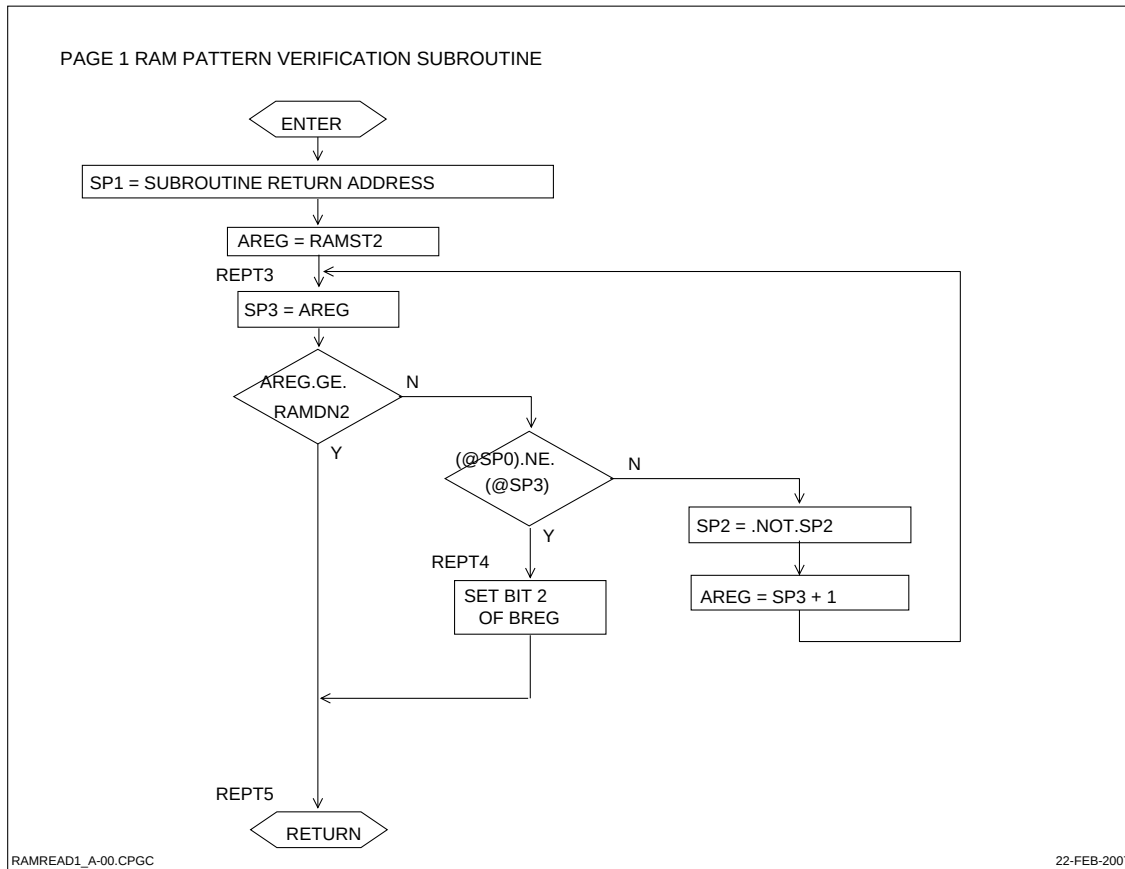
graph TD
    C{{C}} --> QUAD53
    QUAD53["@SP2 = AREG + @SP2  
[+/- OVERFLOW PROTECT]"] --> QUAD59["AREG = @SP2 * SP9"]
    QUAD59 --> QUADR6{"AREG  
EQ.0"}
    QUADR6 -- Y --> QUAD61["AREG = 1"]
    QUADR6 -- N --> QUAD63
    QUAD61 --> QUAD63
    QUAD63["@SP4 = AREG + @SP4  
[+/- OVERFLOW PROTECT]"] --> QUAD67
    B{{B}} --> QUAD67
    QUAD67["AREG = SP10"] --> RETURN{{RETURN}}

```

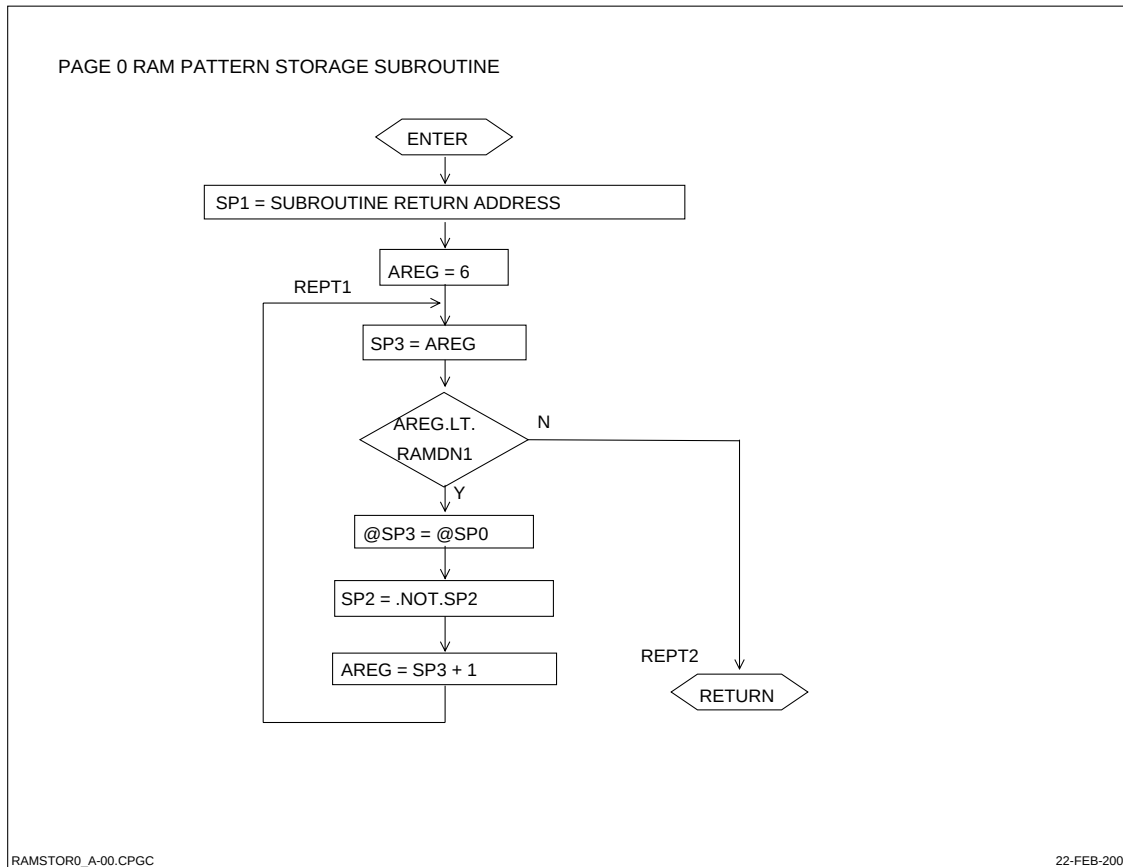
13.00 Figure 557: RAMREAD0_A



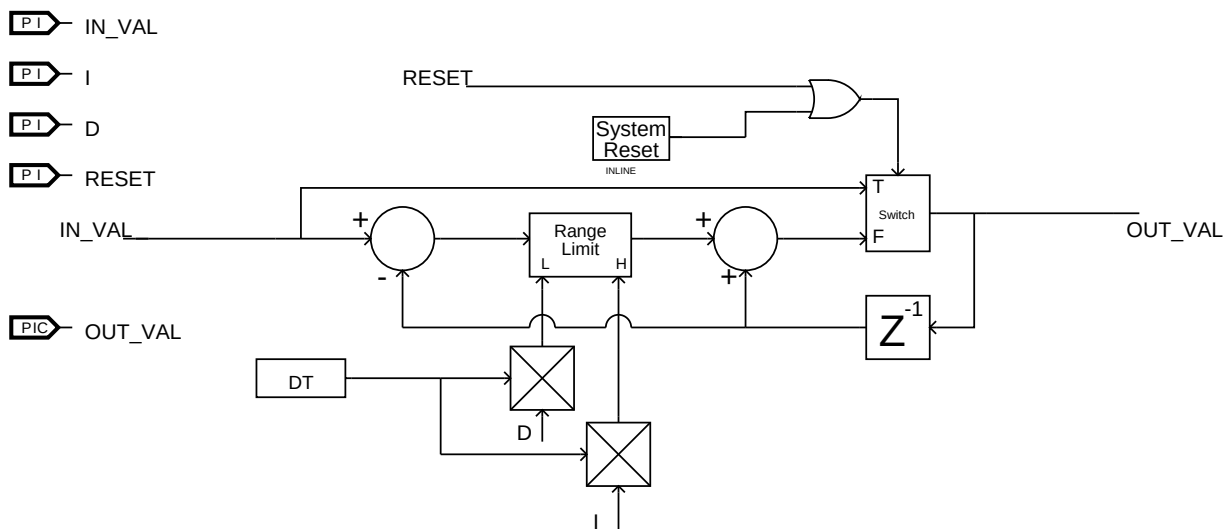
13.00 Figure 558: RAMREAD1_A



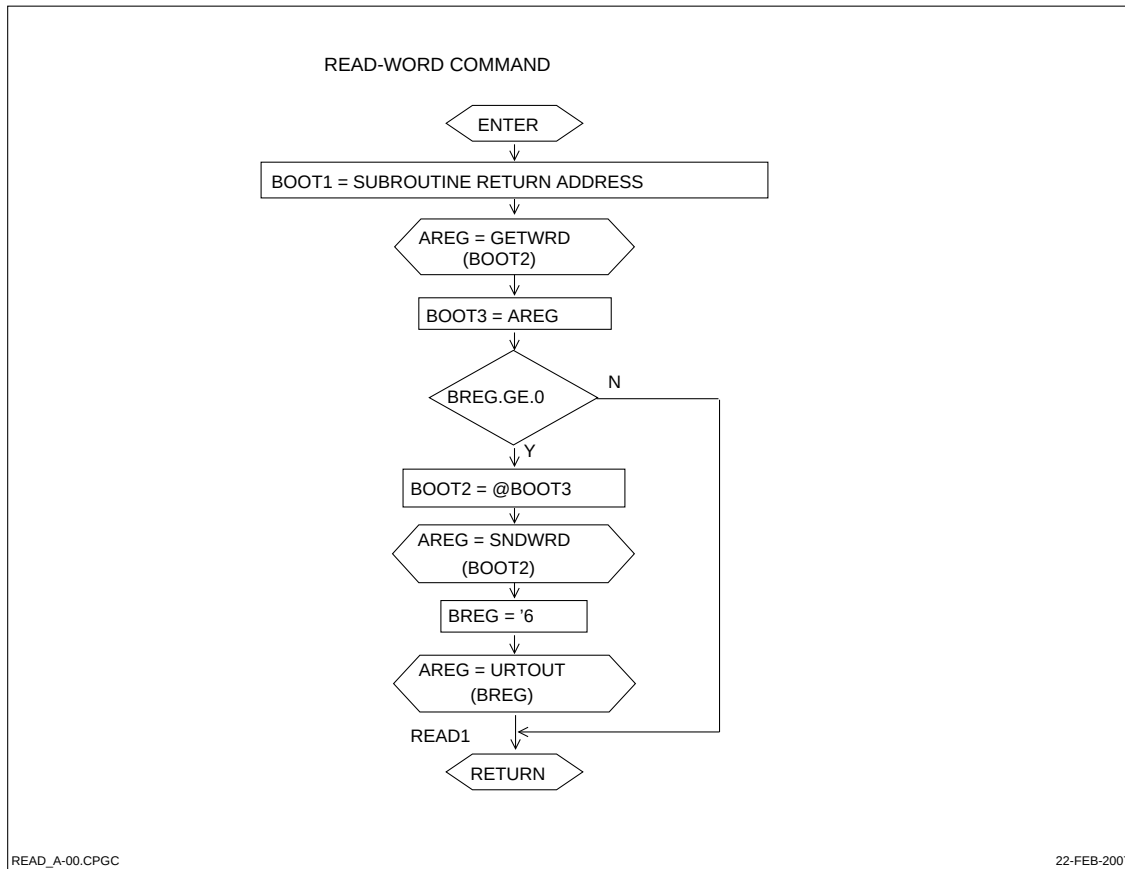
13.00 Figure 559: RAMSTOR0 _A



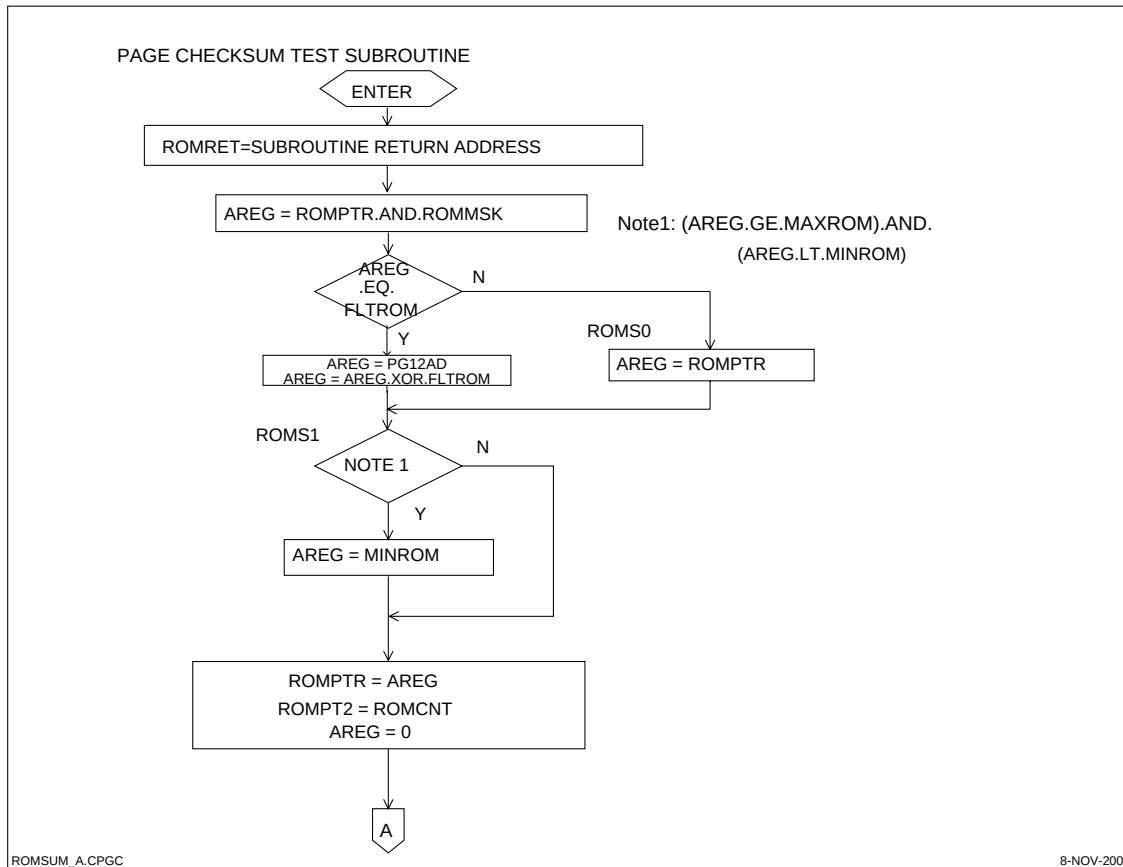
13.00 Figure 561: RATE_LIMIT



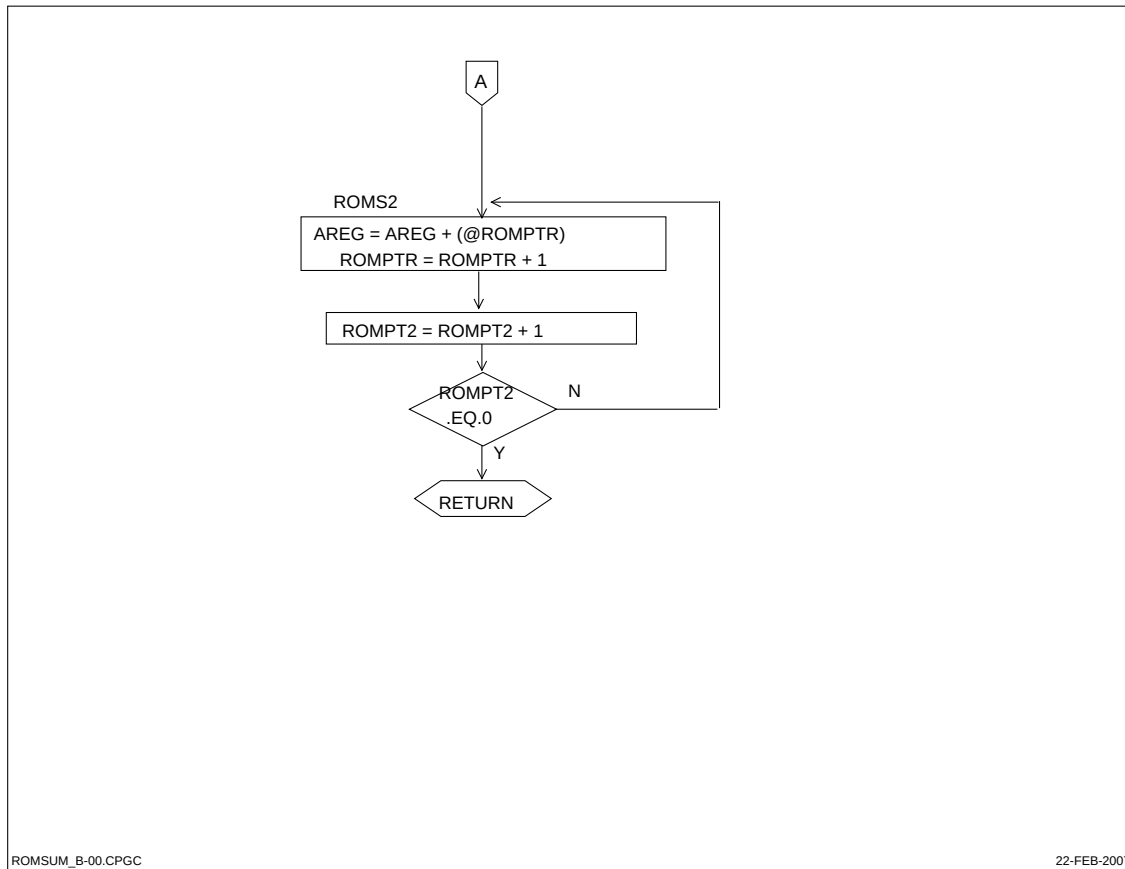
13.00 Figure 562: READ _A



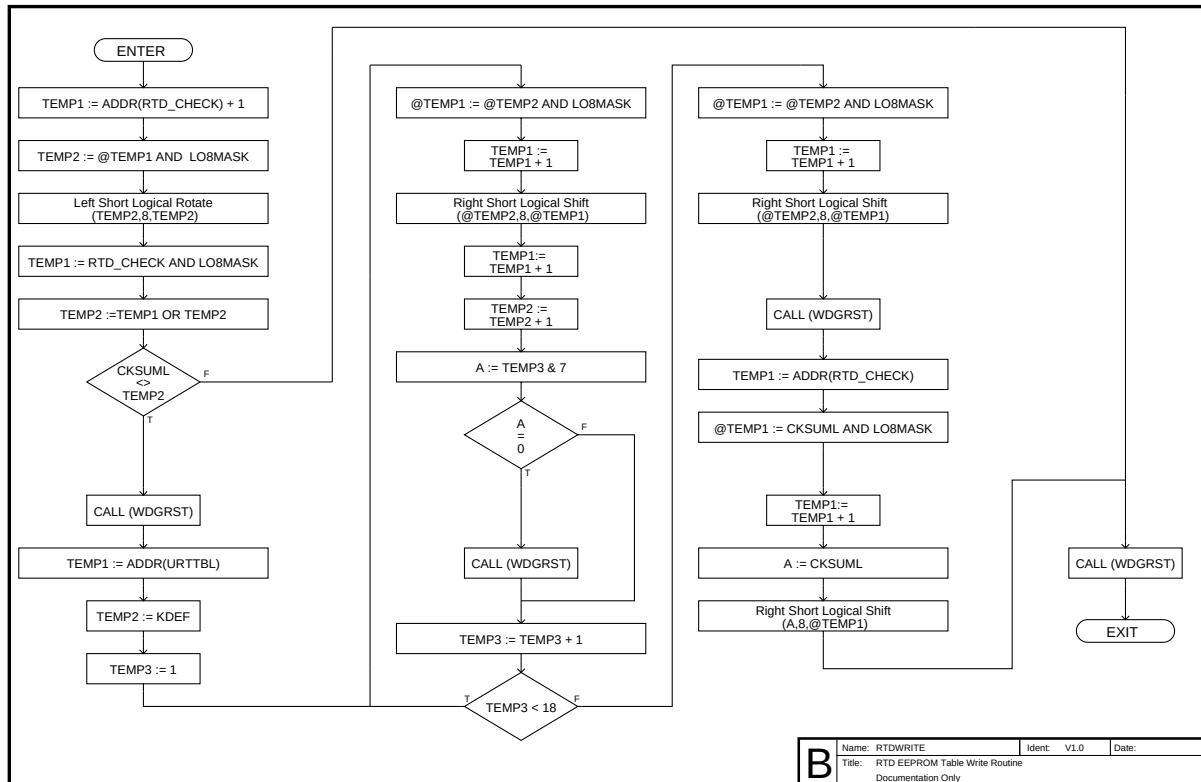
13.00 Figure 563: ROMSUM_A



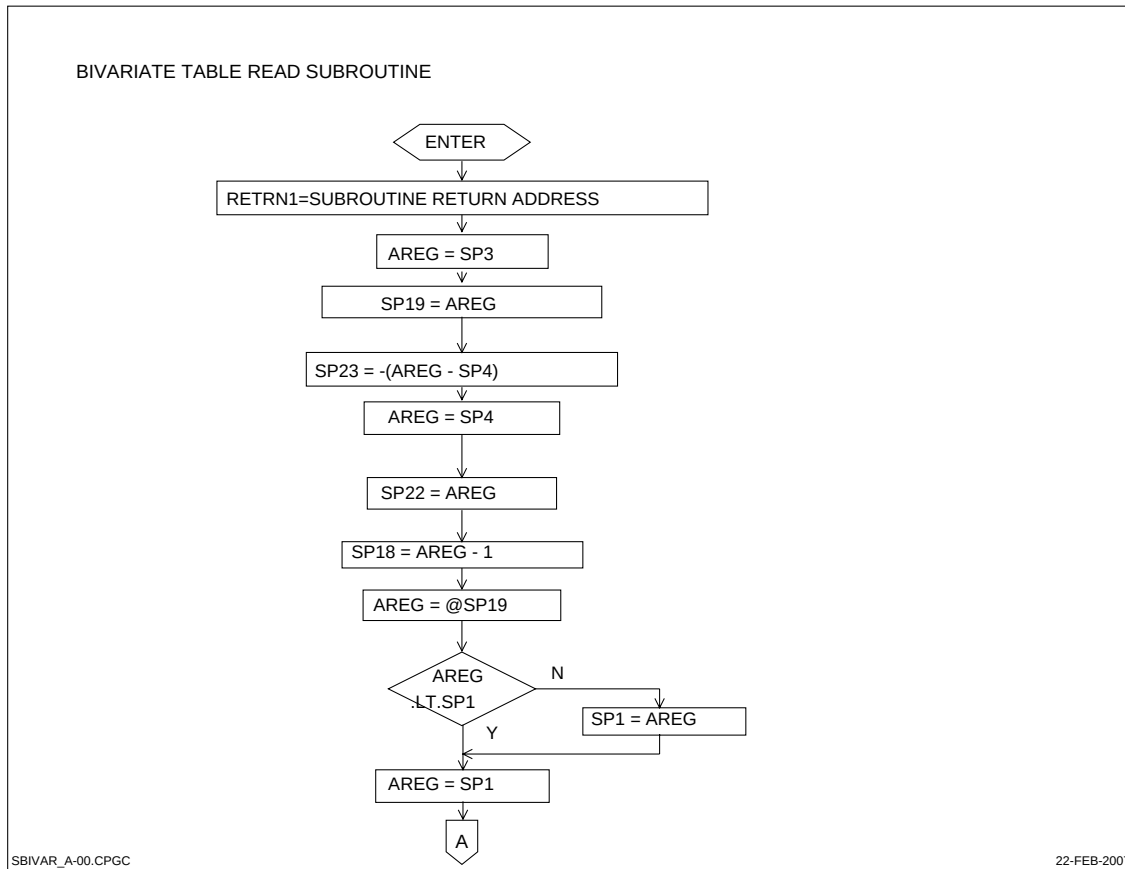
13.00 Figure 564: ROMSUM_B



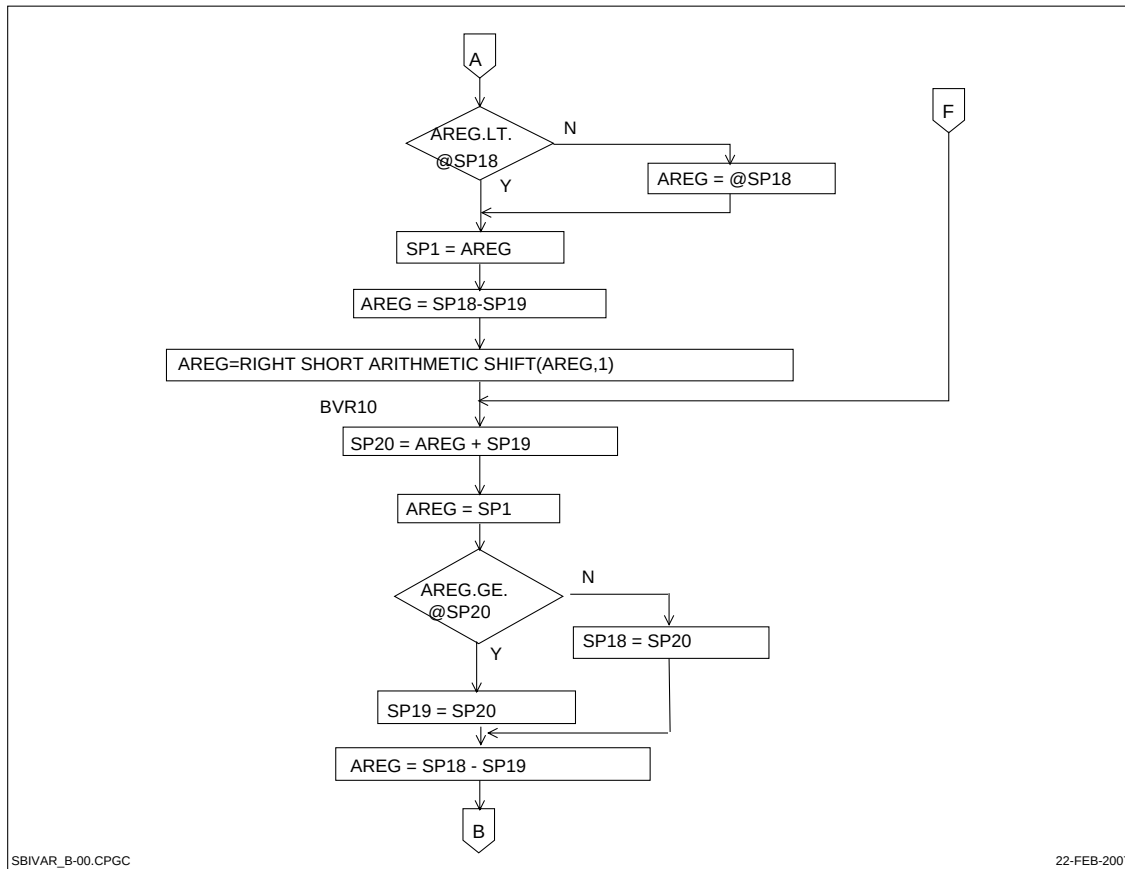
13.00 Figure 565: RTDWRITE



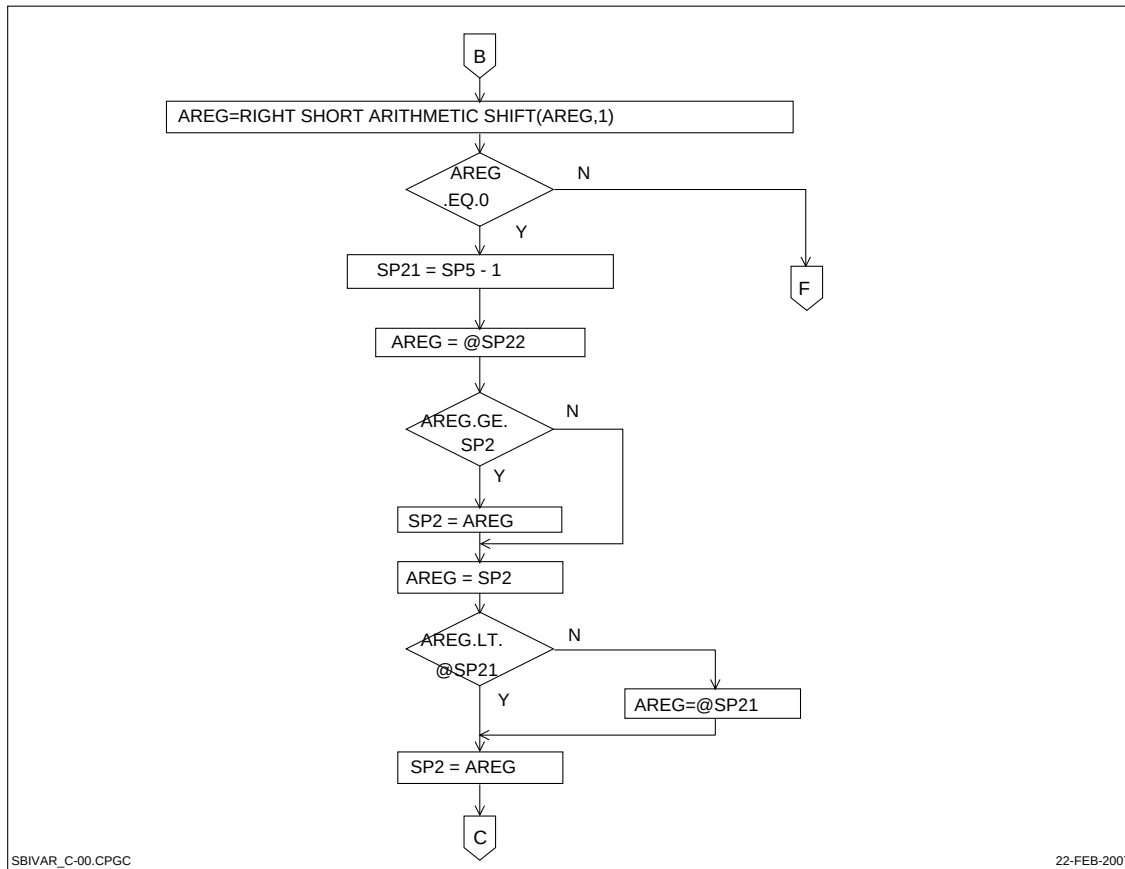
13.00 Figure 566: SBIVAR_A



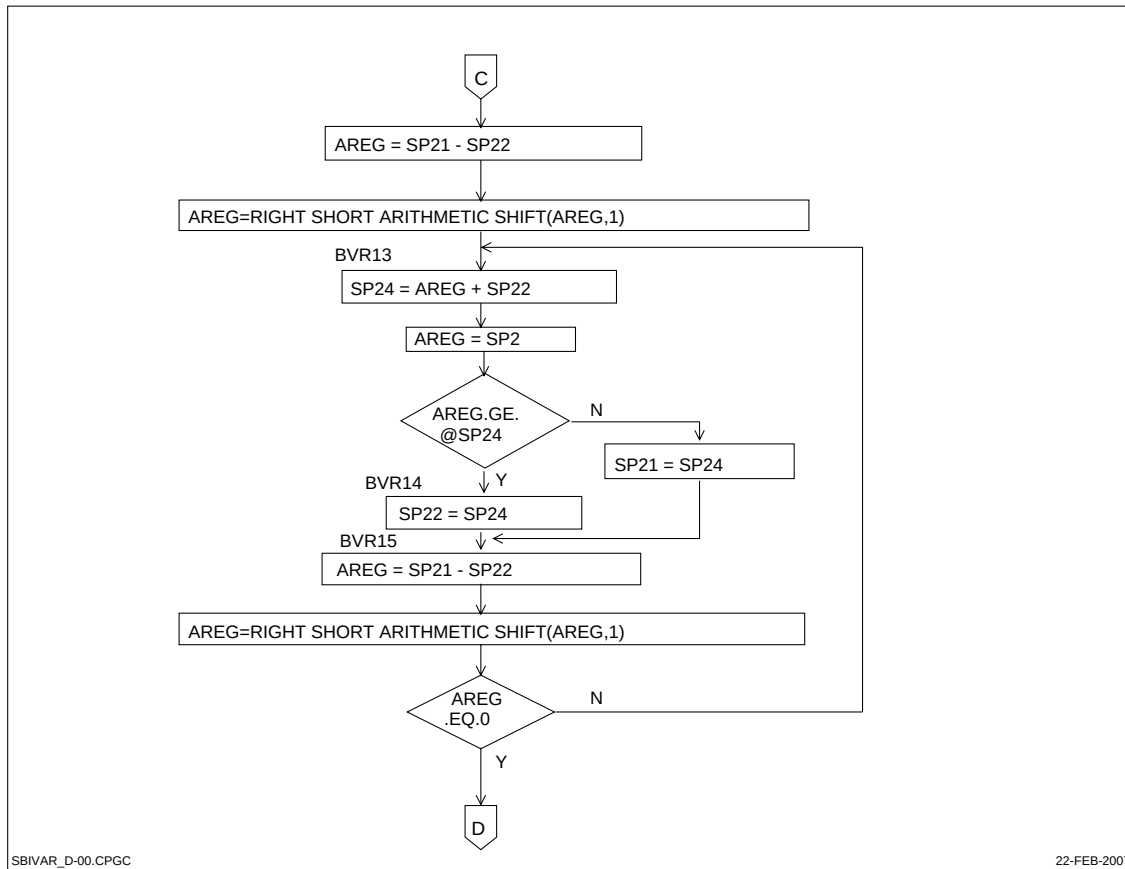
13.00 Figure 567: SBIVAR_B



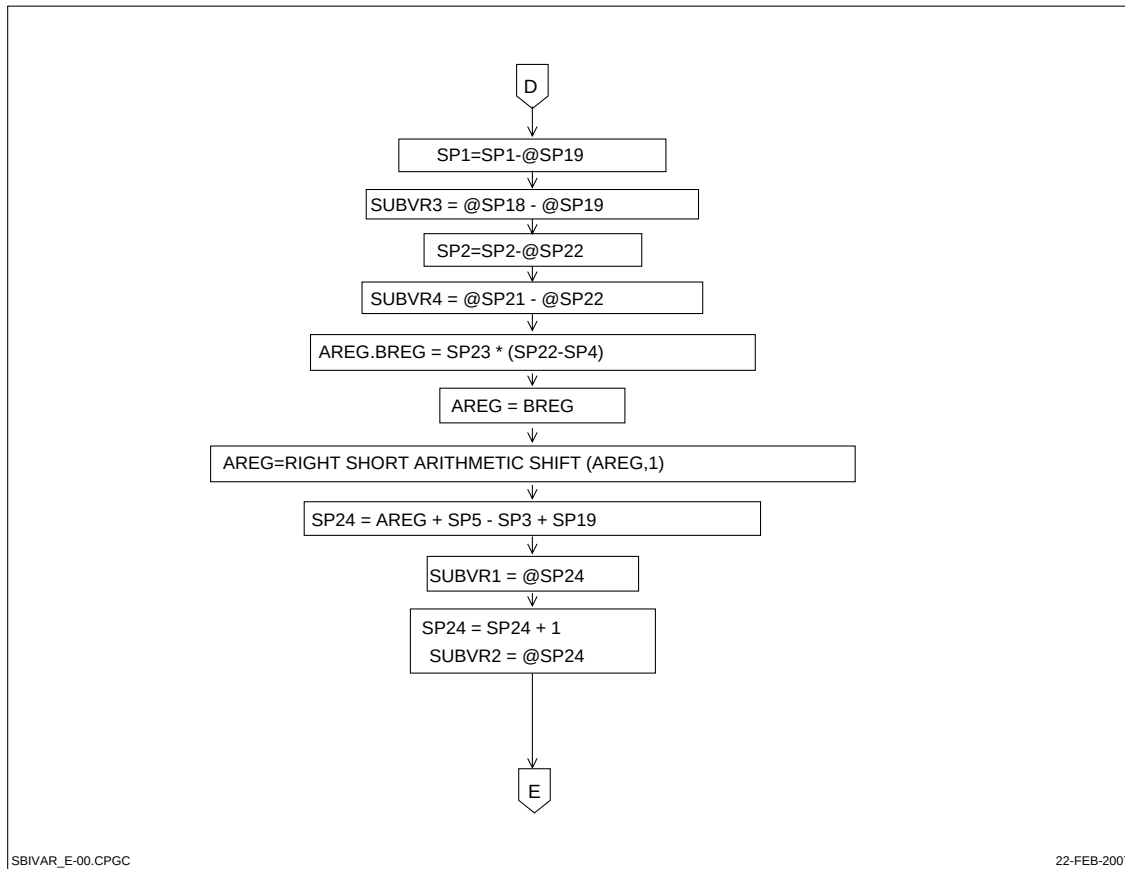
13.00 Figure 568: SBIVAR_C



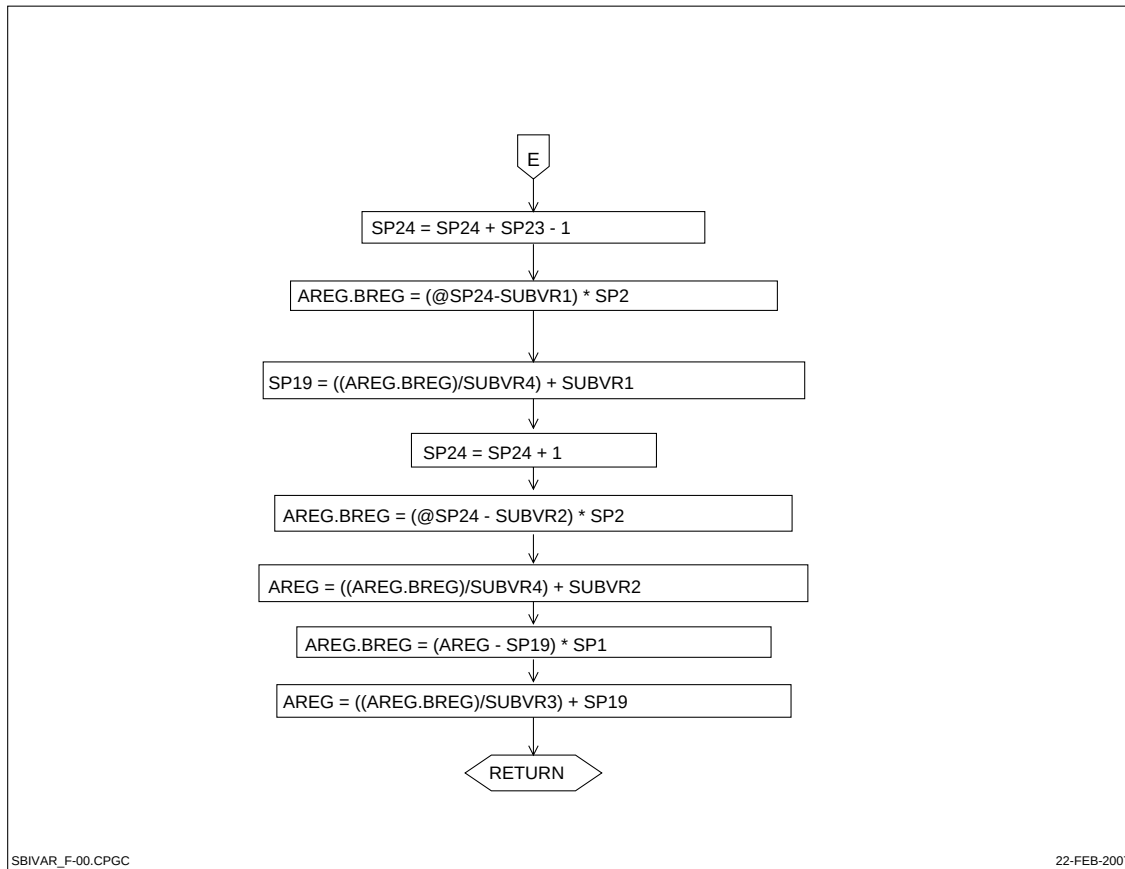
13.00 Figure 569: SBIVAR_D



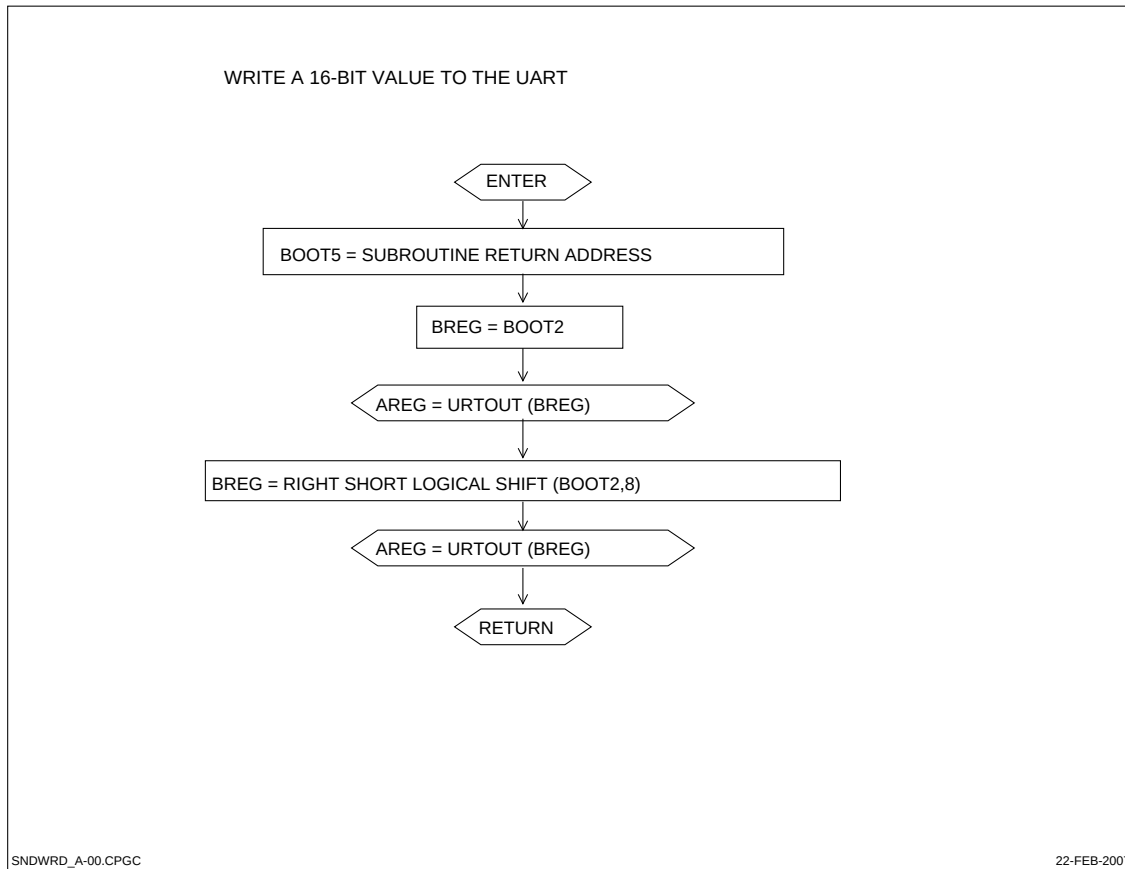
13.00 Figure 570: SBIVAR_E



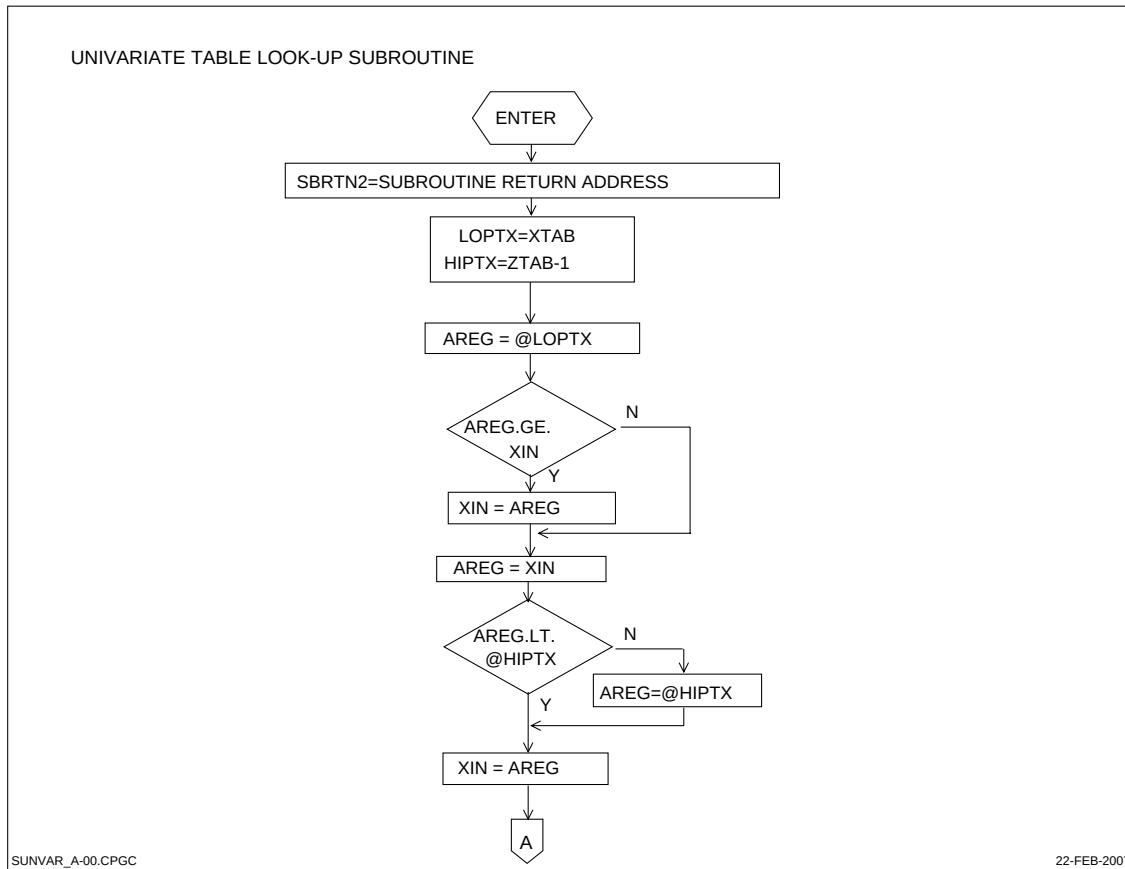
13.00 Figure 571: SBIVAR_F



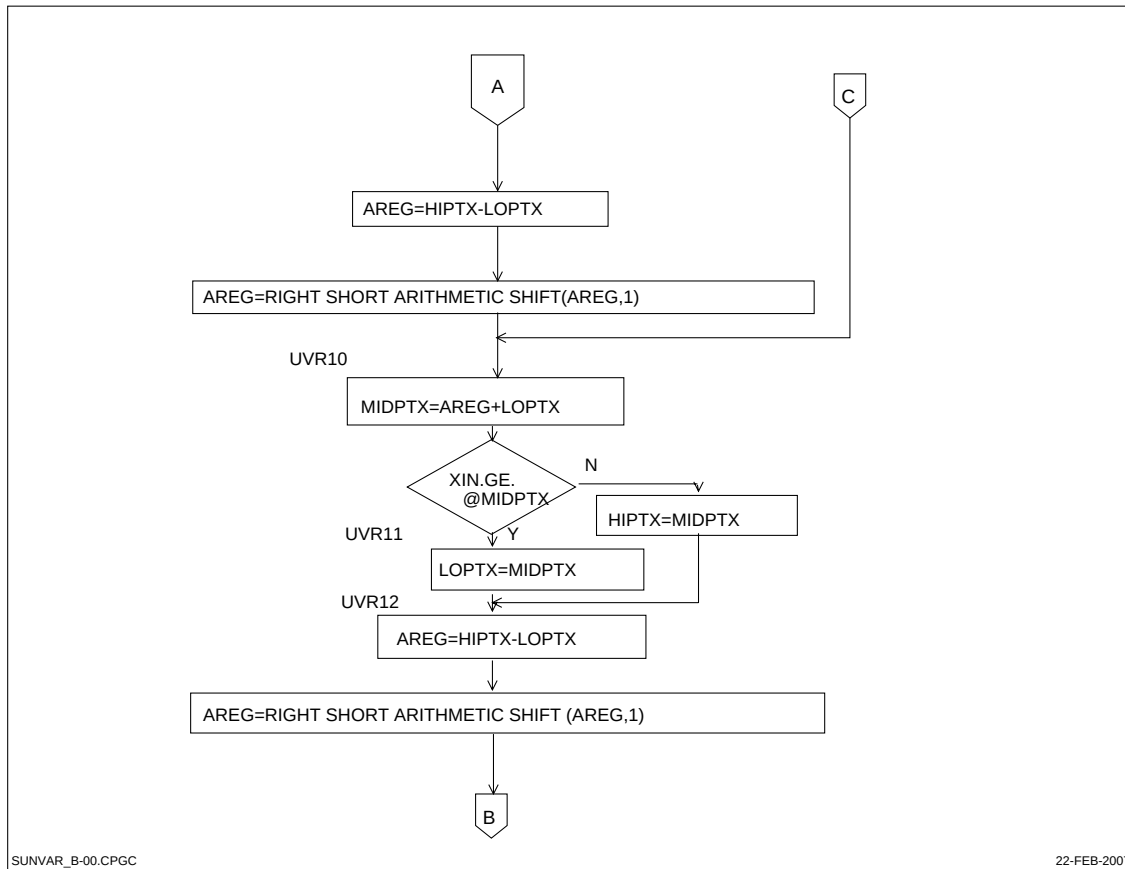
13.00 Figure 572: SNDWRD _A



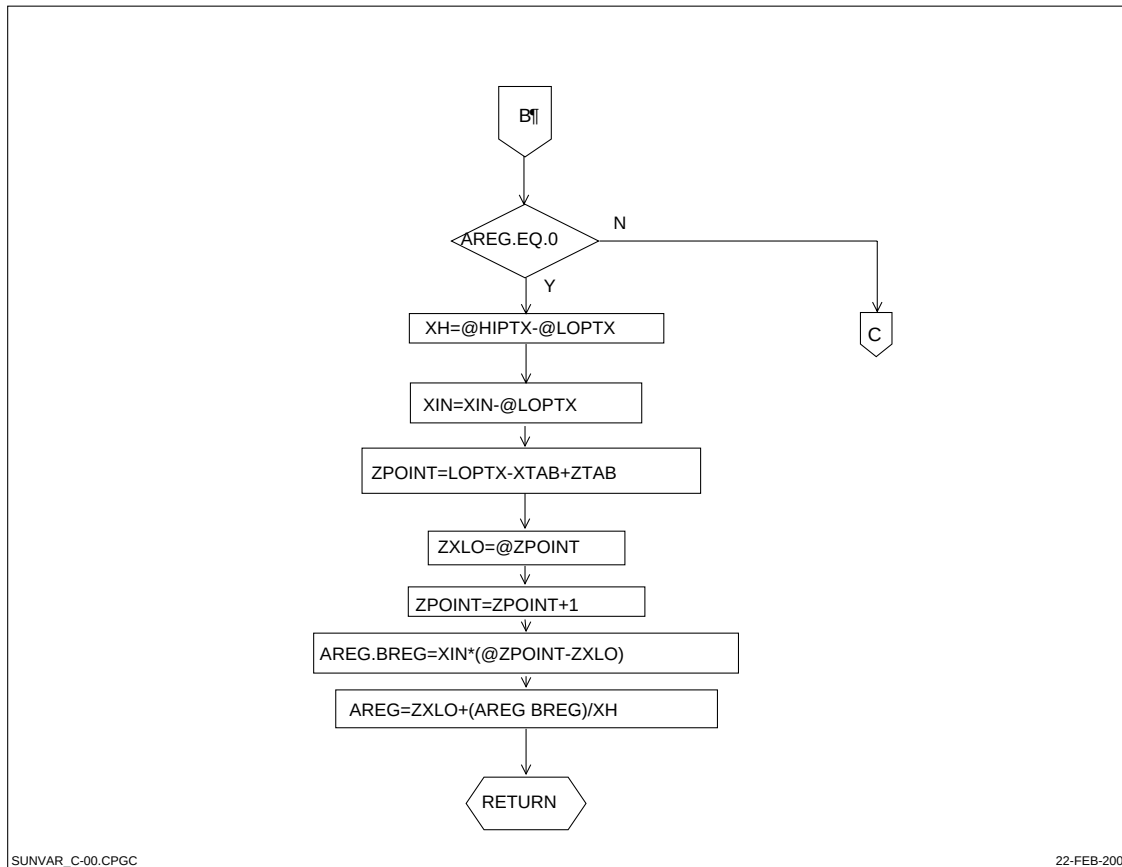
13.00 Figure 573: SUNVAR_A



13.00 Figure 574: SUNVAR_B



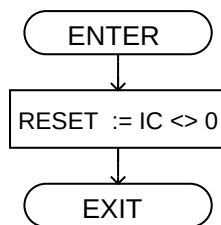
13.00 Figure 575: SUNVAR _C



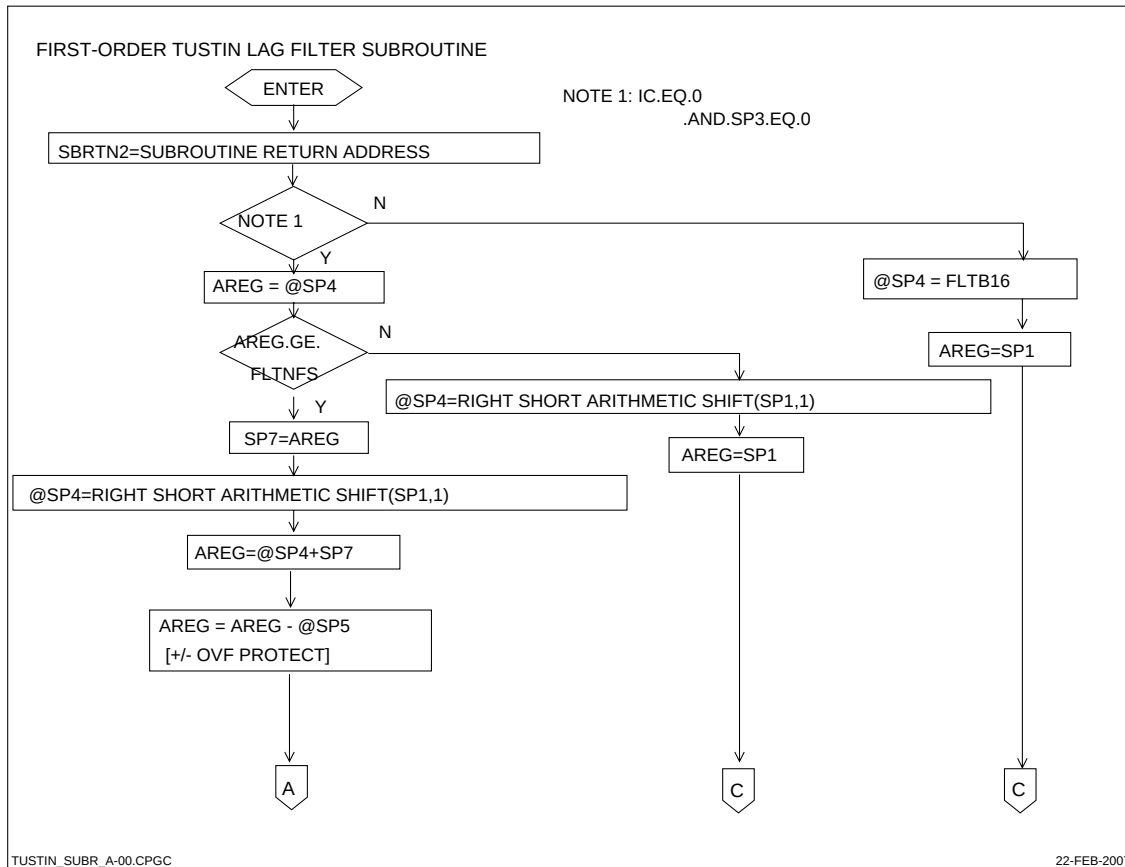
13.00 Figure 576: SYSTEM _RESET

GI IC

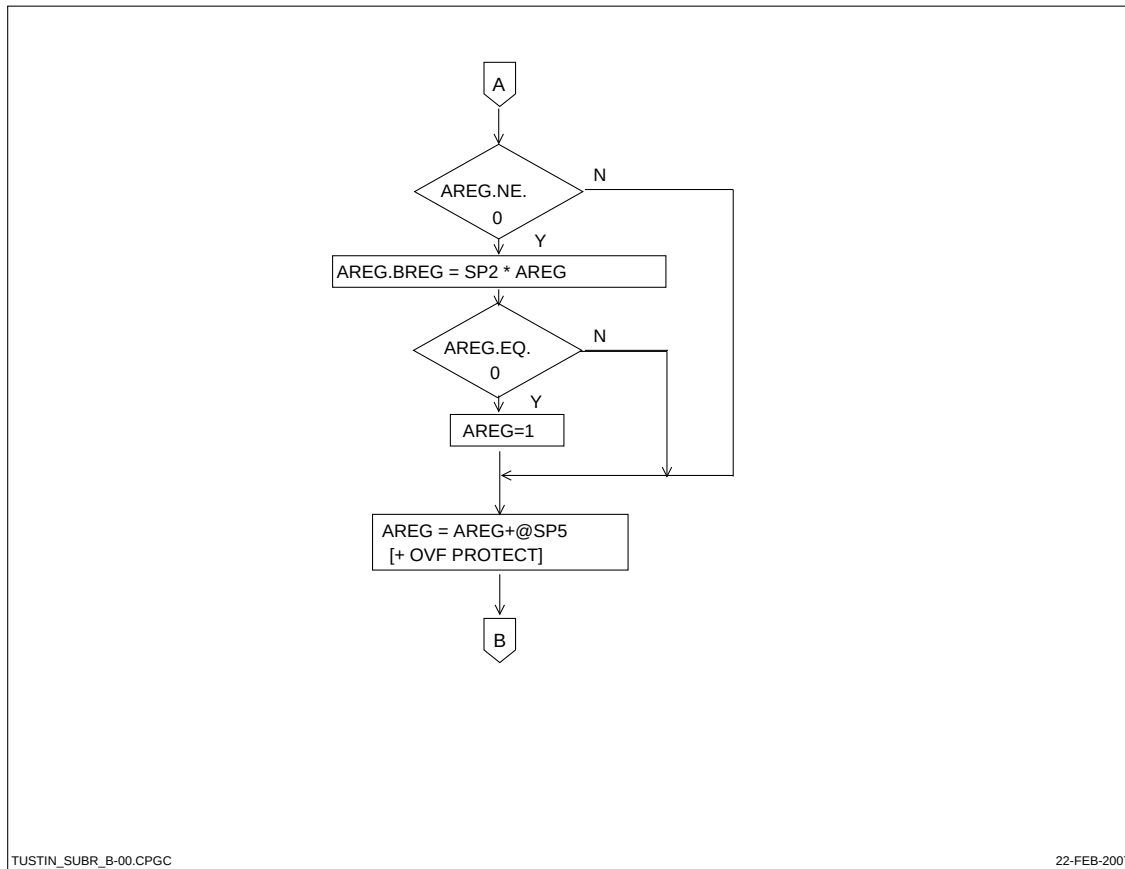
RESET FO



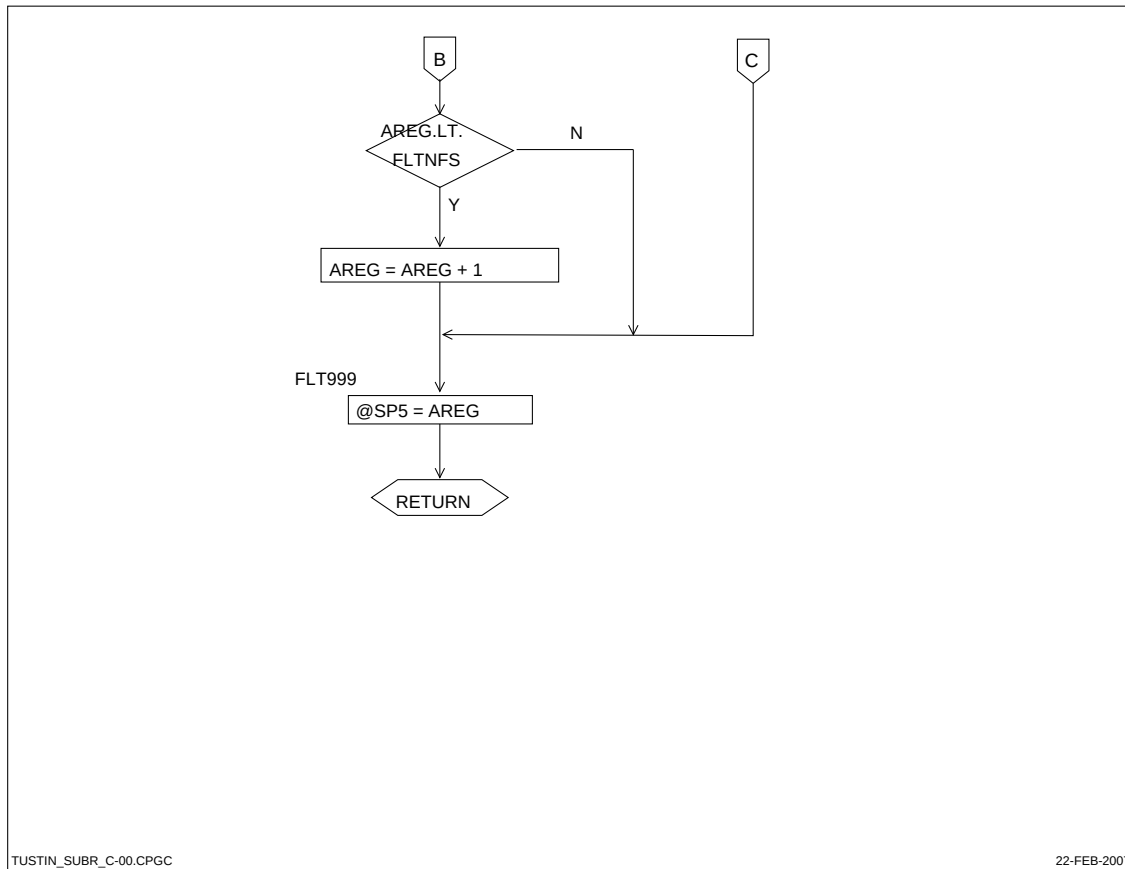
13.00 Figure 577: TUSTIN_SUBR_A



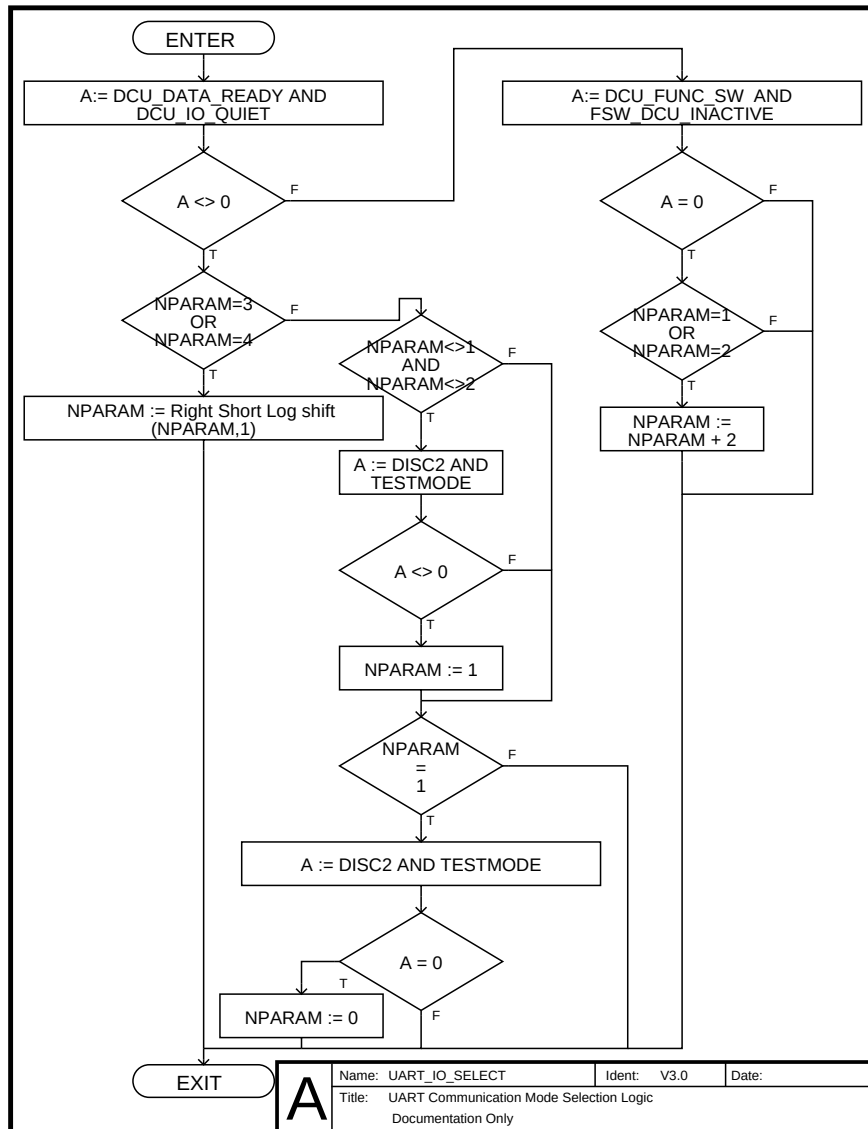
13.00 Figure 578: TUSTIN_SUBR_B



13.00 Figure 579: TUSTIN_SUBR_C

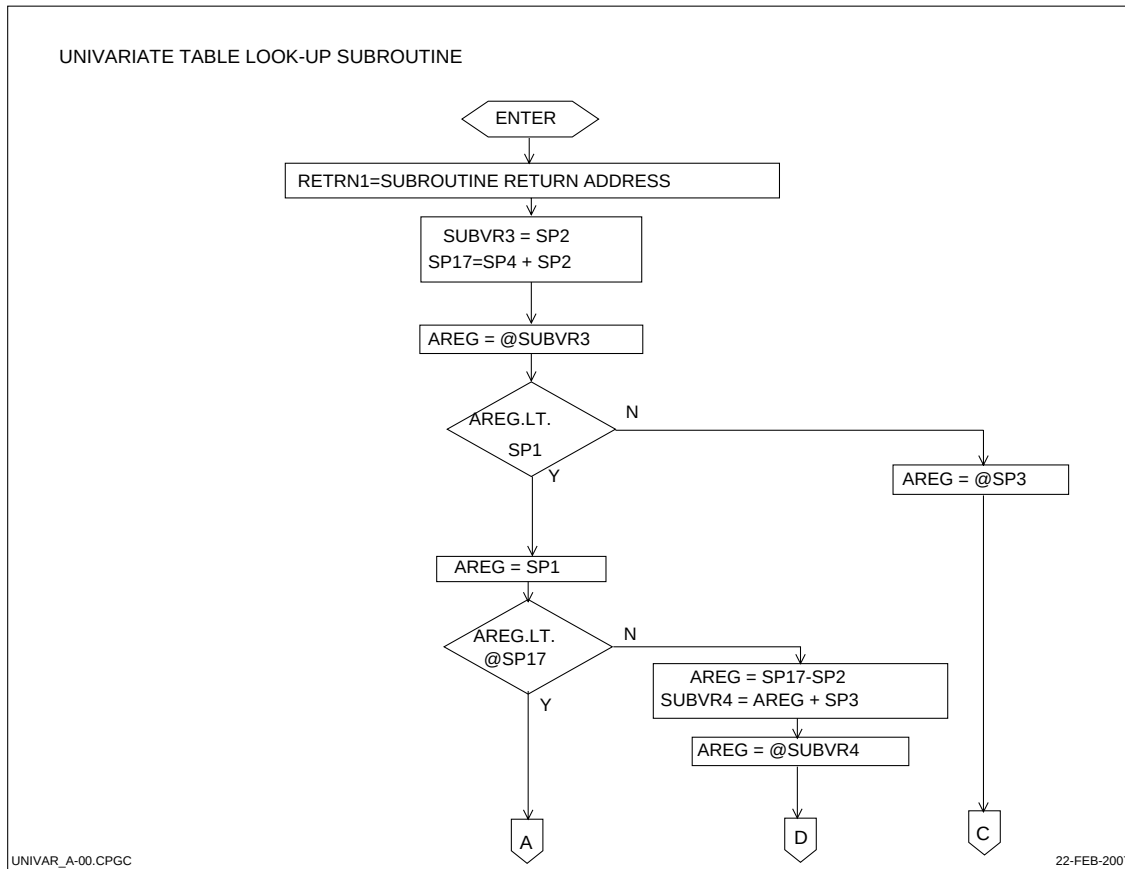


13.00 Figure 580: UART_IO_SELECT

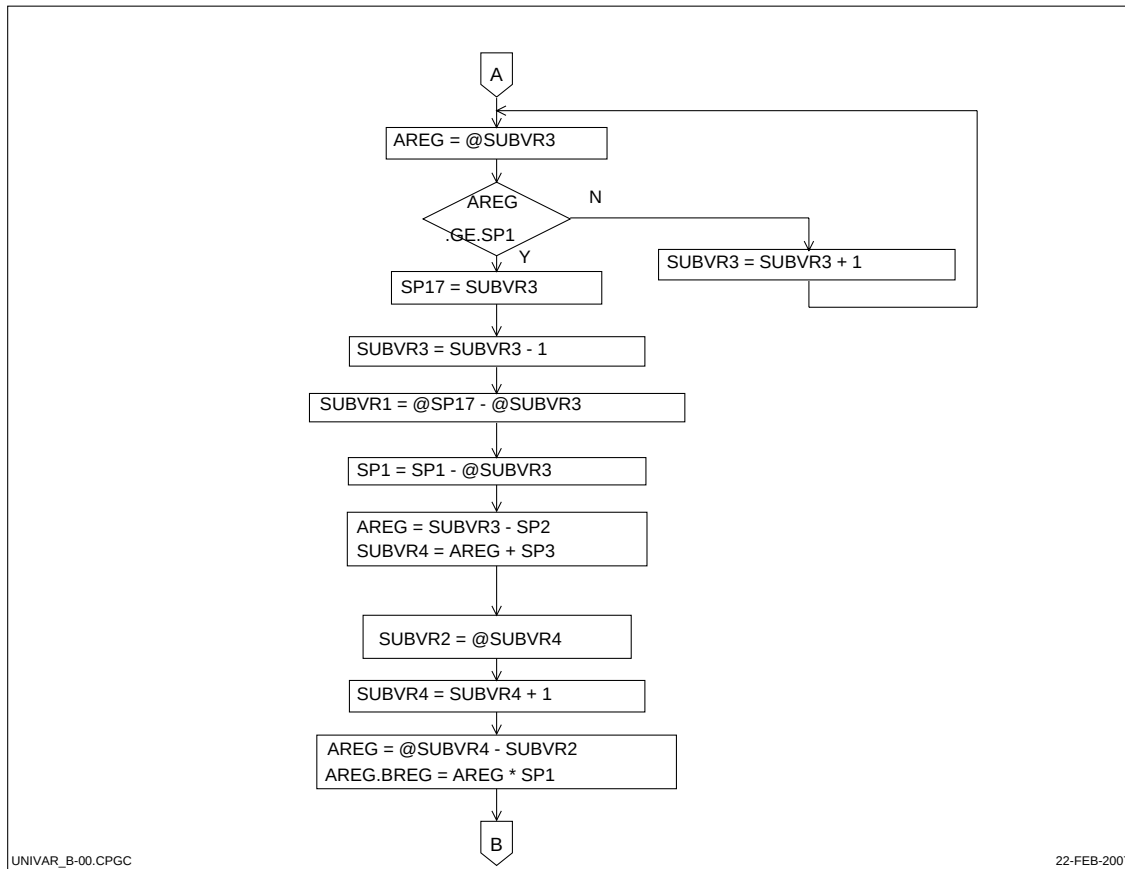


Export Classification: US: EAR 9E991

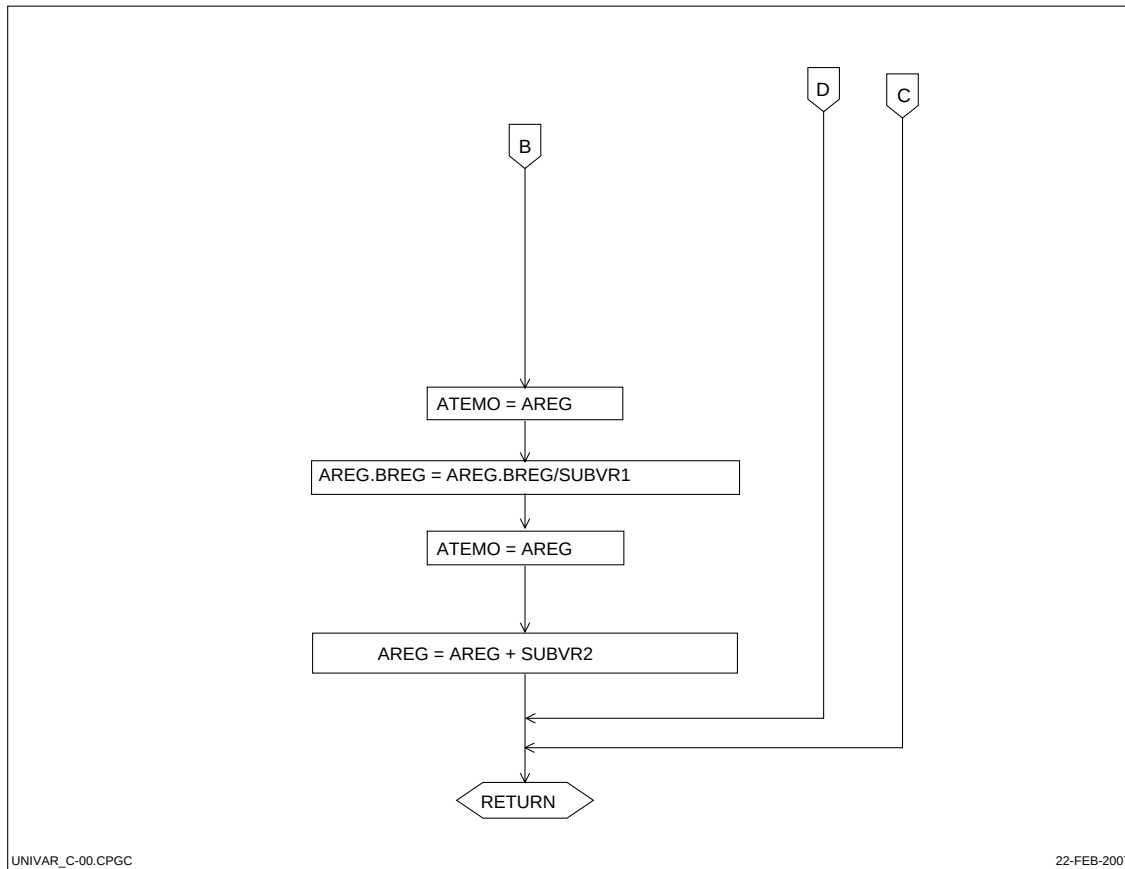
13.00 Figure 581: UNIVAR_A



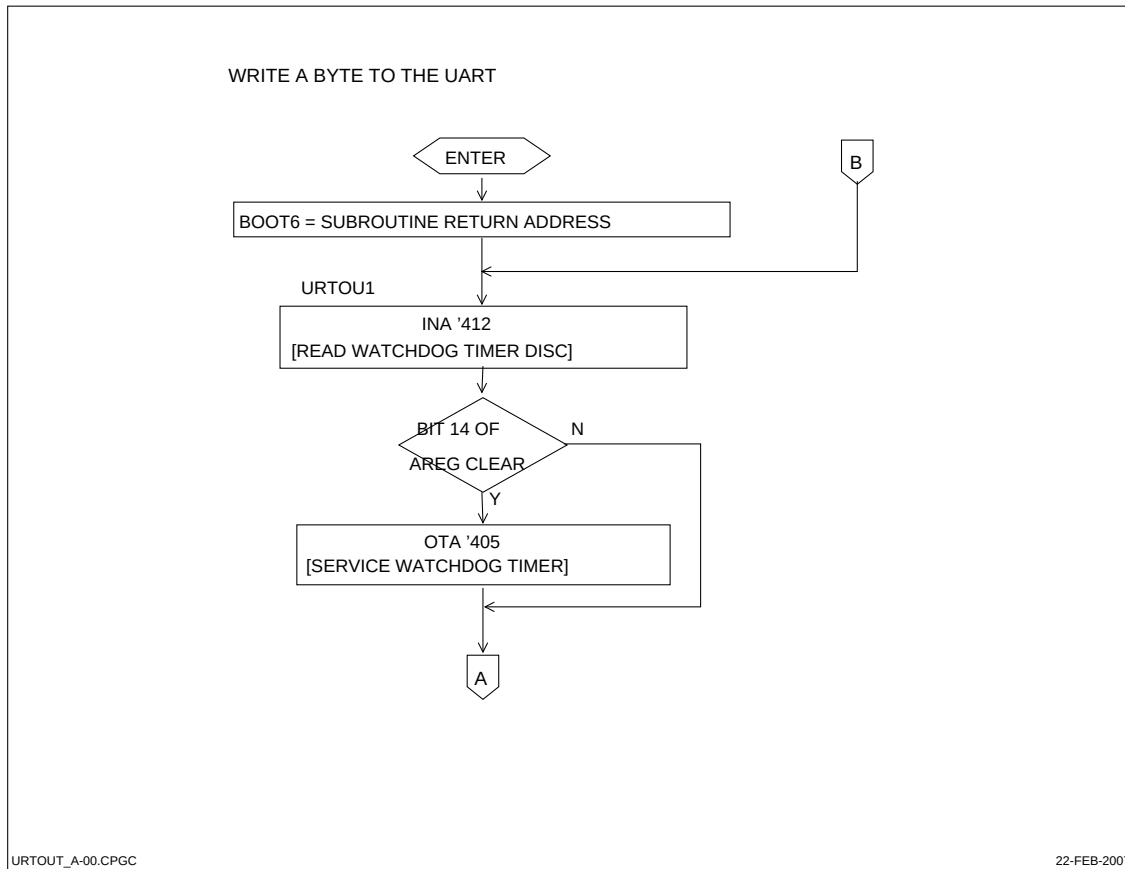
13.00 Figure 582: UNIVAR_B



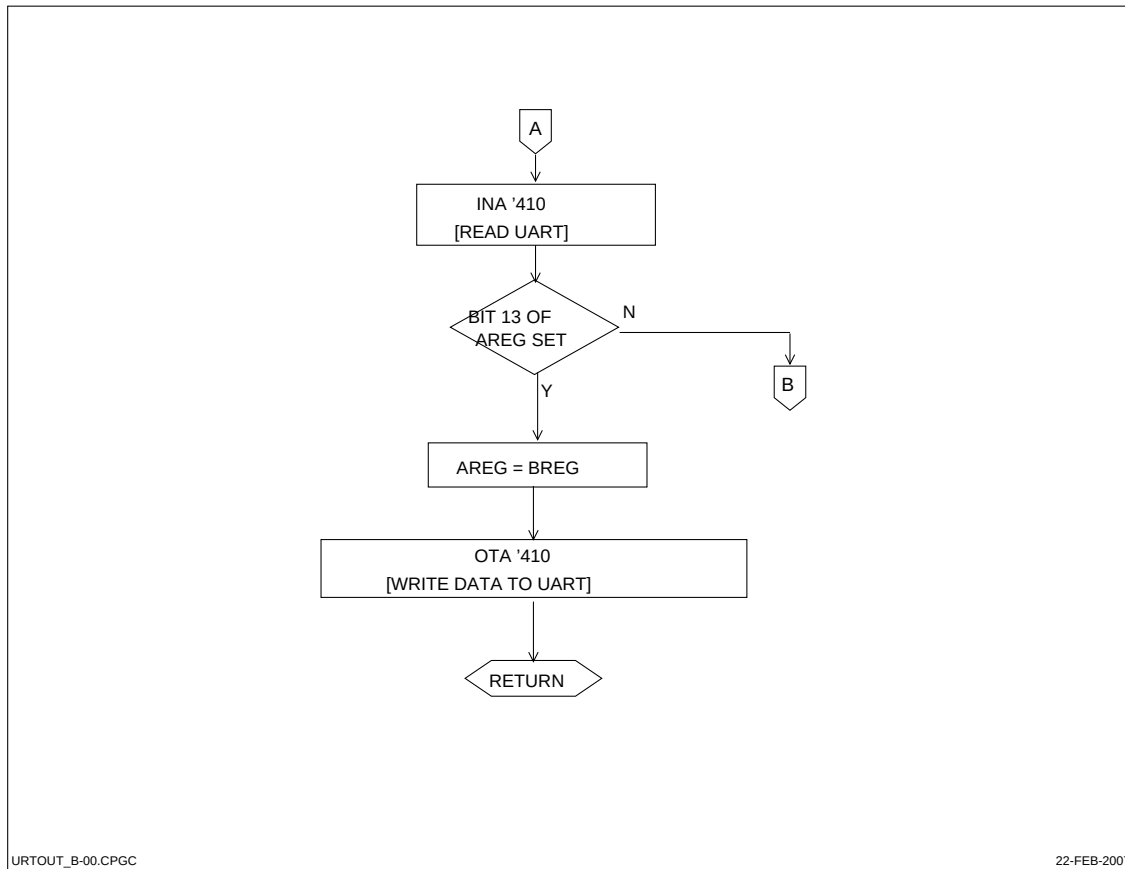
13.00 Figure 583: UNIVAR _C



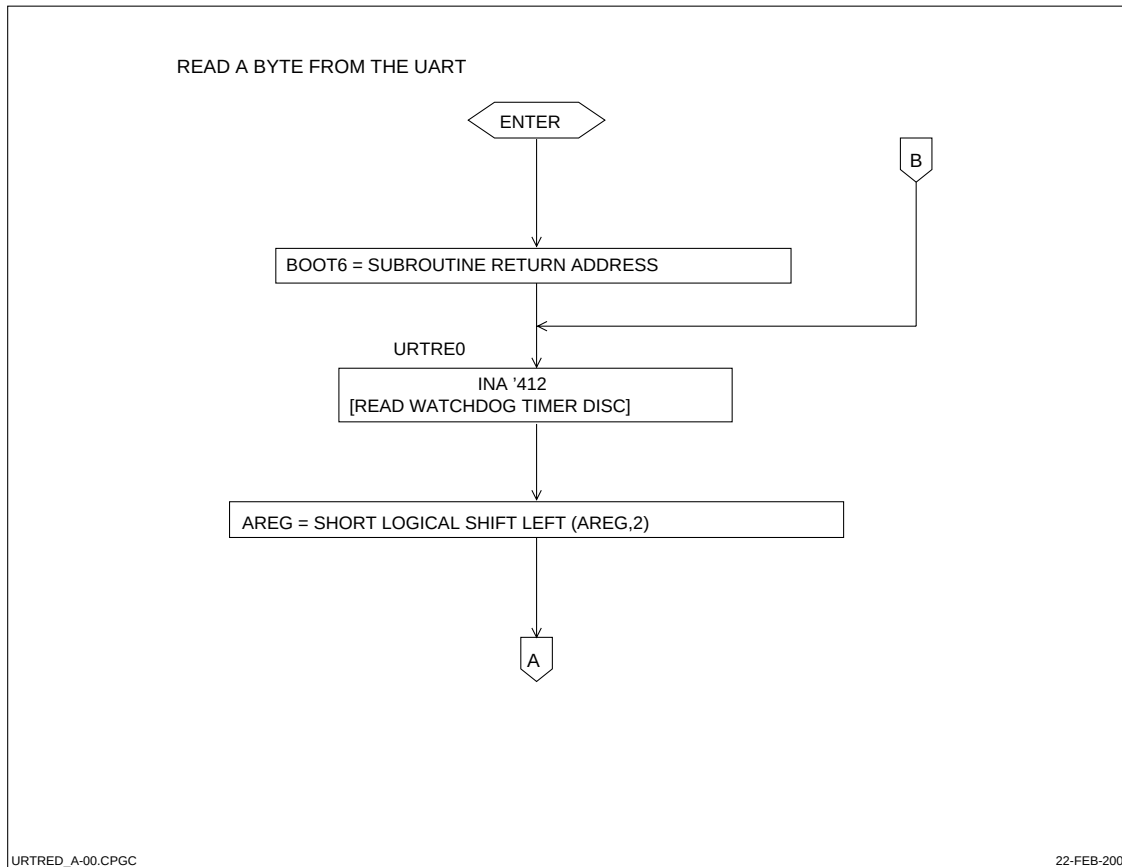
13.00 Figure 584: URTOUT _A



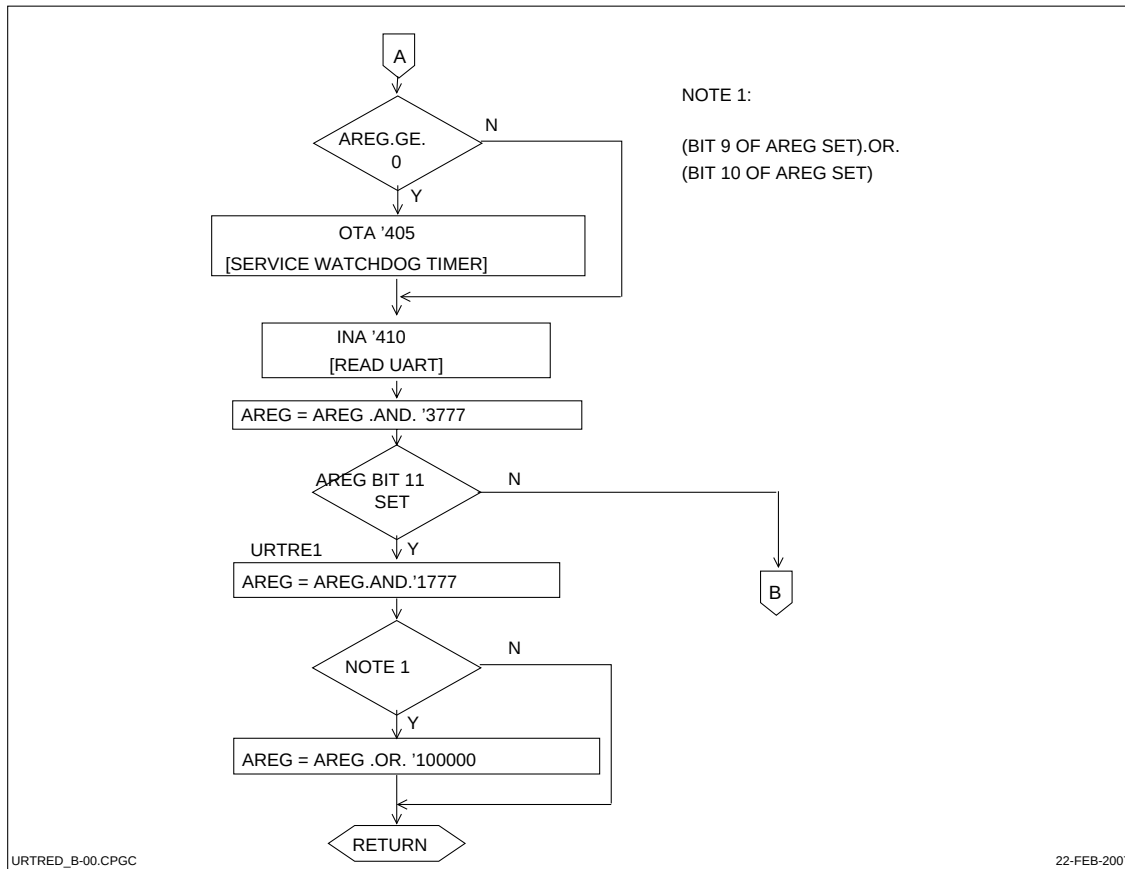
13.00 Figure 585: URTOUT _B



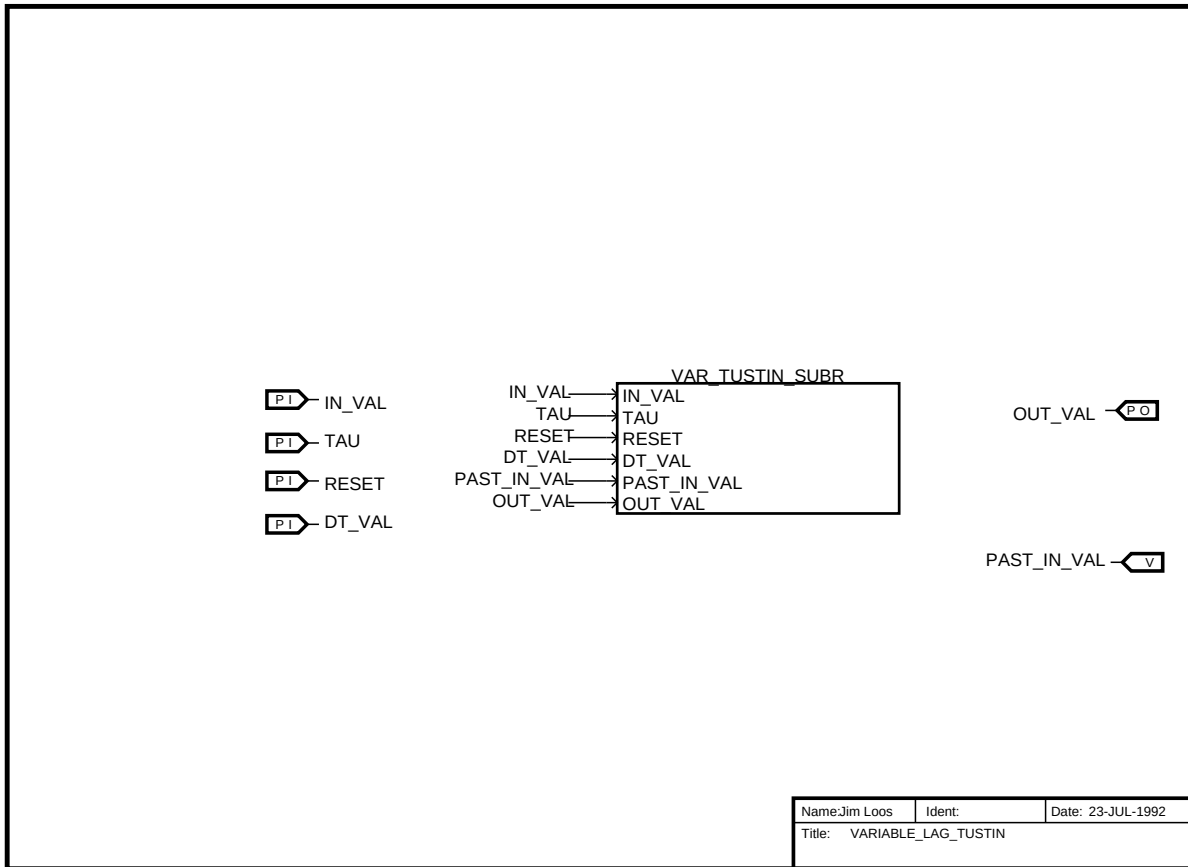
13.00 Figure 586: URTRED _A



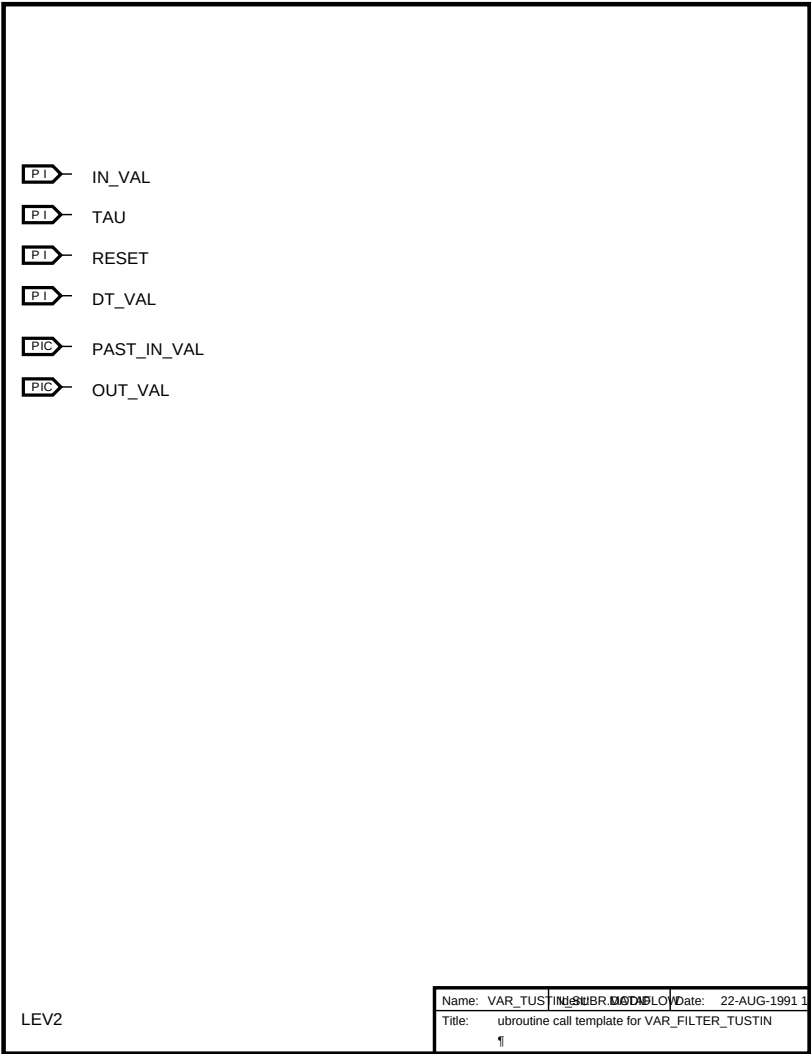
13.00 Figure 587: URTRED _B



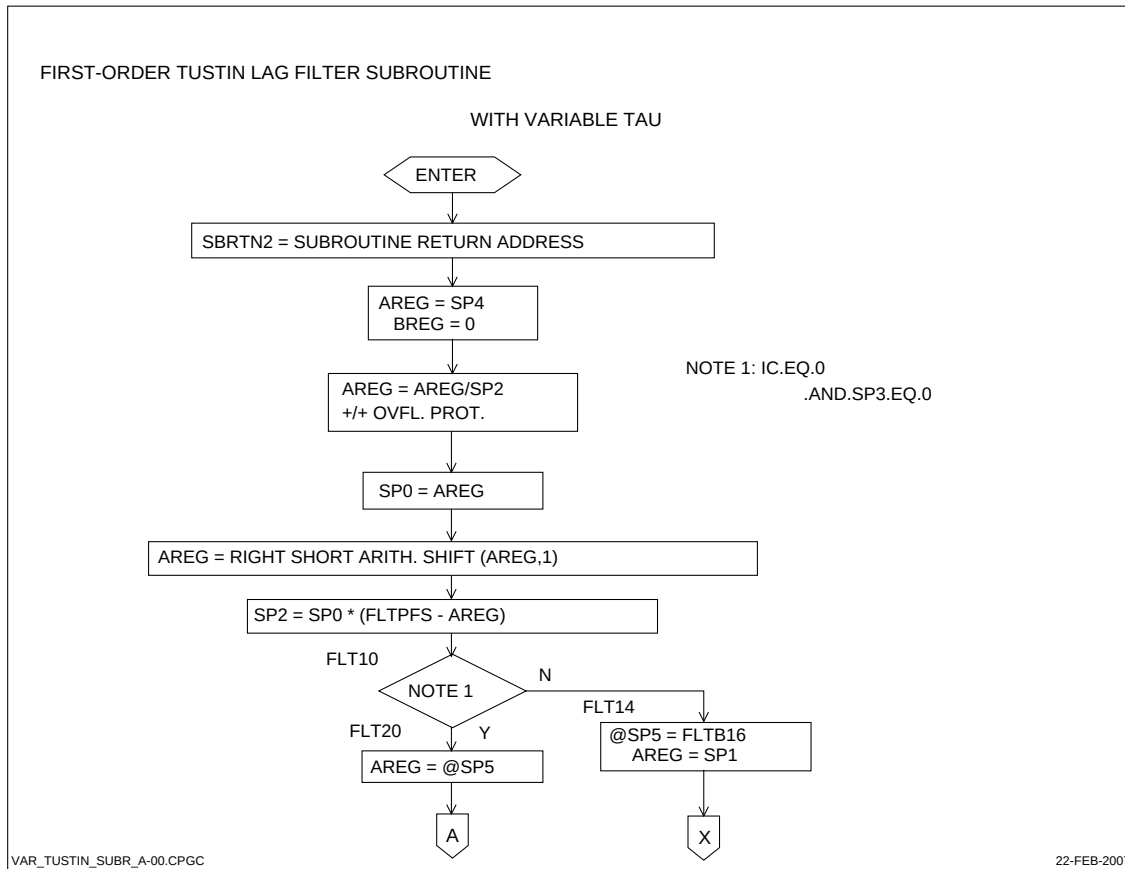
13.00 Figure 588: VARIABLE_LAG_TUSTIN



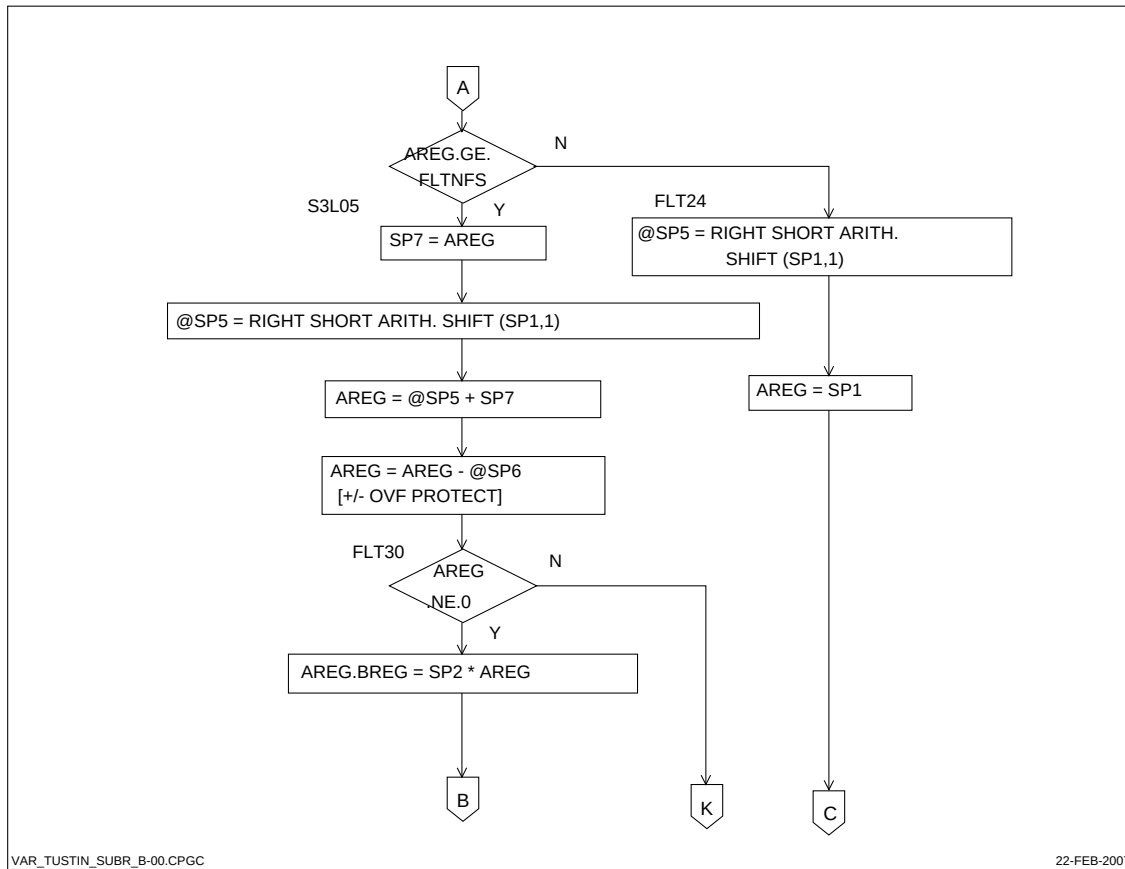
13.00 Figure 589: VAR _TUSTIN _SUBR



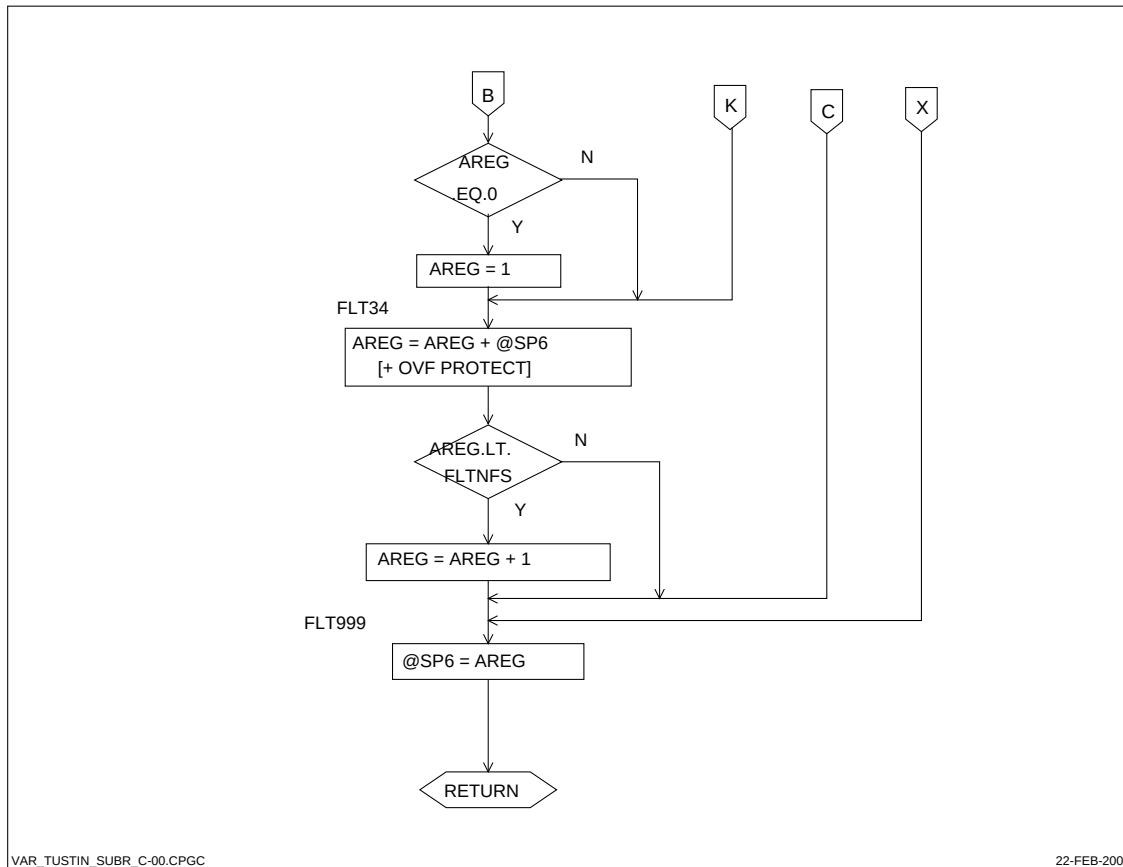
13.00 Figure 590: VAR_TUSTIN_SUBR_A



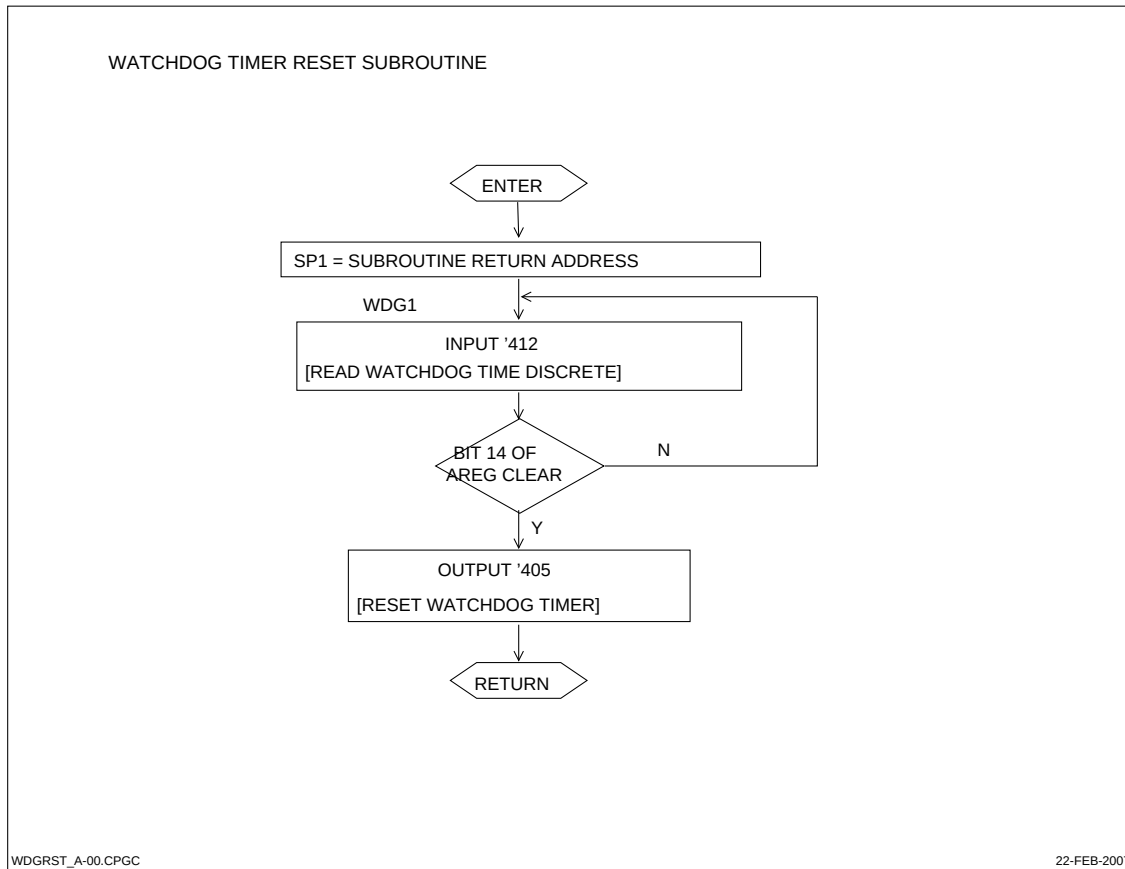
13.00 Figure 591: VAR_TUSTIN_SUBR_B



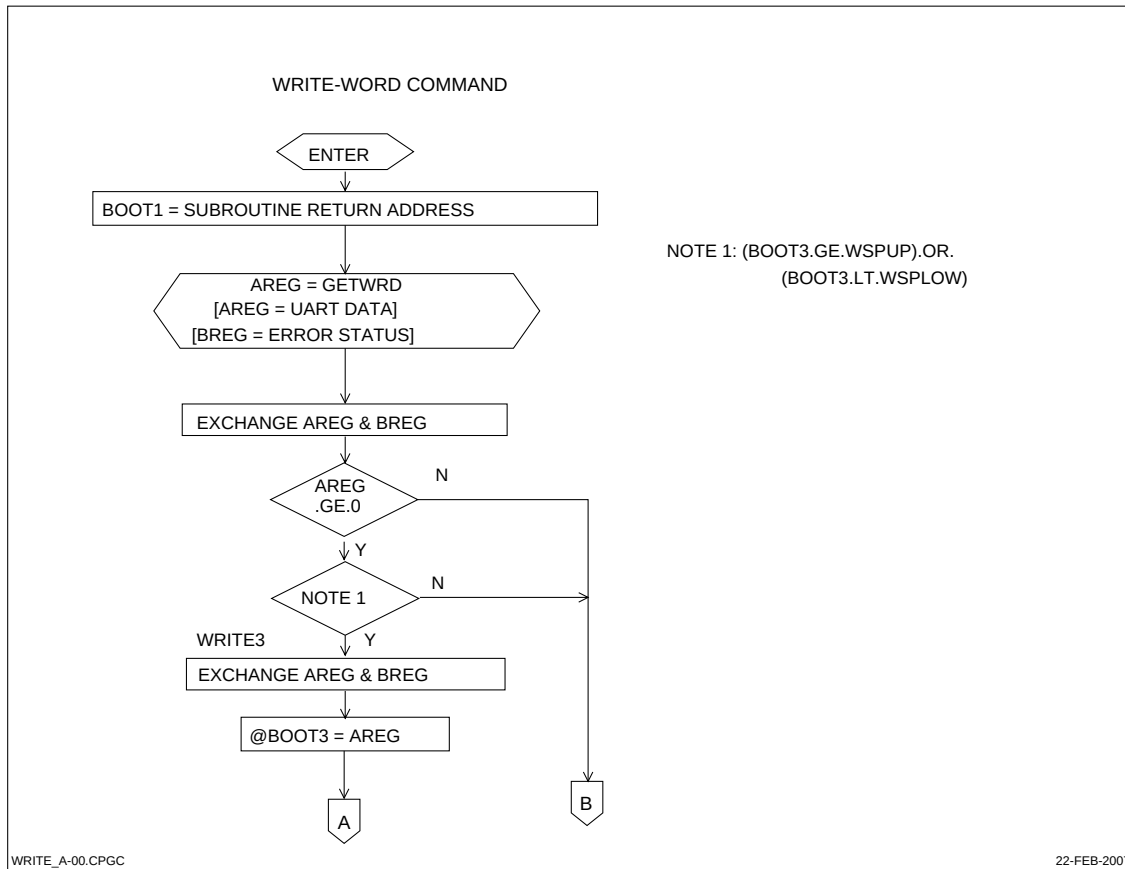
13.00 Figure 592: VAR _TUSTIN _SUBR _C



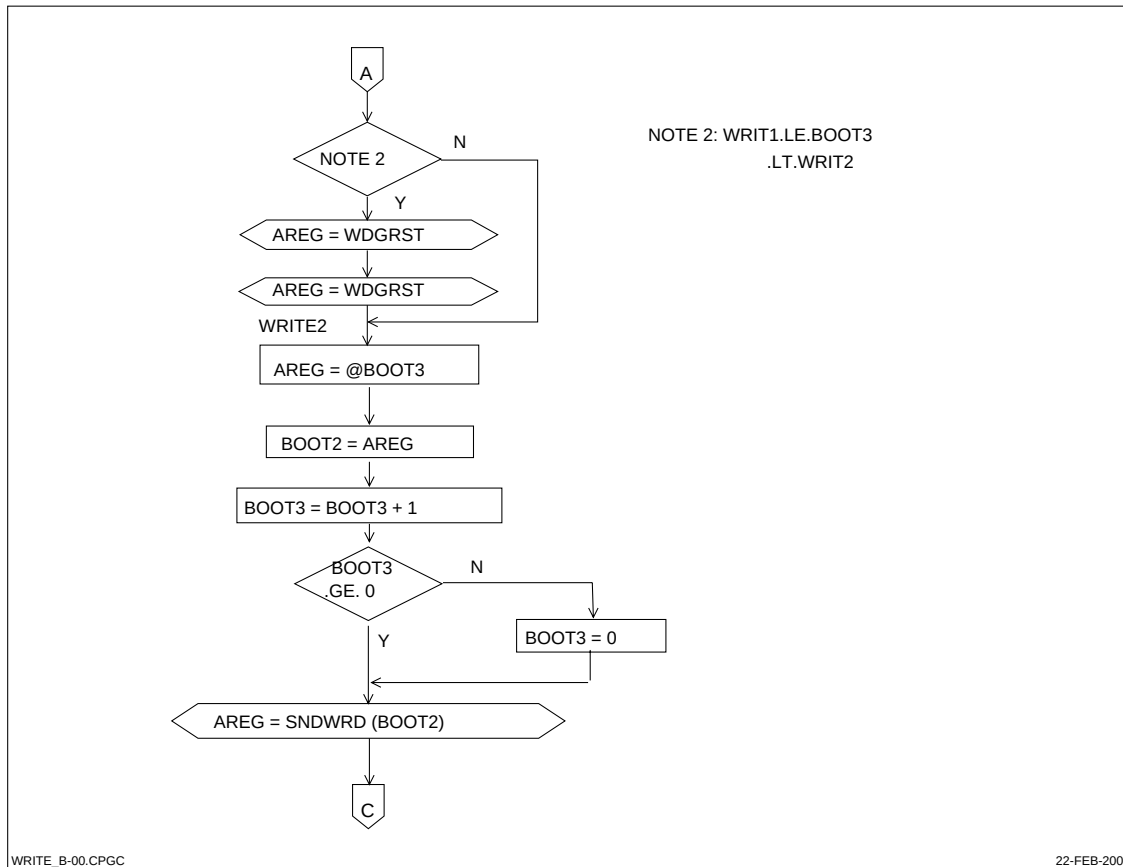
13.00 Figure 593: WDGRST_A



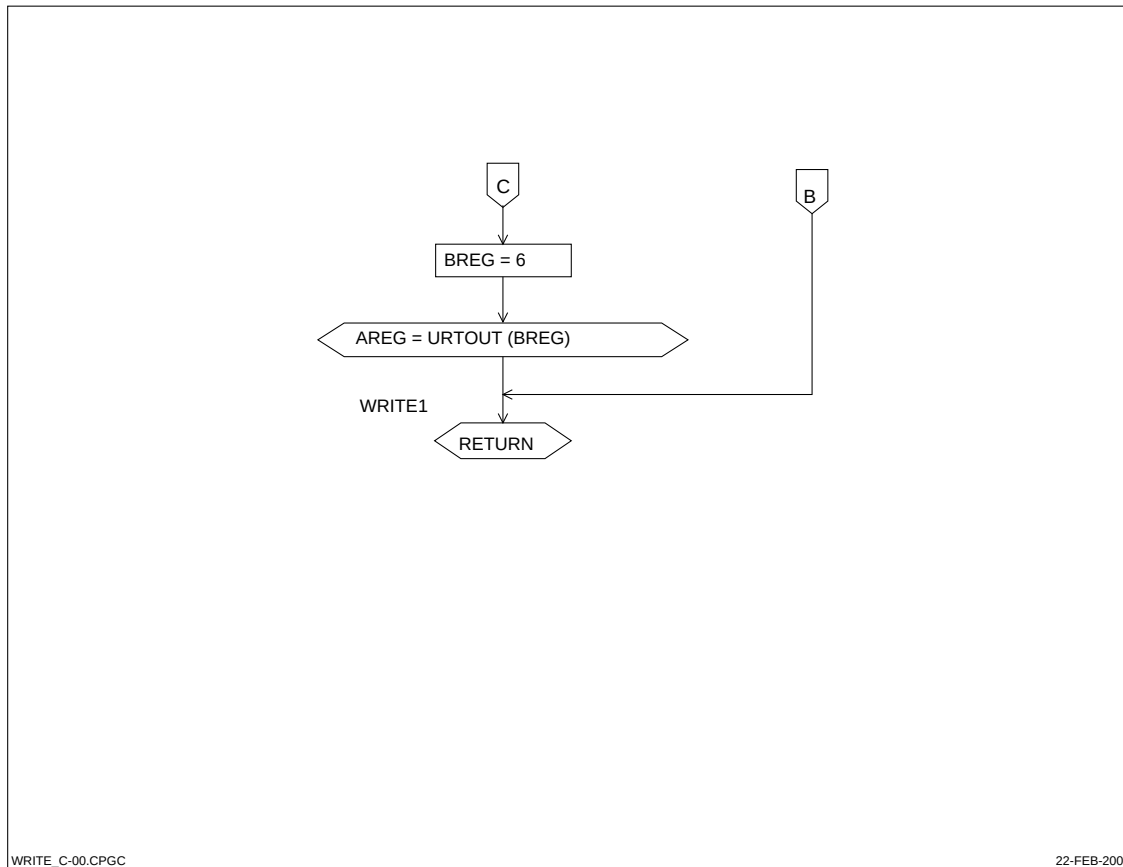
13.00 Figure 594: WRITE_A



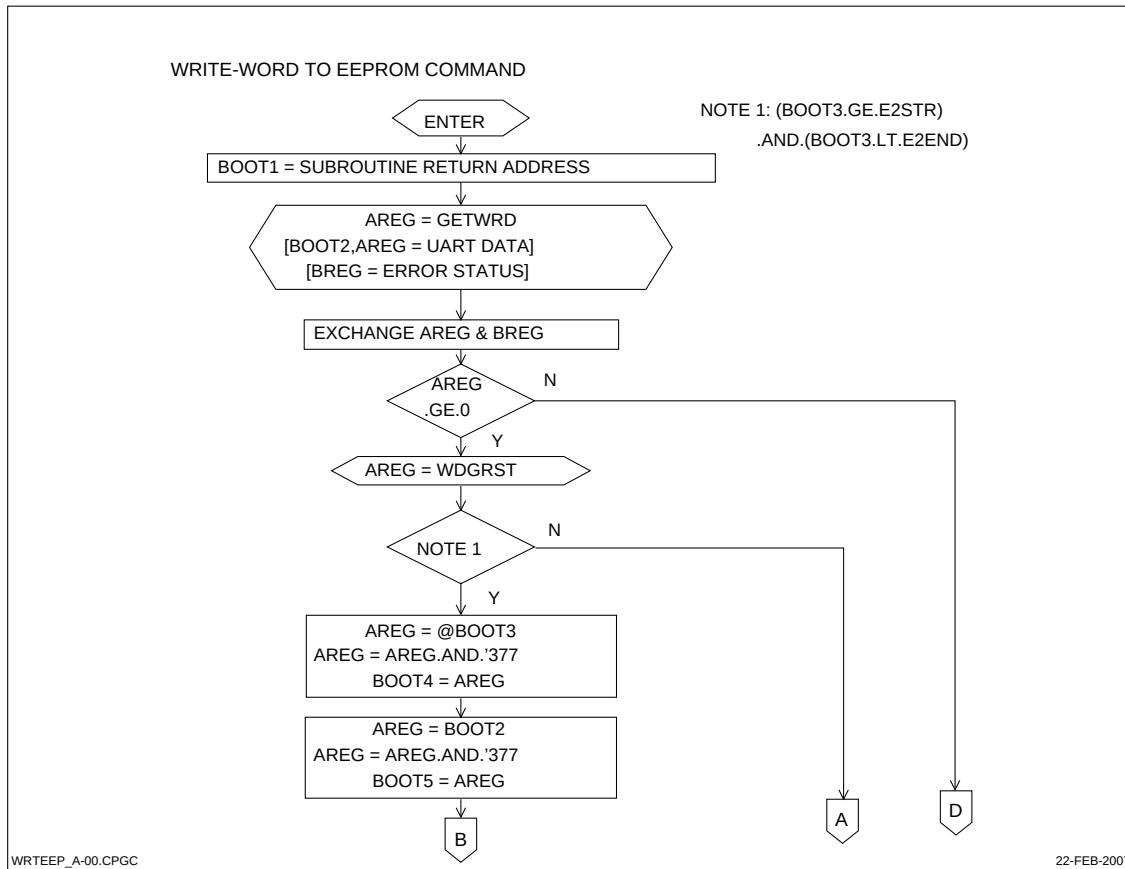
13.00 Figure 595: WRITE_B



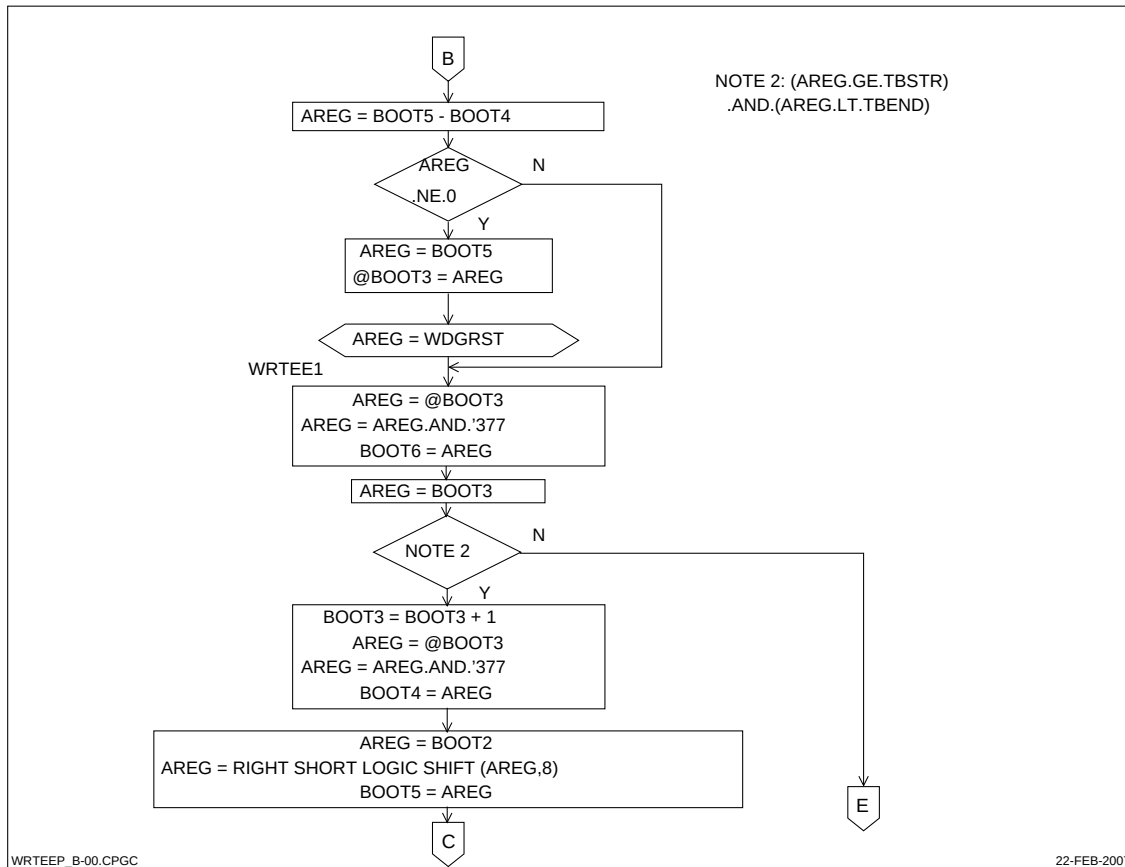
13.00 Figure 596: WRITE_C



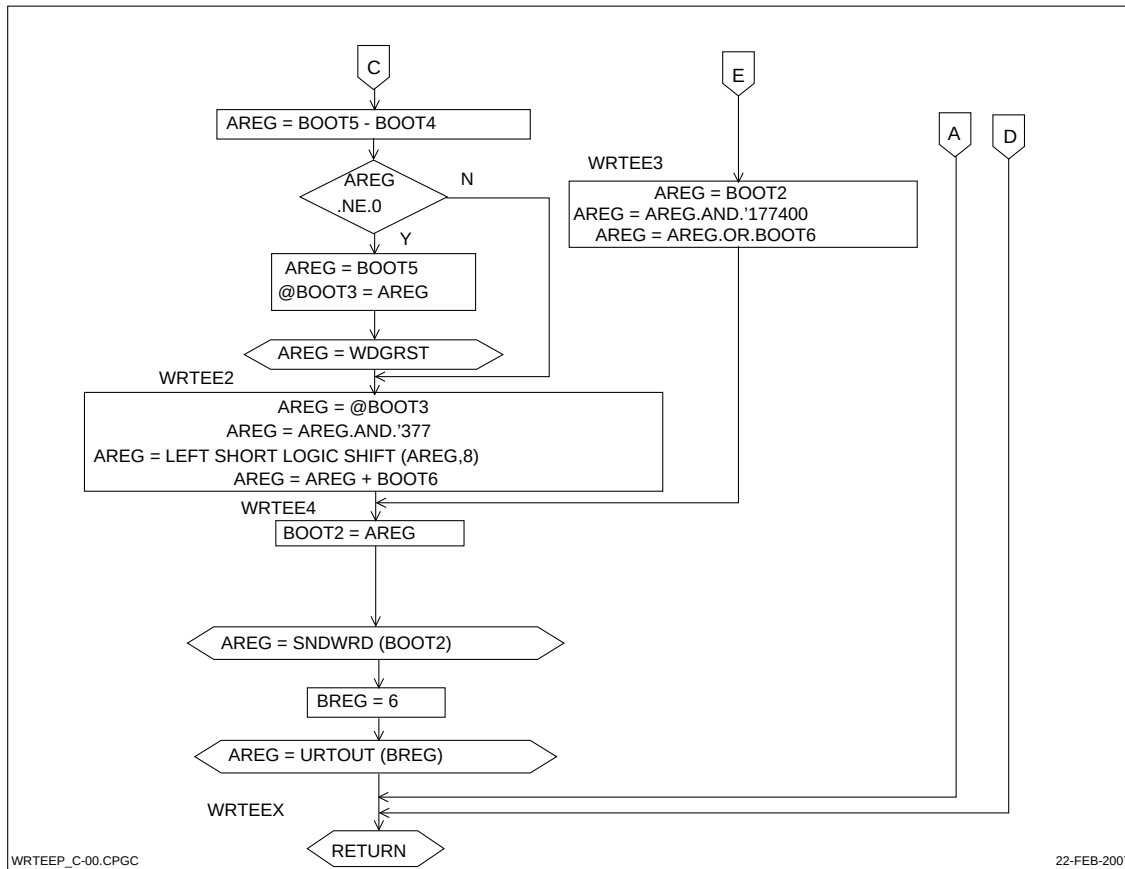
13.00 Figure 597: WRTEEP_A



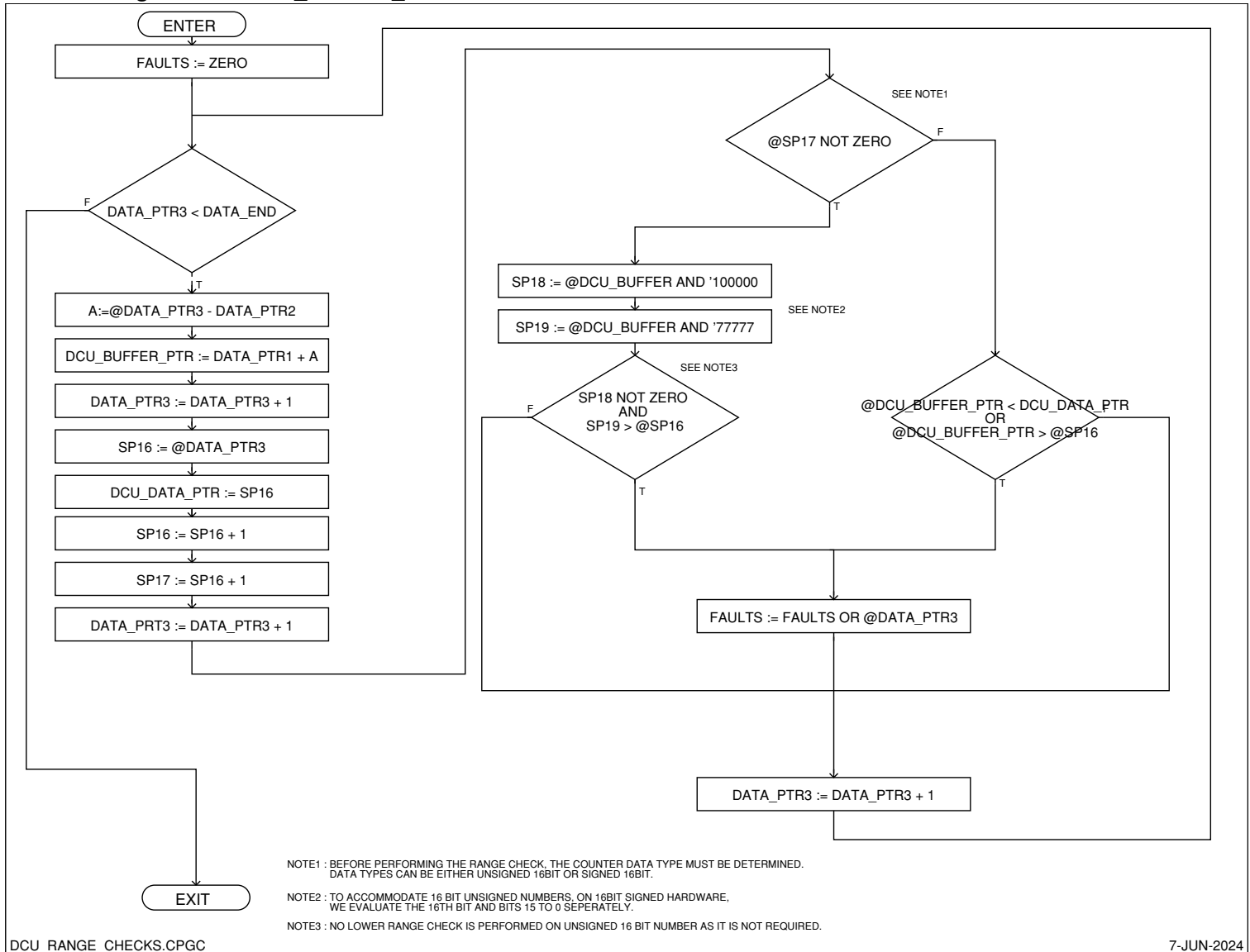
13.00 Figure 598: WRTEEP_B



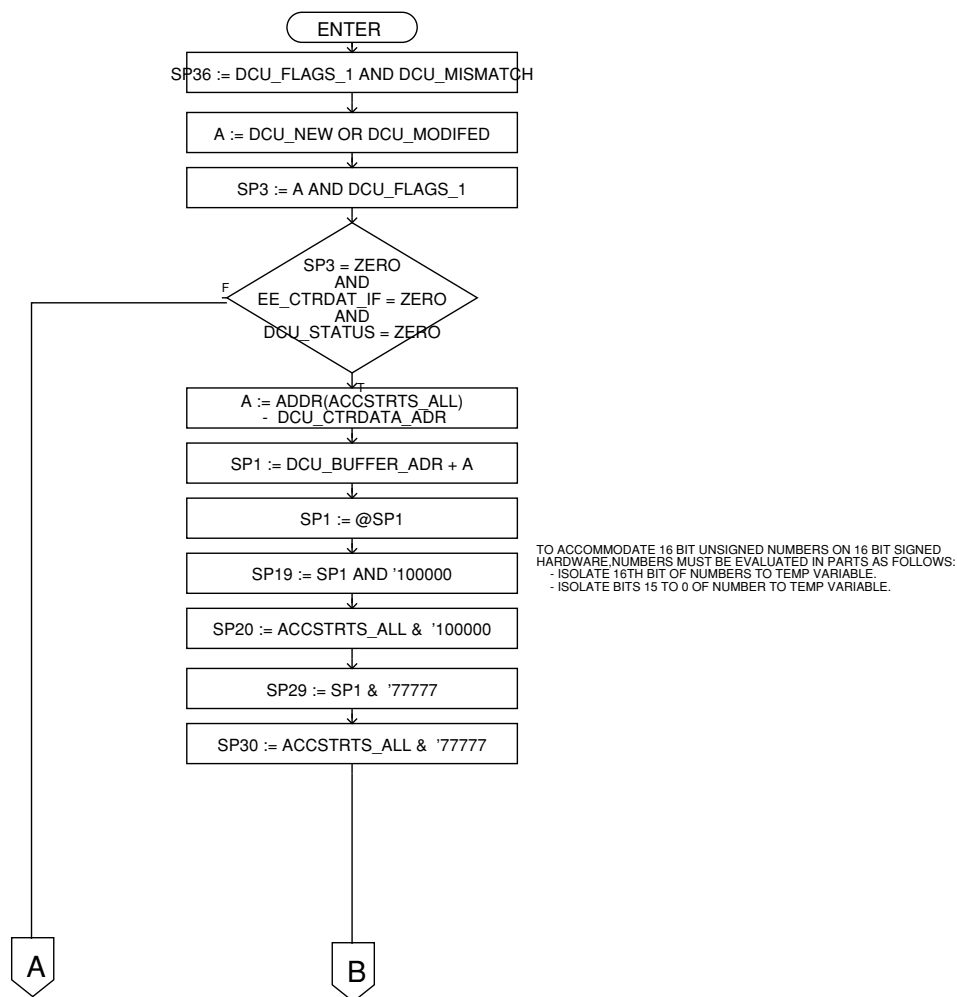
13.00 Figure 599: WRTEEP_C



13.00 Figure 600: DCU_RANGE_CHECKS



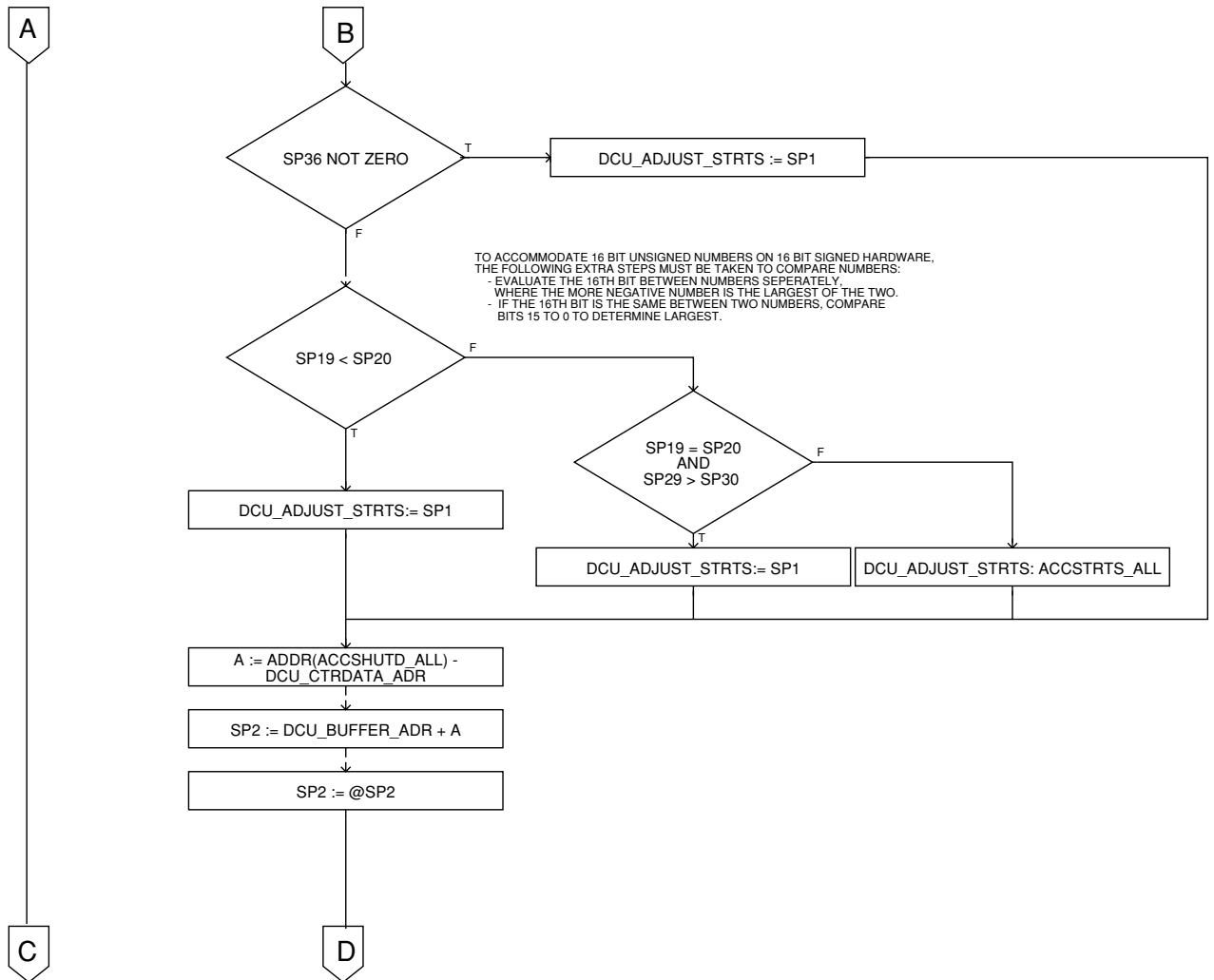
13.00 Figure 601: DCU_IS_OBSOLETE_A



DCU_IS_OBSOLETE_A.CPGC

18-JUN-2024

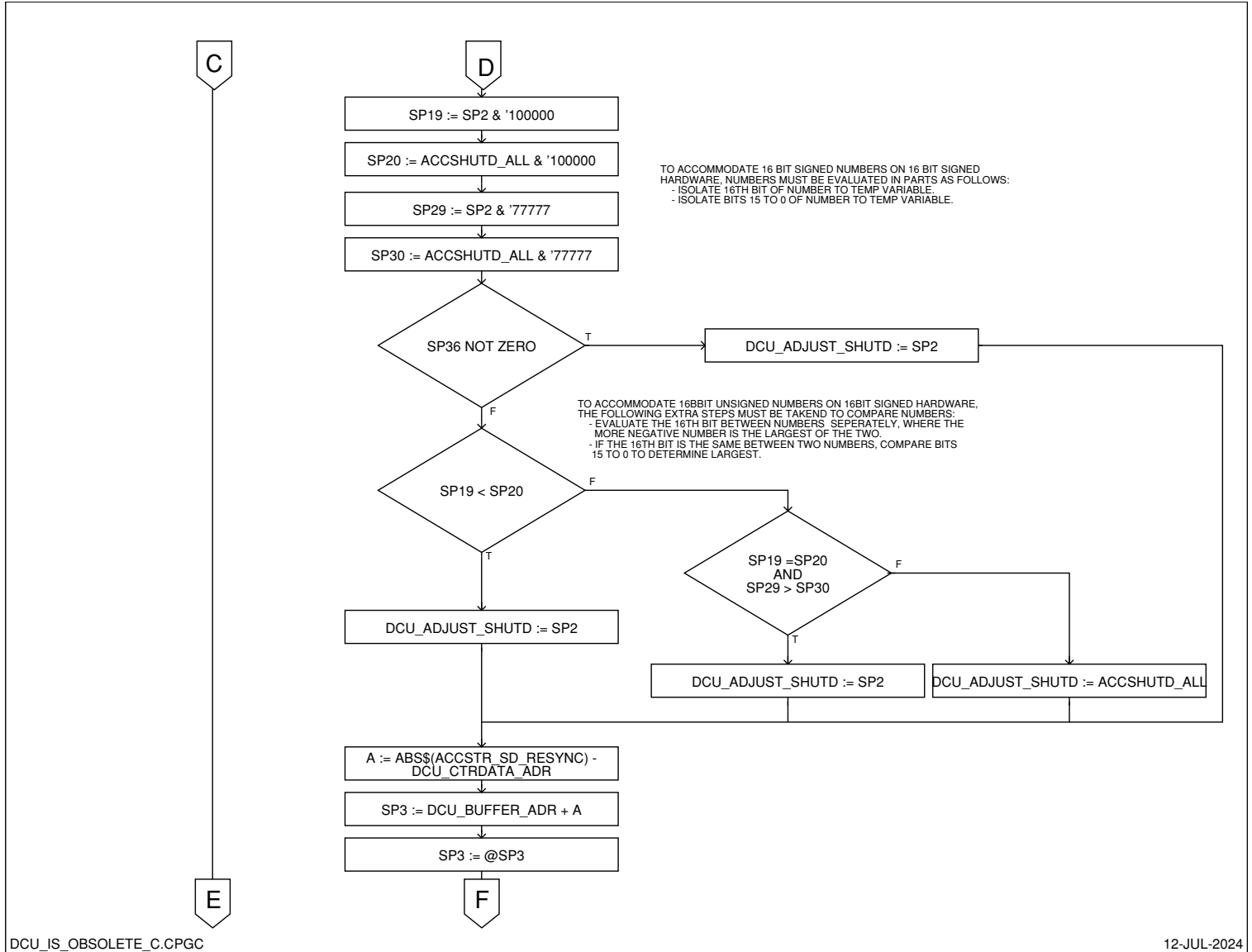
13.00 Figure 602: DCU_IS_OBSOLETE_B



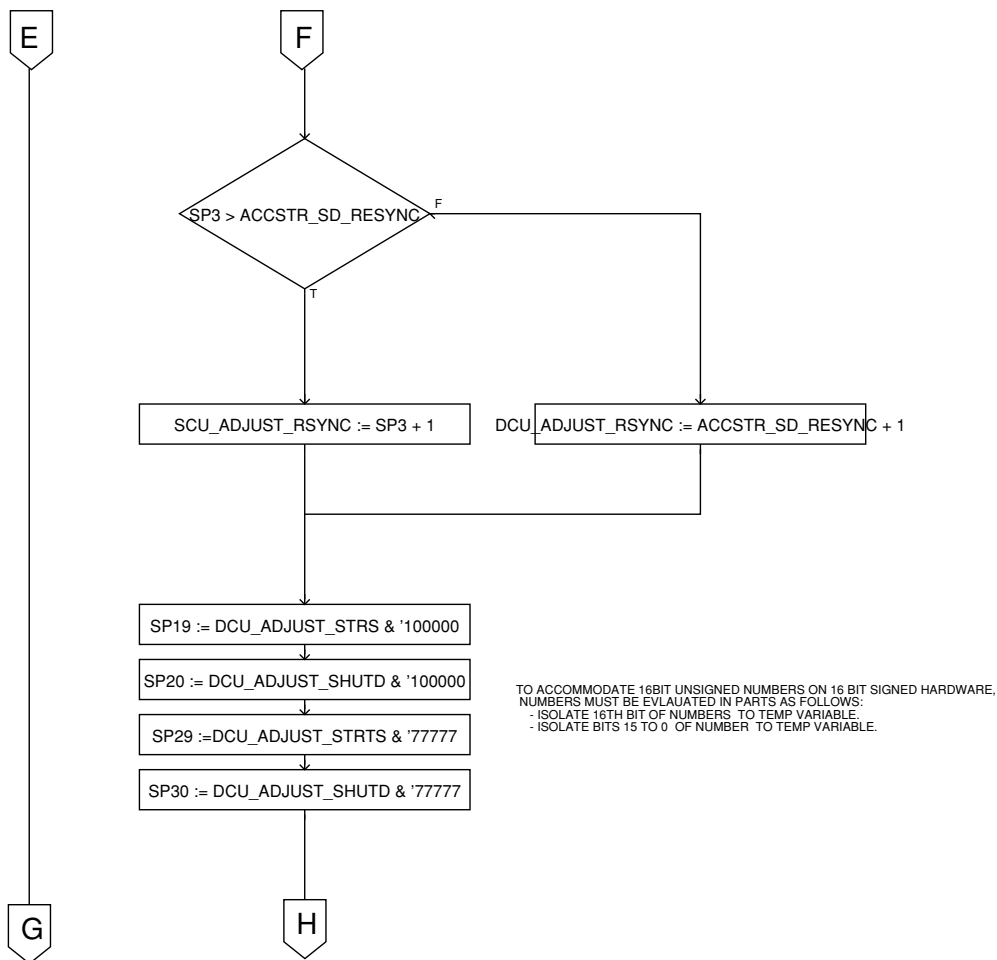
DCU_IS_OBSOLETE_B.CPGC

12-JUL-2024

13.00 Figure 603: DCU_IS_OBSOLETE_C



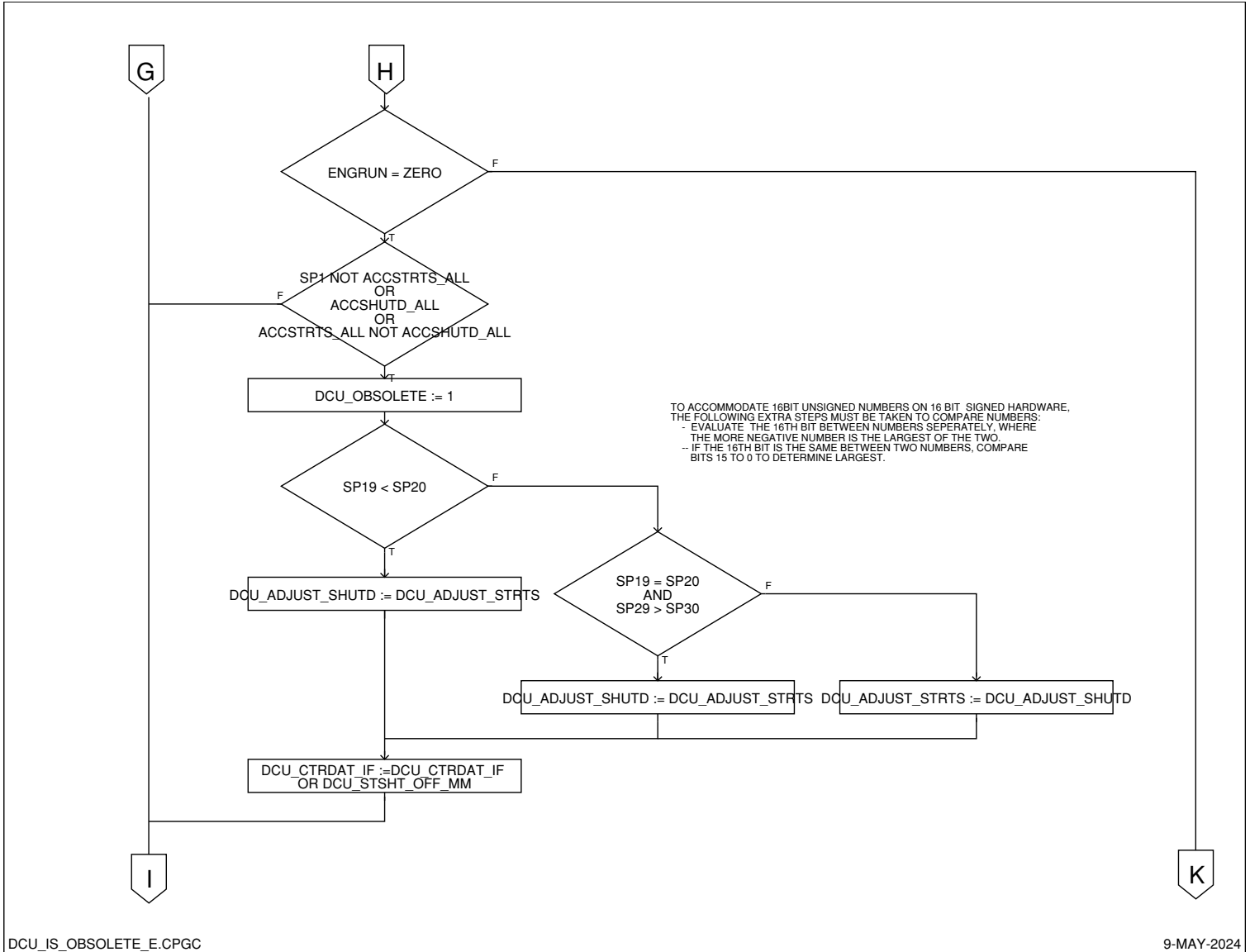
13.00 Figure 604: DCU_IS_OBSOLETE_D



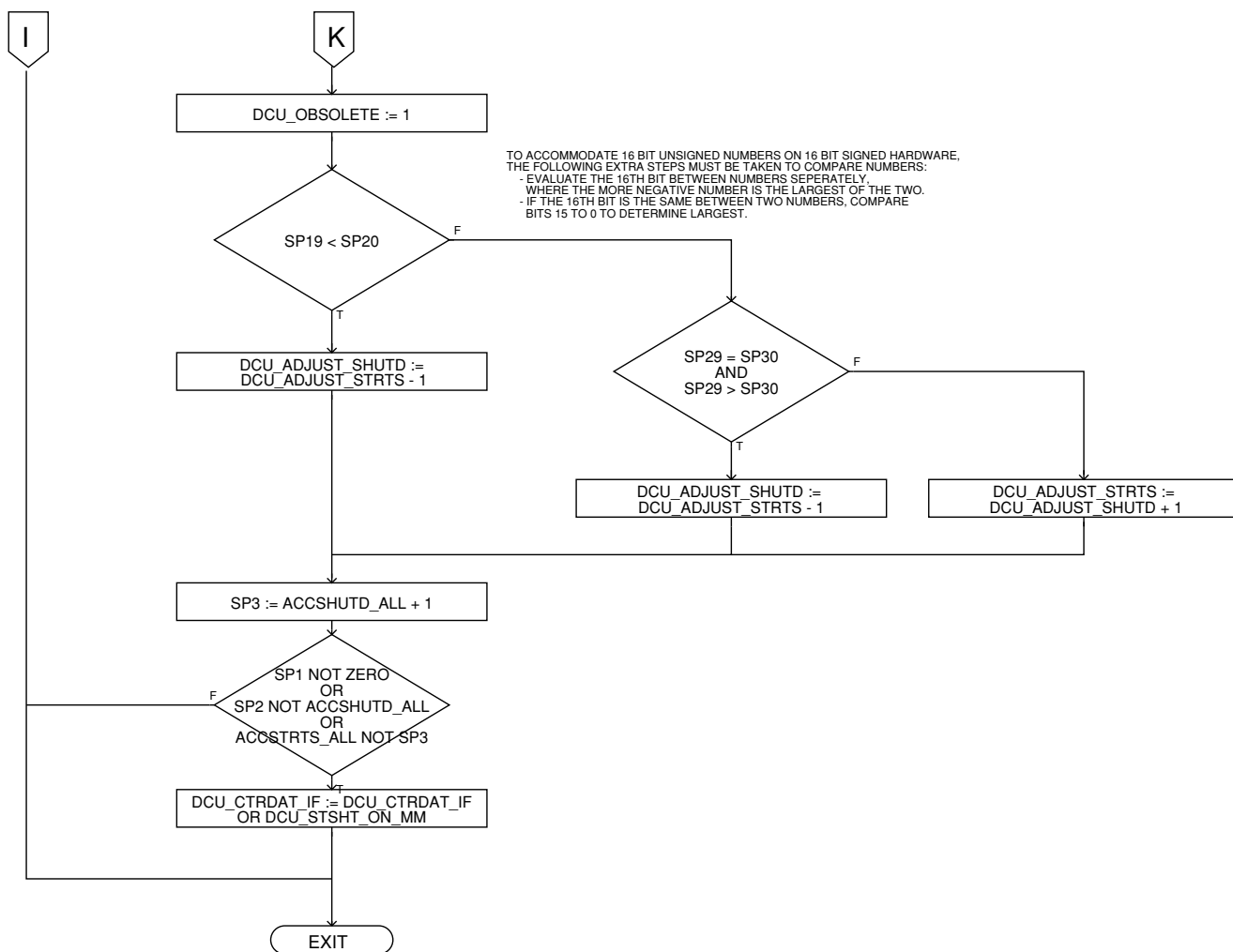
DCU_IS_OBSOLETE_D.CPGC

9-MAY-2024

13.00 Figure 605: DCU_IS_OBSOLETE_E



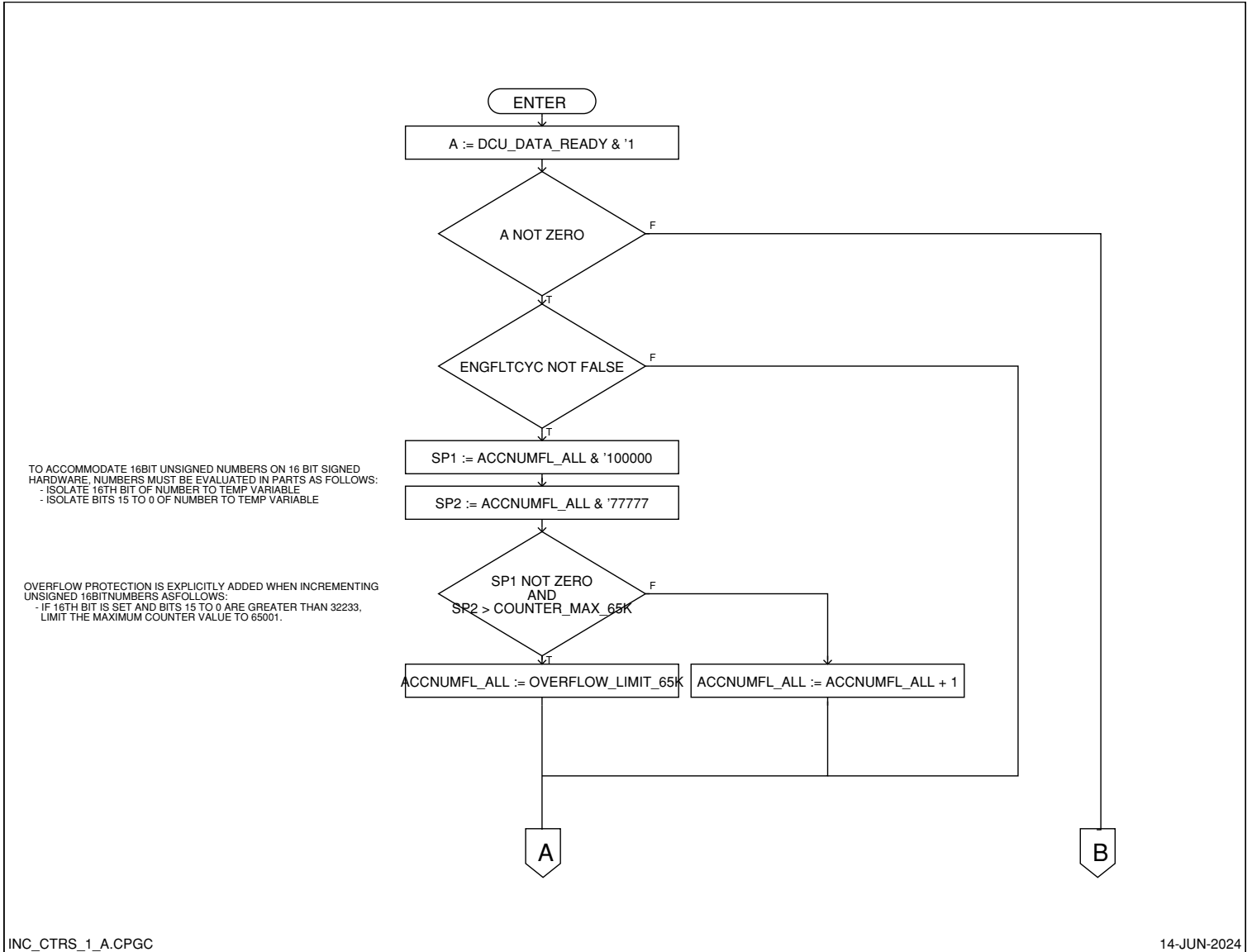
13.00 Figure 606: DCU_IS_OBSOLETE_F



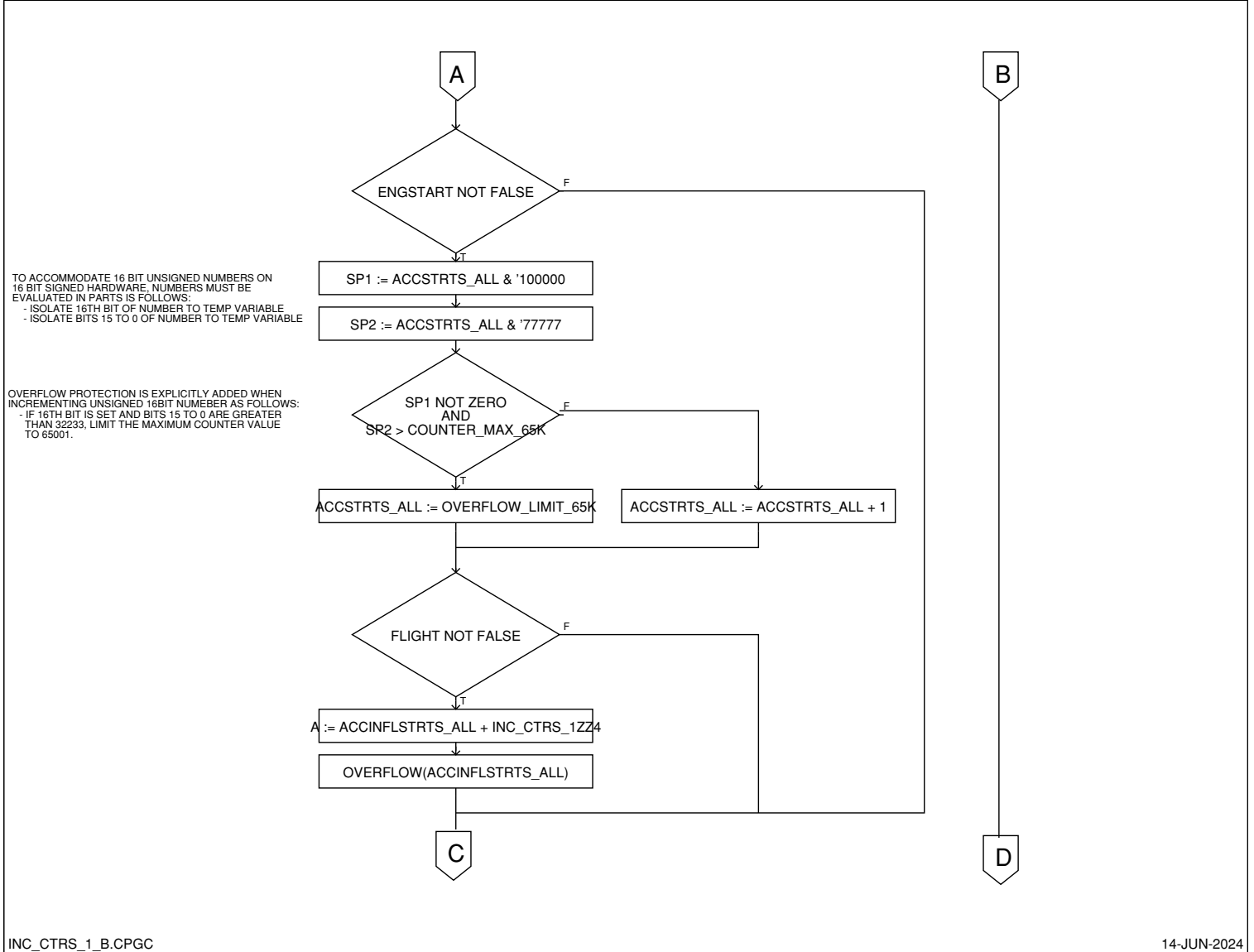
DCU_IS_OBSOLETE_F.CPGC

24-JUL-2024

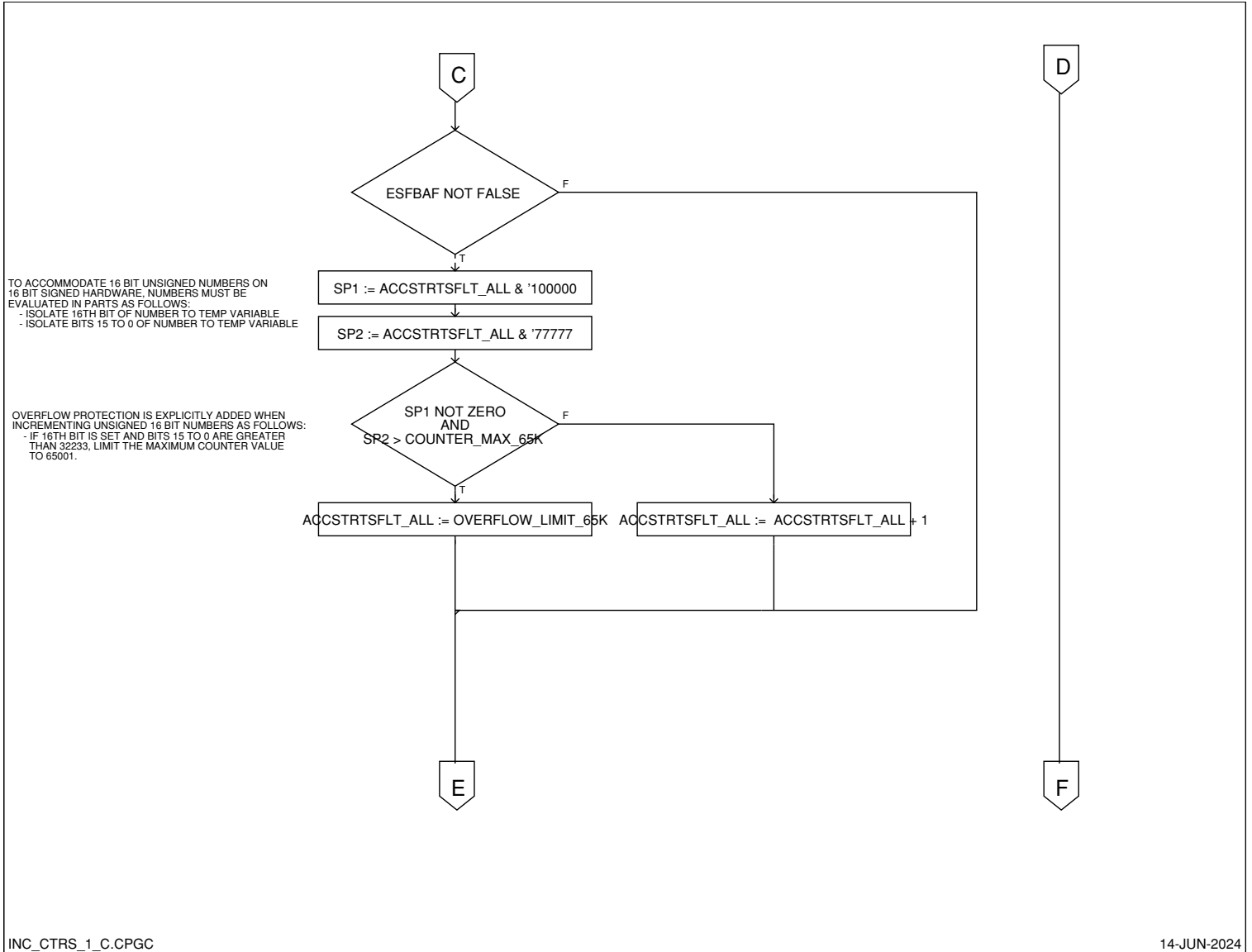
13.00 Figure 607: INC_CTRS_1_A



13.00 Figure 608: INC_CTRS_1_B



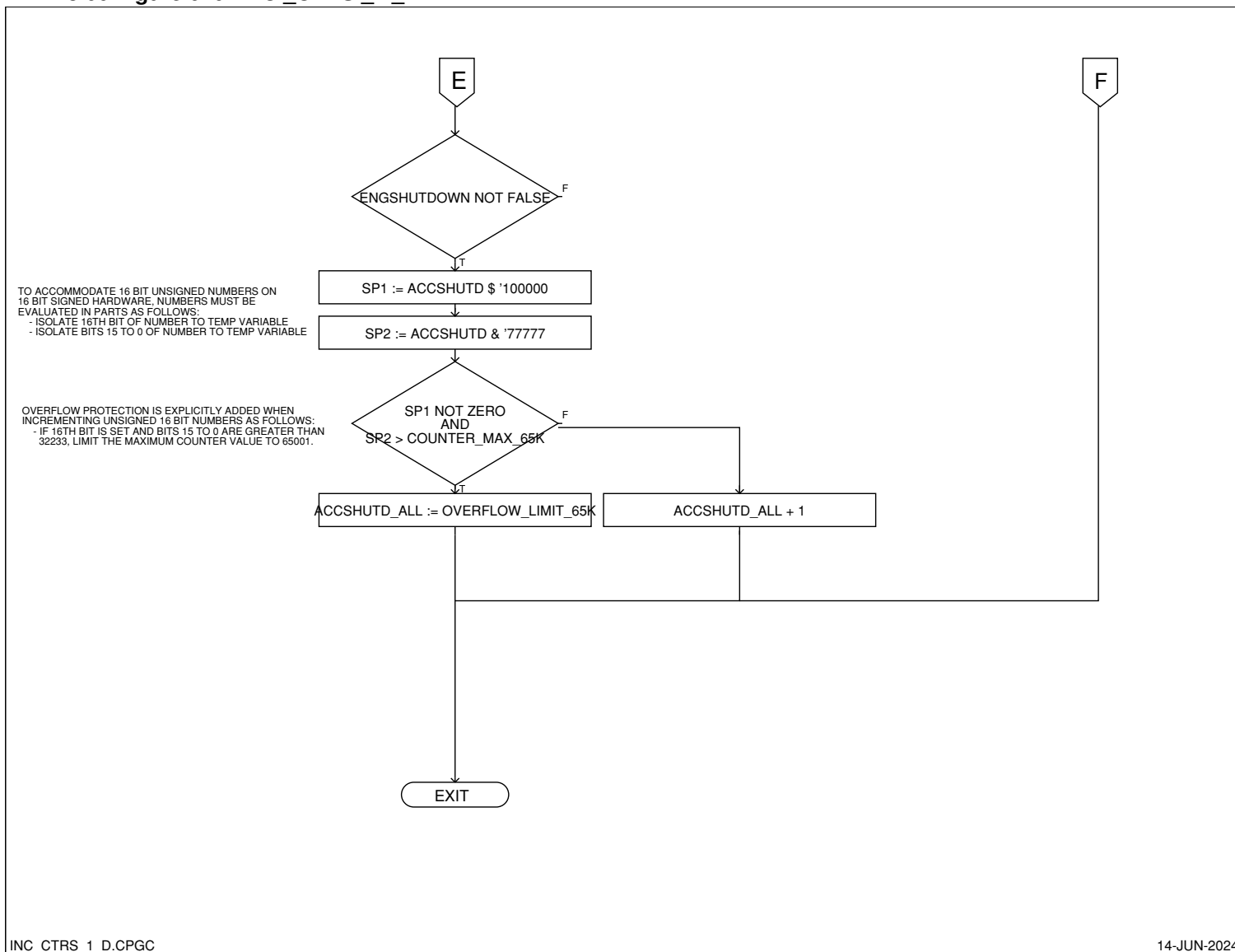
13.00 Figure 609: INC_CTRS_1_C



INC_CTRS_1_C.CPGC

14-JUN-2024

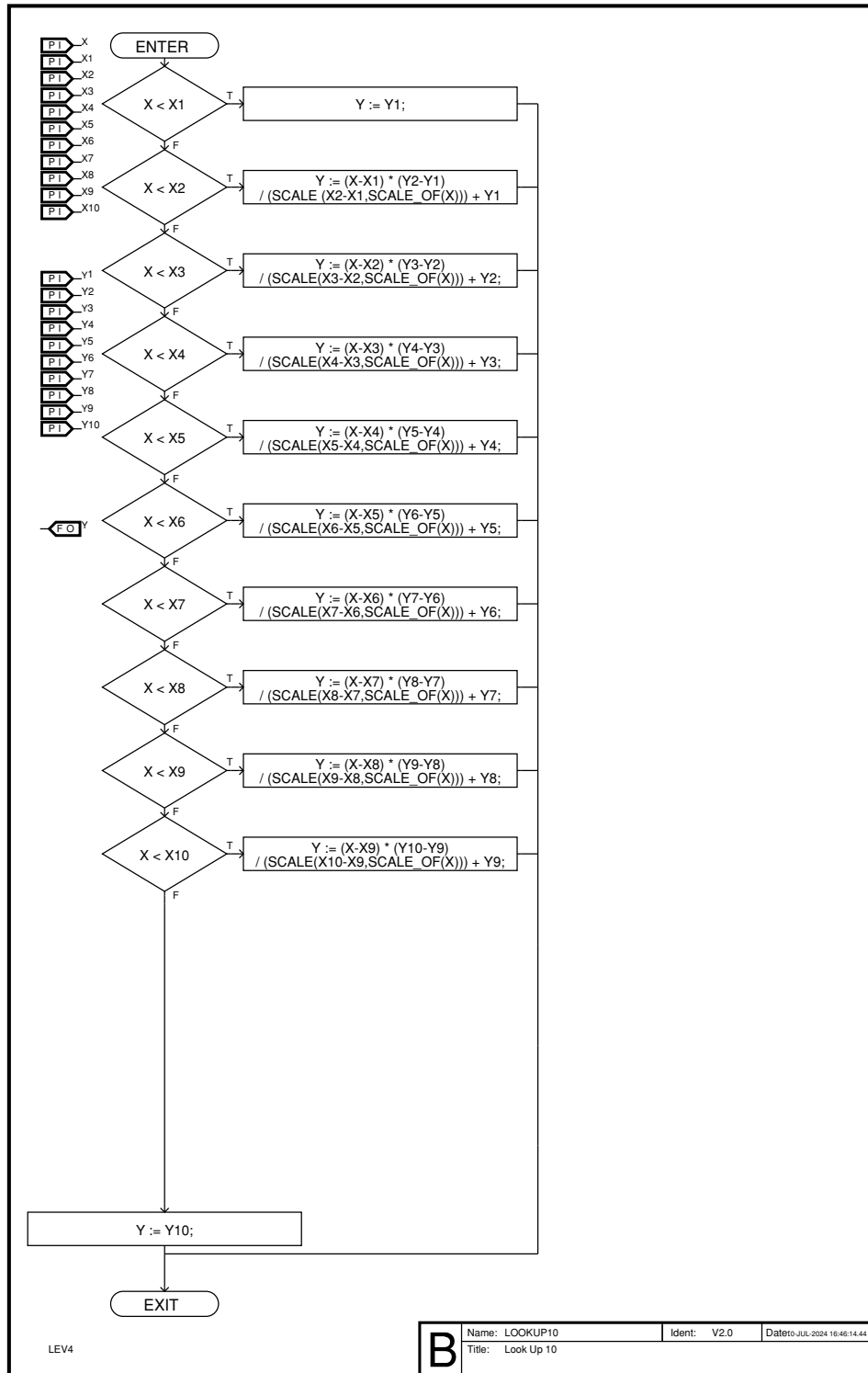
13.00 Figure 610: INC_CTRS_1_D



INC_CTRS_1_D.CPGC

14-JUN-2024

13.00 Figure 611: LOOKUP10



14.00 DEACTIVATED CODE

The table below lists the functions identified as containing deactivated code in software version 127S0100. These functions contain code that is provisional. These portions of the code are never executed by the software version 127S0100.

Deactivated code categories as defined by DO-178C:

Category One: Deactivated code that is not intended to be executed in any current configuration used within any certified product.

Category Two: Deactivated code that is only executed in certain approved configurations of the target computer environment.

For the design models below, timer constructs that have their reset pin tied to the inverted value of the input signal generate unreachable code in the corresponding module.

14.00 Table 1: Deactivated-code

Deactivated Function	Capability	Module	Module Version	Design Model	Design Model Version	Description	DO-178C Deactivated Code Category	Justification
Timer logic for input and reset both True cannot be reached.	C13-Trans_Accel_NH	C13_NHACCELLIM (SRC)	V1.0	C13_NHACCELLIM (DATAFLOW)	V1.0	Reset is desired when input is false. These are all unmodified legacy modules that were developed to DO-178B.	1	By analysis this code cannot be reached. DR 4617 has been generated to eliminate this code at the next module update.
Timer logic for input and reset both True cannot be reached.	C22-Out_HBOV_Man_Mode	C22_HBOVMM_1 (SRC)	V1.0	C22_HBOVMM_1 (DATAFLOW)	V1.0	Reset is desired when input is false. These are all unmodified legacy modules that were developed to DO-178B.	1	By analysis this code cannot be reached. DR 4617 has been generated to eliminate this code at the next module update.

15.00 GLOSSARY

15.00 Table 1: Description-of-Terms-Used-Within-SDD

Term	Definition
ACK	Acknowledge
A/D	Analog-to-Digital
ALT	Altitude
AREG	A Register
ARINC	Aeronautical Radio Inc.
ASCII	American Standard Code for Information Interchange
BREG	B Register
CCAP	Component Capability
CFG	Configuration
CMOS	Complementary Metal-Oxide Semiconductor
CPU	Central Processing Unit
CSCI	Computer Software Configuration Item
CSRD	Control System Requirements Document
DCU	Data Collection Unit
DISC	Discrete
EEC	Electronic Engine Control
EEPROM	Electrically Erasable Programmable Read Only Memory
EEROM	Electrically Erasable Read-Only Memory
ENG	Engine
EXEC	Executive
F/D	Frequency-to-Digital
GPU	Graphical Programming Utility
HLR	High Level Requirement
HSER	Hamilton Sundstrand Engineering Report
I/O	Input/Output
LCF	Low Cycle Fatigue
LEV6	Level 6
LEV7	Level 7
LLR	Low Level Requirement
LRU	Line Replaceable Unit
LSB	Least Significant Byte
MFC	Multi Function Computer
MIN	Minimum
MSB	Most Significant Byte
msec	Millisecond
MUX	Multiplexer
NPT	Power Turbine Speed
PID	Program Identifier
PLA	Power Lever Angle
POR	Power-On Reset
PSR	Power Supply Reset
PTR	Pointer
P&WC	Pratt & Whitney Canada
PWM	Pulse Width Modulator
RAM	Random Access Memory
RCD	Requirements Change Driver

Term	Definition
REF	Reference
REV	Revision
ROM	Read-Only Memory
RTCA	Radio Technical Commission for Aeronautics
RTD	Resistance Temperature Device
RVDT	Rotary Variable Differential Transformer
SCO-D	Software Change Order Design
SCR	Software Change Request
SDD	Software Design Document
sec	Second
SRS	Software Requirements Specification
UART	Universal Asynchronous Receiver Transmitter

16.00 CROSS REFERENCE TO RTCA/DO-178C AND DO-331

The table below identifies the SDD paragraph that shows cross-reference to DO-178C and/or DO-331 guidance, according to Section 11.10 Design Description of the DO-178C/DO-331 documents. The sub-paragraph references in the table are common between DO-178C and DO-331.

16.00 Table 1: CROSS REFERENCE TO RTCA/DO-178C AND DO-331

RTCA/DO-178C & DO-331 Paragraph	Item Description	SDD Section
11.10.a MB.11.10.a	A detailed description of how the software satisfies the specified high-level requirements, including algorithms, data structures, and how software requirements are allocated to processors and tasks.	04.00 Software Implementation
11.10.b MB.11.10.b	The description of the software architecture defining the software structure to implement the requirements.	03.00 Software Structure
11.10.c MB.11.10.c	The input/output description, for example, a data dictionary, both internally and externally throughout the software architecture.	10.00 Data Dictionary Listing
11.10.d MB.11.10.d	The data flow and control flow of the design.	13.00 Design models (GPU Diagrams)
11.10.e MB.11.10.e	Resource limitations, the strategy for managing each resource and its limitations, the margins and the method for measuring those margins, for example, timing and memory.	05.00 Memory and Processing Time Allocation
11.10.f MB.11.10.f	Scheduling procedures and inter-processor/inter-task communication mechanisms, including time-rigid sequencing, preemptive scheduling, Ada rendezvous, and interrupts.	08.00 Module Sequencing
11.10.g MB.11.10.g	Design methods and details for their implementation, for example, software loading, user-modifiable software, or multiple-version dissimilar software.	03.00 Software Structure There is no user-modifiable software nor multiple-version dissimilar software in the system.
11.10.h MB.11.10.h	Partitioning methods and means of preventing partitioning breaches.	The software is not partitioned.
11.10.i MB.11.10.i	Descriptions of the software components, whether they are new or previously developed, and if previously developed, reference to the baseline from which they were taken.	11.00 Modules Affected
11.10.j MB.11.10.j	Derived requirements resulting from the software design process.	06.00 Code to SDD to SRS Traceability 07.00 SRS to SDD to Code Traceability 17.00 Traceability Matrices 18.00 Derived Requirements Justification 19.00 Partially Derived Requirements Justification
11.10.k MB.11.10.k	If the system contains deactivated code, a description of the means to ensure that the code cannot be enabled in the target computer.	14.00 Deactivated Code
11.10.l MB.11.10.l	Rationale for those design decisions that are traceable to safety related system requirements.	08.00 Module Sequencing
11.21.b MB.11.21.b	Trace Data establishes the associations between life cycle data items contents. Trace Data should be provided that demonstrates bi-directional associations between: High-level requirements and Low-level requirements.	06.00 Code to SDD to SRS Traceability 07.00 SRS to SDD to Code Traceability 17.00 Traceability Matrices 18.00 Derived Requirements Justification 19.00 Partially Derived Requirements Justification
11.21.c MB.11.21.c	Trace Data establishes the associations between life cycle data items contents. Trace Data should be provided that demonstrates bi-directional associations between: Low-level requirements and Source Code.	06.00 Code to SDD to SRS Traceability 07.00 SRS to SDD to Code Traceability 17.00 Traceability Matrices 18.00 Derived Requirements Justification 19.00 Partially Derived Requirements Justification

17.00 Traceability Matrices

17.01 LLR to HLR Trace Matrix

LLR	HLR		LLR	HLR
LLR 12490004.000			LLR 12490006.000	CL 12470224.001
LLR 12490006.000	CL 12470225.001		LLR 12490006.000	CL 12470226.000
LLR 12490006.000	CL 12470228.002		LLR 12490006.000	CL 12470229.001
LLR 12490006.000	CL 12470231.001		LLR 12490006.000	CL 12470232.000
LLR 12490006.000	CL 12470233.000		LLR 12490006.000	CL 12470234.000
LLR 12490006.000	CL 12470235.001		LLR 12490006.000	CL 12473029.000
LLR 12490011.001	CL 12470527.002		LLR 12490011.001	CL 12470528.001
LLR 12490011.001	CL 12470529.001		LLR 12490011.001	CL 12470530.001
LLR 12490011.001	CL 12470531.001		LLR 12490011.001	CL 12470532.001
LLR 12490011.001	CL 12470535.001		LLR 12490011.001	CL 12470538.001
LLR 12490011.001	CL 12470541.003		LLR 12490011.001	CL 12470542.000
LLR 12490011.001	CL 12470543.000		LLR 12490020.000	CL 12470553.000
LLR 12490020.000	CL 12470555.001		LLR 12490020.000	CL 12470556.001
LLR 12490020.000	CL 12470557.001		LLR 12490020.000	CL 12472905.000
LLR 12490027.000	CL 12470760.000		LLR 12490027.000	CL 12470761.000
LLR 12490027.000	CL 12470762.000		LLR 12490027.000	CL 12470763.000
LLR 12490027.000	CL 12470764.000		LLR 12490027.000	CL 12470765.000
LLR 12490027.000	CL 12470766.000		LLR 12490027.000	CL 12470767.000
LLR 12490027.000	CL 12470768.000		LLR 12490027.000	CL 12470769.000
LLR 12490027.000	CL 12470770.000		LLR 12490027.000	CL 12470771.000
LLR 12490027.000	CL 12470772.000		LLR 12490027.000	CL 12470773.000
LLR 12490027.000	CL 12470774.000		LLR 12490027.000	CL 12470819.000
LLR 12490029.000	CL 12470774.000		LLR 12490031.000	CL 12470833.000
LLR 12490032.000	CL 12470837.000		LLR 12490033.000	CL 12470838.000
LLR 12490034.000	CL 12470839.000		LLR 12490034.000	CL 12470840.000
LLR 12490035.000	CL 12470835.000		LLR 12490035.000	CL 12470842.000
LLR 12490035.000	CL 12470843.000		LLR 12490035.000	CL 12470844.000
LLR 12490035.000	CL 12470846.000		LLR 12490035.000	CL 12470847.000
LLR 12490036.000	CL 12470834.000		LLR 12490036.000	CL 12470847.000
LLR 12490036.000	CL 12470849.000		LLR 12490036.000	CL 12470853.001
LLR 12490037.000	CL 12470853.001		LLR 12490038.000	CL 12470854.000
LLR 12490038.000	CL 12470855.000		LLR 12490038.000	CL 12471977.002
LLR 12490040.001	CL 12472786.000		LLR 12490042.001	CL 12472786.000
LLR 12490043.001	CL 12472786.000		LLR 12490044.001	CL 12472786.000
LLR 12490047.000	CL 12470821.000		LLR 12490047.000	CL 12470822.000
LLR 12490047.000	CL 12470823.000		LLR 12490047.000	CL 12470824.000
LLR 12490047.000	CL 12470825.000		LLR 12490047.000	CL 12470829.000

Continued on next page

Continued from previous page

LLR 12490047.000	CL 12470830.000
LLR 12490048.000	CL 12470826.000
LLR 12490050.000	CL 12470828.000
LLR 12490051.000	CL 12470832.000
LLR 12490051.000	CL 12470836.000
LLR 12490054.000	CL 12470930.000
LLR 12490056.000	CL 12470930.000
LLR 12490059.000	CL 12472787.000
LLR 12490064.001	Ov 12470016.000
LLR 12490064.001	Ov 12470077.000
LLR 12490064.001	Ov 12470119.000
LLR 12490064.001	Ov 12470158.000
LLR 12490064.001	Ov 12470194.000
LLR 12490064.001	CL 12470224.001
LLR 12490064.001	Ov 12470282.000
LLR 12490064.001	Ov 12470311.000
LLR 12490064.001	Ov 12470332.000
LLR 12490064.001	Ov 12470374.000
LLR 12490064.001	CL 12470525.000
LLR 12490064.001	Ov 12470569.000
LLR 12490064.001	CL 12470930.000
LLR 12490064.001	CL 12471788.000
LLR 12490064.001	CL 12471957.000
LLR 12490064.001	CL 12472028.000
LLR 12490064.001	CL 12472129.000
LLR 12490064.001	CL 12472166.000
LLR 12490064.001	CL 12472660.000
LLR 12490065.001	CL 12471081.000
LLR 12490065.001	CL 12471138.000
LLR 12490065.001	CL 12471180.000
LLR 12490065.001	CL 12471248.000
LLR 12490065.001	CL 12471537.000
LLR 12490065.001	CL 12472283.000
LLR 12490065.001	CL 12472425.000
LLR 12490065.001	CL 12472552.000
LLR 12490065.001	CL 12472674.000
LLR 12490065.001	CL 12472706.000
LLR 12490066.002	CL 12470980.000
LLR 12490066.002	CL 12471179.000
LLR 12490066.002	CL 12471501.000
LLR 12490066.002	CL 12471551.000
LLR 12490066.002	CL 12472404.000
LLR 12490066.002	CL 12472447.000

LLR 12490047.000	CL 12470855.000
LLR 12490049.000	CL 12470827.000
LLR 12490051.000	CL 12470831.000
LLR 12490051.000	CL 12470833.000
LLR 12490053.000	CL 12470930.000
LLR 12490055.000	CL 12470930.000
LLR 12490058.000	CL 12472786.000
LLR 12490061.000	CL 12470930.000
LLR 12490064.001	Ov 12470053.000
LLR 12490064.001	Ov 12470101.000
LLR 12490064.001	Ov 12470144.000
LLR 12490064.001	Ov 12470172.000
LLR 12490064.001	Ov 12470214.000
LLR 12490064.001	Ov 12470262.000
LLR 12490064.001	Ov 12470298.000
LLR 12490064.001	Ov 12470322.000
LLR 12490064.001	Ov 12470344.000
LLR 12490064.001	Ov 12470386.000
LLR 12490064.001	CL 12470553.000
LLR 12490064.001	CL 12470577.000
LLR 12490064.001	CL 12471043.000
LLR 12490064.001	Ov 12471948.000
LLR 12490064.001	CL 12471972.002
LLR 12490064.001	CL 12472105.000
LLR 12490064.001	CL 12472146.000
LLR 12490064.001	CL 12472644.000
LLR 12490065.001	CL 12470930.000
LLR 12490065.001	CL 12471093.000
LLR 12490065.001	CL 12471179.000
LLR 12490065.001	CL 12471247.000
LLR 12490065.001	CL 12471501.000
LLR 12490065.001	CL 12471551.000
LLR 12490065.001	CL 12472404.000
LLR 12490065.001	CL 12472447.000
LLR 12490065.001	CL 12472591.001
LLR 12490065.001	CL 12472675.002
LLR 12490066.002	CL 12470930.000
LLR 12490066.002	CL 12471081.000
LLR 12490066.002	CL 12471180.000
LLR 12490066.002	CL 12471537.000
LLR 12490066.002	CL 12472283.000
LLR 12490066.002	CL 12472425.000
LLR 12490066.002	CL 12472552.000

Continued on next page

Continued from previous page

LLR 12490066.002	CL 12472605.000
LLR 12490067.001	CL 12470908.000
LLR 12490067.001	CL 12470970.000
LLR 12490067.001	CL 12471093.000
LLR 12490067.001	CL 12471180.000
LLR 12490067.001	CL 12471248.000
LLR 12490067.001	CL 12471501.000
LLR 12490067.001	CL 12471551.000
LLR 12490067.001	CL 12472404.000
LLR 12490067.001	CL 12472447.000
LLR 12490067.001	CL 12472552.000
LLR 12490068.001	CL 12470930.000
LLR 12490068.001	CL 12470992.000
LLR 12490068.001	CL 12471179.000
LLR 12490068.001	CL 12471324.000
LLR 12490068.001	CL 12471464.000
LLR 12490068.001	CL 12471537.000
LLR 12490068.001	CL 12471660.000
LLR 12490068.001	CL 12472239.000
LLR 12490068.001	CL 12472270.000
LLR 12490068.001	CL 12472296.000
LLR 12490068.001	CL 12472361.000
LLR 12490068.001	CL 12472404.000
LLR 12490068.001	CL 12472447.000
LLR 12490068.001	CL 12472505.000
LLR 12490071.000	CL 12470930.000
LLR 12490073.000	CL 12470930.000
LLR 12490075.000	CL 12470930.000
LLR 12490077.000	CL 12472713.000
LLR 12490078.000	CL 12472713.000
LLR 12490079.000	CL 12472713.000
LLR 12490080.000	CL 12472713.000
LLR 12490081.000	CL 12472713.000
LLR 12490082.000	CL 12472713.000
LLR 12490083.000	CL 12472737.000
LLR 12490083.000	CL 12472743.000
LLR 12490083.000	CL 12472749.000
LLR 12490087.000	CL 12472713.000
LLR 12490088.000	CL 12472713.000
LLR 12490089.000	CL 12472713.000
LLR 12490090.000	CL 12472713.000
LLR 12490091.000	CL 12472713.000
LLR 12490092.000	CL 12472713.000

LLR 12490066.002	CL 12472706.000
LLR 12490067.001	CL 12470930.000
LLR 12490067.001	CL 12471081.000
LLR 12490067.001	CL 12471179.000
LLR 12490067.001	CL 12471247.000
LLR 12490067.001	CL 12471416.000
LLR 12490067.001	CL 12471537.000
LLR 12490067.001	CL 12472283.000
LLR 12490067.001	CL 12472425.000
LLR 12490067.001	CL 12472528.000
LLR 12490067.001	CL 12472706.000
LLR 12490068.001	CL 12470980.000
LLR 12490068.001	CL 12471081.000
LLR 12490068.001	CL 12471180.000
LLR 12490068.001	CL 12471463.000
LLR 12490068.001	CL 12471501.000
LLR 12490068.001	CL 12471551.000
LLR 12490068.001	CL 12471763.000
LLR 12490068.001	CL 12472253.000
LLR 12490068.001	CL 12472283.000
LLR 12490068.001	CL 12472315.000
LLR 12490068.001	CL 12472385.000
LLR 12490068.001	CL 12472425.000
LLR 12490068.001	CL 12472484.000
LLR 12490068.001	CL 12472552.000
LLR 12490072.000	CL 12470930.000
LLR 12490074.000	CL 12470930.000
LLR 12490077.000	CL 12472707.000
LLR 12490078.000	CL 12472707.000
LLR 12490079.000	CL 12472707.000
LLR 12490080.000	CL 12472707.000
LLR 12490081.000	CL 12472707.000
LLR 12490082.000	CL 12472707.000
LLR 12490083.000	CL 12472734.000
LLR 12490083.000	CL 12472740.000
LLR 12490083.000	CL 12472746.000
LLR 12490087.000	CL 12472707.000
LLR 12490088.000	CL 12472707.000
LLR 12490089.000	CL 12472707.000
LLR 12490090.000	CL 12472707.000
LLR 12490091.000	CL 12472707.000
LLR 12490092.000	CL 12472707.000
LLR 12490094.000	CL 12471083.000

Continued on next page

Continued from previous page

LLR 12490095.000	CL 12471083.000
LLR 12490096.000	CL 12471083.000
LLR 12490096.000	CL 12471085.000
LLR 12490097.000	CL 12471083.000
LLR 12490097.000	CL 12471085.000
LLR 12490103.000	CL 12471082.000
LLR 12490105.000	CL 12471084.000
LLR 12490107.000	CL 12471085.000
LLR 12490111.000	CL 12471095.000
LLR 12490112.000	CL 12471094.000
LLR 12490112.000	CL 12471096.000
LLR 12490113.000	CL 12471095.000
LLR 12490114.000	CL 12471094.000
LLR 12490114.000	CL 12471096.000
LLR 12490121.000	CL 12471140.000
LLR 12490121.000	CL 12471142.000
LLR 12490121.000	CL 12471144.000
LLR 12490123.001	CL 12472722.001
LLR 12490125.000	CL 12472726.000
LLR 12490127.000	CL 12472728.000
LLR 12490133.000	CL 12472711.001
LLR 12490135.001	CL 12472720.001
LLR 12490141.000	CL 12471182.000
LLR 12490141.000	CL 12471184.000
LLR 12490141.000	CL 12471186.000
LLR 12490141.000	CL 12471188.000
LLR 12490141.000	CL 12471190.000
LLR 12490148.000	CL 12471251.000
LLR 12490148.000	CL 12471253.000
LLR 12490148.000	CL 12471255.000
LLR 12490148.000	CL 12471257.000
LLR 12490148.000	CL 12471259.000
LLR 12490148.000	CL 12471261.000
LLR 12490152.001	CL 12473052.000
LLR 12490152.001	CL 12473054.000
LLR 12490156.001	CL 12472893.000
LLR 12490156.001	CL 12473053.000
LLR 12490156.001	CL 12473055.000
LLR 12490157.001	CL 12473052.000
LLR 12490157.001	CL 12473054.000
LLR 12490158.001	CL 12472958.000
LLR 12490158.001	CL 12473053.000
LLR 12490158.001	CL 12473055.000

LLR 12490096.000	CL 12471082.000
LLR 12490096.000	CL 12471084.000
LLR 12490097.000	CL 12471082.000
LLR 12490097.000	CL 12471084.000
LLR 12490102.000	CL 12471082.000
LLR 12490104.000	CL 12471084.000
LLR 12490106.000	CL 12471085.000
LLR 12490111.000	CL 12471094.000
LLR 12490111.000	CL 12471096.000
LLR 12490112.000	CL 12471095.000
LLR 12490113.000	CL 12471094.000
LLR 12490113.000	CL 12471096.000
LLR 12490114.000	CL 12471095.000
LLR 12490121.000	CL 12471139.000
LLR 12490121.000	CL 12471141.000
LLR 12490121.000	CL 12471143.000
LLR 12490121.000	CL 12471145.000
LLR 12490124.001	CL 12472723.001
LLR 12490126.000	CL 12472727.000
LLR 12490128.000	CL 12472725.000
LLR 12490134.001	CL 12472719.001
LLR 12490136.002	CL 12472721.001
LLR 12490141.000	CL 12471183.000
LLR 12490141.000	CL 12471185.000
LLR 12490141.000	CL 12471187.000
LLR 12490141.000	CL 12471189.000
LLR 12490148.000	CL 12471250.000
LLR 12490148.000	CL 12471252.000
LLR 12490148.000	CL 12471254.000
LLR 12490148.000	CL 12471256.000
LLR 12490148.000	CL 12471258.000
LLR 12490148.000	CL 12471260.000
LLR 12490152.001	CL 12472955.000
LLR 12490152.001	CL 12473053.000
LLR 12490152.001	CL 12473055.000
LLR 12490156.001	CL 12473052.000
LLR 12490156.001	CL 12473054.000
LLR 12490157.001	CL 12472959.000
LLR 12490157.001	CL 12473053.000
LLR 12490157.001	CL 12473055.000
LLR 12490158.001	CL 12473052.000
LLR 12490158.001	CL 12473054.000
LLR 12490159.001	CL 12472957.000

Continued on next page

Continued from previous page

LLR 12490159.001	CL 12473052.000
LLR 12490159.001	CL 12473054.000
LLR 12490160.001	CL 12472956.000
LLR 12490160.001	CL 12473053.000
LLR 12490160.001	CL 12473055.000
LLR 12490167.000	CL 12471426.000
LLR 12490167.000	CL 12471428.000
LLR 12490167.000	CL 12471430.000
LLR 12490167.000	CL 12471432.000
LLR 12490167.000	CL 12471434.000
LLR 12490167.000	CL 12471436.000
LLR 12490167.000	CL 12471439.000
LLR 12490167.000	CL 12471441.000
LLR 12490167.000	CL 12471443.000
LLR 12490167.000	CL 12471445.000
LLR 12490167.000	CL 12471447.000
LLR 12490167.000	CL 12472932.000
LLR 12490167.000	CL 12472934.000
LLR 12490167.000	CL 12472940.000
LLR 12490167.000	CL 12472942.000
LLR 12490173.000	CL 12471418.001
LLR 12490173.000	CL 12471471.001
LLR 12490173.000	CL 12472798.000
LLR 12490173.000	CL 12472800.000
LLR 12490173.000	CL 12473004.000
LLR 12490173.000	CL 12473006.000
LLR 12490173.000	CL 12473072.000
LLR 12490173.000	CL 12473074.000
LLR 12490177.000	CL 12471555.001
LLR 12490177.000	CL 12471557.000
LLR 12490177.000	CL 12471559.000
LLR 12490182.000	CL 12471563.000
LLR 12490184.000	CL 12471567.000
LLR 12490186.000	CL 12471571.000
LLR 12490192.000	CL 12471663.002
LLR 12490197.000	CL 12471773.000
LLR 12490197.000	CL 12471775.000
LLR 12490199.000	CL 12471766.000
LLR 12490199.000	CL 12471768.000
LLR 12490199.000	CL 12471770.000
LLR 12490204.000	
LLR 12490206.000	CL 12471798.000
LLR 12490210.000	CL 12471789.000

LLR 12490159.001	CL 12473053.000
LLR 12490159.001	CL 12473055.000
LLR 12490160.001	CL 12473052.000
LLR 12490160.001	CL 12473054.000
LLR 12490167.000	CL 12471421.001
LLR 12490167.000	CL 12471427.000
LLR 12490167.000	CL 12471429.000
LLR 12490167.000	CL 12471431.000
LLR 12490167.000	CL 12471433.000
LLR 12490167.000	CL 12471435.000
LLR 12490167.000	CL 12471438.000
LLR 12490167.000	CL 12471440.000
LLR 12490167.000	CL 12471442.000
LLR 12490167.000	CL 12471444.000
LLR 12490167.000	CL 12471446.000
LLR 12490167.000	CL 12471448.000
LLR 12490167.000	CL 12472933.000
LLR 12490167.000	CL 12472937.000
LLR 12490167.000	CL 12472941.000
LLR 12490167.000	CL 12473047.000
LLR 12490173.000	CL 12471470.002
LLR 12490173.000	CL 12472797.000
LLR 12490173.000	CL 12472799.000
LLR 12490173.000	CL 12473003.000
LLR 12490173.000	CL 12473005.000
LLR 12490173.000	CL 12473049.000
LLR 12490173.000	CL 12473073.000
LLR 12490177.000	CL 12471554.000
LLR 12490177.000	CL 12471556.000
LLR 12490177.000	CL 12471558.001
LLR 12490181.000	CL 12471561.000
LLR 12490183.000	CL 12471565.000
LLR 12490185.000	CL 12471569.000
LLR 12490190.000	CL 12471661.000
LLR 12490192.000	CL 12471664.002
LLR 12490197.000	CL 12471774.000
LLR 12490199.000	CL 12471765.000
LLR 12490199.000	CL 12471767.000
LLR 12490199.000	CL 12471769.000
LLR 12490201.001	CL 12471794.000
LLR 12490205.000	CL 12471797.000
LLR 12490209.000	CL 12471788.000
LLR 12490211.000	CL 12471790.000

Continued on next page

Continued from previous page

LLR 12490212.000	CL 12471794.000
LLR 12490218.000	CL 12471985.001
LLR 12490220.000	CL 12471989.000
LLR 12490224.000	CL 12471975.000
LLR 12490226.001	CL 12471980.001
LLR 12490227.001	CL 12472067.001
LLR 12490227.001	CL 12472069.001
LLR 12490228.001	CL 12471982.001
LLR 12490233.001	CL 12472117.002
LLR 12490235.000	CL 12472119.000
LLR 12490239.000	CL 12472107.000
LLR 12490241.000	
LLR 12490245.000	CL 12472134.000
LLR 12490247.000	CL 12472137.000
LLR 12490251.000	CL 12472130.000
LLR 12490253.000	CL 12472134.000
LLR 12490257.000	CL 12472151.000
LLR 12490259.000	CL 12472156.000
LLR 12490263.000	CL 12472147.000
LLR 12490265.000	CL 12472151.000
LLR 12490269.000	CL 12472171.000
LLR 12490271.000	CL 12472175.000
LLR 12490275.000	CL 12472167.000
LLR 12490277.000	CL 12472171.000
LLR 12490283.000	CL 12471957.000
LLR 12490283.000	CL 12471962.000
LLR 12490284.000	CL 12471960.001
LLR 12490284.000	CL 12471962.000
LLR 12490287.000	CL 12472035.001
LLR 12490288.000	CL 12472035.001
LLR 12490289.000	CL 12472032.000
LLR 12490289.000	CL 12472042.000
LLR 12490289.000	CL 12472046.000
LLR 12490289.000	CL 12472049.000
LLR 12490293.000	CL 12472042.000
LLR 12490293.000	CL 12472046.000
LLR 12490293.000	CL 12472058.000
LLR 12490294.000	CL 12470535.001
LLR 12490294.000	CL 12472796.000
LLR 12490295.000	CL 12472043.000
LLR 12490296.000	CL 12472043.000
LLR 12490297.000	CL 12472043.000
LLR 12490298.000	CL 12472043.000

LLR 12490214.001	CL 12471795.001
LLR 12490219.000	CL 12471987.001
LLR 12490223.001	CL 12471972.002
LLR 12490225.000	CL 12471977.002
LLR 12490227.001	CL 12471981.002
LLR 12490227.001	CL 12472068.001
LLR 12490227.001	CL 12472070.001
LLR 12490230.001	CL 12472114.001
LLR 12490234.000	CL 12472118.000
LLR 12490238.000	CL 12472105.000
LLR 12490240.000	
LLR 12490243.001	CL 12472113.001
LLR 12490246.000	CL 12472136.000
LLR 12490250.000	CL 12472129.000
LLR 12490252.000	CL 12472131.000
LLR 12490255.000	CL 12472134.000
LLR 12490258.000	CL 12472155.000
LLR 12490262.000	CL 12472146.000
LLR 12490264.000	CL 12472148.000
LLR 12490267.000	CL 12472151.000
LLR 12490270.000	CL 12472174.000
LLR 12490274.000	CL 12472166.000
LLR 12490276.000	CL 12472168.000
LLR 12490279.000	CL 12472171.000
LLR 12490283.000	CL 12471961.000
LLR 12490284.000	CL 12471959.000
LLR 12490284.000	CL 12471961.000
LLR 12490284.000	CL 12471963.000
LLR 12490287.000	CL 12472043.000
LLR 12490288.000	CL 12472043.000
LLR 12490289.000	CL 12472035.001
LLR 12490289.000	CL 12472043.000
LLR 12490289.000	CL 12472047.000
LLR 12490293.000	CL 12472028.000
LLR 12490293.000	CL 12472043.000
LLR 12490293.000	CL 12472047.000
LLR 12490293.000	CL 12472796.000
LLR 12490294.000	CL 12472043.000
LLR 12490295.000	CL 12472035.001
LLR 12490296.000	CL 12472035.001
LLR 12490297.000	CL 12472035.001
LLR 12490298.000	CL 12472035.001
LLR 12490303.001	CL 12472059.000

Continued on next page

Continued from previous page

LLR 12490303.001	CL 12472060.001
LLR 12490303.001	CL 12472064.000
LLR 12490312.001	CL 12472660.000
LLR 12490312.001	CL 12472663.001
LLR 12490312.001	CL 12472665.000
LLR 12490317.001	CL 12472644.000
LLR 12490317.001	CL 12472649.001
LLR 12490318.001	CL 12472644.000
LLR 12490318.001	CL 12472647.001
LLR 12490318.001	CL 12472649.001
LLR 12490318.001	CL 12472975.000
LLR 12490321.000	CL 12472345.000
LLR 12490323.000	CL 12472336.000
LLR 12490324.000	CL 12472336.000
LLR 12490326.000	CL 12472336.000
LLR 12490327.000	CL 12472336.000
LLR 12490329.000	CL 12472336.000
LLR 12490333.000	CL 12472336.000
LLR 12490334.000	CL 12472336.000
LLR 12490335.000	CL 12472336.000
LLR 12490336.000	CL 12472336.000
LLR 12490337.000	CL 12472336.000
LLR 12490338.000	CL 12472336.000
LLR 12490339.000	CL 12472336.000
LLR 12490340.000	CL 12472336.000
LLR 12490341.000	CL 12472336.000
LLR 12490343.000	CL 12472336.000
LLR 12490343.000	CL 12472341.000
LLR 12490344.000	CL 12472337.002
LLR 12490345.000	CL 12472336.000
LLR 12490345.000	CL 12472341.000
LLR 12490345.000	CL 12472344.000
LLR 12490346.000	CL 12472336.000
LLR 12490346.000	CL 12472342.000
LLR 12490348.000	CL 12472340.000
LLR 12490350.000	CL 12472337.002
LLR 12490351.000	CL 12472337.002
LLR 12490354.000	CL 12472336.000
LLR 12490356.000	CL 12472336.000
LLR 12490358.000	CL 12472333.000
LLR 12490359.000	CL 12472333.000
LLR 12490361.000	CL 12472336.000
LLR 12490362.000	CL 12472334.000

LLR 12490303.001	CL 12472063.000
LLR 12490303.001	CL 12472065.000
LLR 12490312.001	CL 12472662.001
LLR 12490312.001	CL 12472664.000
LLR 12490312.001	CL 12472666.000
LLR 12490317.001	CL 12472648.001
LLR 12490317.001	CL 12472975.000
LLR 12490318.001	CL 12472646.001
LLR 12490318.001	CL 12472648.001
LLR 12490318.001	CL 12472650.001
LLR 12490321.000	CL 12472339.000
LLR 12490321.000	CL 12472348.000
LLR 12490323.000	CL 12472337.002
LLR 12490325.000	CL 12472336.000
LLR 12490326.000	CL 12472337.002
LLR 12490327.000	CL 12472337.002
LLR 12490329.000	CL 12472337.002
LLR 12490333.000	CL 12472337.002
LLR 12490334.000	CL 12472337.002
LLR 12490335.000	CL 12472337.002
LLR 12490336.000	CL 12472337.002
LLR 12490337.000	CL 12472337.002
LLR 12490338.000	CL 12472337.002
LLR 12490339.000	CL 12472337.002
LLR 12490340.000	CL 12472337.002
LLR 12490341.000	CL 12472337.002
LLR 12490343.000	CL 12472337.002
LLR 12490344.000	CL 12472336.000
LLR 12490344.000	CL 12472341.000
LLR 12490345.000	CL 12472337.002
LLR 12490345.000	CL 12472343.000
LLR 12490346.000	CL 12472334.000
LLR 12490346.000	CL 12472337.002
LLR 12490348.000	CL 12472334.000
LLR 12490350.000	CL 12472336.000
LLR 12490351.000	CL 12472336.000
LLR 12490352.000	CL 12472336.000
LLR 12490355.000	CL 12472337.002
LLR 12490357.000	CL 12472336.000
LLR 12490358.000	CL 12472334.000
LLR 12490359.000	CL 12472334.000
LLR 12490361.000	CL 12472337.002
LLR 12490362.000	CL 12472336.000

Continued on next page

Continued from previous page

LLR 12490362.000	CL 12472337.002
LLR 12490364.000	CL 12472368.000
LLR 12490365.000	CL 12472367.000
LLR 12490367.000	CL 12472362.000
LLR 12490367.000	CL 12472369.000
LLR 12490367.000	CL 12472371.000
LLR 12490367.000	CL 12472374.000
LLR 12490370.000	CL 12472368.000
LLR 12490372.000	CL 12472368.000
LLR 12490375.000	CL 12472368.000
LLR 12490378.000	CL 12472369.000
LLR 12490378.000	CL 12472371.000
LLR 12490379.000	CL 12472369.000
LLR 12490379.000	CL 12472371.000
LLR 12490380.000	CL 12472369.000
LLR 12490380.000	CL 12472371.000
LLR 12490381.000	CL 12472369.000
LLR 12490381.000	CL 12472371.000
LLR 12490382.000	CL 12472368.000
LLR 12490382.000	CL 12472370.000
LLR 12490382.000	CL 12472372.000
LLR 12490383.000	CL 12472370.000
LLR 12490383.000	CL 12472372.000
LLR 12490384.000	CL 12472370.000
LLR 12490384.000	CL 12472372.000
LLR 12490386.000	CL 12472370.000
LLR 12490386.000	CL 12472372.000
LLR 12490387.000	CL 12472370.000
LLR 12490387.000	CL 12472372.000
LLR 12490388.000	CL 12472370.000
LLR 12490388.000	CL 12472372.000
LLR 12490389.000	CL 12472370.000
LLR 12490389.000	CL 12472372.000
LLR 12490390.000	CL 12472370.000
LLR 12490390.000	CL 12472372.000
LLR 12490391.000	CL 12472370.000
LLR 12490391.000	CL 12472372.000
LLR 12490392.000	CL 12472370.000
LLR 12490392.000	CL 12472372.000
LLR 12490393.000	CL 12472366.000
LLR 12490393.000	CL 12472370.000
LLR 12490393.000	CL 12472372.000
LLR 12490394.000	CL 12472370.000

LLR 12490362.000	CL 12472348.000
LLR 12490365.000	CL 12472362.000
LLR 12490366.000	CL 12472368.000
LLR 12490367.000	CL 12472368.000
LLR 12490367.000	CL 12472370.000
LLR 12490367.000	CL 12472372.000
LLR 12490368.000	CL 12472374.000
LLR 12490371.000	CL 12472368.000
LLR 12490373.000	CL 12472368.000
LLR 12490376.000	CL 12472368.000
LLR 12490378.000	CL 12472370.000
LLR 12490378.000	CL 12472372.000
LLR 12490379.000	CL 12472370.000
LLR 12490379.000	CL 12472372.000
LLR 12490380.000	CL 12472370.000
LLR 12490380.000	CL 12472372.000
LLR 12490381.000	CL 12472370.000
LLR 12490381.000	CL 12472372.000
LLR 12490382.000	CL 12472369.000
LLR 12490382.000	CL 12472371.000
LLR 12490383.000	CL 12472369.000
LLR 12490383.000	CL 12472371.000
LLR 12490384.000	CL 12472369.000
LLR 12490384.000	CL 12472371.000
LLR 12490386.000	CL 12472369.000
LLR 12490386.000	CL 12472371.000
LLR 12490387.000	CL 12472369.000
LLR 12490387.000	CL 12472371.000
LLR 12490388.000	CL 12472369.000
LLR 12490388.000	CL 12472371.000
LLR 12490389.000	CL 12472369.000
LLR 12490389.000	CL 12472371.000
LLR 12490390.000	CL 12472369.000
LLR 12490390.000	CL 12472371.000
LLR 12490391.000	CL 12472369.000
LLR 12490391.000	CL 12472371.000
LLR 12490392.000	CL 12472369.000
LLR 12490392.000	CL 12472371.000
LLR 12490393.000	CL 12472364.000
LLR 12490393.000	CL 12472369.000
LLR 12490393.000	CL 12472371.000
LLR 12490394.000	CL 12472369.000
LLR 12490394.000	CL 12472371.000

Continued on next page

Continued from previous page

LLR 12490394.000	CL 12472372.000
LLR 12490398.000	CL 12472369.000
LLR 12490400.000	CL 12472371.000
LLR 12490402.000	CL 12472361.000
LLR 12490404.000	CL 12472368.000
LLR 12490405.000	CL 12472369.000
LLR 12490405.000	CL 12472371.000
LLR 12490406.000	CL 12472368.000
LLR 12490410.001	CL 12472950.000
LLR 12490415.001	CL 12472954.000
LLR 12490423.001	CL 12472948.000
LLR 12490429.001	CL 12472459.000
LLR 12490429.001	CL 12472461.000
LLR 12490429.001	CL 12472463.000
LLR 12490439.000	CL 12472465.000
LLR 12490442.000	CL 12472465.000
LLR 12490444.000	CL 12472469.000
LLR 12490449.000	CL 12472452.000
LLR 12490454.000	CL 12472487.000
LLR 12490456.000	CL 12472487.000
LLR 12490460.000	CL 12472488.000
LLR 12490462.000	CL 12472490.000
LLR 12490464.000	CL 12472492.000
LLR 12490466.000	CL 12472494.000
LLR 12490469.000	CL 12472516.000
LLR 12490471.000	CL 12472516.000
LLR 12490473.000	CL 12472516.000
LLR 12490474.000	CL 12472515.000
LLR 12490475.000	CL 12472509.000
LLR 12490476.000	CL 12472517.000
LLR 12490478.000	CL 12472507.000
LLR 12490483.000	CL 12472510.000
LLR 12490483.000	CL 12472513.000
LLR 12490484.000	CL 12472516.000
LLR 12490486.000	CL 12472516.000
LLR 12490488.000	CL 12472516.000
LLR 12490493.000	CL 12472240.000
LLR 12490501.000	CL 12472271.000
LLR 12490505.000	CL 12472285.000
LLR 12490510.000	CL 12472449.002
LLR 12490513.000	CL 12472285.000
LLR 12490516.000	CL 12472303.000
LLR 12490517.000	CL 12472303.000

LLR 12490397.000	CL 12472368.000
LLR 12490399.000	CL 12472370.000
LLR 12490401.000	CL 12472372.000
LLR 12490403.000	CL 12472368.000
LLR 12490405.000	CL 12472368.000
LLR 12490405.000	CL 12472370.000
LLR 12490405.000	CL 12472372.000
LLR 12490408.001	CL 12472951.000
LLR 12490413.001	CL 12472952.000
LLR 12490421.001	CL 12472949.000
LLR 12490429.001	CL 12472458.000
LLR 12490429.001	CL 12472460.000
LLR 12490429.001	CL 12472462.000
LLR 12490438.000	CL 12472464.000
LLR 12490441.000	CL 12472465.000
LLR 12490443.000	CL 12472466.000
LLR 12490445.000	
LLR 12490450.000	CL 12472453.000
LLR 12490455.000	CL 12472487.000
LLR 12490459.000	CL 12472487.000
LLR 12490461.000	CL 12472489.000
LLR 12490463.000	CL 12472491.000
LLR 12490465.000	CL 12472493.000
LLR 12490468.000	CL 12472516.000
LLR 12490470.000	CL 12472516.000
LLR 12490472.000	CL 12472516.000
LLR 12490474.000	CL 12472514.000
LLR 12490474.000	CL 12472516.000
LLR 12490475.000	CL 12472516.000
LLR 12490478.000	CL 12472506.001
LLR 12490478.000	CL 12472508.000
LLR 12490483.000	CL 12472511.000
LLR 12490483.000	CL 12472516.000
LLR 12490485.000	CL 12472516.000
LLR 12490487.000	CL 12472516.000
LLR 12490489.000	CL 12472516.000
LLR 12490495.000	CL 12472241.000
LLR 12490503.000	CL 12472272.000
LLR 12490506.000	CL 12472285.000
LLR 12490512.000	CL 12472285.000
LLR 12490514.000	CL 12472285.000
LLR 12490516.000	CL 12472305.000
LLR 12490517.000	CL 12472305.000

Continued on next page

Continued from previous page

LLR 12490519.000	CL 12472298.000
LLR 12490521.000	CL 12472298.000
LLR 12490523.000	CL 12472301.000
LLR 12490524.000	CL 12472301.000
LLR 12490527.000	CL 12472302.000
LLR 12490528.000	CL 12472302.000
LLR 12490529.000	CL 12472302.000
LLR 12490530.000	CL 12472300.000
LLR 12490532.000	CL 12472299.000
LLR 12490533.000	CL 12472299.000
LLR 12490534.000	CL 12472299.000
LLR 12490535.000	CL 12472298.000
LLR 12490538.000	CL 12472304.000
LLR 12490541.000	CL 12472303.000
LLR 12490544.000	CL 12472322.000
LLR 12490545.000	CL 12472322.000
LLR 12490547.000	CL 12472322.000
LLR 12490549.000	CL 12472318.004
LLR 12490550.000	CL 12472323.000
LLR 12490554.000	CL 12472317.000
LLR 12490554.000	CL 12472322.000
LLR 12490555.000	CL 12472317.000
LLR 12490556.000	CL 12472318.004
LLR 12490559.000	CL 12472317.000
LLR 12490564.001	CL 12472558.000
LLR 12490564.001	CL 12472564.000
LLR 12490565.000	CL 12472575.001
LLR 12490565.000	CL 12472577.001
LLR 12490565.000	CL 12472579.001
LLR 12490566.001	CL 12472553.000
LLR 12490567.001	CL 12472553.000
LLR 12490571.000	CL 12472607.001
LLR 12490571.000	CL 12472609.001
LLR 12490574.001	CL 12472627.001
LLR 12490578.000	CL 12472631.000
LLR 12490586.001	CL 12472625.001
LLR 12490596.000	CL 12472973.000
LLR 12490597.000	CL 12473021.000
LLR 12490597.000	CL 12473023.000
LLR 12490604.000	CL 12472454.001
LLR 12490604.000	CL 12472949.000
LLR 12490604.000	CL 12472951.000
LLR 12490604.000	CL 12472953.000

LLR 12490519.000	CL 12472305.000
LLR 12490521.000	CL 12472305.000
LLR 12490523.000	CL 12472305.000
LLR 12490524.000	CL 12472305.000
LLR 12490527.000	CL 12472305.000
LLR 12490528.000	CL 12472305.000
LLR 12490529.000	CL 12472305.000
LLR 12490530.000	CL 12472305.000
LLR 12490532.000	CL 12472305.000
LLR 12490533.000	CL 12472305.000
LLR 12490534.000	CL 12472305.000
LLR 12490535.000	CL 12472305.000
LLR 12490538.000	CL 12472305.000
LLR 12490541.000	CL 12472305.000
LLR 12490545.000	CL 12472318.004
LLR 12490546.000	CL 12472322.000
LLR 12490548.000	CL 12472323.000
LLR 12490549.000	CL 12472323.000
LLR 12490551.000	CL 12472323.000
LLR 12490554.000	CL 12472318.004
LLR 12490554.000	CL 12472323.000
LLR 12490556.000	CL 12472317.000
LLR 12490558.000	CL 12472317.000
LLR 12490564.001	CL 12472555.000
LLR 12490564.001	CL 12472561.000
LLR 12490564.001	CL 12472573.000
LLR 12490565.000	CL 12472576.001
LLR 12490565.000	CL 12472578.001
LLR 12490565.000	CL 12472580.001
LLR 12490566.001	CL 12472567.000
LLR 12490567.001	CL 12472570.000
LLR 12490571.000	CL 12472608.000
LLR 12490571.000	CL 12472614.001
LLR 12490576.001	CL 12472629.001
LLR 12490584.000	CL 12472623.000
LLR 12490595.000	CL 12473012.000
LLR 12490597.000	CL 12473020.000
LLR 12490597.000	CL 12473022.000
LLR 12490597.000	CL 12473024.000
LLR 12490604.000	CL 12472948.000
LLR 12490604.000	CL 12472950.000
LLR 12490604.000	CL 12472952.000
LLR 12490604.000	CL 12472954.000

Continued on next page

Continued from previous page

LLR 12490607.000	CL 12470851.001
LLR 12490607.000	CL 12472930.000
LLR 12490607.000	CL 12472943.000
LLR 12490607.000	CL 12472945.000
LLR 12490608.000	CL 12470851.001
LLR 12490610.000	CL 12472930.000
LLR 12490612.000	CL 12472946.000
LLR 12490614.000	CL 12472943.000
LLR 12490616.000	CL 12473014.000
LLR 12490619.000	CL 12471451.001
LLR 12490619.000	CL 12473000.000
LLR 12490619.000	CL 12473002.000
LLR 12490620.000	CL 12472894.000
LLR 12490620.000	CL 12472898.000
LLR 12490620.000	CL 12472900.000
LLR 12490621.000	CL 12472902.000
LLR 12490622.000	CL 12470705.000
LLR 12490622.000	CL 12470707.000
LLR 12490622.000	CL 12470709.000
LLR 12490622.000	CL 12470711.000
LLR 12490622.000	CL 12470713.000
LLR 12490622.000	CL 12472792.000
LLR 12490624.000	CL 12472903.000
LLR 12490624.000	CL 12472910.000
LLR 12490624.000	CL 12472928.000
LLR 12490624.000	CL 12473032.000
LLR 12490624.000	CL 12473034.000
LLR 12490624.000	CL 12473036.000
LLR 12490624.000	CL 12473038.000
LLR 12490624.000	CL 12473040.000
LLR 12490624.000	CL 12473042.000
LLR 12490624.000	CL 12473044.000
LLR 12490625.000	CL 12471503.000
LLR 12490626.000	CL 12472043.000
LLR 12490627.000	CL 12470679.000
LLR 12490627.000	CL 12470681.000
LLR 12490627.000	CL 12470683.000
LLR 12490627.000	CL 12470685.000
LLR 12490627.000	CL 12470687.000
LLR 12490627.000	CL 12470689.000
LLR 12490627.000	CL 12470691.000
LLR 12490627.000	CL 12473056.000
LLR 12490628.000	CL 12470726.000

LLR 12490607.000	CL 12472929.000
LLR 12490607.000	CL 12472931.000
LLR 12490607.000	CL 12472944.000
LLR 12490607.000	CL 12472946.000
LLR 12490609.000	CL 12472931.000
LLR 12490611.000	CL 12472929.000
LLR 12490613.000	CL 12472945.000
LLR 12490615.000	CL 12472944.000
LLR 12490616.000	CL 12473015.000
LLR 12490619.000	CL 12472999.000
LLR 12490619.000	CL 12473001.000
LLR 12490619.000	CL 12473065.000
LLR 12490620.000	CL 12472897.000
LLR 12490620.000	CL 12472899.000
LLR 12490620.000	CL 12472901.000
LLR 12490621.000	CL 12473050.000
LLR 12490622.000	CL 12470706.000
LLR 12490622.000	CL 12470708.000
LLR 12490622.000	CL 12470710.000
LLR 12490622.000	CL 12470712.000
LLR 12490622.000	CL 12472791.000
LLR 12490623.000	CL 12470909.002
LLR 12490624.000	CL 12472909.000
LLR 12490624.000	CL 12472914.000
LLR 12490624.000	CL 12473031.000
LLR 12490624.000	CL 12473033.000
LLR 12490624.000	CL 12473035.000
LLR 12490624.000	CL 12473037.000
LLR 12490624.000	CL 12473039.000
LLR 12490624.000	CL 12473041.000
LLR 12490624.000	CL 12473043.000
LLR 12490624.000	CL 12473046.000
LLR 12490626.000	CL 12472035.001
LLR 12490627.000	CL 12470678.000
LLR 12490627.000	CL 12470680.000
LLR 12490627.000	CL 12470682.000
LLR 12490627.000	CL 12470684.000
LLR 12490627.000	CL 12470686.000
LLR 12490627.000	CL 12470688.000
LLR 12490627.000	CL 12470690.000
LLR 12490627.000	CL 12470692.000
LLR 12490628.000	CL 12470725.000
LLR 12490628.000	CL 12470728.000

Continued on next page

Continued from previous page

LLR 12490628.000	CL 12470730.000
LLR 12490628.000	CL 12470734.000
LLR 12490628.000	CL 12470738.000
LLR 12490629.000	CL 12470631.000
LLR 12490629.000	CL 12470633.000
LLR 12490629.000	CL 12470635.000
LLR 12490629.000	CL 12470637.000
LLR 12490629.000	CL 12470639.000
LLR 12490629.000	CL 12470641.000
LLR 12490629.000	CL 12470643.000
LLR 12490629.000	CL 12473058.000
LLR 12490630.000	CL 12473060.000
LLR 12490630.000	CL 12473063.000
LLR 12490635.000	CL 12472186.001
LLR 12490635.000	CL 12472188.000
LLR 12490640.000	CL 12472199.000
LLR 12490640.000	CL 12472201.000
LLR 12490640.000	CL 12472203.000
LLR 12490645.000	CL 12472214.001
LLR 12490645.000	CL 12472216.000
LLR 12490650.000	CL 12471823.000
LLR 12490650.000	CL 12471825.000
LLR 12490650.000	CL 12471827.000
LLR 12490650.000	CL 12471829.000
LLR 12490655.000	CL 12471928.001
LLR 12490655.000	CL 12471930.000
LLR 12490660.000	CL 12471942.000
LLR 12490660.000	CL 12471944.000
LLR 12490660.000	CL 12471946.000
LLR 12490665.000	CL 12471999.001
LLR 12490665.000	CL 12472001.000
LLR 12490670.000	CL 12472013.000
LLR 12490670.000	CL 12472015.000
LLR 12490670.000	CL 12472017.000
LLR 12490675.000	CL 12472843.000
LLR 12490675.000	CL 12472865.000
LLR 12490677.000	CL 12472592.000
LLR 12490678.000	CL 12471229.001
LLR 12490678.000	CL 12472876.000
LLR 12490678.000	CL 12472896.000
LLR 12490678.000	CL 12473008.000
LLR 12490680.000	CL 12471232.001
LLR 12490680.000	CL 12473009.000

LLR 12490628.000	CL 12470732.000
LLR 12490628.000	CL 12470736.000
LLR 12490628.000	CL 12473057.000
LLR 12490629.000	CL 12470632.000
LLR 12490629.000	CL 12470634.000
LLR 12490629.000	CL 12470636.000
LLR 12490629.000	CL 12470638.000
LLR 12490629.000	CL 12470640.000
LLR 12490629.000	CL 12470642.000
LLR 12490629.000	CL 12470644.000
LLR 12490630.000	CL 12473059.000
LLR 12490630.000	CL 12473062.000
LLR 12490635.000	CL 12472185.000
LLR 12490635.000	CL 12472187.000
LLR 12490635.000	CL 12472189.000
LLR 12490640.000	CL 12472200.001
LLR 12490640.000	CL 12472202.000
LLR 12490645.000	CL 12472213.000
LLR 12490645.000	CL 12472215.000
LLR 12490645.000	CL 12472217.000
LLR 12490650.000	CL 12471824.001
LLR 12490650.000	CL 12471826.000
LLR 12490650.000	CL 12471828.000
LLR 12490655.000	CL 12471927.000
LLR 12490655.000	CL 12471929.000
LLR 12490655.000	CL 12471931.000
LLR 12490660.000	CL 12471943.001
LLR 12490660.000	CL 12471945.000
LLR 12490665.000	CL 12471998.000
LLR 12490665.000	CL 12472000.000
LLR 12490665.000	CL 12472002.000
LLR 12490670.000	CL 12472014.001
LLR 12490670.000	CL 12472016.000
LLR 12490670.000	CL 12472018.000
LLR 12490675.000	CL 12472853.000
LLR 12490676.000	CL 12472592.000
LLR 12490678.000	CL 12471228.001
LLR 12490678.000	CL 12471230.001
LLR 12490678.000	CL 12472895.000
LLR 12490678.000	CL 12473007.000
LLR 12490680.000	CL 12471227.001
LLR 12490680.000	CL 12471233.001
LLR 12490681.000	CL 12472882.000

Continued on next page

Continued from previous page

LLR 12490681.000	CL 12472892.000
LLR 12490682.000	CL 12472891.000
LLR 12490683.000	CL 12472890.000
LLR 12490684.000	CL 12472888.000
LLR 12490685.000	CL 12472889.000
LLR 12490686.000	CL 12472962.000
LLR 12490686.000	CL 12472964.000
LLR 12490686.000	CL 12473070.000
LLR 12490688.000	CL 12472454.001

LLR 12490682.000	CL 12472881.000
LLR 12490683.000	CL 12472880.000
LLR 12490684.000	CL 12472878.000
LLR 12490685.000	CL 12472879.000
LLR 12490686.000	CL 12472961.000
LLR 12490686.000	CL 12472963.000
LLR 12490686.000	CL 12472965.000
LLR 12490686.000	CL 12473071.000
LLR 12490689.000	CL 12472953.000

17.02 HLR to LLR Trace Matrix

HLR	LLR		HLR	LLR
CL 12470008.000			CL 12470009.000	
CL 12470010.000			CL 12470011.000	
CL 12470012.000			CL 12470013.000	
CL 12470014.000			CL 12470025.000	
CL 12470026.000			CL 12470027.000	
CL 12470028.000			CL 12470029.000	
CL 12470030.000			CL 12470031.000	
CL 12470032.000			CL 12470043.000	
CL 12470045.000			CL 12470046.000	
CL 12470047.000			CL 12470048.000	
CL 12470049.000			CL 12470050.000	
CL 12470051.000			CL 12470064.000	
CL 12470065.000			CL 12470066.000	
CL 12470067.000			CL 12470068.000	
CL 12470069.000			CL 12470070.000	
CL 12470071.000			CL 12470072.000	
CL 12470073.000			CL 12470074.000	
CL 12470075.000			CL 12470089.000	
CL 12470090.000			CL 12470091.000	
CL 12470092.000			CL 12470093.000	
CL 12470094.000			CL 12470095.000	
CL 12470096.000			CL 12470097.000	
CL 12470098.000			CL 12470099.000	
CL 12470111.000			CL 12470112.000	
CL 12470113.000			CL 12470114.000	
CL 12470115.000			CL 12470117.000	
CL 12470129.000			CL 12470131.000	
CL 12470132.000			CL 12470133.000	
CL 12470134.000			CL 12470135.000	
CL 12470136.000			CL 12470137.000	
CL 12470138.000			CL 12470139.000	
CL 12470140.000			CL 12470141.000	
CL 12470142.000			CL 12470153.000	
CL 12470154.000			CL 12470155.000	
CL 12470156.000			CL 12470166.000	
CL 12470167.000			CL 12470168.000	
CL 12470169.000			CL 12470170.000	
CL 12470183.000			CL 12470184.000	
CL 12470185.000			CL 12470186.000	

Continued on next page

Continued from previous page

CL 12470187.000	
CL 12470189.000	
CL 12470191.000	
CL 12470204.000	
CL 12470206.000	
CL 12470208.000	
CL 12470210.000	
CL 12470212.000	
CL 12470224.001	LLR 12490064.001
CL 12470226.000	LLR 12490006.000
CL 12470229.001	LLR 12490006.000
CL 12470232.000	LLR 12490006.000
CL 12470234.000	LLR 12490006.000
CL 12470249.000	
CL 12470251.000	
CL 12470253.000	
CL 12470256.000	
CL 12470258.000	
CL 12470260.000	
CL 12470274.000	
CL 12470277.000	
CL 12470279.000	
CL 12470291.000	
CL 12470294.000	
CL 12470296.000	
CL 12470308.000	
CL 12470320.000	
CL 12470407.000	
CL 12470409.000	
CL 12470411.000	
CL 12470413.000	
CL 12470416.000	
CL 12470418.000	
CL 12470420.000	
CL 12470422.000	
CL 12470424.000	
CL 12470426.000	
CL 12470428.000	
CL 12470430.000	
CL 12470433.000	
CL 12470437.000	
CL 12470439.000	
CL 12470441.000	

CL 12470188.000	
CL 12470190.000	
CL 12470192.000	
CL 12470205.000	
CL 12470207.000	
CL 12470209.000	
CL 12470211.000	
CL 12470224.001	LLR 12490006.000
CL 12470225.001	LLR 12490006.000
CL 12470228.002	LLR 12490006.000
CL 12470231.001	LLR 12490006.000
CL 12470233.000	LLR 12490006.000
CL 12470235.001	LLR 12490006.000
CL 12470250.000	
CL 12470252.000	
CL 12470255.000	
CL 12470257.000	
CL 12470259.000	
CL 12470263.000	
CL 12470276.000	
CL 12470278.000	
CL 12470280.000	
CL 12470293.000	
CL 12470295.000	
CL 12470307.000	
CL 12470309.000	
CL 12470363.000	
CL 12470408.000	
CL 12470410.000	
CL 12470412.000	
CL 12470415.000	
CL 12470417.000	
CL 12470419.000	
CL 12470421.000	
CL 12470423.000	
CL 12470425.000	
CL 12470427.000	
CL 12470429.000	
CL 12470431.000	
CL 12470435.000	
CL 12470438.000	
CL 12470440.000	
CL 12470442.000	

Continued on next page

Continued from previous page

CL 12470443.000	
CL 12470445.000	
CL 12470448.000	
CL 12470450.000	
CL 12470453.000	
CL 12470456.000	
CL 12470459.000	
CL 12470462.000	
CL 12470464.000	
CL 12470466.000	
CL 12470468.000	
CL 12470470.000	
CL 12470472.000	
CL 12470485.000	
CL 12470487.000	
CL 12470509.000	
CL 12470511.000	
CL 12470513.000	
CL 12470525.000	LLR 12490064.001
CL 12470528.001	LLR 12490011.001
CL 12470530.001	LLR 12490011.001
CL 12470532.001	LLR 12490011.001
CL 12470535.001	LLR 12490294.000
CL 12470541.003	LLR 12490011.001
CL 12470543.000	LLR 12490011.001
CL 12470553.000	LLR 12490064.001
CL 12470556.001	LLR 12490020.000
CL 12470566.000	
CL 12470576.000	
CL 12470578.000	
CL 12470580.000	
CL 12470589.000	
CL 12470591.000	
CL 12470593.000	
CL 12470595.000	
CL 12470597.000	
CL 12470599.000	
CL 12470612.000	
CL 12470614.000	
CL 12470616.000	
CL 12470618.000	
CL 12470632.000	LLR 12490629.000
CL 12470634.000	LLR 12490629.000

CL 12470444.000	
CL 12470447.000	
CL 12470449.000	
CL 12470452.000	
CL 12470455.000	
CL 12470457.000	
CL 12470460.000	
CL 12470463.000	
CL 12470465.000	
CL 12470467.000	
CL 12470469.000	
CL 12470471.000	
CL 12470484.000	
CL 12470486.000	
CL 12470498.000	
CL 12470510.000	
CL 12470512.000	
CL 12470514.000	
CL 12470527.002	LLR 12490011.001
CL 12470529.001	LLR 12490011.001
CL 12470531.001	LLR 12490011.001
CL 12470535.001	LLR 12490011.001
CL 12470538.001	LLR 12490011.001
CL 12470542.000	LLR 12490011.001
CL 12470553.000	LLR 12490020.000
CL 12470555.001	LLR 12490020.000
CL 12470557.001	LLR 12490020.000
CL 12470567.000	
CL 12470577.000	LLR 12490064.001
CL 12470579.000	
CL 12470581.000	
CL 12470590.000	
CL 12470592.000	
CL 12470594.000	
CL 12470596.000	
CL 12470598.000	
CL 12470611.000	
CL 12470613.000	
CL 12470615.000	
CL 12470617.000	
CL 12470631.000	LLR 12490629.000
CL 12470633.000	LLR 12490629.000
CL 12470635.000	LLR 12490629.000

Continued on next page

Continued from previous page

CL 12470636.000	LLR 12490629.000
CL 12470638.000	LLR 12490629.000
CL 12470640.000	LLR 12490629.000
CL 12470642.000	LLR 12490629.000
CL 12470644.000	LLR 12490629.000
CL 12470658.000	
CL 12470660.000	
CL 12470662.000	
CL 12470664.000	
CL 12470678.000	LLR 12490627.000
CL 12470680.000	LLR 12490627.000
CL 12470682.000	LLR 12490627.000
CL 12470684.000	LLR 12490627.000
CL 12470686.000	LLR 12490627.000
CL 12470688.000	LLR 12490627.000
CL 12470690.000	LLR 12490627.000
CL 12470692.000	LLR 12490627.000
CL 12470706.000	LLR 12490622.000
CL 12470708.000	LLR 12490622.000
CL 12470710.000	LLR 12490622.000
CL 12470712.000	LLR 12490622.000
CL 12470725.000	LLR 12490628.000
CL 12470728.000	LLR 12490628.000
CL 12470732.000	LLR 12490628.000
CL 12470736.000	LLR 12490628.000
CL 12470748.000	
CL 12470761.000	LLR 12490027.000
CL 12470763.000	LLR 12490027.000
CL 12470765.000	LLR 12490027.000
CL 12470767.000	LLR 12490027.000
CL 12470769.000	LLR 12490027.000
CL 12470771.000	LLR 12490027.000
CL 12470773.000	LLR 12490027.000
CL 12470774.000	LLR 12490029.000
CL 12470785.000	
CL 12470805.000	
CL 12470819.000	LLR 12490027.000
CL 12470822.000	LLR 12490047.000
CL 12470824.000	LLR 12490047.000
CL 12470826.000	LLR 12490048.000
CL 12470828.000	LLR 12490050.000
CL 12470830.000	LLR 12490047.000
CL 12470832.000	LLR 12490051.000

CL 12470637.000	LLR 12490629.000
CL 12470639.000	LLR 12490629.000
CL 12470641.000	LLR 12490629.000
CL 12470643.000	LLR 12490629.000
CL 12470657.000	
CL 12470659.000	
CL 12470661.000	
CL 12470663.000	
CL 12470665.000	
CL 12470679.000	LLR 12490627.000
CL 12470681.000	LLR 12490627.000
CL 12470683.000	LLR 12490627.000
CL 12470685.000	LLR 12490627.000
CL 12470687.000	LLR 12490627.000
CL 12470689.000	LLR 12490627.000
CL 12470691.000	LLR 12490627.000
CL 12470705.000	LLR 12490622.000
CL 12470707.000	LLR 12490622.000
CL 12470709.000	LLR 12490622.000
CL 12470711.000	LLR 12490622.000
CL 12470713.000	LLR 12490622.000
CL 12470726.000	LLR 12490628.000
CL 12470730.000	LLR 12490628.000
CL 12470734.000	LLR 12490628.000
CL 12470738.000	LLR 12490628.000
CL 12470760.000	LLR 12490027.000
CL 12470762.000	LLR 12490027.000
CL 12470764.000	LLR 12490027.000
CL 12470766.000	LLR 12490027.000
CL 12470768.000	LLR 12490027.000
CL 12470770.000	LLR 12490027.000
CL 12470772.000	LLR 12490027.000
CL 12470774.000	LLR 12490027.000
CL 12470784.000	
CL 12470796.000	
CL 12470806.000	
CL 12470821.000	LLR 12490047.000
CL 12470823.000	LLR 12490047.000
CL 12470825.000	LLR 12490047.000
CL 12470827.000	LLR 12490049.000
CL 12470829.000	LLR 12490047.000
CL 12470831.000	LLR 12490051.000
CL 12470833.000	LLR 12490031.000

Continued on next page

Continued from previous page

CL 12470833.000	LLR 12490051.000
CL 12470835.000	LLR 12490035.000
CL 12470837.000	LLR 12490032.000
CL 12470839.000	LLR 12490034.000
CL 12470842.000	LLR 12490035.000
CL 12470844.000	LLR 12490035.000
CL 12470846.000	LLR 12490035.000
CL 12470847.000	LLR 12490036.000
CL 12470851.001	LLR 12490607.000
CL 12470853.001	LLR 12490036.000
CL 12470854.000	LLR 12490038.000
CL 12470855.000	LLR 12490047.000
CL 12470879.000	
CL 12470892.000	
CL 12470894.000	
CL 12470896.000	
CL 12470898.000	
CL 12470909.002	LLR 12490623.000
CL 12470918.000	
CL 12470920.000	
CL 12470930.000	LLR 12490054.000
CL 12470930.000	LLR 12490056.000
CL 12470930.000	LLR 12490064.001
CL 12470930.000	LLR 12490066.002
CL 12470930.000	LLR 12490068.001
CL 12470930.000	LLR 12490072.000
CL 12470930.000	LLR 12490074.000
CL 12470946.000	
CL 12470957.000	
CL 12470959.000	
CL 12470979.000	
CL 12470980.000	LLR 12490068.001
CL 12470992.000	LLR 12490068.001
CL 12471001.000	
CL 12471003.000	
CL 12471005.000	
CL 12471007.000	
CL 12471009.000	
CL 12471011.000	
CL 12471017.000	
CL 12471019.000	
CL 12471021.000	
CL 12471028.000	

CL 12470834.000	LLR 12490036.000
CL 12470836.000	LLR 12490051.000
CL 12470838.000	LLR 12490033.000
CL 12470840.000	LLR 12490034.000
CL 12470843.000	LLR 12490035.000
CL 12470845.000	
CL 12470847.000	LLR 12490035.000
CL 12470849.000	LLR 12490036.000
CL 12470851.001	LLR 12490608.000
CL 12470853.001	LLR 12490037.000
CL 12470855.000	LLR 12490038.000
CL 12470867.000	
CL 12470891.000	
CL 12470893.000	
CL 12470895.000	
CL 12470897.000	
CL 12470908.000	LLR 12490067.001
CL 12470917.000	
CL 12470919.000	
CL 12470930.000	LLR 12490053.000
CL 12470930.000	LLR 12490055.000
CL 12470930.000	LLR 12490061.000
CL 12470930.000	LLR 12490065.001
CL 12470930.000	LLR 12490067.001
CL 12470930.000	LLR 12490071.000
CL 12470930.000	LLR 12490073.000
CL 12470930.000	LLR 12490075.000
CL 12470947.000	
CL 12470958.000	
CL 12470970.000	LLR 12490067.001
CL 12470980.000	LLR 12490066.002
CL 12470981.000	
CL 12471000.000	
CL 12471002.000	
CL 12471004.000	
CL 12471006.000	
CL 12471008.000	
CL 12471010.000	
CL 12471012.000	
CL 12471018.000	
CL 12471020.000	
CL 12471022.000	
CL 12471029.000	

Continued on next page

Continued from previous page

CL 12471030.000	
CL 12471032.000	
CL 12471043.000	LLR 12490064.001
CL 12471057.000	
CL 12471059.000	
CL 12471081.000	LLR 12490065.001
CL 12471081.000	LLR 12490067.001
CL 12471082.000	LLR 12490096.000
CL 12471082.000	LLR 12490102.000
CL 12471083.000	LLR 12490094.000
CL 12471083.000	LLR 12490096.000
CL 12471084.000	LLR 12490096.000
CL 12471084.000	LLR 12490104.000
CL 12471085.000	LLR 12490096.000
CL 12471085.000	LLR 12490106.000
CL 12471092.000	
CL 12471093.000	LLR 12490067.001
CL 12471094.000	LLR 12490112.000
CL 12471094.000	LLR 12490114.000
CL 12471095.000	LLR 12490112.000
CL 12471095.000	LLR 12490114.000
CL 12471096.000	LLR 12490112.000
CL 12471096.000	LLR 12490114.000
CL 12471114.000	
CL 12471116.000	
CL 12471118.000	
CL 12471120.000	
CL 12471122.000	
CL 12471124.000	
CL 12471139.000	LLR 12490121.000
CL 12471141.000	LLR 12490121.000
CL 12471143.000	LLR 12490121.000
CL 12471145.000	LLR 12490121.000
CL 12471179.000	LLR 12490065.001
CL 12471179.000	LLR 12490067.001
CL 12471180.000	LLR 12490065.001
CL 12471180.000	LLR 12490067.001
CL 12471182.000	LLR 12490141.000
CL 12471184.000	LLR 12490141.000
CL 12471186.000	LLR 12490141.000
CL 12471188.000	LLR 12490141.000
CL 12471190.000	LLR 12490141.000
CL 12471211.000	

CL 12471031.000	
CL 12471038.000	
CL 12471056.000	
CL 12471058.000	
CL 12471071.000	
CL 12471081.000	LLR 12490066.002
CL 12471081.000	LLR 12490068.001
CL 12471082.000	LLR 12490097.000
CL 12471082.000	LLR 12490103.000
CL 12471083.000	LLR 12490095.000
CL 12471083.000	LLR 12490097.000
CL 12471084.000	LLR 12490097.000
CL 12471084.000	LLR 12490105.000
CL 12471085.000	LLR 12490097.000
CL 12471085.000	LLR 12490107.000
CL 12471093.000	LLR 12490065.001
CL 12471094.000	LLR 12490111.000
CL 12471094.000	LLR 12490113.000
CL 12471095.000	LLR 12490111.000
CL 12471095.000	LLR 12490113.000
CL 12471096.000	LLR 12490111.000
CL 12471096.000	LLR 12490113.000
CL 12471113.000	
CL 12471115.000	
CL 12471117.000	
CL 12471119.000	
CL 12471121.000	
CL 12471123.000	
CL 12471138.000	LLR 12490065.001
CL 12471140.000	LLR 12490121.000
CL 12471142.000	LLR 12490121.000
CL 12471144.000	LLR 12490121.000
CL 12471157.000	
CL 12471179.000	LLR 12490066.002
CL 12471179.000	LLR 12490068.001
CL 12471180.000	LLR 12490066.002
CL 12471180.000	LLR 12490068.001
CL 12471183.000	LLR 12490141.000
CL 12471185.000	LLR 12490141.000
CL 12471187.000	LLR 12490141.000
CL 12471189.000	LLR 12490141.000
CL 12471193.000	
CL 12471227.001	LLR 12490680.000

Continued on next page

Continued from previous page

CL 12471228.001	LLR 12490678.000
CL 12471230.001	LLR 12490678.000
CL 12471233.001	LLR 12490680.000
CL 12471247.000	LLR 12490065.001
CL 12471248.000	LLR 12490065.001
CL 12471250.000	LLR 12490148.000
CL 12471252.000	LLR 12490148.000
CL 12471254.000	LLR 12490148.000
CL 12471256.000	LLR 12490148.000
CL 12471258.000	LLR 12490148.000
CL 12471260.000	LLR 12490148.000
CL 12471264.000	
CL 12471266.000	
CL 12471307.000	
CL 12471341.000	
CL 12471370.000	
CL 12471400.000	
CL 12471418.001	LLR 12490173.000
CL 12471426.000	LLR 12490167.000
CL 12471428.000	LLR 12490167.000
CL 12471430.000	LLR 12490167.000
CL 12471432.000	LLR 12490167.000
CL 12471434.000	LLR 12490167.000
CL 12471436.000	LLR 12490167.000
CL 12471439.000	LLR 12490167.000
CL 12471441.000	LLR 12490167.000
CL 12471443.000	LLR 12490167.000
CL 12471445.000	LLR 12490167.000
CL 12471447.000	LLR 12490167.000
CL 12471451.001	LLR 12490619.000
CL 12471464.000	LLR 12490068.001
CL 12471471.001	LLR 12490173.000
CL 12471501.000	LLR 12490066.002
CL 12471501.000	LLR 12490068.001
CL 12471503.000	LLR 12490625.000
CL 12471521.000	
CL 12471537.000	LLR 12490066.002
CL 12471537.000	LLR 12490068.001
CL 12471539.000	
CL 12471551.000	LLR 12490065.001
CL 12471551.000	LLR 12490067.001
CL 12471554.000	LLR 12490177.000
CL 12471556.000	LLR 12490177.000

CL 12471229.001	LLR 12490678.000
CL 12471232.001	LLR 12490680.000
CL 12471246.000	
CL 12471247.000	LLR 12490067.001
CL 12471248.000	LLR 12490067.001
CL 12471251.000	LLR 12490148.000
CL 12471253.000	LLR 12490148.000
CL 12471255.000	LLR 12490148.000
CL 12471257.000	LLR 12490148.000
CL 12471259.000	LLR 12490148.000
CL 12471261.000	LLR 12490148.000
CL 12471265.000	
CL 12471296.000	
CL 12471324.000	LLR 12490068.001
CL 12471358.000	
CL 12471382.000	
CL 12471416.000	LLR 12490067.001
CL 12471421.001	LLR 12490167.000
CL 12471427.000	LLR 12490167.000
CL 12471429.000	LLR 12490167.000
CL 12471431.000	LLR 12490167.000
CL 12471433.000	LLR 12490167.000
CL 12471435.000	LLR 12490167.000
CL 12471438.000	LLR 12490167.000
CL 12471440.000	LLR 12490167.000
CL 12471442.000	LLR 12490167.000
CL 12471444.000	LLR 12490167.000
CL 12471446.000	LLR 12490167.000
CL 12471448.000	LLR 12490167.000
CL 12471463.000	LLR 12490068.001
CL 12471470.002	LLR 12490173.000
CL 12471501.000	LLR 12490065.001
CL 12471501.000	LLR 12490067.001
CL 12471502.000	
CL 12471520.000	
CL 12471537.000	LLR 12490065.001
CL 12471537.000	LLR 12490067.001
CL 12471538.000	
CL 12471550.000	
CL 12471551.000	LLR 12490066.002
CL 12471551.000	LLR 12490068.001
CL 12471555.001	LLR 12490177.000
CL 12471557.000	LLR 12490177.000

Continued on next page

Continued from previous page

CL 12471558.001	LLR 12490177.000
CL 12471561.000	LLR 12490181.000
CL 12471565.000	LLR 12490183.000
CL 12471569.000	LLR 12490185.000
CL 12471572.000	
CL 12471615.000	
CL 12471632.000	
CL 12471634.000	
CL 12471636.000	
CL 12471638.000	
CL 12471660.000	LLR 12490068.001
CL 12471663.002	LLR 12490192.000
CL 12471682.000	
CL 12471684.000	
CL 12471686.000	
CL 12471688.000	
CL 12471690.000	
CL 12471692.000	
CL 12471694.000	
CL 12471696.000	
CL 12471698.000	
CL 12471700.000	
CL 12471702.000	
CL 12471714.000	
CL 12471716.000	
CL 12471718.000	
CL 12471720.000	
CL 12471722.000	
CL 12471724.000	
CL 12471726.000	
CL 12471728.000	
CL 12471730.000	
CL 12471732.000	
CL 12471734.000	
CL 12471743.000	
CL 12471745.000	
CL 12471747.000	
CL 12471749.000	
CL 12471751.000	
CL 12471753.000	
CL 12471765.000	LLR 12490199.000
CL 12471767.000	LLR 12490199.000
CL 12471769.000	LLR 12490199.000

CL 12471559.000	LLR 12490177.000
CL 12471563.000	LLR 12490182.000
CL 12471567.000	LLR 12490184.000
CL 12471571.000	LLR 12490186.000
CL 12471586.000	
CL 12471631.000	
CL 12471633.000	
CL 12471635.000	
CL 12471637.000	
CL 12471648.000	
CL 12471661.000	LLR 12490190.000
CL 12471664.002	LLR 12490192.000
CL 12471683.000	
CL 12471685.000	
CL 12471687.000	
CL 12471689.000	
CL 12471691.000	
CL 12471693.000	
CL 12471695.000	
CL 12471697.000	
CL 12471699.000	
CL 12471701.000	
CL 12471703.000	
CL 12471715.000	
CL 12471717.000	
CL 12471719.000	
CL 12471721.000	
CL 12471723.000	
CL 12471725.000	
CL 12471727.000	
CL 12471729.000	
CL 12471731.000	
CL 12471733.000	
CL 12471742.000	
CL 12471744.000	
CL 12471746.000	
CL 12471748.000	
CL 12471750.000	
CL 12471752.000	
CL 12471763.000	LLR 12490068.001
CL 12471766.000	LLR 12490199.000
CL 12471768.000	LLR 12490199.000
CL 12471770.000	LLR 12490199.000

Continued on next page

Continued from previous page

CL 12471773.000	LLR 12490197.000
CL 12471775.000	LLR 12490197.000
CL 12471788.000	LLR 12490209.000
CL 12471790.000	LLR 12490211.000
CL 12471794.000	LLR 12490212.000
CL 12471797.000	LLR 12490205.000
CL 12471823.000	LLR 12490650.000
CL 12471825.000	LLR 12490650.000
CL 12471827.000	LLR 12490650.000
CL 12471829.000	LLR 12490650.000
CL 12471928.001	LLR 12490655.000
CL 12471930.000	LLR 12490655.000
CL 12471942.000	LLR 12490660.000
CL 12471944.000	LLR 12490660.000
CL 12471946.000	LLR 12490660.000
CL 12471957.000	LLR 12490283.000
CL 12471960.001	LLR 12490284.000
CL 12471961.000	LLR 12490284.000
CL 12471962.000	LLR 12490284.000
CL 12471972.002	LLR 12490064.001
CL 12471975.000	LLR 12490224.000
CL 12471977.002	LLR 12490225.000
CL 12471981.002	LLR 12490227.001
CL 12471985.001	LLR 12490218.000
CL 12471989.000	LLR 12490220.000
CL 12471999.001	LLR 12490665.000
CL 12472001.000	LLR 12490665.000
CL 12472013.000	LLR 12490670.000
CL 12472015.000	LLR 12490670.000
CL 12472017.000	LLR 12490670.000
CL 12472028.000	LLR 12490064.001
CL 12472032.000	LLR 12490289.000
CL 12472035.001	LLR 12490288.000
CL 12472035.001	LLR 12490295.000
CL 12472035.001	LLR 12490297.000
CL 12472035.001	LLR 12490626.000
CL 12472042.000	LLR 12490293.000
CL 12472043.000	LLR 12490288.000
CL 12472043.000	LLR 12490293.000
CL 12472043.000	LLR 12490295.000
CL 12472043.000	LLR 12490297.000
CL 12472043.000	LLR 12490626.000
CL 12472046.000	LLR 12490293.000

CL 12471774.000	LLR 12490197.000
CL 12471788.000	LLR 12490064.001
CL 12471789.000	LLR 12490210.000
CL 12471794.000	LLR 12490201.001
CL 12471795.001	LLR 12490214.001
CL 12471798.000	LLR 12490206.000
CL 12471824.001	LLR 12490650.000
CL 12471826.000	LLR 12490650.000
CL 12471828.000	LLR 12490650.000
CL 12471927.000	LLR 12490655.000
CL 12471929.000	LLR 12490655.000
CL 12471931.000	LLR 12490655.000
CL 12471943.001	LLR 12490660.000
CL 12471945.000	LLR 12490660.000
CL 12471957.000	LLR 12490064.001
CL 12471959.000	LLR 12490284.000
CL 12471961.000	LLR 12490283.000
CL 12471962.000	LLR 12490283.000
CL 12471963.000	LLR 12490284.000
CL 12471972.002	LLR 12490223.001
CL 12471977.002	LLR 12490038.000
CL 12471980.001	LLR 12490226.001
CL 12471982.001	LLR 12490228.001
CL 12471987.001	LLR 12490219.000
CL 12471998.000	LLR 12490665.000
CL 12472000.000	LLR 12490665.000
CL 12472002.000	LLR 12490665.000
CL 12472014.001	LLR 12490670.000
CL 12472016.000	LLR 12490670.000
CL 12472018.000	LLR 12490670.000
CL 12472028.000	LLR 12490293.000
CL 12472035.001	LLR 12490287.000
CL 12472035.001	LLR 12490289.000
CL 12472035.001	LLR 12490296.000
CL 12472035.001	LLR 12490298.000
CL 12472042.000	LLR 12490289.000
CL 12472043.000	LLR 12490287.000
CL 12472043.000	LLR 12490289.000
CL 12472043.000	LLR 12490294.000
CL 12472043.000	LLR 12490296.000
CL 12472043.000	LLR 12490298.000
CL 12472046.000	LLR 12490289.000
CL 12472047.000	LLR 12490289.000

Continued on next page

Continued from previous page

CL 12472047.000	LLR 12490293.000
CL 12472058.000	LLR 12490293.000
CL 12472060.001	LLR 12490303.001
CL 12472064.000	LLR 12490303.001
CL 12472067.001	LLR 12490227.001
CL 12472069.001	LLR 12490227.001
CL 12472079.000	
CL 12472105.000	LLR 12490238.000
CL 12472108.001	
CL 12472113.001	LLR 12490243.001
CL 12472117.002	LLR 12490233.001
CL 12472119.000	LLR 12490235.000
CL 12472129.000	LLR 12490250.000
CL 12472131.000	LLR 12490252.000
CL 12472134.000	LLR 12490253.000
CL 12472136.000	LLR 12490246.000
CL 12472146.000	LLR 12490064.001
CL 12472147.000	LLR 12490263.000
CL 12472151.000	LLR 12490257.000
CL 12472151.000	LLR 12490267.000
CL 12472155.000	LLR 12490258.000
CL 12472166.000	LLR 12490064.001
CL 12472167.000	LLR 12490275.000
CL 12472171.000	LLR 12490269.000
CL 12472171.000	LLR 12490279.000
CL 12472174.000	LLR 12490270.000
CL 12472185.000	LLR 12490635.000
CL 12472187.000	LLR 12490635.000
CL 12472189.000	LLR 12490635.000
CL 12472200.001	LLR 12490640.000
CL 12472202.000	LLR 12490640.000
CL 12472213.000	LLR 12490645.000
CL 12472215.000	LLR 12490645.000
CL 12472217.000	LLR 12490645.000
CL 12472227.000	
CL 12472229.000	
CL 12472240.000	LLR 12490493.000
CL 12472251.000	
CL 12472254.000	
CL 12472256.000	
CL 12472258.000	
CL 12472260.000	
CL 12472271.000	LLR 12490501.000

CL 12472049.000	LLR 12490289.000
CL 12472059.000	LLR 12490303.001
CL 12472063.000	LLR 12490303.001
CL 12472065.000	LLR 12490303.001
CL 12472068.001	LLR 12490227.001
CL 12472070.001	LLR 12490227.001
CL 12472105.000	LLR 12490064.001
CL 12472107.000	LLR 12490239.000
CL 12472112.002	
CL 12472114.001	LLR 12490230.001
CL 12472118.000	LLR 12490234.000
CL 12472129.000	LLR 12490064.001
CL 12472130.000	LLR 12490251.000
CL 12472134.000	LLR 12490245.000
CL 12472134.000	LLR 12490255.000
CL 12472137.000	LLR 12490247.000
CL 12472146.000	LLR 12490262.000
CL 12472148.000	LLR 12490264.000
CL 12472151.000	LLR 12490265.000
CL 12472152.000	
CL 12472156.000	LLR 12490259.000
CL 12472166.000	LLR 12490274.000
CL 12472168.000	LLR 12490276.000
CL 12472171.000	LLR 12490277.000
CL 12472172.000	
CL 12472175.000	LLR 12490271.000
CL 12472186.001	LLR 12490635.000
CL 12472188.000	LLR 12490635.000
CL 12472199.000	LLR 12490640.000
CL 12472201.000	LLR 12490640.000
CL 12472203.000	LLR 12490640.000
CL 12472214.001	LLR 12490645.000
CL 12472216.000	LLR 12490645.000
CL 12472225.000	
CL 12472228.000	
CL 12472239.000	LLR 12490068.001
CL 12472241.000	LLR 12490495.000
CL 12472253.000	LLR 12490068.001
CL 12472255.000	
CL 12472257.000	
CL 12472259.000	
CL 12472270.000	LLR 12490068.001
CL 12472272.000	LLR 12490503.000

Continued on next page

Continued from previous page

CL 12472283.000	LLR 12490065.001
CL 12472283.000	LLR 12490067.001
CL 12472285.000	LLR 12490505.000
CL 12472285.000	LLR 12490512.000
CL 12472285.000	LLR 12490514.000
CL 12472298.000	LLR 12490519.000
CL 12472298.000	LLR 12490535.000
CL 12472299.000	LLR 12490533.000
CL 12472300.000	LLR 12490530.000
CL 12472301.000	LLR 12490524.000
CL 12472302.000	LLR 12490528.000
CL 12472303.000	LLR 12490516.000
CL 12472303.000	LLR 12490541.000
CL 12472305.000	LLR 12490516.000
CL 12472305.000	LLR 12490519.000
CL 12472305.000	LLR 12490523.000
CL 12472305.000	LLR 12490527.000
CL 12472305.000	LLR 12490529.000
CL 12472305.000	LLR 12490532.000
CL 12472305.000	LLR 12490534.000
CL 12472305.000	LLR 12490538.000
CL 12472315.000	LLR 12490068.001
CL 12472317.000	LLR 12490555.000
CL 12472317.000	LLR 12490558.000
CL 12472318.004	LLR 12490545.000
CL 12472318.004	LLR 12490554.000
CL 12472322.000	LLR 12490544.000
CL 12472322.000	LLR 12490546.000
CL 12472322.000	LLR 12490554.000
CL 12472323.000	LLR 12490549.000
CL 12472323.000	LLR 12490551.000
CL 12472332.000	
CL 12472333.000	LLR 12490359.000
CL 12472334.000	LLR 12490348.000
CL 12472334.000	LLR 12490359.000
CL 12472336.000	LLR 12490323.000
CL 12472336.000	LLR 12490325.000
CL 12472336.000	LLR 12490327.000
CL 12472336.000	LLR 12490333.000
CL 12472336.000	LLR 12490335.000
CL 12472336.000	LLR 12490337.000
CL 12472336.000	LLR 12490339.000
CL 12472336.000	LLR 12490341.000

CL 12472283.000	LLR 12490066.002
CL 12472283.000	LLR 12490068.001
CL 12472285.000	LLR 12490506.000
CL 12472285.000	LLR 12490513.000
CL 12472296.000	LLR 12490068.001
CL 12472298.000	LLR 12490521.000
CL 12472299.000	LLR 12490532.000
CL 12472299.000	LLR 12490534.000
CL 12472301.000	LLR 12490523.000
CL 12472302.000	LLR 12490527.000
CL 12472302.000	LLR 12490529.000
CL 12472303.000	LLR 12490517.000
CL 12472304.000	LLR 12490538.000
CL 12472305.000	LLR 12490517.000
CL 12472305.000	LLR 12490521.000
CL 12472305.000	LLR 12490524.000
CL 12472305.000	LLR 12490528.000
CL 12472305.000	LLR 12490530.000
CL 12472305.000	LLR 12490533.000
CL 12472305.000	LLR 12490535.000
CL 12472305.000	LLR 12490541.000
CL 12472317.000	LLR 12490554.000
CL 12472317.000	LLR 12490556.000
CL 12472317.000	LLR 12490559.000
CL 12472318.004	LLR 12490549.000
CL 12472318.004	LLR 12490556.000
CL 12472322.000	LLR 12490545.000
CL 12472322.000	LLR 12490547.000
CL 12472323.000	LLR 12490548.000
CL 12472323.000	LLR 12490550.000
CL 12472323.000	LLR 12490554.000
CL 12472333.000	LLR 12490358.000
CL 12472334.000	LLR 12490346.000
CL 12472334.000	LLR 12490358.000
CL 12472334.000	LLR 12490362.000
CL 12472336.000	LLR 12490324.000
CL 12472336.000	LLR 12490326.000
CL 12472336.000	LLR 12490329.000
CL 12472336.000	LLR 12490334.000
CL 12472336.000	LLR 12490336.000
CL 12472336.000	LLR 12490338.000
CL 12472336.000	LLR 12490340.000
CL 12472336.000	LLR 12490343.000

Continued on next page

Continued from previous page

CL 12472336.000	LLR 12490344.000
CL 12472336.000	LLR 12490346.000
CL 12472336.000	LLR 12490351.000
CL 12472336.000	LLR 12490354.000
CL 12472336.000	LLR 12490357.000
CL 12472336.000	LLR 12490362.000
CL 12472337.002	LLR 12490326.000
CL 12472337.002	LLR 12490329.000
CL 12472337.002	LLR 12490334.000
CL 12472337.002	LLR 12490336.000
CL 12472337.002	LLR 12490338.000
CL 12472337.002	LLR 12490340.000
CL 12472337.002	LLR 12490343.000
CL 12472337.002	LLR 12490345.000
CL 12472337.002	LLR 12490350.000
CL 12472337.002	LLR 12490355.000
CL 12472337.002	LLR 12490362.000
CL 12472340.000	LLR 12490348.000
CL 12472341.000	LLR 12490344.000
CL 12472342.000	LLR 12490346.000
CL 12472344.000	LLR 12490345.000
CL 12472346.000	
CL 12472348.000	LLR 12490321.000
CL 12472349.000	
CL 12472351.000	
CL 12472361.000	LLR 12490068.001
CL 12472362.000	LLR 12490365.000
CL 12472364.000	LLR 12490393.000
CL 12472367.000	LLR 12490365.000
CL 12472368.000	LLR 12490366.000
CL 12472368.000	LLR 12490370.000
CL 12472368.000	LLR 12490372.000
CL 12472368.000	LLR 12490375.000
CL 12472368.000	LLR 12490382.000
CL 12472368.000	LLR 12490403.000
CL 12472368.000	LLR 12490405.000
CL 12472369.000	LLR 12490367.000
CL 12472369.000	LLR 12490379.000
CL 12472369.000	LLR 12490381.000
CL 12472369.000	LLR 12490383.000
CL 12472369.000	LLR 12490386.000
CL 12472369.000	LLR 12490388.000
CL 12472369.000	LLR 12490390.000

CL 12472336.000	LLR 12490345.000
CL 12472336.000	LLR 12490350.000
CL 12472336.000	LLR 12490352.000
CL 12472336.000	LLR 12490356.000
CL 12472336.000	LLR 12490361.000
CL 12472337.002	LLR 12490323.000
CL 12472337.002	LLR 12490327.000
CL 12472337.002	LLR 12490333.000
CL 12472337.002	LLR 12490335.000
CL 12472337.002	LLR 12490337.000
CL 12472337.002	LLR 12490339.000
CL 12472337.002	LLR 12490341.000
CL 12472337.002	LLR 12490344.000
CL 12472337.002	LLR 12490346.000
CL 12472337.002	LLR 12490351.000
CL 12472337.002	LLR 12490361.000
CL 12472339.000	LLR 12490321.000
CL 12472341.000	LLR 12490343.000
CL 12472341.000	LLR 12490345.000
CL 12472343.000	LLR 12490345.000
CL 12472345.000	LLR 12490321.000
CL 12472347.000	
CL 12472348.000	LLR 12490362.000
CL 12472350.000	
CL 12472360.000	
CL 12472361.000	LLR 12490402.000
CL 12472362.000	LLR 12490367.000
CL 12472366.000	LLR 12490393.000
CL 12472368.000	LLR 12490364.000
CL 12472368.000	LLR 12490367.000
CL 12472368.000	LLR 12490371.000
CL 12472368.000	LLR 12490373.000
CL 12472368.000	LLR 12490376.000
CL 12472368.000	LLR 12490397.000
CL 12472368.000	LLR 12490404.000
CL 12472368.000	LLR 12490406.000
CL 12472369.000	LLR 12490378.000
CL 12472369.000	LLR 12490380.000
CL 12472369.000	LLR 12490382.000
CL 12472369.000	LLR 12490384.000
CL 12472369.000	LLR 12490387.000
CL 12472369.000	LLR 12490389.000
CL 12472369.000	LLR 12490391.000

Continued on next page

Continued from previous page

CL 12472369.000	LLR 12490392.000
CL 12472369.000	LLR 12490394.000
CL 12472369.000	LLR 12490405.000
CL 12472370.000	LLR 12490378.000
CL 12472370.000	LLR 12490380.000
CL 12472370.000	LLR 12490382.000
CL 12472370.000	LLR 12490384.000
CL 12472370.000	LLR 12490387.000
CL 12472370.000	LLR 12490389.000
CL 12472370.000	LLR 12490391.000
CL 12472370.000	LLR 12490393.000
CL 12472370.000	LLR 12490399.000
CL 12472371.000	LLR 12490367.000
CL 12472371.000	LLR 12490379.000
CL 12472371.000	LLR 12490381.000
CL 12472371.000	LLR 12490383.000
CL 12472371.000	LLR 12490386.000
CL 12472371.000	LLR 12490388.000
CL 12472371.000	LLR 12490390.000
CL 12472371.000	LLR 12490392.000
CL 12472371.000	LLR 12490394.000
CL 12472371.000	LLR 12490405.000
CL 12472372.000	LLR 12490378.000
CL 12472372.000	LLR 12490380.000
CL 12472372.000	LLR 12490382.000
CL 12472372.000	LLR 12490384.000
CL 12472372.000	LLR 12490387.000
CL 12472372.000	LLR 12490389.000
CL 12472372.000	LLR 12490391.000
CL 12472372.000	LLR 12490393.000
CL 12472372.000	LLR 12490401.000
CL 12472374.000	LLR 12490367.000
CL 12472383.000	
CL 12472386.000	
CL 12472388.000	
CL 12472390.000	
CL 12472392.000	
CL 12472404.000	LLR 12490065.001
CL 12472404.000	LLR 12490067.001
CL 12472405.000	
CL 12472407.000	
CL 12472409.000	
CL 12472411.000	

CL 12472369.000	LLR 12490393.000
CL 12472369.000	LLR 12490398.000
CL 12472370.000	LLR 12490367.000
CL 12472370.000	LLR 12490379.000
CL 12472370.000	LLR 12490381.000
CL 12472370.000	LLR 12490383.000
CL 12472370.000	LLR 12490386.000
CL 12472370.000	LLR 12490388.000
CL 12472370.000	LLR 12490390.000
CL 12472370.000	LLR 12490392.000
CL 12472370.000	LLR 12490394.000
CL 12472370.000	LLR 12490405.000
CL 12472371.000	LLR 12490378.000
CL 12472371.000	LLR 12490380.000
CL 12472371.000	LLR 12490382.000
CL 12472371.000	LLR 12490384.000
CL 12472371.000	LLR 12490387.000
CL 12472371.000	LLR 12490389.000
CL 12472371.000	LLR 12490391.000
CL 12472371.000	LLR 12490393.000
CL 12472371.000	LLR 12490400.000
CL 12472372.000	LLR 12490367.000
CL 12472372.000	LLR 12490379.000
CL 12472372.000	LLR 12490381.000
CL 12472372.000	LLR 12490383.000
CL 12472372.000	LLR 12490386.000
CL 12472372.000	LLR 12490388.000
CL 12472372.000	LLR 12490390.000
CL 12472372.000	LLR 12490392.000
CL 12472372.000	LLR 12490394.000
CL 12472372.000	LLR 12490405.000
CL 12472374.000	LLR 12490368.000
CL 12472385.000	LLR 12490068.001
CL 12472387.000	
CL 12472389.000	
CL 12472391.000	
CL 12472402.000	
CL 12472404.000	LLR 12490066.002
CL 12472404.000	LLR 12490068.001
CL 12472406.000	
CL 12472408.000	
CL 12472410.000	
CL 12472412.000	

Continued on next page

Continued from previous page

CL 12472413.000	
CL 12472425.000	LLR 12490066.002
CL 12472425.000	LLR 12490068.001
CL 12472427.000	
CL 12472429.000	
CL 12472431.000	
CL 12472433.000	
CL 12472446.000	
CL 12472447.000	LLR 12490066.002
CL 12472447.000	LLR 12490068.001
CL 12472451.000	
CL 12472453.000	LLR 12490450.000
CL 12472454.001	LLR 12490688.000
CL 12472459.000	LLR 12490429.001
CL 12472461.000	LLR 12490429.001
CL 12472463.000	LLR 12490429.001
CL 12472465.000	LLR 12490439.000
CL 12472465.000	LLR 12490442.000
CL 12472469.000	LLR 12490444.000
CL 12472482.000	
CL 12472485.000	
CL 12472487.000	LLR 12490455.000
CL 12472487.000	LLR 12490459.000
CL 12472489.000	LLR 12490461.000
CL 12472491.000	LLR 12490463.000
CL 12472493.000	LLR 12490465.000
CL 12472505.000	LLR 12490068.001
CL 12472507.000	LLR 12490478.000
CL 12472509.000	LLR 12490475.000
CL 12472511.000	LLR 12490483.000
CL 12472514.000	LLR 12490474.000
CL 12472516.000	LLR 12490468.000
CL 12472516.000	LLR 12490470.000
CL 12472516.000	LLR 12490472.000
CL 12472516.000	LLR 12490474.000
CL 12472516.000	LLR 12490483.000
CL 12472516.000	LLR 12490485.000
CL 12472516.000	LLR 12490487.000
CL 12472516.000	LLR 12490489.000
CL 12472528.000	LLR 12490067.001
CL 12472531.000	
CL 12472534.000	
CL 12472537.000	

CL 12472425.000	LLR 12490065.001
CL 12472425.000	LLR 12490067.001
CL 12472426.000	
CL 12472428.000	
CL 12472430.000	
CL 12472432.000	
CL 12472434.000	
CL 12472447.000	LLR 12490065.001
CL 12472447.000	LLR 12490067.001
CL 12472449.002	LLR 12490510.000
CL 12472452.000	LLR 12490449.000
CL 12472454.001	LLR 12490604.000
CL 12472458.000	LLR 12490429.001
CL 12472460.000	LLR 12490429.001
CL 12472462.000	LLR 12490429.001
CL 12472464.000	LLR 12490438.000
CL 12472465.000	LLR 12490441.000
CL 12472466.000	LLR 12490443.000
CL 12472481.000	
CL 12472484.000	LLR 12490068.001
CL 12472487.000	LLR 12490454.000
CL 12472487.000	LLR 12490456.000
CL 12472488.000	LLR 12490460.000
CL 12472490.000	LLR 12490462.000
CL 12472492.000	LLR 12490464.000
CL 12472494.000	LLR 12490466.000
CL 12472506.001	LLR 12490478.000
CL 12472508.000	LLR 12490478.000
CL 12472510.000	LLR 12490483.000
CL 12472513.000	LLR 12490483.000
CL 12472515.000	LLR 12490474.000
CL 12472516.000	LLR 12490469.000
CL 12472516.000	LLR 12490471.000
CL 12472516.000	LLR 12490473.000
CL 12472516.000	LLR 12490475.000
CL 12472516.000	LLR 12490484.000
CL 12472516.000	LLR 12490486.000
CL 12472516.000	LLR 12490488.000
CL 12472517.000	LLR 12490476.000
CL 12472530.000	
CL 12472533.000	
CL 12472536.000	
CL 12472539.000	

Continued on next page

Continued from previous page

CL 12472540.000	
CL 12472543.000	
CL 12472552.000	LLR 12490066.002
CL 12472552.000	LLR 12490068.001
CL 12472553.000	LLR 12490567.001
CL 12472558.000	LLR 12490564.001
CL 12472564.000	LLR 12490564.001
CL 12472570.000	LLR 12490567.001
CL 12472575.001	LLR 12490565.000
CL 12472577.001	LLR 12490565.000
CL 12472579.001	LLR 12490565.000
CL 12472591.001	LLR 12490065.001
CL 12472592.000	LLR 12490677.000
CL 12472607.001	LLR 12490571.000
CL 12472609.001	LLR 12490571.000
CL 12472623.000	LLR 12490584.000
CL 12472627.001	LLR 12490574.001
CL 12472631.000	LLR 12490578.000
CL 12472644.000	LLR 12490317.001
CL 12472646.001	LLR 12490318.001
CL 12472648.001	LLR 12490317.001
CL 12472649.001	LLR 12490317.001
CL 12472650.001	LLR 12490318.001
CL 12472660.000	LLR 12490312.001
CL 12472663.001	LLR 12490312.001
CL 12472665.000	LLR 12490312.001
CL 12472674.000	LLR 12490065.001
CL 12472705.000	
CL 12472706.000	LLR 12490066.002
CL 12472707.000	LLR 12490077.000
CL 12472707.000	LLR 12490079.000
CL 12472707.000	LLR 12490081.000
CL 12472707.000	LLR 12490087.000
CL 12472707.000	LLR 12490089.000
CL 12472707.000	LLR 12490091.000
CL 12472711.001	LLR 12490133.000
CL 12472713.000	LLR 12490078.000
CL 12472713.000	LLR 12490080.000
CL 12472713.000	LLR 12490082.000
CL 12472713.000	LLR 12490088.000
CL 12472713.000	LLR 12490090.000
CL 12472713.000	LLR 12490092.000
CL 12472713.000	LLR 12490092.000
CL 12472720.001	LLR 12490135.001

CL 12472541.000	
CL 12472552.000	LLR 12490065.001
CL 12472552.000	LLR 12490067.001
CL 12472553.000	LLR 12490566.001
CL 12472555.000	LLR 12490564.001
CL 12472561.000	LLR 12490564.001
CL 12472567.000	LLR 12490566.001
CL 12472573.000	LLR 12490564.001
CL 12472576.001	LLR 12490565.000
CL 12472578.001	LLR 12490565.000
CL 12472580.001	LLR 12490565.000
CL 12472592.000	LLR 12490676.000
CL 12472605.000	LLR 12490066.002
CL 12472608.000	LLR 12490571.000
CL 12472614.001	LLR 12490571.000
CL 12472625.001	LLR 12490586.001
CL 12472629.001	LLR 12490576.001
CL 12472644.000	LLR 12490064.001
CL 12472644.000	LLR 12490318.001
CL 12472647.001	LLR 12490318.001
CL 12472648.001	LLR 12490318.001
CL 12472649.001	LLR 12490318.001
CL 12472660.000	LLR 12490064.001
CL 12472662.001	LLR 12490312.001
CL 12472664.000	LLR 12490312.001
CL 12472666.000	LLR 12490312.001
CL 12472675.002	LLR 12490065.001
CL 12472706.000	LLR 12490065.001
CL 12472706.000	LLR 12490067.001
CL 12472707.000	LLR 12490078.000
CL 12472707.000	LLR 12490080.000
CL 12472707.000	LLR 12490082.000
CL 12472707.000	LLR 12490088.000
CL 12472707.000	LLR 12490090.000
CL 12472707.000	LLR 12490092.000
CL 12472713.000	LLR 12490077.000
CL 12472713.000	LLR 12490079.000
CL 12472713.000	LLR 12490081.000
CL 12472713.000	LLR 12490087.000
CL 12472713.000	LLR 12490089.000
CL 12472713.000	LLR 12490091.000
CL 12472719.001	LLR 12490134.001
CL 12472721.001	LLR 12490136.002

Continued on next page

Continued from previous page

CL 12472722.001	LLR 12490123.001
CL 12472725.000	LLR 12490128.000
CL 12472727.000	LLR 12490126.000
CL 12472734.000	LLR 12490083.000
CL 12472740.000	LLR 12490083.000
CL 12472746.000	LLR 12490083.000
CL 12472767.000	
CL 12472780.000	
CL 12472786.000	LLR 12490042.001
CL 12472786.000	LLR 12490044.001
CL 12472787.000	LLR 12490059.000
CL 12472792.000	LLR 12490622.000
CL 12472796.000	LLR 12490294.000
CL 12472798.000	LLR 12490173.000
CL 12472800.000	LLR 12490173.000
CL 12472853.000	LLR 12490675.000
CL 12472876.000	LLR 12490678.000
CL 12472879.000	LLR 12490685.000
CL 12472881.000	LLR 12490682.000
CL 12472888.000	LLR 12490684.000
CL 12472890.000	LLR 12490683.000
CL 12472892.000	LLR 12490681.000
CL 12472894.000	LLR 12490620.000
CL 12472896.000	LLR 12490678.000
CL 12472898.000	LLR 12490620.000
CL 12472900.000	LLR 12490620.000
CL 12472902.000	LLR 12490621.000
CL 12472905.000	LLR 12490020.000
CL 12472910.000	LLR 12490624.000
CL 12472928.000	LLR 12490624.000
CL 12472929.000	LLR 12490611.000
CL 12472930.000	LLR 12490610.000
CL 12472931.000	LLR 12490609.000
CL 12472933.000	LLR 12490167.000
CL 12472937.000	LLR 12490167.000
CL 12472941.000	LLR 12490167.000
CL 12472943.000	LLR 12490607.000
CL 12472944.000	LLR 12490607.000
CL 12472945.000	LLR 12490607.000
CL 12472946.000	LLR 12490607.000
CL 12472948.000	LLR 12490423.001
CL 12472949.000	LLR 12490421.001
CL 12472950.000	LLR 12490410.001

CL 12472723.001	LLR 12490124.001
CL 12472726.000	LLR 12490125.000
CL 12472728.000	LLR 12490127.000
CL 12472737.000	LLR 12490083.000
CL 12472743.000	LLR 12490083.000
CL 12472749.000	LLR 12490083.000
CL 12472779.000	
CL 12472786.000	LLR 12490040.001
CL 12472786.000	LLR 12490043.001
CL 12472786.000	LLR 12490058.000
CL 12472791.000	LLR 12490622.000
CL 12472796.000	LLR 12490293.000
CL 12472797.000	LLR 12490173.000
CL 12472799.000	LLR 12490173.000
CL 12472843.000	LLR 12490675.000
CL 12472865.000	LLR 12490675.000
CL 12472878.000	LLR 12490684.000
CL 12472880.000	LLR 12490683.000
CL 12472882.000	LLR 12490681.000
CL 12472889.000	LLR 12490685.000
CL 12472891.000	LLR 12490682.000
CL 12472893.000	LLR 12490156.001
CL 12472895.000	LLR 12490678.000
CL 12472897.000	LLR 12490620.000
CL 12472899.000	LLR 12490620.000
CL 12472901.000	LLR 12490620.000
CL 12472903.000	LLR 12490624.000
CL 12472909.000	LLR 12490624.000
CL 12472914.000	LLR 12490624.000
CL 12472929.000	LLR 12490607.000
CL 12472930.000	LLR 12490607.000
CL 12472931.000	LLR 12490607.000
CL 12472932.000	LLR 12490167.000
CL 12472934.000	LLR 12490167.000
CL 12472940.000	LLR 12490167.000
CL 12472942.000	LLR 12490167.000
CL 12472943.000	LLR 12490614.000
CL 12472944.000	LLR 12490615.000
CL 12472945.000	LLR 12490613.000
CL 12472946.000	LLR 12490612.000
CL 12472948.000	LLR 12490604.000
CL 12472949.000	LLR 12490604.000
CL 12472950.000	LLR 12490604.000

Continued on next page

Continued from previous page

CL 12472951.000	LLR 12490408.001
CL 12472952.000	LLR 12490413.001
CL 12472953.000	LLR 12490604.000
CL 12472954.000	LLR 12490415.001
CL 12472955.000	LLR 12490152.001
CL 12472957.000	LLR 12490159.001
CL 12472959.000	LLR 12490157.001
CL 12472962.000	LLR 12490686.000
CL 12472964.000	LLR 12490686.000
CL 12472973.000	LLR 12490596.000
CL 12472975.000	LLR 12490318.001
CL 12473000.000	LLR 12490619.000
CL 12473002.000	LLR 12490619.000
CL 12473004.000	LLR 12490173.000
CL 12473006.000	LLR 12490173.000
CL 12473008.000	LLR 12490678.000
CL 12473012.000	LLR 12490595.000
CL 12473015.000	LLR 12490616.000
CL 12473021.000	LLR 12490597.000
CL 12473023.000	LLR 12490597.000
CL 12473029.000	LLR 12490006.000
CL 12473032.000	LLR 12490624.000
CL 12473034.000	LLR 12490624.000
CL 12473036.000	LLR 12490624.000
CL 12473038.000	LLR 12490624.000
CL 12473040.000	LLR 12490624.000
CL 12473042.000	LLR 12490624.000
CL 12473044.000	LLR 12490624.000
CL 12473047.000	LLR 12490167.000
CL 12473050.000	LLR 12490621.000
CL 12473052.000	LLR 12490156.001
CL 12473052.000	LLR 12490158.001
CL 12473052.000	LLR 12490160.001
CL 12473053.000	LLR 12490156.001
CL 12473053.000	LLR 12490158.001
CL 12473053.000	LLR 12490160.001
CL 12473054.000	LLR 12490156.001
CL 12473054.000	LLR 12490158.001
CL 12473054.000	LLR 12490160.001
CL 12473055.000	LLR 12490156.001
CL 12473055.000	LLR 12490158.001
CL 12473055.000	LLR 12490160.001
CL 12473057.000	LLR 12490628.000

CL 12472951.000	LLR 12490604.000
CL 12472952.000	LLR 12490604.000
CL 12472953.000	LLR 12490689.000
CL 12472954.000	LLR 12490604.000
CL 12472956.000	LLR 12490160.001
CL 12472958.000	LLR 12490158.001
CL 12472961.000	LLR 12490686.000
CL 12472963.000	LLR 12490686.000
CL 12472965.000	LLR 12490686.000
CL 12472975.000	LLR 12490317.001
CL 12472999.000	LLR 12490619.000
CL 12473001.000	LLR 12490619.000
CL 12473003.000	LLR 12490173.000
CL 12473005.000	LLR 12490173.000
CL 12473007.000	LLR 12490678.000
CL 12473009.000	LLR 12490680.000
CL 12473014.000	LLR 12490616.000
CL 12473020.000	LLR 12490597.000
CL 12473022.000	LLR 12490597.000
CL 12473024.000	LLR 12490597.000
CL 12473031.000	LLR 12490624.000
CL 12473033.000	LLR 12490624.000
CL 12473035.000	LLR 12490624.000
CL 12473037.000	LLR 12490624.000
CL 12473039.000	LLR 12490624.000
CL 12473041.000	LLR 12490624.000
CL 12473043.000	LLR 12490624.000
CL 12473046.000	LLR 12490624.000
CL 12473049.000	LLR 12490173.000
CL 12473052.000	LLR 12490152.001
CL 12473052.000	LLR 12490157.001
CL 12473052.000	LLR 12490159.001
CL 12473053.000	LLR 12490152.001
CL 12473053.000	LLR 12490157.001
CL 12473053.000	LLR 12490159.001
CL 12473054.000	LLR 12490152.001
CL 12473054.000	LLR 12490157.001
CL 12473054.000	LLR 12490159.001
CL 12473055.000	LLR 12490152.001
CL 12473055.000	LLR 12490157.001
CL 12473055.000	LLR 12490159.001
CL 12473056.000	LLR 12490627.000
CL 12473058.000	LLR 12490629.000

Continued on next page

Continued from previous page

CL 12473059.000	LLR 12490630.000
CL 12473062.000	LLR 12490630.000
CL 12473065.000	LLR 12490619.000
CL 12473071.000	LLR 12490686.000
CL 12473073.000	LLR 12490173.000
CL 12473075.000	

CL 12473060.000	LLR 12490630.000
CL 12473063.000	LLR 12490630.000
CL 12473070.000	LLR 12490686.000
CL 12473072.000	LLR 12490173.000
CL 12473074.000	LLR 12490173.000

18.00 DERIVED REQUIREMENTS JUSTIFICATION

LLR	JUSTIFICATION
LLR 12490413.001	ACCSTRTS_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.
LLR 12490410.001	ACCNUMFL_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.
LLR 12490689.000	ACCSHUTD_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.
LLR 12490415.001	ACCSTRTSFLT_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.
LLR 12490609.000	ACCSTRTS_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

Continued on next page

Continued from previous page

LLR	JUSTIFICATION
LLR 12490611.000	ACCCNUMFL_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.
LLR 12490615.000	ACCSHUTD_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.
LLR 12490614.000	ACCSTRTSFLT_ALL is implemented in software as an unsigned 16 bit number, consequently negatives values are not possible and do not require a lower bound range check. Due to the limitation of signed 16 bit hardware, unsigned 16 bits numbers are evaluated in parts by considering the sixteenth bit and bits fifteen to zero separately. Consequently, when performing the range check, the sixteenth bit is checked to see if its set, and bits fifteen to zero are evaluated to see if they are greater than 32232.

19.00 PARTIALLY DERIVED REQUIREMENTS JUSTIFICATION

LLR	JUSTIFICATION
-----	---------------

20.00 Requirements Allocation Matrix

Requirement	Allocation
LLR 12490004.000	Collins IP
LLR 12490006.000	Collins IP, PWC IP
LLR 12490011.001	Collins IP, PWC IP
LLR 12490020.000	Collins IP
LLR 12490027.000	Collins IP
LLR 12490029.000	Collins IP
LLR 12490031.000	Collins IP
LLR 12490032.000	Collins IP
LLR 12490033.000	Collins IP
LLR 12490034.000	Collins IP
LLR 12490035.000	Collins IP, PWC IP
LLR 12490036.000	Collins IP, PWC IP
LLR 12490037.000	Collins IP
LLR 12490038.000	Collins IP
LLR 12490040.001	Collins IP, PWC IP
LLR 12490042.001	Collins IP, PWC IP
LLR 12490043.001	Collins IP, PWC IP
LLR 12490044.001	Collins IP, PWC IP
LLR 12490047.000	Collins IP, PWC IP
LLR 12490048.000	Collins IP
LLR 12490049.000	Collins IP
LLR 12490050.000	Collins IP
LLR 12490051.000	Collins IP
LLR 12490053.000	Collins IP
LLR 12490054.000	Collins IP
LLR 12490055.000	Collins IP
LLR 12490056.000	Collins IP
LLR 12490058.000	Collins IP
LLR 12490059.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490061.000	Collins IP
LLR 12490064.001	Collins IP, PWC IP
LLR 12490065.001	Collins IP, PWC IP
LLR 12490066.002	Collins IP, PWC IP
LLR 12490067.001	Collins IP, PWC IP
LLR 12490068.001	Collins IP, PWC IP
LLR 12490071.000	Collins IP
LLR 12490072.000	Collins IP
LLR 12490073.000	Collins IP
LLR 12490074.000	Collins IP
LLR 12490075.000	Collins IP
LLR 12490077.000	Collins IP
LLR 12490078.000	Collins IP
LLR 12490079.000	Collins IP
LLR 12490080.000	Collins IP
LLR 12490081.000	Collins IP
LLR 12490082.000	Collins IP
LLR 12490083.000	Collins IP
LLR 12490087.000	Collins IP
LLR 12490088.000	Collins IP
LLR 12490089.000	Collins IP
LLR 12490090.000	Collins IP
LLR 12490091.000	Collins IP
LLR 12490092.000	Collins IP
LLR 12490094.000	Collins IP
LLR 12490095.000	Collins IP
LLR 12490096.000	Collins IP
LLR 12490097.000	Collins IP
LLR 12490102.000	Collins IP
LLR 12490103.000	Collins IP
LLR 12490104.000	Collins IP
LLR 12490105.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490106.000	Collins IP
LLR 12490107.000	Collins IP
LLR 12490111.000	Collins IP
LLR 12490112.000	Collins IP
LLR 12490113.000	Collins IP
LLR 12490114.000	Collins IP
LLR 12490121.000	Collins IP
LLR 12490123.001	PWC IP
LLR 12490124.001	Collins IP, PWC IP
LLR 12490125.000	Collins IP
LLR 12490126.000	Collins IP
LLR 12490127.000	Collins IP
LLR 12490128.000	Collins IP
LLR 12490133.000	Collins IP
LLR 12490134.001	Collins IP, PWC IP
LLR 12490135.001	Collins IP, PWC IP
LLR 12490136.002	Collins IP, PWC IP
LLR 12490141.000	Collins IP
LLR 12490148.000	Collins IP
LLR 12490152.001	Collins IP, PWC IP
LLR 12490156.001	Collins IP, PWC IP
LLR 12490157.001	Collins IP, PWC IP
LLR 12490158.001	Collins IP, PWC IP
LLR 12490159.001	Collins IP, PWC IP
LLR 12490160.001	Collins IP, PWC IP
LLR 12490167.000	Collins IP
LLR 12490173.000	Collins IP, PWC IP
LLR 12490177.000	Collins IP
LLR 12490181.000	Collins IP
LLR 12490182.000	Collins IP
LLR 12490183.000	Collins IP
LLR 12490184.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490185.000	Collins IP
LLR 12490186.000	Collins IP
LLR 12490190.000	Collins IP
LLR 12490192.000	Collins IP
LLR 12490197.000	Collins IP
LLR 12490199.000	Collins IP
LLR 12490201.001	Collins IP, PWC IP
LLR 12490204.000	Collins IP
LLR 12490205.000	Collins IP
LLR 12490206.000	Collins IP
LLR 12490209.000	Collins IP
LLR 12490210.000	Collins IP
LLR 12490211.000	Collins IP
LLR 12490212.000	Collins IP
LLR 12490214.001	Collins IP, PWC IP
LLR 12490218.000	Collins IP
LLR 12490219.000	Collins IP
LLR 12490220.000	Collins IP
LLR 12490223.001	Collins IP, PWC IP
LLR 12490224.000	Collins IP
LLR 12490225.000	Collins IP
LLR 12490226.001	Collins IP, PWC IP
LLR 12490227.001	Collins IP, PWC IP
LLR 12490228.001	Collins IP, PWC IP
LLR 12490230.001	Collins IP, PWC IP
LLR 12490233.001	Collins IP, PWC IP
LLR 12490234.000	Collins IP
LLR 12490235.000	Collins IP
LLR 12490238.000	Collins IP
LLR 12490239.000	Collins IP
LLR 12490240.000	Collins IP
LLR 12490241.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490243.001	Collins IP, PWC IP
LLR 12490245.000	Collins IP
LLR 12490246.000	Collins IP
LLR 12490247.000	Collins IP
LLR 12490250.000	Collins IP
LLR 12490251.000	Collins IP
LLR 12490252.000	Collins IP
LLR 12490253.000	Collins IP
LLR 12490255.000	Collins IP
LLR 12490257.000	Collins IP
LLR 12490258.000	Collins IP
LLR 12490259.000	Collins IP
LLR 12490262.000	Collins IP
LLR 12490263.000	Collins IP
LLR 12490264.000	Collins IP
LLR 12490265.000	Collins IP
LLR 12490267.000	Collins IP
LLR 12490269.000	Collins IP
LLR 12490270.000	Collins IP
LLR 12490271.000	Collins IP
LLR 12490274.000	Collins IP
LLR 12490275.000	Collins IP
LLR 12490276.000	Collins IP
LLR 12490277.000	Collins IP
LLR 12490279.000	Collins IP
LLR 12490283.000	Collins IP
LLR 12490284.000	Collins IP
LLR 12490287.000	Collins IP
LLR 12490288.000	Collins IP
LLR 12490289.000	Collins IP
LLR 12490293.000	Collins IP
LLR 12490294.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490295.000	Collins IP
LLR 12490296.000	Collins IP
LLR 12490297.000	Collins IP
LLR 12490298.000	Collins IP
LLR 12490303.001	Collins IP, PWC IP
LLR 12490312.001	Collins IP
LLR 12490317.001	Collins IP, PWC IP
LLR 12490318.001	Collins IP, PWC IP
LLR 12490321.000	Collins IP
LLR 12490323.000	Collins IP
LLR 12490324.000	Collins IP
LLR 12490325.000	Collins IP
LLR 12490326.000	Collins IP
LLR 12490327.000	Collins IP
LLR 12490329.000	Collins IP
LLR 12490333.000	Collins IP
LLR 12490334.000	Collins IP
LLR 12490335.000	Collins IP
LLR 12490336.000	Collins IP
LLR 12490337.000	Collins IP
LLR 12490338.000	Collins IP
LLR 12490339.000	Collins IP
LLR 12490340.000	Collins IP
LLR 12490341.000	Collins IP
LLR 12490343.000	Collins IP
LLR 12490344.000	Collins IP
LLR 12490345.000	Collins IP
LLR 12490346.000	Collins IP
LLR 12490348.000	Collins IP
LLR 12490350.000	Collins IP
LLR 12490351.000	Collins IP
LLR 12490352.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490354.000	Collins IP
LLR 12490355.000	Collins IP
LLR 12490356.000	Collins IP
LLR 12490357.000	Collins IP
LLR 12490358.000	Collins IP
LLR 12490359.000	Collins IP
LLR 12490361.000	Collins IP
LLR 12490362.000	Collins IP
LLR 12490364.000	Collins IP
LLR 12490365.000	Collins IP
LLR 12490366.000	Collins IP
LLR 12490367.000	Collins IP
LLR 12490368.000	Collins IP
LLR 12490370.000	Collins IP
LLR 12490371.000	Collins IP
LLR 12490372.000	Collins IP
LLR 12490373.000	Collins IP
LLR 12490375.000	Collins IP
LLR 12490376.000	Collins IP
LLR 12490378.000	Collins IP
LLR 12490379.000	Collins IP
LLR 12490380.000	Collins IP
LLR 12490381.000	Collins IP
LLR 12490382.000	Collins IP
LLR 12490383.000	Collins IP
LLR 12490384.000	Collins IP
LLR 12490386.000	Collins IP
LLR 12490387.000	Collins IP
LLR 12490388.000	Collins IP
LLR 12490389.000	Collins IP
LLR 12490390.000	Collins IP
LLR 12490391.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490392.000	Collins IP
LLR 12490393.000	Collins IP
LLR 12490394.000	Collins IP
LLR 12490397.000	Collins IP
LLR 12490398.000	Collins IP
LLR 12490399.000	Collins IP
LLR 12490400.000	Collins IP
LLR 12490401.000	Collins IP
LLR 12490402.000	Collins IP
LLR 12490403.000	Collins IP
LLR 12490404.000	Collins IP
LLR 12490405.000	Collins IP
LLR 12490406.000	Collins IP
LLR 12490408.001	Collins IP, PWC IP
LLR 12490410.001	Collins IP, PWC IP
LLR 12490413.001	Collins IP, PWC IP
LLR 12490415.001	Collins IP, PWC IP
LLR 12490421.001	Collins IP, PWC IP
LLR 12490423.001	Collins IP, PWC IP
LLR 12490429.001	Collins IP, PWC IP
LLR 12490438.000	Collins IP
LLR 12490439.000	Collins IP
LLR 12490441.000	Collins IP
LLR 12490442.000	Collins IP
LLR 12490443.000	Collins IP
LLR 12490444.000	Collins IP
LLR 12490445.000	Collins IP
LLR 12490449.000	Collins IP
LLR 12490450.000	Collins IP
LLR 12490454.000	Collins IP
LLR 12490455.000	Collins IP
LLR 12490456.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490459.000	Collins IP
LLR 12490460.000	Collins IP
LLR 12490461.000	Collins IP
LLR 12490462.000	Collins IP
LLR 12490463.000	Collins IP
LLR 12490464.000	Collins IP
LLR 12490465.000	Collins IP
LLR 12490466.000	Collins IP
LLR 12490468.000	Collins IP
LLR 12490469.000	Collins IP
LLR 12490470.000	Collins IP
LLR 12490471.000	Collins IP
LLR 12490472.000	Collins IP
LLR 12490473.000	Collins IP
LLR 12490474.000	Collins IP
LLR 12490475.000	Collins IP
LLR 12490476.000	Collins IP
LLR 12490478.000	Collins IP
LLR 12490483.000	Collins IP
LLR 12490484.000	Collins IP
LLR 12490485.000	Collins IP
LLR 12490486.000	Collins IP
LLR 12490487.000	Collins IP
LLR 12490488.000	Collins IP
LLR 12490489.000	Collins IP
LLR 12490493.000	Collins IP
LLR 12490495.000	Collins IP
LLR 12490501.000	Collins IP, PWC IP
LLR 12490503.000	Collins IP
LLR 12490505.000	Collins IP
LLR 12490506.000	Collins IP
LLR 12490510.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490512.000	Collins IP
LLR 12490513.000	Collins IP
LLR 12490514.000	Collins IP
LLR 12490516.000	Collins IP
LLR 12490517.000	Collins IP
LLR 12490519.000	Collins IP
LLR 12490521.000	Collins IP
LLR 12490523.000	Collins IP
LLR 12490524.000	Collins IP
LLR 12490527.000	Collins IP
LLR 12490528.000	Collins IP
LLR 12490529.000	Collins IP
LLR 12490530.000	Collins IP
LLR 12490532.000	Collins IP
LLR 12490533.000	Collins IP
LLR 12490534.000	Collins IP
LLR 12490535.000	Collins IP
LLR 12490538.000	Collins IP
LLR 12490541.000	Collins IP
LLR 12490544.000	Collins IP
LLR 12490545.000	Collins IP
LLR 12490546.000	Collins IP
LLR 12490547.000	Collins IP
LLR 12490548.000	Collins IP
LLR 12490549.000	Collins IP
LLR 12490550.000	Collins IP
LLR 12490551.000	Collins IP
LLR 12490554.000	Collins IP
LLR 12490555.000	Collins IP
LLR 12490556.000	Collins IP
LLR 12490558.000	Collins IP
LLR 12490559.000	Collins IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490564.001	Collins IP, PWC IP
LLR 12490565.000	Collins IP
LLR 12490566.001	Collins IP, PWC IP
LLR 12490567.001	Collins IP, PWC IP
LLR 12490571.000	Collins IP
LLR 12490574.001	Collins IP, PWC IP
LLR 12490576.001	Collins IP, PWC IP
LLR 12490578.000	Collins IP
LLR 12490584.000	Collins IP
LLR 12490586.001	Collins IP, PWC IP
LLR 12490595.000	PWC IP
LLR 12490596.000	PWC IP
LLR 12490597.000	PWC IP
LLR 12490604.000	PWC IP
LLR 12490607.000	PWC IP
LLR 12490608.000	PWC IP
LLR 12490609.000	PWC IP
LLR 12490610.000	PWC IP
LLR 12490611.000	PWC IP
LLR 12490612.000	PWC IP
LLR 12490613.000	PWC IP
LLR 12490614.000	PWC IP
LLR 12490615.000	PWC IP
LLR 12490616.000	PWC IP
LLR 12490619.000	PWC IP
LLR 12490620.000	PWC IP
LLR 12490621.000	PWC IP
LLR 12490622.000	PWC IP
LLR 12490623.000	PWC IP
LLR 12490624.000	PWC IP
LLR 12490625.000	PWC IP
LLR 12490626.000	PWC IP

Continued on next page

Continued from previous page

Requirement	Allocation
LLR 12490627.000	PWC IP
LLR 12490628.000	PWC IP
LLR 12490629.000	PWC IP
LLR 12490630.000	PWC IP
LLR 12490635.000	PWC IP
LLR 12490640.000	PWC IP
LLR 12490645.000	PWC IP
LLR 12490650.000	PWC IP
LLR 12490655.000	PWC IP
LLR 12490660.000	PWC IP
LLR 12490665.000	PWC IP
LLR 12490670.000	PWC IP
LLR 12490675.000	PWC IP
LLR 12490676.000	PWC IP
LLR 12490677.000	PWC IP
LLR 12490678.000	PWC IP
LLR 12490680.000	PWC IP
LLR 12490681.000	PWC IP
LLR 12490682.000	PWC IP
LLR 12490683.000	PWC IP
LLR 12490684.000	PWC IP
LLR 12490685.000	PWC IP
LLR 12490686.000	PWC IP
LLR 12490688.000	PWC IP
LLR 12490689.000	PWC IP

[illegible]

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i

Cage Code 00198

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i.

Cage Code 00198

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i.

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i.

Cage Code 00198

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i

Cage Code 00198

Export Classification: US: EAR 9E991, Outside US: EAR 9E991, Canadian EIPA: NSR, DPA:NO
THIS DOCUMENT IS SUBJECT TO THE RESTRICTIONS CONTAINED ON PAGE i

[illegible]